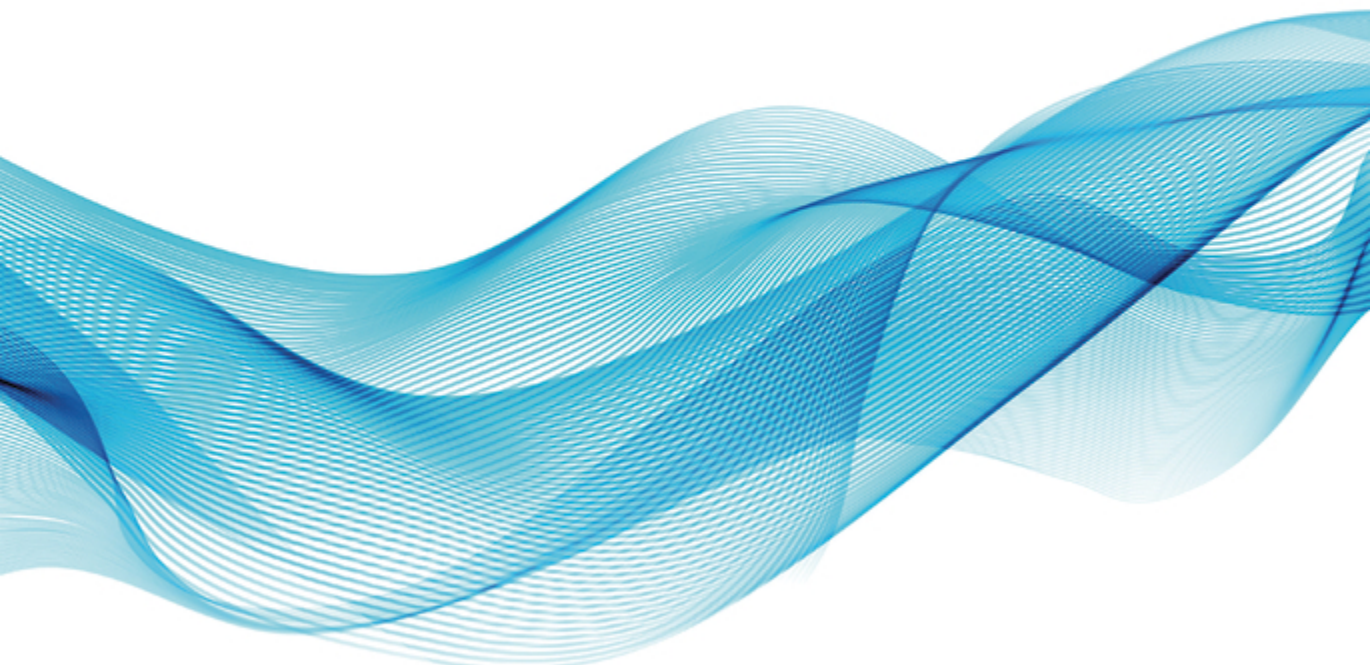
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time series and panel data econometrics



M. HASHEM PESARAN

TIME SERIES AND PANEL DATA ECONOMETRICS

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To my wife and in memory of my parents.

Preface

This book is concerned with recent developments in time series and panel data techniques for the analysis of macroeconomic and financial data. It provides a rigorous, nevertheless user-friendly, account of the time series techniques dealing with univariate and multivariate time series models, as well as panel data models. An overview of econometrics as a subject is provided in Pesaran (1987a) and updated in Geweke, Horowitz, and Pesaran (2008).

It is distinct from other time series texts in the sense that it also covers panel data models and attempts at a more coherent integration of time series, multivariate analysis, and panel data models. It builds on the author's extensive research in the areas of time series and panel data analysis and covers a wide variety of topics in one volume. Different parts of the book can be used as teaching material for a variety of courses in econometrics. It can also be used as a reference manual.

It begins with an overview of basic econometric and statistical techniques and provides an account of stochastic processes, univariate and multivariate time series, tests for unit roots, cointegration, impulse response analysis, autoregressive conditional heteroskedasticity models, simultaneous equation models, vector autoregressions, causality, forecasting, multivariate volatility models, panel data models, aggregation and global vector autoregressive models (*GVAR*). The techniques are illustrated using Microfit 5 (Pesaran and Pesaran (2009)) with applications to real output, inflation, interest rates, exchange rates, and stock prices.

The book assumes that the reader has done an introductory econometrics course. It begins with an overview of the basic regression model, which is intended to be accessible to advanced undergraduates, and then deals with more advanced topics which are more demanding and suited to graduate students and other interested scholars.

The book is organized into six parts:

Part I: Chapters 1 to 7 present the classical linear regression model, describe estimation and statistical inference, and discuss the violation of the assumptions underlying the classical linear regression model. This part also includes an introduction to dynamic economic modelling, and ends with a chapter on predictability of asset returns.

Part II: Chapters 8 to 11 deal with asymptotic theory and present the maximum likelihood and generalized method of moments estimation frameworks.

Part III: Chapters 12 and 13 provide an introduction to stochastic processes and spectral density analysis.

Part IV: Chapters 14 to 18 focus on univariate time series models and cover stationary *ARMA* models, unit root processes, trend and cycle decomposition, forecasting and univariate volatility models.

Part V: Chapters 19 to 25 consider a variety of reduced form and structural multivariate models, rational expectations models, as well as VARs, vector error corrections, cointegrating *VAR*s, *VARX* models, impulse response analysis, and multivariate volatility models.

Part VI: Chapters 26 to 33 considers panel data models both when the time dimension (T) of the panels is short, as well as when panels with N (the cross-section dimension) and T are large. These chapters cover a wide range of panel data models, starting with static panels with homogenous slopes and graduating to dynamic panels with slope heterogeneity, error cross-section dependence, unit roots, and cointegration.

There are also chapters dealing with the aggregation of large dynamic panels and the theory and practice of *GVAR* modelling. This part of the book focuses more on large N and T panels which are less covered in other texts, and draws heavily on my research in this area over the past 20 years starting with Pesaran and Smith (1995).

Appendices A and B present background material on matrix algebra, probability and distribution theory, and Appendix C provides an overview of Bayesian analysis.

This book has evolved over many years of teaching and research and brings together in one place a diverse set of research areas that have interested me. It is hoped that it will also be of interest to others. I have used some of the chapters in my teaching of postgraduate students at Cambridge University, University of Southern California, UCLA, and University of Pennsylvania. Undergraduate students at Cambridge University have also been exposed to some of the introductory material in Part I of the book. It is impossible to name all those who have helped me with the preparation of this volume. But I would like particularly to name two of my Cambridge Ph.D. students, Alexander Chudik and Elisa Tosetti, for their extensive help, particularly with the material in Part VI of the book.

The book draws heavily from my published and unpublished research. In particular:

Chapter 7 is based on Pesaran (2010).

Chapter 25 draws from Pesaran and Pesaran (2010).

Chapter 32 is based on Pesaran (2003) and Pesaran and Chudik (2014) where additional technical details and proofs are provided.

Chapter 31 is based on Breitung and Pesaran (2008) and provides some updates and extensions.

Chapter 33 is based on Chudik and Pesaran (2015b).

I would also like to acknowledge all my coauthors whose work has been reviewed in this volume. In particular, I would like to acknowledge Ron Smith, Bahram Pesaran, Allan Timmermann, Kevin Lee, Yongcheol Shin, Vanessa Smith, Cheng Hsiao, Michael Binder, Richard Smith, Alexander Chudik, Takashi Yamagata, Tony Garratt, Til Schermann, Filippo di Mauro, Stéphane Dées, Alessandro Rebucci, Adrian Pagan, Aman Ullah, and Martin Weale. It goes without saying that none of them is responsible for the material presented in this volume.

Finally, I would like to acknowledge the helpful and constructive comments and suggestions from two anonymous referees which provided me with further impetus to extend the coverage of the material included in the book and to improve its exposition over the past six months. Ron Smith has also provided me with detailed comments and suggestions over a number of successive drafts. I am indebted to him for helping me to see the wood from the trees over the many years that we have collaborated with each other.

Hashem Pesaran

Cambridge and Los Angeles
January 2015

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Part I

Introduction to Econometrics

1 Relationship Between Two Variables

1.1 Introduction

There are a number of ways that a regression between two or more variables can be motivated. It can, for example, arise because we know *a priori* that there exists an exact linear relationship between Y and X, with Y being observed with measurement errors. Alternatively, it could arise if (Y, X) have a bivariate distribution and we are interested in the conditional expectations of Y given X, namely $E(Y \mid X)$, which will be a linear function of X either if the underlying relationship between Y and X is linear, or if Y and X have a bivariate normal distribution. A regression line can also be considered without any underlying statistical model, just as a method of fitting a line to a scatter of points in a two-dimensional space.

1.2 The curve fitting approach

We first consider the problem of regression purely as an act of fitting a line to a scatter diagram. Suppose that T pairs of observations on the variables Y and X, given by (y_1, x_1), (y_2, x_2), ..., (y_T, x_T), are available. We are interested in obtaining the equation of a straight line such that, for each observation x_t, the corresponding value of Y on a straight line in the (Y, X) plane is as 'close' as possible to the observed values y_t.

Immediately, different criteria of 'closeness' or 'fit' present themselves. Two basic issues are involved:

A: How to define and measure the *distance* of the points in the scatter diagram from the fitted line. There are three plausible ways to measure the distance of a point from the fitted line:

(i) perpendicular to x-axis
(ii) perpendicular to y-axis
(iii) perpendicular to the fitted line.

B: How to *add up* all such distances of the sampled observations. Possible weighting (adding-up) schemes are:

(i) simple average of the square of distances
(ii) simple average of the absolute value of distances
(iii) weighted averages either of squared distance measure or absolute distance measures.

The simplest is the combination **A**(i) and **B**(i), which gives the ordinary least squares (*OLS*) estimates of the regression of Y on X. The method of ordinary least squares will be extensively treated in the rest of this Chapter and in Chapter 2. The difference between **A**(i) and **A**(ii) can also be characterized as to which of the two variables, X or Y, is represented on the horizontal axis. The combination **A**(ii) and **B**(i) is also referred to as the 'reverse regression of Y on X'. Other combinations of distance/weighting schemes can also be considered. For example **A**(iii) and **B**(i) is called orthogonal regression, **A**(i) and **B**(ii) yields the absolute minimum distance regression. **A**(i) and **B**(iii) gives the weighted (or absolute distance) least squares (or absolute distance) regression.

1.3 The method of ordinary least squares

Treating X as the regressor and Y as the regressand, then choosing the distance measure, $d_t = \left|y_t - \alpha - \beta x_t\right|$, the *least squares criterion* function to be minimized is[1]

$$Q(\alpha, \beta) = \sum_{t=1}^{T} d_t^2 = \sum_{t=1}^{T} \left(y_t - \alpha - \beta x_t\right)^2.$$

The necessary conditions for this minimization problem are given by

$$\frac{\partial Q(\alpha, \beta)}{\partial \alpha} = \sum_{t=1}^{T} (-2)\left(y_t - \hat{\alpha} - \hat{\beta} x_t\right) = 0, \tag{1.1}$$

$$\frac{\partial Q(\alpha, \beta)}{\partial \beta} = \sum_{t=1}^{T} (-2x_t)\left(y_t - \hat{\alpha} - \hat{\beta} x_t\right) = 0. \tag{1.2}$$

Equations (1.1) and (1.1) are called *normal equations* for the *OLS* problem and can be written as

$$\sum_{t=1}^{T} \hat{u}_t = 0, \tag{1.3}$$

$$\sum_{t=1}^{T} \hat{u}_t x_t = 0, \tag{1.4}$$

[1] The notations $\sum_{t=1}^{T}$ and $\sum_t$ are used later to denote the sum of the terms after the summation sign over $t = 1, 2, \ldots, T$.

where

$$\hat{u}_t = y_t - \hat{\alpha} - \hat{\beta}x_t, \tag{1.5}$$

are the *OLS* residuals. The condition $\sum_{t=1}^{T} \hat{u}_t = 0$ also gives $\bar{y} = \hat{\alpha} + \hat{\beta}\bar{x}$, where $\bar{x} = \sum_{t=1}^{T} x_t/T$ and $\bar{y} = \sum_{t=1}^{T} y_t/T$, and demonstrates that the least squares regression line $\hat{y}_t = \hat{\alpha} + \hat{\beta}x_t$, goes through the sample means of Y and X. Solving (1.3) and (1.4) for $\hat{\beta}$, and hence for $\hat{\alpha}$, we have

$$\hat{\beta} = \frac{\sum_{t=1}^{T} x_t y_t - T\bar{x}\bar{y}}{\sum_{t=1}^{T} x_t^2 - T\bar{x}^2}, \tag{1.6}$$

$$\hat{\alpha} = \bar{y} - \hat{\beta}\bar{x} \tag{1.7}$$

or since

$$\sum_{t=1}^{T} (x_t - \bar{x})\left(y_t - \bar{y}\right) = \sum_{t=1}^{T} x_t y_t - T\bar{x}\bar{y},$$

$$\sum_{t=1}^{T} (x_t - \bar{x})^2 = \sum_{t=1}^{T} x_t^2 - T\bar{x}^2,$$

equivalently

$$\hat{\beta} = \frac{\sum_{t=1}^{T} (x_t - \bar{x})\left(y_t - \bar{y}\right)}{\sum_{t=1}^{T} (x_t - \bar{x})^2} = \frac{S_{XY}}{S_{XX}},$$

where

$$S_{XY} = \frac{\sum_{t=1}^{T} (x_t - \bar{x})\left(y_t - \bar{y}\right)}{T} = S_{YX},$$

$$S_{XX} = \frac{\sum_{t=1}^{T} (x_t - \bar{x})^2}{T}.$$

1.4 Correlation coefficients between *Y* and *X*

There are many measures of quantifying the strength of correlation between two variables. The most popular one is the product moment correlation coefficient which was developed by Karl Pearson and builds on an earlier contribution by Francis Galton. Other measures of correlations include the Spearman rank correlation and Kendall's τ correlation. We now consider each of these measures in turn and discuss their uses and relationships.

1.4.1 Pearson correlation coefficient

The Pearson correlation coefficient is a parametric measure of dependence between two variables, and assumes that the underlying bivariate distribution from which the observations are drawn have moments. For the variables Y and X, and the T pairs of observations $\{(y_1,x_1), (y_2,x_2), \ldots, (y_T,x_T)\}$ on these variables, Pearson or the simple correlation coefficient between Y and X is defined by

$$\hat{\rho}_{YX} = \frac{\sum_{t=1}^{T}(x_t - \bar{x})\left(y_t - \bar{y}\right)}{\left[\sum_{t=1}^{T}(x_t - \bar{x})^2 \sum_{t=1}^{T}\left(y_t - \bar{y}\right)\right]^{1/2}} = \frac{S_{XY}}{(S_{YY}S_{XX})^{\frac{1}{2}}}, \tag{1.8}$$

It is easily seen that $\hat{\rho}_{YX}$ lies between -1 and $+1$. Notice also that the correlation coefficient between Y and X is the same as the correlation coefficient between X and Y, namely $\hat{\rho}_{XY} = \hat{\rho}_{YX}$. In this bivariate case we have the following interesting relationship between $\hat{\rho}_{XY}$ and the regression coefficients of the regression Y on X and the 'reverse' regression of X on Y. Denoting these two regression coefficients respectively by $\hat{\beta}_{Y\cdot X}$ and $\hat{\beta}_{X\cdot Y}$, we have

$$\hat{\beta}_{Y\cdot X}\hat{\beta}_{X\cdot Y} = \frac{S_{YX}S_{XY}}{(S_{XX}S_{YY})} = \hat{\rho}_{YX}^2. \tag{1.9}$$

Hence, if $\hat{\beta}_{Y\cdot X} > 0$ then $\hat{\beta}_{X\cdot Y} > 0$. Since $\hat{\rho}_{XY}^2 \leq 1$, if we assume that $\hat{\beta}_{Y\cdot X} > 0$ it follows that $\hat{\beta}_{X\cdot Y} \leq \frac{1}{\hat{\beta}_{Y\cdot X}}$. If we further assume that $0 < \hat{\beta}_{Y\cdot X} < 1$, then $\hat{\beta}_{X\cdot Y} = \frac{\hat{\rho}_{XY}^2}{\hat{\beta}_{Y\cdot X}} > \hat{\rho}_{XY}^2$.

1.4.2 Rank correlation coefficients

Rank correlation is often used in situations where the available observations are in the form of cardinal numbers, or if they are not sufficiently precise. Rank correlations are also used to avoid undue influences from outlier (extreme tail) observations on the correlation analysis. A number of different rank correlations have been proposed in the literature. In what follows we focus on the two most prominent of these, namely Spearman's rank correlation and Kendall's τ correlation coefficient. A classic treatment of the subject can be found in Kendall and Gibbons (1990).

Spearman rank correlation

Consider the T pairs of observations $\left\{(y_t, x_t)\right.$, for $\left.t = 1, 2, \ldots, T\right\}$ and rank the observations on each of the variables y and x, in an ascending (or descending) order. Denote the rank of these ordered series by $1, 2, \ldots, T$, so that the first observation in the ordered set takes the value of 1, the second takes the value of 2, etc. The Spearman rank correlation, r_s, between y and x is defined by

$$r_s = 1 - \frac{6\sum_{t=1}^{T} d_t^2}{T(T^2-1)}, \tag{1.10}$$

where

$$d_t = Rank(y_t : \mathbf{y}) - Rank(x_t : \mathbf{x}),$$

and $Rank(y_t : \mathbf{y})$ is equal to a number in the range [1 *to* T] determined by the size of y_t relative to the other $T-1$ values of $\mathbf{y} = (y_1, y_2, \ldots, y_T)'$. Note also that by construction $\sum_{t=1}^{T} d_t = 0$, and that $\sum_{t=1}^{T} d_t^2$ can only take even integer values and has a mean equal to $(T^3 - T)/6$. Hence $E(r_s) = 0$. The Spearman rank correlation can also be computed as a simple correlation between $ry_t = Rank(y_t : \mathbf{y})$ and $rx_t = Rank(x_t : \mathbf{x})$. It is easily seen that

$$r_s = \frac{\sum_{t=1}^{T}(ry_t - \overline{ry})(rx_t - \overline{rx})}{\left[\sum_{t=1}^{T}(ry_t - \overline{ry})^2\right]^{1/2}\left[\sum_{t=1}^{T}(rx_t - \overline{rx})^2\right]^{1/2}},$$

where

$$\overline{ry} = \overline{rx} = T^{-1}\sum_{t=1}^{T} ry_t = T^{-1}\sum_{t=1}^{T} rx_t = \frac{T+1}{2}.$$

Kendall's τ correlation

Another rank correlation coefficient was introduced by Kendall (1938). Consider the T pairs of ranked observations (ry_t, rx_t), associated with the quantitative measures (y_t, x_t), for $t = 1, 2, \ldots, T$ as discussed above. Then the two pairs of ranks (ry_t, rx_t) and (ry_s, rx_s) are said to be *concordant* if

$$(rx_t - rx_s)(ry_t - ry_s) > 0, \text{ concordant pairs for all } t \text{ and } s,$$

and *discordant* if

$$(rx_t - rx_s)(ry_t - ry_s) \leq 0, \text{ discordant pairs for all } t \text{ and } s.$$

Denoting the number of concordant pairs by P_T and the number of discordant pairs by Q_T, Kendall's τ correlation coefficient is defined by

$$\tau_T = \frac{2}{T(T-1)}(P_T - Q_T). \tag{1.11}$$

More formally

$$P_T = \sum_{t,s=1}^{T} I\left[(rx_t - rx_s)(ry_t - ry_s)\right],$$
$$Q_T = \sum_{t,s=1}^{T} I\left[-(rx_t - rx_s)(ry_t - ry_s)\right],$$

where $I(A) = 1$ if $A > 0$, and zero otherwise.

1.4.3 Relationships between Pearson, Spearman, and Kendall correlation coefficients

In the case where (y_t, x_t) are draws from a normal distribution we have

$$E(\tau_T) = \frac{2}{\pi}\sin^{-1}(\rho),$$

where ρ is the simple (Pearson) correlation coefficient between y_t and x_t. Furthermore,

$$E(r_s) = \rho_s + \frac{3(\tau - \rho_s)}{T+1},$$

where ρ_s is the population value of Spearman rank correlation. Finally, in the bivariate normal case we have

$$\rho = 2\sin\left(\frac{\pi\rho_s}{6}\right).$$

These relationships suggest the following indirect possibilities for estimation of the simple correlation coefficient, namely

$$\hat{\rho}_1 = \sin\left(\frac{\pi}{2}\tau_T\right),$$
$$\hat{\rho}_2 = 2\sin\left[\frac{\pi}{6}\left(r_s - \frac{3(\tau_T - r_s)}{T+1}\right)\right],$$

as possible alternatives to $\hat{\rho}$, the simple correlation coefficient. See Kendall and Gibbons (1990, p. 169). The alternative estimators, $\hat{\rho}_1$ and $\hat{\rho}_2$, are likely to have some merit over $\hat{\rho}$ in small samples in cases where the population distribution of (y_t, x_t) differs from bivariate normal and/or when the observations are subject to measurement errors.

Tests based on the different correlation measures are discussed in Section 3.4.

1.5 Decomposition of the variance of *Y*

It is possible to divide the total variation of Y into two parts, the variation of the estimated Y and a residual variation. In particular

$$\begin{aligned}\sum_{t=1}^{T}(y_t - \bar{y})^2 &= \sum_{t=1}^{T}\left[(\hat{y}_t - \bar{y}) - (\hat{y}_t - y_t)\right]^2 \\ &= \sum_{t=1}^{T}(\hat{y}_t - \bar{y})^2 + \sum_{t=1}^{T}(\hat{y}_t - y_t)^2 - 2\sum_{t=1}^{T}(\hat{y}_t - y_t)(\hat{y}_t - \bar{y}) \\ &= \sum_{t=1}^{T}(y_t - \hat{y}_t)^2 + \sum_{t=1}^{T}(\hat{y}_t - \bar{y})^2 + 2\sum_{t=1}^{T}\hat{u}_t(\hat{y}_t - \bar{y}).\end{aligned}$$

But, notice that

$$\sum_{t=1}^{T} \hat{u}_t \left(\hat{y}_t - \bar{y}\right) = \sum_{t=1}^{T} \hat{u}_t \left(\hat{\alpha} + \hat{\beta} x_t\right) - \sum_{t=1}^{T} \hat{u}_t \bar{y}$$
$$= \hat{\alpha} \sum_{t=1}^{T} \hat{u}_t + \hat{\beta} \sum_{t=1}^{T} \hat{u}_t x_t - \bar{y} \sum_{t=1}^{T} \hat{u}_t = 0,$$

since from the normal equations (1.3) and (1.4), $\sum_{t=1}^{T} \hat{u}_t = 0$ and $\sum_{t=1}^{T} \hat{u}_t x_t = 0$, then

$$\sum_{t=1}^{T} \left(y_t - y_t\right)^2 = \sum_{t=1}^{T} \left(\hat{y}_t - \bar{y}\right)^2 + \sum_{t=1}^{T} \left(y_t - \hat{y}_t\right)^2. \tag{1.12}$$

This decomposition of the total variations in Y forms the basis of the *analysis of variance*, which is described in the following table.

Source of variation	Sums of squares	Degrees of freedom	Mean square
Explained by the regression line	$\sum_{t=1}^{T} \left(\hat{y}_t - \bar{y}\right)^2$	2	$\frac{\sum_{t=1}^{T} (\hat{y}_t - \bar{y})^2}{2}$
Residual	$\sum_{t=1}^{T} \left(y_t - \hat{y}_t\right)^2$	$T - 2$	$\frac{\sum_{t=1}^{T} (y_t - \hat{y}_t)^2}{T-2}$
Total variation	$\sum_{t=1}^{T} \left(y_t - \bar{y}\right)^2$	T	$\frac{\sum_{t=1}^{T} (y_t - \bar{y})^2}{T}$

Proposition 1 highlights the relation between $\hat{\rho}_{XY}^2$ and the variance decomposition.

Proposition 1

$$\hat{\rho}_{XY}^2 = \frac{S_{XY}^2}{S_{XX} S_{YY}} = 1 - \frac{\sum_{t=1}^{T} \left(y_t - \hat{y}_t\right)^2}{\sum_{t=1}^{T} \left(y_t - \bar{y}\right)^2}. \tag{1.13}$$

Proof Notice that

$$1 - \frac{\sum_t \left(y_t - \hat{y}_t\right)^2}{\sum_t \left(y_t - \bar{y}\right)^2} = \frac{\sum_t \left(y_t - \bar{y}\right)^2 - \sum_t \left(y_t - \hat{y}_t\right)^2}{\sum_t \left(y_t - \bar{y}\right)^2},$$

and using the result in (1.12), we have

$$1 - \frac{\sum_t \left(y_t - \hat{y}_t\right)^2}{\sum_t \left(y_t - \bar{y}\right)^2} = \frac{\sum_t \left(\hat{y}_t - \bar{y}\right)^2}{\sum_t \left(y_t - \bar{y}\right)^2}.$$

Further, since $\hat{y}_t = \hat{\alpha} + \hat{\beta} x_t$, we have

$$\sum_t \left(\hat{y}_t - \bar{y}\right)^2 = \sum_t \left(\hat{\alpha} + \hat{\beta} x_t - \bar{y}\right)^2$$
$$= \sum_t \left[\hat{\beta}\left(x_t - \bar{x}\right) + \hat{\beta}\bar{x} + \hat{\alpha} - \bar{y}\right]^2,$$

By (1.1), $\bar{y} = \hat{\alpha} + \hat{\beta}\bar{x}$. Hence, it follows that

$$\sum_t \left(\hat{y}_t - \bar{y}\right)^2 = \hat{\beta}^2 \sum_t \left(x_t - \bar{x}\right)^2 = \frac{S_{XY}^2}{S_{XX}^2} \cdot S_{XX} = \frac{S_{XY}^2}{S_{XX}},$$
$$1 - \frac{\sum_t \left(y_t - \hat{y}_t\right)^2}{\sum_t \left(y_t - \bar{y}\right)^2} = \frac{S_{YY}^2}{S_{YY} S_{XX}} = \hat{\rho}_{XY}^2.$$

The above result is important since it also provides a natural generalization of the concept of the simple correlation coefficient, $\hat{\rho}_{XY}$, to the multivariate regression case, where it is referred to as the multiple correlation coefficient (see Section 2.10).

1.6 Linear statistical models

So far we have viewed the regression equation as a line fitted to a scatter of points in a two-dimensional space. As such it is purely a descriptive scheme that attempts to summarize the scatter of points by a single regression line. An alternative procedure would be to adopt a statistical model where the regression disturbances, u_t's, are characterized by a probability distribution. Under this framework there are two important statistical models that are used in the literature:

A: Classical linear regression model. This model assumes that the relationship between Y and X is a linear one:

$$y_t = \alpha + \beta x_t + u_t, \tag{1.14}$$

and that the disturbances u_ts satisfy the following assumptions:

(i) *Zero mean*: the disturbances u_t have zero means, i.e., $E(u_t) = 0$.

(ii) *Homoskedasticity:* conditional on x_t the disturbances u_t have constant conditional variance. $Var\left(u_t \mid x_s\right) = \sigma^2$, for all t and s.

(iii) *Non-autocorrelated error*: the disturbances u_t are serially uncorrelated. $\text{Cov}(u_t, u_s) = 0$ for all $t \neq s$.

(iv) *Orthogonality:* the disturbances u_t and the regressor x_t are uncorrelated, or conditional on x_s, u_t has a zero mean (namely $E\left(u_t \mid x_s\right) = 0$, for all t and s).

Assumption (i) ensures that the unconditional mean of y_t is correctly specified by the regression equation. The other assumptions can be relaxed and are introduced to provide a simple model that can be used as a benchmark in econometric analysis.

B: Another way of motivating the linear regression model is to focus on the *joint* distribution of Y and X, and assume that this distribution is normal with constant means, variances and covariances. In this case the regression of Y on X defined as the conditional mean of Y given a particular value of X, say $X = x$ will be a linear function of x. In particular we have:

$$E(Y|X = x_t) = \alpha + \beta x_t, \tag{1.15}$$

and

$$Var(Y|X = x_t) = Var(Y)\left(1 - \rho_{XY}^2\right), \tag{1.16}$$

and where $Var(Y)$ is the unconditional variance of Y and

$$\rho_{XY} = \text{Cov}(Y, X)/\sqrt{Var(X)\,Var(Y)}$$

is the population correlation coefficient between Y and X.

The parameters α and β are related to the moments of the joint distribution of Y and X in the following manner:

$$\alpha = E(Y) - \frac{\text{Cov}(X, Y)}{Var(X)}E(X), \tag{1.17}$$

and

$$\beta = \frac{\text{Cov}(X, Y)}{Var(X)} = \rho_{XY}\sqrt{\frac{Var(Y)}{Var(X)}}. \tag{1.18}$$

Using (1.17) and (1.18), relation (1.15) can also be written as:

$$E(Y|X = x_t) = E(Y) + \frac{\text{Cov}(X, Y)}{Var(X)}\left[x_t - E(X)\right]. \tag{1.19}$$

Model B does not postulate a linear relationship between Y and X, but assumes that (Y, X) have a bivariate normal distribution. In contrast, model A assumes linearity of the relationship between Y and X, but does not necessarily require that the joint probability distribution of (Y, X) be normal. It is clear that under assumption (iv), (1.14) implies (1.15). Also (1.15) can be used to obtain (1.14) by defining u_t to be

$$u_t = y_t - E(Y|X = x_t),$$

or more simply

$$u_t = y_t - E\left(y_t|x_t\right). \tag{1.20}$$

It is in the light of this expression that u_t's are also often referred to as 'innovations' or 'unexpected components' of y_t.

Both the above statistical models are used in the econometric literature. The two models can also be combined to yield the 'classical normal linear regression model' which adds the extra assumption that u_t are normally distributed to the list of the *four* basic assumptions of the classical linear regression model set out above.

Finally, it is worth noting that under the normality assumption using (1.16) we also have

$$Var\,(u_t\,|x_t\,) = \sigma^2 = Var\,(Y)\left(1 - \rho_{YX}^2\right). \tag{1.21}$$

Hence,

$$\rho_{YX}^2 = 1 - \frac{\sigma^2}{Var\,(Y)},$$

which is the population value of the sample correlation coefficient defined by (1.8) and (1.13).

1.7 Method of moments applied to bivariate regressions

The *OLS* estimators can also be motivated by the method of moments originally introduced by Karl Pearson in 1894. Under the method of moments the parameters α and β are estimated by replacing population moments by their sample counterparts. Under Assumptions (i) and (iv) above that the errors, u_t, have zero means and are orthogonal to the regressors, we have the following two moment conditions

$$\begin{aligned} E(u_t) &= E(y_t - \alpha - \beta x_t) = 0, \\ E(x_t u_t) &= E\left[x_t(y_t - \alpha - \beta x_t)\right] = 0, \end{aligned}$$

which can also be written equivalently as

$$\begin{aligned} E(y_t) &= \alpha + \beta E\,(x_t)\,, \\ E(y_t x_t) &= \alpha E(x_t) + \beta E(x_t^2). \end{aligned}$$

It is clear that α and β can now be derived in terms of the population moments, $E(y_t)$, $E(x_t)$, $E(x_t^2)$, and $E(y_t x_t)$, namely

$$\begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \begin{pmatrix} 1 & E\,(x_t) \\ E\,(x_t) & E(x_t^2) \end{pmatrix}^{-1} \begin{pmatrix} E(y_t) \\ E(y_t x_t) \end{pmatrix}. \tag{1.22}$$

The inverse exists if $Var(x_t) = E(x_t^2) - \left[E\,(x_t)\right]^2 > 0$. *The method of moment estimators of* α and β are obtained when the population moments in the above expression are replaced by the sample moments which are given by

$$\begin{aligned} &\hat{E}(y_t) = \bar{y}, \hat{E}\,(x_t) = \bar{x}, \\ &\hat{E}(x_t^2) = T^{-1}\sum_{t=1}^{T} x_t^2, \hat{E}(y_t x_t) = T^{-1}\sum_{t=1}^{T} y_t x_t. \end{aligned}$$

Using these sample moments in (1.22) gives $\hat{\alpha}_{MM}$ and $\hat{\beta}_{MM}$, that are easily verified to be the same as the *OLS* estimators given by (1.7) and (1.6).

In cases where the number of moment conditions exceed the number of unknown parameters, the method of moments is generalized to take account of the additional moment conditions in an efficient manner. The resultant estimator is then referred to as the generalized method of moments (*GMM*), which will be discussed in some detail in Chapter 10.

1.8 The likelihood approach for the bivariate regression model

An alternative estimation approach developed by R. A. Fisher over the period 1912–22 (building on the early contributions of Gauss, Laplace, and Edgeworth) is to estimate the unknown parameters by maximizing their likelihood. The likelihood function is then given by the joint probability distribution of the observations. In the case of the bivariate classical regression model the likelihood is obtained from the joint distribution of $\mathbf{y} = (y_1, y_2, \ldots, y_T)'$, conditional on $\mathbf{x} = (x_1, x_2, \ldots, x_T)'$. To obtain this joint probability distribution, in addition to the assumptions of the classical linear regression, (i)-(iv) given in Section 1.6, we also need to specify the probability distribution of the errors, u_t. Typically, it is assumed that $u_t's$ are normally distributed, and the joint probability distribution of $\mathbf{y}$ conditional on $\mathbf{x}$, is then obtained as (since the Jacobian of the transformation between y_t and u_t is unity)

$$\Pr\left(\mathbf{y}\,|\mathbf{x},\alpha,\beta,\sigma^2\right) = \Pr(u_1, u_2, \ldots, u_T\,|\mathbf{x}).$$

But under the assumption that the errors are normally distributed, the non-autocorrelated error assumption, (iii), implies that the errors are independently distributed and hence we have

$$\Pr\left(\mathbf{y}\,|\mathbf{x},\alpha,\beta,\sigma^2\right) = \Pr(u_1)\Pr(u_2)\ldots.\Pr(u_T).$$

But the probability density function of a $N(0,\sigma^2)$ random variable is given by

$$\Pr(u_t) = (2\pi\sigma^2)^{-1/2}\exp\left(\frac{-1}{2\sigma^2}u_t^2\right).$$

Using this result and noting that $u_t = y_t - \alpha - \beta x_t$, we have

$$\Pr\left(\mathbf{y}\,|\mathbf{x},\alpha,\beta,\sigma^2\right) = (2\pi\sigma^2)^{-T/2}\exp\left[\frac{-\sum_{t=1}^{T}\left(y_t-\alpha-\beta x_t\right)^2}{2\sigma^2}\right].$$

The likelihood of the unknown parameters, which we collect in the 3×1 vector $\boldsymbol{\theta} = (\alpha,\beta,\sigma^2)'$, is the same as the above joint density function, but is viewed as a function of $\boldsymbol{\theta}$ rather than $\mathbf{y}$. Denoting the likelihood function of $\boldsymbol{\theta}$ by $L_T(\boldsymbol{\theta})$ we have

$$L_T(\boldsymbol{\theta}) = (2\pi\sigma^2)^{-T/2}\exp\left[\frac{-\sum_{t=1}^{T}\left(y_t-\alpha-\beta x_t\right)^2}{2\sigma^2}\right]. \tag{1.23}$$

To obtain the maximum likelihood estimator (*MLE*) of $\boldsymbol{\theta}$ it is often more convenient to work with the logarithm of the likelihood function, referred to as the log-likelihood function, which we denote by $\ell_T(\boldsymbol{\theta})$. Using (1.23) we have

$$\ell_T(\boldsymbol{\theta}) = -\frac{T}{2}\log(2\pi\sigma^2) - \frac{\sum_{t=1}^{T}\left(y_t - \alpha - \beta x_t\right)^2}{2\sigma^2}.$$

It is now clear that maximization of $\ell_T(\boldsymbol{\theta})$ with respect to α and β will be the same as minimizing $\sum_{t=1}^{T}\left(y_t - \alpha - \beta x_t\right)^2$ with respect to these parameters, which establish that the *MLE* of α and β is the same as their *OLS* estimators, namely $\hat{\alpha}_{ML} = \hat{\alpha}$, and $\hat{\beta}_{ML} = \hat{\beta}$, where $\hat{\alpha}$, and $\hat{\beta}$ are given by (1.7) and (1.6), respectively. The *MLE* of σ^2 can be obtained by taking the first derivative of $\ell_T(\boldsymbol{\theta})$ with respect to σ^2. We have

$$\frac{\partial \ell_T(\boldsymbol{\theta})}{\partial \sigma^2} = -\frac{T}{2\sigma^2} + \frac{\sum_{t=1}^{T}\left(y_t - \alpha - \beta x_t\right)^2}{2\sigma^4}.$$

Setting $\partial \ell_T(\boldsymbol{\theta})/\partial\sigma^2 = 0$ and solving for $\hat{\sigma}^2_{ML}$ in terms of the *MLE* of α and β now yields

$$\hat{\sigma}^2_{ML} = \frac{\sum_{t=1}^{T}\left(y_t - \hat{\alpha}_{ML} - \hat{\beta}_{ML}x_t\right)^2}{T} = \frac{\sum_{t=1}^{T}\left(y_t - \hat{\alpha} - \hat{\beta}x_t\right)^2}{T} = \frac{\sum_{t=1}^{T}\hat{u}_t^2}{T}, \qquad (1.24)$$

where $\hat{u}_t$ is the *OLS* residual, given be (1.5).

The likelihood approach is used extensively in subsequent chapters. For an analysis of the *MLE* for multiple regression models see (2.4). The general theory of maximum likelihood estimation is provided in Chapter 9.

1.9 Properties of the *OLS* estimators

Under the classical assumptions (i)–(iv) in Section 1.6 above, the *OLS* estimators of α and β possess the following properties:

(i) $\hat{\alpha}$ and $\hat{\beta}$ are *unbiased* estimators. Namely, that $E\left(\hat{\alpha}\right) = a$ and $E\left(\hat{\beta}\right) = \beta$, where α and β are the 'true' values of the regression coefficients.

(ii) Both estimators are *linear functions* of the values of y_t.

(iii) Among the class of linear unbiased estimators, $\hat{\alpha}$ and $\hat{\beta}$ have the *least variances*. This result is known as the *Gauss–Markov theorem*.

In what follows we present a proof of properties (i) to (iii) for $\hat{\beta}$. A similar proof can also be established for $\hat{\alpha}$. Recall that

$$\hat{\beta} = \frac{S_{XY}}{S_{XX}} = \frac{\sum_{t=1}^{T}\left(y_t - \bar{y}\right)(x_t - \bar{x})}{\sum_{t=1}^{T}(x_t - \bar{x})^2}.$$

But the numerator of this ratio can be written as

$$\sum_{t=1}^{T}\left(y_t-\bar{y}\right)\left(x_t-\bar{x}\right)=\sum_{t=1}^{T}y_t\left(x_t-\bar{x}\right)-\sum_{t=1}^{T}\bar{y}\left(x_t-\bar{x}\right),$$

and since $\sum_{t=1}^{T}\bar{y}\left(x_t-\bar{x}\right)=\bar{y}\sum_{t=1}^{T}\left(x_t-\bar{x}\right)=0$, then

$$\sum_{t=1}^{T}\left(y_t-\bar{y}\right)\left(x_t-\bar{x}\right)=\sum_{t=1}^{T}y_t\left(x_t-\bar{x}\right).$$

Hence $\hat{\beta}$ can be written as a weighted linear function of y_t's

$$\hat{\beta}=\sum_{t=1}^{T}w_t y_t, \tag{1.25}$$

where the weights

$$w_t=\frac{x_t-\bar{x}}{\sum_{t=1}^{T}\left(x_t-\bar{x}\right)^2} \tag{1.26}$$

are fixed and add up to zero, namely $\sum_{t=1}^{T}w_t=0$. This establishes property (ii).

Notice that x_t's are taken as given, which is justified if they are strictly exogenous. Further discussion of the concept of strict exogeneity is given in Section 2.2, but in the present context x_t will be strictly exogenous if it is uncorrelated with current, past, as well as future values of the error terms, u_s; more specifically if $Cov(x_t, u_s)=0$, for all values of t and s. Under this assumption, taking conditional expectations of both sides of (1.25), we have:

$$\begin{aligned}E\left(\hat{\beta}\right)&=E\left(\sum_{t=1}^{T}w_t y_t\,|x_1,x_2,\ldots,x_T\right)\\&=\sum_{t=1}^{T}w_t E\left(y_t\,|x_t\right),\end{aligned}$$

But using (1.14) or (1.15), conditional on x_t, we have $E\left(y_t\,|x_t\right)=\alpha+\beta x_t$. Consequently,

$$\begin{aligned}E\left(\hat{\beta}\right)&=\sum_{t=1}^{T}w_t\left(\alpha+\beta x_t\right)\\&=\alpha\sum_{t=1}^{T}w_t+\beta\sum_{t=1}^{T}w_t x_t.\end{aligned} \tag{1.27}$$

However, using (1.26) we have

$$\sum_{t=1}^{T} w_t x_t = \frac{\sum_{t=1}^{T} x_t \left(x_t - \bar{x}\right)}{\sum_{t=1}^{T} \left(x_t - \bar{x}\right)^2},$$

and since

$$\begin{aligned}
\sum_{t=1}^{T} \left(x_t - \bar{x}\right)^2 &= \sum_{t=1}^{T} \left(x_t - \bar{x}\right) \left(x_t - \bar{x}\right) \\
&= \sum_{t=1}^{T} x_t \left(x_t - \bar{x}\right) - \sum_{t=1}^{T} \bar{x} \left(x_t - \bar{x}\right) \\
&= \sum_{t=1}^{T} x_t \left(x_t - \bar{x}\right) - \bar{x} \sum_{t=1}^{T} \left(x_t - \bar{x}\right) \\
&= \sum_{t=1}^{T} x_t \left(x_t - \bar{x}\right),
\end{aligned}$$

it then follows that $\sum_{t=1}^{T} w_t x_t = 1$. We have also seen that $\sum_{t=1}^{T} w_t = 0$, hence it follows from (1.27) that $E\left(\hat{\beta}\right) = \beta$, which establishes that $\hat{\beta}$ is an unbiased estimator, that is, point (i).

The variance of $\hat{\beta}$ can also be computed easily using (1.25). We have

$$\begin{aligned}
Var\left(\hat{\beta}\right) &= \sum_{t=1}^{T} w_i^2 Var\left(y_t \,|x_t\right) \\
&= \sum_{t=1}^{T} w_i^2 Var\left(u_t \,|x_t\right) \\
&= \sigma^2 \sum_{t=1}^{T} w_i^2,
\end{aligned}$$

and using (1.26) yields

$$Var\left(\hat{\beta}\right) = \frac{\sigma^2}{\sum_{t=1}^{T} \left(x_t - \bar{x}\right)^2} = \frac{\sigma^2}{S_{XX}}. \tag{1.28}$$

Similarly, we have

$$Var\left(\hat{\alpha}\right) = \frac{\sigma^2 \sum_{t=1}^{T} x_t^2}{T \sum_{t=1}^{T} \left(x_t - \bar{x}\right)^2}, \tag{1.29}$$

and

$$\text{Cov}\left(\hat{\alpha},\hat{\beta}\right)=\frac{-\sigma^2\bar{x}}{\sum_{t=1}^{T}(x_t-\bar{x})^2}. \tag{1.30}$$

The Gauss–Markov theorem (i.e., property (iii) above) states that among all linear, unbiased estimators of β (or α) the *OLS* estimator, $\hat{\beta}$, has the smallest variance. To prove this result consider *another* linear unbiased estimator of β and denote it by $\tilde{\beta}$. Then by assumption

$$\tilde{\beta}=\sum_{t=1}^{T}\tilde{w}_t y_t,$$

where $\tilde{w}_t$ are fixed weights (which do not depend on y_t) and satisfy the conditions

$$\sum_{t=1}^{T}\tilde{w}_t=0, \tag{1.31}$$

and

$$\sum_{t=1}^{T}\tilde{w}_t x_t=1. \tag{1.32}$$

These two conditions ensure that $\tilde{\beta}$ is an unbiased estimator of β, that is, that $E\left(\tilde{\beta}\right)=\beta$. Suppose now $\tilde{w}_t$ differ from w_t, the *OLS* weights given in (1.26), by the amount δ_t and let

$$\tilde{w}_t=w_t+\delta_t,\qquad t=1,2,\ldots,T, \tag{1.33}$$

where δ_t is the amount of discrepancy between the two weighting schemes. Since $\sum_{t=1}^{T}w_t=\sum_{t=1}^{T}\tilde{w}_t=0$. It follows also that $\sum_{t=1}^{T}\delta_t=0$, and since $\sum_{t=1}^{T}w_t x_t=\sum_{t=1}^{T}\tilde{w}_t x_t=1$, then we should also have $\sum_{t=1}^{T}\delta_t x_t=0$.

The variance of $\tilde{\beta}$ is now given by

$$\begin{aligned}Var\left(\tilde{\beta}\right)&=\sum_{t=1}^{T}\tilde{w}_t^2 Var\left(y_t\,|x_t\right)\\&=\sigma^2\sum_{t=1}^{T}\tilde{w}_t^2,\end{aligned}$$

and using (1.33)

$$Var\left(\tilde{\beta}\right)=\sigma^2\left(\sum_{t=1}^{T}w_i^2+\sum_{t=1}^{T}\delta_t^2+2\sum_{t=1}^{T}w_t\delta_t\right).$$

But, using (1.26),

$$\sum_{t=1}^{T} w_t \delta_t = \frac{\sum_{t=1}^{T} \delta_t (x_t - \bar{x})}{\sum_{t=1}^{T} (x_t - \bar{x})^2}.$$

The numerator of this ratio can be written more fully as

$$\sum_{t=1}^{T} \delta_t (x_t - \bar{x}) = \sum_{t=1}^{T} \delta_t x_t - \bar{x} \sum_{t=1}^{T} \delta_t,$$

which is equal to zero. Recall that $\sum_{t=1}^{T} \delta_t = 0$, and $\sum_{t=1}^{T} \delta_t x_t = 0$. Hence $\sum_{t=1}^{T} w_t \delta_t = 0$, and

$$Var\left(\tilde{\beta}\right) = \sigma^2 \left\{ \sum_{t=1}^{T} w_i^2 + \sum_{t=1}^{T} \delta_t^2 \right\} \geq Var\left(\tilde{\beta}\right),$$

which establishes the Gauss–Markov theorem for $\hat{\beta}$. The equality sign holds if and only if $\delta_t = 0$ for all i. The proof of the Gauss–Markov theorem for the multivariate case is presented in Section 2.7.

1.9.1 Estimation of σ^2

Since $Var\left(\hat{\alpha}\right)$ and $Var\left(\tilde{\beta}\right)$ depend on the unknown parameter, σ^2 (the variance of the disturbance term), in order to obtain estimates of the variances of the *OLS* estimators, it is also necessary to obtain an estimate of σ^2. For this purpose we first note that

$$\sigma^2 = Var\left(u_t \,|x_t\right) = E\left(u_t^2\right).$$

It is, therefore, reasonable to interpret σ^2 as the mean value of the squared disturbances. A moment estimator of σ^2 can then be obtained by the sample average of u_t^2. In practice, however, u_t's are observed indirectly through the estimates of α and β. Hence a feasible estimator of σ^2 can be obtained by replacing α and β in the definition of u_t by their *OLS* estimators. Namely,

$$\tilde{\sigma}^2 = \frac{\sum_{t=1}^{T} \hat{u}_t^2}{T} = \frac{\sum_{t=1}^{T} \left(y_t - \hat{\alpha} - \hat{\beta} x_t\right)^2}{T},$$

which is the same as the *ML* estimator of σ^2 given by (3). When T is large, this provides a reasonable estimator of σ^2. However, in finite samples a more satisfactory estimator of σ^2 can be obtained by dividing the sum of squares of the residuals by $T - 2$ rather than T. Namely,

$$\hat{\sigma}^2 = \frac{\sum_{t=1}^{T} \left(y_t - \hat{\alpha} - \hat{\beta} x_t\right)^2}{T - 2}, \tag{1.34}$$

where '2' is equal to the number of estimated unknown parameters in the simple regression model (here, $\hat{\alpha}$ and $\hat{\beta}$). Unlike $\tilde{\sigma}^2$, the above estimator of σ^2 given by (1.34) is unbiased. Namely $E(\hat{\sigma}^2) = \sigma^2$.

Using the above estimator of σ^2 it is now possible to estimate the variances and covariances of $\hat{\beta}$ given in (1.28). For example we have

$$\widehat{Var}\left(\hat{\beta}\right) = \frac{\hat{\sigma}^2}{\sum_{t=1}^{T}(x_t - \bar{x})^2},$$

and similarly for $\widehat{Var}\left(\hat{\alpha}\right)$ and $\widehat{\text{Cov}}\left(\hat{\alpha}, \hat{\beta}\right)$.

The problem of testing the statistical significance of the estimates and their confidence bands will be addressed in Chapter 3.

1.10 The prediction problem

Suppose T pairs of observations (y_1, x_1), $(y_2, x_2), \ldots (y_T, x_T)$ are available on Y and X and assume that the linear regression of Y on X provides a reasonable model for this T-tuple. The problem of prediction arises when a new observation on X, say x_{T+1}, is considered and it is desired to obtain the 'best' estimate y_{T+1}, the value of Y which corresponds to x_{T+1}. This is called the problem of conditional prediction, namely estimating the value of Y conditional on a *given* value of X. The solution is given by the mathematical expectation of y_{T+1} conditional on the available information, namely $x_1, x_2, \ldots, x_T, x_{T+1}$, and possibly observations on lagged values of Y. In the case of the simple linear regression (1.14) we have

$$E\left(y_{T+1} \left| y_1, y_2, \ldots, y_T; x_1, x_2, \ldots, x_T, x_{T+1}\right.\right) = E\left(y_{T+1} \left| x_{T+1}\right.\right) = \alpha + \beta x_{T+1}.$$

An estimate of this expression gives the *estimate* of the conditional predictor of y_{T+1}. The *OLS* estimate of y_{T+1} is given by

$$\hat{y}_{T+1} = \hat{E}\left(y_{T+1} \left| x_1, x_2, \ldots\right.\right) = \hat{\alpha} + \hat{\beta} x_{T+1}.$$

The variance of the prediction can now be computed as[2]

$$Var\left(\hat{y}_{T+1}\right) = Var\left(\hat{\alpha}\right) + x_{T+1}^2 Var\left(\hat{\beta}\right) + 2x_{T+1}\,\text{Cov}\left(\hat{a}, \hat{\beta}\right).$$

[2] Notice that for the two random variables x and y, and the fixed constants α and β, we have

$$Var\left(\alpha x + \beta y\right) = \alpha^2 Var\left(x\right) + \beta^2 Var\left(y\right) + 2\alpha\beta\,\text{Cov}\left(x, y\right).$$

Now using the results in (1.28) we have:

$$\begin{aligned} Var\left(\hat{y}_{T+1}\right) &= \frac{\sigma^2 \sum_t x_t^2}{T\sum_t (x_t-\bar{x})^2} + \left\{x_{T+1}^2 \frac{\sigma^2}{\sum_t (x_t-\bar{x})^2}\right\} + 2x_{T+1}\left[\frac{-\bar{x}\sigma^2}{\sum_t (x_t-\bar{x})^2}\right] \\ &= \frac{\sigma^2\left[\frac{\sum_t x_t^2}{T} + x_{T+1}^2 - 2\bar{x}x_{T+1}\right]}{\sum_{t=1}^{T}(x_t-\bar{x})^2} \\ &= \frac{\sigma^2}{T}\left[\frac{\sum_t x_t^2 + Tx_{T+1}^2 - 2\left(\sum_t x_t\right)x_{T+1}}{\sum_t (x_t-\bar{x})^2}\right]. \end{aligned}$$

Therefore

$$Var\left(\hat{y}_{T+1}\right) = \frac{\sigma^2}{T\sum_t (x_t-\bar{x})^2}\left[\sum_t (x_t-\bar{x})^2 + T(x_{T+1}-\bar{x})^2\right],$$

or

$$Var\left(\hat{y}_{T+1}\right) = \sigma^2\left[\frac{1}{T} + \frac{(x_{T+1}-\bar{x})^2}{\sum_t (x_t-\bar{x})^2}\right]. \tag{1.35}$$

An estimate of $Var\left(\hat{y}_{T+1}\right)$ is now given by

$$\widehat{Var}\left(\hat{y}_{T+1}\right) = \hat{\sigma}^2\left[\frac{1}{T} + \frac{(x_{T+1}-\bar{x})^2}{\sum_t (x_t-\bar{x})^2}\right]. \tag{1.36}$$

The general theory of prediction under alternative loss functions is discussed in Chapter 17.

1.10.1 Prediction errors and their variance

The error of the conditional forecast of y_{T+1} is defined by

$$\hat{u}_{T+1} = y_{T+1} - \hat{y}_{T+1}.$$

Under the assumption that y_{T+1} is generated according to the simple regression model we have

$$\hat{u}_{T+1} = \alpha + \beta x_{T+1} + u_{T+1} - \hat{\alpha} - \hat{\beta}x_{T+1}.$$

To compute the variance of $\hat{u}_{T+1}$ we first note that both $\hat{\alpha}$ and $\hat{\beta}$ are linear functions of the disturbances over the estimation period (namely $u_1, u_2, \ldots, u_T$) and do not depend on u_{T+1}. Since by assumption u_t's are serially uncorrelated it therefore follows that

$$\begin{aligned} \text{Cov}\left(u_{T+1}, \hat{\alpha} - \alpha\right) &= 0, \\ \text{Cov}\left(u_{T+1}, \hat{\beta} - \beta\right) &= 0. \end{aligned}$$

Hence, conditional on x_{T+1}, u_{T+1} and $\hat{y}_{T+1} = \hat{\alpha} + \hat{\beta}x_{T+1}$ will also be uncorrelated, and

$$Var\left(\hat{u}_{T+1}\right) = Var\left(u_{T+1}\right) + Var\left(\hat{y}_{T+1}\right).$$

Now noting that $Var\left(u_{T+1}\right) = \sigma^2$ and using (1.35) we have

$$Var\left(\hat{u}_{T+1}\right) = \sigma^2\left\{1 + \frac{1}{T} + \frac{(x_{T+1} - \bar{x})^2}{\sum_t (x_t - \bar{x})^2}\right\}. \tag{1.37}$$

This variance can again be estimated by

$$\widehat{Var}\left(\hat{u}_{T+1}\right) = \hat{\sigma}^2\left\{1 + \frac{1}{T} + \frac{(x_{T+1} - \bar{x})^2}{\sum_t (x_t - \bar{x})^2}\right\}.$$

In the case where $\{x_t\}$ has a constant variance, $Var\left(\hat{u}_{T+1}\right)$ converges to σ^2 as $T \to \infty$. The above derivations also clearly show that $Var\left(\hat{u}_{T+1}\right)$ is composed of two parts: one part is due to the inherent uncertainty that surrounds the regression line (i.e., $Var\left(u_t\right) = \sigma^2$), and the other part is due to the sampling variation that is associated with the estimation of the regression parameters, α and β. It is, therefore, natural that as $T \to \infty$, the latter source of variations disappears and we are left with the inherent uncertainty due to the regression, as measured by σ^2.

1.10.2 *Ex ante* predictions

In the case of the linear regression model the *ex ante* prediction of y_{T+1} is obtained without assuming x_{T+1} is known. The prediction is conditional on knowing the past (but not the current) values of x. To obtain *ex ante* prediction of y_{T+1} we therefore also need to predict x_{T+1} conditional on its past values. This requires developing an explicit model for x_t. One popular method of generating *ex ante* forecasts is to assume a univariate time series process for x_ts, and then predict x_{T+1} from information on its lagged values. A simple example of such a time series process is the $AR(1)$ model:

$$x_t = \rho x_{t-1} + \varepsilon_t, \qquad |\rho| < 1,$$

where ε_ts are assumed to have zero means and constant variances. Under this specification the 'optimal' forecast of x_{T+1} (conditional on past values of x's) is given by

$$E\left(x_{T+1} \left| x_1, x_2, \ldots, x_T\right.\right) = \rho x_T,$$

which in turn yields the following *ex ante* forecast of y_{T+1}

$$E\left(y_{T+1} \left| x_1, x_2, \ldots, x_T, y_1, y_2, \ldots, y_T\right.\right) = \alpha + \beta E\left(x_{T+1} \left| x_1, \ldots, x_T\right.\right).$$

An estimate of this forecast is now given by

$$\hat{y}_{T+1} = \hat{E}\left(y_{T+1} \left| x_T\right.\right) = \hat{\alpha} + \hat{\beta}\hat{\rho}x_T,$$

where $\hat{\rho}$ is the *OLS* estimator of ρ, obtained from the regression of x_t on its one-period lagged value. In Chapter 17 we review forecasting within the general context of *ARMA* models, introduced in Chapter 12.

1.11 Exercises

1. Show that the correlation coefficient defined in (1.8) ranges between -1 and 1.
2. In the model $y_t = \alpha + \beta x_t + u_t$ what happens to the *OLS* estimator of β if x_t and/or y_t are standardized by demeaning and scaling by their standard deviations?
3. The following table provides a few key summary statistics for daily rates of change of UK stock index (FTSE) and the GB pound versus US dollar.

Daily UK stock returns and GBP/US$ rate (%)
sample period 2 Jan 1987–16 June 1998

	Stock (FTSE)	**FX (GBP/US$)**
Max	5.69	2.82
Min	−12.11	−3.2861
Mean	0.0396	0.0033
St. dev.	0.8342	0.6200
Skewness	−1.82	−0.27
Kurtosis − 3	26.17	2.55

Using these statistics what do you think are the main differences between these two series and how best these differences are characterized?

4. Consider the following data

Height in centimeters	**Weight in kilograms**
(X)	(Y)
169.6	71.2
166.8	58.2
157.1	56.0
181.1	64.5
158.4	53.0
165.6	52.4
166.7	56.8
156.5	49.2
168.1	55.6
165.3	77.8
$\bar{X} = 165.52$	$\bar{Y} = 59.47$

We obtain

$$S_{XX} = 472.076,$$
$$S_{YY} = 731.961,$$
$$S_{XY} = 274.786.$$

Plot Y against X. Run *OLS* regressions of Y on X and the reverse regression of Y on X. Check that the fitted regression line goes through the means of X and Y.

5. Consider the simple regression model

$$y_t = \alpha + \beta x_t + u_t, \qquad t = 1, 2, \ldots, T$$

where x_t is the explanatory variable and u_t is the unobserved disturbance term.

(a) Explain briefly what is meant by saying that an estimator, $\hat{\beta}$, of β is:
   i. unbiased
   ii. consistent
   iii. maximum likelihood.

(b) Under what assumptions is the *OLS* estimator of β:
   i. the best linear unbiased estimator
   ii. the maximum likelihood estimator

(c) For each of the assumptions you have listed under (b) give an example where the assumption might not hold in economic applications.

(d) In the model above, why do econometricians make assumptions about the distribution of u_t when testing a hypothesis about the value of β?

6. Consider the following two specifications

$$W_i = a + b \log(E_i) + \varepsilon_i,$$
$$\ln(W_i) = \alpha + \beta \log(E_i) + v_i,$$

where $W_i = P F_i / E_i$, is the share of food expenditure of household i, P is the price of food assumed fixed across all households, $E_i = F_i + NF_i$, with F_i and NF_i are respectively food and non-food expenditures of the household, ε_i and v_i are random errors, a, b, α and β are constant coefficients.

(a) How do you use these specifications to compute the elasticity of food expenditure relative to the total expenditure?

(b) Discuss the relative statistical and theoretical merits of these specifications for the analysis of food expenditure.

2 Multiple Regression

2.1 Introduction

This chapter considers the extension of the bivariate regression discussed in Chapter 1 to the case where more than one variable is available to explain/predict y_t, the dependent variable. The topic is known as multiple regression analysis, although only one relationship is in fact considered between y_t and the k explanatory variables, x_{ti}, for $i = 1, 2, \ldots, k$. The problem of multiple regressions where m sets of dependent (or endogenous) variables, y_{tj}, $j = 1, 2, \ldots, m$ are explained in terms of x_{ti}, for $i = 1, 2, \ldots, k$ will be considered in Chapter 19 and is known as multivariate analysis and includes topics such as canonical correlation and factor analysis. This chapter covers standard techniques such as ordinary least squares (*OLS*) and examines the properties of *OLS* estimators under classical assumption, discusses the Gauss–Markov theorem, multiple correlation coefficient, the multicollinearity problem, partitioned regression, introduces regressions that are nonlinear in variables and discusses the interpretation of coefficients.

2.2 The classical normal linear regression model

Consider the general linear regression model

$$y_t = \sum_{j=1}^{k} \beta_j x_{tj} + u_t, \quad \text{for } t = 1, 2, \ldots, T, \tag{2.1}$$

where $x_{t1}, x_{t2}, \ldots, x_{tk}$ are the tth observation on k regressors. If the regression contains an intercept, then one of the k regressors, say the first one x_{t1}, is set equal to unity for all t, namely $x_{t1} = 1$. The parameters $\beta_1, \beta_2, \ldots, \beta_k$ assumed to be fixed (i.e., time invariant) are the regression coefficients, and u_t are the 'disturbances' or the 'errors' of the regression equation. The regression equation can also be written more compactly as

$$y_t = \boldsymbol{\beta}' \mathbf{x}_t + u_t, \quad \text{for } t = 1, 2, \ldots, T, \tag{2.2}$$

where $\boldsymbol{\beta} = (\beta_1, \beta_2, \ldots, \beta_k)'$ and $\mathbf{x}_t = (x_{t1}, x_{t2}, \ldots, x_{tk})'$. Stacking the equations for all the T observation and using matrix notations, (2.1) or (2.2) can be written as (see Appendix A for an introduction to matrices and matrix operations)

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{u}, \tag{2.3}$$

where

$$\mathbf{X} = \begin{pmatrix} x_{11} & x_{12} & \cdots & x_{1k} \\ x_{21} & x_{22} & \cdots & x_{2k} \\ \vdots & \vdots & \ddots & \vdots \\ x_{T1} & x_{T2} & \cdots & x_{Tk} \end{pmatrix}, \quad \mathbf{y} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_T \end{pmatrix}, \quad \mathbf{u} = \begin{pmatrix} u_1 \\ u_2 \\ \vdots \\ u_T \end{pmatrix}.$$

The disturbances u_t (or $\mathbf{u}$) satisfy the following assumptions:

Assumption A1: *Zero mean*: the disturbances u_t have zero means

$$E(\mathbf{u}) = \mathbf{0}, \text{ or } E(u_t) = 0, \quad \text{for all} \quad t.$$

Assumption A2: *Homoskedasticity*: the disturbances u_t have constant conditional variances

$$Var(u_t \,|\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_T) = \sigma^2 > 0, \quad \text{for all } t.$$

Assumption A3: *Non-autocorrelated errors*: the disturbances u_t are serially uncorrelated

$$Cov(u_t, u_s \,|\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_T) = 0, \quad \text{for all} \quad t \neq s.$$

Assumption A4: *Orthogonality*: the disturbances u_t and the regressors $x_{t1}, x_{t2}, \ldots, x_{tk}$ are uncorrelated

$$E(u_t \,|\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_T) = 0, \quad \text{for all } t.$$

Assumption A5: *Normality*: the disturbances u_t are normally distributed.

Assumption A2 implies that the variances of u_ts are constant also unconditionally, since,[1]

$$Var\,(u_t) = Var\,[E(u_t \,|\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_T)] + E\,[Var(u_t \,|\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_T)] = \sigma^2,$$

given that, under A4, $E(u_t \,|\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_T) = 0$. The assumption of constant conditional and unconditional error variances is likely to be violated when dealing with cross-sectional regressions, while that of constant conditional error variances is often violated in analysis of financial and macro-economic times series, such as exchange rates, stock returns and interest rates. However, it is possible for errors to be unconditionally constant (time-invariant) but conditionally

[1] See Appendix B, result (B.22).

time varying. Examples include stationary autoregressive conditional heteroskedastic (ARCH) models developed by Engle (1982) and discussed in detail in Chapters 18 and 25.

In time series analysis the critical assumptions are A3 and A4. Assumption A3 is particularly important when the regression equation contains lagged values of the dependent variable, namely $y_{t-1}, y_{t-2}, \ldots$. However, even if lagged values of y_t are not included among the regressors, the breakdown of assumption A3 can lead to misleading inferences, a problem recognized as early as 1920s by Yule (1926), and known in the econometrics time series literature as the spurious regression problem.[2] The orthogonality assumption, A4, allows the empirical analysis of the relationship between y_t and $x_{t1}, x_{t2}, \ldots, x_{tk}$ to be carried out without fully specifying the stochastic processes generating the regressors, also known as 'forcing' variables. We notice that assumption A1 is implied by A4, if a vector of ones is included among the regressors. It is therefore important that an intercept is always included in the regression model, unless it is found to be statistically insignificant.

As they stand, assumptions A2, A3, and A4 require the regressors to be strictly exogenous, in the sense that the first- and second-order moments of the errors, u_t, $t = 1, 2, \ldots, T$, are uncorrelated with the current, past *and* future values of the regressors (see Section 9.3 for a discussion of strict and weak exogeneity, and their impact on the properties of estimators). This assumption is too restrictive for many applications in economics and in effect treats the regressors as given which is more suitable to outcomes of experimental designs rather than economic observations that are based on survey data of transaction prices and quantities. The strict exogeneity assumption also rules out the inclusion of lagged values of y_t amongst the regressors. However, it is possible to relax these assumptions somewhat so that it is only required that the first- and second-order moments of the errors are uncorrelated with current and past values of the regressors, but allowing for the errors to be correlated with the future values of the regressors. In this less restrictive setting, assumptions A2–A4 need to be replaced by the following assumptions:

Assumption A2(i) *Homoskedasticity*: the disturbances u_t have constant conditional variances

$$Var(u_t \,|\mathbf{x}_\ell\,) = \sigma^2 > 0, \quad \text{for all } \ell \leq t.$$

Assumption A3(i) *Non-autocorrelated errors*: the disturbances u_t are serially uncorrelated

$$Cov(u_t, u_s \,|\mathbf{x}_\ell\,) = 0, \quad \text{for all} \quad t \neq s \text{ and } \ell \leq \min(t, s).$$

Assumption A4(i) *Orthogonality*: the disturbances u_t and the regressors $x_{t1}, x_{t2}, \ldots, x_{tk}$ are uncorrelated

$$E(u_t \,|\mathbf{x}_\ell\,) = 0, \quad \text{for all } \ell \leq t.$$

Under these assumptions the regressors are said to be *weakly exogenous*, and allow for lagged values of y_t to be included in $\mathbf{x}_t$.

Adding assumption A5 to the classical model yields the classical linear normal regression model. This model can also be derived using the *joint* distribution of y_t, $\mathbf{x}_t$, and by assuming

[2] Champernowne (1960) and Granger and Newbold (1974) provide Monte Carlo evidence on the spurious regression problem, and Phillips (1986) establishes a number of theoretical results.

that this distribution is a multivariate *normal* with constant means, variances and covariances. In this setting, the regression of y_t on $\mathbf{x}_t$, defined as the mathematical expectation of y_t conditional on the realized values of the regressors, will be linear in the regressors. The linearity of the regression equation follows from the joint normality assumption and need not hold if this assumption is relaxed. To be more precise suppose that

$$\begin{pmatrix} y_t \\ \mathbf{x}_t \end{pmatrix} \backsim N(\boldsymbol{\mu}, \boldsymbol{\Sigma}), \tag{2.4}$$

where

$$\boldsymbol{\mu} = \begin{pmatrix} \mu_y \\ \boldsymbol{\mu}_x \end{pmatrix}, \text{ and } \Sigma = \begin{pmatrix} \sigma_{yy} & \boldsymbol{\sigma}_{yx} \\ \boldsymbol{\sigma}_{xy} & \boldsymbol{\Sigma}_{xx} \end{pmatrix}.$$

Then using known results from theory of multivariate normal distributions (see Appendix B for a summary and references) we have

$$\begin{aligned} E(y_t | \mathbf{x}_t) &= \mu_y + \boldsymbol{\sigma}_{yx}\boldsymbol{\Sigma}_{xx}^{-1}(\mathbf{x}_t - \boldsymbol{\mu}_x), \\ Var(y_t | \mathbf{x}_t) &= \sigma_{yy} - \boldsymbol{\sigma}_{yx}\boldsymbol{\Sigma}_{xx}^{-1}\sigma_{xy}. \end{aligned}$$

Under this setting, assuming that (2.2) includes an intercept, the regression coefficients $\boldsymbol{\beta}$ will be given by $(\mu_y - \boldsymbol{\sigma}_{yx}\boldsymbol{\Sigma}_{xx}^{-1}\boldsymbol{\mu}_x, \boldsymbol{\sigma}_{yx}\boldsymbol{\Sigma}_{xx}^{-1})'$. It is also easily seen that the regression errors associated with (2.4) are given by

$$u_t = y_t - (\mu_y - \boldsymbol{\sigma}_{yx}\boldsymbol{\Sigma}_{xx}^{-1}\boldsymbol{\mu}_x) - \boldsymbol{\sigma}_{yx}\boldsymbol{\Sigma}_{xx}^{-1}\mathbf{x}_t,$$

and, by construction, satisfy the classical assumptions. But note that no dynamic effects are allowed in the distribution of $(y_t, \mathbf{x}_t')'$.

Both of the above interpretations of the classical normal regression model have been used in the literature (see, e.g., Spanos (1989)). We remark that the normality assumption A5 may be important in small samples, but is not generally required when the sample under consideration is large enough.

All the various departures from the classical normal regression model mentioned here will be analysed in Chapters 3 to 6.

2.3 The method of ordinary least squares in multiple regression

The criterion function in this general case will be

$$Q(\beta_1, \beta_2, \ldots, \beta_k) = \sum_{t=1}^{T}\left(y_t - \sum_{j=1}^{k}\beta_j x_{tj}\right)^2. \tag{2.5}$$

The necessary conditions for the minimization of $Q\left(\beta_1, \beta_2, \ldots, \beta_k\right)$ are given by

$$\frac{\partial Q\left(\beta_1, \beta_2, \ldots, \beta_k\right)}{\partial \beta_s} = -2\sum_{t=1}^{T} x_{ts}\left(y_t - \sum_{j=1}^{k}\hat{\beta}_j x_{tj}\right) = 0, \qquad s = 1, 2, \ldots, k, \tag{2.6}$$

where $\hat{\beta}_j$ is the *OLS* estimator of β_j. The k equations in (2.6) are known as the 'normal' equations. Denoting the residuals by $\hat{u}_t = y_t - \sum_j \hat{\beta}_j x_{tj}$, the normal equations can be written as $\sum_{t=1}^{T} x_{ts}\hat{u}_t = 0$, for $s = 1, 2, \ldots, k$, or, in expanded form

$$\begin{aligned}\sum_{t=1}^{T} x_{ts}y_t &= \sum_{t=1}^{T}\sum_{j=1}^{k}\hat{\beta}_j x_{tj}x_{ts} \\ &= \sum_{j=1}^{k}\hat{\beta}_j\left(\sum_{t=1}^{T} x_{tj}x_{ts}\right).\end{aligned}$$

Without the use of matrix notations, the study of the properties of multiple regression would be extremely tedious. In matrix form, the criterion function (2.5) to be minimized is

$$Q\left(\boldsymbol{\beta}\right) = \left(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\right)'\left(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\right), \tag{2.7}$$

and the first-order conditions become

$$\frac{\partial Q\left(\boldsymbol{\beta}\right)}{\partial \boldsymbol{\beta}} = -2\mathbf{X}'\left(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}\right) = \mathbf{0},$$

which yield the normal equations,

$$\left(\mathbf{X}'\mathbf{X}\right)\hat{\boldsymbol{\beta}} = \mathbf{X}'\mathbf{y}.$$

Suppose now that $\mathbf{X}'\mathbf{X}$ is of full rank, that is $\text{Rank}\left(\mathbf{X}'\mathbf{X}\right) = k$, [or $\text{Rank}(\mathbf{X}) = k$] a necessary condition for this is that $k \leq T$. There should be at least as many observations as there are unknown coefficients. Then

$$\hat{\boldsymbol{\beta}} = \left(\mathbf{X}'\mathbf{X}\right)^{-1}\mathbf{X}'\mathbf{y}. \tag{2.8}$$

In the case where $\text{Rank}(\mathbf{X}) = r < k$, $\hat{\boldsymbol{\beta}} = \left(\mathbf{X}'\mathbf{X}\right)^{-}\mathbf{X}'\mathbf{y}$, where $\left(\mathbf{X}'\mathbf{X}\right)^{-}$ represents the generalized inverse of $\mathbf{X}'\mathbf{X}$. In this case only r linear combinations of the regression coefficients are uniquely determined.

2.4 The maximum likelihood approach

Under the normality assumption A5, the *OLS* estimator can be derived by maximizing the likelihood function associated to model (2.2) (or (2.3)). Let $\boldsymbol{\theta} = \left(\boldsymbol{\beta}', \sigma^2\right)'$, then the likelihood of

a sample of T independent, identically and normally distributed disturbances is[3]

$$L_T(\boldsymbol{\theta}) = \left(2\pi\sigma^2\right)^{-T/2} \exp\left(-\frac{1}{2\sigma^2}\sum_{t=1}^{T} u_t^2\right)$$
$$= \left(2\pi\sigma^2\right)^{-T/2} \exp\left[-\frac{1}{2\sigma^2}\sum_{t=1}^{T}\left(y_t - \boldsymbol{\beta}'\mathbf{x}_t\right)^2\right].$$

Adopting the matrix notation,

$$L_T(\boldsymbol{\theta}) = \left(2\pi\sigma^2\right)^{-T/2} \exp\left[-\frac{1}{2\sigma^2}\left(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\right)'\left(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\right)\right]. \tag{2.9}$$

Taking logs, we obtain the log-likelihood function for the classical linear regression model

$$\ell_T(\boldsymbol{\theta}) = \log L_T(\boldsymbol{\theta}) = -\frac{T}{2}\log\left(2\pi\sigma^2\right) - \frac{1}{2\sigma^2}\sum_{t=1}^{T}\left(y_t - \boldsymbol{\beta}'\mathbf{x}_t\right)^2. \tag{2.10}$$

The necessary conditions for maximizing (2.10) are

$$\begin{pmatrix} \frac{\partial \ell_T(\boldsymbol{\theta})}{\partial \boldsymbol{\beta}} \\ \frac{\partial \ell_T(\boldsymbol{\theta})}{\partial \sigma^2} \end{pmatrix} = \begin{pmatrix} \frac{1}{\sigma^2}\mathbf{X}'\left(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\right) \\ -\frac{T}{2\sigma^2} + \frac{1}{2\sigma^4}\left(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\right)'\left(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\right) \end{pmatrix} = \begin{pmatrix} \mathbf{0} \\ 0 \end{pmatrix}.$$

The values that satisfy these equations are

$$\widetilde{\boldsymbol{\beta}} = \left(\mathbf{X}'\mathbf{X}\right)^{-1}\mathbf{X}'\mathbf{y}, \text{ and } \widetilde{\sigma}^2 = \frac{\sum_t \widetilde{u}_t^2}{T} = \frac{\widetilde{\mathbf{u}}'\widetilde{\mathbf{u}}}{T},$$

where $\widetilde{\mathbf{u}} = \mathbf{y} - \mathbf{X}\widetilde{\boldsymbol{\beta}}$. Notice that the estimator for the slope coefficients is identical to the *OLS* estimator (2.8), while the variance estimator differs from (2.14) by the divisor of T instead of $T - k$. Clearly, the *OLS* estimator inherits all the asymptotic properties of the *ML* estimator. We refer to Chapter 9 for a review of the theory underlying the maximum likelihood approach, and to Chapter 19 for an extension of the above results to the case of multivariate regression.

The likelihood approach also forms the basis of the Bayesian inference where the likelihood is combined with prior distributions on the unknown parameters to obtain posterior probability distributions which is then used for estimation and inference: see Section C.6 in Appendix C.

[3] See also Section 1.8 where the likelihood approach is introduced for the analysis of bivariate regression models.

2.5 Properties of *OLS* residuals

The residual vector is given by

$$\begin{aligned}\hat{\mathbf{u}} &= \mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{y} - \mathbf{X}\left(\mathbf{X}'\mathbf{X}\right)^{-1}\mathbf{X}'\mathbf{y} \\ &= \left[\mathbf{I}_T - \mathbf{X}\left(\mathbf{X}'\mathbf{X}\right)^{-1}\mathbf{X}'\right]\mathbf{y} \\ &= \mathbf{M}\mathbf{y},\end{aligned}$$

where $\mathbf{I}_T$ is an identity matrix of order T, $\mathbf{M} = \mathbf{I}_T - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$, with the property $\mathbf{M}^2 = \mathbf{M}$, which makes $\mathbf{M}$ to be an *idempotent* matrix. Also $\mathbf{M} = \mathbf{I}_T - \mathbf{P}$, where $\mathbf{P} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$ is called the *projection matrix* of the regression (2.3). Note that

$$\begin{aligned}\mathbf{M}\mathbf{X} &= \left[\mathbf{I}_T - \mathbf{X}\left(\mathbf{X}'\mathbf{X}\right)^{-1}\mathbf{X}'\right]\mathbf{X} \\ &= \mathbf{X} - \mathbf{X}\left(\mathbf{X}'\mathbf{X}\right)^{-1}\mathbf{X}'\mathbf{X} \\ &= \mathbf{X} - \mathbf{X} = \mathbf{0}.\end{aligned}$$

Therefore

$$\mathbf{X}'\hat{\mathbf{u}} = \mathbf{X}'\mathbf{M}\mathbf{y} = \mathbf{0}, \tag{2.11}$$

or $\sum_{t=1}^{T} x_{ts}\hat{u}_t = 0$, for $s = 1, 2, \ldots, k$ which are the normal equations of the regression problem. Therefore, the regressors are by construction 'orthogonal' to the vector of *OLS* residuals.

In the case where the regression equation contains an intercept term (i.e., when one of the x_{tj}'s is equal to 1 for all t) we also have

$$\sum_{t=1}^{T}\hat{u}_t = 0 = T\left(\bar{y} - \hat{\beta}_1\bar{x}_1 - \hat{\beta}_2\bar{x}_2 - \ldots - \hat{\beta}_k\bar{x}_k\right) = 0,$$

where $\bar{x}_j$ stands for the sample mean of the jth regressor, x_{tj}. This result follows directly from the normal equations $\sum_{t=1}^{T} x_{ts}\hat{u}_t = 0$, by choosing x_{ts} to be the intercept term, namely setting $x_{ts} = 1$ in $\sum_{t=1}^{T} x_{ts}\hat{u}_t = 0$.

To summarize, the *OLS* residual vector, $\hat{\mathbf{u}}$, has the following properties:

(i) By construction all the regressors are orthogonal to the residual vector, that is, $\mathbf{X}'\hat{\mathbf{u}} = 0$.

(ii) When the regression equation contains an intercept term, the residuals, $\hat{u}_t$, have mean zero exactly, i.e. $\sum_{t=1}^{T}\hat{u}_t = 0$. This result also implies that the regression plane goes through the sample mean of $\mathbf{y}$ and the sample means of all the regressors.

(iii) Even if u_t are homoskedastic and serially uncorrelated, the *OLS* residuals, $\hat{u}_t$, will be *heteroskedastic* and *autocorrelated* in small samples.

Result (iii) follows by noting that

$$\hat{\mathbf{u}} = \mathbf{M}\mathbf{y} = \mathbf{M}\left(\mathbf{X}\boldsymbol{\beta} + \mathbf{u}\right) = \mathbf{M}\mathbf{u},$$

and

$$E\left(\hat{\mathbf{u}}\hat{\mathbf{u}}'\right) = E\left(\mathbf{M}\mathbf{u}\mathbf{u}'\mathbf{M}'\right) = \mathbf{M}E\left(\mathbf{u}\mathbf{u}'\right)\mathbf{M}'.$$

But under the classical assumptions $E\left(\mathbf{u}\mathbf{u}'\right) = \sigma^2\mathbf{I}_T$. Hence

$$E\left(\hat{\mathbf{u}}\hat{\mathbf{u}}'\right) = \mathbf{M}\left(\sigma^2\mathbf{I}_T\right)\mathbf{M}' = \sigma^2\mathbf{M}\mathbf{M}' = \sigma^2\mathbf{M},$$

which is different from an identity matrix and establishes that $\hat{u}_t$ and $\hat{u}_{t'}$ $\left(t \neq t'\right)$ are neither uncorrelated nor homoskedastic. These properties of *OLS* residuals lie at the core of some of the difficulties encountered in practice in developing tests of the classical assumptions based on *OLS* residuals, that perform well in small samples. Fortunately, the serial correlation and heteroskedasticity properties of *OLS* residuals tend to disappear in 'large enough' samples.

2.6 Covariance matrix of $\hat{\boldsymbol{\beta}}$

The covariance matrix of $\hat{\boldsymbol{\beta}}$ is defined as

$$\begin{aligned} Var\left(\hat{\boldsymbol{\beta}}\right) &= E\left\{\left[\hat{\boldsymbol{\beta}} - E\left(\hat{\boldsymbol{\beta}}\right)\right]\left[\hat{\boldsymbol{\beta}} - E\left(\hat{\boldsymbol{\beta}}\right)\right]'\right\} \\ &= \begin{pmatrix} Var\left(\hat{\beta}_1\right) & \text{Cov}\left(\hat{\beta}_1, \hat{\beta}_2\right) & \cdots & \text{Cov}\left(\hat{\beta}_1, \hat{\beta}_k\right) \\ \text{Cov}\left(\hat{\beta}_2, \hat{\beta}_1\right) & Var\left(\hat{\beta}_2\right) & \cdots & \text{Cov}\left(\hat{\beta}_2, \hat{\beta}_k\right) \\ \vdots & & & \\ \text{Cov}\left(\hat{\beta}_k, \hat{\beta}_1\right) & \text{Cov}\left(\hat{\beta}_k, \hat{\beta}_2\right) & \cdots & Var\left(\hat{\beta}_k\right) \end{pmatrix}. \end{aligned} \tag{2.12}$$

The diagonal elements of the matrix $Var\left(\hat{\boldsymbol{\beta}}\right)$ are the variances of the *OLS* estimators, $\hat{\boldsymbol{\beta}} = \left(\hat{\beta}_1, \hat{\beta}_2, \ldots, \hat{\beta}_k\right)'$, and the off-diagonal elements are the covariances.

To obtain the formula for $Var\left(\hat{\boldsymbol{\beta}}\right)$ we first note that

$$\begin{aligned} \hat{\boldsymbol{\beta}} &= \left(\mathbf{X}'\mathbf{X}\right)^{-1}\mathbf{X}'\mathbf{y} = \left(\mathbf{X}'\mathbf{X}\right)^{-1}\mathbf{X}'\left(\mathbf{X}\boldsymbol{\beta} + \mathbf{u}\right) \\ &= \boldsymbol{\beta} + \left(\mathbf{X}'\mathbf{X}\right)^{-1}\mathbf{X}'\mathbf{u}. \end{aligned}$$

But $E(\mathbf{X}'\mathbf{u}\,|\mathbf{X}\,) = \mathbf{X}'E(\mathbf{u}\,|\mathbf{X}\,)$ and, under assumption A4, $E(\mathbf{u}\,|\mathbf{X}\,) = \mathbf{0}$, and hence $E\left(\hat{\boldsymbol{\beta}}\right) = \boldsymbol{\beta}$, namely that $\hat{\boldsymbol{\beta}}$ is an unbiased estimator of $\boldsymbol{\beta}$. Also

$$\hat{\boldsymbol{\beta}} - E\left(\hat{\boldsymbol{\beta}}\right) = \left(\mathbf{X}'\mathbf{X}\right)^{-1}\mathbf{X}'\mathbf{u}.$$

Therefore

$$Var\left(\hat{\boldsymbol{\beta}}\right) = E\left[\left(\mathbf{X}'\mathbf{X}\right)^{-1}\mathbf{X}'\mathbf{u}\mathbf{u}'\mathbf{X}\left(\mathbf{X}'\mathbf{X}\right)^{-1}\right].$$

Again under assumption A4

$$E\left[\left(\mathbf{X}'\mathbf{X}\right)^{-1}\mathbf{X}'\mathbf{u}\mathbf{u}'\mathbf{X}\left(\mathbf{X}'\mathbf{X}\right)^{-1}|\mathbf{X}\right]=\left(\mathbf{X}'\mathbf{X}\right)^{-1}\mathbf{X}'E\left(\mathbf{u}\mathbf{u}'|\mathbf{X}\right)\mathbf{X}\left(\mathbf{X}'\mathbf{X}\right)^{-1},$$

and under assumptions A2 and A3, $E\left(\mathbf{u}\mathbf{u}'|\mathbf{X}\right)=\sigma^2\mathbf{I}_T$. Therefore,

$$E\left[\left(\mathbf{X}'\mathbf{X}\right)^{-1}\mathbf{X}'\mathbf{u}\mathbf{u}'\mathbf{X}\left(\mathbf{X}'\mathbf{X}\right)^{-1}|\mathbf{X}\right]=\sigma^2\left(\mathbf{X}'\mathbf{X}\right)^{-1},$$

and hence

$$Var\left(\hat{\boldsymbol{\beta}}\right)=\sigma^2E\left[\left(\mathbf{X}'\mathbf{X}\right)^{-1}\right]. \tag{2.13}$$

For given values of $\mathbf{X}$ an estimator of $Var\left(\hat{\boldsymbol{\beta}}\right)$ is

$$\widehat{Var}\left(\hat{\boldsymbol{\beta}}\right)=\hat{\sigma}^2\left(\mathbf{X}'\mathbf{X}\right)^{-1},$$

where $\hat{\sigma}^2$ is

$$\hat{\sigma}^2=\frac{\sum_t\hat{u}_t^2}{T-k}=\frac{\hat{\mathbf{u}}'\hat{\mathbf{u}}}{T-k}, \tag{2.14}$$

with k being the number of regressors, including the intercept term. As in the case of the simple regression model, $\hat{\sigma}^2$ is an unbiased estimator of σ^2, namely $E(\hat{\sigma}^2)=\sigma^2$. Unbiasedness of $\hat{\sigma}^2$ is easily established by noting that $\hat{\mathbf{u}}=\mathbf{M}\mathbf{u}$ and hence

$$\begin{aligned}E\left(\hat{\sigma}^2\right)&=\left(\frac{1}{T-k}\right)E\left(\mathbf{u}'\mathbf{M}\mathbf{u}\right)\\&=\left(\frac{1}{T-k}\right)E\left[Tr\left(\mathbf{u}'\mathbf{M}\mathbf{u}\right)\right]=\left(\frac{1}{T-k}\right)E\left[Tr\left(\mathbf{u}\mathbf{M}\mathbf{u}'\right)\right]\\&=\left(\frac{1}{T-k}\right)Tr\left[\mathbf{M}E\left(\mathbf{u}\mathbf{u}'\right)\right]=\left(\frac{1}{T-k}\right)Tr\left(\mathbf{M}\sigma^2\right),\end{aligned}$$

Noting that

$$\begin{aligned}Tr\left(\mathbf{M}\right)&=Tr\left[\mathbf{I}_T-\mathbf{X}\left(\mathbf{X}'\mathbf{X}\right)^{-1}\mathbf{X}'\right]\\&=Tr\left(\mathbf{I}_T\right)-Tr\left[\mathbf{X}\left(\mathbf{X}'\mathbf{X}\right)^{-1}\mathbf{X}'\right]=T-k,\end{aligned}$$

it follows that

$$E\left(\hat{\sigma}^2\right)=\frac{\sigma^2Tr\left(\mathbf{M}\right)}{T-k}=\sigma^2.$$

The estimator of Cov $\left(\hat{\beta}_j, \hat{\beta}_s\right)$ is given by the $(j,s)^{th}$ element of matrix $\hat{\sigma}^2 \left(\mathbf{X}'\mathbf{X}\right)^{-1}$.

Example 1 *Consider the three variable regression model*

$$y_t = \beta_1 + \beta_2 x_{t2} + \beta_3 x_{t3} + u_t, \qquad t = 1, 2, \ldots, T, \tag{2.15}$$

where we have set the first variable, x_{t1}, equal to unity to allow for an intercept in the regression. To simplify the derivations we work with variables in terms of their deviations from their respective sample means. Summing the equation (2.15) over t and dividing by the sample size, T, yields:

$$\bar{y} = \beta_1 + \beta_2 \bar{x}_2 + \beta_3 \bar{x}_3 + \bar{u}, \tag{2.16}$$

where $\bar{y} = \sum_t y_t/T$, $\bar{x}_2 = \sum_t x_{t2}/T$, $\bar{x}_3 = \sum_t x_{t3}/T$, $\bar{u} = \sum_t u_t/T$ are the sample means. Subtracting (2.16) from (2.15) we obtain

$$y_t - \bar{y} = \beta_2 \left(x_{t2} - \bar{x}_2\right) + \beta_3 \left(x_{t3} - \bar{x}_3\right) + \left(u_t - \bar{u}\right).$$

The OLS estimators of β_2 and β_3 are now given by (using (2.8))

$$\begin{pmatrix} \hat{\beta}_2 \\ \hat{\beta}_3 \end{pmatrix} = \begin{pmatrix} S_{22} & S_{23} \\ S_{23} & S_{33} \end{pmatrix}^{-1} \begin{pmatrix} S_{2y} \\ S_{3y} \end{pmatrix},$$

where

$$\begin{aligned} S_{js} = \textstyle\sum_t \left(x_{tj} - \bar{x}_j\right)\left(x_{ts} - \bar{x}_s\right) = \sum_t \left(x_{tj} - \bar{x}_j\right) x_{ts}, \qquad & j, s = 2, 3, \\ S_{jy} = \textstyle\sum_t \left(x_{tj} - \bar{x}_j\right) y_t, \qquad & j = 2, 3, \end{aligned}$$

$$\begin{pmatrix} S_{22} & S_{23} \\ S_{23} & S_{33} \end{pmatrix}^{-1} = \frac{1}{S_{22}S_{33} - S_{23}^2} \begin{bmatrix} S_{33} & -S_{23} \\ -S_{23} & S_{22} \end{bmatrix}.$$

Hence

$$\hat{\beta}_2 = \frac{S_{33}S_{2y} - S_{23}S_{3y}}{S_{22}S_{33} - S_{23}^2}, \tag{2.17}$$

$$\hat{\beta}_3 = \frac{S_{22}S_{3y} - S_{23}S_{2y}}{S_{22}S_{33} - S_{23}^2}. \tag{2.18}$$

The estimator of β_1, the intercept term, can now be obtained recalling that the regression plane goes through the sample means when the equation has an intercept term. Namely

$$\bar{y} = \hat{\beta}_1 + \hat{\beta}_2 \bar{x}_2 + \hat{\beta}_3 \bar{x}_3,$$

and hence

$$\hat{\beta}_1 = \bar{y} - \hat{\beta}_2 \bar{x}_2 - \hat{\beta}_3 \bar{x}_3. \tag{2.19}$$

The estimates of the variances and the covariance of $\hat{\beta}_2$ and $\hat{\beta}_3$ are given by [using (2.12) and (2.13)]

$$\widehat{Cov}\begin{pmatrix} \hat{\beta}_2 \\ \hat{\beta}_3 \end{pmatrix} = \hat{\sigma}^2 \begin{pmatrix} S_{22} & S_{23} \\ S_{23} & S_{33} \end{pmatrix}^{-1},$$

or

$$\widehat{Var}\left(\hat{\beta}_2\right) = \frac{\hat{\sigma}^2 S_{33}}{S_{22}S_{33} - S_{23}^2}, \tag{2.20}$$

$$\widehat{Var}\left(\hat{\beta}_3\right) = \frac{\hat{\sigma}^2 S_{22}}{S_{22}S_{33} - S_{23}^2}, \tag{2.21}$$

and

$$\widehat{Cov}\left(\hat{\beta}_2, \hat{\beta}_3\right) = -\frac{\hat{\sigma}^2 S_{23}}{S_{22}S_{33} - S_{23}^2}. \tag{2.22}$$

Finally,

$$\hat{\sigma}^2 = \frac{\sum_t \hat{u}_t^2}{T-3} = \frac{\sum_t \left(y_t - \hat{\beta}_1 - \hat{\beta}_2 x_{t2} - \hat{\beta}_3 x_{t3}\right)^2}{T-3}. \tag{2.23}$$

Notice that the denominator of $\hat{\sigma}^2$ is $T - 3$, as we have estimated three coefficients, namely the intercept term, β_1, and the two regression coefficients, β_2 and β_3.

2.7 The Gauss–Markov theorem

The Gauss–Markov theorem states that under the classical assumptions A1–A4 the *OLS* estimator (2.8) has the least variance in the class of all linear unbiased estimators of $\boldsymbol{\beta}$, namely it is the best linear unbiased estimator (BLUE). More formally, let $\boldsymbol{\beta}^*$ be an alternative linear unbiased estimator of $\boldsymbol{\beta}$ defined by

$$\boldsymbol{\beta}^* = \hat{\boldsymbol{\beta}} + \mathbf{C}'\mathbf{y}, \tag{2.24}$$

where $\mathbf{C}$ is a $k \times T$ matrix with elements possibly depending on $\mathbf{X}$, but not on $\mathbf{y}$. It is clear that $\boldsymbol{\beta}^*$ is a linear estimator. Also since $\hat{\boldsymbol{\beta}}$ is an unbiased estimator of $\boldsymbol{\beta}$, for $\boldsymbol{\beta}^*$ to be an unbiased estimator we need

$$E\left(\boldsymbol{\beta}^*\right) = E\left(\hat{\boldsymbol{\beta}}\right) + \mathbf{C}'E\left(\mathbf{y}\right) = \boldsymbol{\beta} + \mathbf{C}'E\left(\mathbf{y}\right) = \boldsymbol{\beta},$$

or that $\mathbf{C}'E\left(\mathbf{y}\right) = \mathbf{0}$, which in turn implies that

$$\mathbf{C}'E\left(\mathbf{y}\right) = \mathbf{C}'\left(\mathbf{X}\boldsymbol{\beta} + E\left(\mathbf{u}\right)\right) = \mathbf{C}'\mathbf{X}\boldsymbol{\beta} = 0, \tag{2.25}$$

for all values of $\boldsymbol{\beta}$.

To prove the Gauss–Markov we need to show that subject to the unbiasedness condition (2.25), $Var(\hat{\boldsymbol{\beta}}) \leq Var\left(\boldsymbol{\beta}_*\right)$, in the sense that $Var\left(\boldsymbol{\beta}_*\right) - Var(\hat{\boldsymbol{\beta}})$ is a semi-positive definite matrix. Using (2.3) and (2.8) in (2.24), we have

$$\begin{aligned}\boldsymbol{\beta}^* &= \left[\left(\mathbf{X}'\mathbf{X}\right)^{-1}\mathbf{X}' + \mathbf{C}'\right]\mathbf{y} \\ &= \left[\left(\mathbf{X}'\mathbf{X}\right)^{-1}\mathbf{X}' + \mathbf{C}'\right]\left(\mathbf{X}\boldsymbol{\beta} + \mathbf{u}\right),\end{aligned}$$

or

$$\boldsymbol{\beta}^* - \boldsymbol{\beta} = \mathbf{C}'\mathbf{X}\boldsymbol{\beta} + \left[\left(\mathbf{X}'\mathbf{X}\right)^{-1}\mathbf{X}' + \mathbf{C}'\right]\mathbf{u}.$$

But using (2.25), $\mathbf{C}'\mathbf{X}\boldsymbol{\beta} = \mathbf{0}$ and

$$\boldsymbol{\beta}^* - \boldsymbol{\beta} = \left[\left(\mathbf{X}'\mathbf{X}\right)^{-1}\mathbf{X}' + \mathbf{C}'\right]\mathbf{u}.$$

Hence (for a given set of observations, $\mathbf{X}$)

$$\begin{aligned}Var\left(\boldsymbol{\beta}^*\right) &= E\left[\left(\boldsymbol{\beta}^* - \boldsymbol{\beta}\right)\left(\boldsymbol{\beta}^* - \boldsymbol{\beta}\right)'\right] \\ &= \sigma^2\left[\left(\mathbf{X}'\mathbf{X}\right)^{-1} + \mathbf{C}'\mathbf{C} + \left(\mathbf{X}'\mathbf{X}\right)^{-1}\mathbf{X}'\mathbf{C} + \mathbf{C}'\mathbf{X}\left(\mathbf{X}'\mathbf{X}\right)^{-1}\right].\end{aligned}$$

However, since $\mathbf{C}'\mathbf{X}\boldsymbol{\beta} = \mathbf{0}$ for all parameter values, $\boldsymbol{\beta}$, then we should also have $\mathbf{C}'\mathbf{X} = 0$, and

$$Var\left(\boldsymbol{\beta}^*\right) = \sigma^2\left(\mathbf{X}'\mathbf{X}\right)^{-1} + \sigma^2\left(\mathbf{C}'\mathbf{C}\right).$$

Therefore

$$Var\left(\boldsymbol{\beta}^*\right) - Var\left(\hat{\boldsymbol{\beta}}\right) = \sigma^2\left(\mathbf{C}'\mathbf{C}\right),$$

which is a semi-positive definite matrix.

The Gauss–Markov theorem readily extends to the *OLS* estimator of any linear combination of the parameters, $\boldsymbol{\beta}$. Consider, for example, the linear combination

$$\delta = \boldsymbol{\lambda}'\boldsymbol{\beta},$$

where λ is a $k \times 1$ vector of fixed coefficients. Denote the *OLS* estimator of δ by $\hat{\delta}$, and the alternative linear unbiased estimator by δ^*. We have

$$\hat{\delta} = \boldsymbol{\lambda}'\hat{\boldsymbol{\beta}}, \qquad \delta^* = \boldsymbol{\lambda}'\boldsymbol{\beta}^*,$$

and

$$\begin{aligned} Var\left(\delta^*\right) - Var\left(\hat{\delta}\right) &= \boldsymbol{\lambda}' Var\left(\boldsymbol{\beta}^*\right)\boldsymbol{\lambda} - \boldsymbol{\lambda}' Var\left(\hat{\boldsymbol{\beta}}\right)\boldsymbol{\lambda} \\ &= \boldsymbol{\lambda}'\left[Var\left(\boldsymbol{\beta}^*\right) - Var\left(\hat{\boldsymbol{\beta}}\right)\right]\boldsymbol{\lambda}. \end{aligned}$$

But we have already shown that $Var\left(\boldsymbol{\beta}^*\right) - Var(\hat{\boldsymbol{\beta}})$ is a semi-positive definite matrix. Therefore,

$$Var\left(\delta^*\right) - Var\left(\hat{\delta}\right) \geq 0.$$

A number of other interesting results also follow from this last inequality. Setting $\boldsymbol{\lambda}' = (1, 0, \ldots, 0)$, for example, gives

$$\delta = \boldsymbol{\lambda}'\boldsymbol{\beta} = \beta_1,$$

and establishes that

$$Var\left(\beta_1^*\right) - Var\left(\hat{\beta}_1\right) \geq \mathbf{0}.$$

Similarly, $Var(\beta_j^*) - Var(\hat{\beta}_j) \geq 0$, for $j = 1, 2, \ldots, k$.

It is important to bear in mind that the Gauss–Markov theorem does not apply if the regressors are weakly exogenous even if all the other assumptions of the classical model are satisfied.

2.8 Mean square error of an estimator and the bias-variance trade-off

The Gauss–Markov theorem states that, under the classical assumptions, it is not possible to find linear unbiased estimators of regression coefficients which have smaller variances than the *OLS* estimator, (2.8). However, as shown by James and Stein (1961), it is possible to find other estimators that are biased but have a lower variance than the *OLS* estimator. The trade-off between bias and variance can be formalized if the alternative estimators are compared by their mean square error defined as

$$MSE(\widetilde{\boldsymbol{\beta}}) = E\left[(\widetilde{\boldsymbol{\beta}} - \boldsymbol{\beta}_0)(\widetilde{\boldsymbol{\beta}} - \boldsymbol{\beta}_0)'\right],$$

where $\widetilde{\boldsymbol{\beta}}$ denotes an alternative estimator to the *OLS* estimator, $\hat{\boldsymbol{\beta}}$, and $\boldsymbol{\beta}_0$ is the true value of $\boldsymbol{\beta}$. To see the bias-variance trade-off we first note that

$$E\left[(\widetilde{\boldsymbol{\beta}}-\boldsymbol{\beta}_0)(\widetilde{\boldsymbol{\beta}}-\boldsymbol{\beta}_0)'\right] = E\left\{\left[(\widetilde{\boldsymbol{\beta}}-E(\widetilde{\boldsymbol{\beta}})) - (\boldsymbol{\beta}_0 - E(\widetilde{\boldsymbol{\beta}}))\right]\left[(\widetilde{\boldsymbol{\beta}}-E(\widetilde{\boldsymbol{\beta}})) - (\boldsymbol{\beta}_0 - E(\widetilde{\boldsymbol{\beta}}))\right]'\right\}$$
$$= E\left\{\left[\widetilde{\boldsymbol{\beta}}-E(\widetilde{\boldsymbol{\beta}})\right]\left[\widetilde{\boldsymbol{\beta}}-E(\widetilde{\boldsymbol{\beta}})\right]'\right\} + E\left\{\left[\boldsymbol{\beta}_0 - E(\widetilde{\boldsymbol{\beta}})\right]\left[\boldsymbol{\beta}_0 - E(\widetilde{\boldsymbol{\beta}})\right]'\right\}$$
$$- E\left\{\left[\widetilde{\boldsymbol{\beta}}-E(\widetilde{\boldsymbol{\beta}})\right]\left[\boldsymbol{\beta}_0 - E(\widetilde{\boldsymbol{\beta}})\right]'\right\} - E\left\{\left[\boldsymbol{\beta}_0 - E(\widetilde{\boldsymbol{\beta}})\right]\left[\widetilde{\boldsymbol{\beta}} - E(\widetilde{\boldsymbol{\beta}})\right]'\right\}.$$

But $\boldsymbol{\beta}_0 - E(\widetilde{\boldsymbol{\beta}})$ is a constant (i.e., non-stochastic), and can be taken outside of the expectations operator. Also

$$E\left\{\left[\widetilde{\boldsymbol{\beta}}-E(\widetilde{\boldsymbol{\beta}})\right]\left[\widetilde{\boldsymbol{\beta}}-E(\widetilde{\boldsymbol{\beta}})\right]'\right\} = Var\left(\widetilde{\boldsymbol{\beta}}\right),$$

and by construction

$$E\left[\widetilde{\boldsymbol{\beta}}-E(\widetilde{\boldsymbol{\beta}})\right] = \mathbf{0}.$$

Hence

$$MSE(\widetilde{\boldsymbol{\beta}}) = Var\left(\widetilde{\boldsymbol{\beta}}\right) + \left[\boldsymbol{\beta}_0 - E(\widetilde{\boldsymbol{\beta}})\right]\left[\boldsymbol{\beta}_0 - E(\widetilde{\boldsymbol{\beta}})\right]'.$$

Namely, the $MSE(\widetilde{\boldsymbol{\beta}})$ can be decomposed into a variance term plus the square of the bias. In principle it is clearly possible to find an estimator for $\boldsymbol{\beta}$ with lower variance at the expense of some bias, leading to a reduction in the overall MSE. This result has been used by James and Stein (1961) to propose a biased estimator for $\boldsymbol{\beta}$ such that its MSE is smaller than the MSE of $\hat{\boldsymbol{\beta}}$. Specifically, they considered the estimator

$$\widetilde{\beta}_j = \left[1 - \frac{(k-2)\,\sigma^2}{\hat{\boldsymbol{\beta}}'\,(\mathbf{X}\mathbf{X}')\,\hat{\boldsymbol{\beta}}}\right]\hat{\beta}_j, \quad j = 1, 2, \ldots, k,$$

obtained by minimizing the overall MSE of $\widetilde{\boldsymbol{\beta}}$. James and Stein proved that this estimator, by shrinking the *OLS* estimator towards zero, has a MSE smaller than the MSE of *OLS* estimator when $k > 2$. For further details see, for example, Draper and van Nostrand (1979) and Gruber (1998).

2.9 Distribution of the *OLS* estimator

Under the classical normal assumptions A1–A5, for a given realization of the regressors, $\mathbf{X}$, the *OLS* estimator, $\hat{\boldsymbol{\beta}}$, is a linear function of u_t, for $t = 1, 2, \ldots, T$, and hence is also normally distributed. More specifically, using (2.8), note that

$$\hat{\boldsymbol{\beta}} = \boldsymbol{\beta} + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{u},$$

and since under assumptions A1–A5, $\mathbf{u} \sim N(\mathbf{0}, \sigma^2\mathbf{I}_T)$, then recalling that $\mathbf{X}'\mathbf{X}$ is a positive definite matrix, we have

$$\left(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}\right) \sim N[\mathbf{0}, \sigma^2(\mathbf{X}'\mathbf{X})^{-1}].$$

Equivalently,

$$(\mathbf{X}'\mathbf{X})^{1/2}\left(\frac{\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}}{\sigma}\right) \sim N(\mathbf{0}, \mathbf{I}_k),$$

and

$$\left(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}\right)'\left(\frac{\mathbf{X}'\mathbf{X}}{\sigma^2}\right)\left(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}\right) \sim \chi^2_k, \tag{2.26}$$

where χ^2_k stands for the central chi-square distribution with k degrees of freedom. The above result also follows unconditionally.

Consider now the distribution of $\hat{\sigma}^2$, the unbiased estimator of σ^2, given by (2.14). We note that $\hat{\mathbf{u}} = \mathbf{M}\mathbf{u}$, where $\mathbf{M} = \mathbf{I}_T - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$ is an idempotent matrix with rank $T-k$. Then the singular value decomposition of $\mathbf{M}$ is given by $\mathbf{G}\mathbf{M}\mathbf{G}' = \boldsymbol{\Lambda}$, where $\mathbf{G}$ is an orthonormal matrix such that $\mathbf{G}\mathbf{G}' = \mathbf{I}_T$, and

$$\boldsymbol{\Lambda} = \begin{pmatrix} \mathbf{I}_{T-k} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{pmatrix}.$$

Hence

$$\frac{(T-k)\,\hat{\sigma}^2}{\sigma^2} = \frac{\hat{\mathbf{u}}'\hat{\mathbf{u}}}{\sigma^2} = \frac{\mathbf{u}'\mathbf{M}\mathbf{u}}{\sigma^2} = \boldsymbol{\xi}'\boldsymbol{\Lambda}\boldsymbol{\xi},$$

where $\boldsymbol{\xi} = \sigma^{-1}\mathbf{G}\mathbf{u} \sim N(\mathbf{0}, \mathbf{I}_T)$. Partition $\boldsymbol{\xi}$ conformable to $\boldsymbol{\Lambda}$, and note that

$$\frac{(T-k)\,\hat{\sigma}^2}{\sigma^2} = \sum_{i=1}^{T-k} \xi_i^2,$$

where ξ_i are independently and identically distributed as $N(0, 1)$. Thus

$$\frac{(T-k)\,\hat{\sigma}^2}{\sigma^2} \sim \chi^2_{T-k}. \tag{2.27}$$

Finally, using (2.26) and the above result, we have

$$\frac{T-k}{k}\frac{\left(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}\right)'\left(\frac{\mathbf{X}'\mathbf{X}}{\sigma^2}\right)\left(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}\right)}{\frac{(T-k)\hat{\sigma}^2}{\sigma^2}} = \frac{\left(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}\right)'(\mathbf{X}'\mathbf{X})\left(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}\right)}{k\hat{\sigma}^2} \sim F(k, T-k),$$

where $F(k, T-k)$ stands for the central F-distribution with k and $T-k$ degrees of freedom. This result follows immediately from the definition of F-distribution, which is given by the ratio

of two independent chi-squared variates corrected for their respective degrees of freedom (see Appendix B). In the present application, the two chi-squared distributions are

$$\frac{(T-k)\,\hat{\sigma}^2}{\sigma^2} = \frac{\mathbf{u}'\mathbf{M}\mathbf{u}}{\sigma^2} \sim \chi^2_{T-k},$$

and

$$\begin{aligned}\left(\hat{\boldsymbol{\beta}}-\boldsymbol{\beta}\right)'\left(\frac{\mathbf{X}'\mathbf{X}}{\sigma^2}\right)\left(\hat{\boldsymbol{\beta}}-\boldsymbol{\beta}\right) &= \mathbf{u}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\left(\frac{\mathbf{X}'\mathbf{X}}{\sigma^2}\right)(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{u} \\ &= \frac{\mathbf{u}'(\mathbf{I}_T-\mathbf{M})\mathbf{u}}{\sigma^2} \sim \chi^2_k.\end{aligned}$$

The independence of $\mathbf{u}'\mathbf{M}\mathbf{u}$ and $\mathbf{u}'(\mathbf{I}_T-\mathbf{M})\mathbf{u}$ follows from the fact that $(\mathbf{I}_T-\mathbf{M})\mathbf{M} = \mathbf{M}-\mathbf{M}^2 = \mathbf{M}-\mathbf{M} = \mathbf{0}$.

The above results can be readily adapted for deriving the distribution of linear subsets of $\hat{\boldsymbol{\beta}}$. Suppose we are interested in the distribution of $\mathbf{R}\hat{\boldsymbol{\beta}}$, where $\mathbf{R}$ is an $r \times k$ matrix of fixed constants with rank $r \leq k$. Then

$$\begin{aligned}&\frac{\left(\mathbf{R}\hat{\boldsymbol{\beta}}-\mathbf{R}\boldsymbol{\beta}\right)'\left[\mathbf{R}\left(\mathbf{X}'\mathbf{X}\right)^{-1}\mathbf{R}'\right]^{-1}\left(\mathbf{R}\hat{\boldsymbol{\beta}}-\mathbf{R}\boldsymbol{\beta}\right)}{r\hat{\sigma}^2} \\ &= \frac{\left(\mathbf{R}\hat{\boldsymbol{\beta}}-\mathbf{R}\boldsymbol{\beta}\right)'\left[\mathbf{R}\widehat{Var}(\hat{\boldsymbol{\beta}})\mathbf{R}'\right]^{-1}\left(\mathbf{R}\hat{\boldsymbol{\beta}}-\mathbf{R}\boldsymbol{\beta}\right)}{r} \sim F(r, T-k). \qquad (2.28)\end{aligned}$$

In the case where $r = 1$, the F-test reduces to a t-test. For example, by setting $\mathbf{R} = (1, 0, \ldots, 0)$, the above result implies

$$\frac{(\hat{\beta}_1-\beta_1)^2}{\widehat{Var}(\hat{\beta}_1)} \sim F(1, T-k),$$

which in turn yields the familiar t-test statistic, given by $\left(\hat{\beta}_1-\beta_1\right)/\sqrt{\widehat{Var}(\hat{\beta}_1)} \sim t_{T-k}$.

2.10 The multiple correlation coefficient

By analogy to the case of the simple regression model, the strength of the fit of a multiple regression equation is measured *via* the multiple correlation coefficient, R, defined by the proportion of the total variation of y explained by the regression equation:

$$R^2 = \frac{\sum_t \left(\hat{y}_t-\bar{y}\right)^2}{\sum_t \left(y_t-\bar{y}\right)^2}. \qquad (2.29)$$

As in the case of the simple regression equation, the total variation of y, measured by $S_{yy} = \sum_t (y_t - \bar{y})^2$, can be decomposed into that explained by the regression equation, $\sum_t (\hat{y}_t - \bar{y})^2$, and the rest:[4]

$$\sum_t (y_t - \bar{y})^2 = \sum_t (\hat{y}_t - \bar{y})^2 + \sum_t (y_t - \hat{y}_t)^2 .$$

Hence, R^2 can also be written as

$$R^2 = \frac{\sum_t (y_t - \bar{y})^2 - \sum_t (y_t - \hat{y}_t)^2}{\sum_t (y_t - \bar{y})^2},$$

or

$$\begin{aligned} R^2 &= 1 - \frac{\sum_t (y_t - \hat{y}_t)^2}{\sum_t (y_t - \bar{y})^2} \\ &= 1 - \frac{\sum_t \hat{u}_t^2}{S_{yy}} = 1 - \frac{\hat{\mathbf{u}}'\hat{\mathbf{u}}}{S_{yy}}, \end{aligned} \tag{2.30}$$

which provides an alternative interpretation for R^2 and establishes that $0 \leq R^2 \leq 1$, so long as the underlying regression equation contains an intercept.[5] The limiting value of $R^2 = 1$ indicates perfect fit and arises if and only if $\hat{\mathbf{u}} = \mathbf{0}$ (or $\hat{u}_t = 0$, for $t = 1, 2, \ldots, T$). When T, the sample size, is finite this can only happen if the number of estimated regression coefficients, k, is equal to T. The R^2 statistic is problematic as a measure of quality of the fit of a regression models because it always increases when a new regressor is added to the model. Therefore a high value of R^2 is not by itself indicative of a good fit. An alternative measure of fit which attempts to take account of the number of estimated coefficients is due to Theil. It is called *adjusted* R^2, and written as $\bar{R}^2$:

$$\bar{R}^2 = 1 - \frac{T-1}{T-k} \frac{\sum_t \hat{u}_t^2}{\sum_t (y_t - \bar{y})^2}, \tag{2.31}$$

or equivalently (using (2.14)):

$$\bar{R}^2 = 1 - \frac{\hat{\sigma}^2}{S_{YY}/(T-1)}.$$

This 'adjusted' measure provides a trade-off between fit, as measured by R^2, and parsimony as measured by $T - k$. To make this trade-off more explicit $\bar{R}^2$ is also often defined as

$$1 - \bar{R}^2 = \frac{T-1}{T-k}\left(1 - R^2\right). \tag{2.32}$$

[4] The proof is similar to that presented in Chapter 1 for the bivariate regression model and will not be repeated here.

[5] When the regression equation does not contain an intercept term, R^2 can become negative.

All the above three definitions of $\bar{R}^2$ are algebraically equivalent. Note that, unlike R^2, there is no guarantee for the $\bar{R}^2$ to be non-negative, and hence $\bar{R}$ is not always defined.

In applied econometrics, $\bar{R}^2$ is often used as a criterion of model selection. However, its use can be justified when the regression models under consideration are non-nested, in the sense that none of the models under consideration can be obtained from the others by means of some suitable parametric restrictions. In the case where the models are nested, a more suitable procedure would be to apply classical hypotheses testing procedures and test the models against one another by means of *F*- or *t*-tests. (See Chapter 3 on hypotheses testing in linear regression models.)

Remark 1 *When y_t is trended (upward or downward) it is possible to obtain an R^2 very close to unity, irrespective of whether the trend is deterministic or stochastic. This is because the denominator of R^2, namely $S_{yy} = \sum_t \left(y_t - \bar{y}\right)^2$, implicitly assumes that y_t is stationary with a constant mean and variance (see Chapter 12 for definition of stationarity). In the case of trended variables a more appropriate measure of fit would be to define R^2 with respect to the first differences of y_t, $\Delta y_t = y_t - y_{t-1}$, namely*

$$R^2_{\Delta y} = 1 - \frac{\hat{\mathbf{u}}'\hat{\mathbf{u}}}{\sum_t \left(\Delta y_t - \overline{\Delta y}\right)^2},$$

where $\overline{\Delta y} = \sum_t \Delta y_t / T$. This measure is applicable irrespective of whether y_t is trend-stationary (namely when its deviations from a deterministic trend line are stationary), or first difference stationary. A variable is said to be first difference stationary if it must be first differenced once before it becomes stationary (see Chapter 15 for further details). The following simple relation exists between R^2 and $R^2_{\Delta y}$:

$$1 - R^2_{\Delta y} = \left(\frac{\sum_t \left(y_t - \bar{y}\right)^2}{\sum_t \left(\Delta y_t - \overline{\Delta y}\right)^2}\right)\left(1 - R^2\right).$$

Since in the case of trended y_t, for modest values of T, the sum $\sum_t \left(y_t - \bar{y}\right)^2$ will most certainly be substantially larger than $\sum_t \left(\Delta y_t - \overline{\Delta y}\right)^2$, it then follows that in practice $R^2_{\Delta y}$ will be less than R^2, often by substantial amounts. Also as T tends to infinity R^2 will tend to unity, but $R^2_{\Delta y}$ remain bounded away from unity. An alternative approach to arriving at a plausible measure of fit in the case of trended variables would be to ensure that the dependent variable of the regression is stationary by running regressions of first differences, Δy_t on the regressors, $\mathbf{x}_t$, of interest. But in that case it is important that lagged values of y_t, are also included amongst the regressors, namely a dynamic specification should be considered. This naturally leads to the analysis of error correction specifications to be discussed in Chapters 6, 23, and 24.

2.11 Partitioned regression

Consider the classical linear regression model

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{u}, \tag{2.33}$$

and suppose that $\mathbf{X}$ is partitioned into two sub-matrices $\mathbf{X}_1$ and $\mathbf{X}_2$ of order $T \times k_1$ and $T \times k_2$ such that $k = k_1 + k_2$.[6] Partitioning $\boldsymbol{\beta}$ conformably with $\mathbf{X} = \left(\mathbf{X}_1 \vdots \mathbf{X}_2\right)$ we have

$$\mathbf{y} = \mathbf{X}_1\boldsymbol{\beta}_1 + \mathbf{X}_2\boldsymbol{\beta}_2 + \mathbf{u}. \tag{2.34}$$

Such partitioned regressions arise, for example, when $\mathbf{X}_1$ is composed of seasonal dummy variables or time trends, and $\mathbf{X}_2$ contains the regressors of interest, or the 'focus' regressors. The *OLS* estimators of $\boldsymbol{\beta}_1$ and $\boldsymbol{\beta}_2$ are given by the normal equations

$$\mathbf{X}_1'\mathbf{y} = \left(\mathbf{X}_1'\mathbf{X}_1\right)\hat{\boldsymbol{\beta}}_1 + \left(\mathbf{X}_1'\mathbf{X}_2\right)\hat{\boldsymbol{\beta}}_2, \tag{2.35}$$

$$\mathbf{X}_2'\mathbf{y} = \left(\mathbf{X}_2'\mathbf{X}_1\right)\hat{\boldsymbol{\beta}}_1 + \left(\mathbf{X}_2'\mathbf{X}_2\right)\hat{\boldsymbol{\beta}}_2. \tag{2.36}$$

Solving for $\hat{\boldsymbol{\beta}}_1$ and $\hat{\boldsymbol{\beta}}_2$ we have

$$\hat{\boldsymbol{\beta}}_1 = \left(\mathbf{X}_1'\mathbf{M}_2\mathbf{X}_1\right)^{-1}\mathbf{X}_1'\mathbf{M}_2\mathbf{y}, \tag{2.37}$$

$$\hat{\boldsymbol{\beta}}_2 = \left(\mathbf{X}_2'\mathbf{M}_1\mathbf{X}_2\right)^{-1}\mathbf{X}_2'\mathbf{M}_1\mathbf{y}, \tag{2.38}$$

where

$$\mathbf{M}_j = \mathbf{I}_T - \mathbf{X}_j\left(\mathbf{X}_j'\mathbf{X}_j\right)^{-1}\mathbf{X}_j', \qquad \text{for} \qquad j = 1, 2.$$

The estimators of the 'focus' coefficients, $\hat{\boldsymbol{\beta}}_2$, can also be written as (recall that $\mathbf{M}_j$ are symmetric and idempotent: $\mathbf{M}_j' = \mathbf{M}_j = \mathbf{M}_j^2$):

$$\hat{\boldsymbol{\beta}}_2 = \left[\left(\mathbf{M}_1\mathbf{X}_2\right)'\left(\mathbf{M}_1\mathbf{X}_2\right)\right]^{-1}\left(\mathbf{M}_1\mathbf{X}_2\right)'\mathbf{y},$$

or

$$\hat{\boldsymbol{\beta}}_2 = \left(\widetilde{\mathbf{X}}_2'\widetilde{\mathbf{X}}_2\right)^{-1}\widetilde{\mathbf{X}}_2'\mathbf{y},$$

where $\widetilde{\mathbf{X}}_2 = \mathbf{M}_1\mathbf{X}_2$ and $\widetilde{\mathbf{y}} = \mathbf{M}_1\mathbf{y}$ are the *residual* matrices and vectors of the regressions of $\mathbf{X}_2$ on $\mathbf{X}_1$ and of $\mathbf{y}$ on $\mathbf{X}_2$, respectively. The residuals from the regression of $\widetilde{\mathbf{y}} = \mathbf{y} - \hat{\mathbf{y}}$ on $\widetilde{\mathbf{X}}_2 = \mathbf{X}_2 - \hat{\mathbf{X}}_2$ are also given by $\widetilde{\mathbf{u}} = \widetilde{\mathbf{y}} - \widetilde{\mathbf{X}}_2\hat{\boldsymbol{\beta}}_2$. It is now easily seen that $\widetilde{\mathbf{u}}$ is in fact the same as the *OLS* residual vector from the unpartitioned regression of $\mathbf{y}$ on $\mathbf{X}$.[7] Therefore, a regression of $\mathbf{y}$ on $\widetilde{\mathbf{X}}_2$ yields the same estimate for $\boldsymbol{\beta}_2$ as the standard regression of $\mathbf{y}$ on $\mathbf{X}_1$ and $\mathbf{X}_2$ simultaneously and

[6] See Section A.9 in Appendix A for a description of partitoned matrices and their properties.

[7] Notice that

$$\begin{aligned}\tilde{\mathbf{u}} &= \left[\mathbf{I} - \mathbf{X}_1\left(\mathbf{X}_1'\mathbf{X}_1\right)^{-1}\mathbf{X}_1'\right]\mathbf{y} - \left[\mathbf{I} - \mathbf{X}_1\left(\mathbf{X}_1'\mathbf{X}_1\right)^{-1}\mathbf{X}_1'\right]\mathbf{X}_2\hat{\boldsymbol{\beta}}_2 \\ &= \mathbf{y} - \mathbf{X}_2\hat{\boldsymbol{\beta}}_2 - \mathbf{X}_1\left(\mathbf{X}_1'\mathbf{X}_1\right)^{-1}\left[\mathbf{X}_1'\mathbf{y} - \mathbf{X}_1'\mathbf{X}_2\hat{\boldsymbol{\beta}}_2\right].\end{aligned}$$

(*continued*)

without orthogonalization of the effect of $\mathbf{X}_1$ on $\mathbf{X}_2$. This property is known as the Frisch-Waugh-Lovell theorem, first introduced by Frisch and Waugh (1933), and then by Lovell (1963). For further details see also Davidson and MacKinnon (1993).

The partitioned and the unpartitioned regressions also yield the same results for the variance matrix of $\hat{\boldsymbol{\beta}}_2$. It is, therefore, possible to estimate the coefficients of the 'focus' regressors in two ways. The partitioned method first 'filters' the observations by allowing for the effect of 'non-focus' variables by running regressions of $\mathbf{y}$ on $\mathbf{X}_1$, and $\mathbf{X}_2$ on $\mathbf{X}_1$ and then computes estimates of $\boldsymbol{\beta}_2$ by regression of the filtered variables. In the case where $\mathbf{X}_1$ contains seasonal dummies, the residuals from regression of $\mathbf{y}$ on $\mathbf{X}_1$ represent seasonally adjusted $\mathbf{y}$, and similarly the residuals from regressions of the columns of $\mathbf{X}_2$ on $\mathbf{X}_1$ represent seasonally adjusted $\mathbf{X}_2$. Hence, regression of seasonally adjusted variables yields the same coefficient estimates as running a regression of seasonally unadjusted variables so long as the same seasonal dummies used to adjust $\mathbf{y}$ and $\mathbf{X}_2$ are also included in the unseasonally adjusted regression. The same results also hold for the regressions of detrended and non-detrended variables.

Special care should be exercised when using the above results from partitioned regressions. Firstly, the results do not apply when the seasonal adjustments or detrending are carried out over a time period that differs from the period over which the regression of focus variables are run. Neither do they apply if the seasonal adjustments are carried out by government agencies who often use their own in-house methods. Secondly, the computer results based on regression of seasonally adjusted variables do not generally take account of the loss in degrees of freedom associated with the estimation of seasonal or trend effects. In view of these pitfalls, it is often advisable to base estimation and hypothesis testing on the unpartitioned regression of $\mathbf{y}$ on $\mathbf{X}$. The use of partitioned regressions is helpful primarily for pedagogic purposes.

2.12 How to interpret multiple regression coefficients

The issue of how to interpret the regressions coefficients in a multiple regression model has been recently discussed in Pesaran and Smith (2014). Suppose we are interested in measuring the effects of a unit change in the regressor x_{it} on y_t. The standard procedure is to use the estimated coefficient of x_{it}, namely β_i, on the assumption that the hypothetical change in x_{it}, does not affect $x_{jt}, j \neq i$, namely it assumes that the hypothetical change in x_{it} is accompanied with holding the other regressors constant, the so called *ceteris paribus* assumption. But in almost all economic applications we are not able to control the inputs and the counterfactual exercise by which all other regressors can be held constant might not be relevant. Pesaran and Smith (2014) argue that in time series analysis, rather than focussing on the signs of individual coefficients in multiple regressions holding the other variables constant, we should measure a total impact effect which allows for direct and indirect induced changes that arise due to the historical correlations amongst the regressors. The limitation of the usual *ceteris paribus* approach lies in the fact that it ignores the stochastic interdependence of the regressors which we need to allow for in time series economic applications. Similar issues arise in the derivation of impulse response functions for

But using (2.35), $\mathbf{X}_1'\mathbf{y} - \mathbf{X}_1'\mathbf{X}_2\hat{\boldsymbol{\beta}}_2 = \left(\mathbf{X}_1'\mathbf{X}_1\right)\hat{\boldsymbol{\beta}}_1$, and hence

$$\begin{aligned}\tilde{\mathbf{u}} &= \mathbf{y} - \mathbf{X}_2\hat{\boldsymbol{\beta}}_2 - \mathbf{X}_1\left(\mathbf{X}_1'\mathbf{X}_1\right)^{-1}\left(\mathbf{X}_1'\mathbf{X}_1\right)\hat{\boldsymbol{\beta}}_1 \\ &= \mathbf{y} - \mathbf{X}_1\hat{\boldsymbol{\beta}}_1 - \mathbf{X}_2\hat{\boldsymbol{\beta}}_2 = \hat{\mathbf{u}}.\end{aligned}$$

the analysis of dynamic models and have been discussed by Koop, Pesaran, and Potter (1996) and Pesaran and Shin (1998) and will be addressed in Chapter 24.

To illustrate Pesaran and Smith (2014)'s argument consider the following simple classical linear regression model with two regressors:

$$y_t = \beta_0 + \beta_1 x_{1t} + \beta_2 x_{2t} + u_t.$$

Suppose further that x_{1t} and x_{2t} are random draws from a bivariate normal distribution with the covariance matrix

$$Var\begin{pmatrix} \Delta x_{1t} \\ \Delta x_{2t} \end{pmatrix} = \begin{pmatrix} \sigma_{11} & \sigma_{21} \\ \sigma_{21} & \sigma_{22} \end{pmatrix}.$$

It is now easily seen that

$$E(\Delta x_{2t} \,|\Delta x_{1t}) = \rho_{21} \Delta x_{1t},$$

where $\rho_{21} = \sigma_{21}/\sigma_{11}$. The total effect of a unit change in x_{it} on y_t is therefore given by

$$E(\Delta y_t \,|\Delta x_{1t}) = \left(\beta_1 + \rho_{21}\beta_1\right) \Delta x_{1t},$$

which reduces to the β_1 only if $\sigma_{21} = 0$.

As a second example, suppose that we have a quadratic function of a single regressor, so that the regression model is given by

$$y_t = \beta_0 + \beta_1 x_t + \beta_2 x_t^2 + u_t. \tag{2.39}$$

Here it clearly does not make any sense to ask what is the effect on y_t of a change in x_t, holding x_t^2 fixed. In this case we have

$$E(\Delta y_t \,|\Delta x_t) = \left(\beta_1 + 2\beta_2 x_t\right) \Delta x_t, \tag{2.40}$$

for sufficiently small increments, Δx_t.

Pesaran and Smith (2014) show that the total effect of a unit change in x_{it} on y_t can be estimated consistently by a simple regression of y_t on x_{it}, which is to be contrasted with the *ceteris paribus* effect of unit change in x_{it} on y_t which is given by β_i and requires estimation of the correctly specified multiple regression model.

2.13 Implications of misspecification for the *OLS* estimators

The unbiasedness property of the *OLS* estimators in the classical linear regression model crucially depends on the validity of the classical assumptions and the correct specification of the regression equation. Here we consider the effects of misspecification, that results from adding or omitting a regressor in error, on the *OLS* estimators. In Chapter 3 we consider the implications of such misspecifications for inference on the regression coefficients.

2.13.1 The omitted variable problem

Suppose that y_t's are generated according to the classical linear regression equation

$$y_t = \alpha + \beta_1 x_t + \beta_2 z_t + u_t, \tag{2.41}$$

but the investigator estimates the simple regression equation

$$y_t = \alpha + \beta x_t + \varepsilon_t, \tag{2.42}$$

which omits the regressor z_t. The new error, ε_t, now contains the effect of the omitted variable and the orthogonality assumption that requires x_t and ε_t to be uncorrelated might no longer hold. To see this consider the *OLS* estimator of β in (2.42), which is given by

$$\hat{\beta} = \frac{\sum_{t=1}^{T} (x_t - \bar{x}) \left(y_t - \bar{y}\right)}{\sum_{t=1}^{T} (x_t - \bar{x})^2}.$$

Under the correct model (33.34)

$$y_t - \bar{y} = \beta_1 (x_t - \bar{x}) + \beta_2 (z_t - \bar{z}) + u_t - \bar{u}.$$

Hence

$$\hat{\beta} = \beta_1 + \beta_2 \frac{\sum_t (x_t - \bar{x}) (z_t - \bar{z})}{\sum_t (x_t - \bar{x})^2} + \frac{\sum_t (x_t - \bar{x}) (u_t - \bar{u})}{\sum_t (x_t - \bar{x})^2},$$

and taking expectations conditional on the regressors

$$E\left(\hat{\beta}\right) = \beta_1 + \beta_2 b_{x \bullet z}, \tag{2.43}$$

where $b_{x \bullet z}$ stands for the *OLS* estimator of the regression coefficient of x_t on z_t. In general, therefore, $\hat{\beta}$ is not an unbiased estimator of β_1 [the 'true' regression coefficient of x_t in (2.41)]. The extent of the bias depends on the importance of the z_t variable as measured by β_2 and the degree of the dependence of x_t on z_t. Only in the case where x_t and z_t are uncorrelated $\hat{\beta}$ will yield an unbiased estimator of β_1. See Section 3.13 on the effects of omitting relevant regressors on testing hypothesis involving the regression coefficients.

The omitted regressor bias can be readily generalized to the case where two or more relevant regressors are omitted. The appropriate set up is the partitioned regression equation given in (2.34). Suppose that in that equation the regressors $\mathbf{X}_2$ are incorrectly omitted and $\boldsymbol{\beta}_1$ is estimated by

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}_1' \mathbf{X}_1)^{-1} \mathbf{X}_1' \mathbf{y}.$$

Then, under (2.34), it is easily seen that

$$E(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_1 \,|\mathbf{X}) = (\mathbf{X}_1'\mathbf{X}_1)^{-1}\mathbf{X}_1'\mathbf{X}_2\boldsymbol{\beta}_2 = \mathbf{P}_{12}\boldsymbol{\beta}_2,$$

Also see Exercise 3 at the end of this chapter.

2.13.2 The inclusion of irrelevant regressors

Inclusion of irrelevant regressors in the regression equation is less problematic. For example, suppose that the correct model is

$$y_t = \alpha + \beta x_t + u_t,$$

but we estimate the expanded regression equation by mistake:

$$y_t = \alpha + \beta_1 x_t + \beta_2 z_t + \varepsilon_t.$$

The *OLS* estimator of β_1 in this regression will still be unbiased, but will no longer be an efficient estimator. There will also be the possibility of a multicollinearity problem that can arise if the erroneously included regressor, z_t, is highly correlated with x_t (see Section 15.3.1). In general suppose that the correct regression model is

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{u}, \tag{2.44}$$

but $\boldsymbol{\beta}$ is estimated by running the expanded regression of $\mathbf{y}$ on $\mathbf{X}$ and $\mathbf{Z}$. The *OLS* estimator of the coefficients of $\mathbf{X}$ in this regression, say $\boldsymbol{\beta}_1$, is given by (see also (2.37))

$$\hat{\boldsymbol{\beta}}_1 = \left(\mathbf{X}'\mathbf{M}_z\mathbf{X}\right)^{-1}\mathbf{X}'\mathbf{M}_z\mathbf{y},$$

where $\mathbf{M}_z = \mathbf{I}_T - \mathbf{Z}(\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}$. Under (2.44) we have

$$E\left(\hat{\boldsymbol{\beta}}_1 - \boldsymbol{\beta} \,|\mathbf{X}, \mathbf{Z}\right) = \left(\mathbf{X}'\mathbf{M}_z\mathbf{X}\right)^{-1}\mathbf{X}'\mathbf{M}_z E\left(\mathbf{u} \,|\mathbf{X}, \mathbf{Z}\right).$$

Therefore, so long as $\mathbf{Z}$ as well as $\mathbf{X}$ are strictly exogenous and the orthogonality assumption $E\left(\mathbf{u} \,|\mathbf{X}, \mathbf{Z}\right) = \mathbf{0}$ is satisfied we obtain

$$E\left(\hat{\boldsymbol{\beta}}_1 - \boldsymbol{\beta} \,|\mathbf{X}, \mathbf{Z}\right) = \mathbf{0},$$

or unconditionally

$$E\left(\hat{\boldsymbol{\beta}}_1\right) = \boldsymbol{\beta}.$$

Notice, however, that the additional variables in $\mathbf{Z}$ can not be weakly exogenous. For example, adding lagged values of y_t to the regressors in error can lead to biased estimators.

2.14 Linear regressions that are nonlinear in variables

A linear regression model does not necessarily require the relationship between **y** (the regressand) and **x** (the regressor) to be linear. A simple example of a linear regression model with a nonlinear functional relationship between **y** and **x** is given by the quadratic regression equation:

$$y_i = \alpha + \beta x_i + \gamma x_i^2 + u_i.$$

To transform this nonlinear relation to a linear regression model, set $z_i = x_i^2$ and write the quadratic equation as

$$y_i = \alpha + \beta x_i + \gamma z_i + u_i,$$

which is a linear regression in the two regressors x_i and z_i. Other examples of nonlinear relations that are transformable to linear regressions are general polynomial regressions, logistic models, log-linear, semi-log-linear and inverse models. Here we examine some of these models in more detail.

Example 2 *Consider the following Cobb–Douglas production function*

$$Q_i = AL_i^{\alpha} K_i^{\beta} \exp(u_i),$$

where Q_i is output of firm i, L_i and K_i are the quantities of labour and capital used in the production process, and u_i are independently distributed productivity shocks. Taking logarithms of both sides now yields the linear logarithmic specification

$$\log Q_i = \log A + \alpha \log L_i + \beta \log K_i + u_i,$$

and setting $y_i = \log Q_i$, $x_{1i} = \log L_i$, $x_{2i} = \log K_i$, $a = \log A$, yields

$$y_i = a + \alpha x_{1i} + \beta x_{2i} + u_i,$$

which is a linear regression equation in the two regressors x_{1i} and x_{2i}. The estimate of A can now be obtained by $\hat{A} = \exp(\hat{a})$, where $\hat{a}$ is the OLS estimate of the intercept term in the above regression.

Example 3 (Logistic function with a known saturation level) *The logistic model has the general form*

$$Y_i = \frac{A}{1 + \gamma x_i^{\beta} \exp(u_i)}, \qquad \beta, \gamma > 0, \qquad x_i > 0,$$

where A is the saturation level of Y, which is assumed to be known. We also assume that $A > Y_i$, for all i. This is clearly a nonlinear model in terms of Y and x. To transform this model into a linear regression model in terms of the unknown parameters γ and β, we first note that

$$\gamma x_i^{\beta} \exp(u_i) = \frac{A}{Y_i} - 1,$$

which upon taking logarithms of both sides yields

$$y_i = \log\left(\frac{A - Y_i}{Y_i}\right) = \alpha + \beta \log x_i + u_i,$$

in which $\alpha = \log(\gamma)$. *In the case where A is known the parameters* α *and* β *(and hence* γ *and* β*) can be estimated by the OLS regression of* $y_i = \log\left(\frac{A-Y_i}{Y_i}\right)$ *on log*x_i. *The logistic function, has important applications in econometrics (e.g. ownership of demand for durable goods, TV's, cars etc.) and in population studies.*

Other examples of nonlinear functions that can be transformed into linear regressions include semi-logarithmic model

$$y_i = \alpha + \beta \log x_i + u_i,$$

and the inverse model

$$y_i = \alpha + \frac{\beta}{x_i} + u_i.$$

These models have proved very useful in cross-section studies of household consumption behaviour.

2.15 Further reading

Further reading on multiple regression and on the properties of *OLS* estimator can be found in Wooldridge (2000) and in Greene (2002) (see Chapters 2–4). An interesting geometric interpretation of linear regression, shedding light on the numerical properties of *OLS*, is presented in Davidson and MacKinnon (1993). The latter also provides an in-depth discussion on the Frisch–Waugh–Lovell theorem, and partitioned regression.

2.16 Exercises

1. Suppose that in the classical regression model $y_i = \alpha + \beta x_i + u_i$ the true value of the constant, α, is zero. Compare the variance of the *OLS* estimator for β computed without a constant term with that of the *OLS* estimator for β computed with the constant term.
2. Consider the following linear regression model

$$y_t = \alpha + \beta x_t + \gamma w_t + u_t. \tag{2.45}$$

Suppose that the classical assumptions are applicable to (2.45), but β is estimated by running an *OLS* regression of y_t on a vector of ones and x_t. Denote such an estimator by $\widehat{\beta}$, and show that $\widehat{\beta}$ is a biased estimator of β in (2.45). Derive the formula for the bias of $\widehat{\beta}$ in terms of the correlation coefficient of x_t and w_t, and their variances, namely $\rho_{xw}, \sigma_x^2, \sigma_w^2$.

3. Consider the following partitioned classical linear regression model:

$$\mathbf{y} = \mathbf{X}_1\boldsymbol{\beta}_1 + \mathbf{X}_2\boldsymbol{\beta}_2 + \mathbf{u},$$

where y is a $T \times 1$ vector of observations on the dependent variable, and $\mathbf{X}_1$ and $\mathbf{X}_2$ are $T \times k_1$ and $T \times k_2$ observation matrices on the regressors.

(a) Show that if we omit the variables included in $\mathbf{X}_2$, and estimate $\boldsymbol{\beta}_1$ by running a regression of $\mathbf{y}$ on $\mathbf{X}_1$ only, then $\hat{\boldsymbol{\beta}}_1$ is generally biased with the bias:

$$E(\hat{\boldsymbol{\beta}}_1|\mathbf{X}) - \boldsymbol{\beta}_1 = \mathbf{P}_{12}\boldsymbol{\beta}_2, \quad \text{where } \mathbf{P}_{12} = (\mathbf{X}_1'\mathbf{X}_1)^{-1}\mathbf{X}_1'\mathbf{X}_2,$$

where $\mathbf{X} = (\mathbf{X}_1, \mathbf{X}_2)$.

(b) Interpret the elements of matrix $\mathbf{P}_{12}$. Under what conditions $\hat{\boldsymbol{\beta}}_1$ will be unbiased?

(c) A researcher is estimating the demand equation for furniture using cross-section data. As regressors she uses an intercept term, the relative price of furniture, and omits the relevant income variable. Find an expression for the bias of the *OLS* estimate of the price variable in such a regression. What other regressors should she have considered, and how could their omission have affected her estimate of the price effect?

4. Consider the following linear regression model

$$y_t = \alpha + \beta_1 x_{1t} + \beta_2 x_{2t} + \varepsilon_t, \tag{2.46}$$

and suppose that the observations (y_t, x_{1t}, x_{2t}), for $t = 1, 2, \ldots, T$ are available.

(a) Specify the assumptions under which (2.46) can be viewed as a classical linear regression model. In your response clearly distinguish between the cases where x_{1t} and x_{2t} are fixed in repeated samples, strictly exogenous, and weakly exogenous (see Chapter 9 for definition of strictly exogenous, and weakly exogenous regressors).

(b) Suppose that the classical assumptions are applicable to (2.46), but β_1 is estimated by running an *OLS* regression of y_t on a vector of ones and x_{1t}, and β_2 is estimated by running an *OLS* regression of y_t on a vector of ones and x_{2t}. Denote these estimators by $\widehat{\beta}_{yx_1}$ and $\widehat{\beta}_{yx_2}$. Show that in general $\widehat{\beta}_{yx_1}$ and $\widehat{\beta}_{yx_1}$ are biased estimators of β_1 and β_2 in (2.46).

(c) Denote the *OLS* estimators of β_1 and β_2 in the regression of y_t on x_{1t} and x_{2t} as in (2.46) by $\hat{\beta}_1$ and $\hat{\beta}_2$, respectively. Show that

$$\hat{\beta}_1 = \frac{\widehat{\beta}_{yx_1} - r(s_2/s_1)\widehat{\beta}_{yx_2}}{1 - r^2},$$
$$\hat{\beta}_2 = \frac{\widehat{\beta}_{yx_2} - r(s_1/s_2)\widehat{\beta}_{yx_1}}{1 - r^2},$$

where s_1 and s_2 are the standard deviations of x_{1t} and x_{2t}, respectively, and r denotes the correlation coefficients of x_{1t} and x_{2t}. Discuss the relevance of these results for empirical time series research.

5. Consider the regression model

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{u}, \qquad \mathbf{u} \sim N(\mathbf{0}, \sigma^2\mathbf{I}_T),$$

where $\mathbf{X}$ is a $T \times k$ stochastic matrix of rank k, distributed independently of $\mathbf{u} = (u_1, u_2, \ldots, u_T)'$, and $u_t \backsim IID(0, \sigma^2)$.

(a) Let $\lambda_{\max}(\mathbf{X}'\mathbf{X})$ and $\lambda_{\min}(\mathbf{X}'\mathbf{X})$ denote the largest and the smallest characteristic roots (or eigenvalues) of $\mathbf{X}'\mathbf{X}$. Prove that the following four statements are equivalent:
- $\lambda_{\min}(\mathbf{X}'\mathbf{X})$ tends to infinity
- $\lambda_{\max}\left((\mathbf{X}'\mathbf{X})^{-1}\right)$ tends to zero
- *Trace* $\left((\mathbf{X}'\mathbf{X})^{-1}\right)$ tends to zero
- Every diagonal element of $(\mathbf{X}'\mathbf{X})^{-1}$ tends to zero

(b) Using the results under (*a*), or otherwise show that the *OLS* estimator of $\boldsymbol{\beta}$ is consistent if $\lambda_{\min}(\mathbf{X}'\mathbf{X})$ tends to infinity.

(c) Prove $\hat{\sigma}^2 = \hat{\mathbf{u}}'\hat{\mathbf{u}}/T$ is a consistent estimator of σ^2, where $\hat{\mathbf{u}}$ is the vector of *OLS* residuals.

3 Hypothesis Testing in Regression Models

3.1 Introduction

Statistical hypothesis testing is at the core of the classical theory of statistical inference. Although it is closely related to the problem of estimation, it can be considered almost independently of it. In this chapter, we introduce some key concepts of statistical inference, and show their use to investigate the statistical significance of the (linear) relationships modelled through regression analysis, or to investigate the validity of the classical assumptions in simple and multiple linear regression.

3.2 Statistical hypothesis and statistical testing

A *statistical hypothesis* is an assertion about the distribution of one or more random variables. If the hypothesis completely specifies the probability distribution, it is called a *simple* hypothesis, otherwise it is called a *composite* hypothesis. For example, suppose $x_1, x_2, \ldots, x_T$ are drawn from $N(\theta, 1)$. Then $H : \theta = 0$ is a simple hypothesis, while $H : \theta > 0$ is a composite hypothesis. If one hypothesis can be derived as a limiting sequence of another, we say the two hypotheses are *nested*. If neither hypothesis can be obtained from the other as a limiting process, then we call the hypotheses under consideration *non-nested*. For example, suppose $x_1, x_2, \ldots, x_T$ are drawn from log-normal distribution under H_0, while under H_1 they are drawn from an exponential distribution. Then H_0 and H_1 are non-nested hypotheses. We refer to Chapter 11 for a review of tests for non-nested hypotheses.

3.2.1 Hypothesis testing

A test of a statistical hypothesis H is a rule for *rejecting* H. If the sample space is denoted by $\chi = (x_1, x_2, \ldots, x_T)$, a test procedure decomposes χ into two regions. If $(x_1, x_2, \ldots, x_T) \in C_T$, where C_T is called the critical or *rejection region* of the test, then H is rejected, otherwise H is not rejected. In practice we often map $(x_1, x_2, \ldots, x_T)$ into a test statistic $T(x_1, x_2, \ldots, x_T)$ and consider whether $T(x_1, x_2, \ldots, x_T) \geq C_T$ or not.

The hypothesis being tested (i.e. the maintained hypothesis) is usually denoted by H_0 and is called the *null* hypothesis. The hypothesis against which H_0 is tested is called the *alternative* hypothesis and is usually denoted by H_1.

3.2.2 Types of error and the size of the test

The decision rule yields two types of error:

- The *type I error* is the error involved in rejecting H_0 when it is true
- The *type II error* is the error involved in not rejecting H_0 when it is false

The probability of a type I error is called the *size of the test* and, often denoted by α_T, $\alpha_T \times 100$ per cent, is also called the *significance level* of the test. The probability of the type II error is called the *size of the type II error* and is often denoted by β_T. Ideally, we would like both errors to be as small as possible. However, there is a trade-off between the two, and by reducing the probability of a type I error, we must increase the probability of a type II error.

The *power* of a test is defined as 1 minus the size of the type II error, namely $power_T = 1 - \beta_T$. For a given significance level, α_T, we would like the power of the test, $power_T$, to be as large as possible.

Example 4 (Testing a hypothesis about a mean) *Assume we have a sample of T observations $x_1, x_2, \ldots, x_T$, obtained as random draws from a normal $N(\mu, \sigma^2)$ distribution, with σ^2 known. Suppose that we wish to test $H_0 : \mu = \mu_0$, where μ_0 is a given (assumed) value of μ. To this end, consider the sample mean $\bar{x} = T^{-1} \sum_{i=1}^{T} x_i$. Under the null hypothesis the random variable*

$$z = \frac{\sqrt{T}\bar{x} - \mu_0}{\sigma},$$

is distributed as a $N(0, 1)$ and the critical values of the normal distribution will be applicable. Setting the significance level at 5 per cent, the critical value for a two-sided test (with the alternative being $\mu \neq \mu_0$) is 1.96. Hence, in this case the power of the test is the probability that the absolute value of the test statistic will exceed 1.96 given that the true value of μ is not μ_0. The power clearly depends on the alternative value selected for μ. As expected, the test becomes more powerful the further the true mean is from the hypothesized value. The interval

$$P\left(\bar{x} - 1.96\sigma/\sqrt{T} \leq \mu \leq \bar{x} + 1.96\sigma/\sqrt{T}\right) = 0.95,$$

is called the 95 per cent confidence interval of μ.

Let the critical region of a test be defined by $T(x_1, x_2, \ldots, x_T) \geq C_T$, we have

$$\begin{aligned} \text{Prob. of type I error} &= \Pr\{T(x_1, x_2, \ldots, x_T) \geq C_T \,|H_0\} = \alpha_T, \\ \text{Prob. of type II error} &= \Pr\{T(x_1, x_2, \ldots, x_T) < C_T \,|H_1\} = \beta_T. \end{aligned}$$

Let Π_T denote the power of the test, then

$$\Pi_T = 1 - \beta_T = 1 - \Pr\{T(x_1, x_2, \ldots, x_T) < C_T | H_1\},$$

or equivalently,

$$\Pi_T = \Pr\{T(x_1, x_2, \ldots, x_T) \geq C_T | H_1\}.$$

3.3 Hypothesis testing in simple regression models

In deriving the ordinary least squares (OLS) estimator and its properties in Chapter 2, we have not used Assumption A5 on the normality of u_t. This assumption is useful for hypotheses testing. Consider first the simple regression model

$$y_t = \alpha + \beta x_t + u_t,$$

and assume that Assumptions A1–A4 hold (see Chapter 2), together with Assumption A5, that is, $u_t \sim N(0, \sigma^2)$. Suppose that we are interested in testing the null hypothesis

$$H_0 : \beta = \beta_0,$$

against the two-sided alternative hypothesis

$$H_1 : \beta \neq \beta_0,$$

where β_0 is a given value of β. To construct a test for β, first recall that, from (1.25) and (1.26),

$$\hat{\beta} = \sum_{t=1}^{T} w_t y_t,$$

where

$$w_t = \frac{x_t - \bar{x}}{\sum_{s=1}^{T} (x_s - \bar{x})^2}.$$

Replacing $y_t = \alpha + \beta x_t + u_t$ in the above expression now yields

$$\hat{\beta} = \sum_{t=1}^{T} w_t (\alpha + \beta x_t + u_t),$$

$$\hat{\beta} = \alpha \left(\sum_{t=1}^{T} w_t \right) + \beta \left(\sum_{t=1}^{T} w_t x_t \right) + \sum_{t=1}^{T} w_t u_t,$$

and since $\sum_{t=1}^{T} w_t = 0$, $\sum_{t=1}^{T} w_t x_t = 1$ (see the derivations in Section 1.9), we have

$$\hat{\beta} = \beta + \sum_{t=1}^{T} w_t u_t. \tag{3.1}$$

Noting that the weighted average of normal variates is also normal, it follows that

$$\hat{\beta} \,|\mathbf{x} \sim N\left[\beta, Var\left(\hat{\beta}\right)\right], \tag{3.2}$$

where

$$Var\left(\hat{\beta}\right) = \sigma^2 \sum_{t=1}^{T} w_t^2 = \frac{\sigma^2}{\sum_{t=1}^{T} (x_t - \bar{x})^2}.$$

In the case where σ^2 is known, we can base the test of $H_0 : \beta = 0$, on the following standardized statistic

$$Z_{\hat{\beta}} = \frac{\hat{\beta} - \beta_0}{\sqrt{Var\left(\hat{\beta}\right)}} = \frac{\hat{\beta} - \beta_0}{S.E.\left(\hat{\beta}\right)}, \tag{3.3}$$

where *S.E.* $(\cdot)$ stands for the standard errors. Under the null hypothesis, $Z_{\hat{\beta}} \sim N(0, 1)$ and the critical values of the normal distribution will be applicable.

The appropriate choice of the critical values depends on the distribution of the test statistic, the size of the test (or the level of significance), and whether the alternative hypothesis is two sided, (namely $H_1 : \beta \neq \beta_0$) or one-side, namely whether $H_1 : \beta \geq \beta_0$ or $H_1 : \beta \leq \beta_0$.

In the case where σ^2 is not known, the use of statistic $Z_{\hat{\beta}}$ defined by (3.3) is not feasible and σ^2 needs to be replaced by its estimate. Using the unbiased estimator of σ^2, given by (1.34), namely

$$\hat{\sigma}^2 = \frac{\sum_t \left(y_t - \hat{\alpha} - \hat{\beta} x_t\right)^2}{T - 2},$$

we have the *t*-statistic

$$t_{\hat{\beta}} = \frac{\hat{\beta} - \beta_0}{\sqrt{\widehat{Var}\left(\hat{\beta}\right)}} = \frac{\hat{\beta} - \beta_0}{\hat{\sigma} \Big/ \left\{\sum_t (x_t - \bar{x})^2\right\}^{\frac{1}{2}}},$$

which under the null hypothesis, $H_0 : \beta = \beta_0$ has a *t*-distribution with $T - 2$ degrees of freedom. The $t_{\hat{\beta}}$ statistic is pivotal in the sense that it does not depend on any unknown parameters.

Example 5 *Suppose we are interested to test the hypothesis that the marginal propensity to consume out of disposable income is equal to unity. Using aggregate UK consumption data over the period 1948–89 we obtained the following OLS estimates:*

$$\hat{c}_t = \underset{(2108.9)}{7600.3} + \underset{(0.01169)}{0.87233}\ y_t.$$

The bracketed figures are standard errors. The estimate of the marginal propensity to consume is equal to $\hat{\beta} = 0.87233$. *To test* $H_0 : \beta = 1$ *against* $H_1 : \beta \neq 1$ *we compute the t-statistic*

$$t_{\hat{\beta}} = \frac{\hat{\beta} - \beta_0}{S.E.(\hat{\beta})} = \frac{0.87233 - 1.0}{0.01169} = -10.92.$$

The number of degrees of freedom of this test is equal to $42 - 2 = 40$, *and the* 95 *per cent critical value of the t-distribution with* 40 *degrees of freedom for a two-sided test is equal to* ± 2.021. *Hence since the value of* $t_{\hat{\beta}}$ *for the test of* $\beta = 1$ *against* $\beta \neq 1$ *is well below the critical value of the test (i.e.,* -2.021*) we reject the null hypothesis that* $\beta = 1$.

3.4 Relationship between testing $\beta = 0$, and testing the significance of dependence between *Y* and *X*

Recall that the correlation coefficient between Y and X is estimated by (see Section 1.9)

$$\hat{\rho}^2_{XY} = \frac{S^2_{XY}}{S_{XX}S_{YY}}.$$

But since

$$\hat{\beta} = \frac{S_{XY}}{S_{XX}},$$
$$\widehat{Var}(\hat{\beta}) = \frac{\hat{\sigma}^2}{S_{XX}},$$

we have

$$\hat{\rho}^2_{XY} = \frac{\hat{\beta}^2 S^2_{XX}}{S_{XX}S_{YY}} = \frac{\hat{\beta}^2 S_{XX}}{S_{YY}}. \tag{3.4}$$

The t-statistic for testing $H_0 : \beta = 0$ against $H_1 : \beta \neq 0$ is given by

$$\hat{t}_\beta = \frac{\hat{\beta}}{\sqrt{\widehat{Var}(\hat{\beta})}},$$

or upon using the above results:

$$\hat{t}_{\hat{\beta}}^2 = \frac{\hat{\beta}^2 S_{XX}}{\hat{\sigma}^2}. \tag{3.5}$$

Finally, recall from the decomposition of $S_{YY} = \sum \left(y_t - \bar{y}\right)^2$ in the analysis of variance table that (see Section 1.5)

$$\hat{\rho}_{XY}^2 = 1 - \frac{\sum_t \left(y_t - \hat{y}_t\right)^2}{\sum_t \left(y_t - \bar{y}\right)^2} = 1 - \frac{(T-2)\,\hat{\sigma}^2}{S_{YY}},$$

or

$$\hat{\sigma}^2 = \frac{S_{YY}\left(1 - \hat{\rho}_{XY}^2\right)}{T-2}. \tag{3.6}$$

Consequently, using (3.4) and (3.5) in (3.6) we have

$$t_{\hat{\beta}}^2 = \frac{(T-2)\,\hat{\rho}_{XY}^2}{\left(1 - \hat{\rho}_{XY}^2\right)}. \tag{3.7}$$

Alternatively, $\hat{\rho}_{XY}^2$ can be written as an increasing function of $t_{\hat{\beta}}^2$ for $T > 2$, namely

$$\hat{\rho}_{XY}^2 = \frac{t_{\hat{\beta}}^2}{T - 2 + t_{\hat{\beta}}^2} < 1. \tag{3.8}$$

These results show that in the context of a simple regression model the statistical test of the 'fit' of the model (i.e., $H_0 : \rho_{XY} = 0$ against $H_1 : \rho_{XY} \neq 0$) is the *same* as the test of zero restriction on the slope coefficient of the regression model (i.e., test of $H_0 : \beta = 0$ against $H_1 : \beta \neq 0$). Moreover, the test results under the null hypothesis of a zero relationship between Y and X is equivalent to testing the significance of the reverse regression of X on Y, namely testing $H_0 : \delta = 0$, against $H_1 : \delta \neq 0$, in the reverse regression

$$x_t = a_x + \delta y_t + v_t, \tag{3.9}$$

assuming that the classical assumptions now apply to this model. Of course, it is clear that the classical assumptions cannot apply to the regression of Y on X and to the reverse regression of X on Y at the same time. But testing the null hypothesis that $\beta = 0$ and $\delta = 0$ are equivalent since the null states that there is no relationship between the two variables. However, if the null of no relationship between Y and X is rejected, then to measure the size of the effect of X on Y ($\beta_{X \cdot Y}$) as compared with the size of the effect of Y on X ($\beta_{Y \cdot X}$), will crucially depend on whether the classical assumptions are likely to hold for the regression of Y on X or for the reverse regression of X on Y. As was already established in Chapter 1, $\hat{\beta}_{Y \cdot X}\hat{\beta}_{X \cdot Y} = \hat{\rho}_{YX}^2 = \hat{\rho}_{XY}^2$ (see (1.9)), from

which it follows in general that the estimates of the effects of X on Y and the effects of Y on X do not match, in the sense that $\hat{\beta}_{Y\cdot X}$ is not equal to $1/\hat{\beta}_{X\cdot Y}$, unless $\hat{\rho}^2_{XY} = 1$, which does not apply in practice.

Hence, in order to find the size of the effects the direction of the analysis (whether Y is regressed on X or X regressed on Y) matters crucially. But, if the purpose of the analysis is simply to test for the significance of the statistical relationship between Y and X, the direction of the regression does not matter and it is sufficient to test the null hypothesis of zero correlation (or more generally zero dependence) between Y and X. This can be done using a number of alternative measures of dependence between Y and X. In addition to ρ_{YX}, one can also use Spearman rank correlation and Kendall's τ coefficients defined in Section 1.4. The rank correlation measures are less sensitive to outliers and are more appropriate when the underlying bivariate distribution of (Y and X) show significant departures from Gaussianity and the sample size, T, under consideration is small. But in cases where T is sufficient large (60 or more), and the underlying bivariate distribution has fourth-order moments, then the use of simple correlation coefficient, ρ_{YX}, seems appropriate and tests based on it are likely to be more powerful than tests based on rank correlation coefficients.

Under the null hypothesis that Y and X are independently distributed $\sqrt{T}\hat{\rho}_{YX}$ is asymptotically distributed as $N(0, 1)$, and a test of $\rho_{YX} = 0$ can be based on

$$z_\rho = \sqrt{T}\hat{\rho}_{YX} \rightarrow_d N(0, 1).$$

Fisher has derived an exact sample distribution for $\hat{\rho}_{YX}$ when the observations are from an underlying bivariate normal distribution. But in general no exact sampling distribution is known for $\hat{\rho}_{YX}$ in the case of non-Gaussian processes. In small samples more accurate inferences can be achieved by basing the test of $\rho_{YX} = 0$ on $t_{\hat{\beta}} = \hat{\rho}_{YX}\sqrt{(T-2)/(1-\hat{\rho}^2_{YX})}$ which is distributed approximately as the Student's t with $T - 2$ degrees of freedom. This result follows from the equivalence of testing $\rho_{YX} = 0$ and testing $\beta = 0$ in the simple regression model $y_t = \alpha + \beta x_t + u_t$.

To use the Spearman rank correlation to test the null hypothesis that Y and X are independent, we recall from (1.10) that the Spearmen rank correlation, r_s, between Y and X is defined by

$$r_s = 1 - \frac{6\sum_{t=1}^{T} d_t^2}{T(T^2-1)}, \tag{3.10}$$

where d_t is the difference between the ranks of the two variables. Under the null hypothesis of zero rank correlation between y and x ($\rho_s = 0$, where ρ_s is the rank correlation coefficient in the population from which the sample is drawn) we have

$$Var(r_s) = \frac{1}{T-1}. \tag{3.11}$$

Furthermore, for sufficiently large T, r_s is normally distributed. A more accurate approximate test of $\rho_s = 0$ is given by

$$t_{s,T-2} = \frac{r_s\sqrt{T-2}}{\sqrt{1-r_s^2}}, \tag{3.12}$$

which is distributed (under $\rho_s = 0$) as Student t with $T - 2$ degrees of freedom

Alternatively, Kendall's τ correlation coefficient, defined by (1.11), can be used to test the null hypothesis that Y and X are independent, or in the context of Kendall's measure under the null hypothesis of zero concordance between Y and X in the population. Under the null of zero concordance $E(\tau_T) = 0$ and $Var(\tau_T) = 2(2T+5)/[9T(T-1)]$, and the test can be based on

$$z_\tau = \frac{\sqrt{9T(T-1)}\tau_T}{\sqrt{2(2T+5)}}, \tag{3.13}$$

which is approximately distributed as $N(0, 1)$.

3.5 Hypothesis testing in multiple regression models

Consider now the multiple regression model

$$y_t = \boldsymbol{\beta}'\mathbf{x}_t + u_t, \qquad u_t \sim N\left(0, \sigma^2\right), \tag{3.14}$$

and suppose that we are interested in testing the null hypothesis on the jth coefficient

$$H_0 : \beta_j = \beta_{j0}, \tag{3.15}$$

against the two-sided alternative

$$H_1 : \beta_j \neq \beta_{j0}.$$

Using a similar line of reasoning as above, it is easy to see that conditional on $\mathbf{X}$

$$\hat{\beta}_j \sim N\left(\beta_j, \sigma^2 \left(\mathbf{X}'\mathbf{X}\right)^{-1}_{jj}\right),$$

where $\left(\mathbf{X}'\mathbf{X}\right)^{-1}_{jj}$ is the (j, j) element of the matrix $\left(\mathbf{X}'\mathbf{X}\right)^{-1}$ (see expression (2.13)). Hence, in the case where σ^2 is known, the test can be based on the following standardized statistic

$$Z_{\hat{\beta}_j} = \frac{\hat{\beta}_j - \beta_{j0}}{\sigma\left[\left(\mathbf{X}'\mathbf{X}\right)^{-1}_{jj}\right]^{1/2}},$$

Under the null hypothesis (3.15), $Z_{\hat{\beta}_j} \sim N(0, 1)$ and the critical values of the normal distribution will be applicable. When σ^2 is not known, the unbiased estimator of σ^2, given by (2.14), namely

$$\hat{\sigma}^2 = \frac{\sum_t \hat{u}_t^2}{T-k} = \frac{\hat{\mathbf{u}}'\hat{\mathbf{u}}}{T-k},$$

can be used, where k is the number of regression coefficients (inclusive of an intercept, if any). Replacing σ^2 with $\hat{\sigma}^2$, yields the t-statistic

$$t_{\hat{\beta}_j} = \frac{\hat{\beta}_j - \beta_{j0}}{\hat{\sigma}\left[(\mathbf{X}'\mathbf{X})^{-1}_{jj}\right]^{1/2}},$$

which, under the null hypothesis, H_0 has a t-distribution with $T - k$ degrees of freedom.

3.5.1 Confidence intervals

Knowledge of the distribution of the estimated regression coefficients $\hat{\beta}_1, \hat{\beta}_2, \ldots, \hat{\beta}_k$ can also be used to construct exact confidence intervals for the regression coefficients $\beta_1, \beta_2, \ldots, \beta_k$. Consider the multiple regression model (3.14), and suppose that we are interested in $(1 - \alpha) \times 100$ per cent confidence interval for the regression coefficients. Since $\hat{\beta}_j$ individually have a t-distribution with $T - k$ degree of freedom, then the $(1 - \alpha) \times 100$ per cent confidence interval for β_j is given by

$$\hat{\beta}_j \pm t_\alpha\,(T - k)\,\widehat{S.E.}\left(\hat{\beta}_j\right), \tag{3.16}$$

where $t_\alpha\,(T - k)$ is the $(1 - \alpha) \times 100$ per cent critical value of the t-distribution with $T - k$ degrees of freedom for a two-sided test, and $\widehat{S.E.}\left(\hat{\beta}_j\right)$ is the estimated standard error of $\hat{\beta}_j$.

3.6 Testing linear restrictions on regression coefficients

Consider the linear regression model

$$y_t = \alpha + \beta_1 x_{t1} + \beta_2 x_{t2} + u_t, \tag{3.17}$$

and assume that it satisfies all the classical assumptions. Suppose now that we are interested in testing the hypothesis

$$H_0 : \beta_1 + \beta_2 = 1,$$

against

$$H_1 : \beta_1 + \beta_2 \neq 1.$$

Let

$$\delta = \beta_1 + \beta_2 - 1, \tag{3.18}$$

then the test of H_0 against H_1 simplifies to the test of

$$H_0 : \delta = 0,$$

against

$$H_1 : \delta \neq 0.$$

The *OLS* estimator of δ is given by

$$\hat{\delta} = \hat{\beta}_1 + \hat{\beta}_2 - 1,$$

and the relevant statistic for testing $\delta = 0$ is given by

$$t_{\hat{\delta}} = \frac{\hat{\delta} - 0}{\sqrt{\widehat{Var}\left(\hat{\delta}\right)}} = \frac{\hat{b}_1 + \hat{b}_2 - 1}{\sqrt{\widehat{Var}\left(\hat{\delta}\right)}}.$$

where

$$\widehat{Var}\left(\hat{\delta}\right) = \widehat{Var}\left(\hat{\beta}_1\right) + \widehat{Var}\left(\hat{\beta}_2\right) + 2\widehat{\text{Cov}}\left(\hat{\beta}_1, \hat{\beta}_2\right).$$

The relevant expressions of the variance-covariance matrix of the regression coefficients are given in relations (2.20)–(2.22).

An alternative procedure for testing $\delta = 0$ which does not require knowledge of Cov $\left(\hat{\beta}_1, \hat{\beta}_2\right)$ would be to use (3.18) to solve for β_1 or β_2 in the regression equation (3.17). Solving for β_2, for example, we have

$$y_t = \beta_0 + \beta_1 x_{t1} + \left(\delta - \beta_1 + 1\right) x_{t2} + u_t,$$

or

$$y_t - x_{t2} = \beta_0 + \beta_1 \left(x_{t1} - x_{t2}\right) + \delta x_{t2} + u_t. \tag{3.19}$$

Therefore, the test of $\delta = 0$ against $\delta \neq 0$ can be carried out by means of a simple t-test on the regression coefficient of x_{t2} in the regression of $(y_t - x_{t2})$ on $(x_{t1} - x_{t2})$ and x_{t2}.

Example 6 *This example describes two different methods of testing the hypothesis of constant returns to scale in the context of the Cobb–Douglas (CD) production function*

$$Y_t = AK_t^{\alpha} L_t^{\beta} e^{u_t}, \quad t = 1, 2, \ldots, T, \tag{3.20}$$

where Y_t = Output, K_t = Capital Stock, L_t = Employment. The unknown parameters A, α and β are fixed, and u_ts are serially uncorrelated disturbances with zero means and a constant variance. We also assume that u_ts are distributed independently of K_t and L_t. The constant returns to scale hypothesis postulates that proportionate changes in inputs (K_t and L_t) result in the same proportionate change in output. For example, doubling K_t and L_t should, under the constant returns to scale hypothesis, lead also to the doubling of Y_t. This imposes the following parametric restriction on (3.20):

$$H_0: \quad \alpha + \beta = 1,$$

which we consider as the null hypothesis and derive an appropriate test of it against the two-sided alternative:

$$H_1: \quad \alpha + \beta \neq 1.$$

In order to implement the test of H_0 against H_1, we first take logarithms of both sides of (3.20), which yield the log-linear specification

$$LY_t = a + \alpha LK_t + \beta LL_t + u_t \tag{3.21}$$

where

$$LY_t = \log(Y_t), \quad LK_t = \log(K_t), \quad LL_t = \log(L_t)$$

and $a = \log(A)$. It is now possible to obtain estimates of α and β by running OLS regressions of LY_t on LK_t and LL_t (for $t = 1, 2, \ldots, T$), including an intercept in the regression. Denote the OLS estimates of α and β by $\hat{\alpha}$ and $\widehat{\beta}$, and define a new parameter, δ, as

$$\delta = \alpha + \beta - 1. \tag{3.22}$$

The hypothesis $\alpha + \beta = 1$ against $\alpha + \beta \neq 1$ can now be written equivalently as

$$\begin{aligned} H_0: &\quad \delta = 0, \\ H_1: &\quad \delta \neq 0. \end{aligned}$$

We now consider two alternative methods of testing $\delta = 0$: a direct method and a regression method. The first method directly focuses on the OLS estimates of δ, namely $\hat{\delta} = \hat{\alpha} + \hat{\beta} - 1$, and examines whether this estimate is significantly different from zero. For this we need an estimate of the variance of $\hat{\delta}$. We have

$$V(\hat{\delta}) = V(\hat{\alpha}) + V(\hat{\beta}) + 2\,\mathrm{Cov}\left(\hat{\alpha}, \hat{\beta}\right),$$

where $V(\cdot)$ and $\mathrm{Cov}(\cdot)$ stand for the variance and the covariance operators, respectively. The OLS estimator of $V(\hat{\delta})$ is given by

$$\hat{V}(\hat{\delta}) = \hat{V}(\hat{\alpha}) + \hat{V}(\hat{\beta}) + 2\widehat{\mathrm{Cov}}(\hat{\alpha}, \hat{\beta}).$$

The relevant test-statistic for testing $\delta = 0$ against $\delta \neq 0$ is now given by

$$t_{\hat{\delta}} = \frac{\hat{\delta}}{\sqrt{\hat{V}(\hat{\delta})}} = \frac{\hat{\alpha} + \hat{\beta} - 1}{\sqrt{\hat{V}(\hat{\alpha}) + \hat{V}(\hat{\beta}) + 2\widehat{\mathrm{Cov}}(\hat{\alpha}, \hat{\beta})}}, \tag{3.23}$$

and, under $\delta = 0$, has a t-distribution with $T - 3$ degrees of freedom. An alternative method for testing $\delta = 0$ is the regression method. This starts with (3.21) and replaces β (or α) in terms of δ

and α (or β). Using (3.22) we have

$$\beta = \delta - \alpha + 1.$$

Substituting this in (3.21) for β now yields

$$LY_t - LL_t = a + \alpha(LK_t - LL_t) + \delta LL_t + u_t,$$

or

$$Z_t = a + \alpha W_t + \delta LL_t + u_t, \tag{3.24}$$

where $Z_t = \log(Y_t/L_t) = LY_t - LL_t$ and $W_t = \log(K_t/L_t) = LK_t - LL_t$. A test of $\delta = 0$ can now be carried out by first regressing Z_t on W_t and LL_t (including an intercept term), and then carrying out the usual t-test on the coefficient of LL_t in (3.24). The t-ratio of δ in (3.24) will be identical to $t_{\hat{\delta}}$ defined by (3.23). We now apply the two methods discussed above to the historical data on Y, K, and L used originally by Cobb and Douglas (1928), covering the period 1899–1922. The following estimates of $\hat{\alpha}$, $\hat{\beta}$ and of the variance covariance matrix of $(\hat{\alpha}, \hat{\beta})'$ can be obtained:

$$\hat{\alpha} = 0.23305, \quad \hat{\beta} = 0.80728,$$

$$\begin{bmatrix} \hat{V}(\hat{\alpha}) & \widehat{\mathrm{Cov}}\left(\hat{\alpha}, \hat{\beta}\right) \\ \widehat{\mathrm{Cov}}\left(\hat{\alpha}, \hat{\beta}\right) & \hat{V}(\hat{\beta}) \end{bmatrix} = \begin{bmatrix} 0.004036 & -0.0083831 \\ -0.0083831 & 0.021047 \end{bmatrix}.$$

Using the above results in (3.23) yields

$$t_{\hat{\delta}} = \frac{0.23305 + 0.80728 - 1}{\sqrt{0.004036 + 0.021047 - 2(0.0083831)}} = 0.442. \tag{3.25}$$

Comparing $t_{\hat{\delta}} = 0.442$ and the 5 per cent critical value of the t-distribution with $T - 3 = 24 - 3 = 21$ degrees of freedom (which is equal to 2.080), it is clear that since $t_{\hat{\delta}} = 0.442 < 2.080$, then the hypothesis $\delta = 0$ or $\alpha + \beta = 1$ cannot be rejected at the 5 per cent level. Implementing the regression approach, we estimate (3.24) by OLS and obtain estimates for the coefficients of W_t and LL_t of 0.2330(0.06353) and 0.0403(0.0912), respectively. (The figures in brackets are standard errors.) Note that the t-ratio of the coefficient of the LL variable in this regression is equal to $0.0403/0.0912 = 0.442$, which is identical to $t_{\hat{\delta}}$ as computed in (3.25). It is worth noting that the estimates of α and β, which have played a historically important role in the literature, are very 'fragile', in the sense that they are highly sensitive to the sample period chosen in estimating them. For example, estimating the model (given in (3.21)) over the period 1899–1920 (dropping the observations for the last two years) yields $\hat{\alpha} = 0.0807(0.1099)$ and $\hat{\beta} = 1.0935(0.2241)$.

3.7 Joint tests of linear restrictions

So far we have considered testing a *single* linear restriction on the regression coefficients. Suppose now that we are interested in testing two or more linear restrictions, jointly. One simple example

is the joint test of zero restrictions on the regression coefficients:

$$H_0 : \beta_1 = \beta_2 = 0,$$
$$H_1 : \beta_1 \neq 0 \qquad \text{and/or} \qquad \beta_2 \neq 0.$$

Note that this joint hypothesis is different from testing the following two hypotheses separately.

$$\begin{cases} H_0^I : \beta_1 = 0, \\ H_1^I : \beta_2 \neq 0. \end{cases} \qquad \text{or} \qquad \begin{cases} H_0^{II} : \beta_2 = 0, \\ H_1^{II} : \beta_1 \neq 0. \end{cases}$$

The latter tests are known as separate induced tests and could lead to test outcomes that differ from the outcome of a joint test.

The general procedure for testing joint hypotheses in regression contexts is to construct the *F*-statistic that compares the sum of squares of residuals (SSR) of the regression under the restrictions (i.e., under H_0) with the SSR under the alternative hypothesis, H_1, when the parameter restrictions are not applied. This procedure is valid for a two-sided test. Carrying out one sided tests in the case of joint hypotheses is more complicated and will not be addressed here.

The relevant statistic for the joint test of $r \leq k$ different linear restrictions on the regression coefficients is

$$F = \left(\frac{T-k-1}{r}\right)\left(\frac{SSR_R - SSR_U}{SSR_U}\right), \tag{3.26}$$

where

$SSR_R \equiv$ Restricted sum of squares of errors (residuals)
$SSR_U \equiv$ Unrestricted sum of squares of errors
$k \equiv$ Number of regression coefficients, excluding the intercept term
$T \equiv$ Number of observations
$r \equiv$ Number of independent linear restrictions on the regression coefficients.

Under the null hypothesis, the above statistic, *F*, has an *F*-distribution with *r* and $T - k - 1$ degrees of freedom.

Consider now the application of this general procedure to the problem of testing $\beta_1 = \beta_2 = 0$. The restricted sum of squares of errors (SSR_R) for the problem is obtained by imposing the restrictions $\beta_1 = \beta_2 = 0$ on (3.17) and then by estimating the restricted model

$$y_t = \beta_0 + u_t.$$

This yields $\hat{\beta}_0 = \bar{y}$ and hence

$$SSR_R = \sum_t \left(y_t - \bar{y}\right)^2 = S_{YY}.$$

The unrestricted sum of squares of errors is given by

$$\begin{aligned} SSR_U &= \sum_t \left(y_t - \hat{\beta}_0 - \hat{\beta}_1 x_{t1} - \hat{\beta}_2 x_{t2}\right)^2 \\ &= \sum_t \hat{u}_t^2. \end{aligned}$$

Hence

$$F = \left(\frac{T-3}{2}\right)\left(\frac{S_{YY} - \sum_t \hat{u}_t^2}{\sum_t \hat{u}_t^2}\right),$$

which under the null hypothesis $H_0 : \beta_1 = \beta_2 = 0$, has an F-distribution with 2 and $T-3$ degrees of freedom. The joint hypothesis is rejected if F is larger that the $(1-\alpha)$ per cent critical value of the F-distribution with 2 and $T-3$ degrees of freedom.

3.8 Testing general linear restrictions

All the above tests can be derived as a special case of tests of the following r general linear restrictions

$$\begin{aligned} H_0 &: \mathbf{R}\boldsymbol{\beta} - \mathbf{d}_0 = \mathbf{0}, \\ H_1 &: \mathbf{R}\boldsymbol{\beta} - \mathbf{d}_0 \neq \mathbf{0}, \end{aligned}$$

where $\mathbf{R}$ is an $r \times k$ matrix of known constants with full row rank given by $r \leq k$, and $\mathbf{d}$ is an $r \times 1$ vector of constants. The different hypotheses considered above can be obtained by appropriate choice of $\mathbf{R}$ and $\mathbf{d}_0$. For example, if the object of the exercise is to test the null hypothesis that the first element of $\boldsymbol{\beta}$ is equal to zero, then we need to set $\mathbf{R} = (1, 0, \ldots, 0)$, and $\mathbf{d}_0 = 0$. To test the hypothesis that the sum of the first two elements adds up to 2 and the sum of the second two elements of $\boldsymbol{\beta}$ adds up to 3 we set

$$\mathbf{R} = \begin{pmatrix} 1 & 1 & 0 & 0 & \ldots & 0 \\ 0 & 1 & 1 & 0 & \ldots & 0 \end{pmatrix}, \mathbf{d}_0 = \begin{pmatrix} 2 \\ 3 \end{pmatrix}.$$

The F-statistic for testing H_0 is given by

$$F = \frac{\left(\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{d}_0\right)' \left[\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'\right]^{-1} \left(\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{d}_0\right)}{r\hat{\sigma}^2}, \tag{3.27}$$

where $\hat{\boldsymbol{\beta}}$ is the unrestricted *OLS* estimator of $\boldsymbol{\beta}$, and $\hat{\sigma}^2 = (\mathbf{y} - X\hat{\boldsymbol{\beta}})'(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}})/(T-k)$ is the unbiased estimator of σ^2. Using the distributional results obtained in Chapter 2, in particular the result given by (2.28), it follows that under H_0 the F statistic given by (3.27) has a central F-distribution with r and $T-k$ degrees of freedom. This result of course requires that the classical normal regression assumptions A1–A5 set out in Chapter 2 hold.

3.8.1 Power of the F-test

To obtain the power of the F-test defined by (3.27), consider the alternative hypothesis, H_1, where $\mathbf{R}\boldsymbol{\beta} = \mathbf{d}_1$, and recall that $\mathbf{R}$ is an $r \times k$ matrix of constants with full column rank r. Note that, under H_1,

$$\begin{aligned}\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{d}_0 &= \mathbf{R}\boldsymbol{\beta} - \mathbf{d}_0 + \mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{u} \\ &= \boldsymbol{\delta} + \mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{u} \sim N(\boldsymbol{\delta}, \sigma^2\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'),\end{aligned}$$

where $\boldsymbol{\delta} = \mathbf{d}_1 - \mathbf{d}_0$. Hence

$$X_1 = \frac{\left(\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{d}_0\right)'\left[\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'\right]^{-1}\left(\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{d}_0\right)}{\sigma^2} \sim \chi^2_r(\lambda), \tag{3.28}$$

where $\chi^2_r(\lambda)$ is a non-central chi-square variate with r degrees of freedom and the non-centrality parameter[1]

$$\lambda = \frac{\boldsymbol{\delta}'\left[\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'\right]^{-1}\boldsymbol{\delta}}{\sigma^2} = \boldsymbol{\delta}'\left[\mathbf{R}Var(\hat{\boldsymbol{\beta}})R'\right]^{-1}\boldsymbol{\delta}. \tag{3.29}$$

Furthermore, from (2.27) we know that $X_2 = (T-k)\,\hat{\sigma}^2/\sigma^2 \sim \chi^2_{T-k}$. Using a similar line of reasoning as in Chapter 2, it is easily seen that X_1 (defined by (3.28)) and X_2 are independently distributed, and hence under H_1 the F-statistic given by (3.27) is distributed as a non-central F-distribution with r and $T-k$ degrees of freedom, and the non-centrality parameter, λ, given by (3.29). For given values of r and k, the power of the F test is monotonically increasing in λ. It is clear that the power is higher the greater the distance between the null and the alternative hypotheses as measured by $\boldsymbol{\delta}$, and the greater the precision with which the *OLS* estimators are estimated, as measured by the inverse of $Var(\hat{\boldsymbol{\beta}})$.

3.9 Relationship between the *F*-test and the coefficient of multiple correlation

The relationship between the correlation coefficient and the t-statistic discussed earlier can be readily extended to the multivariate context. Consider the multivariate regression model

$$y_t = \beta_0 + \sum_{j=1}^{k}\beta_j x_{tj} + u_t, \qquad t = 1, 2, \ldots, T,$$

[1] For further information regarding the non-central chi-square distribution see Section B.10.2 in Appendix B.

and suppose we are interested in testing the joint significant of the regressors $x_{t1}, x_{t2}, \ldots, x_{tk}$. The relevant hypothesis is

$$H_0 : \beta_1 = \beta_2, \cdots = \beta_k = 0,$$
$$H_1 : \beta_1 \neq 0, \beta_2 \neq 0, \cdots \beta_k \neq 0.$$

The F-test for this test is given by

$$F = \left(\frac{T-k-1}{k}\right)\left(\frac{S_{YY} - \sum_t \hat{u}_t^2}{\sum_t \hat{u}_t^2}\right),$$

The multiple correlation coefficient is defined by (see (2.30))

$$R^2 = 1 - \frac{\sum_t \hat{u}_t^2}{S_{YY}}.$$

Hence

$$F = \left(\frac{T-k-1}{k}\right)\left(\frac{S_{YY}}{\sum_t \hat{u}_t^2} - 1\right) = \left(\frac{T-k-1}{k}\right)\left(\frac{R^2}{1-R^2}\right),$$

which yields the generalization of the result (3.7) obtained in the case of the simple regression.

3.10 Joint confidence region

To construct a joint confidence region of size $(1-\alpha) \times 100$ for $(\beta_1, \beta_2, \ldots, \beta_k)$, we first note that the combination of the confidence intervals (3.16) constructed for each β_j separately *does not* yield a joint confidence region with the correct size (namely $1-\alpha$). This is because of dependence of the estimated regression coefficients on each other. Only in the case where $\text{Cov}\left(\hat{\beta}_i, \hat{\beta}_j\right) = 0$ for all $i \neq j$, the joint confidence region of $(\beta_1, \beta_2, \ldots, \beta_k)$ coincides with the intersection of the confidence intervals obtained for each regression coefficient separately. The appropriate joint confidence region for $\beta_1, \beta_2, \ldots, \beta_k$ is constructed using the F-statistic.

The $(1-\alpha) \times 100$ per cent joint confidence region for β_1 and β_2 in the three variable regression model (2.15) is an ellipsoid in the β_1 and β_2 plane. The shape and the position of this ellipsoid is determined by the size of the confidence region, $1-\alpha$, the *OLS* estimates $\hat{\beta}_1$ and $\hat{\beta}_2$ and the degree of the statistical dependence between the estimators of β_1 and β_2. In matrix notations the formula for this ellipsoid is given by

$$F_\alpha\,(2, T-3) = \left(\boldsymbol{\beta} - \hat{\boldsymbol{\beta}}\right)' \left[\widehat{\text{Cov}}\left(\hat{\boldsymbol{\beta}}\right)\right]^{-1} \left(\boldsymbol{\beta} - \hat{\boldsymbol{\beta}}\right), \tag{3.30}$$

where $\boldsymbol{\beta} = (\beta_1, \beta_2)'$,

$$\widehat{\text{Cov}}\left(\hat{\boldsymbol{\beta}}\right) = \begin{pmatrix} \widehat{Var}\left(\hat{\beta}_1\right) & \widehat{\text{Cov}}\left(\hat{\beta}_1, \hat{\beta}_2\right) \\ \widehat{\text{Cov}}\left(\hat{\beta}_1, \hat{\beta}_2\right) & \widehat{Var}\left(\hat{\beta}_2\right) \end{pmatrix},$$

and $F_\alpha\,(2, T-3)$ is the $(1-\alpha) \times 100$ per cent critical value of the F-distribution with 2 and $T-3$ degrees of freedom.

3.11 The multicollinearity problem

Multicollinearity is commonly attributed to situations where there is a high degree of intercorrelations among the explanatory variables in a multivariate regression equation. Multicollinearity is particularly prevalent in the case of time series data where there often exists the same common trend in two or more regressors in the regression equation. As a simple example consider the model

$$y_t = \beta_0 + \beta_1 x_{t1} + \beta_2 x_{t2} + u_t, \tag{3.31}$$

and assume for simplicity that (x_{t1}, x_{t2}) have a bivariate distribution with the correlation coefficient, ρ_{12}. That is

$$\rho_{12} = \frac{\text{Cov}\,(x_{t1}, x_{t2})}{[Var\,(x_{t1})\;Var\,(x_{t2})]^{\frac{1}{2}}}.$$

It is clear that as ρ approaches unity *separate* estimation of the slope coefficients β_1 and β_2 becomes more and more problematic. Multicollinearity (namely a value of ρ_{12} near unity in the context of the present example) will be a problem if x_{t1} and x_{t2} are jointly statistically significant but neither is statistically significant when taken individually. Put differently, multicollinearity will be a problem when the hypothesis $\beta_1 = 0$ and $\beta_2 = 0$ can not be rejected when tested separately, while the joint hypothesis that $\beta_1 = \beta_2 = 0$ is rejected. This clearly happens when x_{t1} (or x_{t2}) is an exact linear function of x_{t2} (or x_{t1}). In this case $x_{t2} = \gamma x_{t1}$ and (3.31) reduces to the simple regression equation

$$y_t = \alpha + (\beta_1 + \beta_2 \gamma)\, x_{t1} + u_t,$$

and it is only possible to estimate $\beta_1 + \gamma\beta_2$. Neither β_1 nor β_2 can be estimated (or tested) separately. This is the case of 'perfect multicollinearity' and arises out of faulty specification of the regression equation. One important example is when four seasonal dummies are included in a quarterly regression model that already contains an intercept term. In general the multicollinearity problem is likely to arise when ρ_{12}^2 is close to 1.

The multicollinearity problem is also closely related to the problem of low power when testing hypotheses concerning the values of the regression coefficients separately. It is worth noting that no matter how large the correlation coefficient between x_{t1} and x_{t2}, so long as it is not exactly equal to ± 1, a test of $\beta_1 = 0$ (or $\beta_2 = 0$) will have the correct size. The high degree of

correlation between x_{t1} and x_{t2} causes the power of the test to be rather low and as a result we may end up not rejecting the null hypothesis that $\beta_1 = 0$ even if it is false.

Example 7 *To demonstrate the multicollinearity problem and its relation to the problem of low power, using Microfit 5.0 we generated 1,000 observations on x_1, x_2 and y in the following manner.*

$$\begin{aligned} x_1 &\sim N(0,1), \\ x_2 &= x_1 + 0.15v, \\ v &\sim N(0,1), \\ y &= \beta_0 + \beta_1 x_1 + \beta_2 x_2 + u, \\ u &\sim N(0,1), \end{aligned}$$

with $\beta_0 = \beta_1 = \beta_2 = 1$ and where x_1, v and u were generated as independent standardized normal variates using respectively the 'seeds' of 123, 321 and 4321 in the normal random generator. The Microfit batch file for this exercise is called MULTI.BAT and contains the following instructions:

```
SAMPLE   1   1000;
X1 = NORMAL(123);
V = NORMAL(321);
U = NORMAL(4321);
X2 = X1+0.15*V;
Y = 1 + X1 + X2 + U;
```

Now running the regression of y on x_1 and x_2 (including an intercept term) using only the first fifty observations yields

$$\underset{\quad}{y_t =} \underset{(0.1299)}{0.9047} + \underset{(1.0403)}{1.0950}\, x_{t1} + \underset{(1.0200)}{0.8719}\, x_{t2} + \hat{u}_t \qquad t = 1, 2, \ldots, 50, \tag{3.32}$$

$$R^2 = 0.8498, \qquad \hat{\sigma} = 0.8890, \qquad F_{2,47} = 132.98, \tag{3.33}$$

The standard errors of the parameter estimates are given in brackets, R is the multiple correlation coefficient, $\hat{\sigma}$ is the estimated standard error of the regression equation, and $F_{2,47}$ is the F-statistics for testing the joint hypothesis

$$H_0^J : \beta_1 = \beta_2 = 0,$$

against

$$H_1^J : \beta_1 \neq 0, \quad \beta_2 \neq 0.$$

The t*-statistics for the test of the separate induced tests of*

$$H_0^I : \beta_1 = 0$$

against

$$H_1^I : \beta_1 \neq 0,$$

and of

$$H_0^{II} : \beta_2 = 0,$$

against

$$H_1^{II} : \beta_2 \neq 0,$$

It is firstly clear that since the value of the F*-statistic* ($F_{2,47} = 132.98$) *for the test of* $H_0^J : \beta_1 = \beta_2 = 0$ *is well above the 95 critical value of the* F*-distribution with 2 and 47 degrees of freedom, we conclude that the joint hypothesis* $\beta_1 = \beta_2 = 0$ *is rejected at least at the 95 per cent significance level. Turning now to the tests of* $\beta_1 = 0$ *and* $\beta_2 = 0$ *separately, (i.e. testing the separate induced null hypotheses* H_0^I *and* H_0^{II}*), we note that the* t*-statistics for these hypotheses are equal to* $t_{\hat{\beta}_1} = 1.0950/1.0403 = 1.05$ *and* $t_{\hat{\beta}_2} = 0.8719/1.0200 = 0.85$*, respectively. Neither is statistically significant and the null hypothesis of* $\beta_1 = 0$ *or* $\beta_2 = 0$ *can not be rejected. There is clearly a multicollinearity problem. The joint hypothesis that* β_1 *and* β_2 *are both equal to zero is strongly rejected, but neither of the hypotheses that* β_1 *and* β_2 *are separately equal to zero can be rejected. The sample correlation coefficient of* x_1 *and* x_2 *computed using the first 50 observations is equal to 0.99316 which is apparently too high, given the sample size and the fit of the underlying equation, for the* β_1 *and* β_2 *coefficients to be estimated separately with any degree of precision. In short, the separate induced tests lack the necessary power to allow rejection of* $\beta_1 = 0$ *and* $\beta_2 = 0$ *separately. The relationship between the* F*-statistic used to test the joint hypothesis* $\beta_1 = \beta_2 = 0$*, and the* t*-statistics used to test* $\beta_1 = 0$ *and* $\beta_2 = 0$ *separately, can also be obtained theoretically. Recall from Section 3.7 that*

$$F = \left(\frac{T-3}{2}\right)\left(\frac{S_{YY} - \sum_t \hat{u}_t^2}{\sum_t \hat{u}_t^2}\right). \tag{3.34}$$

Denote the t-statistics for testing $\beta_1 = 0$ *and* $\beta_2 = 0$ *separately by* t_1 *and* t_2*, respectively. Then*

$$t_j^2 = \frac{\hat{\beta}_j^2}{\widehat{Var}\left(\hat{\beta}_j\right)}, \qquad j = 1, 2.$$

But using results in Example 1 (Chapter 2)

$$\widehat{Var}\left(\hat{\beta}_1\right) = \frac{\hat{\sigma}^2 S_{22}}{S_{11}S_{22} - S_{12}^2},$$
$$\widehat{Var}\left(\hat{\beta}_2\right) = \frac{\hat{\sigma}^2 S_{11}}{S_{11}S_{22} - S_{12}^2},$$

where as before $S_{js} = \sum_t (x_{tj} - \bar{x}_j)(x_{ts} - \bar{x}_s)$. *Also since* $y_t - \bar{y} = \hat{\beta}_1 (x_{t1} - \bar{x}_1) + \hat{\beta}_2 (x_{t2} - \bar{x}) + \hat{u}_t$ *we have*[2]

$$S_{YY} = \sum_t (y_t - \bar{y})^2 = \hat{\beta}_1^2 S_{11} + \hat{\beta}_2^2 S_{22} + 2\hat{\beta}_1 \hat{\beta}_2 S_{12} + \sum_t \hat{u}_t^2.$$

Using these results in the expression for the F*-statistic in (3.34) we obtain:*

$$F = \frac{t_1^2 + t_2^2 + 2\rho_{12} t_1 t_2}{2(1 - \rho_{12}^2)}, \tag{3.35}$$

where ρ_{12} *is the sample correlation coefficient between* x_{t1} *and* x_{t2}.[3] *This relationship clearly shows that even for small values of* t_1 *and* t_2 *it is possible to get quite large values of F so long as* ρ_{12} *is chosen to be close enough to 1.*

The above example considers the simple case of a regression model with two explanatory variables. In case of regression models with more than two regressors the detection of the multicollinearity problem becomes more complicated. For example, when there are three regressors with the coefficients β_1, β_2 and β_3, we need to consider all the possible combinations of the coefficients, namely testing them separately: $\beta_1 = 0, \beta_2 = 0, \beta_3 = 0$; in pairs: $\beta_1 = \beta_2 = 0$, $\beta_2 = \beta_3 = 0, \beta_1 = \beta_3 = 0$; and jointly: $\beta_1 = \beta_2 = \beta_3$. Only in the case where the results of separate induced tests, the 'pairs' tests and the joint test are free from contradictions can we be confident that multicollinearity is not a problem.

There exist a number of measures in the literature that purport to detect and measure the seriousness of the multicollinearity problem. One commonly used diagnostic is the *condition number* defined as the square root of the ratio of the largest to the smallest eigenvalue of the matrix $\mathbf{X}'\mathbf{X}$, where the columns of $\mathbf{X}$ have been re-scaled to length 1 (namely, the elements of the jth column of $\mathbf{X}$ have been divided by $s_j = (\sum_{t=1}^{T} x_{tj}^2)^{1/2}$, for $j = 1, 2, \ldots, k$). The condition number detects whether the matrix $\mathbf{X}'\mathbf{X}$ has a small determinant, namely if it is *ill-conditioned*. The larger the condition number, the more ill-conditioned is the matrix, and difficulties can be encountered in calculations involving $(\mathbf{X}'\mathbf{X})^{-1}$. Values of condition number higher than 30 are suggested as indicative of a problem (see Belsley, Kuh, and Welsch (1980) for details). Another diagnostic used to detect multicollinearity is the *variance-inflation factor (VIF)*, defined as $VIF_j = (1 - R_j^2)^{-1}$, for the jth regressor, where R_j^2 is the squared multiple correlation coefficient of the regression of x_{tj} on all other variables in the regression. A high value of VIF_j suggests that x_{tj} is in some collinear relationship with the other regressors. As a rule of thumb, for scaled data, a VIF_j higher than ten indicates severe collinearity (see Kennedy (2003)). We remark that these measures only examine the inter-correlation between the regressors, and at best give a partial picture of the multicollinearity problem, and can often 'lead' to misleading conclusions.

[2] Note that the *OLS* residuals are orthogonal to the regressors.

[3] In the simulation exercise we obtained $t_1 = 1.05$, $t_2 = 0.85$ and $\rho_{12} = 0.99316$. Using these estimates in (3.35) yields $F = 131.50$, which is of the same order of magnitude as the *F*-statistic reported in (3.34). The difference between the two values is due to the error of approximations.

A useful rule of thumb which goes beyond regressor correlations is to compare the squared multiple correlation coefficient of the regression equation, R^2, with R_j^2. Klein (1962) suggests that collinearity is likely to be a problem and could lead to imprecise estimates if $R^2 < R_j^2$, for some $j = 1, 2, \ldots, k$.

Example 8 *To illustrate the problem return to the simulation exercise, and use the first 500 observations (instead of the first 50 observations) in computing the regression of y on x_1 and x_2. The results are*

$$\underset{}{y_t} = \underset{(0.0428)}{0.9307} + \underset{(0.28343)}{1.1045}\, x_{t1} + \underset{(0.27081)}{0.93138}\, x_{t2} + \hat{u}_t \qquad t = 1, 2, \ldots, 500,$$

$$R^2 = 0.8333, \qquad \hat{\sigma} = 0.95664, \qquad F_{2,497} = 1242.3.$$

As compared with the estimates based on the first 50 observations [see (3.32) and (3.33)], these estimates have much smaller standard errors and using the 95 percent significance level we arrive at similar conclusions whether we test $\beta_1 = 0$ and $\beta_2 = 0$ separately or jointly. Yet the sample correlation coefficient between x_{t1} and x_{t2} estimated over the first 500 observations is equal to 0.9895 which is only marginally smaller than the estimate obtained for the first 50 observations. By increasing the sample size from 50 to 500 we have increased the precision with which β_1 and β_2 are estimated and the power of testing $\beta_1 = 0$ and $\beta_2 = 0$ both separately and jointly.

The above illustration also points to the fact that the main cause of the multicollinearity problem is lack of adequate observations (or information), and hence the imprecision with which the parameters of interest are estimated. Assuming the regression model under consideration is correctly specified, the only valid solution to the problem is to increase the information on the basis of which the regression is estimated. The new information could be either in the form of additional observations on y, x_1 and x_2, or it could be some *a priori* information concerning the parameters. The latter fits well with the Bayesian approach, but is difficult to accommodate within the classical framework. There are also other approaches suggested in the literature such as the ridge regression, and the principle component regression to deal with the multicollinearity problem. For a Bayesian treatment of the regression analysis see Section C.6 in Appendix C. However, in using Bayesian techniques to deal with the multicollinearity problem it is important to bear in mind that the posterior means of the regression coefficients are well defined in small samples even if the regressors are highly multicollinear and even if $\mathbf{X}'\mathbf{X}$ is rank deficient. But in such cases the posterior mean of $\boldsymbol{\beta}$ can be very sensitive to the choice of the priors, and unless $T^{-1}\mathbf{X}'\mathbf{X}$ tends to a positive definite matrix the Bayes estimates of $\boldsymbol{\beta}$ could become unstable as $T \to \infty$.

Example 9 *As an example consider the following Fisher type explanation of nominal interests estimated on US quarterly data over the period 1948(1)–1990(4) using the file USGNP.FIT provided in Microfit 5:*

$$\begin{aligned} R_t = {} & \underset{(0.1295)}{-0.0381} + \underset{(0.0754)}{1.2606}\, R_{t-1} \underset{(0.1144)}{-.61573}\, R_{t-2} + \underset{(0.1208)}{0.6073}\, R_{t-3} - \\ & \underset{(0.0782)}{0.3168}\, R_{t-4} + \underset{(0.1075)}{0.13198}\, DM_{t-1} + \underset{(0.1064)}{0.1072}\, DM_{t-2} + \hat{u}_t, \end{aligned}$$

$$R^2 = 0.9520, \qquad \bar{R}^2 = 0.9503, \qquad \hat{\sigma} = 0.7086, \qquad F_{6,165} = 545.83,$$

where R_t = nominal rate of interest, DM_t = the growth of money supply (M_2 definition). In this regression, the coefficients of the lagged interest rate variables are all significant, but neither of the two coefficients of the lagged monetary growth variable is statistically significant. The t-ratios for the coefficients of DM_{t-1} and DM_{t-2} are equal to 1.23 and 1.01, respectively, while the 95 percent critical value of the t-distribution with 165 (namely $T - k = 172 - 7$) degrees of freedom is equal to 1.97. As we have seen above, it would be a mistake to necessarily conclude from this result that monetary growth has no significant impact on the nominal interest rates in the US. The statistical insignificance of the coefficients of DM_{t-1} and DM_{t-2}, when tested separately may be due to the high intercorrelation between the regressors. Also we are not interested in testing the statistical significance of individual coefficients of the past monetary growth rates. What is of interest is the sum of the two coefficients of the lagged monetary growth rates, and not the individual coefficients, separately. Denote the coefficients of DM_{t-1} and DM_{t-2} by γ_1 and γ_2 respectively, and let $\delta = \gamma_1 + \gamma_2$. We have

$$\hat{\delta} = \hat{\gamma}_1 + \hat{\gamma}_2 = 0.1319 + 0.1072 = 0.2391$$

To compute the estimate of the standard error of $\hat{\delta}$ we recall that

$$\widehat{Var}\left(\hat{\delta}\right) = \widehat{Var}\left(\hat{\gamma}_1\right) + \widehat{Var}\left(\hat{\gamma}_2\right) + 2\widehat{\mathrm{Cov}}\left(\hat{\gamma}_1, \hat{\gamma}_2\right),$$

and using the Microfit package we have

$$\widehat{Var}\left(\hat{\gamma}_1\right) = 0.01156, \qquad \widehat{Var}\left(\hat{\gamma}_2\right) = 0.01132, \qquad \widehat{\mathrm{Cov}}\left(\hat{\gamma}_1, \hat{\gamma}_2\right) = -0.00854$$

and hence $\sqrt{\widehat{Var}\left(\hat{\delta}\right)} = 0.0762$, and $t_\delta = 0.2391/0.0762 = 3.14$ which is well above the 95 percent critical value of the t-distribution with 165 degrees of freedom. Therefore, we strongly reject the hypothesis that monetary growth has no effect on the nominal interest in the US. We also note that for every one percent increase in the growth of money supply there is around 0.24 of one percent increase in nominal interest within the space of two quarters. The long-run impact of money supply growth on nominal interest is much larger and depends on the magnitude of the lagged coefficients of the nominal interest rates.

3.12 Multicollinearity and the prediction problem

Consider the following regression model

$$\mathbf{y} = \mathbf{X}_1\boldsymbol{\beta}_1 + \mathbf{X}_2\boldsymbol{\beta}_2 + \mathbf{u} = \mathbf{X}\boldsymbol{\beta} + \mathbf{u},$$

where $\mathbf{y} = (y_1, y_2, \ldots, y_T)'$, $\mathbf{X}_1$ and $\mathbf{X}_2$ are $T \times k_1$ and $T \times k_2$ regressor matrices that are perfectly correlated, namely

$$\mathbf{X}_2 = \mathbf{X}_1\mathbf{A}',$$

and $\mathbf{A}$ is a $k_2 \times k_1$ matrix of fixed constants. Further assume that $\mathbf{X}_1'\mathbf{X}_1$ is a positive definite matrix. Consider now the forecast of y_{T+1} conditional on $\mathbf{x}_{T+1} = (\mathbf{x}_{1T}', \mathbf{x}_{2T}')'$ which is given by[4]

$$\hat{y}_{T+1} = \mathbf{x}_{T+1}'\hat{\boldsymbol{\beta}}_T = \mathbf{x}_{T+1}'(\mathbf{X}'\mathbf{X})^{+}\mathbf{X}'\mathbf{y},$$

where $(\mathbf{X}'\mathbf{X})^{+}$ is the generalized inverse of $\mathbf{X}'\mathbf{X}$, defined by (see also Section A.7)

$$\left(\mathbf{X}'\mathbf{X}\right)(\mathbf{X}'\mathbf{X})^{+}\left(\mathbf{X}'\mathbf{X}\right) = \left(\mathbf{X}'\mathbf{X}\right).$$

It is well known that $(\mathbf{X}'\mathbf{X})^{+}$ is not unique when $\mathbf{X}'\mathbf{X}$ is rank deficient. In what follows we show that $\hat{y}_{T+1}$ is unique despite the non-uniqueness of $(\mathbf{X}'\mathbf{X})^{+}$. Note that

$$\mathbf{X}'\mathbf{X} = \begin{pmatrix} \mathbf{X}_1'\mathbf{X}_1 & \mathbf{X}_1'\mathbf{X}_1\mathbf{A}' \\ \mathbf{A}\mathbf{X}_1'\mathbf{X}_1 & \mathbf{A}\mathbf{X}_1'\mathbf{X}_1\mathbf{A}' \end{pmatrix} = \mathbf{H}\mathbf{X}_1'\mathbf{X}_1\mathbf{H}',$$

where $\mathbf{H}$ is a $k \times k_1$ matrix $(k = k_1 + k_2)$:

$$\mathbf{H} = \begin{pmatrix} \mathbf{I}_{k_1} \\ \mathbf{A} \end{pmatrix}.$$

Also

$$\mathbf{X}'\mathbf{y} = \mathbf{H}\mathbf{X}_1'\mathbf{y}, \text{ and } \mathbf{x}_{T+1}' = \mathbf{x}_{1T}'\mathbf{H}'.$$

Hence

$$\hat{y}_{T+1} = \mathbf{x}_{T+1}'(\mathbf{X}'\mathbf{X})^{+}\mathbf{X}'\mathbf{y} = \mathbf{x}_{1T}'\mathbf{H}'\left(\mathbf{H}\mathbf{X}_1'\mathbf{X}_1\mathbf{H}'\right)^{+}\mathbf{H}\mathbf{X}_1'\mathbf{y}.$$

Since $\mathbf{X}_1'\mathbf{X}_1$ is a symmetric positive definite matrix, then

$$\hat{y}_{T+1} = \mathbf{x}_{1T}'\left(\mathbf{X}_1'\mathbf{X}_1\right)^{-1/2}\left(\mathbf{X}_1'\mathbf{X}_1\right)^{1/2}\mathbf{H}'\left(\mathbf{H}\left(\mathbf{X}_1'\mathbf{X}_1\right)^{1/2}\left(\mathbf{X}_1'\mathbf{X}_1\right)^{1/2}\mathbf{H}'\right)^{+}$$
$$\mathbf{H}\left(\mathbf{X}_1'\mathbf{X}_1\right)^{1/2}\left(\mathbf{X}_1'\mathbf{X}_1\right)^{-1/2}\mathbf{X}_1'\mathbf{y},$$

or

$$\hat{y}_{T+1} = \mathbf{x}_{1T}'\left(\mathbf{X}_1'\mathbf{X}_1\right)^{-1/2}\mathbf{G}'\left(\mathbf{G}\mathbf{G}'\right)^{+}\mathbf{G}\left(\mathbf{X}_1'\mathbf{X}_1\right)^{-1/2}\mathbf{X}_1'\mathbf{y},$$

where

$$\mathbf{G} = \mathbf{H}\left(\mathbf{X}_1'\mathbf{X}_1\right)^{1/2}.$$

Consider now the $k_1 \times k_1$ matrix $\mathbf{G}\left(\mathbf{G}'\mathbf{G}\right)^{+}\mathbf{G}'$ and note that from properties of generalized inverse we have

[4] A general treatment of the prediction problem is given in Chapter 17.

$$\left(\mathbf{GG}'\right)\left(\mathbf{GG}'\right)^{+}\left(\mathbf{GG}'\right)=\left(\mathbf{GG}'\right).$$

Pre- and post-multiplying the above by $\mathbf{G}'$ and $\mathbf{G}$, we have

$$\left(\mathbf{G}'\mathbf{G}\right)\mathbf{G}'\left(\mathbf{GG}'\right)^{+}\mathbf{G}\left(\mathbf{G}'\mathbf{G}\right)=\left(\mathbf{G}'\mathbf{G}\right)\left(\mathbf{G}'\mathbf{G}\right). \tag{3.36}$$

But

$$\mathbf{G}'\mathbf{G}=\left(\mathbf{X}_1'\mathbf{X}_1\right)^{1/2}\mathbf{H}'\mathbf{H}\left(\mathbf{X}_1'\mathbf{X}_1\right)^{1/2},$$

and since

$$\mathbf{H}'\mathbf{H}=\mathbf{I}_{k_1}+\mathbf{A}'\mathbf{A},$$

then

$$\mathbf{G}'\mathbf{G}=\left(\mathbf{X}_1'\mathbf{X}_1\right)^{1/2}\left(\mathbf{I}_{k_1}+\mathbf{A}'\mathbf{A}\right)\left(\mathbf{X}_1'\mathbf{X}_1\right)^{1/2},$$

is a nonsingular matrix (for any $\mathbf{A}$) and has a unique inverse. Using this result in (3.36) it now follows that

$$\mathbf{G}'\left(\mathbf{GG}'\right)^{+}\mathbf{G}=\mathbf{I}_{k_1},$$

and hence

$$\begin{aligned}\hat{y}_{T+1}&=\mathbf{x}_{1T}'\left(\mathbf{X}_1'\mathbf{X}_1\right)^{-1/2}\mathbf{G}'\left(\mathbf{GG}'\right)^{+}\mathbf{G}\left(\mathbf{X}_1'\mathbf{X}_1\right)^{-1/2}\mathbf{X}_1'\mathbf{y}\\&=\mathbf{x}_{1T}'\left(\mathbf{X}_1'\mathbf{X}_1\right)^{-1}\mathbf{X}_1'\mathbf{y},\end{aligned}$$

which is unique and invariant to the choice of the generalized inverse of $\mathbf{X}'\mathbf{X}$.

3.13 Implications of misspecification of the regression model on hypothesis testing

Suppose that y_t is generated according to the classical linear regression equation

$$y_t=\beta_0+\beta_1x_t+\beta_2z_t+u_t, \tag{3.37}$$

but the investigator estimates the simple regression equation

$$y_t=\alpha+\beta x_t+\varepsilon_t, \tag{3.38}$$

which omits the regressor z_t. We have seen in Section 2.13 that omitting a relevant regressor, z_t, may lead to biased estimates, unless the included regressor, x_t, and the omitted variable, z_t, are uncorrelated. However, even in the case x_t and z_t are uncorrelated, $\hat{\beta}$ will not be an efficient

estimator of β_1. This is because the correct estimator of the variance of $\hat{\beta}$ requires knowledge of an estimator of $\sigma_u^2 = Var(u_t)$, namely

$$\hat{\sigma}_u^2 = \frac{\sum_t \hat{u}_t^2}{T-3} = \frac{\sum_t \left(y_t - \hat{\beta}_0 - \hat{\beta}_1 x_t - \hat{\beta}_2 z_t\right)^2}{T-3}.$$

with $\hat{\beta}_0$, $\hat{\beta}_1$, and $\hat{\beta}_2$ being *OLS* estimators of parameters in (3.37), while the regression with the omitted variable only yields an estimator of $\sigma_\varepsilon^2 = Var(\varepsilon_t)$, namely

$$\hat{\sigma}_\varepsilon^2 = \frac{\sum_t \hat{\varepsilon}_t^2}{T-2} = \frac{\sum_t \left(y_t - \hat{\alpha} - \hat{\beta} x_t\right)^2}{T-2},$$

with $\hat{\alpha}$ and $\hat{\beta}$ being *OLS* estimators of parameters in (3.38). Notice that, in general, $\hat{\sigma}_\varepsilon^2 \geq \hat{\sigma}_u^2$, and therefore the variance of $\hat{\beta}$ will be generally larger than the variance of $\hat{\beta}_1$. A similar problem in the estimation of the variance of estimated regression parameters arises when additional irrelevant variables are included in the regression equation.

3.14 Jarque–Bera's test of the normality of regression residuals

In many applications, particularly involving financial time series, it is important to investigate the extent to which regression errors exhibit departures from normality. There are two important ways that error distributions could deviate from normality: skewness and Kurtosis (or tail-fatness)

$$\begin{aligned} \text{Skewness} &= \sqrt{b_1} = m_3/m_2^{3/2}, \\ \text{Kurtosis} &= b_2 = m_4/m_2^2, \end{aligned}$$

where

$$m_j = \frac{\sum_{t=1}^{T} \hat{u}_t^j}{T}, \qquad j = 1, 2, 3, 4,$$

For a normal distribution $\sqrt{b_1} \approx 0$, and $b_2 \approx 3$. The Jarque–Bera's test of the departures from normality is given by (see Jarque and Bera (1980) and Bera and Jarque (1987))

$$\chi_T^2(2) = T\left\{\tfrac{1}{6} b_1 + \tfrac{1}{24}(b_2 - 3)^2\right\},$$

if the regression contains an intercept term (note that in that case $m_1 = 0$). When the regression does not contain an intercept term, then $m_1 \neq 0$, and the test statistic has the additional term

$$Tb_0 = T\left\{3m_1^2/(2m_2) - m_3 m_1/m_2^2\right\},$$

namely

$$\chi^2_T(2) = T\left\{b_0 + \tfrac{1}{6}b_1 + \tfrac{1}{24}(b_2 - 3)^2\right\}.$$

3.15 Predictive failure test

Consider the following linear regression models specified for each of the two sample periods

$$\mathbf{y}_1 = \mathbf{X}_1\boldsymbol{\beta}_1 + \mathbf{u}_1; \qquad \mathbf{u}_1 \sim N(0, \sigma_1^2\mathbf{I}_{T1}), \tag{3.39}$$

$$\mathbf{y}_2 = \mathbf{X}_2\boldsymbol{\beta}_2 + \mathbf{u}_2; \qquad \mathbf{u}_2 \sim N(0, \sigma_2^2\mathbf{I}_{T2}), \tag{3.40}$$

where $\mathbf{y}_r, \mathbf{X}_r, r = 1, 2$, are $T_r \times 1$ and $T_r \times k$ observation matrices on the dependent variable and the regressors for the two sample periods, and $\mathbf{I}_{T_1}$ and $\mathbf{I}_{T_2}$ are identity matrices of order T_1 and T_2, respectively. Combining (3.39) and (3.40) by stacking the observations on the two sample periods now yields

$$\begin{bmatrix} \mathbf{y}_1 \\ \mathbf{y}_2 \end{bmatrix} = \begin{bmatrix} \mathbf{X}_1 & \mathbf{0}_{T_1 \times T_2} \\ \mathbf{X}_2 & \mathbf{I}_{T2} \end{bmatrix} \begin{bmatrix} \boldsymbol{\beta}_1 \\ \boldsymbol{\delta} \end{bmatrix} + \begin{bmatrix} \mathbf{u}_1 \\ \mathbf{u}_2 \end{bmatrix}.$$

The above system of equations may also be written more compactly as

$$\mathbf{y}_0 = \mathbf{X}_0\boldsymbol{\beta}_1 + \mathbf{S}_2\boldsymbol{\delta} + \mathbf{u}_0, \tag{3.41}$$

where $\mathbf{y}_0 = (\mathbf{y}_1', \mathbf{y}_2')', \mathbf{X}_0 = (\mathbf{X}_1', \mathbf{X}_2')'$, and $\mathbf{S}_2$ represents the $(T_1 + T_2) \times T_2$ matrix of T_2 dummy variables, one dummy variable for each observation in the second period. For example, for observation $T_1 + 1$, the first column of $\mathbf{S}_2$ will have unity on its $(T_1 + 1)^{th}$ element and zeros elsewhere. The predictive failure test can now be carried out by testing the hypothesis of $\boldsymbol{\delta} = 0$ against $\boldsymbol{\delta} \neq 0$ in (3.41). This yields the following F-statistic

$$F_{PF} = \frac{(\hat{\mathbf{u}}_0'\hat{\mathbf{u}}_0 - \hat{\mathbf{u}}_1'\hat{\mathbf{u}}_1)/T_2}{\hat{\mathbf{u}}_1'\hat{\mathbf{u}}_1/(T_1 - k)} \sim F(T_2, T_1 - k), \tag{3.42}$$

where

- $\hat{\mathbf{u}}_0$ is the *OLS* residual vector of the regression of $\mathbf{y}_0$ on $\mathbf{X}_0$ (i.e., based on the first and the second sample periods together).
- $\hat{\mathbf{u}}_1$ is the *OLS* residual vector of the regression of $\mathbf{y}_1$ on $\mathbf{X}_1$ (i.e., based on the first sample period).

Under the classical normal assumptions, the predictive failure test statistic, F_{PF}, has an exact F-distribution with T_2 and $T_1 - k$ degrees of freedom.

The *LM* version of the above statistic is computed as

$$\chi^2_{PF} = T_2 F_{PF} \overset{a}{\sim} \chi^2(T_2), \tag{3.43}$$

which is distributed as a chi-squared with T_2 degrees of freedom for large T_1 (see Chow (1960), Salkever (1976), Dufour (1980), and Pesaran, Smith, and Yeo (1985), section III.)

It is also possible to test if the predictive failure is due to particular time period(s) by applying the *t*- or the *F*-tests to one or more elements of $\boldsymbol{\delta}$ in (3.41).

3.16 A test of the stability of the regression coefficients: the Chow test

This test is proposed by Chow (1960) and aims at testing the hypothesis that in (3.39) and (3.40) $\boldsymbol{\beta}_1 = \boldsymbol{\beta}_2$, conditional on equality of variances, that is, $\sigma_1^2 = \sigma_2^2$. In econometrics literature this is known as the Chow test, and is known as the analysis of covariance test in the statistics literature (see Scheffe (1959)). The *F*-version of the Chow test statistic is defined by

$$F_{SS} = \frac{(\hat{\mathbf{u}}_0'\hat{\mathbf{u}}_0 - \hat{\mathbf{u}}_1'\hat{\mathbf{u}}_1 - \hat{\mathbf{u}}_2'\hat{\mathbf{u}}_2)/k}{(\hat{\mathbf{u}}_1'\hat{\mathbf{u}}_1 + \hat{\mathbf{u}}_2'\hat{\mathbf{u}}_2)/(T_1 + T_2 - 2k)} \sim F(k, T_1 + T_2 - 2k), \tag{3.44}$$

where

- $\hat{\mathbf{u}}_0$ is the *OLS* residual vector for the first two sample periods together
- $\hat{\mathbf{u}}_1$ is the *OLS* residual vector for the first sample period
- $\hat{\mathbf{u}}_2$ is the *OLS* residual vector for the second sample period.

The *LM* version of this test statistic is computed as

$$\chi_{SS}^2 = kF_{SS} \overset{a}{\sim} \chi^2(k). \tag{3.45}$$

For more details see, for example, Pesaran, Smith, and Yeo (1985, p. 285).

3.17 Non-parametric estimation of the density function

Suppose $f(y)$ denotes the density function of a variable Y at point y, and $y_1, y_2, \ldots, y_T$ are observations drawn from $f(.)$. Two general approaches have been proposed to estimate $f(.)$. The first is a parametric method, which assumes that the form for $f(.)$ is known (e.g., normal), except for the few parameters that need to be estimated consistently from data (e.g., the mean and variance). In contrast, the non-parametric approach tries to estimate $f(.)$ directly, without strong assumptions on its form. One simple example of such an estimator is the histogram, although it has the drawback of being discontinuous, and not applicable for estimating the distribution of two or more variables. The non-parametric density estimator takes the following general form

$$\hat{f}(y) = \frac{1}{T}\sum_{t=1}^{T} \frac{1}{h_T} K\left(\frac{y - y_t}{h_T}\right),$$

where $K(\cdot)$ is called kernel function, and h_T is the window width, also called the *smoothing* parameter or *bandwidth*. The kernel function needs to satisfy some regularity conditions typical of probability density functions, for example, $K(-\infty) = K(\infty) = 0$, and $\int_{-\infty}^{+\infty} K(x)\, dx = 1$. There exists a vast literature on the choice of this function. One popular choice is the Gaussian kernel, namely

$$K(y) = \frac{1}{\sqrt{2\pi}} e^{-\frac{y^2}{2}}.$$

Another common choice is the *Epanechnikov* kernel

$$K(y) = \begin{cases} \frac{3}{4}\left(1 - \frac{1}{5}y^2\right)/\sqrt{5}, & \text{if } |y| < \sqrt{5}, \\ 0, & \text{otherwise.} \end{cases}$$

As also pointed by Pagan and Ullah (1999), the choice of K is not critical to the analysis, and the optimal kernel in most cases will only yield modest improvements in the performance of $\hat{f}(y)$, over selections such as the Gaussian kernel.

When implementing density estimates, the choice of the window width, h_T, plays an essential role. One crude way of choosing h_T is by a trial-and-error approach, consisting of looking at several different plots of $\hat{f}(y)$ against y, when $\hat{f}(y)$ is computed for different values of h_T. Other more objective and automatic methods for selecting h_T have been proposed in the literature. One popular choice is the *Silverman rule of thumb*, according to which

$$h_{srot} = 0.9 \cdot A \cdot T^{-\frac{1}{5}}, \tag{3.46}$$

where $A = \min(\sigma, R/1.34)$, σ is the standard deviation of the variable y, R is the interquartile range, and T is the number of observations, see Silverman (1986, p. 47). Another very popular method is the *least squares cross-validation* method, according to which the window width is the value, h_{lscv}, that minimizes the following criterion

$$ISE(h_T) = \frac{1}{T^2 h_T} \sum_{t \neq s}^{T} K_2\left(\frac{y_t - y_s}{h_T}\right) - \frac{2}{T} \sum_{t=1}^{T} \hat{f}_{-t}(y_t), \tag{3.47}$$

where $K_2(.)$ is the convolution of the kernel with itself, defined by

$$K_2(y) = \int_{-\infty}^{+\infty} K(t)\, K(y - t)\, dt.$$

and $\hat{f}_{-t}(y_t)$ is the density estimator obtained after omitting the tth observation. We have

$$\frac{1}{T} \sum_{t=1}^{T} \hat{f}_{-t}(y_t) = \frac{1}{T(T-1)h_T} \sum_{t \neq j}^{T} K\left(\frac{y_t - y_j}{h_T}\right).$$

If K is the Gaussian kernel, then K_2 is $N(0, 2)$, or

$$K_2(y) = (4\pi)^{-1/2} e^{-y^2/4},$$

for the Epanechnikov kernel we have

$$K_2(y) = \begin{cases} \frac{3\sqrt{5}}{100}\left(4 - y^2\right), & \text{if } |y| < \sqrt{5} \\ 0, & \text{otherwise.} \end{cases}.$$

For the Gaussian kernel the expression for $ISE(h_T)$ simplifies to (see Bowman and Azzalini (1997, p. 37))

$$ISE(h_T) = \frac{1}{(T-1)}\phi\left(0, \sqrt{2}h_T\right) + \frac{T-2}{T(T-1)^2}\sum_{t\neq j}^{T}\phi\left(y_t - y_j, \sqrt{2}h_T\right) - \frac{2}{T(T-1)}\sum_{t\neq j}^{T}\phi\left(y_t - y_j, h_T\right),$$

where $\phi(y, \sigma)$ denotes the normal density function with mean 0 and standard deviation σ:

$$\phi(y, \sigma) = (2\pi\sigma^2)^{-1/2}\exp\left(\frac{-y^2}{2\sigma^2}\right).$$

In cases where local minima are encountered we select the bandwidth that corresponds to the local minimum with the largest value for h_T. See Bowman and Azzalini (1997, pp. 33–4). See also Pagan and Ullah (1999), Silverman (1986), Jones, Marron, and Sheather (1996), and Sheather (2004) for further details.

3.18 Further reading

Further material on statistical inference and its application to econometrics can be found in Rao (1973) and Bierens (2005). See also Appendix B for a review of key concepts from probability theory and statistics useful for this chapter. For what concerns non-parametric density estimators, further discussion can be found in Horowitz (2009), which contains a treatment of non-parametric methods within econometrics.

3.19 Exercises

1. Consider the model

$$\log Y_t = \beta_0 + \beta_1 \log L_t + \beta_2 \log K_t + u_t,$$

where L_t and K_t are exogenous and the u_t are distributed independently as $N(0, \sigma^2)$ variates. The estimated equation, based on data for 1929–67, is

$$\log Y_t = -3.38766 + 1.4106 \log L_t + 0.4162 \log K_t,$$

where $R^2 = 0.9937$, $\hat{\sigma} = 0.03755$ and $\sum_{t=1}^{T}(y_t - \bar{y})^2 = 0.1241$ (with $y_t = \log Y_t$). The variance-covariance matrix of the least squares estimates of β_1 and β_2 is estimated to be:

$$\begin{pmatrix} 0.007820 & -0.004235 \\ -0.004235 & 0.002549 \end{pmatrix}.$$

(a) Test the hypothesis $H_1 : \beta_1 = 0$ and also test $H_2 : \beta_2 = 0$. Each at the 5 per cent level.

(b) Test the joint hypothesis $H : \beta_1 = \beta_2 = 0$ at the 5 per cent level.

(c) Find the 95 per cent confidence interval for $\beta_1 + \beta_2$ (the return to scale parameter) and test the hypothesis $\beta_1 + \beta_2 = 1$.

(d) Re-estimating the equation with $\beta_1 = 1.5$ and $\beta_2 = 0.5$, the (restricted) sum of squared residuals is 0.0678. Use a 5 per cent level F-test to test the joint hypothesis: $H : \beta_1 = 1.5, \beta_2 = 0.5$.

2. An economist wishes to estimate from aggregate time series the model:

$$q = \alpha_0 + \alpha_1 y + \alpha_2 p_1 + \alpha_3 p_2 + \alpha_4 n + u \tag{3.48}$$

where q is the volume of food consumption, y real disposable income, p_1 an index of the price of food, p_2 an index of all other prices and n population. All variables are in logarithms. He knows that the correlation between p_1 and p_2 is 0.95 and between y and n is 0.93, and decides that the equation suffers from multicollinearity. On asking his colleagues for advice, he gets the following suggestions: Colleague A suggests dropping all variables with t-statistics less than 2. Colleague B says that multicollinearity results from too little data variation and suggests pooling the aggregate time series data with a cross-section budget survey on food consumption. Colleague C recommends that he should reduce the amount he is asking of the data by imposing the restrictions $\alpha_2 + \alpha_3 = 0$ and $1 - \alpha_1 - \alpha_4 = 0$ which are suggested by economic theory. Colleague D says multicollinearity will be reduced by replacing (3.48) by (3.49)

$$Z_1 = \beta_0 + \beta_1 Z_2 + \beta_2 Z_3 + \beta_3 Z_4 + u \tag{3.49}$$

where $Z_1 = q - n, Z_2 = y - n, Z_3 = p_1 - p_2, Z_4 = p_1$, because in (3.49) the correlations between the right hand side variables are lower. Colleague E says that adding lagged values of q to the equation will reduce multicollinearity since it is known that it has a significant effect on food consumption.

(a) Is the economist correct in being sure that (3.48) will necessarily suffer from multicollinearity?

(b) How would you diagnose multicollinearity in (3.48)?

(c) Which of the suggestions would you adopt?

3. Consider the following linear regression model

$$y_t = \beta_0 + \beta_1 x_{1t} + \beta_2 x_{2t} + \varepsilon_t, \quad t = 1, 2, \ldots, T. \tag{3.50}$$

Suppose that the classical assumptions are applicable to (3.50) and $\varepsilon_t \sim IID(0, \sigma^2)$. Denote the *OLS* estimators of β_0, β_1 and β_2 by $\hat{\beta}_0$, $\hat{\beta}_1$ and $\hat{\beta}_2$, respectively, and the estimator of σ^2 by

$$\hat{\sigma}^2 = \frac{1}{T-3}\sum_t (y_t - \hat{\beta}_0 - \hat{\beta}_1 x_{1t} - \hat{\beta}_2 x_{2t})^2.$$

(a) Show that the estimated variances of $\hat{\beta}_1$ and $\hat{\beta}_2$ are given by

$$\widehat{Var}(\hat{\beta}_1) = \frac{\hat{\sigma}^2}{s_1^2(1-r^2)}, \ \widehat{Var}(\hat{\beta}_2) = \frac{\hat{\sigma}^2}{s_2^2(1-r^2)},$$

where s_1 and s_2 are the standard deviations of x_{1t} and x_{2t}, respectively, and r is the correlation coefficient between x_{1t} and x_{2t}.

(b) Suppose that $x_{2t} - \mu_2 = \lambda(x_{1t} - \mu_1)$, where μ_1 and μ_2 are the means of x_{1t} and x_{2t}, respectively, and λ is a fixed non-zero constant. Show that in this case $Var(\hat{\beta}_1 + \lambda\hat{\beta}_2)$ is finite, although $\widehat{Var}(\hat{\beta}_1)$ and $\widehat{Var}(\hat{\beta}_2)$ both blow up individually. How do you interpret this result?

(c) What is meant by the 'multicollinearity problem'? How is it detected? What are the possible solutions. Discuss, in the light of the results in part (b).

4. Define a non-central chi-squared random variable with m degrees of freedom.

(a) Let $\mathbf{x} \sim N(\boldsymbol{\mu}, \boldsymbol{\Omega})$, where $\mathbf{x}$ and $\boldsymbol{\mu}$ are s dimensional vectors, and $\boldsymbol{\Omega}$ is an $s \times s$ positive definite matrix. Show that

$$\mathbf{x}'\boldsymbol{\Omega}^{-1}\mathbf{x} \sim \chi_s^2(\delta^2),$$

where $\chi_s^2(\delta^2)$ is a non-central chi-squared variate with s degrees of freedom and the non-centrality parameter, $\delta^2 = \boldsymbol{\mu}'\boldsymbol{\Omega}^{-1}\boldsymbol{\mu}$.

(b) Consider the regression model

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{u}, \qquad \mathbf{u} \sim N(0, \sigma^2\mathbf{I}_T),$$

where $\mathbf{X}$ is a $T \times k$ non-stochastic matrix of rank $k < T$. Suppose we wish to test

$$H_0 : \mathbf{R}\boldsymbol{\beta} = \mathbf{c},$$

where $\mathbf{R}$ is a known $s \times k$ matrix of rank s and $\mathbf{c}$ is a known $s \times 1$ vector. Use the results in (b) to show that

$$(\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{c})' \left[\sigma^2 \mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'\right]^{-1} (\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{c}) \sim \chi_s^2(\delta^2) \ ,$$

where $\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$ and

$$\delta^2 = (\mathbf{R}\boldsymbol{\beta} - \mathbf{c})' \left[\sigma^2 \mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'\right]^{-1} (\mathbf{R}\boldsymbol{\beta} - \mathbf{c}).$$

Briefly discuss how to test H_0 when σ^2 is known.

(c) Discuss how to test H_0 when σ^2 is not known.

4 Heteroskedasticity

4.1 Introduction

Two important assumptions of the classical linear regression model are that the disturbances of the regression equation have constant variances across statistical units and that they are uncorrelated. These assumptions are needed to prove the Gauss–Markov theorem and to show that the least squares estimator is asymptotically efficient. In this chapter, we consider an important extension of the classical model introduced in Chapter 2 by allowing the disturbances, u_i, to have variances differing across different values of i, namely to be heteroskedastic. Chapter 5 will consider the case of autocorrelated disturbances.

The heteroskedasticity problem frequently arises in cross-section regressions, while it is less common in time-series regressions. Important examples of regressions with heteroskedastic errors include cross-section regressions of household consumption expenditure on household income, cross-country growth regressions, and the cross-section regression of labour productivity on output growth across firms or industries. Heteroskedasticity also arises if regression coefficients vary randomly across observations or when observations on y_i and x_i are average estimates based on stratified sampling where the average estimates are based on different sample sizes. In time series regressions heteroskedasticity can arise either because of structural change or omission of relevant variables from the regression.

4.2 Regression models with heteroskedastic disturbances

Consider the following simple linear regression model

$$y_i = \alpha + \beta x_i + u_i, \tag{4.1}$$

and assume that all the classical assumptions apply to this model except that

$$Var\,(u_i\,|x_i\,) = \sigma_i^2, \qquad i = 1, 2, \ldots, T. \tag{4.2}$$

We first examine the consequences of heteroskedastic errors for the *OLS* estimators of α and β, and their standard errors. The *OLS* estimator of β is given by

$$\hat{\beta} = \frac{\sum_i \left(y_i - \bar{y}\right)(x_i - \bar{x})}{\sum_i (x_i - \bar{x})^2} = \beta + \frac{\sum_i (x_i - \bar{x})\, u_i}{\sum_i (x_i - \bar{x})^2},$$

where $\sum_i$ stands for summation over $i = 1, 2, \ldots, T$. Hence

$$\hat{\beta} = \beta + \sum_i w_i u_i, \tag{4.3}$$

where the weights $w_i = (x_i - \bar{x}) / \sum_j \left(x_j - \bar{x}\right)^2$ are as given by (1.26). Taking expectations of both sides of (4.3) we have

$$E\left(\hat{\beta}\right) = \beta + \sum_i w_i E\left(u_i\right).$$

Note that x_i, and hence w_i are taken as given. Therefore $E\left(\hat{\beta}\right) = \beta$, and the *OLS* estimator of β continues to be unbiased. Consider now the variance of $\hat{\beta}$ in the presence of heteroskedasticity. Taking variances of both sides of (4.3) and noting that u_i is assumed to be uncorrelated we have

$$Var\left(\hat{\beta}\right) = \sum_i w_i^2 Var\left(u_i\right).$$

But under (4.2)

$$Var\left(\hat{\beta}\right) = \sum_i \sigma_i^2 w_i^2. \tag{4.4}$$

Substituting for w_i back in (4.4) now yields

$$Var\left(\hat{\beta}\right) = \frac{\sum_i \sigma_i^2 (x_i - \bar{x})^2}{\left[\sum_i (x_i - \bar{x})^2\right]^2}, \tag{4.5}$$

which differs from the standard *OLS* formula for the variance of the slope coefficient given in (1.28) and reduces to it only when $\sigma_i^2 = \sigma^2$, for all i. Therefore, the presence of heteroskedastic errors biases the estimates of the variances of the *OLS* estimators and hence invalidates the application of the usual t or F tests to the parameters. This result readily generalizes to the multivariate case.

The direction of the bias in the *OLS* variance of $\hat{\beta}$ depends on the pattern of the heteroskedasticity. In the case where σ_i^2 and $(x_i - \bar{x})^2$ are positively correlated, the *OLS* formula underestimates the true variance $\hat{\beta}$ given by (4.5), and its use can thus result in invalid inferences.

There are two general ways of dealing with the heteroskedasticity problem. One possibility is to specify the form of the heteroskedasticity and estimate the regression coefficients and the standard errors of their estimates by taking explicit account of the assumed pattern of heteroskedasticity. An alternative approach is to continue using the *OLS* estimators (i.e., $\hat{\alpha}$ and $\hat{\beta}$), but adjust the variances of $\hat{\alpha}$ and $\hat{\beta}$ for the presence of heteroskedasticity. This latter procedure is particularly of interest when the exact form of the heteroskedasticity (namely σ_i^2; $i = 1, 2, \ldots, T$) is not known. In these circumstances, consistent estimators of the variances and covariances of the *OLS* estimators of the regression coefficients have been suggested in the literature, and are known as heteroskedasticity-consistent estimators (HCV) (see Eicker (1963), Eicker, LeCam, and Neyman (1967), Rao (1970), and White (1980)). In the case of the simple regression the heteroskedasticity-consistent estimator of $Var(\hat{\beta})$ is given by

$$\widehat{HCV}\left(\hat{\beta}\right) = \left(\frac{T}{T-2}\right)\frac{\sum_i \hat{u}_i^2\,(x_i-\bar{x})^2}{\left[\sum_i (x_i-\bar{x})^2\right]^2}, \tag{4.6}$$

where $\hat{u}_i = y_i - \hat{\alpha} - \hat{\beta}x_i$ are the *OLS* residuals. The factor $T/(T-2)$ is asymptotically negligible and is introduced to allow for the loss in degrees of freedom due to the estimation of the parameters, α and β.[1] Notice that apart from the degrees of freedom correction factor, (4.6) gives a consistent estimator of $Var\left(\hat{\beta}\right)$ in (4.5) by replacing σ_i^2 with $\hat{u}_i^2$.

The result (4.6) readily generalizes to the multivariate case, and in matrix notations can be written as

$$\widehat{HCV}\left(\hat{\boldsymbol{\beta}}\right) = \left(\frac{T}{T-k}\right)\left(\mathbf{X}'\mathbf{X}\right)^{-1}\left(\sum_{i=1}^{T}\hat{u}_i^2\mathbf{x}_i\mathbf{x}_i'\right)\left(\mathbf{X}'\mathbf{X}\right)^{-1}, \tag{4.7}$$

where k is the dimension of the coefficient vector $\boldsymbol{\beta}$ in the multivariate regression model,

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{u},$$

$\hat{u}_i$ is the *OLS* residual of the ith observation

$$\hat{u}_i = y_i - \hat{\boldsymbol{\beta}}'\mathbf{x}_i,$$

and $\mathbf{x}_i$ is the vector of the ith observation on the variables (including the intercept term) in the regression equation. See White (1980) and also Sections 5.9 and 5.10 for further discussion on robust estimation and testing in the presence of heteroskedastic and autocorrelated errors.

[1] The small sample correction implicit in the introduction of the degrees of freedom factor in the calculation of the heteroskedasticity-consistent estimators has been suggested in MacKinnon and White (1985).

4.3 Efficient estimation of the regression coefficients in the presence of heteroskedasticity

The above procedure popularized in econometrics by Halbert White deals with the inconsistency of the *OLS* standard errors when heteroskedasticity is present, but does not deal with the inefficiency of the *OLS* estimators. In cases where the form of the heteroskedasticity is known, or the investigator is prepared to formulate it explicitly, it is possible to improve over the *OLS* estimators. Consider, for example, the simple formulation

$$\sigma_i^2 = \sigma^2 z_i^2, \tag{4.8}$$

where σ^2 is a unknown scalar, and z_i are known observations on a variable thought to be distributed independently of u_i. In this case, as heteroskedasticity takes a known form, efficient estimation of α and β can be achieved by the method of weighted least squares. Dividing both sides of (4.1) by z_i we have

$$\frac{y_i}{z_i} = \alpha\left(\frac{1}{z_i}\right) + \beta\left(\frac{x_i}{z_i}\right) + \frac{u_i}{z_i}, \tag{4.9}$$

we obtain

$$y_i^* = \alpha x_{1i} + \beta x_{2i} + \varepsilon_i, \tag{4.10}$$

where $y_i^* = y_i/z_i$, $x_{1i} = 1/z_i$, $x_{2i} = x_i/z_i$ and $\varepsilon_i = u_i/z_i$. Using (4.8), we now have

$$Var\,(\varepsilon_i) = \frac{Var\,(u_i)}{z_i^2} = \frac{\sigma^2 z_i^2}{z_i^2} = \sigma^2.$$

Furthermore, since z_i and x_i are assumed to be distributed independently of u_i, it also follows that x_{1i} and x_{2i} in (4.10) will be distributed independently of ε_i, and (4.10) satisfies all the classical assumptions. Therefore, by the Gauss–Markov theorem, the *OLS* estimators of α and β in the regression of y_i^* on x_{1i} and x_{2i} will be BLUE (see Section 2.7 for a definition of BLUE estimators). These estimators of α and β are also referred to as the weighted or the generalized least square (*GLS*) estimators. The efficiency of the estimators of α and β in the transformed equation (4.9) over their *OLS* counterpart easily follows from the fact that the *GLS* estimators of α and β in (4.10) satisfy the Gauss–Markov theorem, while the *OLS* estimators in (4.1) do not. See Section 5.4 for a direct proof for the general case of a non-diagonal error covariance matrix.

4.4 General models of heteroskedasticity

In general, three types of models for heteroskedasticity have been considered in the literature:

1. The *multiplicative* specification defined by

$$\sigma_i^2 = \sigma^2\,(z_{i1})^{\gamma_1}\,(z_{i2})^{\gamma_2}\cdots\left(z_{ip}\right)^{\gamma_p}, \qquad \sigma^2 > 0, \quad z_{i1} > 0, \quad z_{i2} > 0, \cdots z_{ip} > 0, \tag{4.11}$$

which is a straightforward generalization of the simple model (4.8) considered above.

2. The *additive* specification:

$$\sigma_i^2 = \sigma^2 + \lambda_1 z_{i1} + \lambda_2 z_{i2} + \cdots + \lambda_p z_{ip}, \tag{4.12}$$

defined over values of λ_i and z_{ij} such that $\sigma_i^2 > 0$.

3. The *mean-variance* specification:

$$Var\left(y_i \,|x_i\right) = \sigma_i^2 = \sigma^2 \left\{E\left(y_i \,|x_i\right)\right\}^{2\delta},$$

or for the simple regression model

$$\sigma_i^2 = \sigma^2 \left(\alpha + \beta x_i\right)^{2\delta}. \tag{4.13}$$

This model postulates a nonlinear relation between the conditional mean and the conditional variance of the dependent variable and can be justified on theoretical grounds. For example, in the case of risk-adverse investors, mean return is related to volatility of stock returns measured by conditional variance of returns.

The estimation of the regression model under any of the above specifications of the heteroskedasticity can be carried out by the maximum likelihood (ML) method as illustrated in Chapter 9. However, in many cases estimation involves using some iterative numerical algorithm, as in the following example.

Example 10 *Consider the linear regression model (4.1) with heteroskedastic errors having variances*

$$\sigma_i^2 = \sigma^2 z_i^{\gamma}, \qquad z_i > 0, \tag{4.14}$$

where both σ^2 and γ are unknown parameters. To apply the weighted least squares approach, we first need an estimate of γ, which can be done by the asymptotically efficient ML method. Assume that u_i are normally distributed, then the ln-likelihood function for this problem is

$$\begin{aligned} \ell\left(\boldsymbol{\theta}\right) &= \ln L\left(\boldsymbol{\theta}\right) = \ln\left\{P\left(y_1, y_2, \ldots, y_T\right)\right\} \\ &= \sum_{i=1}^{T} \ln\left\{P\left(y_i\right)\right\}, \end{aligned} \tag{4.15}$$

where $\boldsymbol{\theta} = \left(\alpha, \beta, \gamma, \sigma^2\right)'$, and $P\left(y_i\right)$ is the probability density function of y_i conditional on x_i, given by

$$\begin{aligned} P\left(y_i\right) &= \left(2\pi\sigma_i^2\right)^{-\frac{1}{2}} \exp\left\{-\frac{1}{2\sigma_i^2}\left(y_i - \alpha - \beta x_i\right)^2\right\} \\ &= -\frac{T}{2}\ln\left(2\pi\sigma^2\right) - \frac{\gamma}{2}\sum_i \ln z_i - \frac{1}{2\sigma^2}\sum_i \frac{\left(y_i - \alpha - \beta x_i\right)^2}{z_i^{\gamma}}. \end{aligned}$$

The ML estimators are obtained by maximizing the above function with respect to the unknown parameters α, β, γ and σ^2. For this purpose, the Newton–Raphson method can be used (see Section A.16). The updated relation for the present problem is given by

$$\hat{\boldsymbol{\theta}}^{(i+1)} = \hat{\boldsymbol{\theta}}^{(i)} - \left[\frac{\partial^2 \ell(\boldsymbol{\theta})}{\partial \boldsymbol{\theta} \partial \boldsymbol{\theta}'}\right]^{-1}\Bigg|_{\boldsymbol{\theta}=\hat{\boldsymbol{\theta}}^{(i)}} \frac{\partial \ell(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}\Bigg|_{\boldsymbol{\theta}=\hat{\boldsymbol{\theta}}^{(i)}}, \quad i = 1, 2, \ldots \tag{4.16}$$

where the expressions for $\partial \ell(\boldsymbol{\theta}) / \partial(\boldsymbol{\theta})$ and $\partial^2 \ell(\boldsymbol{\theta}) / \partial \boldsymbol{\theta} \partial \boldsymbol{\theta}'$ are given by

$$\frac{\partial \ell(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}} = \frac{1}{\sigma^2}\begin{bmatrix} \sum_i u_i z_i^{-\gamma} \\ \sum_i x_i u_i z_i^{-\gamma} \\ -\frac{\sigma^2}{2}\sum_i \ln z_i + \frac{1}{2}\sum_i u_i^2 z_i^{-\gamma} \ln z_i \\ -\frac{T}{2} + \frac{1}{2\sigma^2}\sum_i u_i^2 z_i^{-\gamma} \end{bmatrix}, \tag{4.17}$$

and

$$\frac{\partial^2 \ell(\boldsymbol{\theta})}{\partial \boldsymbol{\theta} \partial \boldsymbol{\theta}'} = -\frac{1}{\sigma^2}\begin{bmatrix} \sum_i z_i^{-\gamma} & \sum_i x_i z_i^{-\gamma} & \sum_i u_i z_i^{-\gamma} \ln z_i & \frac{\sum_i u_i z_i^{-\gamma}}{\sigma^2} \\ \sum_i x_i z_i^{-\gamma} & \sum_i x_i^2 z_i^{-\gamma} & \sum_i x_i u_i z_i^{-\gamma} \ln z_i & \frac{\sum_i x_i u_i z_i^{-\gamma}}{\sigma^2} \\ \sum_i u_i z_i^{-\gamma} \ln z_i & \sum_i x_i u_i z_i^{-\gamma} \ln z_i & \frac{1}{2}\sum_i u_i^2 (\ln z_i)^2 z_i^{-\gamma} & \frac{\sum_i u_i^2 z_i^{-\gamma} \ln z_i}{\sigma^2} \\ \frac{\sum_i u_i z_i^{-\gamma}}{\sigma^2} & \frac{\sum_i x_i u_i z_i^{-\gamma}}{\sigma^2} & \frac{\sum_i u_i^2 z_i^{-\gamma} \ln z_i}{\sigma^2} & \left(\frac{\sum_i u_i^2 z_i^{-\gamma}}{\sigma^4} - \frac{T}{2\sigma^2}\right) \end{bmatrix},$$

with $u_i = y_i - \alpha - \beta x_i$. It is often convenient to replace $\partial^2 \ell(\boldsymbol{\theta}) / \partial \boldsymbol{\theta} \partial \boldsymbol{\theta}'$ by its expectations (or its probability limit). In this case we have

$$E\left[\frac{-\partial^2 \ell(\boldsymbol{\theta})}{\partial \boldsymbol{\theta} \partial \boldsymbol{\theta}'}\right] = \frac{1}{\sigma^2}\begin{bmatrix} \sum_i z_i^{-\gamma} & \sum_i x_i z_i^{-\gamma} & 0 & 0 \\ \sum_i x_i z_i^{-\gamma} & \sum_i x_i^2 z_i^{-\gamma} & 0 & 0 \\ 0 & 0 & \frac{\sigma^2}{2}\sum_i (\ln z_i)^2 & \sum_i \ln z_i \\ 0 & 0 & \sum_i \ln z_i & \frac{T}{2\sigma^2} \end{bmatrix}, \tag{4.18}$$

which is Fisher's information matrix for the ML estimator of $\boldsymbol{\theta}$. The asymptotic variance-covariance matrix of the ML estimator of $\boldsymbol{\theta}$ is given by the inverse of Fisher's information matrix given in (4.18) (see Section 9.4). The block diagonality of the information matrix in the case of this example establishes that the ML estimators of the regression coefficients (α and β) and the parameters of the error-variance (σ^2 and γ) are asymptotically independent. Hence, we have:

$$\text{Asy Var}\begin{pmatrix} \hat{\alpha} \\ \hat{\beta} \end{pmatrix} = \sigma^2 \begin{bmatrix} \sum_i z_i^{-\gamma} & \sum_i x_i z_i^{-\gamma} \\ \sum_i x_i z_i^{-\gamma} & \sum_i x_i^2 z_i^{-\gamma} \end{bmatrix}^{-1}, \tag{4.19}$$

which is the same as the variance matrix of α and β in the OLS regression of $z_i^{-\gamma} y_i$ on $z_i^{-\gamma}$ and $x_i z_i^{-\gamma}$. Similarly

$$\text{Asy Var}\begin{pmatrix}\hat{\gamma}\\ \hat{\sigma}^2\end{pmatrix} = \sigma^2\begin{bmatrix}\frac{1}{2}\sigma^2\sum_i(\ln z_i)^2 & \sum_i \ln z_i\\ \sum_i \ln z_i & \frac{T}{2\sigma^2}\end{bmatrix}^{-1},$$

which yields

$$\text{Asy Var}\left(\hat{\gamma}\right) = \frac{2}{\sum_i(\ln z_i)^2 - \frac{4}{T}\left(\sum_i \ln z_i\right)^2}. \tag{4.20}$$

This result can be used, for example, to test the homoskedasticity hypothesis, $H_0 : \gamma = 0$.

Estimation procedures other than *ML* have also been suggested in the literature. For example, in the case of the mean-variance specification, (4.13), the following two-step procedure has often been suggested for the case where $\delta = 1$:

Step I Run the *OLS* regression of y_i on x_i (including an intercept) to obtain $\hat{\alpha}$ and $\hat{\beta}$ and hence the fitted values $\hat{y}_i = \hat{\alpha} + \hat{\beta}x_i$.

Step II Run the *OLS* regression of $y_i/\hat{y}_i$ on $1/\hat{y}_i$ and $x_i/\hat{y}_i$ to obtain new estimates of α and β.

The estimates of α and β obtained in Step II are asymptotically more efficient than the *OLS* estimator. In the case where δ is not known a further step is needed to estimate δ from the regression of $\ln \hat{u}_i^2$ on $\ln \hat{y}_i^2$, including an intercept term. The coefficient of $\ln \hat{y}_i^2$ in this regression provides us with an estimate of δ (say $\hat{\delta}$), which can then be used to compute new estimates of α and β from the *OLS* regression of $y_i\hat{y}_i^{-\hat{\delta}}$ on $\hat{y}_i^{-\hat{\delta}}$ and $\hat{y}_i^{-\hat{\delta}}x_i$. Recall that $\hat{y}_i = \hat{\alpha} + \hat{\beta}x_i$.

4.5 Diagnostic checks and tests of homoskedasticity

There exist three general methods that can be used to check the validity of the homoskedasticity assumption:

(i) Graphical methods
(ii) General non-parametric methods, such as the Goldfeld–Quandt test
(iii) Parametric tests.

4.5.1 Graphical methods

The graphical approach simply involves plotting the squares of the *OLS* residuals against the square of fitted values (i.e., $\hat{y}_i^2$), or against other variables thought to be important in explaining heteroskedasticity of the error variances. Identification of systematic patterns in such graphical displays can be viewed as casual evidence of heteroskedasticity.

4.5.2 The Goldfeld–Quandt test

This test involves grouping the observations into m $(m \geq 2)$ categories and then testing the hypothesis that error variances are equal across these m groups. The test assumes that the error variances, $Var(u_i)$, are homoskedastic within each observation group. In practice, the observations are placed into three categories and the test of the equality of error-variances is carried out across the first and the third category, thus ignoring the observations in the middle category. This is done to reduce the possibility of dependence between the estimates of $Var(u_i)$ over the first group of observations and over the third group of observations. Application of the Goldfeld–Quandt test to the simple regression model comprises the following steps.

Step I: Split the sample of observations into three groups, so that

$$\begin{array}{lll} \text{Group 1:} & y_i = \alpha_I + \beta_I x_i + u_i^I & i = 1, 2, \ldots T_1, \\ \text{Group 2:} & y_i = \alpha_{II} + \beta_{II} x_i + u_i^{II} & i = T_1 + 1, \ldots, T_1 + T_2, \\ \text{Group 3:} & y_i = \alpha_{III} + \beta_{III} x_i + u_i^{III} & i = T_1 + T_2 + 1, \ldots, T, \end{array}$$

where $T = T_1 + T_2 + T_3$.

Step II: Run the *OLS* regressions of y_i on x_i for the first and the third groups separately. Obtain the sums of squares of residuals for these two regressions, and denote them by SSR_I and SSR_{III}, respectively.

Step III: Construct the statistic

$$F = \frac{SSR_I / (T_1 - 2)}{SSR_{III} / (T_3 - 2)} = \frac{\hat{\sigma}_I^2}{\hat{\sigma}_{III}^2},$$

where $\hat{\sigma}_I^2$ and $\hat{\sigma}_{III}^2$ are the unbiased estimates of $Var\left(u_i^I\right)$ and $Var\left(u_i^{III}\right)$, computed using the observations in groups I and III, respectively. It is convenient to compute the above F statistic such that it is larger than unity (by putting the larger estimate of the variance in the numerator), so that the test statistic is more directly comparable to the critical values in F Tables.

Under the null hypothesis of homoskedasticity, the above F-statistic has an F-distribution with $T_1 - 2$ and $T_3 - 2$ degrees of freedom. Large values of F are associated with the rejection of the homoskedasticity assumption, and possible evidence of the heteroskedasticity. The Goldfeld–Quandt test readily generalizes to multivariate regressions and to more than three observation groups.

4.5.3 Parametric tests of homoskedasticity

The starting point of these tests is a parametric formulation of the heteroskedasticity, such as those specified by (4.11), (4.12), and (4.13). In the case of these specifications, the homoskedasticity hypothesis can be formulated in terms of the following null hypotheses:

1. Multiplicative specification:

$$H_0 : \gamma_1 = \gamma_2, \ldots, \gamma_p = 0,$$

2. Additive specification:

$$H_0 : \lambda_1 = \lambda_2 = \cdots = \lambda_p = 0,$$

3. Mean-variance specification:

$$H_0 : \delta = 0.$$

Any one of the three likelihood-based approaches discussed in Section 9.7 can be used to implement the tests. The simplest procedure to compute is the Lagrange multiplier (LM) method, since this method does not require the estimation of the regression model under heteroskedasticity. One popular LM procedure for testing the homoskedasticity assumption is based on an additive version of the mean-variance model, (4.13). The LM statistic is computed as the t-ratio of the slope coefficient in the regression of $\hat{u}_i^2$ on $\hat{y}_i^2$ (including an intercept), where $\hat{u}_i$ are the OLS residuals and $\hat{y}_i$ are the fitted values. Under the null hypothesis of homoskedastic variances, this t-ratio is asymptotically distributed as a standard normal variable. In small samples, however, it is more advisable to use critical values from the t-distribution rather than the critical values from the normal distribution.

An LM test of the homoskedasticity assumption based on the additive specification, (4.12), involves running the following OLS regression:

$$\hat{u}_i^2 = \alpha + \lambda_1 z_{i1} + \lambda_2 z_{i2} + \cdots + \lambda_p z_{ip} + \text{error},$$

and then testing the hypothesis

$$H_0 : \lambda_1 = \lambda_2 = \cdots = \lambda_p = 0,$$

against

$$H_1 : \lambda_i \neq 0, \quad \lambda_2 \neq 0, \cdots, \lambda_p \neq 0,$$

using the F-test or other asymptotically equivalent procedures. For example, denoting the multiple correlation coefficient of the regression of $\hat{u}_i^2$ on $z_{i1}, z_{i2}, \ldots, z_{ip}$ by R, it is easily seen that under $H_0 : \lambda_1 = \lambda_2 = \cdots = \lambda_p = 0$, the statistic $T \cdot R^2$ is asymptotically distributed as a χ^2 with p degrees of freedom. This test is also asymptotically equivalent to the test proposed by Breusch and Pagan (1980), which tests H_0 against the more general alternative specification: $\sigma_i^2 = f\left(\alpha_0 + \lambda_1 z_{i1} + \cdots + \lambda_p z_{ip}\right)$, where $f(\cdot)$ could be any general function. The White (1980) test of homoskedasticity is a particular application of the above test where z_{ij}'s are chosen to be equal to the regressors, their squares and their cross products. For example, in the case of the regression equation

$$y_i = \alpha + \beta_1 x_{i1} + \beta_2 x_{i2} + u_i,$$

White's test set, $p = 5$ and

$$z_{i1} = x_{i1}, \quad z_{i2} = x_{i2}, \quad z_{i3} = x_{i1}^2, \quad z_{i4} = x_{i2}^2, \quad z_{i5} = x_{i1}x_{i2}.$$

A particularly simple example of the above testing procedure involves running the auxiliary regression

$$\hat{u}_i^2 = \text{constant} + \alpha \hat{y}_i^2, \tag{4.21}$$

and then testing $\alpha = 0$, against $\alpha \neq 0$, using the standard t-test.

4.6 Further reading

Further reading on heteroskedasticity-robust inference and testing for heteroskedasticity can be found in Wooldridge (2000, ch. 9), Wooldridge (2010, chs 4 and 5), and Greene (2002, ch. 11). Robust inference under heteroskedasticity in two-stage least squares is treated in Wooldridge (2010, ch. 5). See also Chapter 18, which introduces various autoregressive models for the conditional variance of disturbance terms, in a time series framework.

4.7 Exercises

1. The linear regression model

$$y_t = \sum_{j=1}^{k} b_j x_{tj} + u_t, t = 1, 2, \ldots, T, \tag{4.22}$$

satisfies all the assumptions of the classical normal regression model except for the variance of u_t which is given by: $V(u_t) = \sigma^2 |z_t|$.

   (a) Discuss the statistical problems that arise if relation (4.22) is estimated by *OLS*.
   (b) Set out the computational steps involved in estimating the parameters of (4.22) efficiently.
   (c) How would you test for heteroskedasticity if the form of error variance is not known?

2. Consider the simple regression model

$$\bar{y}_i = \alpha + \beta \bar{x}_i + \varepsilon_i, \text{ for } i = 1, 2, \ldots, N,$$

where $\bar{y}_i$ is the mean expenditure on alcohol in group i, and $\bar{x}_i$ is the mean income of group i. Each group i has N_i members and the model satisfies all the classical assumptions except that the variance of ε_i is equal to σ^2/N_i.

   (a) What are the statistical properties of the *OLS* estimates of α and β?
   (b) Obtain the best linear unbiased estimators of α and β.

3. Using cross-sectional observations on 2049 households, the following linear Engel curve was estimated by *OLS*:

$$\begin{aligned} W_i &= 0.0645 + 623.3/Y_i + \hat{\epsilon}_i, \qquad i = 1, 2, \ldots, 2049 \qquad (4.23) \\ &\quad (0.0035) \quad (17.01) \\ \bar{R}^2 = 0.3959 \quad \hat{\sigma} &= 0.1302 \quad DW = 1.816 \\ \left|\hat{\epsilon}_i\right| &= -0.0785 + 834.7/Y_i, \qquad i = 1, 2, \ldots, 2049 \\ &\quad (0.0054) \quad (26.40) \end{aligned}$$

where $W_i \equiv C_i/Y_i$, $C_i \equiv$ household i's expenditure on food, $Y_i \equiv$ household i's income, $\hat{\epsilon}_i \equiv$ *OLS* residuals from equation (4.23). Standard errors are shown in parentheses.

(a) Test 'Engel's Law' (that the share of food in household expenditure declines with household income).

(b) Test the hypothesis that the error variances in equation (24) are homoskedastic. Interpret your results.

(c) Discuss the relevance of your test for heteroskedasticity. Do the reported summary and diagnostic statistics have any implications for your test of Engel's Law?

4. An investigator, concerned about the possibility of measurement error in C_i and Y_i, decides to test Engel's Law using grouped data. The 2,049 households are grouped into 25 income classes and the following results were obtained by *OLS*:

$$\begin{aligned} \bar{W}_g &= 0.0565 + 674.5/\bar{Y}_g + \hat{\epsilon}_g, \qquad i = 1, 2, \ldots, 25 \qquad (4.24) \\ &\quad (0.0042) \quad (11.00) \\ \bar{R}^2 = 0.994 \quad \hat{\sigma} &= 0.0181 \quad DW = 1.466 \end{aligned}$$

where $\bar{W}_g \equiv \bar{C}_g/\bar{Y}_g$, $\bar{C}_g \equiv$ household expenditure on food for income group g, $\bar{Y}_g \equiv$ mean income for group g, $N_g \equiv$ number of households in income group g.

(a) Test Engel's Law using (4.24). Compare your results with those based on (4.23)

(b) Suppose that grouping the data has dealt with the measurement error problem. Discuss the other econometric problems that may arise with grouped data. Suggest a more efficient way of estimating an Engel curve using grouped data.

5 Autocorrelated Disturbances

5.1 Introduction

This chapter considers extensions of multiple regression analysis to the case where regression disturbances are serially correlated. Serial correlation, or autocorrelation, arises when the regression errors are not independently distributed either due to persistent of observations over time or over space. Our focus in this chapter is on time series observations; the problem of spatial dependence is addressed in Chapter 30. Serially correlated errors may also arise when the dynamics of the interactions between the dependent variable, y_t, and the regressors, $\mathbf{x}_t$, are not adequately taken into account, or if the regression model is misspecified due to the omission of persistent regressors.

5.2 Regression models with non-spherical disturbances

Consider the linear regression model expressed in matrix form (see Section 2.2)

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{u}, \tag{5.1}$$

where we assume

$$E(\mathbf{u}\,|\mathbf{X}) = \mathbf{0}, \quad \text{for all } t, \tag{5.2}$$

$$E(\mathbf{u}\mathbf{u}'\,|\mathbf{X}) = \boldsymbol{\Sigma}, \quad \text{for all}, \quad t \neq s, \tag{5.3}$$

with $\boldsymbol{\Sigma}$ being a $T \times T$ positive definite matrix. Model (5.1), together with assumptions (5.2) and (5.3) is known as the *generalized linear regression model*. Note that the classical linear regression model can be obtained by setting $\boldsymbol{\Sigma} = \sigma^2\mathbf{I}_T$. Another special case of the above specification is the regression model with heteroskedastic disturbances introduced in Chapter 4.

5.3 Consequences of residual serial correlation

Consider the *OLS* estimator of $\boldsymbol{\beta}$ in model (5.1) with errors satisfying (5.2) and (5.3). We have

$$\hat{\boldsymbol{\beta}}_{OLS} = \boldsymbol{\beta} + \left(\mathbf{X}'\mathbf{X}\right)^{-1}\mathbf{X}'\mathbf{u}. \tag{5.4}$$

Hence, under the condition that $E(\mathbf{u}\,|\mathbf{X}) = \mathbf{0}$,

$$\mathrm{E}\left(\hat{\boldsymbol{\beta}}_{OLS}\right) = \mathrm{E}_X\left[\mathrm{E}\left(\hat{\boldsymbol{\beta}}_{OLS}\,|\mathbf{X}\right)\right] = \boldsymbol{\beta}, \tag{5.5}$$

and $\hat{\boldsymbol{\beta}}_{OLS}$ is still an unbiased estimator of $\boldsymbol{\beta}$. The variance of the *OLS* estimator is

$$\begin{aligned} Var\left(\hat{\boldsymbol{\beta}}_{OLS}\right) &= E\left[\left(\hat{\boldsymbol{\beta}}_{OLS} - \boldsymbol{\beta}\right)\left(\hat{\boldsymbol{\beta}}_{OLS} - \boldsymbol{\beta}\right)'\right] \\ &= E\left[\left(\mathbf{X}'\mathbf{X}\right)^{-1}\mathbf{X}'\mathbf{u}\mathbf{u}'\mathbf{X}\left(\mathbf{X}'\mathbf{X}\right)^{-1}\right] \\ &= \left(\mathbf{X}'\mathbf{X}\right)^{-1}\mathbf{X}'\Sigma\mathbf{X}\left(\mathbf{X}'\mathbf{X}\right)^{-1}. \end{aligned}$$

It follows that, if the matrices $Plim_{T\to\infty}\left(T^{-1}\mathbf{X}'\mathbf{X}\right)$ and $Plim_{T\to\infty}\left(T^{-1}\mathbf{X}'\boldsymbol{\Sigma}\mathbf{X}\right)$ are both positive definite matrices with finite elements, then $\hat{\boldsymbol{\beta}}_{OLS}$ is consistent for $\boldsymbol{\beta}$. Further, under normality of $\mathbf{u}$,

$$\hat{\boldsymbol{\beta}}_{OLS} - \boldsymbol{\beta} \sim N\left[\boldsymbol{\beta}, \left(\mathbf{X}'\mathbf{X}\right)^{-1}\mathbf{X}'\Sigma\mathbf{X}\left(\mathbf{X}'\mathbf{X}\right)^{-1}\right].$$

Hence, in the presence of residual serial correlation, the variance of the least squares estimator is not $\sigma^2\left(\mathbf{X}'\mathbf{X}\right)^{-1}$, and statistical inference based on $\hat{\sigma}^2\left(\mathbf{X}'\mathbf{X}\right)^{-1}$ may be misleading.

5.4 Efficient estimation by generalized least squares

Consider the Cholesky decomposition of the inverse of $\boldsymbol{\Sigma}$

$$\boldsymbol{\Sigma}^{-1} = \mathbf{Q}\mathbf{Q}'.$$

Matrix $\mathbf{Q}$ exists if $\boldsymbol{\Sigma}$ is positive definite, but it is not unique. It can be constructed from eigenvalues and eigenvectors of $\boldsymbol{\Sigma}$ (see Section A.5 in Appendix A). Now consider the following transformations

$$\mathbf{y}_* = \mathbf{Q}'\mathbf{y}, \mathbf{X}_* = \mathbf{Q}'\mathbf{X}, \mathbf{u}_* = \mathbf{Q}'\mathbf{u}.$$

The transformed errors, $\mathbf{u}_*$, satisfy

$$E(\mathbf{u}_*\mathbf{u}_*') = E(\mathbf{Q}'\mathbf{u}\,\mathbf{u}'\mathbf{Q}) = \mathbf{Q}'E(\mathbf{u}\mathbf{u}')\mathbf{Q} = \mathbf{Q}'\Sigma\mathbf{Q} = \mathbf{I}_T.$$

It follows that the transformed model

$$\mathbf{y}_* = \mathbf{X}_*\boldsymbol{\beta} + \mathbf{u}_*,$$

satisfies all the classical assumption and establishes that the *OLS* estimator of $\boldsymbol{\beta}$ in the regression of $\mathbf{y}_*$ on $\mathbf{X}_*$ is the best linear unbiased estimation (BLUE). We have

$$\hat{\boldsymbol{\beta}}_{GLS} = \left(\mathbf{X}_*'\mathbf{X}_*\right)^{-1}\mathbf{X}_*'\mathbf{y}_*,$$

and substituting $\mathbf{X}_*$ and $\mathbf{y}_*$ in terms of the original observations we obtain

$$\hat{\boldsymbol{\beta}}_{GLS} = \left(\mathbf{X}'\boldsymbol{\Sigma}^{-1}\mathbf{X}\right)^{-1}\mathbf{X}'\boldsymbol{\Sigma}^{-1}\mathbf{y}.$$

This is known as the generalized least squares (*GLS*) estimator, and is more efficient than the *OLS* estimator. The efficiency of the *GLS* over the *OLS* estimator follows immediately from the fact that the *GLS* estimator satisfies the assumptions of the Gauss–Markov theorem. It is also instructive to give a direct proof of the efficiency of $\hat{\boldsymbol{\beta}}_{GLS}$ over the $\hat{\boldsymbol{\beta}}_{OLS}$ estimator. We first note that

$$Var\left(\hat{\boldsymbol{\beta}}_{GLS}\right) = \left(\mathbf{X}_*'\mathbf{X}_*\right)^{-1} = \left(\mathbf{X}'\Sigma^{-1}\mathbf{X}\right)^{-1},$$

and

$$Var\left(\hat{\boldsymbol{\beta}}_{OLS}\right) = \left(\mathbf{X}'\mathbf{X}\right)^{-1}\mathbf{X}'\Sigma\mathbf{X}\left(\mathbf{X}'\mathbf{X}\right)^{-1}.$$

To prove that $\hat{\boldsymbol{\beta}}_{GLS}$ is at least as efficient as $\hat{\boldsymbol{\beta}}_{OLS}$ it is sufficient to show that $Var\left(\hat{\boldsymbol{\beta}}_{OLS}\right) - Var\left(\hat{\boldsymbol{\beta}}_{GLS}\right)$ is a positive semi-definite matrix. This follows if it is shown that

$$\left[Var\left(\hat{\boldsymbol{\beta}}_{GLS}\right)\right]^{-1} - \left[Var\left(\hat{\boldsymbol{\beta}}_{OLS}\right)\right]^{-1} \geq 0,$$

or equivalently if

$$\mathbf{X}'\Sigma^{-1}\mathbf{X} - \left(\mathbf{X}'\mathbf{X}\right)\left(\mathbf{X}'\Sigma^{-1}\mathbf{X}\right)^{-1}\left(\mathbf{X}'\mathbf{X}\right) \geq 0.$$

Note that the left-hand side of the above inequality can be written as

$$\mathbf{X}'\Sigma^{-\frac{1}{2}}\left[\mathbf{I}_T - \Sigma^{\frac{1}{2}}\mathbf{X}\left(\mathbf{X}'\boldsymbol{\Sigma}^{\frac{1}{2}}\boldsymbol{\Sigma}^{\frac{1}{2}}\mathbf{X}\right)^{-1}\mathbf{X}'\Sigma^{\frac{1}{2}}\right]\Sigma^{-\frac{1}{2}}\mathbf{X},$$

and the condition for the efficiency of the $\hat{\boldsymbol{\beta}}_{GLS}$ over $\hat{\boldsymbol{\beta}}_{OLS}$ becomes

$$\mathbf{Y}'\left[\mathbf{I}_T - \mathbf{Z}\left(\mathbf{Z}'\mathbf{Z}\right)^{-1}\mathbf{Z}'\right]\mathbf{Y} \geq 0,$$

where $\mathbf{Y} = \boldsymbol{\Sigma}^{-\frac{1}{2}}\mathbf{X}$ and $\mathbf{Z} = \boldsymbol{\Sigma}^{\frac{1}{2}}\mathbf{X}$. The desired result now follows by noting that $\mathbf{M}_Z = \mathbf{I}_T - \mathbf{Z}\left(\mathbf{Z}'\mathbf{Z}\right)^{-1}\mathbf{Z}'$ being an idempotent matrix allows us to write $\mathbf{Y}'\mathbf{M}_Z\mathbf{Y}$ as $(\mathbf{M}_Z\mathbf{Y})'(\mathbf{M}_Z\mathbf{Y})$.

The above proof also shows that for the *GLS* estimators to be strictly more efficient than the least squares estimators, it is required that $\mathbf{M}_Z\mathbf{Y}$ be non-zero.

See Section 19.2.1 for a review of the *GLS* estimator in the context of seemingly unrelated regressions.

Example 11 *In general $\boldsymbol{\Sigma}$ is not known, but there are some special cases of interest where $\boldsymbol{\Sigma}$ is known up to a scalar constant. One important example is the simple heteroskedastic model introduced in Chapter 4 where*

$$\boldsymbol{\Sigma} = \sigma^2 \begin{pmatrix} z_1^2 & 0 & \dots & 0 \\ 0 & z_2^2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & z_T^2 \end{pmatrix},$$

and z_t's are observations on a variable distributed independently of the disturbances, $\mathbf{u}$. In this case the GLS estimator reduces to the weighted least squares estimator obtained from the regression of y_t/z_t on x_{it}/z_t, for $i = 1, 2, \dots, k$, the weights being the inverse of z_ts.

5.4.1 Feasible generalized least squares

The *GLS* estimator, $\hat{\boldsymbol{\beta}}_{GLS}$, assumes $\boldsymbol{\Sigma}$ is known. But in general $\boldsymbol{\Sigma}$ is not known, and for this reason $\hat{\boldsymbol{\beta}}_{GLS}$ is known as the 'infeasible' *GLS* estimator. A 'feasible' *GLS* estimator can be obtained by replacing $\boldsymbol{\Sigma}$ with a consistent estimator. However, in cases where $\boldsymbol{\Sigma}$ is unrestricted, there are $T(T+1)/2$ additional unknown parameters to be estimated from T observations and this is clearly not feasible. Therefore, to proceed further, some structure must be imposed on $\boldsymbol{\Sigma}$. In practice, it is typically assumed that $\boldsymbol{\Sigma}$ depends on a fixed number of unknown parameters, $\boldsymbol{\theta}$, which can be consistently estimated by $\hat{\boldsymbol{\theta}}$. Then, under certain regularity conditions, $\hat{\boldsymbol{\Sigma}} = \boldsymbol{\Sigma}(\hat{\boldsymbol{\theta}})$ will be a consistent estimator of $\Sigma(\boldsymbol{\theta})$. The feasible generalized least squares (*FGLS*) estimator is given by

$$\hat{\boldsymbol{\beta}}_{FGLS} = \left(\mathbf{X}'\hat{\boldsymbol{\Sigma}}^{-1}\mathbf{X}\right)^{-1}\mathbf{X}'\hat{\boldsymbol{\Sigma}}^{-1}\mathbf{y}.$$

The estimator $\hat{\boldsymbol{\beta}}_{FGLS}$ is asymptotically equivalent to $\hat{\boldsymbol{\beta}}_{GLS}$ if

$$\underset{T\to\infty}{Plim}\left(T^{-1}\mathbf{X}'\hat{\boldsymbol{\Sigma}}^{-1}\mathbf{X} - T^{-1}\mathbf{X}'\boldsymbol{\Sigma}^{-1}\mathbf{X}\right) = \mathbf{0},$$

and

$$\underset{T\to\infty}{Plim}\left(T^{-1/2}\mathbf{X}'\hat{\boldsymbol{\Sigma}}^{-1}\mathbf{u} - T^{-1/2}\mathbf{X}'\boldsymbol{\Sigma}^{-1}\mathbf{u}\right) = \mathbf{0}.$$

If these conditions are satisfied, then the FGLS estimator, based on $\hat{\boldsymbol{\theta}}$, has the same asymptotic properties as the infeasible *GLS* estimator, $\hat{\boldsymbol{\beta}}_{GLS}$.

5.5 Regression model with autocorrelated disturbances

Consider the regression model

$$y_t = \alpha + \beta x_t + u_t, \tag{5.6}$$

$$u_t = \phi u_{t-1} + \varepsilon_t \tag{5.7}$$

where we assume that $|\phi| < 1$, and ε_t is a white-noise process, namely it is a serially uncorrelated process with a zero mean and a constant variance σ_ε^2

$$\text{Cov}\,(\varepsilon_t, \varepsilon_{t'}) = 0, \qquad \text{for} \qquad t \neq t', \qquad \text{and} \qquad \text{Var}\,(\varepsilon_t) = \sigma_\varepsilon^2. \tag{5.8}$$

We assume that ε_t and the regressors are uncorrelated, namely

$$E(\varepsilon_t \mid x_t, x_{t-1}, \ldots.) = 0. \tag{5.9}$$

Note that condition (5.9) is weaker than the orthogonality assumption (5.3), where it is assumed that ε_t is uncorrelated with future values of $\mathbf{x}_t$, as well as with its current and past values.

By repeated substitution, in (5.7), we have

$$u_t = \varepsilon_t + \phi\varepsilon_{t-1} + \phi^2\varepsilon_{t-2} + \ldots,$$

which is known as the *moving average* form for u_t. From the above expression, under $|\phi| < 1$, each disturbance u_t embodies the entire past history of the ε_t, with the most recent observations receiving greater weight than those in the distant past. Since the successive values of ε_t are uncorrelated, the variance of u_t is

$$\begin{aligned} Var\,(u_t) &= \sigma_\varepsilon^2 + \phi^2\sigma_\varepsilon^2 + \phi^4\sigma_\varepsilon^2 + \ldots. \\ &= \frac{\sigma_\varepsilon^2}{1-\phi^2}. \end{aligned}$$

The covariance between u_t and u_{t-1} is given by

$$Cov\,(u_t, u_{t-1}) = E\,(u_t u_{t-1}) = E\left[(\phi u_{t-1} + \varepsilon_t)\,u_{t-1}\right] = \phi\frac{\sigma_\varepsilon^2}{1-\phi^2}.$$

To obtain the covariance between u_t and u_{t-s}, for any s, first note that by applying repeated substitution equation (5.7) can be written as

$$u_t = \phi^s u_{t-s} + \sum_{i=1}^{s-1} \phi^i \varepsilon_{t-i}.$$

It follows that

$$Cov\,(u_t, u_{t-s}) = E\,(u_t u_{t-s}) = E\left[\left(\phi^s u_{t-s} + \sum_{i=1}^{s-1} \phi^i \varepsilon_{t-i}\right) u_{t-s}\right] = \phi^s \frac{\sigma_\varepsilon^2}{1-\phi^2}.$$

To summarize, the covariance matrix of $\mathbf{u} = (u_1, u_2, \ldots, u_T)'$ is given by

$$\mathbf{\Sigma} = \frac{\sigma_\varepsilon^2}{1-\phi^2}\begin{pmatrix} 1 & \phi & \ldots & \phi^{T-1} \\ \phi & 1 & \ldots & \phi^{T-2} \\ \vdots & \vdots & \vdots & \vdots \\ \phi^{T-1} & \phi^{T-2} & \ldots & 1 \end{pmatrix}.$$

Note that, under $|\phi| < 1$, the values decline exponentially as we move away from the diagonal.

5.5.1 Estimation

Suppose, initially, that the parameter ϕ in (5.7) is known. Then model (5.6) can be transformed so that the transformed equation satisfies the classical assumptions. To do this, first substitute $u_t = y_t - \alpha - \beta x_t$ in $u_t = \phi u_{t-1} + \varepsilon_t$ to obtain:

$$y_t - \phi y_{t-1} = \alpha\,(1-\phi) + \beta\,(x_t - \phi x_{t-1}) + \varepsilon_t. \tag{5.10}$$

Define $y_t^* = y_t - \phi y_{t-1}$, and $x_t^* = x_t - \phi x_{t-1}$, then

$$y_t^* = \alpha\,(1-\phi) + \beta x_t^* + \varepsilon_t, \qquad t = 2, 3, \ldots, T. \tag{5.11}$$

It is clear that in this transformed regression, disturbances ε_t satisfy all the classical assumptions, and efficient estimators of α and β can be obtained by the *OLS* regression of y_t^* on x_t^*.

For the AR(2) error process we need to use the following transformations:

$$x_t^* = x_t - \phi_1 x_{t-1} - \phi_2 x_{t-2}, \qquad t = 3, 4, \ldots, T, \tag{5.12}$$

$$y_t^* = y_t - \phi_1 y_{t-1} - \phi_2 y_{t-2}, \qquad t = 3, 4, \ldots, T. \tag{5.13}$$

The above procedure ignores the effect of initial observations. For example, for the AR(1) case we can allow for the initial observations using:

$$x_1^* = \frac{x_1}{\sqrt{1-\phi^2}}, \tag{5.14}$$

$$y_1^* = \frac{y_1}{\sqrt{1-\phi^2}}. \tag{5.15}$$

Efficient estimators of α and β that take account of initial observations can now be obtained by running the *OLS* regression of

$$\mathbf{y}^* = \left(y_1^*, y_2^*, \ldots, y_T^*\right)', \tag{5.16}$$

on an intercept and

$$\mathbf{x}^* = \left(x_1^*, x_2^*, \ldots, x_T^*\right)'. \tag{5.17}$$

The estimators that make use of the initial observations and those that do not are asymptotically equivalent (i.e., there is little to choose between them when it is known that $|\phi| < 1$ and T is relatively large).

If $\phi = 1$, then y_t^* and x_t^* will be the same as the first differences of y_t and x_t, and β can be estimated by regression of Δy_t on Δx_t. There is no long-run relationship between the levels of y_t and x_t.

When ϕ is *unknown*, ϕ and β can be estimated using the Cochrane and Orcutt (C–O method) two-step procedure. Let $\hat{\phi}(0)$ be the initial estimate of ϕ, then generate the quasi-differenced variables

$$x_t^*(0) = x_t - \hat{\phi}(0)\, x_{t-1}, \tag{5.18}$$

$$y_t^*(0) = y_t - \hat{\phi}(0)\, y_{t-1}. \tag{5.19}$$

Then run a regression of $y_t^*(0)$ on $x_t^*(0)$ to obtain a new estimate of α and β, (say $\hat{\alpha}(1)$ and $\hat{\beta}(1)$, and hence a new estimate of ϕ, given by

$$\hat{\phi}(1) = \frac{\sum_{t=2}^{T} \hat{u}_t(1)\, \hat{u}_{t-1}(1)}{\sum_{t=2}^{T} \hat{u}_t^2(1)},$$

where $\hat{u}_t(1) = y_t - \hat{\alpha}(1) - \hat{\beta}(1)\, x_t$. Generate new transformed observations $x_t^*(1) = x_t - \hat{\phi}(1)\, x_{t-1}$, and $y_t^*(1) = y_t - \hat{\phi}(1)\, y_{t-1}$, and repeat the above steps until the two successive estimates of β are sufficiently close to one another.

5.5.2 Higher-order error processes

The above procedure can be readily extended to general regression models and higher-order error processes. Consider for example the general regression model

$$y_t = \boldsymbol{\beta}' \mathbf{x}_t + u_t, \quad t = 1, 2, \ldots, T,$$

where u_t follows the $AR(2)$ specification:

$$AR(2): u_t = \phi_1 u_{t-1} + \phi_2 u_{t-2} + \epsilon_t, \qquad \epsilon_t \sim N(0, \sigma_\epsilon^2), \quad t = 1, 2, \ldots, T. \tag{5.20}$$

Assuming the error process is stationary and has started a long time prior to the first observation date (i.e., $t = 1$) we have

$$AR(1)\ Case: \qquad Var(u_1) = \frac{\sigma_\epsilon^2}{1-\phi^2},$$

$$AR(2)\ Case: \quad \begin{cases} Var(u_1) = Var(u_2) = \dfrac{\sigma_\epsilon^2(1-\phi_2)}{(1+\phi_2)\left[(1-\phi_2)^2-\phi_1^2\right]}, \\ Cov(u_1,\ u_2) = \dfrac{\sigma_\epsilon^2\phi_1}{(1+\phi_2)\left[(1-\phi_2)^2-\phi_1^2\right]}. \end{cases}$$

The exact *ML* estimation procedure then allows for the effect of initial values on the parameter estimates by adding the logarithm of the density function of the initial values to the log-density function of the remaining observations obtained conditional on the initial values. For example, in the case of the $AR(1)$ model the log-density function of $(u_2,\ u_3, \ldots, u_T)$ conditional on the initial value, u_1, is given by

$$\log\left\{f(u_2, u_3, \ldots, u_T \,|u_1)\right\} = -\frac{(T-1)}{2}\log(2\pi\sigma_\epsilon^2) - \frac{1}{2\sigma_\epsilon^2}\left(\sum_{t=2}^{T} u_t^2\right), \tag{5.21}$$

and

$$\log\left\{f(u_1)\right\} = -\frac{1}{2}\log(2\pi\sigma_\epsilon^2) + \frac{1}{2}\log(1-\phi^2) - \frac{(1-\phi^2)}{2\sigma_\epsilon^2}u_1^2.$$

Combining the above log-densities yields the full (unconditional) log-density function of $(u_1, u_2, \ldots, u_T)$

$$\log\left\{f(u_1, u_2, \ldots, u_T)\right\} = -\frac{T}{2}\log(2\pi\sigma_\epsilon^2) + \frac{1}{2}\log(1-\phi^2) - \frac{1}{2\sigma_\epsilon^2}\left[\sum_{t=2}^{T}(u_t - \phi u_{t-1})^2 + (1-\phi^2)u_1^2\right]. \tag{5.22}$$

Asymptotically, the effect of the distribution of the initial values on the *ML* estimators is negligible, but it could be important in small samples where $\mathbf{x}_t$s are trended and ϕ is suspected to be near but *not* equal to unity. See Pesaran (1972) and Pesaran and Slater (1980) (Chs 2 and 3) for further details. Also see Judge et al. (1985), Davidson and MacKinnon (1993), and the papers by Hildreth and Dent (1974), and Beach and MacKinnon (1978). Strictly speaking, the *ML* estimation will be exact if lagged values of y_t are not included amongst the regressors. For a discussion of the exact *ML* estimation of models with lagged dependent variables and serially correlated errors see Pesaran (1981a).

5.5.3 The *AR*(1) case

For this case, the *ML* estimators can be computed by maximizing the log-likelihood function[1]

$$\ell_{AR1}(\boldsymbol{\theta}) = -\frac{T}{2}\log(2\pi\sigma_\epsilon^2) + \tfrac{1}{2}\log(1-\phi^2) - \frac{1}{2\sigma_\epsilon^2}(\mathbf{y}-\mathbf{X}\boldsymbol{\beta})'\mathbf{R}(\phi)(\mathbf{y}-\mathbf{X}\boldsymbol{\beta}), \tag{5.23}$$

with respect to the unknown parameters $\boldsymbol{\theta} = (\boldsymbol{\beta}', \sigma_\epsilon^2, \phi)'$, where $\mathbf{R}(\phi)$ is the $T \times T$ matrix

$$R(\phi) = \begin{pmatrix} 1 & -\phi & \cdots & 0 & 0 \\ -\phi & 1+\phi^2 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \cdots & 1+\phi^2 & -\phi \\ 0 & 0 & \cdots & -\phi & 1 \end{pmatrix}, \tag{5.24}$$

and $|\phi| < 1$.

The computations are carried out by the 'inverse interpolation' method which is *certain to converge*. See Pesaran and Slater (1980) for further details.

The concentrated log-likelihood function in this case is given by

$$\ell_{AR1}(\phi) = -\frac{T}{2}[1+\log(2\pi)] + \tfrac{1}{2}\log(1-\phi^2) - \frac{T}{2}\log\{\tilde{\mathbf{u}}'R(\phi)\tilde{\mathbf{u}}/T\}, \qquad |\phi| < 1, \tag{5.25}$$

where $\tilde{\mathbf{u}}$ is the $T \times 1$ vector of *ML* residuals:

$$\tilde{\mathbf{u}} = \mathbf{y} - \mathbf{X}[\mathbf{X}'\mathbf{R}(\phi)\mathbf{X}]^{-1}\mathbf{X}'\mathbf{R}(\phi)\mathbf{y}.$$

5.5.4 The *AR*(2) case

For this case, the *ML* estimators are obtained by maximizing the log-likelihood function

$$\ell_{AR2}(\boldsymbol{\theta}) = -\frac{T}{2}\log(2\pi\sigma_\epsilon^2) + \log(1+\phi_2) + \tfrac{1}{2}\log\left[(1-\phi_2)^2 - \phi_1^2\right] - \frac{1}{2\sigma_\epsilon^2}(\mathbf{y}-\mathbf{X}\boldsymbol{\beta})'\mathbf{R}(\boldsymbol{\phi})(\mathbf{y}-\mathbf{X}\boldsymbol{\beta}), \tag{5.26}$$

[1] This result follows readily from (5.22) and can be obtained by substituting $u_t = y_t - \boldsymbol{\beta}'\mathbf{x}_t$ in (5.22).

with respect to $\boldsymbol{\theta} = (\boldsymbol{\beta}', \sigma_\epsilon^2, \phi)'$, where $\phi = (\phi_1, \phi_2)'$

$$R(\boldsymbol{\phi}) = \begin{pmatrix} 1 & -\phi_1 & -\phi_2 & 0 & \dots & 0 & 0 \\ -\phi_1 & 1+\phi_1^2 & -\phi_1+\phi_1\phi_2 & -\phi_2 & \dots & 0 & 0 \\ -\phi_2 & -\phi_1+\phi_1\phi_2 & 1+\phi_1^2+\phi_2^2 & -\phi_1+\phi_1\phi_2 & \dots & 0 & 0 \\ 0 & -\phi_2 & -\phi_1+\phi_1\phi_2 & 1+\phi_1^2+\phi_2^2 & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \dots & 1+\phi_1^2 & -\phi_1 \\ 0 & 0 & 0 & 0 & \dots & -\phi_1 & 1 \end{pmatrix}. \tag{5.27}$$

The estimation procedure imposes the restrictions

$$\begin{aligned} &1+\phi_2 > 0, \\ &1-\phi_2+\phi_1 > 0, \ , \\ &1-\phi_2-\phi_1 > 0. \end{aligned} \tag{5.28}$$

needed if the $AR(2)$ process, (5.20), is to be stationary.

5.5.5 Covariance matrix of the exact *ML* estimators for the *AR*(1) and *AR*(2) disturbances

The estimates of the covariance matrix of the exact *ML* estimators defined in the above subsections are computed on the assumption that the regressors $\mathbf{x}_t$ do not include lagged values of the dependent variable.

For the $AR(1)$ case we have

$$\tilde{V}(\tilde{\boldsymbol{\beta}}) = \hat{\sigma}_\epsilon^2[\mathbf{X}'\mathbf{R}(\tilde{\phi})\mathbf{X}]^{-1}, \tag{5.29}$$

$$\tilde{V}(\tilde{\phi}) = T^{-1}(1-\tilde{\phi}^2), \tag{5.30}$$

where $\mathbf{R}(\tilde{\phi})$ is already defined by (5.24), and $\hat{\sigma}_\epsilon^2$ is given below by (5.39) below.

For the $AR(2)$ case we have

$$\tilde{V}(\tilde{\boldsymbol{\beta}}) = \hat{\sigma}_\epsilon^2[\mathbf{X}'\mathbf{R}(\tilde{\phi}_1,\tilde{\phi}_2)\mathbf{X}]^{-1}, \tag{5.31}$$

$$\tilde{V}(\tilde{\phi}_1) = \tilde{V}(\tilde{\phi}_2) = T^{-1}(1-\tilde{\phi}_2^2), \tag{5.32}$$

$$\widetilde{Cov}(\tilde{\phi}_1,\tilde{\phi}_2) = -T^{-1}\tilde{\phi}_1(1+\tilde{\phi}_2), \tag{5.33}$$

where $\mathbf{R}(\tilde{\phi}_1,\tilde{\phi}_2)$ is defined by (5.27). Here the *ML* estimators are designated by $\sim$.

5.5.6 Adjusted residuals, $R^2, \bar{R}^2$, and other statistics

In the case of the exact *ML* estimators, the 'adjusted' residuals are computed as (see Pesaran and Slater (1980, pp. 49, 136)):

$$\tilde{\epsilon}_1 = \tilde{u}_1\sqrt{\left\{[(1-\tilde{\phi}_2)^2 - \tilde{\phi}_1^2](1+\tilde{\phi}_2)/(1-\tilde{\phi}_2)\right\}}, \tag{5.34}$$

$$\tilde{\epsilon}_2 = \tilde{u}_2\sqrt{(1-\tilde{\phi}_2^2)} - \tilde{u}_1\ \tilde{\phi}_1\sqrt{[(1+\tilde{\phi}_2)/(1-\tilde{\phi}_2)]}, \tag{5.35}$$

$$\tilde{\epsilon}_t = \tilde{u}_t - \tilde{\phi}_1\tilde{u}_{t-1} - \tilde{\phi}_2\tilde{u}_{t-2}, \qquad t = 3, 4, \ldots, T, \tag{5.36}$$

where

$$\tilde{u}_t = y_t - \mathbf{x}_t^{'}\tilde{\boldsymbol{\beta}}, \qquad t = 1, 2, \ldots, T,$$

are the 'unadjusted' residuals, and

$$\tilde{\boldsymbol{\beta}} = \left(\mathbf{X}'\mathbf{R}(\tilde{\boldsymbol{\phi}})\mathbf{X}\right)^{-1}\mathbf{X}'\mathbf{R}(\tilde{\boldsymbol{\phi}})\mathbf{y}. \tag{5.37}$$

Recall that $\tilde{\boldsymbol{\phi}} = (\tilde{\phi}_1, \tilde{\phi}_2)'$. The programme also takes account of the specification of the *AR*-error process in computations of the fitted values. Denoting these adjusted (or conditional) fitted values by $\tilde{y}_t$, we have

$$\tilde{y}_t = \tilde{E}(y_t \,|y_{t-1}, y_{t-2}, \ldots;\ \mathbf{x}_t, \mathbf{x}_{t-1}, \ldots) = y_t - \tilde{\epsilon}_t, \quad t = 1, 2, \ldots, T. \tag{5.38}$$

The standard error of the regression is computed using the formula

$$\hat{\sigma}_\epsilon^2 = \tilde{\mathbf{u}}'\mathbf{R}(\tilde{\boldsymbol{\phi}})\tilde{\mathbf{u}}/(T-k-p), \tag{5.39}$$

where $p = 1$, for the $AR(1)$ case, and $p = 2$ for the $AR(2)$ case. Given the way the adjusted residuals $\tilde{\epsilon}_t$ are defined above we also have

$$\hat{\sigma}_\epsilon^2 = \tilde{\mathbf{u}}'\mathbf{R}(\tilde{\boldsymbol{\phi}})\tilde{\mathbf{u}}/(T-k-p) = \sum_{t=1}^{T}\tilde{\epsilon}_t^2/(T-k-p). \tag{5.40}$$

Notice that this estimator of σ_ϵ^2 differs from the *ML* estimator given by

$$\tilde{\sigma}_\epsilon^2 = \sum_{t=1}^{T}\tilde{\epsilon}_t^2/T,$$

and the estimator adopted in Pesaran and Slater (1980). The difference lies in the way the sum of squares of residuals $\Sigma\tilde{\epsilon}_t^2$, is corrected for the loss in degrees of freedom arising from the estimation of the regression coefficients, $\boldsymbol{\beta}$, and the parameters of the error process, $\boldsymbol{\phi} = (\phi_1, \phi_2)'$.

The R^2, $\bar{R}^2$, and the F-statistic are computed from the adjusted residuals:

$$R^2 = 1 - \left(\sum_{t=1}^{T}\tilde{\epsilon}_t^2 \Big/ \sum_{t=1}^{T}(y_t - \bar{y})^2\right),$$

$$\bar{R}^2 = 1 - (\hat{\sigma}_\epsilon^2/\hat{\sigma}_y^2), \tag{5.41}$$

where $\hat{\sigma}_y$ is the standard deviation of the dependent variable, defined as before by $\hat{\sigma}_y^2 = \sum_{t=1}^{T}(y_t - \bar{y})^2/(T-1)$.

The F-statistics reported following the regression results are computed according to the formula

$$F\text{-statistic} = \left(\frac{R^2}{1-R^2}\right)\left(\frac{T-k-p}{k+p-1}\right) \overset{a}{\sim} F(k+p-1,\ T-k-p) \tag{5.42}$$

with

$$p = 1, \qquad \text{under AR}(1) \text{ error specification}$$

and

$$p = 2, \qquad \text{under AR}(2) \text{ error specification}$$

Notice that R^2 in (5.42) is given by (5.41). The above F-statistic can be used to test the joint hypothesis that except for the intercept term, all the other regression coefficients *and* the parameters of the AR-error process are zero. Under this hypothesis the F-statistic is distributed *approximately* as F with $k+p-1$ and $T-k-p$ degrees of freedom. The chi-squared version of this test can be based on $TR^2/(1-R^2)$, which under the null hypothesis of zero slope and AR coefficients is asymptotically distributed as a chi-squared variate with $k+p-1$ degrees of freedom.

The Durbin–Watson statistic is also computed using the adjusted residuals, $\tilde{\epsilon}_t$:

$$\widetilde{DW} = \frac{\sum_{t=2}^{T}(\tilde{\epsilon}_t - \tilde{\epsilon}_{t-1})^2}{\sum_{t=1}^{T}\tilde{\epsilon}_t^2}.$$

5.5.7 Log-likelihood ratio statistics for tests of residual serial correlation

The log-likelihood ratio statistic for the test of $AR(1)$ against the non-autocorrelated error specification is given by

$$\chi^2_{AR1,OLS} = 2(LL_{AR1} - LL_{OLS}) \overset{a}{\sim} \chi^2_1.$$

The log-likelihood ratio statistic for the test of the AR(2)-error specification against the AR(1)-error specification is given by

$$\chi^2_{AR2,AR1} = 2(LL_{AR2} - LL_{AR1}) \overset{a}{\sim} \chi^2_1.$$

Both of the above statistics are asymptotically distributed, under the null hypothesis, as a chi-squared variate with one degree of freedom.

The log-likelihood values, LL_{AR1} and LL_{AR2}, represent the maximized values of the log-likelihood functions defined by (5.23), and (5.26), respectively. LL_{OLS} denotes the maximized value of the log-likelihood estimator for the *OLS* case.

5.6 Cochrane–Orcutt iterative method

This estimation method employs the Cochrane and Orcutt (1949) iterative procedure to compute *ML* estimators of $\boldsymbol{\beta}$ under the assumption that the disturbances, u_t, follow the $AR(p)$ process

$$u_t = \sum_{i=1}^{p} \phi_i u_{t-i} + \epsilon_t, \quad \epsilon_t \sim N(0, \sigma_\epsilon^2), \quad t = 1, 2, \ldots, T, \tag{5.43}$$

with 'fixed initial' values. The fixed initial value assumption is the same as treating the values, $y_1, y_2, \ldots, y_p$ as given or non-stochastic. This procedure in effect ignores the possible contribution of the distribution of the initial values to the overall log-likelihood function of the model. Once again the primary justification of treating initial values as fixed is asymptotic and is plausible only when (5.43) is stationary and T is reasonably large (see Pesaran and Slater (1980, Section 3.2), and Judge et al. (1985) for further discussion).

The log-likelihood function for this case is defined by

$$LL_{CO}(\boldsymbol{\theta}) = -\frac{(T-p)}{2}\log(2\pi\sigma_\epsilon^2) - \frac{1}{2\sigma_\epsilon^2}\sum_{t=p+1}^{T}\epsilon_t^2 + c, \tag{5.44}$$

where $\boldsymbol{\theta} = (\boldsymbol{\beta}', \sigma_\epsilon^2, \boldsymbol{\phi}')'$ with $\boldsymbol{\phi} = (\phi_1, \phi_2, \ldots, \phi_p)'$. Notice that the constant term c in (5.44) is undefined, and is usually set equal to zero. The Cochrane–Orcutt (C-O) method maximizes $LL_{CO}(\boldsymbol{\theta})$, or equivalently minimizes $\Sigma_{t=p+1}^{T}\epsilon_t^2$ with respect to $\boldsymbol{\theta}$ by the iterative method of 'successive substitution'. Each iteration involves two steps: in the first step, LL_{CO} is maximized with respect to $\boldsymbol{\beta}$, taking $\boldsymbol{\phi}$ as given. In the second step, $\boldsymbol{\beta}$ is taken as given and the log-likelihood function is maximized with respect to $\boldsymbol{\phi}$. In each of these steps, the optimization problem is solved by running *OLS* regressions. To start the iterations $\boldsymbol{\phi}$ is initially set equal to zero. The iterations are terminated if

$$\sum_{i=1}^{p}\left|\tilde{\phi}_{i,(j)} - \tilde{\phi}_{i,(j-1)}\right| < p/1000, \tag{5.45}$$

where $\tilde{\phi}_{i,j}$ and $\tilde{\phi}_{i,(j-1)}$ stand for estimators of ϕ_i in the jth and $(j-1)^{th}$ iterations, respectively. The estimator of σ_ϵ^2 is computed as

$$\hat{\sigma}_\epsilon^2 = \sum_{t=p+1}^{T}\tilde{\epsilon}_t^2/(T-p-k), \tag{5.46}$$

where $\tilde{\epsilon}_t$, the adjusted residuals, are given by

$$\tilde{\epsilon}_t = \tilde{u}_t - \sum_{i=1}^{p} \tilde{\phi}_i \tilde{u}_{t-i}, \qquad t = p+1, p+2, \ldots, T, \tag{5.47}$$

where

$$\tilde{u}_t = y_t - \sum_{i=1}^{k} \tilde{\beta}_i x_{it}, \qquad t = 1, 2, \ldots, T. \tag{5.48}$$

As before, the symbol $\sim$ on top of an unknown parameter stands for *ML* estimators (now under fixed initial values). The estimator of σ_ϵ^2 in (5.46) differs from the *ML* estimator, given by $\tilde{\sigma}_\epsilon^2 = \sum_{t=p+1}^{T} \tilde{\epsilon}_t^2/(T-p)$. The estimator $\hat{\sigma}_\epsilon^2$ allows for the loss of degrees of freedom associated with the estimation of the unknown coefficients, $\boldsymbol{\beta}$, and the parameters of the AR process, $\boldsymbol{\phi}$. Notice also that the estimator of σ_ϵ^2 is based on $T-p$ adjusted residuals, since the initial values $y_1, y_2, \ldots, y_p$ are treated as fixed.

The adjusted fitted values, $\tilde{y}_t$, in the case of this option are computed as

$$\tilde{y}_t = \hat{E}(y_t \,|\, y_{t-1}, y_{t-2}, \ldots; \mathbf{x}_t, \mathbf{x}_{t-1}, \ldots) = y_t - \tilde{\epsilon}_t, \tag{5.49}$$

for $t = p+1, p+2, \ldots, T$. Notice that the initial values $\tilde{y}_1, \tilde{y}_2, \ldots, \tilde{y}_p$, are not defined.

In the case where $p = 1$, Microfit also provides a plot of the concentrated log-likelihood function in terms of ϕ_1, defined by

$$LL_{CO}(\tilde{\phi}_1) = -\frac{(T-1)}{2}[1 + \log(2\pi\tilde{\sigma}_\epsilon^2)], \tag{5.50}$$

where

$$\tilde{\sigma}_\epsilon^2 = \sum_{t=2}^{T} \tilde{\epsilon}_t^2/(T-1),$$

and $\tilde{\epsilon}_t = \tilde{u}_t - \tilde{\phi}_1 \tilde{u}_{t-1}$.

5.6.1 Covariance matrix of the C-O estimators

The estimator of the asymptotic variance matrix of $\tilde{\boldsymbol{\phi}} = (\tilde{\boldsymbol{\beta}}', \tilde{\boldsymbol{\phi}}')'$ is computed as

$$\hat{V}(\tilde{\boldsymbol{\phi}}) = \hat{\sigma}_\epsilon^2 \begin{pmatrix} \tilde{\mathbf{X}}_*'\tilde{\mathbf{X}}_* & \tilde{\mathbf{X}}_*'\mathbf{S} \\ \mathbf{S}'\tilde{\mathbf{X}}_* & \mathbf{S}'\mathbf{S} \end{pmatrix}^{-1}, \tag{5.51}$$

where $\tilde{\mathbf{X}}_*$ is the $(T-p) \times k$ matrix of transformed regressors[2]

[2] A typical element of $\tilde{X}_*$ is given by

$$\tilde{x}_{jt}^* = x_{jt} - \sum_{i=1}^{p} \tilde{\rho}_i x_{j,t-i} \qquad t = p+1, p+2, \ldots, T, \qquad j = 1, 2, \ldots, k.$$

$$\tilde{\mathbf{X}}_* = \sum_{i=1}^{p} \tilde{\phi}_i \mathbf{X}_{-i}, \tag{5.52}$$

and $\mathbf{S}$ is an $(T-p) \times p$ matrix containing the p lagged values of the C-O residuals, $\tilde{u}_t$, namely

$$\mathbf{S} = \begin{pmatrix} \tilde{u}_p & \tilde{u}_{p-1} & \dots & \tilde{u}_1 \\ \tilde{u}_{p+1} & \tilde{u}_p & \dots & \tilde{u}_2 \\ \vdots & \vdots & \vdots & \vdots \\ \tilde{u}_{T-1} & \tilde{u}_{T-2} & \dots & \tilde{u}_{T-p} \end{pmatrix}. \tag{5.53}$$

The unadjusted residuals, $\tilde{u}_t$, are already defined by (5.48). The above estimator of the variance matrix of $\tilde{\boldsymbol{\beta}}$ and $\tilde{\boldsymbol{\phi}}$ is asymptotically valid even if the regression model contains lagged dependent variables.

Example 12 *Consider the regression equation*

$$s_t = \alpha_0 + \alpha_1 s_{t-1} + \alpha_2 \Delta \log y_t + \alpha_3 \left(\Pi_t - \Pi_t^e \right) + u_t, \tag{5.54}$$

where s_t is the saving rate, $\Delta \log y_t$ is the rate of change of real disposable income, Π_t is the rate of inflation, and Π_t^e are the adaptive expectations of Π_t, and

$$u_t = \phi_1 u_{t-1} + \epsilon_t. \tag{5.55}$$

Equation (5.54) is a modified version of the saving function estimated by Deaton (1977).[3] *In the following, we use an approximation of Π_t^e by a geometrically declining distributed lag function of the UK inflation rate (see Lesson 10.12 in Pesaran and Pesaran (2009) for details). Figure 5.1 shows the log-likelihood profile for different values of ϕ_1, in the range $[-0.99, 0.99]$. The log-likelihood function is bimodal at positive and negative values of ϕ_1. The global maximum of the log-likelihood is achieved for $\phi_1 < 0$. Bimodal log-likelihood functions frequently arise in estimation of models with lagged dependent variables subject to a serially correlated error process, particularly in cases where the regressors show a relatively low degree of variability. The bimodal problem is sure to arise if apart from the lagged values of the dependent there are no other regressors in the regression equation. Table 5.1 reports maximum likelihood estimation of the model in the Cochrane–Orcutt method. The iterative algorithm has converged to the correct estimate of ϕ_1 (i.e. $\hat{\phi}_1 = -0.22838$) and refers to the global maximum of the log-likelihood function given by $LL(\hat{\phi}_1 = -0.22838) = 445.3720$. Notice also that the estimation results are reasonably robust to the choice of the initial estimates chosen for ϕ_1, so long as negative or small positive values are chosen. However, if the iterations are started from $\phi_1^{(0)} = 0.5$ or higher, the results in Table 5.2 will be obtained. The iterative process has now converged to $\hat{\phi}_1 = 0.81487$ with the maximized value for the log-likelihood function given by $LL(\hat{\phi}_1 = 0.81487) = 444.3055$, which is a local maximum. (Recall from Table 5.1*

[3] Note, however, that the saving function estimated by Deaton (1977) assumes that the inflation expectations Π_t^e are time invariant.

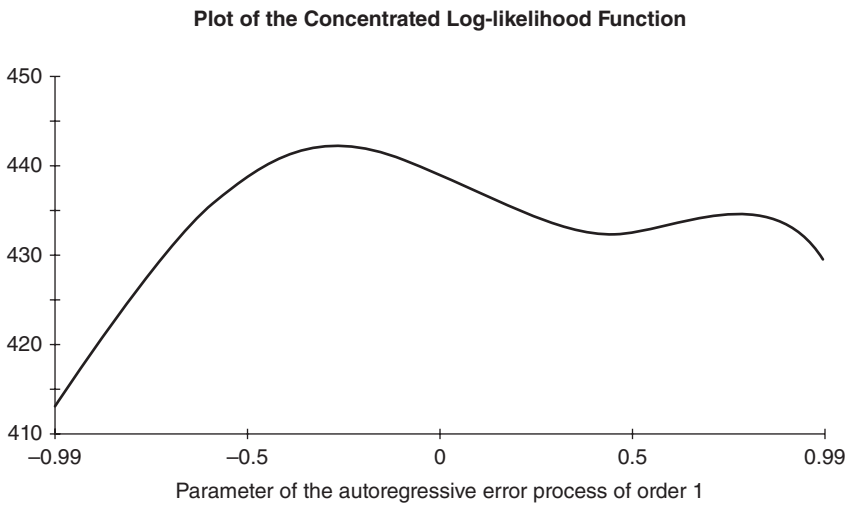


Figure 5.1 Log-likelihood profile for different values of ϕ_1.

that $LL(\hat{\phi}_1 = -0.22838) = 445.3720$.*) This example clearly shows the importance of experimenting with different initial values when estimating regression models (particularly when they contain lagged dependent variables) with serially correlated errors. (For further details see Lesson 11.6 in Pesaran and Pesaran (2009).)*

Table 5.1 Cochrane–Orcutt estimates of a UK saving function

Cochrane–Orcutt method $AR(1)$ converged after 3 iterations			
Dependent variable is S 140 observations used for estimation from 1960Q1 to 1994Q4			
Regressor	Coefficient	Standard Error	T-Ratio[Prob]
INPT	−.0032323	.0041204	−.78448 [.434]
S(−1)	.99250	.040347	24.5989 [.000]
DLY	.66156	.060082	11.0111 [.000]
DPIE	.31032	.093382	3.3231 [.001]
R-Squared	.76673	R-Bar-Squared	.75977
S.E. of Regression	.010004	F-Stat. F(4,134)	110.1102 [.000]
Mean of Dependent Variable	.096441	S.D. of Dependent Variable	.020696
Residual Sum of Squares	.013412	Equation Log-likelihood	445.3720
Akaike Info. Criterion	440.3720	Schwarz Bayesian Criterion	433.0179
DW-statistic	1.9615		
Parameters of the autoregressive error specification			
U= −.22838*U(-1)+E (−2.5135) [.013] t-ratio(s) based on asymptotic standard errors in brackets			

Table 5.2 An example in which the Cochrane–Orcutt method has converged to a local maximum

Cochrane–Orcutt method $AR(1)$ converged after 7 iterations			
Dependent variable is S 140 observations used for estimation from 1960Q1 to 1994Q4			
Regressor	Coefficient	Standard Error	T-Ratio[Prob]
INPT	.075353	.0098576	7.6441 [.000]
S(−1)	.19990	.084385	2.3689 [.019]
DLY	.55758	.052907	10.5388 [.000]
DPIE	.45522	.10271	4.4322 [.000]
R-Squared	.76312	R-Bar-Squared	.75605
S.E. of Regression	.010081	F-Stat. F (4,134)	107.9234 [.000]
Mean of Dependent Variable	.096441	S.D. of Dependent Variable	.020696
Residual Sum of Squares	.013619	Equation Log-likelihood	444.3055
Akaike Info. Criterion	439.3055	Schwarz Bayesian Criterion	431.9514
DW-statistic	2.2421		
Parameters of the autoregressive error specification			
U= .81487*U(−1)+E (16.1214) [.000] t-ratio(s) based on asymptotic standard errors in brackets			

5.7 *ML/AR* estimators by the Gauss–Newton method

This method provides an alternative numerical procedure for the maximization of the log-likelihood function (5.44). In cases where this log-likelihood function has a unique maximum, the Gauss–Newton and the *CO* iterative methods should converge to nearly identical results. But in general this need not be the case. However, the Gauss–Newton method is likely to perform better than the *CO* method when the regression equation contains lagged dependent variables.

The computations using the Gauss–Newton procedure are based on the following iterative relations (also see Section A.16 in Appendix A for further details):

$$\begin{pmatrix} \tilde{\boldsymbol{\beta}} \\ \tilde{\boldsymbol{\phi}} \end{pmatrix}_j = \begin{pmatrix} \tilde{\boldsymbol{\beta}} \\ \tilde{\boldsymbol{\phi}} \end{pmatrix}_{j-1} + \begin{pmatrix} \tilde{\mathbf{X}}_*'\tilde{\mathbf{X}}_* & \tilde{\mathbf{X}}_*'\mathbf{S} \\ \mathbf{S}'\tilde{\mathbf{X}}_* & \mathbf{S}'\mathbf{S} \end{pmatrix}_{j-1}^{-1} \begin{pmatrix} \tilde{\mathbf{X}}_*'\tilde{\boldsymbol{\epsilon}} \\ \mathbf{S}'\tilde{\boldsymbol{\epsilon}} \end{pmatrix}_{j-1}, \tag{5.56}$$

where the subscripts j and $j-1$ refer to the jth and the $(j-1)^{th}$ iterations; and $\tilde{\boldsymbol{\epsilon}} = (\tilde{\epsilon}_{p+1}, \tilde{\epsilon}_{p+2}, \ldots, \tilde{\epsilon}_T)'$, $\tilde{\mathbf{X}}_*$, and $\mathbf{S}$ have the same expressions as those already defined by (5.47), (5.52), and (5.53), respectively. The iterations can start with

$$\tilde{\boldsymbol{\beta}}_{(0)} = \tilde{\boldsymbol{\beta}}_{OLS} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y},$$
$$\tilde{\boldsymbol{\phi}}_{(0)} = 0,$$

and end them if either the number of iterations exceeds, say by 20, or if the condition (5.45) is satisfied.

On exit from the iterations Microfit computes a number of statistics including estimates of σ_ϵ^2, the variance matrices of $\tilde{\boldsymbol{\beta}}$ and $\tilde{\phi}$, R^2, $\bar{R}^2$, etc. using the results already set out in Sections 5.6.1 and 5.5.6.

5.7.1 *AR*(p) error process with zero restrictions

Microfit applies the Gauss–Newton iterative method to compute estimates of the regression equation when the $AR(p)$ error process (5.43) is subject to zero restrictions. Notice that in the restricted case the estimator of the standard error of the regression is given by

$$\hat{\sigma}_\epsilon^2 = \sum_{t=p+1}^{T} \tilde{\epsilon}_t^2/(T-p-r-k), \qquad T > p+r+k, \tag{5.57}$$

where r represents the number of non-zero parameters of the $AR(p)$ process.

Similarly, the appropriate formula for the F-statistic is now given by

$$F = \left(\frac{R^2}{1-R^2}\right)\left(\frac{T-p-k-r}{k+r-1}\right) \overset{a}{\sim} F(k+r-1, T-p-k-r). \tag{5.58}$$

The chi-squared version of this statistic can, as before, be computed by $TR^2/(1-R^2)$, which is asymptotically distributed (under the null hypothesis) as a chi-squared variate with $k+r-1$ degrees of freedom.

5.8 Testing for serial correlation

The Durbin–Watson statistic was the first formal procedure developed for testing for autocorrelation, using least squares residuals (see Durbin and Watson (1950, 1951)). The test statistic is

$$d = \frac{\sum_{t=2}^{T}\left(\hat{u}_t - \hat{u}_{t-1}\right)^2}{\sum_{t=1}^{T}\hat{u}_t^2}, \tag{5.59}$$

where $\hat{u}_t = y_t - \hat{\alpha} - \sum_{i=1}^{k}\hat{\beta}_i x_{it}$, are the *OLS* residuals. Note that the test is not valid if the regression equation does not include an intercept term. It is easy to see that, for large sample size,

$$d \approx 2\left(1-\hat{\phi}\right),$$

where $\hat{\phi} = \sum_{t=2}^{T}\hat{u}_t\hat{u}_{t-1} \Big/ \sum_{t=2}^{T}u_{t-1}^2$. Further, for values of $\hat{\phi}$ near unity, we have $d \approx 0$.

The critical values depend only on T and on k and are available in Appendix C of Microfit 5 (Pesaran and Pesaran (2009)). From the tables provided we get upper (d_u) and lower bound (d_L) values. We have:

- If $d \leq d_L$, then we reject $H_0 : \phi = 0$.
- If $d > d_u$, then we do not reject $H_0 : \phi = 0$.
- If $d_L \leq d \leq d_u$, then the test is inconclusive.
- If $d > 2$ then calculate $d^* = 4 - d$ and apply the above testing procedure to d^*.

One limitation of the Durbin–Watson statistic is that it only deals with the first-order auto regressive error process. Further, it is biased towards acceptance when the regression model contains lagged dependent variables. The h-statistic can be used to deal with the problem of lagged dependent variable. This statistic is defined by

$$h\text{-statistic} = \left(1 - \frac{DW}{2}\right)\sqrt{\frac{T}{1 - T\hat{V}\left(\hat{\lambda}\right)}},$$

where $\hat{V}\left(\hat{\lambda}\right)$ is the estimator of the variance of the *OLS* estimator of λ in the regression

$$y_t = \alpha + \beta x_t + \lambda y_{t-1} + u_t.$$

5.8.1 Lagrange multiplier test of residual serial correlation

The Lagrange multiplier (LM) test is valid irrespective of whether there are lagged dependent variables among the regressors. It is also applicable for testing higher-order AR-error processes. The LM test of residual serial correlation consists of testing the null hypothesis $H_0 : \delta_1 = \delta_2 = \ldots = \delta_p = 0$, against the alternative hypothesis $H_1 : \delta_1 \neq 0$, and/or $\delta_2 \neq 0, \ldots, \delta_p \neq 0$, in the following regression of $\hat{u}_t$ (the *OLS* residuals) on the regressors and the lagged values of the *OLS* residuals, namely

$$\hat{u}_t = \alpha_0 + \sum_{j=1}^{k} \alpha_j x_{tj} + \gamma y_{t-1} + \delta_1 \hat{u}_{t-1} + \delta_2 \hat{u}_{t-2} + \cdots + \delta_p \hat{u}_{t-p} + \text{error}. \tag{5.60}$$

The LM test is also applicable if the regression equation contains higher-order lags of y_t (i.e., when (5.60) contains $y_{t-1}, y_{t-2}, \ldots, y_{t-q}$).

Two versions of this test have been used in the literature; an LM, version and an F version. In the general case the LM version is given by (see Godfrey (1978a, 1978b))

$$\chi^2_{SC}(p) = T\left(\frac{\hat{\boldsymbol{u}}'_{OLS}\mathbf{W}(\mathbf{W}'\mathbf{M}_x\mathbf{W})^{-1}\mathbf{W}'\hat{\boldsymbol{u}}_{OLS}}{\hat{\boldsymbol{u}}'_{OLS}\hat{\boldsymbol{u}}_{OLS}}\right) \overset{a}{\sim} \chi^2_p, \tag{5.61}$$

where $\hat{\boldsymbol{u}}_{OLS}$ is the vector of *OLS* residuals, $\mathbf{X}$ is the observation matrix, possibly containing lagged values of the dependent variable,

$$\mathbf{M}_x = \mathbf{I}_T - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}',$$
$$\hat{\mathbf{u}}_{OLS} = \mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}_{OLS} = (\hat{u}_1, \hat{u}_2, \ldots, \hat{u}_T)',$$

$$\mathbf{W} = \begin{pmatrix} 0 & 0 & \dots & 0 \\ \hat{u}_1 & 0 & \dots & 0 \\ \hat{u}_2 & \hat{u}_1 & \dots & 0 \\ & \hat{u}_2 & \dots & \\ \vdots & \vdots & \vdots & \hat{u}_{T-p-1} \\ \hat{u}_{T-1} & \hat{u}_{T-2} & \dots & \hat{u}_{T-p} \end{pmatrix}, \tag{5.62}$$

and p is the order of the error process, and χ^2_p stands for a chi-squared variate with p degrees of freedom.

The F-version of (5.61) is given by[4]

$$F_{SC}(p) = \left(\frac{T-k-p}{p}\right)\left(\frac{\chi^2_{SC}(p)}{T-\chi^2_{SC}(p)}\right) \overset{a}{\sim} F_{p,T-k-p}, \tag{5.63}$$

where $\chi^2_{SC}(p)$ is given by (5.61). The above statistic can also be computed as the F-statistic for the (joint) test of zero restrictions on the coefficients of $\mathbf{W}$ in the auxiliary regression

$$\mathbf{y} = \mathbf{X}\boldsymbol{\alpha} + \mathbf{W}\boldsymbol{\delta} + \mathbf{v}.$$

The two versions of the test of residual serial correlation, namely $\chi^2_{SC}(p)$ and $F_{SC}(p)$, are asymptotically equivalent.

5.9 Newey–West robust variance estimator

When errors are heteroskedastic and/or autocorrelated, provided that the other standard regularity conditions of the regression model are satisfied, particularly the orthogonality condition $Plim_{T\to\infty}\left(T^{-1}\mathbf{X}'\mathbf{u}\right) = \mathbf{0}$, then the *OLS* estimator of the regression parameters remains consistent and asymptotically normal. However, the standard statistical tests such as the *t*- or the *Wald* tests are no longer valid and inference based on them can be misleading. One possible way of dealing with this problem is to base the computation of the test statistics on the Newey and West (1987) robust standard errors. The Newey–West (*NW*) heteroskedasticity and autocorrelation consistent (*HAC*) variance matrix is a direct generalization of White's estimators described in Section 4.2. The *NW* variance matrix is computed according to the following formula:

$$\hat{V}_{NW}(\hat{\boldsymbol{\beta}}) = \frac{1}{T}\mathbf{Q}_T^{-1}\hat{\mathbf{S}}_T\mathbf{Q}_T^{-1}, \tag{5.64}$$

where

$$\begin{aligned} \mathbf{Q}_T &= \frac{1}{T}\sum_{t=1}^{T}\mathbf{x}_t\mathbf{x}_t', \\ \hat{\mathbf{S}}_T &= \hat{\boldsymbol{\Omega}}_0 + \sum_{j=1}^{m} w(j,m)(\hat{\boldsymbol{\Omega}}_j + \hat{\boldsymbol{\Omega}}_j') \end{aligned} \tag{5.65}$$

[4] For a derivation of the relationship between the *LM*-version, and the *F*-version of the test statistics see, for example, Pesaran (1981a, pp. 78–80).

in which

$$\hat{\boldsymbol{\Omega}}_j = \sum_{t=j+1}^{T} \hat{u}_t \hat{u}_{t-j} \mathbf{x}_t \mathbf{x}'_{t-j},$$

$$\hat{u}_t = y_t - \hat{\boldsymbol{\beta}}' \mathbf{x}_t,$$

and $w(j, m)$ is the kernel or lag window, and m is the 'bandwidth' or the 'window size'. White's heteroskedasticity-consistent estimators considered in Section 4.2 can be computed using the Newey-West estimator by setting the window size, m, equal to zero. In applied work the following are popular choices for the kernel:

- *Uniform (or rectangular) kernel*

$$w(j, m) = 1, \qquad \text{for} \quad j = 1, 2, \ldots, m,$$

- *Bartlett kernel*

$$w(j, m) = 1 - \frac{j}{m+1}, \qquad j = 1, 2, \ldots, m,$$

- *Parzen kernel*

$$\begin{aligned} w(j, m) &= 1 - 6\left(\frac{j}{m+1}\right)^2 + 6\left(\frac{j}{m+1}\right)^3, && 1 \leq j \leq \frac{m+1}{2}, \\ &= 2\left(1 - \frac{j}{m+1}\right)^2 && \frac{m+1}{2} < j \leq m. \end{aligned}$$

In their paper, Newey and West (1987) adopted the Bartlett kernel. The uniform kernel is appropriate when estimating a regression model with moving average errors of known order. This type of model arises in testing the market efficiency hypothesis where the forecast horizon exceeds the sampling interval (see, e.g., Pesaran (1987c, Section 7.6)). In other cases, a Parzen lag window may be preferable. Note that the positive semi-definiteness of the Newey-West variance matrix is only ensured in the case of the Bartlett and Parzen kernels. The choice of the uniform kernel can result in a negative-definite variance matrix, especially if a large value for m is chosen relative to the number of available observations, T. See the discussion in Andrews (1991), and Andrews and Monahan (1992).

The choice of the window size or bandwidth, m, is even more critical for the properties of the *NW* estimator. The maximum lag m must be determined in advance to be sufficiently large so that the correlation between $\mathbf{x}_t u_t$ and $\mathbf{x}_{t-j} u_{t-j}$ for $j \geq m$ is essentially zero. Current practice is to use the smallest integer greater than or equal to $T^{1/3}$, or $T^{1/4}$. Automatic bandwidth selection procedures that asymptotically minimize estimation errors have also been proposed in the literature. See, for example, Newey and West (1994) and Sun, Phillips, and Jin (2008).

One major problem in the use of tests based on the *NW* estimator is the tendency to over-reject the null hypothesis in finite samples. This problem has been documented in several

studies; see, among others, Andrews and Monahan (1992) and den Haan and Levin (1997). To deal with this issue, as discussed below, a number of authors have proposed a stochastic transformation of *OLS* estimates so that the asymptotic distribution of the transformed estimates does not depend on nuisance parameters.

5.10 Robust hypothesis testing in models with serially correlated/heteroskedastic errors

Kiefer, Vogelsang, and Bunzel (2000) proposed an alternative approach which uses a data transformation to deal with the error serial correlation problem and which does not explicitly require a *NW* type variance matrix estimator. The procedure consists of applying a nonsingular stochastic transformation to the *OLS* estimates, so that the asymptotic distribution of the transformed estimates does not depend on nuisance parameters. Let

$$\hat{\mathbf{C}}_t = \frac{1}{T^2}\sum_{t=1}^{T}\hat{\mathbf{s}}_t\hat{\mathbf{s}}_t',$$

where

$$\hat{\mathbf{s}}_t = \sum_{j=1}^{t}\mathbf{x}_t\hat{u}_t,$$

and $\hat{u}_t = y_t - \hat{\boldsymbol{\beta}}'\mathbf{x}_t$. Now define

$$\hat{\boldsymbol{M}} = \left(\frac{1}{T}\sum_{j=1}^{T}\mathbf{x}_t\mathbf{x}_t'\right)^{-1}\hat{\mathbf{C}}_t^{1/2},$$

where $\hat{\mathbf{C}}_t^{1/2}$ represents a lower triangular Cholesky factor of $\hat{\mathbf{C}}_t$. Kiefer, Vogelsang, and Bunzel (2000) establish that as $T \to \infty$,

$$\hat{\boldsymbol{M}}^{-1}\sqrt{T}\left(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}\right) \xrightarrow{d} \mathbf{Z}_k^{-1}\mathbf{b}_k(1), \tag{5.66}$$

where $\mathbf{Z}_k$ is a lower triangular Cholesky factor of $\mathbf{P}_k$ defined by

$$\mathbf{P}_k = \int \left[\mathbf{b}_k(r) - r\mathbf{b}_k(1)\right]\left[\mathbf{b}_k(r) - r\mathbf{b}_k(1)\right]' dr, \tag{5.67}$$

and $\mathbf{b}_k(r)$ denotes a k-dimensional vector of independent standard Wiener processes.[5] This transformation results in a limiting distribution that does not depend on nuisance parameters. However, the distribution of $\mathbf{Z}_k^{-1}\mathbf{b}_k(1)$ is non-standard, although it only depends on k, the number of regression coefficients being estimated. Critical values have been computed by simulation by Kiefer, Vogelsang, and Bunzel (2000). The main advantage of this approach compared

[5] A brief account of Wiener processes is provided in Section B.13 Appendix B.

with standard approaches is that estimates of the variance-covariance matrix are not explicitly required to construct the tests. Further, Kiefer, Vogelsang, and Bunzel (2000) show that tests constructed using their procedure can have better finite sample size properties than tests based on consistent *NW* estimates.

Kiefer and Vogelsang (2002) showed that the above approach is exactly equivalent to using *NW* standard errors with Bartlett kernel, and without truncation (namely, setting $m = T$ in (5.65)). This result suggests that valid tests can be constructed using kernel based estimators with bandwidth $m = T$.

Kiefer and Vogelsang (2005) studied the limiting distribution of robust tests based on the *NW* estimators setting $m = b \cdot T$, where $b \in (0, 1]$ is a constant, labelling the asymptotics obtained under this framework as 'fixed-b asymptotics'. The authors showed that the limiting distribution of the F- and t-statistics based on such *NW* variance estimator are non-standard, and that they depend on the choice of the kernel and on b. Kiefer and Vogelsang (2005) have also analysed the properties of these test statistics *via* a simulation study. Their results indicate a trade-off between size distortions and power with regard to choice of the bandwidth. Smaller bandwidths lead to tests with higher power but at the cost of greater size distortions, whereas larger bandwidths lead to tests with smaller size distortions but lower power. They also found that, among a group of common choice kernels, the Bartlett kernel leads to tests with highest power in their fixed-b framework.

Phillips, Sun, and Jin (2006) suggested a new class of kernel functions obtained by exponentiating a 'mother' kernel (such as the Bartlett or Parzen lag window), but without using lag truncation. When the exponent parameter is not too large, the absence of lag truncation influences the variability of the estimate because of the presence of autocovariances at long lags. Such effects can have the advantage of better reflecting finite sample behavior in test statistics that employ *NW* estimates, and leading to some improvement in test size, as also reported in a simulation study by Phillips, Sun, and Jin (2006).

While this approach works well, Kapetanios and Psaradakis (2007) note that it does not exploit information on the structure of the dependence in the regression errors. However, such information may be used to improve the properties of robust inference procedures. Hence, the authors suggest to employ a feasible *GLS* estimator where the stochastic process generating disturbances is approximated by an autoregessive model with an order that grows at a slower rate than the sample size (see also Amemiya (1973) on such approximation). Specifically, let $\hat{u}_t = y_t - \hat{\boldsymbol{\beta}}' \mathbf{x}_t$, where $\hat{\boldsymbol{\beta}}$ is an initial consistent estimator of $\boldsymbol{\beta}$. For some positive integer p chosen as a function of T so that $p \to \infty$ and $p/T \to 0$ as $T \to \infty$, let $\hat{\boldsymbol{\phi}}_{pp} = \left(\hat{\phi}_{p,1}, \hat{\phi}_{p,2}, \ldots, \hat{\phi}_{p,p}\right)'$ be the pth order *OLS* estimator of the autoregressive coefficients for $\{\hat{u}_t\}$, obtained as the solution to the minimization of

$$\min_{\phi_{p,1},\phi_{p,2},\ldots,\phi_{p,p} \in \mathbb{R}^p} (T-p)^{-1} \sum_{t=p+1}^{T} \left(\hat{u}_t - \phi_{p,1}\hat{u}_{t-1} - \phi_{p,2}\hat{u}_{t-2} \ldots - \phi_{p,p}\hat{u}_{t-p}\right)^2.$$

A feasible *GLS* estimator of $\boldsymbol{\beta}$ may then be obtained as

$$\hat{\boldsymbol{\beta}} = \left(\mathbf{X}'\hat{\boldsymbol{\Phi}}'\hat{\boldsymbol{\Phi}}\mathbf{X}\right)^{-1} \mathbf{X}'\hat{\boldsymbol{\Phi}}'\hat{\boldsymbol{\Phi}}\mathbf{y}, \tag{5.68}$$

where $\hat{\mathbf{\Phi}}$ is the $(T-p)\times T$ matrix defined as

$$\hat{\mathbf{\Phi}}=\begin{pmatrix} -\hat{\phi}_{p,p} & -\hat{\phi}_{p,p-1} & -\hat{\phi}_{p,p-2} & \dots & -\hat{\phi}_{p,1} & 1 & \dots & 0 & 0 \\ 0 & -\hat{\phi}_{p,p} & -\hat{\phi}_{p,p-1} & \dots & -\hat{\phi}_{p,2} & -\hat{\phi}_{p,1} & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & -\hat{\phi}_{p,p} & -\hat{\phi}_{p,p-1} & \dots & -\hat{\phi}_{p,1} & 1 \end{pmatrix}.$$

Note that (5.68) can be obtained by applying *OLS* to the regression of $\hat{y}_t^*=\left(1-\sum_{j=1}^{p}\hat{\phi}_{p,j}L^j\right)y_t$ on $\hat{\boldsymbol{x}}_t^*=\left(1-\sum_{j=1}^{p}\hat{\phi}_{p,j}L^j\right)\mathbf{x}_t$.

One drawback of *NW*-type estimators is that they cannot be employed to obtain valid tests of the significance of *OLS* estimates when there are lagged dependent variables in the regressors, and errors are serially correlated. The problem is that these procedures require the *OLS* estimator to be consistent. However, as formally proved in Section 14.6, in general *OLS* estimators will be inconsistent when errors are autocorrelated and there are lagged values of the dependent variable among the regressors. One possible way of dealing with this problem would be to use an instrumental variables (IV) approach for estimating consistently the parameters of the regression model, and then obtain *IV*-based robust tests. As an alternative, Godfrey (2011) suggested a joint test for misspecification and autocorrelation using the *J*-test approach by Davidson and MacKinnon (1981) and introduced in Chapter 11 (see Section 11.6). Suppose that the validity of M_1 is to be tested using information about M_2, and that the regressors x_t and z_t in models (11.22)–(11.23) both contain at least one lagged value of y_t. Also suppose that the autoregressive or moving average model of order m is used as the alternative to the assumption of independent errors. The author suggested a heteroskedasticity-robust joint test of the $(1+m)$ restrictions $\lambda=\phi_1=\ldots=\phi_m=0$, in the 'artificial' *OLS* regression

$$\mathbf{y}_t=\boldsymbol{\beta}_1'\mathbf{x}_t+\lambda(\hat{\boldsymbol{\beta}}_2'\mathbf{z}_t)+\phi_1\hat{u}_{1,t-1}+\ldots+\phi_m\hat{u}_{1,t-m}+\epsilon_t, \tag{5.69}$$

where $\hat{u}_{1,t-j}$ is a lagged value of the *OLS* residual from estimation of (11.22) when $(t-j)>0$ and is set equal to zero when $(t-j)\leq 0$. More specifically, Davidson and MacKinnon (1981) proposed using the Wald approach, combined with heteroskedasticity-consistent variance (HCV) estimator:

$$\tau_J=\left(\hat{\lambda},\hat{\phi}_1,\ldots,\hat{\phi}_m\right)'\left(\mathbf{R}_1\hat{\mathbf{C}}_J\mathbf{R}_1'\right)^{-1}\left(\hat{\lambda},\hat{\phi}_1,\ldots,\hat{\phi}_m\right), \tag{5.70}$$

where $\mathbf{R}_1=(\mathbf{0},\mathbf{I}_{m+1})$ is a $(1+m)\times k_1$ matrix, $\hat{\lambda},\hat{\phi}_1,\ldots,\hat{\phi}_m$ are *OLS* estimates of $\lambda,\phi_1,\ldots,\phi_m$ in (5.69), and

$$\hat{\mathbf{C}}_J=\left(\sum_{t=1}^{T}\hat{\mathbf{r}}_t\hat{\mathbf{r}}_t'\right)\left(\sum_{t=1}^{T}\hat{u}_{1t}^2\hat{\mathbf{r}}_t\hat{\mathbf{r}}_t'\right)\left(\sum_{t=1}^{T}\hat{\mathbf{r}}_t\hat{\mathbf{r}}_t'\right), \tag{5.71}$$

with $\hat{\mathbf{r}}_t = \left(\mathbf{x}_t', \hat{\boldsymbol{\beta}}_2' \mathbf{z}_t, \hat{u}_{1,t-1}, \ldots, \hat{u}_{1,t-m}\right)'$. Note that, under the null hypothesis, the *OLS* estimators of the artificial alternative regression are consistent and asymptotically normal. It follows that $\tau_J \sim \chi^2_{m+1}$. Recent work has indicated that, when several restrictions are under test, the use of asymptotic critical values with *HCV*-based test statistics produces estimates of null hypothesis rejection probabilities that are too small (see Godfrey and Orme (2004)). To overcome this problem, Davidson and MacKinnon (1981) suggested a bootstrap implementation of the above test, using the wild bootstrap method.

5.11 Further reading

Further reading on multiple regression analysis under autocorrelated disturbances can be found in Wooldridge (2000, ch. 14), and Greene (2002, chs. 10 and 12). A brief review of the literature on robust variance estimators and robust testing can be found in the introduction of the paper by Kiefer and Vogelsang (2005).

5.12 Exercises

1. Consider the generalized linear regression model (5.1), under assumptions (5.2) and (5.3), and suppose that $\boldsymbol{\Sigma}$ is known. Then:

   (a) What is the variance matrix of the *OLS* residual vector $\hat{\mathbf{u}}_{OLS} = \mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}_{OLS}$?
   (b) What is the variance matrix of the *GLS* residual vector $\hat{\mathbf{u}}_{GLS} = \mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}_{GLS}$?
   (c) What is the covariance matrix of *OLS* and *GLS* residual vectors?

2. Consider

$$y_t = \boldsymbol{\beta}' \mathbf{x}_t + u_t,$$

   where

$$u_t = \varepsilon_t + \theta \varepsilon_{t-1},$$

   and $\varepsilon_t \sim IID(0, \sigma^2)$, for $t = 1, 2, \ldots, T$.

   (a) Derive the covariance matrix of $\mathbf{u} = (u_1, u_2, \ldots, u_T)'$.
   (b) Prove that under this model $\hat{\boldsymbol{\beta}}_{OLS}$ is consistent for $\boldsymbol{\beta}$.
   (c) Find an expression for the *GLS* estimator of $\boldsymbol{\beta}$, and its covariance matrix.

3. Consider the simple regression equation

$$y_t = \beta x_{t-1} + u_t, \tag{5.72}$$

where u_t and x_t follow the $AR(1)$ processes

$$u_t = \rho u_{t-1} + \varepsilon_t,$$
$$x_t = \phi x_{t-1} + v_t,$$

with $|\phi\rho| < 1$, and

$$\begin{pmatrix} \varepsilon_t \\ v_t \end{pmatrix} \sim IID\left[\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \sigma_{\varepsilon\varepsilon} & \sigma_{\varepsilon v} \\ \sigma_{\varepsilon v} & \sigma_{vv} \end{pmatrix}\right].$$

(a) Show that (5.72) can be written as

$$y_t = \lambda y_{t-1} + \psi_0 x_{t-1} + \psi_1 x_{t-2} + \varepsilon_t, \tag{5.73}$$

and derive expressions for λ, ψ_0 and ψ_1 in terms of β, ρ and ϕ. In particular, show that $\theta = (\psi_0 + \psi_1)/(1 - \lambda)$ is equal to β. Hence, or otherwise, suggest how to test model (5.73) against model (5.72).

(b) How do you think β should be estimated?

4. Continue with the regression model set out in Question 3 above.

(a) Show that the *OLS* estimator of β is biased, and derive an expression for its bias. Under what conditions the *OLS* estimator of β is unbiased?

(b) Show that $\hat{\beta}_{OLS}\, x_{t-1}$ is an unbiased estimator of $E\left(y_t \,|x_{t-1}\right)$. How do you estimate $E\left(y_t \,|x_{t-1}, x_{t-2}, y_{t-1}\right)$?

6 Introduction to Dynamic Economic Modelling

6.1 Introduction

Dynamic economic models typically arise as a characterization of the path of the economy around its long-run equilibrium (steady state), and involve modelling expectations, learning and adjustment costs. There exist a variety of dynamic specifications used in applied time series econometrics. This chapter reviews a number of single-equation specifications suggested by econometric literature to represent dynamics in regression models. It provides a preliminary introduction to distributed lag models, autoregressive distributed lag models, partial adjustment models, error-correction models, and adaptive and rational expectations models. More general multi-equation dynamic systems will be considered in the second part of the book, where vector autoregressive models with and without weakly exogenous variables and multivariate rational expectations are discussed.

6.2 Distributed lag models

Most linear distributed lag models discussed in the literature belong to the class of rational distributed lag models. Early examples of this type of models include the polynomial and geometric distributed lag models. The rational distributed lag model can also be written in the form of the autoregressive distributed lag $(ARDL)$ models.

Polynomial distributed lag models:
A general specification for the polynomial distributed lag (DL) model is

$$\begin{aligned} y_t &= \alpha + \beta_0 x_t + \beta_1 x_{t-1} + \cdots + \beta_q x_{t-q} + u_t \\ &= \alpha + \beta(L)\, x_t + u_t, \end{aligned} \tag{6.1}$$

where u_t is the unobservable part and is assumed to be serially uncorrelated, and

$$\beta(L) = \beta_0 + \beta_1 L + \ldots + \beta_q L^q,$$

is a lag polynomial and L is the lag operator, defined by

$$L^i x_t = x_{t-i}, \quad i = 0, 1, 2, \ldots. \tag{6.2}$$

The lag coefficients are often restricted to lie on a polynomial of order $r \leq q$. In the case where $r = q$ the distributed lag model is unrestricted.

Rational distributed lag models:
The rational distributed lag model is given by

$$y_t = \alpha + \frac{\beta\,(L)}{\lambda\,(L)} x_t + v_t, \tag{6.3}$$

where $\beta\,(L)$ as defined above and

$$\lambda\,(L) = \lambda_0 + \lambda_1 L + \ldots + \lambda_p L^p.$$

A comprehensive early treatment of rational distributed lag models can be found in Dhrymes (1971). See also Jorgenson (1966).

Autoregressive-distributed lag models of order (p, q):
The $ARDL(p, q)$ is

$$y_t + \lambda_1 y_{t-1} + \lambda_2 y_{t-2} + \cdots + \lambda_p y_{t-p} = \alpha + \beta_0 x_t + \beta_1 x_{t-1} + \cdots + \beta_q x_{t-q} + u_t, \tag{6.4}$$

or

$$\lambda(L) y_t = \alpha + \beta(L) x_t + u_t. \tag{6.5}$$

Note that the rational distributed lag model defined by (6.3) can also be written in the form of an $ARDL(p, q)$ model with moving average errors, namely

$$\lambda(L) y_t = \alpha\lambda(1) + \beta(L) x_t + \lambda(L) v_t,$$

which is the same as (6.4) except that the error term is now given by

$$u_t = \lambda(L) v_t,$$

which is a moving average error model.

Recent developments in time series analysis focus on the $ARDL(p, q)$ specification for two reasons: it is analytically much simpler to work with, as compared with the rational distributed lag models. Secondly, by selecting p and q to be sufficiently large one can provide a reasonable approximation to the rational distributed lag specification if required.

Deterministic trends, or seasonal dummies, can be easily incorporated in the $ARDL$ model. For example, we could have

$$y_t + \lambda_1 y_{t-1} + \lambda_2 y_{t-2} + \cdots + \lambda_p y_{t-p} = a + bt + \beta_0 x_t + \beta_1 x_{t-1} + \cdots + \beta_q x_{t-q} + u_t. \quad (6.6)$$

The *ARDL* model is said to be stable if all the roots of the pth order polynomial equation

$$\lambda(z) = 1 + \lambda_1 z + \lambda_2 z^2 + \ldots + \lambda_p z^p = 0, \quad (6.7)$$

lie outside the unit circle, namely if $|z| > 1$. The process is said to have a unit root if $\lambda(1) = 0$. The explosive case where one or more roots of $\lambda(z) = 0$ lie inside the unit circle is not of practical relevance for the analysis of economic time series and will not be considered.

The simple model (6.4) with one regressor can be extended to the case of k regressors, each with a specific number of lags. Specifically, consider the following $ARDL(p, q_1, q_2, \ldots, q_k)$ model

$$\lambda(L, p) y_t = \sum_{j=1}^{k} \beta_j(L, q_j) x_{tj} + u_t,$$

where

$$\lambda(L, p) = 1 + \lambda_1 L + \lambda_2 L^2 + \ldots + \lambda_p L^p.$$
$$\beta_j(L, q_j) = \beta_{j0} + \beta_{j1} L + \ldots + \beta_{jq_j} L^{q_j}, \quad j = 1, 2, \ldots, k.$$

See Hendry, Pagan, and Sargan (1984) for a comprehensive early review of *ARDL* models.

6.2.1 Estimation of *ARDL* models

The $ARDL(p, q)$ model (6.4) can be estimated by applying the *OLS* method. Specifically, let $\boldsymbol{\theta} = (-\lambda_1, -\lambda_2, \ldots, -\lambda_p, \alpha, \beta_0, \beta_1, \beta_2, \ldots, \beta_q)'$ and $\mathbf{z}_t = (y_{t-1}, y_{t-2}, \ldots, y_{t-p}, 1, x_t, x_{t-1}, x_{t-2}, \ldots, x_{t-q})'$, for $t = 1, 2, \ldots, T$. The *OLS* estimator of $\boldsymbol{\theta}$ is

$$\hat{\boldsymbol{\theta}} = \left(\sum_{t=1}^{T} \mathbf{z}_t \mathbf{z}_t' \right)^{-1} \sum_{t=1}^{T} \mathbf{z}_t y_t. \quad (6.8)$$

Sufficient conditions for $\hat{\boldsymbol{\theta}}$ to be a consistent estimator of $\boldsymbol{\theta}$ are:

1. $T^{-1} \sum_{t=1}^{T} \mathbf{z}_t \mathbf{z}_t'$ converges to a non-stochastic positive definite matrix as $T \to \infty$.
2. $T^{-1} \sum_{t=1}^{T} \mathbf{z}_t u_t$ converges to a zero vector as $T \to \infty$.

The first condition is satisfied if $\mathbf{z}_t$ follows a covariance stationary process. This will be the case if all the roots of the pth order polynomial equation $\lambda(z)$ lie outside the unit circle, and x_t and u_t are covariance stationary processes with absolute summable autocovariances, as defined by

$$x_t = \sum_{i=0}^{\infty} a_i v_{t-i},$$

and

$$u_t = \sum_{i=0}^{\infty} b_i \varepsilon_{t-i},$$

where $\sum_{i=0}^{\infty} |a_i| < K < \infty$, $\sum_{i=0}^{\infty} |b_i| < K < \infty$, and $\{v_t\}$, and $\{\varepsilon_t\}$ are standard white noise processes. However, as we will show in Section 14.6, condition 1 alone does not guarantee consistency of *OLS* estimators and must be accompanied by condition 2.

To select the lag orders p and q, one possibility is to estimate model (6.4) by *OLS* for all possible values of $p = 0, 1, 2, \ldots, m$, $q = 0, 1, 2, \ldots, m$, where m is the maximum lag, and $t = m+1, m+2, \ldots, T$. Hence, one of the $(m+1)^2$ estimated models can be selected by using one of the four model selection criteria described in Chapters 2 and 11, namely the $\bar{R}^2$ criterion, the Akaike information criterion, the Schwarz Bayesian criterion, and the Hannan and Quinn criterion. See Section 6.4 for the derivation of the error-correction representation of *ARDL* models, and Section 6.5 for the computation of the long-run coefficients for the response of y_t to a unit change in x_t.

6.3 Partial adjustment model

The *ARDL* models arise naturally from decision processes where agents try to adjust to a desired or equilibrium value of a decision variable in the presence of adjustment costs. Let y_t^* be the desired value of a decision variable y_t, and assume that the desired value is determined by

$$y_t^* = \alpha + \beta x_t + u_t. \tag{6.9}$$

Further suppose that y_t adjusts to its desired level according to the following first-order partial adjustment equation

$$y_t - y_{t-1} = \lambda \left(y_t^* - y_{t-1}\right), \tag{6.10}$$

where λ is the adjustment coefficient. No adjustment will take place if $\lambda = 0$, and adjustment will be instantaneous if $\lambda = 1$. Using (6.9) to substitute for y_t^* we have

$$y_t = \lambda\alpha + (1 - \lambda) y_{t-1} + \lambda\beta x_t + \lambda u_t, \tag{6.11}$$

which is an $ARDL(1,0)$ specification.

The partial adjustment model can also be derived as a solution to a simple optimization problem. Consider the following minimization problem, where y_t^* is the desired or target value of y_t:

$$\underset{y_t}{Min} \left[\left(y_t - y_t^*\right)^2 + \theta \left(y_t - y_{t-1}\right)^2\right], \qquad \theta > 0. \tag{6.12}$$

In this formulation, θ measures the cost of adjustment relative to the cost of being out of equilibrium. The first-order condition for this optimization problem is

$$\left(y_t - y_t^*\right) + \theta \left(y_t - y_{t-1}\right) = 0, \tag{6.13}$$

which after some algebraic manipulation can also be written as:

$$y_t - y_{t-1} = \frac{1}{1+\theta}\left(y_t^* - y_{t-1}\right), \tag{6.14}$$

comparing this result with (6.10) shows that $\lambda = \frac{1}{1+\theta} > 0$.

6.4 Error-correction models

The error-correction model (*ECM*) was originally introduced into the economic literature by Phillips (1954, 1957). Sargan (1964) provides the first important empirical application of *ECM*, which was later extended by Davidson et al. (1978) to the estimation of the UK consumption function.[1] The *ECM* is a representation of the dynamics of a linear time series in terms of its level and a lag polynomial in the differences. Consider the autoregressive-distributed lag of order (1, 1) namely *ARDL*(1, 1):

$$y_t = \alpha + \beta_0 x_t + \beta_1 x_{t-1} + \lambda y_{t-1} + u_t. \tag{6.15}$$

It is easily seen that this model can be written equivalently as

$$\Delta y_t = \alpha + \beta_0 \Delta x_t - (1-\lambda)\left(y_{t-1} - \theta x_{t-1}\right) + u_t, \tag{6.16}$$

where $\theta = \left(\beta_0 + \beta_1\right) / (1-\lambda)$ is the slope coefficient in the long-run relationship between y_t and x_t (see (6.24) below). Relation (6.16) is known as the error-correction representation of the *ARDL*(1, 1) model (6.15). The term Δx_t is referred to as the *derivative effect*, and $\left(y_{t-1} - \theta x_{t-1}\right)$ as the *error-correction term*.

More general error-correction models can also be obtained from *ARDL*(p, q) specifications, given by (6.4). In general, any lagged variable, y_{t-j}, for $j = 1, 2, \ldots$, can be written as

$$y_{t-j} = y_{t-1} - \left(\Delta y_{t-1} + \Delta y_{t-2} + \ldots + \Delta y_{t-j+1}\right).$$

Similarly

$$x_{t-j} = x_{t-1} - \left(\Delta x_{t-1} + \Delta x_{t-2} + \ldots + \Delta x_{t-j+1}\right).$$

Also

$$\begin{aligned} y_t &= \Delta y_t + y_{t-1}, \\ x_t &= \Delta x_t + x_{t-1}. \end{aligned}$$

Substituting these in (6.4), and after some algebra we have

[1] See Alogoskoufis and Smith (1991) for an historical survey of the evolution of *ECM*s.

$$\Delta y_t = a - \lambda(1) y_{t-1} + \beta(1) x_{t-1} + \sum_{i=1}^{p-1} \lambda_i^* \Delta y_{t-i} + \sum_{i=0}^{q-1} \beta_i^* \Delta x_{t-i} + \varepsilon_t, \tag{6.17}$$

where

$$\lambda(1) = 1 + \lambda_1 + \ldots + \lambda_p,\ \beta(1) = \beta_0 + \beta_1 + \ldots + \beta_q,$$

$$\lambda_i^* = \sum_{j=i+1}^{p} \lambda_j,\ \beta_i^* = -\sum_{j=i+1}^{q} \beta_j.$$

The stability of the *ARDL* model, or the equilibrating properties of the *ECM* representation in the present simple (single equation) case critically depend on the roots of the characteristic equation $\lambda(z) = 0$, defined by (6.7). If the underlying *ARDL* model is stable, $\lambda(1) \neq 0$, the process $\{y_t\}$ is said to be 'mean reverting'. The equilibrium (or steady state) value of y_t depends on the nature of the $\{x_t\}$ process. If x_t is a stationary process with a constant mean, μ_x and a constant variance, σ_x^2, then in the limit y_t will also tend to a constant mean given by

$$\mu_y = \frac{\beta(1)\mu_x + a}{\lambda(1)}.$$

In the case where x_t is trended, or unit root non-stationary (also known as first difference stationary) then y_t is given by

$$y_t = \alpha^* + \left[\frac{\beta(1)}{\lambda(1)}\right] x_t + v_t,$$

where a^* is a constant term and v_t is a stationary random variable. In this case we have

$$\mu_{\Delta y} = \frac{\beta(1)\mu_{\Delta x}}{\lambda(1)},$$

where $\mu_{\Delta y}$, and $\mu_{\Delta x}$ are the means of Δy_t and Δx_t , which are assumed to be stationary. Pesaran, Shin, and Smith (2001) develop a procedure for testing the existence of a level relationship between y_t and x_t in (6.17), when it is not known if the regressor, x_t, is $I(1)$ or $I(0)$. In this case, the distribution of the *F*- or Wald statistics for testing the existence of the level relations in the *ARDL* model are non-standard and must be computed by stochastic simulations. See Section 22.3.

6.5 Long-run and short-run effects

Consider the simple partial adjustment model (6.9). The 'impact' or *short-run* effect of a unit change in x_t on y_t is given by (using (6.11)):

$$\frac{\partial y_t}{\partial x_t} = \lambda\beta. \tag{6.18}$$

The *long-run* effect of a unit change in x upon y can be obtained (when it exists) by replacing all current and lagged values of y and x in (6.11) equal to $\tilde{y}$ and $\tilde{x}$, respectively, namely[2]

$$\tilde{y} = \alpha\lambda + (1-\lambda)\,\tilde{y} + \lambda\beta\tilde{x} + \lambda\tilde{u}, \tag{6.19}$$

and solving for $\tilde{y}$ in terms of $\tilde{x}$:

$$\tilde{y} = \alpha + \beta\tilde{x} + \tilde{u}. \tag{6.20}$$

Hence

$$\frac{\partial \tilde{y}}{\partial \tilde{x}} = \beta.$$

As another example, consider the $ARDL(1, 1)$ model (see Section 6.2)

$$y_t = \alpha + \beta_0 x_t + \beta_1 x_{t-1} + \lambda y_{t-1} + u_t. \tag{6.21}$$

Then setting y_t and y_{t-1} equal to $\tilde{y}$, and x_t and x_{t-1} equal to $\tilde{x}$, we have

$$\tilde{y}\,(1-\lambda) = \alpha + \left(\beta_0 + \beta_1\right)\tilde{x}, \tag{6.22}$$

and solving for $\tilde{y}$

$$\tilde{y} = \frac{\alpha}{1-\lambda} + \frac{\beta_0 + \beta_1}{1-\lambda}\tilde{x}. \tag{6.23}$$

Therefore, the long-run effect of a unit change in x upon y is given by

$$\frac{\partial \tilde{y}}{\partial \tilde{x}} = \frac{\beta_0 + \beta_1}{1-\lambda}. \tag{6.24}$$

In the case where the long run effect is assumed to be equal to unity, then $\frac{\beta_0+\beta_1}{1-\lambda} = 1$, or $1 = \lambda + \beta_0 + \beta_1$. The parametric restriction $\beta_0 + \beta_1 + \lambda = 1$ can be tested by noting that (6.15) may also be written as

$$\Delta y_t = \alpha + \beta_0 \Delta x_t - \left(\beta_0 + \beta_1\right)\left(y_{t-1} - x_{t-1}\right) - \left(1 - \lambda - \beta_0 - \beta_1\right) y_{t-1} + u_t, \tag{6.25}$$

and testing $H_0 : \beta_0 + \beta_1 + \lambda = 1$ against $H_1 : \beta_0 + \beta_1 + \lambda \neq 1$ is the same as testing zero restriction on the coefficients of y_{t-1} in (6.25). Consider now the general $ARDL(p, q)$ model (6.4). The long-run coefficient for the response of y_t to a unit change in x_t is now given by

[2] See Section 22.2 for a definition of long-run relation.

$$\frac{\beta_0 + \beta_1 + \ldots + \beta_q}{1 - \lambda_1 - \lambda_2 \ldots - \lambda_p}. \tag{6.26}$$

The long-run coefficient can be estimated by replacing β_j and λ_j by their *OLS* estimates, and p and q by their estimates $\hat{p}$ and $\hat{q}$ obtained by applying one of the selection criteria introduced in Chapters 2 and 11. The asymptotic standard errors of the long-run coefficient can be computed by using the methodology developed by Bewley (1979) or by applying the Δ-method.

6.6 Concept of mean lag and its calculation

In the general case we have

$$y_t = a_0 + \theta \sum_{i=0}^{\infty} w_i x_{t-i} + v_t,$$

where the weights, w_i, are non-negative and add up to unity: $w_i \geq 0, \sum_i w_i = 1$. The long-run impact of x_t on y_t is given by θ. Using the lag operator L $(Lx_t = x_{t-1})$ we have

$$y_t = a_0 + \theta W(L) x_t + v_t,$$

where

$$W(L) = \sum_{i=0}^{\infty} w_i L^i.$$

The mean lag in this general case is now defined by

$$\left[\frac{\partial W(L)}{\partial L}\right]_{L=1} = W'(1) = \sum_{i=0}^{\infty} i w_i.$$

In the case of the geometric distributed lag model

$$W(L) = \frac{1 - \lambda}{1 - \lambda L},$$

and

$$W'(L) = \frac{(1 - \lambda)\lambda}{(1 - \lambda L)^2},$$

which yields the mean lag of $W'(1) = \lambda/(1 - \lambda)$, as derived above.

Let us go back to the simple ARDL$(1, 0)$ model

$$y_t = \alpha + \beta x_t + \lambda y_{t-1} + u_t.$$

To obtain the mean lag we first write this model in its distributed lag form

$$y_t = \frac{\alpha}{1-\lambda} + \beta \sum_{i=0}^{\infty} \lambda^i x_{t-i} + \sum_{i=0}^{\infty} \lambda^i u_{t-i},$$

or equivalently

$$y_t = \frac{\alpha}{1-\lambda} + \frac{\beta}{1-\lambda}\left[(1-\lambda)\sum_{i=0}^{\infty} \lambda^i x_{t-i}\right] + \sum_{i=0}^{\infty} \lambda^i u_{t-i}.$$

Now the distributed effect of a unit change in x on y over-time are given by $(1-\lambda)$, $(1-\lambda)\lambda$, $(1-\lambda)\lambda^2$,. . . , for 0, 1, 2, . . . lagged values of x_t. Hence,

$$\begin{aligned}
\text{Mean lag} &= (1-\lambda)\left\{(0)(1) + (\lambda)(1) + \left(\lambda^2\right)(2) + \cdots\right\} \\
&= \lambda(1-\lambda)\frac{d}{d\lambda}\left(\sum_{i=0}^{\infty} \lambda^i\right) \\
&= \lambda(1-\lambda)\frac{d}{d\lambda}\left(1 + \lambda + \lambda^2 + \lambda^2 + \cdots\right) \\
&= \lambda(1-\lambda)\frac{d}{d\lambda}\left(\frac{1}{1-\lambda}\right) \\
&= \lambda(1-\lambda)\frac{1}{(1-\lambda)^2} = \frac{\lambda}{1-\lambda}.
\end{aligned}$$

6.7 Models of adaptive expectations

Consider the simple expectations model

$$y_t = \alpha + \beta\, {}_t x_{t+1}^e + u_t, \tag{6.27}$$

where ${}_t x_{t+1}^e$ denotes the expectations of x_{t+1} formed at time t. The adaptive model of expectations formation postulates that expectations are revised with the past error of expectations, namely

$${}_t x_{t+1}^e - {}_{t-1} x_t^e = (1-\mu)\left(x_t - {}_{t-1} x_t^e\right), \tag{6.28}$$

$\mu = 0$ represents instantaneous adjustments, and $\mu = 1$ represents no adjustments at all. Solving for ${}_t x_{t+1}^e$ by backward recursions yields

$${}_t x_{t+1}^e = (1-\mu)\sum_{i=0}^{\infty} \mu^i x_{t-i}.$$

Substituting this result in (6.27) gives the distributed lag model with geometrically declining weights

$$y_t = \alpha + \beta (1 - \mu) \sum_{i=0}^{\infty} \mu^i x_{t-i} + u_t. \tag{6.29}$$

Writing this result in terms of lag operators we have

$$= \alpha + \left[\frac{\beta (1 - \mu)}{1 - \mu L} \right] x_t + u_t,$$

which can also be written as

$$y_t = \mu y_{t-1} + \alpha (1 - \mu) + \beta (1 - \mu) x_t + u_t - \mu u_{t-1}. \tag{6.30}$$

Notice that the basic difference between a partial adjustment model and an adaptive expectations model lies in the autocorrelation patterns of their residuals. Clearly, one can consider a combined partial adjustment/expectations model, namely

$$y_t^* = \alpha + \beta \, {}_t x_{t+1}^e + u_t, \tag{6.31}$$

where ${}_t x_{t+1}^e$ and y_t^* are defined by (6.28) and (6.10), respectively. Using (6.28) and (6.10) we have

$$y_t - (1 - \lambda) y_{t-1} = \alpha\lambda + \lambda\beta (1 - \mu) \sum_{i=1}^{\infty} \mu^i x_{t-i} + \lambda u_t, \tag{6.32}$$

or

$$y_t = (1 - \lambda) y_{t-1} + \alpha\lambda + \lambda\beta (1 - \mu) \sum_{i=0}^{\infty} \mu^i x_{t-i} + \lambda u_t.$$

This geometric distributed lag model can also be written as an *ARDL* model. We have

$$y_t = \alpha\lambda (1 - \mu) + [\mu + (1 - \lambda)] y_{t-1} - \mu (1 - \lambda) y_{t-2} + \lambda\beta (1 - \mu) x_t + v_t, \tag{6.33}$$

where

$$v_t = \lambda (u_t - \mu u_{t-1}).$$

This is an $ARDL(2, 0)$ model with a first-order moving average error process.

6.8 Rational expectations models

Rational expectations (RE) models assume that economic agents (e.g., firms, consumers, etc.) are influenced by their expectations on the unknown, future behaviour of a variable of interest. Under the rational hypothesis (REH), originally advanced by Muth (1961), expectations are

formed by mathematical expectations of variables, conditional on all the available information at the time.

6.8.1 Models containing expectations of exogenous variables

Consider the following model

$$y_t = \alpha + \beta \left({}_t x^e_{t+1}\right) + u_t, \tag{6.34}$$

where x_t is assumed to follow, for example, the $AR(2)$ process:

$$x_t = \mu_1 x_{t-1} + \mu_2 x_{t-2} + \varepsilon_t. \tag{6.35}$$

Under the rational expectations hypothesis we have (see Pesaran (1987c))

$${}_t x^e_{t+1} = E\left(x_{t+1} \,|\Omega_t\right),$$

where Ω_t is the information set, and it is assumed that $\Omega_t = \left\{x_t, x_{t-1}, \ldots, y_t, y_{t-1}, \ldots\right\}$. Hence

$${}_t x^e_{t+1} = \mu_1 x_t + \mu_2 x_{t-1}, \tag{6.36}$$

and using this result in (6.34):

$$y_t = \alpha + \beta\left(\mu_1 x_t + \mu_2 x_{t-1}\right) + u_t, \tag{6.37}$$

or

$$y_t = \alpha + \beta\mu_1 x_t + \beta\mu_2 x_{t-1} + u_t. \tag{6.38}$$

6.8.2 *RE* models with current expectations of endogenous variable

Consider a simple model where current expectations of y_t enter as a regressor

$$\begin{aligned} y_t &= \alpha + \beta \left({}_{t-1} y^e_t\right) + \gamma x_t + u_t, \qquad \beta \neq 1, \\ &= \alpha + \beta E\left(y_t \,|\Omega_{t-1}\right) + \gamma x_t + u_t. \end{aligned} \tag{6.39}$$

Since, by assumption, $E(u_t \,|\Omega_{t-1}) = 0$ we have

$$\begin{aligned} E\left(y_t \,|\Omega_{t-1}\right) &= \alpha + \beta E\left(y_t \,|\Omega_{t-1}\right) + \gamma E\left(x_t \,|\Omega_{t-1}\right), \\ E\left(y_t \,|\Omega_{t-1}\right) &= \frac{\alpha}{1-\beta} + \frac{\gamma}{1-\beta} E\left(x_t \,|\Omega_{t-1}\right), \\ E\left(y_t \,|\Omega_{t-1}\right) &= (1-\beta)^{-1}\left(\alpha + \gamma x^e_t\right). \end{aligned} \tag{6.40}$$

Using this result in (6.39) we obtain

$$\begin{aligned} y_t &= \alpha + \beta (1-\beta)^{-1} \left(\alpha + \gamma x_t^e\right) + \gamma x_t + u_t \\ &= \alpha + \frac{\alpha\beta}{1-\beta} + \frac{\gamma\beta}{1-\beta} x_t^e + \gamma x_t + u_t. \end{aligned} \tag{6.41}$$

Finally

$$y_t = \frac{\alpha}{1-\beta} + \frac{\gamma\beta}{1-\beta} x_t^e + \gamma x_t + u_t. \tag{6.42}$$

Once again we need to model expectations of x_t. Assuming x_t follows an AR(2) process

$$x_t = \mu_1 x_{t-1} + \mu_2 x_{t-2} + \varepsilon_t, \tag{6.43}$$

as before we have $x_t^e = \mu_1 x_{t-1} + \mu_2 x_{t-2}$, and after substituting this result in (6.42) we obtain the solution

$$y_t = \frac{\alpha}{1-\beta} + \frac{\gamma\beta}{1-\beta} \left(\mu_1 x_{t-1} + \mu_2 x_{t-2}\right) + \gamma x_t + u_t. \tag{6.44}$$

6.8.3 *RE* models with future expectations of the endogenous variable

Consider now the following *RE* model with future expectations of the endogenous variable

$$y_t = \alpha + \beta E(y_{t+1} \,|\Omega_t) + \gamma x_t + u_t. \tag{6.45}$$

Let $w_t = \alpha + \beta x_t + u_t$, and rolling the equation one step forward we have

$$y_{t+1} = \beta E(y_{t+2} \,|\Omega_{t+1}) + w_{t+1}. \tag{6.46}$$

Then noting that Ω_{t+1} contains Ω_t and using the law of iterated expectations we have

$$E(y_{t+1} \,|\Omega_t) = \beta E(y_{t+2} \,|\Omega_t) + E(w_{t+1} \,|\Omega_t).$$

Proceeding in a similar manner we have

$$E(y_{t+2} \,|\Omega_t) = \beta E(y_{t+3} \,|\Omega_t) + E(w_{t+2} \,|\Omega_t),$$

and so on. Using these results recursively forward we obtain the forward solution of the model given by

$$y_t = \beta^h E(y_{t+h} \,|\Omega_t) + \sum_{j=0}^{h-1} \beta^j E(w_{t+j} \,|\Omega_t). \tag{6.47}$$

This solution is unique if $|\beta| < 1$, such that

$$\lim_{h\to\infty} \beta^h E(y_{t+h}\,|\Omega_t) = 0, \tag{6.48}$$

and the stochastic process of the forcing variables $\{w_t\}$ is such that

$$\lim_{h\to\infty}\left[\sum_{j=0}^{h-1}\beta^j E(w_{t+j}\,|\Omega_t)\right] = \sum_{j=0}^{\infty}\beta^j E(w_{t+j}\,|\Omega_t),$$

exists. Under these assumptions the unique solution of the RE equation, (6.46), is give by the familiar present value formula

$$y_t = \sum_{j=0}^{\infty}\beta^j E(w_{t+j}\,|\Omega_t).$$

The condition (6.48) is known as the transversality condition and is likely to be met if and only if $|\beta| < 1$. The present value expression exists for a wide class of stochastic processes. It clearly exists if w_t is stationary. It also exists if w_t follows a unit root process.

As an example, suppose that u_t is serially uncorrelated and x_t follows the $AR(1)$ process

$$x_t = \rho x_{t-1} + \varepsilon_t.$$

Then it is easily seen that

$$\begin{aligned} E(w_t\,|\Omega_t) &= \alpha + \gamma x_t + u_t, \\ E(w_{t+j}\,|\Omega_t) &= \alpha + \gamma E(w_{t+j}\,|\Omega_t) = \alpha + \gamma\rho^j x_t, \text{ for } j > 0. \end{aligned}$$

Hence,

$$y_t = \sum_{j=0}^{\infty}\beta^j E(w_{t+j}\,|\Omega_t) = \frac{\alpha}{1-\beta} + \frac{\gamma}{1-\beta\rho}x_t + u_t.$$

This result holds so long as $|\beta\rho| < 1$, which allows for $\rho = 1$.

The above solution is unique even if x_t follows a geometric random walk with a drift, so long as the drift coefficient is not too large. To see this consider the following geometric random process for x_t

$$x_t = x_{t-1}\exp(\mu + \varepsilon_t),$$

where $\varepsilon_t \sim N(0, \sigma_\varepsilon^2)$. In this case

$$x_{t+j} = x_t \exp\left(j\mu + \varepsilon_{t+1} + \varepsilon_{t+2} + \ldots + \varepsilon_{t+j}\right),$$

and making use of results from the moment generating function of normal variates, we have

$$E(x_{t+j}\,|\Omega_t) = x_t \exp\left(j\mu + \frac{j\sigma_\varepsilon^2}{2}\right).$$

Hence, the present value exits if $\left|\beta \exp(\mu + 0.5 * \sigma_\varepsilon^2)\right| < 1$, and the unique solution of the RE model (assuming $|\beta| < 1$) is given by

$$y_t = \frac{\alpha}{1-\beta} + \frac{\gamma}{1-\beta\lambda}x_t + u_t,$$

where $\lambda = \exp\left(\mu + \frac{\sigma_\varepsilon^2}{2}\right)$.

If $|\beta| > 1$ the solution is not unique and depends on $E(y_{t+h}\,|\Omega_t)$ for some h. Note that in this case the present value component of the solution, (6.47), might exist but the transversality condition would not be satisfied. The multiplicity of the solution in the case of $|\beta| > 1$, also known as the irregular case, can be obtained by observing that under the *REH* the expectations error

$$v_{t+1} = y_{t+1} - E(y_{t+1}\,|\Omega_t),$$

is a martingale difference process, since $E(v_{t+1}\,|\Omega_t) = 0$. Eliminating $E(y_{t+1}\,|\Omega_t)$ from (6.45) we have

$$y_t = \alpha + \beta(y_{t+1} - v_{t+1}) + \gamma x_t + u_t,$$

and

$$y_t = \beta^{-1}y_{t-1} - \alpha\beta^{-1} - \gamma\beta^{-1}x_{t-1} - \beta^{-1}u_{t-1} + v_t.$$

The non-uniqueness nature of this solution is due to the presence of the martingale process v_t in the solution. Note that v_t is a function of the information set and can be further characterized as functions of the innovations in the forcing variables. For example,

$$v_t = g_0\left[x_t - E(x_t\,|\Omega_{t-1})\right] + \xi_t,$$

where g_0 is an arbitrary constant and ξ_t is another martingale process, in the sense that $E(\xi_t\,|\Omega_{t-1}) = 0$.

Solution of more complicated *RE* models where there are backward as well as forward components are reviewed in Pesaran (1987c) and Binder and Pesaran (1995). In Chapter 20 we consider *RE* models in a multivariate setting.

6.9 Further reading

Further discussion on autoregressive distributed lag models can be found in Hamilton (1994, ch. 1). A review of non-stochastic difference equations is provided in Appendix A, while a

formal treatment of stochastic processes is given in Chapter 12. More details on univariate rational expectations models are provided in Pesaran (1987c) and Gouriéroux, Monfort, and Gallo (1997).

6.10 Exercises

1. Consider the dynamic model

$$y_t = \lambda y_{t-1} + \beta x_t + u_t, \tag{6.49}$$

where λ, β are parameters and u_t is a white-noise disturbance term (i.e. $E(u_t) = 0$, $E\left(u_t^2\right) = \sigma^2$, and $E\left(u_i u_j\right) = 0$, for all $i \neq j$). Show that (6.49) is equivalent to the model

$$y_t = \beta \sum_{j=0}^{\infty} \lambda^j x_{t-j} + v_t,$$

where v_t is an autoregressive process with parameter λ. Derive the short-run and long-run effects on y of a once and for all change in x.

2. Show that the mean lag for the DL model

$$y_t = \alpha + \beta_0 x_t + \beta_1 x_{t-1} + \beta_2 x_{t-2} + u_t,$$

is given by

$$\frac{\beta_1 + 2\beta_2}{\beta_0 + \beta_1 + \beta_2}.$$

3. Consider the following finite distributed lag models

$$y_t = 2.6 + 0.1x_t + 0.7x_{t-3} + 0.2x_{t-5},$$
$$y_t = 1.5 + 0.2x_{t-2} + 0.4x_{t-3} + 0.1x_{t-4}.$$

How do you interpret the difference between the two mean lags of the above two equations?

4. Consider the following simple model

$$y_t = \lambda y_{t-1} + u_t,$$

with $|\lambda| < 1$, and let $\Omega_t = \left\{y_t, y_{t-1}, \ldots, y_{t-p}\right\}$. Derive the rational expectations at horizon h, namely $\mathrm{E}\left(y_{t+h} \middle| \Omega_t\right)$.

5. Suppose that the desired level of consumption, C_t^* is determined by

$$C_t^* = \alpha + \beta Y_t + \varepsilon_t,$$

where Y_t is real disposable income, ε_t is a error term, and actual (realized) consumption, C_t, evolves gradually in response to desired consumption

$$C_t - C_{t-1} = \Delta C_t = \theta(C_t^* - C_{t-1}), \qquad 0 < \theta < 1. \tag{6.50}$$

(i) Derive the autoregressive lag model

$$C_t = \alpha\theta + \beta\theta Y_t + (1-\theta)C_{t-1} + \nu_t \tag{6.51}$$

where $\nu_t = \theta\varepsilon_t$.

(ii) Discuss the advantages and disadvantages of estimating (6.51) over (6.50).

(iii) Show that the autoregressive lag model (6.51) can be rewritten as a distributed lag model.

7 Predictability of Asset Returns and the Efficient Market Hypothesis

7.1 Introduction

Economists have long been fascinated by the nature and sources of variations in the stock market. By the early 1970s, a consensus had emerged among financial economists suggesting that stock prices could be well approximated by a random walk model and that changes in stock prices were basically unpredictable. Fama (1970) provides an early, definitive statement of this position. Historically, the 'random walk' theory of stock prices was preceded by theories relating movements in the financial markets to the business cycle. A prominent example is the interest shown by Keynes in the variation in stock returns over the business cycle.

The efficient market hypothesis (*EMH*) evolved in the 1960s from the random walk theory of asset prices advanced by Samuelson (1965). Samuelson showed that, in an informationally efficient market, price changes must be unforecastable. Kendall (1953), Cowles (1960), Osborne (1959, 1962), and many others had already provided statistical evidence on the random nature of equity price changes. Samuelson's contribution was, however, instrumental in providing academic respectability for the hypothesis, despite the fact that the random walk model had been around for many years, having been originally discovered by Louis Bachelier, a French statistician, back in 1900.

Although a number of studies found some statistical evidence against the random walk hypothesis, these were dismissed as economically unimportant (they could not generate profitable trading rules in the presence of transaction costs) and statistically suspect (they could be due to data mining). For example, Fama (1965) concluded that 'there is no evidence of important dependence from either an investment or a statistical point of view'. Despite its apparent empirical success, the random walk model was still a statistical statement and not a coherent theory of asset prices. For example, it need not hold in markets populated by risk-averse traders, even under market efficiency.

There now exist many different versions of the *EMH*, and one of the aims of this chapter is to provide a simple framework where alternative versions of the *EMH* can be articulated and discussed. We begin with an overview of the statistical properties of asset returns at

different frequencies (daily, weekly, and monthly), and consider the evidence on return predictability, risk aversion, and market efficiency. We then focus on the theoretical foundation of the *EMH*, and show that market efficiency could co-exist with heterogeneous beliefs and individual 'irrationality', so long as individual errors are cross-sectionally weakly dependent in the sense defined by Chudik, Pesaran, and Tosetti (2011). But at times of market euphoria or gloom these individual errors are likely to become cross sectionally strongly dependent and the collective outcome could display significant departures from market efficiency. Market efficiency could be the norm, but most likely it will be punctuated by episodes of bubbles and crashes. To test for such episodes we argue in favour of compiling survey data on individual expectations of price changes that are combined with information on whether such expectations are compatible with market equilibrium. A trader who believes that asset prices are too high (low) might still expect further price rises (falls). Periods of bubbles and crashes could result if there are sufficiently large numbers of such traders that are prepared to act on the basis of their beliefs. The chapter also considers if periods of market inefficiency can be exploited for profit.

7.2 Prices and returns

7.2.1 Single period returns

Let P_t be the price of a security at date t. The absolute price change over the period $t-1$ to t is given by $P_t - P_{t-1}$, the relative price change by

$$R_t = (P_t - P_{t-1})/P_{t-1},$$

the gross return (excluding dividends) on security by

$$1 + R_t = P_t/P_{t-1},$$

and the log price change by

$$r_t = \Delta \ln(P_t) = \ln(1 + R_t).$$

It is easily seen that for *small relative price changes* the log-price change and the relative price change are almost identical.

In the case of daily observations when dividends are negligible, $100 \cdot R_t$ measures the *per cent return* on the security, and $100 \cdot r_t$ is the continuously compounded return. R_t is also known as discretely compounded return. The continuously compounded return, r_t, is particularly convenient in the case of temporal aggregation (multi-period returns: see Section 7.2.2), while the discretely compounded returns are convenient for use in cross-sectional aggregation, namely aggregation of returns across different instruments in a portfolio. For example, for a portfolio composed of N instruments with weights $w_{i,t-1}$, $(\sum_{i=1}^{N} w_{i,t-1} = 1, w_{i,t-1} \geq 0)$ we have

$$R_{pt} = \sum_{i=1}^{N} w_{i,t-1} R_{it}, \quad \text{(per cent return)},$$

$$r_{pt} = \ln\left(\sum_{i=1}^{N} w_{i.t-1} e^{r_{it}}\right), \quad \text{(continuously compounded)}.$$

Often r_{pt} is approximated by $\sum_{i=1}^{N} w_{i,t-1} r_{it}$.

When dividends are paid out we have

$$\begin{aligned} R_t &= (P_t - P_{t-1})/P_{t-1} + D_t/P_{t-1}, \\ &\approx \Delta \ln(P_t) + D_t/P_{t-1}, \end{aligned}$$

where D_t is the dividend paid out during the holding period.

7.2.2 Multi-period returns

Single-period price changes (returns) can be used to compute multi-period price changes or returns. Denote the return over the most recent h periods by $R_t(h)$ then (abstracting from dividends)

$$R_t(h) = \frac{P_t - P_{t-h}}{P_{t-h}},$$

or

$$1 + R_t(h) = P_t/P_{t-h},$$

and

$$r_t(h) = \ln(P_t/P_{t-h}) = r_t + r_{t-1} + \ldots + r_{t-h+1},$$

where r_{t-i}, $i = 0, 1, 2, \ldots, h-1$ are the single-period returns. For example, weekly returns are defined by $r_t(5) = r_t + r_{t-1} + \ldots + r_{t-4}$. Similarly, since there are 25 business days in one month, then the 1-month return can be computed as the sum of the last 25 1-day returns, or $r_t(25)$.

7.2.3 Overlapping returns

Note that multi-period returns have overlapping daily observations. In the case of weekly returns, $r_t(5)$ and $r_{t-1}(5)$ have the four daily returns, $r_{t-1} + r_{t-2} + r_{t-3} + r_{t-4}$ in common. As a result the multi-period returns will be serially correlated even if the underlying daily returns are not serially correlated. One way of avoiding the overlap problem would be to sample the multi-period returns h periods apart. But this is likely to be inefficient as it does not make use of all available observations. A more appropriate strategy would be to use the overlapping returns but allow for the fact that this will induce serial correlations. For further details see Pesaran, Pick, and Timmermann (2011).

7.3 Statistical models of returns

A simple model of returns (or log price changes) is given by

$$\begin{aligned} r_{t+1} &= \Delta \ln(P_{t+1}) = p_{t+1} - p_t, \\ &= \mu_t + \sigma_t \varepsilon_{t+1}, \qquad t = 1, 2, \ldots, T, \end{aligned} \tag{7.1}$$

where μ_t and σ_t^2 are the conditional mean and the conditional variance of returns (with respect to the information set Ω_t available at time t) and ε_{t+1} represents the unpredictable component of return. Two popular distributions for ε_{t+1} are

$$\varepsilon_{t+1} \mid \Omega_t \sim IID\ \mathcal{Z},$$
$$\varepsilon_{t+1} \mid \Omega_t \sim \left(\sqrt{\frac{\nu-2}{\nu}}\right) IID\ \mathcal{T}_\nu,$$

where $\mathcal{Z} \sim N(0,1)$ stands for a standard normal distribution, and $\mathcal{T}_\nu$ stands for Student's t-distribution with ν degrees of freedom. Unlike the normal distribution that has moments of all orders, $\mathcal{T}_\nu$ only has moments of order $\nu - 1$ and smaller. For the Student's t to have a variance, for example, we need $\nu > 2$.

Since $r_{t+1} = \ln(1 + R_{t+1})$, where $R_{t+1} = (P_{t+1} - P_t)/P_t$, it then follows that under $\varepsilon_{t+1} \mid \Omega_t \sim IID\ \mathcal{Z}$, the price level, P_{t+1} conditional on Ω_t will be lognormally distributed. Note that $\Omega_t = (P_t, P_{t-1}, \ldots.)$ and $\Omega_t = (r_t, r_{t-1}, \ldots.)$ convey the same information and are equivalent. Hence, $P_{t+1} = P_t \exp(r_{t+1})$, and we have[1]

$$\begin{aligned} E(P_{t+1} \mid \Omega_t) &= P_t E(\exp(r_{t+1}) \mid \Omega_t) \\ &= P_t \exp(\mu_t + \frac{1}{2}\sigma_t^2). \end{aligned}$$

Similarly,

$$Var(P_{t+1} \mid \Omega_t) = P_t^2 \exp(2\mu_t + \sigma_t^2)\left[\exp(\sigma_t^2) - 1\right].$$

In practice, it is much more convenient to work with log returns, r_{t+1}, rather than asset prices.

The probability density functions of $\mathcal{Z}$ and $\mathcal{T}_\nu$ are given by

$$f(\mathcal{Z}) = (2\pi)^{-1/2} \exp\left(\frac{-\mathcal{Z}^2}{2}\right), \quad -\infty < \mathcal{Z} < \infty, \tag{7.2}$$

and

$$f(\mathcal{T}_\nu) = \frac{1}{\sqrt{\nu} B(\nu/2, 1/2)} \left(1 + \frac{\mathcal{T}_\nu^2}{\nu}\right)^{-(\nu+1)/2}, \tag{7.3}$$

[1] Using properties of the moment generating function of normal variates, if $x \backsim N(\mu_x, \sigma_x^2)$ then, $E[exp(x)] = exp(\mu_x + .5\sigma_x^2)$.

where $-\infty < \mathcal{T}_v < \infty$, and $B(v/2, 1/2)$ is the beta function defined by

$$B(\alpha, \beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha + \beta)}, \ \Gamma(\alpha) = \int_0^\infty u^{\alpha-1} e^{-u} du.$$

It is easily seen that

$$E(\mathcal{T}_v) = 0, \text{and } Var(\mathcal{T}_v) = \frac{v}{v-2}.$$

A large part of financial econometrics is concerned with alternative ways of modelling the conditional mean (mean returns), μ_t, the conditional variance (asset return volatility), σ_t, and the cumulative probability distribution of the errors, ε_{t+1}. A number of issues need to be addressed in order to choose an adequate model. In particular:

- Is the distribution of returns normal?
- Is the distribution of returns constant over time?
- Are returns statistically independent over time?
- Are squares or absolute values of returns independently distributed over time?
- What are the cross correlation of returns on different instruments?

The above modelling issues can be readily extended to the case where we are concerned with a vector of asset returns, $\mathbf{r}_t = (r_{1t}, r_{2t}, \ldots r_{mt})'$. In this case we also need to model the pair-wise conditional correlations of asset returns, namely

$$Corr(r_{it}, r_{jt} \mid \Omega_t) = \frac{Var(r_{it}, r_{jt} \mid \Omega_t)}{\sqrt{Var(r_{it} \mid \Omega_t) Var(r_{jt} \mid \Omega_t)}}.$$

Typically the conditional variances and correlations are modelled using exponential smoothing procedures or the multivariate generalized autoregressive conditional heteroskedastic models developed in the econometric literature. See Chapters 18 and 25 for further details.

7.3.1 Percentiles, critical values, and Value at Risk

Suppose a random variable r (say daily returns on an instrument) has the probability density function $f(r)$. Then the pth percentile of the distribution of r, denoted by C_p, is defined as that value of return such that p per cent of the returns fall below it. Mathematically we have

$$p = \Pr(r < C_p) = \int_{-\infty}^{C_p} f(r) dr.$$

In the literature on risk management C_p is used to compute 'Value at Risk' or *VaR* for short. For $p = 1\%$, C_p associated with the one-sided critical value of the normal distribution is given by -2.33σ, where σ is the standard deviation of returns (see Chapter 25 for an application of the *VaR* in the context of risk management).

In hypothesis testing C_p is known as the critical value of the test associated with a (one-sided) test of size p. In the case of two-sided tests of size p, the associated critical value is computed as $C_{p/2}$. See Chapter 3.

7.3.2 Measures of departure from normality

The normal probability density function for r_{t+1} conditional on the information at time t, Ω_t, is given by

$$f(r_{t+1}) = (2\pi\sigma_t^2)^{-1/2} \exp\left[-\frac{1}{2\sigma_t^2}(r_{t+1} - \mu_t)^2\right],$$

with $\mu_t = E(r_{t+1} \mid \Omega_t)$ and $\sigma_t^2 = E\left[(r_{t+1} - \mu_t)^2 \mid \Omega_t\right]$ being the conditional mean and variance. If the return process is stationary, unconditionally we also have $\mu = E(r_{t+1})$, and $\sigma^2 = E\left[(r_{t+1} - \mu_t)^2\right]$.

Skewness and tail-fatness measures are defined by

$$\begin{aligned} \text{Skewness} &= \sqrt{b_1} = m_3/m_2^{3/2}, \\ \text{Kurtosis} &= b_2 = m_4/m_2^2, \end{aligned}$$

where

$$m_j = \frac{\sum_{t=1}^{T}(r_t - \bar{r})^j}{T}, \qquad j = 2, 3, 4.$$

For a normal distribution $\sqrt{b_1} \approx 0$, and $b_2 \approx 3$. In particular

$$\hat{\mu} = \bar{r} = \sum_{t=1}^{T} r_t/T, \quad \hat{\sigma} = \sqrt{\frac{\sum_{t=1}^{T}(r_t - \bar{r})^2}{T-1}}.$$

The Jarque–Bera's test statistic for departure from normality is given by, (see Jarque and Bera (1980), and Section 3.14)

$$JB = T\left\{\tfrac{1}{6}b_1 + \tfrac{1}{24}(b_2 - 3)^2\right\}.$$

Under the joint null hypothesis that $b_1 = 0$ and $b_2 = 3$, the JB statistic is asymptotically distributed (as $T \to \infty$) as a chi-squared with 2 degrees of freedom, χ_2^2. Therefore, a value of JB in excess of 5.99 will be statistically significant at the 95 per cent confidence level, and the null hypothesis of normality will be rejected.

7.4 Empirical evidence: statistical properties of returns

Table 7.1 gives a number of statistics for daily returns ($\times 100$) on four main equity index futures, namely S&P 500 (SP), FTSE 100 (FTSE), German DAX (DAX), and Nikkei 225 (NK), over the period 3 Jan 2010–31 Aug 2009 (for a total of 2,519 observations).[2]

The kurtosis coefficients are particularly large for all four equity futures and exceed the benchmark value of 3 for the normal distribution. There is some evidence of positive skewness, but it is of second order importance as compared to the magnitude of excess kurtosis coefficient given by, $b_2 - 3$. The large values of excess kurtosis are reflected in the huge values of the *JB* statistics reported in 7.1. Under the assumption that returns are normally distributed, we would have expected the maximum and minimum of daily returns to fall (with 99 per cent confidence) in the region of $\pm 2.33\times$ S. D., which is ± 3.24 for S&P 500, as compared to the observed values of -9.88 and 14.11. See also Figure 7.1.

The departure from normality is particularly pronounced over the past decade where markets have been subject to two important episodes of financial crises: the collapse of markets in 2000 after the dot-com bubble and the stock market crash of 2008 after the 2007 credit crunch (see Figure 7.2).

However, the evidence of departure from normality can be seen in daily returns even before 2000. For example, over the period 3 Jan 1994–31 Dec 1999 (1565 daily observations) kurtosis coefficient of returns on S&P 500 was 9.5, which is still well above the benchmark value of 3. The recent financial crisis has accentuated the situation but cannot be viewed as the cause of the observed excess kurtosis of equity returns.

Similar results are also obtained if we consider weekly returns. The kurtosis coefficients estimated using weekly returns over the period Jan 2000–31 Aug 2009 (504 weeks) were 12.4, 15.07, 8.9, and 15.2 for S&P 500, FTSE, DAX, and Nikkei, respectively. These are somewhat lower than the estimates obtained using daily observations for S&P 500 and Nikkei, but are quite a bit higher for FTSE. For DAX daily and weekly observations yield a very similar estimate of the kurtosis coefficient.

The kurtosis coefficient of returns for currencies (measured in terms of the US dollar) varies from 4.5 for the euro to 13.8 for the Australian dollar. The estimates computed using daily observations over the period 3 Jan 2000–31 Aug 2009 are summarized in Table 7.2. The currencies

Table 7.1 Descriptive statistics for daily returns on S&P 500, FTSE 100, German DAX, and Nikkei 225

Variables	SP	FTSE	DAX	NK
Maximum	14.11	10.05	12.83	20.70
Minimum	−9.88	−9.24	−8.89	−13.07
Mean ($\bar{r}$)	−0.01	−0.01	−0.01	−0.01
S. D. ($\hat{\sigma}$)	1.39	1.33	1.65	1.68
Skewness ($\sqrt{b_1}$)	0.35	0.06	0.24	0.16
Kurtosis (b_2)	14.3	9.7	8.5	17.8
JB statistic	13453.6	4713.1	3199.2	23000.8

[2] All statistics and graphs have been obtained using *Microfit* 5.0.

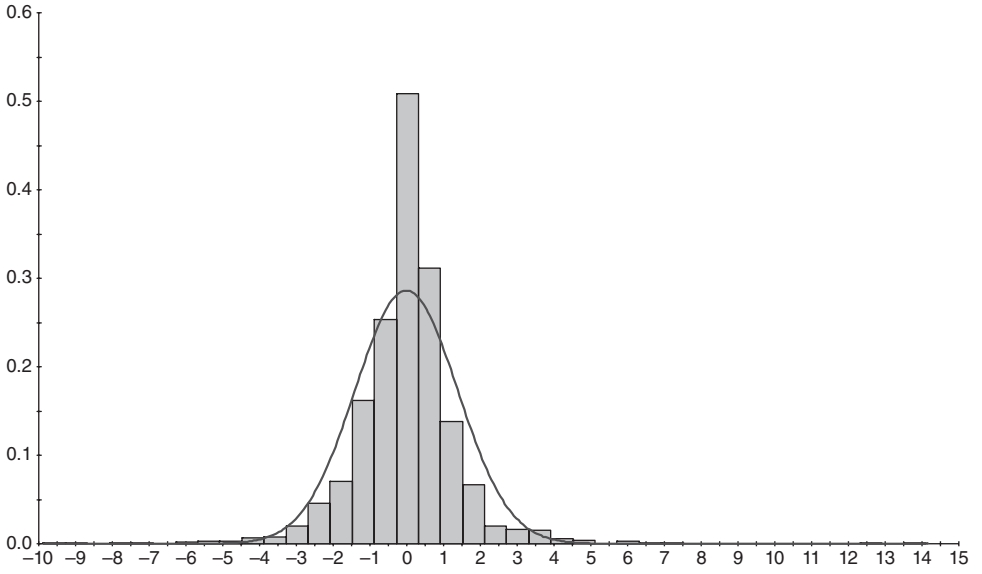


Figure 7.1 Histogram and Normal curve for daily returns on S&P 500 (over the period 3 Jan 2000–31 Aug 2009).

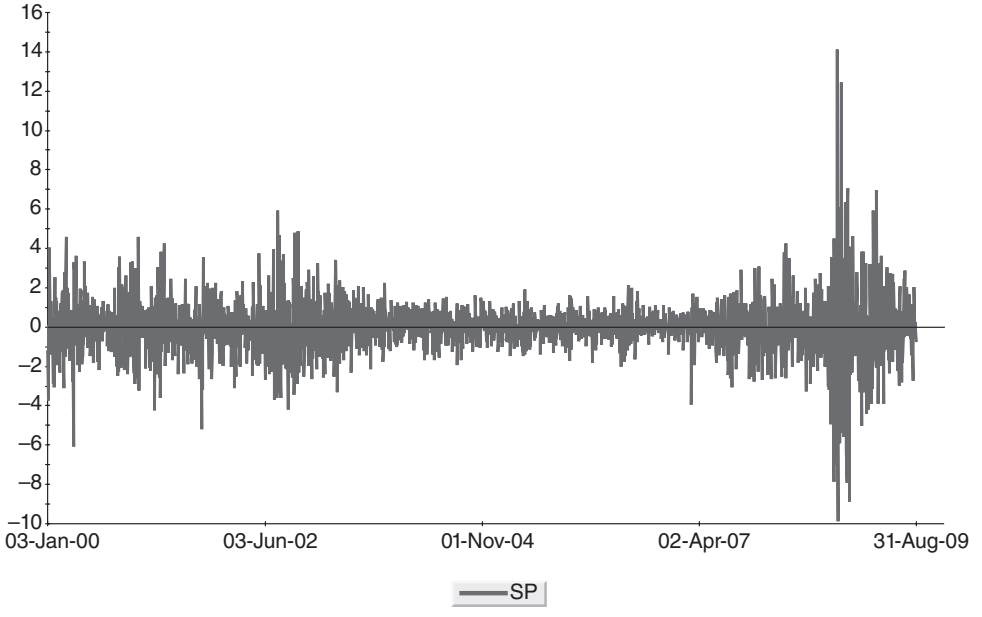


Figure 7.2 Daily returns on S&P 500 (over the period 3 Jan 2000–31 Aug 2009).

Table 7.2 Descriptive statistics for daily returns on British pound, euro, Japanese yen, Swiss franc, Canadian dollar, and Australian dollar

Variables	*JPY*	*EU*	*GBP*	*CHF*	*CAD*	*AD*
Maximum	4.53	3.17	3.41	4.58	5.25	6.21
Minimum	–3.93	–3.01	–5.04	–3.03	–3.71	–9.50
Mean $(\bar{r})$	–.006	.016	.007	.012	.013	.022
S. D. $(\hat{\sigma})$	–.65	.65	.60	.70	.59	.90
Skewness $(\sqrt{b_1})$	–.28	.01	–.35	.12	.09	–.76
Kurtosis (b_2)	5.99	4.5	7.2	4.9	9.1	13.8

Table 7.3 Descriptive statistics for daily returns on US T-Note 10Y, Europe Euro Bund 10Y, Japan Government Bond 10Y, and, UK Long Gilts 8.75-13Y

Variables	BU	BE	BG	BJ
Maximum	3.63	1.48	2.43	1.53
Minimum	–2.40	−1.54	−1.85	−1.41
Mean $(\bar{r})$	.0	.01	.01	.01
S. D. $(\hat{\sigma})$	.43	.32	.35	.24
Skewness $(\sqrt{b_1})$	–.004	−.18	.02	−.18
Kurtosis (b_2)	6.67	4.49	6.02	6.38

considered are the British pound (GBP), euro (EU), Japanese yen (JPY), Swiss franc (CHF), Canadian dollar (CAD), and Australian dollar (AD), all measured in terms of the US dollar.

The returns on government bonds are generally less fat-tailed than the returns on equities and currencies. But their distribution still shows a significant degree of departure from normality.

Table 7.3 reports descriptive statistics on daily returns on the main four government bond futures: US T-Note 10Y (BU), Europe Euro Bund 10Y (BE), Japan Government Bond 10Y (BJ), and UK Long Gilts 8.75-13Y (BG) over the period 03 Jan 2000–31 Aug 2009.

It is clear that, for all three asset classes, there are significant departures from normality which need to be taken into account when analysing financial time series.

7.4.1 Other stylized facts about asset returns

Asset returns are typically uncorrelated over time, are difficult to predict and, as we have seen, tend to have distributions that are fat-tailed. In contrast, the absolute or squares of asset returns (that measure risk), namely $|r_t|$ or r_t^2, are serially correlated and tend to be predictable. It is interesting to note that r_t can be written as

$$r_t = sign(r_t)\,|r_t|\,,$$

where $sign(r_t) = +1$ if $r_t > 0$ and $sign(r_t) = -1$ if $r_t \leq 0$. Since $|r_t|$ is predictable, it is, therefore, the non-predictability of $sign(r_t)$, or the direction of the market, which lies behind the difficulty of predicting returns.

The extent to which returns are predictable depends on the forecast horizon, the degree of market volatility, and the state of the business cycle. Predictability tends to rise during crisis periods. Similar considerations also apply to the degree of fat-tailedness of the underlying distribution and the cross-correlations of asset returns. The return distributions become less fat-tailed as the horizon is increased, and cross-correlations of asset returns become more predictable with the horizon. Cross-correlation of returns also tends to increase with market volatility. The analysis of time variations in the cross correlation of asset returns is discussed in Chapter 25.

In the case of daily returns, equity returns tend to be negatively serially correlated. During normal times they are small and only marginally significant statistically, but they become relatively large and attain a high level of statistical significance during crisis periods. These properties are illustrated in the following empirical application.

The first- and second-order serial correlation coefficients of daily returns on S&P 500 over the period 3 Jan 2000–31 Aug 2007 are −0.015 (0.0224) and −0.0458 (0.0224), respectively, but increase to −0.068 (0.0199) and −0.092 (0.0200) once the sample is extended to the end of August 2009, which covers the 2008 global financial crisis.[3] Similar patterns are also observed for other equity indices. For currencies the evidence is more mixed. In the case of major currencies such as euro and yen, there is little evidence of serial correlation in returns and this outcome does not seem much affected by whether one considers normal or crisis periods. For other currencies there is some evidence of negative serial correlation, particularly at times of crisis. For example, over the period 3 Jan 2000–31 Aug 2009 the first-order serial correlation of daily returns on Australian dollar amounts to −0.056 (0.0199), but becomes statistically insignificant if we exclude the crisis period. There is also very little evidence of serial correlation in daily returns on the four major government bonds that we have been considering. This outcome does not depend on whether the crisis period is included in the sample. Irrespective of whether the underlying returns are serially correlated, their absolute values (or their squares) are highly serially correlated, often over many periods. For example, over the 3 Jan 2000–31 Aug 2009 period the first- and second-order serial correlation coefficients of absolute return on S&P 500 are 0.2644(0.0199), 0.3644(0.0204); for euro they are 0.0483(0.0199) and 0.1125(0.0200), and for US 10Y bond they are 0.0991(0.0199) and 0.1317(0.0201). The serial correlation in absolute returns tends to decay very slowly and continues to be statistically significant event after 120 trading days (see Figure 7.3).

It is also interesting to note that there is little correlation between r_t and $|r_t|$. Based on the full sample ending in August 2009, this correlation is −.0003 for S&P 500, 0.025 for euro, and 0.009 for the US 10Y bond.

7.4.2 Monthly stock market returns

Many of the regularities and patterns documented for returns using daily or weekly observations can also be seen in monthly observations, once a sufficiently long period is considered. For the US stock market long historical monthly data on prices and dividends are compiled by Shiller and can be downloaded from his homepage.[4] An earlier version of this data set has been analysed

[3] The figures in brackets are standard errors.

[4] See <http://www.econ.yale.edu/~shiller/data.htm>.

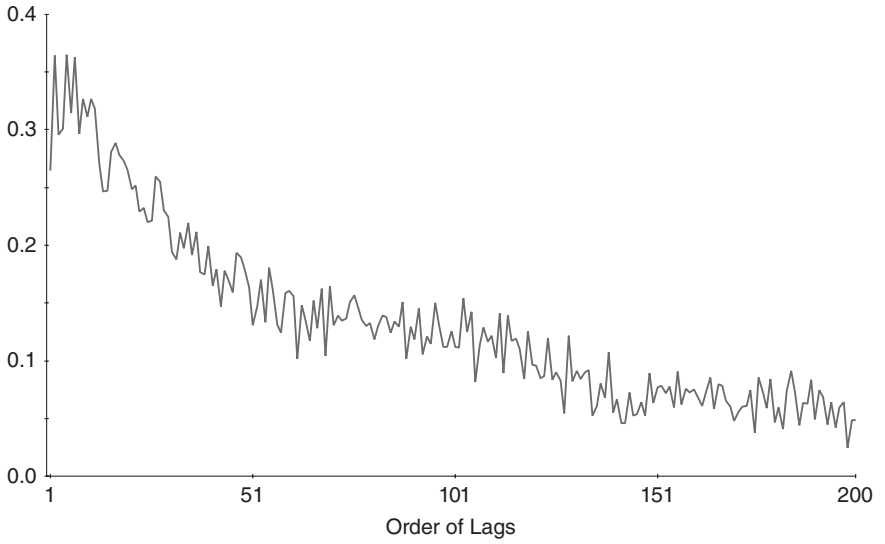


Figure 7.3 Autocorrelation function of the absolute values of returns on S&P 500 (over the period 3 Jan 2000–31 Aug 2009).

in Shiller (2005). Monthly returns on S&P 500 (inclusive of dividends) is computed as

$$RSP_t = 100\left(\frac{SP_t - SP_{t-1} + SPDIV_t}{SP_{t-1}}\right),$$

where SP_t is the monthly spot price index of S&P 500 and $SPDIV_t$ denotes the associated dividends on the S&P 500 index. Over the period 1871m1 to 2009m9 (a total of 1,664 monthly observations) the coefficient of skewness and kurtosis of *RSP* amounted to 1.07 and 23.5 per cents, respectively. The excess kurtosis coefficient of 20.5 is much higher than the figure of 11.3 obtained for the daily observations on SP over the period 3 Jan 2000–31 Aug 2009. Also as before the skewness coefficient is relatively small. However, the monthly returns show a much higher degree of serial correlation and a lower degree of volatility as compared to daily or weekly returns. The correlation coefficients of *RSP* are 0.346 (0.0245) and 0.077 (0.027), and the serial correlation coefficients continue to be statistically significant up to the lag order of 12 months. Also, the pattern of serial correlations in absolute monthly returns, $|RSP_t|$, is not that different from that of the serial correlation in RSP_t , which suggests a lower degree of return volatility (as compared with the volatility of daily or weekly returns) once the effects of mean returns are taken into account.

Similar, but less pronounced, results are obtained if we exclude the 1929 stock market crash and focus on the post-Second World War period. The coefficients of skewness and kurtosis of monthly returns over the period 1948m1 to 2009m9 (741 observations) are –0.49 and 5.2, respectively. The first- and second-order serial correlation coefficients of returns are 0.361 (0.0367) and 0.165 (0.041), respectively. The main difference between these sub-sample estimates and those obtained for the full sample is the much lower estimate for the kurtosis coefficient. But even the lower post 1948 estimates suggest a significant degree of fat-tailedness in the monthly returns.

7.5 Stock return regressions

Consider the linear excess return regression

$$R_{t+1} - r_t^f = a + b_1 x_{1t} + b_2 x_{2t} + \ldots + b_k x_{kt} + \varepsilon_{t+1}, \tag{7.4}$$

where R_{t+1} is the one-period holding return on an stock index, such as FTSE or Dow Jones, defined by

$$R_{t+1} = (P_{t+1} + D_{t+1} - P_t)/P_t, \tag{7.5}$$

P_t is the stock price at the end of the period, D_{t+1} is the dividend paid out over the period t to $t+1$, and x_{it} , $i = 1, 2, \ldots, k$ are the factors/variables thought to be important in predicting stock returns. Finally, r_t^f is the return on the government bond with one-period to maturity (the period to maturity of the bond should be exactly the same as the holding period of the stock). $R_{t+1} - r_t^f$ is known as the excess return (return on stocks in excess of the return on the safe asset). Note also that r_t^f would be known to the investor/trader at the end of period t, before the price of stocks, P_{t+1}, is revealed at the end of period $t+1$.

Examples of possible stock market predictors are past changes in macroeconomic variables such as interest rates, inflation, dividend yield (D_t/P_{t-1}), price earnings ratio, output growth, and term premium (the difference in yield of a high grade and a low grade bond such as AAA rated minus BAA rated bonds).

For individual stocks the relevant stock market regression is the capital asset pricing model (*CAPM*), augmented with potential predictors:

$$R_{i,t+1} = a_i + b_{1i} x_{1t} + b_{2i} x_{2t} + \ldots + b_{ki} x_{kt} + \beta_i R_{t+1} + \varepsilon_{i,t+1}, \tag{7.6}$$

where $R_{i,t+1}$ is the holding period return on asset i (shares of firm i), defined similarly as R_{t+1}. The asset-specific regressions (7.6) could also include firm specific predictors, such as R_{it} or its higher-order lags, book-to-market value or size of firm i. Under market efficiency, as characterized by *CAPM*,

$$a_i = 0, b_{1i} = b_{2i} = \ldots = b_{ki} = 0,$$

and only the 'betas', β_i, will be significantly different from zero. Under *CAPM*, the value of β_i captures the risk of holding the share i with respect to the market.

7.6 Market efficiency and stock market predictability

It is often argued that if stock markets are efficient then it should not be possible to predict stock returns, namely that none of the variables in the stock market regression (7.4) should be statistically significant. Some writers have even gone so far as to equate stock market efficiency with the non-predictability property. But this line of argument is not satisfactory and does not help in furthering our understanding of how markets operate. The concept of market

efficiency needs to be defined separately from predictability. In fact, it is easily seen that stock market returns will be non-predictable only if market efficiency is combined with risk neutrality.

7.6.1 Risk-neutral investors

Suppose there exists a risk-free asset such as a government bond with a known payout. In such a case an investor with an initial capital of $\$A_t$, is faced with two options:

Option 1: hold the risk-free asset and receive

$$\$(1 + r_t^f)A_t,$$

at the end of the next period.

Option 2: switch to stocks by purchasing A_t/P_t shares, hold them for one period and expect to receive

$$\$ \left(A_t/P_t\right) \left(P_{t+1} + D_{t+1}\right),$$

at the end of period $t + 1$.

A risk-neutral investor will be indifferent between the certainty of $\$(1 + r_t^f)A_t$, and her/his expectations of the uncertain payout of option 2. Namely, for such a risk-neutral investor

$$(1 + r_t^f)A_t = E\left[\left(A_t/P_t\right) \left(P_{t+1} + D_{t+1}\right) |\Omega_t\right], \tag{7.7}$$

where Ω_t is the investor's information at the end of period t. This relationship is called the 'arbitrage condition'. Using (7.5) we now have

$$P_{t+1} + D_{t+1} = P_t \left(1 + R_{t+1}\right),$$

and the above arbitrage condition can be simplified to

$$E\left[\left(1 + R_{t+1}\right) |\Omega_t\right] = \left(1 + r_t^f\right),$$

or

$$E\left(R_{t+1} - r_t^f |\Omega_t\right) = 0. \tag{7.8}$$

This result establishes that if the investor forms his/her expectations of future stock (index) returns taking account of all market information efficiently, then the excess return, $R_{t+1} - r_t^f$, should not be predictable using any of the market information that are available at the end of period t. Notice that r_t^f is known at time t and is therefore included in Ω_t. Hence, under the joint hypothesis of market efficiency and risk neutrality we must have $E\left(R_{t+1} |\Omega_t\right) = r_t^f$.

The above set up can also be used to derive conditions under which asset prices can be characterized as a random walk model. Suppose, the risk-free rate, r_t^f, in addition to being known at time t, is also constant over time and given by r^f. Then using (7.7) we can also write

$$P_t = \left(\frac{1}{1+r^f}\right) E\left[(P_{t+1} + D_{t+1}) \,|\Omega_t\right],$$

or

$$P_t = \left(\frac{1}{1+r^f}\right) \left[E\left(P_{t+1} \,|\Omega_t\right) + E\left(D_{t+1} \,|\Omega_t\right)\right].$$

Under the rational expectations hypothesis and assuming that the 'transversality condition'

$$\lim_{j\to\infty} \left(\frac{1}{1+r^f}\right)^j E\left(P_{t+j} \,|\Omega_t\right) = 0,$$

holds we have the familiar result

$$P_t = \sum_{j=1}^{\infty} \left(\frac{1}{1+r^f}\right)^j E\left(D_{t+j} \,|\Omega_t\right), \tag{7.9}$$

that equates the level of stock price to the present discounted stream of the dividends expected to occur to the asset over the infinite future. The transversality condition rules out rational speculative bubbles and is satisfied if the asset prices are not expected to rise faster than the exponential decay rate determined by the discount factor, $0 < 1/(1+r^f) < 1$. It is now easily seen that if D_t follows a random walk so will P_t. For example, suppose

$$D_t = D_{t-1} + \varepsilon_t, \tag{7.10}$$

where ε_t is a white noise process. Then

$$E\left(D_{t+j} \,|\Omega_t\right) = D_t, \text{ for all } j,$$

and

$$P_t = \frac{D_t}{r^f}. \tag{7.11}$$

Therefore, using this result in (7.10) we also have

$$P_t = P_{t-1} + u_t, \tag{7.12}$$

where $u_t = \varepsilon_t / r^f$.

The random walk property holds even if $r^f = 0$, since in such a case it would be reasonable to expect no dividends to be paid out, namely $D_t = 0$. In this case the arbitrage condition becomes

$$E\left(P_{t+1}\,|\Omega_t\right) = P_t, \tag{7.13}$$

which is satisfied by the random walk model but is in fact more general than the random walk model. An asset price that satisfies (7.13) is a martingale process. Random walk processes with zero drift are martingale processes but not all martingale processes are random walks. For example, the price process

$$P_{t+1} = P_t + \lambda\left\{(\Delta P_{t+1})^2 - E\left[(\Delta P_{t+1})^2\,|\Omega_t\right]\right\} + \varepsilon_t,$$

where ε_t is a white noise process is a martingale process with respect to the information set Ω_t, but it is clearly not a random walk process, unless $\lambda = 0$. See Section 15.3.1 for a brief discussion of martingale processes.

Other modifications of the random walk theory are obtained if it is assumed that dividends follow a geometric random walk which is more realistic than the linear dividend model assumed in (7.10). In this case

$$D_{t+1} = D_t \exp(\mu_d + \sigma_d \nu_{t+1}), \tag{7.14}$$

where μ_d and σ_d are mean and standard deviation of the growth rate of the dividends. If it is further assumed that $\nu_{t+1}\,|\Omega_t$ is $N(0, 1)$, we have

$$E\left(D_{t+j}\,|\Omega_t\right) = D_t \exp\left(j\mu_d + \frac{1}{2}j\sigma_d^2\right).$$

Using this result in (7.9) now yields, assuming that $(1 + r^f)^{-1} \exp\left(\mu_d + \frac{1}{2}\sigma_d^2\right) < 1$,

$$P_t = \frac{D_t}{\rho}, \tag{7.15}$$

where

$$\rho = (1 + r^f) \exp\left(-\mu_d - \frac{1}{2}\sigma_d^2\right) - 1.$$

The condition $(1 + r^f)^{-1} \exp\left(\mu_d + \frac{1}{2}\sigma_d^2\right) < 1$ ensures that the infinite sum in (7.9) is convergent and $\rho > 0$. Under this set up $\ln(P_t) = \ln(D_t) - \ln(\rho)$, and

$$\ln(P_t) = \ln(P_{t-1}) + \mu_d + \sigma_d \nu_t, \tag{7.16}$$

which establishes that in this case it is log prices that follow the random walk model. This is a special case of the statistical model of return, (7.1), discussed in Section 7.3, where $\mu_t = \mu_d$, and $\sigma_t = \sigma_d$.

There are, however, three different types of empirical evidence that shed doubt on the empirical validity of the present value model under risk neutrality.

1. The model predicts a constant price-dividend ratio for a large class of the dividend processes, two prominent examples being the linear and the geometric random walk models, (7.10) and (7.14), discussed above. For more general dividend processes the price-dividend ratio,

$\rho_t = P_t/D_t$, could be time varying, but it must be mean-reverting, in the sense that shocks to prices and dividends must eventually cancel out. In reality, the price-dividend ratio varies considerably over time, shows a high degree of persistent, and in general it is not possible to reject the hypothesis that the processes for ρ_t or $\ln(\rho_t)$ contain a unit root. For the Shiller data discussed in 7.4.2 the autocorrelation coefficient of the log dividend to price ratio computed over the period 1871m1 to 2009m9 is 0.994 (0.024) and falls very gradually as its order is increased and amounts to 0.879 (0.111) at the lag order 12. Formal tests of unit root hypothesis are discussed in Chapter 15.

2. We have already established that under risk neutrality excess returns must not be predictable (see equation (7.8)). Yet there is ample evidence of excess return predictability at least in periods of high market volatility. For example, it is possible to explain 15 per cent of the variations in monthly excess returns on S&P 500 over the period 1872m2–2009m9 by running a linear regression of the excess return on a constant and its 12 lagged values—namely by a univariate $AR(12)$ process. This figure rises to 19 per cent if we exclude the 1929 stock market crash and focus on the post 1948 period. See also the references cited in Section 7.7.1. Formal tests of unit root hypothesis are discussed in Chapter 15.

3. To derive the geometric random walk model of asset prices, (7.16), from the present value model under risk neutrality, we have assumed that innovations to the dividend process are normally distributed. This implies that innovations to asset returns must also be normally distributed. But the empirical evidence discussed in Section 7.4 above clearly shows that innovations to asset returns tend to be fat-tailed, and often significantly depart from normality. This anomaly between the theory and the evidence is also difficult to reconcile. Under the present value model prices will have fat-tailed innovations only if the dividends that drive asset prices are also fat-tailed. But under the geometric random walk model for dividends (7.14), $E\left(D_{t+j} \mid \Omega_t\right)$ need not exist if the dividend innovations, ν_t, are fat-tailed. One important example arises when ν_t has the Student t-distribution as defined by (7.3). For the derivation of the present value expression in this case we need $E(\exp(\sigma_d \nu_{t+j}))$, which is the moment generating function of ν_{t+j} evaluated at σ_d. But the Student t-distribution does not have a moment generating function, and hence the present value formula cannot be computed when innovations to the dividends are t distributed.

7.6.2 Risk-averse investors

In addition to the above documented empirical shortcomings, it is also important to note that risk neutrality is a behavioural assumption and need not hold even if all market information is processed efficiently by all the market participants. A more reasonable way to proceed is to allow some or all of the investors to be risk averse. In this more general case the certain pay out, $(1 + r_t^f)A_t$, and the expectations of the uncertain pay out, $E\left[(A_t/P_t)\left(P_{t+1} + D_{t+1}\right) \mid \Omega_t\right]$, will not be the same and differ by a (possibly) time varying risk premium which could also vary with the level of the initial capital, A_t. More specifically, we have

$$E\left[(A_t/P_t)\left(P_{t+1} + D_{t+1}\right) \mid \Omega_t\right] = \left(1 + r_t^f\right) A_t + \lambda_t A_t,$$

where λ_t is the premium per \$ of invested capital required (expected) by the investor. It is now easily seen that

$$E\left(R_{t+1} - r_t^f \mid \Omega_t\right) = \lambda_t,$$

and it is no longer necessarily true that under market efficiency excess returns are non-predictable. The extent to which excess returns can be predicted will depend on the existence of a historically stable relationship between the risk premium, λ_t, and the macro and business cycle indicators such as changes in interest rates, dividends and various business cycle indicators.

In the context of the consumption capital asset pricing model, λ_t is determined by the *ex ante* correlation of excess returns and changes in the marginal utility of consumption. In the case of a representative consumer with the single period utility function, $u(c_t)$, the first-order inter-temporal optimization condition (the Euler equation) is given by

$$E\left[\left(R_{t+1} - r_t^f\right)\frac{u'(c_{t+1})}{u'(c_t)}\,|\Omega_t\right] = 0, \tag{7.17}$$

where c_t denotes the consumer's real consumption in period t. Using the above condition it is now easily seen that[5]

$$\lambda_t = -\frac{Cov\left[R_{t+1}, \frac{u'(c_{t+1})}{u'(c_t)}\,|\Omega_t\right]}{E\left[\frac{u'(c_{t+1})}{u'(c_t)}\,|\Omega_t\right]} = -\frac{Cov\left[R_{t+1}, u'(c_{t+1})\,|\Omega_t\right]}{E\left[u'(c_{t+1})\,|\Omega_t\right]}.$$

For a power utility function, $u(c_t) = (c_t^{1-\gamma} - 1)/(1-\gamma)$, we have $u'(c_{t+1})/u'(c_t) = \exp(-\gamma\Delta \ln(c_{t+1}))$, where $\gamma > 0$ is the coefficient of relative risk aversion. In this case λ_t is given by

$$\lambda_t = \frac{-Cov\left\{R_{t+1}, \exp\left[-\gamma\Delta \ln(c_{t+1})\right]\,|\Omega_t\right\}}{E\left\{\exp\left[-\gamma\Delta \ln(c_{t+1})\right]\,|\Omega_t\right\}}. \tag{7.18}$$

This result shows that the risk premium depends on the covariance of asset returns with the marginal utility of consumption. The premium demanded by the investor to hold the stock is higher if the return on the asset co-varies positively with consumption. The extent of this co-variation depends on the magnitude of the risk aversion coefficient γ. For plausible values of γ (in the range 1 to 3) and historically observed values of the consumption growth, we would expect λ_t to be relatively small, below 1 per cent per annum. However, using annual observations over relatively long periods one obtains a much larger estimate for λ_t. This was first pointed out by Mehra and Prescott (1985) who found that in the 90 years from 1889 to 1978 the average estimate of λ_t in fact amounted to 6.18 per cent per annum, which could only be reconciled with the theory if one was prepared to consider an implausibly large value for the relative risk aversion coefficient (in the regions of 30 or 40). The large discrepancy between the historical estimate of λ_t based on $R_{t+1} - r_t^f$, and the theory-consistent estimate of λ_t based on (7.18) is known as the 'equity premium puzzle'. There have been many attempts in the literature to resolve the puzzle by modifications to the utility function, attitudes towards risk, allowing for the possibility of rare

[5] Let $X_{t+1} = R_{t+1} - r_t^f$ and $Y_{t+1} = u'(c_{t+1})/u'(c_t)$, and write the Euler equation (7.17) as

$$E\left(X_{t+1}Y_{t+1}\,|\Omega_t\right) = 0 = Var\left(X_{t+1}Y_{t+1}\,|\Omega_t\right) + E\left(X_{t+1}\,|\Omega_t\right)E\left(Y_{t+1}\,|\Omega_t\right),$$

then the required results follow immediately, also noting that r_t^f is known at time t and hence has a zero correlation with $u'(c_{t+1})/u'(c_t)$.

events, and the heterogeneity in asset holdings and preferences across consumers. For reviews see Kocherlakota (2003) and Mehra and Prescott (2003).

But even if the mean discrepancy between $E\left(R_{t+1} - r_t^f | \Omega_t\right)$ and λ_t as given by (7.18) is resolved, the differences in the higher moments of historically and theory-based risk premia are likely to be important empirical issues of concern. It seems difficult to reconcile the high volatility of excess returns with the low volatility of consumption growth that is observed historically.

7.7 Return predictability and alternative versions of the efficient market hypothesis

In his 1970 review, Fama distinguishes between three different forms of the *EMH*:

(a) The *weak* form asserts that all price information is fully reflected in asset prices, in the sense that current price changes cannot be predicted from past prices. This weak form was also introduced in an unpublished paper by Roberts (1967).

(b) The *semi-strong* form that requires asset price changes to fully reflect all publicly available information and not only past prices.

(c) The *strong* form that postulates that prices fully reflect information even if some investor or group of investors have monopolistic access to some information.

Fama regarded the strong form version of the *EMH* as a benchmark against which the other forms of market efficiencies are to be compared. With respect to the weak form version he concluded that the test results strongly support the hypothesis, and considered the various departures documented as economically unimportant. He reached a similar conclusion with respect to the semi-strong version of the hypothesis; although as he noted, the empirical evidence available at the time was rather limited and far less comprehensive as compared to the evidence on the weak version.

The three forms of the *EMH* present different degrees whereby public and private information are revealed in transaction prices. It is difficult to reconcile all the three versions to the mainstream asset pricing theory, and as we shall see in Section 7.7.1 a closer connection is needed between market efficiency and the specification of the model economy that underlies it.

7.7.1 Dynamic stochastic equilibrium formulations and the joint hypothesis problem

Evidence on the semi-strong form of the *EMH* was revisited by Fama in a second review of the efficient capital markets published in 1991. By then it was clear that the distinction between the weak and the semi-strong forms of the *EMH* was redundant. The random walk model could not be maintained either—in view of more recent studies, in particular that of Lo and MacKinlay (1988).

A large number of studies in the finance literature had confirmed that stock returns over different horizons (days, weeks, and months) can be predicted to some degree by means of interest rates, dividend yields and a variety of macroeconomic variables exhibiting clear business cycle variations. A number of studies also showed that returns tend to be more predictable the longer

the forecast horizon. While the vast majority of these studies had looked at the US stock market, an emerging literature has also considered the UK stock market. US studies include Balvers, Cosimano, and MacDonald (1990), Breen, Glosten, and Jagannathan (1989), Campbell (1987), Fama and French (1989), and more recently Ferson and Harvey (1993), Kandel and Stambaugh (1996), Pesaran and Timmermann (1994, 1995). See Granger (1992) for a survey of the methods and results in the literature. UK studies after 1991 included Clare, Thomas, and Wickens (1994), Clare, Psaradakis, and Thomas (1995), Black and Fraser (1995), and Pesaran and Timmermann (2000).

Theoretical advances over Samuelson's seminal paper by Leroy (1973), Rubinstein (1976), and Lucas (1978) also made it clear that in the case of risk-averse investors tests of predictability of excess returns could not on their own confirm or falsify the *EMH*. The neoclassical theory cast the *EMH* in the context of dynamic stochastic general equilibrium models and showed that excess returns weighted by marginal utility could be predictable. Only under risk neutrality, where marginal utility was constant, the equilibrium condition implied the non-predictability of excess returns.

As Fama (1991) noted in his second review, the test of the *EMH* involved a joint hypothesis—market efficiency and the underlying equilibrium asset pricing model. He concluded that 'Thus, market efficiency per se is not testable'. (see p. 1575). This, did not, however, mean that market efficiency was not a useful concept; almost all areas of empirical economics are subject to the joint hypotheses problem.

7.7.2 Information and processing costs and the *EMH*

The *EMH*, in the sense of asset 'prices fully reflect all available information', was also criticized by Grossman and Stiglitz (1980) who pointed out that there must be 'sufficient profit opportunities, i.e. inefficiencies, to compensate investors for the cost of trading and information-gathering'.

Only in the extreme and unrealistic case where all information processing and trading costs are zero would one expect prices to fully reflect all available information. But if information acquisition were in fact costless it would have been known even before market prices are established.

As Fama recognized, a weaker and economically more sensible version of the efficiency hypothesis would be needed, namely 'prices reflect information to the point where the marginal benefits of acting on information (the profits to be made) do not exceed the marginal costs'. This in turn makes the task of testing the market efficiency even more complicated and would require equilibrium asset pricing models that allowed for information and trading costs in markets with many different traders and with non-convergent beliefs.

In view of these difficulties, some advocates of the *EMH* have opted for a trade-based notion, and define markets as efficient if it would not be possible for investors 'to earn above-average returns without accepting above-average risks' (Malkiel (2003, p. 60)). This notion can take account of information and transaction costs and does not involve testing joint hypotheses. But this is far removed from the basic idea of markets as efficient allocators of capital investment across countries, industries, and firms.

Beating the market as a test of market efficiency also poses new challenges. Whilst it is certainly possible to construct trading strategies (inclusive of transaction costs) with Sharpe ratios that exceed those of the market portfolios *ex post*, such evidence is unlikely to be convincing to advocates of the *EMH*. It could be argued that they are carried out with the benefit of

hindsight, and are unlikely to be repeated in real time. In this connection, the following considerations would need to be born in mind:

(a) Data mining/data snooping (Pesaran and Timmermann (2005a))
(b) Structural change and model instability (choice of observation window)
(c) The positive relationship that seems to exist between transaction costs and predictability
(d) Market volatility and learning
(e) The 'beat the market' test is not that helpful either in shedding light on the nature and the extent of market inefficiencies. A more structural approach would be desirable.

7.8 Theoretical foundations of the *EMH*

At the core of the *EMH* lie the following three basic premises:

1. *Investor rationality*: it is assumed that investors are rational, in the sense that they correctly update their beliefs when new information is available.
2. *Arbitrage*: individual investment decisions satisfy the arbitrage condition, and trade decisions are made guided by the calculus of the subjective expected utility theory *à la* Savage.
3. *Collective rationality*: differences in beliefs across investors cancel out in the market.

To illustrate how these premises interact, suppose that at the start of period (day, week, month) t there are N_t traders (investors) that are involved in an act of arbitrage between a stock and a safe (risk-free) asset. Denote the one-period holding returns on these two assets by R_{t+1} and r_t^f, respectively. Following a similar line of argument as in section 7.6.2, the arbitrage condition for trader i is given by

$$\hat{E}_i\left(R_{t+1} - r_t^f \,|\Omega_{it}\right) = \lambda_{it} + \delta_{it},$$

where $\hat{E}_i\left(R_{t+1} - r_t^f \,|\Omega_{it}\right)$ is his/her subjective expectations of the excess return, $R_{t+1} - r_t^f$ taken with respect to the information set

$$\Omega_{it} = \Psi_{it} \cup \Phi_t,$$

where Φ_t is the component of the information which is publicly available, $\lambda_{it} > 0$ represents trader's risk premium, and $\delta_{it} > 0$ is her/his information and trading costs per unit of funds invested. In the absence of information and trading costs, λ_{it} can be characterized in terms of the trader's utility function, $u_i(c_{it})$, where c_{it} is his/her real consumption expenditures during the period t to $t+1$, and is given by

$$\lambda_{it} = \hat{E}_i\left(R_{t+1} - r_t^f \,|\Omega_{it}\right) = \frac{-\widehat{\text{Cov}}_i\left(m_{i,t+1}, R_{t+1} \,|\Omega_{it}\right)}{\hat{E}_i\left(m_{i,t+1} \,|\Omega_{it}\right)},$$

where $\hat{Cov}_i(.|\Omega_{it})$ is the subjective covariance operator conditional on the trader's information set, Ω_{it}, $m_{i,t+1} = \beta_i u_i'(c_{i,t+1})/u_i'(c_{it})$, which is known as the 'stochastic discount factor', $u_i'(.)$ is the first derivative of the utility function, and β_i is his/her discount factor.

The expected returns could differ across traders due to the differences in their perceived conditional probability distribution function of $R_{t+1} - r_t^f$, the differences in their information sets, Ω_{it}, the differences in their risk preferences, and/or endowments. Under the rational expectations hypothesis

$$\hat{E}_i\left(R_{t+1} - r_t^f | \Omega_{it}\right) = E\left(R_{t+1} - r_t^f | \Omega_{it}\right),$$

where $E\left(R_{t+1} - r_t^f | \Omega_{it}\right)$ is the 'true' or 'objective' conditional expectations. Furthermore, in this case

$$E\left[\hat{E}_i\left(R_{t+1} - r_t^f | \Omega_{it}\right) | \Phi_t\right] = E\left[E\left(R_{t+1} - r_t^f | \Omega_{it}\right) | \Phi_t\right],$$

and since $\Phi_t \subset \Omega_{it}$ we have

$$E\left[\hat{E}_i\left(R_{t+1} - r_t^f | \Omega_{it}\right) | \Phi_t\right] = E\left(R_{t+1} - r_t^f | \Phi_t\right).$$

Therefore, under the *REH*, taking expectations of the individual arbitrage conditions with respect to the public information set yields

$$E\left(R_{t+1} - r_t^f | \Phi_t\right) = E\left(\lambda_{it} + \delta_{it} | \Phi_t\right),$$

which also implies that $E(\lambda_{it} + \delta_{it} | \Phi_t)$ *must* be the same across all *i*, or

$$E\left(R_{t+1} - r_t^f | \Phi_t\right) = E\left(\lambda_{it} + \delta_{it} | \Phi_t\right) = \rho_t, \text{ for all } i,$$

where ρ_t is an average market measure of the combined risk premia and transaction costs. The *REH* combined with perfect arbitrage ensures that different *traders* have the same expectations of $\lambda_{it} + \delta_{it}$. Rationality and market discipline override individual differences in tastes, information processing abilities and other transaction related costs and renders the familiar representative agent arbitrage condition:

$$E\left(R_{t+1} - r_t^f | \Phi_t\right) = \rho_t. \tag{7.19}$$

This is clearly compatible with trader-specific λ_{it} and δ_{it}, so long as

$$\lambda_{it} = \lambda_t + \varepsilon_{it},\ E(\varepsilon_{it} | \Phi_t) = 0,$$
$$\delta_{it} = \delta_t + \upsilon_{it}, E(\upsilon_{it} | \Phi_t) = 0,$$

where ε_{it} and υ_{it} are distributed with mean zero independently of Φ_t, and λ_t and δ_t are known functions of the publicly available information.

Under this setting, the extent to which excess returns can be predicted will depend on the existence of a historically stable relationship between the risk premium, λ_t, and the macro and business cycle indicators such as changes in interest rates, dividends, and a number of other indicators.

The rational expectations hypothesis is rather extreme which is unlikely to hold at all times in all markets. Even if one assumes that in financial markets learning takes place reasonably fast, there will still be periods of turmoil where market participants will be searching in the dark, trying and experimenting with different models of $R_{t+1} - r_t^f$ often with marked departures from the common rational outcomes, given by $E\left(R_{t+1} - r_t^f \,|\Phi_t\right)$.

Herding and correlated behaviour across some of the traders could also lead to further departures from the equilibrium *RE* solution. In fact, the objective probability distribution of $R_{t+1} - r_t^f$ might itself be affected by market transactions based on subjective estimates $\hat{E}_i\left(R_{t+1} - r_t^f \,|\Omega_{it}\right)$.

Market inefficiencies provide further sources of stock market predictability by introducing a wedge between a 'correct' *ex ante* measure $E\left(R_{t+1} - r_t^f \,|\Phi_t\right)$, and its average estimate by market participants, which we write as $\sum_{i=1}^{N_t} w_{it}\hat{E}_i\left(R_{t+1} - r_t^f \,|\Omega_{it}\right)$, where w_{it} is the market share of the ith trader. Let

$$\bar{\xi}_{wt} = \sum_{i=1}^{N_t} w_{it}\hat{E}_i\left(R_{t+1} - r_t^f \,|\Omega_{it}\right) - E\left(R_{t+1} - r_t^f \,|\Phi_t\right),$$

and note that it can also be written as (since $\sum_{i=1}^{N_t} w_{it} = 1$)

$$\bar{\xi}_{wt} = \sum_{i=1}^{N_t} w_{it}\xi_{it}, \tag{7.20}$$

where

$$\xi_{it} = \hat{E}_i\left(R_{t+1} - r_t^f \,|\Omega_{it}\right) - E\left(R_{t+1} - r_t^f \,|\Phi_t\right), \tag{7.21}$$

measures the degree to which individual expectations differs from the correct (but unobservable) expectations, $E\left(R_{t+1} - r_t^f \,|\Phi_t\right)$. A non-zero ξ_{it} could arise from individual irrationality, but not necessarily so. Rational individuals faced with an uncertain environment, costly information and limitations on computing power could rationally arrive at their expectations of future price changes that with hindsight differ from the correct ones.[6] A non-zero ξ_{it} could also arise due to disparity of information across traders (including information asymmetries), and heterogeneous priors due to model uncertainty or irrationality. Nevertheless, despite such individual deviations, $\bar{\xi}_{wt}$ which measures the extent of market or collective inefficiency, could be quite negligible. When N_t is sufficiently large, individual 'irrationality' can cancel out at the level of

[6] This is in line with the premise of the recent paper by Angeletos, Lorenzoni, and Pavan (2010) who maintain the axiom of rationality, but allow for dispersed information and the possibility of information spillovers in the financial markets to explain market inefficiencies.

the market, so long as ξ_{it}, $i = 1, 2, \ldots, N_t$, are not cross-sectionally strongly dependent, and no single trader dominates the market, in the sense that $w_{it} = O(N_t^{-1})$ at any time.[7] Under these conditions at each point in time, t, the average expected excess returns across the individual traders converges in quadratic means to the expected excess return of a representative trader, namely we have

$$\sum_{i=1}^{N_t} w_{it} \hat{E}_i \left(R_{t+1} - r_t^f \,|\Omega_{it} \right) \overset{q.m.}{\to} E \left(R_{t+1} - r_t^f \,|\Phi_t \right), \text{as } N_t \to \infty.$$

In such periods the representative agent paradigm would be applicable, and predictability of excess return will be governed solely by changes in business cycle conditions and other publicly available information.[8]

However, in periods where traders' individual expectations become strongly correlated (say as the result of herding or common over-reactions to distressing news), $\bar{\xi}_{wt}$ need not be negligible even in thick markets with many traders; and market inefficiencies and profitable opportunities could prevail. Markets could also display inefficiencies without exploitable profitable opportunities if $\bar{\xi}_{wt}$ is non-zero but there is no stable predictable relationship between $\bar{\xi}_{wt}$ and business cycle or other variables that are observed publicly.

The evolution and composition of $\bar{\xi}_{wt}$ can also help in shedding light on possible bubbles or crashes developing in asset markets. Bubbles tend to develop in the aftermath of technological innovations that are commonly acknowledged to be important, but with uncertain outcomes. The emerging common beliefs about the potential advantages of the new technology and the difficulties individual agents face in learning how to respond to the new investment opportunities can further increase the gap between average market expectations of excess returns and the associated objective rational expectations outcome. Similar circumstances can also prevail during a crash phase of the bubble when traders tend to move in tandem trying to reduce their risk exposures all at the same time. Therefore, one would expect that during bubbles and crashes the individual errors, ξ_{it}, to become more correlated, such that the average errors, $\bar{\xi}_{wt}$, are no longer negligible. In contrast, at times of market calm the individual errors are likely to be weakly correlated, with the representative agent rational expectations model being a reasonable approximation.

More formally note that since r_t^f and P_t are known at time t, then

$$\xi_{it} = \hat{E}_i \left(\frac{P_{t+1} + D_{t+1}}{P_t} \,|\Omega_{it} \right) - E \left(\frac{P_{t+1} + D_{t+1}}{P_t} \,|\Phi_t \right).$$

Also to simplify the exposition assume that the length of the period t is sufficiently small so that dividends are of secondary importance and

$$\xi_{it} \approx \hat{E}_i \left[\Delta \ln(P_{t+1}) \,|\Omega_{it} \right] - f_t,$$

[7] Concepts of weak and strong cross-sectional dependence are defined and discussed in Chudik, Pesaran, and Tosetti (2011). See also Chapter 29.

[8] The heterogeneity of expectations across traders can also help in explaining large trading volume observed in the financial markets, a feature which has proved difficult to explain in representative agent asset pricing models. But see Scheinkman and Xiong (2003), who relate the occurrence of bubbles and crashes to changes in trading volume.

where $f_t = E[\Delta \ln(P_{t+1}) | \Phi_t]$ is the unobserved price change expectations. Individual deviations, ξ_{it}, could then become strongly correlated if individual expectations $\hat{E}_i[\Delta \ln(P_{t+1}) | \Omega_{it}]$ differ systematically from f_t. For example, suppose that

$$\hat{E}_i[\Delta \ln(P_{t+1}) | \Omega_{it}] = \theta_{it} \Delta \ln(P_t),$$

but $f_t = 0$, namely in the absence of heterogeneous expectations $\Delta \ln(P_{t+1})$ would be unpredictable with a zero mean. Then it is easily seen that $\bar{\xi}_{wt} = \bar{\theta}_{wt} \Delta \ln(P_t)$, where $\bar{\theta}_{wt} = \Sigma_{i=1}^{N_t} w_{it} \theta_{it}$. It is clear that $\bar{\xi}_{wt}$ need not converge to zero if in period t the majority of market participants believe future price changes are positively related to past price changes, so that $\lim_{N_t \to \infty} \bar{\theta}_{wt} > 0$. In this simple example price bubbles or crashes occur when $\bar{\theta}_{wt}$ becomes positive over a relatively long period.

It should be clear from the above discussion that testing for price bubbles requires disaggregated time series information on individual beliefs and unobserved price change expectations, f_t. Analysis of aggregate time series observations can provide historical information about price reversals and some of their proximate causes. But such information is unlikely to provide conclusive evidence of bubble formation and its subsequent collapse. Survey data on traders' individual beliefs combined with suitable market proxies for f_t are likely to be more effective in empirical analysis of price bubbles.

An individual investor could be asked to respond to the following two questions regarding the current and future price of a given asset:

1. Do you believe the current price is (a) just right (in the sense that the price is in line with market fundamentals), (b) is above the fundamental price, or (c) is below the fundamental price?
2. Do you expect the market price next period to (a) stay about the level it is currently, (b) fall, or (c) rise?

In cases where the market is equilibrating we would expect a close association between the proportion of respondents who select 1a and 2a, 1b and 2b, and 1c and 2c. But in periods of bubbles (crashes) one would expect a large proportion of respondents who select 1b (1c) to also select 2c (2b).

In situations where the equilibrating process is well established and commonly understood, the second question is redundant. For example, if an individual states that the room temperature is too high, it will be understood that he/she would prefer less heating. The same is not applicable to financial markets and hence responses to both questions are needed for a better understanding of the operations of the markets and their evolution over time.

7.9 Exploiting profitable opportunities in practice

In financial markets the *EMH* is respected but not worshipped. It is recognized that markets are likely to be efficient most of the time but not all the time. Inefficiencies could arise particularly during periods of important institutional and technological changes. It is not possible to know when and where market inefficiencies arise in advance—but it is believed that they will arise from time to time. Market traders love volatility as it signals news and change with profit possibilities

to exploit. Identification of exploitable predictability tends to be fully diversified across markets for bonds, equities and foreign exchange. Misalignments across markets for different assets and in different countries often present the most important opportunities. Examples include statistical arbitrage and global macro arbitrage trading rules.

Predictability and market liquidity are often closely correlated; less liquid markets are likely to be more predictable. Market predictability and liquidity need to be jointly considered in developing profitable trading strategies. Return forecasting models used in practice tend to be recursive and adaptive along the lines developed in Pesaran and Timmermann (1995) and recently reviewed in Pesaran and Timmermann (2005a). The recursive modelling (*RM*) approach is also in line with the more recent developments in behavioural finance. The *RM* approach aims at minimizing the effect of hindsight and data snooping (a problem that afflicts all *ex post* return regressions), and explicitly designed to take account of the potential instability of return regressions over time. For example, Pesaran and Timmermann (1995) find that the switching trading rule manages to beat the market only during periods of high volatility where learning might be incomplete and markets inefficient.

Pesaran and Timmermann (2005a) provide a review of the recursive modelling approach, its use in derivation of trading rules and discuss a number of practical issues in their implementation such as the choice of the universe of factors over which to search, choice of the estimation window, how to take account of measurement and model uncertainty, how to cross validate the *RM*, and how and when to introduce model innovations.

The *RM* approach still faces many challenges ahead. As Pesaran and Timmermann (2005a, p. 229) conclude:

> Automated systems reduce, but do not eliminate the need for discretion in real time decision making. There are many ways that automated systems can be designed and implemented. The space of models over which to search is huge and is likely to expand over time. Different approximation techniques such as genetic algorithms, simulated annealing and MCMC algorithms can be used. There are also many theoretically valid model selection or model averaging procedures. The challenge facing real time econometrics is to provide insight into many of these choices that researchers face in the development of automated systems.

Return forecasts need to be incorporated in sound risk management systems. For this purpose point forecasts are not sufficient and joint probability forecast densities of a large number of interrelated asset returns will be required. Transaction and slippage costs need to be allowed for in the derivation of trading rules. Slippage arises when long (short) orders, optimally derived based on currently observed prices, are placed in rising (falling) markets. Slippage can be substantial, and is in addition to the usual transactions costs.

Familiar risk measures such as the Sharpe ratio and the *VaR* are routinely used to monitor and evaluate the potential of trading systems. But due to cash constraint (for margin calls, etc.) it is large drawdowns that are most feared. Prominent recent examples are the downfall of Long Term Capital who experienced substantial drawdowns in 1998 following the Russian financial crisis, and the collapse of Lehman Brothers during the global financial crisis of 2008.

Successful traders might not be (and usually are not) better in forecasting returns than many others in the market. What they have is a sense of 'big' opportunities when they are confident of making a 'kill'.

7.10 New research directions and further reading

We have identified two important sources of return predictability and possible profitable opportunities. One relates to the familiar business cycle effects and involves modelling ρ_t, defined by (7.19), in terms of the publicly available information, Φ_t. The second relates to the average deviations of individual traders's expectations from the 'correct' unknown expectations, as measured by $\bar{\xi}_{wt}$ and defined by (7.20). As noted earlier, this component could vary considerably over time and need not be related to business cycle factors. It tends to be large during periods of financial crisis when correlation of mis-pricing across traders rises, and tends to be negligible during periods of market calm when correlations are low. Over the past three decades much of the research in finance and macroeconomics has focussed on modelling of ρ_t, and by comparison little attention has been paid to $\bar{\xi}_{wt}$. This is clearly an important area for future research. Our discussions also point to a number of related areas for further research. There are

- limits to rational expectations (for an early treatment see Pesaran (1987c); see also the recent paper on survey expectations by Pesaran and Weale (2006)).
- limits to arbitrage due to liquidity requirements and institutional constraints.
- herding and correlated behaviour with noise traders entering markets during bull periods and deserting during bear periods.

Departures from the *EMH* listed above are addressed by behavioural finance, complexity theory, and the Adaptive Markets Hypothesis recently advocated by Lo (2004). Some of the recent developments in behavioural finance are reviewed in Baberis and Thaler (2003). Farmer and Lo (1999) focus on recent research that views the financial markets from a biological perspective and, specifically, within an evolutionary framework in which markets, instruments, institutions, and investors interact and evolve dynamically according to the 'law' of economic selection. Under this view, financial agents compete and adapt, but they do not necessarily do so in an optimal fashion.

Special care should also be exercised in evaluation of return predictability and trading rules. To minimize the effects of hindsight in such analysis recursive modelling techniques discussed in Pesaran and Timmermann (1995, 2000, 2005a) seem much more appropriate than the return regressions on a fixed set of regressors/factors that are estimated *ex post* on historical data.

7.11 Exercises

1. The file FUTURESDATA.fit, provided in Microfit 5, contains daily returns on a number of equity index futures, currencies and government bonds. Use this data set to compute skewness and kurtosis coefficients for daily returns on different assets over the periods before and after 2000. Examine if your results are qualitatively affected by which sub-period is considered.
2. The file UKUS.fit, provided in Microfit 5, contains monthly observations on UK and US economies. Using the available data, investigate the extent to which stock markets in UK and US could have been predicted during 1990s.

3. Consider the present value expression for the asset price P_t

$$P_t = \sum_{j=1}^{\infty} \left(\frac{1}{1+r} \right)^j E\left(D_{t+j} \,|\Omega_t \right),$$

where $r > 0$, and suppose that D_t follows the $AR(2)$ process

$$D_t - \mu = \phi_1 \left(D_{t-1} - \mu \right) + \phi_2 \left(D_{t-2} - \mu \right) + v_t,\ v_t \sim IID(0, \sigma_v^2).$$

(a) Show that

$$P_t = \mu / r + \theta_1 \left(D_t - \mu \right) + \theta_2 \left(D_{t-1} - \mu \right),$$

where

$$\theta_1 = \frac{\phi_1 \beta + \beta^2 \phi_2}{1 - \beta \phi_1 - \beta^2 \phi_2},\ \ \theta_1 = \frac{\phi_2 \beta}{1 - \beta \phi_1 - \beta^2 \phi_2},$$

and $\beta = 1/(1+r)$.

(b) Suppose now that

$$\log(D_t) = \delta(1-\rho) + \rho \log(D_{t-1}) + u_t,\ u_t \sim IIDN(0, \sigma_u^2).$$

Show that

$$E\left(D_{t+j} \,|\Omega_t \right) = \exp\left[\delta(1-\rho^j) \right] \exp\left[\rho^j \log(D_t) \right] \exp\left[\frac{0.5\sigma_u^2 \left(1 - \rho^{2j} \right)}{1 - \rho^2} \right],$$

and hence or otherwise derive the price equation for this process and establish conditions under which the price equation exists. In particular, consider the case where $\rho = 1$.

(c) How would you go about testing the validity of the above price equations?

4. Consider an investor who wishes to allocate the fractions $\mathbf{w}_t = (w_{t1}, w_{t2}, \ldots, w_{tn})'$ of his/her wealth at time t to n risky assets and the remainder to the risk-free asset.

(a) Show that the portfolio return, ρ_{t+1}, is given by

$$\rho_{t+1} = \mathbf{w}_t' \mathbf{r}_{t+1} + (1 - \boldsymbol{\tau}' \mathbf{w}_t) r_f,$$

where $\mathbf{r}_{t+1}$ is an $n \times 1$ vector of returns on the risky assets, r_f is the return on the safe asset, and $\boldsymbol{\tau}$ is an n-dimensional vector of ones.

(b) Suppose that $E(\mathbf{r}_{t+1}|\mathcal{I}_t) = \mu_t$ and $Var(\mathbf{r}_{t+1}|\mathcal{I}_t) = \Sigma_t$, where $\mathcal{I}_t$ is an information set that contains $\mathbf{r}_t$ and its lagged values. Derive $\mathbf{w}_t$ such that $Var\left(\rho_{t+1}|\mathcal{I}_t\right)$ is minimized subject to $E\left(\rho_{t+1}|\mathcal{I}_t\right) = \bar{\mu}_\rho > 0$.

(c) Under the same assumptions as above, now derive $\mathbf{w}_t$ such that $E\left(\rho_{t+1}|\mathcal{I}_t\right)$ is maximized subject to $Var\left(\rho_{t+1}|\mathcal{I}_t\right) = \bar{\sigma}_\rho^2 > 0$.

(d) Compare your answers under (b) and (c) and discuss the results in the light of investor's degree of risk aversion.

Part II

Statistical Theory

8 Asymptotic Theory

8.1 Introduction

Most econometric methods used in applied economics, particularly in time series econometrics, are asymptotic in the sense that they are likely to hold only when the sample size is 'large enough'. In this chapter we briefly review the different concepts of asymptotic convergence used in mathematical statistics and discuss their applications to econometric problems.

8.2 Concepts of convergence of random variables

Consider a sequence of random variables $x_1(\omega), x_2(\omega), \ldots$ defined on the probability space $(\Omega, \mathcal{F}, P)$, where Ω is the sample space with ω representing a point in this space, $\mathcal{F}$ is the event space (defined here as a σ field of subsets of Ω), and P is a probability distribution defined on $\mathcal{F}$. The random variable $x(\omega)$ is a transformation of Ω onto the real line. The sequence of random variables $x_1(\omega), x_2(\omega), \ldots$ is usually denoted by $x_1, x_2, \ldots$ or simple by $\{x_t\}, t = 1, 2 \ldots$.

A sequence of random variables, assuming that it converges, can either converge to a constant or to a random variable. In the case where $\{x_t\}$ converges to a random variable, say x, the distribution function of x is said to asymptotically approximate that of x_t. Three modes of convergence (of random variables) are distinguished in the statistical literature. These are 'convergence in probability', 'convergence with probability 1' or sure convergence, and 'convergence in s-th mean'.

8.2.1 Convergence in probability

Definition 1 *Let $x_1, x_2, \ldots$, and x be random variables defined on a probability space $(\Omega, \mathcal{F}, P)$, then $\{x_t\}$ is said to converge in probability to x if*

$$\lim_{t \to \infty} \Pr(|\, x_t - x \,| < \epsilon) = 1, \text{ for any } \epsilon > 0. \tag{8.1}$$

This mode of convergence is also often denoted by

$$x_t \xrightarrow{p} x,$$

and when x is a fixed constant it is referred to as the probability limit of x_t, written as $Plim(x_t) = x$, as $t \to \infty$. The above concept is readily extended to multivariate cases where $\{\mathbf{x}_t, t = 1, 2 \ldots\}$ denote mdimensional vectors of random variables. Condition (8.1) should now be replaced by

$$\lim_{t\to\infty} \Pr(\| \mathbf{x}_t - \mathbf{x} \| < \epsilon) = 1, \text{ for every } \epsilon > 0,$$

where $\| \cdot \|$ denotes an appropriate norm measuring the discrepancy between $\mathbf{x}_t$ and $\mathbf{x}$. Using the Euclidean norm we have $\| \mathbf{z} \| = \left(\sum_{i=1}^{m} z_i^2\right)^{\frac{1}{2}}$, where $\mathbf{z} = (z_1, z_2, \ldots, z_m)'$.

Example 13 *Suppose x_T is normally distributed with mean $\mu + \frac{k}{T}$ and the variance $\sigma_T^2 = \frac{\sigma^2}{T} > 0$. Show that $\{x_T\}$ converges in probability to μ, a fixed constant. Here we show the convergence of x_t to μ by obtaining $\Pr(| x_T - \mu | < \epsilon)$ directly. However, as it becomes clear later, the result can be established much more easily using general results on convergence in probability. Since $x_T - \mu \sim N\left(\frac{k}{T}, \frac{\sigma^2}{T}\right)$ it is easily seen that*

$$\Pr(| x_T - \mu | < \epsilon) = \Phi\left(\frac{\epsilon - \frac{k}{T}}{\frac{\sigma}{\sqrt{T}}}\right) - \Phi\left(\frac{-\epsilon - \frac{k}{T}}{\frac{\sigma}{\sqrt{T}}}\right), \tag{8.2}$$

where $\Phi(\cdot)$ represents the cumulative distribution function of a standard normal variate. But

$$\lim_{T\to\infty} \Phi\left(\frac{\epsilon - \frac{k}{T}}{\frac{\sigma}{\sqrt{T}}}\right) = \lim_{T\to\infty} \Phi\left(\frac{\epsilon\sqrt{T}}{\sigma}\right) = 1, \text{ for any } \epsilon > 0,$$

and

$$\lim_{T\to\infty} \Phi\left(\frac{-\epsilon - \frac{k}{T}}{\frac{\sigma}{\sqrt{T}}}\right) = \lim_{T\to\infty} \Phi\left(\frac{-\epsilon\sqrt{T}}{\sigma}\right) = 0, \text{ for any } \epsilon > 0.$$

Therefore

$$\lim_{T\to\infty} \Pr(| x_T - \mu | < \epsilon) = 1,$$

as required. For a given value of ϵ, the rate of convergence of x_T to μ depends on k, σ and the shape of the distribution function $\Phi(\cdot)$. The larger the value of σ, the slower will be the rate of convergence of x_T to μ.

8.2.2 Convergence with probability 1

Definition 2 *The sequence of random variables $\{x_t\}$ is said to converge with probability 1 (or almost surely) to x if*

$$\Pr(\lim_{t\to\infty} x_t = x) = 1. \tag{8.3}$$

This is often written as $x_t \xrightarrow{wp1} x$ (or $\xrightarrow{as}$). An equivalent condition for convergence with probability 1 is given by

$$\lim_{t\to\infty} \Pr(|\, x_m - x \,| < \epsilon, \text{ for all } m \geq t) = 1, \text{ for every } \epsilon > 0. \tag{8.4}$$

The equivalence of conditions (8.3) and (8.4) is proved by Halmos (1950) and clearly shows that the concept of convergence in probability defined by (8.1) is a special case of (8.4) (setting $m = t$ in (8.4) delivers (8.1)). But as we shall see below, the reverse is not necessarily true. The concept of convergence with probability 1 is stronger than convergence in probability and is often referred to as the 'strong convergence' as compared to convergence in probability which is referred to as 'weak convergence'.

8.2.3 Convergence in s-th mean

Definition 3 *The sequence of random variables $\{x_t\}$ is said to converge in the s-th mean to x if for $s > 0$,*

$$\lim_{t\to\infty} E \mid x_t - x \mid^s = 0, \tag{8.5}$$

and it is written as $x_t \xrightarrow{s\text{-}th} x$.

The case of $s = 2$ is of particular interest in econometric applications and is referred to as 'convergence in mean square' or 'convergence in quadratic mean' and denoted as $x_t \xrightarrow{q.m.} x$. It is now easily seen that the higher the value of s the more stringent will be the convergence condition. Let $u = \mid x_t - x \mid^r$ and note that $f(u) = u^{\frac{s}{r}}$ is a convex function for $s > r > 0$. Therefore, by Jensen's inequality $E[f(u)] \geq f[E(u)]$ and we have[1]

$$E\left[u^{\frac{s}{r}}\right] \geq [E(u)]^{\frac{s}{r}},$$

or

$$E|x_t - x|^s \geq [E|x_t - x|^r]^{\frac{s}{r}}, \text{ for } s > r > 0,$$

which is also known as Lyapunov's inequality (see, e.g., Billingsley (1999)). Taking limits of both sides of this inequality and assuming that $x_t \xrightarrow{s\text{-th}} x$, then

$$\lim_{t\to\infty} [E|x_t - x|^r]^{\frac{s}{r}} \leq \lim_{t\to\infty} |x_t - x|^s = 0,$$

and hence it follows that:

$$\lim_{t\to\infty} [E|x_t - x|^r]^{\frac{s}{r}} = 0, \text{ or } \lim_{t\to\infty} |x_t - x|^s = 0,$$

[1] For a proof of Jensen's inequality see Section B.12.4.

and therefore $x_t \xrightarrow{s\text{-th}} x$ implies $x_t \xrightarrow{r\text{-th}} x$, for $s > r > 0$.

8.3 Relationships among modes of convergence

We have already seen that convergence with probability 1 implies convergence in probability, but not necessarily *vice versa*. We now consider the relationship between convergence in quadratic mean and convergence in probability. By Chebyshev's inequality we have[2]

$$\Pr(|x_t - x| > \epsilon) \leq \frac{1}{\epsilon^2} E(x_t - x)^2, \tag{8.6}$$

and for any fixed ϵ, taking limits of both sides yields

$$\lim_{t\to\infty} \Pr(|x_t - x| > \epsilon) \leq \frac{1}{\epsilon^2} \lim_{t\to\infty} E(x_t - x)^2.$$

Therefore, we have the following result:

Theorem 2 *Convergence in quadratic mean implies convergence in probability, i.e.,* $x_t \xrightarrow{q.m.} x \Longrightarrow x_t \xrightarrow{p} x$. *More generally, if*

$$x_t \xrightarrow{s\text{-}th} x \Longrightarrow x_t \xrightarrow{p} x, \text{ for any } s > 0.$$

Proof To prove this result, let $z_t = x_t - x$, and write

$$E|z_t|^s = E\left\{|z_t|^s I(|z_t| \leq \epsilon)\right\} + E\left\{|z_t|^s I(|z_t| > \epsilon)\right\},$$

where $I(|z_t| \leq \epsilon)$ is the indicator function, taking the value of unity if $|z_t| > \epsilon$, and zero otherwise. Since $|z_t|$ is non-negative, we have

$$\begin{aligned} E|z_t|^s &\geq E\left\{|z_t|^s I(|z_t| > \epsilon)\right\} \\ &= \int_{|z_t|>\epsilon}^{s} |z_t| f(z_t)\, dz_t \geq \epsilon^s \int_{|z_t|>\epsilon} f(z_t)\, dz_t = \epsilon^s \Pr\{|z_t| > \epsilon\}. \end{aligned}$$

Hence

$$\Pr\{|z_t| > \epsilon\} \leq \frac{E|z_t|^s}{\epsilon^s},\ s > 0, \tag{8.7}$$

or

$$\Pr\{|x_t - x| > \epsilon\} \leq \frac{E|x_t - x|^s}{\epsilon^s},\ s > 0, \tag{8.8}$$

[2] See Appendix B, Section B.12.1 for a proof of Chebyshev's inequality.

which is a generalization of Chebyshev's inequality given by (8.6). This is also known as Markov's inequality (see, for example, Billingsley (1999)). Take limits of both sides of (8.8) as $t \to \infty$, it then follows that for any $\epsilon > 0$, if $E|x_t - x|^s \to 0$, then we must also have $\Pr(|x_t - x| > \epsilon) \to 0$.

Notice, however, as the following example demonstrates, convergence in probability does not necessarily imply convergence in *s*-th mean.

Example 14 *Consider the random variable*

$$x_t = \begin{cases} t, \text{ with probability } \frac{1}{\log t} \\ 0, \text{ with probability } 1 - \frac{1}{\log t} \end{cases}.$$

As $t \to \infty$, the probability that $x_t \to 0$ tends unity, and hence $x_t \xrightarrow{p} 0$. However, as $t \to \infty$

$$E|x_t|^s = \frac{t^s}{\log t} \to \infty, \text{ for } s > 0,$$

which contradicts the necessary condition for x_t to converge to zero in s-th mean.

The relationship between convergence with probability 1 and convergence in quadratic mean is more complex, and involves additional conditions. A useful result is provided in the following theorem:

Theorem 3 *If x_t converges to c (a fixed constant) in quadratic mean in such a way that*

$$\sum_{t=1}^{\infty} E(x_t - c)^2 < \infty, \tag{8.9}$$

then x_t converges to c with probability 1.

For a proof of the above theorem, see Rao (1973).

Condition (8.9) is not necessary for x_t to converge to x with probability 1. In the case of example 13, it is easily seen that

$$E(x_t - \mu)^2 = \frac{\sigma^2}{t} + \frac{k^2}{t^2}, \tag{8.10}$$

and hence $x_t \xrightarrow{qm} \mu$ which in turn implies that $x_t \xrightarrow{p} \mu$. Using (8.2) we have

$$\Pr(|x_m - \mu| < \epsilon) = \Phi\left(\frac{\epsilon\sqrt{m} - \frac{k}{\sqrt{m}}}{\sigma}\right) - \Phi\left(\frac{-\epsilon\sqrt{m} - \frac{k}{\sqrt{m}}}{\sigma}\right),$$

and it readily follows that for all $m \geq t$, $\Pr(|x_m - \mu| < \epsilon)$ tends to zero as $t \to \infty$, and by condition (8.4) then $x_t \overset{wp1}{\to} \mu$. But, using (8.10), we have

$$\sum_{t=1}^{T} E(x_t - \mu)^2 = \sigma^2 \left(1 + \frac{1}{2} + \frac{1}{3} + \ldots + \frac{1}{T}\right) + k^2 \left(1 + \frac{1}{2^2} + \frac{1}{3^2} + \ldots + \frac{1}{T^2}\right).$$

The sequence $1 + \frac{1}{2} + \frac{1}{3} + \ldots + \frac{1}{T}$ diverges since

$$\begin{aligned} 1 + \frac{1}{2} + \left(\frac{1}{3} + \frac{1}{4}\right) \ldots + \frac{1}{T} &> 1 + \frac{1}{2} + \left(\frac{1}{4} + \frac{1}{4}\right) + \left(\frac{1}{4} + \frac{1}{4} + \frac{1}{4} + \frac{1}{4}\right) \ldots + \frac{1}{T} \\ &= \frac{1}{2} + \frac{1}{2} + \frac{1}{2} + \frac{1}{2} + \ldots. \end{aligned}$$

It is clear that the condition $\sum_{t=1}^{\infty} E(x_t - c)^2 < \infty$ is not satisfied in the present example.

8.4 Convergence in distribution

Definition 4 *Let $x_1, x_2, \ldots$ be a sequence of random variables with distribution functions $F_1(\cdot), F_2(\cdot), \ldots$, respectively. Then x_t is said to converge in distribution to x if*

$$\lim_{t \to \infty} F_t(u) = F(u),$$

for all u at which F is continuous.

Convergence in distribution is usually denoted by $x_t \overset{d}{\to} x$, $x_t \overset{L}{\to} x$, $x_t \overset{a}{\sim} x$, or $F_t \Longrightarrow F$.

Definition 5 *If $F_t \Rightarrow F$, and F is continuous, then the convergence is said to be uniform, that is:*

$$\lim_{t \to \infty} \sup_{u} \mid F_t(u) - F(u) \mid = 0.$$

The limiting distribution function, F, is referred to as the asymptotic distribution of x_t, and provides the basis for approximating the distribution of x_t, as t increases without bounds.

In practice when the mean or variance of x_t increases with t, in deriving the asymptotic distribution of x_t it is necessary to consider the limiting distribution of normalized or rescaled random variable, $z_t = \frac{x_t - \mu_t}{\sigma_t}$, where μ_t and σ_t are appropriate constants.

There are three basic approaches for establishing convergence in distribution. These are convergence of characteristic functions, convergence of moments (when they exist), and convergence of density functions. Among these the characteristic function approach is used most often, and is spelled out more fully in the following theorem:

Theorem 4 *Let $\varphi_t(\theta)$ and $\varphi(\theta)$ be the characteristic functions associated with the distribution functions $F_t(\cdot)$ and $F(\cdot)$, respectively. The following statements are equivalent:*

(i) $F_t \Longrightarrow F \left(or\, x_t \xrightarrow{d} x \right)$.

(ii) $\lim_{t\to\infty} \varphi_t(\theta) = \varphi(\theta)$, *for any real* θ.

(iii) $\lim_{t\to\infty} \int g dF_t = \int g dF$, *for any bounded continuous function* g.

For a proof of the above theorem see Rao (1973) and Serfling (1980). An important application of the above theorem is given by the following lemma.

Lemma 5 *If* $F_t \Rightarrow F$, $a_t \to a$, $b_t \to b$, *then* $F_t(a_t x + b_t) \Rightarrow F(ax + b)$.

Proof Let $x_t = a_t x + b_t$ and denote its distribution function and the associated characteristic function by F_t and $\varphi_t(\theta)$, respectively. From the properties of characteristic functions we have

$$\varphi_t(\theta) = E\left(e^{i\theta x_t}\right) = E\left(e^{i\theta(a_t x + b_t)}\right) = e^{i\theta b_t} E\left(e^{i\theta a_t x}\right).$$

Further

$$\lim_{t\to\infty} \varphi_t(\theta) = e^{i\theta b} E\left(e^{i\theta a x}\right),$$

which is the characteristic function of $ax + b$, and consequently $a_t x + b_t \xrightarrow{d} ax + b$, and $F_t(a_t x + b_t) \Longrightarrow F(ax + b)$.

8.4.1 Slutsky's convergence theorems

In econometric applications it is often the case that statistics or estimators of interest can be written as functions of random variables that have a known limiting distribution. The following theorem due to Slutsky is particularly useful in these circumstances:

Theorem 6 *Let* $\{x_t, y_t\}$, $t = 1, 2, \ldots$ *be a sequence of pairs of random variables with* $y_t \xrightarrow{d} y$, *and* $| y_t - x_t | \xrightarrow{p} 0$. *Then the limiting distribution of* x_t *exists and is the same as that of* y, *that is* $x_t \xrightarrow{d} y$.

Proof Let f_t be the distribution function of x_t, and f_y be the distribution function of y. Set $z_t = y_t - x_t$, and let u be the continuity point of $F_y(\cdot)$. Then

$$\begin{aligned} F_t(u) &= \Pr(x_t < u) = \Pr\left(y_t < u + z_t\right), \\ &= \Pr\left(y_t < u + z_t,\ z_t < \epsilon\right) + \Pr\left(y_t < u + z_t,\ z_t \geq \epsilon\right), \end{aligned} \tag{8.11}$$

where $\epsilon > 0$. We now have

$$\Pr\left(y_t < u + z_t;\ z_t < \epsilon\right) \leq \Pr\left(y_t < u + \epsilon\right),$$

and

$$\Pr\left(y_t < u + z_t,\, z_t \geq \epsilon\right) = \Pr\left(z_t \geq \epsilon\right) \Pr\left(y_t < u + z_t \mid z_t \geq \epsilon\right) \leq \Pr\left(z_t \geq \epsilon\right).$$

Substituting these results in (8.11) yields

$$F_t\left(u\right) \leq \Pr\left(y_t < u + \epsilon\right) + \Pr\left(z_t \geq \epsilon\right),$$

and taking limits we have

$$\lim_{t\to\infty} F_t\left(u\right) \leq \lim_{t\to\infty} \Pr\left(y_t < u + \epsilon\right) + \lim_{t\to\infty} \Pr\left(z_t \geq \epsilon\right).$$

But given that $z_t \xrightarrow{p} 0$ by assumption, the second limit on the right hand side of this equality is equal to zero, and since $y_t \xrightarrow{d} y$, we have

$$\lim_{t\to\infty} F_t\left(u\right) \leq F_y\left(u + \epsilon\right).$$

Also carrying out the same operations for $z_t < -\epsilon$ we obtain

$$\lim_{t\to\infty} F_t\left(u\right) \geq F_t\left(u - \epsilon\right).$$

Therefore, since ϵ is an arbitrary positive constant and u is a continuity point of $F_y\left(\cdot\right)$ by letting $\epsilon \to 0$ we obtain

$$\lim_{t\to\infty} F_t\left(u\right) = F_y\left(u\right),$$

as required.

Theorem 7 *If $x_t \xrightarrow{d} x$ and $y_t \xrightarrow{p} c$, where c is a finite constant, then*

(i) $x_t + y_t \xrightarrow{d} x + c$.

(ii) $y_t x_t \xrightarrow{d} cx$.

(iii) $\frac{x_t}{y_t} \xrightarrow{d} \frac{x}{c}$; *if* $c \neq 0$.

Proof

(i) By assumption, we have $y_t - c = (x_t + y_t) - (x_t + c) \xrightarrow{p} 0$. Therefore, from Theorem 6 $x_t + y_t$ and $x_t + c$ have the same limiting distribution. But by assumption we also have $x_t + c \xrightarrow{d} x + c$. Hence it also follows that $x_t + y_t \xrightarrow{d} x + c$.

(ii) Let $z_t = x_t(y_t - c)$, and for arbitrary positive constants ϵ and δ, consider

$$\Pr(|z_t| > \epsilon) = \Pr(|x_t|\,|y_t - c| > \epsilon, |y_t - c| < \frac{\epsilon}{\delta}) + \Pr(|x_t|\,|y_t - c| > \epsilon, |y_t - c| \geq \frac{\epsilon}{\delta})$$
$$\leq \Pr(|x_t| \geq \delta) + \Pr(|y_t - c| \geq \frac{\epsilon}{\delta}).$$

For any fixed δ, taking limits of both sides of the above inequality, and noting that by assumption $y_t \xrightarrow{p} c$ and $x_t \xrightarrow{d} x$, we have

$$\lim_{t\to\infty} \Pr(|z_t| > \epsilon) \leq \lim_{t\to\infty} \Pr(|x_t| > \delta) = \Pr(|x_t| > \delta).$$

But δ is arbitrary and hence $\Pr(|x_t| > \delta)$ can be made as small as desired by choosing a large enough value for δ. Therefore $\lim_{t\to\infty} \Pr(|z_t| > \epsilon) = 0$ and $z_t \xrightarrow{p} 0$. Hence by Theorem 6, $x_t y_t$ and cx_t will have the same asymptotic distribution given by the distribution of cx.

(iii) The proof is similar to that given above for (ii).

The above results readily extend to the multivariate case where x_t is a vector of random variables. In this connection the following theorem is particularly useful.

Theorem 8 *Let $\mathbf{x}_t = (x_{1t}, x_{2t}, \ldots, x_{mt})'$ be a sequence of $m \times 1$ vector of random variables and suppose that*

$$\boldsymbol{\lambda}'\mathbf{x}_t \xrightarrow{d} \boldsymbol{\lambda}'\mathbf{x},$$

when $\boldsymbol{\lambda} = (\lambda_1, \lambda_2, \ldots, \lambda_m)'$ is an arbitrary vector of fixed constants. Then the limiting distribution of x_t exists and is given by the limiting distribution of x.

Proof $\boldsymbol{\lambda}'\mathbf{x}_t \xrightarrow{d} \boldsymbol{\lambda}'\mathbf{x}$ implies that $\boldsymbol{\lambda}'\mathbf{x}_t$ and $\boldsymbol{\lambda}'\mathbf{x}$ have the same characteristic function (see Theorem 4). Denote the characteristic functions of x_t and x by $\phi_t(\theta_1, \theta_2, \ldots \theta_m)$ and $\phi(\theta_1, \theta_2, \ldots \theta_m)$, respectively. Then the characteristic functions of $\boldsymbol{\lambda}'\mathbf{x}_t$ and $\boldsymbol{\lambda}'\mathbf{x}$ are given by $\varphi_t(\lambda_1\theta_1, \lambda_2\theta_2, \ldots \lambda_m\theta_m)$ and $\varphi(\lambda_1\theta_1, \lambda_2\theta_2, \ldots \lambda_m\theta_m)$, and as $t \to \infty$ by assumption

$$\varphi_t(\lambda_1\theta_1, \lambda_2\theta_2, \ldots \lambda_m\theta_m) \to \varphi(\lambda_1\theta_1, \lambda_2\theta_2, \ldots \lambda_m\theta_m),$$

for any real $\boldsymbol{\lambda} = (\lambda_1, \lambda_2, \ldots, \lambda_m)'$. Therefore

$$\varphi_t(\theta_1, \theta_2, \ldots \theta_m) \to \varphi(\theta_1, \theta_2, \ldots \theta_m),$$

which establishes the desired result.

Theorem 9 *(Convergence Properties of Transformed Sequences)*
Suppose $\{\mathbf{x}_t\}$ *and* $\mathbf{x}$ *are* $m \times 1$ *vectors of random variables on a probability space, and let* $\mathbf{g}(\cdot)$ *be a vector-valued function; and assume that* $g(\cdot)$ *is continuous. Then*

(i) $\mathbf{x}_t \xrightarrow{wp1} \mathbf{x} \Rightarrow \mathbf{g}(\mathbf{x}_t) \xrightarrow{wp1} \mathbf{g}(\mathbf{x})$
(ii) $\mathbf{x}_t \xrightarrow{p} \mathbf{x} \Rightarrow \mathbf{g}(\mathbf{x}_t) \xrightarrow{p} \mathbf{g}(\mathbf{x})$
(iii) $\mathbf{x}_t \xrightarrow{d} \mathbf{x} \Rightarrow \mathbf{g}(\mathbf{x}_t) \xrightarrow{d} \mathbf{g}(\mathbf{x})$
(iv) $\mathbf{x}_t - \mathbf{y}_t \xrightarrow{p} \mathbf{0}$ *and* $\mathbf{y}_t \xrightarrow{d} \mathbf{y} \Rightarrow \mathbf{g}(\mathbf{x}_t) - \mathbf{g}(\mathbf{y}) \xrightarrow{d} \mathbf{0}$.

For a proof see, for example, Serfling (1980) and Rao (1973).

Example 15

(a) Suppose that $\mathbf{x}_t \xrightarrow{p} \mathbf{c}$, *then* $\boldsymbol{\lambda}'\mathbf{x}_t \xrightarrow{p} \boldsymbol{\lambda}'\mathbf{c}$
(b) If $\mathbf{x}_t \xrightarrow{d} N(\mathbf{0}, \mathbf{I}_m)$, *then* $\mathbf{x}_t'\mathbf{M}\mathbf{x}_t \xrightarrow{d} \chi_s^2$,
(c) If $\mathbf{x}_t \xrightarrow{d} N(0, 1)$, *then* $\mathbf{x}_t^2 \xrightarrow{d} \chi_1^2$.

where $\mathbf{M}$ *is an important matrix of rank* $s < m$.

8.5 Stochastic orders $O_p(\cdot)$ and $o_p(\cdot)$

The concepts of stochastic 'big oh', denoted by $O_p(\cdot)$ and the stochastic 'small oh', denoted by $o_p(\cdot)$ are analogous to the $O(\cdot)$ and $o(\cdot)$ notations used in comparing the order of magnitude of two deterministic sequences. The $O_p(\cdot)$ represents the idea of ' boundedness ' in probability.

Definition 6 *Let* $\{a_t\}$ *be a sequence of positive numbers and* $\{x_t\}$ *be a sequence of random variables. Then*

(i) $x_t = O_p(a_t)$, *or* $\frac{x_t}{a_t}$ *is bounded in probability, if, for each* $\epsilon > 0$ *there exist the real numbers* M_ϵ *and* N_ϵ *such that*

$$\Pr\left(\frac{|x_t|}{a_t} > M_\epsilon\right) < \epsilon, \text{ for } t > N_\epsilon. \tag{8.12}$$

(ii) $x_t = o_p(a_t)$, *if*

$$\frac{\mathbf{x}_t}{a_t} \xrightarrow{p} 0.$$

The above definition can be generalized for two sequences of random variables $\{x_t\}$ and $\{y_t\}$. The notation $x_t = O_p(y_t)$ denotes that the sequence $\left\{\frac{x_t}{y_t}\right\}$ is $O_p(1)$. Also $x_t = o_p(y_t)$ means that $\frac{x_t}{y_t} \xrightarrow{p} 0$. See, for example, Bierens (2005, p. 157).

One important use of the stochastic order notation is in the Taylor series expansion of functions of random variables. Let $x_t - c = o_p(a_t)$, where $a_t \to 0$ as $t \to \infty$, and assume that $g(x)$ has a kth order Taylor series expansion at c, namely

$$g(x) = G_k(x, c) + o_p\left(|x - c|^k\right).$$

Then we have

$$g(x_t) = G_k(x_t, c) + o_p\left(a_t^k\right). \tag{8.13}$$

The proof is very simple and follows immediately from the fact that if $x_t - c = o_p(a_t)$, then $|x_t - c|^k = o_p\left(a_t^k\right)$.

8.6 The law of large numbers

The law of large numbers is concerned with the convergence of averages of random variables and plays a central role in the proof of consistency of estimators in econometrics. There is a weak form of the law which refers to the convergence in probability of the sum of random variables, and a strong form that refers to convergence with probability 1. We shall first review the various versions of the law for sequences of independently identically distributed (*IID*) random variables. Subsequently we consider the case where the random variables are independently distributed, but allow them to have different distributions.

Theorem 10 (Khinchine) *Suppose that $\{x_t\}$ is a sequence of IID random variables with constant mean; i.e., $E(x_t) = \mu < \infty$. Then*

$$\bar{x}_T = \frac{\sum_{t=1}^{T} x_t}{T} \xrightarrow{p} \mu.$$

Proof Denote the characteristic function (c.f.) of x_t by $\varphi_x(\theta)$. Since x_t are *IID*, then c.f. of $\bar{x}_T$ is given by

$$\varphi_T(\theta) = \left[\varphi_x\left(\frac{\theta}{T}\right)\right]^T.$$

Since $E(x_t) = \mu < \infty$, then

$$\log\left[\varphi_x\left(\frac{\theta}{T}\right)\right] = 1 + \frac{i\mu\theta}{T} + o\left(\frac{\theta}{T}\right),$$

and

$$\log\left[\varphi_T(\theta)\right] = T\log\left[\varphi_x\left(\frac{\theta}{T}\right)\right]$$
$$= T\left[\frac{i\mu\theta}{T} + o\left(\frac{\theta}{T}\right)\right] = i\mu\theta + o(\theta),$$

and hence $\lim_{T\to\infty}\varphi_T(\theta) = e^{i\mu\theta}$, which is the c.f. of a degenerate distribution at μ. It follows that $\bar{x}_T \to \mu$.

This theorem represents the weak law of large numbers $(WLLN)$ for independent random variables, and only requires that the mean of x_t exists.

Theorem 11 (Chebyshev) *Let $E(x_t) = \mu_t$, $V(x_t) = \sigma_t^2$, and $cov(x_t, x_s) = 0, t \neq s$. Then if $\lim_{T\to\infty}\frac{1}{T}\sum_{t=1}^{T}\sigma_t^2 < \infty$, we have $\bar{x}_T - \bar{\mu}_T \xrightarrow{p} 0$, where $\bar{\mu}_T = T^{-1}\sum_{t=1}^{T}\mu_t$.*

Proof Let $y_t = x_t - \mu_t$, and note that

$$V(\bar{y}_T) = V\left(\frac{\sum_{t=1}^{T} y_t}{T}\right) = \frac{1}{T^2}\sum_{t=1}^{T}\sigma_t^2,$$

and hence $\lim_{T\to\infty} V(\bar{y}_T) = 0$, if $\lim_{T\to\infty}\frac{1}{T}\sum_{t=1}^{T}\sigma_t^2 < \infty$. Therefore, by Theorem 2, $\bar{y}_T \xrightarrow{p} 0$, or $\bar{x}_T - \bar{\mu}_T \xrightarrow{p} 0$.

Theorem 10 and Theorem 11 give different conditions for the weak convergence of the sums of random variables. The strong forms of the law of large numbers are given by the following theorems:

Theorem 12 (Kolmogorov) *Let $\{x_t\}$ be a sequence of independent random variables with $E(x_t) = \mu_t < \infty$ and $V(x_t) = \sigma_t^2$, such that*

$$\sum_{t=1}^{\infty}\frac{\sigma_t^2}{t^2} < \infty. \tag{8.14}$$

Then $\bar{x}_T - \bar{\mu}_T \xrightarrow{wp1} 0$. If the independence assumption is replaced by lack of correlation (i.e. $cov(x_t, x_s) = 0, t \neq s$), the convergence of $\bar{x}_T - \bar{\mu}_T$ with probability one requires the stronger condition

$$\sum_{t=1}^{\infty}\frac{\sigma_t^2(\log t)^2}{t^2} < \infty. \tag{8.15}$$

For a proof see Rao (1973), where other forms of the law are also discussed.

Another version of the strong law of large numbers, which is more relevant in econometric applications, is given in Theorem 13.

Theorem 13 *Suppose that $x_1, x_2, \ldots$ are independent random variables, and that $E(x_i) = 0$, $E(x_i^4) \leq K, \forall i$, where K is an arbitrary positive constant. Then $\bar{x}_T$ converges to zero with probability 1.*

Proof We have

$$E(\bar{x}_T^4) = \frac{1}{T^4} E\left(\sum_{i=1}^{T} x_i\right)^4$$
$$= \frac{1}{T^4} E\left(\sum_{i=1}^{T} x_i^4 + 6\sum\sum_{i<j} x_i^2 x_j^2\right)$$

since for distinct i, j, k, and l,

$$E\left(x_i x_j^3\right) = E\left(x_l x_j^2 x_k\right) = E\left(x_i x_j x_k x_l\right) = 0.$$

Also by Lyapounov's inequality (see Section 8.2.3)

$$\left[E\left(x_i^2\right)\right]^{\frac{1}{2}} \leq \left[E\left(x_i^4\right)\right]^{\frac{1}{4}}$$

and using independence of x_i and x_j for $i \neq j$ we have

$$E\left(x_i^2 x_j^2\right) = E\left(x_i^2\right) E\left(x_j^2\right) \leq E\left(x_i^4\right) \leq K.$$

Therefore

$$E(\bar{x}_T^4) \leq \frac{1}{T}\left[Tk + 3T(T-1)K\right] \leq 3kT^{-2},$$

and

$$E\left(\sum_{i=1}^{\infty} \bar{x}_T^4\right) \leq 3k \sum_{T=1}^{\infty} T^{-2} < \infty,$$

which establishes $\sum_{i=1}^{\infty} \bar{x}_T^4 < \infty$, with probability 1.

It is also easily seen that if the zero-mean assumption is replaced by $E(x_i) = \mu < \infty$, in the statement of the theorem, then the theorem still holds, with $\bar{x}_T \xrightarrow{wp1} \mu$ as its conclusion.

Theorem 14 provides the uniform strong law of large numbers.

Theorem 14 (Uniform strong law of large numbers) *Let $\{x_t\}$ be a sequence of independent random variables and suppose that the strong law of large numbers is satisfied, i.e., $\frac{\sum_{t=1}^{T} x_t}{T} \overset{wp1}{\to} x$. Let $g(x,\theta)$ be a function continuous on $R \times \Theta$ where R is the range of x and θ lies in the compact set Θ. If $E\left\{\sup_{\theta\in\Theta}\left|g(x,\theta)\right|\right\} < \infty$ then*

$$\lim_{T\to\infty} \sup_{\theta\in\Theta} \left| \frac{\sum_{t=1}^{T} g(x_t,\theta)}{T} - E\left[g(x,\theta)\right] \right| = 0, \text{ with probability 1.}$$

The above law is uniform since it happens for the supremum of the difference between the average and the expectations. In other words, it holds at θ for which the difference is the greatest.

8.7 Central limit theorems

There are a large number of central limit theorems (CLT) in the statistics literature. They can be classified under four headings: The *CLT* for *IID* random numbers, for independent but not necessarily identically distributed random variables, for dependent but identically distributed random variables, and for dependent and not necessarily identically distributed random variables. A good exposition of the different *CLT* that are of particular use in econometric applications is given in White (2000). Here we state some of the important *CLT*, and comment on their appropriateness in econometrics. The basic *CLT* for *IID* random variables is as follows:

Theorem 15 (Lindberg and Levy) *Let $\{x_t\}$ be a sequence of IID random variables with $E(x_t) = \mu$ and $V(x_t) = \sigma^2$. Then*

$$\frac{\sqrt{T}(\bar{x}_T - \mu)}{\sigma} \overset{d}{\to} N(0,1). \tag{8.16}$$

Proof Let $\varphi_z(\theta)$ be the characteristic function of $z_t = x_t - \mu$, and let $\varphi_T(\theta)$ be the characteristic function of $\frac{\sqrt{T}(\bar{x}_T-\mu)}{\sigma}$. Then using the independence of x_t, we have

$$\varphi_T(\theta) = \left[\varphi_z\left(\frac{\theta}{\sigma\sqrt{T}}\right)\right]^T.$$

Since the mean and variance of z_t exist, and recalling that $E(z_t) = 0$, the Taylor series expansion of $\varphi_z\left(\frac{\theta}{\sigma\sqrt{T}}\right)$ is

$$\varphi_z\left(\frac{\theta}{\sigma\sqrt{T}}\right) = 1 - \frac{\theta^2}{2T} + o\left(\frac{\theta^2}{T}\right).$$

Therefore,

$$\log\left[\varphi_T(\theta)\right] = T\log\left[1 - \frac{\theta^2}{2T} + o\left(\frac{\theta^2}{T}\right)\right],$$

and, from the definition of exponential function, we have

$$\lim_{T\to\infty} \log\left[\varphi_T(\theta)\right] = -\frac{\theta^2}{2}.$$

It follows that $\lim_{T\to\infty} \varphi_T(\theta) = e^{-\frac{1}{2}\theta^2}$, which is the c.f. of a standard normal variate, which implies (8.16).

A comparison of Theorem 10 and Theorem 15 clearly shows the additional assumption needed when moving from the WLLN to the *CLT*, namely that the *CLT* requires the existence of the second moments, while the WLLN for *IID* random variables only needs the existence of first moments.

Theorem 16 (Liapounov) *Let $\{x_t\}$ be a sequence of independent random variables and assume that the following moments of x_t exist*

$$\begin{aligned} E(x_t) &= \mu, \\ E(x_t-\mu_t)^2 &= \sigma_t^2 > 0, \\ E(x_t-\mu_t)^3 &= \alpha_t, \\ E\left|x_t-\mu_t\right|^3 &= \beta_t, \end{aligned}$$

for all t. Then if

$$\lim_{T\to\theta}\left(\frac{B_T}{C_T}\right) = 0,$$

where

$$B_T = \left(\sum_{t=1}^{T}\beta_t\right)^{\frac{1}{2}}, C_T = \left(\sum_{t=1}^{T}\sigma_t^2\right)^{\frac{1}{2}},$$

it follows that

$$\frac{\sqrt{T}\left(\bar{x}_T - \bar{\mu}_T\right)}{\bar{\sigma}_T} \xrightarrow{d} N(0,1),$$

where $\bar{\mu}_T = \frac{\sum_{t=1}^{T}\mu_t}{T}$, and $\bar{\sigma}_T^2 = \frac{1}{T}\sum_{t=1}^{T}\sigma_t^2$.

Theorem 17 (Linberg–Feller) *Let $\{x_t\}$ be a sequence of independent random variables, and assume that $E(x_t) = \mu_t$ and $V(x_t) = \sigma_t^2 > 0$ exist. Then*

$$\frac{\sqrt{T}\left(\bar{x}_T - \bar{\mu}_T\right)}{\bar{\sigma}_T} \xrightarrow{d} N(0,1),$$

and

$$\lim_{T\to\theta}\max_{1\leq t\leq n}\left\{\frac{1}{T}\frac{\sigma_t^2}{\bar{\sigma}_T^2}\right\}=0, \tag{8.17}$$

if and only if, for every $\epsilon>0$,

$$\lim_{T\to\theta}\frac{1}{T}\frac{1}{\bar{\sigma}_T^2}\sum_{t=1}^{T}\int_{|x-\mu_t|>\epsilon\sqrt{T}\bar{\sigma}_T}\left(x-\mu_t\right)^2 dF_t\left(x\right)=0, \tag{8.18}$$

where $\bar{\mu}_T=\frac{\sum_{t=1}^{T}\mu_t}{T}$, $\bar{\sigma}_T^2=\frac{1}{T}\sum_{t=1}^{T}\sigma_t^2$, *and* $F_t\left(x\right)$ *denotes the distribution function of* x_t.

Theorems 16 and 17 are proved in Gnedenko (1962) and Loeve (1977) and cover the case of independent but heterogeneously distributed random variables. They are particularly useful in the case of cross-section observations. Condition (8.18), known as the *Lindberg condition*, is, however, difficult to verify in practice and the following limit theorem is often used instead.

Theorem 18 *Let* $\{x_t\}$ *be a sequence of independent random variables with* $E\left(x_t\right)=\mu_t$, $Var(x_t)=\sigma_t^2>0$ *and* $E\left|x_t-\mu_t\right|^{2+\delta}<\infty$, *for some* $\delta>0$ *and all* t. *If*

$$\lim_{T\to\infty}\frac{1}{T}\sum_{t=1}^{T}\sigma_t^2>0, \tag{8.19}$$

then $\frac{\sqrt{T}(\bar{x}_T-\bar{\mu}_T)}{\bar{\sigma}_T}\xrightarrow{d}N(0,1)$.

8.8 The case of dependent and heterogeneously distributed observations

The assumption of independence is inappropriate for most economic time series, which typically exhibit temporal dependence. In this section, we provide laws of large numbers and central limit theorems that allow the random variables to be dependent and heterogeneously distributed. We initially present a set of laws of large numbers for mixing processes, asymptotically uncorrelated sequences, martingale difference sequences and mixingales, we then turn to central limit theorems for dependent processes.

8.8.1 Law of large numbers

Let $(\Omega,\mathcal{F},P)$ be a probability space, and let $\mathcal{A}$ and $\mathcal{B}$ be two σ-subfields of $\mathcal{F}$. Then

$$\alpha\left(\mathcal{A},\mathcal{B}\right)=\sup_{\substack{A\in\mathcal{A}\\B\in\mathcal{B}}}\left|P\left(A\cap B\right)-P(A)P(B)\right|, \tag{8.20}$$

is known as the *strong mixing coefficient*, and

$$\phi(\mathcal{A},\mathcal{B}) = \sup_{\substack{A\in\mathcal{A},B\in\mathcal{B};\\ P(A)>0}} |P(B|A) - P(B)|, \tag{8.21}$$

as the *uniform mixing coefficient*. The strong mixing is weaker than the uniform mixing concept, since

$$|P(A\cap B) - P(A)P(B)| \leq |P(B|A) - P(B)|.$$

If $\mathcal{A}$ and $\mathcal{B}$ are independent then $\alpha(\mathcal{A},\mathcal{B}) = \phi(\mathcal{A},\mathcal{B}) = 0$. The converse is true in the case of uniform mixing, while it is not true for strong mixing. See Davidson (1994, p. 206).

Definition 7 *Let $\{x_t\}$ be a stochastic sequence and let $\mathcal{F}_{-\infty}^{t} = \sigma(\ldots, x_{t-2}, x_{t-1}, x_t)$ and $\mathcal{F}_{t+m}^{\infty} = \sigma(x_{t+m}, x_{t+m+1}, \ldots)$. The sequence is said to be α-mixing (or strongly mixing) if $\lim_{m\to\infty} \alpha_m = 0$ with*

$$\alpha_m = \sup_t \alpha\left(\mathcal{F}_{-\infty}^{t}, \mathcal{F}_{t+m}^{\infty}\right),$$

where $\alpha(.,.)$ is given by (8.20). The sequence is said to be ϕ-mixing (or uniform mixing) if $\lim_{m\to\infty} \phi_m = 0$ with

$$\phi_m = \sup_t \phi\left(\mathcal{F}_{-\infty}^{t}, \mathcal{F}_{t+m}^{\infty}\right),$$

where $\phi(.,.)$ is given by (8.21).

Definition 8 *$\{x_t\}$ is α-mixing of size $-a$ if $\alpha_m = O\left(m^{-a-\epsilon}\right)$, for some $\epsilon > 0$ and $a \in \mathbb{R}$. $\{x_t\}$ is ϕ-mixing of size $-a$ if $\phi_m = O\left(m^{-a-\epsilon}\right)$, for some $\epsilon > 0$ and $a \in \mathbb{R}$.*

A stochastic process could be dependent but asymptotically uncorrelated. This notion of asymptotic non-correlated process is very useful for characterizing the correlation structure of a sequence of dependent random variables.

Definition 9 *$\{x_t\}$ has asymptotically uncorrelated elements (or is asymptotically uncorrelated) if there exist constants $\{\rho_\tau, \tau \geq 0\}$ such that $0 < \rho_\tau < 1$, $\sum_{\tau=0}^{\infty} \rho_\tau < \infty$ and*

$$Cov(x_t, x_{t+\tau}) \leq \rho_\tau \left[Var(x_t)\, Var(x_{t+\tau})\right]^{1/2}, \text{ for all } \tau > 0,$$

where $Var(x_t) < \infty$ for all t.

Note that, in the above definition, for $\sum_{\tau=0}^{\infty} \rho_\tau < \infty$, it is necessary that $\rho_\tau \to 0$ as $\tau \to \infty$. A sufficient condition would be, for example, that for τ sufficiently large, $\rho_\tau < \tau^{-(1+\delta)}$, for some $\delta > 0$. Whether a stochastic process is asymptotically uncorrelated can easily be verified. For

example, covariance stationary sequences can often be shown to be asymptotically uncorrelated, although an asymptotically uncorrelated sequence need not be covariance stationary.

Theorem 19 (Strong law for mixing processes) *Let $\{x_t\}$ be a α-mixing sequence of size $-r/(r-1)$ with $r > 1$, or a ϕ-mixing sequence of size $-r/(2r-1)$, with $r \geq 1$. If $E|x_t|^{r+\delta} < K < \infty$, for some $\delta > 0$ and all t, then $\bar{x}_t - \bar{\mu}_t \xrightarrow{wp1} 0$.*

Theorem 20 (Strong law for asymptotically uncorrelated processes) *Let $\{x_t\}$ be a sequence of random variables with asymptotically uncorrelated elements, and with means $E(x_t) = \mu_t$, and variances $Var(x_t) = \sigma_t^2 < \infty$. Then $\bar{x}_T \xrightarrow{wp1} \bar{\mu}_T$.*

See White (2000), page 53. Let, for example, $\{x_t\}$ be a covariance stationary process with $E(x_t) = \mu$, and with autocovariances given by $\gamma(j) = E\left(x_t x_{t-j}\right)$. If autocovariances are absolutely summable, namely if

$$\sum_{j=0}^{\infty} \left|\gamma(j)\right| < \infty,$$

then from the above theorem it follows that $\bar{x}_T \xrightarrow{wp1} \mu$.

It is interesting to observe that Theorem 20, compared with Theorem 19, relaxes the dependence restriction from asymptotic independence (mixing) to asymptotic uncorrelatedness. At the same time, the moment requirements have been strengthened from requiring the existence of moments of order $r + \delta$ (with $r \geq 1$ and $\delta > 0$), to requiring the existence of second-order moments.

We now present some results for martingale difference sequences and for L_p-mixingales (see Section 15.3). To this end, the following definition is helpful.

Definition 10 *$\{x_t\}$ is said to be uniformly integrable if, for every $\epsilon > 0$, there exists a constant $M > 0$ such that*

$$E\left[|x_t|\, 1_{[|x_t| \geq M]}\right] < \epsilon, \tag{8.22}$$

for all t, where $1_{\{|x_t| \geq M\}}$ is an indicator function.

The following theorems provide weak and strong laws of large numbers for martingale difference sequences and for mixingales.

Theorem 21 (Weak law for martingale differences) *Let $\{x_t\}$ be a martingale difference sequence with respect to the information set Ω_t. If $\{x_t\}$ is uniformly integrable then $\bar{x}_T \xrightarrow{p} 0$.*

A proof can be found in Davidson (1994, p. 301).

Theorem 22 (Strong law for martingale differences) *Let $\{x_t\}$ be a martingale difference sequence with respect to the information set Ω_t. If, for $1 \leq p \leq 2$, we have*

$$\sum_{t=1}^{\infty} \left(E\left|x_t\right|^p\right) / t^p < \infty,$$

then $\bar{x}_T \xrightarrow{wp1} 0$.

For a proof, see Davidson (1994, p. 315).

Theorem 23 (Weak law for L_1-mixingales) *Let $\{x_t\}$ be a L_1-mixingale with respect to Ω_t. If $\{x_t\}$ is uniformly integrable and there exists a choice for $\{c_t\}$ such that*

$$\lim_{T\to\infty} T^{-1} \sum_{t=1}^{T} c_t < \infty,$$

then $\bar{x}_T \xrightarrow{p} 0$.

See Davidson (1994), page 302, and Hamilton (1994), page 190.

Theorem 24 (Strong law for L_p-mixingales) *Let $\{x_t\}$ be a L_p-mixingale with respect to Ω_t with either (i) $p = 2$ of size $-1/2$, or (ii) $1 < p < 2$, of size -1; if there exists a choice for $\{c_t\}$ such that*

$$\lim_{T\to\infty} T^{-1} \sum_{t=1}^{T} c_t < \infty,$$

then $\bar{x}_T \xrightarrow{wp1} 0$.

See Davidson (1994, p. 319).

Theorem 25 (Strong law for L_p-mixingales) *Let $\{x_t\}$ be a L_p-mixingale with respect to Ω_t, with $1 < p \leq 2$, of size $-\lambda$. If $c_t/a_t = O(t^{\alpha})$, with $\alpha < \min\left\{-1/p, \left(1 - 1/p\right)\lambda - 1\right\}$, then* $\bar{x}_T \xrightarrow{wp1} 0$.

See Davidson and de Jong (1997).

8.8.2 Central limit theorems

We now provide central limit theorems for sequences of mixing random variables, for stationary processes and for martingale difference sequences, and refer to de Jong (1997) for central limit theorems for triangular arrays of L_p-mixingales.

The following provides some central limit theorem results for sequences of mixing random variables.

Theorem 26 *Let $\{x_t\}$ be a sequence of random variables such that $E|x_t|^r < K < \infty$ for some $r \geq 2$, and all t. If $\{x_t\}$ is α-mixing of size $-r/(r-2)$ or ϕ-mixing of size $-r/2(r-1)$, with $r > 2$, and $\bar{\sigma}_T^2 = Var\left(T^{-1/2}\sum_{t=1}^{T} x_t\right) > 0$, then $\sqrt{T}\left(\bar{x}_T - \bar{\mu}_T\right)/\bar{\sigma}_T \xrightarrow{d} N(0,1)$.*

See White (2000), Theorem 5.20. See also Corollary 3.2 in Wooldridge and White (1988), and Theorem 4.2 in McLeish (1975a).

Theorem 27 *Let*

$$x_t = \sum_{j=0}^{\infty} \psi_j \varepsilon_{t-j},$$

where ε_t is a sequence of IID random variables with $E(\varepsilon_t) = 0$, and $E\left(\varepsilon_t^2\right) < \infty$. Assume that $\sum_{j=0}^{\infty}\left|\psi_j\right| < \infty$. Then $\sqrt{T}\bar{x}_T \xrightarrow{d} N\left(0, \sum_{j=-\infty}^{\infty}\gamma(j)\right)$, where $\gamma(j)$ is the jth order autocovariance of x_t.

See Hamilton (1994, p. 195).

Theorem 28 *Let $\{x_t\}$ be a martingale difference sequence with respect to the information set Ω_t. Let $\bar{\sigma}_T^2 = Var\left(\sqrt{T}\bar{x}_T\right) = \frac{1}{T}\sum_{t=1}^{T}\sigma_t^2$. If $E(|x_t|^r) < K < \infty$, $r > 2$ and for all t, and*

$$T^{-1}\sum_{t=1}^{T} x_t^2 - \bar{\sigma}_T^2 \xrightarrow{p} 0,$$

then $\sqrt{T}\bar{x}_T/\bar{\sigma}_T \xrightarrow{d} N(0,1)$.

See White (2000, Corollary 5.26).

See Hall and Heyde (1980), Davidson (1994), White (2000) for further details on asymptotics for dependent and heterogeneously distributed observations.

8.9 Transformation of asymptotically normal statistics

It is often the case that we know the distribution of one or more estimators or test statistics, but what is of interest is a function of these estimators or statistics. As an example consider the simple dynamic model

$$y_t = \alpha x_t + \lambda y_{t-1} + u_t,\ u_t \sim (0, \sigma^2), \tag{8.23}$$

where under standard classical assumptions it can be established that the *OLS* estimator of $\theta = (\alpha, \lambda)'$, say $\hat{\theta}_T$, is asymptotically normally distributed with mean θ_0 (the 'true' value of θ), and a covariance matrix $\sigma_T^2\mathbf{V}$ where $\sigma_T^2 \to 0$ as $T \to \infty$. But, the parameter of interest is the long-run

response of y_t to a unit change in x_t, namely $g(\theta) = \frac{\alpha}{1-\lambda}$, and the asymptotic distribution of $g\left(\hat{\theta}_T\right)$ is required. The following theorem is particularly useful for such problems:

Theorem 29 *Suppose that* $\mathbf{x}_T = (x_{1T}, x_{2T}, \ldots, x_{mT})'$ *is asymptotically distributed as* $N\left(\boldsymbol{\mu}, \sigma_T^2 \mathbf{V}\right)$, *where* $\mathbf{V}$ *is a fixed matrix and the scalar constants* $\sigma_T^2 \to 0$ *as* $T \to \infty$. *Let* $\mathbf{g}(\mathbf{x}) = (g_1(\mathbf{x}), g_2(\mathbf{x}), \ldots, g_p(\mathbf{x}))$, $\mathbf{x} = (x_1, x_2, \ldots x_m)'$, *be a vector-valued function for which each component* $g_1(\mathbf{x})$ *is real-valued, and with non-zero differentials at* $\mathbf{x} = \boldsymbol{\mu}$ *given by the matrix*

$$\underset{m \times p}{\mathbf{G}} = \left[\frac{\partial g_i(\mathbf{x})}{\partial x_j}\right]_{\mathbf{x}=\boldsymbol{\mu}}.$$

It follows that

$$\mathbf{g}(\mathbf{x}_T) \xrightarrow{d} N\left[\mathbf{g}(\boldsymbol{\mu}), \sigma_T^2 \mathbf{G}\mathbf{V}\mathbf{G}'\right].$$

Proof Since $Var(\mathbf{x}_T) = \sigma_T^2 \mathbf{V}$, as $T \to \infty$, and $\sigma_T^2 \to 0$ it follows that $\mathbf{x}_T \xrightarrow{p} \boldsymbol{\mu}$, and

$$\|\mathbf{x}_T - \boldsymbol{\mu}\| = o_p(1).$$

From the Taylor series approximation result for stochastic processes we have

$$\mathbf{g}(\mathbf{x}_T) = \mathbf{g}(\boldsymbol{\mu}) + \mathbf{G}(\mathbf{x}_T - \boldsymbol{\mu}) + \mathbf{z}_t,$$

where $\mathbf{z}_T = o_p(\|\mathbf{x}_T - \boldsymbol{\mu}\|)$. Now using Slutsky's convergence theorem (see Theorem 6), $\frac{\mathbf{g}(\mathbf{x}_T) - \mathbf{g}(\boldsymbol{\mu})}{\sigma_T}$ and $\frac{\mathbf{G}(\mathbf{x}_T - \boldsymbol{\mu})}{\sigma_T}$ will have the same limiting distribution if we show that $Plim_T(\mathbf{z}_T/\sigma_T) = \mathbf{0}$. But since $\mathbf{z}_T = o_p(\|\mathbf{x}_T - \boldsymbol{\mu}\|) = o_p(1)$, then

$$\underset{T \to \infty}{Plim} \frac{\mathbf{z}_T}{\|\mathbf{x}_T - \boldsymbol{\mu}\|} = 0,$$

or

$$\underset{T \to \infty}{Plim} \frac{\mathbf{z}_T/\sigma_T}{\|\mathbf{x}_T - \boldsymbol{\mu}\|/\sigma_T} = 0.$$

However, by assumption $(\mathbf{x}_T - \boldsymbol{\mu})/\sigma_T$ has a finite limiting normal distribution and is bounded stochastically, that is $\|\mathbf{x}_T - \boldsymbol{\mu}\|/\sigma_T = o_p(1)$, and therefore we have $Plim_{T \to \infty}\left(\mathbf{z}_T/\sigma_T^2\right) = 0$. It follows that

$$\frac{\mathbf{g}(\mathbf{x}_T) - \mathbf{g}(\boldsymbol{\mu})}{\sigma_T} \xrightarrow{d} \frac{\mathbf{G}(\mathbf{x}_T - \boldsymbol{\mu})}{\sigma_T} \xrightarrow{d} N\left(\mathbf{0}, \mathbf{G}\mathbf{V}\mathbf{G}'\right).$$

See Serfling (1980, p. 122) for further details.

Example 16 *An application of this theorem to the dynamic regression model (8.23) yields the following asymptotic distribution for the long-run response of y_t with respect to a unit change in x_t:*

$$\frac{\hat{\alpha}_T}{1-\hat{\lambda}_T} \overset{a}{\sim} N\left(\frac{\alpha}{1-\lambda}, \left(\frac{\sigma^2}{T}\mathbf{G}\mathbf{V}\mathbf{G}'\right)\right), \tag{8.24}$$

where

$$\mathbf{V} = \underset{T\to\infty}{Plim}\begin{pmatrix} \sum_{t=1}^{T}\frac{x_t^2}{T} & \sum_{t=1}^{T}\frac{x_t y_{t-1}}{T} \\ \sum_{t=1}^{T}\frac{y_{t-1}x_t}{T} & \sum_{t=1}^{T}\frac{y_{t-1}^2}{T} \end{pmatrix}, \tag{8.25}$$

and

$$\mathbf{G} = \left(\frac{1}{1-\lambda}, \frac{\alpha}{(1-\lambda)^2}\right), \tag{8.26}$$

assuming that $|\lambda| < 1$. As we shall see later in the case where $\{x_t\}$ is covariance-stationary, the probability limits appearing in (8.25) exist and are finite.

Theorem 29 can also be extended to cover cases where the first or even higher-order partial derivatives of $\mathbf{g}(\mathbf{x})$ evaluated at $\boldsymbol{\mu}$ vanish. This extension to the case where the first-order partial derivatives of $\mathbf{g}(\mathbf{x})$ vanish at $\mathbf{x} = \boldsymbol{\mu}$, will be stated without proof in the following theorem:

Theorem 30 *(A generalization of Theorem 29). Suppose that the $m \times 1$ vector $\mathbf{x}_T$ has an asymptotic normal distribution $N\left(\boldsymbol{\mu}, T^{-1}\mathbf{V}\right)$. Let $\mathbf{g}(\mathbf{x})$ be a vector real-valued function of order $p \times 1$ possessing continuous partials of order 2 in the neighbourhood of $\mathbf{x} = \boldsymbol{\mu}$, with all partial derivatives of order 1 vanishing at $\mathbf{x} = \boldsymbol{\mu}$, but with the second-order partial derivatives not all vanishing at $\mathbf{x} = \boldsymbol{\mu}$. Then*

$$T\left[\mathbf{g}(\mathbf{x}_T) - \mathbf{g}(\boldsymbol{\mu})\right] \overset{d}{\to} \frac{1}{2}\mathbf{z}'\left(\mathbf{Q}'\mathbf{B}\mathbf{Q}\right)\mathbf{z}, \tag{8.27}$$

where $\mathbf{z} = \left(z_1, z_2, \ldots z_m\right)' \sim N\left(\mathbf{0}, \mathbf{I}_m\right)$, $\mathbf{V} = \mathbf{Q}\mathbf{Q}'$ and $\mathbf{B}$ is an $m \times m$ matrix with (i,j) elements

$$\mathbf{B} = \left\{\frac{\partial^2 g(\mathbf{x})}{\partial x_i \partial x_j}\right\}_{\mathbf{x}=\boldsymbol{\mu}}. \tag{8.28}$$

See Serfling (1980, pp. 124–5).

In the remainder of this chapter we consider the application of asymptotic theory to the classical regression model in a series of examples. See also Chapter 2.

Example 17 *Consider the classical linear regression model*

$$y_t = \boldsymbol{\beta}'\mathbf{x}_t + u_t, t = 1, 2, \ldots T, \tag{8.29}$$

where $\boldsymbol{\beta}$ is a $k\times 1$ vector of unknown parameters, $\mathbf{x}_t$ is a $k\times 1$ vector of possibly stochastic regressors and u_t are the disturbance terms. We show that the OLS estimator of $\boldsymbol{\beta}$, defined by

$$\hat{\boldsymbol{\beta}} = \left(\sum_{t=1}^{T}\mathbf{x}_t\mathbf{x}_t'\right)^{-1}\left(\sum_{t=1}^{T}\mathbf{x}_t y_t\right),$$

is consistent, if $T^{-1}\sum_{t=1}^{T}\mathbf{x}_t\mathbf{x}_t' \xrightarrow{p} \Sigma_{xx}$, where Σ_{xx} is a nonsingular matrix, and

$$T^{-1}\sum_{t=1}^{T}\mathbf{x}_t u_t \xrightarrow{p} \mathbf{0}.$$

Writing (8.29) in matrix notation we have

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{u},$$

where $\mathbf{y}$ and $\mathbf{u}$ are $T\times 1$ vectors and $\mathbf{X}$ in a $T\times k$ matrix of observations on $\mathbf{x}_t$. Denoting the OLS estimator of $\boldsymbol{\beta}$ by $\hat{\boldsymbol{\beta}}_T$, under (8.29) we have

$$\begin{aligned}\hat{\boldsymbol{\beta}}_T - \boldsymbol{\beta} &= \left(\sum_{t=1}^{T}\mathbf{x}_t\mathbf{x}_t'\right)^{-1}\left(\sum_{t=1}^{T}\mathbf{x}_t u_t\right)\\ &= \left(\mathbf{X}'\mathbf{X}\right)^{-1}\mathbf{X}'\mathbf{u} = \left(\frac{\mathbf{X}'\mathbf{X}}{T}\right)^{-1}\left(\frac{\mathbf{X}'\mathbf{u}}{T}\right).\end{aligned}$$

Since by assumption $T^{-1}\left(\mathbf{X}'\mathbf{X}\right)$ converges in probability to a nonsingular matrix, then by the convergence Theorem 9, we have

$$\begin{aligned}\underset{T\to\infty}{Plim}\left(\hat{\boldsymbol{\beta}}_T - \boldsymbol{\beta}\right) &= \left[\underset{T\to\infty}{Plim}\left(\frac{\mathbf{X}'\mathbf{X}}{T}\right)\right]^{-1}\underset{T\to\infty}{Plim}\left(\frac{\mathbf{X}'\mathbf{u}}{T}\right)\\ &= (\Sigma_{xx})^{-1}\left[\underset{T\to\infty}{Plim}\left(\frac{\mathbf{X}'\mathbf{u}}{T}\right)\right],\end{aligned}$$

and since by assumption $\left(\frac{\mathbf{X}'\mathbf{u}}{T}\right)\xrightarrow{p} 0$, then $\hat{\boldsymbol{\beta}}_T - \boldsymbol{\beta}\xrightarrow{p} 0$, and hence $\hat{\boldsymbol{\beta}}_T$ is a consistent estimator of $\boldsymbol{\beta}$.

Example 18 *Consider the regression model in the above example and suppose now that one of the regressors is a time trend. It is clear that in this case the row and the column of $T^{-1}\mathbf{X}'\mathbf{X}$ associated with the trended variable blow-up as $T\to\infty$, and the convergence theorems cannot be applied to $\hat{\boldsymbol{\beta}}_T-\boldsymbol{\beta}$ directly. Dividing $\mathbf{X}'\mathbf{X}$ by T^{-3} does not resolve the problem either, since the terms $T^{-3}\sum_{t=1}^{T}x_{it}x_{jt}$ converge to zero when at least one of the variables (x_{it} or x_{jt}) is bounded. Suppose that all the regressors are bounded except for x_{kt} which is trended, i.e., $K_L\le|x_{it}|\le K_U$, for $i=1,2,\dots k-1$, and $x_{kt}=t$, for $t=1,2,\dots T$. Then $T^{-1}\sum_{t=1}^{T}x_{it}x_{kt}$, and $T^{-1}\sum_{t=1}^{T}x_{kt}^2$*

blow up as $T \to \infty$, *and while* $T^{-3}\sum_{t=1}^{T} x_{kt}^2 \to \frac{1}{3}$, *the other terms in* $T^{-3}\sum_{t=1}^{T} x_{it}x_{jt}$ *converge to zero, which makes* Σ_{xx} *a singular matrix. The problem lies in the fact that the OLS estimators of the coefficients of the non-trended variables* $(x_{1t}, x_{2t,} \ldots x_{k-1,t,})$ *and that of the trended variable,* $x_{kt,}$, *converge to their 'true' values at different rates. To simplify the exposition let* $k = 2$, *and suppose* x_{1t} *is bounded (with coefficient* β_1*) and* $x_{2t} = t$, *(with coefficient* β_2*). We have*

$$\underset{T\to\infty}{Plim}\left(\frac{\sum_{t=1}^{T} x_{2t}^2}{T^3}\right) = \lim_{T\to\infty}\left(\frac{\sum_{t=1}^{T} t^2}{T^3}\right) = \frac{1}{3}, \tag{8.30}$$

$$\underset{T\to\infty}{Plim}\left(\frac{\sum_{t=1}^{T} x_{1t}^2}{T}\right) < K_U^2, \tag{8.31}$$

$$\underset{T\to\infty}{Plim}\frac{\sum_{t=1}^{T} x_{1t}x_{2t}}{T^2} < K_U, \tag{8.32}$$

and since $Var\left(T^{-2}\sum_{t=1}^{T} x_{2t}u_t\right) = T^{-4}\left(\sum_{t=1}^{T} t^2\right)\sigma^2 \to 0$, *as* $T \to \infty$, *then* $T^{-2}\sum_{t=1}^{T} x_{2t}u_t = T^{-2}\sum_{t=1}^{T} tu_t \xrightarrow{p} 0$.

Similarly, it is easily established that $T^{-1}\sum_{t=1}^{T} x_{1t}u_t \xrightarrow{p} 0$. *Consider now the following expressions for the OLS estimators of* β_1 *and* β_2

$$\hat{\beta}_{1T} - \beta_1 = \frac{\left(\frac{\sum_{t=1}^{T} x_{2t}^2}{T^3}\right)\left(\frac{\sum_{t=1}^{T} x_{1t}u_t}{T}\right) - \left(\frac{\sum_{t=1}^{T} x_{1t}x_{2t}}{T^2}\right)\left(\frac{\sum_{t=1}^{T} x_{2t}u_t}{T^2}\right)}{\Delta_T}, \tag{8.33}$$

$$T(\hat{\beta}_{2T} - \beta_2) = \frac{\left(\frac{\sum_{t=1}^{T} x_{2t}u_t}{T^2}\right)\left(\frac{\sum_{t=1}^{T} x_{1t}^2}{T}\right) - \left(\frac{\sum_{t=1}^{T} x_{1t}u_t}{T}\right)\left(\frac{\sum_{t=1}^{T} x_{1t}x_{2t}}{T^2}\right)}{\Delta_T}, \tag{8.34}$$

where

$$\Delta_T = \left(T^{-3}\sum_{t=1}^{T} x_{2t}^2\right)\left(T^{-1}\sum_{t=1}^{T} x_{1t}^2\right) - \left(T^{-2}\sum_{t=1}^{T} x_{1t}x_{2t}\right)^2.$$

Using the results (8.30)–(8.32) in (8.33) and (8.34) and noting that $T^{-1}\sum_{t=1}^{T} x_{1t}u_t$ *and* $T^{-2}\sum_{t=1}^{T} x_{2t}u_t$ *both converge in probability to zero, we have* $\hat{\beta}_{1T} - \beta_1 \xrightarrow{p} 0$ *and* $T\left(\hat{\beta}_{2T} - \beta_2\right) \xrightarrow{p} 0$. *It is also easily seen that*

$$\hat{\beta}_{1T} - \beta_1 = O_p(T^{-1/2}),$$
$$\hat{\beta}_{2T} - \beta_2 = O_p(T^{-3/2}).$$

Since the rate of convergence of $\hat{\beta}_{2T}$ to β_2 is much faster than is usually encountered in the statistics literature, $\hat{\beta}_{2T}$ is said to be 'super-consistent' for β_2, a concept which we shall encounter later in our discussion of cointegration. See also Theorem 31 below.

Example 19 (Misspecification) *Suppose the correct regression model for y_t is given by*

$$y_t = \boldsymbol{\alpha}'\mathbf{z}_t + \varepsilon_t \tag{8.35}$$

where $\mathbf{z}_t$ is a $s \times 1$ vector of regressors, in general different from $\mathbf{x}_t$, and ε_t are IID random variables, distributed independently of $\mathbf{x}_{t'}$ and $\mathbf{z}_{t'}$ for all t and t'. Namely, $\mathbf{z}_t$ and $\mathbf{x}_t$ are strictly exogenous in the context of (8.35). But $\boldsymbol{\beta}$ is still estimated using (8.29). We show that if $T^{-1}\mathbf{X}'\mathbf{Z} \xrightarrow{p}$ (a finite $k \times s$ matrix) then $\hat{\boldsymbol{\beta}}_T - \boldsymbol{\beta} \xrightarrow{p} \Sigma_{xx}^{-1}\Sigma_{xz}\alpha$, which, in general, differs from α. Under (8.35) we have

$$\begin{aligned}\hat{\boldsymbol{\beta}}_T &= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'(\mathbf{Z}\boldsymbol{\alpha} + \boldsymbol{\varepsilon}), \\ \hat{\boldsymbol{\beta}}_T &= \left(\frac{\mathbf{X}'\mathbf{X}}{T}\right)^{-1}\left(\frac{\mathbf{X}'\mathbf{Z}}{T}\right)\boldsymbol{\alpha} + \left(\frac{\mathbf{X}'\mathbf{X}}{T}\right)^{-1}\left(\frac{\mathbf{X}'\boldsymbol{\varepsilon}}{T}\right),\end{aligned} \tag{8.36}$$

where $\boldsymbol{\varepsilon} = (\varepsilon_1, \varepsilon_2, \ldots, \varepsilon_T)'$. However, by assumption $T^{-1}\mathbf{X}'\mathbf{X} \xrightarrow{p} \Sigma_{xx}$ (a nonsingular matrix), and $T^{-1}\mathbf{X}'\mathbf{Z} \xrightarrow{p} \Sigma_{xz}$ (a finite matrix). Also, since by assumption $\mathbf{x}_t$ is strictly exogenous we have

$$\begin{aligned}Var\left[T^{-1}(\mathbf{X}'\boldsymbol{\varepsilon})\right] &= E\left(T^{-2}\mathbf{X}'\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}'\mathbf{X}\right) \\ &= E\left[T^{-2}E\left(\mathbf{X}'\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}'\mathbf{X} \mid X\right)\right] = \sigma_\varepsilon^2 T^{-1}E\left(\mathbf{X}'\mathbf{X}\right).\end{aligned}$$

But $T^{-1}\mathbf{X}'\mathbf{X} \xrightarrow{p} \lim_{T\to\infty} T^{-1}E\left(\mathbf{X}'\mathbf{X}\right) = \Sigma_{xx}$, and hence $Var\left(T^{-1}\mathbf{X}'\boldsymbol{\varepsilon}\right) \to 0$ as $T \to \infty$. Therefore, $T^{-1}\mathbf{X}'\boldsymbol{\varepsilon} \xrightarrow{q.m.} 0$, which implies that $T^{-1}\mathbf{X}'\boldsymbol{\varepsilon} \xrightarrow{p} 0$ (see Theorem 2). Using these results in (8.36) and taking advantage of the stochastic convergence Theorem 7 and Theorem 9, we have

$$\underset{T\to\infty}{Plim}\left(\hat{\boldsymbol{\beta}}_T\right) = \Sigma_{xx}^{-1}\Sigma_{xz}\alpha + \Sigma_{xx}^{-1}\underset{T\to\infty}{Plim}\left(\frac{\mathbf{X}'\boldsymbol{\varepsilon}}{T}\right),$$

and since $Plim_{T\to\infty}\left(T^{-1}\mathbf{X}'\boldsymbol{\varepsilon}\right) = 0$, we have

$$\underset{T\to\infty}{Plim}\left(\hat{\boldsymbol{\beta}}_T\right) = \boldsymbol{\beta}_*(\boldsymbol{\alpha}) = \Sigma_{xx}^{-1}\Sigma_{xz}\boldsymbol{\alpha}.$$

In the misspecification literature, $\boldsymbol{\beta}_*(\boldsymbol{\alpha})$ or $\boldsymbol{\beta}_*$ for short, is known as the pseudo-true value of $\hat{\boldsymbol{\beta}}_T$, and shows the explicit dependence of the probability limit of $\hat{\boldsymbol{\beta}}_T$ on the parameters of the correctly specified model.[3] *Theil (1957) and Griliches (1957) were the first to discuss the implications of the*

[3] For a general dicussion of pseudo-true values, see Section 11.3.

above result in econometrics. In particular, they discussed the effect of incorrectly deleting or adding regressors on the OLS estimators.

Theorem 31 *Consider the regression model in Example 17 and assume that u_t is IID $\left(0, \sigma^2\right)$ and that*

$$\lim_{T\to\infty}\left\{\frac{\max_{1\le t\le T}\left(x_{it}\right)^2}{\sum_{t=1}^{T} x_{it}^2}\right\} = 0,\ i = 1, 2, \ldots k, \tag{8.37}$$

then

$$\mathbf{A}_T\left(\hat{\boldsymbol{\beta}}_T - \boldsymbol{\beta}\right) \xrightarrow{d} N\left(\mathbf{0}, \sigma^2\mathbf{V}\right), \tag{8.38}$$

where

$$\mathbf{A}_T = \begin{pmatrix} \left(\sum_{t=1}^{T} x_{1t}^2\right)^{\frac{1}{2}} & 0 & \ldots & 0 \\ 0 & \left(\sum_{t=1}^{T} x_{2t}^2\right)^{\frac{1}{2}} & \ldots & 0 \\ \vdots & \vdots & \ldots & \vdots \\ 0 & 0 & \ldots & \left(\sum_{t=1}^{T} x_{kt}^2\right)^{\frac{1}{2}} \end{pmatrix}, \tag{8.39}$$

and

$$\mathbf{V} = \lim_{T\to\infty}\left(\mathbf{A}_T^{-1}\mathbf{X}'\mathbf{X}\mathbf{A}_T^{-1}\right), \tag{8.40}$$

exists and is nonsingular.

A proof of this theorem is given in Amemiya (1985), and makes use of the Linderberg–Feller central limit Theorem 17. What is interesting about this theorem is the fact that it accommodates regression models containing both trended and non-trended variables. It is clear that when the regressors are bounded the condition (8.37) is satisfied. For trended variables, say $x_{it} = t$, we have $\max_{1\le t\le T} x_{it}^2 = T^2$, and

$$\sum_{t=1}^{T} x_{it}^2 = \frac{T(T+1)(2T+1)}{6}.$$

Hence, once again (8.37) is satisfied.

Similar conclusions also follow if higher-order trended variables are considered. In general, the rate of convergence of $\hat{\beta}_{iT}$ to β_i depends on the rate at which $\left(T^{-1}\sum_{t=1}^{T} x_{it}^2\right)^{\frac{1}{2}}$ converges to zero. Namely

$$\hat{\beta}_{iT} - \beta_i = o_p\left[\left(\sum_{t=1}^{T} \frac{x_{it}^2}{T}\right)^{\frac{1}{2}}\right],$$

or

$$\left(\sum_{t=1}^{T} x_{it}^2\right)^{\frac{1}{2}} \left(\hat{\beta}_{iT} - \beta_i\right) = O_p(1).$$

For bounded regressors $T^{-1}\sum_{t=1}^{T} x_{it}^2 = O_p(1)$, and $\hat{\beta}_{iT} - \beta_i = o_p(1)$. When $x_{jt} = t$, then $T^{-1}\sum_{t=1}^{T} x_{jt}^2 = O(T^2)$ and $\hat{\beta}_{jT} - \beta_j = o_p\left(T^{-1}\right)$, a result already demonstrated in Example 18. When $x_{lt} = t^2$, then $T^{-1}\sum_{t=1}^{T} x_{lt}^2 = O(T^4)$, and $\hat{\beta}_{lT} - \beta_l = o_p\left(T^{-2}\right)$, etc. Similarly, we have $\hat{\beta}_{iT} - \beta_i = O_p\left(T^{-1/2}\right)$, $\hat{\beta}_{jT} - \beta_j = O_p\left(T^{-3/2}\right)$, and $\hat{\beta}_{lT} - \beta_l = O_p\left(T^{-5/2}\right)$.

8.10 Further reading

A systematic review of large sample theory and its application to econometrics can be found in White (2000). Further discussion on limit theorems for heterogeneous and dependent random variables can be found in Davidson (1994). For further reading in probability, see Billingsley (1995), as well as the set of references cited in Appendix B.

8.11 Exercises

1. Prove that, if $x_t \xrightarrow{d} x$, and $P(x = c) = 1$, where c is a constant, then $x_t \xrightarrow{d} c$.
2. Show that if x_t is bounded (i.e., $P(|x_t| \leq M) = 1$ for all t, and for some $M < \infty$), then $x_t \xrightarrow{d} x$ implies $\lim_{t\to\infty} E(x_t) = E(x)$.
3. Let $\{X_t\}$ be a sequence of *IID* random variables with $E(X_t) = \mu$, $E(X_t^2) = 1$, third central moment $E(X_t - \mu)^3 = 0$, and fourth central moment $E(X_t - \mu)^4 = 3$.

   (a) Show that

$$E\left(T^{-1}\sum_{t=1}^{T} X_t^2\right) = 1 + \mu^2,$$

   and discuss the estimation method that underlies the following estimate of μ, based on the T observations $x_1, x_2, \ldots, x_T$

$$\hat{\mu}_T = \left[T^{-1}\sum_{t=1}^{T}(x_t^2 - 1)\right]^{1/2}.$$

(b) Compute the mean and the variance of $x_t^2 - 1 - \mu^2$.

(c) Show that $\hat{\mu}_T$ converges in probability to μ.

(d) Consider the following mean value expansion of $\hat{\mu}_T$ around μ

$$\sqrt{n}(\hat{\mu}_T - \mu) = \frac{1}{2\bar{\mu}_T}\left[\frac{1}{\sqrt{T}}\sum_{t=1}^{T}(x_t^2 - 1 - \mu^2)\right],$$

where $\bar{\mu}_T$ lies on the line joining $\hat{\mu}_T$ to μ. Hence, or otherwise, determine the asymptotic distribution of $\sqrt{T}(\hat{\mu}_T - \mu)$. Discuss possible difficulties with your derivation when $\mu = 0$.

4. Let $\{y_t\}$ be a sequence of random variables, where $y_t = 0$ with probability $1 - t^{-2}$, and $y_t = t$ with probability t^{-2}. Let $x_t = y_t - E(y_t)$. Verify whether the Lindberg and Feller *CLT* holds for $\{x_t\}$.
5. Let $\{x_t\}$ be a sequence of random variables with $x_t = \rho x_{t-1} + \varepsilon_t$, where ε_t is $IID(0, \sigma^2)$ and $|\rho| < 1$. Verify that $\{x_t\}$ is asymptotically uncorrelated (see Definition 9).
6. Consider the following regression

$$y_t = \alpha/t + \varepsilon_t, \text{for } t = 1, 2 \ldots, T,$$

where ε_t are *IID* random variables with mean zero and a constant variance. Show that the *OLS* estimator of α need not be consistent.

9 Maximum Likelihood Estimation

9.1 Introduction

One important tool of estimation and hypothesis testing in econometrics is maximum likelihood (ML) estimation and the various testing procedures associated with it, namely, the likelihood ratio (LR), the score or the Lagrange multiplier (LM), and the Wald (W) procedures. In this chapter, we provide an account of the likelihood theory, describe the different testing procedures and the relationships among them and discuss the consequences of misspecification of the likelihood model on the asymptotic properties of the *ML* estimators. The misspecification analysis is particularly important in econometric applications where the maintained model can at best provide a crude approximation of the underlying data generation process.

9.2 The likelihood function

Let $(x_1, x_2, \ldots, x_T)$ be a random sample from a distribution whose density is given by $f(x, \boldsymbol{\theta})$, where $\boldsymbol{\theta} \in \Theta$, is the $p \times 1$ vector of unknown parameters, and $\Theta \in \mathbb{R}^p$ is the parameter space. The *likelihood function* is defined as the joint probability density of $(x_1, x_2, \ldots, x_T)$, but viewed as a function of the unknown parameters, $\boldsymbol{\theta}$. It is usually denoted by $L_T(\boldsymbol{\theta}; x_1, \ldots, x_T)$, or more simply as

$$L_T(\boldsymbol{\theta}, \mathbf{x}) = f(x_1, \ldots, x_T; \boldsymbol{\theta}), \tag{9.1}$$

where $\mathbf{x} = (x_1, x_2, \ldots, x_T)'$, and $f(\mathbf{x}; \boldsymbol{\theta})$ represents the joint density function of the sample $(x_1, x_2, \ldots, x_T)$. It is often convenient to work with the logarithm of the likelihood function, the so-called log-likelihood function

$$\ell_T(\boldsymbol{\theta}) = \log L_T(\boldsymbol{\theta}, \mathbf{x}). \tag{9.2}$$

In the case of *IID* random variables, we have

$$\ell_T(\boldsymbol{\theta}) = \sum_{t=1}^{T} \log f(x_t, \boldsymbol{\theta}). \tag{9.3}$$

The likelihood function gives the probability that a particular set of realizations of x, namely $x_1, \ldots, x_T$, lie in the range x and $x + \Delta x$. R. A. Fisher suggested estimating the unknown parameters, $\boldsymbol{\theta}$, by maximizing the likelihood function with respect to $\boldsymbol{\theta}$. The value of $\boldsymbol{\theta}$, say $\hat{\boldsymbol{\theta}}_T$, at which $\ell_T(\boldsymbol{\theta})$ is *globally* maximized is referred to as the maximum likelihood (ML) estimator of $\boldsymbol{\theta}$.

Example 20 (The Bernoulli distribution) *Consider a random sample of size T which is drawn from the Bernoulli distribution*

$$f(x_t, \theta) = \theta^x (1-\theta)^{1-x},\ 0 < \theta < 1,$$

where θ is a scalar parameter representing the probability of success (or failure). The sample values $x_1, x_2, \ldots, x_T$, will be a sequence of 0 and 1. The log-likelihood function for this problem is given by

$$\ell_T(\theta) = \left(\sum_{t=1}^{T} x_t\right) \ln(\theta) + \left(T - \sum_{t=1}^{T} x_t\right) \ln(1-\theta).$$

The necessary condition for the maximization of the log-likelihood function is given by equating the first derivative of $\ell_T(\theta)$ to zero

$$\begin{aligned} \frac{\partial \ell_T(\theta)}{\partial \theta} &= \left(\sum_{t=1}^{T} x_t\right) \frac{1}{\theta} - \left(T - \sum_{t=1}^{T} x_t\right) \frac{1}{1-\theta} \\ &= \frac{1}{\theta(1-\theta)} \left(\sum_{t=1}^{T} x_t - T\theta\right). \end{aligned}$$

Hence, $\frac{\partial \ell_T(\theta)}{\partial \theta} = 0$, yields the ML estimator $\hat{\theta}_T = T^{-1} \sum_{t=1}^{T} x_t = \bar{x}_T$.

Example 21 (Linear regression with normal errors) *Consider the classical normal regression model*

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{u}, \tag{9.4}$$

$$\mathbf{u} \,|\mathbf{X} \sim N(\mathbf{0}, \sigma^2 \mathbf{I}_T), \tag{9.5}$$

where $\mathbf{y} = (y_1, y_2, \ldots, y_T)'$, $\boldsymbol{\beta} = (\beta_1, \beta_2, \ldots, \beta_k)'$, and $\mathbf{X}$ is a $T \times k$ matrix of observations on the exogenously given regressors. Noting that conditional on $\mathbf{X}$, the $T \times 1$ error vector $\mathbf{u}$ is assumed to be normally distributed, the log-likelihood function for model (9.4)-(9.5) is given by

$$\ell_T(\boldsymbol{\theta}) = -\frac{T}{2}\ln\left(2\pi\sigma^2\right) - \frac{1}{2\sigma^2}\left(\mathbf{y}-\mathbf{X}\boldsymbol{\beta}\right)'\left(\mathbf{y}-\mathbf{X}\boldsymbol{\beta}\right), \tag{9.6}$$

where $\boldsymbol{\theta} = (\boldsymbol{\beta}', \sigma^2)'$. *The maximization of* $\ell_T(\boldsymbol{\theta})$ *with respect to* $\boldsymbol{\beta}$ *will be the same as the minimization of the sum of squares of errors,* $Q(\boldsymbol{\beta}) = (\mathbf{y}-\mathbf{X}\boldsymbol{\beta})'(\mathbf{y}-\mathbf{X}\boldsymbol{\beta})$ *with respect to* $\boldsymbol{\beta}$, *and establishes that in the context of this model the ML and OLS estimators of* $\boldsymbol{\beta}$ *are algebraically the same. We have*

$$\frac{\partial \ell_T(\boldsymbol{\theta})}{\partial \boldsymbol{\beta}} = \frac{1}{\sigma^2}\mathbf{X}'\left(\mathbf{y}-\mathbf{X}\beta\right), \tag{9.7}$$

and similarly

$$\frac{\partial \ell_T(\boldsymbol{\theta})}{\partial \sigma^2} = -\frac{T}{\sigma^2} + \frac{1}{2\sigma^4}\left(\mathbf{y}-\mathbf{X}\boldsymbol{\beta}\right)'\left(\mathbf{y}-\mathbf{X}\boldsymbol{\beta}\right). \tag{9.8}$$

Setting these derivatives equal to zero now yields the following ML estimators

$$\hat{\boldsymbol{\beta}}_T = \left(\mathbf{X}'\mathbf{X}\right)^{-1}\mathbf{X}'\mathbf{y}, \tag{9.9}$$

$$\hat{\sigma}^2_T = \frac{\left(\mathbf{y}-\mathbf{X}\boldsymbol{\beta}\right)'\left(\mathbf{y}-\mathbf{X}\boldsymbol{\beta}\right)}{T}. \tag{9.10}$$

We refer to Section 2.4 for a comparison of ML and OLS estimators.

9.3 Weak and strict exogeneity

Consider the linear regression model with normal errors

$$y_t = \mathbf{x}_t'\boldsymbol{\beta} + u_t,\ u_t \sim IIDN(0, \sigma^2). \tag{9.11}$$

Suppose now $\mathbf{x}_t$ and $\mathbf{y}_t$ are jointly distributed with parameters $\boldsymbol{\theta} = \left(\boldsymbol{\beta}', \sigma^2, \boldsymbol{\gamma}'\right)'$, where $\boldsymbol{\gamma}$ denotes the parameter vector of the probability distribution of $\mathbf{x}_t$. Let $\mathbf{z}_t = \left(\mathbf{y}_t', \mathbf{x}_t'\right)'$. Then the likelihood function of $\boldsymbol{\theta}$ is given by the joint probability distribution of $\mathbf{z}_1, \mathbf{z}_2, \ldots, \mathbf{z}_T$, namely

$$\begin{aligned} L_T(\boldsymbol{\theta}) &= f(\mathbf{z}_1, \mathbf{z}_2, \ldots, \mathbf{z}_T, \boldsymbol{\theta}) \\ &= \Pr(\mathbf{z}_1 \mid \Omega_0, \boldsymbol{\theta}) \Pr(\mathbf{z}_2 \mid \Omega_1, \boldsymbol{\theta}) \ldots \Pr(\mathbf{z}_T \mid \Omega_{T-1}, \boldsymbol{\theta}), \end{aligned} \tag{9.12}$$

where $\Omega_t, t = 0, 1, \ldots,$ is a sequence of non-decreasing σ-fields, containing at least observations on current and past values of $\mathbf{z}_t$.

In general, $P(\mathbf{z}_T \mid \Omega_{T-1})$ can be decomposed as

$$\Pr(\mathbf{z}_T \mid \Omega_{T-1}, \boldsymbol{\theta}) = \Pr(\mathbf{x}_T \mid \Omega_{T-1}, \boldsymbol{\theta}) \Pr\left(\mathbf{y}_T \mid \mathbf{x}_T;\ \Omega_{T-1}, \boldsymbol{\theta}\right),$$

where $\Pr(\mathbf{x}_T \mid \Omega_{T-1}, \boldsymbol{\theta})$ is known as the marginal density and $\Pr\left(\mathbf{y}_T \mid \mathbf{x}_T;\ \Omega_{T-1}, \boldsymbol{\theta}\right)$ as the conditional density. Suppose now that it is possible to write

$$\Pr(\mathbf{z}_T \mid \Omega_{T-1}, \boldsymbol{\theta}) = \Pr(\mathbf{x}_T \mid \Omega_{T-1}, \boldsymbol{\gamma}) \Pr\left(\mathbf{y}_T \mid \mathbf{x}_T;\ \Omega_{T-1}, \boldsymbol{\beta}, \sigma^2\right). \tag{9.13}$$

When $\boldsymbol{\gamma}$ does not depend on $\boldsymbol{\beta}$ and σ^2, then we say that $\mathbf{x}_t$ is weakly exogenous with respect to the estimation of $\boldsymbol{\beta}$ and σ^2. This decomposition, when it holds, allows us to ignore the marginal density of $\mathbf{x}$ in the *ML* estimation of $\boldsymbol{\beta}$ and σ^2. The concept of weak exogeneity holds more generally and has been discussed in detail by Engle, Hendry, and Richard (1983).

Under weak exogeneity, substituting (9.13) in (9.12) and taking logs we obtain

$$\ell_T\left(\beta, \sigma^2\right) = \sum_{t=1}^{T} \ln \Pr\left(y_t \mid \mathbf{x}_t, \Omega_{t-1}, \beta, \sigma^2\right). \tag{9.14}$$

The probability density of $\mathbf{x}_t$, which does not depend on $\boldsymbol{\beta}$ and σ^2, is left out of the log-likelihood function. Under (9.11) and conditional on $\mathbf{x}_t$ we have

$$\Pr\left(y_t \mid \mathbf{x}_t,\ \Omega_{T-1}, \boldsymbol{\beta}, \sigma^2\right) = \left(2\pi\sigma^2\right)^{-\frac{1}{2}} e^{-\frac{1}{2\sigma^2}u_t^2}.$$

Using this result in (9.14)

$$\ell_T\left(\beta, \sigma^2\right) = -\frac{T}{2}\ln\left(2\pi\sigma^2\right) - \frac{1}{2\sigma^2}\sum_{t=1}^{T} u_t^2,$$

or

$$\ell_T\left(\beta, \sigma^2\right) = -\frac{T}{2}\ln\left(2\pi\sigma^2\right) - \frac{1}{2\sigma^2}\left(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\right)'\left(\mathbf{y} - \mathbf{X}\beta\right).$$

Example 22 *Consider the following model for* $\mathbf{x}_t$

$$\mathbf{x}_t = \mathbf{G}\mathbf{x}_{t-1} + \boldsymbol{\lambda} y_{t-1} + \mathbf{v}_t,\ \mathbf{v}_t \sim IIDN(\mathbf{0}, \Sigma_{xx}), \tag{9.15}$$

where $\mathbf{G}$ *and* $\boldsymbol{\lambda}$ *are* $k \times k$ *and* $k \times 1$*, matrices of free coefficients (unrelated to* $\boldsymbol{\beta}$*, and* σ^2*), and* $\mathbf{v}_t$ *is a* $k \times 1$ *vector of disturbances. In this example,* $\boldsymbol{\gamma}$ *is defined in terms of* $\mathbf{G}$*,* $\boldsymbol{\lambda}$ *and* Σ_{xx}*, which do not depend on parameters of the conditional model (*$\boldsymbol{\beta}$ *and* σ^2*) and the joint probability distribution function of* $\mathbf{z}_t = (y_t, \mathbf{x}_t')'$ *decomposes as in (9.13) with* $\boldsymbol{\gamma}$ *unrelated to* $\boldsymbol{\beta}$*, and* σ^2 *if the weak exogeneity condition*

$$E(u_t\mathbf{v}_t \mid \Omega_{T-1}) = \mathbf{0}$$

holds, we have

$$E(u_t\mathbf{x}_{t-s}) = \mathbf{0}, \text{for } s = 0, 1, 2, \ldots.$$

However, due to the feedback effect from y_{t-1} *in (9.15),* $\mathbf{x}_t$ *is not strictly exogenous, and in general*

$$E(u_t \mathbf{x}_{t+s}) \neq \mathbf{0}, \text{for } s = 0, 1, 2, \ldots.$$

For $\mathbf{x}_t$ to be strictly exogenous we need the additional restrictions that there are no lagged feedbacks from y into $\mathbf{x}$, namely we must also have $\boldsymbol{\lambda} = \mathbf{0}$. With this additional set of restrictions it is now easily seen that

$$E(u_t \mathbf{x}_{t-s}) = \mathbf{0}, \text{for } s = 0, 1, 2, \ldots \text{ and}$$
$$E(u_t \mathbf{x}_{t+s}) \neq \mathbf{0}, \text{for } s = 0, 1, 2, \ldots,$$

namely under strict exogeneity u_t is uncorrelated with past as well as future realizations of $\mathbf{x}$. But under weak exogeneity u_t is only uncorrelated with current and past values of $\mathbf{x}$.

In the above example, weak exogeneity is sufficient for asymptotic inference (as $T \to \infty$), but can lead to biased *OLS* estimators. For the *OLS* estimators to be unbiased strict exogeneity is required. To see this note that:

$$E\left(\hat{\boldsymbol{\beta}}_T \mid \mathbf{x}_1, \mathbf{x}_2, \ldots \mathbf{x}_T\right) = \boldsymbol{\beta} + E\left[\left(\sum_{t=1}^{T} \mathbf{x}_t \mathbf{x}_t'\right)^{-1} \sum_{t=1}^{T} \mathbf{x}_t u_t \mid \mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_T\right]$$
$$= \boldsymbol{\beta} + \left(\sum_{t=1}^{T} \mathbf{x}_t \mathbf{x}_t'\right)^{-1} \sum_{t=1}^{T} \mathbf{x}_t E\left(u_t \mid \mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_T\right).$$

But, under strict exogeneity we have

$$E\left(u_t \mid \mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_T\right) = 0, \text{ for all } t.$$

Hence

$$E\left(\hat{\boldsymbol{\beta}}_T \mid \mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_T\right) = \boldsymbol{\beta},$$

and therefore unconditionally we also have

$$E\left(\hat{\boldsymbol{\beta}}_T\right) = \boldsymbol{\beta}.$$

Under strict exogeneity the exact variance-covariance matrix of $\hat{\boldsymbol{\beta}}_T$ can also be derived and is given by

$$Var\left(\hat{\boldsymbol{\beta}}_T\right) = \sigma^2 E\left[\left(\mathbf{X}'\mathbf{X}\right)^{-1}\right],$$

where expectations are taken with respect to the distribution of $\mathbf{x}_t$.

Under weak exogeneity none of the above results hold. Important examples of regression models with weakly exogenous regressors include time series models with lagged dependent variables (for example the time series models discussed in Chapter 14, and *ARDL* models

introduced in Chapter 6), and dynamic panel data models (the models described in Chapter 27).

9.4 Regularity conditions and some preliminary results

Here we develop the theory of the *ML* estimator for the *IID* case. The heterogeneous and the dependent observations case will be dealt with in Section 9.6.

Suppose that the observations $\mathbf{x} = (x_1, x_2, \ldots, x_T)'$ are T independent draws from the density function $f(x, \boldsymbol{\theta}_0)$, where $\boldsymbol{\theta}_0$ is an interior point of $\Theta \in \mathbb{R}^p$. We can think of $\boldsymbol{\theta}_0$ as the 'true' unknown value of $\boldsymbol{\theta} \in \Theta$. For any $\boldsymbol{\theta}$, the log-likelihood function $\ell_T(\boldsymbol{\theta}, \mathbf{x}) = \sum_{t=1}^{T} \log f(x_t, \boldsymbol{\theta})$, is a random variable whose distribution is governed by $f(x_t, \boldsymbol{\theta})$.

Denote the ith element of $\boldsymbol{\theta}$ by θ_i, and consider the following assumptions that are often referred to in the literature as the *regularity conditions*:

Assumption 1 RC1 *For each $\boldsymbol{\theta} \in \Theta$, the derivatives*

$$\frac{\partial \log f(x, \boldsymbol{\theta})}{\partial \theta_i}, \quad \frac{\partial^2 \log f(x, \boldsymbol{\theta})}{\partial \theta_i \partial \theta_j}, \quad \frac{\partial^3 \log f(x, \boldsymbol{\theta})}{\partial \theta_i \partial \theta_j \partial \theta_k},$$

exist, for all x, and $i, j, k = 1, 2, \ldots, p$.

Assumption 2 RC2 *For $\boldsymbol{\theta}_0 \in \Theta$, there exist functions $G(x)$, $H(x)$ and $K(x)$ (possibly depending on $\boldsymbol{\theta}_0$) such that for $\boldsymbol{\theta}$ in the neighbourhood of $\boldsymbol{\theta}_0$, the inequalities*

$$\left| \frac{\partial f(x, \boldsymbol{\theta})}{\partial \theta_i} \right| \leq G_i(x), \quad \left| \frac{\partial^2 f(x, \boldsymbol{\theta})}{\partial \theta_i \partial \theta_j} \right| \leq H_{ij}(x), \quad \left| \frac{\partial^3 f(x, \boldsymbol{\theta})}{\partial \theta_i \partial \theta_j \partial \theta_k} \right| \leq K_{ijk}(x),$$

hold for all x, and $i, j, k = 1, 2, \ldots, p$, and

$$\int G_i(x)dx < \infty, \quad \int H_{ij}(x)dx < \infty, \quad \int K_{ijk}(x)dx < \infty.$$

Assumption 3 RC3 *For $\boldsymbol{\theta} \in \Theta$*

$$\mathbf{B}(\boldsymbol{\theta}) = E\left[\left(\frac{\partial \log f(x, \boldsymbol{\theta})}{\partial \boldsymbol{\theta}} \right) \left(\frac{\partial \log f(x, \boldsymbol{\theta})}{\partial \boldsymbol{\theta}'} \right) \right], \tag{9.16}$$

exists and is positive definite.

Assumption RC1 ensures that $\frac{\partial \log f(x,\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}$ has a Taylor series expansion as a function of $\boldsymbol{\theta}$. The derivative $\frac{\partial \log f(x,\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}$ is known as the score vector. Assumption RC2 allows differentiation of $\int f(x, \boldsymbol{\theta})\, dx$ and $\int \left[\frac{\partial \log f(x,\boldsymbol{\theta})}{\partial \boldsymbol{\theta}} \right] dx$ with respect to $\boldsymbol{\theta}$ under the integral sign. That is, it permits

the order of differentiation and integration to be interchanged. Finally, assumption RC3 requires that $\frac{\partial \log f(x,\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}$ has a finite variance.

Theorem 32 (Score vector) *Under the regularity conditions RC1 to RC3, the score vector* $\mathbf{d}(\theta) = \frac{\partial \log f(x,\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}$ *has mean zero and a finite variance.*

Proof Since $f(x, \boldsymbol{\theta})$ is a probability density function, we have

$$\int f(x, \boldsymbol{\theta})\, dx = 1.$$

Taking the partial derivatives of both sides of this relation with respect to $\boldsymbol{\theta}$ (and noting that the regularity condition RC2 allows the order of differentiation and integration to be interchanged), we have

$$\begin{aligned} \int \frac{\partial f(x, \boldsymbol{\theta})}{\partial \boldsymbol{\theta}} dx &= \mathbf{0}, \\ \int \left\{ \frac{1}{f(x, \boldsymbol{\theta})} \right\} \frac{\partial f(x, \boldsymbol{\theta})}{\partial \boldsymbol{\theta}} f(x, \boldsymbol{\theta})\, dx &= \mathbf{0}, \\ \int \frac{\partial \log f(x, \boldsymbol{\theta})}{\partial \boldsymbol{\theta}} f(x, \boldsymbol{\theta})\, dx &= \mathbf{0}, \end{aligned} \tag{9.17}$$

which may also be written as $\int \mathbf{d}(\boldsymbol{\theta})\, \partial f(x, \boldsymbol{\theta})\, d(\boldsymbol{\theta}) = \mathbf{0}$, or simply $E[\mathbf{d}(\boldsymbol{\theta})] = \mathbf{0}$, where the expectations are taken with respect to the density function $f(x, \boldsymbol{\theta})$. The variance of the score function is given by

$$\begin{aligned} Var[\mathbf{d}(\boldsymbol{\theta})] &= E[\mathbf{d}(\boldsymbol{\theta})\, \mathbf{d}(\boldsymbol{\theta})'] \\ &= E\left[\frac{\partial \log f(x, \boldsymbol{\theta})}{\partial \boldsymbol{\theta}} \cdot \frac{\partial \log f(x, \boldsymbol{\theta})}{\partial \boldsymbol{\theta}'} \right], \end{aligned}$$

which is finite by assumption RC3. Taking partial derivatives of (9.17) with respect to $\boldsymbol{\theta}'$ we have:

$$\int \left[\frac{\partial^2 \log f(x, \boldsymbol{\theta})}{\partial \boldsymbol{\theta} \partial \boldsymbol{\theta}'} + \frac{\partial \log f(x, \boldsymbol{\theta})}{\partial \boldsymbol{\theta}} \cdot \frac{\partial \log f(x, \boldsymbol{\theta})}{\partial \boldsymbol{\theta}'} \right] f(x, \boldsymbol{\theta})\, dx = \mathbf{0},$$

or

$$E\left[\mathbf{d}(\boldsymbol{\theta})\, \mathbf{d}(\boldsymbol{\theta})' \right] = E\left[-\frac{\partial^2 \log f(x, \boldsymbol{\theta})}{\partial \boldsymbol{\theta} \partial \boldsymbol{\theta}'} \right] = \mathbf{B}(\boldsymbol{\theta}). \tag{9.18}$$

Expression (9.18) is known as *Fisher's information matrix*. $E[\mathbf{d}(\boldsymbol{\theta})\, \mathbf{d}(\boldsymbol{\theta})'] = \mathbf{B}(\boldsymbol{\theta})$ is known as the *outer-product form* and $E\left\{ -\frac{\partial^2 \log f(x,\boldsymbol{\theta})}{\partial \boldsymbol{\theta} \partial \boldsymbol{\theta}'} \right\} = A(\boldsymbol{\theta})$ as the *inner product form* of the information matrix. Notice that these forms are equal only under the assumption that $f(x, \boldsymbol{\theta})$ is the

density function with respect to which expectations are taken. In the misspecified case where the density function of x is not the same as $f(x, \boldsymbol{\theta})$, the two forms of the information matrix need not be the same.

Theorem 33 (Cramer–Rao lower bound) *Let $\tilde{\boldsymbol{\theta}}_T$ be an unbiased estimator of $\boldsymbol{\theta}$ based on the sample of observations $\mathbf{x} = (x_1, x_2, \ldots, x_T)'$. Then*

$$Var\left(\tilde{\boldsymbol{\theta}}_T\right) - \mathbf{B}_T(\boldsymbol{\theta})^{-1}, \tag{9.19}$$

is non-negative definite, where $\mathbf{B}_T(\boldsymbol{\theta})$ is the information matrix defined by

$$\mathbf{B}_T(\boldsymbol{\theta}) = E\left[-\frac{\partial^2 \ell_T(\mathbf{x}, \boldsymbol{\theta})}{\partial \boldsymbol{\theta} \partial \boldsymbol{\theta}'}\right].$$

In the case of the IID observations

$$\mathbf{B}_T(\boldsymbol{\theta}) = T\,\mathbf{B}(\boldsymbol{\theta}),$$

where

$$\mathbf{B}(\boldsymbol{\theta}) = E\left[-\frac{\partial^2 \log f(x, \boldsymbol{\theta})}{\partial \boldsymbol{\theta} \partial \boldsymbol{\theta}'}\right].$$

Proof Since by assumption $\tilde{\boldsymbol{\theta}}_T$ is an unbiased estimator of $\boldsymbol{\theta}$, then

$$\int \tilde{\boldsymbol{\theta}}_T f(\mathbf{x}, \boldsymbol{\theta})\, d\mathbf{x} = \boldsymbol{\theta}, \tag{9.20}$$

where $f(\mathbf{x}, \boldsymbol{\theta})$ represents the joint density function of $\mathbf{x} = (x_1, x_2, \ldots, x_T)'$. Taking partial derivatives of both sides of (9.20), (and noting that $\tilde{\boldsymbol{\theta}}_T$ is dependent only on $\mathbf{x}$) we have

$$\int \tilde{\boldsymbol{\theta}}_T \frac{\partial \ell_T(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}} f(\mathbf{x}, \boldsymbol{\theta})\, d\mathbf{x} = \mathbf{I}_p,$$

where $\mathbf{I}_p$ is an identity matrix of order p. But since from Theorem 32, $E[\frac{\partial \ell_T(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}] = \mathbf{0}$, the above relation can also be written as

$$Cov\left(\tilde{\boldsymbol{\theta}}_T, \frac{\partial \ell_T(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}\right) = \mathbf{I}_p. \tag{9.21}$$

Using a multivariate version of the Cauchy–Schwartz inequality, we have[1]

[1] This inequality can be easily derived by first minimizing $Var\left[\boldsymbol{\lambda}'\tilde{\boldsymbol{\theta}}_n + \boldsymbol{\mu}'\frac{\partial \ell(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}\right]$ with respect to fixed λ, and then noting that the minimized value of this variance is non-negative. $\boldsymbol{\lambda}$ and $\boldsymbol{\mu}$ are vectors of constants.

$$Var\left(\tilde{\boldsymbol{\theta}}_T\right) - Cov\left(\tilde{\boldsymbol{\theta}}_T, \frac{\partial \ell_T(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}\right)\left[Var\left(\frac{\partial \ell_T(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}\right)\right]^{-1} Cov\left(\frac{\partial \ell_T(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}, \tilde{\boldsymbol{\theta}}_T\right) \geq \mathbf{0}. \quad (9.22)$$

Using (9.21) in (9.22) now yields[2]

$$Var\left(\tilde{\boldsymbol{\theta}}_T\right) - \left[Var\left(\frac{\partial \ell_T(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}\right)\right]^{-1} \geq \mathbf{0}.$$

In the case of *IID* observations, we have

$$\begin{aligned} Var\left(\frac{\partial \ell_T(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}\right) &= Var\left\{\sum_{i=1}^{T}\left[\frac{\partial \log f(x_i, \boldsymbol{\theta})}{\partial \boldsymbol{\theta}}\right]\right\} \\ &= TVar\left[\frac{\partial \log f(x_i, \boldsymbol{\theta})}{\partial \boldsymbol{\theta}}\right] \\ &= T\mathbf{B}(\boldsymbol{\theta}), \end{aligned}$$

where $\mathbf{B}(\boldsymbol{\theta})$ is given by (9.18). Therefore, as required

$$Var\left(\tilde{\boldsymbol{\theta}}_T\right) - \frac{1}{T}\mathbf{B}(\boldsymbol{\theta})^{-1} \geq \mathbf{0}.$$

In the case where $\boldsymbol{\theta}$ is a scalar, (9.19) simplifies to $Var(\tilde{\theta}_T) \geq \frac{1}{TB(\theta)}$, which is the more familiar form of the Cramer–Rao (C–R) inequality. The importance of the C–R inequality stems from the fact that for unbiased estimators, the estimates that attain the C–R bound are 'efficient' in the sense of having the least variance in the class of unbiased estimators. This result also readily extends to estimators that are asymptotically unbiased and attain the C–R lower bound asymptotically. The following definition is useful for characterizing the properties of *ML* estimators.

Definition 11 (Asymptotic efficiency) *An estimator $\hat{\boldsymbol{\theta}}$ is asymptotically efficient if it is asymptotically unbiased and has asymptotic covariance matrix that is no larger than the asymptotic covariance matrix of any other asymptotically unbiased estimator.*

In the following section, we will show that, under the regularity conditions set out above, the *ML* estimators are asymptotically efficient and achieve the C–R lower bound.

9.5 Asymptotic properties of *ML* estimators

The optimum properties of *ML* estimators are asymptotic, and hold assuming that the underlying probability model is correctly specified.

[2] The notation '≥ 0' stands for a non-negative matrix,

Theorem 34 (Consistency of *ML* estimators) *Under the regularity conditions RC1 to RC3, the ML estimator of $\boldsymbol{\theta}$, namely $\hat{\boldsymbol{\theta}}_T$, converges in probability to $\boldsymbol{\theta}_0$, the true value of $\boldsymbol{\theta}$ under $f(x, \boldsymbol{\theta})$, as $T \to \infty$.*

Proof In the *IID* case, using the law of large numbers due to Khinchine (Theorem 10), it is easily seen that the average log-likelihood function $T^{-1}\ell_T(\boldsymbol{\theta}) = T^{-1}\sum_{t=1}^{T} \log f(x_t, \boldsymbol{\theta})$ converges in probability to $E[\log f(x_t, \boldsymbol{\theta})]$, where expectations are taken under $f(x_t, \boldsymbol{\theta})$; that is

$$\frac{\ell_T(\boldsymbol{\theta})}{T} \xrightarrow{p} E\{\log f(x_t, \boldsymbol{\theta})\} = \int [\log f(x, \boldsymbol{\theta})] f(x, \boldsymbol{\theta}_0)\, dx. \tag{9.23}$$

Khinchine's theorem directly applies to the sum $T^{-1}\sum_{t=1}^{T} \log f(x_t, \boldsymbol{\theta})$, since independence of $x_t's$ implies that $\log f(x_t, \boldsymbol{\theta}), t = 1, 2, \ldots, T$ are also independently distributed with a constant mean, given by $E[\log f(x, \boldsymbol{\theta})]$. Consider now the divergence of $f(x, \boldsymbol{\theta})$ from $f(x, \boldsymbol{\theta}_0)$, measured by the Kullback–Leibler information criterion (KLIC) defined by,

$$I(\boldsymbol{\theta}, \boldsymbol{\theta}_0) = E[\log f(x, \boldsymbol{\theta}) - \log f(x, \boldsymbol{\theta}_0)].$$

Since $log[\frac{f(x,\boldsymbol{\theta})}{f(x,\boldsymbol{\theta}_0)}]$ is a concave function of the ratio $\frac{f(x;\boldsymbol{\theta})}{f(x;\boldsymbol{\theta}_0)}$, then by Jensen's inequality we have (using f and f_0 for $f(x, \boldsymbol{\theta})$ and $f(x, \boldsymbol{\theta}_0)$, respectively)

$$E\left[log\left(\frac{f}{f_0}\right)\right] \leq \log E\left[\left(\frac{f}{f_0}\right)\right]. \tag{9.24}$$

But

$$\begin{aligned} E\left(\frac{f}{f_0}\right) &= \int \frac{f(x, \boldsymbol{\theta})}{f(x, \boldsymbol{\theta}_0)} f(x, \boldsymbol{\theta}_0)\, dx \\ &= \int f(x, \boldsymbol{\theta})\, dx = 1, \end{aligned}$$

and hence using this in (9.24) we obtain

$$E\left[log\left(\frac{f}{f_0}\right)\right] \leq \log(1) = 0,$$

or

$$E[logf(x, \boldsymbol{\theta})] \leq E[logf(x, \boldsymbol{\theta}_0)], \tag{9.25}$$

with equality holding if and only if $\boldsymbol{\theta} = \boldsymbol{\theta}_0$. Therefore, $\boldsymbol{\theta}_0$ is the value of $\boldsymbol{\theta}$ that globally maximizes $E[logf(x, \boldsymbol{\theta})]$. Hence, on the one hand from (9.23) we note that as $T \to \infty, \hat{\boldsymbol{\theta}}_T$ maximizes $E[logf(x, \boldsymbol{\theta})]$, and from (9.25) we have the value of $\boldsymbol{\theta}$ that maximizes $E[logf(x, \boldsymbol{\theta})]$ is the true value, $\boldsymbol{\theta}_0$. Hence, by the continuity of the log-likelihood function in $\boldsymbol{\theta}$, we also have $\hat{\boldsymbol{\theta}}_T$ converging in probability to $\boldsymbol{\theta}_0$. The strong convergence of $\hat{\boldsymbol{\theta}}_T$ to $\boldsymbol{\theta}_0$ also follows

when $T^{-1}\ell_T(\boldsymbol{\theta})$ converges to $E[logf(x,\boldsymbol{\theta})]$ with probability 1. In the *IID* case, the strong laws of large numbers (Theorem 12) is applicable to $T^{-1}\sum_{t=1}^{T}\log f(x_t,\boldsymbol{\theta})$ and hence, $\hat{\boldsymbol{\theta}}_T$ converges to $\boldsymbol{\theta}_0$ with probability 1.

Remark 2 *It is clear that the $\boldsymbol{\theta}$ maximizing $E[logf(x,\boldsymbol{\theta})]$ needs to be unique or should correspond to the global maximum of $E[logf(x,\boldsymbol{\theta})]$. The necessary condition for $E[logf(x,\boldsymbol{\theta})]$ to have a unique maximum is given by*

$$\frac{\partial^2 E[\log f(x,\boldsymbol{\theta})]}{\partial\boldsymbol{\theta}\partial\boldsymbol{\theta}'} = E\left[\frac{\partial^2\log f(x,\boldsymbol{\theta})}{\partial\boldsymbol{\theta}\partial\boldsymbol{\theta}'}\right] < \mathbf{0}, \tag{9.26}$$

the regularity condition RC3 and noting from (9.18) that[3]

$$E\left[\frac{\partial^2\log f(x,\boldsymbol{\theta})}{\partial\boldsymbol{\theta}\partial\boldsymbol{\theta}'}\right] = -\mathbf{A}(\boldsymbol{\theta}).$$

Remark 3 *Whether $E[\log f(x,\boldsymbol{\theta})]$ has local or global maxima is closely related to the problem of local and global identifiability of parameters. When $E[\log f(x,\boldsymbol{\theta})]$ is a concave function of $\boldsymbol{\theta}$ (and hence $\mathbf{A}(\boldsymbol{\theta})$ is positive definite for all $\boldsymbol{\theta}\in\Theta$), then $\boldsymbol{\theta}$ is globally identified, otherwise $\boldsymbol{\theta}$ is at best only locally identified. Parameters $\boldsymbol{\theta}$ are not identified when $\mathbf{A}(\boldsymbol{\theta})$ is rank deficient for all $\boldsymbol{\theta}$. See the discussion of identification of the parameters of the simultaneous equation models in Chapters 20 and 22, and the paper by Rothenberg (1971).*

Remark 4 *In the case where $\log f(x,\boldsymbol{\theta})$ is differentiable, $\hat{\boldsymbol{\theta}}_T$ is obtainable as a root of the equations*

$$\frac{\partial\ell_T(\boldsymbol{\theta})}{\partial\boldsymbol{\theta}}\Big|_{\boldsymbol{\theta}=\hat{\boldsymbol{\theta}}_T} = \mathbf{0}.$$

In cases where $\frac{\partial\ell_T(\boldsymbol{\theta})}{\partial\boldsymbol{\theta}} = \mathbf{0}$ has multiple solutions, it is clear from the proof of the consistency of the ML estimator that it is essential that the global maximum of the log-likelihood function is chosen.

In practice, multiple roots may arise due to a variety of factors. For example, the parameters may not be globally identified, the available sample size may not be large enough, the underlying model may be misspecified, or any combination of these factors. However, if all three regularity conditions are met, one would expect the multiple root problem to disappear as the sample size increases.

The following theorem provides the asymptotic properties of *ML* estimators.

Theorem 35 *Under the regularity conditions RC1 to RC3, the ML estimator has the following asymptotic properties*

(i) ***Asymptotic normality***: $\sqrt{T}\left(\hat{\boldsymbol{\theta}}_T-\boldsymbol{\theta}_0\right)\xrightarrow{d}N\left[\mathbf{0},\mathbf{A}(\boldsymbol{\theta}_0)^{-1}\right]$, *where*

[3] The interchange of the order of differentiation and integration in (9.26) is justified by the regularity conditions RC1 and RC2.

$$\mathbf{A}(\boldsymbol{\theta}_0) = \lim_{T\to\infty} E\left[-\frac{1}{T}\frac{\partial^2 \ell_T(\boldsymbol{\theta}_0)}{\partial\boldsymbol{\theta}\partial\boldsymbol{\theta}'}\right] = \underset{T\to\infty}{Plim}\left[-\frac{1}{T}\frac{\partial^2 \ell_T(\boldsymbol{\theta}_0)}{\partial\boldsymbol{\theta}\partial\boldsymbol{\theta}'}\right].$$

(ii) ***Asymptotic unbiasedness****:* $\lim_{T\to\infty} E\left(\hat{\boldsymbol{\theta}}_T\right) = \boldsymbol{\theta}_0$.

(iii) ***Asymptotic efficiency****:* $\hat{\boldsymbol{\theta}}_T$ *is an asymptotically efficient estimator and achieves the Cramer-Rao lower bound, asymptotically.*

Proof To prove (i), we expand $\frac{\partial \ell_T\left(\hat{\boldsymbol{\theta}}_T\right)}{\partial\boldsymbol{\theta}}$ around $\boldsymbol{\theta}_0$. By the mean value theorem

$$\frac{1}{\sqrt{T}}\frac{\partial \ell_T\left(\hat{\boldsymbol{\theta}}\right)}{\partial\boldsymbol{\theta}} = \frac{1}{\sqrt{T}}\frac{\partial \ell_T(\boldsymbol{\theta}_0)}{\partial\boldsymbol{\theta}} + \left[\frac{1}{T}\frac{\partial^2 \ell_T(\boldsymbol{\theta}_0)}{\partial\boldsymbol{\theta}\partial\boldsymbol{\theta}'}\right]\sqrt{T}\left(\hat{\boldsymbol{\theta}}_T - \boldsymbol{\theta}_0\right) + \boldsymbol{\delta}_T, \tag{9.27}$$

where the ith element of $\boldsymbol{\delta}_T$ (which we denote by δ_{iT}) is given by

$$\delta_{iT} = \sum_{j=1}^{p}\sum_{k=1}^{p}\left[\frac{1}{T}\frac{\partial^3 \ell_T\left(\bar{\boldsymbol{\theta}}_T\right)}{\partial\theta_i\partial\theta_j\partial\theta_k}\right]\sqrt{T}\left(\hat{\theta}_{jT} - \theta_{j0}\right)\left(\hat{\theta}_{kT} - \theta_{k0}\right),$$

and $\bar{\boldsymbol{\theta}}_T$ lies between $\hat{\boldsymbol{\theta}}_T$ and $\boldsymbol{\theta}_0$. In the IID case

$$\frac{1}{T}\left[\frac{\partial^3 \ell_T\left(\bar{\boldsymbol{\theta}}_T\right)}{\partial\theta_i\partial\theta_j\partial\theta_k}\right] \xrightarrow{p} E\left[\frac{\partial^3 \log f(x,\boldsymbol{\theta}_0)}{\partial\theta_i\partial\theta_j\partial\theta_k}\right],$$

which by assumption RC2 are bounded for all x, and $i, j, k = 1, 2, \ldots, p$. The convergence of $\bar{\boldsymbol{\theta}}_T$ to $\boldsymbol{\theta}_0$ in probability follows from the fact that $\bar{\boldsymbol{\theta}}_T$ lies between $\hat{\boldsymbol{\theta}}_T$ and $\boldsymbol{\theta}_0$ and the fact that $\hat{\boldsymbol{\theta}}_T \xrightarrow{p} \boldsymbol{\theta}_0$ (see Theorem 34). Hence, we have

$$\delta_T = o_p\left[\sqrt{T}\left(\hat{\boldsymbol{\theta}}_T - \boldsymbol{\theta}_0\right)\right]. \tag{9.28}$$

Also, applying the law of large numbers to the elements of the inner product matrix

$$-\frac{1}{T}\frac{\partial^2 \ell_T(\boldsymbol{\theta}_0)}{\partial\boldsymbol{\theta}\partial\boldsymbol{\theta}'} = -\frac{1}{T}\sum_{t=1}^{T}\frac{\partial^2 \log f(x_t,\boldsymbol{\theta})}{\partial\boldsymbol{\theta}\partial\boldsymbol{\theta}'},$$

we have

$$-\frac{1}{T}\frac{\partial^2 \ell_T(\boldsymbol{\theta}_0)}{\partial\boldsymbol{\theta}\partial\boldsymbol{\theta}'} \xrightarrow{p} E\left[\frac{\partial^2 \log f(x,\boldsymbol{\theta})}{\partial\boldsymbol{\theta}\partial\boldsymbol{\theta}'}\right] = \mathbf{A}(\boldsymbol{\theta}_0), \tag{9.29}$$

which, by Assumption RC3, is a positive definite matrix. Using (9.28) and (9.29) in (9.27) and invoking Slutsky's convergence theorem (see Theorem 35) we have

$$\sqrt{T}\left(\hat{\boldsymbol{\theta}}_T - \boldsymbol{\theta}_0\right) = A\left(\boldsymbol{\theta}_0\right)^{-1} \frac{1}{\sqrt{T}} \frac{\partial \ell_T\left(\boldsymbol{\theta}_0\right)}{\partial \boldsymbol{\theta}} + o_p\left(1\right). \tag{9.30}$$

It now remains to show that $\frac{1}{\sqrt{T}} \frac{\partial \ell_T(\boldsymbol{\theta}_0)}{\partial \boldsymbol{\theta}}$ tends to a normal distribution. This follows immediately from the application of the Lindberg–Levy theorem (see Theorem 15) to

$$\frac{1}{\sqrt{T}} \frac{\partial \ell_T\left(\boldsymbol{\theta}_0\right)}{\partial \boldsymbol{\theta}} = \frac{1}{\sqrt{T}} \sum_{i=1}^{T} \frac{\partial \log f\left(x_t, \boldsymbol{\theta}_0\right)}{\partial \boldsymbol{\theta}}.$$

Firstly, as the theorem requires, under the regularity conditions, $\frac{\partial \log f(x_t, \boldsymbol{\theta}_0)}{\partial \boldsymbol{\theta}}$ has a constant mean equal to zero, and a finite nonsingular variance given by $\mathbf{A}\left(\boldsymbol{\theta}_0\right)$. Therefore,

$$\frac{1}{\sqrt{T}} \frac{\partial \ell_T\left(\boldsymbol{\theta}_0\right)}{\partial \boldsymbol{\theta}} \xrightarrow{d} N\left(0, \mathbf{A}\left(\boldsymbol{\theta}_0\right)^{-1}\right). \tag{9.31}$$

This result in conjunction with (9.30) now establishes that

$$\sqrt{T}\left(\hat{\boldsymbol{\theta}}_T - \boldsymbol{\theta}_0\right) \xrightarrow{d} N\left(\mathbf{0}, \mathbf{A}\left(\boldsymbol{\theta}_0\right)^{-1}\right). \tag{9.32}$$

As for (ii), using (9.30) it is easily seen that as $Var(\hat{\boldsymbol{\theta}}_T) \rightarrow \mathbf{0}$ as $T \rightarrow \infty$, and hence $\hat{\boldsymbol{\theta}}_T$ converges to $\boldsymbol{\theta}_0$ in mean squared error, and that $\lim_{T\rightarrow\infty} E(\hat{\boldsymbol{\theta}}_T) = \boldsymbol{\theta}_0$. This latter result follows from the fact that $E\|\hat{\boldsymbol{\theta}}_T - \boldsymbol{\theta}_0\|^2$ is bounded for all T, since by regularity conditions RC1 and RC3, the derivatives of the log-likelihood function up to the third-order exist and are bounded by functions that have finite integrals. Hence it follows that $\lim_{T\rightarrow\infty} E(\hat{\boldsymbol{\theta}}_T)$ will be equal to the mean of the asymptotic distribution of $\hat{\boldsymbol{\theta}}_T$ (for a proof see Rao (1973)). As for (iii), it is clear that $\hat{\boldsymbol{\theta}}_T$ asymptotically achieves the Cramer–Rao lower bound (Theorem 33) given by the inverse of the information matrix. Given that $\hat{\boldsymbol{\theta}}_T$ is asymptotically unbiased and achieves the Cramer–Rao lower bound, then it is also asymptotically efficient.

Example 23 *Consider the problem of deriving the asymptotic distribution of the ML estimators of α, β and σ^2 in the following simple nonlinear regression*

$$y_t = \alpha x_t^{\beta} + u_t,\ u_t \sim N\left(0, \sigma^2\right), \tag{9.33}$$

for $t = 1, 2, \ldots, T$. Let $g(x_t, \boldsymbol{\gamma}) = \alpha x_t^{\beta}$, where $\boldsymbol{\gamma} = (\alpha, \beta)'$ and set $\boldsymbol{\theta} = (\boldsymbol{\gamma}', \sigma^2)'$. Conditional on x_t, the log-likelihood function of this model is given by

$$\ell_T(\boldsymbol{\theta}) = -\frac{T}{2}\log\left(2\pi\sigma^2\right) - \frac{1}{2\sigma^2}\sum_{t=1}^{T}\left[y_t - g(x_t, \boldsymbol{\gamma})\right]^2.$$

To apply Theorem 35 to this problem, we need to find the score vector, $\frac{\partial \ell_T(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}$, and the information matrix, $\mathbf{B}(\boldsymbol{\theta}_0)$. We have

$$\begin{aligned}
\frac{\partial \ell_T(\boldsymbol{\theta})}{\partial \alpha} &= \frac{1}{\sigma^2}\sum_{t=1}^{T} x_t^{\beta}\left(y_t - \alpha x_t^{\beta}\right),\\
\frac{\partial \ell_T(\boldsymbol{\theta})}{\partial \beta} &= \frac{\alpha}{\sigma^2}\sum_{t=1}^{T} x_t^{\beta}\left(y_t - \alpha x_t^{\beta}\right)\log x_t,\\
\frac{\partial \ell_T(\boldsymbol{\theta})}{\partial \sigma^2} &= -\frac{T}{\sigma^2} + \frac{1}{2\sigma^4}\sum_{t=1}^{T}\left(y_t - \alpha x_t^{\beta}\right)^2,\\
\frac{\partial \ell_T^2(\boldsymbol{\theta})}{\partial \alpha^2} &= -\frac{1}{\sigma^2}\sum_{t=1}^{T} x_t^{2\beta},\\
\frac{\partial \ell_T^2(\boldsymbol{\theta})}{\partial \beta^2} &= \frac{\alpha}{\sigma^2}\sum_{t=1}^{T} x_t^{\beta}\left(y_t - \alpha x_t^{\beta}\right)\left(\log x_t\right)^2 - \frac{\alpha^2}{\sigma^2}\sum_{t=1}^{T} x_t^{2\beta}\left(\log x_t\right)^2,\\
\frac{\partial \ell_T^2(\boldsymbol{\theta})}{\partial \alpha \partial \beta} &= \frac{\alpha}{\sigma^2}\sum_{t=1}^{T} x_t^{\beta}\left(y_t - \alpha x_t^{\beta}\right)\log x_t - \frac{\alpha}{\sigma^2}\sum_{t=1}^{T} x_t^{2\beta}\log x_t,\\
\frac{\partial \ell_T^2(\boldsymbol{\theta})}{\partial \sigma^2 \partial \alpha} &= -\frac{1}{\sigma^4}\sum_{t=1}^{T} x_t^{\beta}\left(y_t - \alpha x_t^{\beta}\right),\\
\frac{\partial \ell_T^2(\boldsymbol{\theta})}{\partial \sigma^2 \partial \beta} &= -\frac{\alpha}{\sigma^4}\sum_{t=1}^{T} x_t^{\beta}\left(y_t - \alpha x_t^{\beta}\right)\log x_t.
\end{aligned}$$

Taking expectations of the second derivatives conditional on x_t, then unconditionally we have (assuming that (9.33) holds)

$$\begin{aligned}
E\left(-\frac{\partial \ell_T^2(\boldsymbol{\theta})}{\partial \alpha^2}\right) &= \left(\frac{1}{\sigma^2}\right)\sum_{t=1}^{T} E\left(x_t^{2\beta}\right),\\
E\left(-\frac{\partial \ell_T^2(\boldsymbol{\theta})}{\partial \beta^2}\right) &= \frac{\alpha^2}{\sigma^2}\sum_{t=1}^{T} E\left[x_t^{2\beta}\left(\log x_t\right)^2\right],\\
E\left(-\frac{\partial \ell_T^2(\boldsymbol{\theta})}{\partial \alpha \partial \beta}\right) &= \frac{\alpha}{\sigma^2}\sum_{t=1}^{T} E\left[x_t^{2\beta}\log x_t\right],\\
E\left(-\frac{\partial \ell_T^2(\boldsymbol{\theta})}{\partial \sigma^4}\right) &= \frac{T}{2\sigma^4},
\end{aligned}$$

and $E\left(-\frac{\partial \ell_T^2(\boldsymbol{\theta})}{\partial \sigma^2 \partial \alpha}\right) = E\left(-\frac{\partial \ell_T^2(\boldsymbol{\theta})}{\partial \sigma^2 \partial \beta}\right) = 0$. *Hence* $\mathbf{A}(\boldsymbol{\theta}) = E\left(-\frac{1}{T}\frac{\partial \ell_T^2(\boldsymbol{\theta})}{\partial \boldsymbol{\theta} \partial \boldsymbol{\theta}'}\right)$ *is block-diagonal and (assuming that the expectations of* $x_t^{2\beta}$, $x_t^{2\beta}\left(\log x_t\right)$ *and* $x_t^{2\beta}\left(\log x_t\right)^2$ *exist and are finite, is given by*

$$\mathbf{A}(\boldsymbol{\theta}) = \frac{1}{\sigma^2}\begin{bmatrix} \frac{1}{T}\sum_{t=1}^{T} E\left(x_t^{2\beta}\right) & \frac{\alpha}{T}\sum_{t=1}^{T} E\left[x_t^{2\beta}\log x_t\right] & 0 \\ \frac{\alpha}{T}\sum_{t=1}^{T} E\left[x_t^{2\beta}\log x_t\right] & \alpha^2\frac{1}{T}\sum_{t=1}^{T} E\left[x_t^{2\beta}\left(\log x_t\right)^2\right] & 0 \\ 0 & 0 & \frac{1}{2\sigma^2} \end{bmatrix}.$$

In the case where x_t *are identically distributed* $\mathbf{A}(\boldsymbol{\theta})$ *simplifies further. The information matrix for* $\boldsymbol{\gamma}$ *is given by*

$$\mathbf{A}(\boldsymbol{\gamma}) = \frac{1}{\sigma^2}\begin{bmatrix} E\left(x_t^{2\beta}\right) & \alpha E\left[x_t^{2\beta}\log x_t\right] \\ \alpha E\left[x_t^{2\beta}\log x_t\right] & \alpha^2 E\left[x_t^{2\beta}\left(\log x_t\right)^2\right] \end{bmatrix}. \tag{9.34}$$

Hence $\hat{\boldsymbol{\gamma}}_T$*, the ML estimator of* $\boldsymbol{\gamma}$*, is asymptotically distributed with mean* $\boldsymbol{\gamma}_0$ *(the true value of* $\boldsymbol{\gamma}$ *under (9.33)), and the asymptotic covariance matrix given by* $\frac{1}{T}\mathbf{A}(\boldsymbol{\gamma})^{-1}$*. It is clear that for the nonlinear model, (9.33), to be meaningful, the realized values of* x_t *should all be strictly positive. This requirement is, for example, satisfied if we assume that* x_t *has a log-normal distribution. It is also clear that* α *should be strictly non-zero, otherwise the information matrix becomes singular, and the parameter* β *will no longer be identified.*

9.6 *ML* estimation for heterogeneous and the dependent observations

9.6.1 The log-likelihood function for dependent observations

Consider now the case where $(x_1, x_2, \ldots, x_T)$ are not independently or identically distributed. In this case we have (suppressing the dependence of $f(\cdot)$ on $\boldsymbol{\theta}$)

$$f(x_1, x_2, \ldots, x_T) = f(x_1) f(x_2 \mid x_1) f(x_3 \mid x_2, x_1) \ldots f(x_T \mid x_{T-1}, x_{T-2}, \ldots, x_1),$$

where $f(x_T \mid x_{T-1}, x_{T-2}, \ldots, x_1)$ represents the conditional density function of x_T given the realizations $x_1, x_2, \ldots, x_{T-1}$. The above result can also be written more generally as

$$f(\mathbf{x}, \boldsymbol{\theta}) = \prod_{t=1}^{T} f(x_t, \boldsymbol{\theta} \mid \Omega_{t-1}),$$

where $\{\Omega_t\}$ is a sequence of non-decreasing σ-fields, containing at least some observations on current and past values of x_t. This formulation assumes that $\{x_t\}$ is adapted to $\{\Omega_t\}$, namely that for each t, x_t is Ω_t-measurable. The log-likelihood function for this general case can be written as

$$\ell_T(\boldsymbol{\theta}) = \sum_{t=1}^{T} \ln\left[f(x_t, \boldsymbol{\theta} \mid \Omega_{t-1})\right]. \tag{9.35}$$

Example 24 (The *AR* process) *A simple example of dependent observations is the first-order stationary autoregressive process,*

$$x_t = \phi x_{t-1} + \varepsilon_t, \quad |\phi| < 1, \ \varepsilon_t \sim N(0, \sigma^2).$$

For this example $f(x_t \mid x_{t-1}, x_{t-2}, \ldots, x_1) = f(x_{t-1})$, $t = 2, 3, \ldots, T$, *and*

$$f(\mathbf{x}, \boldsymbol{\theta}) = f(x_1, \boldsymbol{\theta}) f(x_2, \boldsymbol{\theta} \mid x_1) f(x_3, \boldsymbol{\theta} \mid x_2) \ldots f(x_T, \boldsymbol{\theta} \mid x_{T-1}), \tag{9.36}$$

where $f(x_1, \boldsymbol{\theta})$ *is the marginal distribution of the initial observations and* $\boldsymbol{\theta} = (\phi, \sigma^2)'$. *Assuming the process is stationary and has started a long time ago, we have*

$$x_1 \sim N\left(0, \frac{\sigma^2}{1-\phi^2}\right),$$

and $x_t \mid x_{t-1} \sim N(\phi x_{t-1}, \sigma^2)$, *for* $t = 2, 3, \ldots, T$. *That is*

$$\log f(x_1) = -\frac{1}{2}\ln\left(\frac{2\pi\sigma^2}{1-\phi^2}\right) - \frac{1-\phi^2}{2\sigma^2}x_1^2,$$
$$\log f(x_t \mid x_{t-1}) = -\frac{1}{2}\ln\left(2\pi\sigma^2\right) - \frac{1}{2\sigma^2}(x_t - \phi x_{t-1})^2, \ t = 2, 3, \ldots, T.$$

Therefore, substituting these in (9.36) we have

$$\ell_T(\boldsymbol{\theta}) = -\frac{T}{2}\ln\left(2\pi\sigma^2\right) + \frac{1}{2}\ln\left(1-\phi^2\right) - \left(\frac{1-\phi^2}{\sigma^2}\right)x_1^2 - \frac{1}{2\sigma^2}\sum_{t=2}^{T}(x_t - \phi x_{t-1})^2.$$

We shall be dealing with more general time series processes in Chapter 12.

9.6.2 Asymptotic properties of *ML* estimators

The properties of *ML* estimators for dependent observations have been widely investigated by Crowder (1976), Heijmans and Magnus (1986a, 1986b, and 1986c). We first introduce some useful definitions.

Definition 12 (Weak consistency) *The sequence* $\{\hat{\boldsymbol{\theta}}_T\}$ *is said to be weakly consistent if*

$$\lim_{T\to\infty} P\left[\hat{\boldsymbol{\theta}}_T \in N(\boldsymbol{\theta}_0)\right] = 1,$$

for every neighbourhood $N(\boldsymbol{\theta}_0)$ *of* $\boldsymbol{\theta}_0$.

See Heijmans and Magnus (1986b, p. 258).

Definition 13 (First-order efficiency) *A consistent estimator, $\hat{\boldsymbol{\theta}}_T$, of $\boldsymbol{\theta}$ is said to be first-order efficient if*

$$\sqrt{T}\left|\hat{\boldsymbol{\theta}}_T - \boldsymbol{\theta} - \mathbf{C}\,\mathbf{h}_T\right| \xrightarrow{p} 0,$$

where $\mathbf{h}_T = \frac{1}{T}\frac{\partial \log f(\mathbf{x},\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}$, and $\mathbf{C}$ is a matrix of constants which may depend on $\boldsymbol{\theta}$.

See Rao (1973, p. 348).

The following theorem establishes consistency of *ML* estimators under some general conditions.

Theorem 36 *Assume that the parameter space, Θ, is compact, and that the likelihood $L(\boldsymbol{\theta}) = f(\mathbf{x},\boldsymbol{\theta})$ is continuous on Θ. A necessary and sufficient condition for weak consistency of the ML estimator $\hat{\boldsymbol{\theta}}_T$ is that for every $\boldsymbol{\theta} \neq \boldsymbol{\theta}_0$, with $\boldsymbol{\theta} \in \Theta$, there exists a neighbourhood of $\boldsymbol{\theta}$, $N(\boldsymbol{\theta})$, satisfying*

$$\lim_{T\to\infty} \Pr\left\{\sup_{\boldsymbol{\theta}\in N(\boldsymbol{\theta})} [(\ell_T(\boldsymbol{\theta}) - \ell_T(\boldsymbol{\theta}_0)) < 0]\right\} = 1, \tag{9.37}$$

If condition (9.37) is replaced by the stronger condition

$$\limsup_{T\to\infty}\left\{\sup_{\boldsymbol{\theta}\in N(\boldsymbol{\theta})} [(\ell_T(\boldsymbol{\theta}) - \ell_T(\boldsymbol{\theta}_0)) < 0]\right\} \text{ with probability 1,} \tag{9.38}$$

then $\hat{\boldsymbol{\theta}}_T$ converges in probability to $\boldsymbol{\theta}_0$ as $T \to \infty$.

For a proof see Theorem 1 in Heijmans and Magnus (1986b). Note that, since an *ML* estimator may not be unique, the above theorem gives sufficient conditions for the consistency of every *ML* estimator. If (9.37) (or (9.38)) is not satisfied, then at least one inconsistent *ML* estimator exists.

The following theorem establishes asymptotic normality. To this end, let $g_T(\boldsymbol{\theta}) = L_T(\boldsymbol{\theta})/L_{T-1}(\boldsymbol{\theta})$, where $L_T(\boldsymbol{\theta})$ and $L_{T-1}(\boldsymbol{\theta})$ are likelihood functions based on T and $T-1$ observations, respectively, and let $\boldsymbol{\xi}_{Tj} = \partial g_T(\boldsymbol{\theta})/\partial\theta_j$, for $j = 1, 2, \ldots, p$.

Theorem 37 *Assume that the ML estimator, $\hat{\boldsymbol{\theta}}_T$, exists asymptotically almost surely, and is weakly consistent. Further, assume that:*

(i) $\ell_T(\boldsymbol{\theta}) = \ln f(\mathbf{x},\boldsymbol{\theta})$ *is twice continuously differentiable, for every fixed T, and $\mathbf{x} \in \mathbb{R}^T$.*

(ii) $E\left(\boldsymbol{\xi}_{Tj}^4\right) < \infty$, *for all T and $j = 1, 2, \ldots, p$.*

(iii) $E\left(\boldsymbol{\xi}_{Tj}|\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_{T-1}\right) = \mathbf{0}$, *for all $T \geq 2$ and $j = 1, 2, \ldots, p$.*

(iv) $Plim_{T\to\infty}\,(1/T)\max_{1\leq t\leq T}\boldsymbol{\xi}_{tj}^2 = \mathbf{0}$, *for all $j = 1, 2, \ldots, p$.*

(v) $\lim_{T\to\infty}\left(1/T^2\right) Var\left\{\sum_{t=1}^{T}\left(\xi_{ti}\xi_{tj}|\mathbf{x}_1,\mathbf{x}_2,\ldots.,\mathbf{x}_{t-1}\right)\right\} = \mathbf{0}$ *for all* $j = 1,2,\ldots,p$.

(vi) $\lim_{T\to\infty}\left(1/T^2\right) Var\left\{\sum_{t=1}^{T}\left[\xi_{ti}\xi_{tj} - E\left(\xi_{ti}\xi_{tj}|\mathbf{x}_1,\mathbf{x}_2,\ldots.,\mathbf{x}_{t-1}\right)\right]\right\} = \mathbf{0}$ *for all* $j = 1,2,\ldots,p$.

(vii) *There exists a finite positive definite* $p \times p$ *matrix,* $\mathbf{A}(\boldsymbol{\theta}_0)$, *such that*

$$\lim_{T\to\infty}(1/T)\,E\left[\frac{\partial \ell_T(\boldsymbol{\theta}_0)}{\partial \boldsymbol{\theta}}\frac{\partial \ell_T(\boldsymbol{\theta}_0)}{\partial \boldsymbol{\theta}}'\right] = \mathbf{A}(\boldsymbol{\theta}_0),$$

$$\lim_{T\to\infty}\left[(1/T)\,\mathbf{R}\,(\boldsymbol{\theta}_0)\right] = \lim_{T\to\infty}\left[(1/T)\,\frac{\partial^2 \ell_T(\boldsymbol{\theta}_0)}{\partial \boldsymbol{\theta}\,\partial \boldsymbol{\theta}'}\right] = -\mathbf{A}(\boldsymbol{\theta}_0).$$

where $\mathbf{R}\,(\boldsymbol{\theta})$ *is the Hessian matrix with elements* $R_{ij}\,(\boldsymbol{\theta})$, *for* $i, j = 1,2,\ldots,p$.

(viii) *For every* $\epsilon > 0$ *there exists a neighbourhood* $N\,(\boldsymbol{\theta}_0)$ *such that, for* $i, j = 1,2,\ldots,p$,

$$\lim_{T\to\infty}\Pr\left\{T^{-1}\sup_{\boldsymbol{\theta}\in N(\boldsymbol{\theta})}\left|R_{ij}\,(\boldsymbol{\theta}) - R_{ij}\,(\boldsymbol{\theta}_0)\right| > \epsilon\right\} = 0.$$

Then the ML estimator $\hat{\boldsymbol{\theta}}_T$ *is first-order efficient and asymptotically normally distributed, i.e.,*

$$\sqrt{T}\left(\hat{\boldsymbol{\theta}}_T - \boldsymbol{\theta}_0\right) \xrightarrow{d} N\left(\mathbf{0}, \mathbf{A}(\boldsymbol{\theta}_0)^{-1}\right).$$

For a proof, see Heijmans and Magnus (1986c), Theorem 2.

Further results on the properties of *ML* estimators in the case of dependent observations can be fund in Heijmans and Magnus (1986c). A discussion of *MLE* for regression models with correlated disturbances can be found in Mardia and Marshall (1984).

9.7 Likelihood-based tests

There are three main likelihood-based test procedures that are commonly used in econometrics for testing linear or nonlinear parametric restrictions on a maintained model. These are:

(i) The *likelihood ratio* (*LR*) approach.

(ii) The *Lagrange multiplier* (*LM*) approach.

(iii) The *Wald* (*W*) approach.

All these three procedures yield asymptotically valid tests, in the sense that they will have the correct size (i.e., the type I error) and possess certain optimal power properties in large samples: they are asymptotically equivalent, although they can lead to different results in small samples. The choice between them is often made on the basis of computational simplicity and ease of use.

9.7.1 The likelihood ratio test procedure

Let $L_T(\boldsymbol{\theta})$ be the likelihood function of the $p \times 1$ vector of unknown parameters, $\boldsymbol{\theta}$, associated with the joint probability distribution of $(y_1, y_2, \ldots, y_T)$, conditional (possibly) on a set of predetermined variables or regressors. Assume also that the hypothesis of interest to be tested can be written as a set of $s \leq p$ independent restrictions (linear and/or nonlinear) on $\boldsymbol{\theta}$. Denote these s restrictions by[4]

$$H_0 : \mathbf{h}(\boldsymbol{\theta}) = \mathbf{0}, \tag{9.39}$$

where $\mathbf{h}(\cdot)$ is an $s \times 1$ twice differentiable function of $\boldsymbol{\theta}$. Consider the two-sided alternative hypothesis

$$H_1 : \mathbf{h}(\boldsymbol{\theta}) \neq \mathbf{0}. \tag{9.40}$$

The likelihood ratio (LR) test statistic is defined by

$$LR = 2\left[\ell_T\left(\hat{\boldsymbol{\theta}}\right) - \ell_T\left(\tilde{\boldsymbol{\theta}}\right)\right], \tag{9.41}$$

where $\ell_T(\boldsymbol{\theta}) = \log L_T(\boldsymbol{\theta})$, $\hat{\boldsymbol{\theta}}$ is the *unrestricted ML* estimator of $\boldsymbol{\theta}$, and $\tilde{\boldsymbol{\theta}}$ is the *restricted ML* estimator of $\boldsymbol{\theta}$. The latter is computed by maximizing $L_T(\boldsymbol{\theta})$ subject to the s restrictions $\mathbf{h}(\boldsymbol{\theta}) = \mathbf{0}$. Under the null hypothesis, H_0, and assuming that certain regularity conditions are met, it is possible to show that

$$LR \xrightarrow{d} \chi^2_s,$$

where s is the number of restrictions imposed (see below for a sketch of the proof). The null hypothesis H_0 is rejected if LR is larger than the appropriate critical value of the chi-squared distribution.

The LR approach requires that the maintained model is estimated both under the null and under the alternative hypotheses. The other two likelihood-based approaches to be presented below require the estimation of the maintained model either under the null or under the alternative hypothesis, but not under both hypotheses.

9.7.2 The Lagrange multiplier test procedure

The Lagrange multiplier (LM) procedure uses the restricted estimators, $\tilde{\boldsymbol{\theta}}$, and requires the computation of the following statistic

$$LM = \left[\frac{\partial \log L_T(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}'}\right]_{\boldsymbol{\theta}=\tilde{\boldsymbol{\theta}}} \left[-\frac{\partial^2 \log L_T(\boldsymbol{\theta})}{\partial \boldsymbol{\theta} \partial \boldsymbol{\theta}'}\right]^{-1}_{\boldsymbol{\theta}=\tilde{\boldsymbol{\theta}}} \left[\frac{\partial \log L_T(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}\right]_{\boldsymbol{\theta}=\tilde{\boldsymbol{\theta}}}, \tag{9.42}$$

[4] The assumption that these restrictions are independent requires that the $s \times p$ matrix of the derivatives $\partial \mathbf{h}/\partial \boldsymbol{\theta}'$ has a full rank, namely that Rank $(\partial \mathbf{h}/\partial \boldsymbol{\theta}') = s$.

where $\partial \log L_T(\boldsymbol{\theta}) / \partial \boldsymbol{\theta}$ and $\partial^2 \log L_T(\boldsymbol{\theta}) / \partial \boldsymbol{\theta} \partial \boldsymbol{\theta}'$ are the first and the second derivatives of the log-likelihood function which are evaluated at $\boldsymbol{\theta} = \tilde{\boldsymbol{\theta}}$, the *restricted* estimator of $\boldsymbol{\theta}$. Recall that it is computed under the null hypothesis, H_0, which defines the set of restrictions to be tested. The *LM* test was originally proposed by Rao and is also referred to as Rao's score test, or simply the 'score test'. Under the null hypothesis, *LM* has a limiting chi-squared distribution with degrees of freedom equal to the number of restrictions, *s*, in the case of H_0 in (9.39).

9.7.3 The Wald test procedure

The Wald test makes use of the unrestricted estimators, $\hat{\boldsymbol{\theta}}$, and is defined by

$$W = \mathbf{h}'\left(\hat{\boldsymbol{\theta}}\right)\left\{\widehat{Var}\left[\mathbf{h}\left(\hat{\boldsymbol{\theta}}\right)\right]\right\}^{-1}\mathbf{h}\left(\hat{\boldsymbol{\theta}}\right), \tag{9.43}$$

where $\widehat{Var}\left[\mathbf{h}\left(\hat{\boldsymbol{\theta}}\right)\right]$ is the estimator of the variance of $\mathbf{h}\left(\hat{\boldsymbol{\theta}}\right)$ and can be estimated consistently by

$$\widehat{Var}\left[\mathbf{h}\left(\hat{\boldsymbol{\theta}}\right)\right] = \left(\frac{\partial \mathbf{h}(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}'}\right)_{\boldsymbol{\theta}=\hat{\boldsymbol{\theta}}}\left[-\frac{\partial^2 \log L_T(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}\partial \boldsymbol{\theta}'}\right]_{\boldsymbol{\theta}=\hat{\boldsymbol{\theta}}}\left(\frac{\partial \mathbf{h}(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}\right)_{\boldsymbol{\theta}=\hat{\boldsymbol{\theta}}}. \tag{9.44}$$

Under H_0, *W* has a chi-squared distribution with degrees of freedom equal to the number of restrictions. However, it is important to note that in small samples the outcome of the Wald test could crucially depend on the particular choice of the algebraic formulation of the nonlinear restrictions used. This has been illustrated by Gregory and Veall (1985), using Monte Carlo simulations.

Asymptotically (namely as the sample size, *T*, is allowed to increase without a bound), all the three test procedures are equivalent. Like the *LR* statistic, under the null hypothesis, the *LM* and the *W* statistics are asymptotically distributed as chi-squared variates with *s* degrees of freedom. We can write

$$LR \overset{a}{\sim} LM \overset{a}{\sim} W,$$

where '$\overset{a}{\sim}$' stands for 'asymptotic equivalence' in distribution functions.

Other versions of the *LM* and the *W* statistics are also available. One possibility would be to replace $\left\{-\frac{\partial^2 \log L_T(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}\partial \boldsymbol{\theta}'}\right\}$ in (9.42) and (9.44) by $T\ Plim\left\{T^{-1}\frac{\partial^2 \log L_T(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}\partial \boldsymbol{\theta}'}\right\}$. This would not affect the asymptotic distribution of the test statistics, but in some cases could simplify their computation.

The literature contains a variety of proofs of the above propositions at different levels of generality. In what follows we provide a sketch of the proof under basic regularity conditions. It is simpler to start with the *LM* test.

Define the Lagrangian function

$$\ell_T(\boldsymbol{\theta}, \boldsymbol{\lambda}) = \ell_T(\boldsymbol{\theta}) + \boldsymbol{\lambda}_T'\mathbf{h}(\boldsymbol{\theta}),$$

where $\boldsymbol{\lambda}_T$ is an $s \times 1$ vector of Lagrangian multipliers. As before, denote $\hat{\boldsymbol{\theta}}_T$ as unrestricted maximum likelihood estimators and $\tilde{\boldsymbol{\theta}}_T$ as restricted maximum likelihood estimators. The idea of the

LM test is, if H_0 is valid, then the restricted estimator should be near to the unrestricted estimator, i.e., $\tilde{\lambda}_T$ should be near to zero.

The first-order conditions of maximum likelihood function yield

$$\frac{\partial \ell_T(\boldsymbol{\theta}, \lambda)}{\partial \boldsymbol{\theta}} = \frac{\partial \ell_T(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}} + \frac{\partial \boldsymbol{h}_T(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}'} \lambda_T = \mathbf{0},$$
$$\frac{\partial \ell_T(\boldsymbol{\theta}, \lambda)}{\partial \lambda} = \boldsymbol{h}_T(\boldsymbol{\theta}) = \mathbf{0}.$$

That is

$$\frac{\partial \ell_T\left(\tilde{\boldsymbol{\theta}}_T\right)}{\partial \boldsymbol{\theta}} + \boldsymbol{H}'\left(\tilde{\boldsymbol{\theta}}_T\right)\tilde{\boldsymbol{\lambda}}_T = \mathbf{0}, \tag{9.45}$$
$$\boldsymbol{h}_T(\tilde{\boldsymbol{\theta}}_T) = \mathbf{0}. \tag{9.46}$$

Since we are interested in the distribution of $\tilde{\boldsymbol{\lambda}}_T$ under the H_0, we take first-order Taylor expansion of (9.45) and (9.46) around $\boldsymbol{\theta}_0$

$$\boldsymbol{d}(\boldsymbol{\theta}_0) + \frac{\partial \ell_T(\boldsymbol{\theta}_0)}{\partial \boldsymbol{\theta} \partial \boldsymbol{\theta}'}\left(\tilde{\boldsymbol{\theta}}_T - \boldsymbol{\theta}_0\right) + \boldsymbol{H}'\left(\tilde{\boldsymbol{\theta}}_T\right)\tilde{\boldsymbol{\lambda}}_T = o_p(1)$$
$$\boldsymbol{H}(\boldsymbol{\theta}_0)\left(\tilde{\boldsymbol{\theta}}_T - \boldsymbol{\theta}_0\right) = o_p(1).$$

Replacing $\tilde{\boldsymbol{\theta}}_T$ by $\boldsymbol{\theta}_0$ under the null hypothesis and putting them in matrix form

$$\begin{bmatrix} \boldsymbol{d}(\boldsymbol{\theta}_0) \\ \mathbf{0} \end{bmatrix} + \begin{bmatrix} \frac{\partial \ell_T(\boldsymbol{\theta}_0)}{\partial \boldsymbol{\theta} \partial \boldsymbol{\theta}'} & \boldsymbol{H}'(\boldsymbol{\theta}_0) \\ \boldsymbol{H}(\boldsymbol{\theta}_0) & \mathbf{0} \end{bmatrix} \begin{bmatrix} \left(\tilde{\boldsymbol{\theta}}_T - \boldsymbol{\theta}_0\right) \\ \tilde{\boldsymbol{\lambda}}_T \end{bmatrix} = o_p(1).$$

This is equivalent to

$$\begin{bmatrix} \frac{1}{\sqrt{T}}\boldsymbol{d}(\boldsymbol{\theta}_0) \\ \mathbf{0} \end{bmatrix} + \begin{bmatrix} \frac{1}{T}\frac{\partial \ell_T(\boldsymbol{\theta}_0)}{\partial \boldsymbol{\theta} \partial \boldsymbol{\theta}'} & \boldsymbol{H}'(\boldsymbol{\theta}_0) \\ \boldsymbol{H}(\boldsymbol{\theta}_0) & \mathbf{0} \end{bmatrix} \begin{bmatrix} \sqrt{T}\left(\tilde{\boldsymbol{\theta}}_T - \boldsymbol{\theta}_0\right) \\ \frac{1}{\sqrt{T}}\tilde{\boldsymbol{\lambda}}_T \end{bmatrix} = o_p(1).$$

Using (9.29) to replace $\frac{1}{T}\frac{\partial \ell_T(\boldsymbol{\theta}_0)}{\partial \boldsymbol{\theta} \partial \boldsymbol{\theta}'}$ by $-\boldsymbol{A}(\boldsymbol{\theta}_0)$, we get

$$\begin{bmatrix} \sqrt{T}\left(\tilde{\boldsymbol{\theta}}_T - \boldsymbol{\theta}_0\right) \\ \frac{1}{\sqrt{T}}\tilde{\boldsymbol{\lambda}}_T \end{bmatrix} \overset{a}{=} \begin{bmatrix} -\boldsymbol{A}(\boldsymbol{\theta}_0) & \boldsymbol{H}'(\boldsymbol{\theta}_0) \\ \boldsymbol{H}(\boldsymbol{\theta}_0) & \mathbf{0} \end{bmatrix}^{-1} \begin{bmatrix} \frac{1}{\sqrt{T}}\boldsymbol{d}(\boldsymbol{\theta}_0) \\ \mathbf{0} \end{bmatrix}. \tag{9.47}$$

We have already seen under regularity conditions,

$$\sqrt{T}\left(\hat{\boldsymbol{\theta}}_T - \boldsymbol{\theta}_0\right) \overset{a}{\sim} N\left(\mathbf{0}, \mathbf{B}(\boldsymbol{\theta}_0)^{-1}\right),$$

and under H_0

$$\sqrt{T}\left(\tilde{\boldsymbol{\theta}}_T - \boldsymbol{\theta}_0\right) \overset{a}{\sim} N\left(\mathbf{0}, \mathbf{B}\left(\boldsymbol{\theta}_0\right)^{-1}\right),$$

Using this and rearrange (9.47), we get

$$\frac{1}{\sqrt{T}}\tilde{\lambda}_T \overset{a}{\sim} N\left\{\mathbf{0}, \left[\boldsymbol{H}\left(\boldsymbol{\theta}_0\right)\boldsymbol{A}\left(\boldsymbol{\theta}_0\right)^{-1}\boldsymbol{H}\left(\boldsymbol{\theta}_0\right)'\right]^{-1}\right\}.$$

Since $\boldsymbol{A}\left(\boldsymbol{\theta}_0\right)$ is nonsingular and $\boldsymbol{H}\left(\boldsymbol{\theta}_0\right)$ is assumed to have full rank s, we assume that $\boldsymbol{H}\left(\boldsymbol{\theta}_0\right)$ $\boldsymbol{A}\left(\boldsymbol{\theta}_0\right)^{-1}\boldsymbol{H}\left(\boldsymbol{\theta}_0\right)'$ invertible. Hence

$$\frac{1}{\sqrt{\boldsymbol{T}}}\tilde{\boldsymbol{\lambda}}_T'\boldsymbol{H}\left(\boldsymbol{\theta}_0\right)\boldsymbol{A}\left(\boldsymbol{\theta}_0\right)^{-1}\boldsymbol{H}\left(\boldsymbol{\theta}_0\right)'\frac{1}{\sqrt{T}}\tilde{\boldsymbol{\lambda}}_T \sim \chi_s^2, \text{ where } s = rank\left(\boldsymbol{H}\left(\boldsymbol{\theta}_0\right)\right) \text{ and } s \leq p.$$

Thus the Lagrange multiplier is distributed as a normal distribution. Since $\tilde{\boldsymbol{\theta}}_T$ is a consistent estimator of $\boldsymbol{\theta}_0$, we replace $\boldsymbol{H}\left(\boldsymbol{\theta}_0\right)$, $\boldsymbol{A}\left(\boldsymbol{\theta}_0\right)$ by those evaluated at $\tilde{\boldsymbol{\theta}}_T$ and using Slusky's theorem, we get the *LM* test statistic

$$\begin{aligned} LM &= \frac{1}{T}\tilde{\boldsymbol{\lambda}}_T'\boldsymbol{H}\left(\tilde{\boldsymbol{\theta}}_T\right)\boldsymbol{A}\left(\tilde{\boldsymbol{\theta}}_T\right)^{-1}\boldsymbol{H}\left(\tilde{\boldsymbol{\theta}}_T\right)'\tilde{\boldsymbol{\lambda}}_T \\ &= \frac{1}{T}\left(\frac{\partial \ell_T\left(\tilde{\boldsymbol{\theta}}_T\right)}{\partial \boldsymbol{\theta}}\right)'\boldsymbol{A}\left(\tilde{\boldsymbol{\theta}}_T\right)^{-1}\left(\frac{\partial \ell_T\left(\tilde{\boldsymbol{\theta}}_T\right)}{\partial \boldsymbol{\theta}}\right) \sim \chi_s^2. \end{aligned} \tag{9.48}$$

Hence under regularity conditions the *LM* test will be asymptotically distributed as χ_s^2 under the null hypothesis. The advantage of the *LM* test is that we only need to estimate the model under the constraints.

The *LR* test focuses on the difference between the restricted and unrestricted values of the log-likelihood function. The statistic is defined as

$$LR = -2\left[\ell_T\left(\tilde{\boldsymbol{\theta}}_T\right) - \ell_T\left(\hat{\boldsymbol{\theta}}_T\right)\right].$$

Note the difference $\ell_T\left(\tilde{\boldsymbol{\theta}}_T\right) - \ell_T\left(\hat{\boldsymbol{\theta}}_T\right)$ is always non-positive, hence *LR* is always non-negative.

Expand the restricted estimator around the unrestricted estimator

$$\ell_T\left(\tilde{\boldsymbol{\theta}}_T\right) = \ell_T\left(\hat{\boldsymbol{\theta}}_T\right) + \frac{\partial \ell_T\left(\hat{\boldsymbol{\theta}}_T\right)}{\partial \boldsymbol{\theta}'}\left(\tilde{\boldsymbol{\theta}}_T - \hat{\boldsymbol{\theta}}_T\right) + \frac{1}{2}\left(\tilde{\boldsymbol{\theta}}_T - \hat{\boldsymbol{\theta}}_T\right)'\frac{\partial \ell_T^2\left(\bar{\boldsymbol{\theta}}_T\right)}{\partial \boldsymbol{\theta}\partial \boldsymbol{\theta}'}\left(\tilde{\boldsymbol{\theta}}_T - \hat{\boldsymbol{\theta}}_T\right),$$

with $\bar{\boldsymbol{\theta}}_T$ lies between $\tilde{\boldsymbol{\theta}}_T$ and $\hat{\boldsymbol{\theta}}_T$. Since $\hat{\boldsymbol{\theta}}_T$ is an unrestricted *ML* estimator, we get $\partial \ell_T\left(\hat{\boldsymbol{\theta}}_T\right) / \partial \boldsymbol{\theta}' = \mathbf{0}$. Hence

$$\begin{aligned} LR &= -2\left[\ell_T\left(\tilde{\boldsymbol{\theta}}_T\right) - \ell_T\left(\hat{\boldsymbol{\theta}}_T\right)\right] \\ &= \sqrt{T}\left(\tilde{\boldsymbol{\theta}}_T - \hat{\boldsymbol{\theta}}_T\right)' \left[-\frac{1}{T}\frac{\partial \ell_T^2\left(\bar{\boldsymbol{\theta}}_T\right)}{\partial \boldsymbol{\theta}\partial \boldsymbol{\theta}'}\right] \sqrt{T}\left(\tilde{\boldsymbol{\theta}}_T - \hat{\boldsymbol{\theta}}_T\right). \end{aligned}$$

Using (9.29) again, we have

$$LR = \sqrt{T}\left(\tilde{\boldsymbol{\theta}}_T - \hat{\boldsymbol{\theta}}_T\right)' \boldsymbol{A}\left(\boldsymbol{\theta}_0\right) \sqrt{T}\left(\tilde{\boldsymbol{\theta}}_T - \hat{\boldsymbol{\theta}}_T\right).$$

We expand the score

$$\frac{1}{\sqrt{T}}\frac{\partial \ell_T\left(\tilde{\boldsymbol{\theta}}_T\right)}{\partial \boldsymbol{\theta}} = \frac{1}{\sqrt{T}}\frac{\partial \ell_T\left(\hat{\boldsymbol{\theta}}_T\right)}{\partial \boldsymbol{\theta}} + \frac{1}{T}\frac{\partial \ell_T^2\left(\boldsymbol{\theta}_T^*\right)}{\partial \boldsymbol{\theta}\partial \boldsymbol{\theta}'}\sqrt{T}\left(\tilde{\boldsymbol{\theta}}_T - \hat{\boldsymbol{\theta}}_T\right),$$

with $\boldsymbol{\theta}_T^*$ lies between $\tilde{\boldsymbol{\theta}}_T$ and $\hat{\boldsymbol{\theta}}_T$. Since $\frac{1}{\sqrt{T}}\frac{\partial \ell_T\left(\hat{\boldsymbol{\theta}}_T\right)}{\partial \boldsymbol{\theta}} = \mathbf{0}$, and $\frac{1}{T}\frac{\partial \ell_T^2(\theta_T^*)}{\partial \boldsymbol{\theta}\partial \boldsymbol{\theta}'} \xrightarrow{p} \boldsymbol{A}\left(\boldsymbol{\theta}_0\right)$, we get

$$\boldsymbol{A}\left(\boldsymbol{\theta}_0\right)\sqrt{T}\left(\tilde{\boldsymbol{\theta}}_T - \hat{\boldsymbol{\theta}}_T\right) = -\frac{1}{\sqrt{T}}\frac{\partial \ell_T\left(\tilde{\boldsymbol{\theta}}_T\right)}{\partial \boldsymbol{\theta}} + o_p\left(1\right).$$

Hence

$$\sqrt{T}\left(\tilde{\boldsymbol{\theta}}_T - \hat{\boldsymbol{\theta}}_T\right) = -\boldsymbol{A}\left(\boldsymbol{\theta}_0\right)^{-1}\frac{1}{\sqrt{T}}\frac{\partial \ell_T\left(\tilde{\boldsymbol{\theta}}_T\right)}{\partial \boldsymbol{\theta}} + o_p\left(1\right).$$

Hence

$$\begin{aligned} LR &= \sqrt{T}\left(\tilde{\boldsymbol{\theta}}_T - \hat{\boldsymbol{\theta}}_T\right)' \boldsymbol{A}\left(\boldsymbol{\theta}_0\right)^{-1} \sqrt{T}\left(\tilde{\boldsymbol{\theta}}_T - \hat{\boldsymbol{\theta}}_T\right) \\ &= \frac{1}{T}\left(\frac{\partial \ell_T\left(\tilde{\boldsymbol{\theta}}_T\right)}{\partial \boldsymbol{\theta}}\right)' \boldsymbol{A}\left(\boldsymbol{\theta}_0\right)^{-1}\left(\frac{\partial \ell_T\left(\tilde{\boldsymbol{\theta}}_T\right)}{\partial \boldsymbol{\theta}}\right) + o_p\left(1\right). \end{aligned} \tag{9.49}$$

Comparing (9.48) and (9.49) we observe LR has the same χ^2 distribution as LM, that is, LR and *LM* are asymptotically equivalent.

The Wald test focuses on the unrestricted estimation. Note that if H_0 is valid, we should have $\boldsymbol{h}\left(\boldsymbol{\theta}\right)$ be near to zero under the unrestricted estimation. Hence by construction we have $\boldsymbol{h}(\tilde{\boldsymbol{\theta}}_T) = 0$. We carry out a Taylor expansion of the unrestricted estimates around $\boldsymbol{\theta}_0$

$$\boldsymbol{h}(\hat{\boldsymbol{\theta}}_T) = \boldsymbol{h}\left(\boldsymbol{\theta}_0\right) + \frac{\partial \boldsymbol{h}\left(\bar{\boldsymbol{\theta}}_T\right)}{\partial \boldsymbol{\theta}'}\left(\hat{\boldsymbol{\theta}}_T - \boldsymbol{\theta}_0\right),\ \bar{\boldsymbol{\theta}}_T \in \left(\hat{\boldsymbol{\theta}}_T, \boldsymbol{\theta}_0\right).$$

Hence

$$\sqrt{T}\left[\boldsymbol{h}(\hat{\boldsymbol{\theta}}_T) - \boldsymbol{h}(\boldsymbol{\theta}_0)\right] = \sqrt{T}\left[\frac{\partial \boldsymbol{h}(\bar{\boldsymbol{\theta}}_T)}{\partial \boldsymbol{\theta}'}\left(\hat{\boldsymbol{\theta}}_T - \boldsymbol{\theta}_0\right)\right].$$

By consistency of *ML* estimators, $\frac{\partial \boldsymbol{h}(\bar{\boldsymbol{\theta}}_T)}{\partial \boldsymbol{\theta}'} \xrightarrow{p} \frac{\partial \boldsymbol{h}(\boldsymbol{\theta}_0)}{\partial \boldsymbol{\theta}'} = \boldsymbol{H}(\boldsymbol{\theta}_0)$. Now under H_0, $\boldsymbol{h}(\boldsymbol{\theta}_0) = \mathbf{0}$, $\sqrt{T}\boldsymbol{h}\left(\hat{\boldsymbol{\theta}}_T\right)$ has the same distribution as $\sqrt{T}\left(\hat{\boldsymbol{\theta}}_T - \boldsymbol{\theta}_0\right)$ since $\frac{\partial \boldsymbol{h}(\bar{\boldsymbol{\theta}}_T)}{\partial \boldsymbol{\theta}'}$ tends to matrix $\boldsymbol{H}(\boldsymbol{\theta}_0)$ in probability. Hence

$$\sqrt{T}\boldsymbol{h}(\hat{\boldsymbol{\theta}}_T) \overset{a}{\sim} N\left\{0, \boldsymbol{H}(\boldsymbol{\theta}_0)\, Var\left[\sqrt{T}\left(\hat{\boldsymbol{\theta}}_T - \boldsymbol{\theta}_0\right)\right]\boldsymbol{H}(\boldsymbol{\theta}_0)'\right\}.$$

Note that $Var\left[\sqrt{T}\left(\hat{\boldsymbol{\theta}}_T - \boldsymbol{\theta}_0\right)\right]$ is just the Fisher information matrix $\mathbf{B}(\boldsymbol{\theta}_0)$, we get

$$W = \boldsymbol{h}(\hat{\boldsymbol{\theta}}_T)'\left[\boldsymbol{H}(\boldsymbol{\theta}_0)\, \widehat{Var}(\hat{\boldsymbol{\theta}}_T)\boldsymbol{H}(\boldsymbol{\theta}_0)'\right]^{-1}\boldsymbol{h}(\hat{\boldsymbol{\theta}}_T) \sim \chi^2_s.$$

The *LM*, *LR*, and Wald tests will converge to the same χ^2_s distribution asymptotically. Hence in large samples, it does not matter which one we choose. The choice could be based on ease of computations. If both $\tilde{\boldsymbol{\theta}}_T$ and $\hat{\boldsymbol{\theta}}_T$ are easy to compute, we can choose the LR test; if $\tilde{\boldsymbol{\theta}}_T$ is easy to compute, we choose the *LM* test; if $\hat{\boldsymbol{\theta}}_T$ is easy to compute, we choose the Wald test.

Although all three statistics are asymptotically equivalent, there is, however, an interesting inequality relationship between them in small samples. We have

$$W \geq LR \geq LM.$$

This result suggests that in finite samples, the *LR* test rejects the null hypothesis less often than the *W* test, but rejects the null more often than the *LM* test. In practice, the real value of these likelihood-based procedures lies in situations where the problem cannot be cast in the classical normal regression model framework, or when the hypotheses under consideration impose nonlinear parametric restrictions on the parameters of the linear regression model. Among the three testing procedures, the *LR* approach seems to be more robust, especially as far as the formulation of the null hypothesis is concerned, and is to be preferred to the other procedures. This is particularly the case when the null hypothesis imposes nonlinear restrictions on the parameters. Often the *LM* procedure is favoured over the other two approaches on grounds of computational ease, as it requires estimation of the *ML* estimators only under the null hypothesis.

The three tests are discussed under the maximum likelihood context, which requires knowledge of density function of the variable. When we need to estimate parameters without specifying density functions, the Generalized Method of Moments (*GMM*) is more robust. The *GMM* approach is discussed in Chapter 10.

Example 25 (Linear regression with normal errors) *The difference between the three test procedures is best demonstrated by means of a simple example. Suppose we are interested in testing the hypothesis*

$$H_0 : \beta = 0, \qquad \textit{against} \qquad H_1 : \beta \neq 0,$$

where β is the slope coefficient in the simple classical normal regression model

$$y_t = \alpha + \beta x_t + u_t, \qquad u_t \sim N\left(0, \sigma^2\right) \qquad t = 1, 2, \ldots, T.$$

The log-likelihood function of this model is given by (9.6) which we reproduce here for convenience

$$\log L_T\left(\boldsymbol{\theta}\right) = -\frac{T}{2}\log\left(2\pi\sigma^2\right) - \frac{1}{2\sigma^2}\sum_{t=1}^{T}\left(y_t - \alpha - \beta x_t\right)^2,$$

where $\boldsymbol{\theta} = \left(\alpha, \beta, \sigma^2\right)'$. The unrestricted ML estimators obtained under H_1,

$$\hat{\boldsymbol{\theta}} = \begin{pmatrix} \hat{\alpha} \\ \hat{\beta} \\ \hat{\sigma}^2 \end{pmatrix} = \begin{pmatrix} \bar{y} - \bar{x}\left(\frac{S_{XY}}{S_{XX}}\right) \\ \frac{S_{XY}}{S_{XX}} \\ \frac{1}{T}\sum_t \left(y_t - \hat{\alpha} - \hat{\beta}x_t\right)^2 \end{pmatrix},$$

where as before $S_{XY} = \sum_t \left(x_t - \bar{x}\right)\left(y_t - \bar{y}\right)$ and $S_{XX} = \sum_t \left(x_t - \bar{x}\right)^2$. The restricted ML estimators are obtained under $H_0 : \beta = 0$ and are given by

$$\tilde{\boldsymbol{\theta}} = \begin{pmatrix} \bar{y} \\ 0 \\ \frac{1}{T}\sum_t \left(y_t - \bar{y}\right)^2 \end{pmatrix}.$$

The maximized values of the log-likelihood function under H_0 and H_1 are given by

$$\log L_T\left(\tilde{\boldsymbol{\theta}}\right) = \frac{-T}{2}\log\left[2\pi\frac{\sum_t \left(y_t - \bar{y}\right)^2}{T}\right] - \frac{T}{2},$$

and

$$\log L_T\left(\hat{\boldsymbol{\theta}}\right) = \frac{-T}{2}\log\left[2\pi\frac{\sum_t \left(y_t - \hat{\alpha} - \hat{\beta}x_t\right)^2}{T}\right] - \frac{T}{2},$$

respectively. Hence, the LR statistic for testing $H_0 : \beta = 0$, against the two-sided alternatives $H_1 : \beta \neq 0$ will be

$$\begin{aligned} LR &= 2\left[\log L_T\left(\hat{\boldsymbol{\theta}}\right) - \log L_T\left(\tilde{\boldsymbol{\theta}}\right)\right] \\ &= -T\log\left[\frac{\sum_t \left(y_t - \hat{\alpha} - \hat{\beta}x_t\right)^2}{\sum_t \left(y_t - \bar{y}\right)^2}\right], \end{aligned}$$

which can also be written as

$$LR = -T \log \left(\frac{SSR_U}{SSR_R} \right), \tag{9.50}$$

where SSR_U and SSR_R are respectively the unrestricted and the restricted sums of squares of residuals.[5] *In the present application there is also a simple relationship between the LR and the F test of linear restrictions discussed in Section 3.7. Recall from (3.26) that the F-statistic for testing $H_0 : \beta = 0$ is given by*

$$F = \left(\frac{T-2}{1} \right) \frac{SSR_R - SSR_U}{SSR_U}.$$

Using this result in (9.50) now yields

$$LR = T \log \left(1 + \frac{F}{T-2} \right). \tag{9.51}$$

Hence there is a monotonic relationship between the exact F-test of $\beta = 0$ and the asymptotic LR test. Also, since in this simple case under H_0, the F-statistic is distributed with 1 and $T-2$ degrees of freedom, we have $F = t^2$, where t has a t-distribution with $T-2$ degrees of freedom, and (9.51) becomes

$$LR = T \log \left(1 + \frac{t^2}{T-2} \right),$$

which also yields

$$LR = -T \log \left(1 - \hat{\rho}_{XY}^2 \right), \tag{9.52}$$

where $\hat{\rho}_{XY}$ is the sample correlation coefficient between Y and X. Hence, not surprisingly, a large value of the LR statistic is associated with a large value of $|\hat{\rho}_{XY}|$. These results are readily generalized to the multivariate case.[6] *Turning now to the LM and W statistics, we need first and second-order derivatives of the log-likelihood function:*

$$\frac{\partial \log L_T(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}} = \begin{pmatrix} \frac{1}{\sigma^2} \sum_t u_t \\ \frac{1}{\sigma^2} \sum_t x_t u_t \\ -\frac{T}{2\sigma^2} + \frac{1}{2\sigma^4} \sum_t u_t^2 \end{pmatrix}. \tag{9.53}$$

[5] The unrestricted estimators of the residuals are $\hat{u}_t = y_t - \hat{\alpha} - \hat{\beta} x_t$ and the restricted ones are $\tilde{u}_t = y_t - \bar{y}$. Hence $\sum_i \left(y_t - \hat{\alpha} - \hat{\beta} x_t \right) = \sum_t \hat{u}_t^2$ is the unrestricted sums of squares of residuals, and $\sum_t \left(y_t - \bar{y} \right) = \sum_t \tilde{u}_t^2$ is the restricted sums of squares of residuals.

[6] In testing the joint hypothesis $H_0 : \beta_1 = \beta_2 = \ldots = \beta_k = 0$ against the alternative, $H_1 : \beta_1 \neq 0, \beta_2 \neq 0, \ldots, \beta_k \neq 0$, in the multivariate regression model $y_t = \alpha + \sum_{i=1}^{k} \beta_i x_{it}$, we have LR$= -T \log \left(1 - R^2 \right)$, where R^2 is the squared multiple correlation coefficient of the regression equation.

Using (9.1) we have

$$\frac{\partial^2 \log L_T(\boldsymbol{\theta})}{\partial \boldsymbol{\theta} \partial \boldsymbol{\theta}'} = \begin{pmatrix} -\frac{T}{\sigma^2} & \frac{\sum_t x_t}{\sigma^2} & \frac{-\sum_t u_t}{\sigma^4} \\ \frac{\sum_t x_t}{\sigma^2} & -\frac{1}{\sigma^2}\sum_t x_t^2 & \frac{-\sum_t x_t u_t}{\sigma^4} \\ \frac{-\sum_t u_t}{\sigma^4} & \frac{-\sum_t x_t u_t}{\sigma^4} & \left(\frac{T}{2\sigma^4} - \frac{1}{\sigma^6}\sum_t u_t^2\right) \end{pmatrix}. \tag{9.54}$$

Evaluating these derivatives under H_0 involves replacing $\boldsymbol{\theta} = \left(\alpha, \beta, \sigma^2\right)'$ with the restricted estimators, namely $\tilde{\alpha} = \bar{y}, \tilde{\beta} = 0$ and $\tilde{\sigma}^2 = T^{-1}\sum_t \left(y_t - \bar{y}\right)^2$. We have

$$\left[\frac{\partial \log L_T(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}\right]_{\boldsymbol{\theta}=\tilde{\boldsymbol{\theta}}} = \begin{pmatrix} \frac{1}{\tilde{\sigma}^2}\sum_t \left(y_t - \bar{y}\right) \\ \frac{1}{\tilde{\sigma}^2}\sum_t \left(y_t - \bar{y}\right) x_t \\ -\frac{T}{2\tilde{\sigma}^2} + \frac{1}{2\tilde{\sigma}^4}\sum_t \left(y_t - \bar{y}\right)^2 \end{pmatrix} = \begin{pmatrix} 0 \\ \frac{1}{\tilde{\sigma}^2}\sum_t x_t \left(y_t - \bar{y}\right) \\ 0 \end{pmatrix},$$

and

$$\left[\frac{\partial^2 \log L_T(\boldsymbol{\theta})}{\partial \boldsymbol{\theta} \partial \boldsymbol{\theta}'}\right]_{\boldsymbol{\theta}=\tilde{\boldsymbol{\theta}}} = \begin{pmatrix} -\frac{T}{\tilde{\sigma}^2} & \frac{T\bar{x}}{\tilde{\sigma}^2} & 0 \\ \frac{T\bar{x}}{\tilde{\sigma}^2} & \frac{-\sum_t x_t^2}{\tilde{\sigma}^2} & \frac{-\sum_t x_t (y_t - \bar{y})}{\tilde{\sigma}^4} \\ 0 & \frac{-\sum_t x_t (y_t - \bar{y})}{\tilde{\sigma}^4} & \frac{-T}{2\tilde{\sigma}^4} \end{pmatrix}.$$

To simplify the derivations we use an asymptotically equivalent version of the estimator of $\partial^2 \log L_T(\boldsymbol{\theta}) / \partial \boldsymbol{\theta} \partial \boldsymbol{\theta}'$ which in effect treats the off diagonal element in the matrix of second derivatives, $T^{-1}\sum_t x_t \left(y_t - \bar{y}\right)$ as being negligible. With this approximation it is now easily seen that

$$LM = \left(0, \quad \frac{-\sum_t x_t \left(y_t - \bar{y}\right)}{\tilde{\sigma}^2}\right) \begin{pmatrix} \frac{T}{\tilde{\sigma}^2} & -\frac{T\bar{x}}{\tilde{\sigma}^2} \\ \frac{-T\bar{x}}{\tilde{\sigma}^2} & \frac{\sum_t x_t^2}{\tilde{\sigma}^2} \end{pmatrix}^{-1} \begin{bmatrix} 0 \\ \frac{\sum_t x_t (y_t - \bar{y})}{\tilde{\sigma}^2} \end{bmatrix},$$

or

$$LM = \frac{\left[\sum_t x_t \left(y_t - \bar{y}\right)\right]^2}{\tilde{\sigma}^2 \sum_t (x_t - \bar{x})^2}.$$

But since $\tilde{\sigma}^2 = T^{-1}\sum_t \left(y_t - \bar{y}\right)^2 = T^{-1} S_{YY}$, and $\sum_t x_t \left(y_t - \bar{y}\right) = S_{XY}$, then

$$LM = \frac{T S_{XY}^2}{S_{XX} S_{YY}} = T\hat{\rho}_{XY}^2. \tag{9.55}$$

This form of the LM statistic is quite common in more complicated applications of the LM principle. For example, in the multivariate case, the LM statistic for the test of $H_0 : \beta_1 = \beta_2 = \cdots = \beta_k = 0$ in $y_t = a + \sum_{i=1}^{k} \beta_i x_{it} + u_t$ is given by TR^2, where R^2 is the square of the coefficient of the multiple correlation of the regression equation.

The Wald statistic for testing $H_0 : \beta = 0$ *is based on the unrestricted estimate of* β*, namely* $h(\theta) = \beta$ *and* $h(\hat{\theta}) = \hat{\beta}$*. Hence,*

$$W = \hat{\beta}^2 \left[\widehat{Var}\left(\hat{\beta}\right)\right]^{-1},$$

where in the ML framework the estimator of the variance of $\hat{\beta}$ *is based on the ML estimator of* σ^2*, i.e.* $\hat{\sigma}^2 = \frac{\sum_\tau \left(y_t - \hat{\alpha} - \hat{\beta} x_t\right)^2}{T}$ *rather than the unbiased estimator which is obtained by dividing the sum of squares of residuals by* $T - 2$*. We have*

$$W = \frac{T\hat{\beta}^2}{\sum_t \left(y_t - \hat{\alpha} - \hat{\beta} x_t\right)^2} = \left(\frac{T}{T-2}\right) t^2,$$

where t is the t-ratio of the slope coefficient. Once again using (3.7) W can be written in terms of ρ^2_{XY}*. That is*

$$W = \frac{T\hat{\rho}^2_{XY}}{\left(1 - \hat{\rho}^2_{XY}\right)}. \tag{9.56}$$

To summarize, for testing $H_0 : \beta = 0$ *in the simple regression model, the LR, LM, and W statistics given by (9.52), (9.55) and (9.56), respectively, can all be written in terms of* $\hat{\rho}^2_{XY}$*. Collecting these results in one place we have*

$$LR = -T \log\left(1 - \hat{\rho}^2_{XY}\right),$$
$$LM = T\hat{\rho}^2_{XY},$$

and

$$W = \frac{T\hat{\rho}^2_{XY}}{1 - \hat{\rho}^2_{XY}}.$$

9.8 Further reading

We refer to Rao (1973) and White (2000) for textbook treatments of the statistical foundations of the maximum likelihood method.

9.9 Exercises

1. Let $x_1, x_2, \ldots, x_T$ be independent random drawings from the exponential distribution

$$p(x_i, \lambda) = \lambda e^{-\lambda x_i},$$

for $x_i > 0$ and $\lambda > 0$.

(a) Derive the maximum likelihood (ML) estimator for λ, $\hat{\lambda}$, and derive the asymptotic variance of $\hat{\lambda}$.
(b) Obtain the likelihood ratio (LR) test statistic for testing $H_0 : \lambda = \lambda_0$ versus $H_1 : \lambda \neq \lambda_0$, and compare it to the Lagrange multiplier (LM) statistic for the same test.

2. Let $y_i, i = 1, 2, \ldots, n$ be non-negative integers independently drawn from a Poisson distribution

$$\Pr(y_i \,|\theta\,) = \frac{\theta^i \exp(-\theta)}{y_i!},\ y_i = 0, 1, 2, \ldots,$$

where ! denotes the factorial operator, so that $y_i! = 1 \times 2 \times \ldots \times (y_i - 1)y_i$.

(a) Write down the log-likelihood function of θ.
(b) Derive the ML estimator of θ and its asymptotic variance.

3. Let $x_1, x_2, \ldots, x_T$ be independent random drawings from $N(\mu, \sigma^2)$. Construct the likelihood ratio test for testing $H_0 : \sigma^2 = 1$ versus $H_1 : \sigma^2 \neq 1$.
4. Consider the linear regression model

$$y_t = \beta_1 x_{1t} + \beta_2 x_{2t} + u_t.$$

Derive the Wald test for testing the null hypothesis $H_0 : \beta_1 = \beta_2$.
5. Consider the partitioned classical linear regression model

$$\begin{aligned} y_t &= \mathbf{x}_t'\boldsymbol{\beta} + u_t, \\ &= \mathbf{x}_{1t}'\boldsymbol{\beta}_1 + \mathbf{x}_{2t}'\boldsymbol{\beta}_2 + u_t, \end{aligned}$$

where $\boldsymbol{\beta}_1$ and $\boldsymbol{\beta}_2$ are k_1 and k_2 vectors of constant parameters and $u_t \sim IIDN(0, \sigma^2)$. Suppose we are interested in testing $H_0 : \boldsymbol{\beta}_2 = \mathbf{0}$ against $H_1 : \boldsymbol{\beta}_2 \neq \mathbf{0}$.

(a) Obtain the Lagrange multiplier (LM), likelihood ratio (LR) and Wald (W) statistics for testing H_0.
(b) Show that the F-test of H_0 may be obtained as a simple transformation of all three statistics mentioned under (a).
(c) Demonstrate the inequality $LM \leq LR \leq W$.

6. Show that when the regression equation (9.11) is augmented with the model of the exogenous variables given by (9.15), the result can be written in the form of the following vector autoregressive (VAR) model (see also Chapter 21)

$$\mathbf{z}_t = \boldsymbol{\Phi}\mathbf{z}_{t-1} + \boldsymbol{\xi}_t.$$

Derive an expression for $\mathbf{z}_t$, $\boldsymbol{\Phi}$ and $\boldsymbol{\xi}_t$. Hence, or otherwise, assuming that $E(\mathbf{v}_t u_t) = \mathbf{0}$, prove that $E(u_t \mathbf{x}_{t-s}) = 0$, for $s = 0, 1, 2, \ldots$. Also noting that

$$E(\mathbf{z}_t u_t) = E\left(\boldsymbol{\xi}_t u_t\right),$$

and

$$E(\mathbf{z}_{t+1} u_t) = \boldsymbol{\Phi} E(\mathbf{z}_t u_t),$$

show that

$$E(u_t \mathbf{x}_{t+1}) = \sigma^2 \boldsymbol{\lambda}.$$

Using a similar line of reasoning obtain an expression for $E(u_t \mathbf{x}_{t+2})$.

10 Generalized Method of Moments

10.1 Introduction

Standard econometric modelling practice has for a long time been based on strong assumptions concerning the data generating process underlying a series. Such assumptions, although often unrealistic, allowed the construction of estimators with optimal theoretical properties. The most prominent example of this perspective is the maximum likelihood (*ML*) method, which requires a complete specification of the model to be estimated, including the probability distribution of the variables of interest. However, in practice, the investigator may not have full knowledge of the probability distribution to commit himself to such a complete specification of the econometric model. The generalized method of moments (*GMM*), discussed in this chapter, is an alternative estimation procedure which is devised for such circumstances. This estimator requires only the specification of a set of moment conditions that are deduced from the assumptions underlying the econometric model to be estimated. The *GMM* is particularly attractive to economists who deal with a variety of moment or orthogonality conditions derived from the theoretical properties of their economic models. This method is also useful in cases where the complexity of the economic model makes it difficult to write down a tractable likelihood function (Bera and Bilias (2002)). Finally, in cases where the distribution of the data is known, the *GMM* may be a convenient method to adopt to avoid the computational complexities often associated with *ML* techniques.

The utilization of moment-based estimation techniques dates back to the work by Karl Pearson on the method of moments (see Pearson (1894)), although it has been the object of renewed interest by econometricians since the seminal paper by Hansen (1982) on *GMM*. Recently, *GMM* techniques have been widely applied to analyse economic and financial data, using time series, cross-sectional or panel data.

In the rest of the chapter we review the estimation theory of the *GMM*. We also describe the instrumental variables (IV) approach within the broader context of the *GMM*.

10.2 Population moment conditions

Suppose that a sample of T observations $(\mathbf{w}_1, \mathbf{w}_2, \ldots, \mathbf{w}_T)$ is drawn from the joint probability distribution function

$$f(\mathbf{w}_1, \mathbf{w}_2, \ldots, \mathbf{w}_T; \boldsymbol{\theta}_0),$$

where $\boldsymbol{\theta}_0$ is a $q \times 1$ vector of true parameters, belonging to the parameter space, $\boldsymbol{\Theta}$. Typically, $\mathbf{w}_t$ would contain one or more endogenous variables and a number of predetermined and/or exogenous variables. Let $\mathbf{m}(.)$ be a r-dimensional vector of functions, then a *population moment condition* takes the form

$$E\left[\mathbf{m}(\mathbf{w}_t, \boldsymbol{\theta}_0)\right] = \mathbf{0}, \text{ for all } t. \tag{10.1}$$

Three cases are possible:

(i) When $q > r$, the parameters in $\boldsymbol{\theta}$ are not identified.
(ii) When $q = r$, the parameters are exactly identified.
(iii) When $q < r$, the parameters are overidentified and the moment conditions need to be restricted to deliver a unique parameter vector in the estimation, which will be done by the means of a weighting matrix.

Estimation can be based on the empirical counterpart of $E\left[\mathbf{m}(\mathbf{w}_t, \boldsymbol{\theta})\right]$, given by the r-dimensional vector of *sample moments*

$$\mathsf{M}_T(\boldsymbol{\theta}) = \frac{1}{T}\sum_{t=1}^{T} \mathbf{m}(\mathbf{w}_t, \boldsymbol{\theta}). \tag{10.2}$$

Example 26 (Linear regression) *Consider the linear regression model $y_t = \mathbf{x}_t'\boldsymbol{\beta}_0 + u_t$, where $\mathbf{x}_t$ is a k-dimensional vector of regressors. Under the classical assumptions, the following population conditions can be found*

$$E\left(\mathbf{x}_t u_t\right) = E\left[\mathbf{x}_t(y_t - \mathbf{x}_t'\boldsymbol{\beta}_0)\right] = \mathbf{0}, t = 1, 2, \ldots, T.$$

Suppose now that the regressors, $\mathbf{x}_t$, are correlated with the error term, u_t, namely $E\left(\mathbf{x}_t u_t\right) \neq \mathbf{0}$. This may arise in a variety of circumstances such as errors-in-variables, simultaneous equations, or rational expectations models. It is well known that in this case the ordinary least squares (OLS) estimator of $\boldsymbol{\beta}_0$ is biased and inconsistent. Assume we can find a m-dimensional vector of so-called instrumental variables, $\mathbf{z}_t$, satisfying the orthogonality conditions

$$E\left(\mathbf{z}_t u_t\right) = E\left[\mathbf{z}_t(y_t - \mathbf{x}_t'\boldsymbol{\beta}_0)\right] = \mathbf{0}, t = 1, 2, \ldots, T. \tag{10.3}$$

The above population moment conditions yield the relation

$$\Sigma_{zy} = \Sigma_{zx}\boldsymbol{\beta}_0,$$

where $\Sigma_{zy} = E\left(\mathbf{z}_t y_t\right)$, *and* $\Sigma_{zx} = E\left(\mathbf{z}_t \mathbf{x}_t'\right)$. *For identification of* $\boldsymbol{\beta}_0$, *it is required that the* $m \times k$ *matrix* Σ_{zx} *be of full rank, k, ensuring that* $\boldsymbol{\beta}_0$ *is the unique solution to (10.3). If* $m = k$, *then* Σ_{zx} *is invertible and* $\boldsymbol{\beta}_0$ *may be determined by*

$$\boldsymbol{\beta}_0 = \Sigma_{zx}^{-1}\Sigma_{zy}.$$

Example 27 (Consumption based asset pricing model) *One of the most cited applications of the GMM principle for estimating econometric models is the Hansen and Singleton (1982) consumption based asset pricing model. This model involves a representative agent who makes investment and consumption decisions at time t to maximize his/her discounted lifetime utility subject to a budget constraint. Assume that the only one asset available as a possible investment yields a pay-off in the following period. The agent wishes to maximize*

$$E\left[\sum_{i=0}^{\infty}\beta^i U\left(c_{t+i}\right)|\Omega_t\right],$$

subject to the budget constraint

$$c_t + q_t = r_t q_{t-1} + w_t,$$

where β *is a discount factor,* $U(.)$ *is a utility function,* c_t *is consumption in period t,* q_t *is the quantity of the asset held in period t which pays* r_t, w_t *is real labour income, and* Ω_t *is the information available at time t. Optimal choice of consumption and investments satisfies*

$$U'(c_t) = \beta E\left[r_t U'(c_{t+1})|\Omega_t\right],$$

which can be rewritten as

$$E\left[\beta\frac{U'(c_{t+1})}{U'(c_t)}r_{t+1}|\Omega_t\right] - 1 = 0.$$

Hansen and Singleton (1982) set $U(c_t) = \left(c_t^{\gamma} - 1\right)/\gamma$ *so that the above equation becomes*

$$E\left[\beta\left(\frac{c_{t+1}}{c_t}\right)^{\gamma-1}r_{t+1}|\Omega_t\right] - 1 = 0.$$

The above moment can be exploited for estimation of the unknown parameters, β *and* γ. *In particular, let* $\mathbf{z}_t$ *be a vector of variables in the agent's information set at time, namely* $\mathbf{z}_t \in \Omega_t$. *For example,*

$\mathbf{z}_t$ may contain lagged values $\frac{c_t}{c_{t-1}}$, $\frac{c_{t-1}}{c_{t-2}}$, as well as a constant. Hansen and Singleton (1982) suggest the following population moment conditions to be used in GMM estimation

$$E\left\{\mathbf{z}_t\left[\beta\left(\frac{c_{t+1}}{c_t}\right)^{\gamma-1}(1+r_{t+1})-1\right]\right\}=\mathbf{0}.$$

See Hansen and Singleton (1982) for further details.

10.3 Exactly *q* moment conditions

When the number of moment conditions is exactly the same as the number of unknown parameters, it seems reasonable to estimate $\boldsymbol{\theta}_0$ by setting the sample mean of $\mathbf{m}(\mathbf{w}_t, \boldsymbol{\theta}_0)$ to zero, that is to define the moment estimator $\hat{\boldsymbol{\theta}}_T$ as the value of $\boldsymbol{\theta}$ that solves the system of equations

$$\mathsf{M}_T(\hat{\boldsymbol{\theta}}_T)=\frac{1}{T}\sum_{t=1}^{T}\mathbf{m}(\mathbf{w}_t,\hat{\boldsymbol{\theta}}_T)=\mathbf{0}.$$

This estimation procedure was introduced by Pearson (1894), and $\hat{\boldsymbol{\theta}}_T$ is usually referred to as the 'methods-of-moments' estimator. Its application yields many familiar estimators, such as the *OLS* or the *IV*.

Example 28 *Suppose we wish to estimate the parameters of a univariate normal distributed random variable, v_t. It is well known that the normal distribution depends only on two parameters, the population mean, μ_0, and the population variance, σ_0^2. These two parameters satisfy the population moment conditions*

$$\begin{aligned}E(v_t)-\mu_0&=0,\\ E\left(v_t^2\right)-\left(\sigma_0^2+\mu_0\right)&=0.\end{aligned}$$

Hence, Pearson's method involves estimating $\left(\mu_0,\sigma_0^2\right)$ by the values $\left(\hat{\mu}_T,\hat{\sigma}_T^2\right)$ which satisfy the analogous sample moment conditions, and therefore are solutions of

$$\frac{1}{T}\sum_{t=1}^{T}v_t-\hat{\mu}_T=0,\ \frac{1}{T}\sum_{t=1}^{T}v_t^2-\left(\hat{\sigma}_T^2+\hat{\mu}_T\right)=0,$$

from which it follows that

$$\hat{\mu}_T=\frac{1}{T}\sum_{t=1}^{T}v_t,\ \hat{\sigma}_T^2=\frac{1}{T}\sum_{t=1}^{T}\left(v_t-\hat{\mu}_T\right)^2.$$

Example 29 (Wright's demand equation) *Wright (1925) considered the following simple simultaneous equations system in agricultural demand and supply*

$$q_t^D = \alpha_0 p_t + u_t^D,$$
$$q_t^S = \boldsymbol{\beta}_0' \mathbf{n}_t + \gamma_0 p_t + u_t^S,$$
$$q_t^D = q_t^S = q_t,$$

where q_t^D and q_t^S represent demand and supply in year t, p_t is the price of the commodity in year t, q_t equals quantity produced, $\mathbf{n}_t$ is a vector of variables affecting supply but not the demand, and u_t^D and u_t^S are zero mean disturbances. The interest is in estimating α_0, given a sample of T observations on (q_t, p_t). OLS regression of q_t on p_t would yield misleading results, given that price and output are simultaneously determined. Wright (1925) suggests solving this problem by taking a variable, z_t^D, which is related to price, but it is such that Cov $\left(z_t^D u_t^D\right) = 0$. One example is the input price or the yield per acre. Then by taking the covariance of z_t^D with both sides of the equation for q_t yields

$$E\left(z_t^D q_t\right) - \alpha_0 E\left(z_t^D p_t\right) = 0.$$

The above expression provides a population moment condition that can be exploited for estimating α_0. Using Pearson's method of moments yields

$$\hat{\alpha}_T = \sum_{t=1}^{T} z_t^D q_t / \sum_{t=1}^{T} z_t^D p_t,$$

which is the IV estimator, with z_t^D as instrument (see also Section 10.8 below).

10.4 Excess of moment conditions

The *GMM* estimator is used when the number of moment conditions, *r*, is larger than the number of unknown parameters, *q*. In principle, it would be possible to use any *q* of the available moment conditions to estimate $\boldsymbol{\theta}$. However, in this case the estimator would depend on the particular subset of moment conditions chosen. Further, ignoring some of the moments would imply a loss of efficiency, at least in sufficiently large samples. A more appropriate approach is an efficient combination of all the available moment conditions. Specifically, the *GMM* estimator of $\boldsymbol{\theta}$, $\hat{\boldsymbol{\theta}}_T$, based on (10.1), is

$$\hat{\boldsymbol{\theta}}_T = \underset{\boldsymbol{\theta} \in \boldsymbol{\Theta}}{\operatorname{argmin}} \{\mathsf{M}_T'(\boldsymbol{\theta}) \mathbf{A}_T \mathsf{M}_T(\boldsymbol{\theta})\}, \tag{10.4}$$

where $\mathbf{A}_T$ is a $r \times r$ positive semi-definite, possibly random, weighting matrix. We assume that $\mathbf{A}_T$ converges to a unique, *positive definite*, non-random matrix, namely

$$\mathbf{A}_T \xrightarrow{p} \mathbf{A}. \tag{10.5}$$

The matrix $\mathbf{A}_T$ imposes weights on the *r* moment conditions in such a way that we obtain a unique parameter vector $\hat{\boldsymbol{\theta}}_T$. Note that the positive semi-definiteness of $\mathbf{A}_T$ implies that $\mathsf{M}_T'(\boldsymbol{\theta}) \mathbf{A}_T \mathsf{M}_T(\boldsymbol{\theta}) \geq 0$ for any $\boldsymbol{\theta}$. The first-order conditions for minimization of (10.4) are

$$\mathbf{D}_T'(\hat{\boldsymbol{\theta}}_T)\mathbf{A}_T\mathsf{M}_T(\hat{\boldsymbol{\theta}}_T) = \mathbf{0},$$

where

$$\mathbf{D}_T(\boldsymbol{\theta}) = \frac{\partial}{\partial \boldsymbol{\theta}'}\mathsf{M}_T(\boldsymbol{\theta}) = \frac{1}{T}\sum_{t=1}^{T}\frac{\partial}{\partial \boldsymbol{\theta}'}\mathbf{m}_T(\boldsymbol{\theta}) \tag{10.6}$$

is a $r \times q$ matrix of first-order derivatives.

10.4.1 Consistency

To establish the theoretical properties of the *GMM* estimator, it is necessary to make some regularity conditions on $\mathbf{w}_t$ and $\mathbf{m}(.)$. Specifically, we assume:

Assumption A1: (i) $E[\mathbf{m}(\mathbf{w}_t, \boldsymbol{\theta}_0)]$ exists and is finite for all $\boldsymbol{\theta} \in \boldsymbol{\Theta}$, and for all t; (ii) If $\mathbf{g}_t(\boldsymbol{\theta}) = E[\mathbf{m}(\mathbf{w}_t, \boldsymbol{\theta})]$, then there exists a $\boldsymbol{\theta}_0 \in \Theta$ such that $\mathbf{g}_t(\boldsymbol{\theta}) = \mathbf{0}$ for all t if and only if $\boldsymbol{\theta} = \boldsymbol{\theta}_0$.

Assumption A2: $\sup_{\boldsymbol{\theta}\in\Theta} \left\| \mathbf{g}_t(\boldsymbol{\theta}) - \mathsf{M}_T(\boldsymbol{\theta}) \right\| \xrightarrow{p} \mathbf{0}$.

Assumption A1, part (ii), implies that $\boldsymbol{\theta}_0$ can be identified by the population moments, $\mathbf{g}_t(\boldsymbol{\theta})$. Note that if for more than one value of $\boldsymbol{\theta}$ we had $\mathbf{g}_t(\boldsymbol{\theta}) = \mathbf{0}$ then we would not be able to identify $\boldsymbol{\theta}_0$ (see also the discussion on observational equivalence in Section 20.9). Assumption A2 requires that $\mathbf{m}(\mathbf{w}_t, \boldsymbol{\theta})$ satisfies a uniform weak law of large numbers, so that the difference between the average sample and population moments converges in probability to zero. Note that this is a high-level assumption that is satisfied under a variety of more primitive conditions. For example, under some general conditions on $\mathbf{m}(.)$, and if $\boldsymbol{\Theta}$ is compact, Assumption A2 is satisfied if $\{\mathbf{w}_t\}_{t=0}^{\infty}$ satisfies a weak law of large number for independent and heterogeneously distributed processes, stationary and ergodic processes, or mixing processes (see Chapter 8 for further details, Section 8.8).

Under Assumption A1 it follows that $\mathsf{M}_T'(\boldsymbol{\theta})\mathbf{A}\mathsf{M}_T(\boldsymbol{\theta}) = 0$ if and only if $\boldsymbol{\theta} = \boldsymbol{\theta}_0$, and $\mathsf{M}_T'(\boldsymbol{\theta})\mathbf{A}\mathsf{M}_T(\boldsymbol{\theta}) > 0$, otherwise. Further, under Assumption A2 it is possible to show that

$$\mathsf{M}_T'(\boldsymbol{\theta})\mathbf{A}_T\mathsf{M}_T(\boldsymbol{\theta}) - \mathsf{M}_T'(\boldsymbol{\theta})\mathbf{A}\mathsf{M}_T(\boldsymbol{\theta}) \xrightarrow{p} 0. \tag{10.7}$$

The facts that the *GMM* estimator, $\hat{\boldsymbol{\theta}}_T$, defined in (10.4), minimizes $\mathsf{M}_T'(\boldsymbol{\theta})\mathbf{A}_T\mathsf{M}_T(\boldsymbol{\theta})$, that $\boldsymbol{\theta}_0$ minimizes $\mathsf{M}_T'(\boldsymbol{\theta})\mathbf{A}\mathsf{M}_T(\boldsymbol{\theta})$, and relation (10.7) implies weak consistency of $\hat{\boldsymbol{\theta}}_T$, namely, $\hat{\boldsymbol{\theta}}_T \xrightarrow{p} \boldsymbol{\theta}_0$.

See Mátyás (1999) and Amemiya (1985, p. 107), for further details.

10.4.2 Asymptotic normality

Assumptions in addition to those required for consistency are needed to derive the asymptotic normality of *GMM* estimators. In particular, we assume:

Assumption A3: $\mathbf{m}(\mathbf{w}_t, \boldsymbol{\theta})$ is continuously differentiable with respect to $\boldsymbol{\theta} \in \Theta$.

Assumption A4: For any sequence $\boldsymbol{\theta}_T^*$, such that $\boldsymbol{\theta}_T^* \to \boldsymbol{\theta}_0$, $\mathbf{D}_T\left(\boldsymbol{\theta}_T^*\right) - \mathbf{D} \xrightarrow{p} \mathbf{0}$, where $\mathbf{D}$ does not depend on $\boldsymbol{\theta}$.

Assumption A5: $\mathbf{m}(\mathbf{w}_t, \boldsymbol{\theta})$ satisfies the central limit theorem, so that

$$\left[\mathbf{S}(\boldsymbol{\theta}_0)\right]^{-1/2} \sqrt{T}\mathsf{M}_T(\boldsymbol{\theta}_0) \xrightarrow{d} N(\mathbf{0}, \mathbf{I}_r), \tag{10.8}$$

where

$$\mathbf{S}(\boldsymbol{\theta}_0) = E\left[T\mathsf{M}_T(\boldsymbol{\theta}_0)\mathsf{M}_T(\boldsymbol{\theta}_0)'\right],$$

is a nonsingular matrix.

Again, note that Assumption A5 is a high level assumption that is satisfied under a number of more primitive conditions. For example, Assumption A5 holds if $\{\mathbf{w}_t\}_{t=0}^{\infty}$ satisfies central limit theorems for independent and heterogeneously distributed processes, stationary and ergodic processes, or mixing processes (see Chapter 8).

To prove asymptotic normality, consider the mean-value expansion of $\mathsf{M}_T(\hat{\boldsymbol{\theta}}_T)$ around $\boldsymbol{\theta}_0$

$$\mathsf{M}_T(\hat{\boldsymbol{\theta}}_T) = \mathsf{M}_T(\boldsymbol{\theta}_0) + \left[\mathbf{D}_T\left(\bar{\boldsymbol{\theta}}\right)\right](\hat{\boldsymbol{\theta}}_T - \boldsymbol{\theta}_0),$$

where $\bar{\boldsymbol{\theta}}$ lies, element by element, between $\boldsymbol{\theta}_0$ and $\hat{\boldsymbol{\theta}}_T$. Premultiplying by $\mathbf{D}_T'(\hat{\boldsymbol{\theta}}_T)\mathbf{A}_T$ gives

$$\mathbf{D}_T'\left(\hat{\boldsymbol{\theta}}_T\right)\mathbf{A}_T\mathsf{M}_T(\hat{\boldsymbol{\theta}}_T) = \mathbf{D}_T'\left(\hat{\boldsymbol{\theta}}_T\right)\mathbf{A}_T\mathsf{M}_T(\boldsymbol{\theta}_0) + \left[\mathbf{D}_T'\left(\hat{\boldsymbol{\theta}}_T\right)\mathbf{A}_T\mathbf{D}_T\left(\bar{\boldsymbol{\theta}}\right)\right](\hat{\boldsymbol{\theta}}_T - \boldsymbol{\theta}_0), \tag{10.9}$$

or

$$\mathbf{D}_T'\left(\hat{\boldsymbol{\theta}}_T\right)\mathbf{A}_T\mathsf{M}_T(\boldsymbol{\theta}_0) + \left[\mathbf{D}_T'\left(\hat{\boldsymbol{\theta}}_T\right)\mathbf{A}_T\mathbf{D}_T\left(\bar{\boldsymbol{\theta}}\right)\right](\hat{\boldsymbol{\theta}}_T - \boldsymbol{\theta}_0) = \mathbf{0},$$

given that the left hand side of (10.9) are the first-order conditions for minimization of $\mathsf{M}_T'(\boldsymbol{\theta})\mathbf{A}_T\mathsf{M}_T(\boldsymbol{\theta})$. Rearranging and multiplying by $\sqrt{T}$ we obtain

$$\sqrt{T}\left[\mathbf{D}_T'\left(\hat{\boldsymbol{\theta}}_T\right)\mathbf{A}_T\mathbf{D}_T\left(\bar{\boldsymbol{\theta}}\right)\right](\hat{\boldsymbol{\theta}}_T - \boldsymbol{\theta}_0) = -\mathbf{D}_T'\left(\hat{\boldsymbol{\theta}}_T\right)\mathbf{A}_T\sqrt{T}\mathsf{M}_T(\boldsymbol{\theta}_0). \tag{10.10}$$

Then, using (10.8),

$$\sqrt{T}(\hat{\boldsymbol{\theta}}_T(\mathbf{A}_T) - \boldsymbol{\theta}_0) \overset{a}{\sim} N\left[\mathbf{0}, (\mathbf{D}'\mathbf{A}\mathbf{D})^{-1}\mathbf{D}'\mathbf{A}\mathbf{S}\mathbf{A}'\mathbf{D}(\mathbf{D}'\mathbf{A}'\mathbf{D})^{-1}\right], \tag{10.11}$$

where $\mathbf{S} = \mathbf{S}(\boldsymbol{\theta}_0)$.

10.5 Optimal weighting matrix

Appropriate choice of the weighting matrix leads to asymptotic efficiency of the *GMM* estimator. We wish to choose matrix $\mathbf{A}$ so that it minimizes the covariance matrix in equation (10.11), namely

$$\mathbf{V_A} = (\mathbf{D}'\mathbf{A}\mathbf{D})^{-1}\mathbf{D}'\mathbf{A}\mathbf{S}\mathbf{A}'\mathbf{D}(\mathbf{D}'\mathbf{A}'\mathbf{D})^{-1}, \tag{10.12}$$

or, equivalently, its trace. The traditional proof of this is rather tedious and, therefore, omitted here. An informed guess for the optimal $\mathbf{A}$, denoted by $\mathbf{A}^*$, would be

$$\mathbf{A}^* = \mathbf{S}^{-1}. \tag{10.13}$$

Plugging this into equation (10.12) yields

$$\mathbf{V_{A^*}} = (\mathbf{D}'\mathbf{S}^{-1}\mathbf{D})^{-1}, \tag{10.14}$$

We can now prove that this is the smallest possible variance matrix. To this end, we will show that the difference between variance in equation (10.12) with any arbitrary $\mathbf{A}$ and the variance in equation (10.14), $\mathbf{V_A} - \mathbf{V_{A^*}}$, is a positive semi-definite matrix. To establish this it is sufficient to show that $\mathbf{V_{A^*}^{-1}} - \mathbf{V_A^{-1}}$ is a positive semi-definite matrix. Using the results from equations (10.12) and (10.14) we have

$$\mathbf{V_{A^*}^{-1}} - \mathbf{V_A^{-1}} = (\mathbf{D}'\mathbf{S}^{-1}\mathbf{D}) - \left[(\mathbf{D}'\mathbf{A}\mathbf{D})^{-1}\mathbf{D}'\mathbf{A}\mathbf{S}\mathbf{A}'\mathbf{D}(\mathbf{D}'\mathbf{A}'\mathbf{D})^{-1}\right]^{-1}.$$

Since $\mathbf{S}$ is positive definite, we can decompose it using the Cholesky decomposition $\mathbf{S} = \mathbf{C}\mathbf{C}'$, so that $\mathbf{S}^{-1}\mathbf{C} = \mathbf{C}^{-1'}$. Now define

$$\mathbf{H} = \mathbf{C}^{-1}\mathbf{D}, \text{ and } \mathbf{B} = \mathbf{C}'\mathbf{A}'.$$

Then

$$\begin{aligned}
\mathbf{V_{A^*}^{-1}} - \mathbf{V_A^{-1}} &= (\mathbf{D}'\mathbf{S}^{-1}\mathbf{D}) - \left[(\mathbf{D}'\mathbf{A}\mathbf{D})^{-1}\mathbf{D}'\mathbf{A}\mathbf{S}\mathbf{A}'\mathbf{D}(\mathbf{D}'\mathbf{A}'\mathbf{D})^{-1}\right]^{-1} \\
&= (\mathbf{D}'\mathbf{C}^{-1'}\mathbf{C}^{-1}\mathbf{D}) - (\mathbf{D}'\mathbf{A}'\mathbf{D})\left(\mathbf{D}'\mathbf{A}\mathbf{C}\mathbf{C}'\mathbf{A}'\mathbf{D}\right)^{-1}(\mathbf{D}'\mathbf{A}\mathbf{D}) \\
&= \mathbf{D}'\mathbf{C}^{-1'}\left[\mathbf{I}_r - \mathbf{C}'\mathbf{A}'\mathbf{D}\left(\mathbf{D}'\mathbf{A}\mathbf{C}\mathbf{C}'\mathbf{A}'\mathbf{D}\right)^{-1}\mathbf{D}'\mathbf{A}\mathbf{C}\right]\mathbf{C}^{-1}\mathbf{D} \\
&= \mathbf{H}'\left[\mathbf{I}_r - \mathbf{B}\mathbf{D}\left(\mathbf{D}'\mathbf{B}\mathbf{B}'\mathbf{D}\right)^{-1}\mathbf{D}'\mathbf{B}'\right]\mathbf{H}.
\end{aligned}$$

Let $\mathbf{F} = \mathbf{B}\mathbf{D}$, it follows that

$$\mathbf{V_{A^*}^{-1}} - \mathbf{V_A^{-1}} = \mathbf{H}'(\mathbf{I} - \mathbf{F}(\mathbf{F}'\mathbf{F})^{-1}\mathbf{F}')\mathbf{H},$$

which is a positive semi-definite matrix, as required.

10.6 Two-step and iterated *GMM* estimators

We now focus on consistent estimation of $\mathbf{S}$, needed both for statistical inference and efficient estimation. Recall that

$$\mathbf{S} = Var\left(\sqrt{T}\mathsf{M}_T(\boldsymbol{\theta}_0)\right) = \frac{1}{T}\sum_{t=1}^{T}\sum_{s=1}^{T} E\left[\mathbf{m}\left(w_t,\boldsymbol{\theta}_0\right)\mathbf{m}\left(w_s,\boldsymbol{\theta}_0\right)'\right]. \tag{10.15}$$

In the case $\mathbf{m}\left(w_t,\boldsymbol{\theta}_0\right)$ is a stationary and ergodic process, under some regularity conditions it can be shown that

$$\mathbf{S} = \boldsymbol{\Gamma}_0\left(\boldsymbol{\theta}_0\right) + \sum_{j=1}^{\infty}\left(\boldsymbol{\Gamma}_j\left(\boldsymbol{\theta}_0\right) + \boldsymbol{\Gamma}_j\left(\boldsymbol{\theta}_0\right)'\right), \tag{10.16}$$

where $\boldsymbol{\Gamma}_j(\boldsymbol{\theta}_0) = E[\mathbf{m}(w_t,\boldsymbol{\theta}_0)\mathbf{m}(w_{t-j},\boldsymbol{\theta}_0)']$. $\mathbf{S}$ is also known as the long-run variance of $\mathbf{m}_t\left(w_t,\boldsymbol{\theta}_0\right)$ (to be distinguished from the contemporaneous variance of $\mathbf{m}_t$ given by $\boldsymbol{\Gamma}_0$). Hence, one estimator for the above expression is the heteroskedasticity autocorrelation consistent (HAC) estimator, taking the form

$$\hat{\mathbf{S}}_T = \hat{\boldsymbol{\Gamma}}_0 + \sum_{j=1}^{m} w(j,m)\left(\hat{\boldsymbol{\Gamma}}_j + \hat{\boldsymbol{\Gamma}}_j'\right),$$

where $\hat{\boldsymbol{\Gamma}}_j = \frac{1}{T}\sum_{t=j+1}^{T}\mathbf{m}(w_t,\hat{\boldsymbol{\theta}}_T)\mathbf{m}(w_{t-j},\hat{\boldsymbol{\theta}}_T)'$, for $j = 0,1,\ldots,m$, and $w(j,m)$ is the kernel or lag window, and m is the bandwidth. The kernel and bandwidth must satisfy certain restrictions to ensure HAC is both consistent and positive semi-definite (see Section 5.9 for details).

Clearly, the calculation of $\hat{\mathbf{S}}_T$ requires knowledge of the unknown parameters, $\boldsymbol{\theta}_0$. A two-step estimation procedure can be followed to compute the optimal *GMM*. This consists of computing an initial, consistent estimator of $\boldsymbol{\theta}_0$, $\hat{\boldsymbol{\theta}}_T^{(1)}$, for example using the non-efficient *GMM* based on an arbitrary choice of $\mathbf{A}_T$. Hence, $\hat{\boldsymbol{\theta}}_T^{(1)}$ is used to compute $\hat{\mathbf{S}}_T$, which is plugged into (10.4) to obtain the asymptotically efficient *GMM* estimator, $\hat{\boldsymbol{\theta}}_T^{(2)}$, in the second step. One common choice of $\mathbf{A}_T$ at the first step is the identity matrix, $\mathbf{I}_q$. Instead of stopping after just two steps, the procedure can be continued so that on the jth step the *GMM* estimation is performed using as weighting matrix $\hat{\mathbf{S}}_T$ computed using $\hat{\boldsymbol{\theta}}_T^{(j-1)}$. Such an estimator is known as the *iterated GMM* estimator.

Monte Carlo studies show that the estimated asymptotic standard errors of the two-step and iterated *GMM* estimators may be severely downward biased in small samples (see e.g., Hansen, Heaton, and Yaron (1996) and Arellano and Bond (1991)). To improve finite sample properties of *GMM*, Hansen, Heaton, and Yaron (1996) suggested the so-called continuous-updating *GMM* (*CUGMM*) estimator, which is the value of $\boldsymbol{\theta}$ that minimizes $\mathsf{M}_T'(\boldsymbol{\theta})\mathbf{S}_T\left(\boldsymbol{\theta}\right)^{-1}\mathsf{M}_T(\boldsymbol{\theta})$, where $\mathbf{S}_T\left(\boldsymbol{\theta}\right)$ is a matrix function of $\boldsymbol{\theta}$ such that $\mathbf{S}_T\left(\boldsymbol{\theta}\right) \rightarrow \mathbf{S}$. Note that this estimator does not depend on an initial choice of the weighting matrix, although it is computationally more

burdensome than the iterated estimator, especially for a large nonlinear model. It is possible to show that this estimator is asymptotically equivalent to the two-step and iterated estimators, but may differ in finite samples. However, Anatolyev (2005) demonstrated analytically that the *CUGMM* estimator can be expected to exhibit lower finite sample bias than its two-step counterpart.

Windmeijer (2005) showed that the extra variation due to the presence of the estimated parameters in the efficient weighting matrix accounts for much of the difference between the finite sample and the estimated asymptotic variance for two-step *GMM* estimators based on moment conditions that are linear in the parameters. In response to this problem Windmeijer (2005) has proposed a finite sample correction for the estimates of the asymptotic variance. In a Monte Carlo study the author shows that such bias correction leads to more accurate inference.

A further complication arises when the number of available moment conditions for *GMM* estimation is large, a case that often occurs in practice. For example, the application of *GMM* techniques for estimation of dynamic panel data (see Chapter 27) leads, when T increases, to a rapid rise in the number of orthogonality conditions available for estimation. Even though using many moment conditions is desirable according to conventional first-order asymptotic theory, it has been found that the two-step *GMM* estimator has a considerable bias in finite samples, in the presence of many moment restrictions (see, e.g., Newey and Smith (2000)). To deal with this problem, Donald, Imbens, and Newey (2009) developed asymptotic mean-square error criteria (MSE) that can be minimized to choose the number of moments to use in the two-step *GMM* estimator. Koenker and Machado (1999) showed that $\frac{r}{T^3} \to 0$ as $T \to \infty$ is a sufficient condition for the limiting distribution of the *GMM* estimator to remain valid.

10.7 Misspecification test

In the case where the number of moment conditions exceeds the number of parameters, $r > q$, the model is overidentified and more orthogonality conditions are used than needed. Hence, a test would be needed to see whether the sample moments are as close to zero as the corresponding moments would suggest. Consider the null hypothesis

$$H_0 : \mathbf{S}^{-1} E\left[\mathbf{m}\left(w_t, \boldsymbol{\theta}_0\right)\right] = \mathbf{0}. \tag{10.17}$$

Hansen (1982) suggests testing the above hypothesis using T times the minimized value of the *GMM* criterion function

$$J = \left[\sqrt{T}\mathsf{M}_T(\hat{\boldsymbol{\theta}}_T)\right]' \hat{\mathbf{S}}_T^{-1} \left[\sqrt{T}\mathsf{M}_T(\hat{\boldsymbol{\theta}}_T)\right]. \tag{10.18}$$

It can be shown that $J \xrightarrow{d} \chi^2(r-q)$. The J-statistic is known as the over-identifying restrictions test, and is widely adopted as a diagnostic tool for models estimated by *GMM*. If the statistic (10.18) exceeds the relevant critical value of a $\chi^2(r-q)$ distribution, then (10.17) must be rejected since at least some of the moment conditions are not supported by the data. The J-test may also be used to investigate whether an additional vector of moments has mean zero and, thus, may be incorporated in the moment conditions in order to improve inference. To this end,

assume that an initial *GMM* estimator, $\tilde{\boldsymbol{\theta}}_T$, based only on the r_1-dimensional vector, $\mathbf{m}_1(\mathbf{w}_t, \boldsymbol{\theta})$, was computed. Then consider

$$J_1 = \left[\sqrt{T}\mathsf{M}_T(\hat{\boldsymbol{\theta}}_T)\right]' \hat{\mathbf{S}}_T^{-1} \left[\sqrt{T}\mathsf{M}_T(\hat{\boldsymbol{\theta}}_T)\right] - \left[\sqrt{T}\mathsf{M}_{1,T}(\tilde{\boldsymbol{\theta}}_T)\right]' \hat{\mathbf{S}}_{1,T}^{-1} \left[\sqrt{T}\mathsf{M}_{1,T}(\tilde{\boldsymbol{\theta}}_T)\right],$$

where $\mathsf{M}_{1,T}(\tilde{\boldsymbol{\theta}}_T) = \frac{1}{T}\sum_{t=1}^{T} \mathbf{m}_1(\mathbf{w}_t, \tilde{\boldsymbol{\theta}}_T)$, and $\hat{\mathbf{S}}_{1,T}$ is the corresponding optimal weight matrix. Under the null hypothesis, $J_1 \xrightarrow{d} \chi^2(r - r_1)$, as $T \to \infty$.

10.8 The generalized instrumental variable estimator

Consider the following linear regression model

$$y_t = \boldsymbol{\beta}'\mathbf{x}_t + u_t, \tag{10.19}$$

where it is known that $E(\mathbf{x}_t u_t) \neq 0$, with $\mathbf{x}_t$ being a k-dimensional vector of regressors, and the errors, u_t, satisfying $E(u_t) = 0$, $E(u_t u_s) = 0$ for $t \neq s$ and $E\left(u_t^2\right) = \sigma^2$. Let $\mathbf{z}_t$ be a r-dimensional vector of instruments, assumed to be correlated with $\mathbf{x}_t$, but independent of u_s for all s and t. Suppose that the number of instruments is larger than the number of parameters to be estimated, i.e., $r > k$. The generalized instrumental variable estimator (GIVE), proposed by Sargan (1958, 1959), combines all available instruments for estimation of $\boldsymbol{\beta}$. Consider the the r population moment conditions

$$E\left[\mathbf{z}_t(y_t - \boldsymbol{\beta}_0'\mathbf{x}_t)\right] = \mathbf{0}. \tag{10.20}$$

For expositional convenience suppose that the variables y_t, $\mathbf{x}_t$, and $\mathbf{z}_t$ have mean $\mathbf{0}$. The covariance matrices of $\mathbf{x}_t$ and $\mathbf{z}_t$ are denoted by

$$\Sigma_{xx} = E\left(\mathbf{x}_t\mathbf{x}_t'\right), \Sigma_{zz} = E\left(\mathbf{z}_t\mathbf{z}_t'\right), \Sigma_{xz} = E\left(\mathbf{x}_t\mathbf{z}_t'\right).$$

The sample moments associated to the population moment conditions, of equation, (10.20), are given by

$$\mathsf{M}_T(\boldsymbol{\theta}) = \frac{1}{T}\sum_{t=1}^{T} \mathbf{z}_t(y_t - \boldsymbol{\beta}'\mathbf{x}_t).$$

Also

$$\begin{aligned}\mathbf{S} &= E\left[\frac{1}{T}\left(\sum_{t=1}^{T}\mathbf{z}_t u_t\right)\left(\sum_{s=1}^{T}\mathbf{z}_s u_s\right)'\right] \\ &= E\left[\frac{1}{T}\sum_{t,s=1}^{T}\left(\mathbf{z}_t\mathbf{z}_s' u_t u_s\right)\right],\end{aligned}$$

which depends on the assumptions made about $\mathbf{u}_t$ and $\mathbf{z}_t$. Given that in this simple application the errors, u_t, are assumed to be homoskedastic and serially uncorrelated, and distributed independently of $\mathbf{z}_s$ for all s, t, we have

$$\begin{aligned}\mathbf{S} &= \frac{1}{T}\sum_{t=1}^{T}\sum_{\tau=1}^{T} E\left(\mathbf{z}_t\mathbf{z}_\tau'\right) E\left(u_t u_\tau\right) \\ &= \frac{1}{T}\sum_{t=1}^{T} E\left(\mathbf{z}_t\mathbf{z}_t'\right)\sigma_u^2 = \sigma_u^2 E\left(\frac{\mathbf{Z}\mathbf{Z}'}{T}\right) = \sigma_u^2\Sigma_{zz},\end{aligned}$$

and

$$\begin{aligned}\mathbf{D} &= E\left(\frac{\partial \mathbf{m}(\mathbf{w}_t, \boldsymbol{\theta}_0)}{\partial \boldsymbol{\theta}'}\right) \\ &= E\left[\frac{1}{T}\frac{\partial}{\partial \boldsymbol{\beta}_0'}\left(\sum_{t=1}^{T}\mathbf{z}_t(y_t - \mathbf{x}_t'\boldsymbol{\beta})\right)\right] \\ &= E\left(\frac{1}{T}\sum_{t=1}^{T}\mathbf{z}_t\mathbf{x}_t'\right) = \Sigma_{zx}.\end{aligned}$$

Denoting the *GIVE* or *GMM* estimator of $\boldsymbol{\beta}$ by $\hat{\boldsymbol{\beta}}_{IV}$, it follows from the two results above that

$$\begin{aligned}Var(\sqrt{T}\hat{\boldsymbol{\beta}}_{IV}) &= (\mathbf{D}'\mathbf{S}^{-1}\mathbf{D})^{-1} \\ &= \sigma^2(\Sigma_{xz}\Sigma_{zz}^{-1}\Sigma_{zx})^{-1}, \\ \widehat{Var}(\sqrt{T}\hat{\boldsymbol{\beta}}_{IV}) &= \sigma^2\left[\frac{\mathbf{X}'\mathbf{Z}}{T}\left(\frac{\mathbf{Z}'\mathbf{Z}}{T}\right)^{-1}\frac{\mathbf{Z}'\mathbf{X}}{T}\right]^{-1}. \end{aligned} \tag{10.21}$$

Furthermore,

$$\hat{\mathbf{A}}_T = \hat{\mathbf{D}}_T'\hat{\mathbf{S}}_T^{-1} = -\hat{\Sigma}_{zx}'\hat{\Sigma}_{zz}^{-1}\frac{1}{\sigma_u^2} = -(\mathbf{Z}'\mathbf{X})'(\mathbf{Z}'\mathbf{Z})^{-1}\frac{1}{\sigma_u^2}, \tag{10.22}$$

and

$$\begin{aligned}\mathsf{M}_T(\boldsymbol{\beta}) &= \frac{1}{T}\sum_{t=1}^{T}\mathbf{z}_t(y_t - \boldsymbol{\beta}\mathbf{x}_t) \\ &= \frac{1}{T}\mathbf{Z}'(\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}).\end{aligned} \tag{10.23}$$

Hence, using the results from equations (10.22) and (10.23) in expression (10.4) we obtain

$$\hat{\boldsymbol{\beta}}_{IV} = \underset{\boldsymbol{\beta}}{\operatorname{argmin}} \left\| \frac{\mathbf{X}'\mathbf{Z}(\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}'(\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})}{T} \right\|. \tag{10.24}$$

Define

$$\mathbf{P}_z = \mathbf{Z}(\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}',$$

then, from equation (10.24), it follows that

$$\mathbf{X}'\mathbf{P}_z\mathbf{Y} - (\mathbf{X}'\mathbf{P}_z\mathbf{X})\hat{\boldsymbol{\beta}}_T = \mathbf{0},$$

which, in turn, implies that

$$\hat{\boldsymbol{\beta}}_{IV} = (\mathbf{X}'\mathbf{P}_z\mathbf{X})^{-1}\mathbf{X}'\mathbf{P}_z\mathbf{Y}, \tag{10.25}$$

which is the *GIVE* (White (1982a)). Using (10.21), an estimator of the variance matrix of $\hat{\boldsymbol{\beta}}_{IV}$ is

$$\widehat{Var}(\hat{\boldsymbol{\beta}}_{IV}) = \hat{\sigma}^2_{IV}(\mathbf{X}'\mathbf{P}_z\mathbf{X})^{-1}, \tag{10.26}$$

where $\hat{\sigma}^2_{IV}$ is the *IV* estimator of σ^2

$$\hat{\sigma}^2_{IV} = \frac{1}{T-K}\hat{\mathbf{u}}'_{IV}\hat{\mathbf{u}}_{IV}, \tag{10.27}$$

and $\hat{\mathbf{u}}_{IV}$ is the *IV* residuals given by

$$\hat{\mathbf{u}}_{IV} = \mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}_{IV}. \tag{10.28}$$

If $u_t \sim ID\left(0, \sigma_i^t\right)$, an heteroskedasticity-consistent estimator of the covariance matrix of $\hat{\boldsymbol{\beta}}_{IV}$ can be computed (see White (1982b) p. 489)

$$\widehat{HCV}(\hat{\boldsymbol{\beta}}_{IV}) = \left(\frac{T}{T-k}\right)\mathbf{Q}_T^{-1}\mathbf{P}'_T\hat{\mathbf{V}}_T\mathbf{P}_T\mathbf{Q}_T^{-1}, \tag{10.29}$$

where

$$\mathbf{Q}_T = \mathbf{X}'\mathbf{P}_z\mathbf{X}, \quad \mathbf{P}_T = (\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}'\mathbf{X}, \quad \hat{\mathbf{V}}_T = \sum_{t=1}^{T}\hat{u}^2_{t,IV}\mathbf{z}_t\mathbf{z}'_t.$$

Note that (10.29) can also be written as

$$\widehat{HCV}(\hat{\boldsymbol{\beta}}_{IV}) = \left(\frac{T}{T-k}\right)(\hat{\mathbf{X}}'\hat{\mathbf{X}})^{-1}\left(\sum_{t=1}^{T}\hat{u}^2_{t,IV}\hat{\mathbf{x}}_t\hat{\mathbf{x}}'_t\right)(\hat{\mathbf{X}}'\hat{\mathbf{X}})^{-1}, \tag{10.30}$$

where $\hat{\mathbf{X}} = \mathbf{P}_z\mathbf{X}$, and $\hat{\mathbf{x}}_t = \mathbf{X}'\mathbf{Z}(\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{z}_t$. It follows that if $\mathbf{Z}$ is specified to include $\mathbf{X}$, then $\hat{\mathbf{X}} = \mathbf{X}, \hat{\mathbf{x}}_t = \mathbf{x}_t, \hat{u}_{t,IV} = \hat{u}_t$, and $\widehat{HCV}(\hat{\boldsymbol{\beta}}_{IV}) = \widehat{HCV}(\hat{\boldsymbol{\beta}}_{OLS})$ (see Section 4.2).

Before concluding we observe that for an instrument to be valid it must be 'relevant' for the endogenous variables included in the regression equation. When instruments are only weakly correlated with the included endogenous variables, we have the problem of 'weak instruments' or 'weak identification', which poses considerable challenges to inference using *GMM* and *IV* methods. Indeed, if instruments are weak, the sampling distributions of *GMM* and *IV* statistics are in general non-normal, and standard *GMM* and *IV* point estimates, hypothesis tests, and confidence intervals are unreliable (see Stock, Wright, and Yogo (2002)).

Example 30 (Instrumental variables estimation of spatial models) *Consider the following spatial model (see Chapter 30 for an introduction to spatial econometrics)*

$$y_i = \rho \sum_{j=1}^{m} w_{ij} y_j + \boldsymbol{\beta}' \mathbf{x}_i + u_i,\ i = 1, 2, \ldots, m,$$

where $u_t \sim IID(0, \sigma^2)$, $\mathbf{x}_i$ *is a vector of strictly exogenous, non-stochastic regressors,* w_{ij} *are known elements of an* $m \times m$ *matrix, known in the literature as spatial weights matrix,* $\mathbf{W}$. *It is assumed that* $w_{ii} = 0$, *the matrix* $\mathbf{W}$ *is row-normalized so that the elements of each row add up to 1, and* $|\rho| < 1$. *In matrix form the above model is*

$$\mathbf{y} = \rho \mathbf{W}\mathbf{y} + \mathbf{X}\boldsymbol{\beta} + \mathbf{u}. \tag{10.31}$$

The variable $\mathbf{W}\mathbf{y}$ *is typically referred to as spatial lag of* $\mathbf{y}$. *Note that, in general, the elements of the spatially lagged dependent vector are correlated with those of the disturbance vector, i.e.,* $E\left(\mathbf{y}'\mathbf{W}'\mathbf{u}\right) \neq 0$. *One implication of this is that the parameters of (10.31) cannot be consistently estimated by OLS. Under some regularity conditions, (10.31) can be rewritten as*

$$\mathbf{y} = \left(\mathbf{I}_m - \rho \mathbf{W}\right)^{-1} \mathbf{X}\boldsymbol{\beta} + \left(\mathbf{I}_m - \rho \mathbf{W}\right)^{-1} \mathbf{u},$$

and its expected value is

$$E\left(\mathbf{y}\right) = \left(\mathbf{I}_m - \rho \mathbf{W}\right)^{-1} \mathbf{X}\boldsymbol{\beta} \tag{10.32}$$

$$= \sum_{j=1}^{\infty} \left(\rho \mathbf{W}\right)^j \mathbf{X}\boldsymbol{\beta}. \tag{10.33}$$

Note that $\mathbf{W}E\left(\mathbf{y}\right)$ *can be seen as formed by a linear combination of the columns of the matrices* $\mathbf{WX}$, $\mathbf{W}^2\mathbf{X}, \mathbf{W}^3\mathbf{X}, \ldots$. *On the basis of this observation, Kelejian and Prucha (1998) have suggested an IV estimator for the parameters* $\boldsymbol{\theta} = \left(\rho, \boldsymbol{\beta}'\right)'$ *in the spatial regression (10.31), using as instruments the matrix* $\mathbf{Z} = (\mathbf{X}, \mathbf{WX})$ *(see (10.25)).*

10.8.1 Two-stage least squares

The IV estimator, $\hat{\boldsymbol{\beta}}_{IV}$, can also be computed using a two-step procedure, known as the two-stage least squares (2*SLS*), where in the first step the fitted values of the *OLS* regression of $\mathbf{X}$ on $\mathbf{Z}$,

$\hat{\mathbf{X}} = \boldsymbol{P}_z\mathbf{X}$ are computed, where $\boldsymbol{P}_z = \mathbf{Z}(\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}$. Then $\hat{\boldsymbol{\beta}}_{IV}$ is obtained by the *OLS* regression of $\mathbf{y}$ on $\hat{\mathbf{X}}$. Notice, however, that such a two-step procedure does not, in general, produce a correct estimator of σ^2, and hence of $\widehat{Var}(\hat{\boldsymbol{\beta}}_{IV})$. This is because the *IV* residuals, $\hat{\mathbf{u}}_{IV}$, defined by (10.28), used in the estimation of $\hat{\sigma}^2_{IV}$ are not the same as the residuals obtained at the second stage of the 2*SLS* method. To see this denote the 2*SLS* residuals by $\hat{\mathbf{u}}_{2SLS}$ and note that

$$\begin{aligned}\hat{\mathbf{u}}_{2SLS} &= \mathbf{y} - \hat{\mathbf{X}}\hat{\boldsymbol{\beta}}_{IV} \\ &= (\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}_{IV}) + (\mathbf{X} - \hat{\mathbf{X}})\hat{\boldsymbol{\beta}}_{IV} \\ &= \hat{\mathbf{u}}_{IV} + (\mathbf{X} - \hat{\mathbf{X}})\hat{\boldsymbol{\beta}}_{IV},\end{aligned} \tag{10.34}$$

where $\mathbf{X} - \hat{\mathbf{X}}$ are the residual matrix $(T \times k)$ of the regressions of $\mathbf{X}$ on $\mathbf{Z}$. Only in the case where $\mathbf{Z}$ is an exact predictor of $\mathbf{X}$, will the two sets of residuals be the same.

10.8.2 Generalized R^2 for *IV* regressions

In the case of *IV* regressions, the goodness of fit measures R^2 and $\bar{R}^2$ introduced in Section 2.10 are not valid. There is no guarantee that R^2 of a regression model estimated by the IV method is positive, and this result does not depend on whether or not an intercept term is included in the regression (see, for example, Maddala (1988, p. 309)). The failure of R^2 based on the IV residuals in providing a valid model selection criterion is due to the dependence of the residual vector on the endogenous variables. In response to this problem, Pesaran and Smith (1994) proposed a measure of fit appropriate for IV regressions, called Generalized R^2, or GR^2, and given by

$$GR^2 = 1 - \frac{\hat{\mathbf{u}}'_{2SLS}\hat{\mathbf{u}}_{2SLS}}{\sum_{t=1}^{T}(y_t - \bar{y})^2}, \tag{10.35}$$

where $\hat{\mathbf{u}}_{2SLS}$, given by (10.34), is the vector of residuals from the second stage in the 2*SLS* procedure. Note also that

$$\hat{\mathbf{u}}_{2SLS} = \hat{\mathbf{u}}_{IV} + (\mathbf{X} - \hat{\mathbf{X}})\hat{\boldsymbol{\beta}}_{IV}. \tag{10.36}$$

A degrees-of-freedom adjusted Generalized R^2 measure is given by

$$\overline{GR}^2 = 1 - \left(\frac{T-1}{T-k}\right)\left(1 - GR^2\right). \tag{10.37}$$

Pesaran and Smith (1994) show that under reasonable assumptions and for large T, the use of GR^2 is a valid discriminator for models estimated by the *IV* method.

10.8.3 Sargan's general misspecification test

This test is proposed in Sargan (1964, pp. 28–9) as a general test of misspecification in the case of *IV* estimation, and is based on the statistic

$$\chi^2_{SM} = \frac{Q(\hat{\boldsymbol{\beta}}_{IV})}{\hat{\sigma}^2_{IV}}, \tag{10.38}$$

where $Q(\hat{\boldsymbol{\beta}}_{IV})$ is the value of *IV* minimand given by

$$\begin{aligned} Q\left(\hat{\boldsymbol{\beta}}_{IV}\right) &= (\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}_{IV})'\mathbf{P}_z(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}_{IV}) \\ &= \mathbf{y}'[\mathbf{P}_z - \mathbf{P}_z\mathbf{X}(\mathbf{X}'\mathbf{P}_z\mathbf{X})^{-1}\mathbf{X}'\mathbf{P}_z]\mathbf{y} \\ &= \hat{\mathbf{y}}'[\mathbf{I} - \hat{\mathbf{X}}(\hat{\mathbf{X}}'\hat{\mathbf{X}})^{-1}\mathbf{X}']\hat{\mathbf{y}}. \end{aligned} \tag{10.39}$$

Under the null hypothesis that the regression equation (10.19) is correctly specified, and that the r instrumental variables $\mathbf{Z}$ are valid instruments, Sargan's misspecification statistic, $\chi^2_{SM} \overset{a}{\sim} \chi^2(r - k)$. It is easily seen that χ^2_{SM} is a special case of the *J*-statistic introduced in Section 10.7.

10.8.4 Sargan's test of residual serial correlation for *IV* regressions

The Lagrange multiplier (LM) test by Breusch and Godfrey (1981) can be used to test for residual serial correlation in *IV* regression (see Section 5.8.1 for a description of the hypotheses to test). In particular, the following *LM*-type test can be used

$$\chi^2_{SC}(p) = T\left[\frac{\hat{\mathbf{u}}'_{IV}\mathbf{W}(\mathbf{W}'\mathbf{M}_{\hat{x}}\mathbf{W})^{-1}\mathbf{W}'\hat{\mathbf{u}}_{IV}}{\hat{\mathbf{u}}'_{IV}\hat{\mathbf{u}}_{IV}}\right] \overset{a}{\sim} \chi^2(p), \tag{10.40}$$

where $\hat{\mathbf{u}}_{IV}$ is the vector of *IV* residuals defined by (10.28)

$$\hat{\mathbf{u}}_{IV} = \mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}_{IV} = (\hat{u}_{1,IV}, \hat{u}_{2,IV}, \ldots, \hat{u}_{T,IV})'.$$

$\mathbf{W}$ is the $T \times p$ matrix consisting of the p lagged valued of $\hat{\mathbf{u}}_{IV}$, namely

$$\mathbf{W} = \begin{pmatrix} 0 & 0 & \ldots & 0 \\ \hat{u}_{1,IV} & 0 & \ldots & 0 \\ \hat{u}_{2,IV} & \hat{u}_{1,IV} & \ldots & 0 \\ \vdots & \vdots & \vdots & \vdots \\ \hat{u}_{T-2,IV} & \hat{u}_{T-3,IV} & \ldots & \hat{u}_{T-p-1,IV} \\ \hat{u}_{T-1,IV} & \hat{u}_{T-2,IV} & \ldots & \hat{u}_{T-p,IV} \end{pmatrix}, \tag{10.41}$$

and

$$\mathbf{M}_{\hat{x}} = \mathbf{I}_T - \mathbf{X}(\hat{\mathbf{X}}'\hat{\mathbf{X}})^{-1}\hat{\mathbf{X}}' - \hat{\mathbf{X}}(\hat{\mathbf{X}}'\hat{\mathbf{X}})^{-1}\mathbf{X}' + \mathbf{X}(\hat{\mathbf{X}}'\hat{\mathbf{X}})^{-1}\mathbf{X}',$$

in which $\hat{\mathbf{X}} = \mathbf{P}_z\mathbf{X}$.[1] Note that when $\mathbf{Z}$ includes $\mathbf{X}$, then $\hat{\mathbf{X}} = \mathbf{X}$, and (10.40) reduces to (5.61). Under the null hypothesis that the disturbances in (10.19) are serially uncorrelated, $\chi^2_{SC}(p) \overset{a}{\sim} \chi^2(p)$.

10.9 Further reading

We refer to Mátyás (1999) and Hall (2005) for a comprehensive textbook treatment of the *GMM*, and Bera and Bilias (2002) and Hall (2010) for surveys of recent developments. See also Chapter 27 containing an application of *GMM* techniques to estimate panel data regression models.

10.10 Exercises

1. Consider a random sample $x_1, x_2, \ldots, x_T$, drawn from a centered Student's t-distribution with ν_0 degrees of freedom, and assume $\nu_0 > 4$ (see expression (B. 39)). Write down two population moment conditions to estimate ν_0, exploiting the second and fourth moments of the distribution. Derive the corresponding sample moments and write down the objective function for *GMM* estimation.
2. Consider the linear regression model $y_t = \boldsymbol{\beta}_0'\mathbf{x}_t + u_t$, where $\mathbf{x}_t$ is a k-dimensional vector of regressors assumed to be orthogonal to the error term. Assume that u_t is conditionally heteroskedastic and serially correlated. Find k population moment conditions for estimation of $\boldsymbol{\beta}_0$ and show that the GMM estimator of $\boldsymbol{\beta}_0$, $\hat{\boldsymbol{\beta}}_T$, is equivalent to the *OLS* estimator. Derive the covariance matrix of $\hat{\boldsymbol{\beta}}_T$.
3. Consider the linear regression model $y_t = \boldsymbol{\beta}_0'\mathbf{x}_t + u_t$, where $\mathbf{x}_t$ is a k-dimensional vector of regressors assumed to be orthogonal to the error term. Assume that $u_t \sim IID(0, \sigma_t^2)$. Find a set of population moment conditions for estimation of $\boldsymbol{\beta}_0$ and $\sigma_t^2, t = 1, 2, \ldots, N$. Derive the corresponding sample moments, write down the objective function for *GMM* estimation, and derive the matrices $\mathbf{D}$ and $\mathbf{S}$.
4. Consider the MA(1) model

$$y_t = \mu_0 + \varepsilon_t + \psi_0\varepsilon_{t-1},$$

where $\varepsilon_t \sim IID(0, \sigma_0^2)$, $|\psi_0| < 1$. Derive a set of population moment conditions based on the above econometric model and the corresponding sample moments.

[1] See Breusch and Godfrey (1981, p. 101), for further details. The statistic in (10.40) is derived from the results in Sargan (1976).

11 Model Selection and Testing Non-Nested Hypotheses

11.1 Introduction

Model selection in econometric analysis involves both statistical and non-statistical considerations. It depends on the objective(s) of the analysis, the nature and the extent of economic theory used, and the statistical adequacy of the model under consideration compared with other econometric models. The various choice criteria discussed below are concerned with the issue of 'statistical fit' and provide different approaches to trading off the 'fit' and 'parsimony' of a given econometric model.

We also contrast model selection with testing of statistical hypotheses that are non-nested, or belong to separate families of distributions, meaning that none of the individual models may be obtained from the remaining models either by imposition of parameter restrictions or through a limiting process. In econometric analysis non-nested models arise naturally when rival economic theories are used to explain the same phenomena such as unemployment, inflation or output growth. Typical examples from economics literature are Keynesian and new classical explanations of unemployment, structural and monetary theories of inflation, alternative theories of investment, and endogenous and exogenous theories of growth.

Non-nested models could also arise when alternative functional specifications are considered such as multinomial probit and logit distribution functions used in the qualitative choice literature, exponential and power utility functions used in the asset pricing models, and a variety of non-nested specifications considered in the empirical analysis of income and wealth distributions. Finally, even starting from the same theoretical paradigm, it is possible for different investigators to arrive at different models if they adopt different conditioning or follow different paths to a more parsimonious model.

More recently, Bayesian and penalized regression techniques have also been used as alternative approaches to the problem of model selection and model combination, in particular when there are a large number of predictors under consideration. We end this chapter with a brief account of these approaches.

11.2 Formulation of econometric models

Before we introduce model selection procedures we first need to provide a formal definition of what an econometric model is.

Suppose the focus of the analysis is to consider the behaviour of the $n \times 1$ vector of random variables $\mathbf{w}_t = (w_{1t}, w_{2t}, \ldots, w_{nt})'$ observed over the period $t = 1, 2, \ldots, T$. A model of $\mathbf{w}_t$, indexed by $\mathfrak{M}_i$, $i = 1, 2, \ldots, m$, is defined by the joint probability distribution function (p.d.f.) of the observations $\mathbf{W} = (\mathbf{w}_1', \mathbf{w}_2', \ldots, \mathbf{w}_T')'$ conditional on the initial values $\mathbf{w}_0$

$$\mathfrak{M}_i : f_i(\mathbf{w}_1, \mathbf{w}_2, \ldots, \mathbf{w}_T | \mathbf{w}_0, \boldsymbol{\varphi}_i) = f_i(\mathbf{W} | \mathbf{w}_0, \boldsymbol{\varphi}_i), \tag{11.1}$$

where $f_i(.)$ is the probability density function of the model (hypothesis) $\mathfrak{M}_i$, and $\boldsymbol{\varphi}_i$ is a $p_i \times 1$ vector of unknown parameters associated with model $\mathfrak{M}_i$.[1]

The models characterized by $f_i(\mathbf{W}|\mathbf{w}_0, \boldsymbol{\varphi}_i)$ are *unconditional* in the sense that probability distribution of $\mathbf{w}_t$ is fully specified in terms of some initial values, $\mathbf{w}_0$, and for a given value of $\boldsymbol{\varphi}_i$.[2] In econometrics the interest often centres on conditional models, where a vector of 'endogenous' variables, $\mathbf{y}_t$, is explained (or modelled) *conditional* on a set of 'exogenous' variables, $\mathbf{x}_t$. Such conditional models can be derived from (11.1) by noting that

$$\begin{aligned} &f_i(\mathbf{w}_1, \mathbf{w}_2, \ldots, \mathbf{w}_T | \mathbf{w}_0, \boldsymbol{\varphi}_i) \\ &= f_i(\mathbf{y}_1, \mathbf{y}_2, \ldots, \mathbf{y}_T | \mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_T, \boldsymbol{\psi}(\boldsymbol{\varphi}_i)) \times f_i(\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_T | \mathbf{w}_0, \boldsymbol{\kappa}(\boldsymbol{\varphi}_i)), \end{aligned} \tag{11.2}$$

where $\mathbf{w}_t = (\mathbf{y}_t', \mathbf{x}_t')'$. The unconditional model $\mathfrak{M}_i$ is decomposed into a conditional model of $\mathbf{y}_t$ given $\mathbf{x}_t$ and a marginal model of $\mathbf{x}_t$. Denoting the former by $\mathfrak{M}_{i,y|x}$ we have

$$\mathfrak{M}_{i,y|x} : f_i(\mathbf{y}_1, \mathbf{y}_2, \ldots, \mathbf{y}_T | \mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_T, \mathbf{w}_0, \boldsymbol{\psi}(\boldsymbol{\varphi}_i)) = f_i(\mathbf{Y} | \mathbf{X}, \mathbf{w}_0, \boldsymbol{\psi}(\boldsymbol{\varphi}_i)), \tag{11.3}$$

where $\mathbf{Y} = (\mathbf{y}_1', \mathbf{y}_2', \ldots, \mathbf{y}_T')'$ and $\mathbf{X} = (\mathbf{x}_1', \mathbf{x}_2', \ldots, \mathbf{x}_T')'$.

Confining attention to the analysis and comparison of conditional models is valid only if the variations in the parameters of the marginal model, $\boldsymbol{\kappa}(\boldsymbol{\varphi}_i)$, do not induce changes in the parameters of the conditional model, $\boldsymbol{\psi}(\boldsymbol{\varphi}_i)$. Namely $\partial \boldsymbol{\psi}(\boldsymbol{\varphi}_i)/\partial \boldsymbol{\kappa}(\boldsymbol{\varphi}_i) = 0$. When this condition holds it is said that $\mathbf{x}_t$ is *weakly exogenous* for $\boldsymbol{\psi}_i = \boldsymbol{\psi}(\boldsymbol{\varphi}_i)$. The parameters of the conditional model, $\boldsymbol{\psi}_i$, are assumed to be the *parameters of interest* to be estimated.[3]

The conditional models $\mathfrak{M}_{i,y|\mathbf{x}}$ $i = 1, 2, \ldots, m$ all are based on the same conditioning variables, $\mathbf{x}_t$, and differ only insofar as they are based upon different p.d.f.. We may introduce an alternative set of models which share the same p.d.f. but differ with respect to the inclusion of exogenous variables. For any model, $\mathfrak{M}_i$ we may partition the set of exogenous variables $\mathbf{x}_t$ according to a simple included/excluded dichotomy. Therefore, $\mathbf{x}_t = (\mathbf{x}_{it}', \mathbf{x}_{it}'^{*})'$ writes the set of

[1] In cases where one or more elements of $\mathbf{z}_t$ are discrete, as in probit or Tobit specifications, cumulative probability distribution functions can be used instead of probability density functions.

[2] Strictly speaking, however, the models defined by (11.1) are conditional on the initial values. This is unlikely to present any difficulties when dealing with ergodic time series models. But in the case of panel data models with short T the formulation of the unconditional model also requires that the distribution of the initial values be specified as well.

[3] See Engle, Hendry, and Richard (1983).

exogenous variables according to a subset $\mathbf{x}_{it}$ which are included in model $\mathfrak{M}_i$, and a subset $\mathbf{x}^*_{it}$ which are excluded. We may then write

$$\begin{aligned} & f_i(\mathbf{Y}|\mathbf{x}_1, \mathbf{x}_2, \ldots \mathbf{x}_T, \mathbf{w}_0, \boldsymbol{\varphi}_i) \\ &= f_i(\mathbf{Y}|\mathbf{x}_{i1}, \mathbf{x}_{i2}, \ldots \mathbf{x}_{iT}, \mathbf{x}^*_{i1}, \mathbf{x}^*_{i2}, \ldots, \mathbf{x}^*_{iT}, \mathbf{w}_0, \boldsymbol{\varphi}_i) \\ &= f_i(\mathbf{Y}|\mathbf{X}_i, \mathbf{w}_0, \boldsymbol{\varphi}_i(\boldsymbol{\psi}_i)) \times f_i(\mathbf{X}^*_i|\mathbf{X}_i, \mathbf{w}_0, \mathbf{c}_i(\boldsymbol{\varphi}_i)), \end{aligned}$$

where $\mathbf{X}_i = (\mathbf{x}'_{i1}, \mathbf{x}'_{i2}, \ldots, \mathbf{x}'_{iT})'$ and $\mathbf{X}^*_i = (\mathbf{x}'^*_{i1}, \mathbf{x}'^*_{i2}, \ldots, \mathbf{x}'^*_{iT})'$. As noted above in the case of models differentiated solely by different p.d.f., a comparison of models based upon the partition of $\mathbf{x}_t$ into $\mathbf{x}_{it}$ and $\mathbf{x}^*_{it}$ should be preceded by determining whether $\partial \boldsymbol{\psi}_i(\boldsymbol{\varphi}_i)/\partial \mathbf{c}'_i(\boldsymbol{\varphi}_i) = 0$.

The above set up allows consideration of rival models that could differ in the conditioning set of variables, $\{\mathbf{x}_{it}, i = 1, 2, \ldots, m\}$ and/or the functional form of their underlying probability distribution functions, $\{f_i(\cdot), i = 1, 2, \ldots, m\}$.

11.3 Pseudo-true values

In practice, econometric models often represent crude approximations to a complex data generating process (DGP). Consider the following two alternative (conditional) models advanced for explanation/prediction of the vector of the dependent variables $\mathbf{y}_t$

$$H_f : \mathcal{F}_\theta = \{f(\mathbf{y}_t|\mathbf{x}_t, \boldsymbol{\Omega}_{t-1}; \boldsymbol{\theta}),\ \boldsymbol{\theta} \in \Theta\}, \tag{11.4}$$

$$H_g : \mathcal{F}_\gamma = \{g(\mathbf{y}_t|\mathbf{z}_t, \boldsymbol{\Omega}_{t-1}; \boldsymbol{\gamma}),\ \boldsymbol{\gamma} \in \Gamma\}, \tag{11.5}$$

where $\boldsymbol{\Omega}_{t-1}$ denotes the set of all past observations on $\mathbf{y}, \mathbf{x}$ and $\mathbf{z}$, $\boldsymbol{\theta}$ and $\boldsymbol{\gamma}$ are respectively k_f and k_g vectors of unknown parameters belonging to the non-empty compact sets Θ and Γ, and where $\mathbf{x}$ and $\mathbf{z}$ represent the conditioning variables. For the sake of notational simplicity we shall also often use $f_t(\boldsymbol{\theta})$ and $g_t(\boldsymbol{\gamma})$ in place of $f(\mathbf{y}_t|\mathbf{x}_t, \boldsymbol{\Omega}_{t-1}; \boldsymbol{\theta})$ and $g(\mathbf{y}_t|\mathbf{z}_t, \boldsymbol{\Omega}_{t-1}; \boldsymbol{\gamma})$, respectively.

Now given the observations $(\mathbf{y}_t, \mathbf{x}_t, \mathbf{z}_t, t = 1, 2, \ldots, T)$ and conditional on the initial values $\mathbf{w}_0$, the maximum likelihood (ML) estimators of $\boldsymbol{\theta}$ and $\boldsymbol{\gamma}$ are given by

$$\hat{\boldsymbol{\theta}}_T = \underset{\theta \in \Theta}{\operatorname{argmax}}\, L_f(\boldsymbol{\theta}), \quad \hat{\boldsymbol{\gamma}}_T = \underset{\gamma \in \Gamma}{\operatorname{argmax}}\, L_g(\boldsymbol{\gamma}), \tag{11.6}$$

where the respective log-likelihood functions are given by

$$L_f(\boldsymbol{\theta}) = \sum_{t=1}^{T} \ln f_t(\boldsymbol{\theta}), \quad L_g(\boldsymbol{\gamma}) = \sum_{t=1}^{T} \ln g_t(\boldsymbol{\gamma}). \tag{11.7}$$

Throughout we shall assume that the conditional densities $f_t(\boldsymbol{\theta})$ and $g_t(\boldsymbol{\gamma})$ satisfy the usual regularity conditions needed to ensure that $\hat{\boldsymbol{\theta}}_T$ and $\hat{\boldsymbol{\gamma}}_T$ have asymptotically normal limiting distributions under the DGP.[4] We allow the DGP to differ from H_f and H_g, and denote it by H_h, thus admitting the possibility that both H_f and H_g could be misspecified and that both are likely

[4] Refer to Chapter 9 for discussion of regularity conditions.

to be rejected in practice. In this setting, $\hat{\boldsymbol{\theta}}_T$ and $\hat{\boldsymbol{\gamma}}_T$ are referred to as quasi-*ML* estimators and their probability limits under H_h, which we denote by $\boldsymbol{\theta}_{h*}$ and $\boldsymbol{\gamma}_{h*}$ respectively, are known as pseudo-true values. These pseudo-true values are defined by

$$\boldsymbol{\theta}_{h*} = \underset{\theta\in\Theta}{\operatorname{argmax}}\, E_h\left[T^{-1}L_f(\boldsymbol{\theta})\right],\ \boldsymbol{\gamma}_{h*} = \underset{\gamma\in\Gamma}{\operatorname{argmax}}\ E_h\left[T^{-1}L_g(\boldsymbol{\gamma})\right], \tag{11.8}$$

where $E_h(\cdot)$ denotes expectations taken under H_h. In the case where $\mathbf{w}_t$ follows a strictly stationary process, (11.8) simplifies to

$$\boldsymbol{\theta}_{h*} = \underset{\theta\in\Theta}{\operatorname{argmax}}\ E_h\left[\ln f_t(\boldsymbol{\theta})\right],\ \boldsymbol{\gamma}_{h*} = \underset{\gamma\in\Gamma}{\operatorname{argmax}}\ E_h\left[\ln g_t(\boldsymbol{\gamma})\right]. \tag{11.9}$$

To ensure global identifiability of the pseudo-true values, it will be assumed that $\boldsymbol{\theta}_{f*}$ and $\boldsymbol{\gamma}_{f*}$ provide *unique* maxima of $E_h\left[T^{-1}L_f(\boldsymbol{\theta})\right]$ and $E_h\left[T^{-1}L_g(\boldsymbol{\gamma})\right]$, respectively. Clearly, under H_f, namely assuming H_f is the DGP, we have $\boldsymbol{\theta}_{f*}=\boldsymbol{\theta}_0$, and $\boldsymbol{\gamma}_{f*} = \boldsymbol{\gamma}_*(\boldsymbol{\theta}_0)$ where $\boldsymbol{\theta}_0$ is the 'true' value of $\boldsymbol{\theta}$ under H_f. Similarly, under H_g we have $\boldsymbol{\gamma}_{g*}=\boldsymbol{\gamma}_0$, and $\boldsymbol{\theta}_{g*}=\boldsymbol{\theta}_*(\boldsymbol{\gamma}_0)$ with $\boldsymbol{\gamma}_0$ denoting the 'true' value of $\boldsymbol{\gamma}$ under H_g. The functions $\boldsymbol{\gamma}_*(\boldsymbol{\theta}_0)$, and $\boldsymbol{\theta}_*(\boldsymbol{\gamma}_0)$ that relate the parameters of the two models under consideration are called the *binding* functions. These functions do not involve the true model, H_h, and only depend on the models H_f and H_g that are under consideration. We now consider some examples of non-nested models.

11.3.1 Rival linear regression models

Consider the following regression models

$$H_f:\ y_t = \boldsymbol{\alpha}'\mathbf{x}_t + u_{tf},\quad u_{tf} \backsim N(0,\sigma^2),\ \infty > \sigma^2 > 0, \tag{11.10}$$

$$H_g:\ y_t = \boldsymbol{\beta}'\mathbf{z}_t + u_{tg},\quad u_{tg} \backsim N(0,\omega^2),\ \infty > \omega^2 > 0. \tag{11.11}$$

The conditional probability density associated with these regression models are given by

$$H_f:\ f(y_t|\mathbf{x}_t;\boldsymbol{\theta}) = (2\pi\sigma^2)^{-1/2}\exp\left[\frac{-1}{2\sigma^2}(y_t-\boldsymbol{\alpha}'\mathbf{x}_t)^2\right], \tag{11.12}$$

$$H_g:\ g(y_t|\mathbf{z}_t;\boldsymbol{\theta}) = (2\pi\omega^2)^{-1/2}\exp\left[\frac{-1}{2\omega^2}(y_t-\boldsymbol{\beta}'\mathbf{z}_t)^2\right], \tag{11.13}$$

where $\boldsymbol{\theta} = (\boldsymbol{\alpha}', \sigma^2)'$, and $\boldsymbol{\gamma} = (\boldsymbol{\beta}', \omega^2)'$. These regression models are non-nested if it is not possible to write $\mathbf{x}_t$ as an exact linear function of $\mathbf{z}_t$ and *vice versa*, or more formally if $\mathbf{x}_t \nsubseteqq \mathbf{z}_t$ *and* $\mathbf{z}_t \nsubseteqq \mathbf{x}_t$. Model H_f is said to be nested in H_g if $\mathbf{x}_t \subset \mathbf{z}_t$ *and* $\mathbf{z}_t \nsubseteqq \mathbf{x}_t$. The two models are observationally equivalent if $\mathbf{x}_t \subset \mathbf{z}_t$ *and* $\mathbf{z}_t \subset \mathbf{x}_t$. Suppose now that neither of these regression models is true and the DGP is given by

$$H_h:\ y_t = \boldsymbol{\delta}'\mathbf{w}_t + u_{th},\quad u_{th} \backsim N(0,\upsilon^2),\ \infty > \upsilon^2 > 0. \tag{11.14}$$

It is then easily seen that, conditional on $\{\mathbf{x}_t, \mathbf{z}_t, \mathbf{w}_t,\ t = 1, 2, \ldots, T\}$,

$$E_h\left\{T^{-1}L_f(\boldsymbol{\theta})\right\} = -\frac{1}{2}\ln(2\pi\sigma^2) - \frac{\upsilon^2}{2\sigma^2} - \frac{\boldsymbol{\delta}'\hat{\Sigma}_{ww}\boldsymbol{\delta} - 2\boldsymbol{\delta}'\hat{\Sigma}_{wx}\alpha + \boldsymbol{\alpha}'\hat{\Sigma}_{xx}\alpha}{2\sigma^2},$$

where

$$\hat{\Sigma}_{ww} = T^{-1}\sum_{t=1}^{T}\mathbf{w}_t\mathbf{w}_t',\ \hat{\Sigma}_{xx} = T^{-1}\sum_{t=1}^{T}\mathbf{x}_t\mathbf{x}_t',\ \hat{\Sigma}_{wx} = T^{-1}\sum_{t=1}^{T}\mathbf{w}_t\mathbf{x}_t'.$$

Maximizing $E_h\{T^{-1}L_f(\boldsymbol{\theta})\}$ with respect to $\boldsymbol{\theta}$ now yields the conditional pseudo-true values:

$$\boldsymbol{\theta}_{h*} = \begin{pmatrix} \alpha_{h*} \\ \sigma^2_{h*} \end{pmatrix} = \begin{pmatrix} \hat{\Sigma}_{xx}^{-1}\hat{\Sigma}_{xw}\boldsymbol{\delta} \\ \upsilon^2 + \boldsymbol{\delta}'(\hat{\Sigma}_{ww} - \hat{\Sigma}_{wx}\hat{\Sigma}_{xx}^{-1}\hat{\Sigma}_{xw})\boldsymbol{\delta} \end{pmatrix}. \tag{11.15}$$

Similarly,

$$\boldsymbol{\gamma}_{h*} = \begin{pmatrix} \beta_{h*} \\ \omega^2_{h*} \end{pmatrix} = \begin{pmatrix} \hat{\Sigma}_{zz}^{-1}\hat{\Sigma}_{zw}\boldsymbol{\delta} \\ \upsilon^2 + \boldsymbol{\delta}'(\hat{\Sigma}_{ww} - \hat{\Sigma}_{wz}\hat{\Sigma}_{zz}^{-1}\hat{\Sigma}_{zw})\boldsymbol{\delta} \end{pmatrix}, \tag{11.16}$$

where

$$\hat{\Sigma}_{zz} = T^{-1}\sum_{t=1}^{T}\mathbf{z}_t\mathbf{z}_t',\ \hat{\Sigma}_{wz} = T^{-1}\sum_{t=1}^{T}\mathbf{w}_t\mathbf{z}_t'.$$

When the regressors are stationary, the unconditional counterparts of the above pseudo-true values can be obtained by replacing $\hat{\Sigma}_{ww}$, $\hat{\Sigma}_{xx}$, $\hat{\Sigma}_{wx}$ etc. with their population values, namely $\Sigma_{ww} = E(\mathbf{w}_t\mathbf{w}_t')$, $\Sigma_{xx} = E(\mathbf{x}_t\mathbf{x}_t')$, $\Sigma_{wx} = E(\mathbf{w}_t\mathbf{x}_t')$ etc. It is clear that the pseudo-true values of the regression coefficients, α_{h*} and β_{h*}, in general differ from the true values given by $\boldsymbol{\delta}$.

11.3.2 Probit versus logit models

Other examples of non-nested models include discrete choice and duration models used in microeconometric research. Although the analyst may utilise both prior knowledge and theory to select an appropriate set of regressors, there is generally little guidance in terms of the most appropriate probability distribution. Non-nested hypothesis testing is particularly relevant to microeconometric research where the same set of regressors is often used to explain individual decisions but based on different functional distributions, such as multinomial probit and logit specifications in the analysis of discrete choice, or exponential and log-normal distributions in the analysis of duration data. In the simple case of a probit (H_f) *versus* a logit model (H_g) we have

$$H_f:\ \Pr(y_t = 1) = \Phi(\boldsymbol{\theta}'\mathbf{x}_t) = \int_{-\infty}^{\boldsymbol{\theta}'\mathbf{x}_t}\frac{1}{\sqrt{2\pi}}\exp\left\{-\tfrac{1}{2}v^2\right\}dv \tag{11.17}$$

$$H_g:\ \Pr(y_t = 1) = \Lambda(\boldsymbol{\gamma}'\mathbf{z}_t) = \frac{e^{\boldsymbol{\gamma}'\mathbf{z}_t}}{1 + e^{\boldsymbol{\gamma}'\mathbf{z}_t}} \tag{11.18}$$

where y_t, $t = 1, 2, \ldots, T$, are independently distributed binary random variables taking the value of 1 or 0. In practice the two sets of regressors $\mathbf{x}_t$ used in the probit and logit specifications are likely to be identical, and it is only the form of the distribution functions that separates the two models. Other functional forms can also be entertained. Suppose, for example, that the true DGP for this simple discrete choice problem is given by the probability distribution function $H(\boldsymbol{\delta}'\mathbf{x}_t)$, then pseudo-true values for $\boldsymbol{\theta}$ and $\boldsymbol{\gamma}$ can be obtained as functions of $\boldsymbol{\delta}$, but only in an implicit form. We first note that the log-likelihood function under H_f, for example, is given by

$$L_f(\boldsymbol{\theta}) = \sum_{t=1}^{T} y_t \log\left[\Phi(\boldsymbol{\theta}'\mathbf{x}_t)\right] + \sum_{t=1}^{T} (1 - y_t) \log\left[1 - \Phi(\boldsymbol{\theta}'\mathbf{x}_t)\right],$$

and hence under the assumed DGP we have

$$\begin{aligned} E_h\left\{T^{-1}L_f(\boldsymbol{\theta})\right\} = {} & T^{-1}\sum_{t=1}^{T} H(\boldsymbol{\delta}'\mathbf{x}_t) \log\left[\Phi(\boldsymbol{\theta}'\mathbf{x}_t)\right] \\ & + T^{-1}\sum_{t=1}^{T} \left[1 - H(\boldsymbol{\delta}'\mathbf{x}_t)\right] \log\left[1 - \Phi(\boldsymbol{\theta}'\mathbf{x}_t)\right]. \end{aligned}$$

Therefore, the pseudo-true value of $\boldsymbol{\theta}$, namely $\boldsymbol{\theta}_*(\boldsymbol{\delta})$ or simply $\boldsymbol{\theta}_*$, satisfies the following equation

$$T^{-1}\sum_{t=1}^{T} \mathbf{x}_t \phi(\boldsymbol{\theta}_*'\mathbf{x}_t) \left[\frac{H(\boldsymbol{\delta}'\mathbf{x}_t)}{\Phi(\boldsymbol{\theta}_*'\mathbf{x}_t)} - \frac{1 - H(\boldsymbol{\delta}'\mathbf{x}_t)}{1 - \Phi(\boldsymbol{\theta}_*'\mathbf{x}_t)}\right] = 0,$$

where $\phi(\boldsymbol{\theta}_*'\mathbf{x}_t) = (2\pi)^{-1/2} \exp\left[\frac{-1}{2}(\boldsymbol{\theta}_*'\mathbf{x}_t)^2\right]$. It is easily established that the solution of $\boldsymbol{\theta}_*$ in terms of $\boldsymbol{\delta}$ is in fact unique, and $\boldsymbol{\theta}_* = \boldsymbol{\delta}$ if and only if $\Phi(\cdot) = H(\cdot)$. Similar results also obtain for the logistic specification.

11.4 Model selection versus hypothesis testing

Hypothesis testing and model selection are different strands in the model evaluation literature. However, these strands differ in a number of important respects which are worth emphasizing here. Model selection begins with a given set of models, $\mathcal{M}$, characterized by the (possibly) conditional probability density functions

$$\mathcal{M} = \left\{f_i(\mathbf{Y}|\mathbf{X}_i, \boldsymbol{\psi}_i), i = 1, 2, \ldots, m\right\},$$

with the aim of *choosing* one of the models under consideration for a particular purpose with a specific loss (utility) function in mind. In essence, model selection is a part of decision-making and as argued in Granger and Pesaran (2000a), ideally it should be fully integrated into the decision-making process. However, most of the current literature on model selection builds on statistical measures of fit such as sums of squares of residuals or more generally maximized log-likelihood values, rather than economic benefit which one would expect to follow from a model

choice. Consequently, model selection seems much closer to hypothesis testing than it actually is in principle.

The model selection process treats all models under consideration symmetrically, while hypothesis testing attributes a different status to the null and to the alternative hypotheses and by design treats the models asymmetrically. Model selection always ends in a definite outcome, namely one of the models under consideration is selected for use in decision-making. Hypothesis testing on the other hand asks whether there is any statistically significant evidence (in the Neyman–Pearson sense) of departure from the null hypothesis in the direction of one or more alternative hypotheses. Rejection of the null hypothesis does not necessarily imply acceptance of any one of the alternative hypotheses; it only warns the investigator of possible shortcomings of the null that is being advocated. Hypothesis testing does not seek a definite outcome and if carried out with due care need not lead to a favourite model. For example, in the case of non-nested hypothesis testing it is possible for all models under consideration to be rejected, or all models to be deemed as observationally equivalent.

Due to its asymmetric treatment of the available models, the choice of the null hypothesis plays a critical role in the hypothesis testing approach. When the models are nested the most parsimonious model can be used as the null hypothesis. But in the case of non-nested models (particularly when the models are globally non-nested) there is no natural null, and it is important that the null hypothesis is selected on *a priori* grounds.[5] Alternatively, the analysis could be carried out with different models in the set treated as the null. Therefore, the results of non-nested hypothesis testing is less clear cut as compared with the case where the models are nested. It is also important to emphasise the distinction between *paired* and joint non-nested hypothesis tests. Letting f_1 denote the null model and $f_i \in \mathcal{M}$, $i = 2, 3, \ldots, m$ index a set of $m - 1$ alternative models, a paired test is a test of f_1 against a *single* member of $\mathcal{M}$, whereas a joint test is a test of f_1 against multiple alternatives in $\mathcal{M}$.

The distinction between model selection and non-nested hypothesis tests can also be motivated from the perspective of Bayesian versus sampling-theory approaches to the problem of inference. For example, it is likely that with a large amount of data the posterior probabilities associated with a particular hypothesis will be close to one. However, the distinction drawn by Zellner (1971) between 'comparing' and 'testing' hypotheses is relevant given that within a Bayesian perspective the progression from a set of prior to posterior probabilities on $\mathcal{M}$, mediated by the Bayes factor, does not necessarily involve a decision to accept or reject the hypothesis. If a decision is required it is generally based upon minimizing a particular expected loss function. Thus, model selection motivated by a decision problem is much more readily reconcilable with the Bayesian rather than the classical approach to model selection.

Finally, the choice between hypothesis testing and model selection clearly depends on the primary objective of the exercise. There are no definite rules. Model selection is more appropriate when the objective is decision-making. Hypothesis testing is better suited to inferential problems where the empirical validity of a theoretical prediction is the primary objective. A model may be empirically adequate for a particular purpose but of little relevance for another use. Only in the unlikely event that the true model is known or knowable will the selected model be universally applicable. In the real world where the truth is elusive and unknowable both approaches to model evaluation are worth pursuing.

[5] The concepts of globally and partially non-nested models are defined in Pesaran (1987b).

11.5 Criteria for model selection

11.5.1 Akaike information criterion (*AIC*)

Let $\ell_T(\widetilde{\boldsymbol{\theta}})$ be the maximized value of the log-likelihood function of an econometric model, where $\widetilde{\boldsymbol{\theta}}$ is the maximum likelihood estimator of $\boldsymbol{\theta}$, based on a sample of size T. The Akaike information criterion (AIC) for this model is defined as

$$AIC_\ell = \ell_T(\widetilde{\boldsymbol{\theta}}) - p, \tag{11.19}$$

where

$$p \equiv \text{ Dimension } (\boldsymbol{\theta}) \equiv \text{ The number of freely estimated parameters.}$$

In the case of single-equation linear (or nonlinear) regression models, the AIC_ℓ can also be written equivalently as

$$AIC_\sigma = \log(\tilde{\sigma}^2) + \frac{2p}{T}, \tag{11.20}$$

where $\tilde{\sigma}^2$ is the *ML* estimator of the variance of regression disturbances, u_t, given by $\tilde{\sigma}^2 = \mathbf{e}'\mathbf{e}/T$ in the case of linear regression models. The two versions of the *AIC* in (11.19) and (11.20) yield identical results. When using (11.19), the model with the highest value of AIC_ℓ is chosen. But when using the criterion based on the estimated standard errors (11.20), the model with the lowest value for AIC_σ is chosen.[6]

11.5.2 Schwarz Bayesian criterion (*SBC*)

The *SBC* provides a large sample approximation to the posterior odds ratio of models under consideration. It is defined by

$$SBC_\ell = \ell_T(\widetilde{\boldsymbol{\theta}}) - \tfrac{1}{2}p\log T. \tag{11.21}$$

In application of the *SBC* across models, the model with the highest *SBC* value is chosen. For regression models an alternative version of (11.21), based on the estimated standard error of the regression, $\tilde{\sigma}$, is given by

[6] For linear regression models, the equivalence of (11.19) and (11.20) follows by substituting for $\ell_n(\tilde{\theta}) = -\frac{n}{2}(1 + \log 2\pi) - \frac{n}{2}\log\tilde{\sigma}^2$ in (11.19):

$$AIC_\ell = -\frac{n}{2}(1 + \log 2\pi) - \frac{n}{2}\log\tilde{\sigma}^2 - p,$$

hence using (11.20)

$$AIC_\ell = -\frac{n}{2}(1 + \log 2\pi) - \frac{n}{2}AIC_\sigma.$$

Therefore, in the case of regression models estimated on the same sample period, the same preference ordering across models will result irrespective of whether AIC_ℓ or AIC_σ criteria are used.

$$SBC_\sigma = \log(\tilde{\sigma}^2) + \left(\frac{\log T}{T}\right) p.$$

According to this criterion, a model is chosen if it has the lowest SBC_σ value. See C.5 in Appendix C for a Bayesian treatment and derivations in the case of linear regression models.

11.5.3 Hannan–Quinn criterion (HQC)

The criterion has been primarily proposed for selection of the order of autoregressive-moving average or vector autoregressive models, and is defined by

$$HQC_\ell = \ell_T(\widetilde{\boldsymbol{\theta}}) - (\log\log T)p,$$

or equivalently (in the case of regression models)

$$HQC_\sigma = \log\tilde{\sigma} + \left(\frac{2\log\log T}{T}\right) p.$$

11.5.4 Consistency properties of the different model selection criteria

Among the above three model selection criteria, the SBC selects the most parsimonious model (a model with the least number of freely estimated parameters) if $T \geq 8$, and AIC selects the least parsimonious model. The HQC lies somewhere between the other two criteria. Useful discussion of these and other model selection criteria can be found in Amemiya (1980), Judge et al. (1985), and Lütkepohl (2005). The last reference is particularly useful for selecting the order of the vector autoregressive models and contains some discussion of the consistency property of the above three model selection criteria. Under certain regularity conditions it can be shown that SBC and HQC are consistent, in the sense that for large enough samples they lead to the correct model choice, assuming of course that the 'true' model does in fact belong to the set of models over which one is searching. The same is not true of the AIC or Theil's $\bar{R}^2$ criteria. This does not, however, mean that the SBC (or HQC) is necessarily preferable to the AIC or the $\bar{R}^2$ criterion, bearing in mind that it is rarely the case that one is sure that the 'true' model is one of the models under consideration.

11.6 Non-nested tests for linear regression models

To introduce non-nested hypothesis testing, consider the following two linear regression models

$$M_1 \quad : \quad \mathbf{y} = \mathbf{X}\boldsymbol{\beta}_1 + \mathbf{u}_1, \qquad \mathbf{u}_1 \sim N(0, \sigma^2\mathbf{I}_T), \tag{11.22}$$

$$M_2 \quad : \quad \mathbf{y} = \mathbf{Z}\boldsymbol{\beta}_2 + \mathbf{u}_2, \qquad \mathbf{u}_2 \sim N(0, \omega^2\mathbf{I}_T), \tag{11.23}$$

where $\mathbf{y}$ is the $T \times 1$ vector of observations on the dependent variable, $\mathbf{X}$ and $\mathbf{Z}$ are $T \times k_1$ and $T \times k_2$ observation matrices for the regressors of models M_1 and M_2, $\boldsymbol{\beta}_1$ and $\boldsymbol{\beta}_2$ are the $k_1 \times 1$ and $k_2 \times 1$ unknown regression coefficient vectors, and $\mathbf{u}_1$ and $\mathbf{u}_2$ are the $T \times 1$ disturbance vectors.

In the context of these regressions, models M_1 and M_2 are non-nested if the regressors of M_1 (respectively M_2) cannot be expressed as an exact linear combination of the regressors of M_2 (respectively M_1). For a formal definition of the concepts of nested and non-nested models, see Pesaran (1987b). An early review of the literature on non-nested hypothesis testing is given in McAleer and Pesaran (1986). A more recent review can be found in Pesaran and Weeks (2001).

11.6.1 *The N-test*

This is the Cox (1961, 1962) test originally derived in Pesaran (1974, pp. 157–8). The Cox statistic for the test of M_1 against M_2 is computed as

$$N_1 = \left[\frac{T}{2}\log\left(\hat{\omega}^2/\hat{\omega}_*^2\right)\right]\Big/\hat{V}_1\,, \tag{11.24}$$

where

$$\begin{aligned}
\hat{\omega}^2 &= \mathbf{e}_2'\mathbf{e}_2/T,\\
\hat{\omega}_*^2 &= (\mathbf{e}_1'\mathbf{e}_1 + \hat{\boldsymbol{\beta}}_1'\mathbf{X}'\mathbf{M}_2\mathbf{X}\hat{\boldsymbol{\beta}}_1)/T,\\
\hat{V}_1^2 &= (\hat{\sigma}^2/\hat{\omega}_*^4)\hat{\boldsymbol{\beta}}_1'\mathbf{X}'\mathbf{M}_2\mathbf{M}_1\mathbf{M}_2\mathbf{X}\hat{\boldsymbol{\beta}}_1,\\
\hat{\sigma}^2 &= \mathbf{e}_1'\mathbf{e}_1/T\hat{\boldsymbol{\beta}}_1 = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'y,\\
\mathbf{M}_1 &= \mathbf{I}_T - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}', \qquad \mathbf{M}_2 = \mathbf{I}_T - \mathbf{Z}(\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}'.
\end{aligned}$$

Similarly, the Cox statistic N_2 is also computed for the test of M_2 against M_1.

Pesaran and Deaton (1978) extend the Cox test to non-nested nonlinear system equation models.

11.6.2 *The NT-test*

This is the *adjusted* Cox test derived in Godfrey and Pesaran (1983, p. 138), which is referred to as the $\tilde{N}$-test (or the NT-test). The NT-statistic for the test of M_1 against M_2 is given by (see equations (20) and (21) in Godfrey and Pesaran (1983)).

$$\tilde{N}_1 = \tilde{T}_1 \Big/ \sqrt{\left\{\tilde{V}_1(\tilde{T}_1)\right\}}\,, \tag{11.25}$$

where

$$\begin{aligned}
\tilde{T}_1 &= \tfrac{1}{2}(T-k_2)\log(\tilde{\omega}^2/\tilde{\omega}_*^2),\\
\tilde{\omega}^2 &= \mathbf{e}_2'\mathbf{e}_2/(T-k_2), \quad \tilde{\sigma}^2 = \mathbf{e}_1'\mathbf{e}_1/(T-k_1),\\
\tilde{\omega}_*^2 &= \left[\tilde{\sigma}^2 Tr(\mathbf{M}_1\mathbf{M}_2) + \hat{\boldsymbol{\beta}}_1'\mathbf{X}'\mathbf{M}_2\mathbf{X}\hat{\boldsymbol{\beta}}_1\right]/(T-k_2),\\
\tilde{V}_1(\tilde{T}_1) &= (\tilde{\sigma}^2/\tilde{\omega}_*^4)\left[\hat{\boldsymbol{\beta}}_1'\mathbf{X}'\mathbf{M}_2\mathbf{M}_1\mathbf{M}_2\mathbf{X}\hat{\boldsymbol{\beta}}_1 + \tfrac{1}{2}\tilde{\sigma}^2 Tr(\mathbf{B}^2)\right],
\end{aligned}$$

$$Tr(\mathbf{B}^2) = k_2 - Tr(\mathbf{A}_1\mathbf{A}_2)^2 - \frac{[k_2 - Tr(\mathbf{A}_1\mathbf{A}_2)]^2}{T - k_1},$$
$$\mathbf{A}_1 = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}', \qquad \mathbf{A}_2 = \mathbf{Z}(\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}'. \tag{11.26}$$

Similarly, the $\tilde{N}$-test statistic, $\tilde{N}_2$, is also computed for the test of M_2 against M_1.

11.6.3 *The W-test*

This is the Wald-type test of M_1 against M_2 proposed in Godfrey and Pesaran (1983), and is based on the statistic

$$W_1 = \frac{(T - k_2)(\tilde{\omega}^2 - \tilde{\omega}_*^2)}{\left[2\tilde{\sigma}^4 Tr(\mathbf{B}^2) + 4\tilde{\sigma}^2 \hat{\boldsymbol{\beta}}_1' \mathbf{X}'\mathbf{M}_2\mathbf{M}_1\mathbf{M}_2\mathbf{X}\hat{\boldsymbol{\beta}}_1\right]^{1/2}}. \tag{11.27}$$

All the notations are as above. Notice that it is similarly possible to compute a statistic, W_2, for the test of M_2 against M_1.

11.6.4 *The J-test*

This test is due to Davidson and MacKinnon (1981), and for the test of M_1 against M_2 is based on the t-ratio of λ in the 'artificial' *OLS* regression

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta}_1 + \lambda(\mathbf{Z}\hat{\boldsymbol{\beta}}_2) + \mathbf{u}.$$

The relevant statistic for the J-test of M_2 against M_1 is the t-ratio of μ in the *OLS* regression

$$\mathbf{y} = \mathbf{Z}\boldsymbol{\beta}_2 + \mu(\mathbf{X}\hat{\boldsymbol{\beta}}_1) + \mathbf{v},$$

where $\hat{\boldsymbol{\beta}}_1 = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'y$, and $\hat{\boldsymbol{\beta}}_2 = (\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}'\mathbf{y}$. The J-test is asymptotically equivalent to the above non-nested tests but, as demonstrated by extensive Monte Carlo experiments in Godfrey and Pesaran (1983), the $\tilde{N}$-test, and the W-test, defined above, are preferable to the J-test in small samples.

11.6.5 *The* JA*-test*

This test is due to Fisher and McAleer (1981), and for the test of M_1 against M_2 is based on the t-ratio of λ in the *OLS* regression

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta}_1 + \lambda(\mathbf{A}_2\mathbf{X}\hat{\boldsymbol{\beta}}_1) + \mathbf{u}.$$

The relevant statistic for the *JA*-test of M_2 against M_1 is the t-ratio of μ in the *OLS* regression

$$\mathbf{y} = \mathbf{Z}\boldsymbol{\beta}_2 + \mu(\mathbf{A}_1\mathbf{Z}\hat{\boldsymbol{\beta}}_2) + \mathbf{v}.$$

The matrices $\mathbf{A}_1$ and $\mathbf{A}_2$ are already defined by (11.26).

11.6.6 *The Encompassing test*

This test has been proposed in the literature by Deaton (1982), Dastoor (1983), Gourieroux, Holly, and Monfort (1982), and Mizon and Richard (1986). In the case of testing M_1 against M_2, the encompassing test is the same as the classical F-test and is computed as the F-statistic for testing $\boldsymbol{\delta} = 0$ in the combined *OLS* regression

$$\mathbf{y} = \mathbf{X}\mathbf{a}_0 + \mathbf{Z}^*\boldsymbol{\delta} + \mathbf{u},$$

where $\mathbf{Z}^*$ denotes the variables in M_2 that cannot be expressed as exact linear combinations of the regressors of M_1. Similarly, it is possible to compute the F-statistic for the test of M_2 against M_1. The encompassing test is asymptotically equivalent to the above non-nested tests under the null hypothesis, but in general it is less powerful for a large class of alternative non-nested models (see Pesaran (1982)).

A Monte Carlo study of the relative performance of the above non-nested tests in small samples can be found in Godfrey and Pesaran (1983).

11.7 Models with different transformations of the dependent variable

Consider the following non-nested models

$$M_f \quad : \quad \mathbf{f}(\mathbf{y}) = \mathbf{X}\boldsymbol{\beta}_1 + \mathbf{u}_1, \qquad \mathbf{u}_1 \sim N(0, \sigma^2\mathbf{I}_T), \tag{11.28}$$

$$M_g \quad : \quad \mathbf{g}(\mathbf{y}) = \mathbf{Z}\boldsymbol{\beta}_2 + \mathbf{u}_2, \qquad \mathbf{u}_2 \sim N(0, \omega^2\mathbf{I}_T), \tag{11.29}$$

where $\mathbf{f}(\mathbf{y})$ and $\mathbf{g}(\mathbf{y})$ are known transformations of the $T \times 1$ vector of observations on the underlying dependent variable of interest, $\mathbf{y}$. Examples of the functions $\mathbf{f}(\mathbf{y})$ and $\mathbf{g}(\mathbf{y})$, are

Linear form	$\mathbf{f}(\mathbf{y}) = \mathbf{y}$,
Logarithmic form	$\mathbf{f}(\mathbf{y}) = \log(\mathbf{y})$,
Ratio form	$\mathbf{f}(\mathbf{y}) = \mathbf{y}/\mathbf{z}$,
Difference form	$\mathbf{f}(\mathbf{y}) = \mathbf{y} - \mathbf{y}(-1)$,
Log-difference form	$\mathbf{f}(\mathbf{y}) = \log\mathbf{y} - \log\mathbf{y}(-1)$,

where $\mathbf{z}$ is a variable of choice. Notice that $\log(\mathbf{y})$ refers to a vector of observations with elements equal to $\log(y_t)$, $t = 1, 2, \ldots, n$. Also $\mathbf{y} - \mathbf{y}(-1)$ refers to a vector with a typical element equal to $y_t - y_{t-1}$, $t = 1, 2, \ldots, T$.

11.7.1 The P_E test statistic

This statistic is proposed by MacKinnon, White, and Davidson (1983) and in the case of testing M_f against M_g is given by the t-ratio of α_f in the auxiliary regression

$$\mathbf{f}(\mathbf{y}) = \mathbf{X}\mathbf{b} + \alpha_f\left\{\mathbf{Z}\hat{\boldsymbol{\beta}}_2 - \mathbf{g}\left[\mathbf{f}^{-1}(\mathbf{X}\hat{\boldsymbol{\beta}}_1)\right]\right\} + \text{Error}. \tag{11.30}$$

Similarly, the P_E statistic for testing M_g against M_f is given by the t-ratio of α_g in the auxiliary regression

$$\mathbf{g}(\mathbf{y}) = \mathbf{Zd} + \alpha_g \left\{ \mathbf{X}\hat{\boldsymbol{\beta}}_1 - \mathbf{f}\left[\mathbf{g}^{-1}(\mathbf{Z}\hat{\boldsymbol{\beta}}_2) \right] \right\} + \text{Error.} \tag{11.31}$$

Functions $f^{-1}(\cdot)$ and $g^{-1}(\cdot)$ represent the inverse functions for $f(\cdot)$ and $g(\cdot)$, respectively, such that $f\left[f^{-1}(y)\right] = y$, and $g\left[g^{-1}(y)\right] = y$. For example, in the case where M_f is linear (i.e., $f(y) = y$) and M_g is log-linear (i.e., $g(y) = \log y$), we have

$$\begin{aligned} f^{-1}(y_t) &= y_t, \\ g^{-1}(y_t) &= \exp(y_t). \end{aligned}$$

In the case where M_f is in first differences (i.e., $f(y_t) = y_t - y_{t-1}$) and M_g is in log-differences (i.e., $g(y_t) = \log(y_t/y_{t-1})$) we have

$$\begin{aligned} f^{-1}(y_t) &= f(y_t) + y_{t-1}, \\ g^{-1}(y_t) &= y_{t-1} \exp\left[g(y_t)\right], \end{aligned}$$

$\hat{\boldsymbol{\beta}}_1$ and $\hat{\boldsymbol{\beta}}_2$ are the *OLS* estimators of $\boldsymbol{\beta}_1$ and $\boldsymbol{\beta}_2$ under M_f and M_g, respectively.

11.7.2 The Bera-McAleer test statistic

The statistic proposed by Bera and McAleer (1989) is for testing linear versus log-linear models, but it can be readily extended to general known one-to-one transformations of the dependent variable of interest, namely y_t. The Bera–McAleer (BM) statistic for test of M_f against M_g can be computed in two steps. First, the residuals $\hat{\boldsymbol{\eta}}_g$ are computed from the regression of $\mathbf{g}[\mathbf{f}^{-1}(\mathbf{X}\hat{\boldsymbol{\beta}}_1)]$ on $\mathbf{Z}$. Hence, the BM statistic for the test of M_f against M_g is computed as the t-ratio of θ_f in the auxiliary regression

$$\mathbf{f}(\mathbf{y}) = \mathbf{Xb} + \theta_f \hat{\boldsymbol{\eta}}_g + \text{Error.} \tag{11.32}$$

The BM statistic for the test of M_g against M_f is given by the t-ratio of θ_g in the auxiliary regression

$$\mathbf{g}(\mathbf{y}) = \mathbf{Zd} + \theta_g \hat{\boldsymbol{\eta}}_f + \text{Error,} \tag{11.33}$$

where $\hat{\boldsymbol{\eta}}_f$ is the residual vector of the regression of $\mathbf{f}\{\mathbf{g}^{-1}(\mathbf{Z}\hat{\boldsymbol{\beta}}_2)\}$ on $\mathbf{X}$.

11.7.3 The double-length regression test statistic

The double-length (DL) regression statistic is proposed by Davidson and MacKinnon (1984) and for the test of M_f against M_g is given by

$$DL_f = 2T - SSR_f, \tag{11.34}$$

where SSR_f denotes the sums of squares of residuals from the DL regression

$$\begin{bmatrix} \mathbf{e}_1/\hat{\sigma} \\ \boldsymbol{\tau} \end{bmatrix} = \begin{bmatrix} -\mathbf{X} \\ \mathbf{0} \end{bmatrix} \mathbf{b} + \begin{bmatrix} \mathbf{e}_1/\hat{\sigma} \\ -\boldsymbol{\tau} \end{bmatrix} \mathbf{c} + \begin{bmatrix} -\mathbf{e}_2 \\ \hat{\sigma}\hat{\mathbf{v}} \end{bmatrix} \mathbf{d} + \text{Error}, \tag{11.35}$$

where

$$\begin{aligned} \mathbf{e}_1 &= \mathbf{f}(\mathbf{y}) - \mathbf{X}\hat{\boldsymbol{\beta}}_1, & \hat{\sigma}^2 &= \mathbf{e}_1'\mathbf{e}_1/(T-k_1), \\ \mathbf{e}_2 &= \mathbf{g}(\mathbf{y}) - \mathbf{Z}\hat{\boldsymbol{\beta}}_2, & \hat{\omega}^2 &= \mathbf{e}_2'\mathbf{e}_2/(T-k_2), \\ \hat{\mathbf{v}} &= (\hat{v}_1, \hat{v}_2, \ldots, \hat{v}_T)', & \hat{v}_t &= g'(y_t)/f'(y_t), \end{aligned}$$

and $\boldsymbol{\tau} = (1, 1, \ldots, 1)'$ is a $T \times 1$ vector of ones, and $g'(y_t)$ and $f'(y_t)$ stand for the derivatives of $g(y_t)$ and $f(y_t)$ with respect to y_t.

To compute the SSR_f statistic we first note that

$$SSR_f = \tilde{\mathbf{y}}'\tilde{\mathbf{y}} - \tilde{\mathbf{y}}'\tilde{\mathbf{X}}(\tilde{\mathbf{X}}'\tilde{\mathbf{X}})^{-1}\tilde{\mathbf{X}}'\tilde{\mathbf{y}},$$

where

$$\tilde{\mathbf{y}} = \begin{bmatrix} \mathbf{e}_1/\hat{\sigma} \\ \boldsymbol{\tau} \end{bmatrix}, \qquad \tilde{\mathbf{X}} = \begin{bmatrix} -\mathbf{X} & \mathbf{e}_1/\hat{\sigma} & -\mathbf{e}_2 \\ \mathbf{0} & -\boldsymbol{\tau} & \hat{\sigma}\hat{\mathbf{v}} \end{bmatrix}.$$

But $\tilde{\mathbf{y}}'\tilde{\mathbf{y}} = \mathbf{e}_1'\mathbf{e}_1/\hat{\sigma}^2 + T = 2T - k_1$,

$$\tilde{\mathbf{y}}'\tilde{\mathbf{X}} = \left[0,\ -k_1,\ \hat{\sigma}\,\boldsymbol{\tau}'\hat{\mathbf{v}} - \frac{\mathbf{e}_1\mathbf{e}_2}{\hat{\sigma}}\right],$$

and

$$\tilde{\mathbf{X}}'\tilde{\mathbf{X}} = \begin{pmatrix} \mathbf{X}'\mathbf{X} & \mathbf{0} & \mathbf{X}'\mathbf{e}_2 \\ \mathbf{0} & 2T - k_1 & \dfrac{-\mathbf{e}_1'\mathbf{e}_2}{\hat{\sigma}} - \hat{\sigma}(\boldsymbol{\tau}'\hat{\mathbf{v}}) \\ \mathbf{e}_2'\mathbf{X} & \dfrac{-\mathbf{e}_1'\mathbf{e}_2}{\hat{\sigma}} - \hat{\sigma}(\boldsymbol{\tau}'\hat{\mathbf{v}}) & \mathbf{e}_2'\mathbf{e}_2 + \hat{\sigma}^2\hat{\mathbf{v}}'\hat{\mathbf{v}} \end{pmatrix}.$$

Using these results, and after some algebra, we obtain:

$$DL_f = \frac{1}{D}\left(k_1^2 R_1 + (2T - k_1)R_3^2 - 2k_1 R_2 R_3\right), \tag{11.36}$$

where

$$\begin{aligned} R_1 &= (\mathbf{e}_2'\mathbf{M}_1\mathbf{e}_2)/\hat{\sigma}^2 + \hat{\mathbf{v}}'\hat{\mathbf{v}},\ R_2 = (\boldsymbol{\tau}'\hat{\mathbf{v}}) + (\mathbf{e}_1'\mathbf{e}_2)/\hat{\sigma}^2, \\ R_3 &= (\boldsymbol{\tau}'\hat{\mathbf{v}}) - (\mathbf{e}_1'\mathbf{e}_2)/\hat{\sigma}^2,\ D = (2T - k_1)R_1 - R_2^2. \end{aligned}$$

A similar statistic is also computed for the test of M_g against M_f.

11.7.4 Simulated Cox's non-nested test statistics

Simulated Cox test statistics, SC_c, was introduced in Pesaran and Pesaran (1993) and subsequently applied to tests of linear versus log-linear models, and first difference versus log-difference stationary models in Pesaran and Pesaran (1995). The numerator of the SC_c statistic for testing M_f against M_g is computed as

$$T^{1/2}T_f(R) = -\tfrac{1}{2}T^{1/2}\log(\hat{\sigma}^2/\hat{\omega}^2) + T^{-1/2}\sum_{t=1}^{T}\log\left[\left|f'(y_t)/g'(y_t)\right|\right] + \tfrac{1}{2}T^{-1/2}(k_1 - k_2) - T^{1/2}C_R(\hat{\boldsymbol{\theta}}, \hat{\boldsymbol{\gamma}}_*(R)), \tag{11.37}$$

where $\hat{\boldsymbol{\theta}} = \left(\hat{\boldsymbol{\beta}}_1', \hat{\sigma}^2\right)'$, R is the number of replications, $\hat{\boldsymbol{\gamma}}_*(R)$ is the simulated pseudo-ML estimator of $\boldsymbol{\gamma} = \left(\boldsymbol{\beta}_2', \omega^2\right)'$ under M_f:

$$\hat{\boldsymbol{\gamma}}_*(R) = R^{-1}\sum_{j=1}^{R}\hat{\boldsymbol{\gamma}}_j, \tag{11.38}$$

where $\hat{\boldsymbol{\gamma}}_j$ is the ML estimator of $\boldsymbol{\gamma}$ computed using the artificially simulated independent observations $\mathbf{Y}_j = (\mathbf{Y}_{j1}, \mathbf{Y}_{j2}, \ldots, \mathbf{Y}_{jT})$ obtained under M_f with $\boldsymbol{\theta} = \hat{\boldsymbol{\theta}}$. $C_R(\hat{\boldsymbol{\theta}}, \hat{\boldsymbol{\gamma}}_*(R))$ is the simulated estimator of the 'closeness' measure of M_f with respect of M_g (see Pesaran (1987b))

$$C_R(\hat{\boldsymbol{\theta}}, \hat{\gamma}_*(R)) = R^{-1}\sum_{j=1}^{R}[L_f(\mathbf{Y}_j, \hat{\boldsymbol{\theta}}) - L_g\left(\mathbf{Y}_j, \hat{\boldsymbol{\gamma}}_*(R)\right)], \tag{11.39}$$

where $L_f(\mathbf{Y}, \boldsymbol{\theta})$ and $L_g(\mathbf{Y}, \boldsymbol{\gamma})$ are the average log-likelihood functions under M_f and M_g, respectively

$$L_f(\mathbf{Y}, \boldsymbol{\theta}) = -\tfrac{1}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}\left\{\sum_{t=1}^{T}\left[f(y_t) - \boldsymbol{\beta}_1'\mathbf{x}_t\right]^2/T\right\} + T^{-1}\sum_{t=1}^{T}\log\left|f'(y_t)\right|, \tag{11.40}$$

$$L_g(\mathbf{Y}, \boldsymbol{\gamma}) = -\tfrac{1}{2}\log(2\pi\omega^2) - \frac{1}{2\omega^2}\left\{\sum_{t=1}^{T}\left[g(y_t) - \boldsymbol{\beta}_2'\mathbf{z}_t\right]^2/T\right\} + T^{-1}\sum_{t=1}^{T}\log\left|g'(y_t)\right|. \tag{11.41}$$

The denominator of the SC_c statistic is computed as

$$V_{*d}^2(R) = (T-1)^{-1}\sum_{t=1}^{T}(d_{*t} - \bar{d}_*)^2, \tag{11.42}$$

where $\bar{d}_* = T^{-1}\sum_{t=1}^{T} d_{*t}$, and

$$d_{*t} = -\tfrac{1}{2}\log\left(\hat{\sigma}^2/\hat{\omega}_*^2(R)\right) - \frac{1}{2\hat{\sigma}^2}e_{t1}^2 + \frac{1}{2\hat{\omega}_*^2(R)}\left[g(y_t) - \mathbf{z}_t'\hat{\beta}_{*2}(R)\right]^2 + \log\left[\left|f'(y_t)/g'(y_t)\right|\right],$$

and

$$e_{t1} = f(y_t) - \mathbf{x}_t'\hat{\boldsymbol{\beta}}_1.$$

Recall also that $\hat{\boldsymbol{\beta}}_{*2}(R)$ and $\hat{\omega}_*^2(R)$ are given by (11.38), where $\hat{\boldsymbol{\gamma}}_*(R) = \left(\hat{\boldsymbol{\beta}}_{*2}'(R), \hat{\omega}_*^2(R)\right)'$.

The standardized Cox statistic for the test of M_f against M_g is given by

$$SC_c(R) = T^{\frac{1}{2}}T_f(R)/V_{*d}(R),$$

where $T^{\frac{1}{2}}T_f(R)$ is defined by (11.37) and $V_{*d}(R)$ by (11.42). A similar statistic is also computed for the test of M_g against M_f. Notice that two other versions of the simulated Cox statistic have been proposed in the literature. The three test statistics have the same numerator and differ by the choice of the estimator of the variance used to standardize the Cox statistic. However, SC_c seems to have much better small sample properties than the other two test statistics.[7]

11.7.5 Sargan and Vuong's likelihood criteria

The Sargan (1964) likelihood criterion simply compares the maximized values of the log-likelihood functions under M_f and M_g[8]

$$LL_{fg} = T\left[L_f(\mathbf{Y}, \hat{\boldsymbol{\theta}}) - L_g(\mathbf{Y}, \hat{\boldsymbol{\gamma}})\right],$$

or using (11.40) and (11.41)

$$LL_{fg} = -\frac{T}{2}\log(\hat{\sigma}^2/\hat{\omega}^2) + \sum_{t=1}^{T}\log\left|f'(y_t)/g'(y_t)\right| + \tfrac{1}{2}(k_1 - k_2). \tag{11.43}$$

One could also apply the known model selection criteria such as *AIC* and *SBC* to the models M_f and M_g (see Section 11.5). For example, in the case of the *AIC* we have

$$AIC(M_f : M_g) = LL_{fg} - (k_1 - k_2).$$

[7] The Monte Carlo results reported in Pesaran and Pesaran (1995) also clearly show that the SC_c and the DL tests are more powerful than the PE or BM tests discussed in Sections 11.7.1 and 11.7.2 above.

[8] Note that throughout $\hat{\sigma}^2 = \mathbf{e}_1'\mathbf{e}_1/(n-k_1)$ and $\hat{\omega}^2 = \mathbf{e}_2'\mathbf{e}_2/(n-k_2)$ are used as estimators of σ^2 and ω^2, respectively.

Vuong's criterion is motivated in the context of testing the hypothesis that M_f and M_g are equivalent, using the Kullback and Leibler (1951) information criterion as a measure of goodness of fit. The Vuong (1989) test criterion for the comparison of M_f and M_g is computed as

$$V_{fg} = \frac{\sum_{t=1}^{T} d_t}{\left[\sum_{t=1}^{T}(d_t - \bar{d})^2\right]^{1/2}}, \tag{11.44}$$

where $\bar{d} = T^{-1}\sum_{t=1}^{T} d_t$, and

$$d_t = -\tfrac{1}{2}\log(\hat{\sigma}^2/\hat{\omega}^2) - \tfrac{1}{2}\left(\frac{e_{t1}^2}{\hat{\sigma}^2} - \frac{e_{t2}^2}{\hat{\omega}^2}\right) + \log\left[\left|f'(y_t)/g'(y_t)\right|\right],$$
$$e_{t1} = f(y_t) - \hat{\boldsymbol{\beta}}_1'\mathbf{x}_t, \qquad e_{t2} = g(y_t) - \hat{\boldsymbol{\beta}}_2'\mathbf{z}_t.$$

Under the null hypothesis that 'M_f and M_g are equivalent', V_{fg} is approximately distributed as a standard normal variate.

Example 31 *Suppose you are interested in testing the following linear form of the inflation augmented ARDL(1, 1) model for aggregate consumption (c_t)*

$$M_1 : c_t = \alpha_0 + \alpha_1 c_{t-1} + \alpha_2 y_t + \alpha_3 y_{t-1} + \alpha_4 \pi_t + u_{1t},$$

against its log-linear form

$$M_2 : \log c_t = \beta_0 + \beta_1 \log c_{t-1} + \beta_2 \log y_t + \beta_3 \log y_{t-1} + \beta_4 \pi_t + u_{2t},$$

where c_t is real non-durable consumption expenditure in the US, y_t is real disposable income, and π_t is the inflation rate in the years 1990 to 1994. Table 11.1 reports the parameter estimates under models M_1 and M_2. The estimates of the parameters of M_1 computed under M_1 are the OLS estimates ($\hat{\alpha}$), while the estimates of the parameters of M_1 computed under M_2 are the pseudo-true estimators ($\hat{\alpha}_* = \hat{\alpha}_*(\hat{\beta})$). If model M_1 is correctly specified, one would expect $\hat{\alpha}$ and $\hat{\alpha}_*$ to be near to one another. The same also applies to the estimates of the parameters of model M_2 (β). The bottom part of Table 11.1 gives a number of non-nested statistics for testing the linear versus the log-linear model and vice versa, computed by simulations, using a number of replications equal to 100. This table also gives the Sargan (1964) and Vuong (1989) likelihood function criteria for the choice between the two models. All the tests reject the linear model against the log-linear model, and none reject the log-linear model against the linear one at the 5 per cent significance level, although the simulated Cox and the double-length tests also suggest rejection of the log-linear model at the 10 per cent significance level. Increasing the number of replications to 500 does not alter this conclusion. The two choice criteria also favour the log-linear specification over the linear specification.

Table 11.1 Testing linear versus log-linear consumption functions

Non-nested tests by simulation

Dependent variable in model MI is C
Dependent variable in model M2 is LOG(C)
136 observations used from 1960Q2 to 1994Q1. Number of replications 100

	Estimates of parameters of MI			Estimates of parameters of M2	
	Under M1	Under M2		Under M2	Under M1
INPT	20.1367	24.7609	INPT	.14429	.098439
C(−1)	.93128	.90987	LC(−1)	.89781	.94227
Y	.092935	.098593	LY	.29526	.27626
Y(−1)	−.076891	−.077510	LY (−1)	−.22532	−.23860
PI	−160.9665	−156.5597	PI	−.23880	−.23041
Standard Error	4.6528	4.8476	Standard Error	.0057948	.0061028
Adjusted Log-L	−399.5709	−404.4709	Adjusted Log-L	−399.0233	−405.2882

Non-nested test statistics and choice criteria

Test Statistic	M1 against M2	M2 against M1
S-Test	−2.5592 [.010]	−1.8879 [.059]
PE-Test	2.3021 [.021]	−.26402 [.792]
BM-Test	2.0809 [.037]	−.50743 [.612]
DL-Test	2.0006 [.045]	1.8004 [.072]

Sargan's Likelihood Criterion for M1 versus M2= −.54764 favours M2
Vuong's Likelihood Criterion for M1 versus M2= −1.7599 [.078] favours M2

S-Test is the SC_c test proposed by Pesaran and Pesaran (1995) and is the simple version of the simulated Cox test statistic.
PE-Test is the PE test due to MacKinnon, White, and Davidson (1983).
BM-Test is due to Bera and McAleer (1989).
DL-Test is the double-length regression test statistic due to Davidson and MacKinnon (1984).

11.8 A Bayesian approach to model combination

The model selection approach aims at choosing a particular model. An alternative procedure would be to combine models by pooling their forecasts. Bayesian model averaging techniques present a natural way forward.[9]

Suppose we are interested in a decision problem that requires probability forecasts of an event defined in terms of one or more elements of $\mathbf{z}_t$, for $t = T+1, T+2, \ldots, T+h$, where $\mathbf{z}_t = (z_{1t}, z_{2t}, \ldots, z_{nt})'$ is an $n \times 1$ vector of the variables of interest and h is the forecast (decision) horizon. Assume also that the data generating process (DGP) is unknown and the forecasts are made considering m different models indexed by i (that could be nested or non-nested). Each model, M_i, $i = 1, 2, \ldots, m$, is characterized by a probability density function of $\mathbf{z}_t$ defined over the estimation period $t = 1, 2, \ldots, T$, as well as the forecast period $t = T+1, T+2, \ldots, T+h$,

[9] The exposition in this section follows Garratt et al. (2003a). Also see Section C.5 in Appendix C on Bayesian model selection.

in terms of a $k_i \times 1$ vector of unknown parameters, $\boldsymbol{\theta}_i$, assumed to lie in the compact parameter space, $\boldsymbol{\Theta}_i$. Model M_i is then defined by

$$M_i: \quad \left\{ f_i\left(\mathbf{z}_1, \mathbf{z}_2, \ldots, \mathbf{z}_T, \mathbf{z}_{T+1}, \mathbf{z}_{T+2}, \ldots, \mathbf{z}_{T+h}; \boldsymbol{\theta}_i\right), \quad \boldsymbol{\theta}_i \in \boldsymbol{\Theta}_i \right\}, \tag{11.45}$$

where $f_i(.)$ is the joint probability density function of past and future values of $\mathbf{z}_t$. Conditional on each model, M_i, being true we shall assume that the true value of $\boldsymbol{\theta}_i$, which we denote by $\boldsymbol{\theta}_{i0'}$ is fixed and remains constant across the estimation and the prediction periods and lies in the interior of $\boldsymbol{\Theta}_i$. We denote the maximum likelihood estimator of $\boldsymbol{\theta}_{i0}$ by $\hat{\boldsymbol{\theta}}_{iT}$, and assume that it satisfies the usual regularity conditions so that

$$\sqrt{T}\left(\hat{\boldsymbol{\theta}}_{iT} - \boldsymbol{\theta}_{i0}\right) | M_i \overset{a}{\sim} N\left(\mathbf{0}, \mathbf{V}_{\theta_i}\right),$$

where $\overset{a}{\sim}$ stands for 'asymptotically distributed as', and $T^{-1}\mathbf{V}_{\theta_i}$ is the asymptotic covariance matrix of $\hat{\boldsymbol{\theta}}_{iT}$ conditional on M_i. Under these assumptions, parameter uncertainty only arises when T is finite. The case where $\boldsymbol{\theta}_{i0}$ could differ across the estimation and forecast periods poses new difficulties and can be resolved in a satisfactory manner if one is prepared to formalize how $\boldsymbol{\theta}_{i0}$ changes over time.

The object of interest is the probability density function of $\mathbf{Z}_{T+1,h} = (\mathbf{z}_{T+1}, \mathbf{z}_{T+2}, \ldots, \mathbf{z}_{T+h})$ conditional on the available observations at the end of period T, $\mathbf{Z}_T = (\mathbf{z}_1, \mathbf{z}_2, \ldots, \mathbf{z}_T)$. This will be denoted by $\Pr\left(\mathbf{Z}_{T+1,h} | \mathbf{Z}_T\right)$. For this purpose, models and their parameters serve as intermediate inputs in the process of characterization and estimation of $\Pr\left(\mathbf{Z}_{T+1,h} | \mathbf{Z}_T\right)$. The Bayesian approach provides an elegant and logically coherent solution to this problem, with a full solution given by the so-called Bayesian model averaging formula (see, e.g., Draper (1995), Hoeting, Madigan, Raftery, and Volinsky (1999)):

$$\Pr\left(\mathbf{Z}_{T+1,h} | \mathbf{Z}_T\right) = \sum_{i=1}^{m} \Pr\left(M_i | \mathbf{Z}_T\right) \Pr(\mathbf{Z}_{T+1,h} | \mathbf{Z}_T, M_i), \tag{11.46}$$

where $\Pr\left(M_i | \mathbf{Z}_T\right)$ is the posterior probability of model M_i

$$\Pr\left(M_i | \mathbf{Z}_T\right) = \frac{\Pr\left(M_i\right) \Pr(\mathbf{Z}_T | M_i)}{\sum_{j=1}^{m} \Pr\left(M_j\right) \Pr(\mathbf{Z}_T | M_j)}, \tag{11.47}$$

$\Pr\left(M_i\right)$ is the prior probability of model M_i, $\Pr(\mathbf{Z}_T | M_i)$ is the integrated or average likelihood

$$\Pr(\mathbf{Z}_T | M_i) = \int_{\boldsymbol{\theta}_i} \Pr\left(\boldsymbol{\theta}_i | M_i\right) \Pr(\mathbf{Z}_T | M_i, \boldsymbol{\theta}_i) d\boldsymbol{\theta}_i, \tag{11.48}$$

$\Pr\left(\boldsymbol{\theta}_i | M_i\right)$ is the prior on $\boldsymbol{\theta}_i$ conditional on M_i, $\Pr(\mathbf{Z}_T | M_i, \boldsymbol{\theta}_i)$ is the likelihood function of model M_i, and $\Pr(\mathbf{Z}_{T+1,h} | \mathbf{Z}_T, M_i)$ is the posterior predictive density of model M_i defined by

$$\Pr(\mathbf{Z}_{T+1,h} | \mathbf{Z}_T, M_i) = \int_{\boldsymbol{\theta}_i} \Pr\left(\boldsymbol{\theta}_i | \mathbf{Z}_T, M_i\right) \Pr(\mathbf{Z}_{T+1,h} | \mathbf{Z}_T, M_i, \boldsymbol{\theta}_i) d\boldsymbol{\theta}_i, \tag{11.49}$$

in which $\Pr(\boldsymbol{\theta}_i|\mathbf{Z}_T, M_i)$ is the posterior probability of $\boldsymbol{\theta}_i$ given model M_i

$$\Pr(\boldsymbol{\theta}_i|\mathbf{Z}_T, M_i) = \frac{\Pr(\boldsymbol{\theta}_i|M_i)\Pr(\mathbf{Z}_T|M_i, \boldsymbol{\theta}_i)}{\sum_{j=1}^{m}\Pr(M_j)\Pr(\mathbf{Z}_T|M_j)}. \tag{11.50}$$

The Bayesian approach requires *a priori* specifications of $\Pr(M_i)$ and $\Pr(\boldsymbol{\theta}_i|M_i)$ for $i = 1, 2, \ldots, m$, and further assumes that one of the m models being considered is the DGP so that $\Pr(\mathbf{Z}_{T+1,h}|\mathbf{Z}_T)$ defined by (11.46) is proper.

The Bayesian model averaging formula also provides a simple 'optimal' solution to the problem of pooling of the point forecasts, $E(\mathbf{Z}_{T+1,h}|\mathbf{Z}_T, M_i)$, studied extensively in the literature, namely (see, for example, Draper (1995))

$$E(\mathbf{Z}_{T+1,h}|\mathbf{Z}_T) = \sum_{i=1}^{m}\Pr(M_i|\mathbf{Z}_T)\,E(\mathbf{Z}_{T+1,h}|\mathbf{Z}_T, M_i),$$

with the variance given by

$$\begin{aligned} V(\mathbf{Z}_{T+1,h}|\mathbf{Z}_T) = &\sum_{i=1}^{m}\Pr(M_i|\mathbf{Z}_T)\,V(\mathbf{Z}_{T+1,h}|\mathbf{Z}_T, M_i) \\ &+ \sum_{i=1}^{m}\Pr(M_i|\mathbf{Z}_T)\left[E(\mathbf{Z}_{T+1,h}|\mathbf{Z}_T, M_i) - E(\mathbf{Z}_{T+1,h}|\mathbf{Z}_T)\right]^2, \end{aligned}$$

where the first term accounts for within model variability and the second term for between model variability.

There is no doubt that the Bayesian model averaging (*BMA*) provides an attractive solution to the problem of accounting for model uncertainty. But its strict application can be problematic, particularly in the case of high-dimensional models. The major difficulties lie in the choice of the space of models to be considered, the model priors $\Pr(M_i)$, and the specification of meaningful priors for the unknown parameters, $\Pr(\boldsymbol{\theta}_i|M_i)$. The computational issues, while still considerable, are partly overcome by Monte Carlo integration techniques. For an excellent overview of these issues, see Hoeting et al. (1999). Also see Fernandez et al. (2001) for specific applications.

Putting the problem of model specification to one side, the two important components of the *BMA* formula are the posterior probability of the models, $\Pr(M_i|\mathbf{Z}_T)$, and the posterior density functions of the parameters, $\Pr(\boldsymbol{\theta}_i|\mathbf{Z}_T, M_i)$, for $i = 1, \ldots, m$.

11.9 Model selection by LASSO

In cases where the number of regressors, k, is large relative to the number of available observations, T, and in some case even larger than T, the standard model selection procedures such as *AIC* and *SBC* are not applicable. In such cases penalized regressions are used. Suppose the observed data are T realizations ($t = 1, 2, \ldots, T$) on the scalar target variable, y_t, and p potential predictor variables $\mathbf{x}_t = (x_{1t}, x_{2t}, \ldots, x_{pt})'$. In the case of linear regressions, we have

$$y_t = \sum_{i=1}^{p}\beta_i x_{it} + u_t.$$

The predictor variables, x_{it}, are typically standardized (to make the scale of β_i comparable across i). It is assumed that the regressors, x_{it}, are strictly exogenous, precluding the inclusion of lagged dependent variables. Most crucially it is assumed that the 'true' regression model is 'sparse' in the sense that only a few β_i's are non-zero and the rest are zero. Lasso (least absolute shrinkage and selection operator) regressions uses $\sum_{i=1}^{p} |\beta_i|$ as the penalty which is bounded by the sparseness assumption. Lasso was originally proposed by Tibshirani (1996) and is closely related to the Ridge regression which uses the penalty term $\sum_{i=1}^{p} \beta_i^2$, which is less restrictive as compared to the Lasso penalty. As shown in Section C.7 of Appendix , the Ridge regression also results from the application of Bayesian analysis to regression models.

The two penalty terms can also be combined. In general, the penalized regressions can be computed by solving the optimization problem

$$\min_{\boldsymbol{\beta}_p} \left\{ \sum_{t=1}^{T} \left(y_t - \sum_{i=1}^{p} \beta_i x_{it} \right)^2 + \lambda \sum_{i=1}^{p} \left[(1-\alpha) |\beta_i| + \alpha \beta_i^2 \right] \right\},$$

where λ and α are called tuning parameters and are typically estimated by cross validation. *OLS* corresponds to the no penalty case of $\lambda = 0$. When $\lambda \neq 0, \alpha = 1$ yields the Ridge regressions, and if $\alpha = 0$ with $\lambda \neq 0$ we obtain the Lasso regression. As originally noted by Tibshirani (1996), Lasso is a selection procedure since Lasso optimization yields corner solutions due to the non-differentiable nature of Lasso's penalty function. Penalized regressions, particularly Lasso, are easy to apply and have been shown to work well in the context of *independently distributed observations*. Although linear in structure, nonlinear effects can also be included as predictors—such as threshold effects. The tuning parameters, λ and α, are estimated by cross-validation. Comprehensive reviews of the penalized regression techniques can be found in Hastie, Tibshirani, and Friedman (2009) and Buhlmann and van de Geer (2012).

In the case of large data sets often encountered in macroeconomics and finance, penalized regressions must be adapted to deal with temporal dependence and possible structural breaks. These are topics for future research, but some progress has been made for the analysis of high dimensional factor-augmented vector autoregressions. For a review of this literature see Chapter 33.

11.10 Further reading

For further discussion of the general principles involved in model selection see Pesaran and Smith (1985) and Pesaran and Weeks (2001). See also Section C.5 in Appendix C.

11.11 Exercises

1. Let $\ell_f(\mathbf{y}, \boldsymbol{\theta})$ and $\ell_g(\mathbf{y}, \boldsymbol{\gamma})$ be the log-likelihood functions under models H_f and H_g, where $\mathbf{y}$ is a $T \times 1$ vector of observations on Y. Define the closeness of model H_f with respect to H_g by

$$I_{fg}(\boldsymbol{\theta}, \boldsymbol{\gamma}) = E_f \left[\ell_f(\mathbf{y}, \boldsymbol{\theta}) - \ell_g(\mathbf{y}, \boldsymbol{\gamma}) \right]. \tag{11.51}$$

(a) Show that, in general, $I_{fg}(\boldsymbol{\theta}, \boldsymbol{\gamma})$ is not the same as $I_{gf}(\boldsymbol{\gamma}, \boldsymbol{\theta})$. Under what conditions $I_{fg}(\boldsymbol{\theta}, \boldsymbol{\gamma}) = 0$?

(b) Suppose that under H_f, $\mathbf{y}$ are draws from the log-normal density

$$f(y, \boldsymbol{\theta}) = y^{-1}(2\pi\theta_2) \exp\left[\frac{-(\ln y - \theta_1)^2}{2\theta_2}\right], \ \theta_2 > 0, y > 0,$$

and under H_g, $\mathbf{y}$ are draws from the exponential

$$g(\mathbf{y}, \gamma) = \gamma^{-1} \exp\left(-y/\gamma\right), \gamma > 0, y > 0.$$

Derive the expression for $I_{fg}(\boldsymbol{\theta},\gamma)$, and show that $I_{fg}(\boldsymbol{\theta}, \gamma) > 0$ for all values of $\boldsymbol{\theta}$ and γ. What is the significance of this result when comparing log-normal and logistic densities?

2. Suppose that it is known that T observations $y_t = 1, 2, \ldots, T$, are generated from the MA(1) process

$$y_t = \varepsilon_t + \theta\varepsilon_{t-1},$$

where $\varepsilon_t \sim IIDN(0, \sigma^2)$.

(a) What is the pseudo-true value of ρ if it is incorrectly assumed that y_t follows the AR(1) process

$$y_t = \rho y_{t-1} + u_t,$$

where $u_t \sim IIDN(0, \omega^2)$.

(b) Derive the divergence measure of the $MA(1)$ process from the $AR(1)$ process and *vice versa*. The divergence measure of one density against another is defined by (11.51).

(c) Discuss alternative testing procedures for testing the $AR(1)$ model against $MA(1)$ and *vice versa*.

3. Consider the following regression models

$$H_f: \ \mathbf{y} = \mathbf{X}\boldsymbol{\alpha} + \mathbf{u}_f, \quad \mathbf{u}_f \sim N(\mathbf{0}, \sigma^2\mathbf{I}_T), \tag{11.52}$$

$$H_g: \ \mathbf{y} = \mathbf{Z}\boldsymbol{\beta} + \mathbf{u}_g, \quad \mathbf{u}_g \sim N(\mathbf{0}, \omega^2\mathbf{I}_T), \tag{11.53}$$

where $\mathbf{y}$ is the $T \times 1$ vector of observations on the dependent variable, $\mathbf{X}$ and $\mathbf{Z}$ are $T \times k_f$ and $T \times k_g$ observation matrices for the regressors of models H_f and H_g, $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$ are the $k_f \times 1$ and $k_g \times 1$ unknown regression coefficient vectors, $\mathbf{u}_f$ and $\mathbf{u}_g$ are the $T \times 1$ disturbance vectors, and $\mathbf{I}_T$ is an identity matrix of order T. Define the combined model as

$$L_\lambda(\mathbf{y}\,|\mathbf{X}, \mathbf{Z}\,) = \frac{L_f^{1-\lambda}(\mathbf{y}\,|\mathbf{X}\,)L_g^{\lambda}(\mathbf{y}\,|\mathbf{Z}\,)}{\int L_f^{1-\lambda}(\mathbf{y}\,|\mathbf{X}\,)L_g^{\lambda}(\mathbf{y}\,|\mathbf{Z}\,)d\mathbf{y}},$$

where $L_f(\mathbf{y}\,|\mathbf{X}\,)$ and $L_g(\mathbf{y}\,|\mathbf{Z}\,)$ are the likelihood functions associated with models H_f and H_g, respectively.

(a) Show that the combined model can be written as

$$H_\lambda:\ \mathbf{y}=\left[\frac{(1-\lambda)\nu^2}{\sigma^2}\right]\mathbf{X}\boldsymbol{\alpha}+\left[\frac{\lambda\nu^2}{\omega^2}\right]\mathbf{Z}\boldsymbol{\beta}+\mathbf{u},\ \ \mathbf{u}\sim N(\mathbf{0},\nu^2\mathbf{I}_T), \tag{11.54}$$

where $\nu^{-2}=(1-\lambda)\sigma^{-2}+\lambda\omega^{-2}$.

(b) Is the mixing parameter λ identified?

4. Consider the 'combined' regression model indexed by κ

$$H_\kappa:\ \mathbf{y}=(1-\kappa)\mathbf{X}\boldsymbol{\alpha}+\kappa\mathbf{Z}\boldsymbol{\beta}+\mathbf{u}.$$

(a) Show that the t-ratio statistic for testing $\kappa=0$, for a given value of $\boldsymbol{\beta}$, is given by

$$t_\kappa(\mathbf{Z}\boldsymbol{\beta})=\frac{\boldsymbol{\beta}'\mathbf{Z}'\mathbf{M}_x\mathbf{y}}{\hat{\sigma}\left(\boldsymbol{\beta}'\mathbf{Z}'\mathbf{M}_x\mathbf{Z}\boldsymbol{\beta}\right)^{1/2}},$$

where

$$\hat{\sigma}^2=\frac{1}{T-k_f-1}\left[\mathbf{y}'\mathbf{M}_x\mathbf{y}-\frac{\left(\boldsymbol{\beta}'\mathbf{Z}'\mathbf{M}_x\mathbf{y}\right)^2}{\boldsymbol{\beta}'\mathbf{Z}'\mathbf{M}_x\mathbf{Z}\boldsymbol{\beta}}\right].$$

(b) Derive an expression for $\sup_{\boldsymbol{\beta}}\{t_\kappa(\mathbf{Z}\boldsymbol{\beta})\}$ and discuss its relevance for testing H_f against H_g, defined by (11.52) and (11.53).

5. Consider the log-normal and the exponential models set out in Question 1 above. Denote the prior densities of the parameters of the two models by $\pi_f(\boldsymbol{\theta})$ and $\pi_g(\gamma)$.

(a) Suppose you are given the observations $\mathbf{y}=(y_1,y_2,\ldots,y_T)'$. Derive the posterior odds of the log-normal against the exponential model assuming that they have the same prior odds.

(b) Compare the Bayesian posterior odds with the values of $E_{fg}(\boldsymbol{\theta}_0,\gamma_*)$ and $E_{fg}(\gamma_0,\boldsymbol{\theta}_*)$ as $T\to\infty$, where $\boldsymbol{\theta}_0(\gamma_0)$ is the true value of $\boldsymbol{\theta}(\gamma)$ under $H_f(H_g)$, and γ_* and $\boldsymbol{\theta}_*$ are the associated pseudo-true values.

Part III

Stochastic Processes

12 Introduction to Stochastic Processes

12.1 Introduction

Any ordered series may be regarded as a time series. The temporal, immutable order imposed on the observations is the critical and distinguishing feature of time series. As a result, time series techniques cannot be generally applied to cross-section observations (such as those over different individuals, firms, countries, or regions) where they cannot be ordered in an immutable (time-invariant) fashion. The origin of modern time series analysis dates back to the pioneering work of Slutsky (1937) and Yule (1926, 1927), on the analysis of the linear combinations of purely random numbers. There are two main approaches to the analysis of time series; the *time* domain and the *spectral* (or frequency domain) approaches. Time domain techniques are prevalent in econometrics, whilst in engineering and oceanography the frequency domain approach dominates. Until 1980s, the analysis of time series has been confined to stationary processes, or processes that can be transformed to stationarity. But important developments have taken place, particularly in the area of non-stationary and nonlinear times series analysis.

12.2 Stationary processes

The notion of stationarity has played an important role in the theory of stochastic processes and time series analysis. Broadly speaking, a process is said to be stationary if its probability distribution remains unchanged as time progresses. This concept is relevant to stochastic processes for which the underlying data generation process is not changing over time, and is more suited to the study of natural rather than social phenomena. Khintchine (1934) was apparently the first to formalize the concept of stationarity for continuous-time processes, and Wold (1938) developed similar ideas in the context of a discrete-time stochastic process.

For a formal definition of stationarity, consider a stochastic process $\{y_t, t \in \mathcal{T}\}$, where t represents time and belongs to a linear index set $\mathcal{T}$. An index set $\mathcal{T}$ is said to be *linear* if for any t and h belonging to $\mathcal{T}$, their sum $t + h$ also belongs to t. The index set $\mathcal{T}$ could be discrete or continuous, one-sided or two-sided. Examples of index sets for a discrete-time stochastic process is $\mathcal{T} = \{1, 2, \ldots\}$, while for a continuous-time process $\mathcal{T} = \{t, t > 0\}$.

Definition 14 (Strict stationarity of order s) *The stochastic process* $\{y_t, t \in \mathcal{T}\}$ *is said to be strictly stationary of order s, if the joint distribution functions of* $\{y_{t_1}, y_{t_2}, \ldots, y_{t_k}\}$ *and* $\{y_{t_1+h}, y_{t_2+h}, \ldots, y_{t_k+h}\}$ *are identical for all values of* $t_1, t_2, \ldots, t_k$, *and h, and all positive integers* $k \leq s$.

Definition 15 (Strict stationarity) *The stochastic process* $\{y_t, t \in \mathcal{T}\}$ *is said to be strictly stationary if it is strictly stationary of order s for any positive integer s.*

In effect, under strict stationarity the process is in 'stochastic equilibrium' and realizations of the process obtained over different time intervals would be similar. This is a counterpart of the concept of static equilibrium in the theory of deterministic processes. One important implication of strict stationarity is that y_t will have the same distribution for all t. The importance of the stationarity property for empirical analysis is closely tied up with the ergodicity property which, loosely speaking, ensures consistent estimation of the unknown parameters of the process from time-averages. For a rigorous account of the ergodicity property and the conditions under which it holds see, for example, Hannan (1970), and Karlin and Taylor (1975). It is clear that if a process is strictly stationary and has second-order moments, then its mean and variance will be time invariant, namely they do not depend on t.

Another important concept is *weak stationarity* (or, simply, stationarity).

Definition 16 (Weak stationarity) *A stochastic process* $\{y_t, t \in \mathcal{T}\}$ *is said to be weakly stationary if it has a constant mean and variance and its covariance function,* $\gamma(t_1, t_2)$, *defined by*

$$\gamma(t_1, t_2) = Cov\left(y_{t_1}, y_{t_2}\right) = E\left(y_{t_1} y_{t_2}\right) - E\left(y_{t_1}\right) E\left(y_{t_2}\right),$$

depends only on the absolute difference $| t_1 - t_2 |$, *namely* $\gamma(t_1, t_2) = \gamma(| t_1 - t_2 |)$.

Therefore, for a weakly stationary process the covariance between any two observations depends only on the length of time separating the observations. A weakly stationary process is also referred to as 'covariance stationary', or 'wide-sense stationary'. This definition is, however, too restrictive for most economic time series that are trended. A related concept which allows for deterministic trends is the *trend-stationary process.*

Definition 17 *The process* y_t *is said to be trend stationary if* $y_t = x_t - d_t$ *is covariance stationary, where* d_t *is the perfectly predictable component of* y_t.

Examples of purely deterministic processes include time trends, seasonal dummies and sinusoid functions, such as $d_t = 1$ for odd (even) value of t, $d_t = a_0 + a_1 t$, or more generally $d_t = f(t)$, where $f(t)$ is a deterministic function of t.

Finally, note that a strictly stationary process with finite second-order moments is weakly stationary, but a weakly stationary process need not be strictly stationary. It is also worth noting that it is possible for a strictly stationary process not to be weakly stationary. This happens when the strictly stationary process does not have a second-order moment.

The simplest form of a covariance (or weakly) stationary process is the 'white noise process'.

Definition 18 *The process* $\{\varepsilon_t\}$ *is said to be a white noise process if it has mean zero, a constant variance, and* ε_t *and* ε_s *are uncorrelated for all* $s \neq t$.

More general stationary processes can be specified by considering linear combinations of white noise processes at different lags or leads.

12.3 Moving average processes

The process $\{y_t\}$ is said to have a finite moving average representation of order q if it can be written as

$$y_t = \sum_{i=0}^{q} a_i \varepsilon_{t-i},\ t = 0, \pm 1, \pm 2, \ldots, \tag{12.1}$$

where $\{\varepsilon_t\}$ is a white noise process with mean 0 and a constant variance σ^2, and $a_q \neq 0$. Recall that a white noise process is also serially uncorrelated, namely $E(\varepsilon_t \varepsilon_{t'}) = 0$, for all $t \neq t'$. It is easily seen that without loss of generality we can set $a_0 = 1$. This process is also referred to as a 'one-sided moving average process,' and distinguished from the two-sided representation

$$y_t = \sum_{i=-q}^{q} a_i \varepsilon_{t-i},\ t = 0, \pm 1, \pm 2, \ldots.$$

But by letting $\eta_t = \varepsilon_{t+q}$, the above two-sided process can be written as the one-sided moving average process

$$y_t = \sum_{i=0}^{2q} a_i^* \eta_{t-i},\ t = 0, \pm 1, \pm 2, \ldots,$$

where $a_i^* = a_{i-q}$. Therefore, in what follows we focus on the one-sided moving average process, (12.1), and simply refer to it as the moving average process of order q, denote by $MA(q)$.

It is often useful to write down the moving average process, (12.1), in terms of polynomial lag operators. Denote a first-order lag operator by L and note that by repeated application of the operator we have $L^i \varepsilon_t = \varepsilon_{t-i}$, where $L^0 = 1$, by convention. Then (12.1) can be written as

$$y_t = \left(\sum_{i=0}^{q} a_i L^i \right) \varepsilon_t = a_q(L) \varepsilon_t.$$

For a finite q an $MA(q)$ process is well defined for all finite values of the coefficients (weights) a_i and is covariance stationary. The autocovariance function of an $MA(q)$ process is given by

$$\begin{aligned} \gamma(h) = E(y_t y_{t+h}) &= \sigma^2 \sum_{i=0}^{q-|h|} a_i a_{i+|h|},\ \text{if } 0 \leq |h| \leq q, \\ &= 0,\ \text{for } |h| > q. \end{aligned} \tag{12.2}$$

Only the first q autocovariances of an $MA(q)$ process are non-zero. This is rather restrictive for many economic and financial time series, but can be relaxed by letting q tend to infinity. However, certain restrictions must be imposed on the coefficients $\{a_i\}$ for the infinite series $\sum_{i=0}^{q} a_i \varepsilon_{t-i}$ to converge to a well defined limit as $q \to \infty$.

Definition 19 *The sequence $\{a_i\}$ is said to be 'absolutely summable' if*

$$\sum_{i=0}^{\infty} \mid a_i \mid < \infty.$$

Definition 20 *The sequence $\{a_i\}$ is said to be 'square summable' if*

$$\sum_{i=0}^{\infty} a_i^2 < \infty.$$

It is easily seen that an absolutely summable sequence is also square summable, but the reverse is not true. For a proof, note that

$$\left(\sum_{i=0}^{\infty} \mid a_i \mid \right)^2 = \sum_{i=0}^{\infty} a_i^2 + 2 \sum_{i>j}^{\infty} \mid a_i \mid\mid a_j \mid > \sum_{i=0}^{\infty} a_i^2.$$

Hence $\left(\sum_{i=0}^{\infty} a_i^2 \right)^{1/2} < \sum_{i=0}^{\infty} \mid a_i \mid$ and $\sum_{i=0}^{\infty} a_i^2$ will be bounded if $\sum_{i=0}^{\infty} \mid a_i \mid < \infty$. To see that the reverse does not hold, note that the sequence $\left\{ \frac{1}{i+1} \right\}$ is square summable, namely

$$\sum_{i=0}^{\infty} \left(\frac{1}{i+1} \right)^2 < \infty,$$

but it is not absolutely summable since the series

$$\sum_{i=0}^{\infty} \frac{1}{i+1} = 1 + \frac{1}{2} + \frac{1}{3} + \ldots,$$

in fact diverges.

Definition 21 *The infinite moving average process is defined as*

$$y_t = \lim_{q \to \infty} \sum_{i=0}^{q} a_i \varepsilon_{t-i}. \tag{12.3}$$

Proposition 38 *The infinite moving average process exists in the mean squared error sense*

$$\lim_{q\to\infty} E\left(\left|y_t - \sum_{i=0}^{q} a_i\varepsilon_{t-i}\right|^2\right) = 0,$$

if the sequence $\{a_i\}$ *is square summable, with* $E(y_t^2) = \sigma^2 \sum_{i=0}^{\infty} a_i^2 < \infty$.[1]

Proposition 39 *The infinite moving average process converges almost surely to* $y_t = \sum_{i=0}^{\infty} a_i\varepsilon_{t-i}$ *if the sequence* $\{a_i\}$ *is absolutely summable.*[2]

Letting $q \to \infty$ in (12.2), the h-order autocovariance of the $MA(\infty)$ process is given by

$$\gamma(h) = \sigma^2 \sum_{i=0}^{\infty} a_i a_{i+|h|}. \tag{12.4}$$

Notice that $\gamma(h) = \gamma(-h)$, and hence $\gamma(h)$ is an even function of h. Scaling the autocovariance function by $\gamma(0)$ we obtain the autocorrelation function of order h, denote by $\rho(h)$

$$\rho(h) = \frac{\gamma(h)}{\gamma(0)} = \frac{\sum_{i=0}^{\infty} a_i a_{i+|h|}}{\sum_{i=0}^{\infty} a_i^2}.$$

Clearly $\rho(0) = 1$. It is also readily seen that $\rho^2(h) \leq 1$, for all h. For a proof first note that

$$0 \leq Var(y_t - \lambda y_{t-h}) = Var(y_t) + \lambda^2 Var(y_{t-h}) - 2\lambda Cov(y_t, y_{t-h}),$$

and since $\{y_t\}$ is covariance stationary then for all h and λ, we have

$$(1 + \lambda^2)\gamma(0) - 2\lambda\gamma(h) \geq 0.$$

Since this inequality holds for all values of λ, it should also hold for $\lambda_* = \gamma(h)/\gamma(0)$, which globally minimizes its left-hand side. Namely, we must also have

$$(1 + \lambda_*^2)\gamma(0) - 2\lambda_*\gamma(h) = \gamma(0) - \frac{\gamma^2(h)}{\gamma(0)} \geq 0,$$

and since $\gamma(0) > 0$, dividing the last inequality by $\gamma(0)$ we obtain $1 - \rho^2(h) \geq 0$, as desired. For a non-zero h the equality holds if and only if y_t is an exact linear function of y_{t-h}.

Finally, we observe that a linear stationary process with absolutely summable coefficients will have absolutely summable autocovariances. Consider the autocovariance function of the $MA(\infty)$ process given by (12.4). Hence

[1] See, for example, Theorem 2.2.3, p. 35, in Fuller (1996).
[2] See, for example, Theorem 2.2.1, p. 31, in Fuller (1996).

$$\sum_{h=0}^{\infty} |\gamma(h)| \leq \sigma^2 \sum_{h=0}^{\infty} \sum_{i=0}^{\infty} \left| a_i a_{i+|h|} \right| \leq \sigma^2 \sum_{i=0}^{\infty} |a_i| \sum_{h=0}^{\infty} \left| a_{i+|h|} \right|.$$

But, for each i,

$$\sum_{h=0}^{\infty} \left| a_{i+|h|} \right| \leq K < \infty,$$

where K is a fixed constant. Hence

$$\sum_{h=0}^{\infty} |\gamma(h)| \leq \sigma^2 K \sum_{i=0}^{\infty} |a_i| < \infty, \tag{12.5}$$

which establishes the desired result.

12.4 Autocovariance generating function

If $\{y_t\}$ is a process with autocovariance function $\gamma(h)$, then its autocovariance generating function is defined by

$$G(z) = \sum_{h=-\infty}^{\infty} \gamma(h) z^h.$$

In the case of stationary processes $\gamma(h) = \gamma(-h)$, and

$$G(z) = \gamma(0) + \sum_{h=1}^{\infty} \gamma(h)(z^h + z^{-h}).$$

Proposition 40 *Consider the infinite order MA process*

$$y_t = \sum_{i=-\infty}^{\infty} a_i \varepsilon_{t-i} = \left(\sum_{i=-\infty}^{\infty} a_i L^i \right) \varepsilon_t = a(L)\varepsilon_t.$$

Suppose that $\{a_i\}$ is an absolutely summable sequence. Then the autocovariance generating function of $\{y_t\}$ is given by

$$G(z) = \sigma^2 a(z) a(z^{-1}),$$

where

$$a(z) = \sum_{i=-\infty}^{\infty} a_i z^i.$$

For a proof first note that

$$y_t y_{t-h} = \left(\sum_{i=-\infty}^{\infty} a_i \varepsilon_{t-i} \right) \left(\sum_{j=-\infty}^{\infty} a_j \varepsilon_{t-h-j} \right)$$
$$= \sum_{i=-\infty}^{\infty} \sum_{j=-\infty}^{\infty} a_i a_j \varepsilon_{t-i} \varepsilon_{t-h-j}.$$

Multiplying both sides of the above relationship by z^h, summing over $h \in (-\infty, \infty)$ and taking expectations we have

$$G(z) = \sum_{h=-\infty}^{\infty} \sum_{i=-\infty}^{\infty} \sum_{j=-\infty}^{\infty} z^h a_i a_j E\left(\varepsilon_{t-i} \varepsilon_{t-h-j}\right).$$

But $E\left(\varepsilon_{t-i} \varepsilon_{t-h-j}\right)$ is non-zero only if $t - i = t - h - j$, or if $i = h + j$. Hence

$$G(z) = \sigma^2 \sum_{h=-\infty}^{\infty} \sum_{j=-\infty}^{\infty} z^h a_{h+j} a_j$$
$$= \sigma^2 \sum_{h=-\infty}^{\infty} \sum_{j=-\infty}^{\infty} z^{h+j} a_{h+j} z^{-j} a_j$$
$$= \sigma^2 \left(\sum_{s=-\infty}^{\infty} a_s z^s \right) \left(\sum_{j=-\infty}^{\infty} a_j z^{-j} \right).$$

The above proof is carried out for a two-sided *MA* process, but it applies equally to one-sided *MA* processes by setting $a_i = 0$ for $i < 0$. Also, there exist important relationships between the autocovariance generating function and the spectral density function which we shall discuss later.

In a number of time series applications, a stationary stochastic process is obtained from another stationary stochastic process through an infinite-order *MA* filtration. Examples includes consumption growth obtained from the growth of real disposable income, long term real interest rate from short term rates, or equity returns derived from dividend growths. The following proposition establishes the conditions under which such infinite-order filtrations exist and are stationary with absolutely summable autocovariances.

Proposition 41 *Consider the following two infinite moving average processes with absolute summable coefficients*

$$y_t = \sum_{i=0}^{\infty} a_i x_{t-i} = \left(\sum_{i=0}^{\infty} a_i L^i \right) x_t = a(L) x_t,$$

and

$$x_t = \sum_{i=0}^{\infty} b_i \varepsilon_{t-i} = \left(\sum_{i=0}^{\infty} b_i L^i \right) \varepsilon_t = b(L)\varepsilon_t,$$

where, as before, $\{\varepsilon_t\}$ *is a white noise process. Then*

$$\begin{aligned} y_t &= a(L)x_t = a(L)b(L)\varepsilon_t \\ &= c(L)\varepsilon_t = \sum_{i=0}^{\infty} c_i \varepsilon_{t-i}, \end{aligned}$$

where $\{c_i\}$ *is an absolute summable sequence, and* $\{y_t\}$ *is a stationary process with absolutely summable autocovariances.*

A proof is straightforward and follows by first observing that, since

$$c(L) = a(L)b(L),$$

then

$$\begin{aligned} c_0 &= a_0 b_0, \\ c_1 &= a_0 b_1 + a_1 b_0, \\ c_2 &= a_0 b_2 + a_1 b_1 + a_2 b_0, \\ &\vdots \\ c_i &= a_0 b_i + a_1 b_{i-1} + \ldots + a_i b_0, \text{ and so on.} \end{aligned}$$

Then

$$\sum_{i=0}^{\infty} c_i = a_0 \left(\sum_{i=0}^{\infty} b_i \right) + a_1 \left(\sum_{i=0}^{\infty} b_i \right) + \ldots,$$

or

$$\sum_{i=0}^{\infty} |c_i| \leq \left(\sum_{i=0}^{\infty} |a_i| \right) \left(\sum_{i=0}^{\infty} |b_i| \right),$$

which establishes the absolute summability of $\{c_i\}$ considering that $\sum_{i=0}^{\infty} |a_i| < K < \infty$, and $\sum_{i=0}^{\infty} |b_i| < K < \infty$. Also since $y_t = \sum_{i=0}^{\infty} c_i \varepsilon_{t-i}$, then by (12.5) it follows that $\{y_t\}$ has absolutely summable autocovariances.

12.5 Classical decomposition of time series

One important aim of time series analysis is to decompose a series into a number of components that can be associated with different types of temporal variations. Time series are often governed by the following four main components:

- long term *trend*
- *seasonal* component
- *cyclical* component
- *residual* component.

These four components are usually combined together using either an additive or a multiplicative model. The latter is often transformed into an additive structure using the log-transformation. Most statistical procedures are concerned with modelling of the cyclical component and usually take trend and seasonal patterns as given or specified *a priori* by the investigator. Further discussion can be found in Mills (1990, 2003).

The meaning and the importance of stationarity can be appreciated in the context of the famous decomposition theorem due to Wold (1938). Wold proved that any stationary process can be decomposed into the sum of a deterministic (perfectly predictable) and a purely non-deterministic (stochastic) component. More formally

Theorem 42 (Wold's decomposition) *Any trend-stationary process* $\{y_t\}$ *can be represented in the form of*

$$y_t = d_t + \sum_{i=0}^{\infty} \alpha_i \varepsilon_{t-i},$$

where $\alpha_0 = 1$, *and* $\sum_{i=0}^{\infty} \alpha_i^2 < K < \infty$. *The term* d_t *is a deterministic component, while* $\{\varepsilon_t\}$ *is a serially uncorrelated process defined by innovations in* y_t

$$\begin{aligned} \varepsilon_t &= y_t - E(y_t \mid \mathcal{I}_{t-1}), t = 1, 2, \ldots, \\ E\left(\varepsilon_t^2 \mid \mathcal{I}_{t-1}\right) &= \sigma^2 > 0, \\ E(\varepsilon_t d_s) &= 0, \textit{for all } s \textit{ and } t, \end{aligned}$$

where $\mathcal{I}_t = (y_t, y_{t-1}, y_{t-2,} \ldots)$.

In the above decomposition, ε_t is the error in the one step ahead forecast of y_t, and is also known as the 'innovation error'. As noted in Definition 17, the deterministic component, d_t, is also known as the perfectly predictable component of y_t, in the sense that $E(d_t \mid \mathcal{I}_{t-1}) = d_t$. Further discussion on Wold's decomposition theorem can be found in Nerlove, Grether, and Carvalo (1979) and in Brockwell and Davis (1991).

12.6 Autoregressive moving average processes

Wold's decomposition theorem also provides the basis for the approximation of trend-stationary processes by finite-order autoregressive-moving average specifications. Under the assumption that the coefficients α_i in $\alpha(z) = \sum_{i=0}^{\infty} \alpha_i z^i$ are absolutely summable, so that $\sum_{i=0}^{\infty} |\alpha_i| < K < \infty$, it is possible to approximate $\alpha(z)$ by a ratio of two finite-order polynomials, $\phi_p(z)/\theta_q(z)$, where $\phi_p(z) = \sum_{i=0}^{p} \phi_i z^i$ and $\theta_q(z) = \sum_{i=0}^{q} \theta_i z^i$, for sufficiently large, but with finite, p and q. This yields the general form of an $ARMA(p, q)$ process which is given by

$$y_t = \sum_{i=1}^{p} \phi_i y_{t-i} + \sum_{i=0}^{q} \theta_i \varepsilon_{t-i}, \qquad \theta_0 = 1. \tag{12.6}$$

It is easily seen that the *MA* part of the process, $u_t = \sum_{i=0}^{q} \theta_i \varepsilon_{t-i}$, is stationary for any finite q, and hence y_t is stationary if the *AR* part of the process is stationary. Consider the process

$$y_t = \sum_{i=1}^{p} \phi_i y_{t-i} + u_t,$$

and note that the general solution of y_t is given by

$$y_t^g = \sum_{i=1}^{p} A_i \lambda_i^t,$$

where A_i, for $i = 1, 2, \ldots, p$ are fixed arbitrary constants, and λ_i, $i = 1, 2, \ldots, p$ are distinct roots of the characteristic equation associated to the above difference equation, namely

$$\lambda^t = \sum_{i=1}^{p} \phi_i \lambda^{t-i}. \tag{12.7}$$

The above general solution assumes that the roots of this characteristic equation are distinct. More complicated solution forms follow when two or more roots are identical, but the main conclusions are unaffected by such complications. For the y_t process to be stationary it is necessary that all the roots of (12.7) lie strictly inside the unit circle. Alternatively, the condition can be written in terms of $z = \lambda^{-1}$, thus requiring that all the roots of

$$1 - \sum_{i=1}^{p} \phi_i z^i = 0, \tag{12.8}$$

lie outside the unit circle. The *ARMA* process is said to be invertible (so that y_t can be solved uniquely in terms of its past values) if all the roots of

$$1 - \sum_{i=1}^{p} \theta_i z^i = 0, \tag{12.9}$$

fall outside the unit circle.

12.6.1 Moving average processes

The $MA(1)$ is given by

$$y_t = \varepsilon_t + \theta \varepsilon_{t-1}, \qquad \varepsilon_t \sim \left(0, \sigma^2\right).$$

Its autocorrelation function is

$$\rho(1) = \frac{\theta}{1+\theta^2}, \quad \text{and,}$$
$$\rho(h) = 0, \text{ for } h > 1.$$

It is easily seen that, for a given value of $\rho(1)$, the moving average parameter, θ, is not unique; for any choice of θ, its inverse also satisfies the relationship, $\rho(1) = \theta/(1+\theta^2) = \theta^{-1}/(1+\theta^{-2})$. Also notice that for obtaining a real-valued solution θ in terms of $\rho(1)$, it must be that $|\rho(1)| \leq \frac{1}{2}$. Similar conditions apply to the more general $MA(q)$ process defined by

$$y_t = \varepsilon_t + \theta_1\varepsilon_{t-1} + \theta_2\varepsilon_{t-2} + \cdots + \theta_p\varepsilon_{t-q}.$$

12.6.2 *AR* processes

First, consider the first-order autoregressive process, denoted by $AR(1)$

$$y_t = \phi y_{t-1} + \varepsilon_t, \qquad \varepsilon_t \sim \left(0, \sigma^2\right).$$

This process is stationary if $|\phi| < 1$. Under this condition y_t can be written as an infinite MA process with absolutely summable coefficients

$$y_t = \sum_{i=0}^{\infty} \phi^i \varepsilon_{t-i} = \left(\frac{1}{1-\phi L}\right)\varepsilon_t.$$

Therefore, using results in Proposition 40, the autocovariance generating function of the $AR(1)$ process is given by

$$\begin{aligned} G(z) &= \sigma^2 \left(\frac{1}{1-\phi z}\right)\left(\frac{1}{1-\phi z^{-1}}\right) \\ &= \sigma^2 \left(1 + \phi z + \phi^2 z^2 + \ldots\right)\left(1 + \phi z^{-1} + \phi^2 z^{-2} + \ldots\right) \\ &= \left(\frac{\sigma^2}{1-\phi^2}\right)\left[1 + \sum_{h=1}^{\infty} \phi^h (z^h + z^{-h})\right]. \end{aligned}$$

Hence

$$\text{Autocovariance function} : \gamma(h) = \frac{\sigma^2 \phi^{|h|}}{1-\phi^2},$$
$$\text{Autocorrelation function} : \rho(h) = \phi^{|h|}.$$

These results readily extend to a general order $AR(p)$ process

$$y_t = \phi_1 y_{t-1} + \phi_2 y_{t-2} + \cdots + \phi_p y_{t-p} + \varepsilon_t, \tag{12.10}$$

or in terms of lag operators

$$\phi(L)y_t = \varepsilon_t,$$

where

$$\phi(L) = 1 - \phi L - \phi_2 L^2 - \cdots - \phi_p L^p.$$

To derive the conditions under which this process is stationary it is convenient to consider its so-called *companion form* as the following first-order vector autoregressive process

$$\mathbf{y}_t = \mathbf{\Phi}\mathbf{y}_{t-1} + \boldsymbol{\xi}_t,$$

where

$$\mathbf{y}_t = \begin{pmatrix} y_t \\ y_{t-1} \\ \vdots \\ y_{t-p+1} \end{pmatrix}, \mathbf{\Phi} = \begin{pmatrix} \phi_1 & \phi_2 & \cdots & \phi_p \\ 1 & 0 & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & \cdots & 1 & 0 \end{pmatrix}, \boldsymbol{\xi}_t = \begin{pmatrix} \varepsilon_t \\ 0 \\ \vdots \\ 0 \end{pmatrix}.$$

Solving for y_t from the initial value y_0 we have

$$\mathbf{y}_t = \mathbf{\Phi}^t \mathbf{y}_0 + \sum_{j=0}^{t-1} \mathbf{\Phi}^j \boldsymbol{\xi}_{t-j}.$$

Therefore, the necessary condition for the y_t process to be stationary is that

$$\lim_{t\to\infty} \mathbf{\Phi}^t = 0.$$

This condition is satisfied if all the eigenvalues of the companion matrix, $\mathbf{\Phi}$, lie inside the unit circle, which is equivalent to the absolute values of all the roots of $\phi(z) = 0$ being strictly larger than unity (see (12.8)). Under this condition the *AR* process has the following infinite-order *MA* representation

$$y_t = \sum_{i=0}^{\infty} \alpha_i \varepsilon_{t-i} = \alpha(L)\varepsilon_t,$$

where

$$\alpha(L)\phi(L) \equiv 1,$$

or, more explicitly,

$$\alpha_i = \phi_1 \alpha_{i-1} + \phi_2 \alpha_{i-2} + \cdots + \phi_p \alpha_{i-p},$$

for $i = 1, 2, \ldots$, with $\alpha_0 = 1$, $\alpha_i = 0$, for $i < 0$. This is a deterministic difference equation and its solution is given in terms of the roots of $\phi(z) = 0$. Under stationarity, the *MA* coefficients, α_i, are bounded by a geometrically declining sequence and $\{\alpha_i\}$ themselves will be an absolutely summable sequence, and the general results of the previous sections will be directly applicable.

These results readily generalize to stationary *ARMA* processes. Further, due to the geometrically declining nature of α_i's it also follows that

$$\sum_{i=1}^{\infty} i\,|\alpha_i| < \infty, \text{ and } \sum_{i=1}^{\infty} i\alpha_i^2 < \infty. \tag{12.11}$$

To see why, suppose that $|\alpha_i| < K\rho^i$, where $0 < \rho < 1$, and K is a positive finite constant. Then

$$\begin{aligned}\sum_{i=1}^{\infty} i\,|\alpha_i| &< K\sum_{i=1}^{\infty} i\rho^i = K\rho\sum_{i=1}^{\infty} i\rho^{i-1} \\ &= K\rho\frac{d}{d\rho}\left(\sum_{i=1}^{\infty}\rho^i\right) = K\rho\frac{d}{d\rho}\left(\frac{\rho}{1-\rho}\right) \\ &= \frac{K\rho}{(1-\rho)^2} < \infty.\end{aligned}$$

The second result in (12.11) can be established similarly.

The autocovariance generating function of the $AR(p)$ process is given by

$$G(z) = \frac{\sigma^2}{\phi(z)\phi(z^{-1})}.$$

This result can now be used to derive the autocovariances of the $AR(p)$ process. But a simpler approach would be to use the Yule–Walker equations, which can be readily obtained by premultiplying (12.10) with $y_t, y_{t-1}, y_{t-2}, \ldots, y_{t-p}$, and then taking expectations. Namely

$$y_{t-h}y_t = \phi_1 y_{t-h}y_{t-1} + \phi_2 y_{t-h}y_{t-2} + \ldots \phi_p y_{t-h}y_{t-p} + y_{t-h}\varepsilon_t, \text{ for } h = 0, 1, 2, \ldots.$$

Taking expectations of both sides of this relation, and recalling that under stationarity $\gamma(h) = \gamma(-h)$, we have

$$\gamma(h) = \phi_1\gamma(h-1) + \phi_2\gamma(h-2) + \ldots + \phi_p\gamma(h-p) + E(y_{t-h}\varepsilon_t), \text{ for } h = 0, 1, 2, \ldots.$$

But using the infinite *MA* representation of the stationary *AR* process we have

$$\begin{aligned}E(y_{t-h}\varepsilon_t) &= \sigma^2, \text{ for } h = 0, \\ &= 0, \text{ for } h > 0.\end{aligned}$$

Therefore, we have

$$\gamma(0) = \phi_1\gamma(1) + \phi_2\gamma(2) + \ldots + \phi_p\gamma(p) + \sigma^2, \tag{12.12}$$

and, for $h = 1, 2, \ldots, p$,

$$\gamma(h) = \phi_1\gamma(h-1) + \phi_2\gamma(h-2) + \ldots + \phi_p\gamma(h-p). \tag{12.13}$$

The system of equations (12.12) and (12.13) is known as the Yule–Walker equations and can be used in two ways: solving for the autocovariances, recursively, from the autoregressive coefficients; and for a consistent estimation of the former in terms of the latter. Writing (12.13) in matrix notation we have

$$\begin{pmatrix} \gamma(0) & \gamma(1) & \ldots & \gamma(p-1) \\ \gamma(1) & \gamma(0) & \ldots & \gamma(p-2) \\ \vdots & \vdots & \ldots & \vdots \\ \gamma(p-1) & \gamma(p-2) & \ldots & \gamma(0) \end{pmatrix} \begin{pmatrix} \phi_1 \\ \phi_2 \\ \vdots \\ \phi_p \end{pmatrix} = \begin{pmatrix} \gamma(1) \\ \gamma(2) \\ \vdots \\ \gamma(p) \end{pmatrix},$$

which can be used to compute the autoregressive coefficients, ϕ_i, in terms of the autocovariances. Alternatively, we have

$$\begin{pmatrix} 1 & -\phi_1 & \cdots & -\phi_p \\ -\phi_1 & 1 & \cdots & -\phi_{p-1} \\ \vdots & \vdots & \vdots & \vdots \\ -\phi_p & -\phi_{p-1} & \cdots & 1 \end{pmatrix} \begin{pmatrix} \gamma(0) \\ \gamma(1) \\ \vdots \\ \gamma(p) \end{pmatrix} = \begin{pmatrix} \sigma^2 \\ 0 \\ \vdots \\ 0 \end{pmatrix},$$

which can be used to solve for autocovariances in terms of the autoregressive coefficients. For example, in the case of the $AR(2)$ process we have

$$\phi_1 = \frac{\gamma(0)\gamma(1) - \gamma(1)\gamma(2)}{\gamma^2(0) - \gamma^2(1)}$$
$$\phi_2 = \frac{\gamma(0)\gamma(2) - \gamma^2(1)}{\gamma^2(0) - \gamma^2(1)},$$

and

$$\gamma(0) = \frac{(1-\phi_2)\sigma^2}{(1+\phi_2)\left[(1-\phi_2)^2 - \phi_1^2\right]},$$
$$\gamma(1) = \frac{\phi_1\sigma^2}{(1+\phi_2)\left[(1-\phi_2)^2 - \phi_1^2\right]},$$
$$\gamma(2) = \frac{\left[\phi_1^2 + \phi_2(1-\phi_2)\right]\sigma^2}{(1+\phi_2)\left[(1-\phi_2)^2 - \phi_1^2\right]}.$$

Using the above expressions for $\gamma(1)$ and $\gamma(2)$ as initial conditions, the difference equation (12.13) can now be used to solve for higher-order autocovariances either recursively or directly in terms of the roots of (12.8). Assuming the roots of (12.8) are distinct (real or complex) and denoting the inverse of these roots by λ_1 and λ_2 we have

$$\rho(h) = \frac{\gamma(h)}{\gamma(0)} = \frac{(1-\lambda_2^2)\lambda_1^{h+1} - (1-\lambda_1^2)\lambda_2^{h+1}}{(1+\lambda_1\lambda_2)(\lambda_1 - \lambda_2)}.$$

It now readily follows that

$$|\rho(h)| \leq K \max\left(|\lambda_1|^h, |\lambda_2|^h\right),$$

for a fixed constant K. Hence,

$$\sum_{h=1}^{\infty} |\rho(h)| < \infty.$$

Notice also that the stability conditions on the roots of $1 - \phi_1 z - \phi_2 z^2 = 0$ are satisfied if

$$\begin{aligned} 1 - \phi_2 - \phi_1 &> 0, \\ 1 - \phi_2 + \phi_1 &> 0, \\ 1 + \phi_2 &> 0. \end{aligned}$$

Under these conditions $\gamma(0) > 0$, as to be expected.

Stationary *ARMA* processes have been used extensively in the time series literature. An important reference in this literature is the book by Box and Jenkins (1970) that contains a unified treatment of the estimation and the model selection problems associated with univariate stationary *ARMA* processes. Other important references are Whittle (1963), Rozanov (1967), Hannan (1970), Anderson (1971), Granger and Newbold (1977), Nerlove, Grether, and Carvalo (1979), Priestley (1981), Brockwell and Davis (1991), Hannan and Deistler (1988), and Harvey (1989). More introductory expositions can be found in Chatfield (2003), Hamilton (1994), and Davidson (2000).

12.7 Further reading

Further discussion on stationary processes can be found in Mills (1990), Hamilton (1994), Davidson (2000), and Chatfield (2003). Estimation of stationary processes is considered in Chapter 14, while the problem of forecasting is discussed in Chapter 17.

12.8 Exercises

1. Which of the following autoregressive processes are stationary?

   (a) $y_t = 3 - 0.30y_{t-1} + 0.04y_{t-2} + \varepsilon_t$,

(b) $y_t = 5 + 3.10y_{t-1} - 0.3y_{t-2} + \varepsilon_t$,

(c) $y_t = 2.5y_{t-1} - 2y_{t-2} + 0.5y_{t-3} + \varepsilon_t$,

where $E(\varepsilon_t^2) = 1, E(\varepsilon_t\varepsilon_s) = 0$, for $t \neq s$.

2. Derive the autocorrelation function of

$$y_t = \phi y_{t-1} + \varepsilon_t + \theta\varepsilon_{t-1},$$

assuming $|\phi| < 1$, and $E(\varepsilon_t^2) = \sigma^2$, and $E(\varepsilon_t\varepsilon_s) = 0$, for $t \neq s$.

3. Find the MA representation of the $AR(2)$ process

$$y_t = \rho_1 y_{t-1} + \rho_2 y_{t-2} + \varepsilon_t.$$

Consider the following *stationary* second-order autoregressive model

$$x_t = \mu + \rho_1 x_{t-1} + \rho_2 x_{t-2} + \varepsilon_t. \tag{12.14}$$

(a) Show that

$$x_t = \frac{\mu}{1 - \rho_1 - \rho_2} + \sum_{i=0}^{\infty} \alpha_i \varepsilon_{t-i},$$

where $\varepsilon_t \sim IID(0, \sigma^2)$, and the coefficients $\{\alpha_i\}$ are absolute summable.

(b) Let

$$y_t = x_t - \frac{\mu}{1 - \rho_1 - \rho_2}.$$

and show that (12.14) can be written as

$$y_t = \rho_1 y_{t-1} + \rho_2 y_{t-2} + \varepsilon_t,$$

(c) Using the above result, or otherwise, derive the following Yule–Walker equations for the $AR(2)$ model in (12.14)

$$\begin{aligned}
\gamma_0 &= \rho_1 \gamma_1 + \rho_2 \gamma_2 + \sigma^2, \\
\gamma_1 &= \rho_1 \gamma_0 + \rho_2 \gamma_1, \\
\gamma_2 &= \rho_1 \gamma_1 + \rho_2 \gamma_0,
\end{aligned}$$

where γ_s is the sth-order autocovariance of x_t.

4. Let

$$y_{1t} = \varepsilon_t - 0.8\varepsilon_{t-1},$$

and

$$y_{2t} = u_t - 0.9u_{t-1},$$

where $\{\varepsilon_t\}$ is a sequence of independent $(0,1)$ random variables distributed independently of $\{u_t\}$ which is a sequence of independent $(0,6)$ random variables. Express $y_t = y_{1t} + y_{2t}$ as a moving average process.

5. Let $\{y_t\}$ be the process

$$(1 - 0.7L^2)y_t = (1 + 0.3L^2)\varepsilon_t,$$

where $\{\varepsilon_t\}$ is a white noise process with zero mean and variance $\sigma^2 = 1$.

   (a) Find the coefficients $\{\psi_j\}$ in the representation $y_t = \sum_{j=0}^{\infty} \psi_j \varepsilon_{t-j}$.
   (b) Find the coefficients $\{\pi_j\}$ in the representation $\varepsilon_t = \sum_{j=0}^{\infty} \pi_j y_{t-j}$.
   (c) Graph the autocorrelation function of $\{y_t\}$.
   (d) Simulate y_t, $t = 1, 2, \ldots, 150$, assuming that ε_t's are normally distributed, and compare the sample autocorrelation function with the population autocorrelation function in (c).

6. Define the following concepts and discuss their relationships if any.

   (a) Strict stationarity
   (b) Covariance stationarity
   (c) Ergodicity in mean.

7. Consider the scalar process

$$x_t = \mu + \alpha(L)\varepsilon_t,\ \varepsilon_t \backsim IID(0, \sigma^2),$$

where

$$\alpha(L) = \alpha_0 + \alpha_1 L + \alpha_2 L^2 + \ldots,$$

$\alpha_0 = 1$ and L is a lag operator, $Lx_t = x$. Derive the conditions under which $\{x_t\}$ is

   (a) Covariance stationary
   (b) Mean ergodic
   (c) Assuming $\{x_t\}$ is covariance stationary write down its autocovariance generating function.

8. Consider the AR(1) process

$$y_t = \phi y_{t-1} + \varepsilon_t,\ \varepsilon_t \sim IIDN(0, \sigma^2),$$

where $|\phi| < 1$.

(a) Obtain the log-likelihood function of the model assuming (i) y_0 is fixed, or (ii) y_0 is stochastic with mean zero and variance $\sigma^2/(1-\phi^2)$.

(b) Show that the *ML* estimator of ϕ is guaranteed to be in the range $|\phi| < 1$, only under the stochastic initial value case.

9. Prove that linear combinations of a finite number of covariance stationary processes is covariance stationary. Under what conditions does this result hold if the number of stationary processes under consideration tends to infinity?

13 Spectral Analysis

13.1 Introduction

Spectral analysis provides an alternative to the time domain approach to time series analysis. This approach views a stochastic process as a weighted sum of the periodic functions $\sin(\cdot)$ and $\cos(\cdot)$ with different frequencies, namely

$$y_t = \mu + \sum_{j=1}^{m} \left[a_j \cos\left(\omega_j t\right) + b_j \sin(\omega_j t) \right], \tag{13.1}$$

where ω denotes *frequency* in the range $(-\pi, \pi)$, ω_j denotes a particular realization of ω, a_j and b_j are the weights attached to different sine and cosine waves, and m is the window size. The above specification explicitly models y_t as a weighted average of sine and cosine functions rather than lagged values of y_t. Any covariance stationary process has both a time domain and a frequency domain representation, and any feature of the data that can be described by one representation can equally be described by the other. The frequency domain approach, or spectral analysis, is concerned with determining the importance of cycles of different frequencies for the variations of y_t over time.

13.2 Spectral representation theorem

We now derive the conditions under which y_t given by (13.1) represents a stationary process. Taking ω_j, for $j = 1, 2, \ldots, m$, as fixed parameters, it is easily seen that y_t is covariance stationary if a_j and b_j are independently distributed across j and of each other with mean zero and constant variances, σ_j^2. Note that (see Chapter 12 for a definition of autocovariance functions)

$$\gamma(0) = E(y_t - \mu)^2 = \sum_{j=1}^{m} \left[E(a_j^2) \cos^2(\omega_j t) + E(b_j^2) \sin^2(\omega_j t) \right]$$

$$= \sum_{j=1}^{m} \sigma_j^2 \left[\cos^2(\omega_j t + \sin^2(\omega_j t)\right]$$
$$= \sum_{j=1}^{m} \sigma_j^2,$$

and similarly

$$\begin{aligned}
\gamma(h) &= E\left[(y_t - \mu)(y_{t+h} - \mu)\right] \\
&= E \sum_{j=1}^{m} \sum_{j'=1}^{m} \left[a_j \cos\left(\omega_j t\right) + b_j \sin(\omega_j t)\right] \left[a_{j'} \cos\left(\omega_{j'}(t+h)\right) + b_{j'} \sin(\omega_{j'}(t+h))\right] \\
&= E \sum_{j=1}^{m} \left[a_j \cos\left(\omega_j t\right) + b_j \sin(\omega_j t)\right] \left[a_j \cos\left(\omega_j (t+h)\right) + b_j \sin(\omega_j (t+h))\right] \\
&= \sum_{j=1}^{m} \sigma_j^2 \left[\cos\left(\omega_j t\right) \cos\left(\omega_j (t+h)\right) + \sin(\omega_j t) \sin(\omega_j (t+h))\right],
\end{aligned}$$

or

$$\gamma(h) = \sum_{j=1}^{m} \sigma_j^2 \cos\left(\omega_j h\right) = \sum_{j=1}^{m} \sigma_j^2 \cos\left(-\omega_j h\right). \tag{13.2}$$

Clearly, $\gamma(h) = \gamma(-h)$, and it readily follows that y_t is a covariance stationary process.

It is also possible to consider the reverse problem and derive the frequency specific variances, σ_j^2, in terms of the autocovariances. In principle, the unknown variances, σ_j^2, associated with the individual frequencies, ω_j, can be estimated from the estimates of the autocovariances, $\gamma(h)$, $h = 0, 1, \ldots$. For example, for $m = 3$ we have

$$\begin{aligned}
\gamma(0) &= \sigma_1^2 + \sigma_2^2 + \sigma_3^2, \\
\gamma(1) &= \sigma_1^2 \cos(\omega_1) + \sigma_2^2 \cos(\omega_2) + \sigma_3^2 \cos(\omega_3), \\
\gamma(2) &= \sigma_1^2 \cos(2\omega_1) + \sigma_2^2 \cos(2\omega_2) + \sigma_3^2 \cos(2\omega_3),
\end{aligned}$$

for given choices of ω_1, ω_2, and ω_3 in the frequency range $(0, \pi)$. However, this is a rather cumbersome approach and other alternative procedures using Fourier transforms of the autocovariance functions have been explored in the literature. This idea is formalized in the following definition.

Definition 22 (Spectral density) *Let $\{y_t\}$ be a stationary stochastic process, and let $\gamma(h)$ be its autocovariance function of order h. The spectral density function associated to $\gamma(h)$ is defined by the infinite-order Fourier transform*

$$f(\omega) = \frac{1}{2\pi} \sum_{h=-\infty}^{\infty} \gamma(h)\, e^{ih\omega}, \quad \omega \in (-\pi, \pi), \tag{13.3}$$

where $e^{ih\omega} = \cos(h\omega) + i \sin(h\omega)$, *and* $i = \sqrt{-1}$ *is a complex number.*

Using the one-to-one relationship that exists between the spectral density function, $f(\omega)$, and the autocovariances $\gamma(h)$, we also have

$$\gamma(h) = \int_{-\pi}^{+\pi} f(\omega)\, e^{i\omega h} d\omega,$$

or, equivalently,

$$\gamma(h) = \int_{-\pi}^{+\pi} f(\omega) \cos(\omega h) d\omega. \tag{13.4}$$

This last result corresponds to $\gamma(h) = \sum_{j=1}^{m} \sigma_j^2 \cos\left(\omega_j h\right)$, obtained using the trigonometric representation (13.1) and (13.2).

13.3 Properties of the spectral density function

The spectral density function satisfies the following properties

1. $f(\omega)$ always exists and is bounded if $\gamma(h)$ is absolutely summable. Since $e^{ih\omega} = \cos(h\omega) + i \sin(h\omega)$, then $\left|e^{ih\omega}\right|^2 = \cos^2(h\omega) + \sin^2(h\omega) = 1$, and

$$\begin{aligned} \left|f(\omega)\right| &\leq \frac{1}{2\pi} \sum_{h=-\infty}^{\infty} \left|\gamma(h)\, e^{ih\omega}\right| \\ &\leq \frac{1}{2\pi} \sum_{h=-\infty}^{\infty} \left|\gamma(h)\right| \left|e^{ih\omega}\right| = \frac{1}{2\pi} \sum_{h=-\infty}^{\infty} \left|\gamma(h)\right| < \infty. \end{aligned}$$

2. $f(\omega)$ is symmetric. This follows from

$$\begin{aligned} f(-\omega) &= \frac{1}{2\pi} \sum_{h=-\infty}^{\infty} \gamma(h)\, e^{-ih\omega} \\ &= \frac{1}{2\pi} \sum_{s=-\infty}^{\infty} \gamma(-s)\, e^{is\omega}, \end{aligned}$$

if we let $h = -s$. Since for stationary processes we have $\gamma(-s) = \gamma(s)$, then

$$f(-\omega) = \frac{1}{2\pi} \sum_{s=-\infty}^{\infty} \gamma(s) e^{is\omega} = f(\omega),$$

hence $f(-\omega) = f(\omega)$. This shows that $f(\omega)$ is symmetric around $\omega = 0$. Thus the spectral density function can also be written as

$$f(\omega) = \frac{1}{2}[f(\omega) + f(-\omega)],$$

or, upon using (13.3), we have (noting that $e^{i\omega h} + e^{-i\omega h} = 2\cos(\omega h)$)

$$\begin{aligned} f(\omega) &= \frac{1}{2}\left[\frac{1}{2\pi}\sum_{h=-\infty}^{\infty}\gamma(h)e^{ih\omega} + \frac{1}{2\pi}\sum_{h=-\infty}^{\infty}\gamma(h)e^{-ih\omega}\right] \\ &= \frac{1}{2}\frac{1}{2\pi}\sum_{h=-\infty}^{\infty}\gamma(h)\left(e^{ih\omega} + e^{-ih\omega}\right) \\ &= \frac{1}{2\pi}\sum_{h=-\infty}^{\infty}\gamma(h)\cos(h\omega), \end{aligned}$$

or

$$f(\omega) = \frac{1}{2\pi}\left[\gamma(0) + 2\sum_{h=1}^{\infty}\gamma(h)\cos(h\omega)\right], \text{ with } \omega \in [0, \pi]. \tag{13.5}$$

Standardizing $f(\omega)$ by $\gamma(0)$ we also have

$$f^*(\omega) = \frac{1}{2\pi}\left[1 + 2\sum_{h=1}^{\infty}\rho(h)\cos(h\omega)\right],$$

where $\rho(h) = \gamma(h)/\gamma(0)$.

3. The spectrum of a stationary process is finite at zero frequency, namely

$$f(0) = \frac{1}{2\pi}\left[\gamma(0) + 2\sum_{h=1}^{\infty}\gamma(h)\right],$$

which is bounded since by assumption the autocovariances, $\gamma(h)$, are absolute summable.

4. Spectral decomposition of the variance. Using (13.4), and the symmetry property of the spectrum we first note that[1]

[1] This result can be obtained directly by deriving the integral $\int_0^{\pi} f(\omega)d\omega$ with $f(\omega)$ given by (13.5).

$$\gamma(0) = 2\int_0^{\pi} f(\omega)\, d\omega.$$

Consider now the frequencies $\omega_j = j\pi/m$, for $j = 0, 1, \ldots, m$ and approximate $\gamma(0)$ by

$$2\left\{\int_{\omega=0}^{\pi/m} f(\omega)\, d\omega + \int_{\omega=\pi/m}^{2\pi/m} f(\omega)\, d\omega + \ldots + \int_{\omega=(m-1)\pi/m}^{\pi} f(\omega)\, d\omega\right\}.$$

Since $f(\omega) \geq 0$, the term $2\,\gamma^{-1}(0)\int_{\omega=(j-1)\pi/m}^{j\pi/m} f(\omega)\, d\omega$ can be viewed as the proportion of the variance explained by the frequency $\omega_j = j\pi/m$. Compare this result with the decomposition $\gamma(0) = \sum_{j=1}^{m}\sigma_j^2$, for the trigonometric representation where σ_j^2 represents the variance associated with frequency ω_j.

13.3.1 Relation between $f(\omega)$ and autocovariance generation function

There is a one-to-one relation between $f(\omega)$ and autocovariance generation function. Recall that the autocovariance generation function is defined by $G(z) = \sum_{h=-\infty}^{\infty}\gamma(h)z^h$. For the general linear stationary process, y_t, defined by

$$y_t = \sum_{i=0}^{\infty} a_i\varepsilon_{t-i},$$

where $\varepsilon_t \sim IID(0, \sigma^2)$, and $\{a_i\}$ is an absolute summable sequence, then by Proposition 40 the autocovariance generating function of $\{y_t\}$ is given by

$$G(z) = \sigma^2 a(z)a(z^{-1}),$$

where $a(z) = \sum_{i=0}^{\infty} a_i z^i$. The spectral density function of $\{y_t\}$ can now be obtained from $G(z)$ by evaluating it at $z = e^{i\omega}$. More specifically

$$f(\omega) = \frac{1}{2\pi}G(e^{i\omega}) = \frac{\sigma^2}{2\pi}a(e^{i\omega})a(e^{-i\omega}). \tag{13.6}$$

We now calculate the spectral density function for various processes.

Suppose first, that y_t is a white noise process. In this case, $\gamma_0 = \sigma^2$ and $\gamma_k = 0$ for $k \neq 0$. It follows that $f(\omega)$ is flat at σ^2/π for all $\omega \in [0, \pi]$. Consider the stationary $AR(1)$ process $y_t = \phi y_{t-1} + \varepsilon_t$, where $|\phi| < 1$, and $\varepsilon_t \sim IID\left(0, \sigma^2\right)$. Since $\gamma(h) = \sigma^2\phi^{|h|}/\left(1-\phi^2\right)$, by direct methods we have

$$\begin{aligned} f(\omega) &= \frac{1}{2\pi}\sum_{h=-\infty}^{\infty}\frac{\sigma^2\phi^{|h|}}{\left(1-\phi^2\right)}e^{ih\omega} \\ &= \frac{1}{2\pi}\sum_{h=-\infty}^{\infty}\frac{\sigma^2\phi^{|h|}}{\left(1-\phi^2\right)}\left(e^{i\omega}\right)^h \end{aligned} \tag{13.7}$$

$$= \frac{1}{2\pi} \sum_{h=-\infty}^{\infty} \gamma(h) z^h,$$

with $z = e^{i\omega}$.

We can also express $f(\omega)$ as a real valued function of ω. From (13.7) we have

$$\begin{aligned} f(\omega) &= \frac{1}{2\pi} \frac{\sigma^2}{(1-\phi^2)} \left[1 + \sum_{h=-\infty}^{-1} \phi^{|h|} e^{-i\omega h} + \sum_{h=1}^{\infty} \phi^h e^{i\omega h} \right] \\ &= \frac{1}{2\pi} \frac{\sigma^2}{(1-\phi^2)} \left[1 + \sum_{h=1}^{\infty} \phi^h \left(e^{i\omega h} + e^{-i\omega h} \right) \right]. \end{aligned} \tag{13.8}$$

Since $\left|\phi e^{i\omega h}\right| = |\phi| < 1$, the infinite series in the above expression converge and we have

$$f(\omega) = \frac{1}{2\pi} \frac{\sigma^2}{(1-\phi^2)} \left(1 + \frac{\phi e^{i\omega}}{1-\phi e^{i\omega}} + \frac{\phi e^{-i\omega}}{1-\phi e^{-i\omega}} \right),$$

or

$$f(\omega) = \frac{1}{2\pi} \frac{\sigma^2}{\left(1-\phi e^{i\omega}\right)\left(1-\phi e^{-i\omega}\right)}.$$

The same result can also be obtained directly using the autocovariance generating function which for the $AR(1)$ process is given by

$$G(z) = \frac{\sigma^2}{(1-\phi z)\left(1-\phi z^{-1}\right)},$$

using (13.6) the spectral density function of the y_t process can be obtained directly as

$$f(\omega) = \frac{1}{2\pi} \frac{\sigma^2}{\left(1-\phi e^{i\omega}\right)\left(1-\phi e^{-i\omega}\right)}, \text{ for } \omega \in [0, \pi],$$

or

$$f(\omega) = \frac{1}{2\pi} \frac{\sigma^2}{\left(1-2\phi\cos(\omega)+\phi^2\right)}. \tag{13.9}$$

Note that, for $\phi > 0, f(\omega)$ is monotonically decreasing in ω over $[0, \pi]$, while for $\phi < 0, f(\omega)$ is monotonically increasing in ω. Using (13.8) and the result

$$e^{i\omega h} + e^{-i\omega h} = 2\cos(\omega h),$$

we also have

$$f(\omega) = \frac{1}{2\pi} \frac{\sigma^2}{\left(1-\phi^2\right)} \left(1 + 2\sum_{h=1}^{\infty} \phi^h \cos(\omega h)\right),$$

which identifies the coefficients of $\cos(\omega h)$, apart from the scaling factor of $1/2\pi$, as the order-h autocovariances of the $\{y_t\}$ process.

The above result readily generalizes to higher-order processes. For example, the spectral density function of the $ARMA(p, q)$ process

$$\phi(L) y_t = \theta(L)\varepsilon_t,$$

is given by

$$f(\omega) = \frac{1}{2\pi} \frac{\sigma^2 \theta(e^{i\omega})\theta(e^{-i\omega})}{\phi(e^{i\omega})\phi(e^{-i\omega})}, \text{ for } \omega \in [0, 2\pi].$$

13.4 Spectral density of distributed lag models

We wish often to study a stochastic process which is generated by another stochastic process. Let $\{y_t\}$ be an infinite moving average process

$$y_t = \sum_{i=0}^{\infty} a_i x_{t-i} = a(L) x_t, \tag{13.10}$$

with $\{x_t\}$ itself a stationary $MA(\infty)$ process

$$x_t = \sum_{j=0}^{\infty} b_j \varepsilon_{t-i} = b(L)\varepsilon_t, \;\; \varepsilon_t \sim IID\left(0, \sigma^2\right), \tag{13.11}$$

where we assume that x_t is absolutely summable. x_t is the input process, y_t is the output process, and (13.10) is the distributed lag or the transfer function. From (13.10) and (13.11) we can write

$$y_t = a(L)\, b(L)\, \varepsilon_t.$$

Let $c(L) = a(L)\, b(L)$, then

$$\begin{aligned} f_y(\omega) &= \frac{\sigma^2}{2\pi} c\left(e^{i\omega}\right) c\left(e^{-i\omega}\right), \;\; \omega \in (0, \pi) \\ &= a\left(e^{i\omega}\right) a\left(e^{-i\omega}\right) f_x(\omega). \end{aligned}$$

Evaluating at $\omega = 0$,

$$f_y(0) = [a(1)]^2 f_x(0). \tag{13.12}$$

If we consider (13.10) as a filter, from (13.12) we observe the above filter changes the memory of the stochastic process. Since $[a(1)]^2 < \infty$, and x_t absolutely summable implies $f_x(0) < \infty$, we get $f_y(0) < \infty$. This shows that if the input is a stationary stochastic process, then the output process is also stationary. On the other hand, $a(1)$ shows the degree and direction of the changes in memory through the filter. If $a(1) > 1$, the filter increases the memory of the output process, the larger the $a(1)$, the more persistent will be the output process. If $a(1) < 1$, then the filter decreases the memory of the output process. If $a(1) = 1$, the filter does not affect the memory of the output process.

13.5 Further reading

For an introductory account of spectral analysis, see Chatfield (2003), and for a more advanced treatment, see Priestley (1981).

13.6 Exercises

1. Derive the spectral density function of the stationary autoregressive process

$$y_t = \phi_1 y_{t-1} + \phi_2 y_{t-2} + \cdots + \phi_p y_{t-p} + \varepsilon_t,$$

   where ε_t are uncorrelated, zero-mean random variables with variance σ^2.
2. Consider the stationary process y_t with spectral density function $f_y(\omega)$ which is bounded for all ω. Derive the spectral density of $\Delta y_t = y_t - y_{t-1}$ and show that it is equal to zero at $\omega = 0$.
3. Consider the following stationary $ARMA(1,1)$ process

$$y_t = \phi y_{t-1} + \varepsilon_t - \theta \varepsilon_{t-1}.$$

   Derive the spectral density function of y_t and discuss its property in the case when (i) $\phi = \theta$, and (ii) $|\phi - \theta| = \epsilon$, where ϵ is a small positive constant.
4. Consider the univariate process $\{y_t\}_{t=1}^{\infty}$ generated from x_t by the following linear filter

$$y_t = (1 - \lambda) \sum_{i=0}^{\infty} \lambda^i x_{t-i}, \quad t = 1, 2, \ldots T, \quad |\lambda| < 1,$$

   where $\{x_t\}_{-\infty}^{\infty}$ is generated according to the MA(1) process

$$x_t = \epsilon_t + \theta \epsilon_{t-1}, \quad t = \ldots, -1, 0, 1, 2 \ldots, \quad |\theta| < 1,$$

and $\{\epsilon_t\}_{-\infty}^{\infty}$ are $IID(0, \sigma^2)$ random variables. The autocovariances of x_t are denoted by $\gamma_s = E[(x_t - \mu_x)(x_{t+s} - \mu_x)]$, $s = \ldots, -1, 0, 1, \ldots$, where μ_x is the mean of x_t.

(a) Show that the spectral density function of x_t, $f_x(\omega)$, is given by the following expression

$$f_x(\omega) = \frac{\sigma^2}{\pi}(1 + 2\theta\ \cos\omega + \theta^2),\ \ 0 \leq \omega < \pi.$$

(b) Show that the spectral density function of y_t, $f_y(\omega)$, is given by

$$f_y(\omega) = \left[\frac{(1-\lambda)^2}{1 - 2\lambda\ \cos\omega + \lambda^2}\right] f_x(\omega),$$

hence show that $f_y(0) = f_x(0)$.

(c) Prove that

$$\tilde{\gamma}_0 = \left(\frac{1-\lambda}{1+\lambda}\right)(\gamma_0 + 2\lambda\gamma_1),$$

where $\tilde{\gamma}_0$ is the variance of y_t.

(d) Derive the conditions under which $\tilde{\gamma}_0 > \gamma_0$. Discuss the relevance of this result for the analysis of the relationship between stock prices and dividends, and income and consumption.

5. Consider the stationary process

$$x_t = \sum_{i=0}^{\infty} \alpha_i \varepsilon_{t-i},$$

where ε_t are *IID* innovation processes and α_i decay exponentially. Derive the spectral density of Δx_t and show that it is zero at zero frequency.

Part IV

Univariate Time Series Models

14 Estimation of Stationary Time Series Processes

14.1 Introduction

We start with the problem of estimating the mean and autocovariances of a stationary process and then consider the estimation of autoregressive and moving average processes as well as the estimation of spectral density functions. We also relate the analysis of this section to the standard *OLS* regression models and show that when the errors are serially correlated, the *OLS* estimators of models with lagged dependent variables are inconsistent, and derive an asymptotic expression for the bias.

14.2 Estimation of mean and autocovariances

We first briefly discuss the estimation of mean, autocovariances and spectral density of a stationary process.

14.2.1 Estimation of the mean

Suppose that the observations $\{y_t, t = 1, 2, \ldots, T\}$ are generated from a stationary process with mean μ. The sample estimate of μ (the sample mean) is given by

$$\hat{\mu}_T = \bar{y}_T = \frac{y_1 + y_2 + \ldots + y_T}{T}.$$

It is easily seen that $\bar{y}_T$ is an unbiased estimator of μ, since under stationarity $E(y_t) = \mu$ for all t. Also, a sufficient condition for $\bar{y}_T$ to be a consistent estimator of μ is given by $\lim_{T\to\infty} Var(\bar{y}_T) \to 0$. If this condition holds we say the process is ergodic in mean. To investigate this condition further let $\mathbf{y} = \left(y_1, y_2, \ldots, y_T\right)'$, and $\boldsymbol{\tau} = (1, 1, \ldots, 1)'$, a $T \times 1$ vector of ones. Then $\bar{y}_T = T^{-1}\left(\boldsymbol{\tau}'\mathbf{y}\right)$, and $Var(\bar{y}_T) = T^{-2}\left[\boldsymbol{\tau}' Var(\mathbf{y})\boldsymbol{\tau}\right]$, where [1]

[1] Note that in this chapter we use $\gamma_h = Cov(y_t, y_{t-h})$ to denote the autocovariance function, previously denoted by $\gamma(h)$ in Chapter 12.

$$Var(\mathbf{y}) = \begin{pmatrix} Var(y_1) & Cov(y_1, y_2) & \dots & Cov(y_1, y_{T-1}) & Cov(y_1, y_T) \\ Cov(y_2, y_1) & Var(y_2) & \dots & Cov(y_2, y_{T-1}) & Cov(y_2, y_T) \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ Cov(y_{T-1}, y_1) & Cov(y_{T-1}, y_2) & \dots & Var(y_{T-1}) & Cov(y_{T-1}, y_T) \\ Cov(y_T, y_1) & Cov(y_T, y_2) & \dots & Cov(y_T, y_{T-1}) & Var(y_T) \end{pmatrix},$$

$$= \begin{pmatrix} \gamma_0 & \gamma_1 & \cdots & \gamma_{T-2} & \gamma_{T-1} \\ \gamma_1 & \gamma_0 & \cdots & \gamma_{T-3} & \gamma_{T-2} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ \gamma_{T-2} & \gamma_{T-3} & \cdots & \gamma_0 & \gamma_1 \\ \gamma_{T-1} & \gamma_{T-2} & \cdots & \gamma_1 & \gamma_0 \end{pmatrix}.$$

Then

$$\begin{aligned} V(\bar{y}_T) &= \left[\boldsymbol{\tau}' Var(\mathbf{y})\boldsymbol{\tau}\right]/T^2 \\ &= \frac{1}{T}\left[\gamma_0 + 2\sum_{h=1}^{T-1}\left(1 - \frac{h}{T}\right)\gamma_h\right]. \end{aligned} \tag{14.1}$$

To ensure $\lim_{T\to\infty} Var(\bar{y}_T) = \lim_{T\to\infty} \left(\boldsymbol{\tau}' Var\left(\mathbf{y}\right)\boldsymbol{\tau}\right)/T^2 \to 0$, it is therefore sufficient that

$$\lim_{T\to\infty}\left[\sum_{h=1}^{T-1}\left(1 - \frac{h}{T}\right)\gamma_h\right] < \infty,$$

which is clearly satisfied if the autocovariances are absolute summable (recall that $\gamma(0) < \infty$). To see this, note that

$$\left|\sum_{h=1}^{T-1}\left(1 - \frac{h}{T}\right)\gamma_h\right| \leq \sum_{h=1}^{T-1}\left|1 - \frac{h}{T}\right|\left|\gamma_h\right| < \sum_{h=1}^{T-1}\left|\gamma_h\right|.$$

However, for a consistent estimation of $\bar{y}_T$ the less stringent condition

$$\lim_{T\to\infty}\sum_{h=0}^{T-1}\gamma_h < \infty, \tag{14.2}$$

would be sufficient. When this condition is met, it is said that y_t is 'ergodic in mean'.

In spectral analysis, the condition for ergodicity in mean is equivalent to the spectrum, $f_y(\omega)$, being bounded at zero frequency. Recall that (see Chapter 13)

$$f_y(0) = \frac{1}{2\pi}\left(\gamma_0 + 2\sum_{h=1}^{\infty}\gamma_h\right) < \infty,$$

holds if $\sum_{h=0}^{\infty} \gamma_h < \infty$. The spectrum at zero frequency measures the extent to which shocks to the process y_t are persistent, and captures the 'long-memory' property of the series. Note also from (14.1) that

$$\lim_{T\to\infty}\left[Var(\sqrt{T}\bar{y}_T)\right] = \lim_{T\to\infty}\left[\gamma_0 + 2\sum_{h=1}^{T-1}\left(1-\frac{h}{T}\right)\gamma_h\right] = 2\pi f_y(0),$$

which relates the asymptotic variance of $\sqrt{T}\bar{y}_T$ to the value of the spectral density at zero frequency.

14.2.2 Estimation of autocovariances

Recall from Chapter 12 that $\gamma_h = E\left[\left(y_T-\mu\right)\left(y_{T-h}-\mu\right)'\right]$. A moment estimator of γ_h is given by[2]

$$\hat{\gamma}_T(h) = \frac{\sum_{t=h+1}^{T}\left(y_t-\bar{y}_T\right)\left(y_{t-h}-\bar{y}_T\right)}{T}. \tag{14.3}$$

Similarly, the autocorrelation of order h, $\rho_h = \gamma_h/\gamma_0$, can be estimated by

$$\hat{\rho}_T(h) = \frac{\hat{\gamma}_T(h)}{\hat{\gamma}(0)} = \frac{\sum_{t=h+1}^{T}\left(y_t-\bar{y}_T\right)\left(y_{t-h}-\bar{y}_T\right)}{\sum_{t=1}^{T}\left(y_t-\bar{y}_T\right)^2}. \tag{14.4}$$

For a fixed h and as $T\to\infty$, we have

$$\hat{\gamma}_T(h) \xrightarrow{p} \gamma_h.$$

This is relatively easy to prove assuming that the underlying process, $\{y_t\}$, is covariance stationary, has finite fourth-order moment, and (see Bartlett (1946))

$$\lim_{H\to\infty}\frac{1}{H}\sum_{h=1}^{H}\gamma_h^2 \to 0. \tag{14.5}$$

To see this, first note that

$$\begin{aligned}\left(y_t-\bar{y}_T\right)\left(y_{t-h}-\bar{y}_T\right) &= \left(y_t-\mu+\mu-\bar{y}_T\right)\left(y_{t-h}-\mu+\mu-\bar{y}_T\right)\\ &= (y_t-\mu)(y_{t-h}-\mu)+\left(\mu-\bar{y}_T\right)\left(y_{t-h}-\mu\right)\\ &\quad+\left(y_t-\mu\right)\left(\mu-\bar{y}_T\right)+\left(\mu-\bar{y}_T\right)^2,\end{aligned}$$

[2] The denominator is T instead of $T-h$ to ensure the positive definiteness of the covariance matrix. See Brockwell and Davis (1991) for a proof.

and

$$\hat{\gamma}_T(h) = T^{-1}\sum_{t=h+1}^{T}\left(y_t - \bar{y}_T\right)\left(y_{t-h} - \bar{y}_T\right) = T^{-1}\sum_{t=h+1}^{T}(y_t - \mu)(y_{t-h} - \mu)$$
$$+ \left(\mu - \bar{y}_T\right)T^{-1}\sum_{t=h+1}^{T}\left(y_t - \mu\right) + \left(\mu - \bar{y}_T\right)T^{-1}\sum_{t=h+1}^{T}\left(y_{t-h} - \mu\right)$$
$$+ \left(1 - \frac{h}{T}\right)\left(\mu - \bar{y}_T\right)^2. \tag{14.6}$$

But we have already established that

$$\bar{y}_T = \mu + O_p(T^{-1/2}),$$

and for any fixed h

$$T^{-1/2}\sum_{t=h+1}^{T}\left(y_t - \mu\right) = O_p(1).$$

Using the above results in (14.6), it now readily follows that

$$\hat{\gamma}_T(h) = T^{-1}\sum_{t=h+1}^{T}(y_t - \mu)(y_{t-h} - \mu) + O_p\left(T^{-1}\right).$$

Therefore, $\lim_{T\to\infty} E\left[\hat{\gamma}_T(h)\right] = \gamma_h$. Also using results in Bartlett (1946) we have

$$\lim_{T\to\infty} Var\left[\hat{\gamma}_T(h) - \gamma_h\right] = 0,$$

under the assumption that (14.5) is satisfied.

Consider a linear stationary process

$$y_t = \mu + \sum_{i=0}^{\infty}\alpha_i\varepsilon_{t-i}, \tag{14.7}$$

where the following conditions hold

(i) $\sum_{i=0}^{\infty}|\alpha_i| < K < \infty$,
(ii) $\varepsilon_t \sim IID(0, \sigma^2)$,
(iii) $E(\varepsilon_t^4) < K < \infty$,

then it is possible to show that

$$\sqrt{T}\hat{\rho}_T(h) = \frac{\sqrt{T}\hat{\gamma}_T(h)}{\hat{\gamma}_T(0)} \overset{a}{\sim} \frac{\sqrt{T}\hat{\gamma}_T(h)}{\gamma_0},$$

and we have

$$\hat{\rho}_T(h) \xrightarrow{p} \rho_h.$$

When condition (14.5) is met, the process is said to be 'ergodic in variance'. It is easily seen that the mean and variance ergodicity conditions defined by (14.2) and (14.5) are satisfied if the process $\{y_t\}$ has absolute summable autocovariances, namely if $\sum_{h=1}^{\infty} |\gamma_h| < \infty$. The fourth-order moment requirement in (iii) can be relaxed at the expense of augmenting the absolute summability condition (i) above with $\sum_{i=0}^{\infty} i\alpha_i^2 < K < \infty$.

As shown in Chapter 12, stationary *ARMA* processes have exponentially decaying autocovariances. Therefore it also follows that stationary *ARMA* processes are mean/variance ergodic and their mean and autocovariances can be consistently estimated by $\bar{y}_T$ and $\hat{\gamma}_T(h)$, for a fixed h as $T \to \infty$. In fact, since the condition $\sum_{i=0}^{\infty} i\alpha_i^2 < K < \infty$ is also met in the case of stationary *ARMA* processes then consistency of the estimates of the autocorrelation coefficients of stationary *ARMA* processes follow even if the errors, ε_t, do not possess the fourth-order moments. The problem with the estimation of ρ_h arises when one considers values of h that are large relative to T, an issue that one encounters when estimating the spectral density function of y_t, as discussed below. See Section 14.9.

Consider now the asymptotic distribution of $\hat{\boldsymbol{\rho}}_{mT} = (\hat{\rho}_T(1), \hat{\rho}_T(2), \ldots, \hat{\rho}_T(m))'$, where m is fixed and $T \to \infty$ and suppose that y_t follows the stationary linear process (14.7) where

(i) $\varepsilon_t \sim IID(0, \sigma^2)$,
(ii) $\sum_{i=0}^{\infty} |\alpha_i| < K < \infty$,
(iii) $\sum_{i=1}^{\infty} i\alpha_i^2 < K < \infty$, or $E(\varepsilon_t^4) < K < \infty$.

Then

$$\sqrt{T}\left(\hat{\boldsymbol{\rho}}_{mT} - \boldsymbol{\rho}_m\right) \overset{a}{\sim} N(\mathbf{0}, \mathbf{W}_m), \tag{14.8}$$

where $\mathbf{W}_m$ is an $m \times m$ matrix of fixed coefficients with its (i, j) element given by

$$w_{m,ij} = \sum_{h=1}^{\infty} \left\{\rho_{h+i} + \rho_{h-i} - 2\rho_i\rho_h\right\}\left\{\rho_{h+j} + \rho_{h-j} - 2\rho_j\rho_h\right\}.$$

For a proof see, for example, Brockwell and Davis (1991), Section 7.3.

Finally, assuming that conditions (i)–(iii) above hold, under the null hypothesis

$$H_0 : \rho_1 = \rho_2 = \ldots = \rho_m = 0,$$

we have $w_{m,ij} = 0$ if $i \neq j$ and $w_{m,ii} = 1$, and it is easily seen from (14.8) that

$$\sqrt{T}\hat{\rho}_T(h) \overset{a}{\sim} N(0, 1), \quad h = 1, 2, \ldots, m.$$

Furthermore, under the null hypothesis H_0 defined above, the statistics $\sqrt{T}\hat{\rho}_T(h)$, $h = 1, 2, \ldots, m$ are asymptotically independently distributed. This result now can be used to derive the Box and Pierce (1970) Q statistics (of order m)

$$Q = T\sum_{h=1}^{m}\hat{\rho}_T^2(h) \overset{a}{\sim} \chi^2(m),$$

or its small-sample modification, the Ljung and Box (1978) statistics (of order m)

$$Q^* = T(T+2)\sum_{h=1}^{m}\frac{1}{T-h}\hat{\rho}_T^2(h) \overset{a}{\sim} \chi^2(m).$$

Under the assumption that y_t are serially uncorrelated, the Box–Pierce and the Ljung–Box statistics are both distributed asymptotically as χ^2 variates with m degrees of freedom. The two tests are asymptotically equivalent, although the Ljung-Box statistic is likely to perform better in small samples. See Kendall, Stuart, and Ord (1983, Chs 48 and 50), for further details.

14.3 Estimation of *MA*(1) processes

Consider the $MA(1)$ process

$$\begin{aligned} y_t &= \varepsilon_t + \theta\varepsilon_{t-1}, \\ |\theta| &< 1, \ \varepsilon_t \sim IIDN\left(0, \sigma^2\right). \end{aligned} \tag{14.9}$$

Denote that parameters by the 2×1 vector $\boldsymbol{\theta} = \left(\theta, \sigma^2\right)'$. There are two basic approaches for estimation of *MA* processes. The method of moments and the maximum likelihood procedure (see Chapters 9 and 10 for a description of these methods).

14.3.1 Method of moments

Recall that

$$\begin{aligned} \gamma_0 &= E(y_t^2) = \sigma^2(1+\theta^2), \\ \gamma_1 &= E(y_t y_{t-1}) = \sigma^2\theta, \\ \gamma_s &= E(y_t y_{t-s}) = 0, \text{ for } s > 2. \end{aligned}$$

The last set of moment conditions does not depend on the parameters and hence is not informative with respect to the estimation of θ and σ^2, although it can be used to test the $MA(1)$ specification. Using the first two moment conditions we have

$$\rho_1 = \frac{\gamma_1}{\gamma_0} = \frac{\theta}{1+\theta^2}.$$

We have already seen that γ_s can be estimated consistently (for a fixed s), by

$$\hat{\gamma}_s = \frac{\sum_{t=s+1}^{T} y_t y_{t-s}}{T}, \tag{14.10}$$

and hence θ can be estimated consistently by finding the solution to the following estimating equation (assuming that such a solution in fact exists)[3]

$$\hat{\rho}_1 \tilde{\theta}^2 - \tilde{\theta} + \hat{\rho}_1 = 0, \tag{14.11}$$

where we have denoted the moment estimator of θ by $\tilde{\theta}$. This quadratic equation in $\tilde{\theta}$ has a real solution if

$$\Delta = 1 - 4\hat{\rho}_1^2 \geq 0, \text{ or if } \left|\hat{\rho}_1\right| \leq 1/2.$$

When this condition is met, (14.11) has a solution that lies in the range $[-1, 1]$. Note that such a solution exists since the product of the two solutions of (14.11) is equal to unity. This solution is the moment estimator of θ. The reason for selecting the solution that lies inside the unit circle is to ensure that the estimated $MA(1)$ process is invertible, namely that it can be written as the infinite-order AR process

$$y_t + \theta y_{t-1} + \theta^2 y_{t-2} + \ldots\ldots = \varepsilon_t.$$

This representation provides a simple solution to the prediction problem, to be discussed later.

The moment estimator of θ, although simple to compute, is not efficient and does not exist if $\left|\hat{\rho}_1\right| \geq 1/2$. Other estimation procedures need to be considered.

14.3.2 Maximum likelihood estimation of *MA*(1) processes

Let $\mathbf{y} = (y_1, y_2, \ldots, y_T)'$ and assume that it follows a multivariate normal distribution with mean zero and covariance matrix, $\mathbf{\Omega}$, where

$$\mathbf{\Omega} = E\left(\mathbf{y}\mathbf{y}'\right) = \begin{pmatrix} \gamma_0 & \gamma_1 & \cdots & \gamma_{T-2} & \gamma_{T-1} \\ \gamma_1 & \gamma_0 & \cdots & \gamma_{T-3} & \gamma_{T-2} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ \gamma_{T-2} & \gamma_{T-3} & \cdots & \gamma_0 & \gamma_1 \\ \gamma_{T-1} & \gamma_{T-2} & \cdots & \gamma_1 & \gamma_0 \end{pmatrix}, \tag{14.12}$$

$$= \sigma^2 \begin{pmatrix} 1+\theta^2 & \theta & 0 & \ldots & 0 \\ \theta & 1+\theta^2 & \theta & \ldots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & \ldots & \ldots & \theta & 1+\theta^2 \end{pmatrix}.$$

[3] To simplify the notation we are using $\hat{\gamma}_h$ and $\hat{\rho}_h$ instead of $\hat{\gamma}_T(h)$, and $\hat{\rho}_T(h)$, respectively.

Hence

$$\boldsymbol{\Omega} = \sigma^2 \boldsymbol{\Omega}^*, \tag{14.13}$$

where

$$\boldsymbol{\Omega}^* = \left(1 + \theta^2\right) \mathbf{I}_T + \theta \mathbf{A},$$

and $\mathbf{A}$ is the $T \times T$ matrix given by

$$\mathbf{A} = \begin{pmatrix} 0 & 1 & 0 & \dots & \dots \\ 1 & 0 & 1 & & \vdots \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ \vdots & & \ddots & \ddots & 1 \\ 0 & \dots & \dots & 1 & 0 \end{pmatrix}.$$

The log-likelihood function of the $MA(1)$ process can now be written as

$$\ell(\boldsymbol{\theta}) = -\frac{T}{2} \log\left(2\pi\sigma^2\right) - \frac{1}{2} \log\left|\boldsymbol{\Omega}^*\right| - \frac{\mathbf{y}' \boldsymbol{\Omega}^{*-1} \mathbf{y}}{2\sigma^2}, \tag{14.14}$$

where $\left|\boldsymbol{\Omega}^*\right|$ denotes the determinant of $\boldsymbol{\Omega}^*$. Note, we cannot ignore $\log\left|\boldsymbol{\Omega}^*\right|$ if $|\theta|$ is very close to unity. This can be illustrated by noting that $1 + \theta + \theta^2 + \ldots = \left(1 - \theta^T\right) / (1 - \theta)$. As $T \to \infty$, $\left(1 - \theta^T\right) / (1 - \theta)$ converges to $1/(1 - \theta)$ if $|\theta|$ is less than 1. However, when $\theta = 1$, then $\left(1 - \theta^T\right) / (1 - \theta) = T$. By the same reasoning, $\left|\boldsymbol{\Omega}^*\right|$ is of order T if $|\theta| = 1$ and hence in general $\log\left|\boldsymbol{\Omega}^*\right|$ cannot be ignored.

The exact form of $\boldsymbol{\Omega}^{*-1} = \left(w^{ij}\right)$ is given by[4]

$$w^{ij} = \frac{(-\theta)^{j-i}}{\left(1 - \theta^2\right)} \frac{\left(1 - \theta^{2i}\right)\left(1 - \theta^{2(T-i+1)}\right)}{\left(1 - \theta^{2T+2}\right)}, \quad j \geq i. \tag{14.15}$$

As is clear from (14.15), the exact inverse is highly nonlinear in θ and its direct use in (14.14) in order to compute the ML estimators involves a great deal of computations. The computation of the exact inverse of $\boldsymbol{\Omega}^*$ can be quite time consuming when T is large and might not be practical for very large T.

In order to facilitate the computations we first reduce $\boldsymbol{\Omega}^*$ to a diagonal form by means of an orthogonal transformation. To do so we note that the characteristic roots of $\mathbf{A}$ are distinct and do not depend on the unknown parameter, θ. Also from (14.13) it readily follows that $\mathbf{A}$ and $\boldsymbol{\Omega}^*$ commute and hence have the same characteristic vectors, and the characteristic roots of $\boldsymbol{\Omega}^*$ can be obtained from those of $\mathbf{A}$.[5] This implies that $\boldsymbol{\Omega}^*$ has the following characteristic vectors

$$\mathbf{h}_j' = \left[\sin\left(\frac{j\pi}{T+1}\right), \sin\left(\frac{2j\pi}{T+1}\right), \ldots, \sin\left(\frac{Tj\pi}{T+1}\right)\right],$$

[4] See, for example, Pesaran (1973).

[5] Matrices $\mathbf{A}$ and $\mathbf{B}$ are said to commute if and only if $\mathbf{AB} = \mathbf{BA}$.

which form an orthogonal set and correspond to the characteristic roots

$$\lambda_j = \theta^2 + 2\theta \cos\left(\frac{j\pi}{T+1}\right) + 1, \tag{14.16}$$

for $j = 1, 2, \ldots, T$. After the necessary normalization of the above characteristic vectors, the orthogonal transformation $\mathbf{H}$ which does not depend on the unknown parameters can be formed

$$\mathbf{H}' = \left(\frac{2}{T+1}\right)^{1/2} (\mathbf{h}_1, \mathbf{h}_2, \ldots, \mathbf{h}_T). \tag{14.17}$$

Then using the theorem on the diagonalization of real symmetric matrices it follows that $\mathbf{\Omega}^{*-1} = \mathbf{H}'\mathbf{\Lambda}^{-1}\mathbf{H}$, where $\mathbf{\Lambda}$ is a diagonal matrix with the characteristic roots $\lambda_j, j = 1, 2, \ldots, T$ as its diagonal elements. With λ_j specified in (14.16), we can calculate *log* $\left|\mathbf{\Omega}^*\right|$ by

$$\left|\mathbf{\Omega}^*\right| = \prod_{j=1}^{T}\left[\theta^2 + 2\theta \cos\left(\frac{j\pi}{T+1}\right) + 1\right],$$

with the equivalent expression

$$\left|\mathbf{\Omega}^*\right| = \left(1 - \theta^{2T+2}\right) / \left(1 - \theta^2\right), \tag{14.18}$$

which can also be obtained by a direct evaluation of $\left|\mathbf{\Omega}^*\right|$.

Substituting $\bar{\mathbf{y}} = \mathbf{H}y$ and (14.18) in (14.14), the log-likelihood function becomes

$$\ell(\boldsymbol{\theta}) = -\frac{T}{2}\log\left(2\pi\sigma^2\right) - \frac{1}{2}\log\left(\frac{1-\theta^{2T+2}}{1-\theta^2}\right) - \frac{1}{2\sigma^2}\bar{\mathbf{y}}'\Lambda^{-1}\bar{\mathbf{y}}. \tag{14.19}$$

In order to maximize the likelihood function given by (14.19), we first obtain the following concentrated log-likelihood function

$$\ell(\theta) = -\frac{T}{2}\log\left[2\pi\hat{\sigma}^2(\boldsymbol{\theta})\right] - \frac{1}{2}\log\left(\frac{1-\theta^{2T+2}}{1-\theta^2}\right) - \frac{T}{2}, \tag{14.20}$$

where

$$\hat{\sigma}^2(\theta) = \left(\bar{\mathbf{y}}'\Lambda^{-1}\bar{\mathbf{y}}\right)/T.$$

Now dealing with the transformed observations, $\bar{\mathbf{y}}$, the problem of maximizing $\ell(\theta)$ defined in (14.19) is much simpler. A grid search method or iterative procedures, such as the Newton-Raphson method, can be used. Here we describe an iterative procedure for the maximization of (14.20) which is certain to converge. For this purpose we first note that

$$\frac{\partial\ell(\theta)}{\partial\theta} = -\frac{T}{2\hat{\sigma}^2(\theta)}\frac{\partial\hat{\sigma}^2(\theta)}{\partial\theta} + \frac{(T+1)\,\theta^{2T+1}}{\left(1-\theta^{2T+2}\right)} - \frac{\theta}{\left(1-\theta^2\right)} = 0, \tag{14.21}$$

where

$$\frac{\partial \hat{\sigma}^2(\theta)}{\partial \theta} = \frac{1}{T}\left(\bar{\mathbf{y}}' \frac{\partial \Lambda^{-1}}{\partial \theta} \bar{\mathbf{y}}\right) + \frac{2}{T}\bar{\mathbf{y}}' \Lambda^{-1} \frac{\partial \bar{\mathbf{y}}}{\partial \theta}.$$

Hence

$$\frac{\partial \hat{\sigma}^2(\theta)}{\partial \theta} = \frac{1}{T}\sum_{j=1}^{T}\left[\theta + \cos\left(\frac{j\pi}{T+1}\right)\right]\lambda_j^{-2}\bar{y}_j^2.$$

Consequently, using the above result in (14.21) and ignoring θ^{2T+2} ($|\theta| < 1$), the first-order condition for the maximization of the log-likelihood function can be written as

$$f(\theta) = \theta\sum_{j=1}^{T}\lambda_j^{-2}\bar{y}_j^2 - T\left(1-\theta^2\right)\sum_{j=1}^{T}\left[\theta + \cos\left(\frac{j\pi}{T+1}\right)\right]\lambda_j^{-2}\bar{y}_j^2 = 0.$$

Now it is easily seen that $f(1)f(-1) = -\left(\sum_{j=1}^{T}\lambda_j^{-2}\bar{y}_j^2\right)^2 < 0$ and hence the equation $f(\hat{\theta}) = 0$, must have a root within the range $|\theta| < 1$. A simple procedure for computing this root is by the iterative method of 'False position'. A description of this method together with the proof of its convergence can be found, for example, in Hartee (1958); also see Pesaran (1973).

14.3.3 Estimation of regression equations with *MA*(q) error processes

The *MA* processes can also be estimated in the case of regression equations

$$y_t = \mathbf{w}_t'\boldsymbol{\beta} + u_t, \qquad t = 1, 2, \ldots, T, \tag{14.22}$$

where

$$u_t = \sum_{i=0}^{q}\theta_i\varepsilon_{t-i}, \qquad \varepsilon_t \sim N(0, \sigma^2), \qquad \theta_0 \equiv 1. \tag{14.23}$$

The log-likelihood function is given by

$$LL_{MA}(\boldsymbol{\theta}) = -\frac{T}{2}(2\pi\sigma^2) - \tfrac{1}{2}\log\left|\boldsymbol{\Omega}^*\right| - \frac{1}{2\sigma^2}(\mathbf{y}-\mathbf{W}\boldsymbol{\beta})'\boldsymbol{\Omega}^{*-1}(\mathbf{y}-\mathbf{W}\boldsymbol{\beta}), \tag{14.24}$$

where $\mathbf{u} = \mathbf{y} - \mathbf{W}\boldsymbol{\beta}$, and $E(\mathbf{u}\mathbf{u}') = \sigma^2\boldsymbol{\Omega}^*$. This yields *exact ML* estimates of the unknown parameters $\boldsymbol{\theta} = (\boldsymbol{\beta}', \theta_1, \theta_2, \ldots, \theta_q, \sigma^2)'$, when the regressors $\mathbf{w}_t$ do not include lagged values of y_t.

The numerical method used to calculate the above maximization problem involves a Cholesky decomposition of the variance-covariance matrix $\boldsymbol{\Omega}^*$. For the $MA(q)$ error specification we have $\boldsymbol{\Omega}^* = \mathbf{H}\mathbf{D}\mathbf{H}'$, with $\mathbf{H}$ being an upper triangular matrix

$$\mathbf{H} = \begin{pmatrix} 1 & h_{11} & h_{21} & \dots & h_{q1} & \dots & & 0 \\ & 1 & h_{12} & h_{22} & \dots & h_{q2} & & 0 \\ & & \cdot & \cdot & \cdot & & \ddots & \vdots \\ & & & \cdot & \cdot & \cdot & & h_{q,T-q} \\ & & & & \cdot & \cdot & & \vdots \\ & & \mathbf{0} & & & \cdot & \cdot & h_{2,T-2} \\ & & & & & & 1 & h_{1,T-1} \\ & & & & & & & 1 \end{pmatrix},$$

and $\mathbf{D}$ is a diagonal matrix with elements w_t, $t = 1, 2, \dots, T$, and

$$\begin{aligned} d_t &= \delta_0 - \sum_{i=1}^{q} h_{it}^2 d_{t+i}, && t = T-1, T-2, \dots, 1, \\ h_{jt} &= d_{t+j}^{-1} \left(\delta_j - \sum_{i=j+1}^{q} h_{it} h_{i-j,t+j} w_{t+i} \right), && \begin{array}{l} t = T-j, T-j-1, \dots, 1, \\ j = q-1, q-2, \dots, 1, \end{array} \\ h_{qt} &= d_{t+q}^{-1} \delta_q, \\ \delta_s &= \begin{cases} \sum_{i=1}^{q} \theta_i \theta_{i-s}, & 0 \le s \le q, \\ 0, & s > q. \end{cases} \end{aligned}$$

The forward filters on y_t and $\mathbf{w}_t$ are given by

$$\begin{aligned} y_t^f &= y_t - \sum_{i=1}^{q} h_{it} y_{t+i}^f, && \text{for} \quad t = T-1, T-2, \dots, 1, \\ \mathbf{w}_t^f &= \mathbf{w}_t - \sum_{i=1}^{q} h_{it} \mathbf{w}_{t+i}^f, && \text{for} \quad t = T-1, T-2, \dots, 1, \end{aligned}$$

and

$$y_t^* = d_t^{-1/2} y_t^f, \; \mathbf{w}_t^* = d_t^{-1/2} \mathbf{w}_t^f.$$

The terminal values for the above recursions are given by

$$\begin{aligned} d_T &= \delta_0 = 1 + \theta_1^2 + \theta_2^2 + \ldots + \theta_q^2, \\ h_{jT} &= h_{j,T-1} = \cdots = h_{j,T-j+1} = 0, \\ y_T^f &= y_T, \\ \mathbf{w}_T^f &= \mathbf{w}_T. \end{aligned}$$

For a given value of $(\theta_1, \theta_2, \ldots, \theta_q)$, the estimator of $\boldsymbol{\beta}$ can be computed by the *OLS* regression of y_t^* on $\mathbf{w}_t^*$. The estimation of $\theta_1, \theta_2, \ldots, \theta_q$ needs to be carried out iteratively. *Microfit 5.0* carries these iterations by the modified Powell method of conjugate directions that does not require derivatives of the log-likelihood function. See Powell (1964), Brent (1973), and Press et al. (1989). The application of the Gauss–Newton method to the present problem requires derivatives of the log-likelihood function which are analytically intractable, and can be very time-consuming if they are to be computed numerically. In the case of pure $MA(q)$ processes, we need to set $\mathbf{w} = (1, 1, \ldots, 1)'$.

14.4 Estimation of *AR* processes

Consider the following $AR(p)$ process

$$y_t = \sum_{i=1}^{p} \phi_i y_{t-i} + \varepsilon_t,\ \varepsilon_t \sim IID(0, \sigma^2),$$

and suppose that the observations $(y_1, y_2, \ldots, y_T)$ are available. As with the *MA* processes, estimation of the unknown parameters $\boldsymbol{\theta} = (\boldsymbol{\phi}', \sigma^2)'$, where $\boldsymbol{\phi} = (\phi_1, \phi_2, \ldots, \phi_p)'$, can be accomplished by the method of moments or by the *ML* procedure.

14.4.1 Yule–Walker estimators

The Yule–Walker (YW) equations derived in Chapter 12 provide the basis of the method of moments. The *YW* equations can be written in matrix form as

$$\begin{pmatrix} \gamma_0 & \gamma_1 & \cdots & \gamma_{p-2} & \gamma_{p-1} \\ \gamma_1 & \gamma_0 & \cdots & \gamma_{p-3} & \gamma_{p-2} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ \gamma_{p-2} & \gamma_{p-3} & \cdots & \gamma_0 & \gamma_1 \\ \gamma_{p-1} & \gamma_{p-2} & \cdots & \gamma_1 & \gamma_0 \end{pmatrix} \begin{pmatrix} \phi_1 \\ \phi_2 \\ \vdots \\ \phi_{p-1} \\ \phi_p \end{pmatrix} = \begin{pmatrix} \gamma_1 \\ \gamma_2 \\ \vdots \\ \gamma_{p-1} \\ \gamma_p \end{pmatrix},$$

or more compactly as

$$\boldsymbol{\Gamma}_p \boldsymbol{\phi} = \boldsymbol{\gamma}_p.$$

Using the moment estimates of γ_s given by (14.10), the YW estimators of $\boldsymbol{\phi}$ are given by

$$\hat{\boldsymbol{\phi}}_{YW} = \hat{\boldsymbol{\Gamma}}_p^{-1} \hat{\boldsymbol{\gamma}}_p,$$

where $\hat{\boldsymbol{\Gamma}}_p$ and $\hat{\boldsymbol{\gamma}}_p$ are the moment estimators of $\boldsymbol{\Gamma}_p$ and $\boldsymbol{\gamma}_p$.

In the case of the $AR(1)$ process, the *YW* estimator simplifies to

$$\hat{\phi}_1 = \frac{\hat{\gamma}_1}{\hat{\gamma}_0} = \frac{\sum_{t=2}^{T} y_t y_{t-1}}{\sum_{t=1}^{T} y_t^2}. \tag{14.25}$$

The *YW* estimator of $\boldsymbol{\phi}$ is consistent and asymptotically efficient and for sufficiently large T is very close to the *ML* estimator of $\boldsymbol{\phi}$, to which we now turn.

14.4.2 Maximum likelihood estimation of *AR*(1) processes

Consider the first-order stationary autoregressive process

$$y_t = \phi y_{t-1} + \varepsilon_t, \quad |\phi| < 1, \ \varepsilon_t \sim IIDN(0, \sigma^2).$$

For this example $f\left(y_t \mid y_{t-1}, y_{t-2}, \ldots, y_1\right) = f\left(y_t \mid y_{t-1}\right), t = 2, 3, \ldots, T$, hence

$$f\left(y, \boldsymbol{\theta}\right) = f\left(y_1\right) f\left(y_2 \mid y_1\right) f\left(y_3 \mid y_2\right) \ldots f\left(y_T \mid y_{T-1}\right), \tag{14.26}$$

where $f\left(y_1\right)$ is the marginal distribution of the initial observations, and $\boldsymbol{\theta} = (\phi, \sigma^2)'$. Assuming the process has started a long time ago, we have

$$y_1 \sim N\left(0, \frac{\sigma^2}{1-\phi^2}\right), \tag{14.27}$$

and $y_t \mid y_{t-1} \sim N(\phi y_{t-1}, \sigma^2)$, for $t = 2, 3, \ldots, T$. To derive the log-likelihood function we need the joint probability density function $f\left(y_1, y_2, \ldots, y_T\right)$ given our assumption concerning the distribution of the errors $(\varepsilon_1, \varepsilon_2, \ldots, \varepsilon_T)$, and the initialization of the *AR* process. This can be achieved using the transformations

$$\begin{aligned}
y_1 &= y_1, \\
y_2 &= \phi y_1 + \varepsilon_2, \\
y_3 &= \phi y_2 + \varepsilon_3, \\
&\vdots \\
y_T &= \phi y_{T-1} + \varepsilon_T.
\end{aligned}$$

Also

$$\begin{aligned}
f\left(y_1, y_2, \ldots, y_T\right) &= f\left(y_2, \ldots, y_T \mid y_1\right) f\left(y_1\right) \\
&= f\left(y_1, \varepsilon_2, \varepsilon_3 \ldots \varepsilon_T\right) |J|,
\end{aligned}$$

where

$$|J| = \left| \frac{\partial\left(y_1, \varepsilon_2, \varepsilon_3 \ldots \varepsilon_T\right)}{\partial\left(y_1, y_2, \ldots, y_T\right)} \right|$$

$$= \begin{vmatrix} \frac{\partial y_1}{\partial y_1} & \frac{\partial y_1}{\partial y_2} & \cdots & \cdots & \frac{\partial y_1}{\partial y_T} \\ \frac{\partial \varepsilon_2}{\partial y_1} & \frac{\partial \varepsilon_2}{\partial y_2} & \cdots & \cdots & \frac{\partial \varepsilon_2}{\partial y_T} \\ \vdots & & \ddots & & \vdots \\ \vdots & & & \ddots & \vdots \\ \frac{\partial \varepsilon_T}{\partial y_1} & \frac{\partial \varepsilon_T}{\partial y_2} & \cdots & \cdots & \frac{\partial \varepsilon_T}{\partial y_T} \end{vmatrix}.$$

For the above transformations we have

$$|J| = \begin{vmatrix} 1 & 0 & \ldots & 0 & 0 \\ -\phi & 1 & \ldots & 0 & 0 \\ 0 & -\phi & \ldots & 0 & 0 \\ \vdots & \vdots & \ldots & \vdots & \vdots \\ 0 & 0 & \ldots & -\phi & 1 \end{vmatrix} = 1.$$

Hence

$$f\left(y_1, y_2, \ldots, y_T \,|\boldsymbol{\theta}\right) = f(y_1 \,|\boldsymbol{\theta}) \prod_{t=2}^{T} f(\varepsilon_t \,|\boldsymbol{\theta}). \tag{14.28}$$

$$\log f\left(y_1, y_2, \ldots, y_T \,|\boldsymbol{\theta}\right) = \log f(y_1 \,|\boldsymbol{\theta}) + \sum_{t=2}^{T} \log f(\varepsilon_t \,|\boldsymbol{\theta}) \tag{14.29}$$

Under the assumption that the $AR(1)$ process has started a long time in the past and is stationary, we have

$$\log f\left(y_1 \,|\boldsymbol{\theta}\right) = -\frac{1}{2} \log\left(\frac{2\pi\sigma^2}{1-\phi^2}\right) - \frac{1-\phi^2}{2\sigma^2} y_1^2. \tag{14.30}$$

Also

$$\log f\left(\varepsilon_t \,|\boldsymbol{\theta}\right) = -\frac{1}{2} \log\left(2\pi\sigma^2\right) - \frac{1}{2\sigma^2}\left(y_t - \phi y_{t-1}\right)^2, t = 2, 3, \ldots, T. \tag{14.31}$$

Therefore, substituting these in (14.29) we have (recalling that $\ell\left(\boldsymbol{\theta}\right) = \log f\left(y_1, y_2, \ldots, y_T \,|\boldsymbol{\theta}\right)$)

$$\begin{aligned} \ell\left(\boldsymbol{\theta}\right) = &-\frac{T}{2} \log\left(2\pi\sigma^2\right) + \frac{1}{2} \log\left(1-\phi^2\right) \\ &- \left(\frac{1-\phi^2}{2\sigma^2}\right) y_1^2 - \frac{1}{2\sigma^2} \sum_{t=2}^{T} \left(y_t - \phi y_{t-1}\right)^2. \end{aligned} \tag{14.32}$$

We need to take care of the initial value if ϕ is very close to 1. The reason is, if $|\phi|$ is sufficiently less than 1, $\log\left[\left(2\pi\sigma^2\right) / \left(1-\phi^2\right)\right]$ is finite hence $\log f\left(y_1\right)$ is finite. The effect of the distribution of the initial value will become smaller and smaller as $T \to \infty$. However, as $|\phi| \to 1$ then,

$\log\left[\left(2\pi\sigma^2\right)/\left(1-\phi^2\right)\right] \to \infty$. Therefore, initial values matter in small samples or when the process is near non-stationary. Only in the case where $|\phi| < 1$, and T is sufficiently large, is one justified to ignore $\log f\left(y_1\right)$.

Rearranging the terms in (14.32), we now obtain

$$\ell\left(\boldsymbol{\theta}\right) = -\frac{T}{2}\log\left(2\pi\sigma^2\right) + \frac{1}{2}\log\left(1-\phi^2\right) - \frac{1}{2\sigma^2}\left[\left(1-\phi^2\right)y_1^2 + \sum_{t=2}^{T}\left(y_t - \phi y_{t-1}\right)^2\right]. \tag{14.33}$$

There is a one-to-one relationship between the above expression and the general log-likelihood specification given by (14.14). It is easily seen that

$$\left(1-\phi^2\right)y_1^2 + \sum_{t=2}^{T}\left(y_t - \phi y_{t-1}\right)^2 = \mathbf{y}'\boldsymbol{\Omega}^{*-1}\mathbf{y},$$

where

$$\boldsymbol{\Omega}^{*} = \left(\frac{1}{1-\phi^2}\right)\begin{pmatrix} 1 & \phi & \dots & \phi^{T-2} & \phi^{T-1} \\ \phi & 1 & \dots & \phi^{T-3} & \phi^{T-2} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ \phi^{T-2} & \phi^{T-3} & \dots & 1 & \phi \\ \phi^{T-1} & \phi^{T-2} & \dots & \phi & 1 \end{pmatrix},$$

and

$$\boldsymbol{\Omega}^{*-1} = \begin{pmatrix} 1 & -\phi & \dots & 0 & 0 \\ -\phi & 1+\phi^2 & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & \dots & \vdots \\ 0 & 0 & \dots & 1+\phi^2 & -\phi \\ 0 & 0 & \dots & -\phi & 1 \end{pmatrix}.$$

Ignoring the log density of the initial observation, $f(y_1)$, the *ML* estimator of ϕ can be computed by finding the solution to the following least squares problem

$$\hat{\phi}_{LS} = \underset{\phi}{\operatorname{argmin}}\left[\sum_{t=2}^{T}\left(y_t - \phi y_{t-1}\right)^2\right],$$

and is given by the *OLS* coefficient of the regression of y_t, on y_{t-1}, for $t = 2, 3, \dots, T$. In the case where y_t has a non-zero mean, the least squares regression must also include an intercept. In such a case the least squares estimator of ϕ is given by

$$\hat{\phi}_{LS} = \frac{\sum_{t=2}^{T}\left(y_t - \bar{y}\right)\left(y_{t-1} - \bar{y}_{-1}\right)}{\sum_{t=2}^{T}\left(y_{t-1} - \bar{y}_{-1}\right)^2}, \tag{14.34}$$

where $\bar{y} = (T-1)^{-1}\sum_{t=2}^{T} y_t$, and $\bar{y}_{-1} = (T-1)^{-1}\sum_{t=2}^{T} y_{t-1}$. It is now easily seen that $\hat{\phi}_{LS}$ is asymptotically equivalent to the Yule–Walker This result holds for more general *AR* processes and extends to the *ML* estimators when a stationary initial value distribution, $f(y_1)$, is added to the log-likelihood function.

14.4.3 Maximum likelihood estimation of *AR*(p) processes

Consider the $AR(p)$ process

$$y_t = \alpha + \sum_{i=1}^{p} \phi_i y_{t-i} + \varepsilon_t,\ \varepsilon_t \sim IIDN(0, \sigma^2).$$

In this case, the log-likelihood function for the sample observations, $\mathbf{y} = (y_1, y_2, \ldots, y_T)'$, is given by

$$\ell(\boldsymbol{\theta}) = \log f\left(y_1, y_2, \ldots, y_p\right) + \sum_{t=p+1}^{T} \log f\left(y_t \mid y_{t-1}, y_{t-2}, \ldots, y_{t-p}\right), \tag{14.35}$$

where $\log f(y_1, y_2, \ldots, y_p)$ is the log-density of the initial observations, $(y_1, y_2, \ldots, y_p)$, and $\boldsymbol{\theta} = (\phi_1, \phi_2, \ldots, \phi_p, \sigma^2)'$. Under $\varepsilon_t \sim IIDN(0, \sigma^2)$

$$\log f\left(y_t \mid y_{t-1}, y_{t-2}, \ldots, y_{t-p}\right) = -\frac{(T-p)}{2}\log\left(2\pi\sigma^2\right) - \frac{1}{2\sigma^2}\sum_{t=p+1}^{T}\left(y_t - \alpha - \sum_{i=1}^{p}\phi_i y_{t-i}\right)^2.$$

When the $AR(p)$ process is stationary, the average log-likelihood function, $T^{-1}\ell(\boldsymbol{\theta})$, converges to the limit

$$\underset{T\to\infty}{Plim}\ T^{-1}\ell(\theta) = -\frac{1}{2}\log\left(2\pi\sigma^2\right) - \frac{1}{2},$$

and does not depend on the density of the initial observations. Namely, for sufficiently large T, the first part of (14.35) can be ignored, and asymptotically the log-likelihood function can be approximated by

$$\ell(\boldsymbol{\theta}) \approx -\frac{(T-p)}{2}\log\left(2\pi\sigma^2\right) - \frac{1}{2\sigma^2}\sum_{t=p+1}^{T}\left(y_t - \alpha - \sum_{i=1}^{p}\phi_i y_{t-i}\right)^2,$$

where α is an intercept, and allows for the possibility of y_t not having mean zero. The *MLE* of $\boldsymbol{\theta} = (\alpha, \boldsymbol{\phi}', \sigma^2)' = (\boldsymbol{\beta}', \sigma^2)'$ based on this approximation can be computed by the *OLS* regression of y_t on $1, y_{t-1}, y_{t-2}, \ldots, y_{t-p}$. We have

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}_p'\mathbf{X}_p)^{-1}\mathbf{X}_p'\mathbf{y}_p, \tag{14.36}$$

where

$$\mathbf{y}_p = \begin{pmatrix} y_{p+1} \\ y_{p+2} \\ \vdots \\ y_T \end{pmatrix}, \mathbf{X}_p = \begin{pmatrix} 1 & y_p & y_{p-1} & \cdots & y_1 \\ 1 & y_{p+1} & y_p & \cdots & y_2 \\ \vdots & & \vdots & & \vdots \\ 1 & y_{T-1} & y_{T-2} & \cdots & y_{T-p} \end{pmatrix}.$$

Also

$$\hat{\sigma}^2 = \frac{\left(\mathbf{y}_p - \mathbf{X}_p\hat{\boldsymbol{\beta}}\right)'\left(\mathbf{y}_p - \mathbf{X}_p\hat{\boldsymbol{\beta}}\right)}{T-p}.$$

14.5 Small sample bias-corrected estimators of ϕ

Although the least squares estimator of $\hat{\phi}_{LS}$, defined by (14.34), has desirable asymptotic properties, it is nevertheless biased in small samples. The bias is due to the fact that the regressors in the *AR* regressions are not strictly exogenous. For a regressor to be strictly exogenous it should be uncorrelated with past and future errors. But it is easily seen that in the case of *AR* regressions, y_{t-1} is uncorrelated with current and future values of ε_t, but not with its past values; namely, y_{t-1} is not strictly exogenous and the standard regression assumptions do not apply to time series regressions.[6] The fact that $E(y_{t-1}\varepsilon_{t-s}) \neq 0$, for $s = 1, 2, \ldots$, can be easily established for the $AR(1)$ model, noting that when the process is stationary

$$y_{t-1} = \mu + \sum_{j=0}^{\infty} \phi^j \varepsilon_{t-1-j},$$

and $E(y_{t-1}\varepsilon_{t-s}) = \sigma^2\phi^{s-1}$, for $s = 1, 2, \ldots$ The small sample bias of $\hat{\phi}_{LS}$ has been derived in the case where $|\phi| < 1$ and the errors, ε_t, are normally distributed by Kendall (1954) and Marriott and Pope (1954), and is shown to be given by[7]

$$E\left(\hat{\phi}_{LS}\right) = \phi - \frac{1+3\phi}{T} + O\left(\frac{1}{T^2}\right). \tag{14.37}$$

Also

$$Var\left(\hat{\phi}_{LS}\right) = \frac{1-\phi^2}{T} + O\left(\frac{1}{T^2}\right).$$

The bias, given by,

[6] See Sections 2.2 and 9.3 on the differences between weak and strict exogeneity assumptions.

[7] Bias corrections for the LS estimates in the case of higher-order *AR* processes are provided in Shaman and Stine (1988).

$$bias\left(\hat{\phi}_{LS}\right) = E\left(\hat{\phi}_{LS}\right) - \phi = -T^{-1}\left(1 + 3\phi\right) + O(T^{-2}), \tag{14.38}$$

can be substantial when T is small and ϕ close to unity. It is also clear that the bias is negative for all positive values of ϕ. For example, when $T = 40$ and $\phi = 0.9$, the bias is or order of $-0.0.925$, which represents a substantial under estimation of ϕ.

To deal with the problem of small sample bias a number of bias-corrected estimators of ϕ are proposed in the literature. Here we draw attention to two of these estimators. The first, initially proposed by Orcutt and Winokur (1969), uses the bias formula, (14.38), to obtain the following biased-corrected estimator

$$\tilde{\phi} = \frac{1}{T-3}\left(1 + T\hat{\phi}_{LS}\right).$$

It is now easily seen, using (14.37), that

$$\begin{aligned} E\left(\tilde{\phi}\right) &= \frac{1}{T-3}\left[1 + TE\left(\hat{\phi}_{LS}\right)\right] \\ &= \phi + O\left(\frac{1}{T^2}\right). \end{aligned}$$

The second approach is known as the (half) Jackknife bias-corrected estimator and was first proposed by Quenouille (1949). It is defined by

$$\breve{\phi} = 2\hat{\phi}_{LS} - \frac{1}{2}\left(\hat{\phi}_{1,LS} + \hat{\phi}_{2,LS}\right),$$

where $\hat{\phi}_{1,LS}$ and $\hat{\phi}_{2,LS}$ are the least squares estimates of ϕ based on the first $T/2$ and the last $T/2$ observations (it is assumed that T is even, if it is not, one of the observations can be dropped). Again using (14.37) we have

$$\begin{aligned} E\left(\hat{\phi}_{1,LS}\right) &= \phi - \frac{1+3\phi}{(T/2)} + O\left(\frac{1}{(T/2)^2}\right) \\ E\left(\hat{\phi}_{1,LS}\right) &= \phi - \frac{1+3\phi}{(T/2)} + O\left(\frac{1}{(T/2)^2}\right). \end{aligned}$$

It is then easily seen that

$$E\left(\breve{\phi}\right) = \phi + O\left(\frac{1}{(T/2)^2}\right) = \phi + O\left(\frac{1}{T^2}\right).$$

Both bias-corrected estimators work well in reducing the bias, but impact the variances of the bias-corrected estimators differently. The Jackknife estimator does not require knowing the expression for the bias in (14.38), and is more generally applicable and can be used with similar effects in the case of higher-order AR processes.

Finally, when using bias-corrected estimators it is important to bear in mind that the variance of the bias-corrected estimator tends to be higher than the variance of the uncorrected estimator. For example, $Var(\tilde{\phi}) = \left(\frac{T}{T-3}\right)^2 Var(\hat{\phi}_{LS}) > Var(\hat{\phi}_{LS})$, and the overall effect of bias-correction on the mean squared error of the estimators generally depends on the true value, ϕ. In the case of $\hat{\phi}_{LS}$ and $\tilde{\phi}$, for example, we have

$$MSE\left(\hat{\phi}_{LS}\right) = \left(bias\ of\ \hat{\phi}_{LS}\right)^2 + Var(\hat{\phi}_{LS}) = \frac{(1+3\phi)^2 + \left(1-\phi^2\right)^2}{T^2} + O\left(\frac{1}{T^3}\right),$$

$$MSE\left(\tilde{\phi}\right) = \left(bias\ of\ \tilde{\phi}\right)^2 + Var(\tilde{\phi}) = \frac{\left(1-\phi^2\right)^2}{(T-3)^2} + O\left(\frac{1}{T^3}\right).$$

Although for most values of ϕ and T, $MSE(\tilde{\phi}) < MSE(\hat{\phi}_{LS})$, there exist combinations of ϕ and T for which $MSE(\tilde{\phi}) > MSE(\hat{\phi}_{LS})$. For example, this is the case when $\phi = 0$ and $T < 10$.

14.6 Inconsistency of the *OLS* estimator of dynamic models with serially correlated errors

In Section 6.2 we introduced sufficient conditions for consistency of the *OLS* estimator of regression parameters in *ARDL* models. We now illustrate possible inconsistency of the *OLS* estimator when errors in the *ARDL* specification are serially correlated. To see this, consider the simple $ARDL(1,0)$ model

$$y_t = \lambda y_{t-1} + \beta x_t + u_t,$$

where, without loss of generality, we assume that y_t and x_t both have zero means. Suppose that $|\lambda| < 1$, and x_t and u_t are covariance stationary processes with absolute summable autocovariances, as defined by

$$x_t = \sum_{i=0}^{\infty} a_i v_{t-i},$$

and

$$u_t = \sum_{i=0}^{\infty} b_i \varepsilon_{t-i},$$

where $\sum_{i=0}^{\infty} |a_i| < K < \infty$, $\sum_{i=0}^{\infty} |b_i| < K < \infty$, and $\{v_t\}$, and $\{\varepsilon_t\}$ are standard white noise processes.

Let $\boldsymbol{\theta} = (\lambda, \beta)'$, and $\mathbf{z}_t = (y_{t-1}, x_t)'$, for $t = 1, 2, \ldots, T$, (see Section 6.2 for further details), and consider the least squares estimator of $\boldsymbol{\theta}$, $\hat{\boldsymbol{\theta}}_{OLS}$, and note that

$$\hat{\boldsymbol{\theta}}_{OLS} - \boldsymbol{\theta}_0 = \left(T^{-1} \sum_{t=1}^{T} \mathbf{z}_t \mathbf{z}_t' \right)^{-1} \left(T^{-1} \sum_{t=1}^{T} \mathbf{z}_t u_t \right), \tag{14.39}$$

where $\boldsymbol{\theta}_0$ is the 'true' value of $\boldsymbol{\theta}$.

Let $\lambda(L) = (1 - \lambda L)^{-1} = \sum_{i=0}^{\infty} \lambda^i L^i$, $a(L) = \sum_{i=0}^{\infty} a_i L^i$, $b(L) = \sum_{i=0}^{\infty} b_i L^i$, and note that when $|\lambda| < 1$, y_t can be solved in terms of current and past values of v_t and ε_t, namely

$$y_t = \beta c(L) v_t + d(L) \varepsilon_t,$$

where

$$c(L) = \lambda(L) a(L), \text{ and } d(L) = \lambda(L) b(L).$$

Both $c(L)$ and $d(L)$ are products of two polynomials with absolute summable coefficients, and therefore also satisfy the absolute summability conditions $\sum_{i=0}^{\infty} |c_i| < \left(\sum_{i=0}^{\infty} |\lambda_i| \right) \left(\sum_{i=0}^{\infty} |a_i| \right) < K < \infty$ and $\sum_{i=0}^{\infty} |d_i| < \left(\sum_{i=0}^{\infty} |\lambda_i| \right) \left(\sum_{i=0}^{\infty} |b_i| \right) < K < \infty$. Hence,

$$T^{-1} \sum_{t=1}^{T} \mathbf{z}_t \mathbf{z}_t' \xrightarrow{p} E(\mathbf{z}_t \mathbf{z}_t') = \begin{pmatrix} \omega_{11} & \omega_{12} \\ \omega_{12} & \omega_{22} \end{pmatrix},$$

where

$$\omega_{11} = \beta^2 \sum_{i=0}^{\infty} c_i^2 + \sum_{i=0}^{\infty} d_i^2,$$
$$\omega_{12} = \beta \sum_{i=0}^{\infty} c_i a_{i+1}, \quad \omega_{22} = \sum_{i=0}^{\infty} a_i^2.$$

Similarly,

$$\left(T^{-1} \sum_{t=1}^{T} \mathbf{z}_t u_t \right) \xrightarrow{p} E(\mathbf{z}_t u_t) = \begin{pmatrix} \delta \\ 0 \end{pmatrix},$$

where

$$\delta = \sum_{i=0}^{\infty} b_i d_{i-1}.$$

Using the above results in (14.39) we have

$$\underset{T \to \infty}{Plim} \left(\hat{\boldsymbol{\theta}}_{OLS} \right) = \boldsymbol{\theta}_0 + \frac{1}{\omega_{11}\omega_{22} - \omega_{12}^2} \begin{pmatrix} \delta \omega_{22} \\ -\delta \omega_{12} \end{pmatrix}.$$

In general, therefore, the *OLS* estimator, $\hat{\boldsymbol{\theta}}_{OLS}$, is inconsistent, unless $\delta = 0$. The direction of inconsistency (the asymptotic bias) depends particularly on the sign of δ. Note that $\omega_{22} > 0$, and also since $E(\mathbf{z}_t\mathbf{z}_t')$ is a positive definite matrix then $\omega_{11}\omega_{22} - \omega_{12}^2 > 0$. Hence

$$\underset{T\to\infty}{Plim}\,(\hat{\boldsymbol{\lambda}}_{OLS}) - \lambda_0 > 0 \text{ if } \delta > 0,$$
$$\underset{T\to\infty}{Plim}\,(\hat{\boldsymbol{\lambda}}_{OLS}) - \lambda_0 < 0 \text{ if } \delta < 0.$$

In most economic applications where $\lambda > 0$ and the errors are positively serially correlated, $\delta > 0$, the *OLS* estimator of λ will be biased upward.

14.7 Estimation of mixed *ARMA* processes

Consider now the $ARMA(p, q)$ process

$$y_t = \sum_{i=1}^{p} \phi_i y_{t-i} + \sum_{i=0}^{q} \theta_i \varepsilon_{t-i},\ \varepsilon_t \sim IID(0, \sigma^2),\ \text{with } \theta_0 = 1. \tag{14.40}$$

Clearly, estimation of this mixed process is subject to similar considerations as encountered in the cases of pure *AR* and pure *MA* processes. Naturally, the estimation process and the associated computations are much more complicated. But *ML* procedure is probably the most straightforward to apply if the contribution of the log density of the initial observations $(y_1, y_2, \ldots, y_p)$ to the log-likelihood function is ignored. In this case we can write the *ARMA* process as a regression equation with an $MA(p)$ error, very much in the form of (14.22), but with $\mathbf{w}_t = (y_{t-1}, y_{t-2}, \ldots, y_{t-p})'$. In this way (14.40) could be written as

$$y_t = \mathbf{w}_t'\boldsymbol{\beta} + \sum_{i=0}^{q} \theta_i \varepsilon_{t-i}, \tag{14.41}$$

where in the present application $\boldsymbol{\beta} = \boldsymbol{\phi} = (\phi_1, \phi_2, \ldots, \phi_p)'$. The *ML* estimation procedure discussed in Section 14.3.3 can now be applied directly to (14.41). Additional exogenous or pre-determined regressors can also be included in $\mathbf{w}_t$.

In estimation of *ARMA* models it is important that the orders of the *ARMA* model (p, q) are chosen to be minimal, in the sense that the order of the *ARMA* model cannot be reduced by elimination of the same lag polynomial factor from both sides of the *ARMA* specification. For example, consider the following $ARMA(p + 1, q + 1)$ specification that can be obtained from (14.40)

$$(1 - \lambda_{p+1}L)\phi_p(L)y_t = (1 - \mu_{p+1}L)\theta_p(L)\varepsilon_t.$$

Suppose now that λ_{p+1} is very close to μ_{p+1}, then the above *ARMA* specification might be indistinguishable from the $ARMA(p, q)$ specification given by (14.40). In the extreme case where

$\lambda_{p+1} = \mu_{p+1}$, the common lag factor, $1 - \lambda_{p+1}L$, can be cancelled from both sides which yields the minimal-order $ARMA(p, q)$ specification. See also Exercise 5 at the end of this chapter.

14.8 Asymptotic distribution of the *ML* estimator

Under the stationary assumption (see Section 12.2), namely if all the roots of

$$1 - \sum_{i=1}^{p} \phi_i z^i = 0,$$

lie outside the unit circle, $|z| > 1$, and for T sufficiently large and p fixed we have

$$\sqrt{T}(\hat{\phi} - \phi) \overset{a}{\sim} N(\mathbf{0}, \sigma^2 \mathbf{V}_p^{-1}),$$

where

$$\mathbf{V}_p = \underset{T\to\infty}{Plim} \left(\frac{\mathbf{Y}_p' \mathbf{Y}_p}{T} \right) = \begin{pmatrix} \gamma_0 & \gamma_1 & \cdots & \gamma_{p-2} & \gamma_{p-1} \\ \gamma_1 & \gamma_0 & \cdots & \gamma_{p-3} & \gamma_{p-2} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ \gamma_{p-2} & \gamma_{p-3} & \cdots & \gamma_0 & \gamma_1 \\ \gamma_{p-1} & \gamma_{p-2} & \cdots & \gamma_1 & \gamma_0 \end{pmatrix} = \mathbf{\Gamma}_p.$$

In the case of the $AR(1)$ specification, we have

$$Asy.Var(\hat{\phi}_1) = \frac{1 - \phi_1^2}{T}, \text{ for } |\phi_1| < 1.$$

Note that $Asy.Var(\hat{\phi}_1) \to 0$, as $T \to \infty$, which establishes that $\hat{\phi}_1$ is a consistent estimator of ϕ_1. It also follows that

$$\lim_{T\to\infty} E\left(\hat{\phi}_1 - \phi_1\right)^2 = 0,$$

namely that $\hat{\phi}_1$ converges to its true value in mean square errors.

Similar results can also be obtained for the *ML* estimators of the *MA* coefficients, $\theta_1, \theta_2, \ldots, \theta_q$, or the mixed *ARMA* coefficients, although they are more complicated to derive.

14.9 Estimation of the spectral density

Consider now the problem of estimating the spectral density, $f(\omega)$, of a stationary process discussed in Chapter 13. An obvious estimator of $f(\omega)$ defined by (13.5) is given by

$$\tilde{f}_T(\omega) = \frac{1}{2\pi}\left(\hat{\gamma}_T(0) + 2\sum_{h=1}^{T-1}\hat{\gamma}_T(h)\cos(\omega h)\right),$$

where the unknown covariances are replaced by their estimators, (14.3). But this is not a good estimator since for large values of h, $\hat{\gamma}_T(h)$ will be based only on a few data points, and hence the condition for consistency of $\tilde{f}_T(\omega)$ will not be satisfied. To avoid this problem the sum in the above expression needs to be truncated. It is also necessary to put less weight on distance autocovariances, namely those with relatively large h, to reduce the possibility of undue influence of these estimators on $\tilde{f}_T(\omega)$. This suggests using a truncated and weighted version of $\tilde{f}_T(\omega)$. At a given frequency, $\omega = \omega_j$ in the range $[0, \pi]$, a consistent estimator of $f(\omega)$ is given by

$$\hat{f}_T(\omega_j) = \frac{1}{2\pi}\left(\hat{\gamma}_T(0) + 2\sum_{h=1}^{K}\lambda_h\hat{\gamma}_T(h)\cos(\omega_j h)\right), \tag{14.42}$$

where $\omega_j = j\pi/K,\ j = 0, 1, \ldots, K$, K is the 'window size', and $\{\lambda_k\}$ are a set of weights called the 'lag window'.

Many different weighting schemes are proposed in the literature. Among these, the most commonly used are

$$\begin{aligned}
\text{Bartlett window}: \quad & \lambda_h = 1 - \frac{h}{K},\ 0 \le h \le K,\\
\text{Tukey window}: \quad & \lambda_h = \frac{1}{2}\left[1 + \cos(\pi h/K)\right],\ 0 \le h \le K,\\
\text{Parzen window}: \quad & \lambda_h = \left\{\begin{array}{c} 1 - 6(h/K)^2 + 6(h/K)^3,\ 0 \le h \le K/2 \\ 2(1 - h/K)^3,\ K/2 \le h \le K \end{array}\right\}.
\end{aligned}$$

We need the window size K to increase with T, but at a lower rate, such that the condition: $K/T \to 0$ as $T \to \infty$, is satisfied. The value for K is often set equal to $2\sqrt{T}$.

In practice, it is more convenient to work with a standardized spectrum, defined by

$$\hat{f}_T^*(\omega) = \hat{f}_T(\omega)/\hat{\gamma}_T(0),$$

or upon using (14.42) we have

$$\hat{f}_T^*(\omega_j) = \frac{1}{2\pi}\left(1 + 2\sum_{h=1}^{K}\lambda_h\hat{\rho}_T(h)\cos(h\omega_j)\right),$$

where $\hat{\rho}_T(h)$ is given by (14.4). The standard errors reported for the estimates of the standardized spectrum can be calculated according to the following formulae, which are valid asymptotically

$$\begin{aligned}
\widehat{s.e.}\left[\hat{f}(\omega_j)\right] &= \sqrt{\tfrac{2}{\nu}}\hat{f}(\omega_j), \text{ for } j = 1, 2, \ldots, K-1,\\
&= \sqrt{\tfrac{4}{\nu}}\hat{f}(\omega_j), \text{ for } j = 0, K,
\end{aligned}$$

where $\nu = 2T/\sum_{k=-K}^{K}(\lambda_k^2)$. For the three different windows, ν is given by:

$$\begin{aligned}\text{Bartlett window:} \quad & \nu = 3\frac{T}{K},\\ \text{Tukey window:} \quad & \nu = \frac{8}{3}\frac{T}{K},\\ \text{Parzen window:} \quad & \nu = 3.71\frac{T}{K}.\end{aligned}$$

Note that the estimates of the standard error of the spectrum at the limit frequencies 0 and π are twice as large as the standard error estimates at the intermediate frequencies. This is particularly relevant to the analysis of persistence as it involves estimation of the spectrum at zero frequency.

Example 32 *Figure 14.1 shows the estimation of the spectral density function for the rate of change of US real GNP, (denoted by DYUS) using quarterly data from 1979q3 to 2013q1, for a total of 135 observations. Estimates of the standardized spectral density function based on Bartlett, Tukey, and Parzen windows are reported. The estimates of the spectral density are scaled and standardized using the unconditional variance of the variable. The window size is set to* $2\sqrt{T} = 23$. *One important feature of this plot is that the contribution to the sample variance of the lowest-frequency component is much larger than the contributions of other frequencies (for example at business cycle frequencies). For further details see Lessons 10.11 and 12.2 in Pesaran and Pesaran (2009).*

An alternative approach for estimating the spectral density is a non-parametric procedure using the *kernel* estimator given by

$$\hat{f}(\omega_j) = \sum_{i=-K}^{K} \lambda(\omega_{j+i}, \omega_j)\hat{f}(\omega_{j+i}),$$

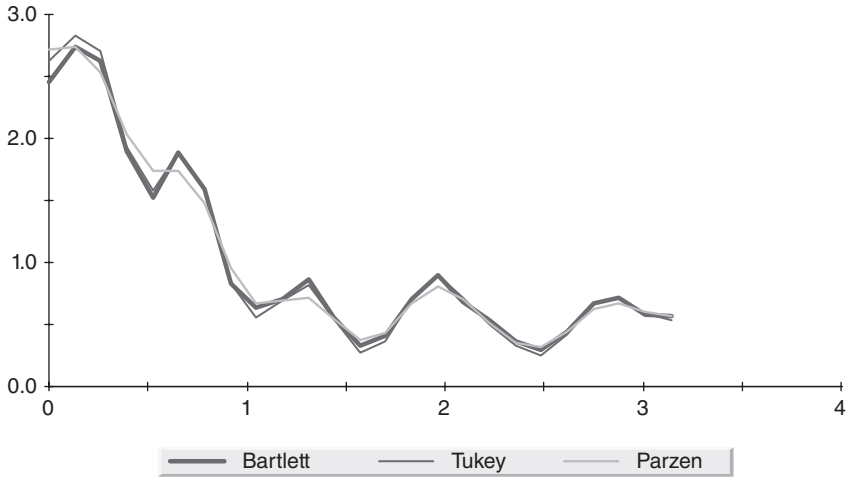


Figure 14.1 Spectral density function for the rate of change of US real GNP.

where $\omega_j = j\pi/K$, K is a *bandwidth* parameter indicating how many frequencies $\{\omega_j, \omega_{j\pm1}, \ldots, \omega_{j\pm K}\}$ are used in estimating the population spectrum, and the *kernel* $\lambda(\omega_{j+i}, \omega_j)$ indicates how much weight each frequency is to be given, where $\sum_{i=-K}^{K} \lambda(\omega_{j+i}, \omega_j) = 1$. Specification of *kernel* $\lambda(\omega_{j+i}, \omega_j)$ can equivalently be described in terms of a weighting sequence $\{\lambda_j,\ j = 1, \ldots, K\}$. One important problem is the choice of the bandwidth parameter, K. As a practical guide, it is often recommended to plot an estimate of the spectrum using several different bandwidths and then rely on subjective judgement to choose the bandwidth that produces the most plausible estimate. More formal statistical procedures for the choice of K have been proposed, among others, by Andrews (1991), and Andrews and Monahan (1992).

For an introductory text on the estimation of the spectrum see Chapter 7 in Chatfield (2003). For more advanced treatments of the subject see Priestley (1981, Ch. 6), and Brockwell and Davis (1991, Ch. 10).

14.10 Exercises

1. Suppose $\{y_t\}$ is a covariance stationary linear process given by

$$y_t = \mu + \sum_{j=0}^{\infty} \alpha_j \varepsilon_{t-j}, \quad \varepsilon_t \backsim IID(0, \sigma^2),$$

where $\sum_{j=0}^{\infty} \mid \alpha_j \mid < \infty$, and $\sum_{j=0}^{\infty} \alpha_j \neq 0$.

(a) Show that the sample mean, $\bar{y}_T = T^{-1} \sum_{t=1}^{T} y_t$ is a consistent estimator of μ.

(b) Consider now the following estimate of the autocovariance function of y_t

$$\hat{\gamma}_s = T^{-1} \sum_{t=1}^{T-s} (y_t - \bar{y}_T)(y_{t+s} - \bar{y}_T),$$

for $0 \leq s \leq T-1$. Show that the matrix $\begin{pmatrix} \hat{\gamma}_0 & \hat{\gamma}_1 \\ \hat{\gamma}_1 & \hat{\gamma}_0 \end{pmatrix}$ is positive definite.

2. Suppose that $\{y_t, t = 1, 2, \ldots, T\}$ is a covariance stationary process having the covariance function γ_s.

(a) Show that the limit of the covariance of y_1 and $\bar{y}_T = T^{-1}(y_1 + y_2 + \ldots + y_T)$ is given by

$$\lim_{T \to \infty} \left(\frac{1}{T} \sum_{j=0}^{T-1} \gamma_j \right) = 0. \tag{14.43}$$

(b) Show that the limit of the variance of $\bar{y}_T$ is given by

$$\lim_{T\to\infty} E[(\bar{y}_T - \mu)^2] = 0. \tag{14.44}$$

where μ is the mean of y_t.

(c) Prove that condition (14.43) is met if and only if condition (14.44) is met.

3. The time series $\{y_t\}$ and $\{x_t\}$ are independently generated according to the following schemes

$$y_t = \lambda y_{t-1} + \varepsilon_{1t},\ |\lambda| < 1,$$
$$x_t = \rho x_{t-1} + \varepsilon_{2t},\ |\rho| < 1,$$

for $t = 1, 2, \ldots, T$, where ε_{1t} and ε_{2t} are non-autocorrelated and distributed independently of each other with zero means and variances equal to σ_1^2 and σ_2^2, respectively. An investigator estimates the simple regression

$$y_t = \beta x_t + u_t,$$

by the *OLS* method. Show that

$$\underset{T\to\infty}{Plim}\,(\hat{\beta}) = 0,$$
$$t_{\hat{\beta}}^2 = \hat{\beta}^2/\hat{V}(\hat{\beta}) = (T-1)r^2/(1-r^2),$$
$$\underset{T\to\infty}{Plim}\,(Tr^2) = (1+\lambda\rho)/(1-\lambda\rho),$$

where $\hat{\beta}$ is the *OLS* estimator of β, $\hat{V}(\hat{\beta})$ is the estimated variance of $\hat{\beta}$, and r is the sample correlation coefficient between x and y, *i.e.*, $r^2 = (\sum_{t=1}^T x_t y_t)^2/(\sum_{t=1}^T x_t^2 \sum_{t=1}^T y_t^2)$. What are the implications of these results for problems of spurious correlation in economic time series analysis?

4. Consider the $AR(1)$ process

$$y_t = \phi y_{t-1} + \varepsilon_t,\ \varepsilon_t \sim IIDN(0, \sigma^2),\ \text{for } t = 1, 2, \ldots, T,$$

where $|\phi| < 1$.

(a) Derive the log-likelihood function of the model: (i) conditional on y_0 being a given fixed constant. (ii) y_0 is normally distributed with mean zero and variance $\sigma^2/(1-\phi^2)$.

(b) Discuss the maximum likelihood estimator of ϕ under (i) and (ii), and show that it is guaranteed to be in the range $|\phi| < 1$, only under (ii).

5. Weekly returns (r_t) on DAX futures contracts are available over the period 14 Jan 1994 – 30 Oct 2009 (825 weeks). A trader interested in predicting future values of r_t proceeds initially by estimating the first- and second-order autocorrelation coefficients of returns and obtains the estimates

$$\hat{\rho}_1 = -0.041, \hat{\rho}_2 = 0.0511.$$

It is known that under $\rho_1 = \rho_2 = 0, \sqrt{T}\hat{\boldsymbol{\rho}} = (\sqrt{T}\hat{\rho}_1, \sqrt{T}\hat{\rho}_2)'$ is asymptotically distributed as $N(\mathbf{0}, \mathbf{I}_2)$, where T is the sample size and $\mathbf{I}_2$ is an identity matrix of order 2. He/she finds these results disappointing and decides to estimate an $ARMA(1,1)$ model for r_t and obtains the following maximum likelihood (ML) estimates

$$r_t = \phi r_{t-1} + \varepsilon_t + \theta \varepsilon_{t-1}.$$

$$\hat{\phi} = \underset{(0.0678)}{-0.8453}, \hat{\theta} = \underset{(0.0767)}{0.7811},$$

$$\widehat{Cov}(\hat{\phi}, \hat{\theta}) = 0.0045,$$

where the figures in brackets are standard errors of the associated *ML* estimates. Knowing that the *ML* estimators are asymptotically normally distributed, the trader argues in favour of the *ARMA* model on the grounds that the parameters of the model, ϕ and θ are statistically highly significant.

(a) Do you agree with the trader's statistical analyses and conclusions? To answer this question we suggest that you carry out the following tests at the 5 per cent significance level

i. Separate and joint tests of $\rho_1 = 0$ and $\rho_2 = 0$.

ii. Separate tests of the following three hypotheses

$$H_0^a : \phi = 0 \text{ against } H_1^a : \phi \neq 0,$$
$$H_0^b : \theta = 0 \text{ against } H_1^b : \theta \neq 0,$$
$$H_0^c : \phi + \theta = 0 \text{ against } H_1^c : \phi + \theta \neq 0.$$

(b) Discuss the general problem of asset return predictability and its relation to the efficient market hypothesis.

15 Unit Root Processes

15.1 Introduction

Within the class of stochastic linear processes discussed in the earlier chapters, the case where one of the roots of the autoregressive representation of the underlying process is unity plays an important role in the analysis of macroeconomic and financial data. In this chapter we compare the properties of unit root processes with the stationary processes and consider alternative ways of testing for unit roots.

15.2 Difference stationary processes

A process $\{y_t\}$ is said to be difference stationary if it is not covariance stationary, but can be transformed into a covariance stationary process by first differencing. A difference stationary process is also known as an *integrated* process. The number of times one needs to difference a process, say d, before it is made into a stationary process is called the order of the integration of the process, and written as $y_t \sim I(d)$. A simple example of a difference stationary process is the random walk model

$$y_t = \mu + y_{t-1} + \varepsilon_t, \ \ \varepsilon_t \sim IID(0, \sigma^2), \ t = 1, 2, \ldots, \tag{15.1}$$

with a given (fixed or stochastic) initial value, y_0. The parameter μ is known as the 'drift' parameter. Solving for y_t in terms of its initial value we obtain

$$y_t = y_0 + t\mu + \varepsilon_1 + \varepsilon_2 + \ldots + \varepsilon_t,$$

and it readily follows that

$$E(y_t|y_0) = y_0 + \mu t, \ \ Var(y_t|y_0) = \sigma^2 t,$$
$$Cov(y_{t_1}, y_{t_2}|y_0) = \sigma^2 \, min(t_1, t_2), \text{ for } t_1, t_2 > 0.$$

Clearly, the random walk model is not covariance stationary, even if we set the drift term μ equal to zero. The coefficients of the innovations, ε_t, are not square summable, and the variance of y_t is trended. But it is easily seen that the first difference of y_t, namely $\Delta y_t = \mu + \varepsilon_t$, is covariance stationary. For this reason the random walk model is also called the first-difference stationary process. Pictorial examples of random walk models are given in Figures 15.1 and 15.2.

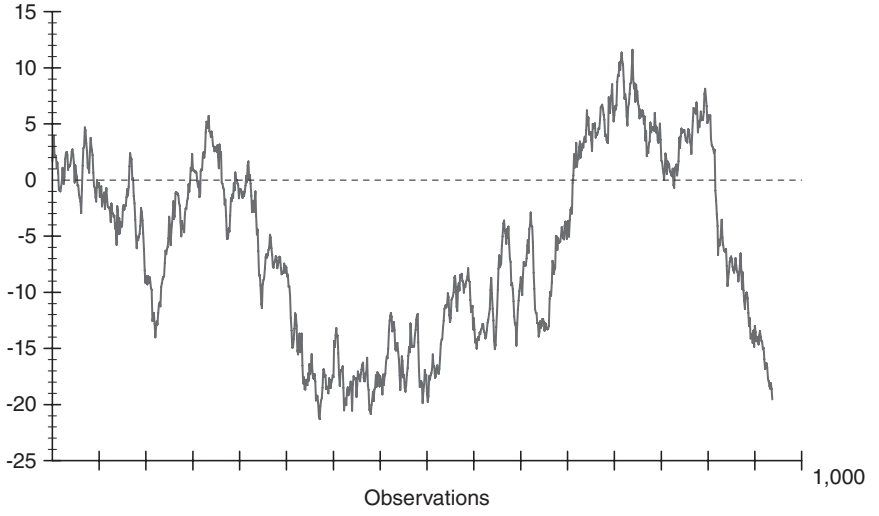


Figure 15.1 A simple random walk model without a drift.

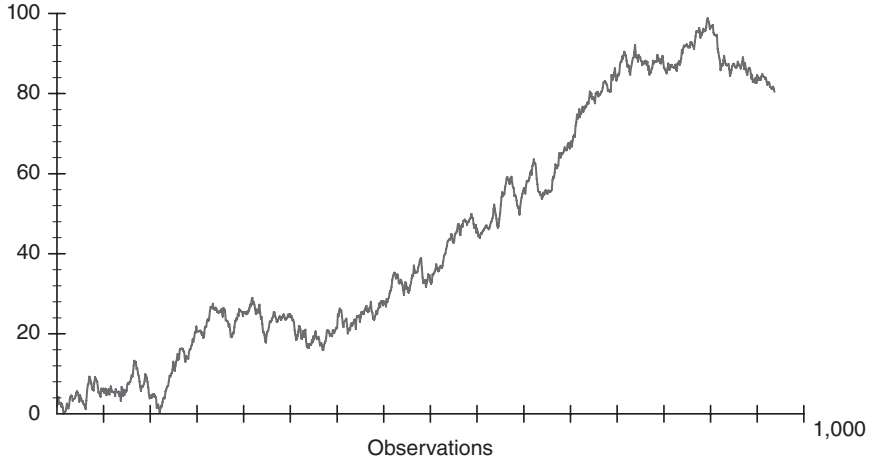


Figure 15.2 A random walk model with a drift, $\mu = 0.1$.

15.3 Unit root and other related processes

A unit root process is a generalization of the random walk model, where the innovations, u_t, in the random walk model

$$y_t = y_{t-1} + u_t, \tag{15.2}$$

are allowed to follow a general linear stationary process

$$u_t = \sum_{i=0}^{\infty} a_i \varepsilon_{t-i}, \tag{15.3}$$

where $\{\varepsilon_t\}$ is mean zero, serially uncorrelated process. Therefore, $\{y_t\}$ is an integrated process of order 1, $I(1)$. Similarly, $\{u_t\}$ is also referred to as an $I(0)$ process.

The $I(1)$ process in (15.2) is a unit root process without a drift, namely $E(y_t) = y_0$. A unit root process with a non-zero drift is defined by

$$\Delta y_t = \mu + u_t, \tag{15.4}$$

where $\mu \neq 0$, is the drift parameter of the process. Under (15.4), y_t is still an $I(1)$ process, but with a drift. The level of the process, y_t, has a trend and $E(y_t) = y_0 + \mu t$, for $t = 0, 1, 2, \ldots$.

15.3.1 Martingale process

The unit root process is a special case of a martingale process.

Definition 23 *Let $\{y_t\}_{t=0}^{\infty}$ be a sequence of random variables, and let Ω_t denote the information set available at date t, which at least contains $\{y_t, y_{t-1}, y_{t-2}, \ldots\}$. If $E(y_t \mid \Omega_{t-1}) = y_{t-1}$ holds then $\{y_t\}$ is a martingale process with respect to Ω_t.*

More formal definition for Ω_t denotes it as a non-decreasing sequence of σ-fields that is generated by $\{y_1, y_2, \ldots, y_t\}$, i.e., $\Omega_t \supseteq \Omega_{t-1} \supseteq \ldots \supseteq \Omega_0$. A process satisfying Definition 23 with Ω_t so defined is said to have unbounded memory. When y_t is the outcome of a game, then the martingale condition, $E(y_{t+1} \mid \Omega_t) = y_t$, is also known as the 'fair game' condition. It is clear that the random walk model

$$y_t = y_{t-1} + \varepsilon_t,$$

where ε_t is a white noise process, is a martingale process since

$$E(y_t \mid \Omega_{t-1}) = y_{t-1} + E(\varepsilon_t \mid \Omega_{t-1}),$$

and $E(\varepsilon_t \mid \Omega_{t-1}) = 0$, by assumption. The main difference between random walk and martingale processes lies in the assumption concerning the innovations ε_t. Under the random walk

model ε_t has a constant variance, whilst when y_t is a martingale process, the innovations ε_t could be conditionally and/or unconditionally heteroskedastic.

Some important properties of the martingale processes are:

1. $E\left(y_{t+j} \mid \Omega_t\right) = y_t$, for all $j \geq 0$. This follows from the law of expected iteration: Suppose there are two sets $S_t \subseteq \Omega_t$, then we have $E\left[E\left(y_t \mid \Omega_{t-1}\right) \mid S_{t-1}\right] = E\left(y_t \mid S_{t-1}\right)$. Apply this to the martingale process at hand

$$E\left[E\left(y_{t+1} \mid \Omega_t\right) \mid \Omega_{t-1}\right] = E\left(y_{t+1} \mid \Omega_{t-1}\right), \tag{15.5}$$

since $E\left(y_{t+1} \mid \Omega_t\right) = y_t$, is then

$$E\left[E\left(y_{t+1} \mid \Omega_t\right) \mid \Omega_{t-1}\right] = E\left(y_t \mid \Omega_{t-1}\right) = y_{t-1} = E\left(y_{t+1} \mid \Omega_{t-1}\right). \tag{15.6}$$

A generalization of (15.6) yields the desired result, $E\left(y_{t+j} \mid \Omega_t\right) = y_t$, for $j \geq 0$.

2. Constant mean. Since

$$E\left(y_t\right) = E\left[E\left(y_t \mid \Omega_{t-1}\right)\right] = E\left(y_{t-1}\right),$$

keep on iterating to obtain

$$E\left(y_{t+j}\right) = E\left(y_{t+j-1}\right) = \ldots = E\left(y_0\right) = \mu.$$

A martingale process has a constant mean, but it can have time-varying variance, that is, a martingale process allows for heteroskedasticity or conditional heteroskedasticity.

15.3.2 Martingale difference process

Definition 24 *Let $\left\{y_t\right\}_{t=1}^{\infty}$ be a sequence of random variables, and let Ω_t denote information available at date t, which at least contains $\left\{y_t, y_{t-1}, y_{t-2}, \ldots\right\}$. If $E\left(y_t \mid \Omega_{t-1}\right) = 0$, then $\left\{y_t\right\}$ is a martingale difference process with respect to Ω_t.*

A martingale difference process is serially uncorrelated. To see this, since $E\left(y_t\right) = E[E\left(y_t \mid \Omega_{t-1}\right)] = 0$, we have

$$Cov\left(y_t, y_{t-1}\right) = E\left(y_t y_{t-1}\right) = E\left[E\left(y_t y_{t-1} \mid \Omega_{t-1}\right)\right] = E\left[y_{t-1} E\left(y_t \mid \Omega_{t-1}\right)\right].$$

But $E\left(y_t \mid \Omega_{t-1}\right) = 0$, by definition. Hence, $Cov\left(y_t, y_{t-1}\right) = 0$. Similarly, it follows that $Cov\left(y_t, y_{t-j}\right) = 0$, for $j > 0$.

Note that the zero correlation property of martingale difference processes does not necessarily imply that the underlying processes are independently distributed. Further distributional assumptions are needed. For example, zero correlation implies independence if y_t is also assumed to be normally distributed. However, in many applications in economics and finance

the normality assumption does not hold and martingale difference processes are more generally applicable.

15.3.3 L_p-mixingales

The one-step ahead unpredictability property of martingale differences is often unrealistic in a time series context. A more general class of processes, known as L_p-mixingales introduced by McLeish (1975b) and Andrews (1988), provide an important generalization of martingale differences to situations where the process is asymptotically unpredictable in the sense formalized in Definition 25.

Definition 25 *Let $\{y_t\}$ be a sequence of random variables with $E(y_t) = 0, t = 1, 2, \ldots$, and let Ω_t be the information set available at time t. The sequence is said to follow an L_p-mixingale with respect to Ω_t if we can find a sequence of deterministic constants $\{c_t\}_{-\infty}^{\infty}$ and $\{\xi_m\}_0^{\infty}$ such that $\xi_m \to 0$ as $m \to \infty$ and*

$$\left\| E\left(y_t|\Omega_{t-m}\right)\right\|_p \leq c_t\xi_m,$$
$$\left\| y_t - E\left(y_t|\Omega_{t+m}\right)\right\|_p \leq c_t\xi_{m+1},$$

for all t, and for all $m \geq 0$.

The mixingale is said to be of size $-\lambda$ if $\xi_m = O\left(m^{-\lambda-\varepsilon}\right)$, for some $\varepsilon > 0$. The size of the mixingale is a useful summary measure of the degree of its dependence. The higher the λ the less dependent the process. A martingale difference process is a mixingale having $\xi_m = 0$ for all $m > 0$.

15.4 Trend-stationary versus first difference stationary processes

If y_t is trend stationary then

1. It will have a finite variance.
2. A shock will have only a *temporary* effect on the value of y_t.
3. The spectrum of y_t exists and will be finite at *zero* frequency.
4. The expected length of time between crossings of the trend of y_t is finite.
5. The autocorrelation coefficients, ρ_s, decrease steadily in magnitude in large enough s.

None of the above properties holds for a model with a unit root. In the case of the *simple* random walk process *without* a drift

$$y_t = y_{t-1} + \sigma\varepsilon_t, \qquad \varepsilon_t \sim IID(0, 1),$$

we have

$$y_t = y_0 + \sigma(\varepsilon_1 + \varepsilon_2 + \cdots + \varepsilon_t),$$

and therefore it follows that for this process

1. The variance of y_t is

$$Var\left(y_t\right) = \sigma^2 t,$$

which is an increasing function of time and hence will not be finite for large enough t.

2. A shock will have a permanent effect on y_t, namely for a non-zero shock of size δ hitting the system at time t

$$\lim_{h\to\infty}\left[E\left(y_{t+h}\,|\varepsilon_t = \delta,\ \Omega_{t-1}\right) - E\left(y_{t+h}\,|\Omega_{t-1}\right)\right] = \delta$$

which is non-zero.

3. The spectrum of y_t has the approximate shape $f(\omega) \sim A\omega^{-2}$, and $f(0) \to \infty$.
4. The expected time between crossings of $y = y_0$ is infinite.
5. $\rho_k \to 1$, for all k as $t \to \infty$.

The trend and difference stationary processes have both played a very important role in the empirical analysis of economic data, with the latter being used particularly in the analysis of financial data. The importance of difference stationary processes for the analysis of economic time series was first emphasized by Nelson and Plosser (1982), who argued that in the case of a majority of aggregate time series such as output, employment, prices and interest rates they are best characterized by a first difference stationary process, rather than by a stationary process round a deterministic trend. The issue of whether economic time series are trend stationary or first difference stationary (also known as the 'unit root' problem) has been the subject of intensive research and controversy.

15.5 Variance ratio test

The variance ratio test has been suggested in the finance literature as a test of the random walk model (15.1) where ε_t's are assumed to be identically and independently distributed with mean 0 and variance σ^2. In this case we have

$$\begin{aligned}
Var\left(y_t - y_{t-1}\right) &= Var\left(\mu + \varepsilon_t\right) = \sigma^2,\\
Var\left(y_{t+1} - y_{t-1}\right) &= Var\left(2\mu + \varepsilon_{t+1} + \varepsilon_t\right) = 2\sigma^2,\\
&\vdots\\
Var\left(y_{t+m-1} - y_{t-1}\right) &= Var\left(m\mu + \varepsilon_{t+m} + \varepsilon_{t+m-1} + \ldots + \varepsilon_t\right) = m\sigma^2.
\end{aligned}$$

Define variance ratio of order m as

$$VR_m = \frac{Var\left(y_{t+m-1} - y_{t-1}\right)}{mVar\left(y_t - y_{t-1}\right)},$$

where $Var\left(y_{t+m-1}-y_t\right)$ is known as the long-difference variance, and $Var\left(y_t-y_{t-1}\right)$ the short-difference variance. Under the random walk hypothesis (with or without a drift), we have $VR_m=1$.

Consider now the properties of VR_m under the alternative hypothesis that y_t is a stationary $AR(1)$ process

$$y_t=\rho y_{t-1}+\varepsilon_t,\quad |\rho|<1. \tag{15.7}$$

For $m=2$, we have

$$VR_2=\frac{Var\left(y_{t+1}-y_{t-1}\right)}{2Var\left(y_t-y_{t-1}\right)}=\frac{Var\left(y_{t+1}\right)-2Cov\left(y_{t+1},y_{t-1}\right)+Var\left(y_{t-1}\right)}{2Var\left(y_t-y_{t-1}\right)}, \tag{15.8}$$

and under the AR (1) model $Var\left(y_{t+1}\right)=Var\left(y_{t-1}\right)=\sigma^2/\left(1-\rho^2\right)$, and $Cov\left(y_{t+1},y_{t-1}\right)=\rho^2\sigma^2/\left(1-\rho^2\right)$. Hence, using these in (15.8) yields

$$VR_2=\frac{2\frac{\sigma^2}{(1-\rho^2)}-2\frac{\rho^2\sigma^2}{(1-\rho^2)}}{2\left[2\frac{\sigma^2}{(1-\rho^2)}-2\frac{\rho\sigma^2}{(1-\rho^2)}\right]}=\frac{1}{2}(1+\rho)<1, \tag{15.9}$$

and reduces to 1, *only* under the random walk model where $\rho=1$.

To derive VR_m in the general case since

$$y_{t+m-1}-y_{t-1}=\Delta y_{t+m}+\Delta y_{t+m-1}+\ldots+\Delta y_t=\sum_{j=0}^{m}\Delta y_{t+j},$$

then

$$Var\left(\frac{y_{t+m-1}-y_{t-1}}{m}\right)=Var\left(\frac{\sum_{j=1}^{m}\Delta y_{t+j}}{m}\right).$$

Now using the general formula for the variance of the sample mean of a stationary process given by (14.1), it readily follows that

$$Var\left(\frac{y_{t+m-1}-y_{t-1}}{m}\right)=\frac{1}{m}\left[\gamma_{\Delta y}(0)+2\sum_{h=1}^{m-1}\left(1-\frac{h}{m}\right)\gamma_{\Delta y}(h)\right],$$

where $\gamma_{\Delta y}(h)$ is the autocovariance function of Δy_t, which is a stationary process. Hence

$$VR_m=\frac{mVar\left(\frac{y_{t+m-1}-y_{t-1}}{m}\right)}{Var(y_t-y_{t-1})}=\frac{mVar\left(\frac{y_{t+m-1}-y_{t-1}}{m}\right)}{\gamma_{\Delta y}(0)}$$

$$= \quad 1 + 2\sum_{h=1}^{m-1}\left(1 - \frac{h}{m}\right)\rho_{\Delta y}(h), \tag{15.10}$$

where $\rho_{\Delta y}(h) = \gamma_{\Delta y}(h)/\gamma_{\Delta y}(0)$ is the autocorrelation function of order s of the first difference process $\{\Delta y_t\}$. It is easily verified that (15.9) is a special case of (15.10), noting that in the case of the $AR(1)$ process given by (15.7) we have

$$\rho_{\Delta y}(1) = \frac{Car\left(y_{t+1} - y_t, y_t - y_{t-1}\right)}{Var\left(y_t - y_{t-1}\right)} = -\frac{1-\rho}{2}.$$

A consistent estimator of VR_m is given by

$$\widehat{VR}_m = 1 + 2\sum_{h=1}^{m-1}\left(1 - \frac{h}{m}\right)\hat{\rho}_{\Delta y}(h),$$

where $\hat{\rho}_{\Delta y}(h)$ is the estimate of $\rho_{\Delta y}(h)$, based on the sample observations, $y_1, y_2, \ldots, y_T$. Specifically,

$$\hat{\rho}_{\Delta y}(h) = \frac{\sum_{t=h+1}^{T} w_t w_{t-h}}{\sum_{t=1}^{T} w_t^2},$$

where $w_t = \Delta y_t$ in the case of models without a deterministic trend, and

$$w_t = \Delta y_t - T^{-1}\sum_{\tau=1}^{T}\Delta y_\tau,$$

for series with linear deterministic trends (or random walk models with a drift).

It is interesting to note that equation (15.10) is closely related to the estimate of the *standardized* spectral density of Δy_t with Bartlett window of size $m-1$, namely (in the case of random walk models without a drift)

$$\hat{f}^*_{\Delta y}(0) = \frac{1}{2\pi}\left(1 + \sum_{h=1}^{m-1}\lambda_h \rho_{\Delta y}(h)\right), \quad \lambda_h = 1 - \frac{h}{m-1}. \tag{15.11}$$

Hence, a test of the random walk model can be carried out by testing the hypothesis that $2\pi\hat{f}^*_{\Delta y}(0) \approx VR_m = 1$. Significant departures of $2\pi f^*_{\Delta y}(0)$ from unity can be interpreted as evidence against the random walk model. The choice of m is guided by the same considerations as with the estimation of the spectral density, namely it should be chosen to be sufficiently large such that $m/T \to 0$ as $T \to \infty$. Popular choices are $m = T^{1/2}$ and $T^{1/3}$.

The above version of random walk hypothesis requires the strict *IID* assumption for the error terms, ε_t. One generalization of the above model relaxes the *IID* assumption and allows ε_ts to have non-identical distributions. This model includes the *IID* case as a special case and allows the unconditional heteroskedasticity in the ε_t's.

A more general version of the random walk hypothesis is obtained by relaxing the independence assumption. This is the same as the $I(1)$ or the first difference stationary model. Under the

null hypothesis that y_t is first difference stationary, $2\pi f^*_{\Delta y}(0)$ can depart from unity. Therefore, in general, it is not possible to base a test of the unit root hypothesis on $2\pi f^*_{\Delta y}(0)$.

15.6 Dickey–Fuller unit root tests

15.6.1 Dickey–Fuller test for models without a drift

Consider the $AR(1)$ model

$$y_t = \mu(1-\phi) + \phi y_{t-1} + \varepsilon_t. \tag{15.12}$$

The absence of drift for the unit root is achieved by the restriction on the intercept. That is, when $|\phi| < 1$, $E(y_t) = \mu$ and when $\phi = 1$, then $E(y_t) = y_0$. Therefore, (15.12) shows the $AR(1)$ without time trend regardless of whether $\phi = 1$ or $|\phi| < 1$. Such consideration is important since the trend characteristics contained in the data are invariant to whether the model used is unit root or not.

The unit root hypothesis is

$$H_0 : \phi = 1,$$

against

$$H_1 : |\phi| < 1.$$

To compute the Dickey–Fuller (DF) test statistic (Dickey and Fuller (1979)) we need first to write (15.12) as

$$\Delta y_t = \mu(1-\phi) - (1-\phi)\, y_{t-1} + \varepsilon_t, \tag{15.13}$$

and then test the null hypothesis that the coefficient of y_{t-1} in the above regression is zero. Letting $\beta = -(1-\phi)$, then (15.13) is

$$\Delta y_t = -\mu\beta + \beta y_{t-1} + \varepsilon_t, \tag{15.14}$$

and

$$H_0 : \beta = 0,$$

against

$$H_1 : \beta < 0.$$

Notice that under H_0 we have $\Delta y_t = \varepsilon_t$.

The DF test statistic is given by the t-ratio of β in (15.14), namely the t-ratio of the level variable, y_{t-1}, in the regression of Δy_t on the intercept term and y_{t-1}, for $t = 1, 2, \ldots, T$. Namely,

$$DF = \frac{\hat{\beta}}{s.e.(\hat{\beta})}, \tag{15.15}$$

with

$$\hat{\beta} = \frac{\sum_{t=1}^{T} \Delta y_t \left(y_{t-1} - \bar{y}_{-1}\right)}{\sum_{t=1}^{T} \left(y_{t-1} - \bar{y}_{-1}\right)^2},$$

where $\bar{y}_{-1} = T^{-1} \sum_{t=1}^{T} y_{t-1}$, is the simple average of $y_0, y_1, \ldots y_{T-1}$ and

$$\hat{V}(\hat{\beta}) = \frac{\hat{\sigma}^2}{\sum_{t=1}^{T} \left(y_{t-1} - \bar{y}_{-1}\right)^2},$$

where $\hat{\sigma}^2$ is the *OLS* estimator of σ^2. Therefore

$$DF = \frac{\hat{\beta}}{s.e.(\hat{\beta})} = \frac{\sum_{t=1}^{T} \Delta y_t \left(y_{t-1} - \bar{y}_{-1}\right)}{\hat{\sigma}[\sum_{t=1}^{T} \left(y_{t-1} - \bar{y}_{-1}\right)^2]^{1/2}}. \tag{15.16}$$

In matrix form,

$$DF = \frac{\Delta \mathbf{y}' \mathbf{M}_\tau \mathbf{y}_{-1}}{\hat{\sigma} \left(\mathbf{y}'_{-1} \mathbf{M}_\tau \mathbf{y}_{-1}\right)^{1/2}}, \tag{15.17}$$

where $\Delta \mathbf{y} = \left(\Delta y_1, \Delta y_2, \ldots, \Delta y_T\right)'$, $\boldsymbol{\tau} = (1, 1, \ldots, 1)'_{T \times 1}$, $\mathbf{M}_\tau = \mathbf{I}_T - \boldsymbol{\tau} \left(\boldsymbol{\tau}' \boldsymbol{\tau}\right)^{-1} \boldsymbol{\tau}'$, $\mathbf{y}_{-1} = \left(y_0, y_1, \ldots, y_{T-1}\right)'$, and $\mathbf{s}_{-1} = (s_0, s_1, s_2, \ldots s_{T-1})'$, with $s_0 = 0$, and $s_t = \sum_{i=1}^{t} \varepsilon_i$. But

$$\mathbf{y}_{-1} = y_0 \boldsymbol{\tau} + \mathbf{s}_{-1}, \tag{15.18}$$

and hence $\mathbf{M}_\tau y_{-1} = y_0 \mathbf{M}_\tau \boldsymbol{\tau} + \mathbf{M}_\tau \mathbf{s}_{-1} = \mathbf{M}_{\boldsymbol{\tau}} \mathbf{s}_{-1}$, since $\mathbf{M}_\tau \boldsymbol{\tau} = \mathbf{0}$. Under H_0, $\Delta \mathbf{y} = \boldsymbol{\varepsilon}$, with $\boldsymbol{\varepsilon} = (\varepsilon_1, \varepsilon_2, \ldots, \varepsilon_T)'$. Using these results in (15.17) we have

$$DF = \frac{\boldsymbol{\varepsilon}' \mathbf{M}_{\boldsymbol{\tau}} \mathbf{s}_{-1}}{\hat{\sigma} \left(\mathbf{s}'_{-1} \mathbf{M}_{\boldsymbol{\tau}} \mathbf{s}_{-1}\right)^{1/2}}. \tag{15.19}$$

Also, under H_0

$$\hat{\sigma}^2 = \sigma^2 + o_p(1).$$

Hence asymptotically,

$$DF \overset{a}{\sim} \frac{\left(\frac{\boldsymbol{\varepsilon}}{\sigma}\right)' \mathbf{M}_{\boldsymbol{\tau}} \left(\frac{\mathbf{s}_{-1}}{\sigma}\right)}{\left[\left(\frac{\mathbf{s}_{-1}}{\sigma}\right)' \mathbf{M}_{\boldsymbol{\tau}} \left(\frac{\mathbf{s}_{-1}}{\sigma}\right)\right]^{1/2}}, \tag{15.20}$$

with $E\left(\frac{\boldsymbol{\varepsilon}}{\sigma}\right) = 0$ and $V\left(\frac{\boldsymbol{\varepsilon}}{\sigma}\right) = 1$. Therefore, for large T, the DF statistic does not depend on σ, and without loss of generality we can set $\sigma = 1$, and write

$$DF \overset{a}{\sim} \frac{\boldsymbol{\varepsilon}' \mathbf{M}_{\boldsymbol{\tau}} \mathbf{s}_{-1}}{\left(\mathbf{s}'_{-1} \mathbf{M}_{\boldsymbol{\tau}} \mathbf{s}_{-1}\right)^{1/2}} = \frac{\left(\boldsymbol{\varepsilon}' \mathbf{M}_{\boldsymbol{\tau}} \mathbf{s}_{-1}\right)/T}{\left(\frac{\mathbf{s}'_{-1} \mathbf{M}_{\boldsymbol{\tau}} \mathbf{s}_{-1}}{T^2}\right)^{1/2}}, \tag{15.21}$$

where $\boldsymbol{\varepsilon} \sim (\mathbf{0}, \mathbf{I}_T)$. It is clear from this result that the asymptotic distribution of the DF statistic does not depend on any nuisance parameters and therefore can be tabulated for reasonably large values of T. We shall return to the mathematical form of the asymptotic distribution of DF below.

15.6.2 Dickey–Fuller test for models with a drift

Consider the case where y_t contains a linear trend (with a restricted coefficient),

$$y_t = \alpha + \mu(1 - \phi)t + \phi y_{t-1} + \varepsilon_t, \tag{15.22}$$

and suppose the following regression is estimated

$$\Delta y_t = \alpha + \mu\,(1 - \phi)\,t - (1 - \phi)\,y_{t-1} + \varepsilon_t,$$

or letting $\beta = -(1 - \phi)$,

$$\Delta y_t = a_0 - \mu\beta t + \beta y_{t-1} + \varepsilon_t, \tag{15.23}$$

and

$$H_0 : \beta = 0,$$

against

$$H_1 : \beta < 0.$$

Equation (15.22) allows the model to share the same trend features, irrespective of whether $|\phi| < 1$ or $\phi = 1$. This follows since under $|\phi| < 1$, we have $E\left(y_t\right) = \alpha/(1 - \phi) + \mu t$, and when $\phi = 1$, we also have $E\left(y_t\right) = y_0 + \alpha t$, namely the mean of y_t follows a linear trend in both cases. The DF statistic is given by the t-ratio of the OLS estimate of β in (15.23), namely the t-ratio of the coefficient associated with the level variable, y_{t-1}, in the regression of Δy_t on the intercept term, a linear time trend and y_{t-1}.

15.6.3 Asymptotic distribution of the Dickey–Fuller statistic

Although the *DF* statistic is a *t*-ratio, due to the random walk nature of the y_t process under $H_0 : \beta = 0$, the distribution of the *DF* statistic is non-standard, in the sense that its limiting distribution is not normal. To derive the limiting distribution of the *DF* statistic, new tools from continuous time analysis of Wiener processes are required.

Definition 26 (Wiener process) *Let $\Delta w(t)$ be the change in $w(t)$ during the time interval dt. Then $w(t)$ is said to follow a Wiener process if*

$$\Delta w(t) = \varepsilon_t \sqrt{dt}, \ \ \varepsilon_t \sim IID(0,1), \tag{15.24}$$

and $w(t)$ denotes the value of the $w(\cdot)$ at date t. Clearly,

$$E(\Delta w(t)) = 0, \text{ and } Var(\Delta w(t)) = dt.$$

Theorem 43 (Donsker's theorem) *Let $a \in [0,1)$, $t \in [0,T]$, and suppose $(J-1)/T \leq a < J/T, J = 1,2,\ldots T$. Define*

$$R_T(a) = \frac{1}{\sqrt{T}} s_{[Ta]}, \tag{15.25}$$

where

$$s_{[Ta]} = \varepsilon_1 + \varepsilon_2 + \ldots. + \varepsilon_{[Ta]},$$

$[Ta]$ denotes the largest integer part of Ta and $s_{[Ta]} = 0$, if $[Ta] = 0$. Then $R_T(a)$ weakly converges to $w(a)$, i.e.,

$$R_T(a) \Rightarrow w(a),$$

where $w(a)$ is a Wiener process.

For a proof, see Billingsley (1999) or Chan and Wei (1988).

Note that when $a = 1, R_T(1) = \frac{1}{\sqrt{T}} S_{[T]} = \frac{1}{\sqrt{T}} (\varepsilon_1 + \varepsilon_2 + \ldots \varepsilon_T)$. Since $\varepsilon_t' s$ are IID, by the central limit theorem, $R_T(1) \Rightarrow N(0,1)$.

Another related theorem is the continuous mapping theorem.

Theorem 44 (Continuous mapping) *Let $a \in [0,1)$, $t \in [0,T]$, and suppose $(J-1)/T \leq a < J/T, J = 1,2,\ldots T$. Define $R_T(a) = \frac{1}{\sqrt{T}} S_{[Ta]}$. Iff $(\cdot)$ is continuous over [0,1), then*

$$f[R_T(a)] \Rightarrow f[w(a)]. \tag{15.26}$$

The following limit results are useful for deriving the limiting distribution of the DF statistic. Let $\bar{s}_T = (s_1 + s_2 + \ldots + s_T)/T$, then

$$\begin{aligned}
\sqrt{T}\bar{s}_T &\approx \int_0^1 R_T(a)\,da \Rightarrow \int_0^1 w(a)\,da, \\
\frac{\sum_{t=1}^T s_t^2}{T^2} &\Rightarrow \int_0^1 w(a)^2\,da, \\
T^{-\frac{5}{2}}\sum_{t=1}^T ts_t &\Rightarrow \int_0^1 aw(a)\,da, \\
T^{-\frac{3}{2}}\sum_{t=1}^T t\varepsilon_t &\Rightarrow \int_0^1 a\,dw(a),
\end{aligned} \tag{15.27}$$

and

$$T^{-1}\sum_{t=1}^T \varepsilon_t s_{t-1} \Rightarrow \int_0^1 w(a)\,dw(a).$$

In addition, we have the following result

$$\int_0^1 w(a)\,d(w(a)) \Rightarrow \frac{1}{2}\left[w(1)^2 - 1\right] = \frac{1}{2}\left(\chi_1^2 - 1\right). \tag{15.28}$$

To establish (15.28), first note that

$$\begin{aligned}
\sum_{t=1}^T s_t^2 &= \sum_{t=1}^T (s_{t-1} + \varepsilon_t)^2, \\
&= \sum_{t=1}^T s_{t-1}^2 + \sum_{t=1}^T \varepsilon_t^2 + 2\sum_{t=1}^T s_{t-1}\varepsilon_t.
\end{aligned}$$

We also have

$$\begin{aligned}
\frac{2\sum_{t=1}^T s_{t-1}\varepsilon_t}{T} &= \frac{\sum_{t=1}^T s_t^2}{T} - \frac{\sum_{t=1}^T s_{t-1}^2}{T} - \frac{\sum_{t=1}^T \varepsilon_t^2}{T} \\
&= \frac{s_T^2}{T} - \frac{\sum_{t=1}^T \varepsilon_t^2}{T}.
\end{aligned}$$

Since $\left(1/\sqrt{T}\right)s_T = R_T(1) \overset{a}{\sim} N(0,1)$, we have $s_T^2/T \overset{a}{\sim} \chi_1^2$, and since $\varepsilon_t \sim IID(0,1)$, by the application of standard law of large numbers, $\sum_{t=1}^T \varepsilon_t^2/T$ converges to its limit 1. Hence, it follows that

$$\int_0^1 w(a)\,dw(a) \Rightarrow \frac{\sum_{t=1}^T s_{t-1}\varepsilon_t}{T} \Rightarrow \frac{1}{2}\left(\chi_1^2 - 1\right).$$

Also

$$\int_0^1 w(a)\,da \Rightarrow N\left(0, \frac{1}{3}\right).$$

More general results can also be obtained for $I(1)$ processes. Suppose that y_t follows the general linear process

$$y_t - y_{t-1} = u_t = \sum_{i=0}^{\infty} a_i \varepsilon_{t-i}, t = 1, 2, \ldots, T,$$

where $y_0 = 0$, $\varepsilon_t \sim IID(0, \sigma^2)$, $\sum_{i=0}^{\infty} |a_i| < \infty$, and $Var(u_t) = \sigma^2 \sum_{i=0}^{\infty} a_i^2 = \sigma_u^2 < \infty$. Let

$$\gamma_{\Delta y}(h) = E(u_t u_{t-h}), \text{ and } \lambda = \sigma a(1).$$

Then the following limits can be established, for $h \geq 0$:[1]

$$T^{-1/2} \sum_{t=1}^{T} u_t \xrightarrow{d} \sigma a(1) w(1) \equiv N[0, \sigma^2 a^2(1)],$$

$$T^{-1/2} \sum_{t=1}^{T} u_{t-h} \varepsilon_t \xrightarrow{d} N\left[0, \sigma^2 \gamma_{\Delta y}(0)\right],$$

$$T^{-1} \sum_{t=1}^{T} u_t u_{t-h} \xrightarrow{p} \gamma_{\Delta y}(h),$$

$$T^{-1} \sum_{t=1}^{T} u_{t-1} \varepsilon_t \xrightarrow{d} \frac{1}{2}\sigma^2 a(1)\left[w^2(1) - 1\right] \equiv \frac{1}{2}\sigma^2 a(1)\left(\chi_1^2 - 1\right),$$

$$T^{-1} \sum_{t=1}^{T} y_{t-1} u_t \xrightarrow{d} \frac{1}{2}\left[\sigma^2 a^2(1) w(1)^2 - \gamma_{\Delta y}(0)\right],$$

$$T^{-1} \sum_{t=1}^{T} y_{t-1} u_{t-h} \xrightarrow{d} \frac{1}{2}\left[\sigma^2 a^2(1) w(1)^2 - \gamma_{\Delta y}(0)\right] + \gamma_{\Delta y}(0) + \gamma_{\Delta y}(1) + \ldots$$
$$+\gamma_{\Delta y}(h-1), \text{ for } h = 1, 2, \ldots.$$

Also

$$T^{-3/2} \sum_{t=1}^{T} y_{t-1} \xrightarrow{d} \sigma a(1) \int_0^1 w(a) da,$$

[1] See, for example, Hamilton (1994, Ch. 17).

$$T^{-3/2}\sum_{t=1}^{T} ty_{t-j} \xrightarrow{d} \sigma a(1)\left[w(1) - \int_0^1 w(a)da\right],$$

$$T^{-2}\sum_{t=1}^{T} y_{t-1}^2 \xrightarrow{d} \sigma^2 a^2(1)\int_0^1 w^2(a)da,$$

$$T^{-5/2}\sum_{t=1}^{T} ty_{t-1} \xrightarrow{d} \sigma a(1)\int_0^1 aw(a)da,$$

$$T^{-3}\sum_{t=1}^{T} ty_{t-1}^2 \xrightarrow{d} \sigma^2 a^2(1)\int_0^1 aw^2(a)da.$$

15.6.4 Limiting distribution of the Dickey–Fuller statistic

The limiting distribution of *DF* depends on whether the *DF* regression contains an intercept or a linear trend, or neither. In the simple case where the *DF* regression does not contain any deterministics (Case I), we have

$$DF \Rightarrow \frac{\int_0^1 w(a)\,dw(a)}{\left[\int_0^1 w(a)^2\,da\right]^{1/2}} = \frac{\frac{1}{2}\left(\chi_1^2 - 1\right)}{\left[\int_0^1 w(a)^2\,da\right]^{1/2}}.$$

If the *DF* regression only contains an intercept (Case II), we have

$$DF \Rightarrow \frac{\int_0^1 w(a)\,dw(a) - w(1)\int_0^1 w(a)da}{\left\{\int_0^1 w^2(a)da - \left[\int_0^1 w(a)da\right]^2\right\}^{1/2}}.$$

Similar expressions can also be obtained for *DF* regression models with a linear trend (Case III).

15.6.5 Augmented Dickey–Fuller test

Augmented *DF* (*ADF*) regression of order p is given by

$$\Delta y_t = \alpha + \mu(1-\phi)t - (1-\phi)y_{t-1} + \sum_{i=1}^{p}\psi_i \Delta y_{t-i} + \varepsilon_t,$$

where p is chosen such that the equation's residuals, ε_t, are serially uncorrelated. In practice model selection criteria such as the Akaike information criterion (*AIC*), or the Schwarz Bayesian Criterion (*SBC*) are used to select p.[2] The *ADF*(p) statistic is given by the t-ratio of y_{t-1} in the above regression. Critical values for *DF* and *ADF* tests are the same. This follows since as $t \to \infty$, the y_{t-1} component dominates the augmentation part, $\sum_{i=1}^{p}\psi_i\Delta y_{t-i}$, which is a

[2] For a discussion of AIC and SBC see Sections 11.5.1 and 11.5.2.

stationary process, and hence its effect diminishes and can be ignored as $T \to \infty$. When using the *ADF* tests the following points are worth bearing in mind:

1. The *ADF* test is not very powerful in finite samples for alternatives $H_1 : \phi < 1$ when ϕ is near unity.
2. There is a size-power trade-off depending on the order of augmentation used in dealing with the problem of residual serial correlation. Therefore, it is often crucial that an appropriate value is chosen for p, the order of augmentation of the test.

Appropriate (asymptotic) critical values of the *DF* test have been tabulated by Fuller (1996) and by MacKinnon (1991). In the following, we describe how critical values can be obtained.

15.6.6 Computation of critical values of the DF statistics

The calculation of the critical values can be achieved by stochastic simulation. Evaluating (15.23) under the H_0 gives $\Delta y_t = \varepsilon_t$. Simulate $\Delta y_t^j = \varepsilon_t^j$, $t = 1, 2, \ldots, T$, $j = 1, 2, \ldots$, where ε_t^j represents a $N(0, 1)$ random draw. More specifically we first generate

$$\begin{aligned} y_1^j &= \varepsilon_1^j, \\ y_2^j &= \varepsilon_1^j + \varepsilon_2^j, \\ &\vdots \\ y_T^j &= \varepsilon_1^j + \varepsilon_2^j + \ldots + \varepsilon_T^j. \end{aligned}$$

Run the *OLS* regression of Δy_t^j on y_{t-1}^j, for $t = 2, 3, \ldots, T$ (with intercepts or linear trends as the case might be) and compute the *OLS* t-ratio of the coefficient of y_{t-1}^j. Repeat this process for $j = 1, 2, \ldots, R$, and then order all the R t-ratios (from lowest to the highest) in the form of a histogram. The α per cent critical value is given by the value of the t-ratio to the left of the histogram which represents α per cent of the replications.

This computation is possible under the assumption that errors ε_t in $y_t = \rho y_{t-1} + \varepsilon_t$ are serially uncorrelated. If such an assumption does not hold, then testing for unit root involves using the *ADF* statistic described in Section 15.6.4.

Other unit root tests have been proposed in the literature. In Section 15.7, we review unit root tests proposed by Phillips and Perron (1988), Elliott, Rothenberg, and Stock (1996), Park and Fuller (1995), and Leybourne (1995). We also describe a test of stationarity developed by Kwiatkowski et al. (1992).

15.7 Other unit root tests

15.7.1 Phillips–Perron test

Phillips and Perron (1988) (*PP*) provide a semi-parametric test of the unit root hypothesis for a single time series. Unlike the *ADF* test, the *PP* test attempts to correct for the effect of residual serial correlation in a simple *DF* regression using non-parametric estimates of the long-run variance (which is also proportional to the spectral density of the residuals at zero frequency).

Models with intercepts but without trend

Consider the simple *DF* regression

$$\Delta y_t = a + by_{t-1} + u_t,\ t = 1, 2, \ldots, T,$$

and denote the residuals from this regression by $\hat{u}_t$ and the t ratio of the *OLS* estimator of b by DF_τ. Compute

$$s_T^2 = \frac{\sum_{t=1}^{T} \hat{u}_t^2}{T}, \quad \hat{\gamma}_j = \frac{\sum_{t=j+1}^{T} \hat{u}_t \hat{u}_{t-j}}{T},$$

and

$$s_{LT}^2 = \hat{\gamma}_0 + 2\sum_{j=1}^{m}\left(1 - \frac{j}{m+1}\right)\hat{\gamma}_j$$

which uses the Bartlett window.[3] The *PP* unit root test is given by

$$Z_{\tau,df} = \left(\frac{s_T}{s_{LT}}\right) DF_\tau - \frac{\frac{1}{2}\left(s_{LT}^2 - s_T^2\right)}{s_{LT}\left[\frac{\sum_{t=1}^{T}\left(y_{t-1}-\bar{y}_{-1}\right)^2}{T^2}\right]^{1/2}}$$

where $\bar{y}_{-1} = \sum_{t=1}^{T-1} y_{t-1}/T$.

Models with an intercept and a linear trend

In this case the underlying *DF* regression is

$$\Delta y_t = a_0 + a_1 t + by_{t-1} + u_t,\ t = 1, 2, \ldots, T,$$

with DF_t given by the t ratio of the *OLS* estimate of y_{t-1}, and

$$\hat{u}_t = \Delta y_t - \hat{a}_0 - \hat{a}_1 t - \hat{b}y_{t-1}.$$

Write the above in matrix notation as

$$\Delta \mathbf{y} = \mathbf{W}\boldsymbol{\theta} + \mathbf{u}$$

where

$$\mathbf{W} = \begin{pmatrix} 1 & 1 & y_0 \\ 1 & 2 & y_1 \\ \vdots & \vdots & \vdots \\ 1 & T-1 & y_{T-2} \\ 1 & T & y_{T-1} \end{pmatrix}.$$

[3] Parzen and Tukey's windows can also be used.

Then the *PP* statistic in this case is given by

$$Z_{t,df} = \left(\frac{s_T}{s_{LT}}\right) DF_t - \frac{T^3\left(s_{LT}^2 - s_T^2\right)}{4\sqrt{3}s_{LT}D_w^{1/2}},$$

where D_w is the determinant of the matrix $\mathbf{W}'\mathbf{W}$.

The asymptotic critical values of DF_τ and DF_t (simple *DF* tests for models with intercept and models with linear trend. respectively) continues to apply to the *PP* tests.

15.7.2 *ADF-GLS* unit root test

The *ADF-GLS* test is proposed by Elliott, Rothenberg, and Stock (1996) and generally has better power characteristics than the standard *ADF* test. Consider the ordered series $y_t, t = 1, 2, \ldots, T$ and make the following transformations

$$\begin{aligned} y_{\rho 1} &= y_1, \\ y_{\rho t} &= y_t - \rho y_{t-1}, \text{ for } t = 2, \ldots, T, \end{aligned}$$

where ρ is a fixed coefficient to be set *a priori* (see below). Similarly

$$\begin{aligned} z_{11}(\rho) &= 1, \\ z_{1t}(\rho) &= (1 - \rho), \text{ for } t = 2, \ldots, T, \end{aligned}$$

and

$$\begin{aligned} z_{21}(\rho) &= 1, \\ z_{2t}(\rho) &= t - \rho(t-1), \text{ for } t = 2, \ldots, T. \end{aligned}$$

Models with intercepts but without trend

Compute the *OLS* regression of $y_{\rho t}$ on $z_{1t}(\rho)$

$$\hat{\beta}_\rho = \frac{y_1 + (1-\rho)\sum_{t=2}^{T}\left(y_t - \rho y_{t-1}\right)}{1 + (1-\rho)^2(T-1)},$$

and then deviations

$$w_t = y_t - \hat{\beta}_\rho, \text{ for } t = 1, 2, \ldots, T,$$

and carry out $ADF(p)$ test applied to w_t. It is recommended that ρ is set to $1 - 7/T$.

Models with an intercept and a linear trend

Compute the *OLS* regression coefficients of $y_{\rho t}$ on $z_{1t}(\rho)$, and $z_{2t}(\rho)$, and denote these coefficients by $\hat{\beta}_{1\rho}$ and $\hat{\beta}_{2\rho}$ and then compute

$$w_t = y_t - \hat{\beta}_{1\rho} - \hat{\beta}_{2\rho}t, \text{ for } t = 1, 2, \ldots, T,$$

and then apply the $ADF(p)$ procedure to w_t. The recommended choice of ρ for this case is $1 - 13.5/T$.

The form of the *ADF-GLS* test can be set out as

$$ADF\text{-}GLS(c_\mu, c_\tau),$$

where

$$\rho = 1 - \frac{c_\mu}{T}, \text{ for models with intercept only,}$$
$$\rho = 1 - \frac{c_\tau}{T}, \text{ for models with intercept and linear trend.}$$

The 5 per cent critical values of *ADF-GLS* test can be found in Pantula, Gonzalez-Farias, and Fuller (1994) and in Elliott, Rothenberg, and Stock (1996) and have been reproduced in Table 15.1.

Table 15.1 The 5 per cent critical values of *ADF-GLS* tests

Deterministics/T	25	50	100	200	250	500	∞
Intercept[1] ($c_\mu = 7$)	−2.56	−2.30	−2.14	**−2.05**	−2.03	**−1.99**	−1.95
Linear trend[2] ($c_\tau = 13.5$)	**−3.55**	−3.19	−3.03	−2.93	**−2.93**	**−2.92**	−2.89

Bold figures are simulated using *Microfit 5.0*, with 10,000 replications.
(1) From Table 2 of Pantula, Gonzalez-Farias, and Fuller (1994)
(2) From Table I in Elliott, Rothenberg, and Stock (1996)

15.7.3 The weighted symmetric tests of unit root

The Weighted Symmetric *ADF* (*WS-ADF*) has been proposed by Park and Fuller (1995) and analyzed further by the means of Monte Carlo simulations by Pantula, Gonzalez-Farias, and Fuller (1994). A detailed discussion is also provided by Fuller (1996, Section 10.1.3).

The weighted symmetric estimates

The *WS-ADF* attempts to increase the power of the test by making use of the fact that any stationary autoregressive process can be given a forward as well as a backward representation. An estimator of the autoregressive parameters that takes account of this property is generally known as *WS* estimators. Consider the *p*th-order (backward) *ADF* regression

$$y_t = \phi y_{t-1} + \sum_{j=1}^{p} \delta_j \Delta y_{t-j} + \varepsilon_t^b,$$

then under stationarity we also have

$$y_t = \phi y_{t+1} - \sum_{j=1}^{p} \delta_j \Delta y_{t+j+1} + \varepsilon_t^f.$$

The *WS* estimator of ϕ is obtained by solving the following weighted least squares problem

$$Q(\phi,\boldsymbol{\delta}) = \sum_{t=p+2}^{T} w_t \left(y_t - \phi y_{t-1} - \sum_{j=1}^{p} \delta_j \Delta y_{t-j} \right)^2 + \sum_{t=p+2}^{T} (1 - w_{t-p}) \left(y_{t-p-1} - \phi y_{t-p} + \sum_{j=1}^{p} \delta_j \Delta y_{t-p+j} \right)^2,$$

or equivalently

$$Q(\phi,\boldsymbol{\delta}) = \sum_{t=p+2}^{T} w_t \left(y_t - \phi y_{t-1} - \sum_{j=1}^{p} \delta_j \Delta y_{t-j} \right)^2 + \sum_{t=1}^{T-p-1} (1 - w_{t+1}) \left(y_t - \phi y_{t+1} + \sum_{j=1}^{p} \delta_j \Delta y_{t+j+1} \right)^2,$$

where

$$w_t = \begin{cases} 0, \text{for } 1 \le t \le p+1, \\ \frac{t-p-1}{T-2p}, \text{for } p+1 < t \le T-p, \\ 1, \text{for } T-p < t \le T, \end{cases}$$

and assuming that $T \ge 2(p+1)$.

Let $\hat{\phi}$ and $\hat{\boldsymbol{\delta}} = (\hat{\delta}_1, \ldots, \hat{\delta}_p)'$ be the estimators of ϕ and $\boldsymbol{\delta}$ that minimize $Q(\phi,\boldsymbol{\delta})$. Then the *WS-ADF*($p$) statistic is given by

$$\frac{\hat{\phi} - 1}{\sqrt{\widehat{Var}\left(\hat{\phi}\right)}},$$

where

$$\widehat{Var}\left(\hat{\phi}\right) = \hat{\sigma}^2 a_{\phi\phi},$$

$$\hat{\sigma}^2 = \frac{Q(\hat{\phi},\hat{\boldsymbol{\delta}})}{T-p-2} \text{ for a model with intercept,}$$

$$\hat{\sigma}^2 = \frac{Q(\hat{\phi},\hat{\boldsymbol{\delta}})}{T-p-3} \text{ for a model with a linear trend,}$$

and $a_{\phi\phi}$ is the element (1,1) in the inverse of $\partial^2 Q(\phi,\boldsymbol{\delta})/\partial\boldsymbol{\theta}\partial\boldsymbol{\theta}'$, where $\boldsymbol{\theta} = (\hat{\phi}, \hat{\boldsymbol{\delta}}')'$.

Explicit solution

Let $\mathbf{z}_{bt} = (y_{t-1}, \Delta y_{t-1}, \ldots, \Delta y_{t-p})'$ and $\mathbf{z}_{ft} = (y_{t+1}, -\Delta y_{t+2}, -\Delta y_{t+3}, \ldots, -\Delta y_{t+p+1})'$, then it is easily seen that

$$\hat{\boldsymbol{\theta}} = \mathbf{A}_T^{-1}\mathbf{b}_T,$$

where

$$\mathbf{A}_T = \sum_{t=p+2}^{T} w_t \mathbf{z}_{bt}\mathbf{z}_{bt}' + \sum_{t=1}^{T-p-1} (1 - w_{t+1})\, \mathbf{z}_{ft}\mathbf{z}_{ft}',$$

$$\mathbf{b}_T = \sum_{t=p+2}^{T} w_t \mathbf{z}_{bt} y_t + \sum_{t=1}^{T-p-1} (1 - w_{t+1})\, \mathbf{z}_{ft} y_t.$$

Also

$$\frac{\partial^2 Q(\phi, \boldsymbol{\delta})}{\partial\theta\,\partial\theta'} = \mathbf{A}_T,$$

and

$$Var\left(\hat{\boldsymbol{\theta}}\right) = \hat{\sigma}^2 \mathbf{A}_T^{-1}.$$

Clearly, $\hat{\boldsymbol{\theta}}$ reduces to the *OLS* estimator if $w_t = 1$ for all t.

Treatment of deterministic components

The submitted series for testing need to be de-meaned or de-trended (depending on the case being considered) before the computations. This can be done by simple regression techniques or by the *GLS* procedures in Elliott, Rothenberg, and Stock (1996).

Critical values

The relevant critical values are given in Fuller (1996, p. 644, Table 10.A.4). The 5 per cent critical values of the *WS-ADF* test for various sample sizes are reproduced in Table 15.2.

Table 15.2 The 5 per cent critical values of *WS-ADF* tests

Deterministics/T	25	50	100	250	500	∞
None	–2.09	–2.13	–2.16	–2.17	–2.18	–2.18
Intercept	–2.60	–2.57	–2.55	–2.54	–2.53	–2.52
Linear trend	–3.37	–3.28	–3.24	–3.21	–3.20	–3.19

15.7.4 Max *ADF* unit root test

This test is proposed by Leybourne (1995) and is given by $Max\left(ADF_f, ADF_r\right)$, where ADF_f is the usual forward *ADF* test statistic and ADF_r is the *ADF* statistic based on the associated reversed data series (after de-meaning or de-trending as the case might be). Let $y_t, t = 1, 2, \ldots, T$, be the original series. Then the reversed series is given by $y_t^r = y_{T-t+1}, t = 1, 2, \ldots, T$. The critical values of *MAX-ADF* tests (for models with intercept only and linear trends) at 10, 5, and 1 per cent significance levels are reproduced in Table 15.3.

Pantula, Gonzalez-Farias, and Fuller (1994) and Leybourne (1995) provide Monte Carlo evidence suggesting that the *MAX-ADF* test and the *WS-ADF* test could be equal to or even more powerful than the *ADF-GLS* test. Another possibility would be to apply the *WS-ADF* procedure (or the *MAX-ADF* procedure) to *GLS* de-meaned or de-trended series. New critical values are however needed for the implementation of such a test.

In addition to the above unit root test, tests that focus on the null of stationarity have also been proposed.

Table 15.3 The critical values of *MAX-ADF* tests

(a) With intercepts

Size/T	25	50	100	200	400
10%	–2.15	–2.14	–2.14	–2.13	–2.13
5%	–2.50	–2.48	–2.45	–2.44	–2.43
1%	–3.25	–3.17	–3.11	–3.06	–3.04

(b) With linear trends

Size/T	25	50	100	200	400
10%	–2.89	–2.87	–2.84	–2.83	–2.82
5%	–3.26	–3.22	–3.16	–3.12	–3.09
1%	–3.99	–3.84	–3.75	–3.72	–3.70

15.7.5 Testing for stationarity

The null hypothesis of *ADF* and Phillips and Perron (1988) tests is the presence of a single unit root in the autoregressive representation of a process. An alternative approach would be to take the 'stationarity' as the null hypothesis Such a test is developed by Kwiatkowski, Phillips, Schmidt, and Shin (1992, KPSS). The test is based on the idea that the variance of the partial sum series

$$s_t = \sum_{j=1}^{t} \hat{e}_j, \text{ where } \hat{e}_j = y_j - \hat{\alpha} - \hat{\beta} j,$$

is relatively small under stationarity as compared with the alternative unit root hypothesis.

The *KPSS* test statistic is defined by

$$\widehat{\zeta}_T = \left[\frac{1}{Ts_T(l)}\right]^2 \sum_{t=1}^{T} s_t^2,$$

where $s_T^2(l)$ is the estimate of the long-run variance of s_t, given by

$$s_T^2(l) = \frac{1}{T}\sum_{t=1}^{T}\hat{e}_t^2 + \frac{2}{T}\left[\sum_{j=1}^{l} w_j \left(\sum_{t=j+1}^{T}\hat{e}_t\hat{e}_{t-j}\right)\right],$$

and

$$w_j = 1 - \frac{j}{l+1}, \quad j = 1, 2, \ldots, l.$$

These weights are the Bartlett's window introduced in Chapter 13. Other choices for w_j are also possible, such as Tukey or Parzen windows. The critical values of the *KPSS* test statistic are reproduced in Table 15.4.

Table 15.4 The critical values of *KPSS* test

	10%	5%	2.5%	1%
Constant no trend	0.35	0.46	0.57	0.74
Constant and trend	0.12	0.15	0.18	0.22

15.8 Long memory processes

Long memory processes lie somewhere between covariance stationary and unit root processes. Covariance stationary processes have absolutely summable autocovariances and fall in the class of *short memory* processes, in the sense that their autocovariances decay relatively fast. The unit root processes stand at the other extreme and, as we have seen, their autocovariances do not decay at all. But, in the case of many time series in economics and finance, the effects of a shock might not be permanent and could take a very long time to vanish. In such cases autocovariances could fail to be absolutely summable. Such covariance stationary processes whose autocovariances are not absolute summable are known as long memory processes.

As a simple example, consider the linear process

$$y_t = \sum_{i=0}^{\infty}\left(\frac{1}{1+i}\right)\varepsilon_{t-i},$$

where $\{\varepsilon_t\}$ is a white noise process. It is easily seen that y_t has mean zero, a finite constant variance, $\sigma^2\sum_{i=0}^{\infty}\left(\frac{1}{1+i}\right)^2 = \sigma^2\left(\pi^2/6\right)$, and

$$\gamma(h) = \gamma(-h) = \sigma^2 \sum_{i=0}^{\infty} \left(\frac{1}{1+i}\right)\left(\frac{1}{1+i+|h|}\right) < K < \infty, \text{for all } h.$$

But these autocovariances are not absolutely summable. This is because

$$\begin{aligned}\sum_{h=0}^{\infty} |\gamma(h)| &= \sigma^2 \sum_{h=0}^{\infty}\sum_{i=0}^{\infty} \left(\frac{1}{1+i}\right)\left(\frac{1}{1+i+|h|}\right) \\ &= \sigma^2 \sum_{i=0}^{\infty} \left(\frac{1}{1+i}\right) \sum_{h=0}^{\infty} \left(\frac{1}{1+i+|h|}\right) = \infty,\end{aligned}$$

since all the elements are positive and for each $i = 0, 1, 2, \ldots$, we have $\sum_{h=0}^{\infty} \left(\frac{1}{1+i+|h|}\right) = \infty$.

Definition 27 *Consider a covariance stationary process, $\{y_t\}$, and let $\gamma(h)$ be its autocovariance function at lag h. Then y_t is said to be a long memory process if*

$$\sum_{h=-\infty}^{\infty} |\gamma(h)| = \infty, \tag{15.29}$$

or alternatively if

$$\gamma(h) \backsim h^{2d-1} g(h), \text{ as } h \to \infty, \tag{15.30}$$

where $g(h)$ is a slowly varying function of h. The constant d is known as the 'long-memory parameter'.

The function $g(.)$ is said to be slowly varying if for any $c > 0$, $g(ch)/g(h)$ converges to unity as $h \to \infty$. An example of slowly varying functions is $\ln(.)$.

Consider the infinite-order moving average process

$$y_t = \lim_{q \to \infty} \sum_{i=0}^{q} a_i \varepsilon_{t-i}.$$

The long memory condition can also be defined in terms of the weights, a_i, in the infinite-order moving average representation of y_t. Note that for such a representation to exist we need $\{a_i\}$ to be square summable and not necessarily absolute summable. The infinite-order moving average representation is said to be a long memory process if

$$a_i \backsim (1+i)^{d-1} \kappa(j), \tag{15.31}$$

where $\kappa(j)$ is a slowly varying function.

The above four definitions of long memory are not necessarily equivalent, unless further restrictions are imposed. But all point to a decay in the dependence of the series on their past which is slower than the exponential decay, but with the decay being sufficiently fast to ensure that the series have a finite variance. In the case where $0 < d < 1/2$, it is easily seen that (15.31) implies (15.30), and (15.30) implies (15.29).[4]

15.8.1 Spectral density of long memory processes

In this case the spectral density of the long memory process is defined by[5]

$$f(\omega) \backsim |\omega|^{-2d} b_f(1/|\omega|), \tag{15.32}$$

where $d > 0$ is the long-memory parameter, and $b_f(\cdot)$ is a slowly varying function. The existence of the spectral density for long memory processes depends on the properties of the slowly varying function $b_f(\cdot)$. It is possible to show that (15.32) has a one-to-one relationship with the following specification of the autocovariance function

$$\gamma(h) \backsim h^{2d-1} b_\gamma(h),$$

so long as $b_f(\cdot)$ and $b_\gamma(\cdot)$ are slowly varying in the sense of Zygmund (1959). For a proof and further details see Palma (2007, Ch. 3).

15.8.2 Fractionally integrated processes

Fractionally integrated processes can be used to represent long memory in auto-correlation analysis. Consider the process $y_t \sim I(d)$. When the order of integration, d, needed to make y_t into a stationary process is not an integer number we have a fractionally integrated process. For example, the process y_t follows an autoregressive fractionally integrated moving average ($ARFIMA$) process if

$$\phi(L)(1-L)^d y_t = \theta(L)\varepsilon_t, \tag{15.33}$$

where L is the usual lag operator, ε_t is a white noise with zero mean and variance σ_ε^2, and d can be any real number.

When $d = 0$, then y_t is stationary, while under $d = 1$ we have $y_t \sim I(1)$. When $-1/2 < d < 1/2$, it is possible to prove that the process is covariance stationary and invertible. Under $d \neq 0$, the above model displays long memory, and can be used to characterise a wide range of long-term patterns. The autocorrelation function of y_t defined by (15.33) declines to zero at a very slow rate. These processes are therefore very useful in the study of economic time series that are known to display rather slow long-term movements as is the case with some inflation and interest rate series. For large h, the autocorrelation function of $ARFIMA$ models can be approximated by

[4] Other notions of long memory or long-range dependence are also proposed in terms of other more general concepts of slowly varying functions. But they will not be pursued here.

[5] For an introduction to spectral density analysis see Chapter 13.

$$\rho(h) \simeq Kh^{2d-1},$$

where K is a constant. For $d < 0.5$, the exponent $2d - 1 < 0$ so that the correlations eventually decay, but at a slow hyperbolic rate compared with the fast exponential decay in the case of standard stationary *ARMA* models.

When $1/2 < d < 1$, a number of studies have shown that the usual unit root tests display a bias in favour of the hypothesis $d = 1$ (see, e.g., Diebold and Rudebush (1991)).

Estimation techniques for fractionally integrated processes include semi-parametric estimators of their spectral density function, such as the methods proposed by Robinson (1995) and Velasco (1999), or parametric methods based on approximation of the likelihood function (see, for example, Fox and Taqqu (1986)). Further details of long memory processes can be found in Robinson (1994) and Baillie (1996).

15.8.3 Cross-sectional aggregation and long memory processes

Long memory processes can arise from cross-section aggregation of covariance stationary processes. As a simple example, consider the following $AR(1)$ micro relations

$$y_{it} = \lambda_i y_{i,t-1} + u_{it},$$

for $i = 1, 2, \ldots, N$, and $t = \ldots - 1, 0, 1, 2, \ldots$, where $|\lambda_i| < 1$. It is clear that for each i, y_{it} is covariance stationary with absolute summable autocovariances. Suppose that $Var(u_{it}) = \sigma_i^2 < \infty$, and λ_i are independently and identically distributed random draws with the distribution function $F(\lambda)$ for λ on the range $[0, 1)$. Consider now the moving average representation of the cross-sectional average $\bar{y}_t = N^{-1} \sum_{i=1}^{N} y_{it}$, and note that

$$\begin{aligned} \bar{y}_t &= N^{-1} \sum_{i=1}^{N} \left(\frac{1}{1 - \lambda_i L} \right) u_{it} = N^{-1} \sum_{i=1}^{N} \sum_{j=0}^{\infty} \lambda_i^j u_{i,t-j} \\ &= \sum_{j=0}^{\infty} N^{-1} \sum_{i=1}^{N} \lambda_i^j u_{i,t-j}. \end{aligned}$$

Under the assumption that for each t, λ_i and u_{it} are independently distributed we have

$$E\left(\bar{y}_1 \,|\mathbf{u}_t, \mathcal{F}_{t-1}\right) = \sum_{j=0}^{\infty} N^{-1} \sum_{i=1}^{N} E\left(\lambda_i^j\right) u_{i,t-j},$$

where $\mathcal{F}_{t-1} = \left(u_{1,t-1}, u_{1,t-2}, \ldots; u_{2,t-1}, u_{2,t-2}, \ldots; u_{N,t-1}, u_{N,t-2}, \ldots\right)$, and $\mathbf{u}_t = (u_{1t}, u_{2t}, \ldots, u_{Nt})'$. But since by assumption λ_i is identically distributed across i, then $E\left(\lambda_i^j\right) = a_j$ and we have

$$E\left(\bar{y}_t \,|\mathbf{u}_t, \mathcal{F}_{t-1}\right) = \sum_{j=0}^{\infty} a_j \bar{u}_{t-j},$$

where $\bar{u}_{t-j} = N^{-1} \sum_{i=1}^{N} u_{i,t-j}$. Hence,

$$\bar{y}_t = \sum_{j=1}^{\infty} a_j \bar{u}_{t-j} + v_t,$$

where $v_t = \bar{y}_t - E\left(\bar{y}_t \,|\mathcal{F}_{t-1}\right)$. It is easily seen that $\bar{u}_{t-j}$, for $j = 1, 2, \ldots$, and v_t are serially uncorrelated with zero means and finite variances. Therefore, $\bar{y}_t$ has a moving average representation with coefficients $a_j = E\left(\lambda_i^j\right)$. The rate of decay of a_j over j depends on the distribution λ_i. For example, if λ_i are random draws from a uniform distribution over $[0, 1)$ we have $a_j = 1/(1+j)$, and the coefficients a_j are not absolute summable, and therefore $\bar{y}_t$ is a long memory process. Similar results follow if it is assumed that λ_i are draws from beta distribution with support that covers unity. Granger (1980) was the first to obtain this result, albeit under a more restrictive set of assumptions. Granger also showed that when λ is type II beta distributed with parameters $p > 0$ and $q > 0$, the sth-order autocovariance of $\bar{y}_t$ is $O(s^{1-q})$, and therefore the aggregate variable behaves as a fractionally integrated process of order $1 - q/2$. For a generalization to multivariate models see Pesaran and Chudik (2014) and Chapter 32.

Finally, it is important to note that the long memory property of the aggregate, $\bar{y}_t$, critically depends on whether the support of the distribution of λ_i covers unity. For example, if λ_i, for $i = 1, 2, \ldots, N$, are draws from uniform distribution over the range $[0, b)$ where $0 \leq b < 1$, the moving average coefficients are given by $a_j = b^j/(1+j)$, and we have[6]

$$\sum_{j=0}^{\infty} \left|a_j\right| = \sum_{j=0}^{\infty} \frac{b^j}{1+j} = \frac{-\ln(1-b)}{b} < \infty, \text{for } b < 1.$$

Hence, $\{a_i\}$ is an absolute summable sequence and the aggregate variable, $\bar{y}_t$, will no longer be a long memory process.

15.9 Further reading

See Hall and Heyde (1980) for further discussion and results on martingale processes, and Davidson (1994, Ch. 16) for a general discussion of the properties of L_p-mixingales. The properties of unit root processes, as well as testing for unit root are broadly reviewed in Hamilton (1994). For further discussions of the time series properties of cross-section aggregates see Chapter 32 and Robinson (1978), Granger (1980), Pesaran (2003), Zaffaroni (2004), and Pesaran and Chudik (2014).

[6] Note that

$$\frac{d}{db}\left[\sum_{j=0}^{\infty} \frac{b^{1+j}}{1+j}\right] = \sum_{j=0}^{\infty} b^j = \frac{1}{1-b},$$

and hence $\sum_{j=0}^{\infty} \frac{b^{1+j}}{1+j} = -\ln(1-b)$.

15.10 Exercises

1. Consider the simple autoregressive distributed lag model

$$y_t = \alpha + \lambda y_{t-1} + \beta x_t + u_t,$$

where

$$x_t = \rho x_{t-1} + \varepsilon_t,$$
$$\begin{pmatrix} u_t \\ \varepsilon_t \end{pmatrix} \sim IID\left[\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \sigma^2 & 0 \\ 0 & \omega^2 \end{pmatrix}\right],$$

for $t = 1, 2, \ldots,$ and given initial values y_0, and x_0.

(a) Show that

$$y_t = \lambda^t y_0 + \alpha\left(\frac{1-\lambda^t}{1-\lambda}\right) + \beta\sum_{j=0}^{t-1}\lambda^j x_{t-j} + \sum_{j=0}^{t-1}\lambda^j u_{t-j},$$

for $t = 1, 2, \ldots, T$.

(b) Hence, or otherwise, derive the mean and variance of y_t.

(c) Show that under the conditions $|\lambda| < 1$ and $|\rho| < 1$, $y_\infty = \lim_{t\to\infty}(y_t)$ has a finite mean and variance, and derive an expression for the variance of y_∞.

(d) Discuss the case where $|\lambda| < 1$, but $\rho = 1$, and consider the limiting properties of y_t.

2. Consider the general linear first difference stationary process

$$\Delta y_t = \mu + A(L)\varepsilon_t, \tag{15.34}$$

where Δ is the first difference operator,

$$A(L) = a_0 + a_1 L + a_2 L^2 + \cdots,$$

is a polynomial in the lag operator L, $(Ly_t = y_{t-1})$, and μ is a scalar constant. The ε_t are mean zero, serially uncorrelated shocks with common variance, σ_ε^2.

(a) Derive the conditions under which (15.34) reduces to the trend-stationary process

$$y_t = \lambda t + B(L)\varepsilon_t. \tag{15.35}$$

(b) Given the observations $(y_1, y_2, \ldots, y_n)$, discuss alternative methods of testing (15.34) against (15.35) and *vice versa*.

(c) What is meant by 'persistence' of shocks in time series models? How useful do you think the concept of 'persistence' is for an understanding of cyclical fluctuations of the US real GNP?

3. Suppose that the time series of interest can be decomposed into a deterministic trend, a random walk component and stationary errors

$$y_t = \alpha + \delta t + \gamma_t + v_t, \qquad (15.36)$$
$$\gamma_t = \gamma_{t-1} + u_t.$$

Assume that v_t are $IIDN(0, \sigma_v^2)$, and that u_t are $IIDN(0, \sigma_u^2)$. Let $\lambda = \sigma_u^2/\sigma_v^2$.

(a) Show that under $\lambda = 0$, y_t reduces to a trend stationary process.

(b) Alternatively, suppose that y_t follows an $ARIMA(0,1,1)$ process of the form

$$y_t = \delta + y_{t-1} + w_t, \qquad (15.37)$$
$$w_t = \varepsilon_t + \theta\varepsilon_{t-1},$$

where ε_t are $IIDN(0, \sigma_\varepsilon^2)$. In this case show that under $\theta = -1$, y_t is a trend stationary process.

(c) Derive a relation between λ and the $MA(1)$ parameter θ, and hence or otherwise show that a test of $\theta = -1$ in (15.37) is equivalent to a test of $\lambda = 0$ in (15.36).

(d) Assume that v_t and u_t are distributed independently, then show that (15.37) as a characterization of (15.36) implies $\theta < 0$.

4. Suppose it is of interest to test the null hypothesis that

$$H_0 : \rho = 1,$$

against

$$H_1 : \rho < 1,$$

in the following univariate first-order autoregressive $(AR(1))$ model

$$y_t = \rho y_{t-1} + \epsilon_t, \quad t = 1, 2, \ldots, T, \qquad (15.38)$$

where $\{\epsilon_t\}_{-\infty}^{\infty}$ is a sequence of IID random variables with mean 0 and variance, σ^2. Let $\hat{\rho}$ be the ordinary least squares (OLS) estimator of ρ defined by

$$\hat{\rho} = \frac{\sum_{t=2}^{T} y_t y_{t-1}}{\sum_{t=2}^{T} y_t^2}.$$

(a) Derive the asymptotic distribution of $T(\hat{\rho} - 1)$ under the null hypothesis.

(b) How does the asymptotic distribution in (a) change if an intercept is included in (15.38)? What is the appropriate way of including such an intercept in the model?

(c) Suppose now that, instead of (15.38), $\{y_t\}$ is generated according to the second-order autoregressive $(AR(2))$ process

$$y_t = \rho_1 y_{t-1} + \rho_2 y_{t-2} + \epsilon_t, \quad t = 1, 2, \ldots, T, \qquad (15.39)$$

where $\{\epsilon_t\}_{-\infty}^{\infty}$ is a sequence of *IID* random variables with mean 0 and variance, σ^2.

(d) How would you test the hypothesis that

$$H_0 : \rho_1 + \rho_2 = 1,$$

against

$$H_1 : \rho_1 + \rho_2 < 1\,?$$

5. Suppose observations on y_t are generated from the process $y_t = \alpha + y_{t-1} + u_t$, where $u_t = \psi(L)\epsilon_t$, $\sum_{j=0}^{\infty} j|\psi_j| < \infty$, and $\epsilon \sim \text{IID}(0, \sigma^2)$ with finite fourth-order moments, and α can be any value including zero. Consider *OLS* estimation of

$$y_t = \alpha + \rho y_{t-1} + \delta t + u_t.$$

Note that the fitted values and estimate of ρ from this regression are identical to those from an *OLS* regression of y_t on a constant , time trend , and $\xi_{t-1} = y_{t-1} - \alpha(t-1)$

$$y_t = \alpha^* + \rho^* \xi_{t-1} + \delta t + u_t.$$

Finally, define $\lambda \equiv \sigma\psi(1)$.

(a) Find the relation between α and α^*, and between δ and δ^*.

(b) Prove that the following holds

$$\begin{pmatrix} T^{\frac{1}{2}}\hat{\alpha}^* \\ T(\hat{\rho}^* - 1) \\ T^{\frac{3}{2}}(\hat{\delta}^* - \alpha) \end{pmatrix} \Rightarrow \begin{pmatrix} \lambda & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & \lambda \end{pmatrix} \begin{pmatrix} 1 & \int W(r)\,dr & \frac{1}{2} \\ \int W(r)\,dr & \int W(r)^2\,dr & \int rW(r)\,dr \\ \frac{1}{2} & \int rW(r)\,dr & \frac{1}{3} \end{pmatrix}^{-1} \times \begin{pmatrix} W(1) \\ \frac{1}{2}\left\{W(1)^2 - \frac{\gamma_0}{\lambda^2}\right\} \\ W(1) - \int W(r)\,dr \end{pmatrix},$$

where γ_0 is the variance of u_t.

6. The following regression equations are estimated by ordinary least squares using US monthly data over the period 1948M1-2009M9.
Model (A)

$$P_t = \underset{(0.8593)}{1.2881} + \underset{(0.002461)}{0.9975}\ P_{t-1} + \hat{\varepsilon}_{pt}$$

$$LL = -2951.6, \bar{R}^2 = 0.9955, \hat{\sigma}_{\varepsilon} = 13.0089$$

Estimated variance-covariance matrix of parameters		
	INPT	P_{t-1}
INPT	.7383	–.001758
P_{t-1}	–.001758	.6058E-5

where INPT denotes the intercept term in the regression.

Model (B)

$$P_t = \underset{(0.8382)}{1.2907} + \underset{(0.03592)}{1.2203}\, P_{t-1} - \underset{(0.03591)}{0.2232}\, P_{t-2} + \hat{\varepsilon}_{pt}$$

$LL = --2932.7, \bar{R}^2 = 0.9957, \hat{\sigma}_\varepsilon = 12.6898$

Estimated variance-covariance matrix of parameters			
	INPT	P_{t-1}	P_{t-2}
INPT	.7026	–.001658	–.1481E-4
P_{t-1}	–.001658	.001290	–.001287
P_{t-2}	–.1481E-4	–.001287	.001289

Model (C)

$$P_t = \underset{(3.8009)}{-2.3591} + \underset{(0.003535)}{0.003483} \cdot t + \underset{(0.003895)}{0.9945}\, P_{t-1} + \hat{\varepsilon}_{pt}$$

$LL = --2951.1, \bar{R}^2 = 0.9955, \hat{\sigma}_\varepsilon = 13.0091$

Estimated variance-covariance matrix of parameters			
	INPT	TREND	P_{t-1}
INPT	14.4466	–.01309	.009418
TREND	–.01309	.1250E-4	–.1067E-4
P_{t-1}	.009418	–.1067E-4	.1517E-4

Model (D)

$$P_t = \underset{(3.7098)}{-3.4460} + \underset{(0.003451)}{0.004523} \cdot t + \underset{(0.03592)}{1.2187}\, P_{t-1} - \underset{(0.03593)}{0.2254}\, P_{t-2} + \hat{\varepsilon}_{pt}$$

$LL = -2931.8, \bar{R}^2 = 0.9957, \hat{\sigma}_\varepsilon = 12.6836$

Estimated variance-covariance matrix of parameters				
	INPT	TREND	P_{t-1}	P_{t-2}
INPT	13.7627	–.01247	.002766	.006224
TREND	–.01247	.1191E-4	–.4222E-5	–.5957E-5
P_{t-1}	.002766	–.4222E-5	.001290	–.001283
P_{t-2}	.006224	–.5957E-5	–.001283	.001290

Model (E)

$$D_t = \underset{(0.009465)}{0.02649} + \underset{(0.001173)}{0.9979}\, D_{t-1} + \hat{\varepsilon}_{dt}$$

$LL = 1048.8, \bar{R}^2 = 0.9990, \hat{\sigma}_\varepsilon = 0.05884$

Estimated variance-covariance matrix of parameters		
	INPT	D_{t-1}
INPT	.8958E-4	–.1081E-4
D_{t-1}	–.1081E-4	.1377E-5

Model (F)

$$D_t = \underset{(0.006437)}{0.01420} + \underset{(0.02533)}{1.7431}\, D_{t-1} - \underset{(0.02529)}{0.7446}\, D_{t-2} + \hat{\varepsilon}_{dt}$$

$LL = 1336.5, \bar{R}^2 = 0.9995, \hat{\sigma}_\varepsilon = 0.03993$

Estimated variance-covariance matrix of parameters			
	INPT	D_{t-1}	D_{t-2}
INPT	.4143E-4	–.1555E-4	.1056E-4
D_{t-1}	–.1555E-4	.6414E-3	–.6403E-3
D_{t-2}	.1056E-4	–.6403E-3	.6398E-3

Model (G)

$$D_t = \underset{(0.01373)}{0.01974} + \underset{(0.00001904)}{0.00001293} \cdot t + \underset{(0.002211)}{0.9966}\, D_{t-1} + \hat{\varepsilon}_{dt}$$

$LL = 1049.0, \bar{R}^2 = 0.9990, \hat{\sigma}_\varepsilon = 0.05896$

Estimated variance-covariance matrix of parameters			
	INPT	TREND	D_{t-1}
INPT	.1885E-3	–.1893E-6	.7812E-5
TREND	–.1893E-6	.3626E-9	–.3568E-7
D_{t-1}	.7812E-5	–.3568E-7	.4888E-5

Model (H)

$$D_t = \underset{(0.009321)}{0.002845} + \underset{(0.00001291)}{0.00002172} \cdot t + \underset{(0.02530)}{1.7419}\, D_{t-1} - \underset{(0.02527)}{0.7456}\, D_{t-2} + \hat{\varepsilon}_{dt}$$

$LL = 1337.9, \bar{R}^2 = 0.9995, \hat{\sigma}_\varepsilon = 0.03988$

Estimated variance-covariance matrix of parameters				
	INPT	TREND	D_{t-1}	D_{t-2}
INPT	.8688E-4	–.8710E-7	–.1088E-4	.1447E-4
TREND	–.8710E-7	.1666E-9	–.8859E-8	–.7523E-8
D_{t-1}	–.1088E-4	–.8859E-8	.6403E-3	–.6383E-3
D_{t-2}	.1447E-4	–.7523E-8	–.6383E-3	.6385E-3

where P_t represents real equity prices, and D_t is real dividends per annum for the S&P 500 portfolio.

(a) Use the above regression results to test the hypothesis of a unit root in price and dividend processes.

(b) Consider the following asset pricing model ($r > 0$)

$$P_t = \left(\frac{1}{1+r}\right) E\left(P_{t+1} + D_{t+1} \,|\mathcal{I}_t\right), \tag{15.40}$$

where $\mathcal{I}_t = (P_t, D_t, P_{t-1}, D_{t-1}, \ldots)$. Suppose that D_t follows the following stationary $AR(\mathrm{p})$ process in ΔD_t

$$\Delta D_t = a_0 + \phi_1 \Delta D_{t-1} + \cdots + \phi_p \Delta D_{t-p} + u_t,$$

where u_t is $\mathrm{IID}(0, \sigma_u^2)$. Show that $P_t = D_t/r + v_t$, is the solution to equation (15.40), where v_t is a stationary process.

(c) Assuming $r = 3\%$ per annum the researcher runs the following additional *OLS* regressions

Model (I)

$$v_t = \underset{(0.4891)}{0.3387} + \underset{(0.003229)}{0.9962}\, v_{t-1} + \hat{\varepsilon}_{vt}$$

$LL = -2955.5, \bar{R}^2 = 0.9922, \hat{\sigma}_\varepsilon = 13.0781$

Estimated variance-covariance matrix of parameters		
	INPT	v_{t-1}
INPT	.23919	–.2954E-3
v_{t-1}	–.2954E-3	.1042E-4

Model (J)

$$v_t = \underset{(0.4756)}{0.3185} + \underset{(0.03586)}{1.2318}\, v_{t-1} - \underset{(0.03585)}{0.2365}\, v_{t-2} + \hat{\varepsilon}_{vt}$$

$LL = -2934.3, \bar{R}^2 = 0.9927, \hat{\sigma}_\varepsilon = 12.7173$

Estimated variance-covariance matrix of parameters			
	INPT	v_{t-1}	v_{t-2}
INPT	.22619	–.3888E-3	.1098E-3
v_{t-1}	–.3888E-3	.0012856	–.0012806
v_{t-2}	.1098E-3	–.0012806	.0012854

Model (K)

$$v_t = \underset{(3.8287)}{-3.7123} + \underset{(0.002995)}{0.003195} \cdot t + \underset{(0.004305)}{0.9931}\, v_{t-1} + \hat{\varepsilon}_{vt}$$

$LL = -2954.9, \bar{R}^2 = 0.9923, \hat{\sigma}_\varepsilon = 13.0769$

Estimated variance-covariance matrix of parameters			
	INPT	TREND	v_{t-1}
INPT	14.6590	–.011372	.010518
TREND	–.011372	.8968E-5	–.8528E-5
v_{t-1}	.010518	–.8528E-5	.1853E-4

Model (L)

$$v_t = \underset{(3.7246)}{-4.7734} + \underset{(0.002913)}{0.004015} \cdot t + \underset{(0.03586)}{1.2301}\, v_{t-1} - \underset{(0.03586)}{0.2386}\, v_{t-2} + \hat{\varepsilon}_{vt}$$

$$LL = -2933.3, \bar{R}^2 = 0.9927, \hat{\sigma}_\varepsilon = 12.7095$$

Estimated variance-covariance matrix of parameters				
	INPT	TREND	v_{t-1}	v_{t-2}
INPT	13.8724	–.010761	.0042571	.0057189
TREND	–.010761	.8486E-5	–.3663E-5	–.4423E-5
v_{t-1}	.0042571	–.3663E-5	.0012856	–.0012771
v_{t-2}	.0057189	–.4423E-5	–.0012771	.0012862

Test the hypothesis that the v_t process contains a unit root. Interpret the result of your tests in relation to the market efficiency hypothesis (see Chapter 7).

16 Trend and Cycle Decomposition

16.1 Introduction

In this chapter we consider alternative approaches proposed in the literature for the decomposition of time series into trend and cyclical components. We focus on univariate techniques, and consider Hodrick–Prescott and band-pass filters, structural time series techniques, and the Beveridge–Nelson decomposition technique specifically designed for the unit root processes. A multivariate version of the Beveridge–Nelson decomposition is considered in Section 22.15, where the role of long run economic theory in such decomposition is also discussed.

16.2 The Hodrick–Prescott filter

The Hodrick–Prescott (HP) filter is a curve fitting procedure proposed by Hodrick and Prescott (1997) to estimate the trend path of a series. More specifically, suppose an observed time series, y_t, is composed of a trend component, y_t^*, and a cyclical component, c_t, as follows

$$y_t = y_t^* + c_t.$$

Hodrick and Prescott (1997) suggested a way to isolate c_t from y_t by the following minimization problem

$$\min_{y_1^*, y_2^*, \ldots y_T^*} \left[\sum_{t=1}^{T} (y_t - y_t^*)^2 + \lambda \sum_{t=2}^{T-1} (\Delta^2 y_{t+1}^*)^2 \right],$$

where λ is a penalty parameter. The first term in the loss function penalizes the variance of c_t, while the second term penalizes the lack of smoothness in y_t^*, with the parameter λ regulating the trade-off between the two sources of variations. Putting it differently, the HP filter identifies the cyclical component c_t from y_t by trading-off the extent to which the trend component, y_t^*, keeps track of the original series, y_t, (goodness of fit) whilst maintaining a desired degree of smoothness

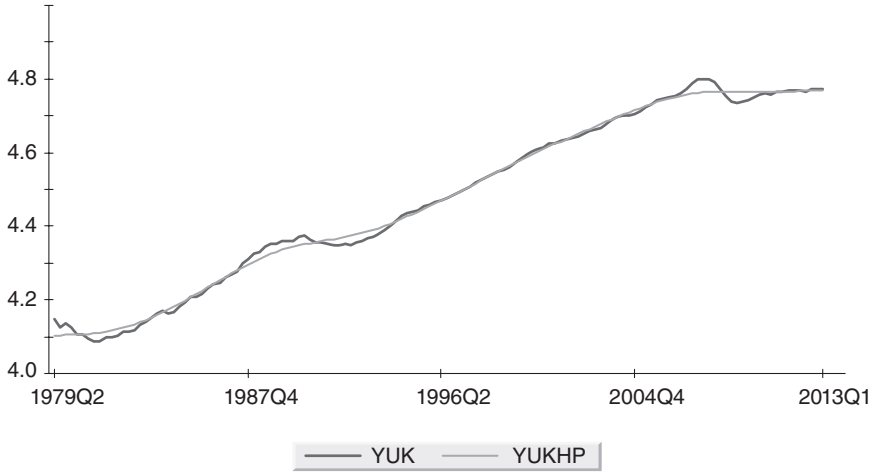


Figure 16.1 Logarithm of UK output and its Hodrick–Prescott filter using $\lambda = 1,600$.

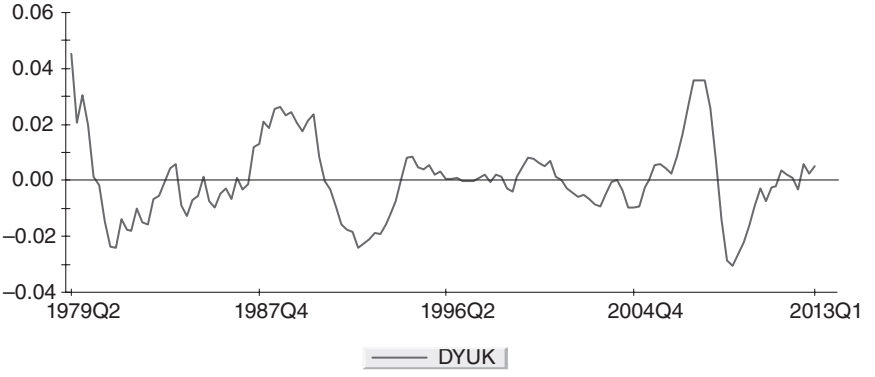


Figure 16.2 Plot of detrended UK output series using the Hodrick–Prescott filter with $\lambda = 1,600$.

in the trend component. Note that as λ approaches 0, the trend component becomes equivalent to the original series, while as λ diverge to ∞, y_t^* becomes a linear trend, since for sufficiently large λ it is optimal to set $\Delta^2 y_{t+1}^* = 0$, which yields, $y_{t+1}^* = d_0 + d_1 t$, where d_0 and d_1 are fixed constants.

The 'smoothing' parameter λ is usually chosen by trial and error, and for quarterly observations it is set to 1,600. A discussion on the value of λ for different observation frequencies can be found in Ravn and Uhlig (2002) and Maravall and Rio (2007).

Example 33 *Figure 16.1 shows the plot of the logarithm of UK real GDP and its trend computed using the HP filter, setting* $\lambda = 1600$*, over the period 1970Q1 to 2013Q1. Figure 16.2 reports the detrended series, computed using Microfit 5.0. The HP detrending procedure in this exercise is quite sensitive to the choice of* λ*, giving much more pronounced cyclical fluctuations for smaller values of* λ*.*

For a discussion of the statistical properties of the *HP* filter, see, for example, Cogley (1995), Söderlind (1994), and Harvey and Jaeger (1993) who show that the use of the *HP* filter can generate spurious cyclical patterns.

16.3 Band-pass filter

The *HP* filter can be seen as a high-pass filter, in that it removes low frequencies and passes through only higher frequencies. In contrast, the filter advanced by Baxter and King (1999) is a band-pass filter, since it allows suppression of both very slow-moving (trend) components and very high-frequency (irregular) components while retaining intermediate (business-cycle) components. Baxter and King (1999) argue that the National Bureau of Economic Research (NBER) definition of a business cycle (see Burns and Mitchell (1946)) requires a two-sided, or band-pass approach, that passes through components of the time series with periodic fluctuations, for example, between six and 32 quarters, while removing components at higher and lower frequencies.

Specifically, the band-pass filter proposed by Baxter and King takes the form of a two-sided moving average

$$y_t^* = \sum_{k=-K}^{K} a_k y_{t-k} = a(L)\, y_t. \tag{16.1}$$

The weights can be derived from the inverse Fourier transform of the frequency response function (see Priestley (1981)). Baxter and King adjust the band-pass filter with a constraint that the sum of the coefficients in (16.1) must be zero. Under this condition, the authors show that $a(L)$ can be factorized as

$$a(L) = (1 - L)\left(1 - L^{-1}\right) a^*(L),$$

with $a^*(L)$ being a symmetric moving average with $K-1$ leads and lags. It follows that the moving average has the characteristic of rendering stationary series that contain quadratic deterministic trends.

When applied to quarterly data, K in the band-pass filter is usually set at $K = 12$, and as a result 24 data points (12 at the start and 12 at the end of the sample) are sacrificed, seriously limiting the usefulness of the filter for the analysis of the current state of the economy. The use of two-sided filters also creates difficulties in forecasting. To avoid some of these difficulties two-sided filters must be applied recursively, rather than to the full sample. Further details are provided in the papers by Baxter and King (1999) and Christiano and Fitzgerald (2003).

16.4 The structural time series approach

In macroeconomic research a crucial step is establishing some 'stylized facts' associated with a set of time series variables. For such facts to be useful, they should be consistent with the stochastic properties of the data and present meaningful information. However, as also pointed out by

Harvey and Jaeger (1993), many stylized facts reported in the literature do not fulfil these criteria. Information based on mechanically detrended series can easily lead the researcher to report spurious cyclical behaviours; analysis based on *ARIMA* models can also be misleading if such models are chosen primarily on grounds of parsimony.

Structural time series models which are linear and time invariant all have a corresponding reduced form *ARIMA* representation which is equivalent in the sense that it will give identical forecasts to the structural form.[1] For example, consider the local trend model,

$$y_t = \mu_t + \varepsilon_t,$$
$$\mu_t = \mu_{t-1} + \eta_t,$$

where ε_t and η_t are uncorrelated white noise disturbances. Taking first differences yields

$$\Delta y_t = \eta_t + \varepsilon_t - \varepsilon_{t-1},$$

which is equivalent to an $MA(1)$ process with a non-positive autocorrelation at lag one. By equating autocorrelations at lag one it is possible to derive the relationship between the moving average parameter and q, the ratio of the variance of η_t to that of ε_t. In more complex models, there may not be a simple correspondence between the structural and reduced form parameters. The key to handling structural time series models is the state space form, with the state of the system representing the various unobserved components such as trends and seasonals. See Harvey and Shephard (1993) for a detailed analysis of structural time series models.

16.5 State space models and the Kalman filter

State space models were originally developed by control engineers in aerospace-related research, and have been used extensively in time series analysis. State space models consist of two sets of linear equations that define how observable and unobservable components evolve over time. In the general state space form, the m-dimensional vector of observable variables, $\mathbf{y}_t$, is related to an r-dimensional vector of (partly) unobservable state variables, $\boldsymbol{\alpha}_t$, through the measurement equation

$$\mathbf{y}_t = \mathbf{Z}_t\boldsymbol{\alpha}_t + \boldsymbol{\varepsilon}_t, \quad t = 1, 2, \ldots, T, \tag{16.2}$$

where $\mathbf{Z}_t$ is an $m \times r$ matrix and $\boldsymbol{\varepsilon}_t$ is an m-dimensional vector of serially uncorrelated disturbances with mean zero and the covariance matrix, $\mathbf{H}_t$; that is $E(\boldsymbol{\varepsilon}_t) = \mathbf{0}$, $E(\boldsymbol{\varepsilon}_t\boldsymbol{\varepsilon}_t') = \mathbf{H}_t$, and $E(\boldsymbol{\varepsilon}_t\boldsymbol{\alpha}_t') = \mathbf{0}$, for $t \neq s$. The state variables are generated by a transition equation

$$\boldsymbol{\alpha}_t = \mathbf{T}_t\boldsymbol{\alpha}_{t-1} + \mathbf{R}_t\boldsymbol{\eta}_t, \tag{16.3}$$

[1] It is worth noting that the word 'structural' in this literature has a very different meaning to what is meant by structural in the literature on simultaneous equation models, and the more recent literature on dynamic stochastic general equilibrium models. For these alternative meanings of 'structural' see Chapters 19 and 20.

where $\mathbf{T}_t$ and $\mathbf{R}_t$ are $r \times r$ coefficient matrices, and $\boldsymbol{\eta}_t$ is an r-dimensional vector of serially uncorrelated disturbances with mean zero and the covariance matrix, $\mathbf{Q}_t$, that is $E(\boldsymbol{\eta}_t) = \mathbf{0}$ and $E(\boldsymbol{\eta}_t \boldsymbol{\eta}_t') = \mathbf{Q}_t$. Note that $\mathbf{Z}_t$, $\mathbf{T}_t$, $\mathbf{R}_t$, and $\mathbf{Q}_t$ are time varying non-stochastic matrices. However, we note that in many applications of state space models these system matrices are assumed to be time invariant, i.e. $\mathbf{Z}_t = \mathbf{Z}$, $\mathbf{T}_t = \mathbf{T}$, $\mathbf{R}_t = \mathbf{R}$, and $\mathbf{Q}_t = \mathbf{Q}$, for $t = 1, 2, \ldots, T$. The specification of the state space system is completed by assuming that the initial state vector, $\boldsymbol{\alpha}_0$, has a mean denoted by $\mathbf{a}_0$, a covariance matrix, $\mathbf{P}_0$, and that the disturbances $\boldsymbol{\varepsilon}_t$ and $\boldsymbol{\eta}_t$ are uncorrelated with the initial state, that is $E(\boldsymbol{\alpha}_0 \boldsymbol{\varepsilon}_t') = \mathbf{0}$, and $E(\boldsymbol{\alpha}_0 \boldsymbol{\eta}_t') = \mathbf{0}$, for $t = 1, 2, \ldots, T$. It is also assumed that the disturbances are uncorrelated with each other at all time periods, that is $E(\boldsymbol{\varepsilon}_t \boldsymbol{\eta}_s') = \mathbf{0}$, for all $t, s = 1, 2, \ldots, T$, though this assumption may be relaxed, the consequence being a slight complication in some of the filtering formulae. A wide range of econometric models can be cast in the state space format.

Example 34 *Consider the process*

$$x_t = \phi_1 x_{t-1} + \phi_2 x_{t-2} + u_t.$$

This can be rewritten in the form (16.2)–(16.3) by setting

$$y_t = \mathbf{Z}\boldsymbol{\alpha}_t + \varepsilon_t,$$
$$\boldsymbol{\alpha}_t = \mathbf{T}\boldsymbol{\alpha}_{t-1} + \boldsymbol{\eta}_t,$$

where

$$y_t = x_t, \ \boldsymbol{\alpha}_t = \begin{pmatrix} x_t \\ x_{t-1} \end{pmatrix},$$

$$\mathbf{Z} = (1, 0), \ \varepsilon_t = 0,$$

$$\mathbf{T} = \begin{pmatrix} \phi_1 & \phi_2 \\ 1 & 0 \end{pmatrix}, \ \boldsymbol{\eta}_t = \begin{pmatrix} u_t \\ 0 \end{pmatrix},$$

with $\mathbf{H} = \mathbf{0}$, *and*

$$\mathbf{Q} = \begin{pmatrix} Var\,(u_t) & 0 \\ 0 & 0 \end{pmatrix}.$$

Example 35 *As another example consider the following time varying coefficient*

$$y_t = \mathbf{x}_t' \boldsymbol{\beta}_t + u_t, for\, t = 1, 2, \ldots, T, \tag{16.4}$$

where $u_t \sim IIDN(0, \sigma_u^2)$, $\mathbf{x}_t$ *is a* $k \times 1$ *vector of exogeneously given regressors,* $\boldsymbol{\beta}_t$ *is a vector of time varying coefficients whose time evolution is given by*

$$\boldsymbol{\beta}_t - \boldsymbol{\beta} = \mathbf{T}(\boldsymbol{\beta}_{t-1} - \boldsymbol{\beta}) + \boldsymbol{\varepsilon}_t, \tag{16.5}$$

$\mathbf{T}$ is an $r \times r$ matrix of fixed coefficients, and $\boldsymbol{\varepsilon}_t \sim IIDN(0, \sigma_\varepsilon^2)$. In this setup, (16.4) is the measurement (or observation) equation, and (16.5) is the state equation, with $\mathbf{T}$ known as the transition matrix from one state to the next. It is easy to show that this model can also be written as a fixed coefficient regression model with heteroskedastic errors.

Kalman (1960) showed that the calculations needed for estimating a state space model could be set out in a recursive form which has proved very convenient computationally. Since then, the so-called Kalman filter has been almost universally adopted in modern control and system theory, and has been useful in handling time series models (Harvey (1989), Durbin and Koopman (2001)). The method based on the Kalman filter has many practical advantages among which are its applicability to cases where there are missing observations, measurement errors and variables which are observed at different frequencies. The optimal forecasts of $\boldsymbol{\alpha}_t$ in the mean squared forecast error sense (see Section 17.2), given information up to period $t-1$, are given by the prediction equations

$$\mathbf{a}_{t|t-1} = \mathbf{T}_t \mathbf{a}_{t-1|t-1}, \tag{16.6}$$

$$\mathbf{P}_{t|t-1} = \mathbf{T}_t \mathbf{P}_{t-1|t-1} \mathbf{T}_t + \mathbf{R}_t \mathbf{Q}_t \mathbf{R}_t', \tag{16.7}$$

where $\mathbf{a}_{t|t-1} = E(\boldsymbol{\alpha}_t | \Omega_{t-1})$, and $\mathbf{P}_{t|t-1} = E\left[\left(\boldsymbol{\alpha}_t - \mathbf{a}_{t|t-1}\right)\left(\boldsymbol{\alpha}_t - \mathbf{a}_{t|t-1}\right)' | \Omega_{t-1}\right]$, is the covariance matrix of the prediction error, $\boldsymbol{\alpha}_t - \mathbf{a}_{t|t-1}$. From (16.2), and using equation (16.6)–(16.7), the best estimate of $\mathbf{y}_t$ is

$$\hat{\mathbf{y}}_{t|t-1} = \mathbf{Z}_t \mathbf{a}_{t|t-1},$$

with prediction error

$$\mathbf{e}_t = \mathbf{y}_t - \hat{\mathbf{y}}_{t|t-1} = \mathbf{Z}_t \left(\boldsymbol{\alpha}_t - \mathbf{a}_{t|t-1}\right) + \boldsymbol{\varepsilon}_t.$$

The covariance matrix of the prediction errors is

$$\mathbf{F}_t = E\left(\mathbf{e}_t \mathbf{e}_t'\right) = \mathbf{Z}_t \mathbf{P}_{t|t-1} \mathbf{Z}_t' + \mathbf{H}_t.$$

Once a new observation, $\mathbf{y}_t$, becomes available, using results on multivariate normal distributions (see Section B.10 in Appendix B), it follows that $\mathbf{a}_{t|t-1}$ and $\mathbf{P}_{t|t-1}$ can be revised using the updating equations

$$\mathbf{a}_{t|t} = \mathbf{a}_{t|t-1} + \mathbf{P}_{t|t-1} \mathbf{Z}_t' \mathbf{F}_t \mathbf{v}_t, \tag{16.8}$$

$$\mathbf{P}_{t|t} = \mathbf{P}_{t|t-1} - \mathbf{P}_{t|t-1} \mathbf{Z}_t' \mathbf{F}_t \mathbf{Z}_t \mathbf{P}_{t|t-1}. \tag{16.9}$$

Note that the term $\mathbf{P}_{t|t-1} \mathbf{Z}_t' \mathbf{F}_t \mathbf{Z}_t$ in equation (16.9) is the weight assigned to the new information available at time t. The Kalman algorithm calculates optimal predictions of $\boldsymbol{\alpha}_t$ in a recursive manner. It starts with the initial values $\boldsymbol{\alpha}_0$ and $\mathbf{P}_0$, and then iterates between (16.6)–(16.7) and (16.8)–(16.9), for $t = 1, 2, \ldots, T$.

If the initial value $\boldsymbol{\alpha}_0$ and the innovations $\boldsymbol{\varepsilon}_t$ and $\boldsymbol{\eta}_t$ are Gaussian processes, we have

$$\mathbf{y}_t|\Omega_{t-1} \sim N\left(\mathbf{Z}_t\mathbf{a}_{t|t-1}, \mathbf{F}_t\right),$$

and the associated log-likelihood function is

$$\ell\left(\boldsymbol{\theta}\right) = -\frac{mT}{2} - \frac{1}{2}\sum_{t=1}^{T}\ln|\mathbf{F}_t| - \frac{1}{2}\sum_{t=1}^{T}\mathbf{e}_t'\mathbf{F}_t^{-1}\mathbf{e}_t,$$

where $\boldsymbol{\theta}$ contains the parameters of interest. Maximization of the above log-likelihood function can be achieved by employing, for example, the Newton-Raphson algorithm or the expectation maximization algorithm introduced by Dempster, Laird, and Rubin (1977).

16.6 Trend-cycle decomposition of unit root processes

Recall that every trend stationary process can be decomposed into a deterministic trend and a stationary component according to Wold's decomposition (see Section 12.5). However, this method is not applicable for non-stationary time series.

16.6.1 Beveridge–Nelson decomposition

Beveridge and Nelson (1981) proposed to decompose unit root processes into a permanent and a transitory component allowing both components to be stochastic. Consider the first difference process $\left\{\Delta y_t\right\}$ with

$$\begin{aligned}\Delta y_t &= \mu + a\left(L\right)\varepsilon_t, \quad \varepsilon_t \sim IID\left(0,\sigma^2\right), \quad t = 1,2,\ldots T,\\ a\left(L\right) &= a_0 + a_1L + a_2L^2 + \ldots,\end{aligned} \tag{16.10}$$

where μ is a drift coefficient and the coefficients $\{a_i\}$ are absolutely summable $\left(\sum_{i=0}^{\infty}|a_i| < \infty\right)$. We can decompose the non-stationary process $\left\{y_t\right\}$ as

$$y_t = z_t + \xi_t, \quad t = 1,2,\ldots T, \tag{16.11}$$

with

$$\begin{aligned}z_t &= z_{t-1} + \mu + u_t \quad u_t \sim IID\left(0,\sigma_u^2\right),\\ \xi_t &= c\left(L\right)v_t, \; v_t \sim IID\left(0,\sigma_v^2\right),\\ c\left(L\right) &= c_0 + c_1L + c_2L^2 + \ldots.\end{aligned} \tag{16.12}$$

Here z_t is a random walk with drift. It is considered as the permanent component of the series since shocks to z_t have permanent effects on y_t, whilst shocks to ξ_t do not have a permanent effect on y_t, namely their effects die out eventually. This occurs because ξ_t, also called the transitory or cyclical component of the series, is a stationary process.

There are two issues that now need to be addressed:

(i) Can we find μ, the sequences $\{u_t\}$, $\{v_t\}$, and $\{c_i\}$ such that the above decomposition is compatible with the original process defined by (16.10)?
(ii) Is the solution for μ, c_i, u_t, and v_t unique?

From (16.11) and (16.10) we have

$$\Delta y_t = \Delta z_t + \Delta \xi_t = \mu + a(L)\,\varepsilon_t.$$

That is,

$$\mu + u_t + (1-L)\,c(L)\,v_t = \mu + a(L)\,\varepsilon_t.$$

Hence

$$u_t + (1-L)\,c(L)\,v_t = a(L)\,\varepsilon_t. \tag{16.13}$$

Whether the decomposition is unique is clearly of interest. Recall that two processes are considered to be observationally equivalent if they have the same autocovariance generating function (see Section 12.4). In the present context, since Δy_t and ξ_t are stationary processes with $a(L)$ and $c(L)$ being absolutely summable, the autocovariance generating functions for the two sides of (16.13) exist and are equal, namely

$$\begin{aligned}&\sigma_u^2 + (1-z)\left(1-z^{-1}\right)c(z)\,c\left(z^{-1}\right)\sigma_v^2 + 2\sigma_{uv}(1-z)\left(1-z^{-1}\right)c(z)\,c\left(z^{-1}\right),\\ &= \sigma^2 a(z)\,a\left(z^{-1}\right),\end{aligned} \tag{16.14}$$

where

$$\begin{pmatrix} u_t \\ v_t \end{pmatrix} \sim IID\left[\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \sigma_u^2 & \sigma_{uv} \\ \sigma_{vu} & \sigma_v^2 \end{pmatrix}\right].$$

Now any u_t and v_t processes satisfying (16.14) will also satisfy (16.10), hence they can be consistent with the original series. A solution clearly exists but it is not unique. To obtain a unique solution, Beveridge and Nelson (1981) (BN) assume that u_t and v_t are perfectly collinear, that is

$$u_t = \lambda v_t.$$

Then (16.13) becomes

$$\left[\lambda + (1-z)\,c(z)\right]\left[\lambda + \left(1-z^{-1}\right)c\left(z^{-1}\right)\right]\sigma_v^2 = \sigma^2 a(z)\,a\left(z^{-1}\right). \tag{16.15}$$

Without loss of generality, setting $\sigma_v^2 = \sigma^2$, we have

$$\left[\lambda + (1-z)\,c(z)\right]\left[\lambda + \left(1-z^{-1}\right)c\left(z^{-1}\right)\right] = a(z)\,a\left(z^{-1}\right). \tag{16.16}$$

We need to solve for λ and $c(z)$ using (16.16). By equating the constant terms and the terms with the same order of z from both sides of (16.16), we obtain

$$c_0 = a_0 - \lambda, \tag{16.17}$$

and

$$c_i = c_{i-1} + a_i, \text{ for } i = 1, 2, \ldots,$$

or

$$c_i = c_0 + \sum_{j=1}^{i} a_j. \tag{16.18}$$

Since (16.16) is satisfied for all z, including $z = 1$, we also have

$$\lambda^2 = a(1)^2,$$

and without loss of generality, we can set $\lambda = a(1) = \sum_{i=0}^{\infty} a_i$. Using this result in (16.17) we now have $c_0 = -\sum_{j=1}^{\infty} a_j$, and in view of (16.18) we obtain

$$c_i = -\sum_{j=i+1}^{\infty} a_j. \tag{16.19}$$

Thus, under the assumption that u_t and v_t are perfectly correlated we have the following unique answer to the decomposition problem

$$y_t = z_t + \xi_t,$$

where

$$z_t = z_{t-1} + \mu + a(1)\varepsilon_t, \tag{16.20}$$

and

$$\xi_t = c(L)v_t = \left(c_0 + c_1 L + c_2 L^2 + \ldots\right)\varepsilon_t, \tag{16.21}$$

with c_i given by (16.19).

As pointed out earlier, the shocks to z_t have permanent effects, but not when ξ_t is shocked. Consequently, $a(1)$ is often referred to as a measure of shock persistence. It captures the amount by which y_t is displaced when z_t is shocked by one unit (or one standard error). For example, in the $ARMA(p, q)$ process, $\phi(L)\Delta y_t = \theta(L)v_t$, the persistence measure, $a(1)$, equals $\theta(1)/\phi(1)$ and can be estimated by first estimating the coefficients of the underlying $ARMA$ model (see Section 12.6 for a description of $ARMA$ processes). Note that if $a(1) = 0$, from (16.12) we have $z_t = z_{t-1} + \mu$, and hence $y_t = z_0 + \mu t + c(L)v_t$, which reduces to Wold's

decomposition. Therefore, testing the hypothesis $a(1) = 0$ is the same as testing for a unit root. For this reason, $a(1)$ is also often referred to as the 'size of unit root'.

Another method of estimating $a(1)$ would be *via* the spectral density approach. Since $\{\Delta y_t\}$ is a stationary process, the related spectral density is

$$f_{\Delta y}(\omega) = \frac{\sigma^2}{2\pi} a\left(e^{iw}\right) a\left(e^{-iw}\right).$$

Evaluating at zero frequency,

$$f_{\Delta y}(0) = \frac{\sigma^2}{2\pi} a(1)\, a(1),$$

which yields

$$a(1) = \frac{\left[2\pi f_{\Delta y}(0)\right]^{1/2}}{\sigma}.$$

Since $a(1)$ is the square root of the standardized spectral density at zero frequency, it follows that identification of $a(1)$ does not depend on particular decomposition advocated by Beverage and Nelson. Thus the non-uniqueness of the BN decomposition does not pose any difficulty for the estimation and interpretation of $a(1)$.

16.6.2 Watson decomposition

As stated above, the unique answer to BN decomposition depends on the assumption that u_t and v_t are perfectly collinear. At another extreme, the Watson decomposition (see Watson (1986)) assumes zero correlation between the two error processes, that is $\sigma_{uv} = 0$. From (16.14) we have

$$\sigma_u^2 + (1-z)(1-z^{-1})c(z)c(z^{-1})\sigma_v^2 = \sigma^2 a(z)a(z^{-1}). \tag{16.22}$$

Evaluating (16.22) at $z = 1$ we obtain

$$\sigma_u^2 = \sigma^2 a(1)^2.$$

Substituting back in (16.22), gives

$$\sigma^2 a(1)^2 + (1-z)(1-z^{-1})c(z)\, c(z^{-1})\sigma_v^2 = \sigma^2 a(z)\, a(z^{-1}). \tag{16.23}$$

Dividing both sides of (16.23) by σ^2

$$a(1)^2 + (1-z)(1-z^{-1})c(z)\, c(z^{-1})\frac{\sigma_v^2}{\sigma^2} = a(z)\, a(z^{-1}).$$

Let $\gamma = \sigma_v^2/\sigma^2$ and normalize on this ratio, namely setting $\gamma = 1$, to obtain

$$a(1)^2 + (1-z)(1-z^{-1})c(z)\, c(z^{-1}) = a(z)\, a(z^{-1}). \tag{16.24}$$

Again, by equating both sides of (16.24), we can get a unique answer to the decomposition.

16.6.3 Stochastic trend representation

The Beveridge–Nelson decomposition discussed above (see Section 16.6.1) can also be obtained as stochastic trend decomposition of unit root processes. Consider the general unit root process

$$\Delta y_t = \mu + \sum_{i=0}^{\infty} a_i \varepsilon_{t-i} = \mu + a(L)\,\varepsilon_t,$$

and $\varepsilon_t \sim IID(0, \sigma^2)$. Then it follows that

$$y_t = y_0^* + \mu t + a(1) \sum_{i=1}^{t} \varepsilon_i + a^*(L)\,\varepsilon_t,$$

which is known as the stochastic trend representation of Δy_t process, and decomposes y_t into a deterministic linear trend, $y_0^* + \mu t$, a stochastic trend component, $a(1)\sum_{i=1}^{t} \varepsilon_i$, and a stationary (cyclical) component, $a^*(L)\,\varepsilon_t$, which satisfies the absolute summability condition, $\sum_{j=0}^{\infty} |a_j^*| < \infty$, if $\sum_{j=0}^{\infty} j\,|a_j| < \infty$. In relation to BN decomposition

$$y_0^* + \mu t + a(1) \sum_{i=1}^{t} \varepsilon_i = z_t,$$

and $a^*(L)\,\varepsilon_t = \xi_t$, where z_t and ξ_t are defined by (16.20) and (16.21), respectively.

To obtain the stochastic trend representation of the unit root process, first note that $a(L)$ can be written as

$$a(L) = a(1) + (1 - L)a^*(L), \tag{16.25}$$

where $a^*(L) = \sum_{i=0}^{\infty} a_i^* L^i$. Therefore

$$\Delta y_t = \mu + a(1)\,\varepsilon_t + (1 - L)a^*(L)\,\varepsilon_t,$$

and

$$\Delta\left(y_t - \zeta_t\right) = \mu + a(1)\,\varepsilon_t,$$

where

$$\zeta_t = a^*(L)\,\varepsilon_t.$$

Hence, iterating from the initial state, $y_0^* = y_0 - \zeta_0$ we have

$$y_t = y_0^* + \mu t + a(1) \sum_{i=1}^{t} \varepsilon_i + a^*(L)\,\varepsilon_t.$$

The coefficients a_i^* can be obtained in terms of a_i using (16.25). Equating powers of L^i in expansions of both sides of (16.25) we have

$$a_i^* = a_{i-1}^* + a_i,\ a_0^* = a_0 - a(1),$$

and hence

$$a_i^* = -\sum_{j=i+1}^{\infty} a_j.$$

Also, since

$$\begin{aligned}\sum_{i=0}^{\infty}\Big|\sum_{j=i+1}^{\infty} a_j\Big| &\le \sum_{i=0}^{\infty}\sum_{j=i+1}^{\infty}|a_j| = \sum_{j=1}^{\infty}|a_j| + \sum_{j=2}^{\infty}|a_j| + \sum_{j=3}^{\infty}|a_j| + \dots \\ &= \sum_{i=1}^{\infty} i\,|a_i|\,.\end{aligned}$$

Then it follows that

$$\sum_{i=0}^{\infty}|a_i^*| = \sum_{i=0}^{\infty}\Big|\sum_{j=i+1}^{\infty} a_j\Big| \le \sum_{i=1}^{\infty} i|a_i| < \infty,$$

which is bounded, by assumption. Hence, $\zeta_t = a^*(L)\,\varepsilon_t$ is covariance stationary.

It is also interesting to note that the stochastic permanent component of the $\{y_t\}$ process, namely $a(1)\sum_{i=1}^{t}\varepsilon_i$, can be viewed as the long-horizon expectations (forecast) of y_t defined by

$$y_t^P = \lim_{h\to\infty} E\left[y_{t+h} - y_0^* - (h+t)\mu\,|\mathcal{I}_t\right] = a(1)\sum_{i=1}^{t}\varepsilon_i,$$

where $\mathcal{I}_t = (y_t, y_{t-1}, \dots)$. This follows by noting that the long-horizon expectations of the mean zero stationary component of y_t, namely $a^*(L)\,\varepsilon_t$, is zero.

A multivariate version of the above trend/cycle decomposition is discussed in Section 22.15.

16.7 Further reading

A textbook treatment of state space models and the Kalman filter can be found in Harvey (1981, 1989), and Durbin and Koopman (2001).

16.8 Exercises

1. Consider the $ARMA(p, q)$ model analysed in Section 12.6

$$\phi(L)y_t = \theta(L)\varepsilon_t,$$

where

$$\begin{aligned}\phi(L) &= 1 - \phi_1 L - \phi_2 L^2 - \ldots - \phi_p L^p,\\ \theta(L) &= 1 - \theta_1 L - \theta_2 L^2 - \ldots - \theta_q L^q,\end{aligned}$$

and $\varepsilon_t \sim IID(0, \sigma^2)$. Suppose that all the roots of $\phi(z) = 0$, lie outside the unit circle and y_t has the infinite-order moving average process

$$y_t = \varepsilon_t + \psi_1\varepsilon_{t-1} + \psi_2\varepsilon_{t-1} + \ldots.. = \psi(L)\varepsilon_t.$$

(a) Show that

$$\begin{aligned}\psi_1 &= \phi_1 - \theta_1,\\ \psi_2 &= \phi_1\psi_1 + \phi_2 - \theta_2,\\ &\vdots\\ \psi_n &= \phi_1\psi_{n-1} + \phi_2\psi_{n-2} + \ldots. + \phi_{n-1}\psi_1 + \phi_n - \theta_n.\end{aligned}$$

(b) Consider the conditional forecasts $y_{t+h|t+s} = E\left(y_{t+h}\,|\mathcal{F}_{t+s}\right)$, where $\mathcal{F}_{t+s} = (y_{t+s}, y_{t+s-1}, \ldots)$, and $s < h$. Show that

$$\begin{aligned}y_{t+h|t+1} &= y_{t+h|t} + \psi_{h-1}\varepsilon_{t+1},\\ y_{t+h|t+1} &= \sum_{i=1}^{h}\phi_i y_{t+h-i|t} + \psi_{h-1}\varepsilon_{t+1}.\end{aligned}$$

(c) Hence, or otherwise, show that the *ARMA* process can be written in the following state-space form

$$\begin{aligned}y_t &= (1, 0, \ldots, 0)\mathbf{s}_t\\ \mathbf{s}_{t+1} &= \mathbf{T}\mathbf{s}_t + \mathbf{R}\varepsilon_{t+1},\end{aligned}$$

where $\mathbf{s}_t$ is an $m \times 1$ vector of conditional forecasts, $y_{t+h|t+1}$ for $h = 1, 2, \ldots, m$, where $m = \max(p, q+1)$. Also derive the $m \times m$ matrices $\mathbf{T}$ and $\mathbf{R}$ in terms of the coefficients of the *ARMA* model.

2. Consider the general linear first difference stationary process

$$\Delta y_t = \mu + a(L)\varepsilon_t, \tag{16.26}$$

where Δ is the first difference operator,

$$a(L) = a_0 + a_1 L + a_2 L^2 + \ldots,$$

is a polynomial in the lag operator, ($Ly_t = y_{t-1}$) and μ is a scalar constant. The ε_t are mean zero, serially uncorrelated shocks with common variance, σ_ε^2.

(a) Show that the $\{y_t\}$ process can be decomposed into a stationary component, x_t, and a random walk component τ_t

$$y_t = x_t + \tau_t, \tag{16.27}$$

where

$$x_t = b(L)\varepsilon_t,$$
$$\tau_t = \mu + \tau_{t-1} + \eta_t,\ \eta_t \sim IID(0,\ \sigma_\eta^2),$$

and $b(L) = b_0 + b_1 L + b_2 L^2 + \ldots$.

(b) Obtain the coefficients $\{b_i\}$ in terms of $\{a_i\}$, and show that

$$\eta_t = \left(\sum_{i=0}^{\infty} a_i\right)\varepsilon_t.$$

(c) Discuss the relevance of the decomposition (16.27) for the impulse response analysis of shocks to y.

3. Suppose that y_t follows the $ARIMA(p, d, q)$ process

$$\phi(L)(1-L)^d y_t = \theta(L)\varepsilon_t,\ \varepsilon_t \sim IID(0, \sigma^2). \tag{16.28}$$

(a) Show that

$$w_t = y_t + A_0 + A_1 t + \cdots + A_{d-1} t^{d-1},$$

where $A_0,\ A_1,\ \cdots,\ A_{d-1}$ are arbitrary random variables, also satisfies (16.28).

(b) For $d = 1$, write down the Beveridge and Nelson (1981) decomposition of y_t, and hence or otherwise show that the Campbell and Mankiw (1987) persistence measure is given by $\theta(1)/\phi(1)$.

(c) Find the lower bounds on the persistence measure for $\phi(1) = 1$, and for $p = 1, 2, 3$, assuming $d = 1$.

4. Consider the $AR(1)$ process with a deterministic trend

$$y_t = a_0 + a_1 t + \rho y_{t-1} + u_t.$$

   (a) Let $x_t = y_t - \delta_0 - \delta_1 t$ and derive δ_0 and δ_1 in terms of the parameters of the $AR(1)$ process such that

$$x_t = \rho x_{t-1} + u_t.$$

   (b) Derive the long horizon forecast of x_t, defined by $E(x_{t+h}|\Omega_t)$, where $\Omega_t = (y_t, y_{t-1}, \ldots)$ for values of ρ inside the unit circle as well as when $\rho = 1$.

   (c) Using the results in (b) above derive the permanent component of y_t, and compare your results with the Beveridge–Nelson decomposition for ρ inside the unit circle as well as when $\rho = 1$.

5. Use quarterly time series observations on US GDP over the period 1979Q1-2013Q2 (provided in the *GVAR* data set https://sites.google.com/site/gvarmodelling/data) to compute the permanent component of the log of US output (y_t) using the Hodrick–Presoctt filter. Compare your results with the long-run forecasts of y_t, namely $E(y_{t+h} | y_t, y_{t-1}, \ldots)$, for h sufficiently large, computed using the following $ARIMA(1, 1, 1)$ specification

$$\Delta y_t = \phi \Delta y_{t-1} + \varepsilon_t + \theta \varepsilon_{t-1}.$$

17 Introduction to Forecasting

17.1 Introduction

This chapter provides an introduction to the theory of forecasting and presents some applications to forecasting univariate processes. It begins with a discussion of alternative criteria of forecast optimality. It distinguishes between point and probability forecasts, one-step and multi-step ahead forecasts, conditional and *ex ante* forecasts. Using a quadratic loss function, point and probability forecasts are derived for univariate time series processes that are optimal in the mean squared forecast error sense. Also, the problem of parameter and model uncertainty in forecasting is discussed, and an overview of the techniques for forecast evaluation is provided.

17.2 Losses associated with point forecasts and forecast optimality

Since errors in forecasts invariably entail costs (or losses), a specification of how costly different mistakes are is needed to guide our procedure. The loss function describes in relative terms how costly any forecast error is given the outcome and possibly other observed data (Elliott and Timmermann (2008)). The most commonly used loss function is the quadratic loss, also known as the mean squared forecast error (*MSFE*) loss.

17.2.1 Quadratic loss function

Let $y^*_{t+1|t}$ be the *point* forecast, associated with the realization y_{t+1}, made at time $t+1$, with respect to the information set $\Omega_t = (y_t, y_{t-1}, \ldots)$.[1] $y^*_{t+1|t}$, is also known as the one-step-ahead forecast of y_{t+1}. The corresponding forecast or prediction error is given by $e_{t+1} = y_{t+1} - y^*_{t+1|t}$, with the quadratic loss function

[1] The information set could also contain data on other variables.

$$L_q(y_{t+1}, y^*_{t+1|t}) = Ae^2_{t+1} = A\left(y_{t+1} - y^*_{t+1|t}\right)^2, \tag{17.1}$$

where A is a positive non-zero constant. The optimal forecast is obtained by minimizing the expected loss conditional on the information available at time t, namely

$$y^*_{t+1|t} = \underset{y^*_{t+1|t}}{\operatorname{argmin}} \left\{ E\left[L_q(y_{t+1}, y^*_{t+1|t}) \mid \Omega_t \right] \right\}.$$

$y^*_{t+1|t}$ is also said to be optimal in the mean squared forecast error sense. Note that (setting $A = 1$ without loss of generality)

$$E\left[L_q(y_{t+1}, y^*_{t+1|t}) \mid \Omega_t \right] = \int_R \left(y_{t+1} - y^*_{t+1|t}\right)^2 f(y_{t+1} \mid \Omega_t) dy_{t+1},$$

where R denotes the range of variation of y_{t+1}, and $f(y_{t+1} \mid \Omega_t)$ is the probability density of y_{t+1} conditional on the information set Ω_t. Suppose now that the probability density function is exogenously given and is not affected by the forecasting exercise (reality is invariant to the way forecasts are formed), then the first-order condition for the above minimization problem is given by

$$\begin{aligned}
\frac{\partial E\left(y_{t+1} - y^*_{t+1|t}\right)^2}{\partial y^*_{t+1|t}} &= \frac{\partial}{\partial y^*_{t+1|t}} \left[\int_R \left(y_{t+1} - y^*_{t+1|t}\right)^2 f(y_{t+1} \mid \Omega_t) dy_{t+1} \right] \\
&= \int_R \frac{\partial}{\partial y^*_{t+1|t}} \left[\left(y_{t+1} - y^*_{t+1|t}\right)^2 f(y_{t+1} \mid \Omega_t) dy_{t+1} \right] \\
&= -2 \int_R \left(y_{t+1} - y^*_{t+1|t}\right) f(y_{t+1} \mid \Omega_t) dy_{t+1} = 0.
\end{aligned} \tag{17.2}$$

Since $y^*_{t+1|t}$, the predicted value, can be viewed as known by the forecaster, the integral in (17.2) can also be written as

$$\int_R y_{t+1} f(y_{t+1} \mid \Omega_t) dy_{t+1} = y^*_{t+1|t} \int_R f(y_{t+1} \mid \Omega_t) dy_{t+1}. \tag{17.3}$$

But since $f(y_{t+1} \mid \Omega_t)$ is a density function then $\int_R y_{t+1} f(y_{t+1} \mid \Omega_t) dy_{t+1} = E\left(y_{t+1} \mid \Omega_t\right)$, and $\int_R f(y_{t+1} \mid \Omega_t) dy_{t+1} = 1$, and from (17.3) we obtain

$$y^*_{t+1|t} = E\left(y_{t+1} \mid \Omega_t\right). \tag{17.4}$$

Thus it is established that $E\left(y_{t+1} \mid \Omega_t\right)$ is the optimal point forecast of y_{t+1} conditional on Ω_t when

(a) the underlying loss function is quadratic
(b) the true conditional density function, $f(y_{t+1} \mid \Omega_t)$, is known
(c) and the act of forecasting will not change the true density function.

This fundamental result will be used in Section 17.6 to construct forecasts of *ARMA* processes.

17.2.2 Asymmetric loss function

An important example of an asymmetric loss function is a simple version of the linear exponential (LINEX) function, first introduced by Varian (1975) and analysed in a Bayesian context by Zellner (1986)

$$L_a\left(y_{t+1}, y^*_{t+1|t}\right) = \frac{2\left[\exp\left(\alpha e_{t+1}\right) - \alpha e_{t+1} - 1\right]}{\alpha^2}, \tag{17.5}$$

where as before, $e_{t+1} = y_{t+1} - y^*_{t+1|t}$, and α is a parameter that controls the degree of asymmetry.

This function has the interesting property that it reduces to the familiar quadratic loss function for $\alpha = 0$. Using L'Hopital's rule

$$\lim_{\alpha \to 0} L_a\left(y_{t+1}, y^*_{t+1|t}\right) = \lim_{\alpha \to 0} \frac{2\left[\exp\left(\alpha e_{t+1}\right) e_{t+1} - e_{t+1}\right]}{2\alpha}.$$

Using the rule again

$$\lim_{\alpha \to 0} L_a\left(y_{t+1}, y^*_{t+1|t}\right) = \lim_{\alpha \to 0} \frac{2\exp\left(\alpha e_{t+1}\right) e^2_{t+1}}{2} = e^2_{t+1}. \tag{17.6}$$

A pictorial representation of the LINEX function for $\alpha = 0.5$ is provided in Figure 17.1.

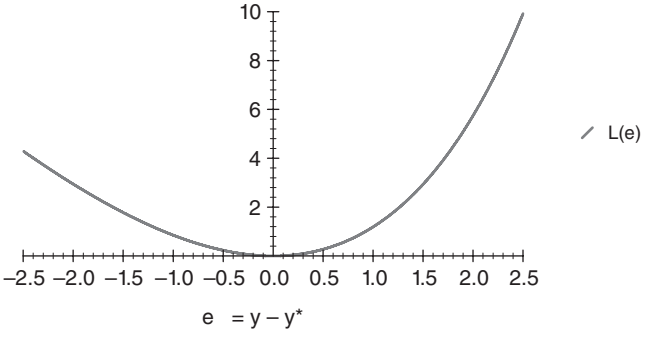


Figure 17.1 The LINEX cost function defined by (17.5) for $\alpha = 0.5$.

For this particular loss function under-predicting is more costly than over-predicting when $\alpha > 0$. The reverse is true when $\alpha < 0$. Again assuming that the forecast will not affect the range of the integral, for the LINEX loss function the optimal forecast, $y^*_{t+1|t}$, can be obtained as the solution of the following equation

$$E\left[\frac{\partial}{\partial y^*_{t+1|t}} L_a\left(y_{t+1}, y^*_{t+1|t}\right) \mid \Omega_t\right] = 0. \tag{17.7}$$

But using (17.5) we have

$$\begin{aligned}\frac{\partial}{\partial y^*_{t+1|t}} L_a\left(y_{t+1}, y^*_{t+1|t}\right) &= \frac{2}{\alpha}\left[\frac{\partial e_{t+1}}{\partial y^*_{t+1|t}} \exp\left(\alpha e_{t+1}\right) + 1\right] \\ &= \frac{2}{\alpha}\left[1 - \exp\alpha\left(y_{t+1} - y^*_{t+1|t}\right)\right],\end{aligned}$$

and using this result in (17.7) it is easily seen that

$$y^*_{t+1|t} = \frac{1}{\alpha}\log\left[E\left(\exp\alpha y_{t+1} \mid \Omega_t\right)\right],$$

where the expectations are taken with respect to the conditional true density function of y_{t+1}. In the case where this density is normal, we have

$$y^*_{t+1|t} = E\left(y_{t+1} \mid \Omega_t\right) + \frac{\alpha}{2} Var\left(y_{t+1} \mid \Omega_t\right),$$

where $E\left(y_{t+1} \mid \Omega_t\right)$ and $Var\left(y_{t+1} \mid \Omega_t\right)$ are the conditional mean and variance of y_t. Notice that the higher the degree of asymmetry in the cost function (as measured by the magnitude of α), the larger will be the discrepancy between the optimal forecast and $E\left(y_{t+1} \mid \Omega_t\right)$. The average realized value of the cost function, evaluated at the optimal forecast, is given by

$$E\left[L_a\left(y_{t+1}, y^*_{t+1|t}\right)\right] = E\left[Var\left(y_{t+1} \mid \Omega_t\right)\right],$$

which, interestingly enough, is independent of α, the degree of asymmetry of the underlying loss function.

17.3 Probability event forecasts

In the context of decision-making, forecasting methods are employed in order to produce better decisions. In this case, point forecasts are usually not sufficient and the full conditional probability distribution of the variables entering the decision problem is required.

As a simple example, consider a situation where there are two 'states' of the world, say 'Bad' and 'Good'. Let $\hat{\pi}_{t+1|t}$ denote the forecast probability made at time t that the Bad event will occur on day $t+1$. Thus the forecast probability of the Good event is $1-\hat{\pi}_{t+1|t}$, while $\hat{\pi}_{t+1|t}$ is an estimate of the probability of the Bad event occurring in day $t+1$, denoted by π_{t+1}. This case of two states with predicted probabilities of $\hat{\pi}_{t+1|t}$ and $1-\hat{\pi}_{t+1|t}$ is the simplest possible example of forecasting a probability distribution (or *predictive distribution*). Consider the variable $z_{t+1} = 1$ if the Bad event occurs and $z_{t+1} = 0$, otherwise. As an alternative to the probability forecasts, a point forecast of z_{t+1}, is $\hat{z}^*_{t+1|t} = 1$ if the Bad state is forecast to occur or otherwise $\hat{z}^*_{t+1|t} = 0$.

In this case, the probability (event) forecast, $\hat{\pi}_{t+1|t}$, can be converted to the point forecast, using a 'rule of thumb' which gives $\hat{z}^*_{t+1|t} = 1$ if $\hat{\pi}_{t+1|t}$ exceeds some specified probability threshold, $\alpha_t \in (0, 1)$.

Hence, the economic forecaster has two alternative forms of forecast to announce, either $\hat{\pi}_{t+1|t}$, which takes some value in the region $0 \leq \hat{\pi}_{t+1|t} \leq 1$, and represents a probability forecast; or $\hat{z}^*_{t+1|t}$ which is an event forecast. The relationship between probability and event forecasts can also be written as $\hat{z}^*_{t+1|t} = I(\hat{\pi}_{t+1|t} - \alpha_t)$, where the indicator function $I(\cdot)$, is defined by $I(\mathfrak{A}) = 1$ if $\mathfrak{A} > 0$, and $I(\mathfrak{A}) = 0$, otherwise. For further discussion see Pesaran and Granger (2000a, 2000b). Two-states decision problems typically arise when the focus of the analysis is correct prediction of the direction of change in the variable under consideration (up, down) (see Pesaran and Timmermann (1992) on this, and also Section 17.12 below).

More generally, let y_t be a variable of interest, and suppose that we are interested in forecasting y_{t+1} at time t, having available an information set Ω_t. Probability event forecast refers to the probability of a particular event taking place, say the probability that the event $\mathfrak{A}_{t+1} = \{b \leq y_{t+1} \leq a\}$ occurs. For example, the probability of inflation (Δp_{t+1}) conditional on the information at time t falling in the range (a_1, a_2), or the probability of a recession defined as two successive negative growth rates (Δy_{t+1})

$$\Pr(\Delta y_{t+1} < 0\,, \Delta y_{t+2} < 0 \mid \Omega_t),$$

or the joint probability of the inflation rate (Δp_{t+1}) falling within a target range and a positive output growth

$$\Pr(a_1 < \Delta p_{t+1} < a_2\,, \Delta y_{t+1} > 0 \mid \Omega_t).$$

Probability forecasts also play an important role in the Value-at-Risk (VaR) analysis in insurance and finance (see Chapter 7). For example, it is often required that return on a given portfolio, r_{t+1} (or insurance claim) satisfies the following VaR probability constraint

$$\Pr(r_{t+1} < -VaR \mid \Omega_t) \leq \alpha,$$

where VaR denotes the maximum permitted loss over the period t to $t + 1$.

A density forecast of the realization of a random variable at some future time is an estimate of the probability distribution of the possible future values of that variable. Thus, density forecasting is concerned with $\hat{f}_t(y_{t+1} \mid \Omega_t)$ for all feasible values of y_{t+1}, or equivalently with its probability distribution function, $\hat{F}_t(y) = \int_{-\infty}^{y} \hat{f}_t(u \mid \Omega_t)du$, for all feasible values of y. It thus provides a complete description of the uncertainty associated with a forecast, and stands in contrast to a point forecast, which by itself contains no description of the associated uncertainty. Notice that probability forecasts can be seen as a special case of density forecasting, since we have

$$\widehat{\Pr}\,(\mathfrak{A}_{t+1} \mid \Omega_t) = \hat{F}_t(a) - \hat{F}_t(b).$$

As explained above, an event forecast can be put in the form of an indicator function and states whether an event is forecast to occur. For example, in the case of $\mathfrak{A}_{t+1} = \{b \leq y_{t+1} \leq a\}$, the event forecast will simply be

$$\hat{I}(\mathfrak{A}_{t+1} \mid \Omega_t) = 1, \text{ if } \mathfrak{A}_{t+1} \text{ is predicted to occur,}$$
$$\hat{I}(\mathfrak{A}_{t+1} \mid \Omega_t) = 0, \text{otherwise.}$$

It is always possible to compute event forecasts from probability event forecasts, but not *vice versa*. This could be done with respect to a probability threshold, p, often taken to be $1/2$ in practice. In the case of the above example we have

$$\hat{I}(\mathfrak{A}_{t+1} \mid \Omega_t) = I\left(\hat{F}_t(a) - \hat{F}_t(b) - p\right).$$

Finally, the main object of interest could be point forecasts, as the mean

$$E\left(y_{t+1} \mid \Omega_t\right) = \int_{-\infty}^{\infty} uf(u \mid \Omega_t)du,$$

or volatility forecasts as the variance

$$Var\left(y_{t+1} \mid \Omega_t\right) = \int_{-\infty}^{\infty} u^2 f(u \mid \Omega_t)du - \left[E\left(y_{t+1} \mid \Omega_t\right)\right]^2.$$

See also Chapter 18.

17.3.1 Estimation of probability forecast densities

In practice, $f(y_{t+1} \mid \Omega_t)$ must be estimated from the available observations contained in Ω_t. This is achieved by first specifying a model of y_{t+1} in terms of its lagged values or other variables deemed relevant to the evolution of y_{t+1}. The model is then estimated and used to compute an estimate of $f(y_{t+1} \mid \Omega_t)$ which we denote by $\hat{f}(y_{t+1} \mid \Omega_t)$. This is an example of the parametric approach to forecasting. Non-parametric forecast procedures have also been suggested in the literature, but will not be considered here.

17.4 Conditional and unconditional forecasts

A conditional forecast of y_{t+1} is typically formed using the information available at time t as well as on assumed values for a set of conditioning variables. Conditional forecasts play an important role in scenario and counter-factual analysis. Consider for example the simple autoregressive distributed lag model

$$y_t = a + \lambda y_{t-1} + \beta x_t + u_t,$$

where u_t is a serially uncorrelated process with mean zero and x_t is a conditioning variable. Then assuming a mean squared loss function, the conditional forecast of y_{t+1} based on $\Omega_t = (y_t, x_t, y_{t-1}, x_{t-1}, \ldots)$ and the value of x_{t+1} is given by

$$E\left(y_{t+1} \mid \Omega_t, x_{t+1}\right) = a + \lambda y_t + \beta x_{t+1}.$$

In contrast, unconditional forecasting does not assume known future values for the conditioning variable, x_t. An unconditional (or *ex ante*) forecast of y_{t+1} is given by

$$E\left(y_{t+1} \mid \Omega_t\right) = a + \lambda y_t + \beta E\left(x_{t+1} \mid \Omega_t\right).$$

Clearly, in this case we also need to specify a model for x_t.

17.5 Multi-step ahead forecasting

Economists are commonly asked to forecast uncertain events multiple periods ahead in time. For example, in a recession state a policy maker may want to know when the economy will recover and so is interested in forecasts of output growth for, say, horizons of 1, 3, 6, 12, and 24 months. Similarly, fixed-income investors are interested in comparing forecasts of spot rates multiple periods ahead against current long-term interest rates in order to arrive at an optimal investment strategy. More formally, multi-step ahead forecasting consists of forecasting y_{T+h} based on information up to period T, given that $T+h$ observations $y_t, t = -h+1, -h+2, \ldots, T$, are available for model estimation. Let $y^*_{T+h|T}$ be the forecast of y_{T+h} formed at time T. The date $T + h$ is also known as the target date, and h as the forecast horizon.

In the general case of h-step ahead forecasts, the quadratic loss function is

$$L_q(y_{T+h}, y^*_{T+h|T}) = (y_{T+h} - y^*_{T+h|T})^2 = e^2_{T+h},$$

where

$$e_{T+h} = y_{T+h} - y^*_{T+h|T},$$

is the forecast error. As with the case of 1-step ahead forecasts outlined in Section 17.2, the value of $y^*_{T+h|T}$ that minimizes the expected loss, $E\left[L_q(y_{T+h}, y^*_{T+h|T}) \mid \Omega_T\right]$, is

$$y^*_{T+h|T} = E(y_{T+h} \mid \Omega_T) = \underset{y^*_{T+h|T}}{\operatorname{argmin}} \left\{E\left[L_q(y_{T+h}, y^*_{T+h|T}) \mid \Omega_T\right]\right\}. \tag{17.8}$$

For the LINEX function

$$L_a\left(y_{T+h}, y^*_{T+h|T}\right) = 2\left[\exp(\alpha e_{T+h}) - \alpha e_{T+h} - 1\right]/\alpha^2, \tag{17.9}$$

the optimal forecast, $y^*_{T+h|T}$, is easily seen to be

$$y^*_{T+h|T} = \alpha^{-1} \log\left\{E\left[\exp(\alpha y_{T+h}) \mid \Omega_T\right]\right\},$$

where the expectations are taken with respect to the conditional true density function of y_{T+h}. In the case where this density is normal, we have

$$y^*_{T+h|T} = E\left(y_{T+h}\,|\Omega_T\right) + \frac{\alpha}{2} Var\left(y_{T+h}\,|\Omega_T\right),$$

where $Var\left(y_{T+h}\,|\Omega_T\right)$ is the conditional variance of y_{T+h}.

17.6 Forecasting with *ARMA* models

Consider the $ARMA(p, q)$ model

$$y_t = \sum_{i=1}^{p} \phi_i y_{t-i} + \sum_{i=0}^{q} \theta_i \varepsilon_{t-i}, \qquad \theta_0 = 1,$$

and suppose that we are interested in forecasting y_{T+h} given the information set $\Omega_T = (y_T, y_{T-1}, \ldots)$. Optimal point or probability forecasts of y_{T+h} can be derived with respect to a given loss function and conditional on the information set Ω_T. In the following, we derive optimal point forecasts of y_{T+h} using the quadratic loss function and result (17.8) for *AR*, *MA*, and *ARMA* models.

17.6.1 Forecasting with *AR* processes

As a simple example consider forecasting with $AR(1)$ model

$$y_t - d_t = \phi(y_{t-1} - d_{t-1}) + \varepsilon_t,$$

where d_t is the deterministic or perfectly predictable component of the process—recall that for d_t we would have $E\left(d_{T+h}\,|\Omega_T\right) = d_{T+h}$. It is now easily seen that

$$y^*_{T+h|T} = E\left(y_{T+h}\,|\Omega_T\right) = d_{T+h} + \phi^h(y_T - d_T), \tag{17.10}$$

and $E\left(y_{T+h}\,|\Omega_T\right)$ converges to its perfectly predictable component (also known in economic applications as the steady state) as $h \to \infty$, if the process $(y_t - d_t)$ is stationary, namely if $|\phi| < 1$. For this reason a (trend-) stationary processes is also known as a *mean reverting* process. Note, however, that the above forecasts are optimal even if the underlying process is nonstationary. For example, if $\phi = 1$ we would have $E\left(y_{T+h}\,|\Omega_T\right) = y_T + (d_{T+h} - d_T)$. But in this case the long-horizon forecast, defined by $\lim_{h\to\infty} E\left(y_{T+h}\,|\Omega_T\right)$, is no longer mean reverting. When parameters are not known and have to be estimated, (17.10) becomes (abstracting from deterministic components)

$$\hat{y}^*_{T+h|T} = \hat{\phi}^h y_T, \tag{17.11}$$

where $\hat{\phi}$ is an estimator of ϕ, for example the Yule-Walker estimator (see formula (14.25) in Chapter 14). Notice that this formula, obtained by minimizing the quadratic loss function, is equivalent to the forecast obtained by using the iterative approach (see, in particular, formula (17.15)).

For higher-order *AR* models, *h*-step ahead forecasts can be obtained recursively. For example, the optimal point forecasts for an $AR(2)$ process are given by

$$\begin{aligned} y^*_{T+1|T} &= \phi_1 y_T + \phi_2 y_{T-1}, \\ y^*_{T+2|T} &= \phi_1 y^*_{T+1|T} + \phi_2 y_T, \\ y^*_{T+j|T} &= \phi_1 y^*_{T+j-1|T} + \phi_2 y^*_{T+j-2|T}, \text{ for } j = 3, 4, \ldots, h. \end{aligned}$$

More generally

$$y^*_{T+j|T} = \sum_{i=1}^{p} \phi_i y^*_{T+j-i|T}, \; j = 1, 2, \ldots, h,$$

with the initial values

$$y^*_{T+j-i|T} = y_{T-i} \text{ for } j - i \leq 0.$$

17.6.2 Forecasting with *MA* processes

Consider now forecasting with *MA* processes. Since

$$\begin{aligned} y_{T+h} &= \varepsilon_{T+h} + \theta_1 \varepsilon_{T+h-1} + \ldots + \theta_q \varepsilon_{T+h-q}, \\ y^*_{T+h|T} &= 0, \text{ if } h > q. \end{aligned}$$

For $h = 1 < q$,

$$y^*_{T+1|T} = \theta_1 \varepsilon_T + \theta_2 \varepsilon_{T-1} \ldots + \theta_q \varepsilon_{T-q+1},$$

for $h = 2 < q$,

$$y^*_{T+2|T} = \theta_2 \varepsilon_T + \theta_3 \varepsilon_{T-1} \ldots + \theta_q \varepsilon_{T-q+2},$$

and so on. To compute the forecasts we now need to estimate $\varepsilon_T, \varepsilon_{T-1}, \ldots$ from the realizations $y_T, y_{T-1}, \ldots$. This can be achieved assuming that the invertibility condition (discussed in Section 12.6) holds. When this condition is met we can obtain ε_T and its lagged values from a truncated version of the infinite *AR* representation of the *MA* process

$$\varepsilon_T = \theta(L)^{-1} y_T = \alpha(L) y_T,$$

where $\theta(L) = \theta_0 + \theta_1 L + \theta_2 L^2 + \ldots + \theta_q L^q$, and $\alpha(L)\theta(L) = 1$. The coefficients of the infinite-order polynomial $\alpha(L) = \alpha_0 + \alpha_1 L + \alpha_2 L^2 + \ldots$, can be obtained recursively using the relations

$$\begin{aligned}
\alpha_0\theta_0 &= 1 \\
\alpha_1\theta_0 + \alpha_0\theta_1 &= 0 \\
&\vdots \\
\alpha_q\theta_0 + \alpha_{q-1}\theta_1 + \ldots + \alpha_0\theta_q &= 0 \\
\alpha_i\theta_0 + \alpha_{i-1}\theta_1 + \ldots + \alpha_{i-q}\theta_q &= 0, \text{ for } i > q,
\end{aligned}$$

where $\theta_0 = 1$.

The above procedures can be adapted to forecasting using *ARMA* models. We have

$$\varepsilon_T = \theta(L)^{-1}\phi(L)y_T = \alpha(L)y_T,$$

and

$$\begin{aligned}
\phi(L)y^*_{T+h|T} &= 0, \text{ for } h > q, \\
\phi(L)y^*_{T+h|T} &= \theta_1\alpha(L)y_T + \theta_2\alpha(L)y_{T-1} + \ldots + \theta_q\alpha(L)y_{T-q+1}, \text{ for } h = 1 < q, \\
\phi(L)y^*_{T+h|T} &= \theta_2\alpha(L)y_T + \theta_3\alpha(L)y_{T-1} + \ldots + \theta_q\alpha(L)y_{T-q+2}, \text{ for } h = 2 < q, \\
&\quad \text{and so on.}
\end{aligned}$$

Also as before, $y^*_{T+j-i|T} = y_{T-i}$ for $j - i \leq 0$.

17.7 Iterated and direct multi-step *AR* methods

Suppose that the observations $y_1, y_2, \ldots, y_T$ follow the stationary $AR(1)$ model

$$y_t = a + \phi y_{t-1} + \varepsilon_t, \quad |\phi| < 1, \varepsilon_t \sim IID(0, \sigma^2_\varepsilon). \tag{17.12}$$

Letting $\mu = a/(1-\phi)$, we can use the equivalent representation

$$\begin{aligned}
y_t &= \mu + u_t, \quad \text{where} \\
u_t &= \sum_{i=0}^{\infty} b_i\varepsilon_{t-i}, \ b_i = \phi^i.
\end{aligned} \tag{17.13}$$

Rewrite (17.12) as

$$\begin{aligned}
y_t &= a\left(\frac{1-\phi^h}{1-\phi}\right) + \phi^h y_{t-h} + v_t, \\
&\equiv a_h + \phi_h y_{t-h} + v_t,
\end{aligned} \tag{17.14}$$

where

$$v_t = \sum_{j=0}^{h-1} \phi^j \varepsilon_{t-j}.$$

Notice that v_t follows an $MA(h-1)$ process even when ε_t is serially uncorrelated due to the data overlap resulting from $h > 1$.

Two basic strategies exist for generating multi-period forecasts in the context of AR models. The first approach, known as the 'iterated' or 'indirect' method, consists of estimating (17.12) for the data observed and then using the chain rule to generate a forecast at the desired horizon, $h \geq 1$. Specifically, the iterated forecast of y_{T+h}, denoted by $\hat{y}_{T+h}$, is given as

$$\hat{y}^*_{T+h|T} = \hat{a}_T \left(\frac{1 - \hat{\phi}_T^h}{1 - \hat{\phi}_T} \right) + \hat{\phi}_T^h y_T, \tag{17.15}$$

where $\hat{a}_T$ and $\hat{\phi}_T$ are the estimators of a and ϕ obtained from the *OLS* regression (17.12) of y_t on an intercept and y_{t-1}, using the observations y_t, $t = -h+1, -h+2, \ldots, T$. We have

$$(T+h-1)^{-1} \sum_{t=-h+2}^{T} y_t = \hat{a}_T + \hat{\phi}_T (T+h-1)^{-1} \sum_{t=-h+2}^{T} y_{t-1},$$

or, equivalently,

$$\bar{y}_{h:T} = \hat{a}_T + \hat{\phi}_T \bar{y}_{h:T,-1}, \tag{17.16}$$

where

$$\bar{y}_{h:T} = (T+h-1)^{-1} \sum_{t=-h+2}^{T} y_t, \quad \bar{y}_{h:T,-1} = (T+h-1)^{-1} \sum_{t=-h+2}^{T} y_{t-1},$$

$$\hat{\phi}_T = \frac{\sum_{t=-h+2}^{T} y_t (y_{t-1} - \bar{y}_{h:T,-1})}{\sum_{t=-h+2}^{T} (y_{t-1} - \bar{y}_{h:T,-1})^2}.$$

Under this approach, the forecasting equation is the same across all forecast horizons; only the number of iterations changes with h. Note also that this method yields identical forecasts as when minimizing the *MSFE* loss function.

An alternative approach, known as the 'direct' method, consists of estimating a model for the variable measured h-periods ahead as a function of current information. Specifically, the direct forecast of y_{T+h}, $\tilde{y}^*_{T+h|T}$, is given by

$$\tilde{y}^*_{T+h|T} = \tilde{a}_{h,T} + \tilde{\phi}_{h,T} y_T, \tag{17.17}$$

where $\tilde{a}_{h,T}$ and $\tilde{\phi}_{h,T}$ are the *OLS* estimators of a_h and ϕ_h obtained directly from (17.14), by regressing y_t on an intercept and y_{t-h} using the same sample observations, y_t, $t = -h+1, -h+2, \ldots, T$, used in the computation of the iterated forecasts. Notice that under this approach, the forecasting model and its estimates will typically vary across different forecast horizons.

We next establish conditions under which both the direct and indirect forecasts are *unconditionally* unbiased:

Proposition 45 *Suppose data is generated by the stationary AR(1) process, (17.12) and define the h-step ahead forecast errors from the iterated and direct methods,*

$$\hat{e}_{T+h|T} = y_{T+h} - \hat{y}^*_{T+h|T},$$

and

$$\tilde{e}_{T+h|T} = y_{T+h} - \tilde{y}^*_{T+h|T},$$

where the iterated h-step forecast, $\hat{y}^*_{T+h|T}$, and the direct forecast, $\tilde{y}^*_{T+h|T}$, are given by (17.15) and (17.17), respectively. Assume that u_t and v_t, defined in (17.13) and (17.14), are symmetrically distributed around zero, have finite second-order moments and expectations of $\hat{\phi}_T$ and $\tilde{\phi}_{h,T}$ exist. Then for any finite T and h we have

$$E(\hat{e}_{T+h|T}) = E(\tilde{e}_{T+h|T}) = 0.$$

The proposition generalizes the known result in the literature for $h = 1$ established, for example, by Fuller (1996) to multi-step ahead forecasts. For $h = 1$, Pesaran and Timmermann (2005b) also show that forecast errors are unconditionally unbiased for symmetrically distributed error processes even in the presence of breaks in the autocorrelation coefficient, ϕ, so long as μ is stable over the estimation sample.

In comparing iterated and direct forecasts, it is worth noting that when ϕ is positive and not too close to unity, for moderately large values of h, $\hat{\phi}^h_T \approx 0$, since $|\hat{\phi}_T| < |\phi| < 1$.[2] It follows that in such cases, $\hat{e}_{T+h|T} = (\mu - \hat{\mu}_T) + v_{T+h} + o(\phi^h)$. Similarly, $\tilde{e}_{T+h|T} = -\bar{v}_T + v_{T+h} + o(\phi^h)$. Hence, for h moderately large and ϕ not too close to the unit circle, a measure of the relative efficiency of the two forecasting methods can be obtained as

$$\frac{E(\hat{e}^2_{T+h|T})}{E(\tilde{e}^2_{T+h|T})} = \frac{E(\hat{\mu}_T - \mu)^2 + E(v^2_{T+h|T}) + o(\phi^h)}{E(\bar{v}^2_T) + E(v^2_{T+h|T}) + o(\phi^h)},$$

since $(\mu - \hat{\mu}_T)$ and $\bar{v}_T$ are uncorrelated with v_{T+h}. But $E(\mu - \hat{\mu}_T)^2 = O(T^{-1})$ and does not depend on h. To derive $E(\bar{v}^2_T)$, recall that $v_t = \sum_{j=0}^{h-1} \phi^j \varepsilon_{t-j}$, and hence after some algebra

$$T\bar{v}_T = \sum_{j=1}^{h-1} \left(\frac{1-\phi^j}{1-\phi} \right) \phi^{h-j} \varepsilon_{-h+j+1} + \left(\frac{1-\phi^h}{1-\phi} \right) \sum_{t=1}^{T-h+1} \varepsilon_t + \sum_{j=1}^{h-1} \left(\frac{1-\phi^j}{1-\phi} \right) \varepsilon_{T-j+1},$$

[2] Recall that the *OLS* estimator of $|\phi|$ is biased downward.

and

$$E(\bar{v}_T^2) = \text{Var}(\bar{v}_T) = \frac{\sigma_\varepsilon^2}{T^2}\left\{\sum_{j=1}^{h-1}\left(\frac{1-\phi^j}{1-\phi}\right)^2\left[1+\phi^{2(h-j)}\right] + \left(\frac{1-\phi^h}{1-\phi}\right)^2 (T-h+1)\right\}.$$

Clearly, $E(\bar{v}_T^2) = O(T^{-1})$ if h is fixed. But it is easily seen that we continue to have $E(\bar{v}_T^2) = O(T^{-1})$ even if $h \to \infty$ so long as $h/T \to \kappa$, where κ is a fixed finite fraction in the range $[0, 1)$. Therefore,

$$\frac{E(\tilde{e}_{T+h|T}^2)}{E(\tilde{e}_{T+h|T}^2)} = 1 + O(T^{-1}) + o(\phi^h),$$

and for sufficiently large T there will be little to choose between the iterated and the direct procedures. From the above result, we should expect to find the greatest difference between the performance of the two forecasting methods in small samples (T) or in situations where h is large, that is, when h/T is large.

Marcellino, Stock, and Watson (2006) compared the performance of iterated and direct approaches by applying simulated out-of-sample methods to 170 US macroeconomic time series spanning 1959–2002. They found that iterated forecasts outperform direct forecasts, particularly if the models can select long lag specifications. Along similar lines, Pesaran, Pick, and Timmermann (2011) conducted a broad-based comparison of iterated and direct multi-period forecasting approaches applied to both univariate and multivariate models in the form of parsimonious factor-augmented vector autoregressions. These authors also accounted for the serial correlation in the residuals of the multi-period direct forecasting models by considering SURE-based estimation methods, and proposed modified Akaike information criteria for model selection. Using the data set studied by Marcellino, Stock, and Watson (2006), Pesaran, Pick, and Timmermann (2011) further show that information in factors helps improve forecasting performance for most types of economic variables, although it can also lead to larger biases. They also show that SURE estimation and finite-sample modifications to the Akaike information criterion can improve the performance of the direct multi-period forecasts.

17.8 Combining forecasts

In many cases, multiple forecasts are available for the same variable of interest. These forecasts may differ because they are based on different information sets, or because the individual models underlying each forecast are different or are subject to misspecification bias (Bates and Granger (1969)). Forecast combination, also known as forecast pooling, refers to a procedure by which two or more individual forecasts are combined from a set of forecasts to produce a single, 'pooled' forecast (Elliott, Granger, and Timmermann (2006)). Empirical literature has shown that forecast combination often produces better forecasts than methods based on individual forecasting models. Indeed, the forecast resulting from a combination of several forecasts may be more robust against misspecification bias and measurement errors in the data sets underlying individual forecasts.

Consider the problem of forecasting at time T the future value of a target variable, y, after h periods, whose realization is denoted y_{T+h}. Suppose we have an m-dimensional vector of alternative forecasts of y_{T+h}, namely $\mathbf{y}^*_{T+h|T} = \left(y^*_{1,T+h|T}, y^*_{2,T+h|T}, \ldots, y^*_{m,T+h|T}\right)'$, where $y^*_{i,T+h|T}$ is the ith forecast of y_{T+h} formed on the basis of information available at time T. Forecast combination consists of aggregating or pooling the forecasts so that the information in the m components of $\mathbf{y}^*_{T+h|T}$ is reduced to a single combined or pooled point forecast,

$$y^C_{T+h|T} = \alpha_{1,T+h|T} y^*_{1,T+h|T} + \alpha_{2,T+h|T} y^*_{2,T+h|T} + \ldots + \alpha_{m,T+h|T} y^*_{m,T+h|T},$$

where $\alpha_{i,T+h|T}$ is the weight attached to the ith forecast, $y^*_{i,T+h|T}$. These weights are typically constrained to be positive and add up to unity, namely $\sum_{i=1}^{m} \alpha_{i,T+h|T} = 1$, with $\alpha_{i,T+h|T} > 0$. The combined forecast is optimal if the weights $\boldsymbol{\alpha}^*_{T+h|T} = \left(\alpha^*_{1,T+h|T}, \alpha^*_{2,T+h|T}, \ldots, \alpha^*_{m,T+h|T}\right)'$ solve the problem

$$\boldsymbol{\alpha}^*_{T+h|T} = \underset{\boldsymbol{\alpha}_{T+h|T}}{\operatorname{argmin}} E\left[L_q\left(y_{T+h}, y^C_{T+h}\right) \middle| \mathbf{y}^*_{T+h|T}\right],$$

assuming the quadratic loss function (17.1). Bates and Granger (1969) have shown that the optimal weights, $\boldsymbol{\alpha}^*_{T+h|T}$, depend on the (unknown) covariance matrix of all forecasts errors, namely $\left(e_{1,T+h}, e_{2,T+h}, \ldots, e_{m,T+h}\right)'$, with $e_{i,T+h} = y_{T+h} - y^*_{i,T+h|T}$. In practice, if the number of forecasts, m, is large, computing the covariance matrix of $\left(e_{1,T+h}, e_{2,T+h}, \ldots, e_{m,T+h}\right)'$ is unfeasible. Even when m is small the estimates of the weights might be unreliable due to short data samples and/or breaks in the underlying forecasting processes. In practice, other (possibly sub-optimal) weighting schemes are used. A prominent example is the equal weights average forecast

$$y^C_{T+h|T} = \frac{1}{m} \sum_{i=1}^{m} y^*_{i,T+h|T},$$

which often works well. Other combinations that are less sensitive to outliers than the simple average combinations are the median or the trimmed mean forecasts. Stock and Watson (2004) have suggested using weights that depend inversely on the historical forecasting performance of individual models. To evaluate the historical forecasting performance, the authors suggest splitting the sample into two sub-samples: the observations prior to date T_0 are used for estimating the individual forecasting models, while $T - T_0$ (with $T - T_0 \geq h$) observations are used for evaluation purposes. Hence, the weights are set to

$$\alpha_{i,T+h|T} = \frac{p_{i,T+h|T}}{\sum_{j=1}^{m} p_{j,T+h|T}}, \text{ with } p_{i,T+h|T} = \sum_{s=T_0}^{T-h} \delta^{T-h-s} \left(y_{s+h} - y^*_{i,s+h|s}\right)^2,$$

where δ is a discount factor. When $\delta = 1$, there is no discounting, while for $\delta < 1$, greater importance is attributed to the recent forecast performance of the individual models. Other possible

choices of weights involve the use of shrinking methods, which shrink the weights towards a value imposed *a priori*.

Also see Sections 17.9, and Section C.4 in Appendix C on a Bayesian approach to forecast combination.

17.9 Sources of forecast uncertainty

Forecast uncertainty reflects the variability of possible outcomes relative to the forecast being made. Clements and Hendry (1998) identify five sources of uncertainties for model-based forecasts:

- Mis-measurement of the data used for forecasting
- Misspecification of the model (or model uncertainty, including policy uncertainty)
- Future changes in the underlying structure of the economy
- The cumulation of future errors, or shocks, to the economy (or future uncertainty)
- Inaccuracies in the estimates of the parameters of a given model (or parameter uncertainty).

Measurement uncertainty and future changes in the underlying structure of the economy pose special problems of their own and will not be addressed in this chapter. We refer to Hendry and Ericsson (2003) for further discussion on these sources of forecast uncertainty. Model uncertainty concerns the 'structural' assumptions underlying a statistical model for the variable of interest. Further details on the problem of model uncertainty can be found in Draper (1995). Future uncertainty refers to the effects of unobserved future shocks on forecasts, while parameter uncertainty is concerned with the robustness of forecasts to the choice of parameter values, assuming a given forecasting model. In the following, we focus on future and parameter uncertainty and consider alternative ways that these types of uncertainty can be taken into account.

The standard textbook approach to taking account of future and parameter uncertainties is through the construction of forecast intervals. For the purpose of exposition, initially we abstract from parameter uncertainty and consider the following simple linear regression model

$$y_t = \mathbf{x}_{t-1}'\boldsymbol{\beta} + u_t, \qquad t = 1, 2, \ldots, T,$$

where $\mathbf{x}_{t-1}$ is a $k \times 1$ vector of predetermined regressors, $\boldsymbol{\beta}$ is a $k \times 1$ vector of fixed but unknown coefficients, and $u_t \sim N(0, \sigma^2)$. The optimal forecast of y_{T+1} at time T (in the mean squared error sense) is given by $\mathbf{x}_T'\boldsymbol{\beta}$. In the absence of parameter uncertainty, the calculation of a probability forecast for a specified event is closely related to the more familiar concept of forecast confidence interval. For example, suppose that we are interested in the probability that the value of y_{T+1} lies below a specified threshold, say a, conditional on $\Omega_T = (y_T, \mathbf{x}_T, y_{T-1}, \mathbf{x}_{T-1}, \ldots)$, the information available at time T. For given values of $\boldsymbol{\beta}$ and σ^2, we have

$$\Pr\left(y_{T+1} < a \mid \Omega_T\right) = \Phi\left(\frac{a - \mathbf{x}_T'\boldsymbol{\beta}}{\sigma}\right),$$

where $\Phi(\cdot)$ is the standard normal cumulative distribution function, while the $(1-\alpha)\%$ forecast interval for y_{T+1} (conditional on Ω_T) is given by $\mathbf{x}_T'\boldsymbol{\beta} \pm \sigma\,\Phi^{-1}\left(1-\frac{\alpha}{2}\right)$.

The two approaches, although related, are motivated by different considerations. The point forecast provides the threshold value $a = \mathbf{x}_T'\boldsymbol{\beta}$ for which $\Pr\left(y_{T+1} < a \mid \Omega_T\right) = 0.5$, while forecast interval provides the threshold values $c_L = \mathbf{x}_T'\boldsymbol{\beta} - \sigma\,\Phi^{-1}\left(1-\frac{\alpha}{2}\right)$, and $c_U = \mathbf{x}_T'\boldsymbol{\beta} + \sigma\,\Phi^{-1}\left(1-\frac{\alpha}{2}\right)$ for which $\Pr\left(y_{T+1} < c_L \mid \Omega_T\right) = \frac{\alpha}{2}$, and $\Pr\left(y_{T+1} < c_U \mid \Omega_T\right) = 1-\frac{\alpha}{2}$. Clearly, the threshold values, c_L and c_U, associated with the $(1-\alpha)\%$ forecast interval may or may not be of interest.[3] Only by chance will the forecast interval calculations provide information in a way which is directly useful in specific decision making contexts.

The relationship between probability forecasts and interval forecasts becomes even more obscure when parameter uncertainty is also taken into account. In the context of the above regression model, the point estimate of the forecast is given by $\hat{y}^*_{T+1|T} = \mathbf{x}_T'\hat{\boldsymbol{\beta}}_T$, where

$$\hat{\boldsymbol{\beta}}_T = \mathbf{Q}_{T-1}^{-1}\mathbf{q}_T,$$

is the *OLS* estimate of $\boldsymbol{\beta}$, with

$$\mathbf{Q}_{T-1} = \sum_{t=1}^{T}\mathbf{x}_{t-1}\mathbf{x}_{t-1}', \text{ and } \mathbf{q}_T = \sum_{t=1}^{T}\mathbf{x}_{t-1}y_t.$$

The relationship between y_{T+1} and its time T predictor can be written as

$$\begin{aligned} y_{T+1} &= \mathbf{x}_T'\boldsymbol{\beta} + u_{T+1} \\ &= \mathbf{x}_T'\hat{\boldsymbol{\beta}}_T + \mathbf{x}_T'(\boldsymbol{\beta} - \hat{\boldsymbol{\beta}}_T) + u_{T+1}, \end{aligned} \tag{17.18}$$

so that the estimated forecast error, e_{T+1}, is given by

$$\hat{e}_{T+1} = y_{T+1} - \hat{y}^*_{T+1|T} = \mathbf{x}_T'(\boldsymbol{\beta} - \hat{\boldsymbol{\beta}}_T) + u_{T+1}.$$

This example shows that the point forecasts, $\mathbf{x}_T'\hat{\boldsymbol{\beta}}_T$, are subject to two types of uncertainties, namely that relating to $\boldsymbol{\beta}$ and that relating to the distribution of u_{T+1}. For any given sample of data, Ω_T, $\hat{\boldsymbol{\beta}}_T$ is known and can be treated as fixed. On the other hand, although $\boldsymbol{\beta}$ is assumed fixed at the estimation stage, it is unknown to the forecaster and, from this perspective, it is best viewed as a random variable at the forecasting stage. Hence, in order to compute probability forecasts which account for future as well as parameter uncertainties, we need to specify the joint probability distribution of $\boldsymbol{\beta}$ and u_{T+1}, conditional on Ω_T. As far as u_{T+1} is concerned, we continue to assume that

$$u_{T+1}|\Omega_T \sim N(0, \sigma^2),$$

[3] The association between probability forecasts and interval forecasts is even weaker when one considers *joint* events. For example, it would be impossible to infer the probability of the joint event of a positive output growth and an inflation rate falling within a pre-specified range from individual, variable-specific forecast intervals. Many different such intervals will be needed for this purpose.

and to keep the exposition simple, for the time being we shall assume that σ^2 is known and that u_{T+1} is distributed independently of $\boldsymbol{\beta}$. For $\boldsymbol{\beta}$, noting that

$$\left(\hat{\boldsymbol{\beta}}_T - \boldsymbol{\beta}\right) | \Omega_T \sim N\left(\mathbf{0}, \sigma^2 \mathbf{Q}_{T-1}^{-1}\right), \tag{17.19}$$

we assume that

$$\boldsymbol{\beta} \, | \Omega_T \sim N\left(\hat{\boldsymbol{\beta}}_T, \sigma^2 \mathbf{Q}_{T-1}^{-1}\right), \tag{17.20}$$

which is akin to a Bayesian approach with non-informative priors for $\boldsymbol{\beta}$. Hence

$$\hat{e}_{T+1} \mid \Omega_T \sim N\left[0, \sigma^2\left(1 + \mathbf{x}_T' \mathbf{Q}_{T-1}^{-1} \mathbf{x}_T\right)\right].$$

The $(1-\alpha)$% forecast interval in this case is given by

$$c_{LT} = \mathbf{x}_T' \hat{\boldsymbol{\beta}}_T - \sigma \left\{1 + \mathbf{x}_T' \mathbf{Q}_{T-1}^{-1} \mathbf{x}_T\right\}^{1/2} \Phi^{-1}\left(1 - \frac{\alpha}{2}\right), \tag{17.21}$$

and

$$c_{UT} = \mathbf{x}_T' \hat{\boldsymbol{\beta}}_T + \sigma \left\{1 + \mathbf{x}_T' \mathbf{Q}_{T-1}^{-1} \mathbf{x}_T\right\}^{1/2} \Phi^{-1}\left(1 - \frac{\alpha}{2}\right). \tag{17.22}$$

When σ^2 is unknown, under the standard non-informative Bayesian priors on (β, σ^2), the appropriate forecast interval can be obtained by replacing σ^2 by its unbiased estimate,

$$\hat{\sigma}_T^2 = (T-k)^{-1} \sum_{t=1}^{T} (y_t - \mathbf{x}_{t-1}' \hat{\boldsymbol{\beta}}_T)'(y_t - \mathbf{x}_{t-1}' \hat{\boldsymbol{\beta}}_T),$$

and $\Phi^{-1}\left(1 - \frac{\alpha}{2}\right)$ by the $(1-\frac{\alpha}{2})$% critical value of the standard t-distribution with $T-k$ degrees of freedom. Although such interval forecasts have been discussed in the econometrics literature, the particular assumptions that underlie them are not fully recognized.

Using this interpretation, the effect of parameter uncertainty on forecasts can also be obtained *via* stochastic simulations, by generating alternative forecasts of y_{T+1} for different values of $\boldsymbol{\beta}$ (and σ^2) drawn from the conditional probability distribution of $\boldsymbol{\beta}$ given by (17.20). Alternatively, one could estimate probability forecasts by focusing directly on the probability distribution of y_{T+1} for a given value of $\mathbf{x}_T$, simultaneously taking into account both parameter and future uncertainties. For example, in the simple case where σ^2 is known, this can be achieved by simulating $\hat{y}_{T+1|T}^{*(j)}, j = 1, 2, \ldots, J$, where

$$y_{T+1|T}^{*(j)} = \mathbf{x}_T' \hat{\boldsymbol{\beta}}^{(j)} + u_{T+1}^{(j)},$$

$\hat{\boldsymbol{\beta}}^{(j)}$ is the jth random draw from $N\left(\hat{\boldsymbol{\beta}}_T, \sigma^2 \mathbf{Q}_{T-1}^{-1}\right)$, and $u_{T+1}^{(j)}$ is the jth random draw from $N\left(0, \sigma^2\right)$, with σ^2 replaced by its unbiased estimator, $\hat{\sigma}_T^2$, defined above.

17.10 A decision-based forecast evaluation framework

As Whittle (1979, p. 177) notes, 'Prediction is not an end in itself, but only a means of optimizing current actions against the prospect of an uncertain future.' To make better decisions we need better forecasts, and to evaluate forecasts we need to know how and by whom forecasts are used. From a user's perspective, the criteria for forecast evaluation must depend on the decision environment.

Consider a single period decision problem faced by an individual with a globally convex cost function, $C(y_t, x_{t+1})$, where x_{t+1} is a state variable of relevance to the decision with the conditional probability distribution function

$$F_t(x) = \Pr(x_{t+1} < x \mid \Omega_t), \tag{17.23}$$

y_t is the decision variable to be chosen by the decision maker, and Ω_t is the information set containing at least observations on current and past values of x_t. To simplify the analysis, we assume that the choice of y_t does not affect $F_t(x)$, although clearly changes in $F_t(x)$ will influence the decisions. In general, the cost and the probability distribution functions, $C(y_t, x_{t+1})$ and $F_t(x)$, also depend on a number of parameters characterizing the degree of risk aversion of the decision maker and his/her (subjective) specification of the future uncertainty characterized by the curvature of the conditional distribution function of x_{t+1}.

Suppose now that, at time t, a forecaster provides the decision maker with the predictive distribution $\hat{F}_t$, being an estimate of $F_t(x)$, and we are interested in computing the value of this forecast to the decision maker. Under the traditional approach, the forecasts $\hat{F}_t$ are evaluated using statistical criteria which are based on the degree of closeness of $\hat{F}_t$ to $F_t(x)$ at different realizations of x. This could involve the first- or higher-order conditional moments of x_{t+1}, the probability that x_{t+1} falls in a particular range, or other event forecasts of interest. However, such evaluation criteria need not be directly relevant to the decision maker. A more appropriate criterion would be the loss function that underlies the decision problem. As we shall see, such decision-based evaluation criteria simplify to the familiar *MSFE* criterion only in special cases.

Under the decision-based approach, we first need to solve for the decision variable y_t based on the predictive distribution function $\hat{F}_t$. For the above simple decision problem, the optimal value of y_t, which we denote by y_t^*, is given by

$$y_t^* = \underset{y_t}{\operatorname{argmin}} \left\{ E_{\hat{F}} \left[C(y_t, x_{t+1}) \mid \Omega_t \right] \right\}, \tag{17.24}$$

where $E_{\hat{F}} \left[C(y_t, x_{t+1}) \mid \Omega_t \right]$ is the conditional expectations operator with respect to the predictive distribution function, $\hat{F}_t$. A 'population average' criterion function for the evaluation of the probability distribution function, $\hat{F}_t$, is given by

$$\Psi_C \left(F_t, \hat{F}_t \right) = E_F \left[C(y_t^*, x_{t+1}) \mid \Omega_t \right], \tag{17.25}$$

where the conditional expectations are taken with respect to $F_t(x)$, the 'true' probability distribution function of x_{t+1} conditional on Ω_t. The above function can also be viewed as the average

cost of making errors when large samples of forecasts and realizations are available, for the same specifications of cost and predictive distribution function.

To simplify notation, we drop the subscript F when the expectations are taken with respect to the true distribution functions. We now turn to some decision problems of particular interest.

17.10.1 Quadratic cost functions and the *MSFE* criteria

In general, the forecast evaluation criterion given by (17.25) depends on the parameters of the underlying cost (loss) function as well as on the difference between $F_t(x)$ and its estimate $\hat{F}_t$. An exception arises when the cost function is quadratic and the constraints (if any) are linear; the so called LQ (linear-quadratic) decision problem. To see this, consider the following quadratic specification for the cost function

$$C(y_t, x_{t+1}) = ay_t^2 + 2by_t x_{t+1} + cx_{t+1}^2, \tag{17.26}$$

where $a > 0$ and $ca - b^2 > 0$, thus ensuring that $C(y_t, x_{t+1})$ is globally convex in y_t and x_{t+1}. Based on the forecasts, $\hat{F}_t$, the unique optimal decision rule for this problem is given by

$$\begin{aligned} y_t^* &= \left(\frac{-b}{a}\right) E_{\hat{F}}(x_{t+1} \mid \Omega_t) \\ &= \left(\frac{-b}{a}\right) \hat{x}_{t+1|t}, \end{aligned}$$

where $\hat{x}_{t+1|t}$ is the one-step forecast of x formed at time t based on the estimate, $\hat{F}_t$. Substituting this result in the utility function, after some simple algebra we have

$$C(y_t^*, x_{t+1}) = \left(c - \frac{b^2}{a}\right) x_{t+1}^2 + \frac{b^2}{a}\left(x_{t+1} - \hat{x}_{t+1|t}\right)^2.$$

Therefore,

$$\Psi_C(F_t, \hat{F}_t) = \left(c - \frac{b^2}{a}\right) E\left(x_{t+1}^2 \mid \Omega_t\right) + \frac{b^2}{a} E\left[\left(x_{t+1} - \hat{x}_{t+1|t}\right)^2 \mid \Omega_t\right].$$

In this case, the evaluation criterion implicit in the decision problem is proportional to the familiar statistical measure $E\left[\left(x_{t+1} - \hat{x}_{t+1|t}\right)^2 \mid \Omega_t\right]$, namely the *MSFE* criterion, which does not depend on the parameters of the underlying cost function.

This is a special result, and does not carry over to the multivariate context even under the LQ set up. To see this, consider the following multivariate version of (17.26)

$$C(\mathbf{y}_t, \mathbf{x}_{t+1}) = \mathbf{y}_t' \mathbf{A} \mathbf{y}_t + 2\mathbf{y}_t' \mathbf{B} \mathbf{x}_{t+1} + \mathbf{x}_{t+1}' \mathbf{C} \mathbf{x}_{t+1}, \tag{17.27}$$

where $\mathbf{y}_t$ is an $m \times 1$ vector of decision variables, $\mathbf{x}_{t+1}$ is a $k \times 1$ vector of state variables, and $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{C}$ are $m \times m$, $m \times k$ and $k \times k$ coefficient matrices. To ensure that $C(\mathbf{y}_t, \mathbf{x}_{t+1})$ is globally convex in its arguments we also assume that the $(m + k) \times (m + k)$ matrix

$$\begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B}' & \mathbf{C} \end{pmatrix},$$

is positive definite and symmetric. As before, due to the quadratic nature of the cost function, the optimal decision depends only on the first conditional moment of the assumed conditional probability distribution function of the state variables and is given by

$$\mathbf{y}_t^* = -\mathbf{A}^{-1}\mathbf{B}\hat{\mathbf{x}}_{t+1|t}, \tag{17.28}$$

where $\hat{\mathbf{x}}_{t+1|t}$ is the point forecast of $\mathbf{x}_{t+1}$ formed at time t, with respect to the conditional probability distribution function, $\hat{F}_t$. Substituting this result in (17.27) and taking conditional expectations with respect to the $\mathbf{F}_t(\mathbf{x})$, the true conditional probability distribution function of $\mathbf{x}_{t+1}$, we have

$$\Psi_C\left(F_t, \hat{F}_t\right) = E[\mathbf{x}_{t+1}'(\mathbf{C}-\mathbf{H})\mathbf{x}_{t+1} \mid \Omega_t] + E\left[(\mathbf{x}_{t+1} - \hat{\mathbf{x}}_{t+1|t})'\mathbf{H}(\mathbf{x}_{t+1} - \hat{\mathbf{x}}_{t+1|t}) \mid \Omega_t\right],$$

where $\mathbf{H} = \mathbf{B}'\mathbf{A}^{-1}\mathbf{B}$**.** Therefore, the implied forecast evaluation criterion is given by

$$E\left[(\mathbf{x}_{t+1} - \hat{\mathbf{x}}_{t+1|t})'\mathbf{H}(\mathbf{x}_{t+1} - \hat{\mathbf{x}}_{t+1|t}) \mid \Omega_t\right],$$

which, through $\mathbf{H}$, depends on the parameters of the underlying cost function. Only in the univariate LQ case can the implied evaluation criterion be cast in terms of a purely statistical criterion function.

The dependence of the evaluation criterion in the multivariate case on the parameters of the cost (or utility) function of the underlying decision model has direct bearing on the non-invariance critique of *MSFE*s to scale-preserving linear transformations discussed by Clements and Hendry (1993). In multivariate forecasting problems, the choice of the evaluation criterion is not as clear cut as in the univariate case even if we confine our attention to *MSFE* type criteria. One possible procedure, commonly adopted, is to use $E\left[(\mathbf{x}_{t+1} - \hat{\mathbf{x}}_{t+1|t})'(\mathbf{x}_{t+1} - \hat{\mathbf{x}}_{t+1|t}) \mid \Omega_t\right]$, or equivalently the trace of the *MSFE* matrix $E\left[(\mathbf{x}_{t+1} - \hat{\mathbf{x}}_{t+1|t})'(\mathbf{x}_{t+1} - \hat{\mathbf{x}}_{t+1|t}) \mid \Omega_t\right]$. Alternatively, the determinant of the *MSFE* matrix has also been suggested. In the context of the LQ decision problem, both of these purely statistical criteria are inappropriate. The trace *MSFE* criterion is justified only when $\mathbf{H}$ is proportional to an identity matrix of order $m + k$.

17.10.2 Negative exponential utility: a finance application

The link between purely statistical and decision-based forecast evaluation criteria becomes even more tenuous for non-quadratic cost or utility functions. One important example is the negative exponential utility function often used in finance for the determination of optimal portfolio weights in asset allocation problems. Consider a risk-averse speculator with a negative exponential utility function who wishes to decide on his/her long ($y_t > 0$), and short positions ($y_t < 0$), in a given security.[4] To simplify the exposition we abstract from transaction

[4] Edison and Cho (1993) consider a utility-based procedure for comparisons of exchange rate volatility models. Skouras (1998) discusses asset allocation decisions and forecasts of a 'risk neutral' investor.

costs. At the end of the period (the start of period $t+1$) the speculator's net worth will be given by

$$W_{t+1} = y_t \rho_{t+1},$$

where ρ_{t+1} is the rate of return on the security. The speculator chooses y_t in order to maximize the expected value of the negative exponential utility function

$$U(y_t, \rho_{t+1}) = -\exp\left(-\lambda y_t \rho_{t+1}\right), \quad \lambda > 0, \tag{17.29}$$

with respect to the publicly available information, Ω_t.

Now suppose that the speculator is told that conditional on Ω_t, excess returns can be forecast using[5]

$$\rho_{t+1} \mid \Omega_t \backsim N(\hat{\rho}_{t+1|t}, \hat{\sigma}^2_{t+1|t}). \tag{17.30}$$

What is the economic value of this forecast to the speculator?

Under (17.30), we have[6]

$$E_{\hat{F}}\left[U(y_t, \rho_{t+1}) \mid \Omega_t\right] = -\exp\left(-\lambda y_t \hat{\rho}_{t+1|t} + \frac{1}{2}\lambda^2 y_t^2 \hat{\sigma}^2_{t+1|t}\right),$$

and

$$\frac{\partial E_{\hat{F}}\left[U(y_t, \rho_{t+1}) \mid \Omega_t\right]}{\partial y_t} = -\left(-\lambda \hat{\rho}_{t+1|t} + \lambda^2 y_t \hat{\sigma}^2_{t+1|t}\right) \exp\left(-\lambda y_t \hat{\rho}_{t+1|t} + \frac{1}{2}\lambda^2 y_t^2 \hat{\sigma}^2_{t+1|t}\right).$$

Setting this derivative equal to zero, we now have the following familiar result for the speculator's optimal decision

$$y_t^* = \frac{\hat{\rho}_{t+1|t}}{\lambda \hat{\sigma}^2_{t+1|t}}. \tag{17.31}$$

Hence

$$U(y_t^*, \rho_{t+1}) = -\exp\left(-\frac{\rho_{t+1}\hat{\rho}_{t+1|t}}{\hat{\sigma}^2_{t+1|t}}\right), \tag{17.32}$$

[5] We assume that y_t is small relative to the size of the market and the choice of y_t does not influence the returns distribution.

[6] In general, where the conditional distribution of returns are not normally distributed we have

$$E_{\hat{F}}\left[U(y_t, \rho_{t+1}) \mid \Omega_t\right] = -M_{\hat{F}}(-\lambda y_t),$$

where $M_{\hat{F}}(\theta)$ is the moment generating function of the assumed conditional distribution of returns. In this more general case, the optimal solution is y_t^* that solves $\partial M_{\hat{F}}(-\lambda y_t)/\partial y_t = 0$.

and the expected economic value of the forecasts in (17.30) is given by

$$\Psi_U\left(F_t, \hat{F}_t\right) = E_F\left[-\exp\left(-\frac{\rho_{t+1}\hat{\rho}_{t+1|t}}{\hat{\sigma}^2_{t+1|t}}\right) \mid \Omega_t\right], \tag{17.33}$$

where expectations are taken with respect to the true distribution of returns, $F_t(\rho)$.[7] This result has three notable features. The decision-based forecast evaluation measure does not depend on the risk-aversion coefficient, λ. It has little bearing on the familiar purely statistical forecast evaluation criteria such as the MSFEs of the mean return, given by $E_F\left[(\rho_{t+1} - \hat{\rho}_{t+1|t})^2 \mid \Omega_t\right]$. Finally, even under Gaussian assumptions the evaluation criterion involves return predictions, $\hat{\rho}_{t+1|t}$, as well as volatility predictions, $\hat{\sigma}_{t+1|t}$.

It is also interesting to note that under the assumption that (17.30) is based on a correctly specified model we have

$$\Psi_U\left(F_t, \hat{F}_t\right) = -\exp\left[-\frac{1}{2}\left(\frac{\hat{\rho}_{t+1|t}}{\hat{\sigma}_{t+1|t}}\right)^2\right],$$

where $\hat{\rho}_{t+1|t}/\hat{\sigma}_{t+1|t}$ is a single-period Sharpe ratio routinely used in the finance literature for the economic evaluation of risky portfolios.

The average loss associated with the error in forecasting can now be computed as

$$\bar{\Psi}_U = -h^{-1}\sum_{t=T}^{T+h-1}\exp\left(-\frac{\rho_{t+1}\hat{\rho}_{t+1|t}}{\hat{\sigma}^2_{t+1|t}}\right),$$

where $t = T+1, T+2, \ldots, T+h$, is the forecast evaluation period.

17.11 Test statistics of forecast accuracy based on loss differential

One possible approach to measure forecast accuracy is to consider statistics that make a comparison between competing forecasts rather than seek absolute standards of forecast accuracy. Suppose that we have two competing forecasting models, that produce the forecasts $y^{*1}_{t+h|t}$ and $y^{*2}_{t+h|t}$, respectively. To determine which model produces better forecasts, we may test the null hypothesis

$$H_0 : E\left[L(y_{t+h}, y^{*1}_{t+h|t})\right] - E\left[L(y_{t+h}, y^{*2}_{t+h|t})\right] = 0, \tag{17.34}$$

[7] Notice that under (17.30) we have

$$\hat{F}_t(\rho) = \int_{-\infty}^{\rho}(2\pi\hat{\sigma}^2_{t,1})^{-1/2}\exp\left[\frac{-(u-\hat{\rho}_{t,1})^2}{2\hat{\sigma}^2_{t,1}}\right]du.$$

where $L(\cdot)$ is a given loss function, against the alternative

$$H_1 : E\left[L(y_{t+h}, y^{*1}_{t+h|t})\right] - E\left[L(y_{t+h}, y^{*2}_{t+h|t})\right] \neq 0. \tag{17.35}$$

Hence, under (17.34) the two forecast models are equally accurate on average, according to a given loss function. If the null hypothesis is rejected, one would choose the model yielding the lower loss. Diebold and Mariano (1995) have proposed a test that is based on the loss-differential

$$d_t = L(y_{t+h}, y^{*1}_{t+h|t}) - L(y_{t+h}, y^{*2}_{t+h|t}).$$

The null of equal predictive accuracy is then $H_0 : E(d_t) = 0$. Given a series of T forecast errors, the Diebold and Mariano (1995) test statistic is

$$DM = \frac{T^{1/2}\bar{d}}{\left(\widehat{Var}\left(\bar{d}\right)\right)^{1/2}}, \tag{17.36}$$

where

$$\bar{d} = \frac{1}{T}\sum_{t=1}^{T} d_t, \tag{17.37}$$

and $\widehat{Var}\left(\bar{d}\right)$ is an estimator of

$$Var\left(\bar{d}\right) = \sum_{j=-\infty}^{\infty} \gamma_j, \quad \text{with } \gamma_j = Cov\left(d_t, d_{t-j}\right). \tag{17.38}$$

Expression (17.38) is used for the variance of $\bar{d}$ because the sample of loss differentials, d_t, is serially correlated for $h > 1$. Under the null of equal predictive ability, and under a set of regularity conditions, it is possible to show that as $T \to \infty$, $DM \overset{a}{\sim} N(0, 1)$. Notice that this result holds for a wide class of loss functions (see McCracken and West (2004)). A number of modifications and extensions of the above test have been suggested in the literature. West (1996) has extended the *DM* test to deal with the case in which forecasts and forecast errors depend on estimated regression parameters. Harvey, Leybourne, and Newbold (1997) have proposed two modifications of the *DM* test. Since the *DM* test could be seriously oversized for moderate numbers of samples observations,[8] the authors suggests the use of the following modified statistic

$$MDM = T^{-1/2}\left[T + 1 - 2h + T^{-1}h(h-1)\right]^{1/2} DM.$$

A further modification of the *DM* test proposed by Harvey, Leybourne, and Newbold (1997) is to compare the statistic with critical values from the t-distribution with $T-1$ degrees of freedom,

[8] See also the Monte Carlo study reported in Diebold and Mariano (1995).

rather than the standard normal. Monte Carlo experiments provided by these authors show substantially better size properties for the *MDM* test when compared to the *DM* test in moderate samples (see also Harvey, Leybourne, and Newbold (1998)).

The statistic (17.36) is based on unconditional expectations of forecasts and forecast errors, and therefore can be seen as a test of unconditional out-of-sample predictive ability. More recently, Giacomini and White (2006) (GW) have focused on a test for the null hypothesis of equal conditional predictive ability, namely

$$H_0 : E\left[L(y_{t+h}, \hat{y}^{*1}_{t+h|t}) \mid \Omega_t\right] - E\left[L(y_{t+h}, \hat{y}^{*2}_{t+h|t}) \mid \Omega_t\right] = 0. \tag{17.39}$$

Notice that, in the above expression expectations are conditional on the information set Ω_t available at time t, and the losses depend on the parameter estimates at time t. One important advantage of the GW test is that it captures the effect of estimation uncertainty together with model uncertainty, and can be used to study forecasts produced by general estimation methods. These advantages come at the cost of having to specify a test function, which helps to predict the loss from a forecast.

17.12 Directional forecast evaluation criteria

Directional evaluation criteria can be used in the context of two-state decision problems discussed in Section 17.3, where the focus of the analysis is correct prediction of the direction of change in the variable under consideration (e.g., up, down). These criteria can be formulated using information from a contingency table in realizations and forecasts, like that reproduced in Table 17.1. The states are denoted by z_{t+1} taking the value of unity if the 'Up' state materializes and 0 otherwise.

Table 17.1 Contingency matrix of forecasts and realizations

		Actual outcomes (z_{t+1})	
		Up ($z_{t+1} = 1$)	Down ($z_{t+1} = 0$)
Forecasts	Up ($\hat{z}_{t+1|t} = 1$)	Hits (N_{uu})	False alarms (N_{ud})
	Down($\hat{z}_{t+1|t} = 0$)	Misses (N_{du})	Correct rejections (N_{dd})

In Table 17.1, the proportion of Ups that were correctly forecast to occur, and the proportion of Downs that were incorrectly forecast are known as the 'hit rate' and the 'false alarm rate' respectively. These can be computed as

$$HI = \frac{N_{\text{uu}}}{N_{\text{uu}} + N_{du}}, \quad F = \frac{N_{ud}}{N_{ud} + N_{dd}}. \tag{17.40}$$

One important evaluation criterion for directional forecasts is the Kuipers score (KS), originally developed for evaluation of weather forecasts. This is defined by

$$KS = HI - F. \tag{17.41}$$

For more details, see Murphy and Dann (1985) and Wilks (1995).

The Henriksson and Merton (1981) (HM) market-timing statistic is based on the conditional probabilities of making correct forecasts. Merton (1981) postulates the following conditional probabilities of taking correct actions

$$p_1(t) = \Pr(\hat{\rho}_{t+1|t} \geq 0 \mid \rho_{t+1} \geq 0),$$
$$p_2(t) = \Pr(\hat{\rho}_{t+1|t} < 0 \mid \rho_{t+1} < 0),$$

where ρ_{t+1} is the (excess) return on a given security, and $\hat{\rho}_{t+1|t}$ is its forecast (see Section 17.10.2) for further details). Assuming that $p_1(t)$ and $p_2(t)$ do not depend on the size of the excess returns, $|\rho_{t+1}|$, Merton (1981) shows that $p_1(t) + p_2(t)$ is a sufficient statistic for the evaluation of the forecasting ability. Together with Henriksson, he then develops a nonparametric statistic for testing the hypothesis

$$H_0 : \; p_1(t) + p_2(t) = 1,$$

or, equivalently,

$$H_0 : \; p_1(t) = 1 - p_2(t),$$

that a market-timing forecast ($\hat{\rho}_{t+1|t} \geq 0$ or $\hat{\rho}_{t+1|t} < 0$) has no economic value against the alternative

$$H_1 : \; p_1(t) + p_2(t) > 1,$$

that has positive economic value. As *HM* point out, their test is essentially a test of the independence between the forecasts and whether the excess return on the market portfolio is positive. In terms of the notation in the above contingency table, the sample estimate of the *HM* statistic, $p_1(t) - (1 - p_2(t))$, is *exactly* equal to the Kuipers score given by (17.41). The hit rate, *HI*, is the sample estimate of $p_1(t)$ and the false alarm rate, *F*, is the sample estimate of $1 - p_2(t)$.

17.12.1 Pesaran–Timmermann test of market timing

An alternative formulation of the *HM* test and its extension has been advanced in Pesaran and Timmermann (1992), where the market-timing test is based on

$$PT = \frac{\hat{P} - \hat{P}_*}{\left[\hat{V}(\hat{P}) - \hat{V}(\hat{P}_*)\right]^{\frac{1}{2}}}, \tag{17.42}$$

where $\hat{P}$ is the proportion of Ups that are correctly predicted, $\hat{P}_*$ is the estimate of the probability of correctly predicting the events assuming predictions and realizations are independently distributed, and $\hat{V}(\hat{P})$ and $\hat{V}(\hat{P}_*)$ are consistent estimates of the variances of $\hat{P}$ and $\hat{P}_*$, respectively. More specifically, suppose we are interested in testing whether one binary variable, $x_t = I(X_t)$ is related to another binary variable, $y_t = I(Y_t)$ using a sample of observations

(y_1, x_1), $(y_2, x_2), \ldots, (y_T, x_T)$. Let $I(A)$ be an indicator function that takes the value of unity if $A > 0$ and zero otherwise. Now we have

$$\hat{P} = T^{-1} \sum_{t=1}^{T} I(Y_t X_t), \quad \hat{P}_* = \bar{y}\bar{x} + (1 - \bar{y})(1 - \bar{x}), \tag{17.43}$$

$$\hat{V}(\hat{P}) = T^{-1} \hat{P}_* (1 - \hat{P}_*), \tag{17.44}$$

$$\hat{V}(\hat{P}_*) = T^{-1}(2\bar{y} - 1)^2 \bar{x}(1 - \bar{x}) + T^{-1}(2\bar{x} - 1)^2 \bar{y}(1 - \bar{y}) + 4T^{-2}\bar{y}\bar{x}(1 - \bar{y})(1 - \bar{x}), \tag{17.45}$$

and $\bar{y} = T^{-1} \sum_{t=1}^{T} y_t$, and $\bar{x} = T^{-1} \sum_{t=1}^{T} x_t$.[9] Under the null hypothesis that y_t and x_t are distributed independently (namely x_t has no power in predicting y_t), PT is asymptotically distributed as a standard normal, $PT \overset{a}{\sim} N(0, 1)$, see Pesaran and Timmermann (1992) for details.

17.12.2 Relationship of the *PT* statistic to the Kuipers score

Granger and Pesaran (2000b) have established the following relationship between the Kuipers score KS (defined by (17.41)) and the PT statistic

$$PT = \frac{\sqrt{h} KS}{\left[\frac{\hat{\pi}_f(1-\hat{\pi}_f)}{\hat{\pi}_a(1-\hat{\pi}_a)}\right]^{1/2}},$$

where $N = N_{\text{uu}} + N_{ud} + N_{du} + N_{dd}$ is the total number of forecasts (provided in Table 17.1), $\hat{\pi}_a = N^{-1}(N_{\text{uu}} + N_{du})$ is the estimate of the probability that the realizations are Up, and $\hat{\pi}_f = (N_{\text{uu}} + N_{ud})/N$ is the estimate of the probability that outcomes are forecast to be Up. The above results also establish the asymptotic equivalence of the HM and PT statistics.

17.12.3 A regression approach to the derivation of the *PT* test

As a step towards allowing for serial dependence in the outcomes, we next show how the PT test can be cast in a regression context. It turns out that the PT statistic can be well approximated by the t-ratio of the coefficient of $x_t = I(X_t)$ in the OLS regression of $y_t = I(Y_t)$ on x_t and an intercept

$$y_t = \alpha + \beta x_t + u_t, \tag{17.46}$$

where $E(u_t | x_t, x_{t-1}, \ldots) = 0$. We deal with the case where u_t could be serially correlated and/or heteroskedastic below.

The t-ratio of the OLS estimator of β in the above regression is given by

$$t_\beta = \frac{r\sqrt{T-2}}{\sqrt{1-r^2}}, \tag{17.47}$$

[9] The PT statistic is undefined when $\bar{y}$ or $\bar{x}$ take the extreme values of zero or unity.

where r is the simple correlation coefficient between y_t and x_t. To establish the relationship between t_β and the PT statistic, note that

$$\begin{aligned} I(Y_t X_t) &= I(Y_t)I(X_t) + [1 - I(Y_t)][1 - I(X_t)] \\ &= 2y_t x_t - y_t - x_t + 1, \end{aligned}$$

and hence

$$\hat{P} = T^{-1}\sum_{t=1}^{T} I(Y_t X_t) = 2T^{-1}\sum_{t=1}^{T} y_t x_t - \bar{y} - \bar{x} + 1.$$

Using the above results in the numerator of the (17.42) we have

$$\hat{P} - \hat{P}_* = 2\left(T^{-1}\sum_{t=1}^{T} y_t x_t - \bar{y}\bar{x}\right) = 2T^{-1}\sum_{t=1}^{T}\left(y_t - \bar{y}\right)(x_t - \bar{x}) = 2S_{yx}.$$

Also, after some algebra, it is easily seen that

$$\hat{V}(\hat{P}) - \hat{V}(\hat{P}_*) = 4T^{-1}\bar{y}(1-\bar{y})\bar{x}(1-\bar{x}) - 4T^{-2}\bar{y}(1-\bar{y})\bar{x}(1-\bar{x}).$$

Ignoring the second term which is of order T^{-2}, and noting that $x_t^2 = x_t$ and $y_t^2 = y_t$, we have

$$S_{xx} = T^{-1}\sum_{t=1}^{T}(x_t - \bar{x})^2 = \bar{x}(1-\bar{x}), S_{xy} = S_{yx} = T^{-1}\sum_{t=1}^{T}(x_t - \bar{x})\left(y_t - \bar{y}\right),$$

$$S_{yy} = T^{-1}\sum_{t=1}^{T}\left(y_t - \bar{y}\right)^2 = \bar{y}(1-\bar{y}).$$

It follows that (up to order T^{-1})

$$PT = \frac{\hat{P} - \hat{P}_*}{\left\{\hat{V}(\hat{P}) - \hat{V}(\hat{P}_*)\right\}^{\frac{1}{2}}} \approx \frac{\sqrt{T}S_{yx}}{\sqrt{S_{yy}S_{xx}}} = \sqrt{T}\, r_{yx}. \tag{17.48}$$

This in turn establishes that the student-t test of $\beta = 0$ in (17.47), and the PT test defined by (17.42), will be asymptotically equivalent. The two test statistics are also likely to be numerically very close in most applications.

17.12.4 A generalized *PT* test for serially dependent outcomes

One shortcoming of directional evaluation criteria described above is the assumption that under the null hypothesis the outcome (and the underlying data generating process) are serially independent. This assumption is clearly restrictive and unlikely to hold for many economic and financial time series. This could be due to the construction of the data—a single isolated quarter

with negative GDP growth is usually not viewed as a recession, nor is the emergence of a short period with negative stock returns sufficient to constitute a bear market—or may reflect the serial dependence properties of the underlying data generating process. For example, the presence of regimes whose dynamics are determined by a Markov process as in Hamilton (1989) might give rise to persistence in output growth. Serial correlation in such variables is likely to generate serial dependence in the qualitative outcomes and could cause distortions in the size of the PT test, typically in the form of over-rejection of the null hypothesis.

In the context of the regression based test (17.47), the serial dependence in outcomes under the null hypothesis translates into serial dependence in the errors, u_t. Due to the discrete nature of the $y_t = I(Y_t)$ series, the pattern of serial dependence in y_t could differ from that of Y_t and additionally y_t could be conditionally heteroskedastic even if Y_t is not and *vice versa*.

In testing $\beta = 0$ in (17.46), serial dependence in the errors, u_t, can be dealt with either parametrically or by using Bartlett weights recommended by Newey and West (1987) in the construction of the test statistic. Consider the t-ratio

$$\tilde{t}_\beta = \frac{\hat{\beta}}{\sqrt{\hat{V}_{NW}\left(\hat{\beta}\right)}}, \tag{17.49}$$

where $\hat{\beta}$ is the *OLS* estimator of β, and $\hat{V}_{NW}\left(\hat{\beta}\right)$ is the $(2,2)$ element of the Newey and West variance estimator (see Section 5.9)

$$\hat{\mathbf{V}}_{NW}(\hat{\boldsymbol{\phi}}) = \frac{1}{(T-2)\,\bar{z}^2\,(1-\bar{z})^2} \begin{pmatrix} \bar{z} & -\bar{z} \\ -\bar{z} & 1 \end{pmatrix} \hat{\mathbf{F}}_h \begin{pmatrix} \bar{z} & -\bar{z} \\ -\bar{z} & 1 \end{pmatrix}, \tag{17.50}$$

$\hat{\boldsymbol{\phi}} = (\hat{\alpha}, \hat{\beta})'$, h is the length of the lag window,

$$\hat{\mathbf{F}}_h = \hat{\boldsymbol{\Omega}}_0 + \sum_{j=1}^{h} \left(1 - \frac{j}{h+1}\right) (\hat{\boldsymbol{\Omega}}_j + \hat{\boldsymbol{\Omega}}_j'),$$

and

$$\hat{\boldsymbol{\Omega}}_j = T^{-1} \sum_{t=j+1}^{T} \hat{u}_t \hat{u}_{t-j} \begin{pmatrix} 1 & z_{t-j} \\ z_t & z_t z_{t-j} \end{pmatrix}.$$

Clearly, other estimates of the variance of $\hat{\beta}$, based on different estimates of the spectral density of $\hat{u}_t = y_t - \hat{\alpha} - \hat{\beta} z_t$ at zero frequency, could be used.

17.13 Tests of predictability for multi-category variables

The above approach can be extended to the case of multi-category variables, possibly serially correlated. A number of applications require generalizing the setup to allow for an arbitrary (but countably finite) number of categories. For example, in the evaluation of predictive performance

when interest lies in testing whether one sequence of discrete random variables ('outcomes', $\{\mathbf{y}_t\}$) is predicted by another sequence of discrete random variables ('forecasts', $\{\mathbf{x}_t\}$). For example, prediction of the direction of change in the variable under consideration may have multiple categories such as 'down', 'unchanged' and 'up'.[10]

Suppose a time series of T observations on some explanatory or predictive variable, x, is arranged into m_x categories (states) while observations on the dependent or realized variable, y, are categorized into m_y groups. Without loss of generality we assume that $m_x \leq m_y$ and that these are finite numbers that remain fixed as $T \to \infty$. Denote the x-categories by x_{jt} so that $x_{jt} = 1$ if the jth category occurs at time t and zero otherwise. Similarly, denote the realized outcomes by y_{it} so $y_{it} = 1$ if category i occurs at time t and zero otherwise. Convert the categorical observations into quantitative measures by assigning the (time-invariant) weights a_i to y_{it} for $i = 1, 2, \ldots, m_y$ and b_j to x_{jt} for $j = 1, 2, \ldots, m_x$ and $t = 1, 2, \ldots, T$ as follows

$$y_t = \sum_{i=1}^{m_y} a_i y_{it}, \text{ and } x_t = \sum_{j=1}^{m_x} b_j x_{jt}.$$

Since the outcome categories are mutually exclusive, the regression of y_t on an intercept and x_t can be written as

$$a_{m_y} + \sum_{i=1}^{m_y-1} \left(a_i - a_{m_y}\right) y_{it} = \alpha + \beta b_{m_x} + \beta \left[\sum_{j=1}^{m_x-1} \left(b_j - b_{m_x}\right) x_{jt}\right] + u_t,$$

or, more compactly,

$$\boldsymbol{\theta}' \mathbf{y}_t = c + \boldsymbol{\gamma}' \mathbf{x}_t + u_t, \tag{17.51}$$

where $\mathbf{y}_t = (\mathbf{y}_{1t}, \mathbf{y}_{2t}, \ldots, \mathbf{y}_{m_y-1,t})'$, $\mathbf{x}_t = (\mathbf{x}_{1t}, \mathbf{x}_{2t}, \ldots, \mathbf{x}_{m_x-1,t})'$, $c = \alpha + \beta b_{m_x} - a_{m_y}$ and

$$\boldsymbol{\theta} = \begin{pmatrix} a_1 - a_{m_y} \\ a_2 - a_{m_y} \\ \vdots \\ a_{m_y-1} - a_{m_y} \end{pmatrix}, \boldsymbol{\gamma} = \begin{pmatrix} \beta\left(b_1 - b_{m_x}\right) \\ \beta\left(b_2 - b_{m_x}\right) \\ \vdots \\ \beta\left(b_{m_x-1} - b_{m_x}\right) \end{pmatrix}.$$

Suppose first that u_t is serially uncorrelated. A test of predictability can now be carried out by testing $H_0 : \boldsymbol{\gamma} = \mathbf{0}$ in (17.51), conditional on a given value of the 'nuisance' parameters, $\boldsymbol{\theta}$. (See Section 17.12.4 for the special case where $m_y = m_x = 2$). For a given value of $\boldsymbol{\theta}$, a standard F-statistic can be employed to test independence of y_t and x_t

$$F(\boldsymbol{\theta}) = \left(\frac{T - m_x}{m_x - 1}\right) \frac{\boldsymbol{\theta}' \mathbf{S}_{yx} \mathbf{S}_{xx}^{-1} \mathbf{S}_{xy} \boldsymbol{\theta}}{\boldsymbol{\theta}' \left(\mathbf{S}_{yy} - \mathbf{S}_{yx} \mathbf{S}_{xx}^{-1} \mathbf{S}_{xy}\right) \boldsymbol{\theta}}, \tag{17.52}$$

[10] Another example arises in the analysis of contagion where positive as well as negative discrete jumps in market returns or spreads could be of interest (see, e.g., Favero and Giavazzi (2002) and Pesaran and Pick (2007)).

where

$$\mathbf{S}_{yx} = \mathbf{S}'_{xy} = T^{-1}\mathbf{Y}'\mathbf{M}_\tau\mathbf{X}, \mathbf{S}_{yy} = T^{-1}\mathbf{Y}'\mathbf{M}_\tau\mathbf{Y}, \mathbf{S}_{xx} = T^{-1}\mathbf{X}'\mathbf{M}_\tau\mathbf{X}.$$

$\mathbf{Y} = (\mathbf{y}_1, \mathbf{y}_2, \ldots, \mathbf{y}_T)'$ and $\mathbf{X} = (\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_T)'$, are the $T \times (m_y - 1)$ and $T \times (m_x - 1)$ observation matrices on the qualitative indicators, respectively, and $\mathbf{M}_\tau = \mathbf{I}_T - \boldsymbol{\tau}(\boldsymbol{\tau}'\boldsymbol{\tau})^{-1}\boldsymbol{\tau}'$, where $\boldsymbol{\tau} = (1, 1, \ldots, 1)'$. Since it is not known *a priori* which element of $\boldsymbol{\theta}$ might be non-zero, we employ the normalizing restriction $\boldsymbol{\theta}'\mathbf{S}_{yy}\boldsymbol{\theta} = 1$. This requires that at least one element of $\boldsymbol{\theta}$ is non-zero. A general approach to dealing with the dependence of $F(\boldsymbol{\theta})$ on the nuisance parameters is to base the test on

$$F_{\max} = \underset{\theta}{\operatorname{argmax}} \left[F(\boldsymbol{\theta})\right],$$

subject to the normalizing restriction that $\boldsymbol{\theta}'\mathbf{S}_{yy}\boldsymbol{\theta} = 1$. This idea has been used in the statistical literature in cases where certain parameters of the statistical model disappear under the null hypothesis (e.g., see Davies (1977)). However, we note that in this specific application of Davies's main idea the nuisance parameters, $\boldsymbol{\theta}$, do not disappear under the null. Using (17.52), the first-order condition for optimization of $F(\boldsymbol{\theta})$ is given by

$$\left(\mathbf{S}_{yx}\mathbf{S}_{xx}^{-1}\mathbf{S}_{xy}\right)\hat{\boldsymbol{\theta}} = \hat{\rho}^2\mathbf{S}_{yy}\hat{\boldsymbol{\theta}}, \tag{17.53}$$

where

$$\hat{\rho}^2 = \frac{F(\hat{\boldsymbol{\theta}})\left(\frac{m_x-1}{T-m_x}\right)}{1 + \left(\frac{m_x-1}{T-m_x}\right)F(\hat{\boldsymbol{\theta}})}. \tag{17.54}$$

The value of $\boldsymbol{\theta}$ that maximizes $F(\boldsymbol{\theta})$ is therefore given by the eigenvector associated with the maximum eigenvalue of

$$\mathbf{S} = \mathbf{S}_{yy}^{-1}\mathbf{S}_{yx}\mathbf{S}_{xx}^{-1}\mathbf{S}_{xy}. \tag{17.55}$$

Denoting the non-zero eigenvalues of $\mathbf{S}$ in descending order by $\hat{\rho}_1^2 \geq \hat{\rho}_2^2 \geq \ldots \geq \hat{\rho}_{m_x-1}^2$, we have (using (17.54))

$$F_{\max} = \frac{(T - m_x)\hat{\rho}_1^2}{(m_x - 1)\left(1 - \hat{\rho}_1^2\right)}, \tag{17.56}$$

which is a generalization of (17.48) and reduces to $t_{\hat{\beta}}^2$ in the case of $m_x = 2$. Note that $\hat{\rho}_i^2$, for $i = 1, 2, \ldots, m_x - 1$ are the squared canonical correlation coefficients between the indicators, $\mathbf{x}_t$, and the realizations, $\mathbf{y}_t$ (see Section 19.6 for a definition). There are $m_x - 1$ such canonical correlations, given by the square roots of the ordered non-zero solutions of the determinantal equation (recall that $m_x \leq m_y$)

$$\left|\mathbf{S}_{yx}\mathbf{S}_{xx}^{-1}\mathbf{S}_{xy} - \rho^2\mathbf{S}_{yy}\right| = 0.$$

These are the same as the $m_x - 1$ non-zero eigenvalues of the matrix $\mathbf{S}$ defined by (17.55). The estimator of $\boldsymbol{\theta}$, denoted by $\hat{\boldsymbol{\theta}}_1$, is given by the eigenvector associated with $\hat{\rho}_1^2$, which satisfies

$$\left(\mathbf{S}_{yx}\mathbf{S}_{xx}^{-1}\mathbf{S}_{xy} - \hat{\rho}_1^2\mathbf{S}_{yy}\right)\hat{\boldsymbol{\theta}}_1 = \mathbf{0}. \tag{17.57}$$

Since $\hat{\rho}_1^2 < 1$ and $F_{\max}$ is a monotonic function of $\hat{\rho}_1^2$, a test of $\boldsymbol{\gamma} = \mathbf{0}$ in (17.51) is thus reduced to testing the statistical significance of the largest canonical correlation between $\mathbf{y}_t$ and $\mathbf{x}_t$. The exact joint probability distribution of the canonical correlations, $1 > \hat{\rho}_1^2 > \hat{\rho}_2^2 > \ldots > \hat{\rho}_{m_x-1}^2$, is provided in Anderson (2003) (see pp. 543–545) for the case where the distribution of $\mathbf{y}_t$ conditional on $\mathbf{x}_t$ is Gaussian. In the present application where the elements of $\mathbf{y}_t$ (conditional on $\mathbf{x}_t$) can be viewed as independent draws from a multinominal distribution, the *exact* distribution of the canonical correlations will be less tractable but can be simulated. Critical values of the test statistic (17.56) are provided in Pesaran and Timmermann (2009).

The null of independence between $\mathbf{x}$ and $\mathbf{y}$ implies not only that $\rho_1 = 0$ but that $\rho_1 = \rho_2 = \ldots = \rho_{m_x-1} = 0$. An alternative to the maximum canonical correlation test is therefore to base a test of $\boldsymbol{\gamma} = \mathbf{0}$ on an average concept of $F(\boldsymbol{\theta})$ given by

$$\bar{F} = \frac{(T - m_x)}{m_x - 1}\sum_{i=1}^{m_x-1}\frac{\hat{\rho}_i^2}{1 - \hat{\rho}_i^2} \approx \frac{(T - m_x)}{m_x - 1}Tr\left(\mathbf{S}\right).$$

This test can also be derived in the context of the reduced rank regression

$$\mathbf{y}_t = \mathbf{a} + \boldsymbol{\Pi}\mathbf{x}_t + \boldsymbol{\varepsilon}_t, \tag{17.58}$$

where in our application the null hypothesis of interest is $\boldsymbol{\Pi} = 0$, or $rank\left(\boldsymbol{\Pi}\right) = 0$.[11] Under some regularity conditions and assuming that under the null hypothesis the outcomes or $\boldsymbol{\varepsilon}_t$ are serially independent an asymptotic test of $\boldsymbol{\Pi} = \mathbf{0}$ is given by

$$(T - m_x)\sum_{i=1}^{m_x-1}\hat{\rho}_i^2 \overset{a}{\sim} \chi^2_{(m_x-1)^2}.$$

$\sum_{i=1}^{m_x-1}\hat{\rho}_i^2$ can also be computed by $Tr(\mathbf{S})$ and for this reason is often called the *trace test*.

It is possible to show that the trace test based on the reduced rank regression (17.58) is identical to the Fisher chi-square test of independence for data arranged in a contingency table. The values or 'labels' assigned to the categories for the X and Y variables may have a specific meaning in some applications but are often arbitrary—think of the convention of labelling recessions as unity and expansions as zeros. It would be unfortunate if such labels had an effect on the outcome of the proposed tests. However, Pesaran and Timmermann (2009) showed that both maximum canonical correlation and trace canonical correlation tests are invariant to the values taken by the m_y categories of Y and the values taken by the m_x categories of X.

[11] For an account of the reduced rank regression technique, see Section 19.7 and references cited therein.

17.13.1 The case of serial dependence in outcomes

We now turn to the general case where the outcome variable is serially correlated and heteroskedastic. To allow for possible serial dependencies in the outcomes we consider the regression model (17.51) and assume that the errors, u_t, follow a stationary first-order autoregressive process

$$u_t = \varphi u_{t-1} + \varepsilon_t,\ |\varphi| < 1, \tag{17.59}$$

where ε_t are serially independent. For this error specification, using (17.51) we have

$$\boldsymbol{\theta}'\mathbf{y}_t = c\,(1-\varphi) + \boldsymbol{\gamma}'\mathbf{x}_t - \varphi\boldsymbol{\gamma}'\mathbf{x}_{t-1} + \varphi\boldsymbol{\theta}'\mathbf{y}_{t-1} + \varepsilon_t.$$

As before, a consistent test of $\boldsymbol{\gamma} = \mathbf{0}$ can now be carried out using the maximum or the average of the canonical correlation coefficients of $\mathbf{Y}$ and $\mathbf{X}$ after filtering both sets of variables for the effects of $\mathbf{y}_{t-1}$ and $\mathbf{x}_{t-1}$. More specifically, we compute the eigenvalues of

$$\mathbf{S}_w = \mathbf{S}_{yy,w}^{-1}\mathbf{S}_{yx,w}\mathbf{S}_{xx,w}^{-1}\mathbf{S}_{xy,w},$$

where

$$\mathbf{S}_{yy,w} = T^{-1}\mathbf{Y}'\mathbf{M}_w\mathbf{Y},\ \mathbf{S}_{xx,w} = T^{-1}\mathbf{X}'\mathbf{M}_w\mathbf{X},\ \text{and}\ \mathbf{S}_{xy,w} = T^{-1}\mathbf{X}'\mathbf{M}_w\mathbf{Y},$$
$$\mathbf{M}_w = \mathbf{I}_T - \mathbf{W}\left(\mathbf{W}'\mathbf{W}\right)^{-1}\mathbf{W}',\ \mathbf{W} = (\boldsymbol{\tau}, \mathbf{X}_{-1}, \mathbf{Y}_{-1}),$$

$\mathbf{X}_{-1}$ and $\mathbf{Y}_{-1}$ are $T \times (m_x - 1)$ and $T \times (m_y - 1)$ observation matrices on $\mathbf{x}_{t-1}$ and $\mathbf{y}_{t-1}$, respectively. It is now easy to show that the trace test based on $\mathbf{S}_w$ is the same as testing $\boldsymbol{\Pi} = \mathbf{0}$ in the dynamically augmented reduced rank regression

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\Pi}' + \mathbf{W}\mathbf{B} + \mathbf{E},$$

where $\mathbf{E}$ is a $T \times (m_y - 1)$ matrix of serially uncorrelated errors. Higher-order error dynamics can be accommodated by including further lags of $\mathbf{y}$ and $\mathbf{x}$ as columns of $\mathbf{W}$. Under $\boldsymbol{\Pi} = \mathbf{0}$, and for sufficiently large T, using results established in Anderson (2003) (see also Section 19.6), we have

$$(T - m_x)\,Tr(\mathbf{S}_w) \overset{a}{\sim} \chi^2_{(m_x-1)^2}.$$

An alternative to dynamically augmenting the reduced rank regression is to adjust the moment matrices used in calculating the variance matrix of $\hat{\boldsymbol{\gamma}}$ to account for heteroskedasticity and autocorrelation in the errors in (17.51). The F-statistic corresponding to (17.52) in this case is given by

$$F(\boldsymbol{\theta}) = \left(\frac{T - m_x}{m_x - 1}\right)\frac{\boldsymbol{\theta}'\mathbf{S}_{yx}\mathbf{H}_{xx}^{-1}(\boldsymbol{\theta})\mathbf{S}_{xy}\boldsymbol{\theta}}{\boldsymbol{\theta}'\left[\mathbf{S}_{yy} - \mathbf{S}_{yx}\mathbf{H}_{xx}^{-1}(\boldsymbol{\theta})\mathbf{S}_{xy}\right]\boldsymbol{\theta}},$$

where

$$\mathbf{H}_{xx}(\boldsymbol{\theta}) = \lim_{T\to\infty} E\left[\frac{1}{T}\sum_{s=1}^{T}\sum_{t=1}^{T}(\mathbf{x}_t - \bar{\mathbf{x}})(\mathbf{x}_s - \bar{\mathbf{x}})' u_t(\boldsymbol{\theta})u_s(\boldsymbol{\theta})\right],$$

$\bar{\mathbf{x}} = \left(\bar{x}_1, \bar{x}_2, \ldots, \bar{x}_{m_x-1}\right)'$, $\bar{\mathbf{y}} = \left(\bar{y}_1, \bar{y}_2, \ldots, \bar{y}_{m_y-1}\right)'$, and under $\boldsymbol{\gamma} = \mathbf{0}$, we have $u_t(\boldsymbol{\theta}) = \boldsymbol{\theta}'(\mathbf{y}_t - \bar{\mathbf{y}})$. Hence

$$\mathbf{H}_{xx}(\boldsymbol{\theta}) = \lim_{T\to\infty} E\left[\frac{1}{T}\sum_{s=1}^{T}\sum_{t=1}^{T}\boldsymbol{\theta}'(\mathbf{y}_t - \bar{\mathbf{y}})(\mathbf{x}_t - \bar{\mathbf{x}})(\mathbf{x}_s - \bar{\mathbf{x}})'(\mathbf{y}_s - \bar{\mathbf{y}})'\boldsymbol{\theta}\right],$$

can be viewed as the long-run variance of $T^{-1/2}\sum_{t=1}^{T}\mathbf{d}_t(\boldsymbol{\theta})$, where $\mathbf{d}_t(\boldsymbol{\theta}) = \boldsymbol{\theta}'(\mathbf{y}_t - \bar{\mathbf{y}})(\mathbf{x}_t - \bar{\mathbf{x}})$. Since elements of $\mathbf{x}_t$ and $\mathbf{y}_t$ are bounded, $\mathbf{H}_{xx}(\boldsymbol{\theta})$ exists under general assumptions concerning the serial dependence and heteroskedasticity of the error terms, as set out in Newey and West (1987).

Unlike the serially independent case, the first-order conditions for maximization of $LM(\boldsymbol{\theta})$ cannot get reduced to solving an eigenvalue problem. An asymptotically equivalent alternative (under $\boldsymbol{\gamma} = \mathbf{0}$) is to use a first-stage consistent estimate of $\mathbf{H}_{xx}(\boldsymbol{\theta})$ that abstracts from the serial dependence of the errors. Such an estimator of $\boldsymbol{\theta}$ is given by (17.57), and the first-stage estimate of $\mathbf{H}_{xx}(\boldsymbol{\theta})$ can be obtained by (using a Bartlett window)

$$\hat{\mathbf{H}}_{xx,h}(\hat{\boldsymbol{\theta}}_1) = \hat{\boldsymbol{\Gamma}}_0 + \sum_{j=1}^{h}\left(1 - \frac{j}{h+1}\right)(\hat{\boldsymbol{\Gamma}}_j + \hat{\boldsymbol{\Gamma}}_j'),$$

$$\hat{\boldsymbol{\Gamma}}_j = T^{-1}\sum_{t=j+1}^{T}\mathbf{d}_t(\hat{\boldsymbol{\theta}}_1)\mathbf{d}_{t-j}'(\hat{\boldsymbol{\theta}}_1),$$

$$\mathbf{d}_t(\hat{\boldsymbol{\theta}}_1) = \hat{\boldsymbol{\theta}}_1'(\mathbf{y}_t - \bar{\mathbf{y}})(\mathbf{x}_t - \bar{\mathbf{x}}).$$

Using this estimator, one can solve the following eigenvalue problem

$$\left(\mathbf{S}_{yx}\hat{\mathbf{H}}_{xx}^{-1}(\hat{\boldsymbol{\theta}}_1)\mathbf{S}_{xy} - \tilde{\rho}_1^2\mathbf{S}_{yy}\right)\tilde{\boldsymbol{\theta}}_1 = \mathbf{0},$$

where $\tilde{\rho}_1^2$ is the largest value of $\tilde{\rho}^2$ that solves

$$\left|\mathbf{S}_{yx}\hat{\mathbf{H}}_{xx}^{-1}(\hat{\boldsymbol{\theta}}_1)\mathbf{S}_{xy} - \tilde{\rho}^2\mathbf{S}_{yy}\right| = 0.$$

Under the null that $\boldsymbol{\gamma} = \mathbf{0}$, and conditional on the initial estimator of $\boldsymbol{\theta}$, $\hat{\boldsymbol{\theta}}_1$, the trace test is now given by

$$(T - m_x)\,Tr\left[\tilde{\mathbf{S}}(\hat{\boldsymbol{\theta}}_1)\right] \overset{a}{\sim} \chi^2_{(m_x-1)^2}, \tag{17.60}$$

where

$$\tilde{\mathbf{S}}(\hat{\boldsymbol{\theta}}_1) = S_{yy}^{-1}\mathbf{S}_{yx}\hat{\mathbf{H}}_{xx}^{-1}(\hat{\boldsymbol{\theta}}_1)\mathbf{S}_{xy}.$$

The estimate of $\boldsymbol{\theta}$ used for the estimation of $\mathbf{H}_{xx}(\boldsymbol{\theta})$ can be iterated upon as required until convergence is achieved, subject to the normalization restriction, $\boldsymbol{\theta}'\mathbf{S}_{yy}\boldsymbol{\theta} = 1$.

See Pesaran and Timmermann (2009) for further details and small sample results from Monte Carlo experiments.

17.14 Evaluation of density forecasts

Statistical techniques for the evaluation of density forecasts have been advanced by Dawid (1984) and further elaborated by Diebold, Gunther, and Tay (1998) in the case of univariate predictive densities, and by Diebold, Hahn, and Tay (1999) and Clements and Smith (2000) for the evaluation of multivariate density forecasts. These techniques are based on the idea of the probability integral transform that originates in the work of Rosenblatt (1952). Suppose that we have the h consecutive pairs $\left\{x_{t+1}, \hat{f}_t(x_{t+1} \mid \Omega_t), t = T+1, \ldots, T+h\right\}$ of the realizations and density forecasts. The probability integral transform for the h sequence is (see Diebold, Gunther, and Tay (1998))

$$U_t = \int_{-\infty}^{x_{t+1}} \hat{f}_t(u \mid \Omega_t)du, t = T, T+1, \ldots, T+h-1. \tag{17.61}$$

Hence, the probability integral transform U_t is the cumulative density function corresponding to the density, evaluated at x_{t+1}. Under the null hypothesis that $\hat{f}_t(x_{t+1}|\Omega_t)$ and $f(x_{t+1}|\Omega_t)$ coincide, the U_t are independent uniform $U(0, 1)$ random variables. Deviations from uniform *IID* will indicate that the forecasts have failed to capture some aspect of the underlying data generating process. Non uniformity may indicate improper distributional assumptions, while the presence of serial correlation in the series U_t may indicate that the dynamics are not adequately captured by the forecast model. Hence, the statistical adequacy of the predictive distributions, $\hat{F}_t$, can be assessed by testing whether $\{U_t, t = T+1, \ldots, T+h\}$ forms an $IIDU(0, 1)$ sequence. (see Diebold, Gunther, and Tay (1998)). This test can be carried out using a number of familiar statistical procedures. The uniformity of the distribution of U_t over t can be evaluated by adopting graphical methods, for example by visually comparing the estimated density (by simple histograms) to $U(0, 1)$. Formal tests of goodness of fit can also be employed, such as the Kolmogorov test, which measures the maximum absolute difference between the empirical distribution function and the uniform distribution. The *IID* property again can be visually evaluated, by examining the correlogram of the series $\left(U_t - \bar{U}\right)$. It is important to note that the *IID* uniform property is a necessary but not a sufficient condition for the optimality of the underlying predictive distribution. For example, expanding the information set Ω_t will lead to a different predictive distribution, but $\hat{f}_t$ continues to satisfy the *IID* uniform property.

In practice, there will be discrepancies between $f(x_{t+1}|\Omega_t)$ and $\hat{f}(x_{t+1}|\Omega_t)$ that could be more serious for some ranges of the state variable, x, and for some decisions as compared with other ranges or decisions. For example, in risk management the extreme values of x (viewed as portfolio

returns) are of interest, while in macroeconomic management values of x (inflation or output growth) the middle of the distribution may be of concern. In general, the forecast density function $\hat{f}(x_{t+1} \mid \Omega_t)$ can at best provide an approximation to the true density and from a decision perspective its evaluation is best carried out following the approach advanced in Section 17.10. Consider the forecast distribution function, $\hat{F}_t$, over the period $t = T, T+1, \ldots, T+h-1$. The sample counterpart of (17.25), namely

$$\overline{\Psi}_h\left(\hat{F}_t\right) = \frac{1}{h}\sum_{t=T}^{T+h-1} C(y_t^*(\hat{F}_t), x_{t+1}), \tag{17.62}$$

may be used, where the dependence of the decision variable, y_t^*, on the choice of the probability forecast distribution, $\hat{F}_t$, is made explicit here. Under certain regularity conditions on the distribution of the state variable and the underlying cost function, $C(\cdot,\cdot)$, it is reasonable to expect that

$$\underset{h\to\infty}{Plim}\left[\overline{\Psi}_h\left(\hat{F}_t\right)\right] = \lim_{h\to\infty}\left[\frac{1}{h}\sum_{t=T}^{T+h-1}\Psi_C\left(F_t, \hat{F}_t\right)\right],$$

namely that $\overline{\Psi}_h\left(\hat{F}_t\right)$ is an asymptotically consistent estimate of $E_F\left[\Psi_C\left(F_t, \hat{F}_t\right)\right]$. The average $\overline{\Psi}_h$ provides an estimate of the realized cost to the decision maker of using $\hat{F}_t$ over the period $t = T, T+1, \ldots, T+h-1$. Clearly, the decision maker will be interested in forecasts that minimize $\overline{\Psi}_h$.

In general, when one does not expect $\hat{F}_t$ to coincide with F_t, a sensible approach to forecast evaluation is to focus on comparisons between two competing forecasts. For example, suppose in addition to $\hat{F}_t$ the alternative predictive distribution function, $\tilde{F}_t$, is also available. A baseline alternative could be the unconditional probability distribution function of the state variable, or some other simple conditional model. Then the average economic loss arising from using the predictive distribution $\hat{F}_t$ as compared with $\tilde{F}_t$ is given by

$$L_h(\hat{F}_t, \tilde{F}_t) = \frac{1}{h}\sum_{t=T}^{T+h-1}\left[C(\mathbf{y}_t^*(\hat{F}_t), \mathbf{x}_{t+1}) - C(\mathbf{y}_t^*(\tilde{F}_t), \mathbf{x}_{t+1})\right],$$

which can also be written as the simple average of the loss-differential series,

$$d_{t+1} = C(\mathbf{y}_t^*(\hat{F}_t), \mathbf{x}_{t+1}) - C(\mathbf{y}_t^*(\tilde{F}_t), \mathbf{x}_{t+1}), \;\; t = T, T+1, \ldots, T+h-1. \tag{17.63}$$

As noted above, (17.63) has been considered in Diebold and Mariano (1995) in the special case where $C(y_t^*(\hat{F}_t), x_{t+1}) = \varphi(x_{t+1} - \hat{x}_{t+1|t})$, $C(y_t^*(\tilde{F}_t), x_{t+1}) = \varphi(x_{t+1} - \tilde{x}_{t+1|t})$, and $\varphi(\cdot)$ is a loss function. But in a decision-based framework it is $C(y_t^*, x_{t+1})$ which determines the appropriate cost-of-error function and is *equally* applicable to the evaluation of point and probability forecasts.

In the case of the illustrative examples of Section 17.10, we have the following expressions for the loss-differential series

(a) For the multivariate quadratic decision model we have[12]

$$d_{t+1} = (\mathbf{x}_{t+1} - \hat{\mathbf{x}}_{t+1|t})'\mathbf{H}(\mathbf{x}_{t+1} - \hat{\mathbf{x}}_{t+1|t}) - (\mathbf{x}_{t+1} - \tilde{\mathbf{x}}_{t+1|t})'\mathbf{H}(\mathbf{x}_{t+1} - \tilde{\mathbf{x}}_{t+1|t}).$$

(b) For the finance example in Section 17.10.2, using (17.32), the loss-differential becomes

$$d_{t+1} = \exp\left(-\frac{\rho_{t+1}\hat{\rho}_{t+1|t}}{\hat{\sigma}^2_{t+1|t}}\right) - \exp\left(-\frac{\rho_{t+1}\tilde{\rho}_{t+1|t}}{\tilde{\sigma}^2_{t+1|t}}\right).$$

17.15 Further reading

We refer to Elliott and Timmermann (2008) for further details on loss functions. For further details on forecast combinations see Bates and Granger (1969), Timmermann (2006), Elliott and Timmermann (2004), and Elliott, Granger, and Timmermann (2006).

17.16 Exercises

1. Compute the 1- and 2- step ahead forecasts for the model

$$y_t = \mu + \phi y_{t-1} + \varepsilon_t,$$

assuming $\mu = 0$ and $\phi = 0.4$, and when μ and ϕ are estimated using the first ten observations from the following data:

Time	y_t
1	101.1
2	103.4
3	103.7
4	104.6
5	104.6
6	105.7
7	107.5
8	108.1
9	108.6
10	108.9
11	109.4
12	109.6

[12] For a fixed choice of $\mathbf{H}$, specified independently of the parameters of the underlying cost function, $C(\mathbf{y}_t, \mathbf{x}_{t+1})$, the loss differential series d_t is not invariant under nonsingular linear transformations of the state variables, $\mathbf{x}_{t+1}$, a point emphasized by Clements and Hendry (1993) and alluded to above. But d_{t+1} is invariant to the nonsingular linear transformations of the state variables once it is recognized that such linear transformations will also alter the weighting matrix $\mathbf{H}$, in line with the transformation of the state variables, thus leaving the loss differential series, d_{t+1}, unaltered.

where ε_t is an *IID* white noise with mean zero and variance one. Use both the iterated and direct methods described in Section 17.7. Compute the corresponding forecast errors under both methods.

2. Compute the h-step ahead optimal forecast (assuming quadratic loss function) for the model

$$y_t - 0.3y_{t-1} = \varepsilon_t + 0.5\varepsilon_{t-1}.$$

3. Consider the following linear exponential (LINEX) cost function in the forecast errors, $e_t = y_t - y^*_{t|t-1}$, where $y^*_{t|t-1}$ is the forecast of y_t formed with the information available at time $t-1$, which we denote by Ω_{t-1}

$$L(e_t) = \frac{\exp(\alpha e_t) - \alpha e_t - 1}{\alpha^2}.$$

   (a) Plot $L(e_t)$ as a function of e_t for $\alpha = 0$ and for $\alpha = 1/2$. What are the main differences between these two plots? How do you interpret parameter α?

   (b) What is the value of $y^*_{t|t-1}$ which minimizes the expected loss, assuming that y_t conditional on Ω_{t-1} is Gaussian with mean $\mu_{t|t-1}$ and variance $h_{t|t-1}$?

   (c) How does the solution obtained under (b) differ from the rational expectations of y_t obtained conditional on Ω_{t-1}?

   (d) Discuss the relevance of the above analysis for tests of the rational expectations hypothesis using survey data where individual forecasters are asked about their expectations of y_t in the future.

4. Consider the $AR(1)$ model

$$y_t = \phi y_{t-1} + u_t,$$

where $u_t \sim IID(0, \sigma^2)$.

   (a) Derive iterated and direct forecasts of y_{t+2} conditional on y_t, and show that they can be estimated as

$$\begin{aligned} \textit{Iterated} \quad &: \quad \hat{y}^{(it)}_{t+2|t} = \hat{\phi}^2 y_t, \\ \textit{Direct} \quad &: \quad \hat{y}^{(d)}_{t+2|t} = \hat{\phi}_2 y_t, \end{aligned}$$

   where $\hat{\phi}$ and $\hat{\phi}_2$ are *OLS* coefficients in the regressions of y_t on y_{t-1} and y_{t-2}, respectively, using the M observations $y_{T-M+1}, y_{T-M+1}, \ldots, y_T$.

   (b) Show that

$$\begin{aligned} E\left(y_{T+2} - \hat{y}^{(it)}_{t+2|t}\right)^2 &= E\left(\phi^2 - \hat{\phi}^2\right)^2 + (1+\phi^2)\sigma^2, \\ E\left(y_{T+2} - \hat{y}^{(d)}_{t+2|t}\right)^2 &= E\left(\phi^2 - \hat{\phi}_2\right)^2 + (1+\phi^2)\sigma^2. \end{aligned}$$

(c) Hence, or otherwise show that

$$\lim_{M\to\infty} E(d_{T+2}) = 0,$$

where d_{T+2} is the loss differential of the two forecasting methods defined by

$$d_{T+2} = \left(y_{T+2} - \hat{y}^{(it)}_{t+2|t}\right)^2 - \left(y_{T+2} - \hat{y}^{(d)}_{t+2|t}\right)^2.$$

(d) Suppose now that there exist iterated and direct forecasts, $\hat{y}^{(it)}_{j,t+2|t}$ and $\hat{y}^{(d)}_{j,t+2|t}$ across N cross-section units $j = 1, 2, \ldots, N$. Develop statistical tests of the two approaches as N and M tend to infinity. Specify your assumptions carefully with justification.

5. Using inflation data over the sample period 1979Q2-2007Q4 from the *GVAR* data, compute one quarter ahead forecasts of output growth for US and UK over the period 2008Q1-2012Q4.

(a) Initially use univariate techniques applied to US and UK output growths separately. The *GVAR* 2013 data vintage can be downloaded from <https://sites.google.com/site/gvarmodelling/data>.

(b) Then use bivariate *VAR* models in US and UK output growth jointly to generate alternative forecasts.

(c) Compare the two sets of forecasts and discuss their relative merit.

18 Measurement and Modelling of Volatility

18.1 Introduction

Volatility as a measure of uncertainty has been used extensively in the theoretical and empirical literature. But 'volatility' as such is not directly observable and like many other economic concepts, such as expectations, demand and supply, it is usually treated as a latent variable and measured indirectly using a number of different proxies. Initially, volatility was measured by standard deviations of price changes computed over time, typically using a rolling window. But it was realized that such an historical measure tends to underestimate sudden changes in volatility and is only suitable when the underlying volatility is relatively stable.

To allow for time variations in volatility, Engle (1982) developed the autoregressive conditional heteroskedastic (*ARCH*) model that relates the (unobserved) volatility to squares of past innovations in price (or output) changes. Such a model-based approach only partly overcomes the deficiency of the historical measure and continues to respond very slowly when volatility undergoes rapid changes, as has been the case during the recent financial crisis. (See, e.g., Hansen, Huang, and Shek (2012)). The use of *ARCH*, or its various generalization (*GARCH*), in macro-econometric modelling is further complicated by the temporal aggregation of daily *GARCH* models for use in quarterly models.

In finance literature, the focus of the volatility measurement has shifted to market-based implied volatility obtained from option prices, and realized measures based on summation of intra-period higher-frequency squared returns. The use of implied volatility in macro-econometric modelling is limited both by availability of option price data and the fact that we still need to aggregate daily implied volatilities to a quarterly measure. By contrast, the idea of realized volatility can be easily adapted for use in macro-econometric models by summing squares of daily returns within a given quarter to construct a quarterly measure of market volatility. The approach can be extended to include intra-daily return observations when available, but this could contaminate the quarterly realized volatility measures with measurement errors of intra-daily returns due to market micro-structure and jumps in intra-daily returns. In addition, intra-daily returns are not available for all markets, and when available tend to cover a relatively short period.

18.2 Realized volatility

Realized volatility (RV) can be computed for any given time interval. In finance, the most popular time interval is a trading day. To construct measures of daily RV, we need information on intra-day price movements. Let $P_t(\tau)$ be the price of a given asset at time τ during day t, $\tau = 1, 2, \ldots, D_t$, where D_t is the number of times that prices are measured during day t. Then daily RV associated with intra-day price changes is computed as

$$RV_t^2 = \frac{1}{D_t} \sum_{\tau=1}^{D_t} (r_t(\tau) - \bar{r}_t)^2 ,$$

where $r_t(\tau) = \Delta \ln P_t(\tau)$ and $\bar{r}_t = D_t^{-1} \sum_{\tau=1}^{D_t} r_t(\tau)$ is the average intra-daily price changes over the day t. In practice, $\bar{r}_t$ is very small and is set to zero. The key issue is the appropriate number of price changes to be included in the computation of RV measures and the implications of large price changes (jumps) for the measurement of volatility. For further details see Andersen et al. (2003) and Barndorff-Nielsen and Shephard (2002).

The same idea can be applied to measuring quarterly RV based on daily price changes. In this case, $r_t(\tau)$ would refer to price changes in day τ of quarter t, and D_t will be the number of trading days within the quarter t. For most quarters we have $D_t = 3 \times 22 = 66$, which is larger than the number of data points typically used in the construction of daily realized market volatility in finance.[1] Similar realized quarterly volatility measures can also be computed for real asset prices with $P_t(\tau)$ in the above expressions replaced by $P_t(\tau)/CPI_t$, where CPI_t is the general price level for quarter t.

18.3 Models of conditional variance

In contrast to realized volatility, which is purely data-based, (conditional) volatility is defined as conditional variance, and denoted by

$$h_t^2 = Var(r_t \,|\Omega_{t-1}),$$

where r_t is the variable under consideration (such as asset returns, inflation rate or output growth), and Ω_t is the information set at time t. Volatility can arise due to a number of factors: over-reaction to news, incomplete learning, parameter variations, and abrupt switches in policy regimes. Econometric analysis of volatility usually focuses on daily, weekly or monthly observations. This chapter provides the technical details of the econometric methods that underlie models of asset return volatility.

18.3.1 RiskMetrics^{TM} (JP Morgan) method

Let

$$z_t = r_t - \bar{r}.$$

[1] In the case of intra-day observations, prices are usually sampled at 10-minute intervals which yield around 48 intra-daily returns in an 8-hour-long trading day.

Under the RiskMetrics approach, the historical volatility of z_t conditional on observations available at time $t-1$ is computed using the exponentially weighted moving average

$$h_t^2 = (1-\lambda)\sum_{\tau=0}^{\infty}\lambda^{\tau}z_{t-\tau-1}^2, \tag{18.1}$$

where λ is known as decay factor (or $1-\lambda$ the decay rate). Note that the weights

$$w_\tau = (1-\lambda)\lambda^{\tau}, \quad \tau = 0, 1, 2, \ldots,$$

add up to unity, and h_t^2 can be computed recursively

$$h_t^2 = \lambda h_{t-1}^2 + (1-\lambda)z_{t-1}^2,$$

which is a restricted version of the $GARCH(1,1)$ model introduced below (see equation (18.5)), with parameters satisfying $\alpha_1 + \phi_1 = 1$. Model (18.1) requires the initialization of the process. For a finite observation window, denoted by H, a more appropriate specification is

$$h_{H,t}^2 = \sum_{\tau=0}^{H} w_{H\tau}z_{t-1-\tau}^2,$$

where

$$w_{H\tau} = \frac{(1-\lambda)\lambda^{\tau}}{1-\lambda^{H+1}}, \quad \tau = 0, 1, \ldots, H. \tag{18.2}$$

Once again, these weights add up to unity. Other weighing schemes have also been considered. In particular, the equal weighted specification

$$h_t^2 = \frac{1}{H+1}\sum_{\tau=0}^{H} z_{t-1-\tau}^2,$$

where $w_{H\tau} = 1/(1+H)$, for all τ, which is a simple moving average specification.

The value chosen for the decay factor, λ, and the size of the observation window, H, are related. For example, for $\lambda = 0.9$, even if a relatively large value is chosen for H, due to the exponentially declining weights attached to past observations only around 110 observations are effectively used in the computation of h_t^2.

18.4 Econometric approaches

Consider the regression model

$$r_t = \boldsymbol{\beta}'\mathbf{x}_{t-1} + \varepsilon_t.$$

Under the classical normal assumptions (A1 to A5) set out in Chapter 2, the disturbances ε_t, in the above regression model have a constant variance both *unconditionally* and *conditionally*. However, in many applications in macroeconomics and finance, the assumption that the conditional variance of ε_t is constant over time is not valid. One possible model capturing such variations over time is the autoregressive conditional heteroskedasticity (*ARCH*) model, where volatility depends on the variability of past observations. In financial econometrics, *ARCH* is a fundamental tool for analyzing the time-variation of conditional variance. The *ARCH* model was introduced into the econometric literature by Engle (1982), and was subsequently generalized by Bollerslev (1986), who proposed the generalized *ARCH* (or *GARCH*) model. Other related models where the conditional variance of ε_t is used as one of the regressors explaining the conditional mean of r_t have also been suggested in the literature, and are known as *ARCH*-in-mean and *GARCH*-in-mean (or *GARCH-M*) models.

18.4.1 *ARCH*(1) and *GARCH*(1,1) specifications

Let $V(\varepsilon_t|\Omega_{t-1}) = h_t^2$. The $ARCH(1)$ model is defined as

$$h_t^2 = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2,\ \alpha_0 > 0. \tag{18.3}$$

It is clear that conditional on the information set, Ω_{t-1}, variance of ε_t is time varying. But this need not hold *unconditionally*. To see this first note that the unconditional variance of ε_t, which we denote by $V(\varepsilon_t)$, can be decomposed as

$$V(\varepsilon_t) = E\left[V(\varepsilon_t|\Omega_{t-1})\right] + V\left[E(\varepsilon_t|\Omega_{t-1})\right],$$

where by assumption $E(\varepsilon_t|\Omega_{t-1}) = 0$, and hence

$$V(\varepsilon_t) = E(\varepsilon_t^2) = E\left[V(\varepsilon_t|\Omega_{t-1})\right] = E\left(h_t^2\right).$$

Taking expectations of both sides of (18.3) we now have

$$E\left(h_t^2\right) = E(\varepsilon_t^2) = \alpha_0 + \alpha_1 E(\varepsilon_{t-1}^2).$$

Let $E(\varepsilon_t^2) = \sigma_t^2$ and write the above as a first-order difference equation in σ_t^2

$$\sigma_t^2 = \alpha_0 + \alpha_1 \sigma_{t-1}^2.$$

Therefore,

$$\sigma_t^2 = \alpha_0\left(1 + \alpha_1 + \alpha_1^2 + \ldots + \alpha_1^{t+M-1}\right) + \alpha_1^{t+M}\sigma_{-M}^2,$$

and provided $|\alpha_1| < 1$, in the limit as $M \to \infty$ we have (for any finite choice of σ_{-M}^2)

$$V(\varepsilon_t) = \sigma^2 = E\left(h_t^2\right) = \frac{\alpha_0}{1-\alpha_1} > 0.$$

Hence, unconditionally the $ARCH(1)$ model is stationary if $|\alpha_1| < 1$.

The above result generalizes. For example, for a pth-order $ARCH(p)$ model, denoted by $ARCH(p)$, we have

$$V(\varepsilon_t|\Omega_{t-1}) = h_t^2 = \alpha_0 + \alpha_1\varepsilon_{t-1}^2 + \cdots + \alpha_p\varepsilon_{t-p}^2, \tag{18.4}$$

with the associated difference equation in σ_t^2

$$\sigma_t^2 = \alpha_0 + \alpha_1\sigma_{t-1}^2 + \cdots + \alpha_p\sigma_{t-p}^2,\ \alpha_0 > 0,$$

and σ_t^2 converges to $\sigma^2 = \alpha_0/(1 - \alpha_1 - \ldots - \alpha_p)$, so long as all the roots of $f(\lambda) = 1 - \sum_{i=1}^{p}\alpha_i\lambda^i = 0$, lie outside the unit circle.

In practice, it is often convenient to write the $ARCH(p)$ specification as

$$\left(h_t^2 - \sigma^2\right) = \alpha_1\left(\varepsilon_{t-1}^2 - \sigma^2\right) + \cdots + \alpha_p\left(\varepsilon_{t-p}^2 - \sigma^2\right),$$

where the stationarity condition $\alpha_0 = \sigma^2(1 - \alpha_1 - \alpha_2 - \ldots - \alpha_p)$ is imposed.

An important extension of the $ARCH(1)$ model, which can also be viewed as restricted form of an infinite-order $ARCH$ model, $ARCH(\infty)$, is the Generalized $ARCH(1, 1)$ or the $GARCH(1, 1)$ model where

$$h_t^2 = \alpha_0 + \alpha_1\varepsilon_{t-1}^2 + \phi_1 h_{t-1}^2, \alpha_0 > 0. \tag{18.5}$$

This process is unconditionally stationary if $\left|\alpha_1 + \phi_1\right| < 1$. Note that

$$E(h_t^2) = \alpha_0 + \alpha_1 E(\varepsilon_{t-1}^2) + \phi_1 E(h_{t-1}^2),$$

or

$$\sigma_t^2 = \alpha_0 + (\alpha_1 + \phi_1)\sigma_{t-1}^2.$$

The unconditional variance exists and is fixed if $\left|\alpha_1 + \phi_1\right| < 1$. The case where $\alpha_1 + \phi_1 = 1$ is known as the Integrated $GARCH(1, 1)$, or I$GARCH(1, 1)$, for short. The RiskMetrics exponentially weighted formulation of h_t^2 for large H is a special case of the I$GARCH(1, 1)$ model where α_0 is set equal to 0. RiskMetrics formulation avoids the variance non-existence problem by focusing on H fixed.

A further generalization of the $GARCH$ model is the asymmetric $GARCH(1, 1)$, where

$$h_t^2 = \alpha_0 + \alpha_1^+ d_{t-1}\varepsilon_{t-1}^2 + \alpha_1^-(1 - d_{t-1})\varepsilon_{t-1}^2 + \phi_1 h_{t-1}^2, \alpha_0 > 0,$$

with $d_t = I(\varepsilon_t)$.

18.4.2 Higher-order *GARCH* models

The various members of the $GARCH$ and $GARCH$-M class of models can be written compactly as

$$y_t = \boldsymbol{\beta}' \mathbf{x}_{t-1} + \gamma h_t^2 + \varepsilon_t, \tag{18.6}$$

where

$$\begin{aligned} h_t^2 &= V(\varepsilon_t \,|\Omega_{t-1}) \,= E(\varepsilon_t^2 \,|\Omega_{t-1}) \\ &= \alpha_0 + \sum_{i=1}^{q} \alpha_i \varepsilon_{t-i}^2 + \sum_{i=1}^{p} \phi_i h_{t-i}^2, \end{aligned} \tag{18.7}$$

and Ω_{t-1} is the information set at time $t-1$, containing at least observations on lagged values of y_t and $\mathbf{x}_t$; namely $\Omega_{t-1} = (\mathbf{x}_{t-1}, \mathbf{x}_{t-2}, \ldots, y_{t-1}, y_{t-2}, \ldots)$. The unconditional variance of ε_t is determined by

$$\sigma_t^2 = \alpha_0 + \sum_{i=1}^{q} \alpha_i \sigma_{t-i}^2 + \sum_{i=1}^{p} \phi_i \sigma_{t-i}^2,$$

and yields a stationary outcome if all the roots of $1 = \sum_{i=1}^{q} \alpha_i \lambda^i + \sum_{i=1}^{p} \phi_i \lambda^i$, lie outside the unit circle. In that case

$$V(\varepsilon_t) = \sigma^2 = \frac{\alpha_0}{1 - \sum_{i=1}^{q} \alpha_i - \sum_{i=1}^{p} \phi_i} > 0. \tag{18.8}$$

Clearly the necessary condition for (18.7) to be covariance stationary is given by

$$\sum_{i=1}^{q} \alpha_i + \sum_{i=1}^{p} \phi_i < 1. \tag{18.9}$$

In addition to the restrictions (18.8) and (18.9), Bollerslev (1986) also assumes that $\alpha_i \geq 0$, $i = 1, 2, \ldots, q$, and $\phi_i \geq 0$, $i = 1, 2, \ldots, q$. Although these additional restrictions are sufficient for the conditional variance to be positive, they are not necessary (see Nelson and Cao (1992)).

18.4.3 Exponential *GARCH*-in-mean model

It is often the case that the conditional variance, h_t^2, is not an even function of the past disturbances, ε_{t-1}, ε_{t-2}, ... The exponential *GARCH* (or *EGARCH*) model proposed by Nelson (1991) aims at capturing this important feature, often observed when analyzing stock market returns. The *EGARCH* model has the following specification

$$\log h_t^2 = \alpha_0 + \sum_{i=1}^{q} \alpha_i \left(\frac{\varepsilon_{t-i}}{h_{t-i}} \right) + \sum_{i=1}^{q} \alpha_i^* \left(\left| \frac{\varepsilon_{t-i}}{h_{t-i}} \right| - \mu \right) + \sum_{i=1}^{p} \phi_i \log h_{t-i}^2, \tag{18.10}$$

where $\mu = E\left(\left|\frac{\varepsilon_t}{h_t}\right|\right)$. The value of μ depends on the density function assumed for the standardized disturbances, $\tilde{\varepsilon}_t = \varepsilon_t / h_t$. We have

$$\mu = \sqrt{\frac{2}{\pi}}, \qquad \text{if} \qquad \tilde{\varepsilon}_t \sim N(0,1),$$

and

$$\mu = \frac{2(v-2)^{\frac{1}{2}}}{(v-1)B\left(\frac{v}{2}, \frac{1}{2}\right)},$$

if ε_t has a standardized t-distribution with v degrees of freedom.

18.4.4 Absolute *GARCH*-in-mean model

The Absolute *GARCH* (*AGARCH*) model, proposed by Heutschel (1991), has the following specification

$$y_t = \boldsymbol{\beta}' \mathbf{x}_{t-1} + \gamma h_t^2 + \varepsilon_t, \tag{18.11}$$

where h_t is given by

$$h_t = \alpha_0 + \sum_{i=1}^{q} \alpha_i |\tilde{\varepsilon}_{t-i}| + \sum_{i=1}^{p} \phi_i h_{t-i} + \boldsymbol{\delta}' \mathbf{w}_t. \tag{18.12}$$

The AGARCH model can also be estimated for different error distributions. The log-likelihood functions for the cases where $\tilde{\varepsilon}_t = \varepsilon_t / h_t$ has a standard normal distribution; but when it has a standardized Student-t-distribution the log-likelihood functions are given by (18.18) and (18.19), where ε_t and h_t are now specified by (18.11) and (18.12), respectively.

18.5 Testing for *ARCH*/*GARCH* effects

The simplest way to test for an *ARCH*(q) effect is to use the Lagrange multiplier test procedure proposed by Engle. The test involves two steps. In the first step, the *OLS* residuals, $\hat{\varepsilon}_t$, from the

regression of y_t on $\mathbf{x}_t$ are computed. In the second step, $\hat{\varepsilon}_t^2$ is regressed on a constant and q of its own lagged values

$$\hat{\varepsilon}_t^2 = \alpha_0 + \alpha_1 \hat{\varepsilon}_{t-1}^2 + \cdots + \alpha_q \hat{\varepsilon}_{t-q}^2 + \text{Error}.$$

A test of the $ARCH(q)$ effect can now be carried out by testing the statistical significance of the slope coefficients

$$\alpha_1 = \alpha_2 = \cdots = \alpha_q = 0,$$

in the above *OLS* regression.

18.5.1 Testing for *GARCH* effects

Suppose we are interested in modeling series $\{z_t\}$, with z_t as daily data defined by $z_t = r_t - \bar{r}$, where r_t is asset return and $\bar{r}$ is the unconditional mean of asset return. The $GARCH(1, 1)$ representation of the volatility is

$$V(z_t \mid \Omega_{t-1}) = h_t^2 = \alpha_0 + \alpha_1 z_{t-1}^2 + \beta_1 h_{t-1}^2.$$

We are interested in testing

$$H_0 : \alpha_1 = 0, \tag{18.13}$$

against

$$H_1 : \alpha_1 \neq 0.$$

Since $GARCH(1, 1)$ can be approximated by $ARCH(\mathrm{q})$

$$V(z_t \mid \Omega_{t-1}) = \tilde{\alpha}_0 + \tilde{\alpha}_1 z_{t-1}^2 + \tilde{\alpha}_2 z_{t-1}^2 + \ldots + \tilde{\alpha}_q z_{t-1}^2,$$

hence testing (18.13) can be shown to be equivalent to testing

$$H_0 : \tilde{\alpha}_1 = \tilde{\alpha}_2 = \ldots = \tilde{\alpha}_q,$$

against

$$H_1 : \tilde{\alpha}_1 \neq 0, \tilde{\alpha}_2 \neq 0, \ldots \tilde{\alpha}_{,q} \neq 0,$$

which can be achieved by using the *LM* test proposed above. Note the *LM* test cannot distinguish between *ARCH* or *GARCH* processes.

Several points need to be kept in mind when we use *GARCH* model:

1. *GARCH* models are not closed under cross-sectional aggregation. This means that if every individual process follows $GARCH(1, 1)$, there is no guarantee that the average of those processes is also $GARCH(1, 1)$.

2. *GARCH* models need restrictions on their coefficients to make sure that the variance is positive.
3. In order to price options, one needs to know how volatile the price is. To achieve this, one has to match the *GARCH* model to some diffusion process. However, *GARCH* models do not fit diffusion processes.

18.6 Stochastic volatility models

Another way to model volatility is by using stochastic volatility models proposed by Melino and Turnbull (1990). The idea is into introduce two shocks into the model

$$z_t = \varepsilon_t e^{\frac{\alpha_t}{2}}, \tag{18.14}$$

$$\alpha_t = \psi \alpha_{t-1} + \xi_t, \tag{18.15}$$

with

$$\begin{pmatrix} \varepsilon_t \\ \xi_t \end{pmatrix} \sim N\left[\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \sigma_\varepsilon^2 & 0 \\ 0 & \sigma_\xi^2 \end{pmatrix}\right].$$

Here α_t is a $AR(1)$ process and the persistence of the shock ξ_t to it is measured by ψ. If we square (18.14) then take logs, we obtain

$$\log z_t^2 = \alpha_t + \log \varepsilon_t^2.$$

The advantage of this model is that we do not need restrictions on parameters since $\log z_t^2$ is well defined and z_t cannot take negative values. On the other hand, this model is computationally demanding as it is nonlinear and non-Gaussian. Even if we assume ε_t is normally distributed, ε_t^2 is a χ_1^2 and its logarithm involves nonlinearities. Prediction also becomes nonlinear and the prediction formula is difficult to derive.

18.7 Risk-return relationships

Suppose ρ_t is the rate of return on a portfolio of N assets with individual returns, r_{it}. Then $\rho_t = \sum_{i=1}^{N} \omega_{i,t-1} r_{it}$, where $\omega_{i,t-1}$ are the weight attached to the ith asset (could be negative if short sale is allowed), and under certain conditions we have

$$E\left(\rho_t \,|\Omega_{t-1}\right) = \gamma V\left(\rho_t \,|\Omega_{t-1}\right),$$

where γ is a measure of risk aversion. Hence

$$\rho_t = \gamma V\left(\rho_t \,|\Omega_{t-1}\right) + \varepsilon_t.$$

But $V\left(\rho_t \,|\Omega_{t-1}\right) = V\left(\varepsilon_t \,|\Omega_{t-1}\right) = h_t^2$, or $\rho_t = \gamma h_t^2 + \varepsilon_t, E\left(\varepsilon_t \,|\Omega_{t-1}\right) = 0$.

This is a simple example of *ARCH*-in-mean model. More generally

$$y_t = \beta_0 + \boldsymbol{\beta}_1' \mathbf{x}_{1,t-1} + \gamma h_t^2 + \varepsilon_t,$$
$$V(\varepsilon_t | \Omega_{t-1}) = h_t^2 = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \cdots + \alpha_p \varepsilon_{t-p}^2.$$

This is an *ARCH*-in-mean model. We can also have *GARCH*-in-mean models etc.

It is also possible to use other variables (in addition to past errors squared) in explanation of volatility. For example, we could have

$$V(\varepsilon_t | \Omega_{t-1}) = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \boldsymbol{\delta}_w' \mathbf{w}_{t-1},$$

where $\mathbf{w}_t$ represents the vector of the additional variables.

18.8 Parameter variations and *ARCH* effects

ARCH effects can be generated through random variations in the coefficients of the *AR* process. For example

$$y_t = \alpha + \varphi_t y_{t-1} + u_t,$$
$$\varphi_t = \varphi + \xi_t,$$

yields

$$y_t = \alpha + \varphi y_{t-1} + \varepsilon_t,$$
$$\varepsilon_t = u_t + \xi_t y_{t-1},$$

and

$$E(\varepsilon_t | \Omega_{t-1}) = 0,$$
$$V(\varepsilon_t | \Omega_{t-1}) = \sigma_u^2 + \sigma_\xi^2 y_{t-1}^2.$$

Assuming u_t and ξ_t are distributed independently and also that they are serially uncorrelated.

18.9 Estimation of *ARCH* and *ARCH*-in-mean models

The *ML* estimation method is often the most appropriate to use. Consider the following generalization of the *GARCH-M* model where, in addition to the past disturbances, other variables could also influence h_t^2

$$h_t^2 = \alpha_0 + \sum_{i=1}^{q} \alpha_i \varepsilon_{t-i}^2 + \sum_{i=1}^{p} \phi_i h_{t-i}^2 + \boldsymbol{\delta}_w' \mathbf{w}_{t-1}, \tag{18.16}$$

where $\mathbf{w}_t$ is a vector of covariance stationary variables in Ω_{t-1}. The unconditional variance of ε_t in this case is given by

$$\sigma^2 = \frac{\alpha_0 + \boldsymbol{\delta}'\boldsymbol{\mu}_w}{1 - \sum_{i=1}^{q} \alpha_i - \sum_{i=1}^{p} \phi_i} > 0, \tag{18.17}$$

where $\boldsymbol{\mu}_w = E(\mathbf{w}_t)$.

The *ML* estimation of the above augmented *GARCH-M* model can be carried out under two different assumptions concerning the conditional distribution of the disturbances, namely Gaussian and standardized *t*-distribution. In both cases the *exact* log-likelihood function depends on the joint density function of the initial observations, $f(y_1, y_2, \ldots, y_q)$, which is non-Gaussian and intractable analytically. In most applications where the sample size is large (as is the case with most financial time series) the effect of the distribution of the initial observations is relatively small and can be ignored.[2]

18.9.1 *ML* estimation with Gaussian errors

The log-likelihood function used in computation of the *ML* estimators for the Gaussian case is given by

$$\begin{aligned}\ell(\boldsymbol{\theta}) = &-\frac{(n-q)}{2}\log(2\pi) - \tfrac{1}{2}\sum_{t=q+1}^{n} \log h_t^2 \\ &- \tfrac{1}{2}\sum_{t=q+1}^{n} h_t^{-2}\varepsilon_t^2,\end{aligned} \tag{18.18}$$

where $\boldsymbol{\theta} = (\boldsymbol{\beta}', \gamma, \alpha_0, \alpha_1, \alpha_2, \ldots, \alpha_q, \phi_1, \phi_2, \ldots, \phi_p, \boldsymbol{\delta}')'$, and ε_t and h_t^2 are given by (18.6) and (18.16), respectively.

18.9.2 *ML* estimation with Student's *t*-distributed errors

Under the assumption that conditional on Ω_{t-1}, the disturbances are distributed as a Student *t*-distribution with ν degrees of freedom $(\nu > 2)$, the log-likelihood function is given by

$$\ell(\boldsymbol{\theta}, \nu) = \sum_{t=q+1}^{n} \ell_t(\boldsymbol{\theta}, \nu), \tag{18.19}$$

where

$$\begin{aligned}\ell_t(\boldsymbol{\theta}, \nu) = &-\log\left[B\left(\tfrac{\nu}{2}, \tfrac{1}{2}\right)\right] - \tfrac{1}{2}\log(\nu - 2) \\ &- \tfrac{1}{2}\log h_t^2 - \left(\frac{\nu+1}{2}\right)\log\left(1 + \frac{\varepsilon_t^2}{h_t^2(\nu-2)}\right),\end{aligned} \tag{18.20}$$

[2] Diebold and Schuermann (1993) examine the quantitative importance of the distribution of the initial observations in the case of simple *ARCH* models and find their effect to be negligible.

and $B\left(\frac{\nu}{2}, \frac{1}{2}\right)$ is the beta function.[3]

The degrees of freedom of the underlying t-distribution, ν, are then estimated along with the other parameters. The Gaussian log-likelihood function (18.18) is a special case of (18.20) and can be obtained from it for large values of ν. In most applications, the two log-likelihood functions give very similar results for values of ν around 20. The t -distribution is particularly appropriate for the analysis of stock returns where the distribution of the standardized residuals, $\hat{\varepsilon}_t/\hat{h}_t$, is often found to have fatter tails than the normal distribution.

The (approximate) log-likelihood function for the *EGARCH* model has the same form as in (18.18) and (18.19) for the Gaussian and Student t-distributions, respectively. Unlike the *GARCH–M* class of models, the *EGARCH–M* model always yields a positive conditional variance, h_t^2, for any choice of the unknown parameters; it is only required that the roots of $1 - \sum_{i=1}^{p} \phi_i z^i = 0$ should all fall outside the unit circle. The unconditional variance of ε_t in the case of the *EGARCH* model does not have a simple analytical form.

The absolute *GARCH* model can also be estimated for different error distributions. The log-likelihood functions for the cases where $\tilde{\varepsilon}_t = \varepsilon_t/h_t$ has a standard normal distribution and when it has a standardized Student t-distribution are given by (18.18) and (18.19), where ε_t and h_t are now specified by (18.11) and (18.12), respectively.

Example 36 (The volatility of asset returns) *In this example, we provide ML estimates of GARCH*$(1, 1)$ *models for* 22 *main industry indices of the Standard & Poor's* 500*. The source of our data is Datastream, which provides 24 S&P 500 industry price indices according to the Global Industry Classification Standard. To ensure a sufficiently long span of daily prices, we exclude the 'Semiconductors & Semiconductor Equipment' and 'Real Estates' from our analysis. The list of the* $N = 22$ *industries included in the analysis is given in Table 18.1. The data set covers the industry indices from 2 Jan. 1995 to 13 October 2003 (*$T = 2291$ *observation). Daily returns are computed as* $r_{it} = 100 * \ln\left(P_{it}/P_{i,t-1}\right)$, $i = 1, \ldots, 22$, *where* P_{it} *is the* i^{th} *price index. The realized returns* $\mathbf{r}_t = (r_{1t}, r_{2t}, \ldots, r_{22,t})'$ *exhibit all the familiar stylized features over our sample period. They are highly cross-correlated, with an average pair-wise cross-correlation coefficient of* 0.5*. A standard factor analysis yields that the two largest estimated eigenvalues are equal to* 11.5 *and* 1.7*, with the remaining being all smaller than unity. The unconditional daily volatility differs significantly across industries and lies in range of* 1.13% *(Food, Beverage & Tobacco) to* 2.39% *(Technology Hardware & Equipment). See Table 18.2. The first-order autocorrelation coefficients of the individual returns are quantitatively very small (ranging from* -0.049 *to* 0.054*) and are statistically significant only in the case of four out of the 22 industries (Automobiles & Components, Health Care Equipment & Services, Diversified Financial, and Utilities). Estimates of univariate GARCH*$(1, 1)$ *models for the returns are summarized in Table 18.3, and support the use of a Student t-distribution with a low degree of freedom for the conditional distribution of the asset returns. The degrees of freedom estimated for the different assets lie in the relatively narrow range of* $(5.2 - 11.7)$ *with an average estimate of* 7.3*, and a mid-point value of* 8.5*.*

[3] Notice that $B(\frac{\nu}{2}, \frac{1}{2}) = \Gamma\left(\frac{\nu+1}{2}\right) \Big/ \Gamma\left(\frac{\nu}{2}\right) \Gamma\left(\frac{1}{2}\right)$. The constant term $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$ is omitted from the expression used by Bollerslev (1987). See his equation (1).

Table 18.1 Standard & Poor 500 industry groups

	Codes	Industries
1	EN	Energy
2	MA	Materials
3	IC	Capital Goods
4	CS	Commercial Services & Supplies
5	TRN	Transportation
6	AU	Automobiles & Components
7	LP	Consumer Durables & Apparel
8	HR	Hotels, Restaurants & Leisure
9	ME	Media
10	MS	Retailing
11	FD	Food & Staples Retailing
12	FBT	Food, Beverage & Tobacco
13	HHPE	Household & Personal Products
14	HC	Health Care Equipment & Services
15	PHB	Pharmaceuticals & Biotechnology
16	BK	Banks
17	DF	Diversified Financials
18	INSC	Insurance
19	IS	Software & Services
20	TEHW	Technology Hardware & Equipment
21	TS	Telecommunication Services
22	UL	Utilities

Note: The codes in the second column are taken from REUTERS for the S&P 500 industry groups according to the Global Industry Classification Standard. 'Real States' and 'Semiconductors & Semiconductor Equipment' industries are excluded.
Source: Datastream.

18.10 Forecasting with *GARCH* models

18.10.1 Point and interval forecasts

Consider the regression model with $GARCH(1, 1)$ variances

$$y_t = \boldsymbol{\beta}' \mathbf{x}_{t-1} + \varepsilon_t$$
$$V(\varepsilon_t | \Omega_{t-1}) = h_t^2 = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \beta_1 h_{t-1}^2.$$

Point forecast of y_{t+1} is given by

$$y_{t+1}^* = \boldsymbol{\beta}' \mathbf{x}_t,$$

and the *GARCH* component does not affect the forecasts, except possibly through its effect on the estimate of $\boldsymbol{\beta}$. The interval forecast for y_{t+1} is also largely unaffected by the *GARCH* structure of the disturbances.

Table 18.2 Summary statistics

Sector	Mean	St.Dev.	Skewness	Kurtosis	Ljung–Box(20)
EN	0.031	1.386	0.049	5.435	40.3
MA	0.015	1.367	0.141	6.347	22.1
IC	0.040	1.395	–0.156	6.784	33.1
CS	0.022	1.318	–0.466	8.777	18.7
TRN	0.027	1.407	–0.501	10.644	28.4
AU	0.011	1.628	–0.172	7.017	36.3
LP	0.015	1.194	–0.099	6.750	25.1
HR	0.034	1.422	–0.393	9.241	16.4
ME	0.030	1.660	–0.056	8.168	37.2
MS	0.057	1.739	0.017	6.120	48.5
FD	0.028	1.328	–0.217	6.597	30.8
FBT	0.032	1.132	0.008	6.312	32.4
HHPE	0.042	1.445	–1.581	30.256	55.1
HC	0.039	1.274	–0.295	7.008	57.4
PHB	0.054	1.472	–0.172	5.821	53.1
BK	0.051	1.590	0.045	5.324	37.0
DF	0.075	1.840	0.036	5.013	47.4
INSC	0.044	1.549	0.415	11.045	38.8
IS	0.062	2.246	0.060	5.019	32.6
TEHW	0.043	2.393	0.165	5.719	30.6
TS	0.000	1.605	–0.072	5.969	22.7
UL	0.005	1.197	–0.363	9.881	25.8

Note: Columns 2 to 4 report the sample mean, standard deviation, skewness and kurtosis. Column 5 reports the Ljung–Box statistic of order 20 for testing autocorrelations in individual asset returns. The critical value of χ^2_{20} at the 1% significance level is 37.56. The sample period is 2nd January 1995 to 13th October 2003.

18.10.2 Probability forecasts

Probability forecasts are particularly relevant when the interest is in forecasting specific events. For example, if the interest is in estimation of the conditional probability that $y_{t+1} > c$, it is easily seen that

$$\Pr(y_{t+1} > c \mid \Omega_{t-1}) = 1 - \Phi\left(\frac{c - \boldsymbol{\beta}'\mathbf{x}_t}{h_{t+1}}\right),$$

assuming that ε_t (conditionally) has a normal distribution. Probability event forecasting, therefore, involves forecasting volatility. See also Chapter 17.

18.10.3 Forecasting volatility

One of the main attractions of the *GARCH* type models is the ease with which one can obtain single-period and multi-period ahead forecasts. For example, in the case of the $GARCH(1,1)$ model of asset returns, $z_t = r_t - \bar{r}, t = 1, 2, \ldots, T$ we have

Table 18.3 Estimation results for univariate $GARCH(1,1)$ models

	Normal innovations			Student t innovations			
Sector	$\hat{\alpha}_{i0}$	$\hat{\alpha}_{i1}$	$\hat{\phi}_{i1}$	$\hat{\alpha}_{i0}$	$\hat{\alpha}_{i1}$	$\hat{\phi}_{i1}$	$\hat{\nu}_i$
EN	0.0241	0.0620	0.9276	0.0287	0.0665	0.9210	10.25
MA	0.0246	0.1016	0.8932	0.0101	0.0625	0.9355	6.71
IC	0.0222	0.0710	0.9216	0.0201	0.0590	0.9329	6.96
CS	0.1265	0.0759	0.8566	0.0559	0.0755	0.8967	5.20
TRN	0.0163	0.0563	0.9399	0.0282	0.0756	0.9135	6.99
AU	0.0352	0.0604	0.9289	0.0371	0.0635	0.9251	6.51
LP	0.0517	0.0697	0.8975	0.0316	0.0586	0.9212	6.51
HR	0.0841	0.0696	0.8943	0.0327	0.0443	0.9402	6.03
ME	0.0328	0.0854	0.9090	0.0199	0.0617	0.9335	6.17
MS	0.0337	0.0560	0.9346	0.0250	0.0470	0.9459	7.29
FD	0.0474	0.0768	0.8996	0.0300	0.0681	0.9179	7.41
FBT	0.0232	0.0693	0.9150	0.0178	0.0568	0.9311	7.06
HHPE	0.0133	0.0741	0.9259	0.0360	0.0678	0.9147	6.70
HC	0.0950	0.1431	0.8086	0.0911	0.1035	0.8443	6.45
PHB	0.0640	0.0701	0.9032	0.0587	0.0736	0.9030	6.95
BK	0.0557	0.0801	0.8998	0.0457	0.0843	0.9008	8.66
DF	0.1033	0.0686	0.9030	0.0890	0.0732	0.9034	8.53
INSC	0.0375	0.0852	0.9044	0.0344	0.0750	0.9149	5.49
IS	0.0915	0.0647	0.9187	0.0639	0.0587	0.9303	10.14
TEHW	0.0730	0.0730	0.9159	0.0532	0.0608	0.9309	11.73
TS	0.0255	0.0478	0.9437	0.0231	0.0438	0.9484	7.09
UL	0.0213	0.1162	0.8735	0.0183	0.1105	0.8838	6.42
Average	0.0501	0.0762	0.9052	0.0387	0.0677	0.9177	7.34

Note: Columns 2 to 4 report the *ML* estimates of the univariate $GARCH(1,1)$ model for each sector $i = 1, 2, \ldots, 22$, assuming Gaussian innovations:

$$h_{it}^2 = a_{i0} + \alpha_{i1} r_{i,t-1}^2 + \phi_{i1} h_{i,t-1}^2.$$

Columns 5 to 8 report the *ML* estimates of the univariate $GARCH(1,1)$ for each sector $i = 1, 2, \ldots, 22$, assuming Student t innovations with ν_i degrees of freedom. All the estimates reported for α_{i0}, α_{i1}, and ϕ_{i1} are statistically significant at the 5% level. The estimation period is 2 Jan. 1995 to 13 Oct. 2003.

$$\hat{h}_{T+1}^2 = \hat{\alpha}_0 + \hat{\alpha}_1 z_T^2 + \hat{\phi}_1 \hat{h}_T^2,$$
$$\hat{h}_{T+h}^2 = \hat{\alpha}_0 + (\hat{\alpha}_1 + \hat{\phi}_1)\hat{h}_{T+h-1}^2, h = 2, 3, \ldots.$$

Notice that for large enough forecast horizon, h, $\hat{h}_{T+h}^2$ tends to the unconditional variance of z_t given by $\hat{h}_{\infty}^2 = \hat{\alpha}_0/(1 - \hat{\alpha}_1 - \hat{\phi}_1)$.

18.11 Further reading

See Gardner Jr (2006) for a review of the research on exponential smoothing. For a useful survey of the literature on *ARCH*/*GARCH* modelling see Bollerslev, Chou, and Kroner (1992) and

Engle, Lillien, and Robins (1987), while textbook treatments can be found in Hamilton (1994), Satchell and Knight (2007), Campbell, Lo, and MacKinlay (1997), and Engle (1995). Shephard (2005) provides selected readings of the literature on stochastic volatility. For an extension of volatility models to the multivariate case see Chapter 25.

18.12 Exercises

1. Consider the generalized autoregressive conditional heteroskedastic ($GARCH$) model

$$y_t = h_t z_t,$$

where

$$z_t \,|\Omega_{t-1} \backsim IIDN(0,1),$$
$$h_t^2 = Var(y_t \,|\Omega_{t-1}) = \bar{\sigma}\,(1-\alpha-\beta) + \alpha y_{t-1}^2 + \beta h_{t-1}^2,$$

and Ω_t is the information set that contains at least y_t and its lagged values.

   (a) Derive the conditions under which $\{y_t\}$ is a stationary process.
   (b) Are the observations, $\{y_t\}$ (i) serially independent, (ii) serially uncorrelated?
   (c) Develop a test of the $GARCH$ effect, and discuss the estimation of the above model by the maximum likelihood method.
   (d) Discuss the relevance of $GARCH$ models for the analysis of financial time series data.

2. The RiskMetrics measure of conditional volatility of r_t, excess return, is given by the exponentially weighted moving average,

$$h_t^2 = (1-\lambda)\sum_{\tau=0}^{\infty} \lambda^{\tau} r_{t-\tau-1}^2,$$

where λ stands for the decay or the forgetting parameter.

   (a) Under what conditions does this measure of volatility coincide with the $GARCH(1,1)$ model?
   (b) What are the main limitations of the RiskMetrics approach?

3. Consider the following regression model with conditionally heteroskedastic errors

$$y_t = \boldsymbol{\gamma}'\mathbf{x}_t + h_t\varepsilon_t,$$

where $\mathbf{x}_t$ is a $k \times 1$ vector of predetermined variables, ε_t is an $IID(0,1)$ process, and

$$h_t^2 = \bar{\sigma}^2(1-\alpha-\beta) + \alpha y_{t-1}^2 + \beta h_{t-1}^2.$$

(a) Suppose that the interest lies in estimating the regression coefficients, $\boldsymbol{\gamma}$. Discuss the relative merits of least squares (LS) and quasi-maximum likelihood (QML) estimators, with the latter obtained assuming (perhaps incorrectly) that the errors, ε_t, are Gaussian. Discuss carefully the assumptions that you must make under the two estimation procedures to ensure that the estimators of $\boldsymbol{\gamma}$ are consistent. What can be said about the relative efficiency of LS and QML estimators.

(b) Derive the asymptotic variance of the LS estimator of $\boldsymbol{\gamma}$.

(c) How would you forecast the mean and variance of y_{T+h} conditional on the information set $\Omega_T = (y_T, \mathbf{x}_T, y_{T-1}, \mathbf{x}_{T-1}, \ldots)$?

4. Consider the dynamic regression model

$$y_t = \lambda y_{t-1} + \beta x_t + u_t,$$
$$x_t = \rho x_{t-1} + v_t,$$

where u_t and v_t are serially uncorrelated with zero means and conditional variances

$$E(u_t^2 \,|\Omega_{t-1}) = a + \phi_{11} u_{t-1}^2 + \phi_{12} v_{t-1}^2,$$
$$E(v_t^2 \,|\Omega_{t-1}) = b + \phi_{21} u_{t-1}^2 + \phi_{22} v_{t-1}^2,$$

and $\Omega_t = (y_t, x_t; y_{t-1}, x_{t-1}; \ldots)$.

(a) Derive conditions under which y_t and x_t are stationary.

(b) Obtain expressions for the unconditional variances of u_t and v_t.

(c) How do you test the hypothesis that $\phi_{11} = \phi_{12} = 0$?

(d) What are the implications of volatilities in u_t and v_t for the estimation and inference regarding λ, β and ρ?

5. Suppose that the errors u_t and v_t in the above question are independently distributed.

(a) Derive expressions for $E(x_t^2 \,|\Omega_{t-1})$ and $E(y_t^2 \,|\Omega_{t-1})$.

(b) How would you go about estimating $E(y_t^2 \,|\Omega_{t-1})$?

(c) Derive expressions for the unconditional variances of y_t and x_t.

Part V

Multivariate Time Series Models

19 Multivariate Analysis

19.1 Introduction

This chapter reviews techniques for the analysis of multivariate systems. It begins with a discussion of the system of regression equations both when the regressors are strictly exogenous (the so called *SURE* model) and when one or more of the regressors are endogenously determined (the classical simultaneous equation system). It provides an overview of two and three stage least squares, and iterated instrumental variables estimators for systems of equations containing endogenous variables. It then considers other statistical techniques for the analysis of multivariate systems and gives an account of principal components analysis and factor models that are useful when introducing econometric techniques for panel data with error cross-sectional dependence (see Chapter 29). We end the chapter with canonical correlation analysis and reduced rank regressions where a sub-set of matrix coefficients in a *SURE* model is assumed to be rank deficient. Such analyses form the basis of cointegration analysis that will be considered in Chapter 22.

19.2 Seemingly unrelated regression equations

Consider the following m 'seemingly' separate linear regression equations

$$\mathbf{y}_i = \mathbf{X}_i\boldsymbol{\beta}_i + \mathbf{u}_i, \quad i = 1, 2, \ldots, m, \tag{19.1}$$

where $\mathbf{y}_i = \left(y_{i1}, y_{i2}, \ldots, y_{iT}\right)'$ is a $T \times 1$ vector of observations on the dependent variable y_{it}, and $\mathbf{X}_i$ is a $T \times k_i$ matrix of observations on the k_i vector of regressors explaining y_{it}, $\boldsymbol{\beta}_i$ is a $k_i \times 1$ vector of unknown coefficients and $\mathbf{u}_i = (u_{i1}, u_{i2}, \ldots, u_{iT})'$ is a $T \times 1$ vector of disturbances or errors, for $i = 1, 2, \ldots, m$. Let $\mathbf{u} = \left(\mathbf{u}_1', \mathbf{u}_2', \ldots, \mathbf{u}_m'\right)'$ be the mT-dimensional vector of disturbances. We assume

Assumption A1: The $mT \times 1$ vector of disturbances, $\mathbf{u}$, has zero conditional mean

$$E\left(\mathbf{u} | \mathbf{X}_1, \mathbf{X}_2, \ldots, \mathbf{X}_m\right) = \mathbf{0}.$$

Assumption A2: The disturbances are uncorrelated across observations

$$E\left(\mathbf{u}_i\mathbf{u}_i' \middle| \mathbf{X}_1, \mathbf{X}_2, \ldots, \mathbf{X}_m\right) = \sigma_{ii}\mathbf{I}_T.$$

In econometric analysis of the system of equations in (19.1), three cases can be distinguished:

1. Contemporaneously uncorrelated disturbances, namely $E\left(\mathbf{u}_i\mathbf{u}_j'\right) = \mathbf{0}$, for $i \neq j$.
2. Contemporaneously correlated disturbances, with identical regressors across all the equations, namely

$$E\left(\mathbf{u}_i\mathbf{u}_j'\right) = \sigma_{ij}\mathbf{I}_T \neq \mathbf{0},$$

where $\mathbf{I}_T$ is an identity matrix of order T and

$$\mathbf{X}_i = \mathbf{X}_j, \text{ for all } i, j.$$

3. Contemporaneously correlated disturbances, with different regressors across the equations, namely

$$E\left(\mathbf{u}_i\mathbf{u}_j'\right) = \sigma_{ij}\mathbf{I}_T \neq \mathbf{0},$$

and

$$\mathbf{X}_i \neq \mathbf{X}_j, \text{ at least for some } i, \text{ and } j.$$

In the rest of this chapter, we briefly review estimation and inference in the above three cases. To this end, it is convenient to stack the different equations in the system in the following manner

$$\begin{pmatrix} \mathbf{y}_1 \\ \mathbf{y}_2 \\ \vdots \\ \mathbf{y}_m \end{pmatrix} = \begin{pmatrix} \mathbf{X}_1 & \mathbf{0} & \ldots & \mathbf{0} \\ \mathbf{0} & \mathbf{X}_2 & \ldots & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{0} & \mathbf{0} & \ldots & \mathbf{X}_m \end{pmatrix} \begin{pmatrix} \boldsymbol{\beta}_1 \\ \boldsymbol{\beta}_2 \\ \vdots \\ \boldsymbol{\beta}_m \end{pmatrix} + \begin{pmatrix} \mathbf{u}_1 \\ \mathbf{u}_2 \\ \vdots \\ \mathbf{u}_m \end{pmatrix}, \tag{19.2}$$

or, more compactly,

$$\mathbf{y} = \mathbf{G}\boldsymbol{\beta} + \mathbf{u}, \tag{19.3}$$

where $\mathbf{y}$, $\mathbf{G}$, $\boldsymbol{\beta}$ and $\mathbf{u}$ have the dimensions $mT \times 1$, $mT \times k$, $k \times 1$, and $mT \times 1$, respectively, and $k = \sum_{i=1}^{m} k_i$.

19.2.1 Generalized least squares estimator

We first consider estimation in the more general case 3 above, and then review cases 1 and 2. Under the assumption that $E\left(\mathbf{u}_i\mathbf{u}_j' \middle| \mathbf{X}_1, \mathbf{X}_2, \ldots, \mathbf{X}_m\right) = \sigma_{ij}\mathbf{I}_T$, we have

$$E\left(\mathbf{u}\mathbf{u}'\middle|\mathbf{X}_1,\mathbf{X}_2,\ldots,\mathbf{X}_m\right)=\boldsymbol{\Omega}=\boldsymbol{\Sigma}\otimes\mathbf{I}_T,$$

where $\boldsymbol{\Sigma}=(\sigma_{ij})$ is an $m\times m$ symmetric positive definite matrix and $\otimes$ stands for the Kronecker matrix multiplication.[1] More specifically, we have

$$\boldsymbol{\Omega}=\boldsymbol{\Sigma}\otimes\mathbf{I}_T=\begin{pmatrix}\sigma_{11}\mathbf{I}_T & \sigma_{12}\mathbf{I}_T & \ldots & \sigma_{1m}\mathbf{I}_T\\ \sigma_{21}\mathbf{I}_T & \sigma_{22}\mathbf{I}_T & \ldots & \sigma_{2m}\mathbf{I}_T\\ \vdots & \vdots & \vdots & \vdots\\ \sigma_{m1}\mathbf{I}_T & \sigma_{m2}\mathbf{I}_T & \ldots & \sigma_{mm}\mathbf{I}_T\end{pmatrix}. \tag{19.4}$$

Note that

$$\boldsymbol{\Omega}^{-1}=\boldsymbol{\Sigma}^{-1}\otimes\mathbf{I}_T.$$

When $\boldsymbol{\Sigma}$ is known, the efficient estimator of $\boldsymbol{\beta}$ is the *GLS* estimator (see Section 4.3) given by

$$\hat{\boldsymbol{\beta}}_{GLS}=\left[\mathbf{G}'\left(\boldsymbol{\Sigma}^{-1}\otimes\mathbf{I}_T\right)\mathbf{G}\right]^{-1}\mathbf{G}'\left(\boldsymbol{\Sigma}^{-1}\otimes\mathbf{I}_T\right)\mathbf{y}, \tag{19.5}$$

with the asymptotic covariance matrix

$$Var(\hat{\boldsymbol{\beta}}_{GLS})=\left[\mathbf{G}'\left(\Sigma^{-1}\otimes\mathbf{I}_T\right)\mathbf{G}\right]^{-1}.$$

But in practice, $\boldsymbol{\Sigma}$ is not known and as a result $\hat{\boldsymbol{\beta}}_{GLS}$ is often referred to as the infeasible *GLS* estimator of $\boldsymbol{\beta}$. A feasible *GLS* estimator is obtained by replacing the unknown elements of $\boldsymbol{\Sigma}$, namely σ_{ij}, with suitable estimators. In the case where m is small relative to T, σ_{ij} can be estimated consistently by

$$\hat{\sigma}_{ij}=\frac{\hat{\mathbf{u}}_i'\hat{\mathbf{u}}_j}{T},\quad i,j=1,2,\ldots,m, \tag{19.6}$$

where $\hat{\mathbf{u}}_i=\mathbf{y}_i-\mathbf{X}_i\hat{\boldsymbol{\beta}}_{i,OLS}$, and $\hat{\boldsymbol{\beta}}_{i,OLS}$ is the ordinary least squares estimator of $\boldsymbol{\beta}_i$. The resultant feasible *GLS* estimator is then given by

$$\hat{\boldsymbol{\beta}}_{FGLS}=\left[\mathbf{G}'\left(\tilde{\Sigma}^{-1}\otimes\mathbf{I}_T\right)\mathbf{G}\right]^{-1}\mathbf{G}'\left(\tilde{\Sigma}^{-1}\otimes\mathbf{I}_T\right)\mathbf{y}.$$

In cases where m is large, the use of $\tilde{\Sigma}$ as an estimator of $\boldsymbol{\Sigma}$ is not recommended and estimation procedures of the type discussed in Chapter 29 are likely to be more appropriate.

[1] For a definition of the Kronecker product and the rules of its operation see Section A.8 in Appendix A.

The case of contemporaneously uncorrelated disturbances

In the case where $E(\mathbf{u}_i\mathbf{u}_j') = \mathbf{0}$, for $i \neq j$, there is nothing to be gained by considering the equations in (19.1) as a system, and the application of single equation methods to the individual relations in (19.1) will yield efficient estimators. To show this, note that if $\sigma_{ij} = 0$ for $i \neq j$, we have $\Sigma = \mathbf{I}_N$ in (19.4). Given the properties of inverse matrices we have

$$\hat{\boldsymbol{\beta}}_{GLS} = \begin{pmatrix} (\mathbf{X}_1'\mathbf{X}_1)^{-1} & \mathbf{0} & \dots & \mathbf{0} \\ \mathbf{0} & (\mathbf{X}_2'\mathbf{X}_2)^{-1} & \dots & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{0} & \mathbf{0} & \dots & (\mathbf{X}_m'\mathbf{X}_m)^{-1} \end{pmatrix} \begin{pmatrix} \mathbf{X}_1'\mathbf{y}_1 \\ \mathbf{X}_2'\mathbf{y}_2 \\ \vdots \\ \mathbf{X}_m'\mathbf{y}_m \end{pmatrix} = \begin{pmatrix} \hat{\boldsymbol{\beta}}_{1,OLS} \\ \hat{\boldsymbol{\beta}}_{2,OLS} \\ \vdots \\ \hat{\boldsymbol{\beta}}_{m,OLS} \end{pmatrix},$$

where $\hat{\boldsymbol{\beta}}_{i,OLS} = (\mathbf{X}_i'\mathbf{X}_i)^{-1}\mathbf{X}_i'\mathbf{y}_i$. Therefore, the *GLS* estimator (19.5) reduces to *OLS* applied one equation at a time.

The case of identical regressors

The case of identical regressors, namely when $\mathbf{X}_i = \mathbf{X}_j = \mathbf{X}$, for all i and j, is quite common. See, for example, vector autoregressive models described in Chapter 21. In this case, there is no efficiency gain in estimating the equations in (19.1) as a system. Note that when $\mathbf{X}_i = \mathbf{X}_j = \mathbf{X}$ the $\mathbf{G}$ matrix in (19.3) can be written as $\mathbf{G} = \mathbf{I}_m \otimes \mathbf{X}$. Now making use of the properties of products and inverse matrices (reviewed in Section A.8 of Appendix A) the *GLS* estimator, (19.5) can be written equivalently as

$$\begin{aligned} \hat{\boldsymbol{\beta}}_{GLS} &= \left[(\mathbf{I}_m \otimes \mathbf{X}')(\boldsymbol{\Sigma}^{-1} \otimes \mathbf{I}_T)(\mathbf{I}_m \otimes \mathbf{X})\right]^{-1}(\mathbf{I}_m \otimes \mathbf{X}')(\boldsymbol{\Sigma}^{-1} \otimes \mathbf{I}_T)\mathbf{y} \\ &= (\boldsymbol{\Sigma}^{-1} \otimes \mathbf{X}'\mathbf{X})^{-1}(\boldsymbol{\Sigma}^{-1} \otimes \mathbf{X}')\mathbf{y} = \left[\boldsymbol{\Sigma} \otimes (\mathbf{X}'\mathbf{X})^{-1}\right](\boldsymbol{\Sigma}^{-1} \otimes \mathbf{X}')\mathbf{y} \\ &= \left[\mathbf{I}_m \otimes (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\right]\mathbf{y} = (\hat{\boldsymbol{\beta}}'_{1,OLS}, \hat{\boldsymbol{\beta}}'_{2,OLS}, \dots, \hat{\boldsymbol{\beta}}'_{m,OLS})'. \end{aligned} \tag{19.7}$$

Notice that (19.7) is now an m-dimensional vector containing single-equation *OLS* estimators. Therefore, when all equations have the *same* regressors in common, then the *GLS* reduces to the least squares procedure applied to one equation at a time.

19.2.2 System estimation subject to linear restrictions

Consider now the problem of estimating the system of equations (19.1) where the coefficient vectors $\boldsymbol{\beta}_i$, $i = 1, 2, \dots, m$, are subject to the following $r \times 1$ linear restrictions

$$\mathbf{R}\boldsymbol{\beta} = \mathbf{b}, \tag{19.8}$$

where $\mathbf{R}$ and $\mathbf{b}$ are $r \times k$ matrix and $r \times 1$ vector of known constants, and as in Section 19.2, $\boldsymbol{\beta} = (\boldsymbol{\beta}_1', \boldsymbol{\beta}_2', \dots, \boldsymbol{\beta}_m')'$, is a $k \times 1$ vector of unknown coefficients, with $k = \sum_{i=1}^{m} k_i$.

In what follows we distinguish between the cases where the restrictions are applicable to the coefficients $\boldsymbol{\beta}_i$ in each equation separately, and when there are cross-equation restrictions. In the former case, the matrix $\mathbf{R}$ is block diagonal, namely

$$\mathbf{R} = \begin{pmatrix} \mathbf{R}_1 & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{0} & \mathbf{R}_2 & \cdots & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{0} & \mathbf{0} & \cdots & \mathbf{R}_m \end{pmatrix}, \tag{19.9}$$

where $\mathbf{R}_i$ is the $r_i \times k_i$ matrix of known constants applicable to $\boldsymbol{\beta}_i$ only, with $rank(\mathbf{R}_i) = r_i < k_i$. In the more general case, where the restrictions involve coefficients from different equations, $\mathbf{R}$ is not block-diagonal.

Computations of the *ML* estimators of $\boldsymbol{\beta}$ in (19.1) subject to the restrictions in (19.8) can be carried out in the following manner. Initially suppose $\boldsymbol{\Sigma}$ is known and define the $mT \times mT$ matrix $\mathbf{P} = (\mathbf{P}_\sigma \otimes \mathbf{I}_T)$ such that $\mathbf{P}_\sigma \boldsymbol{\Sigma} \mathbf{P}'_\sigma = \mathbf{I}_m$, and hence

$$\mathbf{P} (\boldsymbol{\Sigma} \otimes \mathbf{I}_T) \mathbf{P}' = \mathbf{I}_{mT}, \tag{19.10}$$

where $\mathbf{I}_{mT}$ is an identity matrix of order mT. Such a matrix always exists since $\boldsymbol{\Sigma}$ is a symmetric positive definite matrix. Then compute the transformations

$$\mathbf{G}_* = \mathbf{P}\mathbf{G}, \quad \mathbf{y}_* = \mathbf{P}\mathbf{y}, \tag{19.11}$$

where $\mathbf{G}$ and $\mathbf{y}$ are given by (19.2) and (19.3). Now using familiar results the from estimation of linear regression models subject to linear restrictions, we have (see, for example, Section 1.4 in Amemiya (1985))

$$\widetilde{\boldsymbol{\beta}} = \left(\mathbf{G}'_* \mathbf{G}_*\right)^{-1} \mathbf{G}'_* \mathbf{y}_* - \left(\mathbf{G}'_* \mathbf{G}_*\right)^{-1} \mathbf{R}' \widetilde{\mathbf{q}}, \tag{19.12}$$

where

$$\widetilde{\mathbf{q}} = \left[\mathbf{R} \left(\mathbf{G}'_* \mathbf{G}_*\right)^{-1} \mathbf{R}'\right] \left[\mathbf{R} \left(\mathbf{G}'_* \mathbf{G}_*\right)^{-1} \mathbf{G}'_* \mathbf{y}_* - \mathbf{b}\right]. \tag{19.13}$$

In practice, since $\boldsymbol{\Sigma}$ is not known we need to estimate it. Starting with unrestricted *SURE*, or other initial estimates of $\boldsymbol{\beta}_i$ (say $\hat{\boldsymbol{\beta}}_{i,OLS}$) an initial estimate of $\boldsymbol{\Sigma} = (\sigma_{ij})$ can be obtained. Using the *OLS* estimates of $\boldsymbol{\beta}_i$, the initial estimates of σ_{ij} are given by

$$\hat{\sigma}_{ij,OLS} = \frac{\hat{\mathbf{u}}'_{i,OLS} \hat{\mathbf{u}}_{j,OLS}}{T}, \quad i, j = 1, 2, \ldots, m,$$

where

$$\hat{\mathbf{u}}_{i,OLS} = \mathbf{y}_i - \mathbf{X}_i \hat{\boldsymbol{\beta}}_{i,OLS}, \quad i, j = 1, 2, \ldots, m.$$

With the help of these initial estimates, constrained estimates of $\boldsymbol{\beta}_i$ can be computed using (19.12). Starting from these new estimates of $\boldsymbol{\beta}_i$, another set of estimates for σ_{ij} can then be computed. This process can be repeated until the convergence criteria in (19.21) are met.

The covariance matrix of $\widetilde{\boldsymbol{\beta}}$ in this case is given by

$$\widehat{\text{Var}}(\widetilde{\boldsymbol{\beta}}) = \left(\hat{\mathbf{G}}_*'\hat{\mathbf{G}}_*\right)^{-1} - \left(\hat{\mathbf{G}}_*'\hat{\mathbf{G}}_*\right)^{-1}\mathbf{R}'\left[\mathbf{R}\left(\hat{\mathbf{G}}_*'\hat{\mathbf{G}}_*\right)^{-1}\mathbf{R}'\right]^{-1}\mathbf{R}\left(\hat{\mathbf{G}}_*'\hat{\mathbf{G}}_*\right)^{-1}.$$

Notice that

$$\hat{\mathbf{G}}_*'\hat{\mathbf{G}}_* = \mathbf{G}'\hat{\mathbf{P}}'\hat{\mathbf{P}}\mathbf{G} = \mathbf{G}'\left(\hat{\boldsymbol{\Sigma}}^{-1}\otimes\mathbf{I}_T\right)\mathbf{G}.$$

The (i,j) element of $\hat{\boldsymbol{\Sigma}}$ is computed differently depending on whether matrix $\mathbf{R}$ in (19.9) is block diagonal or not. When $\mathbf{R}$ is block diagonal, σ_{ij} is estimated by

$$\hat{\sigma}_{ij} = \frac{\mathbf{u}_i'\mathbf{u}_j}{\sqrt{(T-s_i)(T-s_j)}}, \qquad i,j = 1,2,\ldots,m, \tag{19.14}$$

where $s_i = k_i - Rank(\mathbf{R}_i) = k_i - r_i$. When $\mathbf{R}$ is not block diagonal, σ_{ij} is estimated by

$$\tilde{\sigma}_{ij} = \frac{\mathbf{u}_i'\mathbf{u}_j}{T}, \qquad i,j = 1,2,\ldots,m. \tag{19.15}$$

In the case where $\mathbf{R}$ is not block diagonal, an appropriate degrees of freedom correction is not available, and hence the *ML* estimator of σ_{ij} is used in the computation of the covariance matrix of the *ML* estimators of $\boldsymbol{\beta}$.

19.2.3 Maximum likelihood estimation of *SURE* models

Let

$$\boldsymbol{\theta} = \left(\boldsymbol{\beta}_1', \boldsymbol{\beta}_2', \ldots, \boldsymbol{\beta}_m', \sigma_{11}, \sigma_{12}, \ldots, \sigma_{1m}; \sigma_{22}, \sigma_{23}, \ldots, \sigma_{2m}; \ldots; \sigma_{mm}\right)',$$

be the $\sum_{i=1}^{m} k_i + m(m+1)/2$ dimensional vector of unknown parameters of equation (19.1). Under the assumption that $\mathbf{u}_t \sim IIDN(\mathbf{0}, \Sigma)$, $t = 1,2,\ldots,T$, the log-likelihood function of the stacked system (19.3) can be written as

$$\ell(\boldsymbol{\theta}) = -\frac{Tm}{2}\log(2\pi) - \tfrac{1}{2}\log|\boldsymbol{\Omega}| - \tfrac{1}{2}\left(\mathbf{y}-\mathbf{G}\boldsymbol{\beta}\right)'\boldsymbol{\Omega}^{-1}\left(\mathbf{y}-\mathbf{G}\boldsymbol{\beta}\right).$$

Since

$$\boldsymbol{\Omega}^{-1} = \boldsymbol{\Sigma}^{-1}\otimes\mathbf{I}_T, \;\; |\boldsymbol{\Omega}| = |\boldsymbol{\Sigma}\otimes\mathbf{I}_T| = |\boldsymbol{\Sigma}|^T|\mathbf{I}_T|^m = |\boldsymbol{\Sigma}|^T,$$

then

$$\ell(\boldsymbol{\theta}) = -\frac{Tm}{2}\log(2\pi) - \frac{T}{2}\log|\boldsymbol{\Sigma}| - \tfrac{1}{2}\left(\mathbf{y}-\mathbf{G}\boldsymbol{\beta}\right)'\left(\boldsymbol{\Sigma}^{-1}\otimes\mathbf{I}_T\right)\left(\mathbf{y}-\mathbf{G}\boldsymbol{\beta}\right). \tag{19.16}$$

Denoting the *ML* estimator of $\boldsymbol{\theta}$ by $\widetilde{\boldsymbol{\theta}} = \left(\widetilde{\boldsymbol{\beta}}_1', \widetilde{\boldsymbol{\beta}}_2', \ldots, \widetilde{\boldsymbol{\beta}}_m', \tilde{\sigma}_{11}, \tilde{\sigma}_{12}, \ldots, \tilde{\sigma}_{1m}; \tilde{\sigma}_{22}, \tilde{\sigma}_{23}, \ldots, \tilde{\sigma}_{2m}; \ldots; \tilde{\sigma}_{mm}\right)'$, it is easily then seen that

$$\tilde{\sigma}_{ij} = \frac{\left(\mathbf{y}_i - \mathbf{X}_i\widetilde{\boldsymbol{\beta}}_i\right)'\left(\mathbf{y}_j - \mathbf{X}_j\widetilde{\boldsymbol{\beta}}_j\right)}{T}, \tag{19.17}$$

and

$$\widetilde{\boldsymbol{\beta}} = \left[\mathbf{G}'\left(\widetilde{\boldsymbol{\Sigma}}^{-1} \otimes \mathbf{I}_T\right)\mathbf{G}\right]^{-1}\mathbf{G}'\left(\widetilde{\boldsymbol{\Sigma}}^{-1} \otimes \mathbf{I}_T\right)\mathbf{y}. \tag{19.18}$$

The covariance matrix of $\widetilde{\boldsymbol{\beta}}$ can be estimated as

$$\widehat{\text{Var}}(\widetilde{\boldsymbol{\beta}}) = \left[\mathbf{G}'\left(\widetilde{\boldsymbol{\Sigma}}^{-1} \otimes \mathbf{I}_T\right)\mathbf{G}\right]^{-1}, \tag{19.19}$$

where $\widetilde{\boldsymbol{\Sigma}}^{-1} = \left(\tilde{\sigma}_{ij}\right)$ with

$$\tilde{\sigma}_{ij} = \frac{\widetilde{\mathbf{u}}_i'\widetilde{\mathbf{u}}_j}{T}, \quad i, j = 1, 2, \ldots, m, \tag{19.20}$$

where $\widetilde{\mathbf{u}}_i = \mathbf{y}_i - \mathbf{X}_i\widetilde{\boldsymbol{\beta}}_i$.

The computation of the *ML* estimators $\widetilde{\boldsymbol{\beta}} = \left(\widetilde{\boldsymbol{\beta}}_1', \widetilde{\boldsymbol{\beta}}_2', \ldots, \widetilde{\boldsymbol{\beta}}_m'\right)'$, and $\tilde{\sigma}_{ij}$, $i, j = 1, 2, \ldots, m$, can be carried out by iterating between (19.17) and (19.18) starting from the *OLS* estimators of $\boldsymbol{\beta}_i$, namely $\hat{\boldsymbol{\beta}}_{i,OLS} = \left(\mathbf{X}_i'\mathbf{X}_i\right)^{-1}\mathbf{X}_i'\mathbf{y}_i$. This iterative procedure is continued until a pre-specified convergence criterion is met. For example, the stopping rule could be the following

$$\sum_{\ell=1}^{k_i}\left|\beta_{i\ell}^{(r)} - \beta_{i\ell}^{(r-1)}\right| < k_i \times 10^{-4}, \quad i = 1, 2, \ldots, m, \tag{19.21}$$

where $\beta_{i\ell}^{(r)}$ stands for the estimate of the ℓth element of $\boldsymbol{\beta}_i$ at the rth iteration.

The maximized value of the system log-likelihood function is given by

$$\ell(\widetilde{\boldsymbol{\theta}}) = -\frac{Tm}{2}\log(2\pi) - \tfrac{T}{2}\log\left|\widetilde{\boldsymbol{\Sigma}}\right|. \tag{19.22}$$

Example 37 (Grunfeld's investment equation I) *In an important study of investment demand, Grunfeld (1960) and Grunfeld and Griliches (1960) estimated investment equations for ten firms in the US economy over the period 1935–1954. Here we estimate investment equations for five of these firms by the SURE method, namely for General Motors (GM), Chrysler (CH), General Electric (GE), Westinghouse (WE) and US Steel (USS). This smaller data set is also analysed in Greene (2002). For example, GMI refers to General Motors' gross investment, WEF to the market value of Westinghouse, and CHC to the stock of plant and equipment of Chrysler. The SURE model to be estimated is given by*

$$I_{it} = \beta_{i1} + \beta_{i2} F_{it} + \beta_{i3} C_{it} + u_{it}, \tag{19.23}$$

for i = GM, CH, GE, WE, and USS, and $t = 1935, 1936, \ldots, 1954$. The results for Chrysler reported in Table 19.1 (in the table the above variables are denoted by adding the prefix CH to the variable names). Except for the intercept term, the results in this table are comparable with the SURE estimates for the same equations reported in Table 14.3 in Greene (2002). See also Example 57.

Table 19.1 *SURE* estimates of the investment equation for the Chrysler company

Seemingly unrelated regressions estimation **The estimation method converged after 18 iterations**			
Dependent variable is CHI 20 observations used for estimation from 1935 to 1954			
Regressor	Coefficient	Standard Error	T-Ratio[Prob]
CONST	2.3783	12.6160	.18852 [.853]
CHF	.067451	.018550	3.6362 [.002]
CHC	.30507	.028274	10.7898 [.000]
R-Squared	.91057	R-Bar-Squared	.90004
S.E. of Regression	13.5081	F-Stat. F (2,17)	86.5413 [.000]
Mean of Dependent Variable	86.1235	S.D. of Dependent Variable	42.7256
Residual Sum of Squares	3102.0	Equation Log-likelihood	−78.8193
DW-statistic	1.8851	System Log-likelihood	−459.0922
System AIC	−474.0922	System SBC	−481.5602

19.2.4 Testing linear/nonlinear restrictions

Under fairly general conditions (for a fixed m and as $T \to \infty$), the *ML* estimators, $\widetilde{\boldsymbol{\beta}} = (\widetilde{\boldsymbol{\beta}}_1', \widetilde{\boldsymbol{\beta}}_2', \ldots, \widetilde{\boldsymbol{\beta}}_m')'$, are asymptotically normally distributed with mean $\widetilde{\boldsymbol{\beta}}$ and the covariance matrix given by (19.19). It is therefore possible to test linear or nonlinear restrictions on the elements of $\boldsymbol{\beta}$ using the Wald procedure. Notice that the restrictions to be tested could involve coefficients from different equations (i.e., they could be cross-equation restrictions). To be more precise, suppose it is of interest to test the following $r \times 1$ general nonlinear restrictions on $\boldsymbol{\beta}$

$$H_0 : \mathbf{h}(\boldsymbol{\beta}) = \mathbf{0},$$
$$H_1 : \mathbf{h}(\boldsymbol{\beta}) \neq \mathbf{0},$$

where $\mathbf{h}(\boldsymbol{\beta})$ is the known $r \times 1$ vector function of $\boldsymbol{\beta}$, with continuous partial derivatives. The Wald statistic for testing the null $H_0 : \mathbf{h}(\boldsymbol{\beta}) = \mathbf{0}$ against the two-sided alternatives, $H_1 : \mathbf{h}(\boldsymbol{\beta}) \neq \mathbf{0}$ is given by

$$W = \mathbf{h}(\widetilde{\boldsymbol{\beta}})' \left[\mathbf{H}(\widetilde{\boldsymbol{\beta}}) \widehat{Cov}(\widetilde{\boldsymbol{\beta}}) \mathbf{H}'(\widetilde{\boldsymbol{\beta}})\right]^{-1} \mathbf{h}(\widetilde{\boldsymbol{\beta}}), \tag{19.24}$$

where $\mathbf{H}(\widetilde{\boldsymbol{\beta}})$ is given by $\partial \mathbf{h}(\boldsymbol{\beta})/\partial \boldsymbol{\beta}'$ at $\boldsymbol{\beta} = \widetilde{\boldsymbol{\beta}}$. It will be assumed that $Rank(\mathbf{H}(\boldsymbol{\beta})) = r$. Under H_0, W statistic is asymptotically distributed as chi-square with r degrees of freedom.

Example 38 *Consider the investment equations for five US firms estimated in Example 37. We are now interested in testing the hypothesis that the coefficients of F_{it}, the market value of the firms, are the same across all the five companies. In terms of the coefficients of the equations in (19.23), the relevant null hypothesis is*

$$H_0 : \beta_{i2} = \beta_2, \quad for \quad i = GM,\ CH,\ GE,\ WE,\ USS.$$

These four restrictions clearly involve coefficients from all the five equations. We report the test results in Table 19.2. The LR statistic for testing these restrictions is 20.46 which is well above the 95 per cent critical value of the chi-squared distribution with 4 degrees of freedom, and we therefore strongly reject the slope homogeneity hypothesis.

Table 19.2 Testing the slope homogeneity hypothesis

Wald test of restriction(s) imposed on parameters
The underlying estimated model is: GMI const gmf gmc; CHI const chf chc; GEI const gef gec ; WEI const wef wec; USSI const ussf ussc 20 observations used for estimation from 1935 to 1954
List of restriction(s) for the Wald test: A2=B2; B2=C2; C2=D2; D2=E2
Wald Statistic CHSQ(4)= 20.4580 [. 000]

19.2.5 *LR* statistic for testing whether $\boldsymbol{\Sigma}$ is diagonal

Suppose it is of interest to test the hypothesis

$$\begin{aligned} H_0 : \quad \sigma_{12} = \quad & \sigma_{13} = \cdots = \quad \sigma_{1m} = 0, \\ & \sigma_{23} = \cdots = \quad \sigma_{2m} = 0, \\ & \ddots \\ & \qquad\qquad\quad \sigma_{mm} = 0, \end{aligned}$$

against the alternative that one or more of the off-diagonal elements of $\boldsymbol{\Sigma}$ are non-zero, namely the hypothesis that the errors from different regressions are uncorrelated. One possibility would be to use the log-likelihood ratio statistic which is given by

$$LR = 2\left[\ell(\widetilde{\boldsymbol{\theta}}) - \sum_{i=1}^{m} \ell_i(\hat{\boldsymbol{\theta}}_{i,OLS})\right], \tag{19.25}$$

where $\ell(\widetilde{\boldsymbol{\theta}})$ is given by (19.22) and $\ell_i(\hat{\boldsymbol{\theta}}_{i,OLS})$ is the log-likelihood function of the ith equation evaluated at the *OLS* estimators. Using (19.22), we have

$$LR = T\left[\sum_{i=1}^{m} \log \tilde{\sigma}_{ii}^2 - \log\left|\widetilde{\boldsymbol{\Sigma}}\right|\right], \tag{19.26}$$

where

$$\tilde{\sigma}_{ii} = T^{-1}\left(\mathbf{y}_i - \mathbf{X}_i\hat{\boldsymbol{\beta}}_{i,OLS}\right)'\left(\mathbf{y}_i - \mathbf{X}_i\hat{\boldsymbol{\beta}}_{i,OLS}\right).$$

Under H_0, LR is asymptotically distributed as a χ^2 with $m(m-1)/2$ degrees of freedom.

Alternatively the *LM* statistic proposed by Breusch and Pagan (1980) can be used which is given by

$$LM = T\sum_{i=2}^{m}\sum_{j=1}^{i-1}\hat{\rho}_{ij}^2,$$

where $\hat{\rho}_{ij} = \tilde{\sigma}_{ij,OLS}/\left\{\tilde{\sigma}_{ii,OLS}\tilde{\sigma}_{jj,OLS}\right\}^{\frac{1}{2}}$ is the pair-wise correlation coefficient of the residuals from regression equations i and j. This statistic is also asymptotically distributed as a χ^2 with $m(m-1)/2$ degrees of freedom, for a fixed m and as $T \to \infty$.

These tests of cross-equation error uncorrelatedness are asymptotically equivalent for a fixed m and as $T \to \infty$. They tend, however, to over-reject when m is relatively large and should be used when m is small and T large. In cases where m is also large, a bias-corrected version of the *LM* test is proposed by Pesaran, Ullah, and Yamagata (2008). See also Section 29.7.

Example 39 *We now test the hypothesis of a diagonal error covariance matrix in the context of the investment equation estimated in Example 37. For this purpose, we need to estimate the five individual equations separately by the OLS method, and then employ the log-likelihood ratio procedure. The maximized log-likelihood values for the five equations estimated separately are for General Motors (−117.1418), Chrysler (−78.4766), General Electric (−93.3137), Westinghouse (−73.2271) and US Steel (−119.3128), respectively, yielding the restricted log-likelihood value of*

$$-481.472\,(= -117.1418 - 78.4766 - 93.3137 - 73.2271 - 119.3128).$$

The maximized log-likelihood value for the unrestricted system (namely, when the error covariance matrix is not restricted) is given at the bottom right-hand corner of Table 19.1, under 'System Log-likelihood' (= −459.0922). Therefore, the log-likelihood ratio statistic for testing the diagonality of the error covariance matrix is given by $LR = 2(-459.0922 + 481.472) = 44.76$, *which is asymptotically distributed as a chi-squared variate with* $5\,(5-1)/2 = 10$ *degrees of freedom. The 95 per cent critical value of the chi-squared distribution with* 10 *degrees of freedom is* 19.31. *We therefore reject the hypothesis that the error covariance matrix of the five investment equations is diagonal, which provides support for the application of the SURE technique to this problem.*

Table 19.3 Estimated system covariance matrix of errors for Grunfeld–Griliches investment equations

	GMI	CHI	GEI	WEI	USSI
GMI	8600.8	−389.2322	644.4398	139.1155	−3394.4
CHI	−389.2322	182.4680	13.6558	22.1336	544.8885
GEI	644.4398	13.6558	873.1736	259.9662	1663.1
WEI	139.1155	22.1336	259.9662	121.7357	868.3544
USSI	−3394.4	544.8885	1663.1	868.3544	11401.0

Table 19.3 reports the estimated error covariance matrix. The covariance estimates on the off-diagonal elements are quite large relative to the respective diagonal elements.

19.3 System of equations with endogenous variables

This approach is useful when the dependent variables in some or all equations of system (19.1) also appear as regressors (see Zellner and Theil (1962)). In this case, we have the following system of simultaneous equations

$$\begin{aligned} \mathbf{y}_i &= \mathbf{X}_i\boldsymbol{\beta}_i + \mathbf{Y}_i\boldsymbol{\gamma}_i + \mathbf{u}_i, \\ &= \mathbf{W}_i\boldsymbol{\delta}_i + \mathbf{u}_i,\ i = 1, 2, \ldots, m, \end{aligned} \tag{19.27}$$

where $\mathbf{y}_i$ is a $T \times 1$ vector of observations on the 'normalized' endogenous variable of the ith equation in the system, $\mathbf{X}_i$ are the $T \times k_i$ vector of observations on the exogenous variables in the ith equation, $\mathbf{Y}_i$ is the $T \times p_i$ vector of endogenous variables in the ith equation whose coefficients are not normalized, $\mathbf{W}_i$ is $T \times s_i$ where $s_i = k_i + p_i$, and $\boldsymbol{\delta}_i = \left(\boldsymbol{\beta}_i', \boldsymbol{\gamma}_i'\right)'$. From above, it follows that the ith equation has $p_i + 1$ endogenous and k_i exogenous variables. It is further assumed that $E(\mathbf{u}_i\mathbf{u}_i') = \sigma_{ii}\mathbf{I}_T$.

The choice of the variable to normalize (or, equivalently, the choice of the left-hand-side variable) can be important in practice and is assumed to be guided by economic theory or other forms of *a priori* information. The order condition for identification of parameters of equation i is given by $k - k_i \geq p_i$, namely the number of excluded exogenous variables in equation i must be at least as large as the number of included endogenous variables less than one (the normalization constant applied to $\mathbf{y}_i$).

Denote the union intersection of the exogenous regressors of the system by the $T \times k$ matrix $\mathbf{X}$ (with $k \leq \sum_{i=1}^{m} k_i$)

$$\mathbf{X} = \bigcup_{i=1}^{m} \mathbf{X}_i. \tag{19.28}$$

Each $\mathbf{X}_i$ is a subset of $\mathbf{X}$. Similarly, each $\mathbf{Y}_i$ is a subset of the $T \times m$ matrix $\mathbf{Y} = (\mathbf{y}_1, \mathbf{y}_2, \ldots, \mathbf{y}_m)$, with $p_i < m$. It will also prove helpful to define the (known) selection matrices, $\mathbf{H}_i$ and $\mathbf{G}_i$ such that

$$\mathbf{X}_i = \mathbf{X}\mathbf{H}_i, \text{ and } \mathbf{Y}_i = \mathbf{Y}\mathbf{G}_i. \tag{19.29}$$

Finally, to derive the relationship between the structural form parameters, $(\boldsymbol{\beta}_i, \boldsymbol{\gamma}_i)$ and the reduced form parameters (to be defined below) we first note that (19.27) can be written as

$$\mathbf{y}_i = (\mathbf{X}\mathbf{H}_i)\,\boldsymbol{\beta}_i + (\mathbf{Y}\mathbf{G}_i)\,\boldsymbol{\gamma}_i + \mathbf{u}_i,$$

for $i = 1, 2, \ldots, m$. Stacking these equations as columns we have

$$\mathbf{Y} = \mathbf{X}\mathbf{H}_\beta + \mathbf{Y}\mathbf{G}_\gamma + \mathbf{U},$$

where

$$\begin{aligned} \mathbf{H}_\beta &= (\mathbf{H}_1\boldsymbol{\beta}_1, \mathbf{H}_2\boldsymbol{\beta}_2, \ldots, \mathbf{H}_m\boldsymbol{\beta}_m), \\ \mathbf{G}_\gamma &= \left(\mathbf{G}_1\boldsymbol{\gamma}_1, \mathbf{G}_2\boldsymbol{\gamma}_2, \ldots, \mathbf{G}_m\boldsymbol{\gamma}_m\right), \\ \mathbf{U} &= (\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_m). \end{aligned} \tag{19.30}$$

Hence, the reduced form model (associated to the structural model (19.27)) is

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\Pi} + \mathbf{V}, \tag{19.31}$$

where $\boldsymbol{\Pi} = (\pi_{ij})$ is the $k \times m$ matrix of reduced form coefficients defined by

$$\boldsymbol{\Pi} = \mathbf{H}_\beta \left(\mathbf{I}_m - \mathbf{G}_\gamma\right)^{-1}, \text{and } \mathbf{V} = \mathbf{U}\left(\mathbf{I}_m - \mathbf{G}_\gamma\right)^{-1}. \tag{19.32}$$

19.3.1 Two- and three-stage least squares

Under the exogeneity assumption, $E(\mathbf{U}|\mathbf{X}) = \mathbf{0}$, the reduced form parameters, $\boldsymbol{\Pi}$, in (19.31) can be estimated consistently by *OLS*, even though such estimates will not be efficient noting that $\boldsymbol{\Pi}$ is subject to the restrictions given by (19.32). The *OLS* estimator of $\boldsymbol{\Pi}$ is given by

$$\hat{\boldsymbol{\Pi}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y},$$

assuming that $\mathbf{X}'\mathbf{X}$ is a positive definite matrix. Using these estimates in (19.29), $\mathbf{Y}_i$ (which enters on the right-hand side of (19.27)) can then be consistently estimated by $\hat{\mathbf{Y}}_i = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-\mathbf{1}}\mathbf{X}'\mathbf{Y}_i$. Using these estimates the familiar two-stage least squares (2*SLS*) estimator of $\boldsymbol{\delta}_i$ can then be written as

$$\hat{\boldsymbol{\delta}}_{i,2SLS} = \left(\hat{\mathbf{W}}_i'\hat{\mathbf{W}}_i\right)^{-1}\hat{\mathbf{W}}_i'\mathbf{y}_i,$$

where

$$\hat{\mathbf{W}}_i = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{W}_i. \tag{19.33}$$

The $2SLS$ is consistent if $T^{-1}\hat{\mathbf{W}}_i'\hat{\mathbf{W}}_i$ tends to a positive definite matrix. The order condition $k - k_i \geq p_i$ is necessary but not sufficient. To see this note that

$$T^{-1}\hat{\mathbf{W}}_i'\hat{\mathbf{W}}_i = T^{-1}\mathbf{W}_i'\mathbf{P}_X\mathbf{W}_i,\ T^{-1}\hat{\mathbf{W}}'_i\mathbf{y}_i = \mathbf{W}_i'\mathbf{P}_X\left(\mathbf{W}_i\boldsymbol{\delta}_i + \mathbf{u}_i\right),$$

where $\mathbf{P}_X = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$. Hence

$$\hat{\boldsymbol{\delta}}_{i,2SLS} - \boldsymbol{\delta}_i = \left(T^{-1}\mathbf{W}_i'\mathbf{P}_X\mathbf{W}_i\right)\left(T^{-1}\mathbf{W}_i'\mathbf{P}_X\mathbf{u}_i\right).$$

But

$$T^{-1}\mathbf{W}_i'\mathbf{P}_X\mathbf{W}_i = \left(T^{-1}\mathbf{W}_i'\mathbf{X}\right)\left(T^{-1}\mathbf{X}'\mathbf{X}\right)^{-1}\left(T^{-1}\mathbf{X}'\mathbf{W}_i\right),$$
$$T^{-1}\mathbf{W}_i'\mathbf{P}_X\mathbf{u}_i = \left(T^{-1}\mathbf{W}_i'\mathbf{X}\right)\left(T^{-1}\mathbf{X}'\mathbf{X}\right)^{-1}\left(T^{-1}\mathbf{X}'\mathbf{u}_i\right).$$

Under the assumptions that $T^{-1}\mathbf{X}'\mathbf{X}$ tends to a positive definite matrix, and $T^{-1}\mathbf{X}'\mathbf{u}_i \rightarrow_p \mathbf{0}$, then it follows that $\hat{\boldsymbol{\delta}}_{i,2SLS}$ is a consistent estimator of $\boldsymbol{\delta}_i$ if there exists T_0 such that for all $T > T_0$ the $k \times \left(k_i + p_i\right)$ matrix $T^{-1}\mathbf{X}'\mathbf{W}_i$ is full rank. Since $rank\left(T^{-1}\mathbf{X}'\mathbf{W}_i\right) \leq \min(k, k_i + p_i)$, the order condition is met if $k_i + p_i \leq k$. When the rank condition is satisfied and in addition $E(\mathbf{u}_i\mathbf{u}_i') = \sigma_{ii}\mathbf{I}_T$, it also follows that

$$\sqrt{T}\left(\hat{\boldsymbol{\delta}}_{i,2SLS} - \boldsymbol{\delta}_i\right) \rightarrow_d N(\mathbf{0}, \Sigma_{\delta_i}),$$

where $\boldsymbol{\Sigma}_{\delta_i}$ is consistently estimated by

$$\hat{\Sigma}_{\delta_i} = \hat{\sigma}_{ii}\left(T^{-1}\mathbf{W}_i'\mathbf{P}_X\mathbf{W}_i\right)^{-1},$$

with $\hat{\sigma}_{ii} = T^{-1}\left(\mathbf{y}_i - \mathbf{W}_i\hat{\boldsymbol{\delta}}_{i,2SLS}\right)'\left(\mathbf{y}_i - \mathbf{W}_i\hat{\boldsymbol{\delta}}_{i,2SLS}\right)$. Some of the assumptions of the $2SLS$ can be relaxed using the generalized method of moments estimator. See Chapter 10, in particular Section 10.8.

The three-stage least squares $(3SLS)$ can then be computed using $\hat{\mathbf{W}}_i$ in the following system of equations by the *SURE* procedure

$$\mathbf{y}_i = \hat{\mathbf{W}}_i\boldsymbol{\delta}_i + \boldsymbol{\xi}_i, \text{ for } i = 1, 2, \ldots, m. \tag{19.34}$$

To obtain an explicit expression for the $3SLS$ estimator stack the m equations as

$$\mathbf{y} = \hat{\mathbf{W}}\boldsymbol{\delta} + \boldsymbol{\xi}, \tag{19.35}$$

where

$$\mathbf{y} = \begin{pmatrix} \mathbf{y}_1 \\ \mathbf{y}_2 \\ \vdots \\ \mathbf{y}_m \end{pmatrix}, \hat{\mathbf{W}} = \begin{pmatrix} \hat{\mathbf{W}}_1 & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{0} & \hat{\mathbf{W}}_2 & \cdots & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{0} & \mathbf{0} & \cdots & \hat{\mathbf{W}}_m \end{pmatrix},$$

$$\boldsymbol{\delta} = \begin{pmatrix} \boldsymbol{\delta}_1 \\ \boldsymbol{\delta}_2 \\ \vdots \\ \boldsymbol{\delta}_m \end{pmatrix}, \boldsymbol{\xi} = \begin{pmatrix} \boldsymbol{\xi}_1 \\ \boldsymbol{\xi}_2 \\ \vdots \\ \boldsymbol{\xi}_m \end{pmatrix}.$$

Then

$$\hat{\boldsymbol{\delta}}_{3SLS} = \left[\hat{\mathbf{W}}'\left(\hat{\boldsymbol{\Sigma}}^{-1} \otimes \mathbf{I}_T\right)\hat{\mathbf{W}}\right]^{-1}\hat{\mathbf{W}}'\left(\hat{\boldsymbol{\Sigma}}^{-1} \otimes \mathbf{I}_T\right)\mathbf{y}, \tag{19.36}$$

with

$$\hat{\boldsymbol{\Sigma}} = (\hat{\sigma}_{ij}),\ \hat{\sigma}_{ij} = \frac{\left(\mathbf{y}_i - \mathbf{W}_i\hat{\boldsymbol{\delta}}_{i,2SLS}\right)'\left(\mathbf{y}_j - \mathbf{W}_j\hat{\boldsymbol{\delta}}_{j,2SLS}\right)}{T}. \tag{19.37}$$

$$\hat{\boldsymbol{\delta}}_{i,2SLS} = \left(\hat{\mathbf{W}}_i'\hat{\mathbf{W}}_i\right)\hat{\mathbf{W}}_i'\mathbf{y}_i. \tag{19.38}$$

The covariance matrix of the $3SLS$ estimator is given by

$$Var\left(\hat{\boldsymbol{\delta}}_{3SLS}\right) = \left[\hat{\mathbf{W}}'\left(\hat{\boldsymbol{\Sigma}}^{-1} \otimes \mathbf{I}_T\right)\hat{\mathbf{W}}\right]^{-1}. \tag{19.39}$$

The estimates $\hat{\boldsymbol{\Sigma}}$ and $\hat{\boldsymbol{\delta}}_{3SLS}$ can be updated iteratively until convergence is achieved, as in the *SURE* estimation (see Section 19.2.3).

19.3.2 Iterated instrumental variables estimator

A special case occurs when the number of exogenous variables (or potential instruments) exceeds the number of available observations, namely, $k_i \geq T$. In this case, the $2SLS$ estimator coincides with the *OLS* estimator, which is inconsistent. This problem can be avoided by using a subset of potential instruments, say the $T \times r_i$ matrix $\mathbf{Z}_i$ composed of $\mathbf{X}_i$ and a subset of $(\mathbf{X}_1, \mathbf{X}_2, \ldots, \mathbf{X}_{i-1}, \mathbf{X}_{i+1}, \ldots, \mathbf{X}_m)$. The investigator must specify the instruments $\mathbf{Z}_i$, $i = 1, 2, \ldots, m$, or equivalently the selection $k \times r_i$ $(r_i < T)$ matrices, $\mathbf{L}_i$ such that[2]

$$\mathbf{Z}_i = \mathbf{X}\mathbf{L}_i.$$

Note that the elements of $\mathbf{L}_i$ are known. The *IV* estimator of $\boldsymbol{\delta}_i$ with $\mathbf{Z}_i$ as instruments is given by

[2] The sub-set selection can be carried out using reduced rank regression techniques reviewed in Section 19.7.

$$\hat{\boldsymbol{\delta}}_{i,IV} = \left(\hat{\mathbf{W}}_{zi}'\hat{\mathbf{W}}_{zi}\right)^{-1}\hat{\mathbf{W}}_{zi}'\mathbf{y}_i,$$

where

$$\hat{\mathbf{W}}_{zi} = \mathbf{Z}_i(\mathbf{Z}_i'\mathbf{Z}_i)^{-1}\mathbf{Z}_i'\mathbf{W}_i.$$

Using $\hat{\boldsymbol{\delta}}_{i,IV}$, $i = 1, 2, \ldots, m$, an initial *IV* estimator of $\boldsymbol{\Sigma}$ with elements

$$\tilde{\sigma}_{ij} = \frac{\left(\mathbf{y}_i - \mathbf{W}_i\hat{\boldsymbol{\delta}}_{i,IV}\right)'\left(\mathbf{y}_j - \mathbf{W}_j\hat{\boldsymbol{\delta}}_{j,IV}\right)}{T},$$

can be computed. Using these estimates an initial system *IV* estimator of $\boldsymbol{\delta}$ is given by

$$\hat{\boldsymbol{\delta}}_{(1)IV} = \left[\hat{\mathbf{W}}_z'\left(\tilde{\boldsymbol{\Sigma}}^{-1} \otimes \mathbf{I}_T\right)\hat{\mathbf{W}}_z\right]^{-1}\hat{\mathbf{W}}_z'\left(\tilde{\boldsymbol{\Sigma}}^{-1} \otimes \mathbf{I}_T\right)\mathbf{y}, \tag{19.40}$$

where

$$\hat{\mathbf{W}}_z = \begin{pmatrix} \hat{\mathbf{W}}_{z1} & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{0} & \hat{\mathbf{W}}_{z2} & \cdots & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{0} & \mathbf{0} & \cdots & \hat{\mathbf{W}}_{zm} \end{pmatrix}.$$

This estimate can be updated using $\boldsymbol{\Pi} = \mathbf{H}_\beta\left(\mathbf{I}_m - \mathbf{G}_\gamma\right)^{-1}$, where $\mathbf{H}_\beta$ and $\mathbf{G}_\gamma$ are estimated using $\hat{\boldsymbol{\delta}}_{(1)IV}$, namely $\hat{\boldsymbol{\Pi}}_{(1)IV} = \mathbf{H}_{\beta(1),IV}\left(\mathbf{I}_m - \mathbf{G}_{\gamma(1),IV}\right)^{-1}$. Then compute

$$\hat{\mathbf{W}}_{i(1)} = \left(\mathbf{X}_i, \boldsymbol{\Pi}_{i,(1)IV}\mathbf{X}\right), \tag{19.41}$$

where $\boldsymbol{\Pi}_{i,(1)IV}$ is the ith row of $\hat{\boldsymbol{\Pi}}_{(1)IV}$. The second iteration of system *IV* estimator is given by

$$\hat{\boldsymbol{\delta}}_{(2)IV} = \left[\hat{\mathbf{W}}_{(1)}'\left(\tilde{\boldsymbol{\Sigma}}_{(1)}^{-1} \otimes \mathbf{I}_T\right)\hat{\mathbf{W}}_{(1)}\right]^{-1}\hat{\mathbf{W}}_{(1)}'\left(\tilde{\boldsymbol{\Sigma}}_{(1)}^{-1} \otimes \mathbf{I}_T\right)\mathbf{y}, \tag{19.42}$$

where

$$\hat{\mathbf{W}}_{(1)} = \begin{pmatrix} \hat{\mathbf{W}}_{1(1)} & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{0} & \hat{\mathbf{W}}_{2(1)} & \cdots & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{0} & \mathbf{0} & \cdots & \hat{\mathbf{W}}_{m(1)} \end{pmatrix},$$

and the (i, j) element of $\tilde{\boldsymbol{\Sigma}}_{(1)}$ is given by

$$\tilde{\sigma}_{ij(1)} = \frac{\left(\mathbf{y}_i - \mathbf{W}_i\hat{\boldsymbol{\delta}}_{i,(1)IV}\right)'\left(\mathbf{y}_j - \mathbf{W}_j\hat{\boldsymbol{\delta}}_{j,(1)IV}\right)}{T}.$$

The iterations can be continued to obtain a fully iterated system *IV* estimator.

19.4 Principal components

Principal components (PC) analysis is a technique for reducing the dimensionality of a data set consisting of a number of interrelated variables, whilst simultaneously preserving as much of the variation present in the data as possible. This is achieved by transforming the original variables to a fewer number of uncorrelated variables, known as the principal components, which are ordered so that the first few retain most of the variation present in all of the original variables.

Let $\mathbf{Y} = \left(\mathbf{y}_1, \mathbf{y}_2, \ldots, \mathbf{y}_T\right)'$ with $\mathbf{y}_t = \left(y_{1t}, y_{2t}, \ldots, y_{mt}\right)'$ be the $T \times m$ matrix of T observations on m variables, and assume that $T > m$. To simplify the analysis, also suppose that the elements of $\mathbf{y}_t$ have zero means and consider the $m \times m$ sample covariance matrix

$$\mathbf{S}_T = \frac{\mathbf{Y}'\mathbf{Y}}{T}.$$

Let $\hat{\mathbf{c}}_1 = (\hat{c}_{11}, \hat{c}_{12}, \ldots, \hat{c}_{1m})'$ be an m-dimensional real valued vector. The first principal component is defined by taking the linear combination of the elements of $\mathbf{y}_t$

$$\hat{p}_{1t} = \hat{\mathbf{c}}_1'\mathbf{y}_t = \hat{c}_{11}y_{1t} + \hat{c}_{12}y_{2t} + \ldots + \hat{c}_{1m}y_{mt}, \;\; t = 1, 2, \ldots, T,$$

having maximum variance subject to the constrain, $\hat{\mathbf{c}}_1'\hat{\mathbf{c}}_1 = 1$. Note that $\hat{\mathbf{p}}_1 = \left(\hat{p}_{11}, \hat{p}_{12}, \ldots, \hat{p}_{1T}\right)'$ is a T-dimensional vector, where the generic element, $\hat{p}_{1t}$, is called *score* for the tth observation on the first principal component, $\hat{\mathbf{p}}_1$. To derive the linear combination, $\mathbf{c}'\mathbf{y}_t$, which yields maximum variance subject to the normalization restriction, $\mathbf{c}'\mathbf{c} = 1$, we solve the following constrained optimization problem

$$\max_{\mathbf{c},\lambda} \left[\mathbf{c}'\boldsymbol{\Sigma}_{yy}\mathbf{c} - \lambda(\mathbf{c}'\mathbf{c} - 1)\right],$$

where $\boldsymbol{\Sigma}_{yy}$ is the population variance of $\mathbf{y}_t$, namely $Var(\mathbf{y}_t) = \boldsymbol{\Sigma}_{yy}$, and λ is a Lagrange multiplier for the restriction. The first-order conditions for the above optimization problem are given by

$$\begin{aligned} \left(\boldsymbol{\Sigma}_{yy} - \lambda\right)\mathbf{c} &= \mathbf{0}, \\ \mathbf{c}'\mathbf{c} &= 1. \end{aligned}$$

For a non-trivial solution, with $\mathbf{c} \neq \mathbf{0}$, we have

$$\begin{aligned} \mathbf{c}'\left(\boldsymbol{\Sigma}_{yy} - \lambda\right)\mathbf{c} &= \mathbf{c}'\boldsymbol{\Sigma}_{yy}\mathbf{c} - \lambda\mathbf{c}'\mathbf{c} \\ &= \mathbf{c}'\boldsymbol{\Sigma}_{yy}\mathbf{c} - \lambda = 0. \end{aligned}$$

Therefore, the first (population) PC, denoted by $\mathbf{c}_1$, is given by a suitably normalized eigenvector associated to the largest eigenvalue, denoted by λ_1, of $\boldsymbol{\Sigma}_{yy}$. The estimate of $\mathbf{c}_1$, denoted by $\hat{\mathbf{c}}_1$, is based on the sample estimate of $\boldsymbol{\Sigma}_{yy}$, which is given by $\mathbf{S}_T = T^{-1}\mathbf{Y}'\mathbf{Y}$.[3]

[3] In cases where m is large, one could also base the estimate of $\mathbf{c}_1$ on a regularized estimate of $\boldsymbol{\Sigma}_{yy}$.

Similarly, the second principal component is defined as the linear combination of the elements of $\mathbf{y}_t$, $\hat{p}_{2t} = \hat{\mathbf{c}}_2'\mathbf{y}_t$, having maximum variance, subject to the constraints $\hat{\mathbf{c}}_2'\hat{\mathbf{c}}_2 = 1$, and $Cov(\hat{p}_{1t}, \hat{p}_{2t}) = 0$. Again, we can compute T linear combinations to obtain the vector $\hat{\mathbf{p}}_2 = \left(\hat{p}_{21}, \hat{p}_{22}, \ldots, \hat{p}_{2T}\right)'$. The kth principal component is defined as the linear combination of the elements of $\mathbf{y}_t$, $\hat{p}_{kt} = \hat{\mathbf{c}}_k'\mathbf{y}_t$, having maximum variance, subject to the constraints $\hat{\mathbf{c}}_k'\hat{\mathbf{c}}_k = 1$, and $Cov(\hat{p}_{kt}, \hat{p}_{ht}) = 0$, for $h = 1, 2, \ldots, k - 1$. In this way, we can obtain m principal components. Let

$$\lambda_1 \geq \lambda_2 \geq \ldots \geq \lambda_m \geq 0,$$

be the m eigenvalues of $\mathbf{S}_T$, in a descending order. It is possible to prove that the vector of coefficients $\hat{\mathbf{c}}_k$ for the kth principal component, $\hat{\mathbf{p}}_k$, is given by the eigenvector of $\mathbf{S}_T$ corresponding to λ_k, satisfying

$$\hat{\mathbf{c}}_k'\hat{\mathbf{c}}_k = 1, \ k = 1, \ldots, m,$$
$$\hat{\mathbf{c}}_k'\hat{\mathbf{c}}_h = 0, \ k \neq h.$$

Since the sample covariance matrix $\mathbf{S}_T$ is non-negative definite, it has spectral decomposition (see Section A.5 in Appendix A). Using such decomposition, it is easy to prove that

$$E\left(\hat{\mathbf{p}}_k'\hat{\mathbf{p}}_k\right) = \lambda_k,$$

where λ_k is the kth largest eigenvalue of $\mathbf{S}_T$. If $m > T$, eigenvalues and principal components can be computed using the $T \times T$ matrix $m^{-1}\mathbf{Y}'\mathbf{Y}$.

It is also possible to estimate principal components for $\mathbf{Y}$, once these have been filtered by a set of variables, contained in a $T \times s$ matrix $\mathbf{X}$, that might influence $\mathbf{Y}$. In this case, principal components are computed from eigenvectors and eigenvalues of

$$\mathbf{S}_T = \frac{\mathbf{Y}'\mathbf{M}_X\mathbf{Y}}{T},$$

where $\mathbf{M}_X = \mathbf{I}_T - \mathbf{X}\left(\mathbf{X}'\mathbf{X}\right)^{-1}\mathbf{X}'$, and $\mathbf{I}_T$ is a $T \times T$ identity matrix. For example, in the case where the means of y_{it} are unknown, $\mathbf{X}$ can be chosen to be a vector of ones, namely by setting $\mathbf{M}_X = \mathbf{M}_\tau = \mathbf{I}_T - \boldsymbol{\tau}(\boldsymbol{\tau}'\boldsymbol{\tau})^{-1}\boldsymbol{\tau}$, where $\boldsymbol{\tau}$ is a $T \times 1$ vector of ones.

There are a number of methods that can be used to select, $k < T$, the number of *PC*'s or factors. The simplest and most popular procedures are the Kaiser (1960) criterion and the scree test. To use the Kaiser criterion the observations are standardized so that the variables have unit variances (in sample), and hence $\sum_{i=1}^{m} \lambda_i = m$ (when $T > m$). According to this criterion one would then retain only factors with eigenvalues greater than 1. In effect only factors that explain as much as the equivalent of one original variable are retained.

The scree test is based on a graphical method, first proposed by Cattell (1966). A simple line plot of the eigenvalues is used to identify a visual break in this plot. There is no formal method for identifying the threshold, and a certain degree of personal judgement is required.

For comprehensive treatments of the *PC* literature see Chapter 11 of Anderson (2003), and Jolliffe (2004).

19.5 Common factor models

The common factor model was originally developed in the psychometric literature to measure cognitive abilities on the basis of observation on a large number of individual characteristics (Spearman (1904); Garnett (1920)). Instead of directly shrinking the observations, as is done under PC analysis, the factor modelling approach reduces the number of parameters to be estimated by relating the observations to a fewer number of unobserved latent variables, known as factors. A static factor model is defined by

$$y_{it} = \gamma_{i1} f_{1t} + \gamma_{i2} f_{2t} + \ldots + \gamma_{ik} f_{kt} + u_{it} = \boldsymbol{\gamma}_i' \mathbf{f}_t + u_{it}, \tag{19.43}$$

where the variable y_{it}, observed over $i = 1, 2, \ldots, m$, and $t = 1, 2, \ldots, T$, is explained in terms of k unobserved common factors, $\mathbf{f}_t = \left(f_{1t}, f_{2t}, \ldots, f_{kt}\right)'$. The variable y_{it} could measure cognitive ability of type t for an individual i, or could represent an activity variable of type i measured at time t, or interest rates on bonds of different maturity i observed at time t.

The factors influence the individual units through the parameters $\boldsymbol{\gamma}_i = \left(\gamma_{i1}, \gamma_{i2}, \ldots, \gamma_{ik}\right)'$, known as factor loadings. The factors are assumed to be pervasive (or strong) in the sense that almost all units are affected by variations in the factors. To ensure that the factors are pervasive, it is typically assumed that $m^{-1} \sum_{i=1}^{m} \boldsymbol{\gamma}_i \boldsymbol{\gamma}_i'$ is a positive definite matrix for any m and as $m \to \infty$. The remainder (error) terms, u_{it}, are assumed to be independently distributed over i and t, with zero mean and variance σ_i^2, as well as being independently distributed of $\mathbf{f}_{t'}$, for all i, t and t'. Such a factor model is known as the exact factor model to be distinguished from the approximate factor model where the idiosyncratic errors, u_{it}, are allowed to be weakly cross-sectionally correlated. Analysis of the approximate factor model requires both m and T to be large. For a formal characterization of weak and strong cross-sectional dependence and further details, see Chapter 29.

Before we discuss the estimation of the factor model, it is important to note from (19.43) that without further restrictions the unobserved components, $\boldsymbol{\gamma}_i$ and $\mathbf{f}_t$, cannot be separately identified. For exact identification, we need $k(k+1)/2$ restrictions which can be specified either in terms of $Var\left(\mathbf{f}_t\right)$ or $Var(\boldsymbol{\gamma}_i)$ which are set to $\mathbf{I}_k$. In what follows, we assume that $Var\left(\mathbf{f}_t\right) = \boldsymbol{\Sigma}_f$ is time invariant, and without loss of generality set $Var\left(\mathbf{f}_t\right) = \mathbf{I}_k$, and consider its sample variant, $T^{-1} \sum_{t=1}^{T} \mathbf{f}_t \mathbf{f}_t' = \mathbf{I}_k$. Stacking the T observations on each i, we have

$$\mathbf{y}_{i\cdot} = \mathbf{F} \boldsymbol{\gamma}_i + \mathbf{u}_{i\cdot}, \text{ for } i = 1, 2, \ldots, m,$$

where $\mathbf{y}_{i\cdot} = (y_{i1}, y_{i2}, \ldots, y_{iT})'$, $\mathbf{F} = (\mathbf{f}_1, \mathbf{f}_2, \ldots, \mathbf{f}_T)'$, and $\mathbf{u}_{i\cdot} = (u_{i1}, u_{i2}, \ldots, u_{iT})'$. Assuming that $\mathbf{F}$ is known, the above system of equations has the same format as the SURE model with $\mathbf{X}_i = \mathbf{F}$ for all i. Then, by the result in Section 19.2.1, the *GLS* estimator of $\boldsymbol{\gamma}_i$ is the same as the *OLS* estimator and is given by

$$\hat{\boldsymbol{\gamma}}_i(\mathbf{F}) = (\mathbf{F}'\mathbf{F})^{-1} \mathbf{F}' \mathbf{y}_{i\cdot}, \text{ for } i = 1, 2, \ldots, m, \tag{19.44}$$

or $\hat{\mathbf{\Gamma}}'(\mathbf{F}) = [\hat{\boldsymbol{\gamma}}_1(\mathbf{F}), \hat{\boldsymbol{\gamma}}_2(\mathbf{F}), \ldots, \hat{\boldsymbol{\gamma}}_m(\mathbf{F})] = (\mathbf{F}'\mathbf{F})^{-1}\mathbf{F}'\mathbf{Y}$, where $\mathbf{Y} = (\mathbf{y}_{1\cdot}, \mathbf{y}_{2\cdot}, \ldots, \mathbf{y}_{m\cdot})$. For identification of $\mathbf{F}$, the normalization restrictions $T^{-1}\mathbf{F}'\mathbf{F} = \mathbf{I}_k$ are imposed which yield

$$\hat{\mathbf{\Gamma}}(\mathbf{F}) = T^{-1}\mathbf{Y}'\mathbf{F}. \tag{19.45}$$

It is clear that for a given $\mathbf{F}$, $\hat{\mathbf{\Gamma}}(\mathbf{F})$ is a consistent estimator of $\mathbf{\Gamma}$, which is also robust to cross-correlations of $\mathbf{u}_{i\cdot}$ and $\mathbf{u}_{j\cdot}$.

Similarly, for a given $\mathbf{\Gamma}$, the observations can be stacked over i, which gives

$$\mathbf{y}_{\cdot t} = \mathbf{\Gamma}\boldsymbol{f}_t + \mathbf{u}_{\cdot t},$$

where $\mathbf{y}_{\cdot t} = (y_{1t}, y_{2t}, \ldots, y_{mt})'$, and $\mathbf{u}_{\cdot t} = (u_{1t}, u_{2t}, \ldots, u_{mt})'$. Again using the results in Section 19.2.1, we note that for a given $\mathbf{\Gamma}$, a consistent and efficient estimator of $\mathbf{f}_t$ is given by

$$\hat{\mathbf{f}}_t(\mathbf{\Gamma}) = \left(\mathbf{\Gamma}'\mathbf{\Gamma}\right)^{-1}\mathbf{\Gamma}'\mathbf{y}_{\cdot t}. \tag{19.46}$$

To ensure that these estimates of $\mathbf{f}_t$ satisfy the normalization restrictions we must have

$$T^{-1}\sum_{t=1}^{T}\hat{\mathbf{f}}_t(\mathbf{\Gamma})\hat{\mathbf{f}}_t'(\mathbf{\Gamma}) = \boldsymbol{I}_k = \left(\mathbf{\Gamma}'\mathbf{\Gamma}\right)^{-1}\mathbf{\Gamma}'\mathbf{S}_T\mathbf{\Gamma}\left(\mathbf{\Gamma}'\mathbf{\Gamma}\right)^{-1},$$

which we can write as

$$\mathbf{P}'\mathbf{S}_T\mathbf{P} = \mathbf{I}_k, \tag{19.47}$$

with

$$\mathbf{S}_T = T^{-1}\sum_{t=1}^{T}\mathbf{y}_{\cdot t}\mathbf{y}_{\cdot t}', \text{ and } \mathbf{P}' = \left(\mathbf{\Gamma}'\mathbf{\Gamma}\right)^{-1}\mathbf{\Gamma}'.$$

Therefore, $\mathbf{P}$ is the $m \times k$ matrix of the PCs of the sample $m \times m$ covariance matrix, $\mathbf{S}_T$, and the factor estimates, $\hat{\mathbf{f}}_t(\mathbf{P}) = \mathbf{P}'\mathbf{y}_{\cdot t}$, are formed as linear combinations of the observations (over i), with the weights in these linear combinations given by the first $k < m$ PCs of $\mathbf{Y}'\mathbf{Y}/T$. Using the factor estimates, the loadings $\boldsymbol{\gamma}_i$ can then be estimated by running *OLS* regressions of y_{it} (for each i) on the estimated factors, $\hat{\mathbf{f}}_t$. To summarize, the unobserved factors and the associated loadings can be consistently estimated by

$$\hat{\mathbf{f}}_t = \hat{\mathbf{P}}'\mathbf{y}_{\cdot t}, \tag{19.48}$$

$$\hat{\mathbf{\Gamma}} = T^{-1}\mathbf{Y}'\hat{\mathbf{F}}, \tag{19.49}$$

where $\hat{\mathbf{P}}$ is a $T \times k$ matrix of the first k PCs of $\mathbf{Y}'\mathbf{Y}/T$, namely the eigenvectors corresponding to the k largest eigenvalues of the $T \times T$ matrix $\mathbf{Y}'\mathbf{Y}/T$, and $\hat{\mathbf{F}} = (\hat{\mathbf{f}}_1, \hat{\mathbf{f}}_2, \ldots, \hat{\mathbf{f}}_T)'$. See also Section 19.4.

The PC estimators of the factors and their loadings can also be motivated by the following minimization problem

$$\min_{\substack{\mathbf{\Gamma} \\ \mathbf{f}_t; t=1,2,..,T}} \left[\sum_{t=1}^{T} (\mathbf{y}_{\cdot t} - \mathbf{\Gamma} \boldsymbol{f}_t)' (\mathbf{y}_{\cdot t} - \mathbf{\Gamma} \boldsymbol{f}_t) \right] = \min_{\substack{\mathbf{F} \\ \boldsymbol{\gamma}_i; i=1,2,..,m}} \left[\sum_{i=1}^{m} (\mathbf{y}_{i\cdot} - \mathbf{F} \boldsymbol{\gamma}_i)' (\mathbf{y}_{i\cdot} - \mathbf{F} \boldsymbol{\gamma}_i) \right], \tag{19.50}$$

subject to the $k(k+1)/2$ normalization constraints $T^{-1} \sum_{t=1}^{T} \mathbf{f}_t \mathbf{f}_t' = \mathbf{I}_k$. The first-order conditions for this minimization problem are given by

$$\mathbf{F}'(\mathbf{y}_{i\cdot} - \mathbf{F} \boldsymbol{\gamma}_i) = 0, \text{ for } i = 1, 2, \ldots, m, \tag{19.51}$$

$$\mathbf{\Gamma}'(\mathbf{y}_{\cdot t} - \mathbf{\Gamma} \boldsymbol{f}_t) = 0, \text{ for } t = 1, 2, \ldots, T. \tag{19.52}$$

Recalling that $T^{-1}\mathbf{F}'\mathbf{F} = \mathbf{I}_k$, the estimates of the factor loadings are given by $\hat{\boldsymbol{\gamma}}_i = (\mathbf{F}'\mathbf{F})^{-1} \mathbf{F}'\mathbf{y}_{i\cdot} = T^{-1}\mathbf{F}'\mathbf{y}_{i\cdot}$, or $\hat{\mathbf{\Gamma}} = T^{-1}\mathbf{Y}'\hat{\mathbf{F}}$, which is the same as those given by (19.49). Also using (19.52) we have $\hat{\mathbf{f}}_t = (\mathbf{\Gamma}'\mathbf{\Gamma})^{-1}\mathbf{\Gamma}'\mathbf{y}_{\cdot t}$, which is the same as (19.46). Therefore, minimization of (19.50) with respect to $\mathbf{\Gamma}$ and $\mathbf{F}$ simultaneously yields the same solution as the sequential optimization followed earlier. Both approaches result in (19.48) and (19.49) as the solutions.

19.5.1 *PC* and cross-section average estimators of factors

The sampling properties of the above *PC* estimators of the factors and their loadings depends on whether the factor model is exact or approximate and whether m and T are both large. In the case where m is fixed and $T \to \infty$, the *PC* estimators are consistent only under the exact factor model and homoskedastic errors (namely if $Var(u_{it}) = \sigma_i^2 = \sigma^2$, for all i). But when both m and T are large then the *PC* estimators provide consistent estimators so long as the errors, u_{it}, are weakly cross-sectionally dependent. In some settings the factors can also be approximated by weighted averages of y_{it} where the weights are fixed, as compared to the *PC* estimator that uses endogenous weights that are nonlinear functions of the observations. To see this, consider the following single factor model

$$y_{it} = \gamma_i f_t + u_{it}, \tag{19.53}$$

where

$$\gamma_i = \gamma + \eta_i, \text{with } \eta_i \sim IID(\mu_\gamma, \sigma_\gamma^2), \tag{19.54}$$

and η_i and f_t are distributed independently of u_{it}. Let $\bar{\gamma}_{mw} = \sum_{i=1}^{m} w_i \gamma_i$, where the weights w_i add up to unity, $\sum_{i=1}^{m} w_i = 1$, and are granular in the sense that $w_i = O\left(m^{-1}\right)$, and $\sum_{i=1}^{m} w_i^2 = O\left(m^{-1}\right)$.[4] Suppose that $\bar{\gamma}_{mw} \neq 0$ and $\gamma \neq 0$. Then

$$\bar{y}_{tw} = \bar{\gamma}_{mw} f_t + \bar{u}_{tw}, \tag{19.55}$$

[4] See Section 29.2 for further details.

and f_t can be consistently estimated by $\bar{y}_{tw} = \sum_{t=1}^{T} w_i y_{it}$ (up to the scaling factor $\bar{\gamma}_m$) so long as $\bar{u}_{tw} = O_p(m^{-1/2})$. The restriction that the scaling factor, $\bar{\gamma}_{mw} \neq 0$ serves as the identifying restriction, very much in the same way that $Var(f_t) = 1$ is used as the identifying restriction under the factor model. But condition $\bar{\gamma}_{mw} \neq 0$ is clearly more restrictive than assuming $Var(f_t) \neq 0$, although in most economic applications condition $\bar{\gamma}_{mw} \neq 0$ is likely to be satisfied, since otherwise $\bar{y}_{tw}$ tends to a non-stochastic constant (in the above example to *zero*) which is contrary to what we observe about the highly cyclical and volatile nature of economic and financial aggregates.

Consider now the *PC* estimator of f_t which is given by $\hat{f}_{t,T}^{PC} = \sum_{i=1}^{m} p_{iT} y_{it}$, where $\mathbf{p}_T = (p_{1T}, p_{2T}, \ldots, p_{mT})'$ is the eigenvector associated with the largest eigenvalue of $\mathbf{S}_T = T^{-1} \sum_{t=1}^{T} \mathbf{y}_{\cdot t} \mathbf{y}'_{\cdot t}$. It is clear that both estimators of f_t are cross-sectional weighted averages of the observations. The main difference between the two estimators lies in the choice of the weights. In construction of $\bar{y}_{tw}$ the weights w_i are predetermined and can be typically taken to be $w_i = 1/m$. In contrast, the weights in the *PC* estimator, $\hat{f}_{t,T}^{PC}$, are endogenously obtained as nonlinear functions of the observations, y_{it}. In small samples, the two estimators could have different degrees of correlations with f_t, but when m and T are sufficiently large both estimators ($\bar{y}_{tw}$ and $\hat{f}_{t,T}^{PC}$) become perfectly correlated with f_t and hence with one another. The cross-section average (*CS*) estimator, $\bar{y}_{tw}$, becomes perfectly correlated with f_t even if T is small, but the validity of the *PC* estimator requires that both m and T be large. But as we have noted above, the advantage of the PC estimator over the *CS* estimator is that it is valid even if $\bar{\gamma}_{mw} \to 0$, as $m \to \infty$.

The relationship between the *CS* and *PC* estimators of f_t can be better understood in the case of an exact factor model where $u'_{it}s$ in (19.53) are cross sectionally independently distributed with a common variance, σ_u^2. In this case, and imposing the normalizing restriction $T^{-1} \sum_{t=1}^{T} f_t^2 = 1$, we have

$$\begin{aligned}\mathbf{S}_T &= T^{-1} \sum_{t=1}^{T} \mathbf{y}_{\cdot t} \mathbf{y}'_{\cdot t} = T^{-1} \sum_{t=1}^{T} \left(\boldsymbol{\gamma} f_t + \mathbf{u}_t\right) \left(\boldsymbol{\gamma} f_t + \mathbf{u}_t\right)' \\ &= \mathbf{S} + O_p(T^{-1/2}),\end{aligned}$$

where $\mathbf{S} = \boldsymbol{\gamma}\boldsymbol{\gamma}' + \sigma^2 \mathbf{I}_m$. Therefore[5]

$$\hat{f}_{t,T}^{PC} = \frac{\mathbf{p}'_T \mathbf{y}_{\cdot t}}{\mathbf{p}'_T \mathbf{p}_T} = \frac{\mathbf{p}' \mathbf{y}_{\cdot t}}{\mathbf{p}' \mathbf{p}} + O_p(T^{-1/2}), \tag{19.56}$$

where $\mathbf{p}$ is the first eigenvector of $\mathbf{S}$. Now let $\lambda_{\max}$ be the largest eigenvalue of $\mathbf{S}$ then

$$\left(\boldsymbol{\gamma}\boldsymbol{\gamma}' + \sigma^2 \mathbf{I}_m\right) \mathbf{p} = \lambda_{\max} \mathbf{p},$$

and hence

$$\boldsymbol{\gamma}\boldsymbol{\gamma}' \mathbf{p} = \left(\lambda_{\max} - \sigma^2\right) \mathbf{p}.$$

[5] Recall that $\mathbf{p}_T$ is the first eigenvector of $\mathbf{S}_T$, normalized to have a unit length. Here we scale the PC estimator by $\mathbf{p}'_T \mathbf{p}_T$ which ensures that the PC estimator and f_t have the same scale.

Thus $\mathbf{p}$ is also the first eigenvector of $\boldsymbol{\gamma}\boldsymbol{\gamma}'$ associated to $\left(\lambda_{\max} - \sigma^2\right)$, and since $\boldsymbol{\gamma}\boldsymbol{\gamma}'$ has rank unity, then $\mathbf{p} = \boldsymbol{\gamma}$, and $\lambda_{\max} = \sigma^2 + \boldsymbol{\gamma}'\boldsymbol{\gamma}$. Using this result in (19.56) and using (19.54) we have

$$\begin{aligned}\hat{f}_{t,T}^{PC} &= \left(\boldsymbol{\gamma}'\boldsymbol{\gamma}\right)^{-1}\boldsymbol{\gamma}'\mathbf{y}_{\cdot t} + O_p(T^{-1/2}) \\ &= \left(\frac{\mu_\gamma}{m^{-1}\boldsymbol{\gamma}'\boldsymbol{\gamma}}\right)\bar{y}_t + \frac{m^{-1}\sum_{i=1}^{m}\eta_i y_{it}}{m^{-1}\boldsymbol{\gamma}'\boldsymbol{\gamma}} + O_p(T^{-1/2}).\end{aligned}$$

But under (19.54)

$$\begin{aligned}m^{-1}\boldsymbol{\gamma}'\boldsymbol{\gamma} &= \mu_\gamma^2 + \sigma_\gamma^2 + O_p(m^{-1/2}), \\ m^{-1}\sum_{i=1}^{m}\eta_i y_{it} &= \sigma_\gamma^2 f_t + O_p(m^{-1/2}),\end{aligned}$$

and hence

$$\hat{f}_{t,T}^{PC} = \left(\frac{\mu_\gamma}{\mu_\gamma^2 + \sigma_\gamma^2}\right)\bar{y}_t + \left(\frac{\sigma_\gamma^2}{\mu_\gamma^2 + \sigma_\gamma^2}\right)f_t + O_p(m^{-1/2}) + O_p(T^{-1/2}).$$

Also using (19.55) we have

$$\bar{y}_t = \mu_\gamma f_t + O_p(m^{-1/2}).$$

It is clear that when $\mu_\gamma \neq 0$, then f_t and $\bar{y}_t$ will be perfectly correlated if m is sufficiently large even if T is small. But when $\mu_\gamma = 0$, we have $\bar{y}_t = O_p(m^{-1/2})$ and, as noted earlier, $\bar{y}_t \to_p 0$, and f_t cannot be identified by $\bar{y}_t$. In this case $\hat{f}_{t,T}^{PC}$ identifies f_t if $\sigma_\gamma^2 > 0$. Using the above results it is easily seen that

$$\bar{y}_t = \mu_\gamma \hat{f}_{t,T}^{PC} + O_p(m^{-1/2}) + O_p(T^{-1/2}),$$

which establishes that in the limit as m and $T \to \infty$, $\bar{y}_t$ and $\hat{f}_{t,T}^{PC}$ become perfectly correlated.

It is possible to improve on $\bar{y}_t$ as a proxy for f_t by using the weighted cross-section average $\bar{y}_{\delta t} = m^{-1}\sum_{i=1}^{m}\hat{\delta}_i y_{it}$, where $\hat{\delta}_i$ is the coefficient of $\bar{y}_t$ in a regression of y_{it} on $\bar{y}_t$, namely, $\hat{\delta}_i = \sum_{t=1}^{T}\bar{y}_t y_{it} / \sum_{t=1}^{T}\bar{y}_t^2$. It is now easily seen that

$$\bar{y}_{\delta t} = m^{-1}\sum_{i=1}^{m}\hat{\delta}_i y_{it} = m^{-1}\sum_{i=1}^{m}\hat{\delta}_i\left(\gamma_i f_t + u_{it}\right)$$

$$= a_T \left(m^{-1} \sum_{i=1}^{m} \gamma_i^2 \right) f_t + a_T \left(m^{-1} \sum_{i=1}^{m} \gamma_i u_{it} \right) +$$
$$+ \left(T^{-1} \sum_{t=1}^{T} \bar{y}_t^2 \right)^{-1} (mT)^{-1} \sum_{\tau=1}^{T} \sum_{i=1}^{m} \bar{y}_\tau u_{i\tau} u_{it},$$

where

$$a_T = \left(T^{-1} \sum_{t=1}^{T} \bar{y}_t^2 \right)^{-1} \left(T^{-1} \sum_{t=1}^{T} \bar{y}_t f_t \right).$$

It is now clear that when T is fixed and $a_T \neq 0$, then $\bar{y}_{\delta t}$ becomes proportional to f_t if

$$\lim_{m \to \infty} \left(m^{-1} \sum_{i=1}^{m} \gamma_i^2 \right) > 0, \tag{19.57}$$

$$\lim_{m \to \infty} \left(m^{-1} \sum_{i=1}^{m} \gamma_i u_{it} \right) = c, \tag{19.58}$$

$$\lim_{m \to \infty} \left((mT)^{-1} \sum_{\tau=1}^{T} \sum_{i=1}^{m} \bar{u}_\tau u_{i\tau} u_{it} \right) = c_T, \tag{19.59}$$

where c represents a generic constant. Condition (19.57) is standard in the factor literature. Condition (19.58) is less restrictive than assuming γ_i and u_{it} are uncorrelated, which is typically assumed in the literature. Condition (19.59) is more complicated to relate to the literature, but allows for weak cross-sectional dependence in the idiosyncratic errors. Note that by the Cauchy–Schwarz inequality

$$E \left(m^{-1} \sum_{i=1}^{m} \bar{u}_\tau u_{i\tau} u_{it} \right) \leq \left[E \left(\bar{u}_\tau^2 \right) \right]^{1/2} \left[E \left(m^{-1} \sum_{i=1}^{m} u_{i\tau} u_{it} \right)^2 \right]^{1/2},$$

and condition (19.59) is met if $E\left(\bar{u}_\tau^2\right) < K$ and $E\left(m^{-1} \sum_{i=1}^{m} u_{i\tau} u_{it}\right)^2 < K$. These conditions are satisfied if $u_{it}'s$ have fourth-order moments and are weakly cross-correlated.

To investigate how quickly the correlation between $\bar{y}_t$ and $\hat{f}_{t,T}^{PC}$ tends to unity when m and $T \to \infty$, we carried out a limited number of Monte Carlo experiments using (19.53) as the DGP, with $\gamma_i \sim IIDN(1,1)$; $f_t \sim IIDN(0,1)$; $u_{it} \sim IIDN(0,1)$, for m and $T = 30, 50, 100, 200, 1000$. The squared pair-wise correlation coefficients of $f_t, \hat{f}_{t,T}^{PC}, \bar{y}_t$ and $\bar{y}_{\delta t}$, averaged across 2,000 replications, are summarized in the top part of the following Table 19.4.[6] We have also carried out experiments with spatially correlated errors generated as[7]

[6] I would like to thank Alex Chudik for carrying out the Monte Carlo experiments reported in this sub-section.

[7] For a discussion of spatial models, see Chapter 30.

$$\mathbf{u}_t = a_u \mathbf{H}_u \mathbf{u}_t + \mathbf{e}_t, \tag{19.60}$$

where the elements of $\mathbf{e}_t$ are drawn as $IIDN\left(0, \sigma_e^2\right)$,

$$\mathbf{H}_u = \begin{pmatrix} 0 & \frac{1}{2} & 0 & \cdots & 0 & 0 \\ \frac{1}{2} & 0 & 1 & \cdots & 0 & 0 \\ 0 & 1 & 0 & \ddots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 0 & \frac{1}{2} \\ 0 & 0 & 0 & \cdots & \frac{1}{2} & 0 \end{pmatrix},$$

the spatial autoregressive parameter is set to $a_u = 0.6$, and σ_e^2 is set to ensure that $N^{-1}\sum_{i=1}^{N} Var\left(u_{it}\right) = 1$. Experiments with spatially correlated errors are reported in the bottom part of Table 19.4.

The results clearly show that all three estimators of f_t are highly correlated with the unobserved factor and this correlation is almost perfect for values of m above 100 (for all values of T) when the idiosyncratic errors are independently distributed. But when the errors are weakly (spatially) dependent then the value of m needed to get an almost perfect fit is around 200. It is also clear that T does not matter and for a given m the correlations are hardly affected by increasing T. Finally, although the simple average estimator, $\bar{y}_t$, performs well when m is sufficiently large, for small values of m the weighted average estimator, $\bar{y}_{\delta t}$, is to be preferred. It is also interesting that $\bar{y}_{\delta t}$ performs very similarly to the PC estimator, $\hat{f}_{t,T}^{PC}$, which also performs well even when T is small.

Finally, we also carried out the same experiments but with $E(\gamma_i) = 0$. The results are summarized in Table 19.5. As to be expected, the simple average estimator, $\bar{y}_t$, performs poorly. However, an iterated version of the weighted average estimator, $\bar{y}_{\delta t}$, performs well and very similarly to the PC estimator even if $E(\gamma_i) = 0$. The rth iterated estimator is computed as $\bar{y}_{\delta t}^{(r)} = m^{-1}\sum_{i=1}^{m} \hat{\delta}_i^{(r)} y_{it}$, where $\hat{\delta}_i^{(r)}$ is the coefficient of $\bar{y}_{\delta t}^{(r-1)}$ in the OLS regression of y_{it} on $\bar{y}_{\delta t}^{(r-1)}$, with $\bar{y}_{\delta t}^{(1)} = \bar{y}_{\delta t}$. The results reported in Table 19.5 set $r = 2$. Further iterations did not make much difference.

19.5.2 Determining the number of factors in a large m and large T framework

Formal criteria for selecting the number of factors in a factor model where both m and T are large have been proposed by Bai and Ng (2002), Kapetanios (2004), Hallin and Liska (2007), and Onatski (2009, 2010). In particular, Bai and Ng (2002) consider the following two classes of criteria

$$PC(h) = V\left(h, \hat{\mathbf{F}}^{(h)}\right) + h \times g\left(m, T\right),$$
$$IC(h) = \ln\left[V\left(h, \hat{\mathbf{F}}^{(h)}\right)\right] + h \times g\left(m, T\right),$$

Table 19.4 Monte Carlo findings for squared correlations of the unobserved common factor and its estimates: Experiments with $E\left(\gamma_i\right) = 1$

	$\rho^2\left(f_t, \hat{f}_{t,T}^{PC}\right), \times 100$					$\rho^2\left(f_t, \bar{y}_t\right), \times 100$					$\rho^2\left(f_t, \bar{y}_{\delta t}\right), \times 100$				
(m/T)	30	50	100	200	1000	30	50	100	200	1000	30	50	100	200	1000
	Experiments with IID idiosyncratic errors														
30	98.16	98.21	98.25	98.25	98.27	96.28	96.31	96.41	96.36	96.42	98.16	98.21	98.25	98.25	98.27
50	98.92	98.97	98.97	98.97	98.98	97.87	97.93	97.89	97.92	97.91	98.92	98.97	98.97	98.97	98.98
100	99.46	99.49	99.50	99.49	99.49	98.93	98.96	98.98	98.97	98.98	99.46	99.49	99.50	99.49	99.49
200	99.73	99.74	99.75	99.75	99.75	99.47	99.49	99.49	99.50	99.49	99.73	99.74	99.75	99.75	99.75
500	99.95	99.95	99.95	99.95	99.95	99.90	99.90	99.90	99.90	99.90	99.95	99.95	99.95	99.95	99.95
	Experiments with spatially correlated idiosyncratic errors														
30	96.34	96.35	96.43	96.46	96.49	89.54	89.32	89.59	89.67	89.84	96.14	96.18	96.28	96.32	96.37
50	97.77	97.85	97.87	97.87	97.88	93.44	93.68	93.71	93.71	93.65	97.68	97.78	97.82	97.83	97.84
100	98.88	98.91	98.94	98.94	98.95	96.70	96.80	96.82	96.83	96.84	98.85	98.89	98.92	98.93	98.94
200	99.44	99.46	99.46	99.47	99.47	98.33	98.38	98.39	98.41	98.41	99.43	99.45	99.46	99.47	99.47
1000	99.89	99.89	99.89	99.89	99.89	99.67	99.68	99.68	99.68	99.68	99.89	99.89	99.89	99.89	99.89

Notes: $\hat{f}_{t,T}^{PC}$ is the principal component estimator of f_t, $\bar{y}_t = m^{-1}\sum_{i=1}^{m} y_{it}$, and $\bar{y}_{\delta t} = m^{-1}\sum_{i=1}^{m} \hat{\delta}_i y_{it}$, where $\hat{\delta}_i$ is given by a regression of y_{it} on $\bar{y}_t$. DGP is $y_{it} = \gamma_i f_t + u_{it}$, for $i = 1, 2, \ldots, m$, $t = 1, 2, \ldots, T$, where $\gamma_i \sim IIDN\left(1, 1\right)$, $f_t \sim IIDN\left(0, 1\right)$, and errors are generated either as $u_{it} \sim IIDN\left(0, 1\right)$ (top panel), or from a spatial autoregressive (SAR) process with SAR parameter 0.6. $\rho\left(x_t, y_t\right)$ denotes correlation between x_t and y_t. Findings in this table are based on $R = 2000$ Monte Carlo replications.

Table 19.5 Monte Carlo findings for squared correlations of the unobserved common factor and its estimates: Experiments with $E(\gamma_i) = 0$

	$\rho^2\left(f_t, \hat{f}_{t,T}^{PC}\right), \times 100$					$\rho^2\left(f_t, \bar{y}_t\right), \times 100$					$\rho^2\left(f_t, \bar{y}_{\delta t}\right), \times 100$					$\rho^2\left(f_t, \bar{y}_{\delta t}^{(2)}\right), \times 100$				
(m/T)	30	50	100	200	1000	30	50	100	200	1000	30	50	100	200	1000	30	50	100	200	1000
	Experiments with IID idiosyncratic errors																			
30	96.24	96.40	96.49	96.52	96.60	36.04	35.19	34.18	34.23	34.09	87.48	89.17	90.51	91.53	92.59	95.09	95.75	96.03	96.31	96.42
50	97.80	97.85	97.91	97.95	97.97	36.58	35.00	34.36	34.71	35.06	91.01	92.63	94.04	94.59	95.36	97.22	97.26	97.61	97.84	97.94
100	98.92	98.96	98.97	98.97	98.99	35.94	35.18	35.56	34.00	34.67	92.62	94.91	95.90	96.82	97.57	98.21	98.81	98.82	98.93	98.95
200	99.45	99.48	99.49	99.49	99.50	35.85	35.05	35.31	34.84	35.39	94.09	96.27	97.34	97.92	99.03	99.17	99.42	99.46	99.49	99.50
500	99.89	99.90	99.90	99.90	99.90	37.52	35.16	34.44	35.05	34.98	96.16	96.95	97.98	99.26	99.63	99.74	99.85	99.86	99.90	99.90
	Experiments with spatially correlated idiosyncratic errors																			
30	95.86	96.01	96.18	96.25	96.28	20.48	19.83	19.17	18.11	18.22	72.62	74.44	75.83	75.73	76.18	90.40	91.86	92.72	92.95	93.13
50	97.66	97.77	97.79	97.82	97.85	20.87	19.94	19.12	19.03	18.31	79.42	82.80	83.92	83.50	84.87	94.84	96.07	96.44	96.24	96.96
100	98.84	98.92	98.95	98.95	98.97	20.67	18.75	18.72	18.07	18.44	85.62	87.91	89.80	89.98	92.57	97.57	98.14	98.23	98.58	98.62
200	99.43	99.46	99.48	99.49	99.49	20.72	19.56	18.67	17.96	18.60	89.56	91.30	92.94	94.84	96.21	98.75	98.93	99.22	99.34	99.46
1000	99.89	99.89	99.90	99.90	99.90	20.28	19.32	19.20	18.64	18.32	91.54	94.97	96.61	97.70	99.02	99.40	99.75	99.80	99.88	99.89

Notes: $\hat{f}_{t,T}^{PC}$ is the principal component estimator of f_t, $\bar{y}_t = m^{-1}\sum_{i=1}^{m} y_{it}$, and $\bar{y}_{\delta t} = m^{-1}\sum_{i=1}^{m}\hat{\delta}_i y_{it}$, where $\hat{\delta}_i$ is given by a regression of y_{it} on $\bar{y}_t$. The rth iterated estimator is computed as $\bar{y}_{\delta t}^{(r)} = m^{-1}\sum_{i=1}^{m}\hat{\delta}_i^{(r)} y_{it}$, where $\hat{\delta}_i^{(r)}$ is the coefficient of $\bar{y}_{\delta t}^{(r-1)}$ in the *OLS* regression of y_{it} on $\bar{y}_{\delta t}^{(r-1)}$, with $\bar{y}_{\delta t}^{(1)} = \bar{y}_{\delta t}$. DGP is $y_{it} = \gamma_i f_t + u_{it}$, for $i = 1, 2, \ldots, m$, $t = 1, 2, \ldots, T$, where $\gamma_i \sim IIDN(0, 1)$, $f_t \sim IIDN(0, 1)$, and errors are generated either as $u_{it} \sim IIDN(0, 1)$ (top panel), or from a spatial autoregressive (SAR) process with SAR parameter 0.6. $\rho(x_t, y_t)$ denotes correlation between x_t and y_t. Findings in this table are based on $R = 2000$ Monte Carlo replications.

where

$$V\left(h,\hat{\mathbf{F}}^{(h)}\right)=\min_{\mathbf{\Gamma}}\frac{1}{NT}\sum_{i=1}^{m}\sum_{t=1}^{T}\left(y_{it}-\boldsymbol{\gamma}_i^{(h)\prime}\hat{\mathbf{f}}_t^{(h)}\right),$$

$\boldsymbol{\gamma}_i^{(h)}=\left(\gamma_{i1},\gamma_{i2},\ldots,\gamma_{ih}\right)'$ and $\hat{\mathbf{f}}_t^{(h)}=\left(\hat{f}_{t1},\hat{f}_{t2},\ldots,\hat{f}_{th}\right)'$, where factors are estimated by principal components, and $g(.)$ is a penalty function due to over-fitting, satisfying the following conditions

$$\begin{aligned}(a)\quad &: \quad g(m,T)\to 0,\ \text{as}\ m,T\to\infty,\\ (b)\quad &: \quad C_{NT}^2\times g(m,T)\to\infty,\ \text{as}\ m,T\to\infty,\end{aligned}$$

with $C_{mT}=\min\left(\sqrt{m},\sqrt{T}\right)$. The authors prove that, under some regularity conditions, the criteria $PC(h)$ and $IC(h)$ will consistently estimate k. Bai and Ng (2002) also propose the following specific formulations of $g(m,T)$

$$\begin{aligned}PC_{p_1}(h)&=V\left(h,\hat{\mathbf{F}}^{(h)}\right)+h\hat{\sigma}^2\left(\frac{m+T}{mT}\right)\ln\left(\frac{mT}{m+T}\right),\\ PC_{p_2}(h)&=V\left(h,\hat{\mathbf{F}}^{(h)}\right)+h\hat{\sigma}^2\left(\frac{m+T}{mT}\right)\ln\left(C_{mT}^2\right),\\ PC_{p_3}(h)&=V\left(h,\hat{\mathbf{F}}^{(h)}\right)+h\hat{\sigma}^2\left[\frac{\ln\left(C_{mT}^2\right)}{C_{mT}^2}\right],\end{aligned}$$

where $\hat{\sigma}^2=(mT)^{-1}\sum_{i=1}^{m}\sum_{t=1}^{T}e_{it}^2$, and

$$\begin{aligned}IC_{p_1}(h)&=\ln\left[V\left(h,\hat{\mathbf{F}}^{(h)}\right)\right]+h\left(\frac{m+T}{mT}\right)\ln\left(\frac{mT}{m+T}\right),\\ IC_{p_2}(h)&=\ln\left[V\left(h,\hat{\mathbf{F}}^{(h)}\right)\right]+h\left(\frac{m+T}{mT}\right)\ln\left(C_{mT}^2\right),\\ IC_{p_3}(h)&=\ln\left[V\left(h,\hat{\mathbf{F}}^{(h)}\right)\right]+h\left[\frac{\ln\left(C_{mT}^2\right)}{C_{mT}^2}\right].\end{aligned}$$

In practice, Bai and Ng suggest replacing $\hat{\sigma}^2$ with $V\left(k_{\max},\hat{\mathbf{F}}^{(k_{\max})}\right)$, where $k_{\max}$ is the maximum number of selected factors. Note that, in the IC criteria, scaling by $\hat{\sigma}^2$ is implicitly performed by the logarithmic transformation of $V\left(h,\hat{\mathbf{f}}_h\right)$ and is thus not required in the penalty function.

In a Monte Carlo exercise, Onatski (2010) shows that Bai and Ng (2002) information criteria perform rather poorly, unless N and T are quite large. Further, the performance of these criteria deteriorates considerably as the variances of the idiosyncratic components increase, or when such components are cross-sectionally (weakly) correlated. In particular, Onatski observes an overestimation of the number of factors when the idiosyncratic errors are contemporaneously correlated. One explanation for this result is that, in this case, some linear combinations of the

idiosyncratic errors may have a non-trivial effect on a sizeable portion of the data. Hence, the explanatory power of such linear combination rises and Bai and Ng (2002) criteria have difficulty in distinguishing these linear combinations from $\mathbf{f}_t$.

Onatski (2010) proposes an estimator of the number of factors based on the empirical distribution of eigenvalues of the sample covariance matrix. Let λ_i be the ith largest eigenvalue of $T^{-1}\mathbf{YY}'$, and consider

$$\hat{k}_\delta = \#\left\{i \leq m : \lambda_i > (1+\delta)\hat{v}\right\},$$

where δ is a positive scalar, and $\hat{v} = w\lambda_{k_{\max}} + (1-w)\lambda_{2k_{\max}+1}$, $w = 2^{2/3}/\left(2^{2/3}-1\right)$. Onatski (2010) proves that $\hat{k}_\delta$ is consistent for k when $\delta \sim m^{-\alpha}$ for any scalar α satisfying a set of conditions. See Onatski (2010) for details.

Factor models are used extensively in panel data models to characterize strong cross-sectional dependence. See Chapter 29.

19.6 Canonical correlation analysis

Canonical correlations (CC) measure the degree of correlation between two sets of variables. Let $\mathbf{Y}$ $(T \times m_y)$ be a matrix of T observations on m_y random variables, and $\mathbf{X}$ $(T \times m_x)$ be a matrix of T observations on m_x random variables, and suppose that $T > \max\left(m_y, m_x\right)$. CC is concerned with finding linear combinations of the $\mathbf{Y}$ variables and linear combinations of the $\mathbf{X}$ variables that are most highly correlated. In particular, let

$$u_{it} = \boldsymbol{\alpha}'_{(i)}\mathbf{y}_t \text{ and } v_{it} = \boldsymbol{\gamma}'_{(i)}\mathbf{x}_t,\ i = 1, 2, \ldots, m = \min\left\{m_y, m_x\right\},$$

where $\mathbf{y}_t = (y_{1t}, y_{2t}, \ldots, y_{m_y,t})'$, $\mathbf{x}_t = (x_{1t}, x_{2t}, \ldots, x_{m_x,t})'$, and $\boldsymbol{\alpha}_{(i)}$ and $\boldsymbol{\gamma}_{(j)}$ are the associated $m_y \times 1$ and $m_x \times 1$ loading vectors, respectively. The first canonical correlation of $\mathbf{y}_t$ and $\mathbf{x}_t$ is given by those values $\boldsymbol{\alpha}_{(1)}$ and $\boldsymbol{\gamma}_{(1)}$ that maximize the correlation of u_{1t} and v_{1t}. These variables are known as canonical variates. The second canonical correlation refers to $\boldsymbol{\alpha}_{(2)}$ and $\boldsymbol{\gamma}_{(2)}$ such that u_{2t} and v_{2t} have maximum correlation subject to the restriction that they are uncorrelated with u_{1t} and v_{1t}. The loadings are typically normalized so that the canonical variates have unit variances, namely $\boldsymbol{\alpha}'\boldsymbol{\Sigma}_{yy}\boldsymbol{\alpha} = 1$, and $\boldsymbol{\gamma}'\boldsymbol{\Sigma}_{xx}\boldsymbol{\gamma} = 1$. The optimization problem can be set as

$$\max_{\boldsymbol{\alpha},\boldsymbol{\gamma}}\left[\boldsymbol{\alpha}'\boldsymbol{\Sigma}_{yx}\boldsymbol{\gamma} - \frac{1}{2}\rho_1(\boldsymbol{\alpha}'\boldsymbol{\Sigma}_{yy}\boldsymbol{\alpha} - 1) - \frac{1}{2}\rho_2\left(\boldsymbol{\gamma}'\boldsymbol{\Sigma}_{xx}\boldsymbol{\gamma} - 1\right)\right],$$

where $\boldsymbol{\Sigma}_{yx}$ is the population covariance matrix of $\mathbf{y}_t$ and $\mathbf{x}_t$, $\boldsymbol{\Sigma}_{yy}$ and $\boldsymbol{\Sigma}_{xx}$, are the population variance matrices of $\mathbf{y}_t$ and $\mathbf{x}_t$, respectively, and ρ_1 and ρ_2 are Lagrange multipliers. The first-order conditions for this optimization are given by

$$\begin{aligned}\boldsymbol{\Sigma}_{yx}\boldsymbol{\gamma} - \rho_1\boldsymbol{\Sigma}_{yy}\boldsymbol{\alpha} &= \mathbf{0},\\ \boldsymbol{\Sigma}_{xy}\boldsymbol{\alpha} - \rho_2\boldsymbol{\Sigma}_{xx}\boldsymbol{\gamma} &= \mathbf{0}.\end{aligned}$$

Therefore, we have $\boldsymbol{\alpha}'\boldsymbol{\Sigma}_{yx}\boldsymbol{\gamma} = \rho_1\boldsymbol{\alpha}'\boldsymbol{\Sigma}_{yy}\boldsymbol{\alpha} = \rho_1$, and $\boldsymbol{\gamma}'\boldsymbol{\Sigma}_{xy}\boldsymbol{\alpha} = \rho_2\boldsymbol{\gamma}'\boldsymbol{\Sigma}_{xx}\boldsymbol{\gamma} = \rho_2$, and since $\boldsymbol{\alpha}'\Sigma_{yx}\boldsymbol{\gamma} = \boldsymbol{\gamma}'\boldsymbol{\Sigma}_{xy}\boldsymbol{\alpha}$, then we must have $\rho_1 = \rho_2 = \rho$, and the above first-order conditions can be written as

$$\begin{pmatrix} -\rho\boldsymbol{\Sigma}_{yy} & \boldsymbol{\Sigma}_{yx} \\ \boldsymbol{\Sigma}_{xy} & -\rho\boldsymbol{\Sigma}_{xx} \end{pmatrix}\begin{pmatrix} \boldsymbol{\alpha} \\ \boldsymbol{\gamma} \end{pmatrix} = \mathbf{0}.$$

Hence, a non-trivial solution of $(\boldsymbol{\alpha}', \boldsymbol{\gamma}')'$ is obtained only for values of ρ that ensure

$$\begin{vmatrix} -\rho\boldsymbol{\Sigma}_{yy} & \boldsymbol{\Sigma}_{yx} \\ \boldsymbol{\Sigma}_{xy} & -\rho\boldsymbol{\Sigma}_{xx} \end{vmatrix} = 0.$$

Now assuming that $\boldsymbol{\Sigma}_{xx}$ and $\boldsymbol{\Sigma}_{yy}$ are nonsingular, using standard results on the determinant of partitioned matrices we have (see Section A.9 in Appendix A)

$$\begin{aligned} \begin{vmatrix} -\rho\boldsymbol{\Sigma}_{yy} & \boldsymbol{\Sigma}_{yx} \\ \boldsymbol{\Sigma}_{xy} & -\rho\boldsymbol{\Sigma}_{xx} \end{vmatrix} &= \left|\boldsymbol{\Sigma}_{yy}\right| \left|\rho^2\boldsymbol{\Sigma}_{xx} - \boldsymbol{\Sigma}_{xy}\boldsymbol{\Sigma}_{yy}^{-1}\boldsymbol{\Sigma}_{yx}\right| \\ &= \left|\boldsymbol{\Sigma}_{xx}\right| \left|\rho^2\boldsymbol{\Sigma}_{yy} - \boldsymbol{\Sigma}_{yx}\boldsymbol{\Sigma}_{xx}^{-1}\boldsymbol{\Sigma}_{xy}\right| = 0. \end{aligned}$$

Therefore, the value of ρ^2 is given by the non-zero eigenvalues of $\boldsymbol{\Sigma}_{xx}^{-1}\boldsymbol{\Sigma}_{xy}\boldsymbol{\Sigma}_{yy}^{-1}\boldsymbol{\Sigma}_{yx}$ or $\boldsymbol{\Sigma}_{yy}^{-1}\boldsymbol{\Sigma}_{yx}\boldsymbol{\Sigma}_{xx}^{-1}\boldsymbol{\Sigma}_{xy}$.

The sample counterpart of ρ^2 can be computed using

$$\mathbf{S}_{yy} = T^{-1}(\mathbf{Y}'\mathbf{Y}),\ \mathbf{S}_{xx} = T^{-1}(\mathbf{X}'\mathbf{X}) \text{ and } \mathbf{S}_{yx} = T^{-1}(\mathbf{Y}'\mathbf{X}), \tag{19.61}$$

as estimators of $\boldsymbol{\Sigma}_{yy}$, $\boldsymbol{\Sigma}_{xx}$, and $\boldsymbol{\Sigma}_{yx}$, respectively. More specifically, set

$$\mathbf{S}_{yxy} = \mathbf{S}_{yy}^{-1}\mathbf{S}_{yx}\mathbf{S}_{xx}^{-1}\mathbf{S}_{xy}, \text{ if } m_y \leq m_x,$$

and

$$\mathbf{S}_{xyx} = \mathbf{S}_{xx}^{-1}\mathbf{S}_{xy}\mathbf{S}_{yy}^{-1}\mathbf{S}_{yx}, \text{ if } m_x < m_y,$$

and let $\rho_1^2 \geq \rho_2^2 \geq \ldots \geq \rho_{m_y}^2 \geq 0$ be the eigenvalues of $\mathbf{S}_{yxy}$. Then the kth squared canonical correlation of $\mathbf{Y}$ and $\mathbf{X}$ is given by the kth largest eigenvalue of matrix $\mathbf{S}_{yxy}$, ρ_k^2. These coefficients measure the strength of the overall relationships between the two canonical variates, or weighted sums of $\mathbf{Y}$ and $\mathbf{X}$.

The canonical variates, u_{kt} and v_{kt}, associated with the kth squared canonical correlation, ρ_k^2 is given by

$$u_{kt} = \boldsymbol{\alpha}_{(k)}'\mathbf{y}_t \text{ and } v_{kt} = \boldsymbol{\gamma}_{(k)}'\mathbf{x}_t,$$

where

$$\begin{pmatrix} -\rho_k \mathbf{S}_{yy} & \mathbf{S}_{yx} \\ \mathbf{S}_{xy} & -\rho_k \mathbf{S}_{xx} \end{pmatrix} \begin{pmatrix} \boldsymbol{\alpha}_{(k)} \\ \boldsymbol{\gamma}_{(k)} \end{pmatrix} = \mathbf{0}.$$

But it is easily seen that

$$\left(\mathbf{S}_{yx}\mathbf{S}_{xx}^{-1}\mathbf{S}_{xy} - \rho_k^2 \mathbf{S}_{yy}\right)\boldsymbol{\alpha}_{(k)} = 0,$$
$$\left(\mathbf{S}_{xy}\mathbf{S}_{yy}^{-1}\mathbf{S}_{yx} - \rho_k^2 \mathbf{S}_{xx}\right)\boldsymbol{\gamma}_{(k)} = 0,$$

and hence $\boldsymbol{\alpha}_{(k)}$ can be computed as the eigenvector associated with the kth largest root of $\mathbf{S}_{yxy} = \mathbf{S}_{yy}^{-1}\mathbf{S}_{yx}\mathbf{S}_{xx}^{-1}\mathbf{S}_{xy}$, and $\boldsymbol{\gamma}_{(k)}$ can be computed as the eigenvector associated with the kth largest root of $\mathbf{S}_{xyx} = \mathbf{S}_{xx}^{-1}\mathbf{S}_{xy}\mathbf{S}_{yy}^{-1}\mathbf{S}_{yx}$. These eigenvectors are normalized such that

$$\boldsymbol{\alpha}'_{(k)}\mathbf{S}_{yy}\boldsymbol{\alpha}_{(k)} = 1,\ \boldsymbol{\gamma}'_{(k)}\mathbf{S}_{xx}\boldsymbol{\gamma}_{(k)} = 1,\ \text{and}\ \boldsymbol{\alpha}'_{(k)}\mathbf{S}_{yx}\boldsymbol{\gamma}_{(k)} = \rho_k.$$

Under the null hypothesis $H_0 : Cov(\mathbf{X}, \mathbf{Y}) = \mathbf{0}$, the statistic

$$T \times Trace\left(\mathbf{S}_{yxy}\right) \overset{a}{\sim} \chi^2_{(m_y-1)(m_x-1)}.$$

The above analysis can be extended to control for a third set of variables that might influence $\mathbf{Y}$ and $\mathbf{X}$. Consider the $T \times m_z$ observation matrix $\mathbf{Z}$, and suppose that $T > \max\left(m_y, m_x, m_z\right)$. Let $\mathbf{M}_z = \mathbf{I}_T - \mathbf{Z}\left(\mathbf{Z}'\mathbf{Z}\right)^{-1}\mathbf{Z}'$. Compute

$$\hat{\mathbf{Y}} = \mathbf{M}_z\mathbf{Y},\ \hat{\mathbf{X}} = \mathbf{M}_z\mathbf{X}.$$

Then in this case, the $\mathbf{S}$ matrix is given by

$$\mathbf{S}_{\hat{y}\hat{x}\hat{y}} = \left(\frac{\hat{\mathbf{Y}}'\hat{\mathbf{Y}}}{T}\right)^{-1}\left(\frac{\hat{\mathbf{Y}}'\hat{\mathbf{X}}}{T}\right)\left(\frac{\hat{\mathbf{X}}'\hat{\mathbf{X}}}{T}\right)^{-1}\left(\frac{\hat{\mathbf{X}}'\hat{\mathbf{Y}}}{T}\right),$$

if $m_y \leq m_x$ and

$$\mathbf{S}_{\hat{x}\hat{y}\hat{x}} = \left(\frac{\hat{\mathbf{X}}'\hat{\mathbf{X}}}{T}\right)^{-1}\left(\frac{\hat{\mathbf{X}}'\hat{\mathbf{Y}}}{T}\right)\left(\frac{\hat{\mathbf{Y}}'\hat{\mathbf{Y}}}{T}\right)^{-1}\left(\frac{\hat{\mathbf{X}}'\hat{\mathbf{Y}}}{T}\right),$$

if $m_y > m_x$.

Similarly, the covariates in this case are defined by

$$u_{kt} = \boldsymbol{\alpha}'_{(k)}\hat{\mathbf{y}}_t,\ \text{and}\ v_{kt} = \boldsymbol{\gamma}'_{(k)}\hat{\mathbf{x}}_t,$$

where $\boldsymbol{\alpha}_{(k)}$ is the eigenvector of $\mathbf{S}_{\hat{y}\hat{x}\hat{y}}$ associated with its kth largest eigenvalue, and $\boldsymbol{\gamma}_{(k)}$ is the eigenvector of $\mathbf{S}_{\hat{x}\hat{y}\hat{x}}$ associated with its kth largest eigenvalue. Note that by construction $\widehat{Corr}(u_{kt}, v_{kt}) = \rho_k$, $\widehat{Var}(u_{kt}) = \widehat{Var}(v_{kt}) = 1$, for $k = 1, 2, \ldots, \min(m_y, m_x)$.

See Anderson (2003, Ch. 12) for further details.

19.7 Reduced rank regression

The reduced rank regression is due to Anderson (1951). To introduce this method it is useful to rearrange the elements in (19.2) as follows

$$\mathbf{Y} = \mathbf{XB} + \mathbf{U}, \tag{19.62}$$

where $\mathbf{Y}$ is a $T \times m$ matrix, $\mathbf{X}$ is $T \times km$, and $\mathbf{U}$ is $T \times m$, and $\mathbf{B}$ is a $km \times m$ matrix of unknown parameters. We assume that $E(\mathbf{u}_i | \mathbf{X}_1, \mathbf{X}_2, \ldots, \mathbf{X}_m) = \mathbf{0}$, and $E\left(\mathbf{u}_i \mathbf{u}_j' \middle| \mathbf{X}_1, \mathbf{X}_2, \ldots, \mathbf{X}_m\right) = \sigma_{ij}\mathbf{I}_T$, with σ_{ij} elements of the $m \times m$ matrix, $\boldsymbol{\Sigma}$, and $T > km$. The reduced rank regression (RRR) method imposes rank restrictions on the matrix coefficient $\mathbf{B}$, namely

$$rank(\mathbf{B}) = r < m, \tag{19.63}$$

where r is an integer. The above rank restriction has the interpretation that fewer than m linear combinations of the $\mathbf{X}$ variables are relevant to the explanation of the dependent variables (Tso (1981)). Under the reduced rank hypothesis (19.63), the coefficient matrix $\mathbf{B}$ can be expressed as the product of two matrices of lower dimensions, namely $\mathbf{B} = \mathbf{CD}$, with $\mathbf{C}$ and $\mathbf{D}$ having dimensions $km \times r$ and $r \times m$ respectively, so that model (19.62) can now be written as

$$\mathbf{Y} = \mathbf{XCD} + \mathbf{U}.$$

Under the rank deficiency condition, the *OLS* method is not valid since it ignores the cross-equation restrictions on the elements of $\mathbf{B}$ imposed by the rank deficiency. This is clarified in the following example.

Example 40 *Consider model (19.62) where we assume identical regressors, $k = 2$, and $Rank(\mathbf{B}) = 1$. We have*

$$\mathbf{B}_{2\times 2} = \begin{pmatrix} \beta_{11} & \beta_{12} \\ \beta_{21} & \beta_{22} \end{pmatrix}.$$

Given that $Rank(\mathbf{B}) = 1$, the determinant of $\mathbf{B}$ is 0, and we have the following nonlinear restriction on the elements of $\mathbf{B}$: $\beta_{11}\beta_{22} - \beta_{12}\beta_{21} = 0$.

The log-likelihood function of (19.62) is given by (see Anderson (1951), Tso (1981))

$$\ell(\boldsymbol{\theta}) = -\frac{Tm}{2}\log(2\pi) - \frac{T}{2}\log|\boldsymbol{\Sigma}| - \frac{1}{2}Tr\left(\mathbf{U}\boldsymbol{\Sigma}^{-1}\mathbf{U}'\right), \tag{19.64}$$

with $\boldsymbol{\theta} = \left[vec(\mathbf{C})', vec(\mathbf{D})', vech(\boldsymbol{\Sigma})'\right]'$, and $\mathbf{U} = \mathbf{Y} - \mathbf{XCD}$. The maximum likelihood estimator of $\boldsymbol{\Sigma}$ conditional on $\mathbf{C}$ and $\mathbf{D}$ is given by

$$\boldsymbol{\Sigma}(\mathbf{C}, \mathbf{D}) = T^{-1}\mathbf{U}'\mathbf{U} = T^{-1}(\mathbf{Y} - \mathbf{XCD})'(\mathbf{Y} - \mathbf{XCD}), \tag{19.65}$$

which, if substituted in (19.64), reduces the problem of maximizing (19.64) to the problem of finding the minimum of

$$q(\mathbf{C}, \mathbf{D}) = T^{-1}\left|(\mathbf{Y} - \mathbf{XCD})'(\mathbf{Y} - \mathbf{XCD})\right|. \tag{19.66}$$

We observe that the above optimization problem does not lead to a unique solution for $\mathbf{C}$ and $\mathbf{D}$. In fact, for any $r \times r$ nonsingular matrix, $\mathbf{G}$,

$$\mathbf{B} = \mathbf{CD} = (\mathbf{CG})\left(\mathbf{G}^{-1}\mathbf{D}\right) = \mathbf{C}^*\mathbf{D}^*,$$

with $\mathbf{C}^* = \mathbf{CG}$, and $\mathbf{D}^* = \mathbf{G}^{-1}\mathbf{D}$, and therefore $q(\mathbf{C}, \mathbf{D}) = q(\mathbf{C}^*, \mathbf{D}^*)$. It follows that r^2 identifying restrictions are needed. Tso (1981) suggests the following restrictions

$$\mathbf{P}'\mathbf{P} = \mathbf{I}_r, \text{ where } \underset{T\times r}{\mathbf{P}} = \underset{T\times(mk)}{\mathbf{X}}\underset{(mk)\times r}{\mathbf{C}}, \tag{19.67}$$

which orthonormalizes $\mathbf{X}'\mathbf{X}$, namely $\mathbf{C}$ is such that $\mathbf{C}'\mathbf{X}'\mathbf{XC} = \mathbf{P}'\mathbf{P} = \mathbf{I}_r$. Under (19.67), (19.66) becomes

$$q(\mathbf{P}, \mathbf{D}) = T^{-1}\left|(\mathbf{Y} - \mathbf{PD})'(\mathbf{Y} - \mathbf{PD})\right|. \tag{19.68}$$

Noting that

$$\mathbf{Y} - \mathbf{PD} = \left(\mathbf{I}_T - \mathbf{PP}'\right)\mathbf{Y} + \mathbf{P}\left(\mathbf{P}'\mathbf{Y} - \mathbf{D}\right),$$

we can rewrite (19.68) as

$$q(\mathbf{P}, \mathbf{D}) = T^{-1}\left|\mathbf{Y}'\left(\mathbf{I}_T - \mathbf{PP}'\right)\mathbf{Y} + \left(\mathbf{P}'\mathbf{Y} - \mathbf{D}\right)'\left(\mathbf{P}'\mathbf{Y} - \mathbf{D}\right)\right|. \tag{19.69}$$

Given that when $\mathbf{D} = \mathbf{P}'\mathbf{Y} = \mathbf{C}'\mathbf{X}'\mathbf{Y}$, (19.69) attains its minimum, we are only left with the problem of minimizing the following expression

$$\tilde{q}(\mathbf{P}) = T^{-1}\left|\mathbf{Y}'\left(\mathbf{I}_T - \mathbf{PP}'\right)\mathbf{Y}\right|,$$

over all matrices $\mathbf{P}$ satisfying (19.67). Assume that $\mathbf{X}$ and $\mathbf{Y}$ are full column ranks and consider the following decompositions of $\mathbf{X}$ and $\mathbf{Y}$ (see Section A.2 in Appendix A for a description of matrix decompositions)

$$\mathbf{X} = \mathbf{RS}, \text{ and } \mathbf{Y} = \mathbf{VQ}, \tag{19.70}$$

where $\mathbf{R}$ and $\mathbf{V}$ are $T \times km$ and $T \times m$ orthogonal matrices, and $\mathbf{S}$ and $\mathbf{Q}$ are $km \times km$ and $m \times m$ invertible matrices. Noting that, given (19.70), $\mathbf{P} = \mathbf{RSC} = \mathbf{RF}$ with $\mathbf{F} = \mathbf{SC}$ being a $km \times r$ matrix such that $\mathbf{F}'\mathbf{F} = \mathbf{I}_r$, we have

$$\begin{aligned}\tilde{q}(\mathbf{P}) = \tilde{q}(\mathbf{RF}) &= T^{-1}\left|\mathbf{Q}'\mathbf{V}'\left(\mathbf{I}_T - \mathbf{RFF}'\mathbf{R}'\right)\mathbf{VQ}\right| \\ &= T^{-1}|\mathbf{Q}|^2\left|\mathbf{I}_{km} - \mathbf{HFF}'\mathbf{H}'\right| = T^{-1}|\mathbf{Q}|^2\left|\mathbf{F}'\left(\mathbf{I}_{km} - \mathbf{HH}'\right)\mathbf{F}\right|, \end{aligned} \tag{19.71}$$

where $\mathbf{H} = \mathbf{R}'\mathbf{V}$ is $km \times m$. Expression (19.71) reaches its minimum when the columns of $\mathbf{F}$ are given by the eigenvectors associated to the r largest eigenvalues of $\mathbf{HH}'$, namely $\lambda_k\left(\mathbf{HH}'\right)$, $k = 1, 2, \ldots, r$ (see Tso (1981) for a proof). Since $\mathbf{S}$ and $\mathbf{Q}$ are invertible (then $\mathbf{S}\left(\mathbf{S}'\mathbf{S}\right)^{-1}\mathbf{S}'$ and $\mathbf{Q}\left(\mathbf{Q}'\mathbf{Q}\right)^{-1}\mathbf{Q}'$ are equal to identity matrices)

$$\lambda_k\left(\mathbf{HH}'\right) = \lambda_k\left(\mathbf{R}'\mathbf{VV}'\mathbf{R}\right) = \lambda_k\left[\mathbf{S}\left(\mathbf{S}'\mathbf{S}\right)^{-1}\mathbf{S}'\mathbf{R}'\mathbf{VQ}\left(\mathbf{Q}'\mathbf{Q}\right)^{-1}\mathbf{Q}'\mathbf{V}'\mathbf{R}\right],$$

and using properties of eigenvalues (see Section A.2 in Appendix A) we have

$$\lambda_k\left(\mathbf{HH}'\right) = \lambda_k\left[\left(\mathbf{S}'\mathbf{R}'\mathbf{RS}\right)^{-1}\mathbf{S}'\mathbf{R}'\mathbf{VQ}\left(\mathbf{Q}'\mathbf{V}'\mathbf{VQ}\right)^{-1}\mathbf{Q}'\mathbf{V}'\mathbf{RS}\right].$$

But, using the transformations in (19.70), we have

$$\begin{aligned}\mathbf{S}'\mathbf{R}'\mathbf{RS} = \mathbf{X}'\mathbf{X}, \mathbf{S}'\mathbf{R}'\mathbf{VQ} = \mathbf{X}'\mathbf{Y}, \\ \mathbf{Q}'\mathbf{V}'\mathbf{VQ} = \mathbf{Y}'\mathbf{Y},\end{aligned}$$

and hence

$$\lambda_k\left(\mathbf{HH}'\right) = \lambda_k\left(\mathbf{S}_{xx}^{-1}\mathbf{S}_{xy}\mathbf{S}_{yy}^{-1}\mathbf{S}_{yx}\right),$$

where $\mathbf{S}_{xx}$, $\mathbf{S}_{xy}$, and $\mathbf{S}_{yy}$ are defined by (19.61). It follows that $\lambda_k\left(\mathbf{HH}'\right)$ corresponds to the kth largest squared canonical correlations between the variables in $\mathbf{Y}$ and $\mathbf{X}$ (see Section 19.6). Estimates of $\mathbf{C}$ are given by $\hat{\mathbf{C}} = \mathbf{S}^{-1}\mathbf{F}$, while estimates of $\mathbf{D}$ and $\boldsymbol{\Sigma}$, in terms of $\hat{\mathbf{C}}$, can be obtained as

$$\begin{aligned}\hat{\mathbf{D}} &= \left(\hat{\mathbf{C}}'\mathbf{X}'\mathbf{X}\hat{\mathbf{C}}\right)^{-1\prime}\mathbf{X}\hat{\mathbf{C}}\mathbf{Y} = \mathbf{S}_{yx}\hat{\mathbf{C}}\left(\hat{\mathbf{C}}'\mathbf{S}_{xx}\hat{\mathbf{C}}\right)^{-1}, \\ \hat{\boldsymbol{\Sigma}} &= T^{-1}\left(\mathbf{Y} - \mathbf{X}\hat{\mathbf{C}}\hat{\mathbf{D}}\right)'\left(\mathbf{Y} - \mathbf{X}\hat{\mathbf{C}}\hat{\mathbf{D}}\right).\end{aligned}$$

See Anderson (1951) and Tso (1981) for further details. As we will see in Chapter 22, the *RRR* method is particularly useful in the analysis of cointegrated variables (see also Johansen (1991)).

19.8 Further reading

A general account of principal components and canonical correlation analysis can be found in Anderson (2003), while textbook treatments of system estimation are available in Judge et al. (1985) and in Greene (2002).

19.9 Exercises

1. Consider the following system of regression equations

$$\mathbf{y}_i = \mathbf{X}_i\boldsymbol{\beta}_i + \mathbf{u}_i, \text{ for } i = 1, 2, \ldots, m,$$

where $\boldsymbol{\beta}_i$ is a $k_i \times 1$ vector of unknown coefficients, $\mathbf{y}_i$ is the $T \times 1$ vector of observations on the ith endogenous variable, $\mathbf{X}_i$ is $T \times k_i$ matrix of observations on the regressors of the ith equation and $\mathbf{u}_i$ is $T \times 1$ vector of errors

$$E(\mathbf{u}_i) = \mathbf{0}, E(\mathbf{u}_i\mathbf{u}_j') = \sigma_{ij}\mathbf{I}_T,$$
$$E\left(\mathbf{u}_i|\mathbf{X}_j\right) = \mathbf{0}, \text{ for all } i \text{ and } j,$$

where $\mathbf{I}_T$ is an identity matrix of order T.

(a) Derive the generalized least squares (GLS) estimator of $\boldsymbol{\beta}_i$.

(b) What is a feasible GLS estimator of $\boldsymbol{\beta}_i$ and how does it relate to the full information maximum likelihood estimator of $\boldsymbol{\beta}_i$?

(c) For a fixed m, show that the GLS is as efficient as the OLS applied to each equation.

(d) Establish the conditions under which GLS and OLS are algebraically identical.

2. Consider the regression model

$$\mathbf{y} = \mathbf{W}\boldsymbol{\beta} + \mathbf{u},$$

where W is a $T \times k$ stochastic matrix of rank k, possibly correlated with the $T \times 1$ vector of disturbance. Suppose the $T \times s$ data matrix $\mathbf{Z}$ is available which is asymptotically uncorrelated with $\mathbf{u}$.

(a) Using the information available to you, derive the instrumental variable (IV) estimator of $\boldsymbol{\beta}$

i. when $s = k$,

and

ii. when $s > k$.

(b) Derive necessary and sufficient conditions under which the IV estimators are consistent and asymptotically efficient with respect to the available set of instruments.

(c) What is a suitable statistic for testing the validity of the *IV* estimators when $s > k$? Comment on the usefulness of such a test.

3. Consider the following simultaneous equation structural model

$$y_{1t} = \alpha y_{2t} + u_{1t},$$
$$y_{2t} = \beta y_{1t} + \theta x_t + u_{2t},$$

where $\mathbf{y}_t = (y_{1t}, y_{2t})'$ is the 2×1 vector of endogenous variables, and x_t is the only exogenous variable of the model. The 2×1 vector of errors $\mathbf{u}_t = (u_{1t}, u_{2t})'$ is serially uncorrelated with mean zero and the positive definite covariance matrix

$$Cov\,(\mathbf{u}_t) = \begin{pmatrix} \sigma_{11} & \sigma_{12} \\ \sigma_{21} & \sigma_{22} \end{pmatrix}.$$

(a) Discuss the conditions under which α is identified. Can β be identified as well?

(b) Show that the structural model has the following reduced form representation (assuming that $\alpha\beta \neq 1$)

$$y_{1t} = \left(\frac{\alpha\theta}{1-\alpha\beta}\right) x_t + \frac{u_{1t} + \alpha u_{2t}}{1-\alpha\beta},$$
$$y_{2t} = \left(\frac{\theta}{1-\alpha\beta}\right) x_t + \frac{\beta u_{1t} + u_{2t}}{1-\alpha\beta}.$$

(c) Show that the *OLS* estimator of α based on the observations $\{\mathbf{y}_t, x_t;$ for $t = 1, 2, \ldots, T\}$ is biased. Under what conditions does this bias vanish asymptotically (as $T \to \infty$)?

(d) Consider now the IV (or two-stage) estimator of α

$$\hat{\alpha}_{IV} = \left(\sum_{t=1}^{T} x_t y_{2t}\right)^{-1} \sum_{t=1}^{T} x_t y_{1t}.$$

Show that $\hat{\alpha}_{IV}$ is asymptotically unbiased and consistent if θ is a fixed non-zero constant and $T^{-1}\sum_{t=1}^{T} x_t^2 > 0$, for all T, and as $T \to \infty$.

(e) Suppose now that θ varies with T such that $\theta_T = \delta/\sqrt{T}$, with δ now being a fixed non-zero constant. What are the implications of this specifications for bias, consistency, and the asymptotic distribution of $\hat{\alpha}_{IV}$?

4. Consider the factor model

$$y_{it} = \sum_{j=1}^{k} \gamma_{ij} f_{jt} + u_{it}, \text{ for } i = 1, 2, \ldots, m, \text{ and } t = 1, 2, \ldots, T, \tag{19.72}$$

where the factors, $f_{jt}, j = 1, 2, \ldots, k$ are mutually uncorrelated with unit variances, and distributed independently of u_{it}.

(a) Show that

$$Var(\mathbf{y}_t) = \mathbf{\Gamma}\mathbf{\Gamma}' + \mathbf{\Sigma}_u,$$

where $\mathbf{\Sigma}_u = Var(\mathbf{u}_t)$, $\mathbf{y}_t = (y_{1t}, y_{2t}, \ldots, y_{mt})'$, $\mathbf{u}_t = (u_{1t}, u_{2t}, \ldots, u_{mt})'$, $\mathbf{\Gamma} = (\boldsymbol{\gamma}_1, \boldsymbol{\gamma}_2, \ldots, \boldsymbol{\gamma}_k)$, and $\boldsymbol{\gamma}_j = (\gamma_{1j}, \gamma_{2j}, \ldots, \gamma_{mj})'$.

(b) Discuss the estimation of the factors by principal components.

(c) Show that in the case where $\mathbf{\Sigma}_u = \sigma_u^2 \mathbf{I}_m$, then for sufficiently large T the factors can be consistently estimated by $\hat{f}_{jt} = \mathbf{p}_j' \mathbf{y}_t$, where $\mathbf{p}_j, j = 1, 2, \ldots, k$, are the orthonormalized eigenvectors of $\mathbf{\Gamma}\mathbf{\Gamma}'$.

5. Consider the multifactor model given by (19.72) and suppose that y_{it} denotes the return on security i during period t. Consider the portfolio return $\rho_{\omega t} = \boldsymbol{\omega}' \mathbf{y}_t$, where $\boldsymbol{\omega} = (\omega_1, \omega_2, \ldots, \omega_m)'$ is a vector of granular weights such $\omega_i = O(1/m)$.

(a) Show that

$$Var\left(\rho_{\omega t}\right) \leq \boldsymbol{\omega}'\boldsymbol{\omega} \left[\lambda_1(\mathbf{\Gamma}\mathbf{\Gamma}') + \lambda_1(\mathbf{\Sigma}_u)\right],$$

where $\lambda_1(\mathbf{\Gamma}\mathbf{\Gamma}')$, and $\lambda_1(\mathbf{\Sigma}_u)$ are the largest eigenvalues of $\mathbf{\Gamma}\mathbf{\Gamma}'$ and $\mathbf{\Sigma}_u$, respectively.

(b) Suppose that u_{it}, for $i = 1, 2, \ldots, m$, are weakly cross-sectionally dependent. Investigate the conditions under which the risk of holding the portfolio, $\boldsymbol{\omega}' \mathbf{y}_t$, cannot be fully diversified.

20 Multivariate Rational Expectations Models

20.1 Introduction

All economic and financial decisions are subject to major uncertainties. How to model uncertainty and expectations has been controversial, although since the pioneering contributions of Muth, Lucas, and Sargent, the rational expectations hypothesis (*REH*) has come to dominate economics and finance as the favoured approach to expectations formation. According to the *REH*, subjective characterization of uncertainty as conditional probability distributions will coincide (through learning) with the associated objective outcomes. The *REH* is mathematically elegant and allows model-consistent solutions, and fits nicely within the equilibrium economic theory. Almost all dynamic stochastic general equilibrium (*DSGE*) models used in macroeconomics and finance are solved under the *REH*. It is with this in mind that we devote this chapter to the solution, identification, and estimation of rational expectations models. But readers should be aware of the limitations of the *REH*, as set out in Pesaran (1987c).

We begin with an overview of solution techniques, distinguishing between *RE* models with and without feedbacks from the decision (or target) variables to the state variables. We also consider models with and without lagged values of the decision variables. In the case of *RE* models with feedbacks we argue that it is best to cast the *RE* models as a closed dynamic system before solving them. We then consider identification of structural parameters of *DSGE* models and estimation of *RE* models in general.

20.2 Rational expectations models with future expectations

Most macroeconomic *DSGE* models are constructed by linearizing an underlying nonlinear model around its steady state, where $\boldsymbol{\theta}$ is a vector of deep parameters of this underlying model. Consider a linearized rational expectations model for an $m \times 1$ vector of variables of interest, $\mathbf{y}_t$, $t = 1, 2, \ldots, T$. These would usually be measured as deviations from their steady states. A first-order multivariate rational expectations (*RE*) model with future expectations can be written as

$$\mathbf{y}_t = \mathbf{A}E(\mathbf{y}_{t+1}|\Omega_t) + \mathbf{w}_t, \tag{20.1}$$

where $\mathbf{w}_t = (w_{1t}, w_{2t}, \ldots, w_{mt})'$ is an m-dimensional vector of state variables, and $\mathbf{A}$ is an $m \times m$ dimensional matrix of fixed coefficients, $E(y_{t+1}|\Omega_t)$ is the vector of expectations of the future endogenous variables conditional on Ω_t, where Ω_t is the information set available at time t, and contains all the data on the past history of the variables entering in the model, as well as any other information that may be available. Hence, we assume Ω_t represents a non-decreasing set at time t, containing at least current and lagged values of $\mathbf{y}_t$ and $\mathbf{w}_t$: $\Omega_t = \{\mathbf{y}_t, \mathbf{y}_{t-1}, \ldots; \mathbf{w}_t, \mathbf{w}_{t-1}, \ldots; \ldots\}$.

In the *RE* literature, $\mathbf{w}_t$ is viewed as the 'forcing variable' of the *RE* model. This vector can include a linear combination of strictly exogenous regressors, $\mathbf{x}_t$, as well as purely deterministic processes such as intercept, linear trends or seasonal dummy variables (see Section 12.2). For example, we could have $\mathbf{w}_t = \mathbf{B}\mathbf{x}_t + \mathbf{u}_t$, where $\mathbf{x}_t$ is a $k \times 1$ vector of observed exogenous variables, $\mathbf{B}$ is an $m \times k$ matrix of fixed coefficients, and $\mathbf{u}_t$ is an $m \times 1$ vector of unobserved errors that could possibly be serially correlated. We do not require $\mathbf{w}_t$ to be Gaussian and do not rule out possible patterns of conditional heteroskedasticity in $\mathbf{x}_t$ and/or $\mathbf{u}_t$. The $\mathbf{w}_t$ process could also be a nonlinear function of a set of exogenous variables. However, here we assume that there are no feedbacks from lagged values of $\mathbf{y}_t$ onto $\mathbf{w}_t$.

We next derive a solution for equation (20.1). We recall that a solution is a sequence of functions of variables in Ω_t satisfying (20.1) for all possible realizations of these variables.

20.2.1 Forward solution

The solution of equation (20.1) can be found by applying the forward approach, which is an extension of the univariate method described in Section 6.8 to a multivariate context. Writing (20.1) for period $t + 1$ and taking conditional expectations of both sides with respect to Ω_t, we have

$$E(\mathbf{y}_{t+1}|\Omega_t) = \mathbf{A}E\left[E(\mathbf{y}_{t+2}|\Omega_{t+1})|\Omega_t\right] + E(\mathbf{w}_{t+1}|\Omega_t).$$

But since Ω_t is a non-decreasing information set, by the law of iterated expectations

$$E\left[E(\mathbf{y}_{t+2}|\Omega_{t+1})|\Omega_t\right] = E(\mathbf{y}_{t+2}|\Omega_t),$$

and we obtain

$$E(\mathbf{y}_{t+1}|\Omega_t) = \mathbf{A}E(\mathbf{y}_{t+2}|\Omega_t) + E(\mathbf{w}_{t+1}|\Omega_t).$$

Substituting this result in (20.1) now yields

$$\mathbf{y}_t = \mathbf{A}^2E(\mathbf{y}_{t+2}|\Omega_t) + \mathbf{A}E(\mathbf{w}_{t+1}|\Omega_t) + \mathbf{w}_t.$$

Similarly,

$$E(\mathbf{y}_{t+2}|\Omega_t) = \mathbf{A}E(\mathbf{y}_{t+3}|\Omega_t) + E(\mathbf{w}_{t+2}|\Omega_t),$$

and so on. Using these results recursively forward we obtain

$$\mathbf{y}_t = \mathbf{A}^h E(\mathbf{y}_{t+h}|\Omega_t) + \sum_{j=0}^{h-1} \mathbf{A}^j E\left(\mathbf{w}_{t+j}|\Omega_t\right). \tag{20.2}$$

A unique solution exits if it is possible to eliminate the effect of future expectations, $\mathrm{E}(\mathbf{y}_{t+h}|\Omega_t)$, on $\mathbf{y}_t$. Suppose that all eigenvalues of $\mathbf{A}$ are distinct and consider the spectral decomposition of $\mathbf{A}$ given by $\mathbf{A} = \mathbf{PDP}^{-1}$, where $\mathbf{D}$ is a diagonal matrix formed from the eigenvalues of $\mathbf{A}$, and columns of matrix $\mathbf{P}$ are formed from the associated eigenvectors of $\mathbf{A}$, and $\mathbf{A}^h = \mathbf{PD}^h\mathbf{P}^{-1}$.[1] Using this decomposition (20.2) can be written as

$$\tilde{\mathbf{y}}_t = \mathbf{D}^h E(\tilde{\mathbf{y}}_{t+h}|\Omega_t) + \sum_{j=0}^{h-1} \mathbf{D}^j E\left(\tilde{\mathbf{w}}_{t+j}|\Omega_t\right),$$

where $\tilde{\mathbf{y}}_t = \mathbf{P}^{-1}\mathbf{y}_t = (\tilde{y}_{1t}, \tilde{y}_{2t}, \ldots, \tilde{y}_{mt})'$, and $\tilde{\mathbf{w}}_t = \mathbf{P}^{-1}\mathbf{w}_t$. Hence,

$$\tilde{y}_{it} = \lambda_i^h E(\tilde{y}_{i,t+h}|\Omega_t) + \sum_{j=0}^{h-1} \lambda_i^j E\left(\tilde{w}_{i,t+j}|\Omega_t\right), \text{ for } i = 1, 2, \ldots, m,$$

where $\lambda_i, i = 1, 2, \ldots, m$ are the distinct eigenvalues of $\mathbf{A}$. It is now clear that if all eigenvalues of $\mathbf{A}$ have an absolute value smaller than unity (namely $|\lambda_i| < 1$), then as $h \to \infty$, $\lambda_i^h \to 0$, and the solution to $\tilde{y}_{it}$ will not depend on the future expectations of $\mathbf{y}_t$ so long as for all h the future expectations, $E(\tilde{y}_{i,t+h}|\Omega_t)$, are bounded or satisfy the transversality conditions

$$\lim_{h\to\infty} \lambda_i^h E(\tilde{y}_{i,t+h}|\Omega_t) = 0, \text{ for } i = 1, 2, \ldots, m.$$

In matrix notation, we have

$$\lim_{h\to\infty} \mathbf{A}^h \mathrm{E}(\mathbf{y}_{t+h}|\Omega_t) = \mathbf{0}. \tag{20.3}$$

Finally, assuming that the process of the forcing variables is stable, the unique solution of (20.1) is given by

$$\mathbf{y}_t = \sum_{j=0}^{\infty} \mathbf{A}^j E(\mathbf{w}_{t+j}\,|\Omega_t\,). \tag{20.4}$$

This solution does not require the $\mathbf{w}_t$ process to be stationary and allows the forcing variables to contain unit roots. For example, suppose that $\mathbf{w}_t$ follows the first-order vector autoregressive process

[1] In the case where one or more eigenvalues of $\mathbf{A}$ are the same, one needs to use the Jordan form where the diagonal matrix $\mathbf{D}$ is replaced by an upper (lower) triangular matrix having eigenvalues of $\mathbf{A}$ on its main diagonal.

$$\mathbf{w}_t = \mathbf{\Phi}\mathbf{w}_{t-1} + \mathbf{v}_t,$$

where $\mathbf{v}_t$ are serially uncorrelated innovations. For this process, $E(\mathbf{w}_{t+j}\,|\Omega_t) = \mathbf{\Phi}^j\mathbf{w}_t$, and assuming that all eigenvalues of $\mathbf{A}$ lie inside the unit circle we have

$$\mathbf{y}_t = \left(\sum_{j=0}^{\infty}\mathbf{A}^j\mathbf{\Phi}^j\right)\mathbf{w}_t. \tag{20.5}$$

It is now easily seen that for a finite m, the solution exists if the product of the largest eigenvalues of $\mathbf{A}$ and $\mathbf{\Phi}$ strictly lies within the unit circle. Therefore, one or more eigenvalues of $\mathbf{\Phi}$ could be equal to unity if all the eigenvalues of $\mathbf{A}$ are less than unity in absolute value.

In cases where one or more eigenvalues of $\mathbf{A}$ lie on or outside the unit circle, the solution to the *RE* model is not unique and depends on arbitrary martingale processes. In the extreme case where all eigenvalues of $\mathbf{A}$ fall outside the unit circle the general solution can be written as

$$\mathbf{y}_t = \mathbf{A}^{-t}\mathfrak{m}_t - \sum_{j=0}^{t-1}\mathbf{A}^{-j}\mathbf{w}_{t-j}, \text{ for } t \geq 1,$$

where $\mathfrak{m}_t$ is a martingale vector process with m arbitrary martingale components such that $E(\mathfrak{m}_{t+1}\,|\Omega_t) = \mathfrak{m}_t$ (see Section 15.3.1). In the more general case where m_1 of the roots of $\mathbf{A}$ fall on or outside the unit circle and the rest fall inside, the solution will depend on m_1 arbitrary martingale processes.

20.2.2 Method of undetermined coefficients

Under the above conditions the unique solution can also be obtained by the method of undetermined coefficients. This method, proposed by Whiteman (1983) and Salemi (1986), starts with a 'guess' linear solution in terms of $\mathbf{w}_t$ and its lagged values. The order of the lags will be determined by the order of the *VAR* process for $\mathbf{w}_t$. If $\mathbf{w}_t$ follows a *VAR*(p) process, a guess for the solution will be in the form of the distributed lag function in $\mathbf{w}_t$ with order $(p-1)$. In the simple case where $p = 1$, the guess solution is given by

$$\mathbf{y}_t = \mathbf{G}\mathbf{w}_t,$$

where $\mathbf{G}$ is the matrix of unknown coefficients to be obtained in terms of $\mathbf{A}$ and $\mathbf{\Phi}$. We first note that for this solution $E(\mathbf{y}_{t+1}\,|\Omega_t) = \mathbf{G}E(\mathbf{w}_{t+1}\,|\Omega_t) = \mathbf{G}\mathbf{\Phi}\mathbf{w}_t$. Using $\mathbf{y}_t = \mathbf{G}\mathbf{w}_t$ and $E(\mathbf{y}_{t+1}\,|\Omega_t) = \mathbf{G}\mathbf{\Phi}\mathbf{w}_t$ in (20.1) we have

$$\mathbf{G}\mathbf{w}_t = \mathbf{A}\mathbf{G}\mathbf{\Phi}\mathbf{w}_t + \mathbf{w}_t,$$

and the unknown coefficient matrix, $\mathbf{G}$, must satisfy the system of equations (known as '*Sylverster equations*')

$$\mathbf{G} = \mathbf{A}\mathbf{G}\mathbf{\Phi} + \mathbf{I}_m, \tag{20.6}$$

where $\mathbf{I}_m$ is an identity matrix of order m. To obtain the solution (20.5) matrix $\mathbf{G}$ can be solved in terms of $\mathbf{A}$ and $\mathbf{\Phi}$ iteratively. Consider the recursive system of equations with $\mathbf{G}^{(0)} = \mathbf{I}_m$

$$\mathbf{G}^{(s)} = \mathbf{A}\mathbf{G}^{(s-1)}\mathbf{\Phi} + \mathbf{I}_m, \text{ for } s = 1, 2, \ldots.$$

Then $\mathbf{G}^{(1)} = \mathbf{A}\mathbf{\Phi} + \mathbf{I}_m$, $\mathbf{G}^{(2)} = \mathbf{A}(\mathbf{A}\mathbf{\Phi} + \mathbf{I}_m)\mathbf{\Phi} + \mathbf{I}_m = \mathbf{A}^2\mathbf{\Phi}^2 + \mathbf{A}\mathbf{\Phi} + \mathbf{I}_m, \ldots$. The limit as $s \to \infty$ will be given by $\sum_{j=0}^{\infty} \mathbf{A}^j\mathbf{\Phi}^j$.

Alternatively, (20.6) can be solved directly by writing it as

$$vec(\mathbf{G}) = vec(\mathbf{A}\mathbf{G}\mathbf{\Phi}) + vec(\mathbf{I}_m),$$

where $vec(\mathbf{A})$ denotes a vector composed of the stacked columns of $\mathbf{A}$. But (see, e.g., Magnus and Neudecker (1999, p. 30, Theorem 2))

$$vec(\mathbf{A}\mathbf{G}\mathbf{\Phi}) = \left(\mathbf{\Phi}' \otimes \mathbf{A}\right) vec(\mathbf{G})$$

where $\otimes$ denotes the Kronecker matrix product. Hence

$$\left[\mathbf{I}_{2m} - \left(\mathbf{\Phi}' \otimes \mathbf{A}\right)\right] vec(\mathbf{G}) = vec(\mathbf{I}_m).$$

Since it is assumed that all eigenvalues of $\mathbf{A}$ lie within the unit circle, then $\left[\mathbf{I}_{2m} - \left(\mathbf{\Phi}' \otimes \mathbf{A}\right)\right]$ will be a nonsingular matrix if none of the eigenvalues of $\mathbf{\Phi}'$ lie outside the unit circle.[2] Therefore, in the case where the solution is unique and stable we have

$$vec(\mathbf{G}) = \left[\mathbf{I}_{m^2} - \left(\mathbf{\Phi}' \otimes \mathbf{A}\right)\right]^{-1} vec(\mathbf{I}_m).$$

The above solution strategy can be readily extended to the case where $\mathbf{w}_t$ follows higher-order processes, or when $\mathbf{w}_t$ contains serially correlated unobserved components, as in the following example.

Example 41 *Suppose that* $\mathbf{w}_t = \mathbf{B}\mathbf{x}_t + \mathbf{u}_t$ *where* $\mathbf{x}_t$ *is a second-order process*

$$\mathbf{x}_t = \mathbf{\Phi}_1\mathbf{x}_{t-1} + \mathbf{\Phi}_2\mathbf{x}_{t-2} + \mathbf{v}_t,$$

and

$$\mathbf{u}_t = \mathbf{R}\mathbf{u}_{t-1} + \boldsymbol{\eta}_t,$$

where $\mathbf{v}_t$ *and* $\boldsymbol{\eta}_t$ *are serially uncorrelated with zero means. Using (20.4)*

[2] This happens because the eigenvalues of Kronecker products of two matrices are given by the products of their respective eigenvalues.

$$\mathbf{y}_t = \sum_{j=0}^{\infty} \mathbf{A}^j \mathbf{B} E(\mathbf{x}_{t+j} \,|\Omega_t) + \left(\sum_{j=0}^{\infty} \mathbf{A}^j \mathbf{R}^j \right) \mathbf{u}_t.$$

Since $\mathbf{x}_t$ *is a second-order process then the unique solution (when it exists) will have the general form*

$$\mathbf{y}_t = \mathbf{G}_1 \mathbf{x}_t + \mathbf{G}_2 \mathbf{x}_{t-1} + \mathbf{H}\mathbf{u}_t.$$

To determine $\mathbf{G}_1$, $\mathbf{G}_2$, *and* $\mathbf{H}$ *we first note that*

$$E(\mathbf{y}_{t+1} \,|\Omega_t) = (\mathbf{G}_1 \boldsymbol{\Phi}_1 + \mathbf{G}_2)\, \mathbf{x}_t + \mathbf{G}_1 \boldsymbol{\Phi}_2 \mathbf{x}_{t-1} + \mathbf{H}\mathbf{R}\mathbf{u}_t.$$

Substituting this result in (20.1) and equating the relevant coefficient matrices we obtain

$$\begin{aligned} \mathbf{G}_1 &= \mathbf{A}\mathbf{G}_1 \boldsymbol{\Phi}_1 + \mathbf{A}\mathbf{G}_2 + \mathbf{B}, \\ \mathbf{G}_2 &= \mathbf{A}\mathbf{G}_1 \boldsymbol{\Phi}_2, \\ \mathbf{H} &= \mathbf{A}\mathbf{H}\mathbf{R} + \mathbf{I}_m, \end{aligned}$$

which yield the solution

$$\begin{aligned} vec(\mathbf{G}_1) &= \left[\mathbf{I}_{2m} - \left(\boldsymbol{\Phi}_1' \otimes \mathbf{A} \right) - \left(\boldsymbol{\Phi}_2' \otimes \mathbf{A}^2 \right) \right]^{-1} vec\left(\mathbf{A}\boldsymbol{\Phi}_2 + \mathbf{B} \right), \\ vec(\mathbf{G}_2) &= \left(\boldsymbol{\Phi}_2' \otimes \mathbf{A} \right) vec(\mathbf{G}_1), \\ vec(\mathbf{H}) &= \left[\mathbf{I}_{m^2} - \left(\mathbf{R}' \otimes \mathbf{A} \right) \right]^{-1} vec\left(\mathbf{I}_m \right). \end{aligned}$$

For further details on alternative methods of solving *RE* model with strictly exogenous variables, see Pesaran (1981b) and Pesaran (1987c). See also Whiteman (1983), Salemi (1986) and Uhlig (2001).

20.3 Rational expectations models with forward and backward components

Consider now the following more general model

$$\mathbf{y}_t = \mathbf{A}\mathbf{y}_{t-1} + \mathbf{B} E(\mathbf{y}_{t+1} | \Omega_t) + \mathbf{u}_t, \tag{20.7}$$

where $\mathbf{y}_t$ is an m-dimensional vector of observable variables, $\mathbf{u}_t$ is an m-dimensional vector of forcing variables, and $\mathbf{A}$ and $\mathbf{B}$ are $m \times m$ matrices of fixed coefficients. Note that $\mathbf{y}_t$ in (20.7) simultaneously depends on its past values and future expected values. The restriction to one-lag one-lead form is for simplicity, and more lags and leads can be accommodated in this framework by expanding the $\mathbf{y}_t$ vector appropriately (see Section 20.5). In this sense, equation (20.7) accomodates all possible linear rational expectations models.

20.3.1 Quadratic determinantal equation method

The quadratic determinantal equation method (QDE) method was proposed in Binder and Pesaran (1995) and Binder and Pesaran (1997), and involves a transformation that reduces (20.7) to an *RE* model containing only expected future values. The idea is to find an $m \times m$ matrix $\mathbf{C}$ such that the quasi-difference transformation

$$\mathbf{Y}_t = \mathbf{y}_t - \mathbf{C}\mathbf{y}_{t-1}, \tag{20.8}$$

obeys an equation of the form (20.1). Using the fact that $\mathrm{E}(\mathbf{Y}_{t+1} | \Omega_t) = E\left(\mathbf{y}_{t+1} | \Omega_t\right) - \mathbf{C}\mathbf{y}_t$, so that $E\left(\mathbf{y}_{t+1} | \Omega_t\right) = E\left(\mathbf{Y}_{t+1} | \Omega_t\right) + \mathbf{C}\mathbf{y}_t$, and substituting (20.8) back into (20.7) we obtain

$$\begin{aligned}\mathbf{Y}_t &= -\mathbf{C}\mathbf{y}_{t-1} + \mathbf{A}\mathbf{y}_{t-1} + \mathbf{B}\left[E\left(\mathbf{Y}_{t+1} | \Omega_t\right) + \mathbf{C}\mathbf{y}_t\right] + \mathbf{u}_t \\ &= -\mathbf{C}\mathbf{y}_{t-1} + \mathbf{A}\mathbf{y}_{t-1} + \mathbf{B}E\left(\mathbf{Y}_{t+1} | \Omega_t\right) + \mathbf{BC}\left(\mathbf{Y}_t + \mathbf{C}\mathbf{y}_{t-1}\right) + \mathbf{u}_t.\end{aligned}$$

Collecting the terms we obtain

$$\left(\mathbf{I}_m - \mathbf{BC}\right)\mathbf{Y}_t = \mathbf{B}E\left(\mathbf{Y}_{t+1} | \Omega_t\right) + \left(\mathbf{BC}^2 - \mathbf{C} + \mathbf{A}\right)\mathbf{y}_{t-1} + \mathbf{u}_t. \tag{20.9}$$

This equation characterizes the matrix $\mathbf{C}$ introduced in (20.8) as the solution of the quadratic equation

$$\mathbf{BC}^2 - \mathbf{C} + \mathbf{A} = \mathbf{0}_m, \tag{20.10}$$

where $\mathbf{0}_m$ is an $m \times m$ matrix of zeros. Assuming $(\mathbf{I}_m - \mathbf{BC})$ is nonsingular,[3] premultiply both sides of (20.9) by $(\mathbf{I}_m - \mathbf{BC})^{-1}$ to obtain

$$\mathbf{Y}_t = \mathbf{F}E\left(\mathbf{Y}_{t+1} | \Omega_t\right) + \mathbf{W}_t, \tag{20.11}$$

where

$$\begin{aligned}\mathbf{F} &= \left(\mathbf{I}_m - \mathbf{BC}\right)^{-1}\mathbf{B}, \\ \mathbf{W}_t &= \left(\mathbf{I}_m - \mathbf{BC}\right)^{-1}\mathbf{u}_t.\end{aligned}$$

The new equation system (20.11) does not depend on lagged values of the transformed variable, and can be solved using the martingale difference approach (see Section 20.7.4 on this). Binder and Pesaran (1995) and Binder and Pesaran (1997) have shown that there will be a unique solution if there exists a real matrix solution to equation (20.10) such that all eigenvalues of $\mathbf{C}$ lie inside or on the unit circle, and all eigenvalues of $\mathbf{F}$ lie strictly inside the unit circle. In such a case, the unique solution is given by

$$\mathbf{y}_t = \mathbf{C}\mathbf{y}_{t-1} + \sum_{h=0}^{\infty} \mathbf{F}^h \,\mathrm{E}\left(\mathbf{W}_{t+h} | \Omega_t\right). \tag{20.12}$$

[3] Notice that the nonsingularity of $(\mathbf{I}_n - \mathbf{BC})$ does not necessarily require $\mathbf{B}$ to be nonsingular. Binder and Pesaran (1997) provide sufficient conditions under which $(\mathbf{I}_n - \mathbf{BC})$ is nonsingular (see their Proposition 2).

The infinite sum in the solution can be solved for different choices of the $\mathbf{u}_t$ process. For example, if $\mathbf{u}_t$ follows a $VAR(1)$ given by

$$\mathbf{u}_t = \mathbf{R}\mathbf{u}_{t-1} + \boldsymbol{\varepsilon}_t, \tag{20.13}$$

we have

$$E\left(\mathbf{W}_{t+h}|\Omega_t\right) = \left(\mathbf{I}_m - \mathbf{BC}\right)^{-1}\mathbf{R}^h\mathbf{u}_t.$$

Hence,

$$\mathbf{y}_t = \mathbf{C}\mathbf{y}_{t-1} + \left[\sum_{h=0}^{\infty}\mathbf{F}^h\ \left(\mathbf{I}_m - \mathbf{BC}\right)^{-1}\mathbf{R}^h\right]\mathbf{u}_t,$$

or

$$\mathbf{y}_t = \mathbf{C}\mathbf{y}_{t-1} + \mathbf{G}\mathbf{u}_t. \tag{20.14}$$

As before, $\mathbf{G}$ can also be obtained using the method of undetermined coefficients, noting that $\mathbf{C}$ satisfies the quadratic matrix equation, (20.10). We first note that

$$\begin{aligned} E\left(\mathbf{y}_{t+1}|\Omega_t\right) &= \mathbf{C}\mathbf{y}_t + \mathbf{G}\mathbf{R}\mathbf{u}_t = \mathbf{C}\left(\mathbf{C}\mathbf{y}_{t-1} + \mathbf{G}\mathbf{u}_t\right) + \mathbf{G}\mathbf{R}\mathbf{u}_t \\ &= \mathbf{C}^2\mathbf{y}_{t-1} + \left(\mathbf{C}\mathbf{G} + \mathbf{G}\mathbf{R}\right)\mathbf{u}_t. \end{aligned}$$

Using the above result in (20.7) we have

$$\mathbf{C}\mathbf{y}_{t-1} + \mathbf{G}\mathbf{u}_t = \mathbf{A}\mathbf{y}_{t-1} + \mathbf{B}\left[\mathbf{C}^2\mathbf{y}_{t-1} + \left(\mathbf{C}\mathbf{G} + \mathbf{G}\mathbf{R}\right)\mathbf{u}_t\right] + \mathbf{u}_t.$$

Since $\mathbf{C}$ satisfies (20.10), it therefore follows that

$$\begin{aligned} \mathbf{G} &= \mathbf{B}\left(\mathbf{C}\mathbf{G} + \mathbf{G}\mathbf{R}\right) + \mathbf{I}_m, \\ \mathbf{G} &= \left(\mathbf{I}_m - \mathbf{BC}\right)^{-1}\mathbf{B}\mathbf{G}\mathbf{R} + \left(\mathbf{I}_m - \mathbf{BC}\right)^{-1}, \end{aligned} \tag{20.15}$$

which can be solved for

$$\begin{aligned} vec(\mathbf{G}) &= \left[\mathbf{I}_{m^2} - \left(\mathbf{R}' \otimes \left(\mathbf{I}_m - \mathbf{BC}\right)^{-1}\mathbf{B}\right)\right]^{-1} vec\left[\left(\mathbf{I}_m - \mathbf{BC}\right)^{-\mathbf{1}}\right] \\ &= \left[\mathbf{I}_{m^2} - \left(\mathbf{R}' \otimes \mathbf{F}\right)\right]^{-1} vec\left[\left(\mathbf{I}_m - \mathbf{BC}\right)^{-\mathbf{1}}\right]. \end{aligned}$$

This solution exists if all the eigenvalues of $\mathbf{F}$ lie inside the unit circle and all the roots of $\mathbf{R}$ lie on or inside the unit circle. These conditions ensure that $\mathbf{I}_{m^2} - \left(\mathbf{R}' \otimes \mathbf{F}\right)$ is a nonsingular matrix. The solution in terms of the innovations to the forcing variables can now be written as

$$\mathbf{y}_t = \mathbf{C}\mathbf{y}_{t-1} + \mathbf{G}\mathbf{R}\mathbf{u}_{t-1} + \mathbf{G}\boldsymbol{\varepsilon}_t.$$

Also note that when $\mathbf{R} = \mathbf{0}$, then $\mathbf{G} = \left(\mathbf{I}_m - \mathbf{BC}\right)^{-1}$.

Example 42 *Consider the following new Keynesian Phillips curve (NKPC) with a backward component*

$$\pi_t = \beta_b \pi_{t-1} + \beta_f E(\pi_{t+1} \mid \Omega_t) + \gamma x_t + u_t, \tag{20.16}$$

where π_t is the rate of inflation, x_t is a measure of output gap, and u_t is a serially uncorrelated 'supply' shock with mean zero. The theory also predicts that $\beta_f, \beta_b > 0$. The solution of the model depends on the process generating x_t and u_t, and the backward (β_b) and the forward (β_f) coefficients. Following the QDE approach let $y_t = \pi_t - \lambda \pi_{t-1}$ and write (20.16) as

$$y_t = \left(\frac{\beta_f}{1-\beta_f \lambda}\right) E(y_{t+1} \mid \Omega_t) + \left(\frac{1}{1-\beta_f \lambda}\right)(\gamma x_t + u_t), \tag{20.17}$$

where λ is the root of the quadratic equation

$$\beta_f \lambda^2 - \lambda + \beta_b = 0. \tag{20.18}$$

Denote the roots of this equation by λ_b and λ_f and note that $\lambda_b + \lambda_f = \frac{1}{\beta_f}$, and hence

$$\beta_f^{-1}\left(1 - \beta_f \lambda_b\right) = \beta_f^{-1} - \lambda_b = \lambda_f.$$

For a unique stable solution we need to select λ such that $\left|\beta_f^{-1}\left(1-\beta_f \lambda\right)\right| < 1$. Set $\lambda = \lambda_b$, and using the above result note that $\beta_f^{-1}\left(1-\beta_f \lambda_b\right) = \lambda_f^{-1}$. The solution will be unique if $\left|\lambda_f\right| > 1$. Using the results in Section 20.2.1 the unique solution of y_t is given by

$$y_t = \left(\frac{1}{1-\beta_f \lambda_b}\right) \sum_{j=0}^{\infty} \lambda_f^{-j} E\left(\gamma x_{t+j} + u_{t+j} \mid \Omega_t\right).$$

Since $\pi_t = y_t + \lambda_b \pi_{t-1}$, then the unique solution of the NKPC will be

$$\pi_t = \lambda_b \pi_{t-1} + \left(\frac{\gamma}{1-\beta_f \lambda_b}\right) \sum_{j=0}^{\infty} \lambda_f^{-j} E\left(x_{t+j} \mid \Omega_t\right) + \left(\frac{1}{1-\beta_f \lambda_b}\right) u_t,$$

where $|\lambda_b| < 1$. A sufficient condition for the quadratic equation to have one root, λ_b, inside the unit circle and the other root, λ_f, outside the unit circle is given by (note that $\lambda_b \lambda_f = \beta_b / \beta_f$)

$$(1-\lambda_b)(\lambda_f - 1) = \frac{1}{\beta_f} - \frac{\beta_b}{\beta_f} - 1 = \frac{1-\beta_b-\beta_f}{\beta_f} \geq 0.$$

For an economically meaningful solution, the roots must be real and this is ensured if $\beta_f\beta_b \leq 1/4$. *In the boundary case where* $\beta_f + \beta_b = 1$, *then* $\lambda_b = 1$, *and* $\lambda_f = \frac{1}{\beta_f} - 1$, *and a unique solution follows if* $|\lambda_f| = \left|\frac{1-\beta_f}{\beta_f}\right| > 1$, *or if* $\beta_f < 1/2$. *Therefore, in the case where* $\beta_f + \beta_b = 1$, *and* $\beta_f < 1/2$, *the unique solution of the NKPC is given by*

$$\pi_t = \pi_{t-1} + \left(\frac{\gamma}{1-\beta_f}\right)\sum_{j=0}^{\infty}\left(\frac{\beta_f}{1-\beta_f}\right)^j E\left(x_{t+j}|\Omega_{t-1}\right) + \left(\frac{1}{1-\beta_f}\right)u_t.$$

Since by design the output gap, x_t, *is a stationary process, then inflation will be* $I(1)$ *if* $\beta_b + \beta_f = 1$. *If both roots,* λ_b *and* λ_f, *fall inside the unit circle a general solution is given by*

$$\pi_t = \beta_f^{-1}\pi_{t-1} - \beta_f^{-1}\beta_b\pi_{t-2} - \beta_f^{-1}\gamma x_{t-1} + \mathfrak{m}_t - \beta_f^{-1}u_{t-1}, \tag{20.19}$$

where

$$\mathfrak{m}_{t+1} = \pi_{t+1} - E(\pi_{t+1} \mid \Omega_t),$$

is an arbitrary martingale difference process, namely $E(\mathfrak{m}_{t+1} \mid \Omega_t) = 0$. *When* $\beta_b + \beta_f > 1$, *(20.19) is a stable solution but it is not unique; there is a multiplicity of solutions indexed by* $\mathfrak{m}_t$. *Different stable solutions can be obtained for different choices of the martingale difference process,* $\mathfrak{m}_t$. *One possible choice for* $\mathfrak{m}_t$ *is the bubble free linear specification in terms of innovations to the forcing variable*

$$\mathfrak{m}_t = g\left[x_t - E(x_t \mid \Omega_{t-1})\right],$$

where g is an arbitrary constant. This in itself gives a multiplicity of solutions, depending on the choice of g. Finally, the NKPC does not have any stable solutions if both roots fall outside the unit circle.

20.4 Rational expectations models with feedbacks

The *RE* models considered so far do not allow for feedbacks from past values of $\mathbf{y}_t$ to the model's forcing variables, $\mathbf{w}_t$ or $\mathbf{u}_t$. As an example, consider the following *RE* model

$$\mathbf{y}_t = \mathbf{A}\mathbf{y}_{t-1} + \mathbf{B}E(\mathbf{y}_{t+1}|\Omega_t) + \mathbf{u}_t, \tag{20.20}$$

where

$$\mathbf{u}_t = \mathbf{R}\mathbf{u}_{t-1} + \mathbf{S}\mathbf{y}_{t-1} + \boldsymbol{\varepsilon}_t, \tag{20.21}$$

$\mathbf{S}$ is a non-zero matrix of fixed coefficients and captures the degree of feedbacks from $\mathbf{y}_{t-1}$ back into $\mathbf{u}_t$. It is clear that the solution approaches of the previous section do not apply here directly.

But *RE* models with feedbacks can be written in the form of a larger *RE* model, with no feedbacks. Let $\mathbf{z}_t = \left(\mathbf{y}_t', \mathbf{u}_t'\right)'$ and write the above set of equations as

$$\begin{pmatrix} \mathbf{I}_m & -\mathbf{I}_m \\ \mathbf{0} & \mathbf{I}_m \end{pmatrix} \mathbf{z}_t = \begin{pmatrix} \mathbf{A} & \mathbf{0} \\ \mathbf{S} & \mathbf{R} \end{pmatrix} \mathbf{z}_{t-1} + \begin{pmatrix} \mathbf{B} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{pmatrix} \mathrm{E}(\mathbf{z}_{t+1}|\Omega_t) + \begin{pmatrix} \mathbf{0} \\ \boldsymbol{\varepsilon}_t \end{pmatrix},$$

or, more compactly,

$$\mathbf{z}_t = \widetilde{\mathbf{A}}\mathbf{z}_{t-1} + \widetilde{\mathbf{B}}E(\mathbf{z}_{t+1}|\Omega_t) + \mathbf{v}_t, \tag{20.22}$$

where

$$\widetilde{\mathbf{A}} = \begin{pmatrix} \mathbf{A}+\mathbf{S} & \mathbf{R} \\ \mathbf{S} & \mathbf{R} \end{pmatrix}, \widetilde{\mathbf{B}} = \begin{pmatrix} \mathbf{B} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{pmatrix}, \text{ and } \mathbf{v}_t = \begin{pmatrix} \boldsymbol{\varepsilon}_t \\ \boldsymbol{\varepsilon}_t \end{pmatrix}.$$

In the enlarged *RE* model (20.22), there are no longer any feedbacks, and the solution methods of previous sections can be readily applied to it. In particular, it is easily seen that this model has the unique solution

$$\mathbf{z}_t = \widetilde{\mathbf{C}}\mathbf{z}_{t-1} + (\mathbf{I}_{2m} - \tilde{\mathbf{B}}\tilde{\mathbf{C}})^{-1}\mathbf{v}_t,$$

where $\widetilde{\mathbf{C}}$ is such that

$$\widetilde{\mathbf{B}}\widetilde{\mathbf{C}}^2 - \widetilde{\mathbf{C}} + \widetilde{\mathbf{A}} = \mathbf{0}_{2m\times 2m}, \tag{20.23}$$

assuming that all eigenvalues of $\widetilde{\mathbf{C}}$ lie inside or on the unit circle and $\mathbf{I}_{2m} - \widetilde{\mathbf{B}}\widetilde{\mathbf{C}}$ is nonsingular. To determine $\widetilde{\mathbf{C}}$ let

$$\underset{2m\times 2m}{\widetilde{\mathbf{C}}} = \begin{pmatrix} \mathbf{C} & \mathbf{D} \\ \mathbf{S} & \mathbf{R} \end{pmatrix},$$

where $\mathbf{C}$ and $\mathbf{D}$ satisfy the following set of matrix equations

$$\mathbf{B}\mathbf{C}^2 - \mathbf{C} + \mathbf{A} + (\mathbf{I} + \mathbf{B}\mathbf{D})\mathbf{S} = \mathbf{0}_{m\times m}, \tag{20.24}$$

and

$$(\mathbf{I}_m - \mathbf{B}\mathbf{C})\mathbf{D} = (\mathbf{I} + \mathbf{B}\mathbf{D})\mathbf{R}. \tag{20.25}$$

Also

$$\mathbf{I}_{2m} - \tilde{\mathbf{B}}\tilde{\mathbf{C}} = \begin{pmatrix} \mathbf{I}_m - \mathbf{B}\mathbf{C} & -\mathbf{B}\mathbf{D} \\ \mathbf{0} & \mathbf{I}_m \end{pmatrix}$$

$$(\mathbf{I}_{2m} - \tilde{\mathbf{B}}\tilde{C})^{-1} = \begin{pmatrix} (\mathbf{I}_m - \mathbf{BC})^{-1} & (\mathbf{I}_m - \mathbf{BC})^{-1}\mathbf{BD} \\ \mathbf{0} & \mathbf{I}_m \end{pmatrix}.$$

Using the above result, the unique solution of $\mathbf{y}_t$ is given by (assuming that regularity conditions are satisfied)

$$\mathbf{y}_t = \mathbf{C}\mathbf{y}_{t-1} + \mathbf{D}\mathbf{u}_{t-1} + (\mathbf{I}_m - \mathbf{BC})^{-1}(\mathbf{I}_m + \mathbf{BD})\,\boldsymbol{\varepsilon}_t, \tag{20.26}$$

which upon using (20.25) and after some algebra can be written equivalently as

$$\mathbf{y}_t = \mathbf{C}\mathbf{y}_{t-1} + \mathbf{G}(\mathbf{R}\mathbf{u}_{t-1} + \boldsymbol{\varepsilon}_t), \tag{20.27}$$

where

$$\mathbf{G} = (\mathbf{I}_m - \mathbf{BC})^{-1}(\mathbf{I}_m + \mathbf{BD}).$$

The solution of the *RE* model in this case requires solving the nonlinear matrix equations given by (20.24) and (20.25) for $\mathbf{C}$ and $\mathbf{D}$.

The above solution form can also be used to derive $\mathbf{C}$ and $\mathbf{G}$ directly, using the method of undetermined coefficients. Note that if $\mathbf{y}_t = \mathbf{C}\mathbf{y}_{t-1} + \mathbf{G}(\mathbf{R}\mathbf{u}_{t-1} + \boldsymbol{\varepsilon}_t)$ is to be a solution of (20.20) and (20.21) we must have

$$\mathbf{C}\mathbf{y}_{t-1} + \mathbf{G}(\mathbf{R}\mathbf{u}_{t-1} + \boldsymbol{\varepsilon}_t) = \mathbf{A}\mathbf{y}_{t-1} + \mathbf{B}\left(\mathbf{C}\mathbf{y}_t + \mathbf{G}\mathbf{R}\mathbf{u}_t\right) + \mathbf{u}_t,$$

and hence

$$\begin{aligned}(\mathbf{C} - \mathbf{A})\mathbf{y}_{t-1} + \mathbf{G}(\mathbf{R}\mathbf{u}_{t-1} + \boldsymbol{\varepsilon}_t) = \mathbf{BC}\left[\mathbf{C}\mathbf{y}_{t-1} + \mathbf{G}(\mathbf{R}\mathbf{u}_{t-1} + \boldsymbol{\varepsilon}_t)\right] \\ + (\mathbf{BGR} + \mathbf{I}_m)\left(\mathbf{S}\mathbf{y}_{t-1} + \mathbf{R}\mathbf{u}_{t-1} + \boldsymbol{\varepsilon}_t\right).\end{aligned}$$

Equating coefficient matrices of $\mathbf{y}_{t-1}$, $\mathbf{u}_{t-1}$ and $\boldsymbol{\varepsilon}_t$ from both sides we have

$$\begin{aligned}\mathbf{BC}^2 - \mathbf{C} + \mathbf{A} + (\mathbf{I}_m + \mathbf{BGR})\mathbf{S} &= \mathbf{0}_{m\times m}, \\ (\mathbf{I}_m - \mathbf{BC})\mathbf{GR} &= (\mathbf{I}_m + \mathbf{BGR})\mathbf{R}, \\ \mathbf{I}_m + \mathbf{BGR} &= (\mathbf{I}_m - \mathbf{BC})\mathbf{G},\end{aligned}$$

which simplify to

$$\begin{aligned}\mathbf{BC}^2 - \mathbf{C} + \mathbf{A} + (\mathbf{I}_m - \mathbf{BC})\mathbf{GS} &= \mathbf{0}_{m\times m}, \\ \mathbf{I}_m + \mathbf{BGR} &= (\mathbf{I}_m - \mathbf{BC})\mathbf{G}.\end{aligned}$$

These two sets of matrix equations can now be solved iteratively for $\mathbf{C}$ and $\mathbf{G}$.

20.5 The higher-order case

Models with more lags and leads can be accommodated in framework (20.7) by expanding the $\mathbf{y}_t$ vector appropriately. Consider the following general model

$$\mathbf{y}_t = \sum_{j=1}^{p} \mathbf{A}_{j0}\mathbf{y}_{t-j} + \sum_{j=0}^{p}\sum_{h=1}^{H} \mathbf{A}_{jh}\,\mathrm{E}(\mathbf{y}_{t+h-j}|\Omega_{t-j}) + \mathbf{v}_t, \tag{20.28}$$

where $\mathbf{y}_t$ is an m-dimensional vector of observable variables, $\mathbf{v}_t$ is an m-dimensional vector of disturbances, and $\mathbf{A}_{jh}$, $j = 0, 1, \ldots, p$, $h = 0, 1, \ldots, H$, are $m \times m$-dimensional matrices of fixed coefficients.

Equation (20.28) can always be expressed in form (20.7), thus containing only a vector of one-period lagged dependent variables, and a vector of one-step ahead future expectations of the dependent variable, by defining the auxiliary matrices and vectors

$$\mathbf{z}_t = \left(\mathbf{Y}_t', \quad \mathbf{Y}_{t-1}', \quad \ldots \quad, \mathbf{Y}_{t-p+1}'\right)', \; \mathbf{Y}_t' = \left(\mathbf{y}_t', \quad E(\mathbf{y}_{t+1}'|\Omega_t), \quad \ldots \quad, E(\mathbf{y}_{t+H}'|\Omega_t)\right)',$$

$$\mathbf{A} = -\mathbf{D}_0^{-1}\mathbf{D}_1, \; \mathbf{B} = -\mathbf{D}_0^{-1}\mathbf{D}_{-1}, \mathbf{u}_t = \mathbf{D}_0^{-1}\widetilde{\boldsymbol{\vartheta}}_t,$$

$$\widetilde{\boldsymbol{\vartheta}}_t = \left(\boldsymbol{\vartheta}_t', \quad \mathbf{0}_{m\times 1}', \quad \ldots \quad, \mathbf{0}_{m\times 1}'\right)', \; \boldsymbol{\vartheta}_t = \left(\mathbf{v}_t', \quad \mathbf{0}_{m\times 1}', \quad \ldots \quad, \mathbf{0}_{m\times 1}'\right)',$$

where $\mathbf{z}_t$, $\mathbf{u}_t$ and $\widetilde{\boldsymbol{\vartheta}}_t$ are of dimension $m(H+1)p \times 1$, $\boldsymbol{\vartheta}_t$ is of dimension $m(H+1) \times 1$, and $\mathbf{A}$ and $\mathbf{B}$ are square matrices of dimension $m(H+1)p$, with $\mathbf{D}_i$, $i = -1, 0, 1$, defined by

$$\mathbf{D}_{-1} = \begin{pmatrix} \boldsymbol{\Gamma}_{-1} & \mathbf{0}_m & \cdots & \mathbf{0}_m \\ \mathbf{0}_m & \mathbf{0}_m & \cdots & \mathbf{0}_m \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{0}_m & \mathbf{0}_m & \cdots & \mathbf{0}_m \end{pmatrix}, \; \mathbf{D}_0 = \begin{pmatrix} \boldsymbol{\Gamma}_0 & \boldsymbol{\Gamma}_1 & \cdots & \boldsymbol{\Gamma}_{p-1} \\ \mathbf{0}_m & \mathbf{I}_m & \cdots & \mathbf{0}_m \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{0}_m & \mathbf{0}_m & \cdots & \mathbf{I}_n \end{pmatrix},$$

$$\mathbf{D}_1 = \begin{pmatrix} \mathbf{0}_m & \mathbf{0}_m & \cdots & \mathbf{0}_m & \boldsymbol{\Gamma}_p \\ -\mathbf{I}_m & \mathbf{0}_m & \cdots & \mathbf{0}_m & \mathbf{0}_m \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ \mathbf{0}_m & \mathbf{0}_m & \cdots & -\mathbf{I}_m & \mathbf{0}_m \end{pmatrix},$$

and $\boldsymbol{\Gamma}_j$, $j = -1, 0, \ldots, p$, are square matrices of dimension m, defined by

$$\boldsymbol{\Gamma}_0 = \begin{pmatrix} \mathbf{I}_m & -\mathbf{A}_{01} & \cdots & -\mathbf{A}_{0H} \\ \mathbf{0}_m & \mathbf{I}_m & \cdots & \mathbf{0}_m \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{0}_m & \mathbf{0}_m & \cdots & \mathbf{I}_m \end{pmatrix}, \; \boldsymbol{\Gamma}_i = \begin{pmatrix} -\mathbf{A}_{i0} & -\mathbf{A}_{i1} & \cdots & -\mathbf{A}_{iH} \\ \mathbf{0}_m & \mathbf{0}_m & \cdots & \mathbf{0}_m \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{0}_m & \mathbf{0}_m & \cdots & \mathbf{0}_m \end{pmatrix},$$

and

$$\mathbf{\Gamma}_{-1} = \begin{pmatrix} \mathbf{0}_m & \mathbf{0}_m & \cdots & \mathbf{0}_m & \mathbf{0}_m \\ -\mathbf{I}_m & \mathbf{0}_m & \cdots & \mathbf{0}_m & \mathbf{0}_m \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ \mathbf{0}_m & \mathbf{0}_m & \cdots & -\mathbf{I}_m & \mathbf{0}_m \end{pmatrix}.$$

Using the above auxiliary vectors and matrices, we obtain the canonical form

$$\mathbf{z}_t = \mathbf{A}\mathbf{z}_{t-1} + \mathbf{B}E(\mathbf{z}_{t+1}|\Omega_t) + \mathbf{u}_t. \tag{20.29}$$

See, for example, Broze, Gouriéroux, and Szafarz (1995) and Binder and Pesaran (1995) for further details.

Example 43 *Consider the following rational expectations model*

$$\mathbf{y}_t = \mathbf{A}_1 E(\mathbf{y}_{t+1}|\Omega_t) + \mathbf{A}_2 E(\mathbf{y}_{t+2}|\Omega_t) + \mathbf{B}\mathbf{x}_t + \boldsymbol{\varepsilon}_t, \tag{20.30}$$

where $\mathbf{y}_t$ *is an* $m \times 1$ *dimensional vector,* $\boldsymbol{\varepsilon}_t$ *are serially uncorrelated and* $\mathbf{x}_t$ *is a* $k \times 1$ *vector of exogenous variables following the VAR(1) process*

$$\mathbf{x}_t = \mathbf{R}\mathbf{x}_{t-1} + \mathbf{u}_t. \tag{20.31}$$

Let $\mathbf{Y}_t = (\mathbf{y}_t, E(\mathbf{y}_{t+1}|\Omega_t))'$ *and note that (20.30) can be written as*

$$\mathbf{Y}_t = \mathbf{A}E(\mathbf{Y}_{t+1}|\Omega_t) + \mathbf{W}_t,$$

where

$$\mathbf{A} = \begin{pmatrix} \mathbf{A}_1 & \mathbf{A}_2 \\ \mathbf{I}_m & \mathbf{0} \end{pmatrix}, \text{ and } \mathbf{W}_t = \begin{pmatrix} \mathbf{B}\mathbf{x}_t + \boldsymbol{\varepsilon}_t \\ \mathbf{0} \end{pmatrix}.$$

Suppose now that all the eigenvalue of $\mathbf{A}$ *lie within the unit circle and the standard transversality condition is met. Using the method of undetermined coefficients, the unique solution of the RE model is given by*

$$\mathbf{y}_t = \mathbf{C}\mathbf{x}_t + \boldsymbol{\varepsilon}_t,$$

where $\mathbf{C}$ *satisfies the following equations*

$$\mathbf{C} = \mathbf{A}_1\mathbf{C}\mathbf{R} + \mathbf{A}_2\mathbf{C}\mathbf{R}^2 + \mathbf{B},$$

or

$$vec(\mathbf{C}) = (\mathbf{R}' \otimes \mathbf{A}_1)\, vec(\mathbf{C}) + \left[\left(\mathbf{R}^2\right)' \otimes \mathbf{A}_2 \right] vec(\mathbf{C}) + vec(\mathbf{B}),$$

and finally

$$vec(\mathbf{C}) = \left\{\mathbf{I}_{km} - (\mathbf{R}' \otimes \mathbf{A}_1) - \left[\left(\mathbf{R}^2\right)' \otimes \mathbf{A}_2\right]\right\}^{-1} vec(\mathbf{B}). \tag{20.32}$$

20.5.1 Retrieving the solution for y_t

Consider $\mathbf{z}_t$ in model (20.29), and its solution (20.12) obtained by applying the quadratic determinantal equation method

$$\mathbf{z}_t = \mathbf{C}\mathbf{z}_{t-1} + \mathbf{h}_t, \tag{20.33}$$

where $\mathbf{C}$ and $\mathbf{h}_t$ can be obtained from a backward recursion as set out in Binder and Pesaran (1997). We now address the problem of how to retrieve $\mathbf{y}_t$ from this solution. To simplify the exposition we set $p = 2$ and $H = 1$. Let

$$\mathbf{D} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Note that $\mathbf{D}^{-1} = \mathbf{D}$. Hence, (20.33) can be written as

$$\mathbf{D}\mathbf{z}_t = \mathbf{G}\mathbf{D}\mathbf{z}_{t-1} + \mathbf{D}\tilde{\mathbf{h}}_t,$$

where $\mathbf{G} = \mathbf{DCD}$, and $\tilde{\mathbf{h}}_t = \mathbf{D}\mathbf{h}_t$. But

$$\mathbf{D}\mathbf{z}_t = \begin{pmatrix} \mathbf{y}_t \\ \mathbf{y}_{t-1} \\ E(\mathbf{y}_{t+1}|\Omega_t) \\ E(\mathbf{y}_t|\Omega_{t-1}) \end{pmatrix} = \begin{pmatrix} \mathbf{q}_{1t} \\ \mathbf{q}_{2t} \end{pmatrix},$$

and

$$\begin{aligned} \mathbf{q}_{1t} &= \mathbf{G}_{11}\mathbf{q}_{1,t-1} + \mathbf{G}_{12}\mathbf{q}_{2,t-1} + \tilde{\mathbf{h}}_{1t}, \\ \mathbf{q}_{2t} &= \mathbf{G}_{21}\mathbf{q}_{1,t-1} + \mathbf{G}_{22}\mathbf{q}_{2,t-1} + \tilde{\mathbf{h}}_{2t}. \end{aligned}$$

Hence, if all eigenvalues of $\mathbf{G}_{22}$ fall within the unit circle, the solution for $\mathbf{q}_{1t}$ will be given by

$$\mathbf{q}_{1t} = \mathbf{G}_{11}\mathbf{q}_{1,t-1} + \mathbf{G}_{12}\left(\mathbf{I} - \mathbf{G}_{22}L\right)^{-1}\left(\mathbf{G}_{21}\mathbf{q}_{1,t-2} + \tilde{\mathbf{h}}_{2,t-1}\right) + \tilde{\mathbf{h}}_{1t}.$$

The solution for $\mathbf{y}_t$ can be obtained from the above equations. Clearly, the solution for $\mathbf{y}_t$ involves infinite moving average components, unless $\mathbf{G}_{12} = \mathbf{0}$. In the case where $\mathbf{G}_{12} \neq \mathbf{0}$, the presence of the infinite-order moving average term in the solution complicates the problems of

identification and estimation of *RE* models, and raises the issue of whether the solution can be approximated by finite-order *VARMA* processes.

20.6 A 'finite-horizon' *RE* model

One important special case occurs when economic agents face only a finite horizon. Finite-horizon *RE* models have widespread applicability in economics including, for example, the finite-lifetime life-cycle model of consumption, asset pricing models and models involving non-linear adjustment costs such as the Hayashi (1982) formulation of the neoclassical model of investment.

A general formulation for a model with finite and shifting planning horizon and fixed terminal point is:

$$\mathbf{y}_{t+\tau} = \mathbf{A}\mathbf{y}_{t+\tau-1} + \mathbf{B}E(\mathbf{y}_{t+\tau+1}|\Omega_{t+\tau}) + \mathbf{w}_{t+\tau}, \quad \tau = 0, 1, \ldots, T-t, \tag{20.34}$$

where $E\left(\mathbf{y}_{t+\tau+j-i}|\Omega_{t+\tau-i}\right)$ is defined for $t+\tau+j-i > T$. Binder and Pesaran (2000) have presented efficient methods for the solution of model (20.34), and showed that this is linked to the problem of solving sparse linear equations systems with a block tridiagonal coefficients matrix structure.

See Binder and Pesaran (2000) and Gilli and Pauletto (1997).

20.6.1 A backward recursive solution

In the case of finite horizon, multivariate *RE* models do not have a time-invariant solution, and the standard methods described above for the solution of infinite-horizon models are not applicable. One approach to the solution of (20.34) would be to use backward recursions starting from time T. At time T, the solution for $\mathbf{y}_T$, given $\mathbf{y}_{T-1}$ and the terminal condition $E\left(\mathbf{y}_{T+1}|\Omega_T\right)$, is given by (20.34) for $\tau = T-t$, namely

$$\mathbf{y}_T = \mathbf{A}\mathbf{y}_{T-1} + \mathbf{B}E\left(\mathbf{y}_{T+1}|\Omega_T\right) + \mathbf{w}_T. \tag{20.35}$$

Proceeding recursively backward, we can obtain $\mathbf{y}_{T-1}$ as a function of $\mathbf{y}_{T-2}$, the terminal condition $E\left(\mathbf{y}_{T+1}|\Omega_T\right)$, and of $E\left(\mathbf{w}_T|\Omega_{T-1}\right)$ and $\mathbf{w}_{T-1}$. Combining (20.34) for $\tau = T-t-1$ with (20.35), one readily obtains

$$\mathbf{y}_{T-1} = (\mathbf{I}_m - \mathbf{B}\mathbf{A})^{-1}\left[\mathbf{A}\mathbf{y}_{T-2} + \mathbf{B}^2E\left(\mathbf{y}_{T+1}|\Omega_T\right) + \mathbf{B}E\left(\mathbf{w}_T|\Omega_{T-1}\right) + \mathbf{w}_{T-1}\right]. \tag{20.36}$$

Proceeding to period $T-2$, combining (20.34) for $\tau = T-t-2$ with (20.36), the solution for $\mathbf{y}_{T-2}$ is given by

$$\begin{aligned}\mathbf{y}_{T-2} = {} & \left[\mathbf{I}_m - \mathbf{B}\left(\mathbf{I}_m - \mathbf{B}\mathbf{A}\right)^{-1}\mathbf{A}\right]^{-1} \\ & \times \left[\begin{array}{c} \mathbf{A}\mathbf{y}_{T-3} + \mathbf{B}\left(\mathbf{I}_m - \mathbf{B}\mathbf{A}\right)^{-1}\mathbf{B}^2E\left(\mathbf{y}_{T+1}|\Omega_T\right) + \mathbf{B}\left(\mathbf{I}_m - \mathbf{B}\mathbf{A}\right)^{-1}\mathbf{B}E\left(\mathbf{w}_T|\Omega_{T-2}\right) \\ + \mathbf{B}\left(\mathbf{I}_m - \mathbf{B}\mathbf{A}\right)^{-1}E\left(\mathbf{w}_{T-1}|\Omega_{T-2}\right) + \mathbf{w}_{T-2} \end{array}\right].\end{aligned}$$

The pattern of these backward recursions should be apparent. Along the same lines of reasoning, the solution for $\mathbf{y}_{t+\tau}$ to (20.34) is given by

$$\mathbf{y}_{t+\tau} = \mathbf{\Phi}_{\tau}^{-1}\mathbf{A}\mathbf{y}_{t+\tau-1} + \mathbf{\Phi}_{\tau}^{-1}E\left(\mathbf{\Psi}_{t+\tau}|\Omega_{t+\tau}\right), \qquad \tau = 0, 1, \ldots, T-t, \tag{20.37}$$

where

$$\mathbf{\Phi}_{T-t} = \mathbf{I}_m, \quad \mathbf{\Phi}_{T-t-i} = \mathbf{I}_m - \mathbf{B}\mathbf{\Phi}_{T-t-i+1}^{-1}\mathbf{A}, \qquad i = 1, 2, \ldots, T-t,$$

and

$$\mathbf{\Psi}_T = \mathbf{B}E\left(\mathbf{y}_{T+1}|\Omega_T\right)+\mathbf{w}_T, \quad \mathbf{\Psi}_{T-i} = \mathbf{B}\mathbf{\Phi}_{T-t-i+1}^{-1}\mathbf{\Psi}_{T-i+1}+\mathbf{w}_{T-i}, \qquad i = 1, 2, \ldots, T-t.$$

The matrices $\mathbf{\Phi}_{T-t-i}$ are assumed to be nonsingular for $i = 1, 2, \ldots, T-t$. Note that the solution in all periods is a linear combination of the initial and terminal values, and the conditional expectations of the forcing variables. As the forcing variables were assumed to be adapted to the information sets $\{\Omega_{t+\tau}\}$, then so will the solution $\left\{\mathbf{y}_{t+\tau}\right\}$.

20.7 Other solution methods

In this section we review a number of other methods that have been advanced in the literature for solution of *RE* models.

20.7.1 Blanchard and Kahn method

Early work by Blanchard and Kahn (1980) considers solutions for a model of the type

$$\mathbf{\Gamma}_0 E\left(\mathbf{z}_{t+1}|\Omega_t\right) = \mathbf{\Gamma}_1\mathbf{z}_t + \mathbf{\Psi}\mathbf{v}_t, \tag{20.38}$$

where $\mathbf{z}_t$ is a vector of endogenous variables, and $\mathbf{v}_t$ is a vector of strictly exogenous variables. Their procedure consists of partitioning $\mathbf{z}_t = \left(\mathbf{y}_t', \mathbf{x}_t'\right)'$, where $\mathbf{y}_t$ is an $m \times 1$ vector of non-predetermined variables, and $\mathbf{x}_t$ is a $k \times 1$ vector of predetermined variables. According to Blanchard and Kahn a variable x_{jt} is 'predetermined' if it is a function of variables known at time t, such that $\mathrm{E}(x_{j,t+1}|\Omega_t) = x_{j,t+1}$. Blanchard and Kahn (1980) show that the existence and uniqueness of a stable solution for model (20.38) is related to a certain rank condition, that associates how many non-predetermined variables exist in the system relative to the number of unstable canonical variables.[4]

Under the assumption that matrix $\mathbf{\Gamma}_0$ is nonsingular, let $\mathbf{G} = \mathbf{\Gamma}_0^{-1}\mathbf{\Gamma}_1$, and consider the Jordan matrix decomposition of $\mathbf{G}$, namely $\mathbf{G} = \mathbf{C}^{-1}\mathbf{J}\mathbf{C}$, where $\mathbf{J}$ is the upper triangular matrix with the eigenvalues of $\mathbf{G}$ ordered by decreasing absolute value as its diagonal elements, and zeros or ones on the superdiagonal.[5] Hence, premultiply both sides of (20.38) by $\mathbf{C}\mathbf{\Gamma}_0^{-1}$ to obtain the

[4] We will also show that under this rank condition, (20.38) can be written as a special case of the canonical model given by (20.7).

[5] Namely $\mathbf{J}$ is arranged in Jordan blocks. See Broze, Gouriéroux, and Szafarz (1995) for definition of Jordan canonical form, Jordan blocks, and canonical variables.

equivalent dynamic system

$$E(\mathbf{z}_{t+1}^{*}|\Omega_t) = \mathbf{J}\mathbf{z}_t^{*} + \boldsymbol{\Psi}_t^{*}\mathbf{v}_t, \tag{20.39}$$

where $\mathbf{z}_t^{*} = \mathbf{C}\mathbf{z}_t$, and $\boldsymbol{\Psi}^{*} = \mathbf{C}\boldsymbol{\Gamma}_0^{-1}\boldsymbol{\Psi}$. Consider now the decompositions

$$\mathbf{J} = \begin{pmatrix} \mathbf{J}_u & \mathbf{0} \\ \mathbf{0} & \mathbf{J}_s \end{pmatrix}, \ \mathbf{z}_t^{*} = \begin{pmatrix} \mathbf{u}_t \\ \mathbf{s}_t \end{pmatrix}, \ \boldsymbol{\Psi}^{*} = \begin{pmatrix} \boldsymbol{\Psi}_u^{*} \\ \boldsymbol{\Psi}_s^{*} \end{pmatrix},$$

where $\mathbf{J}_u$ contains the unstable eigenvalues of $\mathbf{G}$ with absolute value greater than unity, $\mathbf{J}_s$ contains the stable eigenvalues of $\mathbf{G}$ with absolute value less than unity,[6] and $\mathbf{u}_t$ and $\mathbf{s}_t$ are the canonical variables associated with the eigenvalues in $\mathbf{J}_u$ and $\mathbf{J}_s$, respectively. Substituting the above results in (20.39) we now have

$$E\begin{pmatrix} \mathbf{u}_{t+1} \\ \mathbf{s}_{t+1} \end{pmatrix} \Big| \Omega_t \Big) = \begin{pmatrix} \mathbf{J}_u & \mathbf{0} \\ \mathbf{0} & \mathbf{J}_s \end{pmatrix} \begin{pmatrix} \mathbf{u}_t \\ \mathbf{s}_t \end{pmatrix} + \begin{pmatrix} \boldsymbol{\Psi}_u^{*} \\ \boldsymbol{\Psi}_s^{*} \end{pmatrix} \mathbf{v}_t. \tag{20.40}$$

Since none of the eigenvalues of $\mathbf{J}_u$ are zero, $\mathbf{J}_u$ is nonsingular and we have

$$\mathbf{u}_t = \mathbf{J}_u^{-1} E\left(\mathbf{u}_{t+1} \,|\Omega_t\right) - \mathbf{J}_u^{-1}\boldsymbol{\Psi}_u^{*}\mathbf{v}_t,$$

which is identical to (20.1) with $\mathbf{A}$ set to $\mathbf{J}_u^{-1}$, $\mathbf{y}_t$ to $\mathbf{u}_t$, and $\mathbf{w}_t$ to $-\mathbf{J}_u^{-1}\boldsymbol{\Psi}_u^{*}\mathbf{v}_t$. Hence, it can be solved by the forward approach since all eigenvalues of $\mathbf{J}_u^{-1}$ fall within the unit circle, yielding the unique solution[7]

$$\mathbf{u}_t = -\sum_{h=0}^{\infty} \mathbf{J}_u^{-h}\boldsymbol{\Psi}_u^{*} E\left(\mathbf{v}_{t+h}|\Omega_t\right), \tag{20.41}$$

assuming that the standard transversality conditions hold.

Consider now the transformations

$$\begin{pmatrix} \mathbf{u}_t \\ \mathbf{s}_t \end{pmatrix} = \begin{pmatrix} \mathbf{C}_{uy} & \mathbf{C}_{ux} \\ \mathbf{C}_{sy} & \mathbf{C}_{sx} \end{pmatrix} \begin{pmatrix} \mathbf{y}_t \\ \mathbf{x}_t \end{pmatrix}, \ \begin{pmatrix} \mathbf{y}_t \\ \mathbf{x}_t \end{pmatrix} = \begin{pmatrix} \mathbf{R}_{yu} & \mathbf{R}_{ys} \\ \mathbf{R}_{xu} & \mathbf{R}_{xs} \end{pmatrix} \begin{pmatrix} \mathbf{u}_t \\ \mathbf{s}_t \end{pmatrix}, \tag{20.42}$$

with $\mathbf{R} = \mathbf{C}^{-1}$, from which it follows that

$$\mathbf{u}_t = \mathbf{C}_{uy}\mathbf{y}_t + \mathbf{C}_{ux}\mathbf{x}_t.$$

The above equations link $\mathbf{y}_t$ to the canonical variables $\mathbf{u}_t$ (that evolve according to (20.41)) and the predetermined variables, $\mathbf{x}_t$. In the case where $\mathbf{C}_{uy}$ is nonsingular we have

$$\mathbf{y}_t = \mathbf{C}_{uy}^{-1}\mathbf{u}_t - \mathbf{C}_{uy}^{-1}\mathbf{C}_{ux}\mathbf{x}_t. \tag{20.43}$$

[6] The diagonal elements of $\mathbf{J}_s$ and $\mathbf{J}_u$ are also given by the roots of the determinant equation $|\boldsymbol{\Gamma}_0 z - \boldsymbol{\Gamma}_1| = 0$. This will be useful when introducing the King and Watson (1998) method.

[7] See Section 20.2.1 for a description of the forward method and the assumptions required for obtaining a unique solution.

Noting that from (20.40) and (20.42)

$$\begin{aligned}\mathbf{x}_{t+1} &= \mathbf{R}_{xu}E\left(\mathbf{u}_{t+1}|\Omega_t\right) + \mathbf{R}_{xs}E\left(\mathbf{s}_{t+1}|\Omega_t\right)\\ E\left(\mathbf{s}_{t+1}|\Omega_t\right) &= \mathbf{J}_s\mathbf{s}_t + \boldsymbol{\Psi}_s^*\mathbf{v}_t,\end{aligned}$$

we obtain that the predetermined variables, $\mathbf{x}_t$, in (20.43) evolve according to

$$\mathbf{x}_{t+1} = \left(\mathbf{R}_{xs}\mathbf{J}_s\mathbf{R}_{xs}^{-1}\right)\mathbf{x}_t + \mathbf{R}_{xu}E\left(\mathbf{u}_{t+1}|\Omega_t\right) + \mathbf{R}_{xs}\mathbf{J}_s\mathbf{C}_{sy}\mathbf{C}_{uy}^{-1}\mathbf{u}_t + \mathbf{R}_{xs}\mathbf{D}_s^*\mathbf{v}_t, \tag{20.44}$$

where $\mathbf{R}_{xs} = \mathbf{C}_{sx} - \mathbf{C}_{sy}\mathbf{C}_{uy}^{-1}\mathbf{C}_{ux}$, which is a nonsingular matrix. Recall that by assumption $\mathbf{C}$ and $\mathbf{C}_{uy}$ are nonsingular matrices. Equations (20.43) and (20.44) can then be used recursively to solve for $\mathbf{y}_t$, $\mathbf{x}_t$ and $\mathbf{z}_t$, given the initial values, $\mathbf{y}_0$ and $\mathbf{x}_0$, and the unique solution of $\mathbf{u}_t$ as given above.

20.7.2 King and Watson method

King and Watson (1998) consider solutions for (20.38) where the matrix $\boldsymbol{\Gamma}_0$ is allowed to be singular. Indeed, many economic models do not fit into the Blanchard and Kahn (1980) framework, which requires $\boldsymbol{\Gamma}_0$ to be nonsingular. Rewrite (20.38) as

$$E\left[\left(\boldsymbol{\Gamma}_0 L^{-1} - \boldsymbol{\Gamma}_1\right)\mathbf{z}_t|\Omega_t\right] = \boldsymbol{\Psi}\mathbf{v}_t, \tag{20.45}$$

where L^{-1} is the forward operator, i.e. $L^{-1}\mathbf{z}_t = \mathbf{z}_{t+1}$. To ensure a unique solution King and Watson assume that $|\boldsymbol{\Gamma}_0\lambda - \boldsymbol{\Gamma}_1| \neq 0$, for all values of λ. King and Watson (1998) show that, under this condition, model (20.38) (or (20.45)) can be written equivalently as

$$\begin{pmatrix}\mathbf{G} & \mathbf{0} & \mathbf{0}\\ \mathbf{0} & \mathbf{I}_u & \mathbf{0}\\ \mathbf{0} & \mathbf{0} & \mathbf{I}_s\end{pmatrix}\mathrm{E}\begin{pmatrix}\mathbf{q}_{t+1}\\ \mathbf{u}_{t+1}\\ \mathbf{s}_{t+1}\end{pmatrix}\Big|\Omega_t = \begin{pmatrix}\mathbf{I}_q & \mathbf{0} & \mathbf{0}\\ \mathbf{0} & \mathbf{J}_u & \mathbf{0}\\ \mathbf{0} & \mathbf{0} & \mathbf{J}_s\end{pmatrix}\begin{pmatrix}\mathbf{q}_t\\ \mathbf{u}_t\\ \mathbf{s}_t\end{pmatrix} + \begin{pmatrix}\boldsymbol{\Psi}_\iota^*\\ \boldsymbol{\Psi}_u^*\\ \boldsymbol{\Psi}_s^*\end{pmatrix}\mathbf{v}_t,$$

where $\mathbf{G}$ is an upper triangular matrix with zeros on the main diagonal, and $\mathbf{I}_q$, $\mathbf{I}_u$, and $\mathbf{I}_s$ are identity matrices of orders conformable to $\mathbf{q}_t$, $\mathbf{u}_t$ and $\mathbf{s}_t$. This representation contains the same variables identified by Blanchard and Kahn (1980) (see, in particular, equations (20.40)), but also the new set of variables, $\mathbf{q}_t$. These are the canonical variables associated to the roots of the polynomial $|\boldsymbol{\Gamma}_0\lambda - \boldsymbol{\Gamma}_1|$ that are infinite, or explosive, under singularity of the matrix $\boldsymbol{\Gamma}_0$ (see Section 20.7.1 for the definition of canonical variables). A solution for $\mathbf{q}_t$ can be obtained by noting that

$$\mathbf{q}_t = E\left[\left(\mathbf{G}L^{-1} - \mathbf{I}\right)^{-1}\boldsymbol{\Psi}_\iota^*\mathbf{v}_t|\Omega_t\right] = -\sum_{h=0}^{\infty}\mathbf{G}^h\boldsymbol{\Psi}_\iota^*E\left(\mathbf{v}_{t+h}|\Omega_t\right),$$

where L^{-1} is the forward operator, i.e. $L^{-1}\mathbf{v}_t = \mathbf{v}_{t+1}$. Let $\boldsymbol{\upsilon}_t = \left(\mathbf{q}_t', \mathbf{u}_t'\right)'$. The non-predetermined variables, $\mathbf{y}_t$, are described by the equations (see also transformations (20.42))

$$\boldsymbol{\upsilon}_t = \mathbf{C}_{\upsilon y}\mathbf{y}_t + \mathbf{C}_{\upsilon x}\mathbf{x}_t,$$

where $\mathbf{C}_{\upsilon y}$ and $\mathbf{C}_{\upsilon x}$ have a number of rows equal to the sum of the number of elements in $\mathbf{q}_t$ and $\mathbf{u}_t$. Under the condition that $\mathbf{C}_{\upsilon y}$ is nonsingular, we can write a solution for the non-predetermined variables

$$\mathbf{y}_t = \mathbf{C}_{\upsilon y}^{-1}\boldsymbol{\upsilon}_t + \mathbf{C}_{\upsilon y}^{-1}\mathbf{C}_{\upsilon x}\mathbf{x}_t, \tag{20.46}$$

which is a generalization of (20.43) to include $\mathbf{q}_t$. The solution for $\mathbf{x}_{t+1}$ can then be obtained following the steps outlined in Section 20.7.1 for the Blanchard and Kahn (1980) method. Note that under nonsingularity of $\mathbf{C}_{\upsilon y}$, and using (20.46), it is possible to express (20.38) in the general form (20.7) (see Section 20.7.1 for details in the case of the Blanchard and Kahn (1980) method).

20.7.3 Sims method

Sims (2001) proposes a solution method for models of the type

$$\boldsymbol{\Gamma}_0\mathbf{z}_t = \boldsymbol{\Gamma}_1\mathbf{z}_{t-1} + \boldsymbol{\Psi}\mathbf{v}_t + \boldsymbol{\Pi}\boldsymbol{\eta}_t, \tag{20.47}$$

where in the vector $\mathbf{z}_t$ some variables may enter as actual values and others as expectations, such as $\mathrm{E}\left(y_{j,t+1}|\Omega_t\right)$. In the above model, the matrix $\boldsymbol{\Gamma}_0$ is allowed to be singular, $\mathbf{v}_t$ is a random, exogenous and potentially serially correlated process, and $\boldsymbol{\eta}_t$ satisfies $\mathrm{E}\left(\boldsymbol{\eta}_{t+1}|\Omega_t\right) = \mathbf{0}$, for all t. We note that in the Sims (2001) approach the vector of expectations revisions, $\boldsymbol{\eta}_t$, is determined endogenously as part of the solution.

This method is based on the generalized Schur decomposition of matrices $\boldsymbol{\Gamma}_0$ and $\boldsymbol{\Gamma}_1$

$$\begin{aligned}\mathbf{Q}'\boldsymbol{\Lambda}_0\mathbf{Z}' &= \boldsymbol{\Gamma}_0,\\ \mathbf{Q}'\boldsymbol{\Lambda}_1\mathbf{Z}' &= \boldsymbol{\Gamma}_1,\end{aligned}$$

where $\mathbf{Q}'\mathbf{Q} = \mathbf{I}$, $\mathbf{Z}'\mathbf{Z} = \mathbf{I}$ and $\boldsymbol{\Lambda}_0$, $\boldsymbol{\Lambda}_1$ are upper triangular. An important property of this decomposition, which always exists, is that it produces the so-called generalized eigenvalues, defined as ratios of the diagonal elements of $\boldsymbol{\Lambda}_0$ and $\boldsymbol{\Lambda}_1$. Let $\mathbf{z}_t^* = \mathbf{Z}'\mathbf{z}_t$, and premultiply (20.47) by $\mathbf{Q}$ to obtain

$$\boldsymbol{\Lambda}_0\mathbf{z}_t^* = \boldsymbol{\Lambda}_1\mathbf{z}_{t-1}^* + \mathbf{Q}\left(\boldsymbol{\Psi}\mathbf{v}_t + \boldsymbol{\Pi}\boldsymbol{\eta}_t\right). \tag{20.48}$$

The above system can be rearranged so that the lower right blocks of $\boldsymbol{\Lambda}_0$ and $\boldsymbol{\Lambda}_1$ contain the generalized eigenvalues exploding to infinity. Partition $\mathbf{z}_t^*$ as follows

$$\begin{pmatrix}\boldsymbol{\Lambda}_{0,11} & \boldsymbol{\Lambda}_{0,12}\\ \mathbf{0} & \boldsymbol{\Lambda}_{0,22}\end{pmatrix}\begin{pmatrix}\mathbf{z}_{1t}^*\\ \mathbf{z}_{2t}^*\end{pmatrix} = \begin{pmatrix}\boldsymbol{\Lambda}_{1,11} & \boldsymbol{\Lambda}_{1,12}\\ \mathbf{0} & \boldsymbol{\Lambda}_{1,22}\end{pmatrix}\begin{pmatrix}\mathbf{z}_{1,t-1}^*\\ \mathbf{z}_{2,t-1}^*\end{pmatrix} + \begin{pmatrix}\mathbf{x}_{1t}\\ \mathbf{x}_{2t}\end{pmatrix}, \tag{20.49}$$

where $\mathbf{z}_t^* = \left(\mathbf{z}_{1t}^{*\prime}, \mathbf{z}_{2t}^{*\prime}\right)'$, $\mathbf{x}_t = \left(\mathbf{x}_{1t}', \mathbf{x}_{2t}'\right)' = \mathbf{Q}\left(\boldsymbol{\Psi}\mathbf{v}_t + \boldsymbol{\Pi}\boldsymbol{\eta}_t\right)$, and $\mathbf{z}_{2t}^*$ is the vector of unstable variables associated with the explosive generalized eigenvalues. Note that $\mathbf{z}_{2t}^*$ does not depend on

$\mathbf{z}_{1t}^*$. Letting $\mathbf{M} = \mathbf{\Lambda}_{1,22}^{-1}\mathbf{\Lambda}_{0,22}$, and assuming $lim_{h\to\infty}\mathbf{M}^h\mathbf{z}_{2,t+h}^* = \mathbf{0}$, solve forward the equations for $\mathbf{z}_{2t}^*$ to obtain

$$\begin{aligned}\mathbf{z}_{2t}^* &= -\sum_{h=1}^{\infty}\mathbf{M}^{h-1}\mathbf{\Lambda}_{1,22}^{-1}\mathbf{x}_{2,t+h} \\ &= -\sum_{h=1}^{\infty}\mathbf{M}^{h-1}\mathbf{\Lambda}_{1,22}^{-1}\mathbf{Q}_2\left(\mathbf{\Psi}\mathbf{v}_{t+h} + \mathbf{\Pi}\boldsymbol{\eta}_{t+h}\right),\end{aligned} \tag{20.50}$$

which relates $\mathbf{z}_{2t}^*$ to future values of $\mathbf{v}_{t+h}$ and $\boldsymbol{\eta}_{t+h}$. This means that knowing $\mathbf{z}_{2t}^*$ requires that all future events be known at time t. Since taking expectations conditional on the information available at time t does not change the left-hand side of the above equation, we obtain

$$\mathbf{z}_{2t}^* = -E\left[\sum_{h=1}^{\infty}\mathbf{M}^{h-1}\mathbf{\Lambda}_{1,22}^{-1}\mathbf{Q}_2\mathbf{\Pi}\left(\mathbf{\Psi}\mathbf{v}_{t+h} + \mathbf{\Pi}\boldsymbol{\eta}_{t+h}\right)|\Omega_t\right]. \tag{20.51}$$

The fact that the right-hand side of equation (20.50) never deviates from its expected value implies that the vector of expectations revisions, $\boldsymbol{\eta}_t$, must fluctuate as a function of current and future values of $\mathbf{v}_t$ to guarantee that equality (20.51) holds. In particular, equality in (20.51) is satisfied if and only if $\boldsymbol{\eta}_t$ satisfies

$$\mathbf{Q}_2\mathbf{\Pi}\boldsymbol{\eta}_{t+1} = \mathbf{\Lambda}_{1,22}\sum_{h=1}^{\infty}\mathbf{M}^{h-1}\mathbf{\Lambda}_{1,22}^{-1}\mathbf{Q}_2\mathbf{\Psi}\left[E\left(\mathbf{v}_{t+h}|\Omega_{t+1}\right) - E\left(\mathbf{v}_{t+h}|\Omega_t\right)\right].$$

Hence, the stability of the system crucially depends on the existence of expectations revisions $\boldsymbol{\eta}_t$ to offset the effect that the fundamental shocks $\mathbf{v}_t$ have on $\mathbf{z}_{2t}^*$.

Sims (2001) also proves that a necessary and sufficient condition to have a unique solution is that the row space of $\mathbf{Q}_1\mathbf{\Pi}$ should be contained in that of $\mathbf{Q}_2\mathbf{\Pi}$. In this case, we can write

$$\mathbf{Q}_1\mathbf{\Pi} = \mathbf{\Theta}\mathbf{Q}_1\mathbf{\Pi},$$

for some matrix $\mathbf{\Theta}$. Premultiplying (20.49) by $\mathbf{I} - \mathbf{\Theta}$ yields a new set of equations, free of references to $\boldsymbol{\eta}_t$, that can be combined with (20.50) to give

$$\begin{pmatrix}\mathbf{\Lambda}_{0,11} & \mathbf{\Lambda}_{0,12} - \mathbf{\Theta}\mathbf{\Lambda}_{0,22} \\ \mathbf{0} & \mathbf{I}\end{pmatrix}\begin{pmatrix}\mathbf{z}_{1t}^* \\ \mathbf{z}_{2t}^*\end{pmatrix} = \begin{pmatrix}\mathbf{\Lambda}_{1,11} & \mathbf{\Lambda}_{1,12} - \mathbf{\Theta}\mathbf{\Lambda}_{1,22} \\ \mathbf{0} & \mathbf{0}\end{pmatrix}\times \begin{pmatrix}\mathbf{z}_{1,t-1}^* \\ \mathbf{z}_{2,t-1}^*\end{pmatrix} + \begin{pmatrix}\mathbf{Q}_1 - \mathbf{\Theta}\mathbf{Q}_1 \\ \mathbf{0}\end{pmatrix}\mathbf{\Psi}\mathbf{v}_t - \begin{pmatrix}\mathbf{0} \\ \sum_{h=1}^{\infty}\mathbf{M}^{h-1}\mathbf{\Lambda}_{1,22}^{-1}\mathbf{Q}_2\mathbf{\Psi}E\left(\mathbf{v}_{t+h}|\Omega_t\right)\end{pmatrix}.$$

Hence, when the matrix $\mathbf{\Theta}$ exists, the term involving $\boldsymbol{\eta}_t$ drops out, and the reduced form of the *RE* model can be written as

$$\mathbf{z}_t = \mathbf{\Phi}_1 \mathbf{z}_{t-1} + \mathbf{\Phi}_v \mathbf{v}_t + \mathbf{\Phi}_z \sum_{h=1}^{\infty} \mathbf{\Phi}_f^{h-1} \mathbf{\Phi}_v E\left(\mathbf{v}_{t+h} \,|\Omega_t\right), \tag{20.52}$$

where the matrices $\mathbf{\Phi}_1$, $\mathbf{\Phi}_v$, $\mathbf{\Phi}_z$ and $\mathbf{\Phi}_f$ are a function of parameters in (20.49) (see Sims (2001) for a description of the elements in system (20.52)).

20.7.4 Martingale difference method

This method, proposed by Broze, Gouriéroux, and Szafarz (1990), consists of replacing rational expectations by their realization plus the realization of a martingale difference process. Suppose in equation (20.7) we replace $\mathrm{E}(\mathbf{y}_{t+1}|\Omega_t)$ with $\mathbf{y}_{t+1} - \boldsymbol{\xi}_{t+1}$, where each component of $\boldsymbol{\xi}_{t+1}$ is a martingale difference process with respect to the information set Ω_t. We obtain

$$\mathbf{y}_t - \mathbf{A}\mathbf{y}_{t-1} - \mathbf{B}\mathbf{y}_{t+1} = \mathbf{u}_t - \mathbf{B}\boldsymbol{\xi}_{t+1}. \tag{20.53}$$

The characteristic polynomial of the above equation is

$$\mathbf{\Xi}(L) = -\mathbf{B} + \mathbf{I}_m L - \mathbf{A}L^2.$$

Premultiplying both sides of (20.53) lagged by one period by the adjoint matrix of $\mathbf{\Xi}(L)$, $\mathcal{Y}(L)$, we have

$$L^{m_1} \mathbf{\Phi}(L)\, \mathbf{y}_t = -\mathcal{Y}(L)\, \mathbf{B}\boldsymbol{\xi}_{t+1} + \mathcal{Y}(L)\, \mathbf{u}_t,$$

where $L^{m_1}\mathbf{\Phi}(L) = \det\left[\mathbf{\Xi}(L)\right]$, and m_1 equals the number of zero roots of $\mathbf{\Xi}(L)$. Multiplying both sides by L^{-m_1}, and substituting the expression of $\mathbf{\Xi}(L)$ in the first term of the right-hand side we obtain

$$\mathbf{\Phi}(L)\, \mathbf{y}_t - L^{-m_1}\mathcal{Y}(L)\, \mathbf{\Xi}(L)\, \boldsymbol{\xi}_t = -\mathcal{Y}(L)\left(\mathbf{I}_m L - \mathbf{A}L^2\right)\boldsymbol{\xi}_{t+m_1} + \mathcal{Y}(L)\, \mathbf{u}_{t+m_1-1},$$

or, more compactly

$$\mathbf{\Phi}(L)\left(\mathbf{y}_t - \boldsymbol{\xi}_t\right) = -\mathcal{Y}(L)\left(\mathbf{I}_m L - \mathbf{A}L^2\right)\boldsymbol{\xi}_{t+m_1} + \mathcal{Y}(L)\, \mathbf{u}_{t+m_1-1}. \tag{20.54}$$

Note that now the left-hand side of equation (20.54) only depends on information known at time $t-1$ (recall that $\mathbf{y}_t - \boldsymbol{\xi}_t = \mathrm{E}(\mathbf{y}_t|\Omega_{t-1})$). The same line of reasoning can be applied to the right-hand side of (20.54), and we must therefore have

$$\begin{aligned} &E\left[-\mathcal{Y}(L)\left(\mathbf{I}_m L - \mathbf{A}L^2\right)\boldsymbol{\xi}_{t+m_1} + \mathcal{Y}(L)\, \mathbf{u}_{t+m_1-1}|\Omega_{t-1}\right] \\ &\quad = -\mathcal{Y}(L)\left(\mathbf{I}_m L - \mathbf{A}L^2\right)\boldsymbol{\xi}_{t+m_1} + \mathcal{Y}(L)\, \mathbf{u}_{t+m_1-1}. \end{aligned} \tag{20.55}$$

Solutions to the *RE* model (20.7) can thus be computed by finding the martingale difference processes, $\boldsymbol{\xi}_{t+m_1}$, that satisfy (20.55), and then solving the corresponding difference equation systems (20.53) for $\mathbf{y}_t$ in terms of $\mathbf{u}_t$ and $\boldsymbol{\xi}_t$. Generally there will be an infinite number of bounded solutions, and the number of martingale difference processes that can be chosen arbitrarily may

be derived using the restrictions implied by (20.55). For further details see Broze, Gouriéroux, and Szafarz (1990).

Remark 5 *The choice amongst the alternative solution methods depends on the nature of the RE model and the type of solution sought. For example, the undetermined coefficients method is appropriate when it is known that the solution is unique. The Blanchard and Kahn method only applies when the coefficients of the future expectations are nonsingular which could be highly restrictive in practice. The King and Watson solution strategy relaxes the restrictive nature of the Blanchard and Kahn's approach but does not allow characterizations of all the possible solutions in the general case. The same also applies to Sims' method. In contrast, QDE and the martingale difference methods can be used to develop all the solutions of the RE models in a transparent manner. In the case where a unique solution exists, the numerical accuracy and speed of alternative solution methods are compared by Anderson (2008).*

20.8 Rational expectations *DSGE* models

20.8.1 A general framework

Most macroeconomic *DSGE* models are constructed by linearizing an underlying nonlinear rational expectations, *RE*, model around its steady state. A typical log-linearized *RE* model can be written as

$$\begin{aligned} \mathbf{A}_0(\boldsymbol{\theta})\mathbf{y}_t &= \mathbf{A}_1(\boldsymbol{\theta})E_t(\mathbf{y}_{t+1}) + \mathbf{A}_2(\boldsymbol{\theta})\mathbf{y}_{t-1} + \mathbf{A}_3(\boldsymbol{\theta})\mathbf{x}_t + \mathbf{u}_t, \\ \mathbf{x}_t &= \boldsymbol{\Phi}_x\mathbf{x}_{t-1} + \mathbf{v}_t,\ \mathbf{u}_t = \boldsymbol{\Phi}_u\mathbf{u}_{t-1} + \boldsymbol{\varepsilon}_t, \end{aligned} \tag{20.56}$$

where $\mathbf{y}_t$ is an $m \times 1$ vector of deviations from the steady states, $\mathbf{x}_t$ is a $k \times 1$ vector of observed exogenous variables, $\mathbf{u}_t$ is an $m \times 1$ vector of unobserved variables, and $\boldsymbol{\varepsilon}_t$ is the $m \times 1$ vector of structural shocks, assumed to be serially uncorrelated with mean zero and the covariance matrix, $E(\boldsymbol{\varepsilon}_t\boldsymbol{\varepsilon}_t') = \boldsymbol{\Omega}(\boldsymbol{\theta})$. For Bayesian or maximum likelihood estimation, $\boldsymbol{\varepsilon}_t$ is also typically assumed to be normally distributed. The expectations $E_t(\mathbf{y}_{t+1}) = E(\mathbf{y}_{t+1} \mid \Im_t)$ are assumed to be rationally formed with respect to the information set, $\Im_t = (\mathbf{y}_t, \mathbf{x}_t, \mathbf{y}_{t-1}, \mathbf{x}_{t-1}, \ldots)$. To simplify the exposition it is assumed that both the exogenous and unobserved variables follow $VAR(1)$ processes. The parameters of interest are the $q \times 1$ vector of structural parameters, $\boldsymbol{\theta}$, and the remaining (reduced form) parameters $\boldsymbol{\Phi}_x$ and $\boldsymbol{\Phi}_u$ are assumed as given. It is also assumed that there are no feedbacks from $\mathbf{y}_t$ to $\mathbf{x}_t$ or $\mathbf{u}_t$. To identify the structural shocks it is common in the literature to assume that $\boldsymbol{\Omega}(\boldsymbol{\theta}) = \mathbf{I}_m$.

If $\mathbf{A}_0(\boldsymbol{\theta})$ is nonsingular, then (20.56) can be written as

$$\begin{aligned} \mathbf{y}_t &= \mathbf{A}_0(\boldsymbol{\theta})^{-1}\mathbf{A}_1(\theta)E_t(\mathbf{y}_{t+1}) + \mathbf{A}_0(\boldsymbol{\theta})^{-1}\mathbf{A}_2(\boldsymbol{\theta})\mathbf{y}_{t-1} \\ &\quad + \mathbf{A}_0(\boldsymbol{\theta})^{-1}\mathbf{A}_3(\boldsymbol{\theta})\mathbf{x}_t + \mathbf{A}_0(\boldsymbol{\theta})^{-1}\mathbf{u}_t. \end{aligned} \tag{20.57}$$

The solution of this model is discussed in Section 20.3, and assuming that the solution is unique it takes the form

$$\mathbf{y}_t = \mathbf{C}(\boldsymbol{\theta})\mathbf{y}_{t-1} + \mathbf{G}_x(\boldsymbol{\theta}, \boldsymbol{\phi}_x)\mathbf{x}_t + \mathbf{G}_u(\boldsymbol{\theta}, \boldsymbol{\phi}_u)\mathbf{u}_t, \tag{20.58}$$

where $\boldsymbol{\phi}_i = vec(\boldsymbol{\Phi}_i)$, $i = \mathbf{x}, \mathbf{u}$. The matrices $\mathbf{G}_i(\boldsymbol{\theta}, \boldsymbol{\phi}_i)$, $i = \mathbf{x}, \mathbf{u}$, can be obtained using the method of undetermined coefficients. Notice that the coefficient matrix, $\mathbf{C}(\boldsymbol{\theta})$, for the lagged dependent variable vector is just a function of $\boldsymbol{\theta}$, and does not depend on $\boldsymbol{\phi}_x$ or $\boldsymbol{\phi}_u$.

If $\boldsymbol{\Phi}_u = \mathbf{0}$, this is just a *VAR* with exogenous variables and the likelihood function for the reduced form parameters is easily obtained. In the general case where the unobserved components of the model are serially correlated, the rational expectations solution will involve moving average components and it is more convenient to write the model as a state space model where Kalman filtering techniques can be used to evaluate the likelihood function. In such cases the reduced form parameters may not be identified.

20.8.2 *DSGE* models without lags

Abstracting from lagged values and exogenous regressors and for notational simplicity not making the dependence on $\boldsymbol{\theta}$ explicit, (20.56) simplifies to

$$\begin{aligned} \mathbf{A}_0\mathbf{y}_t &= \mathbf{A}_1 E_t(\mathbf{y}_{t+1}) + \boldsymbol{\varepsilon}_t, \\ E(\boldsymbol{\varepsilon}_t) &= 0,\ E(\boldsymbol{\varepsilon}_t\boldsymbol{\varepsilon}_t') = \boldsymbol{\Sigma}_\varepsilon. \end{aligned} \tag{20.59}$$

If $\mathbf{A}_0$ is nonsingular using (20.59) we have

$$\mathbf{y}_t = \mathbf{A}_0^{-1}\mathbf{A}_1 E_t(\mathbf{y}_{t+1}) + \mathbf{A}_0^{-1}\boldsymbol{\varepsilon}_t = \mathbf{Q}E_t(\mathbf{y}_{t+1}) + \mathbf{A}_0^{-1}\varepsilon_t. \tag{20.60}$$

The regular case, where there is a unique stationary solution, arises if all eigenvalues of $\mathbf{Q} = \mathbf{A}_0^{-1}\mathbf{A}_1$ lie within the unit circle (see Section 20.2). In this case, the unique solution of the model is given by

$$\mathbf{y}_t = \sum_{j=0}^{\infty} \mathbf{Q}^j \mathbf{A}_0^{-1} E_t(\boldsymbol{\varepsilon}_{t+j}). \tag{20.61}$$

Since $E_t(\boldsymbol{\varepsilon}_{t+j}) = \mathbf{0}$, for $j \geq 0$, then $E_t(\mathbf{y}_{t+1}) = 0$ and the solution simplifies to

$$\mathbf{A}_0\mathbf{y}_t = \boldsymbol{\varepsilon}_t, \tag{20.62}$$

or

$$\mathbf{y}_t = \mathbf{A}_0^{-1}\boldsymbol{\varepsilon}_t = \mathbf{u}_t, E(\mathbf{u}_t\mathbf{u}_t') = \boldsymbol{\Sigma}_u = \mathbf{A}_0^{-1}\boldsymbol{\Sigma}_\varepsilon\mathbf{A}_0'^{-1}. \tag{20.63}$$

Notice that (20.63) provides us with a likelihood function which does not depend on $\mathbf{A}_1$ and, therefore, the parameters that are unique to $\mathbf{A}_1$ (i.e., the coefficients that are specific to the forward variables) are not identified. Furthermore, the *RE* model is observationally equivalent to a model without forward variables which takes the form of (20.62). Since what can be estimated from the data, namely $\boldsymbol{\Sigma}_u$, is not a function of $\mathbf{A}_1$, all possible choices of $\mathbf{A}_1$ are observationally equivalent in the sense that they lead to the same observed data covariance matrix.

Although the coefficients in the forward solution (20.61) are functions of $\mathbf{A}_1$, this does not identify them because $E_t(\boldsymbol{\varepsilon}_{t+j}) = \mathbf{0}$. Elements of $\mathbf{A}_1$ could be identified by certain sorts of *a priori* restrictions, but these are likely to be rather special, rather limited in number and cannot be tested.

If the parameters of the *DSGE* model were thought to be known *a priori* from calibration, there would be no identification problem and the structural errors ε_{it} could be recovered and used, for instance, in calculating impulse response functions, IRFs (see Chapter 24). However, suppose someone else believed that the true model was just a set of random errors $\mathbf{y}_t = \mathbf{u}_t$, with different IRFs. There is no information in the data that a proponent of the *DSGE* could use to persuade another person that the *DSGE* model was correct relative to the random error model.

The above result generalizes to higher-order *RE* models. Consider, for example, the model

$$\mathbf{A}_0\mathbf{y}_t = \sum_{i=1}^{p} \mathbf{A}_i E_t(\mathbf{y}_{t+i}) + \boldsymbol{\varepsilon}_t.$$

Once again the unique stable solution of this model is given by $\mathbf{A}_0\mathbf{y}_t = \boldsymbol{\varepsilon}_t$, and none of the elements of $\mathbf{A}_1, \mathbf{A}_2, \ldots, \mathbf{A}_p$ that are variation free with respect to the elements of $\mathbf{A}_0$ are identified.

Example 44 *Consider the following standard three equation NK-DSGE model used in Benati (2010) involving only current and future variables*

$$R_t = \psi \pi_t + \varepsilon_{1t}, \tag{20.64}$$

$$y_t = E(y_{t+1} \mid \Omega_t) - \sigma\,[R_t - E(\pi_{t+1} \mid \Omega_t)] + \varepsilon_{2t}, \tag{20.65}$$

$$\pi_t = \beta E(\pi_{t+1} \mid \Omega_t) + \gamma y_t + \varepsilon_{3t}. \tag{20.66}$$

Equation (20.64) is a Taylor rule determining the interest rate, R_t, (20.66) a Phillips curve determining inflation, π_t, and (20.65) is an IS curve determining output, y_t, all measured as deviations from their steady states. The errors, which are assumed to be white noise, are a monetary policy shock, ε_{1t}, a demand shock, ε_{2t}, and a supply or cost shock, ε_{3t}, which we collect in the vector $\boldsymbol{\varepsilon}_t = (\varepsilon_{1t}, \varepsilon_{2t}, \varepsilon_{3t})'$. These are also usually assumed to be orthogonal. This system is highly restricted, with many parameters set to zero a priori. For instance, output does not appear in the Taylor rule and the coefficient of future output is assumed to be equal to unity. Let $\mathbf{y}_t = (R_t, \pi_t, y_t)'$ and

$$\mathbf{A}_0 = \begin{pmatrix} 1 & -\psi & 0 \\ \sigma & 0 & 1 \\ 0 & 1 & -\gamma \end{pmatrix}, \mathbf{A}_1 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & \sigma & 1 \\ 0 & \beta & 0 \end{pmatrix}. \tag{20.67}$$

and note that

$$\mathbf{y}_t = \mathbf{A}E(\mathbf{y}_{t+1}|\Omega_t) + \mathbf{w}_t,$$

where $\mathbf{w}_t = \mathbf{A}_0^{-1}\boldsymbol{\varepsilon}_t$,

$$\mathbf{A}_0^{-1} = \frac{1}{\gamma\sigma\psi + 1}\begin{pmatrix} 1 & \gamma\psi & \psi \\ -\gamma\sigma & \gamma & 1 \\ -\sigma & 1 & -\sigma\psi \end{pmatrix},$$

$$\mathbf{A} = \mathbf{A}_0^{-1}\mathbf{A}_1 = \frac{1}{\gamma\sigma\psi + 1}\begin{pmatrix} 1 & \gamma\psi & \psi \\ -\gamma\sigma & \gamma & 1 \\ -\sigma & 1 & -\sigma\psi \end{pmatrix}\begin{pmatrix} 0 & 0 & 0 \\ 0 & \sigma & 1 \\ 0 & \beta & 0 \end{pmatrix}$$

$$= \frac{1}{\gamma\sigma\psi + 1}\begin{pmatrix} 0 & \psi(\beta + \gamma\sigma) & \gamma\psi \\ 0 & \beta + \gamma\sigma & \gamma \\ 0 & \sigma(1 - \beta\psi) & 1 \end{pmatrix}.$$

The two non-zero eigenvalues of $\mathbf{A}$ *are*

$$\lambda_1 = \frac{1}{2(\gamma\sigma\psi + 1)}(1 + \beta + \gamma\sigma + \kappa),\ \lambda_2 = \frac{1}{2(\gamma\sigma\psi + 1)}(1 + \beta + \gamma\sigma - \kappa),$$

where $\kappa = \sqrt{\beta^2 - 2\beta + \gamma^2\sigma^2 + 2\gamma\sigma + 2\gamma\sigma\beta - 4\gamma\sigma\beta\psi + 1}$. *Assuming that* $|\lambda_j| < 1$ *for* $j = 1, 2$ *and under serially uncorrelated errors, the solution of the above model is given by the forward solution which in this case reduces to*

$$\mathbf{y}_t = \mathbf{A}_0^{-1}\boldsymbol{\varepsilon}_t, \tag{20.68}$$

which does not depend on $\mathbf{A}_1$. *This solution is also obtainable from (20.64), (20.66), and (20.65) by setting all expectational variables to zero. Writing the solution in full we have*

$$\begin{aligned} R_t &= \psi\pi_t + \varepsilon_{1t}, \\ y_t &= -\sigma R_t + \varepsilon_{2t}, \\ \pi_t &= \gamma y_t + \varepsilon_{3t}, \end{aligned}$$

which does not depend on β. *As we shall see later, this has implications for identification and estimation of* β. *(See Example 46). This example illustrates some of the features of DSGE models. First, the RE model parameter matrices,* $\mathbf{A}_0$ *and* $\mathbf{A}_1$, *are written in terms of deep parameters,* $\boldsymbol{\theta} = (\gamma, \sigma, \psi, \beta)'$. *Second, the parameters which appear only in* $\mathbf{A}_1$ *do not enter the RE solution and, thus, do not enter the likelihood function. In this example,* β *does not appear in the likelihood function, though* σ *which appears in* A_1 *does appear in the likelihood function because it also appears in* $\mathbf{A}_0$. *Third, the restrictions necessary to ensure regularity (i.e.,* $|\lambda_i| < 1$ *for* $i = 1, 2$*), imply bounds involving the structural parameters, including the unidentified* β. *Thus, the parameter space is not variation free. Fourth, if* β *is fixed at some pre-selected value for the discount rate (as would be done by a calibrator), then the model is identified.*

20.8.3 *DSGE* models with lags

In order to reproduce the dynamics that are typically observed with macroeconomic data, most empirical *DSGE* models include lagged values of endogenous or exogenous (observed or unobserved) variables. For instance Clarida, Gali, and Gertler (1999) assume that the errors in the IS and Phillips curve equations follow $AR(1)$ processes and derive an optimal feedback policy for the interest rate based on the forecasts from these autoregressions. In this case, there is a predictable component in expected inflation because of the serial correlation in the equation errors.

Consider the special case of (20.56), where $\mathbf{A}_3 = \mathbf{\Phi}_u = \mathbf{0}$ so that the model only contains lagged endogenous variables

$$\mathbf{A}_0 y_t = \mathbf{A}_1 E_t(\mathbf{y}_{t+1}) + \mathbf{A}_2 \mathbf{y}_{t-1} + \varepsilon_t. \tag{20.69}$$

In this case the unique solution is given by

$$\mathbf{y}_t = \mathbf{C}\mathbf{y}_{t-1} + \mathbf{A}_0^{-1}\boldsymbol{\varepsilon}_t, \tag{20.70}$$

where, as shown in Section 20.3.1, $\mathbf{C}$ solves the quadratic matrix equation $\mathbf{A}_1\mathbf{C}^2 - \mathbf{A}_0\mathbf{C} + \mathbf{A}_2 = \mathbf{0}$. The solution is unique and stationary if all the eigenvalues of $\mathbf{C}$ and $(\mathbf{I}_m - \mathbf{A}_1\mathbf{C})^{-1}\mathbf{A}_1$ lie strictly inside the unit circle. Therefore, the *RE* solution is observationally equivalent to the non-*RE* simultaneous equations model (*SEM*)

$$\mathbf{A}_0\mathbf{y}_t = \mathbf{A}_2\mathbf{y}_{t-1} + \boldsymbol{\varepsilon}_t,$$

where, in the case of the *SEM*, $\mathbf{C} = \mathbf{A}_0^{-1}\mathbf{A}_2$.

Again whereas the order condition for identification of the *SEM* requires m^2 restrictions, the *RE* model requires $2m^2$ restrictions. Not only is the *RE* model observationally equivalent to a purely backward looking *SEM*, it is observationally equivalent (in the sense of having the same reduced form), to any other model of expectations formation where in (20.69) $E_t(\mathbf{y}_{t+1})$ is replaced by $\mathbf{D}\mathbf{y}_{t-1}$. More specifically, knowing the form of the solution, (20.70), does not, on its own, provide information on the cross-equation parametric restrictions. In either case, the identifying cross-equation restrictions are lost.

Thus, in models with lags, the same problem of observational equivalence between *RE* and other models recurs. One may be able to distinguish the reduced forms of particular *RE* models from other observationally equivalent models, because the *RE* models impose particular types of cross-equation restriction on the reduced form, which arise from the nature of the rational expectations. But such restrictions are subject to the objection made by Sims (1980), who criticized identification by 'incredible' dynamic restrictions on the coefficients and lag lengths. *RE* models, which depend on restrictions on the form of the dynamics, such as $AR(1)$ errors, are equally vulnerable to such objections.

Example 45 *Consider the new Keynesian Phillips curve (NKPC) model in example 42, but to simplify the discussion of identification, abstract from the backward component and write the NKPC as*

$$\pi_t = \beta_f E_{t-1}\pi_{t+1} + \gamma x_t + \varepsilon_t, \tag{20.71}$$

where inflation, π_t, is determined by expected inflation and an exogenous driving process, such as the output gap, x_t. β_f and γ are fixed parameters and ε_t is a martingale difference process and $E_{t-1}\pi_{t+1} = E(\pi_{t+1} \mid \Im_{t-1})$, where $\Im_{t-1}$ is the information set available at time $t-1$. Note that in this model expectations are conditioned on $\Im_{t-1}$, rather than on $\Im_t$. It is assumed that there is no feedback from π_t to x_t, and x_t follows a stationary AR(2) process

$$x_t = \rho_1 x_{t-1} + \rho_2 x_{t-2} + v_t, \ \sim IID(0, \sigma_v^2). \tag{20.72}$$

The RE solution of the NKPC is given in this case by

$$\pi_t = \alpha_1 x_{t-1} + \alpha_2 x_{t-2} + \varepsilon_{\pi t}, \ \varepsilon_{\pi t} \sim IID(0, \sigma_{\varepsilon\pi}^2), \tag{20.73}$$

where

$$\alpha_1 = \frac{\gamma\left(\rho_1 + \beta_f \rho_2\right)}{1 - \beta_f \rho_1 - \beta_f^2 \rho_2}, \ \alpha_2 = \frac{\gamma \rho_2}{1 - \beta_f \rho_1 - \beta_f^2 \rho_2}. \tag{20.74}$$

The reduced form parameters $\boldsymbol{\phi} = (\rho_1, \rho_2, \alpha_1, \alpha_2)' = (\rho', \alpha')'$ can be obtained by estimating the system of equations (20.72) and (20.73) in $y_t = (\pi_t, x_t)'$. Assuming that x_t is weakly exogenous $\boldsymbol{\phi}$ is identified and can be estimated by OLS on each equation. Identification of the structural parameters $\boldsymbol{\theta} = (\beta_f, \gamma, \rho_1, \rho_2)'$ will then involve inverting the mapping from $\boldsymbol{\phi}$ to $\boldsymbol{\theta}$, given by (20.74). As noted originally in Chapter 7 of Pesaran (1987c) and emphasized recently by Mavroeidis (2005) and Nason and Smith (2008), among others, identification of the structural parameters critically depends on the process generating x_t. Assuming that $\rho_2 \neq 0, \gamma \neq 0$ and the denominator in (20.74), $1 - \beta_f \rho_1 - \beta_f^2 \rho_2 \neq 0$ then

$$\beta_f = \frac{\alpha_1 \rho_2 - \alpha_2 \rho_1}{\rho_2 \alpha_2}, \text{ for } \rho_2 \alpha_2 \neq 0,$$

$$\gamma = \frac{\alpha_1\left(1 - \beta_f \rho_1 - \beta_f^2 \rho_2\right)}{\rho_1 + \beta_f \rho_2}, \text{ for } 1 - \beta_f \rho_1 - \beta_f^2 \rho_2 \neq 0.$$

Within the classical framework, the matrix of derivatives of the reduced form parameters with respect to the structural parameters, plays an important role in identification. In this example the relevant part of this matrix is the derivatives of $\boldsymbol{\alpha} = (\alpha_1, \alpha_1)'$ with respect to $\boldsymbol{\theta}$, which can be obtained using (20.74)

$$R(\theta) = \frac{\partial \boldsymbol{\alpha}}{\partial \boldsymbol{\theta}'} = \frac{1}{1 - \beta_f \rho_1 - \beta_f^2 \rho_2} \begin{bmatrix} \gamma\left(\rho_2 + \frac{\rho_1 + 2\beta_f \rho_2}{1 - \beta \rho_1 - \beta^2 \rho_2}\right) & \rho_1 + \beta_f \rho_2 \\ \gamma \frac{\rho_2\left(\rho_1 + 2\beta_f \rho_2\right)}{1 - \beta_f \rho_1 - \beta_f^2 \rho_2} & \rho_2 \end{bmatrix}.$$

A 'yes/no' answer to the question of whether a particular value of $\boldsymbol{\theta}$ is identified is given by investigating if the rank of $R(\boldsymbol{\theta})$, evaluated at that particular value, is full. Therefore, necessary conditions for identification are $1 - \beta_f \rho_1 - \beta_f^2 \rho_2 \neq 0, \gamma \neq 0$ and $\rho_2 \neq 0$. This matrix will

also play a role in the Bayesian identification analysis. The weakly identified case can arise if $1 - \beta_f \rho_1 - \beta_f^2 \rho_2 \neq 0, \gamma \neq 0$, *but* ρ_2 *is replaced by* $\rho_{2T} = \delta/\sqrt{T}$.

20.9 Identification of *RE* models: a general treatment

In order to consistently estimate the parameters of a *RE* model from the time series observations, these must be identified. Therefore, identification is fundamental to the empirical analysis of structural models, and must be addressed before estimation and hypothesis testing.

Suppose that the purpose of modelling is to explain the observations $\mathbf{y}_t$ $t = 1, 2, \ldots, T$, on the m-dimensional vector of endogenous variables $\mathbf{y}$ conditional on the occurrence of the k-dimensional exogenous variables $\mathbf{x}$. The model for $\mathbf{y}$ is defined by the joint probability distribution function of $\boldsymbol{Y} = \left(\mathbf{y}_1, \ldots, \mathbf{y}_T\right)$, conditional on $\boldsymbol{X} = (\mathbf{x}_1, \ldots, \mathbf{x}_T)$, namely $f(\boldsymbol{Y} \,|\boldsymbol{X}, \boldsymbol{\theta})$, for all admissible values of the p-dimensional unknown parameter vector, $\boldsymbol{\theta} = (\theta_1, \theta_2, \ldots, \theta_p)'$. We assume that the probability distribution function is known to the researcher except for the parameter vector, $\boldsymbol{\theta}$. The set of admissible values of $\boldsymbol{\theta}$, namely $\boldsymbol{\Theta} \subset \mathbb{R}^p$, is known as the parameter space.

In the context of the model $f(\boldsymbol{Y} \,|\boldsymbol{X}, \boldsymbol{\theta})$, two parameter points $\boldsymbol{\theta}_0$ and $\boldsymbol{\theta}_1$ are said to be observationally equivalent if $f(\boldsymbol{Y} \,|\boldsymbol{X}, \boldsymbol{\theta}_0) = f(\boldsymbol{Y} \,|\boldsymbol{X}, \boldsymbol{\theta}_1)$ for all values of $\boldsymbol{Y}$ and $\boldsymbol{X}$. The parameter vector $\boldsymbol{\theta}$ is said to be identified at $\boldsymbol{\theta} = \boldsymbol{\theta}_0$ if and only if for all $\boldsymbol{\theta} \in \boldsymbol{\Theta}$, and $\boldsymbol{\theta} \neq \boldsymbol{\theta}_0$ implies $f(\boldsymbol{Y} \,|\boldsymbol{X}, \boldsymbol{\theta}) \neq f(Y \,|X, \boldsymbol{\theta}_0)$ for all $\boldsymbol{Y}$ and $\boldsymbol{X}$. If $\boldsymbol{\theta}$ is identified for all $\boldsymbol{\theta}_0 \in \boldsymbol{\Theta}$ then it is said that it is globally identified. Equivalently, $\boldsymbol{\theta}$ is said to be globally identified at $\boldsymbol{\theta} = \boldsymbol{\theta}_0$, if there is no other $\boldsymbol{\theta} \in \boldsymbol{\Theta}$ which is observationally equivalent to $\boldsymbol{\theta}_0$. It is often difficult to establish global identification. A weaker notion, known as local identification, is also used. $\boldsymbol{\theta}$ is said to be locally identified at $\boldsymbol{\theta}_0$ if there exists an open neighbourhood of $\boldsymbol{\theta}_0$ containing no other $\boldsymbol{\theta} \in \boldsymbol{\Theta}$ which is observationally equivalent to $\boldsymbol{\theta}_0$. The identification problem consists of finding conditions on $f(\boldsymbol{Y} \,|\boldsymbol{X}, \boldsymbol{\theta})$ and $\boldsymbol{\Theta}$ that are necessary and sufficient for the identification of the parameters in $\boldsymbol{\Theta}$. Rothenberg (1971) proved that, under some regularity conditions, local identification of $\boldsymbol{\theta}_0$ for a given model occurs if and only if the information matrix evaluated at $\boldsymbol{\theta}_0$ is nonsingular (see Sections 4.2 and 9.4 for details on Fisher's information matrix). Rothenberg (1971) also provided some criteria for local identification of parameters when these satisfy a set of constraints, and when reduced form parameters help to establish the identification of structural parameters.

Early work on identification of multivariate *RE* models have been carried out by Wallis (1980), Pesaran (1981b), Wegge and Feldman (1983), and Pesaran (1987c). Most of these studies focused on models containing exogenous variables in the system, and current expectations (namely, including the term $E\left(\mathbf{y}_t | \Omega_{t-1}\right)$). In particular, Wallis (1980) first provided sufficient conditions for global identification of a simultaneous equations system with current expectations, given in terms of rank conditions on the reduced form matrices. Pesaran (1981b) and Wegge and Feldman (1983) extended this work by allowing for more general identification restrictions, and for models with lagged exogenous variables. Further, they presented a rank condition for identification of a simultaneous equations system, expressed in terms of the structural parameters, rather then of the reduced form parameters, as in Wallis (1980). Pesaran (1987c) provided a rank condition for identification of a single equation in a system with future expectations. A central issue in the early literature was the problem of observational equivalence between two distinct models, namely, when two distinct models generate exactly the same probability distribution and likelihood function for the data (see also Sargent (1976) on this).

Identification conditions were given by determining when the mapping from reduced-form to structural parameters is unique.

The focus of the more recent discussion on identification of *RE* models has been on closed systems with no exogenous variables. Specifically, consider the following structural *RE* model with lagged values:

$$\mathcal{A}_0(\boldsymbol{\theta})\,\mathbf{y}_t = \mathcal{A}_1(\boldsymbol{\theta})\,\mathbf{y}_{t-1} + \mathcal{A}_2(\boldsymbol{\theta})\,E(\mathbf{y}_{t+1}|\Omega_t) + \mathcal{A}_3(\boldsymbol{\theta})\,\mathbf{v}_t, \tag{20.75}$$

where the elements of the matrices $\mathcal{A}_0(\boldsymbol{\theta})$, $\mathcal{A}_1(\boldsymbol{\theta})$, $\mathcal{A}_2(\boldsymbol{\theta})$, and $\mathcal{A}_3(\boldsymbol{\theta})$ are functions of the structural parameters $\boldsymbol{\theta}$, and $\mathbf{v}_t$ is such that $E(\mathbf{v}_t) = \mathbf{0}$, and $E(\mathbf{v}_t\mathbf{v}_t') = \mathbf{I}_m$. Assuming that a unique solution exists, we have seen that it can be cast in the form

$$\mathbf{y}_t = \boldsymbol{\Phi}\mathbf{y}_{t-1} + \mathbf{u}_t, \tag{20.76}$$

where

$$\boldsymbol{\Phi} = \mathbf{C}(\boldsymbol{\theta}), \text{ and } \mathbf{u}_t = \mathbf{G}(\boldsymbol{\theta})\mathbf{v}_t.$$

The solution of the *RE* model can, therefore, be viewed as a restricted form of the *VAR* model popularized in econometrics by Sims (1980). (20.76) can also be viewed as the reduced form model associated with the structural model (20.75). Identification of structural parameters, $\boldsymbol{\theta}$, can be investigated by considering the mapping from the reduced form parameters, $\boldsymbol{\Phi}$ and $Var(\mathbf{u}_t) = \boldsymbol{\Sigma}_u$ to $\boldsymbol{\theta}$. Identification of *RE* models is complicated by the fact that this mapping is often highly nonlinear.

Example 46 *Consider the static DSGE model given in Example 44, and note that under the assumption that* $\boldsymbol{\varepsilon}_t \sim IIDN(0, \boldsymbol{\Sigma}_\varepsilon)$ *the log-likelihood function of the model is given by*

$$\ell_T(\boldsymbol{\theta}) \propto -\frac{T}{2}\ln|\boldsymbol{\Sigma}_\varepsilon| - \frac{1}{2}\sum_{t=1}^{T}\mathbf{y}_t'\mathbf{A}_0'\boldsymbol{\Sigma}_\varepsilon^{-1}\mathbf{A}_0\mathbf{y}_t,$$

where

$$\mathbf{A}_0 = \begin{pmatrix} 1 & -\psi & 0 \\ \sigma & 0 & 1 \\ 0 & 1 & -\gamma \end{pmatrix},$$

and $\boldsymbol{\Sigma}_\varepsilon$ *is a diagonal matrix with* $diag(\boldsymbol{\Sigma}_\varepsilon) = (\sigma_{\varepsilon 1}^2, \sigma_{\varepsilon 2}^2, \sigma_{\varepsilon 3}^2)'$. *It is clear that the likelihood function does not depend on* β, *and hence* β *is not identified. In fact all parameters of* $\mathbf{A}_1$ *(defined by (20.67)) that do not appear as an element of* $\mathbf{A}_0$ *are potentially unidentifiable.*

20.9.1 Calibration and identification

During the 1990s interest in identification waned, partly because of the shift in focus to calibration, where it is assumed that the parameters are known *a priori*, perhaps from microeconometric

evidence. Kydland and Prescott (1996) argue that the task of computational experiments of the sort they conduct is to derive the quantitative implications of the theory rather than to measure economic parameters, one of the primary objects of econometric analysis. Calibration is generally based on estimates from microeconomic studies, or cross-country estimates. As pointed out by Canova and Sala (2009), if the calibrated parameters do not enter in the solution of the model (and therefore do not appear in the likelihood function), then estimates of the remaining parameters will be unaffected.

Over the past ten years it has become more common to estimate, rather than calibrate, dynamic stochastic general equilibrium (*DSGE*) models, often using Bayesian techniques (see, among many others, De Jong, Ingram, and Whiteman (2000), Smets and Wouters (2003), Smets and Wouters (2007) and An and Schorfheide (2007)). In this context, the issue of identification has attracted renewed attention. Questions have been raised about the identification of particular equations of the standard new Keynesian *DSGE* model, such as the Phillips curve (Mavroeidis (2005), Nason and Smith (2008), Kleibergen and Mavroeidis (2009), Dées, di Mauro, Pesaran, and Smith (2009), and others), or the Taylor rule, Cochrane (2011). There have also been questions about the identification of *DSGE* systems as a whole. Canova and Sala (2009, p. 448) conclude: 'it appears that a large class of popular *DSGE* structures are only very weakly identified'. Iskrev (2010a), concludes 'the results indicate that the parameters of the Smets and Wouters (2007) model are quite poorly identified in most of the parameter space'. Other recent papers which consider determining the identification of *DSGE* systems are Iskrev (2010b), Iskrev and Ratto (2010), who provide rank and order conditions for local identification based on the spectral density matrix.

Whereas papers like Iskrev (2010b) and Iskrev (2010a) and Komunjer and Ng (2011) provide classical procedures for determining identification based on the rank of particular matrices, Koop, Pesaran, and Smith (2013) propose Bayesian indicators. A Bayesian approach to identification is useful both because the *DSGE* models are usually estimated by Bayesian methods and since the issues raised by identification are rather different in a Bayesian context. Given an informative choice of the prior, such that a well-behaved marginal prior exists for the parameter of interest, then there is a well-defined posterior distribution, whether or not the parameter is identified. In a Bayesian context lack of identification manifests itself in rendering the Bayesian inference sensitive to the choice of the priors even for sufficiently large sample sizes. If the parameter is not identified, one cannot learn about the parameter directly from the data and, even with an infinite sample of data, the posterior would be determined by the priors.

Within a Bayesian context, learning is interpreted as a changing posterior distribution, and a common practice in *DSGE* estimation is to judge identification by a comparison of the prior and posterior distributions for a parameter. Among many others, Smets and Wouters (2007, p. 594) compare prior and posteriors and note that the mean of the posterior distribution is typically quite close to the mean of the prior distribution and later note that 'It appears that the data are quite informative on the behavioral parameters, as indicated by the lower variance of the posterior distribution relative to the prior distribution.'

As we discuss, not only can the posterior distribution differ from the prior even when the parameter is unidentified, but in addition a changing posterior need not be informative about identification. This can happen because, for instance, the requirement for a determinate solution of a *DSGE* model puts restrictions on the joint parameter space, which may create dependence between identified and unidentified parameters, even if their priors are independent.

What proves to be informative in a Bayesian context is the rate at which learning takes place (posterior precision increases) as more data become available.

Koop, Pesaran, and Smith (2013) suggest two Bayesian indicators of identification. The first, like the classical procedures, indicates non-identification while the second, which is likely to be more useful in practice, indicates either non-identification or weak identification. Like most of the literature the analysis is local in the sense that identification at a given point in the feasible parameter space is investigated. Although these indicators can be applied to any point in the parameter space, in the Bayesian context prior means seem a natural starting point. If the parameters are identified at their prior means then other points could be investigated.

The first indicator, referred to as the 'Bayesian comparison indicator', is based on Proposition 2 of Poirier (1998) and considers identification of the $q_1 \times 1$ vector of parameters, $\boldsymbol{\theta}_1$, assuming that the remaining $q_2 \times 1$ vector of parameters, $\boldsymbol{\theta}_2$, is identified. It compares the posterior distribution of $\boldsymbol{\theta}_1$ with the posterior expectation of its prior distribution conditional on θ_2, and concludes that $\boldsymbol{\theta}_1$ is unidentified if the two distributions coincide. This contrasts with the direct comparison of the prior of $\boldsymbol{\theta}_1$ with its posterior, which could differ even if $\boldsymbol{\theta}_1$ is unidentified. Like the classical indicators based on the rank of a matrix, this Bayesian indicator provides a yes/no answer, though in practice the comparison will depend on the numerical accuracy of the MCMC procedures used to compute the posterior distributions.

The application of the Bayesian comparison indicator to *DSGE* models can be problematic, since it is often difficult to suitably partition the parameters of the model such that there exists a sub-set which is known to be identified. Furthermore, in many applications the main empirical issue of interest is not a yes/no response to an identification question, but whether a parameter of the model is weakly identified, in the sense discussed, for example, by Stock, Wright, and Yogo (2002), and Andrews and Cheng (2012), in the classical literature. Accordingly, Koop, Pesaran, and Smith (2013) also propose a second indicator, which they refer to as the 'Bayesian learning rate indicator', that examines the rate at which the posterior precision of a given parameter gets updated with the sample size, T, using simulated data. For identified parameters the posterior precision increases at the rate T. But for parameters that are either not identified or weakly identified the posterior precision may be updated but its rate of update will be slower than T. Implementation of this procedure requires simulating samples of increasing size and does not require the size of the available realized data to be large. In a recent paper Caglar, Chadha, and Shibayama (2012) apply the learning rate indicator to examine the identification of the parameters of the Bayesian *DSGE* model of Smets and Wouters (2007), and find that many parameters of this widely used model do not appear to be well identified.

We shall return to Bayesian estimation of *RE* models below. But first we consider estimation of *RE* models by *ML* and *GMM* approaches.

20.10 Maximum likelihood estimation of *RE* models

ML estimation of rational expectations models can be carried out by full information or limited information methods. The goal of full information methods is to estimate the entire model by exploiting all its cross-equation restrictions. This estimation method is efficient and produces estimates for all the parameters in the model. These methods are maximum likelihood and its Bayesian counterparts. To apply these methods, the econometrician needs to specify the entire structure of the model, including the distribution of shocks. Consider the structural *RE* model defined by (20.75) which we reproduce here for convenience

$$\mathcal{A}_0(\boldsymbol{\theta})\,\mathbf{y}_t = \mathcal{A}_1(\boldsymbol{\theta})\,\mathbf{y}_{t-1} + \mathcal{A}_2(\boldsymbol{\theta})\,E(\mathbf{y}_{t+1}|\Omega_t) + \mathcal{A}_3(\boldsymbol{\theta})\,\mathbf{v}_t,$$

where as pointed out above the structural errors, $\mathbf{v}_t$, are typically assumed to be uncorrelated, and without loss of generality its variance matrix is set to an identity matrix, $E(\mathbf{v}_t\mathbf{v}_t') = \boldsymbol{\Sigma}_v = \mathbf{I}_m$. Note that under nonsingularity of $\mathcal{A}_0(\boldsymbol{\theta})$ we have

$$\mathbf{y}_t = \mathbf{A}(\boldsymbol{\theta})\,\mathbf{y}_{t-1} + \mathbf{B}(\boldsymbol{\theta})E(\mathbf{y}_{t+1}|\Omega_t) + \mathbf{u}_t$$

$\mathbf{A}(\boldsymbol{\theta}) = \mathcal{A}_0(\boldsymbol{\theta})^{-1}\mathcal{A}_1(\boldsymbol{\theta})$, $\mathbf{B}(\boldsymbol{\theta}) = \mathcal{A}_0(\boldsymbol{\theta})^{-1}\mathcal{A}_2(\boldsymbol{\theta})$, and $\mathbf{u}_t = \mathcal{A}_0(\boldsymbol{\theta})^{-1}\mathcal{A}_3(\boldsymbol{\theta})\,\mathbf{v}_t$. Matrices $\mathbf{A}$ and $\mathbf{B}$ are functions of the structural parameters, $\boldsymbol{\theta}$. We can consider two alternative assumptions on $\mathbf{u}_t$, depending on its serial correlation structure. First we assume that $\mathbf{u}_t \sim IIDN(\mathbf{0}, \boldsymbol{\Sigma}_u)$. In this case, the solution is

$$\mathbf{y}_t = \mathbf{C}(\boldsymbol{\theta})\mathbf{y}_{t-1} + [\mathbf{I}_m - \mathbf{B}(\boldsymbol{\theta})\mathbf{C}(\boldsymbol{\theta})]^{-1}\,\mathbf{u}_t,$$

where $\mathbf{C}(\boldsymbol{\theta})$ satisfies (20.10) and

$$\mathbf{u}_t(\boldsymbol{\theta}) = [\mathbf{I}_m - \mathbf{B}(\boldsymbol{\theta})\mathbf{C}(\boldsymbol{\theta})]\left[\mathbf{y}_t - \mathbf{C}(\boldsymbol{\theta})\mathbf{y}_{t-1}\right].$$

It is easily seen that likelihood function, $L_T(\boldsymbol{\theta})$, for this model is

$$L_T(\boldsymbol{\theta}\,|\mathbf{Y}) = (2\pi)^{\frac{-mT}{2}}\,|\boldsymbol{\Sigma}_u(\boldsymbol{\theta})|^{-T/2}\exp\left[-\frac{1}{2}\sum_{t=1}^{T}\mathbf{u}_t'(\boldsymbol{\theta})\boldsymbol{\Sigma}_u^{-1}(\boldsymbol{\theta})\mathbf{u}_t(\boldsymbol{\theta})\right], \tag{20.77}$$

where $\mathbf{Y} = (\mathbf{y}_0, \mathbf{y}_1, \mathbf{y}_2, \ldots, \mathbf{y}_T)$, $\boldsymbol{\Sigma}_u(\boldsymbol{\theta}) = \mathcal{A}_0(\boldsymbol{\theta})^{-1}\left[\mathcal{A}_3(\boldsymbol{\theta})\,\mathcal{A}_3'(\boldsymbol{\theta})\right]\mathcal{A}_0'(\boldsymbol{\theta})^{-1}$.

Suppose now that $\mathbf{u}_t$ follows a $VAR(1)$ model, namely

$$\mathbf{u}_t = \mathbf{R}\mathbf{u}_{t-1} + \boldsymbol{\varepsilon}_t, \tag{20.78}$$

where $\boldsymbol{\varepsilon}_t \sim IIDN(\mathbf{0}, \boldsymbol{\Sigma}_\varepsilon)$, and we assume all the roots of $\mathbf{R}$ lie inside the unit circle. In this case, the unique solution (20.12) *via* the *QDE* method can be written as

$$\mathbf{y}_t = \mathbf{C}(\boldsymbol{\theta})\mathbf{y}_{t-1} + \mathbf{G}(\boldsymbol{\theta})\mathbf{R}\mathbf{u}_{t-1} + \boldsymbol{\varepsilon}_t, \tag{20.79}$$

where $\mathbf{C}(\boldsymbol{\theta})$ and $\mathbf{G}(\boldsymbol{\theta})$ satisfy equations (20.10) and (20.15). Noting that, under the regularity conditions, $\mathbf{u}_t = (\mathbf{I}_m - \mathbf{R}L)^{-1}\,\boldsymbol{\varepsilon}_t$, and an infinite moving average term arises in (20.79). Its likelihood function can be computed by applying the Kalman filter (see Section 16.5 for further details on state space models and the Kalman filter). Let

$$\mathbf{z}_t = \mathbf{S}\begin{pmatrix}\mathbf{y}_t\\ \mathbf{u}_t\end{pmatrix}, \tag{20.80}$$

where $\mathbf{S}$ is a selection matrix picking the observables from the variables of the model. Then (20.79)–(20.80) represent a state space system. Let $\hat{\mathbf{z}}_{t|t-1}$ be the one-step ahead forecasts of the state vector $\mathbf{z}_t$, (given the information up to time $t-1$). Then the one-step ahead prediction error $\boldsymbol{\nu}_t = \mathbf{z}_t - \hat{\mathbf{z}}_{t|t-1}$ has zero mean and covariance matrix $\mathbf{F}_t$, and the log-likelihood function of the sample is

$$\ell_T(\boldsymbol{\theta}) = \frac{-mT}{2}\ln(2\pi) - \frac{1}{2}\sum_{t=1}^{T}\ln|\mathbf{F}_t| - \frac{1}{2}\sum_{t=1}^{T}\boldsymbol{v}_t'\mathbf{F}_t^{-1}\boldsymbol{v}_t. \tag{20.81}$$

For further details see Pesaran (1987c) and Binder and Pesaran (1995).

Under both of the above specifications, *ML* estimation of the structural parameters, $\boldsymbol{\theta}$, can be computed using suitable numerical algorithms. In practice, the maximization of $\ell_T(\boldsymbol{\theta})$ is complicated when one or more of the structural parameters are not well identified, and can tend to result in log-likelihood profiles that are flat over certain regions of the parameter space. It is therefore important that identification of the parameters is adequately investigated before estimation.

20.11 *GMM* estimation of *RE* models

The parameters of the *RE* model can also be estimated by the *GMM* method discussed in Chapter 10. The starting point of the *GMM* approach is to replace the unobserved expectations with their actual future values minus the expectational errors, in a similar way to the martingale difference solution procedure discussed in Section 20.7.4. For example, to apply the *GMM* procedure to (20.75)

$$\mathcal{A}_0(\boldsymbol{\theta})\,\mathbf{y}_t - \mathcal{A}_1(\boldsymbol{\theta})\,\mathbf{y}_{t-1} - \mathcal{A}_2(\boldsymbol{\theta})\,\mathbf{y}_{t+1} = \mathcal{A}_3(\boldsymbol{\theta})\,\mathbf{v}_t - \mathcal{A}_2(\boldsymbol{\theta})\,\boldsymbol{\xi}_{t+1},$$

where the expectational errors, $\boldsymbol{\xi}_{t+1} = \mathbf{y}_{t+1} - \mathrm{E}(\mathbf{y}_{t+1}|\Omega_t)$ are by assumption uncorrelated with past observations $\mathbf{y}_{t-1}, \mathbf{y}_{t-2}, \ldots$. Also considering that the structural errors are serially uncorrelated, it follows that the composite errors, $\mathcal{A}_3(\boldsymbol{\theta})\,\mathbf{v}_t - \mathcal{A}_2(\boldsymbol{\theta})\,\boldsymbol{\xi}_{t+1}$, are also uncorrelated with past observations. Hence, the *GMM* procedure can be based on the following moment conditions

$$\mathrm{E}\left[y_{i,t-s}\left(\mathcal{A}_0(\boldsymbol{\theta})\,\mathbf{y}_t - \mathcal{A}_1(\boldsymbol{\theta})\,\mathbf{y}_{t-1} - \mathcal{A}_2(\boldsymbol{\theta})\,\mathbf{y}_{t+1}\right)\right] = 0, \text{ for each } i = 1, 2, \ldots, m \text{ and } s = 1, 2, \ldots, p, \tag{20.82}$$

where p denotes the number of lagged variables used in the construction of the moment conditions. The choice of p is determined by the number of unknown parameters, $\boldsymbol{\theta}$, but otherwise is arbitrary. The orthogonality condition (20.82) suggests that a consistent estimate of the parameters can be obtained by applying the IV method using $\mathbf{y}_{t-2}, \mathbf{y}_{t-3}, \ldots, \mathbf{y}_{t-p}$ (as well as current and past values of any exogenous variables present in the model) as instruments. However, the validity of the *GMM* procedure critically depends on whether the parameters $\boldsymbol{\theta}$ are identified. This aspect is clearly illustrated in Example 47.

Example 47 *Consider the following simple model*

$$y_t = \beta_f E(\mathbf{y}_{t+1}|\Omega_t) + \gamma x_t + u_t, \tag{20.83}$$

$$x_t = \rho x_{t-1} + v_t, \quad |\rho| < 1, \tag{20.84}$$

where x_t is a scalar exogenous variable, and v_t has mean zero and variance σ_v^2. Suppose that $\mathbf{z}_t = (x_{t-1}, x_t)'$ is chosen as a vector of instrumental variables. Clearly, the components of this vector satisfy the orthogonality condition (20.82). Further, using (20.84), noting that $E\left(x_t x_{t-j}\right) = \sigma_v^2 \rho^j/(1-\rho^2)$, for $|\rho| < 1$, it follows that the matrix

$$\mathbf{\Sigma}_{zz} = E\left(\mathbf{z}_t \mathbf{z}_t'\right) = \frac{\sigma_v^2}{1-\rho^2}\begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix},$$

is non-singular, as required by the econometric theory on GMM (see Section 10.8). However, a further condition for consistency of the IV estimator is that $\frac{1}{T}\sum_{t=1}^{T} \mathbf{z}_t \mathbf{h}_t'$, where $\mathbf{h}_t = \left(y_{t+1}, x_t\right)'$, converges in probability to a constant matrix of rank 2 (on this, see the discussion on identification conditions provided in Hall (2005, Ch. 2)). Suppose that $\left|\beta_f\right| < 1$, then the unique stationary solution for y_t is given by

$$y_t = \theta x_t + u_t,$$

where $\theta = \gamma/\left(1-\beta_f \rho\right)$. In this case it is easily seen that the matrix

$$\mathbf{\Sigma}_{zh} = E\left(\mathbf{z}_t \mathbf{h}_t'\right) = \frac{\sigma_v^2}{1-\rho^2}\begin{pmatrix} \theta\rho^2 & \rho \\ \theta\rho & 1 \end{pmatrix}$$

has rank 1. Therefore, when $\left|\beta_f\right| < 1$, the matrix $\mathbf{\Sigma}_{zh}$ is not full rank and the application of the IV method fails to yield consistent estimates of β_f and γ. This is not surprising, given that the RE model (20.83)-(20.84) is observationally equivalent to the non-RE model $y_t = \delta x_t + u_t$, and therefore β_f and γ are not identifiable. See also the discussion in Pesaran (1987c, Ch. 7).

The *GMM* method is particularly convenient for estimation of unknown parameters of the nonlinear rational expectations equations obtained as first-order conditions of intertemporal optimization problems encountered by agents and firms. However, such applications can be problematic since in practice it might not be possible to ascertain whether the underlying parameters are identifiable. Example 47 clearly illustrates the danger of indiscriminate application of *GMM* method to *RE* models. This example illustrates that prior to estimation, identification of the parameters should be checked.

A textbook treatment of *GMM* can be found in Hall (2005), where *GMM* inference techniques are also illustrated through some empirical examples from macroeconomics (see, in particular, Ch. 9).

20.12 Bayesian analysis of *RE* models

Given the difficulties associated with the *ML* and *GMM* estimation of *RE* models and recent advances in computing technology, *DSGE* models are increasingly estimated and analysed using Bayesian techniques, where prior information is used to compensate for the weak nature of identification of *DSGE* models or the flat nature of the likelihood function over important part of the parameter space.

In Bayesian analysis the likelihood function, $L_T\left(\boldsymbol{\theta};\mathbf{y}\right)$, given by (20.77), is combined with the prior probability distribution, $P(\boldsymbol{\theta})$, to obtain the posterior distribution

$$P_T(\boldsymbol{\theta}\,|\mathbf{Y}) = P(\boldsymbol{\theta})L_T\left(\boldsymbol{\theta};\mathbf{Y}\right).$$

When the structural parameters, $\boldsymbol{\theta}$, are identified, the posterior probability distribution, $P_T(\boldsymbol{\theta}|\mathbf{Y})$, will become dominated by the likelihood function and the posterior mean will converge in probability to the *ML* estimator of $\boldsymbol{\theta}$. In cases where one or more elements of $\boldsymbol{\theta}$ are not identified, or are only weakly identified, this convergence need not take place or might require T to be very large, which is not the case in most empirical macro economic applications. In such cases the Bayesian inference could be quite sensitive to the choice of the priors and it is important that the robustness of the results to the choice of priors are investigated.

In the context of *DSGE* models, the errors are typically assumed to be Gaussian and the log-likelihood is given by (20.77), or by (20.81) if the errors are serially correlated. For the priors many specifications have been considered in the literature including improper priors, conjugate priors, Minnesota priors (Litterman (1980, 1986) and Doan, Litterman, and Sims (1984)), and priors more recently proposed by Sims and Zha (1998) that are intended to simplify the computations of the posterior distribution in the case of structural *VAR*s. Since the *DSGE* model can be viewed as a restricted *VAR* model, the prior specification must also satisfy the *DSGE* parameter restrictions. To this end Markov chain Monte Carlo (*MCMC*) methods are employed to generate random samples for the purpose of numerical evaluation of the posterior distributions that are otherwise impossible to evaluate analytically or by standard quadrature or Monte Carlo methods. An introduction to the *MCMC* algorithm is provided in Chib (2011). Empirical applications of Bayesian *DSGE* models can be carried out using standard software packages such as Dynare (<http://www.dynare.org/>).

Example 48 *As an example of Bayesian estimation consider the NKPC in Example 45, and note that*

$$\mathbf{y}_t = \mathbf{D}\mathbf{z}_t + \mathbf{v}_t,$$

where $\mathbf{y}_t = (\pi_t, x_t)'$, $\mathbf{v}_t = (v_{\pi t}, v_{xt})'$, $\mathbf{z}_t = (x_t, x_{t-1}, x_{t-2})'$ *and*

$$\mathbf{D} = \begin{pmatrix} \alpha_1 & \alpha_2 & 0 \\ 0 & \rho_1 & \rho_2 \end{pmatrix}.$$

The reduced form parameters in $\mathbf{D}$, *namely* $\boldsymbol{\alpha} = (\alpha_1, \alpha_2)'$ *and* $\boldsymbol{\rho} = (\rho_1, \rho_2)'$ *are related to the structural parameters,* $\boldsymbol{\psi} = (\beta_f, \gamma)'$ *through* α_1 *and* α_2 *which we reproduce below for convenience*

$$\alpha_1(\boldsymbol{\psi}, \boldsymbol{\rho}) = \frac{\gamma\left(\rho_1 + \beta_f\rho_2\right)}{1 - \beta_f\rho_1 - \beta_f^2\rho_2}, \;\; \alpha_2(\boldsymbol{\psi}, \boldsymbol{\rho}) = \frac{\gamma\rho_2}{1 - \beta_f\rho_1 - \beta_f^2\rho_2}. \tag{20.85}$$

Under the assumption that $\mathbf{v}_t \sim IIDN(\mathbf{0}, \boldsymbol{\Sigma}_v)$, *and given the observations,* $\mathbf{Y} = (\mathbf{y}_1, \mathbf{y}_2, \ldots, \mathbf{y}_T)$, *the log-likelihood function of the model is given by*

$$\ell_T(\boldsymbol{\theta}) = -\left(\frac{2T}{2}\right)\ln(2\pi) - \frac{T}{2}\ln|\boldsymbol{\Sigma}_v| - \frac{1}{2}\sum_{t=1}^{T}[\mathbf{y}_t - \mathbf{D}(\boldsymbol{\psi}, \boldsymbol{\phi})\mathbf{z}_t]'\,\boldsymbol{\Sigma}_v^{-1}\,[\mathbf{y}_t - \mathbf{D}(\boldsymbol{\psi}, \boldsymbol{\phi})\mathbf{z}_t],$$

where $\boldsymbol{\theta} = \left(\boldsymbol{\psi}', \boldsymbol{\rho}', vech(\boldsymbol{\Sigma}_v)'\right)'$. *For the posterior distribution of* $\boldsymbol{\theta}$ *we have*

$$P_T(\boldsymbol{\theta}\,|\mathbf{Y}) \propto P(\boldsymbol{\theta})\exp\{\ell_T(\boldsymbol{\theta})\},$$

where $P(\boldsymbol{\theta})$ *denotes the prior distribution of* $\boldsymbol{\theta}$. *To compute the posterior distribution of a parameter of interest, such as* γ *or* β_f, *numerical methods are typically used to integrate out the effects of the other parameters. In small samples, the outcome can critically depend on the choice of* $P(\boldsymbol{\theta})$. *The rank condition for identification for this example is Rank* $\left(\frac{\partial \boldsymbol{\alpha}}{\partial \boldsymbol{\psi}'}\right) = 2$, *where*

$$\frac{\partial \boldsymbol{\alpha}}{\partial \boldsymbol{\psi}'} = \frac{1}{1-\beta_f\rho_1-\beta_f^2\rho_2}\begin{bmatrix} \gamma\left(\rho_1 + \frac{\rho_1+2\beta_f\rho_2}{1-\beta_f\rho_1-\beta_f^2\rho_2}\right) & \rho_1+\beta_f\rho_2 \\ \gamma\frac{\rho_2\left(\rho_1+2\beta_f\rho_2\right)}{1-\beta_f\rho_1-\beta_f^2\rho_2} & \rho_2 \end{bmatrix}.$$

In the case where ρ_2 *is close to zero the parameters* γ *and* β *could be weakly identified and their posterior distribution might depend on the chosen prior distribution,* $P(\boldsymbol{\theta})$, *even for sufficiently large samples.*

Various econometric and computational issues involved in the application of Bayesian techniques to *DSGE* models are surveyed in Karagedikli et al. (2010) and Del Negro and Schorfheide (2011). An overview of Bayesian analysis is provided in Appendix C.

20.13 Concluding remarks

In this chapter we have provided a review of how multivariate *RE* models are solved. The issues of identification and estimation of these models are also discussed briefly, distinguishing between *ML*, *GMM*, and Bayesian procedures. But it is important to recognize that, although less demanding than the perfect foresight assumption, the *REH* still requires economic agents to know or fully learn the 'true' conditional probability distributions of the state variables they encounter in their decision-making process. This could be a demanding requirement, particularly in applications where the source of uncertainty is behavioural or arises from strategic considerations, namely where agents need to guess about the motives and actions of others, first articulated in economics by Keynes in his famous beauty contest example (see, also Pesaran (1987c)). In such cases, other forms of expectations formation models, such as adaptive schemes, might be more appropriate.

20.14 Further reading

Solution techniques for multivariate rational expectations models are reviewed and discussed in Binder and Pesaran (1995, 1997). Del Negro and Schorfheide (2011) provide a review of the Bayesian analysis of *DSGE* models.

20.15 Exercises

1. Consider the new Keynesian Phillips curve

$$\pi_t = \alpha E(\pi_{t+1} | \Omega_t) + \beta x_t + u_t, \tag{20.86}$$

$$x_t = \rho x_{t-1} + v_t, \tag{20.87}$$

where π_t is the rate of inflation, x_t is a measure of output gap, $E(\pi_{t+1} | \Omega_t)$ is the expectations of π_{t+1} conditional on the information set Ω_t which is assumed to contain at least current and past values of inflation and the output gap.

$$\begin{pmatrix} u_t \\ v_t \end{pmatrix} \sim IID\left[\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \sigma_u^2 & 0 \\ 0 & \sigma_v^2 \end{pmatrix}\right].$$

   (a) Show that (20.86) has a unique solution if $|\alpha| < 1$.

   (b) Assuming $|\alpha| < 1$ and $|\rho| \leq 1$ derive the unique solution of π_t and show that α and β are not identified. Discuss a simple extension of (20.87) that ensures identification of α and β.

   (c) Suppose now that there are feedbacks from lagged inflation to the output gap of the following simple form

$$x_t = \theta \pi_{t-1} + v_t.$$

   Derive the conditions under which the solution for π_t is unique in this case.

2. Consider the following asset pricing model

$$p_t = \left(\frac{1}{1+r}\right) E\left(p_{t+1} + d_{t+1} | \Omega_t\right) + u_t,$$

where r is the real rate of discount, p_t is the real share price, d_t is real dividends paid per share, u_t is a serially uncorrelated process, and Ω_t is the information available at time t. Suppose that d_t follows a random walk with a non-zero drift, μ

$$d_t = \mu + d_{t-1} + \varepsilon_t, \varepsilon_t \sim IID(0, \sigma^2).$$

   (a) Derive the solution of p_t and discuss the conditions under which the solution is unique.

   (b) Suppose that there are feedbacks from $p_{t-1} - p_{t-2}$ into d_t so that

$$d_t - d_{t-1} = \mu + \alpha(p_{t-1} - p_{t-2}) + \varepsilon_t.$$

Compare your solution with the case without feedbacks and discuss the conditions under which the solution with feedbacks is unique.

3. Consider a firm that faces the following instantaneous cost function

$$c(\mathbf{y}_t) = (\mathbf{y}_t - \mathbf{y}_t^*)' \mathbf{H}(\mathbf{y}_t - \mathbf{y}_t^*) + \Delta\mathbf{y}_t' \mathbf{G}\Delta\mathbf{y}_t, \tag{20.88}$$

where $\mathbf{y}_t$ is an $m \times 1$ vector of target variables (such as employment, sales and stock levels), $\mathbf{y}_t^*$ is the desired level of $\mathbf{y}_t$, $\mathbf{H}$ and $\mathbf{G}$ are $m \times m$ matrices with known coefficients, β $(0 \leq \beta < 1)$ is a known discount factor. Suppose further that

$$\begin{aligned} \mathbf{y}_t^* &= \mathbf{\Psi}\mathbf{x}_t + \mathbf{u}_t, \\ \mathbf{x}_t &= \mathbf{\Phi}\mathbf{x}_{t-1} + \boldsymbol{\varepsilon}_t, \end{aligned}$$

where $\mathbf{\Psi}$ is an $m \times k$ matrix of known constants, $\mathbf{x}_t$ is a $k \times 1$ vector of observed forcing variables assumed to follow a first-order vector autoregression. Finally, $\mathbf{u}_t$ and $\boldsymbol{\varepsilon}_t$ are unobserved serially uncorrelated error processes.

(a) Derive the first-order (Euler) conditions for the following minimization problem

$$Min_{\mathbf{y}_t, \mathbf{y}_{t+1}, \ldots} E\left[\sum_{j=0}^{\infty} \beta^j c(\mathbf{y}_{t+j}) \,|\mathcal{I}_t\right],$$

where $\mathcal{I}_t$ is the information set available to the firm at time t and contains at least current and past values of $\mathbf{y}_t$ and $\mathbf{x}_t$. Show that these conditions can be written as the following canonical form rational expectations model

$$\mathbf{y}_t = \mathbf{A}\mathbf{y}_{t-1} + \mathbf{B}E\left(\mathbf{y}_{t+1} \,|\mathcal{I}_t\right) + \mathbf{w}_t,$$

where

$$\begin{aligned} \mathbf{A} &= (\mathbf{H} + (1+\beta)\mathbf{G})^{-1}\mathbf{G}, \ \mathbf{B} = \beta\mathbf{A}, \\ \mathbf{w}_t &= (\mathbf{H} + (1+\beta)\mathbf{G})^{-1}\mathbf{H}\mathbf{y}_t^*. \end{aligned}$$

Establish the conditions under which this optimization problem has a unique solution.

(b) Write down the solution in the case where $\mathbf{I}_k - \mathbf{\Phi}$ is rank deficient.

(c) Discuss the conditions under which $\mathbf{A}$ and β can be identified from time series observations on $\mathbf{x}_t$ and $\mathbf{y}_t$.

4. Consider the following rational expectations model

$$y_t = \alpha E\left(y_{t+1} \,|\mathcal{I}_t\right) + \gamma x_t + u_t,$$

where $\mathcal{I}_t$ is the information set at time t, $|\alpha| < 1$, $u_t \sim IID(0, \sigma_u^2)$, and x_t follows the $AR(1)$ process

$$x_t = \rho x_{t-1} + v_t.$$

(a) Show that

$$\begin{aligned} y_t &= \alpha y_{t+1} + \gamma x_t + \xi_t \\ &= \boldsymbol{\beta}' \mathbf{w}_t + \xi_t, \end{aligned}$$

where $\xi_t = u_t - \alpha v_{t+1}$, v_{t+1} is the expectations error of y_{t+1}, $\boldsymbol{\beta} = (\alpha, \gamma)'$ and $\mathbf{w}_t = (y_{t+1}, x_t)'$. Derive $Cov(y_{t+1}, \xi_t)$ and $Cov(x_t, \xi_t)$ and hence or otherwise establish that the least squares estimate of $\boldsymbol{\beta}$ will be inconsistent.

(b) Suppose now that $\boldsymbol{\beta}$ is estimated by IV using $\mathbf{z}_t = (x_{t-1}, x_t)'$ as instrument. Show that although $Cov(x_{t-1}, \xi_t) = 0 = Cov(x_t, \xi_t)$, the IV estimator of $\boldsymbol{\beta}$ is also inconsistent by showing that $E(\mathbf{z}_t \mathbf{w}_t')$ is a rank-deficient matrix.

(c) Can you use other instruments to achieve consistency? Discuss.

21 Vector Autoregressive Models

21.1 Introduction

This chapter provides an overview of vector autoregressive (VAR) models, with a particular focus on estimation and hypothesis testing.

21.2 Vector autoregressive models

The simplest multivariate system is the vector autoregressive model of order 1, also known as $VAR(1)$. The $m \times 1$ vector of random variables, $\mathbf{y}_t$, is said to follow a $VAR(1)$ process with the $m \times m$ coefficients matrix $\mathbf{\Phi}$, if

$$\mathbf{y}_t = \mathbf{\Phi}\mathbf{y}_{t-1} + \mathbf{u}_t, \quad t = 1, 2, \ldots,$$

where $\mathbf{u}_t$ is an $m \times 1$ vector of disturbances satisfying the following assumptions

Assumption B1: $E(\mathbf{u}_t) = \mathbf{0}$, for all t.
Assumption B2: $E(\mathbf{u}_t\mathbf{u}_t') = \mathbf{\Sigma}$, where $\mathbf{\Sigma}$ is an $m \times m$ positive definite matrix.
Assumption B3: $E(\mathbf{u}_t\mathbf{u}_{t'}') = \mathbf{0}$, for all $t \neq t'$.

A straightforward generalization of the $VAR(1)$ model is given by the $VAR(p)$ specification defined as

$$\mathbf{y}_t = \mathbf{\Phi}_1\mathbf{y}_{t-1} + \mathbf{\Phi}_2\mathbf{y}_{t-2} + \ldots + \mathbf{\Phi}_p\mathbf{y}_{t-p} + \mathbf{u}_t, \tag{21.1}$$

where $\mathbf{\Phi}_i$ are $m \times m$ matrices. The above model can be extended to include deterministic components such as intercepts, linear trends or seasonal dummy variables, as well as weakly exogenous $I(1)$ variables. The inclusion of deterministic components in the model will be discussed in Section 21.4, while the extension to weakly exogenous variables will be reviewed in Chapter 23.

21.2.1 Companion form of the *VAR(p)* model

We can rewrite the $VAR(p)$ representation of (21.1) in the companion form

$$\begin{pmatrix} \mathbf{y}_t \\ \mathbf{y}_{t-1} \\ \vdots \\ \mathbf{y}_{t-p+2} \\ \mathbf{y}_{t-p+1} \end{pmatrix} = \begin{pmatrix} \mathbf{\Phi}_1 & \mathbf{\Phi}_2 & \dots & \mathbf{\Phi}_{p-1} & \mathbf{\Phi}_p \\ \mathbf{I}_m & \mathbf{0} & \dots & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{I}_m & \dots & \mathbf{0} & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ \mathbf{0} & \mathbf{0} & \dots & \mathbf{I}_m & \mathbf{0} \end{pmatrix} \begin{pmatrix} \mathbf{y}_{t-1} \\ \mathbf{y}_{t-2} \\ \vdots \\ \mathbf{y}_{t-p+1} \\ \mathbf{y}_{t-p} \end{pmatrix} + \begin{pmatrix} \mathbf{u}_t \\ \mathbf{0} \\ \vdots \\ \mathbf{0} \\ \mathbf{0} \end{pmatrix},$$

which is a $VAR(1)$ model, but in the $mp \times 1$ vector of random variables $\mathbf{Y}_t = (\mathbf{y}_t', \mathbf{y}_{t-1}', \dots, \mathbf{y}_{t-p+1}')'$, namely

$$\mathbf{Y}_t = \mathbf{\Phi}\mathbf{Y}_{t-1} + \mathbf{U}_t, \tag{21.2}$$

where $\mathbf{\Phi}$ is now the $mp \times mp$ companion coefficient matrix,

$$\mathbf{\Phi} = \begin{pmatrix} \mathbf{\Phi}_1 & \mathbf{\Phi}_2 & \dots & \mathbf{\Phi}_{p-1} & \mathbf{\Phi}_p \\ \mathbf{I}_m & \mathbf{0} & \dots & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{I}_m & \dots & \mathbf{0} & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ \mathbf{0} & \mathbf{0} & \dots & \mathbf{I}_m & \mathbf{0} \end{pmatrix}, \tag{21.3}$$

and $\mathbf{U}_t = \left(\mathbf{u}_t', \mathbf{0}, \dots, \mathbf{0}\right)'$ is the $mp \times 1$ vector of error terms.

21.2.2 Stationary conditions for *VAR(p)*

The vector difference equation (21.2) can be solved iteratively from $t = -M+p$, taking $\mathbf{Y}_{-M+p}$ (or equivalently $\mathbf{y}_{-M+1}, \mathbf{y}_{-M+2}, \dots, \mathbf{y}_{-M+p}$) as given to obtain

$$\mathbf{Y}_t = \mathbf{\Phi}^{t+M-p}\mathbf{Y}_{-M+p} + \sum_{j=0}^{t+M-p-1} \mathbf{\Phi}^j \mathbf{U}_{t-j}. \tag{21.4}$$

The solution of $\mathbf{y}_t$ is then obtained as

$$\mathbf{y}_t = \begin{pmatrix} \mathbf{I}_m & \mathbf{0} & \mathbf{0} & \dots & \mathbf{0} \end{pmatrix}_{m \times mp} \mathbf{Y}_t. \tag{21.5}$$

Clearly if $\mathbf{Y}_t$ is stationary, then $\mathbf{y}_t$ will also be stationary.

The solution (21.4) is covariance stationary if all the eigenvalues of $\mathbf{\Phi}$, defined by λ that satisfy the equation $\left|\mathbf{\Phi} - \lambda \mathbf{I}_{mp}\right| = 0$, lie inside the unit circle (namely if $|\lambda| < 1$) and $M \to \infty$. In such a case $\mathbf{Y}_t$ converges in mean squared errors to $\sum_{j=0}^{\infty} \mathbf{\Phi}^j \mathbf{U}_{t-j}$, which can be obtained from (21.4) by letting $M \to \infty$, namely by assuming that the process has been in operation for a sufficiently long time before its realization, $\mathbf{Y}_t$, being observed at time t. Also, when all eigenvalues

of $\mathbf{\Phi}$ lie inside the unit circle, the norm of $\mathbf{\Phi}$ defined by $\left[Tr\left(\mathbf{\Phi}^j\mathbf{\Phi}'^j\right)\right]^{1/2} \to 0$ exponentially in j and the process $\mathbf{Y}_t = \sum_{j=0}^{\infty}\mathbf{\Phi}^j\mathbf{U}_{t-j}$ will also be absolute summable, in the sense that the sum of absolute values of the elements of $\mathbf{\Phi}^j$, for $j = 0, 1, 2, \ldots$ converge. It is then easily seen that $\mathbf{Y}_t$ (or $\mathbf{y}_t$) will have finite moments of order ℓ, assuming $\mathbf{u}_t$ has finite moments of the same order. In particular, by Assumption B1 to B3, since $\mathbf{u}_t \sim IID(\mathbf{0}, \mathbf{\Sigma})$, then

$$E(\mathbf{Y}_t) = \mathbf{0}, \text{ and } Var(\mathbf{Y}_t) = \sum_{j=0}^{\infty}\mathbf{\Phi}^j\mathbf{\Sigma}\mathbf{\Phi}'^j < \infty.$$

Finally, noting that

$$\left|\mathbf{I}_{mp} - \lambda\mathbf{\Phi}\right| = \left|\lambda^p\mathbf{I}_m - \mathbf{\Phi}_1\lambda^{p-1} - \mathbf{\Phi}_2\lambda^{p-2} - \ldots\mathbf{\Phi}_p\right|,$$

the stability condition of the $\mathbf{Y}_t$ (or $\mathbf{y}_t$) can be written equivalently in terms of the roots of the determinantal equation

$$\left|\mathbf{I}_m - \mathbf{\Phi}_1 z - \mathbf{\Phi}_2 z^2 - \ldots\mathbf{\Phi}_p z^p\right| = 0. \tag{21.6}$$

In this formulation the $\mathbf{Y}_t$ process will be stable and covariance stationary if all the roots of (21.6) lie outside the unit circle ($|z| > 1$).

21.2.3 Unit root case

If one or more of the roots of (21.6) lie on the unit root circle, then

$$\left|\mathbf{I}_{mp} - \mathbf{\Phi}\right| = \left|\mathbf{I}_m - \mathbf{\Phi}_1 - \mathbf{\Phi}_2 - \ldots\mathbf{\Phi}_p\right| = 0, \tag{21.7}$$

and $\mathbf{Y}_t$ (or $\mathbf{y}_t$) is said to be a unit root process. It is, however, important to note that the above condition does not rule out the possibility of $\mathbf{y}_t$ being integrated of order 2 or more.[1] To ensure that $\mathbf{y}_t \sim I(1)$, namely that $\Delta\mathbf{y}_t \sim I(0)$, further restrictions are required. See, for example, Johansen (1991, p. 1559, Theorem 4.1) and Johansen (1995, p.49, Theorem 4.2).

Define the long-run multiplier matrix as $\mathbf{\Pi} = \mathbf{I}_m - \mathbf{\Phi}_1 - \mathbf{\Phi}_2 - \ldots\mathbf{\Phi}_p$, when (21.6) holds for all $|z| > 1$ we have $\left|\mathbf{I}_m - \mathbf{\Phi}_1 - \mathbf{\Phi}_2 - \ldots\mathbf{\Phi}_p\right| \neq 0$, and $\mathbf{\Pi}$ will be full rank, namely $Rank(\mathbf{\Pi}) = m$. On the other hand, if (21.7) is true, we must have $Rank(\mathbf{\Pi}) = r < m$, with $m - r$ being the number of unit roots in the system. See also Chapter 22.

21.3 Estimation

We now focus on estimation of the parameters of (21.1). To this end, it is convenient to rewrite (21.1) as follows

$$\begin{aligned}\mathbf{y}_t &= \mathbf{\Phi}_1\mathbf{y}_{t-1} + \mathbf{\Phi}_2\mathbf{y}_{t-2} + \ldots + \mathbf{\Phi}_p\mathbf{y}_{t-p} + \mathbf{u}_t \\ &= \mathbf{A}'\mathbf{g}_t + \mathbf{u}_t,\end{aligned}$$

[1] Recall that a process is said to be integrated of order q if it must be differenced q times before stationarity is achieved.

where

$$\mathbf{A}' = \begin{pmatrix} \mathbf{\Phi}_1 & \mathbf{\Phi}_2 & \cdots & \mathbf{\Phi}_p \end{pmatrix}, \ \mathbf{g}_t' = \begin{pmatrix} \mathbf{y}_{t-1}' & \mathbf{y}_{t-2}' & \cdots & \mathbf{y}_{t-p}' \end{pmatrix}. \tag{21.8}$$

We make the following additional assumptions:

Assumption B5: The augmented $VAR(p)$ model, (21.1), is stable. That is, all the roots of the determinantal equation

$$\left|\mathbf{I}_m - \mathbf{\Phi}_1\lambda - \mathbf{\Phi}_2\lambda^2 - \cdots - \mathbf{\Phi}_p\lambda^p\right| = 0, \tag{21.9}$$

fall outside the unit circle.

Assumption B6: The $m \times 1$ vector of disturbances have a multivariate normal distribution.

Assumption B7: The observations $\mathbf{g}_t$, for $t = 1, 2, \ldots, T$, are not perfectly collinear.

Assumption B8: The initial values $\mathbf{Y}_0 = \left(\mathbf{y}_0', \mathbf{y}_{-1}', \ldots, \mathbf{y}_{-p+1}'\right)'$ are given.

Since the system of equations (21.1) is in the form of a *SURE* model with all the equations having the same set of regressors, $\mathbf{g}_t$, in common, it then follows that when $\mathbf{u}_t$s are Gaussian the *ML* estimators of the unknown coefficients can be computed by *OLS* regressions of $\mathbf{y}_t$ on $\mathbf{g}_t$ (see Section 19.2.1). Writing (21.1) in matrix notation we have

$$\mathbf{y}_t = \mathbf{A}'\mathbf{g}_t + \mathbf{u}_t, \tag{21.10}$$

where $\mathbf{A}$ and $\mathbf{g}_t$ are defined in (21.8). Hence, the *ML* estimators of $\mathbf{A}$ and $\mathbf{\Sigma}$ are given by

$$\hat{\mathbf{A}}' = \left(\sum_{t=1}^{T} \mathbf{y}_t\mathbf{g}_t'\right)\left(\sum_{t=1}^{T} \mathbf{g}_t\mathbf{g}_t'\right)^{-1}, \tag{21.11}$$

and

$$\widetilde{\mathbf{\Sigma}} = T^{-1}\sum_{t=1}^{T}(\mathbf{y}_t - \hat{\mathbf{A}}'\mathbf{g}_t)(\mathbf{y}_t - \hat{\mathbf{A}}'\mathbf{g}_t)'. \tag{21.12}$$

The maximized value of the system's log-likelihood function is then given by

$$\ell\left(\hat{\mathbf{A}}, \widetilde{\mathbf{\Sigma}}\right) = -\frac{Tm}{2}\left(1 + \log 2\pi\right) - \frac{T}{2}\log\left|\widetilde{\mathbf{\Sigma}}\right|. \tag{21.13}$$

21.4 Deterministic components

Model (21.1) can be generalized by allowing $\mathbf{y}_t$ to depend on deterministic components, such as intercepts, linear trends or seasonal dummy variables. A convenient approach to deal with deterministic components is to introduce a new variable $\mathbf{x}_t$ defined as

$$\mathbf{x}_t = \mathbf{y}_t - \mathbf{d}_t,$$

where $\mathbf{d}_t$ is an $m \times 1$ vector of deterministic (or perfectly predictable) components of $\mathbf{y}_t$, that we assume satisfy

Assumption B4: $E(\mathbf{u}_t | \mathbf{d}_s) = 0$, for all t and s.

For the $VAR(1)$ model in $\mathbf{x}_t$ we now have

$$\mathbf{y}_t - \mathbf{d}_t = \mathbf{\Phi}\left(\mathbf{y}_{t-1} - \mathbf{d}_{t-1}\right) + \mathbf{u}_t,\ t = -M, -M+1, \ldots, -1, 0, 1, 2, \ldots,$$

or, equivalently,

$$\mathbf{y}_t = \mathbf{\Phi}\mathbf{y}_{t-1} + \mathbf{d}_t - \mathbf{\Phi}\mathbf{d}_{t-1} + \mathbf{u}_t. \tag{21.14}$$

Obvious examples of $\mathbf{d}_t$ are $\mathbf{d}_t = \mathbf{a}$, a vector of fixed constants, linear time trends, $\mathbf{d}_t = \mathbf{a}_0 + \mathbf{a}_1 t$, or seasonal dummies. The above formulation is preferable to the alternative specification with the deterministic components, $\mathbf{d}_t$, added as in

$$\mathbf{y}_t = \mathbf{\Phi}\mathbf{y}_{t-1} + \mathbf{d}_t + \mathbf{u}_t,\ t = -M, -M+1, \ldots, -1, 0, 1, 2, \ldots. \tag{21.15}$$

This is because the trend properties of $\mathbf{y}_t$ are determined by $\mathbf{d}_t$ under (21.14), while if specification (21.15) is adopted the trend property of $\mathbf{y}_t$ would critically depend on the number of eigenvalues of $\mathbf{\Phi}$ (if any) that fall on the unit circle. For example, setting $\mathbf{d}_t = \mathbf{a}_0 + \mathbf{a}_1 t$, and assuming $\mathbf{\Phi} = \mathbf{I}_m$, under (21.14) we have

$$\mathbf{y}_t = \mathbf{y}_0 + \mathbf{a}_1 t + \mathbf{s}_t,$$

where $\mathbf{s}_t = \sum_{j=1}^{t} \mathbf{u}_j$. Under (21.15) we have

$$\mathbf{y}_t = \mathbf{y}_0 + \mathbf{a}_0 t + \frac{t(t+1)}{2}\mathbf{a}_1 + \mathbf{s}_t.$$

Only when all eigenvalues of $\mathbf{\Phi}$ fall within the unit circle are the two specifications stochastically equivalent. In this case the two specifications, (21.14) and (21.15), can be solved as

$$\mathbf{y}_t = \mathbf{d}_t + (\mathbf{I}_m - \mathbf{\Phi}L)^{-1}\mathbf{u}_t,$$

and

$$\mathbf{y}_t = (\mathbf{I}_m - \mathbf{\Phi}L)^{-1}\mathbf{d}_t + (\mathbf{I}_m - \mathbf{\Phi}L)^{-1}\mathbf{u}_t,$$

respectively. Since $\mathbf{d}_t$ is a deterministic process it then follows that both processes have the same covariance matrix. The two processes have the same means when $\mathbf{d}_t = \mathbf{a}_0 + \mathbf{a}_1 t$. To see this note that

$$
\begin{aligned}
(\mathbf{I}_m - \mathbf{\Phi}L)^{-1}(\mathbf{a}_0 + \mathbf{a}_1 t) &= \left(\mathbf{I}_m + \mathbf{\Phi}L + \mathbf{\Phi}^2L^2 + \ldots.\right)(\mathbf{a}_0 + \mathbf{a}_1 t) \\
&= \left(\mathbf{I}_m + \mathbf{\Phi}L + \mathbf{\Phi}^2L^2 + \ldots.\right)\mathbf{a}_0 + \left(\mathbf{I}_m + \mathbf{\Phi}L + \mathbf{\Phi}^2L^2 + \ldots.\right)\mathbf{a}_1 t \\
&= \mathbf{b}_0 + \mathbf{b}_1 t,
\end{aligned}
$$

where

$$
\begin{aligned}
\mathbf{b}_0 &= (\mathbf{I}_m - \mathbf{\Phi})^{-1}\mathbf{a}_0 - \left(\mathbf{\Phi} + 2\mathbf{\Phi}^2 + 3\mathbf{\Phi}^3 + \ldots\right)\mathbf{a}_1, \\
\mathbf{b}_1 &= (\mathbf{I}_m - \mathbf{\Phi})^{-1}\mathbf{a}_1.
\end{aligned}
$$

Hence, in the stationary case both specifications will have the same linear trends.

21.5 *VAR* order selection

The order of the *VAR* model (21.1), p, can be selected either with the help of model selection criteria such as the Akaike information criterion (*AIC*) and the Schwarz Bayesian criterion (*SBC*), or by means of a sequence of log-likelihood ratio tests. The values of the *AIC* and *SBC* for model (21.1) are given by

$$
AIC_p = -\frac{Tm}{2}\left(1 + \log 2\pi\right) - \frac{T}{2}\log\left|\widetilde{\mathbf{\Sigma}}_p\right| - ms, \tag{21.16}
$$

and

$$
SBC_p = -\frac{Tm}{2}\left(1 + \log 2\pi\right) - \frac{T}{2}\log\left|\widetilde{\mathbf{\Sigma}}_p\right| - \frac{ms}{2}\log(T), \tag{21.17}
$$

where $s = mp$, and $\widetilde{\mathbf{\Sigma}}_p$ is defined by (21.12). The AIC_p and SBC_p can be computed for $p = 0, 1, 2, \ldots, P$, where P is the maximum order for the *VAR* model chosen by the user.

The log-likelihood ratio statistic for testing the hypothesis that the order of the *VAR* is p against the alternative that it is P (with $P > p$) are given by

$$
LR_{P,p} = T\left(\log\left|\widetilde{\mathbf{\Sigma}}_p\right| - \log\left|\widetilde{\mathbf{\Sigma}}_P\right|\right). \tag{21.18}
$$

For $p = 0, 1, 2, \ldots, P-1$, where P is the maximum order for the *VAR* model selected by the user, $\widetilde{\mathbf{\Sigma}}_p$ is defined by (21.12), and $\widetilde{\mathbf{\Sigma}}_0$ refers to the *ML* estimator of the system covariance matrix of $\mathbf{y}_t$.

Under the null hypothesis, the *LR* statistic in (21.18) is asymptotically distributed as a chi-squared variate with $m^2(P-p)$ degrees of freedom.

In small samples the use of the *LR* statistic, (21.18), tends to result in over-rejection of the null hypothesis. In an attempt to take some account of this small sample problem, in practice the following degrees of freedom adjusted *LR* statistics can be computed

$$
LR^*_{P,p} = (T - mP)\left(\log\left|\widetilde{\mathbf{\Sigma}}_p\right| - \log\left|\widetilde{\mathbf{\Sigma}}_P\right|\right), \tag{21.19}
$$

for $p = 0, 1, 2, \ldots, P - 1$. These adjusted *LR* statistics have the same asymptotic distribution as the unadjusted statistics given by (21.18).

Example 49 *We now consider the problem of selecting the order of a trivariate VAR model in the output growths of USA, Japan and Germany, estimated over the period 1964(3)-1992(4). Table 21.1 reports the log-likelihood, Akaike and the Schwarz criteria, the LR and adjusted LR statistics for the seven VAR models VAR(p), for $p = 0, 1, 2, \ldots, 6$. As expected, the maximized values of the log-likelihood function given under the column headed LL increase with p. However, the Akaike and the Schwarz criteria select the orders 1 and 0, respectively. The log-likelihood ratio statistics reject order 0, but do not reject a VAR of order 1. In the light of these results we choose the VAR(1) model. Note that it is quite usual for the SBC to select a lower order VAR as compared with the AIC. Having chosen the order of the VAR it is prudent to examine the residuals of individual equations for serial correlation. Tables 21.2 to 21.4 show the regression results for the US, Japan, and Germany, respectively. There is no evidence of residual serial correlation in the case of the US and Germany's output equations, but there is statistically significant evidence of residual serial correlation in the case of Japan's output equation. There is also important evidence of departures from normality in the case of output equations for the US and Japan. A closer examination of the residuals of these equations suggests considerable volatility during the early 1970s as a result of the abandonment of the Bretton Wood system and the quadrupling increase in oil prices. It is therefore likely that the remaining serial correlation in the residuals of Japan's output equation may be due to these unusual events. Such a possibility can be handled by introducing a dummy variable for the oil shock in the VAR model.*

Table 21.1 Selecting the order of a trivariate *VAR* model in output growths

Test statistics and choice criteria for selecting the order of the *VAR* model

Based on 114 observations from 1964Q3 to 1992Q4. Order of *VAR* = 6
List of variables included in the unrestricted *VAR*:
DLYUSA DLYJAP DLYGER
List of deterministic and/or exogenous variables:
CONST

order	LL	AIC	SBC		LR test	Adjusted LR test
6	1128.4	1071.4	993.3935		——	——
5	1125.2	1077.2	1011.5	CHSQ(9)=	6.4277 [.696]	5.3564 [.802]
4	1120.4	1081.4	1028.1	CHSQ(18)=	15.8583 [.602]	13.2152 [.779]
3	1112.8	1082.8	1041.8	CHSQ(27)=	31.1322 [.266]	25.9435 [.522]
2	1108.1	1087.1	1058.4	CHSQ(36)=	40.5268 [.277]	33.7724 [.575]
1	1101.5	1089.5	1073.1	CHSQ(45)=	53.7352 [.175]	44.7793 [.481]
0	1084.2	1081.2	1077.1	CHSQ(54)=	88.2988 [.002]	73.5824 [.039]

AIC = Akaike information criterion SBC = Schwarz Bayesian criterion

21.6 Granger causality

The concept of Granger causality is based on the idea that the cause occurs before the effect, hence if an event X is the cause of another event Y, then X should proceed Y (Granger (1969)).

Table 21.2 US output growth equation

OLS* estimation of a single equation in the unrestricted *VAR			
Dependent variable is DLYUSA **114 observations used for estimation from 1964Q3 to 1992Q4**			
Regressor	Coefficient	Standard Error	T-Ratio [Prob]
DLYUSA(−1)	.28865	.095475	3.0233 [.003]
DLYJAP(−1)	.029389	.080403	.36552 [.715]
DLYGER(−1)	.072501	.085957	.84346 [.401]
CONST	.0039865	.0014390	2.7704 [.007]
R-Squared	.10155	R-Bar-Squared	.077044
S.E. of Regression	.0092075	F-Stat. F(3,110)	4.1442 [.008]
Mean of Dependent Variable	.0068049	S.D. of Dependent Variable	.0095841
Residual Sum of Squares	.0093256	Equation Log-likelihood	374.6786
Akaike Info. Criterion	370.6786	Schwarz Bayesian Criterion	356.2062
DW-statistic	2.0058	System Log-likelihood	1101.5

Diagnostic Tests

Test Statistics	**LM Version**	**F Version**
A:Serial Correlation	CHSQ(4) = 2.5571 [.634]	F(4,106) = .60806 [.658]
B:Functional Form	CHSQ(1) = .28508 [.593]	F(1,109) = .27326 [.602]
C:Normality	CHSQ(2) = 9.0063 [.011]	Not applicable
D:Heteroskedasticity	CHSQ(1) = 2.1578 [.142]	F(1,112) = 2.1609 [.144]

A:Lagrange multiplier test of residual serial correlation; B:Ramsey's RESET test using the square of the fitted values
C:Based on a test of skewness and kurtosis of residuals; D:Based on the regression of squared residuals on squared fitted values

More specifically, a variable X is said to 'Granger-cause' a variable Y if past and present values of X contain information that helps predict future values of Y better than using the information contained in past and present values of Y alone.

Consider two stationary processes $\{y_t\}$, and $\{x_t\}$. Let $y^*_{T+h|T}$ be the forecast of y_{T+h} formed at time T, using the information set Ω_T, and $\tilde{y}^*_{T+h|T}$ be the forecast based on the information set $\tilde{\Omega}_T$, containing all the information in Ω_T except for that on past and present values of the process $\{x_t\}$. Let $L_q(.,.)$ be a quadratic loss function (see Section 17.5). The process $\{x_t\}$ is said to Granger-cause $\{y_t\}$ if

$$E\left[L_q(y_{T+h}, y^*_{T+h|T})\right] < E\left[L_q(y_{T+h}, \tilde{y}^*_{T+h|T})\right], \text{ for at least one } h = 1, 2, \ldots.$$

Hence, if $\{x_t\}$ fails to Granger-cause $\{y_t\}$, for all $h > 0$, the mean square forecast error based on $y^*_{T+h|T}$ is the same as that based on $\tilde{y}^*_{T+h|T}$.[2]

[2] See Dufour and Renault (1998) for futher details on causality in Granger's sense.

Table 21.3 Japanese output growth equation

OLS* estimation of a single equation in the unrestricted *VAR			
Dependent variable is DLYJAP **114 observations used for estimation from 1964Q3 to 1992Q4**			
Regressor	Coefficient	Standard Error	T-Ratio [Prob]
DLYUSA(−1)	.065140	.10992	.59262 [.555]
DLYJAP(−1)	.21958	.092566	2.3721 [.019]
DLYGER(−1)	.23394	.098960	2.3640 [.020]
CONST	.0081150	.0016567	4.8984 [.000]
R-Squared	.12935	R-Bar-Squared	.10560
S.E. of Regression	.010600	F-Stat. F(3,110)	5.4473 [.002]
Mean of Dependent Variable	.013128	S.D. of Dependent Variable	.011209
Residual Sum of Squares	.012361	Equation Log-likelihood	358.6189
Akaike Info. Criterion	354.6189	Schwarz Bayesian Criterion	349.1465
DW-statistic	2.1004	System Log-likelihood	1101.5

Diagnostic Tests

Test Statistics	**LM Version**	**F Version**
A:Serial Correlation	CHSQ(4) = 13.5014 [.009]	F(4,106) = 3.5601 [.009]
B:Functional Form	CHSQ(1) = 1.3036 [.254]	F(1,109) = 1.2608 [.264]
C:Normality	CHSQ(2) = 39.9038 [.000]	Not applicable
D:Heteroskedasticity	CHSQ(1) = 2.8502 [.091]	F(1,112) = 2.8720 [.093]

See the notes to Table 21.2.

There are other notions of 'cause' and 'effect' discussed in philosophy and statistics that we shall not be considering here. But even if we confine our analysis to the causality in Granger's sense, we need to be careful that all proximate third-party channels that influence the interactions of Y and X are taken into account. The most obvious case arises when there exists a third variable Z which influences both Y and X, but with differential time delays. As an example, suppose that changes in Z affect X before affecting Y. Then in the absence of an *a priori* knowledge of Z, it might be falsely concluded that X is the cause of Y. In practice, the presence Z (not known to the investigator) might not be an issue if the profiles of the effects of Z on Y, and Z on X are stable over time. But the situation could be very different if (for some reason unknown to the investigator) the time delays in the way Z affects Y and X change.

Other examples arise in situations where economic agents possess forecasting skills. Consider an individual who decides to take an umbrella when leaving home depending on whether the forecast is rain or shine. Suppose further that this individual is reasonably good at predicting the weather over the course of a day. Let X be an indicator variable that takes the value of 1 if the individual carries an umbrella and zero otherwise, and Y takes the value of 1 if it rains and zero otherwise. Then it is clear that a crude application of the Granger causality test to X and Y, can lead to the misleading conclusion that the decision to take the umbrella causes rain! However, if a third variable Z, which captures the forecasting skill of the individual, is included in the analysis, we will soon learn that the correlation between Y and X represents the extent to which the individual is good at forecasting the weather.

Table 21.4 Germany's output growth equation

OLS* estimation of a single equation in the unrestricted *VAR			
Dependent variable is DLYGER **114 observations used for estimation from 1964Q3 to 1992Q4**			
Regressor	Coefficient	Standard Error	T-Ratio [Prob]
DLYUSA(−1)	.19307	.10327	1.8695 [.064]
DLYJAP(−1)	.23045	.086968	2.6499 [.009]
DLYGER(−1)	−.03178	.092975	−0.3419 [.733]
CONST	.0025928	.0015565	1.6685 [.099]
R-Squared	.10562	R-Bar-Squared	.081232
S.E. of Regression	.0099594	F-Stat. F(3,110)	4.3303 [.006]
Mean of Dependent Variable	.0067337	S.D. of Dependent Variable	.010390
Residual Sum of Squares	.010911	Equation Log-likelihood	365.7306
Akaike Info. Criterion	361.7306	Schwarz Bayesian Criterion	356.2582
DW-statistic	2.0799	System Log-likelihood	1101.5

Diagnostic Tests

Test Statistics	**LM Version**	**F Version**
A:Serial Correlation	CHSQ(4) = 7.1440 [.128]	F(4,106) = 1.7717 [.140]
B:Functional Form	CHSQ(1) = .30999 [.578]	F(1,109) = .29721 [.587]
C:Normality	CHSQ(2) = .56351 [.754]	Not applicable
D:Heteroskedasticity	CHSQ(1) = .52835 [.467]	F(1,112) = .52150 [.472]

See the notes to Table 21.2.

In short, we need to consider all possible variables that interact to bring about an outcome before making any definite conclusions about causality from the application of Granger non-causality tests. In what follows we discuss a useful generalization that allows for additional factors in the application of the Granger test. But the fundamental problem remains that no matter how exhaustive we are in our analysis of Granger non-causality, we still need to be aware of other possible casual links that we might have inadvertently overlooked.

21.6.1 Testing for block Granger non-causality

A generalization in a multivariate context can be obtained by considering a $m \times 1$ vector $\mathbf{y}_t = \left(\mathbf{y}_{1t}', \mathbf{y}_{2t}'\right)'$, where $\mathbf{y}_{1t}$ and $\mathbf{y}_{2t}$ are $m_1 \times 1$ and $m_2 \times 1$, $(m_1 + m_2 = m)$ variables. Partition the system of equations (21.1) (or equivalently (21.10)) into the two sub-systems

$$\mathbf{Y}_1 = \mathbf{G}_1\mathbf{A}_{11} + \mathbf{G}_2\mathbf{A}_{12} + \mathbf{U}_1, \tag{21.20}$$

$$\mathbf{Y}_2 = \mathbf{G}_1\mathbf{A}_{21} + \mathbf{G}_2\mathbf{A}_{22} + \mathbf{U}_2, \tag{21.21}$$

where $\mathbf{Y}_1$ and $\mathbf{Y}_2$ are $T \times m_1$ and $T \times m_2$ matrices of observations on $\mathbf{y}_{1t}$ and $\mathbf{y}_{2t}$ respectively, $\mathbf{G}_1$ and $\mathbf{G}_2$ are $T \times pm_1$ and $T \times pm_2$ matrices of observations on the p lagged values of $\mathbf{y}_{1,t-\ell}$, and $\mathbf{y}_{2,t-\ell}$, for $t = 1, 2, \ldots, T$, $\ell = 1, 2, \ldots, p$, respectively. The process $\mathbf{y}_{2t}$ does not Granger-cause $\mathbf{y}_{1t}$ if the $m_1 m_2 p$ restrictions $\mathbf{A}_{12} = \mathbf{0}$ hold.

The log-likelihood ratio statistic for testing these restrictions is computed as

$$LR_G\left(\mathbf{A}_{12}=\mathbf{0}\right)=2\left(\log\left|\widetilde{\boldsymbol{\Sigma}}_R\right|-\log\left|\widetilde{\boldsymbol{\Sigma}}\right|\right),$$

where $\widetilde{\boldsymbol{\Sigma}}$ is *ML* estimator of $\boldsymbol{\Sigma}$ for the unrestricted full system

$$\mathbf{Y}=\mathbf{AG}+\mathbf{U},$$

with

$$\mathbf{Y}=\begin{pmatrix}\mathbf{Y}_1\\ \mathbf{Y}_2\end{pmatrix},\ \mathbf{A}=\begin{pmatrix}\mathbf{A}_{11} & \mathbf{A}_{12}\\ \mathbf{A}_{21} & \mathbf{A}_{22}\end{pmatrix},\ \mathbf{G}=\begin{pmatrix}\mathbf{G}_1\\ \mathbf{G}_2\end{pmatrix},\ \mathbf{U}=\begin{pmatrix}\mathbf{U}_1\\ \mathbf{U}_2\end{pmatrix}.$$

See formula (21.12). $\widetilde{\boldsymbol{\Sigma}}_R$ is the *ML* estimator of $\boldsymbol{\Sigma}$ when the restrictions $\mathbf{A}_{12}=\mathbf{0}$ are imposed. Under the null hypothesis that $\mathbf{A}_{12}=\mathbf{0}$, LR_G is asymptotically distributed as a chi-squared variate with m_1m_2p degrees of freedom.

Since under $\mathbf{A}_{12}=\mathbf{0}$ the system of equations (21.20)–(21.21) are block recursive, $\widetilde{\boldsymbol{\Sigma}}_R$ can be computed in the following manner:

1. Run *OLS* regressions of $\mathbf{Y}_1$ on $\mathbf{G}_1$ and compute the $T\times m_1$ matrix of residuals, $\widehat{\mathbf{U}}_1$.
2. Run the *OLS* regressions

$$\mathbf{Y}_2=\mathbf{G}_1\mathbf{A}_{21}^*+\mathbf{G}_2\mathbf{A}_{22}^*+\widehat{\mathbf{U}}_1\mathbf{A}_{24}^*+\mathbf{V}_2, \tag{21.22}$$

and compute the $T\times m_2$ matrix of residuals:

$$\widehat{\mathbf{U}}_2=\mathbf{Y}_2-\mathbf{G}_1\widehat{\mathbf{A}}_{21}^*-\mathbf{G}_2\widehat{\mathbf{A}}_{22}^*,$$

where $\widehat{\mathbf{A}}_{21}^*$ and $\widehat{\mathbf{A}}_{22}^*$ are the *OLS* estimators of $\mathbf{A}_{21}^*$ and $\mathbf{A}_{22}^*$, in (21.22). Define

$$\widehat{\mathbf{U}}=\left(\widehat{\mathbf{U}}_1:\widehat{\mathbf{U}}_2\right).$$

Then

$$\widetilde{\boldsymbol{\Sigma}}_R=T^{-1}\left(\widehat{\mathbf{U}}'\widehat{\mathbf{U}}\right). \tag{21.23}$$

21.7 Forecasting with multivariate models

The *VAR* model can be used easily to generate one or more step ahead forecasts of a number of variables simultaneously. For example, consider the following *VAR*(1) model

$$\mathbf{y}_t=\boldsymbol{\Phi}\mathbf{y}_{t-1}+\mathbf{u}_t,\ \ t=1,2,\ldots. \tag{21.24}$$

Following the discussions in Chapter 17, conditional on $\Omega_T=(\mathbf{y}_T,\mathbf{y}_{T-1},\ldots)$, the point forecast of $\mathbf{y}_{t+h}$ is given

$$\mathbf{y}_{T+h|T}^*=E\left(\mathbf{y}_{T+h}\mid\Omega_T\right), \tag{21.25}$$

which is optimal with respect to the following quadratic loss function

$$L_q(\mathbf{y}_{T+h}, \mathbf{y}^*_{T+h|T}) = \left(\mathbf{y}_{T+h} - \mathbf{y}^*_{T+h|T}\right)' \mathbf{A} \left(\mathbf{y}_{T+h} - \mathbf{y}^*_{T+h|T}\right),$$

where $\mathbf{A}$ is a symmetric positive definite matrix. The elements of $\mathbf{A}$ measure the relative importance of the forecast errors of the different variables in $\mathbf{y}_t$ and their correlations. It is important to note that the optimal forecast does not depend on $\mathbf{A}$. For the $VAR(1)$ specification in (21.24) we have

$$\mathbf{y}^*_{T+h|T} = \mathbf{\Phi}^h \mathbf{y}_T.$$

For the $VAR(p)$ specification defined by (21.1) the h-step ahead forecasts, $\mathbf{y}^*_{T+h|T}$, can be obtained recursively by noting that

$$\mathbf{y}^*_{T+j|T} = \sum_{i=1}^{p} \mathbf{\Phi}_i \mathbf{y}^*_{T+j-i|T},\ j = 1, 2, \ldots, h,$$

with initial values $\mathbf{y}^*_{T+j-i|T} = \mathbf{y}_{T-i}$ for $j - i \leq 0$. See Chapter 23 for a discussion of forecasting in the case of cointegrating *VAR*s possibly in the presence of weakly exogenous variables.

Example 50 *Table 21.5 reports the multivariate, multi-step ahead forecasts and forecast errors of output growths for the VAR(1) model estimated in Example 49 (see Tables 21.2–21.4), for the four quarters of 1993. As can be seen from the summary statistics, the size of the forecast errors and the in-sample residuals are very similar. A similar picture also emerges by plotting in-sample fitted values and out-of-sample forecasts (see Figure 21.1). It is, however, important to note that the US growth experience in 1993 may not have been a stringent enough test of the forecast performance of the VAR, as the US output growth has been positive in all four quarters. A good test of forecast performance is to see whether the VAR model predicts the turning points of the output movements. Similarly, forecasts of output growths for Japan and Germany can also be computed. For Japan the root mean sum of squares of the forecast errors over the 1993(1)–1993(4) period turned out to be 1.48 per cent, which is slightly higher than the value of 1.02 per cent obtained for the root mean sum of squares of residuals over the estimation period. It is also worth noting that the growth forecasts for Japan miss the two negative quarterly output growths that occurred in the second and fourth quarters of 1993. A similar conclusion is also reached in the case of output growth forecasts for Germany.*

21.8 Multivariate spectral density

Spectral density in the multivariate situation is considered as an extension to the univariate case (see Chapter 13 for details). We start with the multivariate linear stationary process

$$\mathbf{y}_t = \mathbf{A}(L)\,\mathbf{u}_t, \quad \mathbf{u}_t \sim IID(\mathbf{0}, \mathbf{\Sigma}), \quad t = 1, 2, \ldots T, \tag{21.26}$$

where L is the lag operators on $\mathbf{A}$, $\mathbf{y}_t$ is a vector of $m \times 1$ time series, $\mathbf{u}_t$ is a vector of $m \times 1$ error terms with mean zero and variance-covariance matrix $\mathbf{\Sigma}$. $\mathbf{A}(L)$ is a $m \times m$ matrix for

Table 21.5 Multivariate dynamic forecasts for US output growth (*DLYUSA*)

Multivariate dynamic forecasts for the level of DLYUSA
Unrestricted vector autoregressive model

Based on 116 observations from 1964Q1 to 1992Q4. Order of *VAR* = 1
List of variables included in the unrestricted *VAR*:
DLYUSA DLYJAP DLYGER
List of deterministic and/or exogenous variables:
CONST D74

Observation	Actual	Prediction	Error
1993Q1	.001950	.011379	−.0094288
1993Q2	.004702	.008416	−.0037135
1993Q3	.007072	.007688	−.6162E-3
1993Q4	.016841	.007538	.0093022

Summary statistics for residuals and forecast errors

	Estimation Period 1964Q1 to 1992Q4	**Forecast Period 1993Q1 to 1993Q4**
Mean	−.0000	−.0011141
Mean Absolute	.0067166	.0057652
Mean Sum Squares	.8082E-4	.4740E-4
Root Mean Sum Squares	.0089902	.0068848

coefficients, and it is square summable (refer to Appendix A for details for the square summability of a matrix).

The autocovariance generating function for (21.26) is

$$\mathbf{G}_{\mathbf{y}}(z) = \mathbf{A}(z)\,\boldsymbol{\Sigma}\boldsymbol{A}'\left(z^{-1}\right),$$

and the corresponding spectral density function is given by

$$\mathbf{F}_{\mathbf{y}}(\omega) = \frac{1}{2\pi}\mathbf{A}\left(e^{i\omega}\right)\boldsymbol{\Sigma}\boldsymbol{A}'\left(e^{-i\omega}\right),\ 0 \leq \omega \leq \pi.$$

Evaluating at zero frequency, we obtain

$$\mathbf{F}_{\mathbf{y}}(0) = \frac{1}{2\pi}\mathbf{A}(1)\boldsymbol{\Sigma}\boldsymbol{A}'(1).$$

Example 51 *Consider VAR*(1) *model*

$$\mathbf{y}_t = \boldsymbol{\Phi}\mathbf{y}_{t-1} + \mathbf{u}_t. \tag{21.27}$$

When (21.27) is stationary, we have $\mathbf{A}(L) = (\mathbf{I}_m - \boldsymbol{\Phi}L)^{-1}$, *where* $\boldsymbol{\Phi}$ *is an* $m \times m$ *coefficient matrix. Hence the spectral density of the VAR* (1) *is*

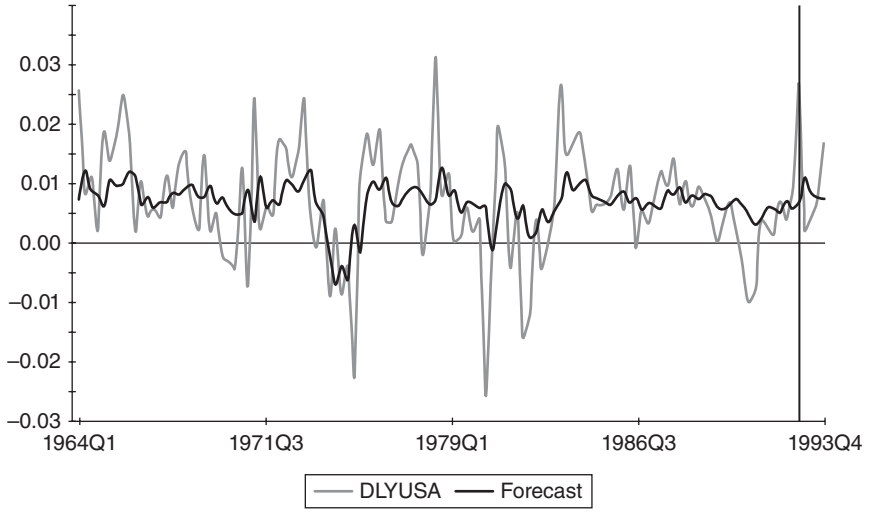


Figure 21.1 Multivariate dynamic forecasts of US output growth (*DLYUSA*).

$$\mathbf{F}_{\mathbf{y}}(\omega) = \frac{1}{2\pi}\left(\mathbf{I}_m - \mathbf{\Phi}e^{i\omega}\right)^{-1}\mathbf{\Sigma}\left(\mathbf{I}_m - \mathbf{\Phi}'e^{i\omega}\right)^{-1}.$$

Evaluating at zero frequency, we obtain

$$\mathbf{F}_{\mathbf{y}}(0) = \frac{1}{2\pi}\left(\mathbf{I}_m - \mathbf{\Phi}\right)^{-1}\mathbf{\Sigma}\left(\mathbf{I}_m - \mathbf{\Phi}'\right)^{-1}.$$

For an introductory text on the estimation of the spectrum see Chatfield (2003). For more advanced treatments of the subject see Priestley (1981) and Brockwell and Davis (1991). See also Chapter 13.

21.9 Further reading

Detailed treatments of multivariate time series analysis can be found in Hamilton (1994), Lütkepohl and Kratzig (2004), and Juselius (2007).

21.10 Exercises

1. Consider the bivariate vector autoregressive model

$$\begin{pmatrix} y_{1t} \\ y_{2t} \end{pmatrix} = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}\begin{pmatrix} y_{1,t-1} \\ y_{2,t-1} \end{pmatrix} + \begin{pmatrix} u_{1t} \\ u_{2t} \end{pmatrix},$$
$$\mathbf{y}_t = \mathbf{A}\mathbf{y}_{t-1} + \mathbf{u}_t, \quad \mathbf{u}_t \backsim IIDN(\mathbf{0}, \mathbf{\Sigma}),$$

where

$$\mathbf{\Sigma} = \begin{pmatrix} \sigma_{11} & \sigma_{12} \\ \sigma_{12} & \sigma_{22} \end{pmatrix}.$$

(a) Derive the conditional mean and variance of y_{1t} with respect to y_{2t}, and lagged values of y_{1t} and y_{2t}.

(b) Show that the univariate representation of y_{1t} (or y_{2t}) is an autoregressive moving average process of order (1,1).

2. Consider the second-order vector autoregressive ($VAR(2)$) model in the m-dimensional vector $\mathbf{y}_t$

$$\mathbf{y}_t = \mathbf{\Phi}_1 \mathbf{y}_{t-1} + \mathbf{\Phi}_2 \mathbf{y}_{t-2} + \mathbf{u}_t, \tag{21.28}$$

where $\mathbf{\Phi}_i$, for $i = 1, 2$ are $m \times m$ matrices of fixed coefficients, and $\mathbf{u}_t$ is a mean zero, serially uncorrelated vector of disturbances with a common positive definite variance–covariance matrix, $\mathbf{\Sigma}$. Derive the conditions under which the $VAR(2)$ model defined in (21.28) is stationary.

3. Consider the first-order vector autoregressive model

$$\mathbf{y}_t = \mathbf{A}\mathbf{y}_{t-1} + \mathbf{u}_t,$$

where $\mathbf{y}_t$ is an $m \times 1$ vector of observed variables, and $\mathbf{u}_t \sim IID(\mathbf{0}, \mathbf{\Sigma})$, with $\mathbf{\Sigma}$ being an $m \times m$ positive definite matrix.

(a) Show that $\mathbf{y}_t$ is covariance stationary if all eigenvalues of $\mathbf{A}$ lie inside the unit circle.

(b) Derive the point forecasts of $\mathbf{y}_{T+1}, \mathbf{y}_{T+2}, \ldots, \mathbf{y}_{T+h}$, based on observations $\mathbf{y}_1, \mathbf{y}_2, \ldots, \mathbf{y}_T$, and show that the j-step ahead forecast errors

$$\boldsymbol{\xi}_{T+j} = \mathbf{y}_{T+j} - E\left(\mathbf{y}_{T+j} | \mathbf{y}_T, \mathbf{y}_{T-1}, \cdots, \mathbf{y}_1\right), \text{ for } j = 1, 2, \ldots, h$$

are serially correlated.

(c) Suppose $\mathbf{y}_t$ is covariance stationary. Show that

$$\lim_{j \to \infty} \left[Var\left(\boldsymbol{\xi}_{T+j}\right) \right] = Var(\mathbf{y}_T).$$

(d) Discuss the relevance of this result for multi-period ahead forecasting.

4. Suppose that the m-dimensional random variable, $\mathbf{y}_t = (y_{1t}, y_{2t}, \ldots, y_{mt})'$, follows a $VAR(1)$ process.

(a) Show that y_{1t} follows a univariate $ARMA(m, m-1)$ process. Start by first proving this result for $m = 2$.

(b) Derive the pair-wise correlation of y_{it} and y_{jt} across all i and j, and show that the univariate representations form a system of seemingly unrelated autoregressions.

(c) Set $m = 2$ and compare the forecast performance of y_{1t} based on a $VAR(1)$ in (y_{1t}, y_{2t}) with forecasts obtained from univariate $ARMA(2, 1)$.

5. Consider the $VAR(1)$ model

$$\mathbf{y}_{1t} = \mathbf{A}_{11}\mathbf{y}_{1,t-1} + \mathbf{A}_{12}\mathbf{y}_{2,t-1} + \mathbf{u}_{1t},$$
$$\mathbf{y}_{2t} = \mathbf{A}_{21}\mathbf{y}_{1,t-1} + \mathbf{A}_{22}\mathbf{y}_{2,t-1} + \mathbf{u}_{2t},$$

where $\mathbf{y}_{1t}$ and $\mathbf{y}_{2t}$ are $m_1 \times 1$ and $m_2 \times 1$ are vectors of random variable, and the $m \times 1$ error vector $\mathbf{u}_t = (\mathbf{u}'_{1t}\, \mathbf{u}'_{2t})'$ is $IID(\mathbf{0}, \mathbf{\Sigma})$.

(a) Given the set of observations, $\mathbf{y}_t$ for $t = 1, 2, \ldots, T$, test the hypothesis that $\mathbf{y}_{2t}$ 'Granger causes' $\mathbf{y}_{1t}$ and not *vice versa*.

(b) Discuss the pros and cons of Granger causality tests. Illustrate your response by an empirical application based on the *GVAR* data set which can be downloaded from <https://sites.google.com/site/gvarmodelling/data>.

(c) Consider now the possibility that $\mathbf{y}_{1t}$ and $\mathbf{y}_{2t}$ are also affected by a third set of variables, $\mathbf{y}_{3t}$, not already included in $\mathbf{y}_t = (\mathbf{y}'_{1t}, \mathbf{y}'_{2t})'$. How does this affect your analysis? Again illustrate your response empirically.

22 Cointegration Analysis

22.1 Introduction

In this chapter we provide an overview of the econometric methods used in long-run structural macroeconometric modelling. In what follows we first introduce the concept of cointegration for a set of time series variables. We then turn our attention to cointegration within a *VAR* framework and review the literature on identification, estimation and hypothesis testing in cointegrated systems. We discuss estimation of cointegrating relations under general linear restrictions, and review tests of the over-identifying restrictions on the cointegrating vectors. We also comment on the small sample properties of some of the test statistics discussed in the chapter, and discuss a bootstrap approach for obtaining critical values. We conclude the chapter by reviewing the multivariate version of the Beveridge-Nelson decomposition, extended to include possible restrictions on the intercepts and/or trend coefficients, as well as the existence of long-run relationships.

22.2 Cointegration

The concept of cointegration was first introduced by Granger (1986) and more formally developed in Engle and Granger (1987). Two or more variables are said to be cointegrated if they are individually integrated (or have a random walk component), but there exist linear combinations of them which are stationary. More formally, consider m time series variables $y_{1t}, y_{2t}, \ldots, y_{mt}$ *known* to be non-stationary with unit roots, integrated of order one, namely (see Section 15.2)

$$y_{it} \sim I(1), \qquad i = 1, 2, \ldots, m.$$

The $m \times 1$ vector time series $\mathbf{y}_t = \left(y_{1t}, y_{2t}, \ldots, y_{mt}\right)'$ is said to be cointegrated if there exists an $m \times r$ matrix ($r \geq 1$) such that

$$\underset{r \times m}{\boldsymbol{\beta}'} \quad \underset{m \times 1}{\mathbf{y}_t} \quad = \quad \underset{r \times 1}{\boldsymbol{\xi}_t} \quad \sim I(0),$$

where the integer r denotes the number of cointegrating vectors, also known as dimension of the cointegration space. Cointegration means that, although each individual series is $\mathrm{I}(1)$, there exist some relations linking the individual series together, represented by the linear combinations, $\boldsymbol{\beta}'\mathbf{y}_t$, which are $I(0)$. The cointegrating relations summarized in the $r \times 1$ vector $\boldsymbol{\beta}'\mathbf{y}_t$ are also known as *long-run relations* (Johansen (1991)).

Example 52 *Many examples of cointegrating relations exist in the literature. In finance, under the expectations hypothesis, interest rates of different maturities are cointegrated. In macroeconomics examples of cointegration include the purchasing power parity hypothesis, the Fisher equation (that relates nominal interest rate to the expected rate of inflation), and the uncovered interest parity. For further details see Garratt et al. (2003b). Here we derive the cointegrating relationship that exists between equity prices and dividends in a simple model where equity prices are assumed to be determined by the discounted stream of dividends that are expected to occur to the equity*

$$P_t = \sum_{i=1}^{\infty} \left(\frac{1}{1+r}\right)^i E\left(D_{t+i} \mid \Omega_t\right),$$

where $\Omega_t = (P_t, D_t, P_{t-1}, D_{t-1}, \ldots)$ is the information set, and assuming that r is constant over time. To model P_t we first need to model the dividend process, D_t. If D_t is a unit root process, then P_t will also be a unit root process. A bilinear version of the above model is

$$P_t = \sum_{i=1}^{\infty} \left(\frac{1}{1+r}\right)^i E\left(D_{t+i} \mid \Omega_t\right) + u_t,$$
$$D_t = D_{t-1} + \sum_{i=0}^{\infty} \alpha_i \varepsilon_{t-i},$$

where u_t could characterize the influence of noise traders or the effects of other similar factors on equity prices. We shall assume that u_t and ε_t are white noise processes, and that $\{\alpha_i\}$ is absolute summable, so that $w_t = \sum_{i=0}^{\infty} \alpha_i \varepsilon_{t-i}$ is covariance stationary. $P_t^* = \sum_{i=1}^{\infty} \left(\frac{1}{1+r}\right)^i E\left(D_{t+i} \mid \Omega_t\right)$ is often referred to as the 'fundamental' price. To derive P_t^* we first note that

$$\sum_{j=1}^{\infty} \lambda^j D_{t+j} = \lambda \sum_{j=1}^{\infty} \lambda^{j-1} D_{t+j-1} + \sum_{j=1}^{\infty} \lambda^j w_{t+j},$$

where $\lambda = 1/(1+r)$. Taking expectations conditional on Ω_t it is then easily seen that

$$P_t^* = \lambda\left(D_t + P_t^*\right) + \xi_t,$$

where

$$\xi_t = E\left(\sum_{j=1}^{\infty} \lambda^j w_{t+j} \mid \Omega_t\right),$$

which is the expected value of the discounted stream of a covariance stationary process, and itself will be stationary (recall that $|\lambda| < 1$).[1] *Hence*

$$P_t^* = \left(\frac{\lambda}{1-\lambda}\right) D_t + \left(\frac{1}{1-\lambda}\right) \xi_t,$$

and noting that $\lambda/(1-\lambda) = 1/r$, we have

$$P_t = \frac{D_t}{r} + \left(\frac{1+r}{r}\right) \xi_t + u_t.$$

Therefore

$$P_t - D_t/r \sim I(0),$$

and P_t and D_t are cointegrated with the cointegrating vector $\boldsymbol{\beta}' = (1, -\frac{1}{r})$.

22.3 Testing for cointegration: single equation approaches

Engle and Granger residual-based tests

Engle and Granger (1987) suggest a two-step method to test for cointegration. In the first step, residuals from an *OLS* regression of y_{1t} on the rest of the variables, namely, $y_{2t}, y_{3t}, \ldots y_{mt}$, are computed. In the second step Dickey–Fuller and augmented Dickey–Fuller statistics are applied to these residuals, assuming no intercept and no linear trend. If the null of a unit root in the residuals is rejected, the test outcome is interpreted as evidence in favour of cointegration. Note that other standard unit root tests, such as those described in Chapter 15, can also be used in the second step.

Because unit root tests are applied to residuals from regressions that are spurious under the null hypothesis that y_{it} are $I(1)$, the asymptotic distribution of the *ADF* test based on the residuals will be different from that in the standard unit root case. Hence, the associated critical values used to interpret the unit root statistic differ from those used in the standard unit root tests, employed in Chapter 15. Engle and Yoo (1987) provide the asymptotic distribution of *DF* and ADF statistics under the null hypothesis that the data follow a vector random walk driven by *IID* random innovations. Phillips and Ouliaris (1990) relax the independence assumption and compute the asymptotic null distribution of the DF tests and other residuals-based tests. Critical values for a set of residual-based statistics can be found in MacKinnon (1996). Since residuals from regressing y_{1t} on $y_{2t}, y_{3t}, \ldots y_{mt}$, are not the same as residuals from regressing y_{2t} on $y_{1t}, y_{3t}, \ldots y_{mt}$, the test can be repeated re-ordering the variables and running *OLS* regressions of y_{2t} (say) on the rest of the variables. For large enough samples, the results should not depend on such re-ordering of the variables, but in practice the situation may be very different.

Residual-based tests have a number of shortcomings. First, in small samples the test results crucially depend on the choice of the left-hand side variable in the first step. Tests for cointegration

[1] See Appendix D in Pesaran (1987c) for exact derivation of ξ_t in terms of the dividends innovations, ε_t.

that are invariant to the ordering of variables based on the full information maximum likelihood have been proposed by Johansen (1991) (see Section 22.10). Another shortcoming of residual-based tests is that they do not allow for more than one cointegrating relation. Further, these tests do not make the best use of available data, and have generally low power. We refer to Pesavento (2007) for a comparison of residuals-based tests under a set of local alternatives.

22.3.1 Bounds testing approaches to the analysis of long-run relationships

Another important difficulty with residual-based tests of cointegration lies in the fact that the investigator must know with certainty that the underlying regressors in the model are $I(1)$. However, given the generally low power of unit root tests, testing whether the underlying variables are $I(1)$ may introduce an additional degree of uncertainty into the analysis. One approach that overcomes this problem has been suggested by Pesaran, Shin, and Smith (2001), and consists of estimating an error correction form of an autoregressive distributed lag ($ARDL$) model in the variables under consideration. Suppose we are interested in testing the existence of a long-run relationship between y_t, x_{1t} and x_{2t} and we are not sure whether these variables are $I(1)$ or $I(0)$. The Pesaran, Shin, and Smith (2001) approach consists of the following steps

Step 1: Estimate the error correction model

$$\begin{aligned}\Delta y_t = a_0 &+ \sum_{i=1}^{p} \psi_i \Delta y_{t-i} \\ &+ \sum_{i=0}^{p} \phi_{i1} \Delta x_{1,t-i} + \sum_{i=0}^{p} \phi_{i2} \Delta x_{2,t-i} \\ &+ \delta_1 y_{t-1} + \delta_2 x_{1,t-1} + \delta_3 x_{2,t-1} + u_t. \end{aligned} \tag{22.1}$$

Step 2: Compute the usual Wald or F-statistics for testing the null hypothesis

$$H_0 : \delta_1 = \delta_2 = \delta_3 = 0.$$

The distribution of this test statistic is non-standard, and the relevant critical value bounds have been tabulated by Pesaran, Shin, and Smith (2001). The critical values differ depending on whether the regression equation has a trend or not.

Step 3: Compare the Wald or F-statistic computed in Step 2 with the upper and lower critical value bounds for a given significance level, denoted by F_U and F_L. Then:

- If $F > F_U$, then reject $\delta_1 = \delta_2 = \delta_3 = 0$, and hence conclude that potentially there exists a long-run relationship between y_t, x_{1t} and x_{2t}.
- If $F < F_L$, then conclude that a long-run relationship between the variables does not seem to exist.
- If $F_L < F < F_U$, then the inference is inconclusive.

Hence, if the computed Wald or *F*-statistics fall outside the critical value bounds, a conclusive decision results without needing to know the order of the integration of the underlying variables. If, however, the Wald or *F*-statistics fall within these bounds, inference would be inconclusive. In such circumstances, more needs to be found out about the order of integration of the underlying variables.

It is also possible to carry out a bounds *t*-test only on the coefficient of the lagged dependent variable, namely testing $\delta_1 = 0$, against $\delta_1 \neq 0$ in the error correction model (22.1). Such test is also non-standard, and the appropriate critical values are tabulated in Pesaran, Shin, and Smith (2001). Once it is established that the linear relationship between the variables is not 'spurious' the parameters of the long-run relationship can be estimated using the *ARDL* procedure, discussed in Chapter 6 (see, in particular, Section 6.5). Pesaran and Shin (1999) show that the *ARDL* approach to estimation of long-run relations continues to be applicable even if the variables under consideration are $I(1)$ and cointegrated. They also provide Monte Carlo evidence on the comparative small sample performance of the *ARDL* and the fully modified *OLS* (*FM-OLS*) approach proposed by Phillips and Hansen (1990), showing that in general the former performs better than the latter. For proofs and further details see Pesaran and Shin (1999). In what follows we provide a brief account of the *FM-OLS* approach for completeness.

22.3.2 Phillips–Hansen fully modified *OLS* estimator

Consider the following linear regression model

$$y_t = \beta_0 + \boldsymbol{\beta}_1' \mathbf{x}_t + u_t,\ t = 1, 2, \ldots, T, \tag{22.2}$$

where the $k \times 1$ vector of $I(1)$ regressors, $\mathbf{x}_t$, are not themselves cointegrated. Therefore, $\mathbf{x}_t$ has a first difference stationary process given by

$$\Delta \mathbf{x}_t = \boldsymbol{\mu} + \mathbf{v}_t,\ \ t = 2, 3, \ldots, T, \tag{22.3}$$

in which $\boldsymbol{\mu}$ is a $k \times 1$ vector of drift parameters and $\mathbf{v}_t$ is a $k \times 1$ vector of $I(0)$, or stationary variables. It is also assumed that $\boldsymbol{\xi}_t = (u_t, \mathbf{v}_t')'$ is strictly stationary with zero mean and a finite positive-definite covariance matrix, $\boldsymbol{\Sigma}$.

The computation of the fully modified *OLS* (*FM-OLS*) estimator of $\boldsymbol{\beta} = (\beta_0, \boldsymbol{\beta}_1')'$ is carried out in two stages. In the first stage y_t is corrected for the long-run correlation of u_t and $\mathbf{v}_t$. For this purpose let $\hat{u}_t$ be the *OLS* residual vector computed using (22.2), and write

$$\hat{\boldsymbol{\xi}}_t = \begin{pmatrix} \hat{u}_t \\ \hat{\mathbf{v}}_t \end{pmatrix},\ t = 2, 3, \ldots, T, \tag{22.4}$$

where $\hat{\mathbf{v}}_t = \Delta \mathbf{x}_t - \hat{\boldsymbol{\mu}}$, for $t = 2, 3, \ldots, T$, and $\hat{\boldsymbol{\mu}} = (T-1)^{-1} \sum_{t=2}^{T} \Delta \mathbf{x}_t$. A consistent estimator of the long-run variance of $\boldsymbol{\xi}_t$ is given by

$$\widehat{\boldsymbol{\Omega}} = \widehat{\boldsymbol{\Sigma}} + \widehat{\boldsymbol{\Lambda}} + \widehat{\boldsymbol{\Lambda}}' = \begin{pmatrix} \widehat{\Omega}_{11} & \widehat{\boldsymbol{\Omega}}_{12} \\ 1 \times 1 & 1 \times k \\ \widehat{\boldsymbol{\Omega}}_{21} & \widehat{\boldsymbol{\Omega}}_{22} \\ k \times 1 & k \times k \end{pmatrix}, \tag{22.5}$$

where

$$\widehat{\boldsymbol{\Sigma}} = \frac{1}{T-1} \sum_{t=2}^{T} \hat{\boldsymbol{\xi}}_t \hat{\boldsymbol{\xi}}_t', \tag{22.6}$$

and

$$\widehat{\boldsymbol{\Lambda}} = \sum_{s=1}^{m} \omega(s, m) \widehat{\boldsymbol{\Gamma}}_s, \tag{22.7}$$

in which

$$\widehat{\boldsymbol{\Gamma}}_s = T^{-1} \sum_{t=1}^{T-s} \hat{\boldsymbol{\xi}}_t \hat{\boldsymbol{\xi}}_{t+s}', \tag{22.8}$$

and $\omega(s, m)$ is the lag window with horizon (or truncation) m. For a choice of lag window such as Bartlett, Tukey or Parzen see Section 5.9.

Now let

$$\widehat{\boldsymbol{\Delta}} = \widehat{\boldsymbol{\Sigma}} + \widehat{\boldsymbol{\Lambda}} = \begin{pmatrix} \widehat{\boldsymbol{\Delta}}_{11} & \widehat{\boldsymbol{\Delta}}_{12} \\ \widehat{\boldsymbol{\Delta}}_{21} & \widehat{\boldsymbol{\Delta}}_{22} \end{pmatrix}, \tag{22.9}$$

$$\widehat{\mathbf{Z}} = \widehat{\boldsymbol{\Delta}}_{21} - \widehat{\boldsymbol{\Delta}}_{22} \widehat{\boldsymbol{\Omega}}_{22}^{-1} \widehat{\boldsymbol{\Omega}}_{21}, \tag{22.10}$$

$$\hat{y}_t^* = y_t - \widehat{\boldsymbol{\Omega}}_{12} \widehat{\boldsymbol{\Omega}}_{22}^{-1} \hat{\mathbf{v}}_t, \tag{22.11}$$

and

$$\underset{(k+1) \times k}{\mathbf{D}} = \begin{pmatrix} \mathbf{0} \\ 1 \times k \\ \mathbf{I}_k \\ k \times k \end{pmatrix}. \tag{22.12}$$

In the second stage, the *FM-OLS* estimator of $\boldsymbol{\beta}$ is computed as

$$\widehat{\boldsymbol{\beta}}_* = (\mathbf{W}'\mathbf{W})^{-1} (\mathbf{W}' \hat{\mathbf{y}}^* - T \mathbf{D} \widehat{\mathbf{Z}}), \tag{22.13}$$

where $\hat{\mathbf{y}}^* = \left(\hat{y}_1^*, \hat{y}_2^*, \ldots, \hat{y}_T^*\right)'$, $\mathbf{W} = (\boldsymbol{\tau}_T, \mathbf{X})$, and $\boldsymbol{\tau}_T = (1, 1, \ldots, 1)'$.

A consistent estimator of the variance matrix of $\widehat{\boldsymbol{\beta}}_*$ defined in (22.13) is given by

$$\widehat{\mathbf{V}}(\widehat{\boldsymbol{\beta}}_*) = \hat{\omega}_{11.2} (\mathbf{W}'\mathbf{W})^{-1}, \tag{22.14}$$

where

$$\hat{\omega}_{11.2} = \widehat{\Omega}_{11} - \widehat{\mathbf{\Omega}}_{12}\widehat{\mathbf{\Omega}}_{22}^{-1}\widehat{\mathbf{\Omega}}_{21}. \tag{22.15}$$

22.4 Cointegrating *VAR*: multiple cointegrating relations

Consider the following $VAR(p)$

$$\mathbf{y}_t = \mathbf{\Phi}_1\mathbf{y}_{t-1} + \mathbf{\Phi}_2\mathbf{y}_{t-2} + \ldots + \mathbf{\Phi}_p\mathbf{y}_{t-p} + \mathbf{u}_t,\ \mathbf{u}_t \sim IID(\mathbf{0}, \mathbf{\Sigma}), \tag{22.16}$$

where p, the order of the *VAR*, is assumed to be known, and the initial values, $\mathbf{y}_0, \mathbf{y}_1, \ldots, \mathbf{y}_{-p+1}$, are assumed to be given. Cointegration within the *VAR* model (22.16) can be introduced by considering its error correction representation. Rewrite (22.16) as

$$\begin{aligned}\Delta\mathbf{y}_t + \mathbf{y}_{t-1} = \mathbf{\Phi}_1\mathbf{y}_{t-1} + \mathbf{\Phi}_2(\mathbf{y}_{t-1} - \Delta\mathbf{y}_{t-1}) + \ldots \\ + \mathbf{\Phi}_p(\mathbf{y}_{t-1} - \Delta\mathbf{y}_{t-1} - \ldots \Delta\mathbf{y}_{t-p+1}) + \mathbf{u}_t,\end{aligned}$$

so that the vector error correction (VEC) model of (22.16) is

$$\Delta\mathbf{y}_t = -\mathbf{\Pi}\mathbf{y}_{t-1} + \sum_{j=1}^{p-1}\mathbf{\Gamma}_j\Delta\mathbf{y}_{t-j} + \mathbf{u}_t, \tag{22.17}$$

with $\mathbf{\Pi} = \mathbf{I}_m - \mathbf{\Phi}_1 - \mathbf{\Phi}_2 - \ldots - \mathbf{\Phi}_p$, and $\mathbf{\Gamma}_j = -\sum_{i=j+1}^{p}\mathbf{\Phi}_i$, for $j = 1, 2, \ldots, p-1$. Suppose now $\mathbf{y}_t \sim I(1)$, then the left-hand side of (22.17) is $I(0)$, and on the right-hand side both $\Delta\mathbf{y}_{t-j}$ and $\mathbf{u}_t$ are $I(0)$. Since $I(1) + I(0) = I(1)$, (22.17) holds if and only if $\mathbf{\Pi}\mathbf{y}_{t-1}$ is $I(0)$. Now let $\mathbf{d}_t = \mathbf{\Pi}\mathbf{y}_{t-1} \sim I(0)$. In the case where $Rank(\mathbf{\Pi}) = m$, $\mathbf{\Pi}$ is nonsingular and we have $\mathbf{y}_{t-1} = \mathbf{\Pi}^{-1}\mathbf{d}_t \sim I(0)$, that is, $\mathbf{y}_{t-1}$ is a stationary process, which contradicts the assumption that $\mathbf{y}_t \sim I(1)$. Therefore, given $\mathbf{y}_t \sim I(1)$ and $\mathbf{\Pi}\mathbf{y}_{t-1} \sim I(0)$ holds, we must have $Rank(\mathbf{\Pi}) = r < m$. This introduces us to the concept of cointegration.

Definition 28 *If $\mathbf{y}_{t-1} \sim I(1)$ and the linear combinations of $\mathbf{y}_{t-1}$, namely $\mathbf{\Pi}\mathbf{y}_{t-1}$, are covariance stationary, namely if $\mathbf{\Pi}\mathbf{y}_{t-1} \sim I(0)$, we say the VAR model (22.16) is cointegrated. Denoting $Rank(\mathbf{\Pi}) = r < m$, r is the dimension of the cointegration space.*

When $Rank(\mathbf{\Pi}) = r < m$, we can write $\mathbf{\Pi}$ as

$$\mathbf{\Pi} = \boldsymbol{\alpha}\boldsymbol{\beta}', \tag{22.18}$$

where $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$ are $m \times r$ matrices of full column ranks, namely $Rank(\boldsymbol{\beta}) = Rank(\boldsymbol{\alpha}) = r$. Then

$$\mathbf{\Pi}\mathbf{y}_{t-1} = \boldsymbol{\alpha}(\boldsymbol{\beta}'\mathbf{y}_{t-1}) \sim I(0),$$

and the *VECM* can be written as

$$\Delta \mathbf{y}_t = -\boldsymbol{\alpha}\boldsymbol{\beta}'\mathbf{y}_{t-1} + \sum_{j=1}^{p-1} \boldsymbol{\Gamma}_j \Delta \mathbf{y}_{t-j} + \mathbf{u}_t. \tag{22.19}$$

Since $\boldsymbol{\alpha}$ is full rank, we have

$$\boldsymbol{\beta}'\mathbf{y}_{t-1} \sim I(0),$$

where $\boldsymbol{\beta}'\mathbf{y}_t$ is the $r \times 1$ vector of cointegrating relations, also known as the *long-run relations*.

Example 53 *Cointegration can also be defined in terms of the spectral density of first differences of the variables evaluated at zero frequency. Consider the $m \times 1$ vector of $I(1)$ processes, $\mathbf{y}_t \sim I(1)$, such that*

$$\Delta \mathbf{y}_t = \mathbf{A}(L)\,\mathbf{u}_t, \tag{22.20}$$

is stationary, with $\mathbf{u}_t \sim IID(\mathbf{0}, \boldsymbol{\Sigma})$, where $\boldsymbol{\Sigma}$ is a positive definite matrix. Since $\Delta\mathbf{y}_t$ is stationary its spectral density exists and when evaluated at zero frequency can be written as (see Section 21.8)

$$\mathbf{F}_{\Delta \mathbf{y}}(0) = \frac{1}{2\pi}\mathbf{A}(1)\,\boldsymbol{\Sigma}\mathbf{A}'(1). \tag{22.21}$$

Suppose now that $\boldsymbol{\xi}_t = \boldsymbol{\beta}'\mathbf{y}_t \sim I(0)$, and note that the spectral density of $\Delta\boldsymbol{\xi}_t$ at zero frequency must be zero, due to over-differencing of a stationary process (see Exercise 5 in Chapter 13). Hence

$$\boldsymbol{\beta}'\mathbf{F}_{\Delta \mathbf{y}}(0)\,\boldsymbol{\beta} = \frac{1}{2\pi}\boldsymbol{\beta}'\mathbf{A}(1)\,\boldsymbol{\Sigma}\mathbf{A}'(1)\,\boldsymbol{\beta} = \mathbf{0}, \tag{22.22}$$

where $\boldsymbol{\Sigma}$ is a positive definite matrix. It then follows that we must have $\mathbf{A}'(1)\,\boldsymbol{\beta} = \mathbf{0}$, which is possible if and only if rank$[\mathbf{A}(1)] = m - r$, where r is the number of cointegrating relations. Therefore, cointegration is present when the spectral density of $\Delta\mathbf{y}_t$, evaluated at zero frequency, is rank deficient. This suggests that non-parametric methods may be used to test for cointegration (see, e.g., Breitung (2000)).

22.5 Identification of long-run effects

In general, $\boldsymbol{\beta}$, as defined in (22.18) or (22.19), is not uniquely determined. To see this, consider a linear transformation of $\boldsymbol{\beta}$, $\widetilde{\boldsymbol{\beta}} = \boldsymbol{\beta}\mathbf{Q}$, with $\mathbf{Q}$ being a nonsingular $r \times r$ matrix. Then

$$\boldsymbol{\Pi}\mathbf{y}_{t-1} = (\boldsymbol{\alpha}\mathbf{Q}^{\prime-1})(\mathbf{Q}'\boldsymbol{\beta}'\mathbf{y}_{t-1}).$$

Therefore, if $\boldsymbol{\beta}'\mathbf{y}_{t-1} \sim I(0)$, so will $\widetilde{\boldsymbol{\beta}}'\mathbf{y}_{t-1} \sim I(0)$, in the sense that as far as the cointegration property of $\mathbf{y}_t$ is concerned, the r columns of $\boldsymbol{\beta}$ and $\widetilde{\boldsymbol{\beta}}$ will both be equally valid as cointegrating vectors. The data allows us to estimate $\boldsymbol{\Pi}$, $\boldsymbol{\Gamma}_j$ and $\mathbf{u}_t$, but we can not estimate $\boldsymbol{\beta}$ uniquely

from the observations. This is called the 'long-run identification problem'. To exactly identify the long-run (or cointegrating) coefficients, we need r^2 exact- or just-identifying restrictions, r restrictions on each of the r cointegrating relations. Note that it is not possible to distribute the r^2 just-identifying restrictions unevenly across the r cointegrating relations.

Example 54 *Consider r=2, m=5, and*

$$\mathbf{x}_t = \begin{pmatrix} p_t \\ p_t^* \\ e_t \\ r_t \\ r_t^* \end{pmatrix},$$

where p_t and p_t^* are domestic and foreign log prices, e_t is the exchange rate at time t, and r_t and r_t^* are domestic and foreign interests, respectively. Denote

$$\boldsymbol{\xi}_t = \boldsymbol{\beta}'\mathbf{x}_t = \begin{pmatrix} \xi_{1t} \\ \xi_{2t} \end{pmatrix},$$

with

$$\xi_{1t} = \beta_{11}p_t + \beta_{12}p_t^* + \beta_{13}e_t + \beta_{14}r_t + \beta_{15}r_t^*, \tag{22.23}$$
$$\xi_{2t} = \beta_{21}p_t + \beta_{22}p_t^* + \beta_{23}e_t + \beta_{24}r_t + \beta_{25}r_t^*. \tag{22.24}$$

By economic theory we know there exist two cointegration relations

$$p_t - p_t^* - e_t \sim I(0), \tag{22.25}$$
$$r_t - r_t^* \sim I(0). \tag{22.26}$$

To distinguish between (22.23) and (22.24) and identify the cointegrating vectors, we need to impose two restrictions on the coefficients of each of the two cointegrating vectors. To identify (22.23), we need to impose the restriction $\beta_{11} = 1$, and either $\beta_{14} = 0$ or $\beta_{15} = 0$. Similarly to identify (22.24) we need to impose $\beta_{24} = 1$, and either $\beta_{21} = 0$ or $\beta_{22} = 0$. Therefore, a possible set of exact identifying restrictions for this example is

$$\begin{cases} \beta_{14} = 0, \beta_{11} = 1 \\ \beta_{21} = 0, \beta_{24} = 1 \end{cases},$$

which involves $r^2 = 4$ restrictions. Note that the economic theory imposes ten restrictions

$$\boldsymbol{\beta}_1 = \begin{pmatrix} 1 \\ -1 \\ -1 \\ 0 \\ 0 \end{pmatrix} \quad \boldsymbol{\beta}_2 = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \\ -1 \end{pmatrix},$$

with four of the restrictions used for exact identification. Hence, we are left with six over-identifying restrictions, which is in line with using the formula $mr - r^2 = 5 \times 2 - 2^2 = 6$.

22.6 System estimation of cointegrating relations

Consider the *VAR* (1) model

$$\mathbf{y}_t = (\mathbf{I}_m - \mathbf{\Pi})\,\mathbf{y}_{t-1} + \mathbf{u}_t, \tag{22.27}$$

and its *VEC* representation

$$\Delta\mathbf{y}_t = -\mathbf{\Pi}\mathbf{y}_{t-1} + \mathbf{u}_t \tag{22.28}$$

$$= -\boldsymbol{\alpha}\left(\boldsymbol{\beta}'\mathbf{y}_{t-1}\right) + \mathbf{u}_t. \tag{22.29}$$

When $\mathbf{\Pi}$ is of full rank m, then $\mathbf{\Pi}$ and the other parameters of (22.28) are identified under fairly general conditions, and can be consistently estimated by *OLS* (see Chapter 21). However, if the rank of $\mathbf{\Pi}$ is $r < m$, then $\mathbf{\Pi}$ is subject to $(m-r)^2$ nonlinear restrictions, and therefore uniquely determined in terms of the $m^2 - (m-r)^2 = 2mr - r^2$ underlying unknown parameters.

The cointegrating *VAR* analysis is concerned with the estimation of *VAR*(1) (22.28) (or (22.29)) when the multiplier matrix, $\mathbf{\Pi}$, is rank deficient. As pointed out in Section 19.7, under the rank deficiency, the *OLS* method is not valid. In addition, since (22.27) is a system of $m \times 1$ equations, the *OLS* method will not be appropriate if we have contemporaneously correlated disturbances with different regressors across the equations, that is, the equations are 'seemingly' unrelated. Estimation of (22.28) can be approached by applying the reduced rank regression method, which consists of carrying a canonical correlation analysis between the variables in $\Delta\mathbf{y}_t$ and $\mathbf{y}_{t-1}$ (see Sections 19.6 and 19.7, Anderson (1951) and Johansen (1991)).

Conditional on the initial values, $\mathbf{y}_0$, and assuming $\mathbf{u}_t \sim IIDN(\mathbf{0}, \mathbf{\Sigma})$, where $\mathbf{\Sigma}$ is a symmetric positive definite matrix, the log-likelihood function of (22.28) is given by

$$\ell(\boldsymbol{\theta};r) = -\frac{Tm}{2}\log 2\pi - \frac{T}{2}\log|\mathbf{\Sigma}| - \frac{1}{2}\sum_{t=1}^{T}\mathbf{u}_t'\mathbf{\Sigma}^{-1}\mathbf{u}_t, \tag{22.30}$$

with $\boldsymbol{\theta} = \left(vec(\boldsymbol{\alpha})', vec(\boldsymbol{\beta})', vech(\mathbf{\Sigma})'\right)'$, $\mathbf{u}_t = \Delta\mathbf{y}_t + \mathbf{\Pi}\mathbf{y}_{t-1}$, and r is the assumed rank of $\mathbf{\Pi} = \boldsymbol{\alpha}\boldsymbol{\beta}'$. Taking $\boldsymbol{\beta}$ as given, $\boldsymbol{\alpha}$ can be estimated by least squares, namely[2]

$$\hat{\boldsymbol{\alpha}} = -\left(\sum_{t=1}^{T}\Delta\mathbf{y}_t\mathbf{y}_{t-1}'\right)\boldsymbol{\beta}\left(\sum_{t=1}^{T}\boldsymbol{\beta}'\mathbf{y}_{t-1}\mathbf{y}_{t-1}'\boldsymbol{\beta}\right)^{-1} = -\mathbf{S}_{01}\boldsymbol{\beta}\left(\boldsymbol{\beta}'\mathbf{S}_{11}\boldsymbol{\beta}\right)^{-1},$$

where

$$\mathbf{S}_{01} = \frac{1}{T}\sum_{t=1}^{T}\Delta\mathbf{y}_t\mathbf{y}_{t-1}' = \frac{1}{T}(\mathbf{Y} - \mathbf{Y}_{-1})'\mathbf{Y}_{-1}, \tag{22.31}$$

[2] This is because for a given $\boldsymbol{\beta}$ the r regressors $\boldsymbol{\beta}'\mathbf{y}_{t-1}$ in the SURE system of equations (22.29) are the same, and hence *OLS* and *MLE* estimators of $\boldsymbol{\alpha}$ will coincide. See sub-section 19.2.1.

$$\mathbf{S}_{11} = \frac{1}{T}\sum_{t=1}^{T}\mathbf{y}_{t-1}\mathbf{y}_{t-1}' = \frac{1}{T}\left(\mathbf{Y}_{-1}'\mathbf{Y}_{-1}\right). \tag{22.32}$$

$\mathbf{Y}$ and $\mathbf{Y}_{-1}$ are the $T \times m$ matrix of observations on $\mathbf{y}_t$ and its lagged value, $\mathbf{y}_{t-1}$. Further, we have

$$\hat{\boldsymbol{\Sigma}}\left(\boldsymbol{\beta}\right) = T^{-1}\sum_{t=1}^{T}\mathbf{u}_t\left(\hat{\boldsymbol{\alpha}}\right)\mathbf{u}_t'\left(\hat{\boldsymbol{\alpha}}\right) = \mathbf{S}_{00} - \mathbf{S}_{01}\boldsymbol{\beta}\left(\boldsymbol{\beta}'\mathbf{S}_{11}\boldsymbol{\beta}\right)^{-1}\boldsymbol{\beta}'\mathbf{S}_{01}',$$

with

$$\mathbf{u}_t\left(\hat{\boldsymbol{\alpha}}\right) = \Delta\mathbf{y}_t + \hat{\boldsymbol{\alpha}}\beta'\mathbf{y}_{t-1} = \Delta\mathbf{y}_t - \mathbf{S}_{01}\boldsymbol{\beta}\left(\boldsymbol{\beta}'\mathbf{S}_{11}\boldsymbol{\beta}\right)^{-1}\boldsymbol{\beta}'\mathbf{y}_{t-1},$$
$$\mathbf{S}_{00} = \frac{1}{T}\sum_{t=1}^{T}\Delta\mathbf{y}_t\Delta\mathbf{y}_t' = \frac{1}{T}(\mathbf{Y} - \mathbf{Y}_{-1})'(\mathbf{Y} - \mathbf{Y}_{-1}).$$

Assume that T is sufficiently large such that matrices $\mathbf{S}_{00}$ and $\mathbf{S}_{11}$ are nonsingular. Then the concentrated log-likelihood function which is given by

$$\ell^c\left(\boldsymbol{\beta};r\right) = -\frac{Tm}{2}\log 2\pi - \frac{T}{2}\log\left|\hat{\boldsymbol{\Sigma}}\left(\boldsymbol{\beta}\right)\right| - \frac{1}{2}\sum_{t=1}^{T}\mathbf{u}_t'\left(\hat{\boldsymbol{\alpha}}\right)\left[\hat{\boldsymbol{\Sigma}}(\beta)\right]^{-1}\mathbf{u}_t\left(\hat{\boldsymbol{\alpha}}\right),$$

can be written as

$$\ell^c\left(\boldsymbol{\beta};r\right) = -\frac{Tm}{2}\left(1 + \log 2\pi\right) - \frac{T}{2}\log\left|\mathbf{S}_{00} - \mathbf{S}_{01}\boldsymbol{\beta}\left(\boldsymbol{\beta}'\mathbf{S}_{11}\boldsymbol{\beta}\right)^{-1}\boldsymbol{\beta}'\mathbf{S}_{01}'\right|. \tag{22.33}$$

However, it is easily seen that[3]

$$\left|\mathbf{S}_{00} - \mathbf{S}_{01}\boldsymbol{\beta}\left(\boldsymbol{\beta}'\mathbf{S}_{11}\boldsymbol{\beta}\right)^{-1}\boldsymbol{\beta}'\mathbf{S}_{01}'\right| = \frac{|\mathbf{S}_{00}|\left|\boldsymbol{\beta}'\mathbf{A}_T\boldsymbol{\beta}\right|}{\left|\boldsymbol{\beta}'\mathbf{B}_T\boldsymbol{\beta}\right|}, \tag{22.34}$$

where

$$\mathbf{B}_T = \mathbf{S}_{11}, \text{ and } \mathbf{A}_T = \mathbf{S}_{11} - \mathbf{S}_{10}\mathbf{S}_{00}^{-1}\mathbf{S}_{01}. \tag{22.35}$$

Substituting (22.34) in (22.33) yields the concentrated log-likelihood

$$\ell^c\left(\boldsymbol{\beta};r\right) = -\frac{Tm}{2}\left(1 + \log 2\pi\right) - \frac{T}{2}\log|\mathbf{S}_{00}| - \frac{T}{2}\left[\log\left|\boldsymbol{\beta}'\mathbf{A}_T\boldsymbol{\beta}\right| - \log\left|\boldsymbol{\beta}'\mathbf{B}_T\boldsymbol{\beta}\right|\right]. \tag{22.36}$$

[3] Note that

$$|\mathbf{G} + \mathbf{XHY}| = |\mathbf{G}||\mathbf{H}||\mathbf{H}^{-1} + \mathbf{YG}^{-1}\mathbf{X}|,$$

where $\mathbf{H}$ and $\mathbf{G}$ are $n \times n$ and $m \times m$ nonsingular matrices, and $\mathbf{X}$ and $\mathbf{Y}$ are $m \times n$ and $n \times m$ matrices.

It is clear that the maximization of $\ell^c(\boldsymbol{\beta}; r)$ with respect to $\boldsymbol{\beta}$ is equivalent to the minimization of the ratio

$$q(\boldsymbol{\beta}) = \frac{\left|\boldsymbol{\beta}'\mathbf{A}_T\boldsymbol{\beta}\right|}{\left|\boldsymbol{\beta}'\mathbf{B}_T\boldsymbol{\beta}\right|},$$

with respect to $\boldsymbol{\beta}$. Also, none of these two optimization problems will lead to a unique solution for $\boldsymbol{\beta}$. It is easily seen $q(\boldsymbol{\beta}\mathbf{Q}) = q(\boldsymbol{\beta})$ holds for any *arbitrary* $r \times r$ nonsingular matrix, $\mathbf{Q}$. Therefore, as also explained in Section 19.7, r^2 just-identifying restrictions are needed for exact identification. For computational purposes Johansen (1991) employs the following restrictions

$$\boldsymbol{\beta}'\mathbf{B}_T\boldsymbol{\beta} = \mathbf{I}_r,$$

and further assumes that the different columns of $\boldsymbol{\beta}$ are orthogonal to each other. These restrictions together impose the required r^2 exact-identifying restrictions; with the restrictions $\boldsymbol{\beta}'\mathbf{B}_T\boldsymbol{\beta} = \mathbf{I}_r$ providing $r(r+1)/2$ restrictions and the orthogonality of the cointegrating vectors supplying the remaining needed $r(r-1)/2$ restrictions.

Hence, *ML* estimates of $\boldsymbol{\beta}$ (and the maximized log-likelihood function) can be obtained by noting that when $\mathbf{A}_T$ and $\mathbf{B}_T$ are positive definite matrices, the minimized value of $q(\boldsymbol{\beta}) = \left|\boldsymbol{\beta}'\mathbf{A}_T\boldsymbol{\beta}\right| / \left|\boldsymbol{\beta}'\mathbf{B}_T\boldsymbol{\beta}\right|$, denoted by $q(\hat{\boldsymbol{\beta}})$, is given by[4]

$$q(\hat{\boldsymbol{\beta}}) = q(\hat{\boldsymbol{\beta}}\mathbf{Q}) = \prod_{i=1}^{r} \hat{\rho}_i,$$

where $\hat{\rho}_1 < \hat{\rho}_2 < \ldots < \hat{\rho}_r$ are the r smallest eigenvalues of $\mathbf{A}_T$ with respect to $\mathbf{B}_T$, given by the solution to the following determinantal equation in ρ

$$|\mathbf{A}_T - \rho\mathbf{B}_T| = 0.$$

But substituting for $\mathbf{A}_T$ and $\mathbf{B}_T$ from (22.35) we have

$$\left|\lambda\mathbf{S}_{11} - \mathbf{S}_{10}\mathbf{S}_{00}^{-1}\mathbf{S}_{01}\right| = 0,$$

where $\lambda = 1 - \rho$. Also since $\mathbf{S}_{11}$ is a nonsingular matrix, then $\hat{\lambda}_i$ can be computed as the ith largest eigenvalue of[5]

$$\mathbf{S}_{10}\mathbf{S}_{00}^{-1}\mathbf{S}_{01}\mathbf{S}_{11}^{-1}.$$

Also, up to a nonsingular $r \times r$ matrix $\mathbf{Q}$, $\hat{\boldsymbol{\beta}}$ is by the r eigenvectors, $\hat{\mathbf{v}}_i$, $i = 1, 2, \ldots, r$ associated with the eigenvalues $\hat{\lambda}_1, \hat{\lambda}_2, \ldots, \hat{\lambda}_r$, defined by

[4] See, for example, Lemma A.8, in Johansen (1995, p. 224).

[5] Note that $\hat{\lambda}_i$ can also be viewed as the ith canonical correlation of $\Delta\mathbf{y}_t$ and $\mathbf{y}_{t-1}$. See Section 19.6.

$$\left(\mathbf{S}_{10}\mathbf{S}_{00}^{-1}\mathbf{S}_{01}\mathbf{S}_{11}^{-1}\right)\hat{\mathbf{v}}_i = \hat{\lambda}_i\hat{\mathbf{v}}_i, i = 1, 2, \ldots, r, \tag{22.37}$$

The maximized log-likelihood function is therefore given by

$$\ell^c(r) = -\frac{Tm}{2}\left(1 + \log 2\pi\right) - \frac{T}{2}\log|\mathbf{S}_{00}| - \frac{T}{2}\sum_{i=1}^{r}\log\left(1 - \hat{\lambda}_i\right). \tag{22.38}$$

Note that the maximized value of the log-likelihood $\ell^c(r)$ is only a function of the cointegration rank r through the eigenvalues $\{\hat{\lambda}_i\}_{i=1}^r$ defined by (22.37). For further details see Johansen (1991) and Pesaran and Shin (2002).

22.7 Higher-order lags

Results on estimation and testing can be easily extended to allow for inclusion of lagged values of $\Delta\mathbf{y}_t$ in the error correction model. Consider the $VEC(p-1)$ model

$$\Delta\mathbf{y}_t = -\boldsymbol{\alpha}\boldsymbol{\beta}'\mathbf{y}_{t-1} + \sum_{j=1}^{p-1}\Gamma_j\Delta\mathbf{y}_{t-j} + \mathbf{u}_t, \text{ for } t = 1, 2, \ldots, T, \tag{22.39}$$

which corresponds to an underlying $VAR(p)$ specification. Writing the model in matrix notation we have the following system of regression equations

$$\Delta\mathbf{Y} = -\mathbf{Y}_{-1}\boldsymbol{\Pi}' + \mathbf{X}\Gamma + \mathbf{U}, \tag{22.40}$$

where $\boldsymbol{\Pi} = \boldsymbol{\alpha}\boldsymbol{\beta}', \mathbf{Y} = \left(\mathbf{y}_1, \mathbf{y}_2, \ldots, \mathbf{y}_T\right)', \mathbf{X} = (\Delta\mathbf{Y}_{-1}, \Delta\mathbf{Y}_{-2}, \ldots, \Delta\mathbf{Y}_{-p+1}), \mathbf{U} = (\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_T)'$, and $\boldsymbol{\Gamma} = (\boldsymbol{\Gamma}_1, \boldsymbol{\Gamma}_2, \ldots, \boldsymbol{\Gamma}_{p-1})'$ is an $m(p-1) \times m$ matrix of unknown coefficients. Further, $\Delta\mathbf{Y} = \mathbf{Y} - \mathbf{Y}_{-1}$, and $\Delta\mathbf{Y}_{-1}, \Delta\mathbf{Y}_{-2}, \ldots, \Delta\mathbf{Y}_{-p+1}$ refer to $T \times m$ matrices of lagged observations on $\Delta\mathbf{Y}$.

Conditional on the p initial values, $\mathbf{y}_{-p+1}, \ldots, \mathbf{y}_0$, the log-likelihood function of (22.40) can be written as

$$\ell(\boldsymbol{\theta}; r) = -\frac{Tm}{2}\log 2\pi - \frac{T}{2}\log|\boldsymbol{\Sigma}| - \frac{1}{2}\mathrm{Tr}\left(\boldsymbol{\Sigma}^{-1}\mathbf{U}'\mathbf{U}\right),$$

where $\mathbf{U} = \Delta\mathbf{Y} + \mathbf{Y}_{-1}\boldsymbol{\Pi}' + \mathbf{X}\boldsymbol{\Gamma}$, and $\boldsymbol{\theta} = \left(vec(\boldsymbol{\alpha})', vec(\boldsymbol{\beta})', vec(\boldsymbol{\Gamma})', vech(\boldsymbol{\Sigma})'\right)'$. The results obtained in Section 22.6 still hold for this more general case. One only needs to replace cross-product sample moment matrices $\mathbf{S}_{01}$ and $\mathbf{S}_{11}$, defined by (22.31) and (22.32), by

$$\mathbf{S}_{ij} = \frac{1}{T}\sum_{t=1}^{T}\mathbf{r}_{it}\mathbf{r}_{jt}', \text{ for } i, j = 0, 1,$$

where $\mathbf{r}_{0t}$ and $\mathbf{r}_{1t}$ are the residual vectors from the OLS regressions of $\Delta\mathbf{y}_t$ and $\mathbf{y}_{t-1}$ on $\left(\Delta\mathbf{y}_{t-1}, \Delta\mathbf{y}_{t-2}, \ldots, \Delta\mathbf{y}_{t-p+1}\right)$, respectively. The rest of the analysis will be unaffected.

It is also possible to include intercepts, linear (deterministic) trends and $I(1)$ weakly exogenous variables in the model. For the inclusion of intercepts, or linear deterministic trends, see Sections 22.8 and 22.9. See Chapter 23 for the inclusion of weakly exogenous variables in the *VAR* model.

22.8 Treatment of trends in cointegrating *VAR* models

Consider the following $VAR(p)$ model where intercepts and linear trends are included in deviations from $\mathbf{y}_t$,

$$\mathbf{\Phi}(L)(\mathbf{y}_t - \boldsymbol{\mu} - \boldsymbol{\gamma} t) = \mathbf{u}_t, \quad t = 1, 2, \ldots, \tag{22.41}$$

where $\boldsymbol{\mu}$ and $\boldsymbol{\gamma}$ are m-dimensional vectors of unknown coefficients, and $\mathbf{\Phi}(L) \equiv \mathbf{I}_m - \sum_{i=1}^{p} \mathbf{\Phi}_i L^i$ is an $m \times m$ matrix lag polynomial of order p. It is convenient to re-express the lag polynomial $\mathbf{\Phi}(L)$ in a form which arises in the vector error correction model

$$\mathbf{\Phi}(L) \equiv -\mathbf{\Pi} L + \mathbf{\Gamma}(L)(1 - L). \tag{22.42}$$

In (22.42), we have defined the long-run multiplier matrix

$$\mathbf{\Pi} \equiv -\Big(\mathbf{I}_m - \sum_{j=1}^{p} \mathbf{\Phi}_j\Big), \tag{22.43}$$

and the short-run response matrix lag polynomial $\mathbf{\Gamma}(L) \equiv \mathbf{I}_m - \sum_{i=1}^{p-1} \mathbf{\Gamma}_i L^i$, $\mathbf{\Gamma}_j = -\sum_{i=j+1}^{p} \mathbf{\Phi}_i$, $j = 1, \ldots, p-1$. Hence, the $VAR(p)$ model (22.41) may be rewritten in the following form

$$\mathbf{\Phi}(L)\mathbf{y}_t = \mathbf{a}_0 + \mathbf{a}_1 t + \mathbf{u}_t, t = 1, 2, \ldots, \tag{22.44}$$

where

$$\mathbf{a}_0 \equiv -\mathbf{\Pi}\boldsymbol{\mu} + (\mathbf{\Gamma} + \mathbf{\Pi})\,\boldsymbol{\gamma}, \ \mathbf{a}_1 \equiv -\mathbf{\Pi}\boldsymbol{\gamma}, \tag{22.45}$$

and the sum of the short-run coefficient matrices, $\mathbf{\Gamma}$, is given by

$$\mathbf{\Gamma} \equiv \mathbf{I}_m - \sum_{j=1}^{p-1} \mathbf{\Gamma}_j. \tag{22.46}$$

The cointegration rank hypothesis is defined by

$$H_r : Rank\,(\mathbf{\Pi}) = r, \quad r = 0, 1, \ldots, m. \tag{22.47}$$

Under H_r we may express

$$\boldsymbol{\Pi} = \boldsymbol{\alpha}\boldsymbol{\beta}', \tag{22.48}$$

where $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$ are $m \times r$ matrices of full column rank. Correspondingly, we may define the $m \times (m-r)$ matrices of full column rank $\boldsymbol{\alpha}_\perp$ and $\boldsymbol{\beta}_\perp$ whose columns form bases for the null spaces (kernels) of $\boldsymbol{\alpha}'$ and $\boldsymbol{\beta}'$, respectively. In particular, $\boldsymbol{\alpha}'\boldsymbol{\alpha}_\perp = \mathbf{0}$ and $\boldsymbol{\beta}'\boldsymbol{\beta}_\perp = \mathbf{0}$. We make the following assumptions.

Assumption 1: The $m \times m$ matrix polynomial $\boldsymbol{\Phi}(z) = \mathbf{I}_m - \sum_{i=1}^{p} \boldsymbol{\Phi}_i z^i$ is such that the roots of the determinantal equation $|\boldsymbol{\Phi}(z)| = 0$ satisfy $|z| > 1$ or $z = 1$.

Assumption 2: The $(m-r) \times (m-r)$ matrix $\boldsymbol{\alpha}'_\perp \boldsymbol{\Gamma} \boldsymbol{\beta}_\perp$ is full rank.

Assumption 1 rules out the possibility that the random process $\{(\mathbf{y}_t - \boldsymbol{\mu} - \boldsymbol{\gamma} t)\}_{t=1}^{\infty}$ admits explosive roots or seasonal unit roots except at zero frequency. Under Assumption 1, Assumption 2 is necessary and sufficient for the processes $\{\boldsymbol{\beta}'_\perp(\mathbf{y}_t - \boldsymbol{\mu} - \boldsymbol{\gamma} t)\}_{t=1}^{\infty}$ and $\{\boldsymbol{\beta}'(\mathbf{y}_t - \boldsymbol{\mu} - \boldsymbol{\gamma} t)\}_{t=1}^{\infty}$ to be integrated of orders one and zero respectively.[6] Moreover, Assumption 2 specifically excludes the process $\{(\mathbf{y}_t - \boldsymbol{\mu} - \boldsymbol{\gamma} t)\}_{t=1}^{\infty}$ being integrated of order two, or $I(2)$. Together these assumptions allow us to write the solution of (22.41) as an infinite-order moving average representation, given below. See Johansen (1991, Theorem 4.1, p. 1559) and Johansen (1995, Theorem 4.2, p. 49).

The differenced process $\{\Delta \mathbf{y}_t\}_{t=1}^{\infty}$ may be expressed as the infinite vector moving average process

$$\Delta \mathbf{y}_t = \mathbf{C}(L)(\mathbf{a}_0 + \mathbf{a}_1 t + \mathbf{u}_t) = \mathbf{b}_0 + \mathbf{b}_1 t + \mathbf{C}(L)\mathbf{u}_t, \; t = 1, 2, \ldots, \tag{22.49}$$

where $\mathbf{b}_0 \equiv \mathbf{C}\mathbf{a}_0 + \mathbf{C}^*\mathbf{a}_1$, $\mathbf{b}_1 \equiv \mathbf{C}\mathbf{a}_1$. The matrix lag polynomial $\mathbf{C}(L)$ is given by[7]

$$\mathbf{C}(L) \equiv \mathbf{I}_m + \sum_{j=1}^{\infty} \mathbf{C}_j L^j = \mathbf{C} + (1-L)\mathbf{C}^*(L), \; \mathbf{C}^*(L) \equiv \sum_{j=0}^{\infty} \mathbf{C}_j^* L^j,$$
$$\mathbf{C} \equiv \sum_{j=0}^{\infty} \mathbf{C}_j, \; \mathbf{C}^* \equiv \sum_{j=0}^{\infty} \mathbf{C}_j^*. \tag{22.50}$$

Now, as $\mathbf{C}(L)\boldsymbol{\Phi}(L) = \boldsymbol{\Phi}(L)\mathbf{C}(L) = (1-L)\mathbf{I}_m$, then $\boldsymbol{\Pi}\mathbf{C} = \mathbf{0}$ and $\mathbf{C}\boldsymbol{\Pi} = \mathbf{0}$, and in particular, $\mathbf{C} = \boldsymbol{\beta}_\perp(\boldsymbol{\alpha}'_\perp \boldsymbol{\Gamma} \boldsymbol{\beta}_\perp)^{-1}\boldsymbol{\alpha}'_\perp$. Re-expressing (22.49) in levels,

$$\mathbf{y}_t = \mathbf{y}_0 + \mathbf{b}_0 t + \mathbf{b}_1 \frac{t(t+1)}{2} + \mathbf{C}\mathbf{s}_t + \mathbf{C}^*(L)(\mathbf{u}_t - \mathbf{u}_0), \tag{22.51}$$

[6] See Johansen (1995), Definitions 3.2 and 3.3, p.35. That is, defining the difference operator $\Delta \equiv (1-L)$, the processes $\{\boldsymbol{\beta}'_\perp[\Delta(\mathbf{y}_t - \boldsymbol{\mu} - \boldsymbol{\gamma} t)]\}_{t=1}^{\infty}$, and $\{\boldsymbol{\beta}'(\mathbf{y}_t - \boldsymbol{\mu} - \boldsymbol{\gamma} t)\}_{t=1}^{\infty}$ admit stationary and invertible *ARMA* representations; see also Engle and Granger (1987, p. 252, Definition).

[7] The matrices $\{\mathbf{C}_i\}$ can be obtained from the recursions $\mathbf{C}_i = \sum_{j=1}^{p} \mathbf{C}_{i-j}\boldsymbol{\Phi}_j$, $i > 1$, $\mathbf{C}_0 = \mathbf{I}_m$, $\mathbf{C}_1 = -(\mathbf{I}_m - \boldsymbol{\Phi}_1)$, defining $\mathbf{C}_i = \mathbf{0}$, for $i < 0$. Similarly, for the matrices $\{\mathbf{C}_j^*\}$, $\mathbf{C}_j^* = \mathbf{C}_j + \mathbf{C}_{j-1}^*$, $j > 0$, $\mathbf{C}_0^* = \mathbf{I}_m - \mathbf{C}$.

where $\mathbf{s}_t \equiv \sum_{s=1}^{t} \mathbf{u}_s$, $t = 1, 2, \ldots$.

Adopting the $VAR(p)$ formulation (22.41) rather than the more usual (22.44), in which $\mathbf{a}_0$ and $\mathbf{a}_1$ are unrestricted, reveals immediately from (22.51) that the restrictions (22.45) on $\mathbf{a}_1$ induce $\mathbf{b}_1 = \mathbf{0}$ and ensure that the nature of the deterministic trending behaviour of the level process $\{\mathbf{y}_t\}_{t=1}^{\infty}$ remains invariant to the rank r of the long-run multiplier matrix $\mathbf{\Pi}$; that is, it is the deterministic trend of $\mathbf{y}_t$ which will be *linear* for all values of r, the rank of $\boldsymbol{\beta}$. Hence, the infinite moving average representation for the level process $\{\mathbf{y}_t\}_{t=1}^{\infty}$ is[8]

$$\mathbf{y}_t = \boldsymbol{\mu} + \boldsymbol{\gamma} t + \mathbf{C}\mathbf{s}_t + \mathbf{C}^*(L)\mathbf{u}_t, \tag{22.52}$$

where we have used the initialization $\mathbf{y}_0 \equiv \boldsymbol{\mu} + \mathbf{C}^*(L)\mathbf{u}_0$.[9] See also Johansen (1994) and Johansen (1995, Section 5.7, p. 80–84).[10] If, however, $\mathbf{a}_1$ were not subject to the restrictions (22.45), the quadratic trend term would be present in the level equation (22.51) apart from in the full rank stationary case $H_m : Rank[\mathbf{\Pi}] = m$ or $\mathbf{C} = \mathbf{0}$. However, $\mathbf{b}_1$ would be unconstrained under the null hypothesis of *no* cointegration; that is, $H_0 : Rank[\mathbf{\Pi}] = 0$, and $\mathbf{C}$ full rank. In the general case $H_r : Rank[\mathbf{\Pi}] = r$ of (22.47), this would imply different deterministic trending behaviour for $\{\mathbf{y}_t\}_{t=1}^{\infty}$ for differing values of the cointegrating rank r, with the number of independent quadratic deterministic trends, $m - r$, decreasing as r increases.

The above analysis further reveals that because cointegration is only concerned with the elimination of stochastic trends it does not rule out the possibility of deterministic trends in the cointegrating relations. Pre-multiplying both sides of (22.52) by the cointegrating matrix $\boldsymbol{\beta}'$, we obtain the cointegrating relations

$$\boldsymbol{\beta}'\mathbf{y}_t = \boldsymbol{\beta}'\boldsymbol{\mu} + (\boldsymbol{\beta}'\boldsymbol{\gamma})t + \boldsymbol{\beta}'\mathbf{C}^*(L)\mathbf{u}_t, \ t = 1, 2, \ldots, \tag{22.53}$$

which are trend-stationary. The restrictions $\boldsymbol{\beta}'\boldsymbol{\gamma} = \mathbf{0}$ in (22.53) are known as 'co-trending' restrictions (see Park (1992) and Ogaki (1992)). In general, we have $\boldsymbol{\beta}'\boldsymbol{\gamma} = \mathbf{0}$ if and only if in (22.44) $\mathbf{a}_1 = \mathbf{0}$ (Park (1992)). In this case, the representation for the $VAR(p)$ model, (22.44), and the cointegrating regression, (22.53), will contain no deterministic trends. However, the co-trending restrictions may not prove to be satisfactory in practice. It is therefore important that the linear combinations, $\boldsymbol{\beta}'\boldsymbol{\gamma}$ in (22.53) or, equivalently, $\mathbf{a}_1 = -\mathbf{\Pi}\boldsymbol{\gamma}$ in (22.44) is estimated along with the other parameters of the model.

22.9 Specification of the deterministics: five cases

Consider the following general $VEC(p-1)$ model

$$\Delta\mathbf{y}_t = \mathbf{a}_0 + \mathbf{a}_1 t - \mathbf{\Pi}\mathbf{y}_{t-1} + \sum_{j=1}^{p-1} \mathbf{\Gamma}_j \Delta\mathbf{y}_{t-j} + \mathbf{u}_t. \tag{22.54}$$

[8] From (22.42), as $\mathbf{C}(L)\mathbf{\Phi}(L) = (1-L)\mathbf{I}_m$ and, in particular, $\mathbf{C}\mathbf{\Pi} = \mathbf{0}$, $\mathbf{C}\mathbf{\Gamma} - \mathbf{C}^*\mathbf{\Pi} = \mathbf{I}_m$.

[9] Notice that the levels equation (22.51) could also have been obtained directly from (22.41) by noting $\Delta(\mathbf{y}_t - \boldsymbol{\mu} - \boldsymbol{\gamma} t) = \mathbf{C}(L)\mathbf{u}_t$, $t = 1, 2, \ldots$.

[10] As the cointegration rank hypothesis (22.47) may be alternatively and equivalently expressed as $H_r' : Rank[\mathbf{C}] = m - r$, $r = 0, \ldots, m$, it is interesting to note that, from (22.44) and (22.45), there are r linearly independent deterministic trends and, from (22.51), $m - r$ independent stochastic trends, $\mathbf{C}\mathbf{s}_t$, the combined total of which is m.

Given the above discussion, we can differentiate between five cases of interest:

Case I (no intercepts and no trends) $\mathbf{a}_0 = \mathbf{0}$ and $\mathbf{a}_1 = \mathbf{0}$.

This corresponds to a model with no deterministic components. In particular, model (22.54) becomes

$$\Delta \mathbf{y}_t = -\mathbf{\Pi} \mathbf{y}_{t-1} + \sum_{j=1}^{p-1} \Gamma_j \Delta \mathbf{y}_{t-j} + \mathbf{u}_t. \tag{22.55}$$

Case II (restricted intercepts and no trends) $\mathbf{a}_0 = \mathbf{\Pi}\boldsymbol{\mu}$ and $\mathbf{a}_1 = \mathbf{0}$.

In this case there are no linear trends in the data, and the constant term is restricted to appear in the cointegrating relation, so that (22.54) becomes

$$\Delta \mathbf{y}_t = -\mathbf{\Pi} \left(\mathbf{y}_{t-1} - \boldsymbol{\mu} \right) + \sum_{j=1}^{p-1} \Gamma_j \Delta \mathbf{y}_{t-j} + \mathbf{u}_t. \tag{22.56}$$

Case III (unrestricted intercepts and no trends) $\mathbf{a}_0 \neq \mathbf{0}$ and $\mathbf{a}_1 = \mathbf{0}$.

This case allows for linear trends in the data and non-zero intercept in the cointegrating relations. The model estimated is

$$\Delta \mathbf{y}_t = \mathbf{a}_0 + \mathbf{\Pi} \mathbf{y}_{t-1} + \sum_{j=1}^{p-1} \Gamma_j \Delta \mathbf{y}_{t-j} + \mathbf{u}_t. \tag{22.57}$$

Case IV (unrestricted intercepts and restricted trends) $\mathbf{a}_0 \neq \mathbf{0}$ and $\mathbf{a}_1 = \mathbf{\Pi}\boldsymbol{\gamma}$.

In this case trend coefficients are restricted to appear in the cointegrating relations. Thus

$$\Delta \mathbf{y}_t = \mathbf{a}_0 + \left[\mathbf{y}_{t-1} - \boldsymbol{\gamma} (t-1) \right] \mathbf{\Pi} + \sum_{j=1}^{p-1} \Gamma_j \Delta \mathbf{y}_{t-j} + \mathbf{u}_t. \tag{22.58}$$

Case V (unrestricted intercepts and trends) $\mathbf{a}_0 \neq \mathbf{0}$ and $\mathbf{a}_1 \neq \mathbf{0}$.

Here, there are no restrictions, and the model estimated is

$$\Delta \mathbf{y}_t = \mathbf{a}_0 + \mathbf{a}_1 t - \mathbf{\Pi} \mathbf{y}_{t-1} + \sum_{j=1}^{p-1} \Gamma_j \Delta \mathbf{y}_{t-j} + \mathbf{u}_t. \tag{22.59}$$

The maximum likelihood estimation for the above cases can be carried out using, instead of $\mathbf{S}_{01}$ and $\mathbf{S}_{11}$ defined by (22.31) and (22.32), the matrices

$$\mathbf{S}_{ij} = \frac{1}{T} \sum_{t=1}^{T} \mathbf{r}_{it} \mathbf{r}'_{jt}, \text{ for } i, j = 0, 1,$$

where $\mathbf{r}_{0t}$ and $\mathbf{r}_{1t}$, respectively, are the residual vectors computed using the following regressions:

Case I: $(\mathbf{a}_0 = \mathbf{a}_1 = \mathbf{0})$

$\mathbf{r}_{0t}$ is the residual vector from the *OLS* regressions of $\Delta\mathbf{y}_t$ on $\left(\Delta\mathbf{y}_{t-1}, \Delta\mathbf{y}_{t-2}, \ldots, \Delta\mathbf{y}_{t-p+1}\right)$, and $\mathbf{r}_{1t}$ is the residual vector from the *OLS* regressions of $\mathbf{y}_{t-1}$ on $\left(\Delta\mathbf{y}_{t-1}, \Delta\mathbf{y}_{t-2}, \ldots, \Delta\mathbf{y}_{t-p+1}\right)$.

Case II: $(\mathbf{a}_1 = \mathbf{0}, \mathbf{a}_0 = \boldsymbol{\Pi}\boldsymbol{\mu})$

$\mathbf{r}_{0t}$ is the residual vector from the *OLS* regressions of $\Delta\mathbf{y}_t$ on $\left(\Delta\mathbf{y}_{t-1}, \Delta\mathbf{y}_{t-2}, \ldots, \Delta\mathbf{y}_{t-p+1}\right)$, and $\mathbf{r}_{1t}$ is the residual vector from the *OLS* regressions of $\begin{pmatrix} 1 \\ \mathbf{y}_{t-1} \end{pmatrix}$ on $\left(\Delta\mathbf{y}_{t-1}, \Delta\mathbf{y}_{t-2}, \ldots, \Delta\mathbf{y}_{t-p+1}\right)$.

Case III: $(\mathbf{a}_1 = \mathbf{0}, \mathbf{a}_0 \neq \mathbf{0})$

$\mathbf{r}_{0t}$ is the residual vector from the *OLS* regressions of $\Delta\mathbf{y}_t$ on $\left(1, \Delta\mathbf{y}_{t-1}, \Delta\mathbf{y}_{t-2}, \ldots, \Delta\mathbf{y}_{t-p+1}\right)$, and $\mathbf{r}_{1t}$ is the residual vector from the *OLS* regressions of $\mathbf{y}_{t-1}$ on $\left(1, \Delta\mathbf{y}_{t-1}, \Delta\mathbf{y}_{t-2}, \ldots, \Delta\mathbf{y}_{t-p+1}\right)$.

Case IV: $(\mathbf{a}_0 \neq \mathbf{0}, \mathbf{a}_1 = \boldsymbol{\Pi}\boldsymbol{\gamma})$

$\mathbf{r}_{0t}$ is the residual vector from the *OLS* regressions of $\Delta\mathbf{y}_t$ on $\left(1, \Delta\mathbf{y}_{t-1}, \Delta\mathbf{y}_{t-2}, \ldots, \Delta\mathbf{y}_{t-p+1}\right)$, and $\mathbf{r}_{1t}$ is the residual vector from the *OLS* regressions of $\begin{pmatrix} t \\ \mathbf{y}_{t-1} \end{pmatrix}$ on $\left(1, \Delta\mathbf{y}_{t-1}, \Delta\mathbf{y}_{t-2}, \ldots, \Delta\mathbf{y}_{t-p+1}\right)$.

Case V: $(\mathbf{a}_0 \neq \mathbf{0}, \mathbf{a}_1 \neq \mathbf{0})$

$\mathbf{r}_{0t}$ is the residual vector from the *OLS* regressions of $\Delta\mathbf{y}_t$ on $\left(1, t, \Delta\mathbf{y}_{t-1}, \Delta\mathbf{y}_{t-2}, \ldots, \Delta\mathbf{y}_{t-p+1}\right)$, and $\mathbf{r}_{1t}$ is the residual vector from the *OLS* regressions of $\mathbf{y}_{t-1}$ on $\left(1, t, \Delta\mathbf{y}_{t-1}, \Delta\mathbf{y}_{t-2}, \ldots, \Delta\mathbf{y}_{t-p+1}\right)$.

The rest of the analysis will be unaffected.

22.10 Testing for cointegration in *VAR* models

22.10.1 Maximum eigenvalue statistic

Residual-based and other cointegration tests described in Section 22.3 only consider the case where $r = 0$ against the alternative that $r > 0$. They are not suited to cases where there are multiple cointegrating relations and we are interested in estimating r. In such cases we need to follow Johansen (1991) and base the cointegration tests on the *VAR* model given by (22.39). Suppose it is of interest to test the null hypothesis of r cointegrating relations

$$H_r : \text{ Rank}(\Pi) = r < m, \tag{22.60}$$

against the alternative hypothesis

$$H_{r+1} : \text{ Rank}(\Pi) = r + 1, \; r = 0, 1, 2, \ldots, m - 1.$$

The log-likelihood ratio statistic for testing the null of r cointegrating relations against the alternative that there are $r+1$ of them is defined by

$$LR\big(H_r \mid H_{r+1}\big) = 2\Big[\ell^c\big(\hat{\boldsymbol{\beta}}; r+1\big) - \ell^c\big(\hat{\boldsymbol{\beta}}; r\big)\Big],$$

where $\ell^c\big(\hat{\boldsymbol{\beta}}; r+1\big)$ and $\ell^c\big(\hat{\boldsymbol{\beta}}; r\big)$ refer to the maximized log-likelihood values under H_{r+1} and H_r, respectively. Hence, by substituting the expression for the maximized concentrated likelihood (see equation (22.38) for the $VAR(1)$) we obtain

$$LR\left(H_r \mid H_{r+1}\right) = -T\log\big(1-\hat{\lambda}_{r+1}\big), \tag{22.61}$$

where $\hat{\lambda}_r$ is defined in (22.37). See Johansen (1991) for further details.

22.10.2 Trace statistic

Suppose now that the interest is in testing the null hypothesis

$$H_r: \ \text{Rank}\,(\Pi) = r < m,$$

against the alternative of trend-stationary, that is

$$H_m: \ \text{Rank}\,(\Pi) = m,$$

for $r = 0, 1, 2, \ldots, m-1$. The log-likelihood ratio statistic for this test is given by

$$LR\big(H_r \mid H_m\big) = 2[\ell^c\big(\hat{\boldsymbol{\beta}}; m\big) - \ell^c\big(\hat{\boldsymbol{\beta}}; r\big)],$$

or

$$LR\left(H_r \mid H_m\right) = -T\sum_{i=r+1}^{m}\log\big(1-\hat{\lambda}_i\big), \tag{22.62}$$

where $\hat{\lambda}_{r+1} > \hat{\lambda}_{r+2} > \ldots > \hat{\lambda}_m$ are the smallest $m-r$ eigenvalues of $\mathbf{S}_{10}\mathbf{S}_{00}^{-1}\mathbf{S}_{01}\mathbf{S}_{11}^{-1}$.

We note that, unlike residual-based cointegration tests, tests based on the maximum eigenvalue or trace statistics are invariant to the ordering of variables, namely, they are not affected when the variables in $\mathbf{y}_t$ are re-ordered or replaced by other linear combinations.

Next we derive the asymptotic distribution of the Trace statistic for model (22.27).

22.10.3 The asymptotic distribution of the trace statistic

For simplicity, assume the model has no intercepts or linear trends. Suppose that $\mathbf{u}_t$ satisfies the following additional assumptions,

Assumption 3: The error process $\mathbf{u}_t = (u_{1t}, u_{2t}, \ldots, u_{mt})'$ is such that

(a) $E\left[\mathbf{u}_t | \left\{\mathbf{y}_{t-i}\right\}_{i=1}^{t-1}, \mathbf{y}_0\right] = \mathbf{0}$ and $Var\left[\mathbf{u}_t | \left\{\mathbf{y}_{t-i}\right\}_{i=1}^{t-1}, \mathbf{y}_0\right] = \mathbf{\Sigma}$ for all t, where $\mathbf{\Sigma}$ is a positive definite symmetric matrix.

(b) $sup_t \mathrm{E}(\|\mathbf{u}_t\|^s) < \infty$, for some $s > 2$.

Assumption 3(a) states that the error process $\{\mathbf{u}_t\}_{t=-\infty}^{\infty}$ is a martingale difference sequence with constant conditional variance; hence, $\{\mathbf{u}_t\}_{t=-\infty}^{\infty}$ is an uncorrelated process. Assumption 3 is required for the multivariate invariance principle to hold (see Appendix B, Section B.13.1).

Consider the trace statistic defined by (22.62) under $r = 0$

$$LR\left(H_0 \mid H_{m_y}\right) = -T\sum_{i=1}^{m}\log\left(1-\hat{\lambda}_i\right) = T\sum_{i=1}^{m}\hat{\lambda}_i + o_p(1),$$

and note that

$$\sum_{i=1}^{m}\hat{\lambda}_i = Tr\left(\mathbf{S}_{10}\mathbf{S}_{00}^{-1}\mathbf{S}_{01}\mathbf{S}_{11}^{-1}\right).$$

Under the null of no cointegration the $VAR(1)$ model (without intercepts or linear trends) implies

$$\mathbf{y}_t = \mathbf{y}_0 + \sum_{j=1}^{t}\mathbf{u}_j = \mathbf{y}_0 + \mathbf{s}_t,$$

and similarly

$$\mathbf{y}_{t-1} = \mathbf{y}_0 + \sum_{j=1}^{t-1}\mathbf{u}_j = \mathbf{y}_0 + \mathbf{s}_{t-1}.$$

Hence

$$T^{-1}\mathbf{S}_{11} = \frac{1}{T^2}\sum_{t=1}^{T}\mathbf{y}_{t-1}\mathbf{y}_{t-1}' = \frac{\mathbf{y}_0\mathbf{y}_0'}{T^2} + \frac{1}{T^2}\sum_{t=1}^{T}\mathbf{s}_{t-1}\mathbf{s}_{t-1}' + \frac{1}{T^2}\mathbf{y}_0\sum_{t=1}^{T}\mathbf{s}_{t-1}' + \left(\frac{1}{T^2}\sum_{t=1}^{T}\mathbf{s}_{t-1}\right)\mathbf{y}_0'.$$

Using the probability limits in Appendix B (see in particular equations (B.52)–(B.54)) as $T \to \infty$,

$$\frac{1}{T^2}\sum_{t=1}^{T}\mathbf{s}_{t-1} \to \mathbf{0}, \quad \frac{\mathbf{y}_0\mathbf{y}_0'}{T^2} \to \mathbf{0},$$

$$T^{-2}\sum_{t=1}^{T}\mathbf{s}_t\mathbf{s}_t' \Rightarrow \int_0^1 \mathbf{W}(a)\mathbf{W}(a)'da.$$

Hence, under $r = 0$ we have

$$T^{-1}\mathbf{S}_{11} \Rightarrow \int_0^1 \mathbf{W}(a)\mathbf{W}(a)'da,$$

where $\mathbf{W}(a)$ is an m-dimensional Brownian motion with the covariance matrix, $\mathbf{\Sigma}$. Similarly, it is possible to show that

$$\mathbf{S}_{01} = \frac{1}{T}\sum_{t=1}^{T} \Delta\mathbf{y}_t\mathbf{y}_{t-1}' = \frac{1}{T}\sum_{t=1}^{T} \mathbf{u}_t\left(\mathbf{y}_0 + \mathbf{s}_{t-1}\right)' \Rightarrow \int_0^1 d\mathbf{W}(a)\mathbf{W}(a)',$$

$$\mathbf{S}_{00} = \frac{1}{T}\sum_{t=1}^{T} \Delta\mathbf{y}_t\Delta\mathbf{y}_t' \xrightarrow{p} \mathbf{\Sigma}.$$

It follows that

$$LR\left(H_0 \mid H_m\right) = -T\sum_{i=1}^{m}\log\left(1-\hat{\lambda}_i\right)$$
$$\Rightarrow Tr\left\{\left[\int_0^1 \mathbf{W}(a)\mathbf{W}(a)'da\right]^{-1}\left[\int_0^1 \mathbf{W}(a)d\mathbf{W}'(a)\right]\mathbf{\Sigma}^{-1}\left[\int_0^1 d\mathbf{W}(a)\mathbf{W}(a)'\right]\right\}.$$

Denote the standard Brownian motion by $\mathbf{B}(a) = \mathbf{\Sigma}^{-1/2}\mathbf{W}(a)$. Then

$$-T\sum_{i=1}^{m}\log\left(1-\hat{\lambda}_i\right)$$
$$\Rightarrow Tr\left\{\left[\int_0^1 d\mathbf{B}(a)\mathbf{B}(a)'\right]\left[\int_0^1 \mathbf{B}(a)\mathbf{B}(a)'da\right]^{-1}\left[\int_0^1 \mathbf{B}(a)d\mathbf{B}(a)'\right]\right\}.$$

This is a multivariate generalization of the Dickey–Fuller distribution used to test the unit root hypothesis (for the basic case of no intercept or trend) where $m = 1$. Note that the asymptotic distribution of the trace statistic does not depend on $\mathbf{\Sigma}$, and depends only on the dimension of $\mathbf{y}_t$, m. It is also easily seen that a test based on $-T\sum_{i=1}^{m}\log\left(1-\hat{\lambda}_i\right)$ will be consistent, in the sense that the power of the test will tend to unity as $T \to \infty$, if $r > 0$.

The critical values for the maximum eigenvalue and the trace statistics defined by (22.61) and (22.62), respectively, depend on m and whether the *VECM* contains intercepts and/or trends and whether these are restricted. These critical values are available in MacKinnon, Haug, and Michelis (1999) (see also Osterwald-Lenum (1992)).

Monte Carlo simulation results indicate that these cointegrating rank test statistics generally tend to under-reject in small samples (see Pesaran, Shin, and Smith (2000)). Appropriate critical values can be computed by adopting a bootstrap approach, as outlined in Section 22.12. We also refer to Lütkepohl, Saikkonen, and Trenkler (2001) for a comparison of the properties of maximum eigenvalue and trace tests under a set of local alternatives.

22.11 Long-run structural modelling

As we have seen, the estimation of the *VECM* (22.54) subject to rank restrictions on the long-run multiplier matrix, $\mathbf{\Pi}$, does not generally lead to a unique choice for the cointegrating relations. The identification of $\boldsymbol{\beta}$ (in $\mathbf{\Pi} = \boldsymbol{\alpha}\boldsymbol{\beta}'$) requires at least r restrictions per each of the r cointegrating relations. In the simple case where $r = 1$, the one restriction needed to identify the cointegrating relation can be viewed as a 'normalizing' restriction which could be applied to the coefficient of any one of the integrated variables that enter the cointegrating relation, so long as it is *a priori* known that the coefficient which is being normalized is not zero. Therefore, the choice of the normalization is not innocuous and is itself based on the *a priori* identifying information that the variable associated with the coefficient being normalized belongs to the cointegrating relation. However, in the more general case where $r > 1$, the number of such 'normalizing' restrictions is just equal to r which needs to be supplemented with further $r^2 - r$ *a priori* restrictions; preferably obtained from a suitable long-run economic theory.

Identification schemes have been proposed by Johansen (1991), Phillips (1991), Phillips (1995), and Pesaran and Shin (2002). In what follows, we introduce a framework for identification of cointegrated systems when the cointegrating coefficients are subject to restrictions obtained from economic theory or other relevant *a priori* information. See Pesaran and Shin (2002) for further details.

22.11.1 Identification of the cointegrating relations

The structural estimation of the cointegrating relations requires maximization of the concentrated log-likelihood function (22.36) subject to appropriate just-identifying or overidentifying restrictions on $\boldsymbol{\beta}$. The just-identifying restrictions utilized by Johansen (1991) make use of the observation matrices $\mathbf{A}_T$ and $\mathbf{B}_T$ defined by (22.35), and are often referred to as 'empirical' or 'statistical' identifying restrictions. This is in contrast to *a priori* restrictions imposed on $\boldsymbol{\beta}$ which are independent of particular values of $\mathbf{A}_T$ and $\mathbf{B}_T$. Johansen's estimates of $\boldsymbol{\beta}$, which we denote by $\hat{\boldsymbol{\beta}}_J$, are obtained as the first r eigenvectors of $\mathbf{B}_T - \mathbf{A}_T$ with respect to $\mathbf{B}_T$, satisfying the following 'normalization' and 'orthogonalization' restrictions

$$\hat{\boldsymbol{\beta}}_J' \mathbf{B}_T \hat{\boldsymbol{\beta}}_J = \mathbf{I}_r, \tag{22.63}$$

and

$$\hat{\boldsymbol{\beta}}_{iJ}' \left(\mathbf{B}_T - \mathbf{A}_T\right) \hat{\boldsymbol{\beta}}_{jJ} = 0, \qquad i \neq j, \quad i, j = 1, 2, \ldots, r, \tag{22.64}$$

where $\hat{\boldsymbol{\beta}}_{iJ}$ represents the ith column of $\hat{\boldsymbol{\beta}}_J$. The conditions (22.63) and (22.64) together exactly impose r^2 just-identifying restrictions on $\boldsymbol{\beta}$. It is, however, clear that the r^2 restrictions in (22.63) and (22.64) are adopted for their mathematical convenience and not because they are meaningful from the perspectives of any long-run economic theory.

A more satisfactory procedure would be to directly estimate the concentrated log-likelihood function (22.36) subject to exact or over-identifying *a priori* restrictions obtained from the long-run equilibrium properties of a suitable underlying economic model (on this see, Pesaran (1997)). We can formulate the following general linear restrictions on the elements of $\boldsymbol{\beta}$

$$\mathbf{R}\,\text{vec}(\boldsymbol{\beta}) = \mathbf{b}, \tag{22.65}$$

where $\mathbf{R}$ and $\mathbf{b}$ are $k \times rm$ matrix and $k \times 1$ vector of known constants (with $\text{Rank}(\mathbf{R}) = k$), and $\text{vec}(\boldsymbol{\beta})$ is the $rm \times 1$ vector of long-run coefficients, which stacks the r columns of $\boldsymbol{\beta}$ into a vector. If the matrix $\mathbf{R}$ is block-diagonal then (22.65) can be written as

$$\mathbf{R}_i\boldsymbol{\beta}_i = \mathbf{b}_i\,,\ \ i = 1, 2, \ldots, r, \tag{22.66}$$

where $\boldsymbol{\beta}_i$ is the ith cointegrating vector, and $\mathbf{R}_i$ is the ith block in matrix $\mathbf{R}$, and $\mathbf{b}_i$ is defined by $\mathbf{b}' = \left(\mathbf{b}_1', \mathbf{b}_2', \ldots, \mathbf{b}_r'\right)$. In this case the necessary and sufficient conditions for identification of the cointegrating vectors are given by

$$\text{Rank}\,(\mathbf{R}_i\boldsymbol{\beta}) = r, \qquad i = 1, 2, \ldots, r. \tag{22.67}$$

This result also implies that there must be at least r independent restrictions on each of the r cointegrating vectors.

The identification condition in the case where $\mathbf{R}$ is not block diagonal is given by

$$\text{Rank}\,\{\mathbf{R}\,(\mathbf{I}_r \otimes \boldsymbol{\beta})\} = r^2. \tag{22.68}$$

A necessary condition for (22.68) to hold is given by the order condition $k \geq r^2$. Three cases of interest can be distinguished:

1. $k < r^2$, the under-identified case,
2. $k = r^2$, the exactly identified case,
3. $k > r^2$, the over-identified case.

22.11.2 Estimation of the cointegrating relations under general linear restrictions

We now focus on cases when the long-run restrictions are exactly identified (i.e. $k = r^2$), and when there are over-identifying restriction on the cointegrating vectors (i.e. $k > r^2$).

Exactly identified case $\left(k = r^2\right)$

In this case the *ML* estimator of $\boldsymbol{\beta}$ that satisfies the restrictions (22.65) is readily computed using the Johansen's estimates, $\hat{\boldsymbol{\beta}}_J$. We have

$$\text{vec}(\hat{\boldsymbol{\beta}}) = \left(\mathbf{I}_r \otimes \hat{\boldsymbol{\beta}}_J\right)\left[\mathbf{R}\left(\mathbf{I}_r \otimes \hat{\boldsymbol{\beta}}_J\right)\right]^{-1}\mathbf{b}, \tag{22.69}$$

Pesaran and Shin (2002) proved that this estimator satisfies the restriction (22.65), and is invariant to nonsingular transformations of the cointegrating space spanned by columns of $\hat{\boldsymbol{\beta}}$.

Over-identified case $\left(k > r^2\right)$

In this case, there are $k - r^2$ additional restrictions that need to be taken into account at the estimation stage. This can be done by maximizing the concentrated log-likelihood function (22.36),

subject to the restrictions given by (22.65). We assume that the normalization restrictions on each of the r cointegrating vectors is also included in $\mathbf{R}\,\text{vec}\,(\boldsymbol{\beta}) = \mathbf{b}$. The Lagrangian function for this problem is given by

$$\begin{aligned}\Lambda\,(\boldsymbol{\theta}, \boldsymbol{\lambda}) &= \frac{1}{T}\ell^c(\boldsymbol{\theta}; r) - \frac{1}{2}\boldsymbol{\lambda}'(\mathbf{R}\boldsymbol{\theta} - \mathbf{b}) \\ &= \text{constant} - \frac{1}{2}\left\{\log\left|\boldsymbol{\beta}'\mathbf{A}_T\boldsymbol{\beta}\right| - \log\left|\boldsymbol{\beta}'\mathbf{B}_T\boldsymbol{\beta}\right|\right\} - \frac{1}{2}\boldsymbol{\lambda}'(\mathbf{R}\boldsymbol{\theta} - \mathbf{b}),\end{aligned}$$

where $\boldsymbol{\theta} = \text{vec}(\boldsymbol{\beta})$, $\boldsymbol{\lambda}$ is a $k \times 1$ vector of Lagrange multipliers, and $\mathbf{A}_T$ and $\mathbf{B}_T$ are defined in (22.35). Then the first-order conditions for this optimization problem are given by

$$\mathbf{d}\left(\tilde{\boldsymbol{\theta}}\right) = \mathbf{R}'\tilde{\boldsymbol{\lambda}}, \tag{22.70}$$

$$\mathbf{R}\tilde{\boldsymbol{\theta}} = \mathbf{b}, \tag{22.71}$$

where $\tilde{\boldsymbol{\theta}}$ and $\tilde{\boldsymbol{\lambda}}$ stand for the restricted *ML* estimators, and $\mathbf{d}(\tilde{\boldsymbol{\theta}})$ is the score function defined by

$$\mathbf{d}(\tilde{\boldsymbol{\theta}}) = \left\{\left[\left(\tilde{\boldsymbol{\beta}}'\mathbf{A}_T\tilde{\boldsymbol{\beta}}\right)^{-1} \otimes \mathbf{A}_T\right] - \left[\left(\tilde{\boldsymbol{\beta}}'\mathbf{B}_T\tilde{\boldsymbol{\beta}}\right)^{-1} \otimes \mathbf{B}_T\right]\right\}\tilde{\boldsymbol{\theta}}. \tag{22.72}$$

Computation of $\tilde{\boldsymbol{\theta}}$ can be obtained by numerical methods such as the Newton Raphson procedure.

Evidence on the small sample properties of alternative methods of estimating the cointegrating relations is provided in Gonzalo (1994), who shows that the Johansen maximum likelihood approach is to be preferred to the other alternatives proposed in the literature.

22.11.3 Log-likelihood ratio statistics for tests of over-identifying restrictions on the cointegrating relations

Consider now the problem of testing over-identifying restrictions on the coefficients of the cointegrating (or long-run) relations. Suppose there are r cointegrating relations and the interest is to test the restrictions

$$\mathbf{R}\,\text{vec}(\boldsymbol{\beta}) = \mathbf{b}, \tag{22.73}$$

where $\mathbf{R}$ is a $k \times mr$ matrix and b is a $k \times 1$ vector of known constants such that $\text{Rank}(\mathbf{R}) = k > r^2$. As before let $\boldsymbol{\theta} = \text{vec}(\boldsymbol{\beta})$ and decompose the k restriction defined by (22.73) into r^2 and $k - r^2$ set of restrictions

$$\underset{r^2 \times rm}{\mathbf{R}_A}\ \ \underset{rm \times 1}{\boldsymbol{\theta}} = \underset{r^2 \times 1}{\mathbf{b}_A}, \tag{22.74}$$

$$\underset{(k - r^2) \times rm}{\mathbf{R}_B}\ \ \underset{rm \times 1}{\boldsymbol{\theta}} = \underset{(k - r^2) \times 1}{\mathbf{b}_B}, \tag{22.75}$$

where $\mathbf{R} = \left(\mathbf{R}_A', \mathbf{R}_B'\right)'$, and $\mathbf{b} = (\mathbf{b}_A, \mathbf{b}_B)$, such that $\text{Rank}(\mathbf{R}_A) = r^2$, $\text{Rank}(\mathbf{R}_B) = k - r^2$, and $\mathbf{b}_A \neq 0$. Without loss of generality the restrictions characterized by (22.74) can be viewed as the just-identifying restrictions, and the remaining restriction defined by (22.75) will then constitute the $k - r^2$ over-identifying restrictions. Let $\hat{\boldsymbol{\theta}}$ be the *ML* estimators of $\boldsymbol{\theta}$ obtained subject to the r^2 exactly-identifying restrictions, and $\tilde{\boldsymbol{\theta}}$ be the *ML* estimators of $\boldsymbol{\theta}$ obtained under all the k restriction in (22.73). Then the log-likelihood ratio statistic for testing the over-identifying restrictions is given by

$$LR\left(\mathbf{R} \,|\mathbf{R}_A\right) = 2\left[\ell^c\left(\hat{\boldsymbol{\theta}}; r\right) - \ell^c\left(\tilde{\boldsymbol{\theta}}; r\right)\right], \tag{22.76}$$

where $\ell^c\left(\hat{\boldsymbol{\theta}}; r\right)$ is given by (22.38) and represents the maximized value of the log-likelihood function under the just-identifying restriction, (say $\mathbf{R}_A\boldsymbol{\theta} = \mathbf{b}_A$), and $\ell^c\left(\tilde{\boldsymbol{\theta}}; r\right)$ is the maximized value of the log-likelihood function under the k just- and over-identifying restrictions given by (22.73).

Pesaran and Shin (2002) proved that, under the null hypothesis that the restrictions (22.73) hold, the log-likelihood ratio statistic $LR\left(\mathbf{R} \,|\mathbf{R}_A\right)$ defined by (22.76) is asymptotically distributed as a χ^2 variate with degrees of freedom equal to the number of the over-identifying restrictions, namely $k - r^2 > 0$.

The above testing procedure is also applicable when interest is on testing restrictions on a single cointegrating vector of a subset of cointegrating vectors. For this purpose, one simply needs to impose just-identifying restrictions on all the vectors except for the vector(s) that are to be subject to the over-identifying restrictions. The resultant test statistic will be invariant to the nature of the just-identifying restrictions. Note that this test of the over-identifying restrictions on the cointegrating relations pre-assumes that the variables $\mathbf{y}_t$ are $I(1)$, and that the number of cointegrating relations, r, is correctly chosen. See Pesaran and Shin (2002).

22.12 Small sample properties of test statistics

The distributions of the maximal eigenvalue and trace statistics ((22.61) and (22.62)) and the log-likelihood ratio tests of the over-identifying restrictions ((22.76)) are appropriate only asymptotically. Moreover, Monte Carlo results show that these asymptotic tests are valid only when T is reasonably large, and m and p relatively small. This suggests that in practice care should be exercised in interpreting the test statistics obtained.

In some cases, it is advisable to use bootstrapped critical values instead of the asymptotic ones. Suppose that the *VEC* model of (22.16) has been estimated subject to the just- or over-identifying restrictions suggested by economic theory. Using the observed initial values for each variable, it is possible to generate S new samples of data (of the same size as the original) under the hypothesis that the estimated version of (22.16) is the true data generating process. For each of the S replications of the data, the tests of the cointegrating rank and of the over-identifying restrictions can be carried out and, hence, distributions of the test statistics are obtained which take into account the small sample of data available when calculating the statistics. Working at the α per cent level of significance, critical values which take into account the small sample properties of the tests can be obtained from the simulated distribution of the bootstrapped test statistics, by ordering them and then selecting the appropriate critical value that match the desired α per cent level.

More specifically, suppose that the model in (22.16) has been estimated under the exact- or over-identifying restrictions given by (22.65). We therefore have estimates of the cointegrating vectors, $\hat{\boldsymbol{\beta}}$, of the short-run parameters, $\hat{\boldsymbol{\alpha}}$, $\hat{\boldsymbol{\Gamma}}_i$, and the elements of the covariance matrix, $\hat{\boldsymbol{\Sigma}}$. Taking the p lagged values of the $\mathbf{y}_t$ observed just prior to the sample as fixed, for the sth replication, we can recursively simulate the values of $\Delta\mathbf{y}_t^{(s)}$, $s = 1, 2, \ldots, S$, using

$$\Delta\mathbf{y}_t^{(s)} = -\hat{\boldsymbol{\alpha}}\hat{\boldsymbol{\beta}}'\mathbf{y}_{t-1}^{(s)} + \sum_{i=1}^{p-1}\hat{\boldsymbol{\Gamma}}_i\Delta\mathbf{y}_{t-i} + \mathbf{u}_t^{(s)},\ t = 1, 2, \ldots, T. \tag{22.77}$$

The simulated errors, $\mathbf{u}_t^{(s)}$, can be obtained in two alternative ways, so that the contemporaneous correlations that exist across the errors in the different equations of the *VAR* model are taken into account and maintained. The first is a *parametric* method where the errors are drawn from an assumed probability distribution function. Alternatively, one could employ a *non-parametric* procedure. The latter is slightly more complicated and is based on re-sampling techniques in which the simulated errors are obtained by a random draw from the in-sample estimated residuals (e.g., Hall (1992)).

22.12.1 Parametric approach

Under the parametric approach the errors are drawn, for example, from a multivariate distribution with zero means and the covariance matrix, $\hat{\boldsymbol{\Sigma}}^{(s)}$. To obtain the simulated errors for m variables over, say, h periods, we first generate mh draws from an assumed *IID* distribution which we denote by $\left\{\boldsymbol{\epsilon}_t^{(s)}\right\}$, for $t = 1, 2, \ldots, h$. These are then used to obtain $\mathbf{u}_t^{(s)}$ computed as $\mathbf{u}_t^{(s)} = \hat{\mathbf{P}}^{(s)}\boldsymbol{\epsilon}_t^{(s)}$, where $\hat{\mathbf{P}}^{(s)}$ is the lower (upper) triangular Choleski factor of $\hat{\boldsymbol{\Sigma}}^{(s)}$ such that $\hat{\boldsymbol{\Sigma}}^{(s)} = \hat{\mathbf{P}}^{(s)}\hat{\mathbf{P}}^{(s)\prime}$, and $\hat{\boldsymbol{\Sigma}}^{(s)}$ is the estimate of $\boldsymbol{\Sigma}$ in the sth replication of the bootstrap procedure set out above. In the absence of parameter uncertainty, we obtain $\mathbf{u}_t^{(s)} = \hat{\mathbf{P}}\epsilon_t^{(s)}$, where $\hat{\mathbf{P}}$ is the lower triangular Choleski factor of $\hat{\boldsymbol{\Sigma}}$.

22.12.2 Non-parametric approach

The most obvious non-parametric approach to generating the simulated errors, which we denote 'Method 1', is simply to take h random draws with replacements from the in-sample residual vectors. The simulated errors thus obtained clearly have the same distribution and covariance structure as that observed in the original sample. However, this procedure is subject to the criticism that it could introduce dependence across the different random samples since the pseudo-random draws are made from the same set of T-dimensional vector of residuals. An alternative non-parametric method for generating simulated errors, 'Method 2', makes use of the Choleski decomposition of the estimated covariance employed in the parametric approach. For a given choice of $\hat{\mathbf{P}}^{(s)}$ a set of mT transformed error terms $\left\{\boldsymbol{\epsilon}_1^{(s)}, \boldsymbol{\epsilon}_2^{(s)}, \ldots, \boldsymbol{\epsilon}_T^{(s)}\right\}$ are computed such that $\boldsymbol{\epsilon}_t^{(s)} = \left(\hat{\mathbf{P}}^{(s)}\right)^{-1}\mathbf{u}_t^{(s)}$, $t = 1, 2, \ldots, T$. The mT individual error terms are uncorrelated with each other, but retain the distributional information contained in the original observed errors. A set of mh simulated errors is then obtained by drawing with replacement from these transformed

residuals, denoted by $\{\boldsymbol{\zeta}_1^{(s)}, \boldsymbol{\zeta}_2^{(s)}, \ldots, \boldsymbol{\zeta}_h^{(s)}\}$ these are then used to obtain $\{\mathbf{u}_1^{(s)}, \mathbf{u}_2^{(s)}, \ldots, \mathbf{u}_h^{(s)}\}$, where $\mathbf{u}_t^{(s)} = \hat{\mathbf{P}}_t^{(s)} \boldsymbol{\zeta}_t^{(s)}$.

Given that the $\hat{\mathbf{P}}^{(s)}$ matrix is used to generate the simulated errors, it is clear that $\mathbf{u}_t^{(s)}$ has the same covariance structure as the original estimated errors. Being based on errors drawn at random from the transformed residuals, these simulated errors will display the same distributional features. Further, given that the re-sampling occurs from the mT transformed error terms, Method 2 also has the advantage over Method 1 that the dependence introduced through sampling with replacement is likely to be less problematic.

The two non-parametric approaches described above have the advantage over the parametric approach in that they make no distributional assumptions on the error terms, and are better able to capture the uncertainties arising from (possibly rare) extreme observations. However, they suffer from the fact that they require random sampling with replacement, which inevitably introduces dependence across the simulated errors.

Having generated the $\Delta\mathbf{y}_t^{(s)}$, $t = 1, 2, \ldots, T$, and making use of the observed $\Delta\mathbf{x}_t$, it is straightforward to estimate the *VEC* of (22.16) subject to just-identifying restrictions and then subject to the over-identifying restrictions of (22.65) to obtain a sequence of log-likelihood ratio test statistics, $LR^{(s)}$, each testing the validity of the over-identifying restrictions in the sth simulated dataset, $s = 1, 2, \ldots, S$. These statistics can be sorted in an ascending order and the critical value associated with the desired level of significance obtained. Since the simulated data has been generated under (22.16) which incorporates the over-identifying restrictions, then the use of the simulated critical values is likely to be more appropriate than the asymptotic critical values for testing the over-identifying restrictions. Hence, for example, the value of $LR^{(s)}$ which exceeds 95 per cent of the observed statistics represents the appropriate 95 per cent critical value for the test of the validity of the over-identifying restrictions.

Finally, it is worth bearing in mind that the maximum likelihood estimation of the *VECM* can be time-consuming, especially if one is to be sure that all of the estimates relate to global and not local maxima of the underlying likelihood function. In practice, the choice of an optimization algorithm is likely to be important in this exercise, and the simulated annealing algorithm discussed in Goffe, Ferrier, and Rogers (1994) can prove useful in this respect.

22.13 Estimation of the short-run parameters of the *VEC* model

Having computed the *ML* estimates of the cointegrating vectors $\hat{\boldsymbol{\beta}}$, obtained under the exact and/or over-identifying restrictions given by (23.28), the *ML* estimates of the short-run parameters $(\boldsymbol{\alpha}, \boldsymbol{\Gamma}_1, \ldots, \boldsymbol{\Gamma}_{p-1})$ in (22.39) can be computed by the *OLS* regressions of $\Delta\mathbf{y}_t$ on $(\hat{\boldsymbol{\xi}}_t, \Delta\mathbf{y}_{t-1}, \ldots, \Delta\mathbf{y}_{t-p+1})$, where $\hat{\boldsymbol{\xi}}_t = \hat{\boldsymbol{\beta}}'\mathbf{y}_{t-1}$ is the *ML* estimator of $\boldsymbol{\xi}_t = \boldsymbol{\beta}'\mathbf{y}_{t-1}$. Notice that $\hat{\boldsymbol{\beta}}$ is super-consistent (T-consistent), while the *ML* estimators of the short-run parameters are $\sqrt{T}$-consistent.

It is worth emphasizing that, having established the form of the long-run relations, then standard *OLS* regression methods and standard testing procedures can be applied. All the right-hand side variables in the error correction regression models are stationary and are the same across all the equations in the *VECM*. In these circumstances, *OLS* is the appropriate estimation procedure and diagnostic statistics for residual serial correlation, normality, heteroskedasticity and

functional form misspecifications can be readily computed, based on these *OLS* regressions, in the usual manner. Further discussion of the validity of standard diagnostic test procedures when different estimation procedures are adopted in models involving unit roots and cointegrating relations is provided in Gerrard and Godfrey (1998). This is an important observation because it simplifies estimation and diagnostic testing procedures. Moreover, it makes clear that the modelling procedure is robust to uncertainties surrounding the order of integration of particular variables. It is often difficult to establish the order of integration of particular variables using the techniques and samples of data which are available, and it would be problematic if the modelling procedure required all the variables in the model to be integrated of a particular order. However, the observations above indicate that, so long as the $r \times 1$ cointegrating relations, $\hat{\boldsymbol{\xi}}_t = \hat{\boldsymbol{\beta}}' \mathbf{y}_{t-1}$, are stationary, the conditional *VEC* model, estimated and interpreted in the usual manner, will be valid even if it turns out that some or all of the variables in $\mathbf{y}_{t-1}$ are $I(0)$ and not $I(1)$ after all.

22.14 Analysis of stability of the cointegrated system

Having estimated the system of equations in the cointegrating *VAR*, we will typically need to check on the stability of the system as a whole, and more particularly to check that the disequilibria from the cointegrating relations are in fact mean-reverting. Although such a mean-reverting property is intrinsic to the modelling framework when the cointegration restrictions are not rejected, it is possible that the estimated model does not display this property in practice or that, if it does, the speed with which the system reverts back to its equilibrium is very slow. Summary statistics that shed light on the convergence property of the error correction terms, $\hat{\boldsymbol{\xi}}_t = \hat{\boldsymbol{\beta}}' \mathbf{y}_{t-1}$, will therefore be of some interest.

In the empirical applications of cointegration analysis where $r = 1$, the rate of convergence of $\hat{\boldsymbol{\xi}}_t$ to its equilibrium is ascertained from the signs of the estimates of the error correction coefficients, $\boldsymbol{\alpha}$. However, as we shall demonstrate below, this procedure is not generally applicable. Consider the simple two variable error correction model

$$\begin{pmatrix} \Delta y_{1t} \\ \Delta y_{2t} \end{pmatrix} = - \begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix} (\beta_1 y_{1,t-1} + \beta_2 y_{2,t-1}) + \begin{pmatrix} u_{1t} \\ u_{2t} \end{pmatrix}, \tag{22.78}$$

in which the variables y_{1t} and y_{2t} are cointegrated with cointegrating vector $\boldsymbol{\beta} = (\beta_1, \beta_2)'$. Denoting $\xi_{t+1} = \beta_1 y_{1t} + \beta_2 y_{2t}$, and pre-multiplying both sides of (22.78) by $\boldsymbol{\beta}'$, we obtain

$$\Delta \xi_{t+1} = -(\boldsymbol{\beta}'\boldsymbol{\alpha})\xi_t + \boldsymbol{\beta}'\mathbf{u}_t,$$

where $\boldsymbol{\alpha} = (\alpha_1, \alpha_2)'$ and $\mathbf{u}_t = (u_{1t}, u_{2t})'$, or

$$\xi_{t+1} = (1 - \boldsymbol{\beta}'\boldsymbol{\alpha})\xi_t + \boldsymbol{\beta}'\mathbf{u}_t. \tag{22.79}$$

Since $\boldsymbol{\beta}'\mathbf{u}_t$ is $\mathrm{I}(0)$, then, the stability of this equation requires $|1 - \boldsymbol{\beta}'\boldsymbol{\alpha}| = |1 - \beta_1\alpha_1 - \beta_2\alpha_2| < 1$, or $\beta_1\alpha_1 + \beta_2\alpha_2 > 0$, and $\beta_1\alpha_1 + \beta_2\alpha_2 < 2$. It is clear that these conditions depend on the adjustment parameters from both equations (α_1 and α_2) as well as the parameters of the cointe-

grating vector, and the estimate of α_1 alone will not allow us to sign the expressions $\beta_1\alpha_1+\beta_2\alpha_2$ and $\beta_1\alpha_1+\beta_2\alpha_2-2$. Hence, for example, restricting α_1 to lie in the range $(0, 2)$ ensures the stability of (22.79) only under the normalization $\beta_1 = 1$, and in the simple case where $\alpha_2 = 0$.[11]

More generally, we can rewrite (22.39) as an infinite-order difference equation in an $r \times 1$ vector of (stochastic) disequilibrium terms, $\boldsymbol{\xi}_t = \boldsymbol{\beta}'\mathbf{y}_{t-1}$. Under our assumption that all the variables in $\mathbf{y}_t$ are $I(1)$, and all the roots of $\left|\mathbf{I}_m - \sum_{i=1}^{p-1} \boldsymbol{\Gamma}_i z^i\right| = 0$, fall outside the unit circle, we have the following expression for $\Delta\mathbf{y}_t$

$$\Delta\mathbf{y}_t = \boldsymbol{\Gamma}(L)^{-1}\left(-\boldsymbol{\alpha}\boldsymbol{\xi}_t + \mathbf{u}_t\right),\ t = 1, 2, \ldots, T, \tag{22.80}$$

where $\boldsymbol{\Gamma}(L) = \mathbf{I}_m - \sum_{i=1}^{p-1} \boldsymbol{\Gamma}_i L^i$. Defining $\boldsymbol{\Theta}(L) = \boldsymbol{\Gamma}(L)^{-1} = \sum_{i=0}^{\infty} \boldsymbol{\Theta}_i L^i$, then it is easily seen that the following recursive relations hold

$$\boldsymbol{\Theta}_n = \boldsymbol{\Gamma}_1\boldsymbol{\Theta}_{n-1} + \boldsymbol{\Gamma}_2\boldsymbol{\Theta}_{n-2} + \cdots + \boldsymbol{\Gamma}_{p-1}\boldsymbol{\Theta}_{n-p+1},\ n = 1, 2, \ldots,$$

where $\boldsymbol{\Theta}_0 = \mathbf{I}_m$, and $\boldsymbol{\Theta}_n = \mathbf{0}$ for $n < 0$. Pre-multiplying (22.80) by $\boldsymbol{\beta}'$, then we have

$$\Delta\boldsymbol{\xi}_{t+1} = -\boldsymbol{\beta}'\left(\mathbf{I}_m + \sum_{i=1}^{\infty} \boldsymbol{\Theta}_i L^i\right)\boldsymbol{\alpha}\boldsymbol{\xi}_t + \boldsymbol{\beta}'\left(\mathbf{I}_m + \sum_{i=1}^{\infty} \boldsymbol{\Theta}_i L^i\right)(\mathbf{a}_0 + \mathbf{a}_1 t + \mathbf{u}_t), \tag{22.81}$$

or

$$\boldsymbol{\xi}_{t+1} = \left[\left(\mathbf{I}_r - \boldsymbol{\beta}'\boldsymbol{\alpha}\right) - \sum_{i=1}^{\infty}\left(\boldsymbol{\beta}'\Theta_i\boldsymbol{\alpha}\right)L^i\right]\boldsymbol{\xi}_t + \left(\boldsymbol{\beta}' + \sum_{i=1}^{\infty}\boldsymbol{\beta}'\Theta_i L^i\right)(\mathbf{a}_0 + \mathbf{a}_1 t + \mathbf{u}_t). \tag{22.82}$$

This shows that, in general, when $p \geq 2$, the error correction variables, $\boldsymbol{\xi}_{t+1}$, follow infinite-order *VARMA* processes, and there exists no simple rule involving $\boldsymbol{\alpha}$ alone that could ensure the stability of the dynamic processes in $\boldsymbol{\xi}_{t+1}$. This result also highlights the deficiency of residual-based approaches to testing for cointegration described in Section 22.3, where finite-order ADF regressions are fitted to the residuals even if the order of the underlying *VAR* is 2 or more.

However, given the assumption that none of the roots of $\left|\mathbf{I}_m - \sum_{i=1}^{p-1} \boldsymbol{\Gamma}_i z^i\right| = 0$, fall on or inside the unit circle, it is easily seen that the matrices $\boldsymbol{\Theta}_i$, $i = 0, 1, 2, \ldots$, are absolute summable, and therefore a suitably truncated version of $\sum_{i=1}^{\infty}\left(\boldsymbol{\beta}'\boldsymbol{\Theta}_i\boldsymbol{\alpha}\right)L^i$ can provide us with an adequate approximation in practice. Using an ℓ-order truncation we have

$$\boldsymbol{\xi}_{t+1} \approx \sum_{i=1}^{\ell} \mathbf{D}_i\boldsymbol{\xi}_{t-i+1} + \mathbf{v}_t,\ t = 1, 2, \ldots, T, \tag{22.83}$$

where

$$\mathbf{D}_1 = \mathbf{I}_r - \boldsymbol{\beta}'\boldsymbol{\alpha},\ \mathbf{D}_i = -\boldsymbol{\beta}'\Theta_{i-1}\boldsymbol{\alpha},\ i = 2, 3, \ldots, \ell, \tag{22.84}$$

[11] When $\alpha_2 = 0$, y_{2t} is said to be long-run forcing for y_{1t}. See Chapter 23.

$$\mathbf{v}_t = \left(\boldsymbol{\beta}' + \sum_{i=1}^{\ell} \boldsymbol{\beta}'\Theta_i L^i\right)(\mathbf{a}_0 + \mathbf{a}_1 t + \mathbf{u}_t).$$

To explicitly evaluate the stability of the cointegrated system, we rewrite (22.83) more compactly as

$$\check{\boldsymbol{\xi}}_{t+1} = \mathbf{D}\check{\boldsymbol{\xi}}_t + \check{\mathbf{v}}_t, \ t = 1, 2, \ldots, T, \tag{22.85}$$

where

$$\underset{r\ell\times 1}{\check{\boldsymbol{\xi}}_t} = \begin{pmatrix} \boldsymbol{\xi}_t \\ \boldsymbol{\xi}_{t-1} \\ \boldsymbol{\xi}_{t-2} \\ \vdots \\ \boldsymbol{\xi}_{t-\ell+1} \end{pmatrix}, \ \underset{r\ell\times r\ell}{\mathbf{D}} = \begin{pmatrix} \mathbf{D}_1 & \mathbf{D}_2 & \cdots & \mathbf{D}_{\ell-1} & \mathbf{D}_\ell \\ \mathbf{I}_r & \mathbf{0} & \cdots & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{I}_r & \cdots & \mathbf{0} & \mathbf{0} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ \mathbf{0} & \mathbf{0} & \cdots & \mathbf{I}_r & \mathbf{0} \end{pmatrix}, \ \underset{r\ell\times 1}{\check{\mathbf{v}}_t} = \begin{pmatrix} \mathbf{v}_t \\ \mathbf{0} \\ \mathbf{0} \\ \vdots \\ \mathbf{0} \end{pmatrix}. \tag{22.86}$$

The above cointegrated system is stable if all the roots of $\left|\mathbf{I}_r - \mathbf{D}_1 z - \cdots - \mathbf{D}_\ell z^\ell\right| = 0$, lie outside the unit circle, or if all the eigenvalues of $\mathbf{D}$ have modulus less than unity.[12]

22.15 Beveridge–Nelson decomposition in *VAR*s

The Beveridge–Nelson trend/cycle decomposition allows partitioning a vector of random variables in the sum of a stationary process, called transitory or cyclical component, and a permanent component, which may be further sub-divided into a deterministic (trend) and a stochastic part (Evans and Reichlin (1994), Mills (2003), Robertson, Garratt, and Wright (2006), and Garratt et al. (2006)). In this section we consider a modification of the multivariate Beveridge–Nelson decomposition (see Beveridge and Nelson (1981) and Engle and Granger (1987)), and extend it to include possible restrictions in the intercept and/or trend, as well as the existence of long-run relationships in the variables under consideration. The univariate Beveridge–Nelson decomposition is already discussed in Section 16.6.

Consider an $m \times 1$ vector of random variables $\mathbf{y}_t$, partitioned into a permanent, $\mathbf{y}_t^P$, and a cyclical component, $\mathbf{y}_t^C$. Since the cyclical part is assumed to be stationary it therefore must satisfy the condition

$$\lim_{h\to\infty} E\left(\mathbf{y}_{t+h}^C | \Omega_t\right) = \mathbf{0}, \tag{22.87}$$

where Ω_t denotes the information available at time t, taken to be $\left\{\mathbf{y}_t, \mathbf{y}_{t-1}, \ldots, \mathbf{y}_0\right\}$. Hence, denote

[12] Notice that the stability analysis is not affected by the presence of deterministic and stationary exogenous variables in the system.

$$\mathbf{y}_{dt}^{P} = \mathbf{g}_0 + \mathbf{g}t,$$

be the deterministic part of $\mathbf{y}_t$, where $\mathbf{g}_0$ is an $m \times 1$ vectors of fixed intercepts, $\mathbf{g}$ is an $m \times 1$ vector of (restricted) trend growth rates, t is a deterministic trend term. From (22.87) it follows that (see Garratt, Robertson, and Wright (2005))

$$\mathbf{y}_{st}^{P} = \lim_{h\to\infty} E\left[\mathbf{y}_{t+h} - \mathbf{y}_{d,t+h}^{P} | \Omega_t\right].$$

The above result forms the basis of the trend/cycle decomposition of $\mathbf{y}_t$ described in Garratt et al. (2006). Suppose that $\mathbf{y}_t$ has the following vector error correction representation with unrestricted intercept and restricted trend

$$\Delta\mathbf{y}_t = \mathbf{a} - \boldsymbol{\alpha}\boldsymbol{\beta}^{'}\left[\mathbf{y}_{t-1} - \boldsymbol{\gamma}(t-1)\right] + \sum_{i=1}^{p-1}\boldsymbol{\Gamma}_i\Delta\mathbf{y}_{t-i} + \mathbf{u}_t. \tag{22.88}$$

Denote the deviation of the variables in $\mathbf{y}_t$ from their deterministic components as $\tilde{\mathbf{y}}_t$, namely

$$\tilde{\mathbf{y}}_t = \mathbf{y}_t - \mathbf{g}_0 - \mathbf{g}t. \tag{22.89}$$

Then in terms of $\tilde{\mathbf{y}}_t$ we have

$$\Delta\tilde{\mathbf{y}}_t = \mathbf{a} - \boldsymbol{\alpha}\boldsymbol{\beta}^{'}\mathbf{g}_0 - \left(\mathbf{I}_m - \sum_{i=1}^{p-1}\boldsymbol{\Gamma}_i\right)\mathbf{g} - \boldsymbol{\alpha}\boldsymbol{\beta}'\left(\mathbf{g} - \boldsymbol{\gamma}\right)(t-1) - \boldsymbol{\alpha}\boldsymbol{\beta}^{'}\tilde{\mathbf{y}}_t + \sum_{i=1}^{p-1}\boldsymbol{\Gamma}_i\Delta\tilde{\mathbf{y}}_{t-i} + \mathbf{u}_t.$$

Since $\tilde{\mathbf{y}}_t$ has no deterministic components by construction, it must be that

$$\mathbf{a} = \boldsymbol{\alpha}\boldsymbol{\beta}^{'}\mathbf{g}_0 + \left(\mathbf{I}_m - \sum_{i=1}^{p-1}\boldsymbol{\Gamma}_i\right)\mathbf{g}, \tag{22.90}$$

and

$$\boldsymbol{\beta}'\mathbf{g} = \boldsymbol{\beta}'\boldsymbol{\gamma}. \tag{22.91}$$

Hence, under the above restrictions

$$\Delta\tilde{\mathbf{y}}_t = -\boldsymbol{\alpha}\boldsymbol{\beta}'\tilde{\mathbf{y}}_{t-1} + \sum_{i=1}^{p-1}\boldsymbol{\Gamma}_i\Delta\tilde{\mathbf{y}}_{t-i} + \mathbf{u}_t, \tag{22.92}$$

or, equivalently,

$$\tilde{\mathbf{y}}_t = \sum_{i=1}^{p}\boldsymbol{\Phi}_i\tilde{\mathbf{y}}_{t-i} + \mathbf{u}_t, \tag{22.93}$$

where

$$\mathbf{\Phi}_1 = \mathbf{I}_m + \mathbf{\Gamma}_1 - \boldsymbol{\alpha}\boldsymbol{\beta}', \mathbf{\Phi}_i = \mathbf{\Gamma}_i - \mathbf{\Gamma}_{i-1},\ i = 2, \ldots, p-1,\ \mathbf{\Phi}_p = -\mathbf{\Gamma}_{p-1}.$$

In the general case where one or more elements of $\tilde{\mathbf{y}}_t$ are $I(1)$, it is not possible to invert the polynomial operator, $\mathbf{I}_m - \sum_{i=1}^{p} \mathbf{\Phi}_i L^i$, to derive $\tilde{\mathbf{y}}_t$ in terms of the shocks, $\mathbf{u}_t$. However, since it is assumed that the order of integration of the variables is at most $I(1)$, it follows that $\Delta\tilde{\mathbf{y}}_t$ will follow a general stationary process irrespective of the $I(0)/I(1)$ properties of the underlying variables. More specifically, we have

$$\Delta\tilde{\mathbf{y}}_t = \mathbf{C}(L)\mathbf{u}_t, \tag{22.94}$$

where $\mathbf{C}(L) = \mathbf{C}_0 + \mathbf{C}_1 L + \mathbf{C}_2 L^2 + \ldots$, such that $\{\mathbf{C}_i\}$ are absolute summable matrices. In the case where $\tilde{\mathbf{y}}_t$ is stationary, we must have $\mathbf{C}(1) = \mathbf{0}$. In general,

$$\mathbf{C}(L) = \mathbf{C}(1) + (1 - L)\mathbf{C}^*(L), \tag{22.95}$$

where $\mathbf{C}^*(1) \neq \mathbf{0}$, and $\left\{\mathbf{C}_i^*\right\}$ are absolute summable matrices. Using (22.93) and (22.94) we first note that

$$\left(\mathbf{I}_m - \sum_{i=1}^{p} \mathbf{\Phi}_i L^i\right)\mathbf{C}(L) = 1 - L,$$

which yields the

$$\mathbf{C}_0 = \mathbf{I}_m, \mathbf{C}_1 = -(\mathbf{I} - \mathbf{\Phi}_1),$$
$$\mathbf{C}_2 = \mathbf{\Phi}_1\mathbf{C}_1 + \mathbf{\Phi}_2\mathbf{C}_0,\ \mathbf{C}_{p-1} = \sum_{i=1}^{p-1} \mathbf{\Phi}_i\mathbf{C}_{p-1-i},$$

or more generally

$$\mathbf{C}_j = \sum_{i=1}^{p} \mathbf{\Phi}_i\mathbf{C}_{j-i}, \text{for } j = p, p+1, \ldots.$$

Also using (22.95) we have

$$\mathbf{C}_0^* = \mathbf{C}_0 - \mathbf{C}(1) = -\left(\mathbf{C}_1 + \mathbf{C}_2 + \mathbf{C}_3 + \ldots\right),$$
$$\mathbf{C}_i^* = \mathbf{C}_{i-1}^* + \mathbf{C}_i,\ \text{for } i = 2, 3, \ldots.$$

Now using (22.95) in (22.94) we have

$$\Delta\tilde{\mathbf{y}}_t = \mathbf{C}(1)\mathbf{u}_t + \mathbf{C}^*(L)\Delta\mathbf{u}_t,$$

and cumulating the above from some initial state $\tilde{\mathbf{y}}_0 = \mathbf{y}_0 - \mathbf{g}_0$, we have

$$\tilde{\mathbf{y}}_t = \tilde{\mathbf{y}}_0 + \mathbf{C}(1)\sum_{i=1}^{t}\mathbf{u}_i + \mathbf{C}^*(L)(\mathbf{u}_t - \mathbf{u}_0).$$

Hence, using (22.89), in terms of $\mathbf{y}_t$, we have

$$\mathbf{y}_t = \mathbf{y}_0 + \mathbf{g}t + \mathbf{C}(1)\sum_{i=1}^{t}\mathbf{u}_i + \mathbf{C}^*(L)(\mathbf{u}_t - \mathbf{u}_0). \tag{22.96}$$

This is the multivariate version of the univariate Beveridge–Nelson decomposition discussed in Section (16.6).

Using the above decomposition the stochastic and the cyclical components are defined, respectively, by

$$\begin{aligned}\mathbf{y}_{st}^{P} &= \mathbf{C}(1)\sum_{i=1}^{t}\mathbf{u}_i, \\ \mathbf{y}_t^{C} &= \mathbf{C}^*(L)(\mathbf{u}_t - \mathbf{u}_0) + \mathbf{y}_0.\end{aligned} \tag{22.97}$$

To see this, recall that $\mathbf{y}_{st}^{P}$ is defined by the long-term expectations

$$\mathbf{y}_{st}^{P} = \lim_{h\to\infty} E\left[\mathbf{y}_{t+h} - \mathbf{g}_0 - \mathbf{g}(t+h)|\Omega_t\right].$$

But using (22.96), we have

$$\mathbf{y}_{t+h} - \mathbf{g}_0 - \mathbf{g}(t+h) = \mathbf{C}(1)\sum_{i=1}^{t+h}\mathbf{u}_i + \boldsymbol{\xi}_{t+h},$$

where $\mathbf{g}_0 = \mathbf{y}_0 - \mathbf{C}^*(L)\mathbf{u}_0$, and $\boldsymbol{\xi}_{t+h} = \mathbf{C}^*(L)\mathbf{u}_{t+h}$. Since $\left\{\mathbf{C}_i^*\right\}$ are absolute summable matrices, and the error vectors, $\mathbf{u}_t$, are serially uncorrelated stationary processes with zero means, then $\boldsymbol{\xi}_{t+h}$ is also a stationary process, and hence

$$\lim_{h\to\infty} E\left(\boldsymbol{\xi}_{t+h}|\Omega_t\right) = 0.$$

As a result

$$\mathbf{y}_{st}^{P} = \lim_{h\to\infty} E\left[\mathbf{C}(1)\sum_{i=1}^{t+h}\mathbf{u}_i|\Omega_t\right] = \mathbf{C}(1)\sum_{i=1}^{t}\mathbf{u}_i,$$

as required.

As for the estimation of the various components, note that $\mathbf{y}_{st}^{P}$ can easily be estimated since the coefficients for $\mathbf{C}_i$ can be derived recursively in terms of $\boldsymbol{\Phi}_i$, which in turn can be obtained from the $\boldsymbol{\Gamma}_i$. Once $\mathbf{y}_{st}^{P}$ has been estimated, consider the difference

$$\hat{\mathbf{w}}_t = \mathbf{y}_t - \hat{\mathbf{C}}(1)\sum_{i=1}^{t}\hat{\mathbf{u}}_i,$$

and notice that this is also equal to

$$\hat{\mathbf{w}}_t = \mathbf{y}_0 + \hat{\mathbf{g}}t + \hat{\mathbf{y}}_t^C.$$

Hence, to obtain $\hat{\mathbf{g}}$ and $\hat{\mathbf{y}}_t^C$, one can perform a seemingly unrelated (*SURE*) regression of $\hat{\mathbf{w}}_t$ on an intercept and a time trend t, subject to the restrictions

$$\hat{\boldsymbol{\beta}}'\hat{\mathbf{g}} = \hat{\boldsymbol{\beta}}'\hat{\boldsymbol{\gamma}}, \tag{22.98}$$

where $\hat{\boldsymbol{\gamma}}$ and $\hat{\boldsymbol{\beta}}$ have already been estimated, under the assumption that the cointegrating vectors are exactly identified. Residuals obtained from such a regression will be an estimate of the cyclical component $\mathbf{y}_t^C$. In the case of a cointegrating *VAR* with no intercept and no trends, we have

$$\mathbf{w}_t = \mathbf{y}_t - \hat{\mathbf{C}}(1)\sum_{i=1}^{t}\hat{\mathbf{u}}_i = \mathbf{y}_0 + \hat{\mathbf{y}}_t^C,$$

and the deterministic component is given by $\mathbf{y}_0$. In the case of a cointegrating *VAR* with restricted intercepts and no trends, consistent estimates of $\mathbf{g}$ and $\mathbf{y}_t^C$ can be obtained by running the *SURE* regressions of $\mathbf{w}_t$ on an intercept, subject to the restrictions

$$\hat{\boldsymbol{\beta}}'\mathbf{g}_0 = \hat{\boldsymbol{\beta}}'\hat{\mathbf{a}},$$

where, once again, $\hat{\boldsymbol{\beta}}$ and $\hat{\mathbf{a}}$ have already been estimated from the *VECM* model.

In the case of a cointegrating *VAR* with unrestricted intercepts and no trends, $\mathbf{g} = \mathbf{0}$ and $\mathbf{g}_0$ can be consistently estimated by computing the sample mean of $\mathbf{w}_t$ (or by running *OLS* regressions of $\mathbf{w}_t$ on intercepts). Finally, for a cointegrating *VAR* with unrestricted intercepts and trends, consistent estimates of $\mathbf{g}$ can be obtained by running *OLS* regressions of $\mathbf{w}_t$ on an intercept and a linear trend. The cyclical component $\hat{\mathbf{y}}_t^C$ in all cases is the residual from the above regressions.

22.16 The trend-cycle decomposition of interest rates

We now show how the Beveridge–Nelson decomposition can be used to find the permanent and transitory components for the domestic and foreign interest rates in the UK. Let r_t and r_t^* be the domestic (UK) and foreign interest rates respectively, and consider the following simple error-correction model

$$\Delta r_t = a(r_{t-1} - r_{t-1}^*) + \varepsilon_t,$$
$$\Delta r_t^* = b(r_{t-1} - r_{t-1}^*) + \varepsilon_t^*.$$

The above two equations can be written more compactly as

$$\mathbf{y}_t = \mathbf{A}\mathbf{y}_{t-1} + \mathbf{u}_t, \tag{22.99}$$

where $\mathbf{y}_t = \left(r_t, r_t^*\right)'$, $\mathbf{u}_t = \left(\varepsilon_t, \varepsilon_t^*\right)'$ and

$$\mathbf{A} = \begin{pmatrix} 1+a & -a \\ b & 1-b \end{pmatrix}.$$

Solving the difference equation (22.99) by recursive substitution we have

$$\mathbf{y}_{t+h} = \mathbf{A}^h\mathbf{y}_t + \mathbf{A}^{h-1}\mathbf{u}_{t+1} + \mathbf{A}^{h-2}\mathbf{u}_{t+2} + \ldots + \mathbf{u}_{t+h},$$

and hence

$$E\left(\mathbf{y}_{t+h}|\Omega_t\right) = \mathbf{A}^h\mathbf{y}_t.$$

Since in this example there are no deterministic variables such as intercept or trend, the permanent component of $\mathbf{y}_t$ is given by

$$\mathbf{y}_t^P = \mathbf{y}_{st}^P = \lim_{h\to\infty} E\left(\mathbf{y}_{t+h}|\Omega_t\right) = \left(\lim_{h\to\infty} \mathbf{A}^h\right)\mathbf{y}_t = \mathbf{A}^\infty\mathbf{y}_t. \tag{22.100}$$

If we instead use the common component moving average representation, we have

$$\mathbf{y}_t = \widetilde{\mathbf{y}}_0 + \mathbf{C}(1)\mathbf{s}_{ut} + \mathbf{C}^*(L)\mathbf{u}_t,$$

where $\widetilde{\mathbf{y}}_0 = \mathbf{y}_0 - \mathbf{C}^*(L)\mathbf{u}_0$, $\mathbf{s}_{ut} = \sum_{i=1}^t \mathbf{u}_i$, and

$$\mathbf{C}(1) = \sum_{i=0}^{\infty} \mathbf{C}_i,$$
$$\mathbf{C}^*(L) = \sum_{i=0}^{\infty} \mathbf{C}_i^* L^i,$$

with

$$\mathbf{C}_0 = \mathbf{I}_2, \mathbf{C}_i = -(\mathbf{I}_2 - \mathbf{A})\mathbf{A}^{i-1} \text{ for } i = 1, 2, \ldots,$$
$$\mathbf{C}_i^* = \mathbf{C}_{i-1}^* + \mathbf{C}_i.$$

Also, recall that $\mathbf{C}^*(L)\mathbf{u}_t$ is the stationary component of $\mathbf{y}_t$. Hence,

$$\begin{aligned} E\left(\mathbf{y}_{t+h}|\Omega_t\right) &= \mathbf{y}_0 + E\left[\mathbf{C}(1)\mathbf{s}_{u,t+h}|\Omega_t\right] + E\left[\mathbf{C}^*(L)\mathbf{u}_{t+h}|\Omega_t\right] \\ &= \tilde{\mathbf{y}}_0 + \mathbf{C}(1)\mathbf{s}_{ut} + E\left[\mathbf{C}^*(L)\mathbf{u}_{t+h}|\Omega_t\right], \end{aligned}$$

and since $\mathbf{C}^*(L)\mathbf{u}_{t+h}$ is stationary, then

$$\lim_{h\to\infty} E\left(\mathbf{y}_{t+h}|\Omega_t\right) = \mathbf{y}_0 + \mathbf{C}(1)\mathbf{s}_{ut},$$

but noting that

$$\mathbf{C}(1) = \mathbf{I}_2 - (\mathbf{I}_2 - \mathbf{A})(\mathbf{I}_2 + \mathbf{A} + \mathbf{A}^2 + \ldots) = \lim_{h\to\infty} \mathbf{A}^h = \mathbf{A}^\infty.$$

Hence,

$$\lim_{h\to\infty} \mathrm{E}\left(\mathbf{y}_{t+h}|\Omega_t\right) = \tilde{\mathbf{y}}_0 + \mathbf{A}^\infty \mathbf{s}_{ut} = \mathbf{y}_0 + \mathbf{A}^\infty(\mathbf{u}_1 + \mathbf{u}_2 + \ldots + \mathbf{u}_t). \tag{22.101}$$

This result looks very different from that in (22.100) obtained using the direct method. However, note that

$$\mathbf{y}_t = \mathbf{A}^t\mathbf{y}_0 + \sum_{j=0}^{t-1} \mathbf{A}^{j-1}\mathbf{u}_{t-j}.$$

Pre-multiplying both sides by $\mathbf{A}^h$ and letting $h \to \infty$ we have

$$\Big(\lim_{h\to\infty} \mathbf{A}^h\Big)\mathbf{y}_t = \Big(\lim_{h\to\infty} \mathbf{A}^{t+h}\Big)\mathbf{y}_0 + \sum_{j=0}^{t-1}\Big(\lim_{h\to\infty} \mathbf{A}^{h+j-1}\Big)\mathbf{u}_{t-j},$$

and since $\lim_{h\to\infty} \mathbf{A}^{t+h} = \mathbf{A}^\infty$, for any t, then

$$\begin{aligned}\mathbf{A}^\infty\mathbf{y}_t &= \mathbf{A}^\infty\mathbf{y}_0 + \mathbf{A}^\infty\Big(\sum_{j=0}^{t-1}\mathbf{u}_{t-j}\Big)\\ &= \tilde{\mathbf{y}}_0 + \mathbf{A}^\infty(\mathbf{u}_1 + \mathbf{u}_2 + \ldots + \mathbf{u}_t),\end{aligned}$$

implying that (22.100) and (22.101) are equivalent.

In this example, $\mathbf{A}^\infty$ can be obtained explicitly. It is easily seen that the eigenvalues of $\mathbf{A}$ are $\lambda_1 = 1$ and $\lambda_2 = 1 + a - b$. Hence, the Jordan form of $\mathbf{A}$ is given by

$$\mathbf{A}^h = \mathbf{Q}\begin{pmatrix} 1 & 0 \\ 0 & a-b+1 \end{pmatrix}^h \mathbf{Q}^{-1},$$

where

$$\mathbf{Q} = \begin{pmatrix} 1 & 1 \\ 1 & b/a \end{pmatrix},\ \mathbf{Q}^{-1} = \begin{pmatrix} \frac{b}{-a+b} & -\frac{a}{-a+b} \\ -\frac{a}{-a+b} & \frac{a}{-a+b} \end{pmatrix}.$$

Assuming that $0 < b - a < 2$, then $|\lambda_2| < 1$, and we have

$$\lim_{h\to\infty} \mathbf{A}^h = \mathbf{Q}\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}\mathbf{Q}^{-1} = \frac{1}{b-a}\begin{pmatrix} b & -a \\ b & -a \end{pmatrix}.$$

Therefore, the stochastic component of $\mathbf{y}_t$, $\mathbf{y}_t^P$, is given by

$$\mathbf{y}_t^P = \frac{1}{b-a}\begin{pmatrix} br_t - ar_t^* \\ br_t - ar_t^* \end{pmatrix}. \tag{22.102}$$

Clearly, r_t and r_t^* have the same stochastic components. Furthermore, the cycles for r_t and r_t^* are given by

$$\tilde{r}_t = r_t - \frac{br_t - ar_t^*}{b-a} = \frac{-a(r_t - r_t^*)}{b-a}, \tag{22.103}$$

$$\tilde{r}_t^* = r_t^* - \frac{br_t - ar_t^*}{b-a} = \frac{b(r_t - r_t^*)}{b-a}. \tag{22.104}$$

Using UK data over the period 1979–2003, $\hat{a} = -0.13647$ and $\hat{b} = 0.098014$ (Dées et al. (2007)). *Microfit* can be used to check that equations (22.102) and (22.103)–(22.104) provide the stochastic and cyclical components of r_t and r_t^* in the BN decomposition (see Lesson 17.6 in Pesaran and Pesaran (2009) and Dées et al. (2007)).

22.17 Further reading

An excellent survey of the early developments in the literature on cointegration can be found in Banerjee et al. (1993), and Watson (1994). For more recent developments and further references to the literature on long-run structural modelling see Lütkepohl (2005), Pesaran and Smith (1998), Lütkepohl and Kratzig (2004), and Juselius (2007).

22.18 Exercises

1. Consider the following standard asset pricing model

$$p_t = \sum_{i=1}^{\infty} \beta^i E(d_{t+i} \mid \Omega_t),$$

where Ω_t is the information available at time t, $\beta = (1+r)^{-1}$, $r > 0$ is the discount rate, p_t is the real share price, d_t is real dividends paid per share, $E(d_{t+i} \mid \Omega_t)$ is the conditional mathematical expectations of d_{t+i} with respect to Ω_t. Suppose that d_t follows a random walk model with a non-zero drift, μ

$$d_t = \mu + d_{t-1} + \varepsilon_t, \quad \varepsilon_t \backsim IID(0, \sigma^2).$$

(a) Show that p_t is integrated of order 1 (namely $I(1)$), and that p_t and d_t are cointegrated.

(b) Derive the cointegrating vector associated with (p_t, d_t).

(c) Write down the error-correction representation of asset prices, p_t, and discuss its relationship to the random walk theory of asset prices.

2. Consider the $VAR(2)$ model in the m-dimensional vector $\mathbf{y}_t$

$$\mathbf{y}_t = \boldsymbol{\mu} + \boldsymbol{\Phi}_1 \mathbf{y}_{t-1} + \boldsymbol{\Phi}_2 \mathbf{y}_{t-2} + \mathbf{u}_t, \tag{22.105}$$

where $\boldsymbol{\mu}$ is an m-dimensional vector of fixed constants, $\boldsymbol{\Phi}_i$ $i = 1, 2$ are $m \times m$ matrices of fixed coefficients, and $\mathbf{u}_t$ is a mean zero, serially uncorrelated vector of disturbances with a common positive definite variance–covariance matrix, $\boldsymbol{\Sigma}$.

(a) Derive the conditions under which the $VAR(2)$ model defined in (22.105) is stationary.

(b) Suppose now that one or more elements of $\mathbf{y}_t$ is $I(1)$. Derive suitable restrictions on the intercepts, $\boldsymbol{\mu}$, such that despite the $I(1)$ nature of the variables in (22.105), $\mathbf{y}_t$ has a fixed mean. Discuss the importance of such restrictions for the analysis of cointegration.

(c) Write down the error-correction form of (22.105), and use it to motivate and describe Johansen's method of testing for cointegration.

3. Consider the first difference stationary multivariate system

$$\Delta \mathbf{x}_t = \mathbf{A}(L)\mathbf{u}_t, \tag{22.106}$$

where $\mathbf{x}_t$ is $m \times 1$, $\mathbf{u}_t \sim IID(\mathbf{0}, \boldsymbol{\Sigma})$, $\boldsymbol{\Sigma}$ is an $m \times m$ nonsingular matrix,

$$\mathbf{A}(L) = \sum_{i=0}^{\infty} \mathbf{A}_i L^i, \ \mathbf{A}_0 = \mathbf{I}_m,$$

and $\mathbf{A}_i$ are absolute summable, $m \times m$ matrices of fixed constants.

(a) What is meant by cointegration in the above system?

(b) Show that the necessary and sufficient condition for $\mathbf{x}_t$ to be cointegrated is given by $Rank[\mathbf{A}(1)] = s < m$. What is the number of cointegrating vectors?

4. Consider the bivariate error-correction model

$$\Delta y_t = -\phi_y(y_{t-1} - \theta x_{t-1}) + \psi_{yy}\Delta y_{t-1} + \psi_{yx}\Delta x_{t-1} + u_{yt},$$
$$\Delta x_t = -\phi_x(y_{t-1} - \theta x_{t-1}) + \psi_{xy}\Delta y_{t-1} + \psi_{xx}\Delta x_{t-1} + u_{xt},$$

where $\mathbf{u}_t = (u_{yt}, u_{xt})'$, is a serially uncorrelated process with zero mean and the constant variance-covariance matrix $\mathbf{\Sigma} = (\sigma_{ij})$, with $\sigma_{ij} \neq 0$.

(a) State the necessary and sufficient conditions under which y_t and x_t are cointegrated. Provide examples of y_t and x_t from macroeconomics and finance where cointegration conditions are expected to hold, based on long-run economic theory.

(b) Write down the above error correction model in the form of the following $VAR(2)$ specification

$$\mathbf{z}_t = \mathbf{\Phi}_1\mathbf{z}_{t-1} + \mathbf{\Phi}_2\mathbf{z}_{t-2} + \mathbf{u}_t,$$

where $\mathbf{z}_t = (y_t, x_t)'$. Show that $\mathbf{I}_2 - \mathbf{\Phi}_1 - \mathbf{\Phi}_2$ is rank deficient, where $\mathbf{I}_2$ is an identity matrix of order 2.

(c) How do you test the hypothesis that 'x_t does not Granger cause y_t' in the context of the above model? In your response distinguish between cases where y_t and x_t are cointegrated from the non-cointegrated case.

(d) How do you test for cointegration using *VAR* representation given under (b)?

5. Consider the following multi-variate model in the $m \times 1$ vector of random variables, $\mathbf{x}_t$

$$\Delta\mathbf{x}_t = \boldsymbol{\alpha}\mathbf{z}_{t-1} + \boldsymbol{\varepsilon}_t, \quad \boldsymbol{\varepsilon}_t \sim IIDN(\mathbf{0}, \mathbf{\Omega}), \tag{22.107}$$

where Ω is a positive definite matrix

$$\mathbf{z}_t = \boldsymbol{\beta}'\mathbf{x}_t,$$

and $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$ are $m \times r$ matrices of full column rank, $r < m$.

(a) Show that $\mathbf{x}_t$ is integrated of order 1 and cointegrated if and only if all the eigenvalues of the $r \times r$ matrix $\mathbf{H} = \boldsymbol{\beta}'\boldsymbol{\alpha}$ lie in the range $(-2, 0)$.

(b) Suppose $T + 1$ observations $\mathbf{x}_0, \mathbf{x}_1, \mathbf{x}_2 \ldots \mathbf{x}_T$ are available. Show that the concentrated log likelihood function of (22.107) in terms of $\boldsymbol{\beta}$ can be written as

$$\ell(\boldsymbol{\beta}) \propto \frac{T}{2}\log\left|\mathbf{S}_{00} - \mathbf{S}_{01}\boldsymbol{\beta}\left(\boldsymbol{\beta}'\mathbf{S}_{11}\boldsymbol{\beta}\right)^{-1}\boldsymbol{\beta}'\mathbf{S}_{01}'\right|,$$

where

$$\mathbf{S}_{01} = \frac{1}{T}\sum_{t=1}^{T}\Delta\mathbf{x}_t\mathbf{x}_{t-1}',$$

$$\mathbf{S}_{11} = \frac{1}{T}\sum_{t=1}^{T}\mathbf{x}_{t-1}\mathbf{x}_{t-1}'.$$

(c) Using the concentrated log-likelihood function or otherwise, derive the conditions under which $\boldsymbol{\beta}$ is exactly identified, and discuss alternative procedures suggested in the literature for identification of $\boldsymbol{\beta}$.

23 *VARX* Modelling

23.1 Introduction

This chapter generalizes the cointegration analysis of Chapter 22 and provides a brief account of the econometric issues involved in the modelling approach advanced by Pesaran, Shin, and Smith (2000). We start by describing a general *VARX* model, which allows for the possibility of distinguishing between endogenous and weakly exogenous I(1) variables, and consider its efficient estimation. In this framework, we also prove that weak exogeneity is sufficient for consistent estimation of the long-run parameters of interest that enter the conditional model. We then turn our attention to the analysis of cointegrating *VARX* models, present cointegrating rank tests, derive their asymptotic distribution, and discuss testing the over-identifying restrictions on the cointegrating vectors. We also consider the problem of forecasting using a *VARX* model, and conclude with an empirical application to the UK economy as discussed in Garratt et al. (2003b). The methods discussed in this chapter are used to estimate country-specific modules in the *GVAR* approach outlined in Chapter 33.

23.2 *VAR* models with weakly exogenous I(1) variables

Let $\mathbf{z}_t = \left(\mathbf{y}_t', \mathbf{x}_t'\right)'$, where $\mathbf{y}_t$ and $\mathbf{x}_t$ are two random vectors of dimensions $m_y \times 1$ and $m_x \times 1$, respectively. Weakly exogenous $I(1)$ variables can be introduced in the context of the following simple error correction model for $\mathbf{z}_t$ with no deterministic components

$$\begin{pmatrix} \Delta \mathbf{y}_t \\ \Delta \mathbf{x}_t \end{pmatrix} = \begin{pmatrix} -\boldsymbol{\Pi}_y \\ -\boldsymbol{\Pi}_x \end{pmatrix} \begin{pmatrix} \mathbf{y}_{t-1} \\ \mathbf{x}_{t-1} \end{pmatrix} + \begin{pmatrix} \mathbf{u}_{yt} \\ \mathbf{u}_{xt} \end{pmatrix}, \tag{23.1}$$

where $\mathbf{u}_t = \left(\mathbf{u}_{yt}', \mathbf{u}_{xt}'\right)'$ is a vector of serially uncorrelated errors distributed independently of $\mathbf{x}_t$ with zero mean and the constant positive definite variance–covariance matrix

$$E\left(\mathbf{u}_t \mathbf{u}_t'\right) = \boldsymbol{\Sigma} = \begin{pmatrix} \boldsymbol{\Sigma}_{yy} & \boldsymbol{\Sigma}_{yx} \\ \boldsymbol{\Sigma}_{xy} & \boldsymbol{\Sigma}_{xx} \end{pmatrix}.$$

For the purpose of exposition, in this section we assume $\mathbf{u}_t \sim IIDN(\mathbf{0}, \boldsymbol{\Sigma})$. Further, the analysis that follows is conducted given the initial values, $\mathbf{z}_0$.

Using known results on conditional distribution of multivariate normal (see Appendix B, Section B.10.3), the partition in (23.1) allows us to express $\mathbf{u}_{yt}$ conditional on $\mathbf{u}_{xt}$ as

$$\mathbf{u}_{yt} = \boldsymbol{\Sigma}_{yx}\boldsymbol{\Sigma}_{xx}^{-1}\mathbf{u}_{xt} + \boldsymbol{\upsilon}_t, \tag{23.2}$$

where $\boldsymbol{\upsilon}_t \sim IIDN(\mathbf{0}, \boldsymbol{\Sigma}_{\upsilon\upsilon})$, with $\boldsymbol{\Sigma}_{\upsilon\upsilon} = \boldsymbol{\Sigma}_{yy} - \boldsymbol{\Sigma}_{yx}\boldsymbol{\Sigma}_{xx}^{-1}\boldsymbol{\Sigma}_{xy}$, and $\boldsymbol{\upsilon}_t$ is uncorrelated with $\mathbf{u}_{xt}$ by construction. Substitution of (23.2) in (23.1) provides a conditional model for $\Delta\mathbf{y}_t$ in terms of $\mathbf{z}_{t-1}$ and $\Delta\mathbf{x}_{t-1}$

$$\begin{aligned}
\Delta\mathbf{y}_t &= -\boldsymbol{\Pi}_y\mathbf{z}_{t-1} + \boldsymbol{\Sigma}_{yx}\boldsymbol{\Sigma}_{xx}^{-1}(\Delta\mathbf{x}_t + \boldsymbol{\Pi}_x\mathbf{z}_{t-1}) + \boldsymbol{\upsilon}_t \\
&= \left(\boldsymbol{\Sigma}_{yx}\boldsymbol{\Sigma}_{xx}^{-1}\boldsymbol{\Pi}_x - \boldsymbol{\Pi}_y\right)\mathbf{z}_{t-1} + \boldsymbol{\Sigma}_{yx}\boldsymbol{\Sigma}_{xx}^{-1}\boldsymbol{\Delta}\mathbf{x}_t + \boldsymbol{\upsilon}_t \\
&= -\boldsymbol{\Pi}_{yy.x}\mathbf{z}_{t-1} + \boldsymbol{\Lambda}\Delta\mathbf{x}_t + \boldsymbol{\upsilon}_t,
\end{aligned} \tag{23.3}$$

where $\boldsymbol{\Pi}_{yy.x} = \boldsymbol{\Pi}_y - \boldsymbol{\Sigma}_{yx}\boldsymbol{\Sigma}_{xx}^{-1}\boldsymbol{\Pi}_x$, and $\boldsymbol{\Lambda} = \boldsymbol{\Sigma}_{yx}\boldsymbol{\Sigma}_{xx}^{-1}$. Following Pesaran, Shin, and Smith (2000), we assume that the process $\{\mathbf{x}_t\}_{t=1}^{\infty}$ is weakly exogenous with respect to the matrix of long-run multiplier parameters $\boldsymbol{\Pi}$, namely

$$\boldsymbol{\Pi}_x = \mathbf{0}, \tag{23.4}$$

so that,

$$\boldsymbol{\Pi}_{yy.x} = \boldsymbol{\Pi}_y. \tag{23.5}$$

Since under $\boldsymbol{\Pi}_x = \mathbf{0}$, the exogenous variables, $\mathbf{x}_t$, are $I(1)$, $\mathbf{x}_t$ are also referred to as $I(1)$ weakly exogenous variables in the conditional model of $\mathbf{y}_t$. Note that the weak exogeneity restriction (23.4) implies that $\{\mathbf{x}_t\}_{t=1}^{\infty}$ is integrated of order 1, and that it is *long-run forcing* for $\left\{\mathbf{y}_t\right\}_{t=1}^{\infty}$ (see Granger and Lin (1995)). Strictly speaking one can also consider a generalization of this concept to the case where $\boldsymbol{\Pi}_x$ is non-zero, but rank deficient.

Under the above restrictions, the conditional model for $\mathbf{y}_t$ can be written as

$$\Delta\mathbf{y}_t = -\boldsymbol{\Pi}_y\mathbf{z}_{t-1} + \boldsymbol{\Lambda}\Delta\mathbf{x}_t + \boldsymbol{\upsilon}_t, \tag{23.6}$$

where by construction $\Delta\mathbf{x}_t$ and $\boldsymbol{\upsilon}_t$ are uncorrelated.

We now provide a formal proof that the weak exogeneity of $\mathbf{x}_t$ with respect to the long-run coefficients, $\boldsymbol{\beta}$, is sufficient for consistent estimation of the remaining parameters of interest that enter the conditional model. To this end, as before, we assume that $\mathbf{u}_t \sim IIDN(\mathbf{0}, \boldsymbol{\Sigma})$, and make use of the Engle, Hendry, and Richard (1983) likelihood framework. Consider the following $VARX(2)$ specification

$$\Delta\mathbf{z}_t = -\boldsymbol{\Pi}\mathbf{z}_{t-1} + \boldsymbol{\Gamma}_1\Delta\mathbf{z}_{t-1} + \mathbf{u}_t. \tag{23.7}$$

Let $\boldsymbol{\psi} = (vec(\boldsymbol{\alpha})', vec(\boldsymbol{\beta})', vec(\boldsymbol{\Gamma}_1)', vech(\boldsymbol{\Sigma})')'$ be the parameters of interest and note that the log-likelihood function of (23.7) for the sample of observations over $t = 1, 2, \ldots, T$, is given by

$$\ell(\boldsymbol{\psi}) = -\frac{T}{2}\ln|\boldsymbol{\Sigma}| - \frac{1}{2}\sum_{t=1}^{T}\mathbf{u}_t'\boldsymbol{\Sigma}^{-1}\mathbf{u}_t.$$

Also it is easily seen that

$$\mathbf{u}_t'\boldsymbol{\Sigma}^{-1}\mathbf{u}_t = \mathbf{u}_{yt}'\boldsymbol{\Sigma}^{yy}\mathbf{u}_{yt} + 2\mathbf{u}_{yt}'\boldsymbol{\Sigma}^{yx}\mathbf{u}_{xt} + \mathbf{u}_{xt}'\boldsymbol{\Sigma}^{xx}\mathbf{u}_{xt},$$

where

$$\boldsymbol{\Sigma}^{-1} = \begin{pmatrix} \boldsymbol{\Sigma}^{yy} & \boldsymbol{\Sigma}^{yx} \\ \boldsymbol{\Sigma}^{xy} & \boldsymbol{\Sigma}^{xx} \end{pmatrix},$$

and

$$|\boldsymbol{\Sigma}| = |\boldsymbol{\Sigma}_{xx}|\left|\boldsymbol{\Sigma}_{yy} - \boldsymbol{\Sigma}_{yx}\boldsymbol{\Sigma}_{xx}^{-1}\boldsymbol{\Sigma}_{xy}\right|.$$

Following the discussion below equation (23.2) (see also Appendix B, Section B.10), we have

$$\mathbf{u}_t'\boldsymbol{\Sigma}^{-1}\mathbf{u}_t = \boldsymbol{\upsilon}_t\boldsymbol{\Sigma}_{\upsilon\upsilon}^{-1}\boldsymbol{\upsilon}_t + \mathbf{u}_{xt}'\boldsymbol{\Sigma}_{xx}^{-1}\mathbf{u}_{xt},$$

where $\boldsymbol{\upsilon}_t = \mathbf{u}_{yt} - \boldsymbol{\Sigma}_{yx}\boldsymbol{\Sigma}_{xx}^{-1}\mathbf{u}_{xt}$. Also, partitioning $\boldsymbol{\Gamma}_1$, as $\boldsymbol{\Gamma}_1' = (\boldsymbol{\Gamma}_{y1}', \boldsymbol{\Gamma}_{x1}')$, we have

$$\boldsymbol{\upsilon}_t = \mathbf{u}_{yt} - \boldsymbol{\Lambda}\mathbf{u}_{xt} = \Delta\mathbf{y}_t - \boldsymbol{\Pi}_y\mathbf{z}_{t-1} - \boldsymbol{\Lambda}\Delta\mathbf{x}_t - \boldsymbol{\Psi}_1\Delta\mathbf{z}_{t-1},$$

where $\boldsymbol{\Lambda} = \boldsymbol{\Sigma}_{yx}\boldsymbol{\Sigma}_{xx}^{-1}$, and $\boldsymbol{\Psi}_1 = \boldsymbol{\Gamma}_{y1} - \boldsymbol{\Sigma}_{yx}\boldsymbol{\Sigma}_{xx}^{-1}\boldsymbol{\Gamma}_{x1}$. Hence, under $\boldsymbol{\Pi}_x = \mathbf{0}$ (see (23.4)), the log-likelihood function can be decomposed as

$$\ell(\boldsymbol{\psi}) = \ell_1(\boldsymbol{\theta}) + \ell_2(\boldsymbol{\Sigma}_{xx}, \boldsymbol{\Gamma}_{x1}),$$

where

$$\begin{aligned}\ell_1(\boldsymbol{\theta}) =& -\frac{T}{2}\ln|\boldsymbol{\Sigma}_{vv}| \\ & -\frac{1}{2}\sum_{t=1}^{T}\left(\Delta\mathbf{y}_t - \boldsymbol{\Pi}_y\mathbf{z}_{t-1} - \boldsymbol{\Lambda}\Delta\mathbf{x}_t - \boldsymbol{\Psi}_1\Delta\mathbf{z}_{t-1}\right)' \\ & \boldsymbol{\Sigma}_{vv}^{-1}\left(\Delta\mathbf{y}_t - \boldsymbol{\Pi}_y\mathbf{z}_{t-1} - \boldsymbol{\Lambda}\Delta\mathbf{x}_t - \boldsymbol{\Psi}_1\Delta\mathbf{z}_{t-1}\right),\end{aligned}$$

and

$$\ell_2(\boldsymbol{\Sigma}_{xx}, \boldsymbol{\Gamma}_{1x}) = -\frac{T}{2}\ln|\boldsymbol{\Sigma}_{xx}| - \frac{1}{2}\sum_{t=1}^{T}\left(\Delta\mathbf{x}_t - \boldsymbol{\Gamma}_{x1}\Delta\mathbf{z}_{t-1}\right)'\boldsymbol{\Sigma}_{xx}^{-1}\left(\Delta\mathbf{x}_t - \boldsymbol{\Gamma}_{x1}\Delta\mathbf{z}_{t-1}\right).$$

Hence, under $\boldsymbol{\Pi}_x$, the parameters of interest, $\boldsymbol{\theta}$, that enter the conditional model, $\ell_1(\boldsymbol{\theta})$, are variation free with respect to the parameters of the marginal model, $\ell_2(\boldsymbol{\Sigma}_{xx}, \boldsymbol{\Gamma}_{x1})$, and the *ML* esti-

mators of $\boldsymbol{\theta}$ based on the conditional model will be identical to the *ML* estimators computed indirectly (as set out above) using the full model, $\ell(\boldsymbol{\psi})$.

23.2.1 Higher-order lags

Consider now the following *VECM* in $\mathbf{z}_t$ with $p-1$ lagged changes

$$\Delta \mathbf{z}_t = -\boldsymbol{\Pi}\mathbf{z}_{t-1} + \sum_{i=1}^{p-1} \boldsymbol{\Gamma}_i \Delta \mathbf{z}_{t-i} + \mathbf{u}_t, \tag{23.8}$$

where the matrices $\{\boldsymbol{\Gamma}_i\}_{i=1}^{p-1}$ are the short-run responses and $\boldsymbol{\Pi}$ is the long-run multiplier matrix. The analysis is now conducted given the initial values $\mathbf{Z}_0 \equiv (\mathbf{z}_{-p+1}, \ldots, \mathbf{z}_0)$. Following a similar line of reasoning as above, the conditional model for $\Delta \mathbf{y}_t$ in terms of $\mathbf{z}_{t-1}$, $\Delta \mathbf{x}_t$, $\Delta \mathbf{z}_{t-1}$, $\Delta \mathbf{z}_{t-2}$, . . ., can be obtained as

$$\Delta \mathbf{y}_t = -\boldsymbol{\Pi}_{yy,x}\mathbf{z}_{t-1} + \boldsymbol{\Lambda}\Delta \mathbf{x}_t + \sum_{i=1}^{p-1} \boldsymbol{\Psi}_i \Delta \mathbf{z}_{t-i} + \boldsymbol{\upsilon}_t, \tag{23.9}$$

where $\boldsymbol{\Pi}_{yy,x} \equiv \boldsymbol{\Pi}_y - \boldsymbol{\Sigma}_{yx}\boldsymbol{\Sigma}_{xx}^{-1}\boldsymbol{\Pi}_x$, $\boldsymbol{\Lambda} = \boldsymbol{\Sigma}_{yx}\boldsymbol{\Sigma}_{xx}^{-1}$, $\boldsymbol{\Psi}_i \equiv \boldsymbol{\Gamma}_{yi} - \boldsymbol{\Sigma}_{yx}\boldsymbol{\Sigma}_{xx}^{-1}\boldsymbol{\Gamma}_{xi}$, $i = 1, 2, \ldots, p-1$. Hence, under restrictions (23.4), we obtain the following system of equations

$$\Delta \mathbf{y}_t = -\boldsymbol{\Pi}_y \mathbf{z}_{t-1} + \boldsymbol{\Lambda}\Delta \mathbf{x}_t + \sum_{i=1}^{p-1} \boldsymbol{\Psi}_i \Delta \mathbf{z}_{t-i} + \boldsymbol{\upsilon}_t, \tag{23.10}$$

$$\Delta \mathbf{x}_t = \sum_{i=1}^{p-1} \boldsymbol{\Gamma}_{xi} \Delta \mathbf{z}_{t-i} + \mathbf{a}_{x0} + \mathbf{u}_{xt}. \tag{23.11}$$

Equation (23.11) describes the dynamics of the weakly exogenous variables, and is also called the *marginal model*. Note from (23.11) that restriction (23.4) implies that the elements of the vector process $\{\mathbf{x}_t\}_{t=1}^{\infty}$ are not cointegrated among themselves. However, it does not preclude $\{\mathbf{y}_t\}_{t=1}^{\infty}$ being Granger-causal for $\{\mathbf{x}_t\}_{t=1}^{\infty}$ in the short run, in the sense that $\Delta \mathbf{y}_{t-1}, \Delta \mathbf{y}_{t-2}, \ldots$ could help in predicting $\Delta \mathbf{x}_t$, even if its lagged values are included in the regression model.

Finally, we note that the cointegration rank hypothesis (22.60) is restated in the context of (23.6) as

$$H_r : Rank(\boldsymbol{\Pi}_y) = r, \; r = 0, 1, \ldots, m_y. \tag{23.12}$$

Under (22.60), we may express

$$\boldsymbol{\Pi}_y = \boldsymbol{\alpha}_y \boldsymbol{\beta}', \tag{23.13}$$

where the $m_y \times r$ loadings matrix $\boldsymbol{\alpha}_y$ and the $m \times r$ matrix (with $m = m_y + m_x$) of cointegrating vectors $\boldsymbol{\beta}$ are each full column rank and identified up to an arbitrary $r \times r$ nonsingular matrix.[1] (See Section 23.5).

[1] That is, $(\boldsymbol{\alpha}_y \mathbf{K}^{-1})(\mathbf{K}\boldsymbol{\beta}') = \tilde{\boldsymbol{\alpha}}_y \tilde{\boldsymbol{\beta}}'$ for any $r \times r$ nonsingular matrix $\mathbf{K}$.

23.3 Efficient estimation

In this section we focus on maximum likelihood estimation of the cointegrating matrix $\boldsymbol{\beta}$ in the context of model (23.10). Let $m = m_y + m_x$. If T observations are available, stacking the *VECM* (23.10) results in

$$\Delta \mathbf{Y} = \boldsymbol{\Psi} \Delta \mathbf{Z}_{-} - \boldsymbol{\Pi}_y \mathbf{Z}_{-1} + \mathbf{V}, \tag{23.14}$$

where $\Delta \mathbf{Y} \equiv (\Delta \mathbf{y}_1, \ldots, \Delta \mathbf{y}_T)$, $\Delta \mathbf{X} \equiv (\Delta \mathbf{x}_1, \ldots, \Delta \mathbf{x}_T)$, $\Delta \mathbf{Z}_{-i} \equiv (\Delta \mathbf{z}_{1-i}, \ldots, \Delta \mathbf{z}_{T-i})$, $i = 1, 2, \ldots, p-1$, $\boldsymbol{\Psi} \equiv (\boldsymbol{\Lambda}, \boldsymbol{\Psi}_1, \ldots, \boldsymbol{\Psi}_{p-1})$, $\Delta \mathbf{Z}_{-} \equiv (\Delta \mathbf{X}', \Delta \mathbf{Z}'_{-1}, \ldots, \Delta \mathbf{Z}'_{1-p})'$, $\mathbf{Z}_{-1} \equiv (\mathbf{z}_0, \ldots, \mathbf{z}_{T-1})$, and $\mathbf{V} \equiv (\boldsymbol{\upsilon}_1, \ldots, \boldsymbol{\upsilon}_T)$.

The log-likelihood function of the structural *VECM* model (23.14) is given by

$$\ell(\boldsymbol{\theta}) = -\frac{m_y T}{2} \ln 2\pi - \frac{T}{2} \ln \left| \boldsymbol{\Sigma}_{\upsilon\upsilon}^{-1} \right| - \frac{1}{2} Trace(\boldsymbol{\Sigma}_{\upsilon\upsilon}^{-1} \mathbf{V}\mathbf{V}'), \tag{23.15}$$

with $\boldsymbol{\theta} = \left(vec(\boldsymbol{\Psi})', vec(\boldsymbol{\alpha}_y)', vec(\boldsymbol{\beta})', vech(\boldsymbol{\Sigma}_{\upsilon\upsilon})'\right)'$. Concentrating out $\boldsymbol{\Sigma}_{\upsilon\upsilon}^{-1}$, $\boldsymbol{\Psi}$ and $\boldsymbol{\alpha}_y$ in (23.15) results in the concentrated log-likelihood function

$$\ell^c(\boldsymbol{\beta}) = -\frac{m_y T}{2}(1 + \ln 2\pi) - \frac{T}{2} \ln \left| T^{-1} \Delta \hat{\mathbf{Y}} \left(\mathbf{I}_T - \hat{\mathbf{Z}}'_{-1} \boldsymbol{\beta} \left(\boldsymbol{\beta}' \hat{\mathbf{Z}}_{-1} \hat{\mathbf{Z}}'_{-1} \boldsymbol{\beta} \right)^{-1} \boldsymbol{\beta}' \hat{\mathbf{Z}}_{-1} \right) \Delta \hat{\mathbf{Y}}' \right|, \tag{23.16}$$

where $\Delta \hat{\mathbf{Y}}$ and $\hat{\mathbf{Z}}_{-1}$ are respectively the *OLS* residuals from regressions of $\Delta \mathbf{Y}$ and $\mathbf{Z}_{-1}$ on $\Delta \mathbf{Z}_{-}$. Defining the sample moment matrices

$$\mathbf{S}_{YY} \equiv T^{-1} \Delta \hat{\mathbf{Y}} \Delta \hat{\mathbf{Y}}', \ \mathbf{S}_{YZ} \equiv T^{-1} \Delta \hat{\mathbf{Y}} \hat{\mathbf{Z}}'_{-1}, \ \mathbf{S}_{ZZ} \equiv T^{-1} \hat{\mathbf{Z}}_{-1} \hat{\mathbf{Z}}'_{-1}, \tag{23.17}$$

the maximization of the concentrated log-likelihood function $\ell^c(\boldsymbol{\beta})$ of (23.16) reduces to the minimization of

$$\left| \mathbf{S}_{YY} - \mathbf{S}_{YZ} \boldsymbol{\beta} \left(\boldsymbol{\beta}' \mathbf{S}_{ZZ} \boldsymbol{\beta} \right)^{-1} \boldsymbol{\beta}' \mathbf{S}_{ZY} \right| = \frac{|\mathbf{S}_{YY}| \left| \boldsymbol{\beta}' \left(\mathbf{S}_{ZZ} - \mathbf{S}_{ZY} \mathbf{S}_{YY}^{-1} \mathbf{S}_{YZ} \right) \boldsymbol{\beta} \right|}{\left| \boldsymbol{\beta}' \mathbf{S}_{ZZ} \boldsymbol{\beta} \right|},$$

with respect to $\boldsymbol{\beta}$. The solution $\hat{\boldsymbol{\beta}}$ to this minimization problem, that is, the maximum likelihood (*ML*) estimator for $\boldsymbol{\beta}$, is given by the eigenvectors corresponding to the r largest eigenvalues $\hat{\lambda}_1 > \ldots > \hat{\lambda}_r > 0$ of

$$\left| \hat{\lambda} \mathbf{S}_{ZZ} - \mathbf{S}_{ZY} \mathbf{S}_{YY}^{-1} \mathbf{S}_{YZ} \right| = 0. \tag{23.18}$$

See Section 22.6, and pp.1553–1554 in Johansen (1991). The *ML* estimator $\hat{\boldsymbol{\beta}}$ is identified up to post-multiplication by an $r \times r$ nonsingular matrix; that is, r^2 just-identifying restrictions on $\boldsymbol{\beta}$ are required for exact identification. The resultant maximized concentrated log-likelihood function $\ell^c(\boldsymbol{\beta})$ at $\hat{\boldsymbol{\beta}}$ of (23.16) is

$$\ell^c(r) = -\frac{m_y T}{2}(1 + \ln 2\pi) - \frac{T}{2}\ln|\mathbf{S}_{YY}| - \frac{T}{2}\sum_{i=1}^{r}\ln\left(1 - \hat{\lambda}_i\right). \tag{23.19}$$

Note that the maximized value of the log-likelihood $\ell^c(r)$ is only a function of the cointegration rank r (and m_y and m_x) through the eigenvalues $\{\hat{\lambda}_i\}_{i=1}^{r}$ defined by (23.18). Also, see Boswijk (1995) and Harbo et al. (1998).

23.3.1 The five cases

Consider the following general model

$$\Delta\mathbf{y}_t = \mathbf{a}_0 + \mathbf{a}_1 t + \mathbf{\Lambda}\Delta\mathbf{x}_t + \sum_{i=1}^{p-1}\mathbf{\Psi}_i\Delta\mathbf{z}_{t-i} - \mathbf{\Pi}_y\mathbf{z}_{t-1} + \boldsymbol{\upsilon}_t.$$

In these more general cases we need to modify the definitions of $\Delta\hat{\mathbf{Y}}$ and $\hat{\mathbf{Z}}_{-1}$ and, consequently, the sample moment matrices $\mathbf{S}_{YY}$, $\mathbf{S}_{YZ}$ and $\mathbf{S}_{ZZ}$ given by (23.17) (see also Section 22.9). Let $\mathbf{1}_T = (1, 1, \ldots, 1)'$ and $\boldsymbol{\tau}_T = (1, 2, \ldots, T)'$. We have:

Case I $(\mathbf{a}_0 = \mathbf{0}$ and $\mathbf{a}_1 = \mathbf{0})$

$\Delta\hat{\mathbf{Y}}$ and $\hat{\mathbf{Z}}_{-1}$ are the *OLS* residuals from the regression of $\Delta\mathbf{Y}$ and $\mathbf{Z}_{-1}$ on $\Delta\mathbf{Z}_-$.

Case II $(\mathbf{a}_0 = -\mathbf{\Pi}_y\boldsymbol{\mu}$ and $\mathbf{a}_1 = \mathbf{0})$

$\Delta\hat{\mathbf{Y}}$ and $\hat{\mathbf{Z}}_{-1}$ are the *OLS* residuals from the regression of $\Delta\mathbf{Y}$ and $\mathbf{Z}^*_{-1}$ on $\Delta\mathbf{Z}_-$, where $\mathbf{Z}^*_{-1} = (\mathbf{1}_T, \mathbf{Z}'_{-1})'$.

Case III $(\mathbf{a}_0 \neq \mathbf{0}$ and $\mathbf{a}_1 = \mathbf{0})$

$\Delta\hat{\mathbf{Y}}$ and $\hat{\mathbf{Z}}_{-1}$ are the *OLS* residuals from the regression of $\Delta\mathbf{Y}$ and $\mathbf{Z}_{-1}$ on $(\mathbf{1}_T, \Delta\mathbf{Z}'_-)'$.

Case IV $(\mathbf{a}_0 \neq \mathbf{0}$ and $\mathbf{a}_1 = -\mathbf{\Pi}_y\boldsymbol{\gamma})$

$\Delta\hat{\mathbf{Y}}$ and $\hat{\mathbf{Z}}_{-1}$ are the *OLS* residuals from the regression of $\Delta\mathbf{Y}$ and $\mathbf{Z}^*_{-1}$ on $(\mathbf{1}_T, \Delta\mathbf{Z}'_-)'$, where $\mathbf{Z}^*_{-1} = (\boldsymbol{\tau}_T, \mathbf{Z}'_{-1})'$.

Case V $(\mathbf{a}_0 \neq \mathbf{0}$ and $\mathbf{a}_1 \neq \mathbf{0})$

$\Delta\hat{\mathbf{Y}}$ and $\hat{\mathbf{Z}}^*_{-1}$ are the *OLS* residuals from the regression of $\Delta\mathbf{Y}$ and $\mathbf{Z}^*_{-1}$ on $(\mathbf{1}_T, \boldsymbol{\tau}_T, \Delta\mathbf{Z}'_-)'$, where $\mathbf{Z}^*_{-1} = (\boldsymbol{\tau}_T, \mathbf{Z}'_{-1})'$.

Tests of the cointegrating rank are obtained along exactly the same lines as those in Section 22.10. Estimation of the *VECM* subject to exact- and over-identifying long-run restrictions can be carried out by maximum likelihood methods as outlined above, applied to (23.10) subject to the appropriate restrictions on the intercepts and trends, subject to $\text{Rank}(\mathbf{\Pi}_y) = r$, and subject to k general linear restrictions. Having computed *ML* estimates of the cointegrating vectors, the short-run parameters of the conditional *VECM* can be computed by *OLS* regressions.

While estimation and inference on the parameters of (23.10) can be conducted without a reference to the marginal model (23.11), for forecasting and impulse response analysis the processes driving the weakly exogenous variables must be specified. In other words, one needs to take into account the possibility that changes in one variable may have an impact on the weakly exogenous variables and that these effects will continue and interact over time.

23.4 Testing weak exogeneity

The main assumption underlying the estimation of *VARX* models is the weak exogeneity of x_{it} with respect to the long-run parameters of the conditional model. Weak exogeneity can be tested along the lines described in Johansen (1992) and Harbo et al. (1998). This involves a test of the joint significance of the estimated error correction terms in auxiliary equations for x_{it}. In particular, for each element of $\mathbf{x}_t$, say x_{it}, the following regression is carried out

$$\Delta x_{it} = \sum_{j=1}^{r} \gamma_{ij} ECM_{j,t-1} + \sum_{k=1}^{s} \varphi_{ik} \Delta y_{i,t-k} + \sum_{m=1}^{n} \vartheta_{im} \Delta x_{i,t-m} + \epsilon_{it},$$

where $ECM_{j,t-1}, j = 1, 2, \ldots, r$, are the estimated error correction terms corresponding to the r cointegrating relations. The statistic for testing the weak exogeneity of x_{it} is the standard F statistic for testing the joint hypothesis that $\gamma_{ij} = 0, j = 1, 2, \ldots, r$, in the above regression.

23.5 Testing for cointegration in *VARX* models

We now consider testing the null hypothesis of cointegration rank r, H_r of (23.12), against the alternative hypothesis

$$H_{r+1} : Rank\left(\mathbf{\Pi}_y\right) = r + 1, \; r = 0, \ldots, m_y - 1,$$

in the structural *VECM* (23.10). To this end, we weaken the independent normal distributional assumption of previous sections on the error process $\{\mathbf{u}_t\}_{t=-\infty}^{\infty}$ and make the following assumptions:

Assumption 1: The error process $\{\mathbf{u}_t\}$ is such that

(a) (i) $\mathrm{E}\left[\mathbf{u}_t | \{\mathbf{z}_{t-i}\}_{i=1}^{t-1}, \mathbf{Z}_0\right] = \mathbf{0}$; (ii) $\mathrm{Var}\left[\mathbf{u}_t | \{\mathbf{z}_{t-i}\}_{i=1}^{t-1}, \mathbf{Z}_0\right] = \mathbf{\Sigma}$, with $\mathbf{\Sigma}$ positive definite;
(b) (i) $\mathrm{E}\left[\boldsymbol{\upsilon}_t | \mathbf{x}_t, \{\mathbf{z}_{t-i}\}_{i=1}^{t-1}, \mathbf{Z}_0\right] = \mathbf{0}$; (ii) $\mathrm{Var}\left[\boldsymbol{\upsilon}_t | \mathbf{x}_t, \{\mathbf{z}_{t-i}\}_{i=1}^{t-1}, \mathbf{Z}_0\right] = \mathbf{\Sigma}_{\upsilon\upsilon}$, where $\boldsymbol{\upsilon}_t = \mathbf{u}_{yt} - \mathbf{\Sigma}_{yx}\mathbf{\Sigma}_{xx}^{-1}\mathbf{u}_{xt}$, and $\mathbf{\Sigma}_{\upsilon\upsilon} \equiv \mathbf{\Sigma}_{yy} - \mathbf{\Sigma}_{yx}\mathbf{\Sigma}_{xx}^{-1}\mathbf{\Sigma}_{xy}$;
(c) $\sup_t \mathrm{E}(\|\mathbf{u}_t\|^s) < \infty$ for some $s > 2$.

Assumption 1 states that the error process $\{\mathbf{u}_t\}_{t=-\infty}^{\infty}$ is a martingale difference sequence with constant conditional variance; hence, $\{\mathbf{u}_t\}_{t=-\infty}^{\infty}$ is an uncorrelated process. Therefore, the *VECM* (23.8) represents a conditional model for $\Delta\mathbf{z}_t$ given $\{\Delta\mathbf{z}_{t-i}\}_{i=1}^{p-1}$ and $\mathbf{z}_{t-1}, t = 1, 2, \ldots$. Under Assumption 1(b)(i), $E\left[\mathbf{u}_{yt} | \mathbf{x}_t, \{\mathbf{z}_{t-i}\}_{i=1}^{t-1}, \mathbf{Z}_0\right] = \mathbf{\Sigma}_{yx}\mathbf{\Sigma}_{xx}^{-1}\mathbf{u}_{xt}$ while 1(b)(ii) ensures that $Var\left[\mathbf{u}_{yt} | \mathbf{x}_t, \{\mathbf{z}_{t-i}\}_{i=1}^{t-1}, \mathbf{Z}_0\right] = \mathbf{\Sigma}_{\upsilon\upsilon}$. Therefore, under this assumption, (23.10) can be interpreted as a conditional model for $\Delta\mathbf{y}_t$ given $\Delta\mathbf{x}_t$, $\{\Delta\mathbf{z}_{t-i}\}_{i=1}^{p-1}$ and $\mathbf{z}_{t-1}, t = 1, 2, \ldots$. Hence, (23.10) remains appropriate for conditional inference. Moreover, the error process $\{\boldsymbol{\upsilon}_t\}_{t=-\infty}^{\infty}$ is also a martingale difference process with constant conditional variance and is uncorrelated with the $\{\mathbf{u}_{xt}\}_{t=-\infty}^{\infty}$ process. Thus, Assumptions 1(a)(ii) and 1(b)(ii) rule out any conditional heteroskedasticity. Assumption 1(c) is standard and, together with Assumption 1(a), is required for

the multivariate invariance principle stated in (23.20) below, while Assumption 1(b) together with Assumption 1(c) implies the multivariate invariance principle (23.21) below.

Define the partial sum process

$$\mathbf{s}_T^u(a) \equiv T^{-1/2} \sum_{s=1}^{[Ta]} \mathbf{u}_s,$$

where $[Ta]$ denotes the integer part of Ta, $a \in [0, 1]$. Under Assumption 1 (see also Assumptions 1–2 in Chapter 22), $\mathbf{s}_T^u(a)$ satisfies the multivariate invariance principle (see Section B.13 in Appendix B and Phillips and Durlaf (1986))

$$\mathbf{s}_T^u(a) \Rightarrow \mathbf{W}_m(a), \quad a \in [0, 1], \tag{23.20}$$

where $\mathbf{W}_m(.)$ denotes an m-dimensional Brownian motion with the variance matrix $\mathbf{\Sigma}$. We partition $\mathbf{s}_T^u(a) = (\mathbf{s}_T^y(a)', \mathbf{s}_T^x(a)')'$ conformably with $\mathbf{z}_t = (\mathbf{y}_t', \mathbf{x}_t')'$ and the Brownian motion $\mathbf{W}_m(a) = (\mathbf{W}_{m_y}(a)', \mathbf{W}_{m_y}(a)')'$ likewise, $a \in [0, 1]$. Define $\mathbf{s}_T^{\upsilon}(a) \equiv T^{-1/2} \sum_{s=1}^{[Ta]} \boldsymbol{\upsilon}_s, a \in [0, 1]$. Hence, as $\boldsymbol{\upsilon}_t = \mathbf{u}_{yt} - \mathbf{\Sigma}_{yx}\mathbf{\Sigma}_{xx}^{-1}\mathbf{u}_{xt}$,

$$\mathbf{s}_T^{\upsilon}(a) \Rightarrow \mathbf{W}_{m_y}^*(a), \tag{23.21}$$

where $\mathbf{W}_{m_y}^*(a) \equiv \mathbf{W}_{m_y}(a) - \mathbf{\Sigma}_{yx}\mathbf{\Sigma}_{xx}^{-1}\mathbf{W}_{m_x}(a)$ is a Brownian motion with variance matrix $\mathbf{\Sigma}_{\upsilon\upsilon}$ which is independent of $\mathbf{W}_{m_x}(a)$, $a \in [0, 1]$. See also Harbo et al. (1998).

Under restriction (23.4), the $m \times (m - r)$ matrix $\boldsymbol{\alpha}_\perp \equiv diag(\boldsymbol{\alpha}_y^\perp, \boldsymbol{\alpha}_x^\perp)$, where $\boldsymbol{\alpha}_x^\perp$, an $m_x \times m_x$ nonsingular matrix, is a basis for the orthogonal complement of the $m \times r$ loadings matrix $\boldsymbol{\alpha} = (\boldsymbol{\alpha}_y', \mathbf{0}')'$. Hence, we define the $(m - r)$-dimensional standard Brownian motion $\mathbf{B}_{m-r}(a) \equiv (\mathbf{B}_{m_y}(a)', \mathbf{B}_{m_x}(a)')'$ partitioned into the m_y- and m_x-dimensional subvector independent standard Brownian motions $\mathbf{B}_{m_y-r}(a) \equiv (\boldsymbol{\alpha}_y^{\perp\prime}\mathbf{\Sigma}_{\upsilon\upsilon}\boldsymbol{\alpha}_y^\perp)^{-1/2}\boldsymbol{\alpha}_y^{\perp\prime}\mathbf{W}_{m_y}^*(a)$ and $\mathbf{B}_{m_x}(a) \equiv (\boldsymbol{\alpha}_x^{\perp\prime}\mathbf{\Sigma}_{xx}\boldsymbol{\alpha}_x^\perp)^{-1/2}\boldsymbol{\alpha}_x^{\perp\prime}\mathbf{W}_{m_y}(a)$, $a \in [0, 1]$. See Pesaran, Shin, and Smith (2000) for further details. We also need to introduce the following associated de-meaned $(m-r)$-vector standard Brownian motion

$$\tilde{\mathbf{B}}_{m-r}(a) \equiv \mathbf{B}_{m-r}(a) - \int_0^1 \mathbf{B}_{m-r}(a)da, \tag{23.22}$$

and de-meaned and de-trended $(m - r)$-vector standard Brownian motion

$$\hat{\mathbf{B}}_{m-r}(a) \equiv \tilde{\mathbf{B}}_{m-r}(a) - 12\left(a - \frac{1}{2}\right)\int_0^1 \left(a - \frac{1}{2}\right)\tilde{\mathbf{B}}_{m-r}(a)da, \tag{23.23}$$

and their respective partitioned counterparts $\tilde{\mathbf{B}}_{m-r}(a) = (\tilde{\mathbf{B}}_{m_y-r}(a)', \tilde{\mathbf{B}}_{m_x}(a)')'$, and $\hat{\mathbf{B}}_{m-r}(a) = (\hat{\mathbf{B}}_{m_y-r}(a)', \hat{\mathbf{B}}_{m_x}(a)')'$, $a \in [0, 1]$.

23.5.1 Testing H_r against H_{r+1}

The log-likelihood ratio statistic for testing $H_r : Rank\left(\mathbf{\Pi}_y\right) = r$, against $H_{r+1} : Rank\left(\mathbf{\Pi}_y\right) = r + 1$ is given by

$$LR(H_r|H_{r+1}) = -T\ln(1-\hat{\lambda}_{r+1}), \tag{23.24}$$

where $\hat{\lambda}_r$ is the rth largest eigenvalue from the determinantal equation (23.18), $r = 0, \ldots, m_y-1$, with the appropriate definitions of $\Delta\hat{\mathbf{Y}}$ and $\hat{\mathbf{Z}}_{-1}^*$ and, thus, the sample moment matrices $\mathbf{S}_{YY}$, $\mathbf{S}_{YZ}$ and $\mathbf{S}_{ZZ}$ given by (23.17) to cover Cases I–V. Under Assumption 1 the limit distribution of $LR(H_r|H_{r+1})$ of (23.24) for testing H_r against H_{r+1} is given by the distribution of the maximum eigenvalue of

$$\int_0^1 d\mathbf{B}_{m_y-r}(a)\mathbf{F}_{m_y-r}(a)'\left(\int_0^1 \mathbf{F}_{m_y-r}(a)\mathbf{F}_{m-r}(a)'da\right)^{-1}\int_0^1 \mathbf{F}_{m_y-r}(a)d\mathbf{W}_{m_y-r}(a)', \tag{23.25}$$

where

$$\mathbf{F}_{m_y-r}(a) = \left\{\begin{array}{cc} \mathbf{B}_{m_y-r}(a) & \text{Case I} \\ (\mathbf{B}_{m_y-r}(a)', 1)' & \text{Case II} \\ \tilde{\mathbf{B}}_{m_y-r}(a) & \text{Case III} \\ (\tilde{\mathbf{B}}_{m_y-r}(a)', a-\frac{1}{2})' & \text{Case IV} \\ \hat{\mathbf{B}}_{m_y-r}(a) & \text{Case V} \end{array}\right\},\ a \in [0,1], \tag{23.26}$$

$r = 0, \ldots, m_y - 1$, where Cases I–V are defined in Section 23.3.1.

23.5.2 Testing H_r against H_{m_y}

The log-likelihood ratio statistic for testing $H_r : Rank\left(\mathbf{\Pi}_y\right) = r$, against $H_{m_y} : Rank\left(\mathbf{\Pi}_y\right) = m_y$ is given by

$$LR(H_r|H_{m_y}) = -T\sum_{i=r+1}^{m_y}\ln(1-\hat{\lambda}_i), \tag{23.27}$$

where $\hat{\lambda}_i$ is the ith largest eigenvalue from the determinantal equation (23.18). Under Assumption 1 the limit distribution of $LR(H_r|H_{m_y})$ of (23.27) for testing H_r against H_{m_y} is given by the distribution of

$$Trace\left\{\int_0^1 d\mathbf{W}_{m_y-r}(a)\mathbf{F}_{m_y-r}(a)'\left(\int_0^1 \mathbf{F}_{m_y-r}(a)\mathbf{F}_{m_y-r}(a)'da\right)^{-1}\int_0^1 \mathbf{F}_{m_y-r}(a)d\mathbf{W}_{m_y-r}(r)'\right\},$$

where $\mathbf{F}_{m_y-r}(a)$, $a \in [0,1]$, is defined in (23.26) for Cases I–V, $r = 0, \ldots, m_y - 1$.

23.5.3 Testing H_r in the presence of $I(0)$ weakly exogenous regressors

The (log-) likelihood ratio tests described above require that the process $\{\mathbf{x}_t\}_{t=1}^{\infty}$ is integrated of order one as noted below (23.11), and the weak exogeneity restriction, (23.4), is satisfied. However, many applications will include current and lagged values of weakly exogenous regressors which are integrated of order zero as explanatory variables in (23.6). In such circumstances, the above results are no longer applicable, and the limiting distributions of the (log-) likelihood

ratio tests for cointegration will now depend on nuisance parameters. However, the above analysis may easily be adapted to deal with this difficulty.

Let $\{\mathbf{w}_t\}_{t=1}^{\infty}$ denote a k_w-vector process of weakly exogenous explanatory variables which is integrated of order zero. Therefore, the partial sum vector process $\left\{\sum_{s=1}^{t}\mathbf{w}_s\right\}_{t=1}^{\infty}$ is integrated of order one. Defining $\sum_{s=1}^{t}\mathbf{w}_s$ as a sub-vector of $\mathbf{x}_t$ with the corresponding sub-vector of $\Delta\mathbf{x}_t$ as $\mathbf{w}_t, t = 1, 2, \ldots$, allows the above analysis to proceed unaltered. With these re-definitions of $\mathbf{x}_t$ and $\Delta\mathbf{x}_t$ to include $\sum_{s=1}^{t}\mathbf{w}_s$ and $\mathbf{w}_t, t = 1, 2, \ldots$, respectively, the partial sum $\sum_{s=1}^{t}\mathbf{w}_s$ will now appear in the cointegrating relations (22.53) and the lagged level term $\mathbf{z}_{t-1}$ in (23.6) although economic theory may indicate its absence; that is, the corresponding (k_w, r) block of the cointegrating matrix $\boldsymbol{\beta}$ is null. This constraint on the cointegrating matrix $\boldsymbol{\beta}$ is straightforwardly tested using a likelihood ratio statistic which will possess a limiting chi-squared distribution with rk_w degrees of freedom under H_r. See Rahbek and Mosconi (1999) for further discussion.

The asymptotic critical values for the (log-) likelihood ratio cointegration rank statistics (23.24) and (23.27) are available in Pesaran, Shin, and Smith (2000). However, as also explained in Section 22.10, these distributions are appropriate only asymptotically. When the sample is small or when the order of the *VARX* or the number of variables in the *VARX* is large, it is advisable to compute critical values by the bootstrap approach, as outlined in Section 22.12.

23.6 Identifying long-run relationships in a cointegrating *VARX*

Typically, the applied econometrician will be interested not only in the number of cointegrating relations that might exist among the variables but also the specification of the identifying (and possibly over-identifying) restrictions on the cointegrating relations. We have already seen that Johansen (1988, 1991) have provided procedures for estimating $\boldsymbol{\alpha}_y$ and $\boldsymbol{\beta}$, using 'statistical' over-identifying restrictions. The more satisfactory approach promoted in Pesaran and Shin (2002) is to estimate the cointegrating relations under a general set of structural long-run restrictions provided by *a priori* economic theory. This approach, described in Section 22.11, can be employed also in the context of the *VARX* model (23.1). Suppose that we are considering an example of a model with unrestricted intercepts and restricted trends (Case IV), and the cointegrating vectors, $\boldsymbol{\beta}$, are subject to the following k general linear restrictions, including cross-equation restrictions

$$\mathbf{R}\,\mathrm{vec}(\boldsymbol{\beta}) = \mathbf{b}, \tag{23.28}$$

where $\mathbf{R}$ and $\mathbf{b}$ are a $k \times (m+1)r$ matrix of full row rank and a $k \times 1$ vector of known constants, respectively, and $vec(\boldsymbol{\beta})$ is the $(m+1)r \times 1$ vector of long-run coefficients, which stacks the r columns of $\boldsymbol{\beta}$ into a vector. As in the case of *VAR* models, three cases can be distinguished:

(i) $k < r^2$: the under-identified case
(ii) $k = r^2$: the exactly-identified case
(iii) $k > r^2$: the over-identified case.

Let $\hat{\boldsymbol{\theta}}$ be the (unrestricted) *ML* estimators of $\boldsymbol{\theta}$ obtained subject to the r^2 exactly-identifying restrictions (say, $\mathbf{R}_A\boldsymbol{\theta} = \mathbf{b}_A$), and $\widetilde{\boldsymbol{\theta}}$ be the restricted *ML* estimators of $\boldsymbol{\theta}$ obtained subject to the

full k restrictions (namely, $\mathbf{R}\boldsymbol{\theta} = \mathbf{b}$), respectively. Then, following similar lines of reasoning as in Section 22.11, the $k - r^2$ over-identifying restrictions on $\boldsymbol{\theta}$ can be tested using the log-likelihood ratio statistic given by

$$LR = 2\left[\ell^c\left(\hat{\boldsymbol{\theta}}; r\right) - \ell^c\left(\widetilde{\boldsymbol{\theta}}; r\right)\right], \tag{23.29}$$

where $\ell^c\left(\hat{\boldsymbol{\theta}}; r\right)$ and $\ell^c\left(\widetilde{\boldsymbol{\theta}}; r\right)$ represent the maximized values of the log-likelihood function obtained under $\mathbf{R}_A\boldsymbol{\theta} = \mathbf{b}_A$ and $\mathbf{R}\boldsymbol{\theta} = \mathbf{b}$, respectively. Pesaran and Shin (2002) prove that the log-likelihood ratio statistic for testing $\mathbf{R}\boldsymbol{\theta} = \mathbf{b}$ given by (23.29) has a χ^2 distribution with $k - r^2$ degrees of freedom, asymptotically.

Also critical values for the above tests can be computed using the bootstrap approach of the type described in Section 22.12.

23.7 Forecasting using *VARX* models

Forecasting with a *VARX* model of the type given by (23.10) requires the specification of the processes driving the weakly exogenous variables, $\mathbf{x}_t$. Indeed, one needs to take into account the possibility of feedbacks from changes in one or more of the endogenous variables to the weakly exogenous variables, which in turn affect the current and future values of the endogenous variables. This last point is worth emphasizing and applies to any analysis involving counterfactuals such as impulse response analysis and forecasting exercises. Macro-modellers frequently consider the dynamic response of a system to a change in an exogenous variable by considering the effects of a once-and-for-all increase in that variable. This (implicitly) imposes restrictions on the processes generating the exogenous variable, assuming that there is no serial correlation in the variable and that a shock to one exogenous variable can be considered without having to take account of changes in other exogenous variables. These counter-factual exercises might be of interest. But, generally speaking, one needs to take into account the possibility that changes in one exogenous variable will have an impact on other exogenous variables and that these effects might continue and interact over time. This requires an explicit analysis of the dynamic processes driving the exogenous variables, as captured by the marginal model in (33.3). For this, we require the full-system *VECM*, obtained by augmenting the conditional model for $\Delta\mathbf{y}_t$, (23.10), with the marginal model for $\Delta\mathbf{x}_t$, (23.11). The combined model can be written as (See also (23.13)).

$$\Delta\mathbf{z}_t = -\boldsymbol{\alpha}\boldsymbol{\beta}'\mathbf{z}_{t-1} + \sum_{i=1}^{p-1}\boldsymbol{\Gamma}_i\Delta\mathbf{z}_{t-i} + \mathbf{H}\boldsymbol{\zeta}_t, \tag{23.30}$$

where

$$\boldsymbol{\alpha} = \begin{pmatrix}\boldsymbol{\alpha}_y \\ \mathbf{0}\end{pmatrix},\ \boldsymbol{\Gamma}_i = \begin{pmatrix}\boldsymbol{\Psi}_i + \boldsymbol{\Lambda}\boldsymbol{\Gamma}_{xi} \\ \boldsymbol{\Gamma}_{xi}\end{pmatrix}, \tag{23.31}$$

$$\boldsymbol{\zeta}_t = \begin{pmatrix}\boldsymbol{\upsilon}_t \\ \mathbf{u}_{xt}\end{pmatrix},\ \mathbf{H} = \begin{pmatrix}\mathbf{I}_{m_y} & \boldsymbol{\Lambda} \\ \mathbf{0} & \mathbf{I}_{m_x}\end{pmatrix},\ Cov(\boldsymbol{\zeta}_t) = \boldsymbol{\Sigma}_{\zeta\zeta} = \begin{pmatrix}\boldsymbol{\Sigma}_{\upsilon\upsilon} & \mathbf{0} \\ \mathbf{0} & \boldsymbol{\Sigma}_{xx}\end{pmatrix}. \tag{23.32}$$

The complete system, (23.30), can be written equivalently as

$$\mathbf{z}_t = \sum_{i=1}^{p} \mathbf{\Phi}_i \mathbf{z}_{t-i} + \mathbf{H}\boldsymbol{\zeta}_t, \tag{23.33}$$

where $\mathbf{\Phi}_1 = \mathbf{I}_m - \boldsymbol{\alpha}\boldsymbol{\beta}' + \mathbf{\Gamma}_1$, $\mathbf{\Phi}_i = \mathbf{\Gamma}_i - \mathbf{\Gamma}_{i-1}, i = 2, \ldots, p-1$, $\mathbf{\Phi}_p = -\mathbf{\Gamma}_{p-1}$.

The above reduced form equation can now be used for forecasting or impulse response analyses. The forecasts can be evaluated in terms of their root mean squared forecast error (*RMSFE*), which constitutes a widely used loss function. See Section 17.2 for a discussion on loss functions. Let $\mathbf{z}_{t+h}$ be the level of the variable that we wish to forecast, and denote the forecast of this variable formed at time t by $\hat{\mathbf{z}}(t+h, t)$. This forecasts can be computed recursively as

$$\hat{\mathbf{z}}(t+\tau, t) = \sum_{i=1}^{p} \hat{\mathbf{\Phi}}_i \hat{\mathbf{z}}(t+\tau-i, t), \text{for } \tau = 1, 2, \ldots, h,$$

with $\hat{\mathbf{z}}(t-i, t) = \mathbf{z}_{t-i}$, for $i = 0, 1, \ldots, p-1$**.** Further, define the h-step ahead forecast changes as $\hat{\mathbf{x}}_t(h) = \hat{\mathbf{z}}(t+h, t) - \mathbf{z}_t$ and the associated h-step ahead realized changes as $\mathbf{x}_t(h) = \mathbf{z}_{t+h} - \mathbf{z}_t$. The h-step ahead forecast error is then computed as

$$\mathbf{e}_t(h) = \mathbf{x}_t(h) - \hat{\mathbf{x}}_t(h) = \mathbf{z}_{t+h} - \hat{\mathbf{z}}(t+h, t).$$

The elements of $\mathbf{e}_t(h)$ can be then be used in forecast evaluation exercises.

23.8 An empirical application: a long-run structural model for the UK

In this section we discuss the empirical application in Garratt et al. (2003b), and consider a small macroeconomic model for the UK economy, show how this can be cast in the form of (23.10), and provide *ML* estimation using quarterly time series data. The model comprises six domestic variables whose developments are widely regarded as essential to a basic understanding of the behaviour of the UK macroeconomy; namely, aggregate output, the ratio of domestic to foreign price level, the rate of domestic inflation, the nominal short-term interest rate, the exchange rate, and the real money balances. The model also contains foreign output, foreign short-term interest rate, and oil prices. For estimation and testing purposes, we use quarterly data over the period 1965q1–1999q4, the data set used by Garratt et al. (2003b).

The long-run relationships of the core model (adopting a log-linear approximation) take the following form

$$p_t - p_t^* - e_t = b_{10} + b_{11}t + \xi_{1,t+1}, \tag{23.34}$$

$$r_t - r_t^* = b_{20} + \xi_{2,t+1}, \tag{23.35}$$

$$y_t - y_t^* = b_{30} + \xi_{3,t+1}, \tag{23.36}$$

$$h_t - y_t = b_{40} + b_{41}t + \beta_{42}r_t + \beta_{43}y_t + \xi_{4,t+1}, \tag{23.37}$$

$$r_t - \Delta p_t = b_{50} + \xi_{5,t+1}, \tag{23.38}$$

where $p_t = \ln(P_t), p_t^* = \ln(P_t^*), e_t = \ln(E_t), y_t = \ln(Y_t/P_t), y_t^* = \ln(Y_t^*/P_t^*), r_t = \ln(1+R_t)$, $r_t^* = \ln(1+R_t^*), h_t - y_t = \ln(H_{t+1}/P_t) - \ln(Y_t/P_t) = \ln(H_{t+1}/Y_t)$ and $b_{50} = \ln(1+\rho)$. The variables P_t and P_t^* are the domestic and foreign price indices, Y_t and Y_t^* are per-capita domestic and foreign outputs, R_t is the nominal interest rate on domestic assets held from the beginning to the end of period t, R_t^* is the nominal interest rate paid on foreign assets during period t, E_t is the effective exchange rate, defined as the domestic price of a unit of foreign currency at the beginning of period t, and $H_t = \tilde{H}_t/POP_{t-1}$ with $\tilde{H}_t$ the stock of high-powered money.

Equations (23.34), (23.35) and (23.38) describe a set of arbitrage conditions, included in many macroeconomic models in one form or another. These are the (relative) purchasing power parity (*PPP*), the uncovered interest parity (*UIP*), and the Fisher inflation parity (*FIP*) relationships. Equation (23.36) is an output gap relation, while (23.37) is a long-run condition that is derived from the solvency constraints to which the economy is subject (see Garratt et al. (2003b) for details).

We have allowed for intercept and trend terms (when appropriate) in order to ensure that (long-run) reduced form disturbances, $\xi_{i,t+1}, i = 1, 2, \ldots, 5$, have zero means.

The five long-run relations of the core model, (23.34)–(23.38), can be written more compactly as

$$\boldsymbol{\xi}_t = \boldsymbol{\beta}'\mathbf{z}_{t-1} - \mathbf{b}_0 - \mathbf{b}_1(t-1), \tag{23.39}$$

where

$$\mathbf{z}_t = \left(p_t^o, e_t, r_t^*, r_t, \Delta p_t, y_t, p_t - p_t^*, h_t - y_t, y_t^*\right)'. \tag{23.40}$$

$$\mathbf{b}_0 = (b_{01}, b_{02}, b_{03}, b_{04}, b_{05})', \ \mathbf{b}_1 = (b_{11}, 0, 0, b_{41}, 0),$$

$$\boldsymbol{\xi}_t = (\xi_{1t}, \xi_{2t}, \xi_{3t}, \xi_{4t}, \xi_{5t})',$$

$$\boldsymbol{\beta}' = \begin{pmatrix} 0 & -1 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & -1 \\ 0 & 0 & 0 & -\beta_{42} & 0 & -\beta_{43} & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & -1 & 0 & 0 & 0 & 0 \end{pmatrix}, \tag{23.41}$$

and p_t^o is the logarithm of oil prices. In modelling the short-run dynamics, we follow Sims (1980) and others and assume that departures from the long-run relations, $\boldsymbol{\xi}_t$, can be approximated by a linear function of a *finite* number of past changes in $\mathbf{z}_{t-1}$. For estimation purposes we also partition $\mathbf{z}_t = (p_t^o, \mathbf{y}_t')'$ where $\mathbf{y}_t = (e_t, r_t^*, r_t, \Delta p_t, y_t, p_t - p_t^*, h_t - y_t, y_t^*)'$. Here, p_t^o is considered to be a 'long-run forcing' variable for the determination of $\mathbf{y}_t$, in the sense that changes in p_t^o have a direct influence on $\mathbf{y}_t$, but changes in p_t^o are not affected by the presence of $\boldsymbol{\xi}_t$, which measures the extent of disequilibria in the UK economy. The treatment of oil prices as 'long-run forcing' represents a generalization of the approach to modelling oil price effects in some previous applications of cointegrating *VAR* analyses (e.g., Johansen and Juselius (1992)), where the oil price change is treated as a strictly exogenous $I(0)$ variable. The approach taken in the previous literature excludes the possibility that there might exist cointegrating relationships which involve the oil price level, while the approach taken here allows the validity of the hypothesized restriction to be tested, and for the restriction to be imposed if it is not rejected. Note that foreign output and

interest rates are treated as endogenous, to allow for the possibility of feedbacks. This involves loss of efficiency in estimation if they were in fact long-run forcing or strictly exogenous.

Under the assumption that oil prices are long-run forcing for $\mathbf{y}_t$, the cointegrating properties of the model can be investigated without having to specify the oil price equation. However, specification of an oil price equation is required for the analysis of the short-run dynamics. We shall adopt the following general specification for the evolution of oil prices

$$\Delta p_t^o = \delta_o + \sum_{i=1}^{s-1} \delta_{oi} \Delta \mathbf{z}_{t-i} + u_{ot}, \tag{23.42}$$

where u_{ot} represents a serially uncorrelated oil price shock with a zero mean and a constant variance. The above specification ensures oil prices are long-run forcing for $\mathbf{y}_t$ since it allows lagged changes in the endogenous and exogenous variables of the model to influence current oil prices but rules out the possibility that error correction terms, $\boldsymbol{\xi}_t$, have any effects on oil price changes. These assumptions are weaker than the requirement of 'Granger non-causality' often invoked in the literature.

Assuming that the variables in $\mathbf{z}_t$ are difference-stationary, our modelling strategy is now to embody $\boldsymbol{\xi}_t$ in an otherwise unrestricted $VAR(s-1)$ in $\mathbf{z}_t$. Under the assumption that oil prices are long-run forcing, it is efficient (for estimation purposes) to base our analysis on the following *conditional* error correction model

$$\Delta \mathbf{y}_t = \mathbf{a}_y - \boldsymbol{\alpha}_y \boldsymbol{\xi}_t + \sum_{i=1}^{s-1} \boldsymbol{\Gamma}_{yi} \Delta \mathbf{z}_{t-i} + \boldsymbol{\psi}_{yo} \Delta p_t^o + \mathbf{u}_{yt}, \tag{23.43}$$

where $\mathbf{a}_y$ is an 8×1 vector of fixed intercepts, $\boldsymbol{\alpha}_y$ is an 8×5 matrix of error-correction coefficients (also known as the loading coefficient matrix), $\{\boldsymbol{\Gamma}_{yi}, i = 1, 2, \ldots, s-1\}$ are 8×9 matrices of short-run coefficients, $\boldsymbol{\psi}_{yo}$ is an 8×1 vector representing the impact effects of changes in oil prices on $\Delta \mathbf{y}_t$, and $\mathbf{u}_{yt}$ is an 8×1 vector of disturbances assumed to be $IID(0, \boldsymbol{\Sigma}_y)$, with $\boldsymbol{\Sigma}_y$ being a positive definite matrix, and by construction uncorrelated with u_{ot}. Using equation (23.39), we now have

$$\Delta \mathbf{y}_t = \mathbf{a}_y - \boldsymbol{\alpha}_y \mathbf{b}_0 - \boldsymbol{\alpha}_y \left[\boldsymbol{\beta}' \mathbf{z}_{t-1} - \mathbf{b}_1 (t-1) \right] + \sum_{i=1}^{s-1} \boldsymbol{\Gamma}_{yi} \Delta \mathbf{z}_{t-i} + \boldsymbol{\psi}_{yo} \Delta p_t^o + \mathbf{u}_{yt}, \tag{23.44}$$

where $\boldsymbol{\beta}' \mathbf{z}_{t-1} - \mathbf{b}_1 (t-1)$ is a 5×1 vector of error correction terms. The above specification embodies the economic theory's long-run predictions by construction, in contrast to the more usual approach where the starting point is an unrestricted *VAR* model, with some vague priors about the nature of the long-run relations.

Estimation of the parameters of the core model, (23.44), can be carried out using the long-run structural modelling approach described in Sections 23.3 and 23.5. With this approach, having selected the order of the underlying *VAR* model (using model selection criteria such as the Akaike information criterion (*AIC*) or the Schwarz Bayesian criterion (*SBC*)), we test for the number of cointegrating relations among the 9 variables in $\mathbf{z}_t$. When performing this task, and in all subsequent empirical analyses, we work with a *VARX* model with unrestricted intercepts and

restricted trend coefficients (Case IV). In terms of (23.44), we allow the intercepts to be freely estimated but restrict the trend coefficients so that $\boldsymbol{\alpha}_y \mathbf{b}_1 = \boldsymbol{\Pi}_y \boldsymbol{\gamma}$, where $\boldsymbol{\Pi}_y = \boldsymbol{\alpha}_y \boldsymbol{\beta}'$ and $\boldsymbol{\gamma}$ is an 9×1 vector of unknown coefficients. We then compute *ML* estimates of the model parameters subject to exact and over-identifying restrictions on the long-run coefficients. Assuming that there is empirical support for the existence of five long-run relationships, as suggested by theory, exact identification in our model requires five restrictions on each of the five cointegrating vectors (each row of $\boldsymbol{\beta}$), or a total of 25 restrictions on $\boldsymbol{\beta}$. These represent only a subset of the restrictions suggested by economic theory, as characterized in (23.41). Estimation of the model subject to all the (exact- and over-identifying) restrictions given in (23.41) enables a test of the validity of the over-identifying restrictions, and hence the long-run implications of the economic theory, to be carried out.

23.8.1 Estimation and testing of the model

We assume our variables are I(1), and refer to Garratt et al. (2003b) for a detailed analysis of the non-stationarity properties of these variables. The first stage of our modelling sequence is to select the order of the underlying *VAR* in these variables. Here we find that a *VAR* of order two appears to be appropriate when using the *AIC* as the model selection criterion, but that the *SBC* favours a *VAR* of order one. We proceed with the cointegration analysis using a *VARX*(2,2), on the grounds that the consequences of over-estimation of the order of the *VAR* is much less serious than under-estimating it (see Kilian (1997)).

Using a *VARX*(2,2) model with unrestricted intercepts and restricted trend coefficients, and treating the oil price variable, p_t^o, as weakly exogenous for the long-run parameters, we computed Johansen's 'trace' and 'maximal eigenvalue' statistics. These statistics, together with their associated 90 per cent and 95 per cent critical values, are reported in Table 23.1.

The maximal eigenvalue statistic indicates the presence of just two cointegrating relationships at the 95 per cent significance level, which does not support our *a priori* expectations of five cointegrating vectors. However, as shown by Cheung and Lai (1993), the maximum eigenvalue test is generally less robust to the presence of skewness and excess kurtosis in the errors than the trace test. Given that we have evidence of non-normality in the residuals of the *VAR* model used to compute the test statistics, we therefore believe it is more appropriate to base our cointegration tests on the trace statistics. As it happens, the trace statistics reject the null hypotheses that $r = 0, 1, 2, 3$ and 4 at the 5 per cent level of significance but cannot reject the null hypothesis that $r = 5$. This is in line with our *a priori* expectations based on the long-run theory discussed above. Hence we proceed under the assumption that there are five cointegrating vectors.

With five cointegrating relations we require five restrictions on each of the five relationships to exactly identify them. In view of the underlying long-run theory (see Garratt et al. (2003b) for description of the derivation of long-run, steady state relations of the UK core macroeconomic model) we impose the following 25 exact-identifying restrictions on the cointegrating matrix

$$\boldsymbol{\beta}' = \begin{pmatrix} \beta_{11} & \beta_{12} & 0 & 0 & \beta_{15} & 0 & 1 & \beta_{18} & 0 \\ \beta_{21} & 0 & \beta_{23} & 1 & \beta_{25} & 0 & 0 & 0 & \beta_{29} \\ \beta_{31} & 0 & 0 & 0 & 0 & 1 & \beta_{37} & \beta_{38} & \beta_{39} \\ \beta_{41} & 0 & 0 & \beta_{44} & \beta_{45} & \beta_{46} & 0 & 1 & 0 \\ \beta_{51} & 0 & 0 & \beta_{54} & -1 & 0 & 0 & \beta_{58} & \beta_{59} \end{pmatrix}, \tag{23.45}$$

Table 23.1 Cointegration rank statistics for the UK model

(a) Trace statistic

H_0	H_1	Test statistic	95% Critical values	90% Critical values
$r = 0$	$r = 1$	324.75	199.12	192.80
$r \leq 1$	$r = 2$	221.16	163.01	157.02
$r \leq 2$	$r = 3$	161.88	128.79	123.33
$r \leq 3$	$r = 4$	116.14	97.83	93.13
$r \leq 4$	$r = 5$	78.94	72.10	68.04
$r \leq 5$	$r = 6$	48.71	49.36	46.00
$r \leq 6$	$r = 7$	22.46	30.77	27.96
$r \leq 7$	$r = 8$	6.70	15.44	13.31

(b) Maximum eigenvalue statistic

H_0	H_1	Test statistic	95% Critical values	90% Critical values
$r = 0$	$r = 1$	103.59	58.08	55.25
$r \leq 1$	$r = 2$	59.27	52.62	49.70
$r \leq 2$	$r = 3$	45.75	46.97	44.01
$r \leq 3$	$r = 4$	37.20	40.89	37.92
$r \leq 4$	$r = 5$	30.23	34.70	32.12
$r \leq 5$	$r = 6$	26.25	28.72	26.10
$r \leq 6$	$r = 7$	15.76	22.16	19.79
$r \leq 7$	$r = 8$	6.70	15.44	13.31

Notes: The underlying *VARX* model is of order 2 and contains unrestricted intercepts and restricted trend coefficients, with p_t^o treated as an exogenous $I(1)$ variable. The statistics refer to Johansen's log-likelihood-based trace and maximal eigenvalue statistics and are computed using 140 observations for the period 1965q1–1999q4. The asymptotic critical values are taken from Pesaran, Shin, and Smith (2000).

that corresponds to $\mathbf{z}_t = \left(p_t^o, e_t, r_t^*, r_t, \Delta p_t, y_t, p_t - p_t^*, h_t - y_t, y_t^*\right)'$. The first vector (the first row of $\boldsymbol{\beta}'$) relates to the *PPP* relationship defined by (23.34) and is normalized on $p_t - p_t^*$; the second relates to the *IRP* relationship defined by (23.35) and is normalized on r_t; the third relates to the 'output gap' relationship defined by (23.36) and is normalized on y_t;[2] the fourth is the money market equilibrium condition defined by (23.37) and is normalised on $h_t - y_t$.; and the fifth is the real interest rate relationship defined by (23.38), normalised on Δp_t.

Having exactly identified the long-run relations, we then test the over-identifying restrictions predicted by the long-run theory. There are 20 unrestricted parameters in (23.45), and two in (23.41), yielding a total of 18 over-identifying restrictions. In addition, working with a cointegrating *VAR* with restricted trend coefficients, there are potentially five further parameters on the trend terms in the five cointegrating relationships. The imposition of zeros on the trend coefficients in the *IRP*, *FIP* or output gap relationships provides a further three over-identifying restrictions. The absence of a trend in the *PPP* relationship is also consistent with the theory, as is the restriction that $\beta_{46} = 0$ (so that equation (23.37) is effectively a relationship explaining the velocity of circulation of money). These final two restrictions, together with those which are

[2] Our use of the term 'output gap relationship' to describe (23.36) should not be confused with the more usual use of the term which relates, more specifically, to the difference between a country's actual and potential output levels (although clearly the two uses of the term are related).

intrinsic to the theory, mean that there are just two parameters to be freely estimated in the cointegrating relationships and provide a total of 23 over-identifying restrictions on which the core model is based and with which the validity of the economic theory can be tested.

The log-likelihood ratio (LR) statistic for jointly testing the 23 over-identifying restrictions takes the value 71.49. In view of the relatively large dimension of the underlying *VAR* model, the number of restrictions considered and the available sample size, we proceed to test the significance of this statistic using critical values which are computed by means of bootstrap techniques. (See Section 22.12 for a discussion of bootstrap techniques applied to cointegrating *VAR* models.) In the present application, the bootstrap exercise is based on 3,000 replications of the LR statistic testing the 23 restrictions. For each replication, an artificial data set is generated (of the same length as the original data set) on the assumption that the estimated version of the core model is the true data-generating process, using the observed initial values of each variable, the estimated model, and a set of random innovations.[3] The test of the over-identifying restrictions is carried out on each of the replicated data sets and the empirical distribution of the test statistic is derived across all replications. This shows that the relevant critical values for the joint tests of the 23 over-identifying restrictions are 67.51 at the 10 per cent significance level and 73.19 at the 5 per cent level. Therefore, LR statistic of 71.49 is not sufficiently large to justify the rejection of the over-identifying restrictions implied by the long-run theory.

ML estimation of the five error correction terms yields

$$(p_t - p_t^*) - e_t = 4.588 + \hat{\xi}_{1,t+1}, \tag{23.46}$$

$$r_t - r_t^* = 0.0058 + \hat{\xi}_{2,t+1}, \tag{23.47}$$

$$y_t - y_t^* = -0.0377 + \hat{\xi}_{3,t+1}, \tag{23.48}$$

$$h_t - y_t = 0.5295 - \underset{(22.2844)}{56.0975}\, r_t - \underset{(0.0012)}{0.0073}\, t + \hat{\xi}_{4,t+1}, \tag{23.49}$$

$$r_t - \Delta p_t = 0.0036 + \hat{\xi}_{5,t+1}. \tag{23.50}$$

The bracketed figures are asymptotic standard errors. The first equation, (23.46), describes the *PPP* relationship and the failure to reject this in the context of our core model provides an interesting empirical finding. Of course, there has been considerable interest in the literature examining the co-movements of exchange rates and relative prices, and the empirical evidence on *PPP* appears to be sensitive to the data set used and the way in which the analysis is conducted. For example, the evidence of a unit root in the real exchange rate found by Darby (1983) and Huizinga (1988) contradicts *PPP* as a long-run relationship, while Grilli and Kaminsky (1991) and Lothian and Taylor (1996) have obtained evidence in favour of rejecting the unit root hypothesis in real exchange rates using longer annual series.

The second cointegrating relation, defined by (23.47), is the *IRP* condition. This includes an intercept, which can be interpreted as the deterministic component of the risk premia associated with bonds and foreign exchange uncertainties. Its value is estimated at 0.0058, implying a risk premium of approximately 2.3 per cent per annum. The empirical support we find for the *IRP* condition, namely that $r_t - r_t^* \backsim I(0)$, is in accordance with the results obtained in the literature,

[3] In light of the evidence of non-normality of residuals, in this exercise we apply the non-parametric bootstrap (see Section 22.12). The cointegrating matrix subject to the over-identifying restrictions is estimated on each replicated data set using the simulated annealing routine by Goffe, Ferrier, and Rogers (1994). Also see Section A.16.3 in Appendix A.

and is compatible with *UIP*, defined by (23.35). However, under the *UIP* hypothesis it is also required that a regression of $r_t - r_t^*$ on $\Delta \ln(E_{t+1})$ has a unit coefficient, but this is not supported by the data.

The third long-run relationship, given by (23.48), is the output gap (*OG*) relationship with per capita domestic and foreign output (measured by the Organisation for Economic Co-operation and Development (OECD) total output) levels moving in tandem in the long run. It is noteworthy that the co-trending hypothesis cannot be rejected; that is, the coefficient of the deterministic trend in the output gap equation is zero. This suggests that average long-run growth rate for the UK is the same as that in the rest of the OECD. This finding seems, in the first instance, to contradict some of the results obtained in the literature on the cointegrating properties of real output across countries. Campbell and Mankiw (1989), and Cogley (1990), for example, consider cointegration among international output series and find little evidence that outputs of different *pairs* of countries are cointegrated. However, our empirical analysis, being based on a single foreign output index, does not necessarily contradict this literature, which focuses on pair-wise cointegration of output levels. The hypothesis advanced here, that y_t and y_t^* are cointegrated, is much less restrictive than the hypothesis considered in the literature that *all* pairs of output variables in the OECD are cointegrated.

For the money market equilibrium (*MME*) condition, given by (23.49), we could not reject the hypothesis that the elasticity of real money balances with respect to real output is equal to unity, and therefore (23.49) in fact represents an M0 velocity equation. The *MME* condition, however, contains a deterministic downward trend, representing the steady decline in the money–income ratio in the UK over most of the period 1965–99, arising primarily from the technological innovations in financial inter-mediation. There is also strong statistical evidence of a negative interest rate effect on real money balances. This long-run specification is comparable with the recent research on the determinants of the UK narrow money velocity reported in, for example, Breedon and Fisher (1996).

Finally, the fifth equation, (23.50), defines the *FIP* relationship, where the estimated constant implies an annual real rate of return of approximately 1.67 per cent. While the presence of this relationship might appear relatively non-contentious, there is empirical work in which the relationship appears not to hold; see, for example MacDonald and Murphy (1989) and Mishkin (1992). The results support the *FIP* relationship and again highlight the important role played by the *FIP* relationship in a model of the macroeconomy which can incorporate interactions between variables omitted from more partial analyses.

The estimates of the long-run relations and short-run dynamics of the model are provided in Table 23.2. The estimates of the error correction coefficients (also known as the loading coefficients) show that the long-run relations make an important contribution in most equations and that the error correction terms provide for a complex and statistically significant set of interactions and feedbacks across commodity, money and foreign exchange markets. The results in Table 23.2 also show that the core model fits the data well and has satisfactory diagnostic statistics.

23.9 Further Reading

For further details on methods for cointegrating *VARX* models see Harbo et al. (1998), Pesaran, Shin, and Smith (2000), Pesaran (2006), and Garratt et al. (2006). Applications of the models

Table 23.2 Reduced form error correction specification for the UK model

Equation	$\Delta(p_t\text{-}p_t^*)$	Δe_t	Δr_t	Δr_t^*	Δy_t	Δy_t^*	$\Delta(h_t\text{-}y_t)$	$\Delta(\Delta p_t)$
$\hat{\xi}_{1,t}$	$-.015^{\dagger}$	$.060^{\dagger}$	.002	.002	$.017^{\dagger}$	$.021^{\dagger}$	$-.024^{*}$	$-.005$
	(.007)	(.029)	(.002)	(.001)	(.008)	(.004)	(.013)	(.004)
$\hat{\xi}_{2,t}$	$-.840^{\dagger}$	1.42	.049	$.130^{*}$	$1.34^{\dagger}$	$.891^{\dagger}$	$-.721$	$-.811^{\dagger}$
	(.301)	(1.28)	(.107)	(.043)	(.353)	(.181)	(.576)	(.297)
$\hat{\xi}_{3,t}$	$.062^{\dagger}$	$-.210^{*}$	$-.013$	$-.006$	$-.165^{\dagger}$	$-.021$	$.106^{*}$	.034
	(.029)	(.121)	(.010)	(.004)	(.034)	(.017)	(.055)	(.028)
$\hat{\xi}_{4,t}$	$.018^{\dagger}$	$-.029$	$-.003^{*}$	$-.001^{*}$	$-.027^{\dagger}$	$-.016^{\dagger}$	$-.003$	$.009^{*}$
	(.005)	(.020)	(.002)	(.001)	(.005)	(.003)	(.009)	(.005)
$\hat{\xi}_{5,t}$	$-.149^{*}$	$-.244$	$-.054^{*}$	$-.024^{\dagger}$	$-.099$	$-.119^{\dagger}$	$.408^{\dagger}$	$.451^{\dagger}$
	(.083)	(.353)	(.028)	(.012)	(.098)	(.050)	(.159)	(.082)
$\Delta(p_{t-1} - p_{t-1}^*)$	$.459^{\dagger}$	.150	$-.039$	$-.028^{\dagger}$	$-.136$	$-.013$	.046	$.436^{\dagger}$
	(.095)	(.404)	(.032)	(.014)	(.111)	(.057)	(.182)	(.094)
Δe_{t-1}	$.051^{\dagger}$	$.216^{\dagger}$	$-.005$	$-.001$	.021	.013	.007	$-.022$
	(.022)	(.092)	(.007)	(.003)	(.025)	(.013)	(.042)	(.021)
Δr_{t-1}	$.416^{\dagger}$	-1.31	.125	$-.067$	.467	.204	$-.677$	$.974^{\dagger}$
	(.294)	(1.25)	(.098)	(.042)	(.345)	(.177)	(.562)	(.290)
Δr_{t-1}^*	-0.810	2.75	$-.606^{\dagger}$	$.430^{\dagger}$	.306	.573	$-.267$	.166
	(.617)	(2.62)	(.205)	(.088)	(.723)	(.371)	(1.18)	(.606)
Δy_{t-1}	.083	.072	.017	.015	$-.044$	.031	$-.168$	$.356^{\dagger}$
	(.089)	(.381)	(.030)	(.013)	(.105)	(.053)	(.172)	(.089)
Δy_{t-1}^*	.010	$-.630$	$-.050$	$.040^{*}$	$-.073$	.069	$.602^{*}$	$-.010$
	(.161)	(.683)	(.054)	(.023)	(.188)	(.097)	(.307)	(.158)
$\Delta(h_{t-1} - y_{t-1})$	.116	.331	.026	.006	.069	$-.014$	$-.253^{\dagger}$	$.140^{\dagger}$
	(.054)	(.228)	(.018)	(.008)	(.063)	(.032)	(.103)	(.053)
$\Delta(\Delta p_{t-1})$	$-.151^{\dagger}$	.321	.016	.010	.125	$-.082^{*}$	.012	$-.244^{\dagger}$
	(.073)	(.302)	(.024)	(.011)	(.086)	(.044)	(.140)	(.072)
Δp_t^o	$-.018^{\dagger}$	$-.024$	.001	$.001^{\dagger}$	$-.010^{\dagger}$	.0001	$.024^{\dagger}$	.003
	(.004)	(.018)	(.001)	(.0005)	(.005)	(.002)	(.008)	(.004)
Δp_{t-1}^o	$.010^{\dagger}$	$-.013$	$-.002$	$-.0001$	.006	.002	-.011	$.016^{\dagger}$
	(.005)	(.019)	(.002)	(.0001)	(.005)	(.003)	(.009)	(.004)
$\overline{R}^2$	.484	.070	.115	.345	.260	.367	.257	.445
Benchmark $\overline{R}^2$	.316	.026	.007	.213	.022	.196	.00	.191
$\hat{\sigma}$	.007	.032	.002	.001	.009	.004	.014	.007
$\chi^2_{SC}[4]$	2.79	0.96	2.43	$17.13^{\dagger}$	6.71	.79	$8.37^{\dagger}$	5.63
$\chi^2_{FF}[1]$	$8.57^{\dagger}$	0.13	$4.34^{\dagger}$	$6.70^{\dagger}$	0.04	$5.28^{\dagger}$	.033	0.01
$\chi^2_{N}[2]$	$12.53^{\dagger}$	$13.98^{\dagger}$	$17.15^{\dagger}$	$19.9^{\dagger}$	$112.4^{\dagger}$	10.84	$31.45^{\dagger}$	$118.9^{\dagger}$
$\chi^2_{H}[1]$	$6.13^{\dagger}$	1.97	$4.53^{\dagger}$	$5.2^{\dagger}$	0.88	0.93	0.19	$4.55^{\dagger}$

Notes: Standard errors are given in parenthesis. '$*$' indicates significance at the 10% level, and '†' indicates significance at the 5% level. The diagnostics are chi-squared statistics for serial correlation (SC), functional form (FF), normality (N) and heteroskedasticity (H). The benchmark $\overline{R}^2$ statistics are computed based on univariate *ARMA*(s,q), s,q=0,1,...,4, specifications with s- and q-order selected by *AIC*.

and methods described in this chapter can be found in Assenmacher-Wesche and Pesaran (2008) for the Swiss economy, and in Garratt et al. (2006) and Garratt et al. (2003b) for the UK.

23.10 Exercises

1. Suppose that $\mathbf{z}_t = (y_t, x_t)'$ is jointly determined by the following vector autoregressive model of order 1, $VAR(1)$,

$$\mathbf{z}_t = \mathbf{\Phi}\mathbf{z}_{t-1} + \mathbf{e}_t,$$

where $\mathbf{\Phi} = (\phi_{ij})$ is a 2×2 matrix of unknown parameters, and $\mathbf{e}_t = \left(e_{yt}, e_{xt}\right)'$ is 2-dimensional vector of reduced form errors. Denoting the covariance of e_{yt} and e_{xt} by $\omega Var\left(e_{xt}\right)$

(a) show that

$$e_{yt} = E\left(e_{yt} \left| e_{xt}\right.\right) + u_t = \omega e_{xt} + u_t,$$

where u_t is uncorrelated with e_{xt}, and therefore the first equation in the *VAR* can be written as the following *ARDL* model

$$y_t = \varphi y_{t-1} + \beta_0 x_t + \beta_1 x_{t-1} + u_t,$$

with

$$\varphi = \phi_{11} - \omega\phi_{21},\ \beta_0 = \omega,\ \beta_1 = \phi_{12} - \omega\phi_{22}.$$

(b) Further show that

$$y_t = \theta x_t + \alpha\left(L\right)\Delta x_t + \tilde{u}_t,$$

where

$$\theta = \frac{\beta_0 + \beta_1}{1 - \varphi},$$

$\tilde{u}_t = (1 - \varphi L)^{-1} u_t$, $\alpha\left(L\right) = \sum_{\ell=0}^{\infty} \alpha_\ell L^\ell$, with $\alpha_\ell = \sum_{s=\ell+1}^{\infty} \delta_s$, for $\ell = 0, 1, 2, \ldots$, and $\delta\left(L\right) = \sum_{\ell=0}^{\infty} \delta_\ell L^\ell = (1 - \varphi L)^{-1}\left(\beta_0 + \beta_1 L\right)$.

(c) Under what condition is $(1, -\theta)'$ is a cointegrating vector?

(d) Discuss alternative approaches to the estimation of θ, distinguishing the cases where x_t is $I(0)$ and $I(1)$.

2. Consider the *VARX* model

$$\begin{aligned}\mathbf{y}_t &= \mathbf{A}\mathbf{y}_{t-1} + \mathbf{B}\mathbf{x}_t + \mathbf{u}_{yt},\\ \mathbf{x}_t &= \mathbf{C}_0\mathbf{x}_{t-1} + \mathbf{C}_1\mathbf{y}_{t-1} + \mathbf{u}_{xt},\end{aligned}$$

where $\mathbf{u}_t = (\mathbf{u}_{yt}', \mathbf{u}_{xt}')' \sim IID(\mathbf{0}, \mathbf{\Sigma}_{\mathrm{u}})$, $\mathbf{A}$, $\mathbf{B}$, $\mathbf{C}_0$ and $\mathbf{C}_1$ are $m_y \times m_y$, $m_y \times m_x$, $m_x \times m_x$ and $m_x \times m_y$ matrices of fixed constants.

(a) Show that

$$E\left(\mathbf{y}_t \left| \mathbf{y}_{t-1}, \mathbf{x}_{t-1}\right.\right) = \mathbf{A}\mathbf{y}_{t-1} + \mathbf{H}\mathbf{x}_{t-1},$$

and

$$E\left(\mathbf{y}_t \left| \mathbf{y}_{t-1}, \mathbf{x}_t\right.\right) = \mathbf{A}\mathbf{y}_{t-1} + \mathbf{G}\mathbf{x}_t,$$

and derive $\mathbf{H}$ and $\mathbf{G}$ in terms of the parameters of the underlying model.

(b) Derive and compare the mean squared forecast errors of predicting $\mathbf{y}_{T+1}$ conditional on $(\mathbf{y}_T, \mathbf{x}_T)$ and $(\mathbf{y}_T, \mathbf{x}_{T+1})$, with respect to a quadratic loss function.

3. Consider the following *VARX* model

$$\mathbf{y}_t = \mathbf{\Phi}\mathbf{y}_{t-1} + \mathbf{C}\mathbf{x}_t + \mathbf{u}_t,$$
$$\mathbf{x}_t = \mathbf{\Psi}\mathbf{x}_{t-1} + \mathbf{v}_t,$$

where $\mathbf{y}_t = \left(y_{1t}, y_{2t}, \ldots, y_{mt}\right)'$, $\mathbf{x}_t = (x_{1t}, x_{2t}, \ldots, x_{kt})'$, $\mathbf{\Phi}$, $\mathbf{C}$, and $\mathbf{\Psi}$ are fixed-coefficient matrices, and $\boldsymbol{\xi}_t = (\mathbf{u}_t', \mathbf{v}_t')' \sim IID(\mathbf{0}, \mathbf{\Sigma})$, where $\mathbf{\Sigma}$ is a positive definite matrix.

(a) Suppose that all eigenvalues of $\mathbf{\Phi}$ lie within the unit circle. Derive the Beveridge–Nelson decomposition of $\mathbf{y}_t$ when the eigenvalues of $\mathbf{\Psi}$ lie inside the unit circle, and when $\mathbf{I}_k - \mathbf{\Psi}$ is rank deficient.

(b) Repeat the exercise under (a) assuming that $\mathbf{I}_k - \mathbf{\Psi}$ and $\mathbf{I}_m - \mathbf{\Phi}$ are both rank deficient.

(c) Derive the long horizon forecasts of $\mathbf{y}_t$ assuming that $\mathbf{\Psi}$ has some roots on the unit circle.

(d) How do you estimate $lim_{h\to\infty} E\left(\mathbf{y}_{t+h} \left| \mathbf{y}_t, \mathbf{x}_t \right.\right)$?

4. Use quarterly time series data on the US and UK economies, which can be downloaded from <https://sites.google.com/site/gvarmodelling/data>, to estimate a *VARX* model for the UK economy conditional on US real equity prices and long term interest rates, taking as endogenous the following UK variables: real output, inflation, real equity prices, short-term and long-term interest rates.

(a) Test for the presence of a long-run relation between UK inflation and UK long-term interest rate.

(b) Test for the presence of a long-run relation between UK and US long-term interest rates.

(c) Discuss the pros and cons of this *VARX* model with an alternative specification where conditioning variables also include euro area variables.

24 Impulse Response Analysis

24.1 Introduction

We first introduce impulse response analysis and forecast error variance decomposition for unrestricted *VAR* models, and discuss the orthogonalized and generalized impulse response functions. We then consider the identification problem of short-run effects in a structural *VAR* model. We review Sims' approach, and then investigate the identification problem of a structural model when one or more of the structural shocks have permanent effects.

24.2 Impulse response analysis

An impulse response function measures the time profile of the effect of shocks at a given point in time on the (expected) future values of the variables. The best way to think of an impulse response function is to view it as the outcome of a *conceptual* experiment, whereby interest is on the effect of shock(s) hitting the economy at time t on the future state of the economy at time $t+n$, given the history of the economy, and the types of shocks that are likely to hit the economy in the future (at the interim times $t+1, t+2, \ldots, t+n$).

The main issues are:

1. What types of shocks hit the system at time t?
2. What was the state of the system at time $t-1$, before the shocks hit?
3. What types of shocks are expected to hit the system from time $t+1$ to $t+n$?

24.3 Traditional impulse response functions

Consider an $m \times 1$ vector of random variables, $\mathbf{y}_t$. The impulse response function of $\mathbf{y}_t$ at horizon n is defined by

$$\begin{aligned}&\mathbf{I}_y\,(n, \boldsymbol{\delta}, \Omega_{t-1})\\&= E\left(\mathbf{y}_{t+n}\,|\mathbf{u}_t = \boldsymbol{\delta},\ \mathbf{u}_{t+1} = \mathbf{0}, \cdots, \mathbf{u}_{t+n} = \mathbf{0}; \Omega_{t-1}\right)\\&-E\left(\mathbf{y}_{t+n}\,|\mathbf{u}_t = \mathbf{0},\ \mathbf{u}_{t+1} = \mathbf{0}, \cdots, \mathbf{u}_{t+n} = \mathbf{0}; \Omega_{t-1}\right),\end{aligned}$$

where $\mathbf{u}_t$ is the vector of variables being shocked, $\boldsymbol{\delta}$ the vector of shocks, and Ω_t is the information set at time t, containing the information available up to time t.

Example 55 *Consider the model*

$$y_t = \rho y_{t-1} + u_t \qquad |\rho| < 1,$$

In this case $m = 1$, and it is easily seen that the impulse response function of y_t is

$$I_y\,(n, \delta, \Omega_{t-1}) = \rho^n \delta, \quad n = 0, 1, 2, \ldots.$$

Now consider the model

$$\Delta y_t = \phi \Delta y_{t-1} + u_t.$$

Its impulse response function is

$$I_y\,(n, \delta, \Omega_{t-1}) = \delta\left(\frac{1 - \phi^{n+1}}{1 - \phi}\right).$$

24.3.1 Multivariate systems

Consider the following $VAR(1)$ model

$$\mathbf{y}_t = \boldsymbol{\Phi}\mathbf{y}_{t-1} + \mathbf{u}_t,$$

assuming that the process is stationary, we can write $\mathbf{y}_t$ in terms of the shocks, $\mathbf{u}_t$, and their lagged values (see Chapter 21)

$$\mathbf{y}_t = \mathbf{u}_t + \boldsymbol{\Phi}\mathbf{u}_{t-1} + \boldsymbol{\Phi}^2\mathbf{u}_{t-1} + \ldots.$$

More generally, we have

$$\begin{aligned}\underset{m \times 1}{\mathbf{y}_t} &= \sum_{i=0}^{\infty} \underset{m \times m}{\mathbf{A}_i}\ \mathbf{u}_{t-i},\\ \underset{m \times 1}{\mathbf{u}_t} &\sim (\mathbf{0}, \underset{m \times m}{\boldsymbol{\Sigma}}).\end{aligned}$$

Then

$$\mathbf{I}_x\,(n, \boldsymbol{\delta}, \Omega_{t-1}) = \mathbf{A}_n\boldsymbol{\delta},$$

where $\boldsymbol{\delta}$ is now an $m \times 1$ vector. For the $VAR(1)$ model we have $\mathbf{A}_n = \boldsymbol{\Phi}^n$.

24.4 Orthogonalized impulse response function

We now describe the traditional approach to impulse response analysis, introduced by Sims (1980) for the analysis of *VAR* models. Consider the $VAR(p)$ model

$$\mathbf{y}_t = \mathbf{\Phi}_1 \mathbf{y}_{t-1} + \mathbf{\Phi}_2 \mathbf{y}_{t-2} + \cdots + \mathbf{\Phi}_p \mathbf{y}_{t-p} + \mathbf{u}_t, \tag{24.1}$$

$\mathbf{y}_t$ is an $m \times 1$ vector, $\mathbf{\Phi}_i$ are $m \times m$ matrices of coefficients and

$$\mathbf{u}_t \sim IID(\mathbf{0}, \mathbf{\Sigma}), \tag{24.2}$$

where $\mathbf{\Sigma}$ is the covariance matrix of the errors. First write the *VAR* model in the form of an infinite-order moving average (MA) representation

$$\mathbf{y}_t = \sum_{j=0}^{\infty} \mathbf{A}_j \mathbf{u}_{t-j}, \tag{24.3}$$

where the matrices $\mathbf{A}_j$ are determined by the recursive relations

$$\mathbf{A}_j = \mathbf{\Phi}_1 \mathbf{A}_{j-1} + \mathbf{\Phi}_2 \mathbf{A}_{j-2} + \cdots + \mathbf{\Phi}_p \mathbf{A}_{j-p}, \ \ j = 1, 2, \ldots, \tag{24.4}$$

with

$$\mathbf{A}_0 = \mathbf{I}_m, \text{ and } \mathbf{A}_j = 0, \text{ for } j < 0.$$

Sims' approach employs the following Cholesky decomposition of $\mathbf{\Sigma}$

$$\mathbf{\Sigma} = \mathbf{P}\mathbf{P}', \tag{24.5}$$

where $\mathbf{P}$ is a lower-triangular matrix (see also Section 24.10). Then rewrite the moving average representation of $\mathbf{y}_t$ as

$$\mathbf{y}_t = \sum_{j=0}^{\infty} \left(\mathbf{A}_j \mathbf{P}\right) \left(\mathbf{P}^{-1} \mathbf{u}_{t-j}\right) = \sum_{j=0}^{\infty} \mathbf{B}_j \boldsymbol{\eta}_{t-j}, \tag{24.6}$$

where

$$\mathbf{B}_j = \mathbf{A}_j \mathbf{P}, \qquad \text{and} \qquad \boldsymbol{\eta}_t = \mathbf{P}^{-1} \mathbf{u}_t.$$

It is now easily seen that

$$E\left(\boldsymbol{\eta}_t \boldsymbol{\eta}_t'\right) = \mathbf{P}^{-1} E\left(\mathbf{u}_t \mathbf{u}_t'\right) \mathbf{P}'^{-1} = \mathbf{P}^{-1} \mathbf{\Sigma} \mathbf{P}'^{-1} = \mathbf{I}_m,$$

and the new errors $\boldsymbol{\eta}_t = \left(\eta_{1t}, \eta_{2t}, \ldots \eta_{mt}\right)'$ in (24.6) are contemporaneously uncorrelated, namely $Var\left(\eta_{it}, \eta_{jt}\right) = 0$, for $i \neq j$. The orthogonalized impact of a unit shock at time t to the ith equation on $\mathbf{y}$ at time $t+n$ is given by

$$\mathbf{B}_n \mathbf{e}_i, \quad n = 0, 1, 2, \ldots, \tag{24.7}$$

where $\mathbf{e}_i$ is an $m \times 1$ selection vector

$$\mathbf{e}_i = \begin{pmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} \leftarrow i^{th} \text{ position.} \tag{24.8}$$

Written more compactly, the orthogonalized impulse response function of a unit (one standard error) shock to the ith variable on the jth variable is given by

$$OI_{ij,n} = \mathbf{e}_j' \mathbf{A}_n \mathbf{P} \mathbf{e}_i, \quad i, j, = 1, 2, \ldots, m. \tag{24.9}$$

These orthogonalized impulse responses are not unique and depend on the particular ordering of the variables in the *VAR*. The orthogonalized responses are invariant to the ordering of the variables only if $\boldsymbol{\Sigma}$ is diagonal. The non-uniqueness of the orthogonalized impulse responses is also related to the non-uniqueness of the matrix $\mathbf{P}$ in the Cholesky decomposition of $\boldsymbol{\Sigma}$ in (24.5). For more details see Lütkepohl (2005).

24.4.1 A simple example

Consider the following $VAR(1)$ model in the two variables y_{1t} and y_{2t}

$$\begin{pmatrix} y_{1t} \\ y_{2t} \end{pmatrix} = \begin{pmatrix} \phi_{11} & \phi_{12} \\ \phi_{21} & \phi_{22} \end{pmatrix} \begin{pmatrix} y_{1,t-1} \\ y_{2,t-1} \end{pmatrix} + \begin{pmatrix} u_{1t} \\ u_{2t} \end{pmatrix}.$$

The linear correlation between u_{1t} and u_{2t} can be characterized by

$$u_{1t} = \left(\frac{\sigma_{12}}{\sigma_{22}}\right) u_{2t} + \eta_{1t},$$

where $\sigma_{12} = var(u_{1t}, u_{2t})$, $\sigma_{22} = var(u_{2t})$, and the new error, η_{1t}, has a zero correlation with u_{2t}. Using this relationship we have

$$\begin{aligned} y_{1t} = {} & \phi_{11} y_{1,t-1} + \phi_{12} y_{2,t-1} + \\ & (\sigma_{12}/\sigma_{22}) \left(y_{2t} - \phi_{21} y_{1,t-1} - \phi_{22} y_{2,t-1}\right) + \eta_{1t}, \end{aligned}$$

or

$$y_{1t} = (\sigma_{12}/\sigma_{22})\, y_{2t} + (\phi_{11} - \frac{\sigma_{12}}{\sigma_{22}}\phi_{21})y_{1,t-1} + (\phi_{12} - \frac{\sigma_{12}}{\sigma_{22}}\phi_{22})y_{2,t-1} + \eta_{1t}. \tag{24.10}$$

In this formulation y_{1t} is contemporaneously related to y_{2t} if $\sigma_{12} \neq 0$. Therefore, in general a 'unit' change in u_{2t}, through changing y_{2t}, will have a contemporaneous impact on y_{1t} and, *vice versa*, a unit change in u_{1t} will have a contemporaneous impact on y_{2t}.

Under orthogonalized impulse response analysis the system is constrained such that the contemporaneous value of y_{1t} does not have a contemporaneous effect on y_{2t}. But the contemporaneous value of y_{2t} does affect both y_{1t} and y_{2t}. Namely, a recursive structure is assumed for the contemporaneous relationship between y_{1t} and y_{2t}. In the present example this is achieved by combining (24.10) with the second equation in the *VAR*, namely

$$y_{2t} = \phi_{21}y_{1,t-1} + \phi_{22}y_{2,t-1} + u_{2t}.$$

By construction η_{1t} and u_{2t} are orthogonal. Hence shocking η_{1t} will move y_{1t} on impact but leaves y_{2t} unchanged. By contrast, a shock to u_{2t} of size, say $\sqrt{\sigma_{22}}$, will move y_{2t} directly by the amount of the shock, $\sqrt{\sigma_{22}}$, and through equation (24.10) will cause y_{1t} to move on impact by the amount of $\left(\frac{\sigma_{12}}{\sigma_{22}}\right)\sqrt{\sigma_{22}} = \frac{\sigma_{12}}{\sqrt{\sigma_{22}}}$.

This system can also be presented by an (upper) triangular form with the orthogonalized shocks, η_{1t} and u_{2t}

$$\begin{bmatrix} 1 & -\left(\frac{\sigma_{12}}{\sigma_{22}}\right) \\ 0 & 1 \end{bmatrix} \begin{pmatrix} y_{1t} \\ y_{2t} \end{pmatrix} = \begin{bmatrix} (\phi_{11} - \frac{\sigma_{12}}{\sigma_{22}}\phi_{21}) & (\phi_{12} - \frac{\sigma_{12}}{\sigma_{22}}\phi_{22}) \\ \phi_{21} & \phi_{22} \end{bmatrix} \begin{pmatrix} y_{1,t-1} \\ y_{2,t-2} \end{pmatrix} + \begin{pmatrix} \eta_{1t} \\ u_{2t} \end{pmatrix}.$$

Writing the above more compactly, we have

$$\mathbf{A}_0\mathbf{y}_t = \mathbf{A}_1\mathbf{y}_{t-1} + \boldsymbol{\varepsilon}_t,$$

where $\mathbf{A}_0$ is an upper triangular matrix, and the shocks in $\boldsymbol{\varepsilon}_t = (\eta_{1t}, u_{2t})'$ are orthogonal by construction. This is the identification scheme of Sims which treats the shocks in $\boldsymbol{\varepsilon}_t$ as structural. It is also worth noting that out of the two reduced form errors, u_{1t}, and u_{2t}, it is u_{2t} which is viewed as structural, and this is made possible by restricting the second equation in the above system not to contain contemporaneous effects from y_{1t}, and by assuming that the shocks η_{1t} and u_{2t} (in $\boldsymbol{\varepsilon}_t$) to be orthogonal.

A generalization of the above identification scheme when there are m equations, with $m > 2$ is provided in Section 24.10, where we show that in order to identify a shock as structural it is not necessary that all the shocks be orthogonal and/or $\mathbf{A}_0$ to be lower triangular. But first we develop the concept of the generalized impulse response function which allows the analysis of systems with non-orthogonalized shocks.

24.5 Generalized impulse response function (*GIRF*)

The main idea behind the generalized *IR* function is to circumvent the problem of the dependence of the orthogonalized impulse responses to the ordering of the variables in the *VAR*. The concept of the generalized impulse response function, advanced in Koop, Pesaran, and Potter (1996) was originally intended to deal with the problem of impulse response analysis in the case of nonlinear dynamic systems, but it can also be readily applied to multivariate time series models such as *VAR*.

The generalized *IR* analysis deals explicitly with three main issues that arise in impulse response analysis:

1. How was the dynamical system hit by shocks at time t? Was it hit by a variable-specific shock or system-wide shocks?
2. What was the state of the system at time $t-1$, before the system was hit by shocks? Was the trajectory of the system in an upward or in a downward phase?
3. How would one expect the system to be shocked in the future, namely over the interim period from $t+1$, to $t+n$?

In the context of the *VAR* model, the *GIRF* for a *system-wide shock*, $\mathbf{u}_t^0$, is defined by

$$GI_{\mathbf{y}}\left(n, \mathbf{u}_t^0, \Omega_{t-1}^0\right) = E\left(\mathbf{y}_{t+n} | \mathbf{u}_t = \mathbf{u}_t^0, \Omega_{t-1}^0\right) - E\left(\mathbf{y}_{t+n} | \Omega_{t-1}^0\right), \tag{24.11}$$

where $E(\cdot|\cdot)$ is the conditional mathematical expectation taken with respect to the *VAR* model, and Ω_{t-1}^0 is a particular historical realization of the process at time $t-1$. In the case of the *VAR* model having the infinite moving average representation (24.3) we have

$$GI_{\mathbf{y}}\left(n, \mathbf{u}_t^0, \Omega_{t-1}^0\right) = \mathbf{A}_n \mathbf{u}_t^0, \tag{24.12}$$

which is independent of the 'history' of the process. This history invariance property of the impulse response function (also shared by the traditional methods of impulse response analysis) is, however, specific to linear systems and does not carry over to nonlinear dynamic models.

In practice, the choice of the vector of shocks, $\mathbf{u}_t^0$, is arbitrary; one possibility would be to consider a large number of likely shocks and then examine the empirical distribution function of $\mathbf{A}_n \mathbf{u}_t^0$ for all these shocks. In the case where $\mathbf{u}_t^0$ is drawn from the same distribution as $\mathbf{u}_t$, namely a multivariate normal with zero means and a constant covariance matrix $\mathbf{\Sigma}$, we have the analytical result that

$$GI_{\mathbf{y}}\left(n, \mathbf{u}_t^0, \Omega_{t-1}^0\right) \sim N\left(\mathbf{0}, \mathbf{A}_n \Sigma \mathbf{A}_n'\right). \tag{24.13}$$

The diagonal elements of $\mathbf{A}_n \Sigma \mathbf{A}_n'$, when appropriately scaled, are the 'persistence profiles' proposed in Lee and Pesaran (1993), and applied in Pesaran and Shin (1996) to analyse the speed of convergence to equilibrium in cointegrated systems. It is also worth noting that when the underlying *VAR* model is stable, the limit of the persistence profile as $n \to \infty$ tends to the spectral density function of $\mathbf{y}_t$ at zero frequency (apart from a multiple of π).

Consider now the effect of a *variable-specific shock* on the evolution of $\mathbf{y}_{t+1}, \mathbf{y}_{t+2}, \ldots, \mathbf{y}_{t+n}$, and suppose that the *VAR* model is perturbed by a shock of size $\delta_i = \sqrt{\sigma_{ii}}$ to its ith equation at time t. By the definition of the generalized IR function we have

$$GI_{\mathbf{y}}\left(n, \delta_i, \Omega_{t-1}^0\right) = E\left(\mathbf{y}_t | u_{it} = \delta_i, \Omega_{t-1}^0\right) - E\left(\mathbf{y}_t | \Omega_{t-1}^0\right). \tag{24.14}$$

Once again using the infinite moving average representation (24.3), we obtain

$$GI_{\mathbf{y}}\left(n, \delta_i, \Omega_{t-1}^0\right) = \mathbf{A}_n E\left(\mathbf{u}_t \,|u_{it} = \delta_i\right), \tag{24.15}$$

which is history invariant (i.e. does not depend on Ω_{t-1}^0). The computation of the conditional expectations $E\left(\mathbf{u}_t \,|u_{it} = \delta_i\right)$ depends on the nature of the multivariate distribution assumed for the disturbances, $\mathbf{u}_t$. In the case where $\mathbf{u}_t \sim IIDN\left(\mathbf{0}, \mathbf{\Sigma}\right)$, we have

$$E\left(\mathbf{u}_t \,|u_{it} = \delta_i\right) = \begin{pmatrix} \sigma_{1i}/\sigma_{ii} \\ \sigma_{2i}/\sigma_{ii} \\ \vdots \\ \sigma_{mi}/\sigma_{ii} \end{pmatrix} \delta_i, \tag{24.16}$$

where as before $\mathbf{\Sigma} = (\sigma_{ij})$. Hence, for a 'unit shock' defined by $\delta_i = \sqrt{\sigma_{ii}}$, we have

$$GI_{\mathbf{y}}\left(n, \delta_i = \sqrt{\sigma_{ii}}, \Omega_{t-1}^0\right) = \frac{\mathbf{A}_n \mathbf{\Sigma} \mathbf{e}_i}{\sqrt{\sigma_{ii}}}, \quad i, j, = 1, 2, \ldots, m, \tag{24.17}$$

where $\mathbf{e}_i$ is a selection vector defined by (24.8). The *GIRF* of a unit shock to the ith equation in the *VAR* model (24.1) on the jth variable at horizon n is given by the jth element of (24.17), or expressed more compactly by

$$GI_{ij,n} = \frac{\mathbf{e}_j' \mathbf{A}_n \Sigma \mathbf{e}_i}{\sqrt{\sigma_{ii}}}, \quad i, j, = 1, 2, \ldots, m. \tag{24.18}$$

Unlike the orthogonalized impulse responses in (24.9), the generalized IR in (24.18) are invariant to the ordering of the variables in the *VAR* . It is also interesting to note that the two impulse responses coincide only for the first variable in the *VAR*, or when $\mathbf{\Sigma}$ is a diagonal matrix. See Pesaran and Shin (1998) for further details and derivations.

24.6 Identification of a single structural shock in a structural model

Now consider the structural simultaneous equation model in m variables

$$\mathbf{A}_0 \mathbf{y}_t = \mathbf{A}_1 \mathbf{y}_{t-1} + \boldsymbol{\varepsilon}_t,$$

where $\mathbf{y}_t$ is an $m \times 1$ vector of endogenous variables, $\mathbf{A}_0$ and $\mathbf{A}_1$ are matrices that contain the structural coefficients, and $\boldsymbol{\varepsilon}_t$ is an $m \times 1$ vector of structural shocks in the sense that $\mathbf{A}_0$ and $\mathbf{A}_1$ are invariant to shocks to one or more elements of $\boldsymbol{\varepsilon}_t$. Here, however, unlike the usual assumptions made in the literature, we do not require the structural shocks to be orthogonal. In particular, we assume that $\boldsymbol{\varepsilon}_t \sim IID(\mathbf{0}, \boldsymbol{\Sigma}_\varepsilon)$, where $\boldsymbol{\Sigma}_\varepsilon$ is unrestricted.

Consider the problem of deriving the generalized impulse response function of a unit shock to the composite structural shock, $\boldsymbol{\varepsilon}_{ct} = \mathbf{a}'\boldsymbol{\varepsilon}_t$, where $\mathbf{a} = (a_1, a_2, \ldots, a_m)'$, and the size of the unit shock is given by $\delta_c = \sqrt{\mathbf{a}'\boldsymbol{\Sigma}_\varepsilon \mathbf{a}}$. We have

$$g_y(h, \delta_c) = E\left(\mathbf{y}_{t+h} \left| \mathbf{a}'\boldsymbol{\varepsilon}_t = \delta_c, \Im_{t-1}\right.\right) - E\left(\mathbf{y}_{t+h} \left| \Im_{t-1}\right.\right),$$

where $\Im_{t-1}$ is the information set at time. It is now easily seen that

$$\begin{aligned}
\mathbf{g}_y(h, \delta_c) &= \mathbf{A}_0^{-1}\mathbf{A}_1\mathbf{g}_y(h-1, \delta_c), \text{ for } h > 0, \\
\mathbf{g}_y(0, \delta_c) &= A_0^{-1}E\left(\boldsymbol{\varepsilon}_t \left| \mathbf{a}'\boldsymbol{\varepsilon}_t = \delta_c\right.\right) = \delta_c^{-1}\mathbf{A}_0^{-1}\boldsymbol{\Sigma}_\varepsilon \mathbf{a}.
\end{aligned}$$

But in terms of the parameters of the reduced form model

$$\mathbf{y}_t = \boldsymbol{\Phi}\mathbf{y}_{t-1} + \mathbf{u}_t,$$

where as before $\mathbf{u}_t \sim IID(\mathbf{0}, \boldsymbol{\Sigma})$, $\boldsymbol{\Sigma} = \mathbf{A}_0^{-1}\boldsymbol{\Sigma}_\varepsilon \mathbf{A}_0^{\prime -1}$, and $\boldsymbol{\Phi} = \mathbf{A}_0^{-1}\mathbf{A}_1$, we have

$$\begin{aligned}
\mathbf{g}_y(h, \delta_c) &= \boldsymbol{\Phi}^h \mathbf{g}_y(\mathbf{0}, \delta_c), \\
\mathbf{g}_y(0, \delta_c) &= \delta_c^{-1}\mathbf{A}_0^{-1}\boldsymbol{\Sigma}_\varepsilon \mathbf{a} = \delta_c^{-1}\boldsymbol{\Sigma}\mathbf{A}_0'\mathbf{a}.
\end{aligned}$$

Note that $\boldsymbol{\Phi}$ and $\boldsymbol{\Sigma}$ can be identified from the reduced form model, and the scaling parameter, δ_c, is given. Hence, for the identification of the effects of the composite shock we require identification of $\mathbf{A}_0'\mathbf{a}$, and not all the elements of $\mathbf{A}_0$.

Suppose now that we are interested in identifying the effects of the ith structural shock, ε_{it}. In this case we need to set $\mathbf{a} = \mathbf{e}_i = (0, 0, \ldots, 0, 1, 0, \ldots, 0, 0)'$ which is a vector of zeros with the exception of its ith element, and $\delta_c = \sqrt{\sigma_{ii,\varepsilon}}$. It is now easily seen that $\mathbf{A}_0'\mathbf{a} = \mathbf{A}_0'\mathbf{e}_i$ which is equal to the ith row of $\mathbf{A}_0$, which we denote by the $m \times 1$ vector, $\mathbf{a}_{0i}$, and note that $\mathbf{A}_0' = (\mathbf{a}_{01}, \mathbf{a}_{02}, \ldots., \mathbf{a}_{0m})$. Hence, to identify the effects of the ith structural shock we only need to identify the elements of the ith row of $\mathbf{A}_0$ even if the structural shocks are not orthogonal. An important example where the effects of the ith structural shock are identified arises if there are no contemporaneous effects in the ith structural equation. This is equivalent to placing the ith endogenous variable first in the list of the variables when applying Sim's orthogonalization procedure, with this important difference that under the above setup we do not require ε_{it} to be orthogonal to the other structural shocks.

It is clear that other more general assumptions concerning the ith row of $\mathbf{A}_0$ can be entertained. For example, it is possible to identify other elements of $\mathbf{a}_{0i}$ by standard exclusion restrictions. The point is that we do not need to make assumptions that identify the effects of *all* the structural shocks, as is often done in the literature, if we are interested in identifying the effects of the ith shock only.

24.7 Forecast error variance decompositions

The forecast error variance decomposition provides a decomposition of the variance of the forecast errors of the variables in the *VAR* at different horizons.

24.7.1 Orthogonalized forecast error variance decomposition

In the context of the orthogonalized *MA* representation of the *VAR* model given by (24.6), the n-step ahead forecast error is given by

$$\boldsymbol{\xi}_t(n) = \sum_{\ell=0}^{n} \mathbf{B}_\ell \boldsymbol{\eta}_{t+n-\ell}.$$

Since all the elements of $\boldsymbol{\eta}_{t+n-\ell}$ are pair-wise orthogonal at all leads and lags, knowing the jth orthogonalized shocks now and in future will have no contemporaneous effect on the other variables, but could affect the ability to forecast the other variables in future periods. More formally, the forecast errors conditional on the shocks $\eta_{j,t+n-\ell}$, for $\ell = 0, 1, 2, \ldots, n$, are given by

$$\boldsymbol{\xi}_t^{(j)}(n) = \sum_{\ell=0}^{n} \mathbf{B}_\ell \Big(\boldsymbol{\eta}_{t+n-\ell} - \mathbf{e}_j \eta_{j,t+n-\ell} \Big),$$

where by construction $\mathbf{e}_j' \boldsymbol{\xi}_t^{(j)}(n) = 0$ for $n = 0$ (note that $\mathbf{B}_0 = \mathbf{I}_m$). It is now easily seen that [noting that $E\left(\boldsymbol{\eta}_{t+n-\ell}\boldsymbol{\eta}_{t+n-\ell'}'\right) = \mathbf{0}$ if $\ell \neq \ell'$ and $E\left(\boldsymbol{\eta}_{t+n-\ell}\boldsymbol{\eta}_{t+n-\ell'}'\right) = \mathbf{I}_m$ if $\ell = \ell'$] we have

$$Var\left[\boldsymbol{\xi}_t(n)\right] = \sum_{\ell=0}^{n} \mathbf{B}_\ell \mathbf{B}_\ell',$$

and

$$Var\left[\boldsymbol{\xi}_t^{(j)}(n)\right] = \sum_{\ell=0}^{n} \mathbf{B}_\ell \mathbf{B}_\ell' - \sum_{\ell=0}^{n} \mathbf{B}_\ell \mathbf{e}_j \mathbf{e}_j' \mathbf{B}_\ell'.$$

Hence, improvements in the n-step ahead forecasts (in the mean squared error sense) of knowing the values of the jth orthogonlized shocks are given by

$$Var\left[\boldsymbol{\xi}_t^{(j)}(n)\right] - Var\left[\boldsymbol{\xi}_t(n)\right] = \sum_{\ell=0}^{n} \mathbf{B}_\ell \mathbf{e}_j \mathbf{e}_j' \mathbf{B}_\ell'.$$

Scaling the ith element of this matrix by the ith element of $\sum_{\ell=0}^{n} \mathbf{B}_\ell \mathbf{B}_\ell'$ yields the proportion of the forecast error variance of the ith variable that can be predicted by knowing the values of the jth orthogonalized shocks, namely

$$\theta_{ij,n} = \frac{\sum_{\ell=0}^{n} \left(\mathbf{e}_i' \mathbf{B}_\ell \mathbf{e}_j\right)^2}{\sum_{\ell=0}^{n} \mathbf{e}_i' \mathbf{B}_\ell \mathbf{B}_\ell' \mathbf{e}_i} = \frac{\mathbf{e}_i' \left(\sum_{\ell=0}^{n} \mathbf{B}_\ell \mathbf{e}_j \mathbf{e}_j' \mathbf{B}_\ell' \right) \mathbf{e}_i}{\mathbf{e}_i' \left(\sum_{\ell=0}^{n} \mathbf{B}_\ell \mathbf{B}_\ell' \right) \mathbf{e}_i}, \quad i = 1, 2, \ldots, m. \tag{24.19}$$

In the multivariate time series literature, $\theta_{ij,n}$ for $i = 1, 2, \ldots, m$ is known as the forecast error variance decomposition of the ith variable in the *VAR*. Since $\sum_{j=1}^{m} \mathbf{e}_j \mathbf{e}_j' = \mathbf{I}_m$, it is then easily seen that $\sum_{j=1}^{m} \theta_{ij,n} = 1$, which follows due to the orthogonal nature of the shocks, η_{jt}. Also since $\mathbf{B}_\ell = \mathbf{A}_\ell \mathbf{P}$, where $\mathbf{P}$ is defined by the Cholesky decomposition of $\boldsymbol{\Sigma}$, (24.5), we can also write $\theta_{ij,n}$ as

$$\theta_{ij,n} = \frac{\sum_{\ell=0}^{n} \left(\mathbf{e}_i' \mathbf{A}_\ell \mathbf{P} \mathbf{e}_j\right)^2}{\sum_{\ell=0}^{n} \mathbf{e}_i' \mathbf{A}_\ell \boldsymbol{\Sigma} \mathbf{A}_\ell' \mathbf{e}_i}, \quad j = 1, 2, \ldots, m. \tag{24.20}$$

$\theta_{ij,n}$ can also be viewed as measuring the proportion of the n-step ahead forecast error variance of variable i, which is accounted for by the orthogonalized innovations in variable j. For further details, see, for example, Lütkepohl (2005). As with the orthogonalized impulse response function, the orthogonalized forecast error variance decompositions in (24.20) are not invariant to the ordering of the variables in the *VAR*.

24.7.2 Generalized forecast error variance decomposition

An alternative procedure to the orthogonalized forecast error variance decomposition would be to consider the proportion of the variance of the n-step forecast errors of $\mathbf{y}_t$ that are explained by conditioning on the non-orthogonalized shocks, $\mathbf{u}_t, \mathbf{u}_{t+1}, \ldots, \mathbf{u}_{t+n}$, but explicitly allowing for the contemporaneous correlations between these shocks and the shocks to the other equations in the system.

Using the MA representation (24.3), the forecast error of predicting $\mathbf{y}_{t+n}$ conditional on the information at time $t-1$ is given by

$$\underset{m \times 1}{\boldsymbol{\xi}_t(n)} = \sum_{\ell=0}^{n} \mathbf{A}_\ell \mathbf{u}_{t+n-\ell}, \tag{24.21}$$

with the *total* forecast error covariance matrix

$$\text{Var}\left[\boldsymbol{\xi}_t(n)\right] = \sum_{\ell=0}^{n} \mathbf{A}_\ell \boldsymbol{\Sigma} \mathbf{A}_\ell'. \tag{24.22}$$

Consider now the forecast error covariance matrix of predicting $\mathbf{y}_{t+n}$ conditional on the information at time $t-1$, and given values of the shocks to the jth equation, $u_{jt}, u_{j,t+1}, \ldots, u_{j,t+n}$.

Using (24.3) we have[1]

$$\underset{m \times 1}{\boldsymbol{\xi}_t^{(j)}(n)} = \sum_{\ell=0}^{n} \mathbf{A}_\ell \left[\mathbf{u}_{t+n-\ell} - E\left(\mathbf{u}_{t+n-\ell} \left| u_{j,t+n-\ell} \right. \right) \right]. \tag{24.23}$$

As in the case of the generalized IR, assuming $\mathbf{u}_t \sim IID(\mathbf{0}, \boldsymbol{\Sigma})$ we have

$$E\left(\mathbf{u}_{t+n-\ell} \left| u_{j,t+n-\ell} \right. \right) = \left(\sigma_{jj}^{-1} \boldsymbol{\Sigma} \mathbf{e}_j \right) u_{j,t+n-\ell} \qquad \text{for} \quad \begin{array}{l} \ell = 0, 1, 2, \ldots, n, \\ j = 1, 2, \ldots, m. \end{array}$$

Substituting this result back in (24.23)

$$\boldsymbol{\xi}_t^{(j)}(n) = \sum_{\ell=0}^{n} \mathbf{A}_\ell \left(\mathbf{u}_{t+n-\ell} - \sigma_{jj}^{-1} \boldsymbol{\Sigma} \mathbf{e}_j u_{j,t+n-\ell} \right),$$

and taking unconditional expectations, yields

$$\text{Var}\left[\boldsymbol{\xi}_t^{(j)}(n) \right] = \sum_{\ell=0}^{n} \mathbf{A}_\ell \boldsymbol{\Sigma} \mathbf{A}_\ell' - \sigma_{jj}^{-1} \left(\sum_{\ell=0}^{n} \mathbf{A}_\ell \boldsymbol{\Sigma} \mathbf{e}_j \mathbf{e}_j' \boldsymbol{\Sigma} \mathbf{A}_\ell' \right). \tag{24.24}$$

Therefore, using (24.22) and (24.24) it follows that the decline in the n-step forecast error variance of $\boldsymbol{\xi}_t$ obtained as a result of conditioning on the future shocks to the ith equation is given by

$$\begin{aligned} \boldsymbol{\Delta}_{jn} &= \text{Var}\left[\boldsymbol{\xi}_t(n) \right] - \text{Var}\left[\boldsymbol{\xi}_t^{(j)}(n) \right] \\ &= \sigma_{jj}^{-1} \sum_{\ell=0}^{n} \mathbf{A}_\ell \boldsymbol{\Sigma} \mathbf{e}_j \mathbf{e}_j' \boldsymbol{\Sigma} \mathbf{A}_\ell'. \end{aligned} \tag{24.25}$$

Scaling the ith diagonal element of $\boldsymbol{\Delta}_{jn}$, namely $\mathbf{e}_i' \boldsymbol{\Delta}_{jn} \mathbf{e}_i$, by the n-step ahead forecast error variance of the ith variable in $\mathbf{y}_t$, we have the following generalized forecast error variance decomposition

$$\boldsymbol{\Psi}_{ij,n} = \frac{\sigma_{jj}^{-1} \sum_{\ell=0}^{n} \left(\mathbf{e}_i' \mathbf{A}_\ell \boldsymbol{\Sigma} \mathbf{e}_j \right)^2}{\sum_{\ell=0}^{n} \mathbf{e}_i' \mathbf{A}_\ell \boldsymbol{\Sigma} \mathbf{A}_\ell' \mathbf{e}_i}. \tag{24.26}$$

Note that the denominator of this measure is the ith diagonal element of the total forecast error variance formula in (24.22) and is the same as the denominator of the orthogonalized forecast error variance decomposition formula (24.20). Also $\boldsymbol{\theta}_{ij,n} = \boldsymbol{\Psi}_{ij,n}$ when y_{it} is the first variable in the *VAR*, and/or $\boldsymbol{\Sigma}$ is diagonal. However, in general the two decompositions differ.

[1] Note that since $\mathbf{u}_t$s are serially uncorrelated, $E\left(\mathbf{u}_{t+n-\ell} \left| u_{it},\ u_{i,t+1}, \ldots, u_{i,t+n} \right. \right) = E\left(\mathbf{u}_{t+n-\ell} \left| u_{i,t+n-\ell} \right. \right), \ell = 0, 1, 2, \ldots, n.$

For computational purposes it is worth noting that the numerator of (24.26) can also be written as the sum of squares of the generalized responses of the shocks to the ith equation on the jth variable in the model, namely $\sum_{\ell=0}^{n}\left(GI_{ij,\ell}\right)^2$, where $GI_{ij,\ell}$ is given by (24.18).

24.8 Impulse response analysis in *VARX* models

Impulse response analysis can be conducted following the lines of the arguments set out in Sections 24.4–24.5, but *applied to the full system in* (23.30) presented in Section 23.7. In particular, consider

$$\Delta \mathbf{z}_t = -\boldsymbol{\alpha}\boldsymbol{\beta}'\mathbf{z}_{t-1} + \sum_{i=1}^{p-1}\boldsymbol{\Gamma}_i\Delta\mathbf{z}_{t-i} + \mathbf{H}\boldsymbol{\zeta}_t, \tag{24.27}$$

and rewrite it the $VAR(p)$

$$\mathbf{z}_t = \sum_{i=1}^{p}\boldsymbol{\Phi}_i\mathbf{z}_{t-i} + \mathbf{a}_0 + \mathbf{a}_1 t + \mathbf{H}\boldsymbol{\zeta}_t, \tag{24.28}$$

where $\boldsymbol{\Phi}_1 = \mathbf{I}_m - \boldsymbol{\alpha}\boldsymbol{\beta}' + \boldsymbol{\Gamma}_1$, $\boldsymbol{\Phi}_i = \boldsymbol{\Gamma}_i - \boldsymbol{\Gamma}_{i-1}$, $i = 2, \ldots, p-1$, $\boldsymbol{\Phi}_p = -\boldsymbol{\Gamma}_{p-1}$, and $m = m_y + m_x$.

Equation (24.28) can be used for forecasting and for impulse response analysis. The generalized impulse response function are derived from the moving average representation of equation (24.28). An infinite moving representation of $\Delta\mathbf{z}_t$ in (24.28) can be written as

$$\Delta \mathbf{z}_t = \mathbf{C}(L)\left(a_0 + a_1 t + \mathbf{H}\boldsymbol{\zeta}_t\right),$$

where

$$\begin{aligned}
\mathbf{C}(L) &= \sum_{j=0}^{\infty}\mathbf{C}_j L^j = \mathbf{C}(1) + (1-L)\mathbf{C}^*(L),\\
\mathbf{C}^*(L) &= \sum_{j=0}^{\infty}\mathbf{C}_j^* L^j, \text{ and } \mathbf{C}_j^* = -\sum_{i=j+1}^{\infty}\mathbf{C}_i,\\
\mathbf{C}_i &= \boldsymbol{\Phi}_1\mathbf{C}_{i-1} + \boldsymbol{\Phi}_2\mathbf{C}_{i-2} + \ldots + \boldsymbol{\Phi}_p\mathbf{C}_{i-p}, \text{ for } i = 2, 3, \ldots,
\end{aligned} \tag{24.29}$$

with $\mathbf{C}_0 = \mathbf{I}_m$, $\mathbf{C}_1 = \boldsymbol{\Phi}_1 - \mathbf{I}_m$ and $\mathbf{C}_i = 0$, for $i < 0$. Cumulating forward one obtains the level *MA* representation,

$$\mathbf{z}_t = \mathbf{z}_0 + \mathbf{b}_0 t + \mathbf{C}(1)\sum_{j=1}^{t}\mathbf{H}\boldsymbol{\zeta}_j + \mathbf{C}^*(L)\mathbf{H}(\boldsymbol{\zeta}_t - \boldsymbol{\zeta}_0),$$

where $\mathbf{b}_0 = \mathbf{C}(1)\mathbf{a}_0 + \mathbf{C}^*(1)\mathbf{a}_1$ and $\mathbf{C}(1)\mathbf{\Pi}\boldsymbol{\gamma} = \mathbf{0}$, with $\boldsymbol{\gamma}$ being an arbitrary $m \times 1$ vector of fixed constants. The latter relation applies because the trend coefficients are restricted to lie in the cointegrating space.

The generalized and orthogonalized impulse response functions of individual variables $\mathbf{z}_{t+n} = (\mathbf{y}'_{t+n}, \mathbf{x}'_{t+n})'$ at horizon n to a unit change in the error, ζ_{it}, measured by one standard deviation, $\sqrt{\sigma_{\zeta,ii}}$ are

$$GI\left(n, \mathbf{z} : \zeta_i\right) = \frac{1}{\sqrt{\sigma_{\zeta,ii}}}\tilde{\mathbf{C}}_n\mathbf{H}\mathbf{\Sigma}_{\zeta\zeta}\mathbf{e}_i,\ n = 0, 1, \ldots,\ i = 1, 2, \ldots, m, \tag{24.30}$$

$$OI\left(n, \mathbf{z} : \zeta_i^*\right) = \tilde{\mathbf{C}}_n\mathbf{H}\mathbf{P}_\zeta\mathbf{e}_i, n = 0, 1, \ldots,\ i = 1, 2, \ldots, m, \tag{24.31}$$

where $\boldsymbol{\zeta}_t$ is $IID\left(\mathbf{0}, \mathbf{\Sigma}_{\zeta\zeta}\right)$, ζ_i^* is an orthogonalized residual, $\sigma_{\zeta,ij}$ is $\left(i, j\right)^{th}$ element of $\mathbf{\Sigma}_{\zeta\zeta}$, $\tilde{\mathbf{C}}_n = \sum_{j=0}^{h}\mathbf{C}_j$, with $\mathbf{C}_j$'s given by the recursive relations (24.29), $\mathbf{H}$ and $\mathbf{\Sigma}_{\zeta\zeta}$ are given in (23.32), $\mathbf{e}_i$ is a selection vector of zeros with unity as its ith element, $\mathbf{P}_\zeta$ is a lower triangular matrix obtained by the Cholesky decomposition of $\mathbf{\Sigma}_{\zeta\zeta} = \mathbf{P}_\zeta\mathbf{P}'_\zeta$.

Similarly, the generalized and orthogonalized impulse response functions for the cointegrating relations with respect to a unit change in the error, ζ_{it} are given by

$$GI\left(n, \boldsymbol{\xi} : \zeta_i\right) = \frac{1}{\sqrt{\sigma_{\zeta,ii}}}\boldsymbol{\beta}'\tilde{\mathbf{C}}_n\mathbf{H}\mathbf{\Sigma}_{\zeta\zeta}\mathbf{e}_i,\ n = 0, 1, \ldots,\ i = 1, 2, \ldots, m, \tag{24.32}$$

$$OI\left(n, \boldsymbol{\xi} : \zeta_i^*\right) = \boldsymbol{\beta}'\tilde{\mathbf{C}}_n\mathbf{H}\mathbf{P}_\zeta\mathbf{e}_i,\ n = 0, 1, \ldots,\ i = 1, 2, \ldots, m, \tag{24.33}$$

where $\boldsymbol{\xi}_t = \boldsymbol{\beta}'\mathbf{z}_{t-1}$.

While the impulse responses show the effect of a shock to a particular variable, the persistence profile, as developed by Lee and Pesaran (1993) and Pesaran and Shin (1996), shows the effects of system-wide shocks on the cointegrating relations. In the case of the cointegrating relations the effects of the shocks (irrespective of their sources) will eventually disappear. Therefore, the shape of the persistence profiles provides valuable information on the speed of convergence of the cointegrating relations towards equilibrium. The persistence profile for a given cointegrating relation defined by the cointegrating vector $\boldsymbol{\beta}_j$ in the case of a *VARX* model is given by

$$h(\boldsymbol{\beta}'_j\mathbf{z}, n) = \frac{\boldsymbol{\beta}'_j\widetilde{\mathbf{C}}_n\mathbf{H}\mathbf{\Sigma}_{\zeta\zeta}\mathbf{H}'\widetilde{\mathbf{C}}'_n\boldsymbol{\beta}_j}{\boldsymbol{\beta}'_j\mathbf{H}\mathbf{\Sigma}_{\zeta\zeta}\mathbf{H}'\boldsymbol{\beta}_j}, n = 0, 1, \ldots, j = 1, \ldots r, \tag{24.34}$$

where $\boldsymbol{\beta}$, $\widetilde{\mathbf{C}}_n$, $\mathbf{H}$ and $\mathbf{\Sigma}_{\zeta\zeta}$ are as defined above.

24.8.1 Impulse response analysis in cointegrating *VAR*s

$$GI(n, \mathbf{y} : \varepsilon_i) = \left(\frac{1}{\sqrt{\omega_{ii}}}\right)\tilde{\mathbf{C}}_n\mathbf{A}_0^{-1}\mathbf{\Omega}\mathbf{e}_i,$$

where $\tilde{\mathbf{C}}_n = \sum_{j=0}^{n}\mathbf{C}_j$. Also

$$GI(n, \mathbf{y} : u_i) = \left(\frac{1}{\sqrt{\sigma_{ii}}}\right) \tilde{\mathbf{C}}_n \boldsymbol{\Sigma} \boldsymbol{e}_i.$$

In particular,

$$GI(\infty, \mathbf{y} : \varepsilon_i) = \frac{1}{\sqrt{\omega_{ii}}} \mathbf{C}(1) \mathbf{A}_0^{-1} \boldsymbol{\Omega} \mathbf{e}_i$$

and

$$GI(\infty, \mathbf{y} : u_i) = \frac{1}{\sqrt{\sigma_{ii}}} \mathbf{C}(1) \boldsymbol{\Sigma} \mathbf{e}_i$$

and unlike the stationary case shocks will have permanent effects on the $I(1)$ variables, though not on the cointegrating relations.

For the cointegrating relations $\boldsymbol{\xi}_t = \boldsymbol{\beta}' \mathbf{y}_t$, we have

$$GI(n, \boldsymbol{\xi} : \varepsilon_i) = \left(\frac{1}{\sqrt{\omega_{ii}}}\right) \boldsymbol{\beta}' \tilde{\mathbf{C}}_n \mathbf{A}_0^{-1} \boldsymbol{\Omega} \mathbf{e}_i.$$

Since $\boldsymbol{\beta}' \tilde{\mathbf{C}}_\infty = \boldsymbol{\beta}' \mathbf{C}(1) = \mathbf{0}$, it then follows that $GI(\infty, \boldsymbol{\xi} : \varepsilon_i) = \mathbf{0}$.

$$GI(n, \boldsymbol{\xi} : u_i) = \left(\frac{1}{\sqrt{\sigma_{ii}}}\right) \boldsymbol{\beta}' \tilde{\mathbf{C}}_n \boldsymbol{\Sigma} \mathbf{e}_i.$$

24.8.2 Persistence profiles for cointegrating relations

Pesaran and Shin (1996) suggest using the persistence profiles to measure the speed of convergence of the cointegrating relations to equilibrium (see also Section 22.14 on this). The (unscaled) persistence profile of the cointegrating relations is given by

$$\boldsymbol{\beta}' \tilde{\mathbf{C}}_n \Sigma \tilde{\mathbf{C}}_n' \boldsymbol{\beta}, \; n = 0, 1, \ldots.$$

The profiles tend to zero as $n \to \infty$, and provide a useful graphical representation of the extent to which the cointegrating (equilibrium) relations adjust to system-wide shocks. The persistence profiles are uniquely determined.

24.9 Empirical distribution of impulse response functions and persistence profiles

The simulation methods described in Section 22.12 can be implemented to compute the empirical distribution of generalized (orthogonalized) impulse response functions and persistence profiles based on a vector error correction model.

Consider equation (24.27), and the impulse response functions (24.30)–(24.33) of both individual variables and cointegrating relations and persistence profiles (24.34). Suppose that

the *ML* estimators of $\mathbf{\Phi}_i$, $i = 1, 2, \ldots, p$, $\mathbf{a}_0$, $\mathbf{a}_1$, $\mathbf{H}$ and $\mathbf{\Sigma}_{\zeta\zeta}$ are given and denoted by $\hat{\mathbf{\Phi}}_i$, $i = 1, \ldots, p$, $\hat{\mathbf{a}}_0$, $\hat{\mathbf{a}}_1$, $\hat{\mathbf{H}}$ and $\hat{\mathbf{\Sigma}}_{\zeta\zeta}$, respectively. To allow for parameter uncertainty, we use the bootstrap procedure and simulate S (*in-sample)* values of $\mathbf{z}_t$, $t = 1, 2, \ldots, T$, denoted by $\mathbf{z}_t^{(s)}$, $s = 1, 2, \ldots, S$, where

$$\mathbf{z}_t^{(s)} = \sum_{i=1}^{p} \hat{\mathbf{\Phi}}_i \mathbf{z}_{t-i}^{(s)} + \hat{\mathbf{a}}_0 + \hat{\mathbf{a}}_1 t + \hat{\mathbf{H}} \boldsymbol{\zeta}_t^{(s)}, \; t = 1, 2, \ldots, T, \tag{24.35}$$

realizations are used for the initial values, $\mathbf{z}_{-1}, \ldots, \mathbf{z}_{-p}$, and $\boldsymbol{\zeta}_t^{(s)}$s can be drawn either by parametric or nonparametric methods (see Section 22.12).

Having obtained the S set of simulated in-sample values, $\left(\mathbf{z}_1^{(s)}, \mathbf{z}_2^{(s)}, \ldots, \mathbf{z}_T^{(s)}\right)$, the $VAR(p)$ model, (24.27), is re-estimated S times to obtain the *ML* estimates, $\hat{\mathbf{\Phi}}_i^{(s)}$, $\hat{\mathbf{a}}_0^{(s)}$, $\hat{\mathbf{a}}_1^{(s)}$, $\hat{\mathbf{H}}^{(s)}$ and $\hat{\mathbf{\Sigma}}_{\zeta\zeta}^{(s)}$, for $i = 1, 2, \ldots, p$, and $s = 1, 2, \ldots, S$. For each of these bootstrap replications, we then obtain the estimates of $GI^{(s)}\left(n, \mathbf{z}^{(s)} : \zeta_i^{(s)}\right)$, $GI^{(s)}\left(n, \boldsymbol{\xi}^{(s)} : \zeta_i^{(s)}\right)$, $OI^{(s)}\left(n, \mathbf{z}^{(s)} : \zeta_i^{*(s)}\right)$, $OI^{(s)}\left(n, \boldsymbol{\xi}^{(s)} : \zeta_i^{*(s)}\right)$, $h^{(s)}\left(\boldsymbol{\beta}_j' \mathbf{z}^{(s)}, n\right)$. Therefore, using the S set of simulated estimates, we will obtain both empirical mean and confidence intervals of impulse response functions and persistence profiles.

24.10 Identification of short-run effects in structural *VAR* models

Consider the following $VAR(p)$

$$\mathbf{y}_t = \mathbf{\Phi}_1 \mathbf{y}_{t-1} + \mathbf{\Phi}_2 \mathbf{y}_{t-2} + \ldots + \mathbf{\Phi}_p \mathbf{y}_{t-p} + \mathbf{u}_t, \tag{24.36}$$

and its $VECM(p-1)$ representation

$$\Delta \mathbf{y}_t = -\boldsymbol{\alpha}\boldsymbol{\beta}' \mathbf{y}_{t-1} + \sum_{j=1}^{p-1} \mathbf{\Gamma}_j \Delta \mathbf{y}_{t-j} + \mathbf{u}_t, \tag{24.37}$$

with $\mathbf{\Pi} = \mathbf{I}_m - \mathbf{\Phi}_1 - \mathbf{\Phi}_2 - \ldots - \mathbf{\Phi}_p$, and $\mathbf{\Gamma}_j = -\sum_{i=j+1}^{p} \mathbf{\Phi}_i$, for $j = 1, 2, \ldots, p-1$. We have seen in Chapter 22 that the $VECM(p-1)$ model, (24.37), under $Rank(\boldsymbol{\beta}) = r < m$, is subject to long-run identification problems. We now consider the problem of identification of short run effects. To this end premultiply both sides of (24.37) by a nonsingular $m \times m$ matrix $\mathbf{A}_0$ to obtain

$$\mathbf{A}_0 \Delta \mathbf{y}_t = -\mathbf{A}_0 \boldsymbol{\alpha}\boldsymbol{\beta}' \mathbf{y}_{t-1} + \sum_{j=1}^{p-1} \mathbf{A}_0 \mathbf{\Gamma}_j \Delta \mathbf{y}_{t-j} + \mathbf{A}_0 \mathbf{u}_t. \tag{24.38}$$

The structural shocks are given by $\boldsymbol{\varepsilon}_t = \mathbf{A}_0 \mathbf{u}_t$, and their identification require knowing $\mathbf{A}_0$. But knowing $\mathbf{A}_0$ does not help in identification of $\boldsymbol{\beta}$, since irrespective of the value of $\mathbf{A}_0$ we have $\mathbf{A}_0 \boldsymbol{\alpha} \mathbf{Q}^{'-1} \mathbf{Q}' \boldsymbol{\beta}'$ and replacing $\mathbf{A}_0$ with another nonsingular matrix will not help in restricting

the $r \times r$ nonsingular matrix $\mathbf{Q}$. Similarly, identification of $\boldsymbol{\beta}$, which is the same as selecting a specific $\mathbf{Q}$ matrix, will not help in identification of $\mathbf{A}_0$. In essence, we have two types of identification problem. The first involves the identification of the coefficients, $\mathbf{A}_0$, while the second concerns the identification of the long-run coefficients, $\boldsymbol{\beta}$, which arises only when the $\mathbf{y}_t$ is $I(1)$. In a *VAR* model without any exogenous variables the estimation of $\mathbf{A}_0$ from the reduced form parameters requires the imposition of m^2 restrictions. Sims (1980) argued against restricting the short-run coefficients such as $\boldsymbol{\alpha}$ or $\boldsymbol{\Gamma}_j, j = 1, 2, \ldots, p$, and instead proposed placing the restrictions directly on elements of $\mathbf{A}_0$, and the variance covariance of the structural shocks, $\mathbf{v}_t = \mathbf{A}_0\mathbf{u}_t$, which is given by

$$E\left(\mathbf{v}_t\mathbf{v}_t'\right) = \boldsymbol{\Sigma}_v = \mathbf{A}_0\boldsymbol{\Sigma}\mathbf{A}_0'.$$

Sims (1980) identification procedure leads to the 'orthogonalized impulses' (see also Section 24.4). Sims assumed that:

(i) The 'structural' shocks, $\mathbf{v}_t$, are orthogonal and hence $\boldsymbol{\Sigma}_v$ is diagonal

$$\boldsymbol{\Sigma}_v = \begin{pmatrix} \sigma_{11} & 0 & \cdots & 0 \\ 0 & \sigma_{22} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma_{mm} \end{pmatrix} = \mathbf{A}_0\boldsymbol{\Sigma}\mathbf{A}_0'.$$

Due to the symmetric nature of $\boldsymbol{\Sigma}_v$ and $\boldsymbol{\Sigma}$ and since the elements of $\boldsymbol{\Sigma}$ are estimated without any restrictions, then the above imposes $m\,(m-1)\,/2$ restrictions on the elements of $\mathbf{A}_0$.

(ii) The contemporaneous coefficient matrix, $\mathbf{A}_0$ is (lower) triangular

$$\mathbf{A}_0 = \begin{pmatrix} a_{0,11} & 0 & \cdots & 0 \\ a_{0,21} & a_{0,22} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ a_{0,m1} & a_{0.m2} & \cdots & a_{0.mm} \end{pmatrix},$$

with the normalizations $a_{0,ii} = 1, i = 1, 2, \ldots, m$. This gives $m + m\,(m-1)\,/2 = m(m+1)/2$ restrictions. Therefore, in total we have $m(m-1)/2 + m(m+1)/2 = m^2$ restrictions, as required for just (or exact) identification of the structural parameters $\mathbf{A}_0$.

The above identification scheme, in addition to requiring structural shocks to be orthogonal (which is a questionable requirement), also imposes a recursive casual ordering on the variables in the *VAR*. For example, different structural models will result from different ordering of the variables in the *VAR*. It is in fact the same as the recursive ordering proposed by Wold whereby the first structural shock, v_{1t}, is assumed to be proportional to the first reduced-form error (u_{1t}), the second structural shock, v_{2t}, is related linearly to the first two reduced form errors (u_{1t} and u_{2t}) the third structural error, v_{3t} is linearly related to (u_{1t}, u_{2t}, and u_{3t}), and so on. As

Wold showed, the parameters of these linear relations can be estimated consistently in a recursive manner.

Other identification schemes have followed the work by Sims (1980). One prominent example is the identification scheme developed in Blanchard and Quah (1989), who distinguished between permanent and transitory shocks and attempted to identify the structural models through long-run restrictions. For example, Blanchard and Quah argued that the effect of a demand shock on real output should be temporary (namely it should have a zero long-run impact), whilst a supply shock should have a permanent effect. This approach is known as 'structural *VAR*' (*SVAR*) and has been used extensively in the literature.

24.11 Structural systems with permanent and transitory shocks

We now consider the identification problem of a structural model when one or more of the structural shocks are permanent.[2] We first introduce a simple structural model and then explore the implications for the identification of the structural shocks provided by a permanent/transitory decomposition.

24.11.1 Structural *VAR*s (*SVAR*)

Consider the structural $VAR(2)$ system

$$\mathbf{A}_0\mathbf{y}_t = \mathbf{A}_1\mathbf{y}_{t-1} + \mathbf{A}_2\mathbf{y}_{t-2} + \boldsymbol{\varepsilon}_t,$$

where $\mathbf{A}_i$ are $m \times m$ matrices of unknown coefficients, $\mathbf{A}_0$ is nonsingular, and $\boldsymbol{\varepsilon}_t$ is an $m \times 1$ vector of structural shocks with mean zero and $E\left(\boldsymbol{\varepsilon}_t\boldsymbol{\varepsilon}_t'\right) = \mathbf{I}_m$. To ensure that $\mathbf{y}_t$ does not contain $I(2)$ variables we also assume that all the eigenvalues of $\mathbf{A}_0^{-1}\mathbf{A}_2$ lie inside the unit circle. The above *SVAR* specification can be transformed to

$$\mathbf{A}_0\Delta\mathbf{y}_t = -\mathbf{A}(1)\mathbf{y}_{t-1} - \mathbf{A}_2\Delta\mathbf{y}_{t-1} + \boldsymbol{\varepsilon}_t,$$

where $\mathbf{A}(1) = \mathbf{A}_0 - \mathbf{A}_1 - \mathbf{A}_2$, with the associated reduced form model given by

$$\begin{aligned}\Delta\mathbf{y}_t &= -\mathbf{A}_0^{-1}\mathbf{A}(1)\mathbf{y}_{t-1} - \mathbf{A}_0^{-1}\mathbf{A}_2\Delta\mathbf{y}_{t-1} + \mathbf{A}_0^{-1}\boldsymbol{\varepsilon}_t,\\ &= -\boldsymbol{\Pi}\mathbf{y}_{t-1} + \boldsymbol{\Psi}\Delta\mathbf{y}_{t-1} + \mathbf{e}_t.\end{aligned}$$

Now suppose that there are $r < m$ cointegrating relations in this system, so that $\boldsymbol{\Pi}$ is rank deficient and $\boldsymbol{\Pi} = \boldsymbol{\alpha}\boldsymbol{\beta}'$, where $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$ are $m \times r$ full column rank matrices. Then

$$\Delta\mathbf{y}_t = -\boldsymbol{\alpha}\boldsymbol{\beta}'\mathbf{y}_{t-1} + \boldsymbol{\Psi}\Delta\mathbf{y}_{t-1} + \mathbf{u}_t, \tag{24.39}$$

[2] This section is based on the paper by Pagan and Pesaran (2008).

with

$$\boldsymbol{\Psi} = -\mathbf{A}_0^{-1}\mathbf{A}_2, \tag{24.40}$$

and

$$\mathbf{A}_0\Delta\mathbf{y}_t = -\boldsymbol{\alpha}^*\boldsymbol{\beta}'\mathbf{y}_{t-1} - \mathbf{A}_2\Delta\mathbf{y}_{t-1} + \boldsymbol{\varepsilon}_t, \tag{24.41}$$

is the structural *VEC* (*SVEC*) model, where $\boldsymbol{\alpha}^* = \mathbf{A}_0\boldsymbol{\alpha}$. The central task in *SVEC* (and *SVAR*) systems is to estimate the m^2 coefficients of $\mathbf{A}_0$, m of which can be fixed by suitable normalization restrictions. The remaining $m(m-1)$ coefficients need to be identified by means of *a priori* restrictions inspired by economic reasoning. A number of different identification schemes are possible depending on the nature of the available *a priori* information. Each identification scheme produces a set of instruments for $\Delta\mathbf{y}_t$ and so enables the consistent estimation of the unknown parameters in $\mathbf{A}_0$. Notice from (24.41) that, if one or more elements of $\boldsymbol{\alpha}^*$ are known and we are able to estimate $\boldsymbol{\beta}$ consistently, then $\boldsymbol{\beta}'\mathbf{y}_{t-1}$ can be used as instruments. This idea will be described and illustrated in the following Section.

24.11.2 Permanent and transitory structural shocks

Suppose that the first $m-r$ shocks in $\boldsymbol{\varepsilon}_t$, denoted by $\boldsymbol{\varepsilon}_{1t}$, are known to be permanent and the remaining r shocks, $\boldsymbol{\varepsilon}_{2t}$, are transitory (see Section 16.6 for definitions of permanent and transitory shocks). Such a decomposition is possible since it is assumed that there are r cointegrating relations amongst the m, $I(1)$ variables in $\mathbf{y}_t$ (see, e.g., Lütkepohl (2005, Ch. 9)). Consider the following common trends representation of (24.39) (see Section 22.15, and Johansen (1995, Theorem 4.2))

$$\mathbf{y}_t = \mathbf{y}_0 + \mathbf{F}\sum_{j=1}^{t}\mathbf{u}_j + \sum_{i=0}^{\infty}\mathbf{F}_i^*\mathbf{u}_{t-i}, \tag{24.42}$$

where $\mathbf{F} = \boldsymbol{\beta}_\perp\left[\boldsymbol{\alpha}'_\perp(\mathbf{I}_m - \boldsymbol{\Psi})\boldsymbol{\beta}_\perp\right]^{-1}\boldsymbol{\alpha}'_\perp$, with $\boldsymbol{\alpha}'_\perp\boldsymbol{\alpha} = \mathbf{0}$ and $\boldsymbol{\beta}'\boldsymbol{\beta}_\perp = \mathbf{0}$, so that ($\boldsymbol{\Psi}$ is defined by (24.40))

$$\mathbf{F}\boldsymbol{\alpha} = \mathbf{0}_{m\times r}, \text{ and } \boldsymbol{\beta}'\mathbf{F} = \mathbf{0}_{r\times m}. \tag{24.43}$$

Writing the permanent component in terms of the structural shocks we have

$$\begin{aligned}\mathbf{y}_t &= \mathbf{y}_0 + \mathbf{F}\sum_{j=1}^{t}\mathbf{A}_0^{-1}\boldsymbol{\varepsilon}_j + \sum_{i=0}^{\infty}\mathbf{F}_i^*\mathbf{u}_{t-i}\\ &= \mathbf{y}_0 + \mathbf{F}\mathbf{A}_0^{-1}\begin{pmatrix}\sum_{j=1}^{t}\boldsymbol{\varepsilon}_{1j}\\ \sum_{j=1}^{t}\boldsymbol{\varepsilon}_{2j}\end{pmatrix} + \sum_{j=0}^{\infty}\mathbf{F}_j^*\mathbf{u}_{t-j}.\end{aligned}$$

In order for $\boldsymbol{\varepsilon}_{2j}$ to have only transitory effects we must have

$$\mathbf{F}\mathbf{A}_0^{-1}\begin{pmatrix}\mathbf{0}_{(m-r)\times r}\\ \mathbf{I}_r\end{pmatrix}=\mathbf{0}. \tag{24.44}$$

These restrictions are necessary and sufficient and apply irrespective of whether the transitory shocks are correlated or not. However, using (24.43) it follows that

$$\mathbf{A}_0^{-1}\begin{pmatrix}\mathbf{0}_{(m-r)\times r}\\ \mathbf{I}_r\end{pmatrix}=\boldsymbol{\alpha}\mathbf{Q}, \tag{24.45}$$

where $\mathbf{Q}$ is an arbitrary $r\times r$ nonsingular matrix.[3] Hence, after multiplying both sides of (24.45) by $\mathbf{A}_0$ we have

$$\begin{pmatrix}\mathbf{0}_{m-r}\\ \mathbf{I}_r\end{pmatrix}=\mathbf{A}_0\boldsymbol{\alpha}\mathbf{Q}=\boldsymbol{\alpha}^*\mathbf{Q}=\begin{pmatrix}\boldsymbol{\alpha}_1^*\mathbf{Q}\\ \boldsymbol{\alpha}_2^*\mathbf{Q}\end{pmatrix}. \tag{24.46}$$

This in turn implies that $\boldsymbol{\alpha}_1^*=\mathbf{0}_{(m-r)\times r}$, namely *the structural equations for which there are known permanent shocks must have no error correction terms present in them*, thereby freeing up the latter to be used as instruments in estimating their parameters. More specifically, the identification of the first $m-r$ structural shocks as permanent imposes $r(m-r)$ restrictions on the structural parameters. Also $\boldsymbol{\alpha}_2^*=\mathbf{Q}^{-1}$ is an arbitrary nonsingular $r\times r$ matrix.

The restrictions $\boldsymbol{\alpha}_1^*=\mathbf{0}_{(m-r)\times r}$ can then be exploited by noting that the r lagged error correction terms, $\boldsymbol{\beta}'\mathbf{y}_{t-1}$, are available to be used as instruments for estimating the structural parameters of the first $m-r$ equations in (24.41). More specifically, under $\boldsymbol{\alpha}_1^*=\mathbf{0}_{(m-r)\times r}$ the first $m-r$ equations can be written as

$$\mathbf{A}_{11}^0\Delta\mathbf{y}_{1t}+\mathbf{A}_{12}^0\Delta\mathbf{y}_{2t}=-\mathbf{A}_{11}^2\Delta\mathbf{y}_{1,t-1}-\mathbf{A}_{12}^2\Delta\mathbf{y}_{2,t-1}+\boldsymbol{\varepsilon}_{1t}, \tag{24.47}$$

and it is clear that the $r\times 1$ error correction terms, $\boldsymbol{\xi}_{t-1}=\boldsymbol{\beta}'\mathbf{y}_{t-1}$, that do not appear in these equations, but are included in the remaining r equations of (24.41) can be used as instruments for the $m-r$ equations in (24.47). These instruments are clearly uncorrelated with the error terms $\boldsymbol{\varepsilon}_{1t}$, whilst at the same time being correlated with $\Delta\mathbf{y}_{1t}$ and $\Delta\mathbf{y}_{2t}$ since $\boldsymbol{\alpha}_2^*$ is a nonsingular matrix. Note also that since instrumental variable estimators are unaffected by nonsingular transformations of the instruments, for the purpose of estimating the structural parameters of the first $m-r$ equations ($\mathbf{A}_{11}^0$ and $\mathbf{A}_{12}^0$) the error correction terms, $\boldsymbol{\xi}_{t-1}$ (or $\boldsymbol{\beta}$), need only be identified up to a nonsingular transformation.

Further discussion on the implications of the permanent/transitory decomposition of shocks for identification can be found in Pagan and Pesaran (2008).

[3] Note that premultiplying both sides of (24.45) by $\mathbf{F}$ yields (24.44).

24.12 Some applications

24.12.1 Blanchard and Quah (1989) model

We now consider an application of the above framework to the model analysed by Blanchard and Quah (1989). Blanchard and Quah (1989) consider a two equation system in GNP (y_t) and unemployment rate (un_t), which are assumed to be $I(1)$ and $I(0)$, respectively. They further assume that there is one permanent (supply) and one transitory (demand) shock. These are denoted by ε_{1t} and ε_{2t}, respectively. Although there is no cointegration in this case, our methodological approach can be applied by treating un_t, the $I(0)$ variable as if it 'cointegrates' with itself.

Let us set up a pseudo cointegrating vector of the form $\boldsymbol{\beta} = \left(0, \beta_2\right)'$ which produces the lagged 'EC term' given by $\beta_2 un_{t-1}$. According to the above results, the equation with the permanent shock will have the form (normalizing on Δy_t)

$$\Delta y_t = \alpha_{12}^0(\beta_2 un_t) + \alpha_{11}^1 \Delta y_{t-1} + \alpha_{12}^1 \Delta(\beta_2 un_{t-1}) + \varepsilon_{1t},$$

and the second equation (normalizing on un_t) will be

$$un_t = \alpha_{21}^0 \Delta y_t + \alpha_{22}^0 \beta_2 un_{t-1} + \alpha_{21}^1 \Delta y_{t-1} + \alpha_{22}^1(\beta_2 \Delta un_{t-1}) + \varepsilon_{2t}.$$

It is clear that in this setup $\beta_2 un_{t-1}$ does not enter the first equation and can therefore be used as an instrument for un_t in it. So long as $\beta_2 \neq 0$, the value of β_2 does not matter as the instrumental variable estimator is invariant to it. However, unlike the cointegration case where β_2 could be estimated super-consistently, this is not possible when un_t is $I(0)$, so that we would need to treat un_{t-1} as a regressor in the second equation. That means un_{t-1} is not available as an instrument for Δy_t. But the residuals from the first equation form a suitable instrument. This instrumental variable interpretation of Blanchard and Quah is due to Shapiro and Watson (1988). The problem with this procedure is that un_{t-1} is often a very poor instrument for Δun_t and this can lead to highly non-normal densities for the instrumental variables estimator. Using the same data as Blanchard and Quah this is shown in Fry and Pagan (2005).

24.12.2 Gali's *IS-LM* model

Gali (1992) presents a model in four $I(1)$ variables: log of GNP (y_t), inflation rate (π_t), growth rate of the money supply (Δm_t) and nominal interest rate (i_t). This model is meant to be an analogue of the *IS-LM* system. He assumes that there are two cointegrating vectors among these four variables, $\xi_{1t} = \Delta m_t - \pi_t$ and $\xi_{2t} = i_t - \pi_t$ so that

$$\boldsymbol{\beta}' = (\boldsymbol{\beta}_1', \boldsymbol{\beta}_2') = \begin{pmatrix} 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & -1 \end{pmatrix}, \text{with } \boldsymbol{\beta}_2' = \begin{pmatrix} -1 & 0 \\ 1 & -1 \end{pmatrix}.$$

Gali works with an *SVAR* in Δy_t, Δi_t, ξ_{1t} and ξ_{2t} rather than the *SVECM* that is implied by the assumptions that there are $I(1)$ variables and cointegration.

The implied *SVAR* for the first equation has the form

$$\Delta y_t = \alpha_{12}^0 \Delta i_t + \alpha_{13}^0 \Delta \xi_{1t} + \alpha_{14}^0 \Delta \xi_{2t} + \alpha_{11}^1 \Delta y_{t-1} + \alpha_{12}^1 \Delta i_{t-1} + \alpha_{13}^1 \Delta \xi_{1,t-1} + \alpha_{14}^1 \Delta \xi_{2,t-1} + \varepsilon_{1t}.$$

It is clear that we can use $\xi_{1,t-1}$ and $\xi_{1,t-1}$ as instruments for $\Delta \xi_{jt}$. But still we need another instrument for Δi_t. To this end Gali assumes that the long-run effect of the second permanent shock upon y_t is zero, which yields the restriction $\alpha_{12}^0 = -\alpha_{12}^1$, and so the equation for Δi_t can be re-expressed in terms of $\Delta^2 i_t$, allowing Δi_{t-1} to be used as an instrument.

The second equation has the form

$$\Delta i_t = \alpha_{21}^0 \Delta y_t + \alpha_{23}^0 \Delta \xi_{1t} + \alpha_{24}^0 \Delta \xi_{2t} + \alpha_{21}^1 \Delta y_{t-1} + \alpha_{22}^1 \Delta i_{t-1} + \alpha_{23}^1 \Delta \xi_{1,t-1} + \alpha_{24}^1 \Delta \xi_{2,t-1} + \varepsilon_{2t}.$$

We can still use the lagged ECM terms as instruments. Assuming that the shocks are uncorrelated, we can also use the residuals from the first equation as instruments. Gali adopts the latter but not the former as instruments.

24.13 Identification of monetary policy shocks

We now discuss the problem of identification of the monetary policy shocks within a model for the UK macroeconomy, described in Garratt et al. (2003b), and link it to impulse response analysis of the monetary policy shocks. The model proposed in Garratt et al. (2003b) comprises six domestic variables whose developments are widely regarded as essential to a basic understanding of the behaviour of the UK macroeconomy; namely, aggregate output, the ratio of domestic to foreign price levels, inflation, the nominal interest rate, the exchange rate, and real money balances. The model also contains foreign output, foreign interest rates and oil prices (see Section 23.8 for further details). For identification of the monetary policy shocks, we need to formally articulate the decision problem of the monetary authorities.

Assume monetary authorities try to influence the market interest rate, r_t, by setting the base rate, r_t^b that they control. Further, assume that the term premium, $r_t - r_t^b$, is determined by

$$\begin{aligned} r_t - r_t^b &= \rho_{t-1} + a_{rr^*}\left[r_t^* - E\left(r_t^* \mid \Omega_{t-1}\right)\right] + a_{re}\left[e_t - E\left(e_t \mid \Omega_{t-1}\right)\right] \\ &\quad + a_{ro}\left[p_t^o - E\left(p_t^o \mid \Omega_{t-1}\right)\right] + \varepsilon_{rt}, \\ E\left(\varepsilon_{rt} \mid \Omega_{t-1}\right) &= 0, \\ E\left(r_t - r_t^b \mid \Omega_{t-1}\right) &= \rho_{t-1}. \end{aligned}$$

Under expectations formation mechanisms consistent with the reduced form *VECM*, the expectational variables $E(\Delta r_t^* \mid \Omega_{t-1})$, $E(\Delta e_t \mid \Omega_{t-1})$, and $E(\Delta p_t^o \mid \Omega_{t-1})$ can be replaced by the error correction terms $\boldsymbol{\beta}'\boldsymbol{\xi}_{t-1} - \mathbf{b}_1(t-1)$ and the lagged changes $\Delta \mathbf{z}_{t-i}$, $i = 1, 2, \ldots, s-1$. This would yield

$$\Delta r_t - a_{rr^*}\Delta r_t^* - a_{re}\Delta e_t - a_{ro}\Delta p_t^o = r_t^b - r_{t-1} + \rho_{t-1} + \boldsymbol{\phi}_r^*\left[\boldsymbol{\beta}'\mathbf{z}_{t-1} - \mathbf{b}_1(t-1)\right] + \sum_{i=1}^{s-1} \boldsymbol{\phi}_{zi}^{*'} \Delta \mathbf{z}_{t-i} + \varepsilon_{rt},$$

where the parameters $\boldsymbol{\phi}_r^*$ and $\boldsymbol{\phi}_{zi}^*$ are functions of a_{rr^*}, a_{re}, a_{ro} and the coefficients in the rows of the reduced form model associated with Δr_t^*, Δe_t and Δp_t^o. Suppose now that r_t^b is set by solving the optimization problem

$$\min_{r_t^b} \{E\left[C(\mathbf{w}_t,\ r_t) \mid \Omega_{t-1}\right]\}, \tag{24.48}$$

where $C(\mathbf{w}_t, r_t)$ is the loss function of the monetary authorities, assumed to be quadratic so that

$$C(\mathbf{w}_t,\ r_t) = \frac{1}{2}(\mathbf{w}_t - \mathbf{w}_t^{\dagger})'\mathbf{H}(\mathbf{w}_t - \mathbf{w}_t^{\dagger}) + \frac{1}{2}\theta(r_t - r_{t-1})^2, \tag{24.49}$$

where $\mathbf{w}_t = (y_t, \Delta p_t)'$ and $\mathbf{w}_t^{\dagger} = (y_t^{\dagger}, \pi_t^{\dagger})'$ are the target variables and their desired values, respectively. The outcome is the reaction function

$$\begin{aligned} &\Delta r_t - a_{rr^*}\Delta r_t^* - a_{re}\Delta e_t - a_{ro}\Delta p_t^o \\ = \ & a_r + \boldsymbol{\lambda}_r\left[\boldsymbol{\beta}'\mathbf{z}_{t-1} - \mathbf{b}_1(t-1)\right] + \sum_{i=1}^{s-1}\mathbf{g}_i'\Delta\mathbf{z}_{t-i} + \varepsilon_{rt}, \end{aligned} \tag{24.50}$$

where the monetary policy shock is identified by ε_{rt}. Note that changes in the preference parameters of the monetary authorities affect the magnitude and the speed with which interest rates respond to economic disequilibria, but such changes have no effect on the long-run coefficients, $\boldsymbol{\beta}$. It is also easily shown that, while changes in the trade-off parameter matrix, $\mathbf{H}$, affect all the short-run coefficients of the interest rate equation, changes to the desired target values affect only the intercept term, a_r.

The structural interest rate equation (24.50) can now be used, in conjunction with certain other *a priori* restrictions, to derive the impulse response functions of the monetary policy shocks, ε_{rt}. See Garratt et al. (2003b) for further details.

24.14 Further reading

For more detailed discussion of persistence profiles and impulse response analysis see Pesaran, Pierse, and Lee (1993), Lee and Pesaran (1993), Koop, Pesaran, and Potter (1996), and Pesaran and Shin (1996).

24.15 Exercises

1. Consider the $VAR(2)$ model

$$\mathbf{x}_t = \boldsymbol{\Phi}_1\mathbf{x}_{t-1} + \boldsymbol{\Phi}_2\mathbf{x}_{t-2} + \boldsymbol{\varepsilon}_t, \quad \boldsymbol{\varepsilon}_t \sim IID(\mathbf{0}, \boldsymbol{\Sigma}),$$

in the $m \times 1$ vector of random variables, $\mathbf{x}_t$, and $\boldsymbol{\Sigma}$ is the covariance matrix of the errors with a typical element, σ_{ij}.

(a) Derive the conditions under which this process is stationary, and show that it has the following moving average representation

$$\mathbf{x}_t = \sum_{j=0}^{\infty} \mathbf{A}_j \boldsymbol{\varepsilon}_{t-j}.$$

(b) Derive the coefficient matrices $\mathbf{A}_i$ in terms of $\boldsymbol{\Phi}_1$ and $\boldsymbol{\Phi}_2$.

(c) Using the above result write down the orthogonalized (OIR) and generalized impulse (GIR) response functions of one standard error shock (i.e. $\sqrt{\sigma_{ii}}$) to the error of the ith equation , $\varepsilon_{it} = \mathbf{s}_i' \boldsymbol{\varepsilon}_t$, where $\mathbf{s}_i$ is an $m \times 1$ selection vector.

(d) What are the main differences between *OIR* and *GIR* functions?

2. Consider the infinite-order vector moving average (MA) representation

$$\mathbf{x}_t = \mathbf{A}(L)\boldsymbol{\varepsilon}_t,$$

where $\mathbf{x}_t$ is an $m \times 1$ vector of random variables, $\boldsymbol{\varepsilon}_t \sim IID(\mathbf{0}, \boldsymbol{\Sigma})$, and

$$\mathbf{A}(L) = \mathbf{A}_0 + \mathbf{A}_1 L + \mathbf{A}_2 L^2 + \ldots,$$

and suppose that $\|\mathbf{A}_h\| < K\lambda^h$, where K is a fixed positive constant and $0 \leq \lambda < 1$, and $\|\mathbf{A}_h\|$ represent a matrix norm.

(a) Show that there exist the infinite-order polynomials $\mathbf{B}(L)$ and $\mathbf{G}(L)$ such that

$$\mathbf{x}_t = \mathbf{B}(L)\mathbf{u}_t,$$

is observationally equivalent to $\mathbf{x}_t = \mathbf{A}(L)\boldsymbol{\varepsilon}_t$, where $\mathbf{B}(L) = \mathbf{A}(L)\mathbf{G}(L^{-1}) = \mathbf{B}_0 + \mathbf{B}_1 L + \mathbf{B}_2 L^2 + \ldots$.

$$\mathbf{u}_t = \mathbf{G}'(L)\boldsymbol{\varepsilon}_t,$$

$\mathbf{G}(L)$ is square summable, and $\|\mathbf{B}_h\| < K\mu^h, 0 \leq \mu < 1$.

(b) As an example, consider the univariate MA(1) process

$$x_t = (1 + \theta L)\varepsilon_t, \text{ with } |\theta| < 1,$$

and show that the alternative specification

$$\begin{aligned} x_t &= (\theta + L)u_t, \text{ with } |\theta| > 1, \\ u_t &= \theta^{-1}\varepsilon_t + \left(1 - \frac{1}{\theta^2}\right)\varepsilon_{t-1} + \left(\frac{-1}{\theta}\right)\left(1 - \frac{1}{\theta^2}\right)\varepsilon_{t-2} \\ &\quad + \left(\frac{-1}{\theta}\right)^2\left(1 - \frac{1}{\theta^2}\right)\varepsilon_{t-3} + \ldots, \end{aligned}$$

is observationally equivalent to $x_t = (1 + \theta L)\varepsilon_t$, with $|\theta| < 1$.

(c) Discuss the implications of the above results for impulse response analysis.

3. Consider the following stationary $VAR(1)$ model in $\mathbf{z}_t = \left(y_t, x_t\right)'$

$$\mathbf{z}_t = \mathbf{\Phi}\mathbf{z}_{t-1} + \mathbf{e}_t,$$

where $\mathbf{\Phi} = (\phi_{ij})$ is a 2×2 matrix of unknown parameters, and $\mathbf{e}_t = \left(e_{yt}, e_{xt}\right)'$ is a 2-dimensional vector of reduced form errors. Define the effects of a permanent shock to the x_t process on itself and on y_t in the long run by

$$g_x = \lim_{s\to\infty} E\left(x_{t+s} \middle| \mathcal{I}_{t-1}, e_{x,t+h} = \sigma_x, \text{for } h = 0, 1, 2, \ldots\right),$$

and

$$g_y = \lim_{s\to\infty} E\left(y_{t+s} \middle| \mathcal{I}_{t-1}, e_{x,t+h} = \sigma_x, \text{for } h = 0, 1, 2, \ldots\right),$$

(a) Show that

$$\mathbf{g} = \begin{pmatrix} g_y \\ g_x \end{pmatrix} = (\mathbf{I}_2 - \mathbf{\Phi})^{-1} \begin{pmatrix} \omega \\ 1 \end{pmatrix} \sigma_x = \begin{pmatrix} -\dfrac{\omega + \phi_{12} - \omega\phi_{22}}{\phi_{11} + \phi_{22} - \phi_{11}\phi_{22} + \phi_{12}\phi_{21} - 1} \\ -\dfrac{\omega\phi_{21} - \phi_{11} + 1}{\phi_{11} + \phi_{22} - \phi_{11}\phi_{22} + \phi_{12}\phi_{21} - 1} \end{pmatrix} \sigma_x,$$

where ω is defined by $E\left(e_{yt} \middle| e_{xt}\right) = \omega e_{xt}$.

(b) Further show that

$$\theta = \frac{g_y}{g_x} = \frac{\omega + \phi_{12} - \omega\phi_{22}}{1 - (\phi_{11} - \omega\phi_{21})},$$

and interpret the meaning you might attach to θ.

(c) How would you go about estimating θ?

(d) Suppose now that one of the eigenvalues of $\mathbf{\Phi}$ is unity. How do you characterize the long run effects of a shock to x_t on y_t?

4. Assume that $\mathbf{y}_t$ and $\mathbf{x}_t$ are $m \times 1$ vector of random variables that follow the following $VAR(1)$ processes

$$\mathbf{y}_t = \mathbf{\Phi}\mathbf{y}_{t-1} + \mathbf{u}_t,$$
$$\mathbf{x}_t = \mathbf{\Psi}\mathbf{x}_{t-1} + \boldsymbol{\varepsilon}_t.$$

Suppose further that only observations on $\mathbf{z}_t = \mathbf{y}_t - \mathbf{x}_t$, for $t = 1, 2, \ldots, T$, are available.

(a) Show that

$$(\mathbf{I}_m - \mathbf{\Phi}L)(\mathbf{I}_m - \mathbf{\Psi}L)\mathbf{z}_t = (\mathbf{I}_m - \mathbf{\Psi}L)\mathbf{u}_t - (\mathbf{I}_m - \mathbf{\Phi}L)\boldsymbol{\varepsilon}_t,$$

if and only if $\mathbf{\Psi}$ and $\mathbf{\Phi}$ commute.

(b) Derive the generalized impulse response functions of a unit (one standard error) shock to the ith element of $\mathbf{u}_t$ on $\mathbf{z}_t$ process, assuming that

$$Var\begin{pmatrix} \mathbf{u}_t \\ \boldsymbol{\varepsilon}_t \end{pmatrix} = \begin{pmatrix} \boldsymbol{\Sigma}_{uu} & \boldsymbol{\Sigma}_{u\varepsilon} \\ \boldsymbol{\Sigma}_{\varepsilon u} & \boldsymbol{\Sigma}_{\varepsilon\varepsilon} \end{pmatrix}.$$

(c) Discuss estimation of the impulse response function under (b), again assuming that only observations on $\mathbf{z}_t$ are available.

(d) How do your responses to the above questions are altered if $\boldsymbol{\Phi}$ and $\boldsymbol{\Psi}$ do not commute?

25 Modelling the Conditional Correlation of Asset Returns

25.1 Introduction

Modelling of conditional volatilities and correlations across asset returns is part of portfolio decision making and risk management. In risk management the Value at Risk (*VaR*) of a given portfolio can be computed using univariate volatility models, but a multivariate model is needed for portfolio decisions.[1] Even in risk management the use of a multivariate model would be desirable when a number of alternative portfolios of the same universe of m assets are under consideration. By using the same multivariate volatility model marginal contributions of different assets towards the overall portfolio risk can be computed in a consistent manner. Multivariate volatility models are also needed for determination of hedge ratios and leverage factors.

There exists a growing literature on multivariate volatility modelling. A general class of such models is the multivariate generalized autoregressive conditional heteroskedastic (*MGARCH*) specification (Engle and Kroner (1995)). However, the number of unknown parameters in the unrestricted *MGARCH* model rises exponentially with m and its estimation will not be possible even for a modest number of assets. To deal with the curse of dimensionality the dynamic conditional correlations (*DCC*) model is proposed by Engle (2002) which generalizes an earlier specification in Bollerslev (1990) by allowing for time variations in the correlation matrix. This is achieved parsimoniously by separating the specification of the conditional volatilities from that of the conditional correlations. The latter are then modelled in terms of a small number of unknown parameters, which avoids the curse of the dimensionality. *DCC* is an attractive estimation procedure which is reasonably flexible in modeling individual volatilities and can be applied to portfolios with a large number of assets. Pesaran and Pesaran (2010) propose a *DCC* model combined with a multivariate t-distribution assumption on the distribution of asset returns. Indeed, in many applications in finance the t-distribution seems more appropriate to capture the fat-tailed nature of the distribution of asset returns. The authors suggest a simultaneous approach for estimating the parameters, including the degree-of-freedom parameter of the multivariate t-distribution, of a *t-DCC* model.

[1] See Chapter 18 for a review of univariate models of conditional volatility.

25.2 Exponentially weighted covariance estimation

Let $\mathbf{r}_t = (r_{1t}, \ldots, r_{mt})'$ be an $m \times 1$ vector of asset returns at close of day t, with conditional mean and variance

$$\boldsymbol{\mu}_{t-1} = E(\mathbf{r}_t | \Omega_{t-1}),$$
$$\boldsymbol{\Sigma}_{t-1} = Var(\mathbf{r}_t | \Omega_{t-1}),$$

where Ω_{t-1} is the information set available at close of day $t-1$, and $\boldsymbol{\Sigma}_{t-1}$ is assumed to be non-singular. Here we are not concerned with how mean returns are predicted and take $\boldsymbol{\mu}_{t-1}$ as given and equal to a zero vector.[2]

25.2.1 One parameter exponential-weighted moving average

To estimate the time varying conditional covariance matrix, one approach would be to use an exponentially decreasing weighting scheme. The one-parameter Exponential-Weighted Moving Average $(EWMA)$ can be written for a given window of size n as

$$\boldsymbol{\Sigma}_{t-1} = \lambda \boldsymbol{\Sigma}_{t-2} + \frac{(1-\lambda)}{(1-\lambda^n)} \mathbf{r}_{t-1} \mathbf{r}'_{t-1} - \frac{(1-\lambda)}{(1-\lambda^n)} \lambda^{n-1} \mathbf{r}_{t-n-1} \mathbf{r}'_{t-n-1}, \tag{25.1}$$

for a constant parameter $0 < \lambda < 1$, and a window of size n. Typically, the initialization of the recursion in (25.1) is based on estimates of the unconditional variances using a pre-sample of data. For the $(i,j)^{th}$ entry of Σ_{t-1} we have

$$\sigma_{ij,t-1} = \frac{(1-\lambda)}{(1-\lambda^n)} \sum_{s=1}^{n} \lambda^{s-1} r_{i,t-s} r_{j,t-s}.$$

The Riskmetrics specification discussed in Chapter 18 is characterized by the fact that n and λ are fixed *a priori*. The choice of λ depends on the frequency of the returns. For daily returns the values of $\lambda = 0.94, 0.95$, and 0.96, have often been used. There is an obvious trade-off between λ and n, with a small λ yielding similar results to a small n. Note that for $\boldsymbol{\Sigma}_{t-1}$ to be non-singular requires $n \geq m$, and it is therefore advisable that a relatively large value is selected for n.

25.2.2 Two parameters exponential-weighted moving average

Practitioners and academics have often pointed out that the effects of shocks on conditional variances and conditional correlations could decay at different rates, with correlations typically responding at a slower pace than volatilities. This suggests using two different parameter values for the decay coefficients, one for volatilities and the other for correlations. This yields the two-parameter Exponential-Weighted Moving Average, EWMA (n, λ, ν). Therefore, the diagonal elements of (25.1) define conditional variances $\sigma_{ii,t-1}$, $i = 1, 2, \ldots, m$, the square-roots of

[2] Although the estimation of $\boldsymbol{\mu}_{t-1}$ and $\boldsymbol{\Sigma}_{t-1}$ are inter-related, in practice mean returns are predicted by least squares techniques (such as recursive estimation or recursive modelling) which do not take account of the conditional volatility. This might involve some loss in efficiency of estimating $\boldsymbol{\mu}_{t-1}$, but considerably simplifies the estimation of the return distribution needed in portfolio decisions and risk management.

which form the diagonal matrix $\mathbf{D}_{t-1}$. The covariances are based on the same recursion as (25.1) but using a smoothing parameter, ν, generally different from λ ($\nu \leq \lambda$) yielding

$$\sigma_{ij,t-1} = \frac{(1-\nu)}{(1-\nu^n)} \sum_{s=1}^{n} \nu^{s-1} r_{i,t-s} r_{j,t-s}, \text{ for } i \neq j.$$

We assume that the same window size, n, applies to variance and covariance recursions. The ratio

$$\rho_{ij,t-1} = \sigma_{ij,t-1} / \sqrt{\sigma_{ii,t-1}\, \sigma_{jj,t-1}} \tag{25.2}$$

represents the $(i,j)^{th}$ entry of the correlation matrix $\mathbf{R}_{t-1}$ with $\mathbf{\Sigma}_{t-1} = \mathbf{D}_{t-1}^{1/2} \mathbf{R}_{t-1} \mathbf{D}_{t-1}^{1/2}$. The parameters ν and λ are not estimated but calibrated *a priori*, as for the one-parameter EWMA model.

25.2.3 Mixed moving average (*MMA*(n,ν))

This is a generalization of the equal-weighted *MA* model discussed above. Under this specification, the conditional variances are computed as in the equal-weighted MA model, the square root of which yields the diagonal matrix $\mathbf{D}_{t-1}$. Then one estimates the conditional covariances using a Riskmetrics type filter as

$$\sigma_{ij,t-1} = \frac{(1-\nu)}{(1-\nu^n)} \sum_{s=1}^{n} \nu^{s-1} \varepsilon_{i,t-s} \varepsilon_{j,t-s},$$

which after normalization according to (25.2), yields the conditional correlation matrix, $\mathbf{R}_{t-1}$, and hence $\mathbf{\Sigma}_{t-1} = \mathbf{D}_{t-1}^{1/2} \mathbf{R}_{t-1} \mathbf{D}_{t-1}^{1/2}$.

25.2.4 Generalized exponential-weighted moving average (*EWMA*(n,p,q,ν))

This is a generalization of the two-parameter EWMA. In the first stage m different univariate $GARCH(p_i, q_i)$, for $i = 1, 2, \ldots, m$, volatility models are estimated for each r_{it} by *QMLE* (see Chapter 18). The conditional covariances are then obtained using the Riskmetrics filter (25.1), with the parameters n and ν fixed *a priori*. The results are then normalized using (25.2), with the resultant variances and correlations recombined according to $\mathbf{\Sigma}_{t-1} = \mathbf{D}_{t-1}^{1/2} \mathbf{R}_{t-1} \mathbf{D}_{t-1}^{1/2}$.

The above multivariate volatility models are either *ad hoc* or are formed by combining the univaraite volatility specification with some *ad hoc* specification of the correlations across the different asset returns. We now turn to system approaches to modelling the multivariate nature of interactions across different returns using the maximum likelihood framework. To deal with the curse of dimensionality, specification of the volatility of individual returns is separated from modelling of the correlations which are specified to depend on only a small number of unknown parameters.

25.3 Dynamic conditional correlations model

This approach begins by expressing the conditional covariance matrix, $\mathbf{\Sigma}_{t-1}$, in terms of the decomposition

$$\mathbf{\Sigma}_{t-1} = \mathbf{D}_{t-1}\mathbf{R}_{t-1}\mathbf{D}_{t-1}, \tag{25.3}$$

where

$$\mathbf{D}_{t-1} = \begin{pmatrix} \sigma_{1,t-1} & 0 & \dots & 0 \\ 0 & \sigma_{2,t-1} & \dots & \vdots \\ \vdots & \vdots & \ddots & 0 \\ 0 & 0 & \dots & \sigma_{m,t-1} \end{pmatrix},$$

$$\mathbf{R}_{t-1} = \begin{pmatrix} 1 & \rho_{12,t-1} & \cdots & \rho_{1m,t-1} \\ \rho_{21,t-1} & 1 & \cdots & \rho_{2m,t-1} \\ \vdots & \vdots & \ddots & \vdots \\ \rho_{m1,t-1} & \rho_{m2,t-1} \cdots & \cdots & 1 \end{pmatrix}.$$

$\mathbf{D}_{t-1}$ is an $m \times m$, diagonal matrix with elements $\sigma_{i,t-1}, i = 1, 2, \dots, m$, denoting the conditional volatilities of assets returns, and $\mathbf{R}_{t-1}$ is the symmetric $m \times m$ matrix of pair-wise conditional correlations. More specifically, the conditional volatility for the ith asset return is defined as

$$\sigma^2_{i,t-1} = Var\left(r_{it} \mid \Omega_{t-1}\right),$$

and the conditional pair-wise return correlation between the ith and the jth asset is

$$\rho_{ij,t-1} = \rho_{ji,t-1} = \frac{Cov\left(r_{it}, r_{jt} \mid \Omega_{t-1}\right)}{\sigma_{i,t-1}\sigma_{j,t-1}}.$$

Clearly, $-1 \leq \rho_{ij,t-1} \leq 1$, and $\rho_{ij,t-1} = 1$, for $i = j$.

Bollerslev (1990) considers (25.3) with a constant correlation matrix $\mathbf{R}_{t-1} = \mathbf{R}$. Engle (2002) allows for $\mathbf{R}_{t-1}$ to be time varying and proposes a class of multivariate *GARCH* models labelled as dynamic conditional correlation (*DCC*) models.

The decomposition of $\mathbf{\Sigma}_{t-1}$ in (25.3) allows separate specification for the conditional volatilities and conditional cross-asset returns correlations. For example, one can utilize the *GARCH* (1,1) model for $\sigma^2_{i,t-1}$, namely

$$\sigma^2_{i,t-1} = \bar{\sigma}^2_i \left(1 - \lambda_{1i} - \lambda_{2i}\right) + \lambda_{1i}\sigma^2_{i,t-2} + \lambda_{2i}r^2_{i,t-1}, \tag{25.4}$$

where $\bar{\sigma}^2_i$ is the unconditional variance of the ith asset return. Note that in (25.4) we allow the parameters $\lambda_{1i}, \lambda_{2i}$ to differ across assets. An alternative approach to model (25.4) would be to use the conditionally heteroskedastic factor model discussed, for example, in Sentana (2000) where the vector of unobserved common factors is assumed to be conditionally heteroskedastic.

Parsimony is achieved by assuming that the number of the common factors is much less than the number of assets under consideration.

Under the restriction $\lambda_{1i} + \lambda_{2i} = 1$, unconditional variance does not exist. In this case we have the integrated *GARCH* (*IGARCH*) model used extensively in the professional financial community [3]

$$\sigma^2_{i,t-1} = (1 - \lambda_i) \sum_{s=1}^{\infty} \lambda_i^{s-1} r^2_{i,t-s} \qquad 0 < \lambda_i < 1, \tag{25.5}$$

or, written recursively,

$$\sigma^2_{i,t-1} = \lambda_i \sigma^2_{i,t-2} + (1 - \lambda_i)\, r^2_{i,t-1}.$$

For cross-asset correlations, Engle proposes the use of the following exponential smoother applied to the 'standardized returns'

$$\tilde{\rho}_{ij,t-1} = \frac{q_{ij,t-1}}{\sqrt{q_{ii,t-1} q_{jj,t-1}}},$$

where $q_{ij,t-1}$ are given by

$$q_{ij,t-1} = \bar{\rho}_{ij}(1 - \phi_1 - \phi_2) + \phi_1 q_{ij,t-2} + \phi_2 \tilde{r}_{i,t-1} \tilde{r}_{j,t-1}. \tag{25.6}$$

In (25.6), $\bar{\rho}_{ij}$ is the $(i,j)^{th}$ unconditional correlation, ϕ_1, ϕ_2 are parameters such that $\phi_1 + \phi_2 < 1$, and $\tilde{r}_{i,t-1}$ are the standardized assets returns. Under $\phi_1 + \phi_2 < 1$, the process is mean reverting. In the case $\phi_1 + \phi_2 = 1$, we have

$$q_{ij,t-1} = \phi q_{ij,t-2} + (1 - \phi)\, \tilde{r}_{i,t-1} \tilde{r}_{j,t-1}.$$

In practice, the hypothesis that $\phi_1 + \phi_2 < 1$ needs to be tested.

Returns in (25.6) are standardized to achieve normality. Transformation to Gaussianity is important since the use of correlation as a measure of dependence can be misleading in the case of (conditionally) non-Gaussian returns (see Embrechts, Hoing, and Juri (2003) on this). Engle (2002) proposes the following standardization for returns

$$\tilde{r}_{i,t-1} = \tilde{r}^{exp}_{i,t-1} = \frac{r_{it}}{\sigma_{i,t-1}}, \tag{25.7}$$

where $\sigma_{i,t-1}$ is given either by (25.4) or, in the case of non-mean reverting volatilities, by (25.5). We refer to (25.7) as the 'exponentially weighted returns'.

An alternative way of standardizing returns is to use a measure of realized volatility (Pesaran and Pesaran (2010))

[3] See, e.g., Litterman and Winkelmann (1998).

$$\tilde{r}_{i,t-1} = \tilde{r}^{devol}_{i,t-1} = \frac{r_{it}}{\sigma^{realized}_{i,t-1}}, \tag{25.8}$$

where $\sigma^{realized}_{i,t-1}$ is a proxy for the realized volatility of the ith return during day t. The use of $\tilde{r}_{it}$ is data intensive and requires intra-daily observations. Although intra-daily observations are becoming increasingly available across a large number of assets, it would still be desirable to work with a version of $\tilde{r}_{it}$ that does not require intra-daily observations, but is nevertheless capable of rendering the devolatized returns approximately Gaussian. One of the main reasons for the non-Gaussian behavior of daily returns is the presence of jumps in the return process as documented for a number of markets in the literature (see, e.g., Barndorff Nielsen and Shephard (2002)). The standardized return (25.7) does not deal with such jumps, since the jump process that affects the numerator of $\tilde{r}^{exp}_{i,t-1}$ in day t does not enter the denominator of $\tilde{r}^{exp}_{i,t-1}$ which is based on past returns and excludes the day t return, r_t. The problem is accentuated due to the fact that jumps are typically independently distributed over time. The use of realized volatility ensures that the numerator and the denominator of the devolatized returns, $\tilde{r}_{it}$, are both affected by the same jumps in day t.

Pesaran and Pesaran (2010) have suggested the following approximation for the realized volatility

$$\tilde{\sigma}^2_{it}(p) = \frac{\sum_{s=0}^{p-1} r^2_{i,t-s}}{p}. \tag{25.9}$$

The lag-order, p, needs to be chosen carefully. We refer to returns (25.8) where the realized volatility is estimated using (25.9) as 'devolatized returns'. In a series of papers Andersen, Bollerslev and Diebold show that daily returns on foreign exchange and stock returns standardized by realized volatility are approximately Gaussian (see, e.g., Andersen, Bollerslev, Diebold, and Ebens (2001), and Andersen et al. (2001)).

Note that $\tilde{\sigma}^2_{it}(p)$ is not the same as the rolling historical estimate of σ_{it} defined by

$$\hat{\sigma}^2_{it}(p) = \frac{\sum_{s=1}^{p} r^2_{i,t-s}}{p}.$$

Specifically,

$$\tilde{\sigma}^2_{it}(p) - \hat{\sigma}^2_{it}(p) = \frac{r^2_{it} - r^2_{i,t-p}}{p}.$$

It is the inclusion of the current squared returns, r^2_{it}, in the estimation of $\tilde{\sigma}^2_{it}$ that seems to be critical in the transformation of r_{it} (which is non-Gaussian) into $\tilde{r}_{it}$ which seems to be approximately Gaussian.

25.4 Initialization, estimation, and evaluation samples

Estimation and evaluation of the dynamic conditional correlation (DCC) model given by (25.4) and (25.6) is done in a recursive manner.

Suppose daily observations are available on m daily returns in the $m \times 1$ vector $\mathbf{r}_t$ over the period $t = 1, 2, \ldots, T, T+1, \ldots, T+N$. The sample period can be divided into three subperiods, choosing s, T_0 and T such that $p < T_0 < s < T$. We call

- Initialization sample: $\mathcal{S}_0 = \{\mathbf{r}_t, t = 1, 2, \ldots, T_0\}$. The first T_0 observations are used for initialization of the recursions in (25.4) and (25.6).
- Estimation sample: $\mathcal{S}_{est} = \{\mathbf{r}_t, t = s, s+1, \ldots, T\}$. A total of $T - s + 1$ observations are used for estimation of (25.4) and (25.6) (see Section 25.5).
- Evaluation sample: $\mathcal{S}_{eval} = \{\mathbf{r}_t, t = T+1, T+2, \ldots, T+N\}$. The last N observations are used for testing the validity of the model (see Section 25.6).

This decomposition allows the size of the estimation window to vary by moving the index s along the time axis in order to accommodate estimation of the unknown parameters using expanding or rolling observation windows, with different estimation update frequencies. For example, for an expanding estimation window we set $s = T_0 + 1$. For a rolling window of size W we need to set $s = T + 1 - W$. The whole estimation process can then be rolled into the future with an update frequency of h by carrying the estimations at $T + h, T + 2h, \ldots$, using either expanding or rolling estimation samples from $t = s$.

25.5 Maximum likelihood estimation of *DCC* model

ML estimation of the *DCC* model can be carried out under two different assumptions concerning the conditional distribution of assets returns, the multivariate Gaussian distribution and the multivariate Student's t-distribution.

In its most general formulation (the mean reverting specifications given by (25.4) and (25.6)) the *DCC* model contains $2m + 2$ unknown parameters; $2m$ coefficients $\boldsymbol{\lambda}_1 = (\lambda_{11}, \lambda_{12}, \ldots, \lambda_{1m})'$ and $\boldsymbol{\lambda}_2 = (\lambda_{21}, \lambda_{22}, \ldots, \lambda_{2m})'$ that enter the individual asset returns volatilities, and the coefficients ϕ_1 and ϕ_2 that enter the conditional correlations. In the case of t-distributed returns, a further parameter, the degrees of freedom of the multivariate Student t-distribution, ν, needs to be estimated.

The intercepts $\bar{\sigma}_i^2$ and $\bar{\rho}_{ij}$ in (25.4) and (25.6) refer to the unconditional volatilities and return correlations and can be estimated as

$$\bar{\sigma}_i^2 = \frac{\sum_{t=1}^{T} r_{it}^2}{T}, \tag{25.10}$$

$$\bar{\rho}_{ij} = \frac{\sum_{t=1}^{T} r_{it} r_{jt}}{\sqrt{\sum_{t=1}^{T} r_{it}^2} \sqrt{\sum_{t=1}^{T} r_{jt}^2}}. \tag{25.11}$$

In the non-mean reverting case these intercept coefficients disappear, but for initialization of the recursive relations (25.4) and (25.6) it is still advisable to use unconditional estimates of the correlation matrix and asset returns volatilities.

25.5.1 *ML* estimation with Gaussian returns

Denote the unknown coefficients by $\boldsymbol{\theta} = \left(\boldsymbol{\lambda}_1', \boldsymbol{\lambda}_2', \phi_1, \phi_2\right)'$. Based on a sample of observations on returns, $\mathbf{r}_1, \mathbf{r}_2, \ldots, \mathbf{r}_t$, available at time t, the time t log-likelihood function based on the decomposition (25.3) is given by

$$l_t(\boldsymbol{\theta}) = \sum_{\tau=s}^{t} f_\tau(\boldsymbol{\theta}),$$

where $s \leq t$ is the start date of the estimation window and

$$\begin{aligned} f_\tau(\boldsymbol{\theta}) = &-\frac{m}{2}\ln(\pi) - \frac{1}{2}\ln \mid R_{\tau-1}(\boldsymbol{\theta}) \mid - \ln \mid D_{\tau-1}(\boldsymbol{\lambda}_1, \boldsymbol{\lambda}_2) \mid \\ &- \ln\left[\mathbf{e}_\tau' D_{\tau-1}^{-1}(\boldsymbol{\lambda}_1, \boldsymbol{\lambda}_2) R_{\tau-1}^{-1}(\boldsymbol{\theta}) D_{\tau-1}^{-1}(\boldsymbol{\lambda}_1, \boldsymbol{\lambda}_2) \mathbf{e}_\tau\right], \end{aligned}$$

with $\mathbf{e}_\tau = \mathbf{r}_\tau - \boldsymbol{\mu}_{\tau-1}$. For estimation of the unknown parameters, Engle (2002) shows that the log-likelihood function of the *DCC* model can be maximized using a two-step procedure. In the first step, m univariate *GARCH* models are estimated separately. In the second step using standardized residuals, computed from the estimated volatilities from the first stage, the parameters of the conditional correlations are then estimated. The two-step procedure can then be iterated if desired for full maximum likelihood estimation. Note that under Engle's specification $\mathbf{R}_{t-1}$ depends on $\boldsymbol{\lambda}_1$ and $\boldsymbol{\lambda}_2$ as well as on ϕ_1 and ϕ_2.

This procedure has two main drawbacks. First, the Gaussian assumption in general does not hold for daily returns (see Chapter 7) and its use can under-estimate the portfolio risk. Second, the two-stage approach is likely to be inefficient even under Gaussianity.

For further details on *ML* estimation using Gaussian returns, see Engle (2002).

25.5.2 *ML* estimation with Student's *t*-distributed returns

Denote the unknown coefficients by $\boldsymbol{\theta} = \left(\boldsymbol{\lambda}_1', \boldsymbol{\lambda}_2', \phi_1, \phi_2, \nu\right)'$, where ν are the (unknown) degrees of freedom of the t-distribution. The time t log-likelihood function based on the decomposition (25.3) is given by

$$l_t(\boldsymbol{\theta}) = \sum_{\tau=s}^{t} f_\tau(\boldsymbol{\theta}), \tag{25.12}$$

where (see B.45 in Appendix B)

$$
\begin{aligned}
f_\tau(\boldsymbol{\theta}) = &-\frac{m}{2}\ln(\pi) - \frac{1}{2}\ln\left|R_{\tau-1}(\boldsymbol{\theta})\right| - \ln\left|D_{\tau-1}(\boldsymbol{\lambda}_1,\boldsymbol{\lambda}_2)\right| \\
&+ \ln\left[\Gamma\left(\frac{m+\nu}{2}\right)/\Gamma\left(\frac{\nu}{2}\right)\right] - \frac{m}{2}\ln(\nu-2) \\
&- \left(\frac{m+\nu}{2}\right)\ln\left[1 + \frac{\mathbf{e}_\tau' D_{\tau-1}^{-1}(\boldsymbol{\lambda}_1,\boldsymbol{\lambda}_2) R_{\tau-1}^{-1}(\boldsymbol{\theta}) D_{\tau-1}^{-1}(\boldsymbol{\lambda}_1,\boldsymbol{\lambda}_2)\mathbf{e}_\tau}{\nu-2}\right],
\end{aligned}
\tag{25.13}
$$

and $\mathbf{e}_\tau = \mathbf{r}_\tau - \boldsymbol{\mu}_{\tau-1}$. Note that

$$
\ln\left|D_{\tau-1}(\boldsymbol{\lambda}_1,\boldsymbol{\lambda}_2)\right| = \sum_{i=1}^{m}\ln\left[\sigma_{i,\tau-1}(\lambda_{1i},\lambda_{2i})\right].
$$

Under the specification based on devolatized returns, $\mathbf{R}_{t-1}$ does not depend on $\boldsymbol{\lambda}_1$ and $\boldsymbol{\lambda}_2$, but depends on ϕ_1 and ϕ_2, and p, the lag order used in the devolatization process. Under the specification based on exponentially weighted returns, $\mathbf{R}_{t-1}$ depends on $\boldsymbol{\lambda}_1$ and $\boldsymbol{\lambda}_2$ as well as on ϕ_1 and ϕ_2.

The *ML* estimate of $\boldsymbol{\theta}$ based on the sample observations, $\mathbf{r}_s, \mathbf{r}_2, \ldots, \mathbf{r}_T$, can now be computed by maximization of $l_t(\boldsymbol{\theta})$ with respect to $\boldsymbol{\theta}$, which we denote by $\hat{\boldsymbol{\theta}}_t$. More specifically

$$
\hat{\boldsymbol{\theta}}_t = \underset{\boldsymbol{\theta}}{\operatorname{argmax}}\left\{l_t(\boldsymbol{\theta})\right\},
\tag{25.14}
$$

for $t = T, T+h, T+2h, \ldots, T+N$, where h is the (estimation) update frequency, and N is the length of the evaluation sample (see Section 25.4). The standard errors of the *ML* estimates are computed using the asymptotic formula

$$
\widehat{Cov}(\hat{\boldsymbol{\theta}}_t) = \left\{\sum_{\tau=s}^{t}\left[\frac{-\partial^2 f_\tau(\boldsymbol{\theta})}{\partial\boldsymbol{\theta}\partial\boldsymbol{\theta}'}\right]_{\boldsymbol{\theta}=\hat{\boldsymbol{\theta}}_t}\right\}^{-1}.
$$

In practice the simultaneous estimation of all the parameters of the DDC model could be problematic, since it can encounter convergence problems, or could lead to a local maxima of the likelihood function. When the returns are conditionally Gaussian one could simplify (at the expense of some loss of estimation efficiency) the computations by adopting Engle's two-stage estimation procedure. But in the case of t-distributed returns the use of such a two-stage procedure could lead to contradictions. For example, estimation of separate t-*GARCH*(1, 1) models for individual asset returns can lead to different estimates of ν, while the multivariate t-distribution requires ν to be the same across all assets.[4]

[4] Marginal distributions associated with a multivariate t-distribution with ν degrees of freedom are also t-distributed with the same degrees of freedom.

25.6 Simple diagnostic tests of the *DCC* model

In the following, we assume that the $m \times 1$ vector of returns $\mathbf{r}_t$ follows a multivariate Student's t-distribution, though the same line of reasoning applies in the case of Gaussian returns. Consider a portfolio based on m assets with returns $\mathbf{r}_t$, using an $m \times 1$ vector of predetermined weights, $\mathbf{w}_{t-1}$. The return on this portfolio is given by

$$\rho_t = \mathbf{w}_{t-1}'\mathbf{r}_t. \tag{25.15}$$

Suppose that we are interested in computing the capital Value at Risk (VaR) of this portfolio expected at the close of business on day $t-1$ with probability $1-\alpha$, which we denote by $VaR(\mathbf{w}_{t-1}, \alpha)$. For this purpose we require that

$$\Pr\left[\mathbf{w}_{t-1}'\mathbf{r}_t < -VaR(\mathbf{w}_{t-1}, \alpha) \mid \Omega_{t-1}\right] \leq \alpha.$$

Under our assumptions, conditional on Ω_{t-1}, $\mathbf{w}_{t-1}'\mathbf{r}_t$ has a Student t-distribution with mean $\mathbf{w}_{t-1}'\boldsymbol{\mu}_{t-1}$, variance $\mathbf{w}_{t-1}'\boldsymbol{\Sigma}_{t-1}\mathbf{w}_{t-1}$, and degrees of freedom v. Hence

$$z_t = \sqrt{\frac{v}{v-2}}\left(\frac{\mathbf{w}_{t-1}'\mathbf{r}_t - \mathbf{w}_{t-1}'\boldsymbol{\mu}_{t-1}}{\sqrt{\mathbf{w}_{t-1}'\boldsymbol{\Sigma}_{t-1}\mathbf{w}_{t-1}}}\right),$$

conditional on Ω_{t-1} will also have a Student t-distribution with v degrees of freedom. It is easily verified that $E(z_t|\Omega_{t-1}) = 0$, and $V(z_t|\Omega_{t-1}) = v/(v-2)$. Denoting the cumulative distribution function of a Student's t with v degrees of freedom by $F_v(z)$, $VaR(\mathbf{w}_{t-1}, \alpha)$ will be given as the solution to

$$F_v\left(\frac{-VaR(\mathbf{w}_{t-1},\alpha) - \mathbf{w}_{t-1}'\boldsymbol{\mu}_{t-1}}{\sqrt{\frac{v-2}{v}\left(\mathbf{w}_{t-1}'\boldsymbol{\Sigma}_{t-1}\mathbf{w}_{t-1}\right)}}\right) \leq \alpha.$$

But since $F_v(z)$ is a continuous and monotonic function of z we have

$$\frac{-VaR(\mathbf{w}_{t-1},\alpha) - \mathbf{w}_{t-1}'\boldsymbol{\mu}_{t-1}}{\sqrt{\frac{v-2}{v}\left(\mathbf{w}_{t-1}'\boldsymbol{\Sigma}_{t-1}\mathbf{w}_{t-1}\right)}} = F_v^{-1}(\alpha) = -c_\alpha,$$

where c_α is the α per cent critical value of a Student t-distribution with v degrees of freedom. Therefore,

$$VaR(\mathbf{w}_{t-1},\alpha) = \tilde{c}_\alpha\sqrt{\left(\mathbf{w}_{t-1}'\boldsymbol{\Sigma}_{t-1}\mathbf{w}_{t-1}\right)} - \mathbf{w}_{t-1}'\boldsymbol{\mu}_{t-1}, \tag{25.16}$$

where $\tilde{c}_\alpha = c_\alpha\sqrt{\frac{v-2}{v}}$.

Following Engle and Manganelli (2004), a simple test of the validity of t-DCC model can be conducted recursively using the indicator statistics

$$d_t = I\left[\mathbf{w}'_{t-1}\mathbf{r}_t + VaR(\mathbf{w}_{t-1},\alpha)\right], \tag{25.17}$$

where $I(A)$ is an indicator function, equal to unity if $A > 0$ and zero otherwise. These indicator statistics can be computed in-sample or preferably can be based on a recursive out-of-sample one-step ahead forecast of $\mathbf{\Sigma}_{t-1}$ and $\boldsymbol{\mu}_{t-1}$, for a given (predetermined set of portfolio weights, $\mathbf{w}_{t-1}$). In such an out–of-sample exercise the parameters of the mean returns and the volatility variables ($\boldsymbol{\beta}$ and $\boldsymbol{\theta}$, respectively) could either be kept fixed at the start of the evaluation sample or changed with an update frequency of h periods (for example with $h = 5$ for weekly updates, or $h = 20$ for monthly updates). For the evaluation sample, $\mathcal{S}_{\text{eval}} = \{\mathbf{r}_t, t = T+1, T+2, \ldots, T+N\}$, the mean hit rate is given by

$$\hat{\pi}_N = \frac{1}{N}\sum_{t=T+1}^{T+N} d_t. \tag{25.18}$$

Under the *t-DCC* specification, $\hat{\pi}_N$ will have mean $1-\alpha$ and variance $\alpha(1-\alpha)/N$, and the standardized statistic,

$$z_\pi = \frac{\sqrt{N}\left[\hat{\pi}_N - (1-\alpha)\right]}{\sqrt{\alpha(1-\alpha)}}, \tag{25.19}$$

will have a standard normal distribution for a sufficiently large evaluation sample size, N. This result holds irrespective of whether the unknown parameters are estimated recursively or fixed at the start of the evaluation sample. In such cases the validity of the test procedure requires that $N/T \to 0$ as $(N,T) \to \infty$. For further details on this statistic, see Pesaran and Timmermann (2005a).

The z_π statistic provides evidence on the performance of Σ_{t-1} and $\boldsymbol{\mu}_{t-1}$ in an average (unconditional) sense. An alternative conditional evaluation procedure can be based on probability integral transforms

$$\hat{U}_t = F_v\left(\frac{\mathbf{w}'_{t-1}\mathbf{r}_t - \mathbf{w}'_{t-1}\hat{\boldsymbol{\mu}}_{t-1}}{\sqrt{\frac{v-2}{v}\mathbf{w}'_{t-1}\hat{\mathbf{\Sigma}}_{t-1}\mathbf{w}_{t-1}}}\right),\ t = T+1, T+2, \ldots, T+N. \tag{25.20}$$

Under the null hypothesis of correct specification of the *t*-DCC model, the probability transform estimates, $\hat{U}_t$, are serially uncorrelated and uniformly distributed over the range $(0,1)$. Both of these properties can be readily tested. The serial correlation property of $\hat{U}_t$ can be tested by Lagrange Multiplier tests using *OLS* regressions of $\hat{U}_t$ on an intercept and the lagged values $\hat{U}_{t-1}, \hat{U}_{t-2}, \ldots, \hat{U}_{t-s}$, where the maximum lag length, s, can be selected by using the AIC criterion. The uniformity of the distribution of $\hat{U}_t$ over t can be tested using the Kolmogorov–Smirnov statistic defined by $KS_N = \sup_x \left|F_{\hat{U}}(x) - U(x)\right|$, where $F_{\hat{U}}(x)$ is the empirical cumulative distribution function (CDF) of the $\hat{U}_t$, for $t = T+1, T+2, \ldots, T+N$, and $U(x) = x$ is the CDF of *IIDU*$[0,1]$. Large values of the Kolmogorov-Smirnov statistic, KS_N, indicate that the sample CDF is not similar to the hypothesized uniform CDF.[5]

[5] For details of the Kolmogorov-Smirnov test and its critical values see, e.g., Neave and Worthington (1992, pp. 89–93).

25.7 Forecasting volatilities and conditional correlations

Having obtained the recursive *ML* estimates, $\hat{\boldsymbol{\theta}}_t$, given by (25.14), the following one step-ahead forecasts can be obtained. For volatilities we have

$$\widehat{V\left(r_{i,T+1} \mid \Omega_T\right)} = \hat{\sigma}^2_{i,T} = \bar{\sigma}^2_{i,T}\left(1 - \hat{\lambda}_{1i,T} - \hat{\lambda}_{2i,T}\right) + \hat{\lambda}_{1i,T}\hat{\sigma}^2_{i,T-1} + \hat{\lambda}_{2i,T}r^2_{iT},$$

where $\bar{\sigma}^2_{i,T}$ is the estimate of the unconditional mean of r^2_{it}, computed as

$$\bar{\sigma}^2_{i,T} = T^{-1}\sum_{\tau=1}^{T} r^2_{i\tau},$$

$\hat{\lambda}_{1i,T}$ and $\hat{\lambda}_{2i,T}$ are the *ML* estimates of λ_{1i} and λ_{2i} computed using the observations over the estimation sample $\mathcal{S}_{est} = \{\mathbf{r}_t, t = s, s+1, \ldots, T\}$, and $\hat{\sigma}^2_{i,T-1}$ is the *ML* estimate of $\sigma^2_{i,T-1}$, based on the estimates $\bar{\sigma}^2_{i,T-1}$, $\hat{\lambda}_{1i,T-1}$ and $\hat{\lambda}_{2i,T-1}$.

Similarly, the one step-ahead forecast of $\rho_{ij,T}$ (using either exponentially weighted returns (25.7) or devolatilized returns (25.8)) is given by

$$\hat{\rho}_{ij,T}(\phi) = \frac{\hat{q}_{ij,T}}{\sqrt{\hat{q}_{ii,T}\hat{q}_{jj,T}}},$$

where

$$\hat{q}_{ij,T} = \bar{\rho}_{ij,T}(1 - \hat{\phi}_{1T} - \hat{\phi}_{2T}) + \hat{\phi}_{1T}\hat{q}_{ij,T-1} + \hat{\phi}_{2,T}\tilde{r}_{i,T}\tilde{r}_{j,T}.$$

As before, $\hat{\phi}_{1T}$ and $\hat{\phi}_{2T}$ are the *ML* estimates of ϕ_{1T} and ϕ_{2T} computed using the estimation sample, and $\hat{q}_{ij,T-1}$ is the *ML* estimate of $q_{ij,T-1}$, based on the estimates $\bar{\rho}_{ij,T-1}$, $\hat{\phi}_{1T-1}$ and $\hat{\phi}_{2T-1}$.

25.8 An application: volatilities and conditional correlations in weekly returns

We estimate alternative versions of the *t-DCC* model for a portfolio composed of weekly returns on:

- 6 currencies: British pound (GBP), euro (EU), Japanese yen (JPY), Swiss franc (CHF), Canadian dollar (CAD), and Australian dollar (AD).
- 4 government bonds: US T-Note 10Y (BU), Europe euro bund 10Y (BE), Japan government bond 10Y (JGB), and UK long gilts 8.75-13Y (BG).
- 7 equity index futures S&P 500 (SP), FTSE 100 (FTSE), German DAX (DAX), French CAC40 (CAC), Swiss Market Index (SM), Australia SPI200 (AUS), Nikkei 225 (NK).

The weekly returns are computed from daily prices obtained from Datastream and cover the period from 7 Jan 94 to 30 Oct 2009.

25.8.1 Devolatized returns and their properties

Table 25.1 provides summary statistics for the weekly returns (r_{it}, in percent) and the devolatized weekly returns $\tilde{r}_{it} = r_{it}/\tilde{\sigma}_{it}(p)$, where in the absence of intra-daily observations $\tilde{\sigma}^2_{it}(p)$ is defined by (25.9), with $p = 13$ weeks. The choice of $p = 13$ was guided by some experimentation with the aim of transforming r_{it} into an approximately Gaussian process. A choice of p well above 13 does not allow the (possible) jumps in r_{it} to become adequately reflected in $\tilde{\sigma}_{it}(p)$, and a value of p well below 13 transforms r_{it} to an indicator looking function. In the extreme case where $p = 1$ we have $\tilde{r}_{it} = 1$, if $r_{it} > 0$, and $\tilde{r}_{it} = -1$, if $r_{it} < 0$, and $\tilde{r}_{it} = 0$, if $r_{it} = 0$. We did not experiment with other values of p for the sample under consideration and set $p = 13$ for all 17 assets.

For the non-devolatized returns, the results are as to be expected from previous studies. The returns seem to be symmetrically distributed with kurtosis in some cases well in excess of 3 (the value for the Gaussian distribution). The excess kurtosis is particularly large for equities, mostly around 5 or more. For currencies, the kurtosis coefficient is particularly large for yen, British pound, and Singapore dollar. By comparison the weekly returns on government bonds are less fat-tailed with kurtosis coefficients only marginally above 3. In contrast, none of the 17 devolatized returns shows any evidence of excess kurtosis. For example, for equities the excess kurtosis of weekly returns on SP, FTSE and Nikkei falls from 8.01, 10.40, 9.65 to –0.124, –0.132 and –0.147, respectively after the returns are devolatized. For currencies. the excess kurtosis of the

Table 25.1 Summary statistics for raw weekly returns and devolatized weekly returns over 1 April 1994 to 20 October 2009

	Returns				Devolatized returns			
Asset	Mean	S.D.	Skewness	Ex. Kurtosis	Mean	S.D.	Skewness	Ex. Kurtosis
Currencies								
Australian dollar	0.044	1.690	–1.163	7.886	0.059	1.005	–0.214	–0.112
British pound	0.019	1.297	–0.831	5.348	0.037	1.013	–0.148	–0.197
Canadian dollar	0.035	1.136	–0.739	7.443	0.031	1.023	–0.040	–0.266
Swiss franc	0.053	1.517	0.210	1.071	0.044	0.994	0.146	–0.299
Euro	0.039	1.381	–0.043	1.424	0.044	1.012	–0.008	–0.281
Yen	0.031	1.669	1.326	9.462	–0.009	1.016	0.328	0.139
Bonds								
Euro Bunds	0.070	0.755	–0.378	0.910	0.123	1.000	–0.210	–0.205
UK Gilt	0.051	0.893	–0.013	1.744	0.068	1.008	–0.015	–0.290
Japan JGB	0.072	0.578	–0.436	2.323	0.152	1.007	–0.364	0.022
US T-Note	0.077	0.894	–0.359	0.954	0.084	1.004	–0.243	–0.188
Equities								
S&P 500	0.094	2.575	–0.749	8.018	0.054	1.011	–0.314	–0.124
Nikkei	–0.017	3.175	–0.979	9.645	–0.005	0.996	–0.235	–0.147
FTSE	0.060	2.535	–0.858	10.399	0.042	1.002	–0.264	–0.132
CAC	0.107	3.116	–0.656	5.473	0.043	1.003	–0.216	–0.478
DAX	0.113	3.398	–0.559	5.673	0.055	1.008	–0.312	–0.220
SM	0.137	2.819	–0.734	10.174	0.077	1.005	–0.349	0.077
AUS	0.083	2.118	–0.670	4.698	0.066	1.001	–0.224	–0.253

weekly returns on AD, BP, and JY falls from 7.89, 5.35, and 9.46 to –0.112, –0.020, and 0.139, respectively. Out of the four ten year government bonds only the weekly returns on Japanese government bond show some degree of excess kurtosis which is eliminated once the returns are devolatized. It is also interesting to note that the standard deviations of the devolatized returns are now all very close to unity, that allows for a more direct comparison of the devolatized returns across assets.

25.8.2 *ML* estimation

It is well established that daily or weekly returns are approximately mean zero serially uncorrelated processes and for the purpose of risk analysis it is reasonable to assume that $\boldsymbol{\mu}_{t-1} = \mathbf{0}$. Using the *ML* procedure described above, initially we estimate a number of *DCC* models on the 17 weekly returns over the period 27 May 1994 to 28 Dec 2007 (710 observations). We then use the post estimation sample observations from 4 January, 2008 to 30 October, 2009 for the evaluation of the estimated volatility models using the *VaR* and distribution free diagnostics.[6] We also provide *separate t-DCC* models for currencies, bonds and equities for purposes of comparison.

We begin with the unrestricted version of the $DCC(1,1)$ model with asset-specific volatility parameters $\boldsymbol{\lambda}_1 = (\lambda_{11}, \lambda_{12}, \ldots, \lambda_{1m})'$, $\boldsymbol{\lambda}_2 = (\lambda_{21}, \lambda_{22}, \ldots, \lambda_{2m})'$, and common conditional correlation parameters, ϕ_1 and ϕ_2, and the degrees-of-freedom parameter, ν, under conditionally t distributed returns (note that $m = 17$). We did not encounter any convergence problems, and obtained the same *ML* estimates when starting from different initial parameter values. But to achieve convergence in some applications we had to experiment with different initial values. In particular we found the initial values $\lambda_{1i} = 0.95, \lambda_{2i} = 0.05, \phi_1 = 0.96, \phi_2 = 0.03$ and $\nu = 12$ to work relatively well. Also the sum of unrestricted estimates of λ_1 and λ_2 for the Canadian dollar exceeded 1, and to ensure a non-explosive outcome we estimated its volatility equation subject to the restriction $\lambda_{1,CD} + \lambda_{2,CD} = 1$.

To evaluate the statistical significance of the multivariate t-distribution for the analysis of return volatilities, in Table 25.2 we first provide the maximized log-likelihood values under multivariate normal and t-distributions for currencies, bonds and equities separately, as well as for all the 17 assets jointly. These results are reported for both the standardized and devolatized returns.

Table 25.2 Maximized log-likelihood values of *DCC* models estimated with weekly returns over 27 May 1994 to 28 December 2007

	Standardized returns			Devolatized returns		
Assets	Normal	t-distribution	D.F.	Normal	t-distribution	D.F.
Currencies (6)	–5783.7	–5689.8	9.62 (1.098)	–5790.6	–5694.1	9.24 (0.94)
Bonds (4)	–2268.5	–2243.5	11.28 (2.00)	–2270.7	–2246.9	11.35 (5.53)
Equities (7)	–9500.1	–9380.7	7.96 (0.74)	–9504.4	–9383.2	7.79 (0.72)
All 17	–17509.2	–17244.8	11.84 (0.90)	–17510.4	–17250.4	12.11 (0.92)

Note: D.F. is the estimated degrees of freedom of the multivariate t-distribution. Standard errors of the estimates are given in parentheses.

[6] The *ML* estimation and the computation of the diagnostic statistics are carried out using Microfit 5. See Pesaran and Pesaran (2009).

It is firstly clear from these results that the normal-*DCC* specifications are strongly rejected relative to the *t-DCC* models for all asset categories. The maximized log-likelihood values for the *t-DCC* models are significantly larger than the ones for the normal-*DCC* models. The estimated degrees of freedom of the multivariate *t*-distribution for different asset classes are quite close and range from 8 (for equities) to 11 (for bonds), all well below the value of 30 and above what one would expect for a multivariate normal distribution. For the full set of 17 assets the estimate of ν is closer to 12. There seems to be a tendency for the estimate of ν to rise as more assets are included in the *t-DCC* model.

The above conclusions are robust to the way returns are scaled for computation of cross asset return correlations. The maximized log-likelihoods for the standardized and devolatized returns are very close although, due to the non-nested nature of the two return transformations, no definite conclusions can be reached as to their relative merits. The specifications where the returns are standardized by the conditional volatilities tend to fit better (give higher log-likelihood values). But this is to be expected since the maximization of the log-likelihood function in this case is carried out with respect to the parameters of the scaling factor, unlike the case where scaling is carried out with respect to the realized volatilities which do not depend on the unknown parameters of the likelihood function. In what follows we base our correlation analysis on the devolatized returns on the grounds of their approximate Gaussianity, as argued in Section 25.3.

25.8.3 Asset-specific estimates

Table 25.3 presents the *ML* estimates of the *t-DCC* model including all the 17 assets computed over the period 27 May 94–28 Dec 07 (710 weekly returns). The asset-specific estimates of the volatility decay parameters are all highly significant, with the estimates of λ_{1i}, $i = 1, 2, \ldots, 17$ falling in the range of 0.818 (for Japanese government bond) to 0.986 (for Canadian dollar).[7] The average estimate of $\boldsymbol{\lambda}_1$ across assets is 0.924, which is somewhat smaller than the values in the range of 0.95 to 0.97 recommended by Riskmetrics for computation of *daily* volatilities using their exponential smoothing procedure. This is not surprising, since one would expect the exponential smoothing parameter for computing the volatility of weekly returns to be smaller than the one used for computing the volatility of daily returns.

There are, however, notable differences across asset groups with λ_{i1} estimated to be larger for currencies as compared to the estimates for equities and bonds. The average estimate of λ_1 across currencies is 0.95 as compared to 0.93 for equities and 0.88 for bonds. The correlation parameters, ϕ_1 and ϕ_2, are very precisely estimated and $\hat{\phi}_1 + \hat{\phi}_2 = 0.9846(0.0028)$, which suggest very slow but statistically significant mean reverting conditional correlations.

The sum of the estimates of λ_{1i} and λ_{2i} is very close to unity, but the hypothesis that $\lambda_{1i} + \lambda_{2i} = 1$ (the integrated *GARCH* hypothesis) against the one-sided alternative $\lambda_{1i} + \lambda_{2i} < 1$ is rejected for 10 out of the 17 assets at the 5 per cent significance level; the exceptions being British pound, Swiss franc, Nikkei, S&P 500, and Australian SPI200. In order to ensure a non-explosive outcome for Canadian dollar, as noted earlier, estimation is carried out subject to the restriction $\lambda_{1,CD} + \lambda_{2,CD} = 1$. If the test is carried out at the 1 per cent significance level, the integrated *GARCH* hypothesis is rejected only in the case of the *JGB* (Japanese government bond).

The integrated *GARCH* (*IGARCH*) hypothesis is implicit in the approach advocated by Riskmetrics, but as shown by Zaffaroni (2008), this can lead to inconsistent estimates. However,

[7] Recall that for Canadian dollar the volatility model is estimated subject to the restriction $\lambda_{1,CD} + \lambda_{2,CD} = 1$.

Table 25.3 *ML* estimates of t-DCC model estimated with weekly returns over the period 27 May 94–28 Dec 07

	***ML* Estimates**		
Asset	$\hat{\lambda}_1$	$\hat{\lambda}_2$	$1-\hat{\lambda}_1-\hat{\lambda}_2$
Currencies			
Australian dollar	0.9437 (0.0201)	0.0361 (0.0097)	0.0201 (0.0140)[1.44]
British pound	0.9862 (0.0110)	0.0124 (0.0056)	0.0014 (0.0081)[0.18]
Canadian dollar	0.9651 (0.0102)	0.0349 (0.0102)	0 (N/A)[N/A]
Swiss franc	0.9365 (0.0517)	0.0303 (0.0157)	0.0332 (0.0378)[0.88]
Euro	0.9222 (0.0264)	0.0487 (0.0133)	0.0291 (0.0154)[1.89]
Yen	0.9215 (0.0235)	0.0586 (0.0151)	0.01992 (0.0107)[1.86]
Bonds			
Euro Bunds	0.9031 (0.0237)	0.0703 (0.0149)	0.0266 (0.0118)[2.26]
UK Gilt	0.9062 (0.0304)	0.0774 (0.0224)	0.0164 (0.0091)[1.80]
Japan JGB	0.8179 (0.0369)	0.1444 (0.0268)	0.0377 (0.0141)[2.74]
US T-Note	0.9072 (0.0249)	0.0714 (0.0165)	0.0216 (0.0115)[1.87]
Equities			
CAC	0.9252 (0.0118)	0.0674 (0.0099)	0.0074 (0.0033)[2.23]
DAX	0.9267 (0.0117)	0.0653 (0.0095)	0.0080 (0.0039)[2.03]
Nikkei	0.9552 (0.0305)	0.0402 (0.0210)	0.0046 (0.0109)[0.42]
S&P 500	0.9326 (0.0194)	0.0582 (0.0150)	0.0091 (0.0060)[1.53]
FTSE	0.9298 (0.0144)	0.0589 (0.0109)	0.0112 (0.0052)[2.16]
SM	0.9066 (0.0225)	0.0774 (0.0165)	0.0160(0.0076)[2.11]
AUS	0.9393 (0.0295)	0.0370 (0.0128)	0.0237(0.0194)[1.22]
$\hat{\nu}=12.11(0.9233)$,	$\hat{\phi}_1=0.9673\ (0.0037)$,	$\hat{\phi}_2=0.0172\ (0.0012)[5.49]$	

Note: Standard errors of the estimates are given in parentheses; *t*-statistics are given is in brackets; λ_{1i} and λ_{2i} are the asset-specific volatility parameters; and ϕ_1 and ϕ_2 are the common conditional correlation parameters.

in the present applications, the unrestricted parameter estimates and those obtained under *IGARCH* are very close and one can view the restrictions $\lambda_{1i}+\lambda_{2i}=1$ as a first-order approximation that avoids explosive outcomes. We also note that the diagnostic test results, to be reported in Section 25.8.5, are not qualitatively affected by the imposition of the restrictions, $\lambda_{1i}+\lambda_{2i}=1$.

Finally, it is worth noting that there is statistically significant evidence of parameter heterogeneity across assets, which could lead to misleading inference if these differences are ignored.

25.8.4 Post estimation evaluation of the *t-DCC* model

The evaluation sample, 04 Jan 08–30 Oct 09, covers the recent periods of financial crisis and includes 96 weeks of post estimation sample of portfolio returns. The parameter values are estimated using the sample 27 May 04–28 Dec 07 and then fixed throughout the evaluation sample. To evaluate the *t-DCC* model we first consider the tests based on probability integral transforms (PIT), $\hat{U}_t$, defined by (25.20). We have already seen that under the null hypothesis the *t-DCC* model is correctly specified, $\hat{U}_t$'s are serially uncorrelated and uniformly distributed over the range (0, 1). To compute $\hat{U}_t$ we consider an equal-weighted portfolio, with all elements of $\mathbf{w}$ in

(25.15) set to 1/17, and use the risk tolerance probability of $\alpha = 1\%$, which is the value typically assumed in practice. We consider two versions of the *t-DCC* model: a version with no restrictions on λ_{1i} and λ_{2i} (except for $i = CD$), and an integrated version where $\lambda_{1i} + \lambda_{2i} = 1$, for all i.

Using the Lagrange multiplier statistic to test the null hypothesis that $\hat{U}_t$'s are serially uncorrelated, we obtained the values of $\chi^2_{12} = 4.74$ and $\chi^2_{12} = 5.31$ for the unrestricted and the restricted *t-DCC* specifications. These statistics are computed assuming a maximum lag order of 12, and are asymptotically distributed as chi-squared variates with twelve degrees of freedom. It is clear that both specifications of the *t-DCC* model pass this test.

Next we apply the Kolmogorov-Smirnov statistic to $\hat{U}_t$'s to test the null hypothesis that the PIT values are draws from a uniform distribution. The *KS* statistics for the unrestricted and the restricted versions amount to 0.0646 and 0.0454, respectively. Both these statistics are well below the *KS* critical value of 0.1388 (at the 5 per cent level).[8] Therefore, the null hypothesis that the sample *CDF* of $\hat{U}_t$'s is similar to the hypothesized uniform *CDF* cannot be rejected.

It is interesting that neither of the tests based on $\hat{U}_t$'s are capable of detecting the effects of the financial turmoil that occured in 2008. A test based on the violations of the *VaR* constraint is likely to be more discriminating, since it focusses on the tail properties of the return distributions. For a tolerance probability of $\alpha = 0.01$, we would expect only one violation of the *VaR* constraint in 100 observations (our evaluation sample contains 96 observations). The unrestricted specification results in three violations of the *VaR* constraint, and the restricted specification in four violations. Both specifications violate the *VaR* constraint in the weeks starting on 5 Sep 08, 3 Oct 08, and 10 Oct 08. The restricted version also violates the *VaR* in the week starting in 18 Jan 08. The test statistics associated with these violations are -2.09 and -3.12 which are normally distributed. Thus both specifications are rejected by the *VaR* violation test.[9] Not surprisingly, the rejection of the test is due to the unprecedented market volatility during the weeks in September and October of 2008. This period covers the Fed's take over of FannieMae <http://www.fanniemae.com/portal/index.html> and Freddie Mac <http://www.freddiemac.com/>, the collapse of Lehman Brothers, and the downgrading of the AIG's credit rating. In fact, during the two weeks starting on 3 Oct 08, the S&P 500 dropped by 29.92 per cent, which is larger than the 20 per cent market decline experienced during the October Crash of 1987.

25.8.5 Recursive estimates and the *VaR* diagnostics

We now consider whether the excess *VaR* violations documented above could have been avoided if the parameter estimates of the *t-DCC* model were updated at regular intervals. To simplify the computations we focused on the *IGARCH* version of the model and re-estimated all its parameters (including the degree-of-freedom parameter, ν) every 13 weeks (or four times in a year). Using the recursive estimates of the PIT, U_t, and the *VaR* indicator d_t we obtained similar results for the post 2007 period. The *KS* statistic for the recursive estimates is 0.0518 as compared with the 5 per cent critical value of 0.1381 and does not reject the null hypothesis that the

[8] See Table 1 in Massey (1951).

[9] We also carried out the *VaR* diagnostic test for the higher risk tolerance value of $\alpha = 5\%$, but did not find statistically significant evidence against the *t*-DCC specifications. For both versions of the model the *VaR* constraint was violated 8 times, 3 more than one would have expected, giving $\hat{\pi} = 0.9167$ and $z_\pi = -1.50$ which is not significant at the 5% level. It is, however, interesting that all the eight violations occurred in 2008 with five of them occurring over the crisis months of 5 Sep 08–21 Nov 08.

the recursive *PIT* values are draws from a uniform distribution. We also could not find any evidence of serial correlation in the *PIT* values. But as before, the violations of the *VaR* constraint were statistically significant with $z_\pi = -3.09$. The violations occur exactly on the same dates as when the parameters were fixed at the end of 2007. Updating the parameter estimates of the *t-DCC* model seem to have little impact on the diagnostic test outcomes.

25.8.6 Changing volatilities and correlations

The time series plots of volatilities are displayed in Figures 25.1–25.3 for returns on currencies, bonds and equities, respectively. Conditional correlations of the euro with other currencies, US 10 year bond futures with other bond futures, and S&P futures with other equity future indices are shown in Figures 25.4 to 25.6, respectively. To reduce the impact of the initialization on the plots of volatilities and conditional correlations, initial estimates for 1994 are not shown. These figures clearly show the declining trends in volatilities over the 2003–06 period just before the financial crisis which led to an unprecedented rise in volatilities, particularly in the currency and equity markets. It is, however, interesting to note that return correlations have been rising historically and seem to be only marginally accentuated by the recent crisis. This trend could reflect the advent of the euro and a closer integration of the world economy, particularly in the euro area. Return correlations across asset types have also been rising, although to a lesser extent. An overall measure of the extent of the correlations across all the 17 assets under consideration is given by the maximum eigenvalue of the 17 by 17 matrix of asset return correlations. Figure 25.7 displays the conditional estimates of this eigenvalue over time and clearly shows the sharp rise in asset return correlations particularly over the years 2008 and 2009.

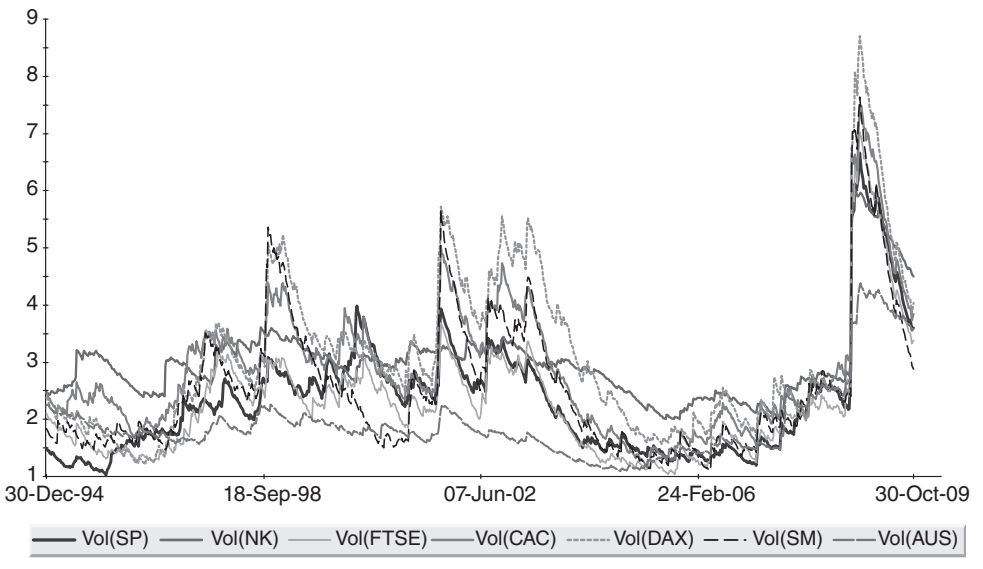


Figure 25.1 Conditional volatilities of weekly currency returns.

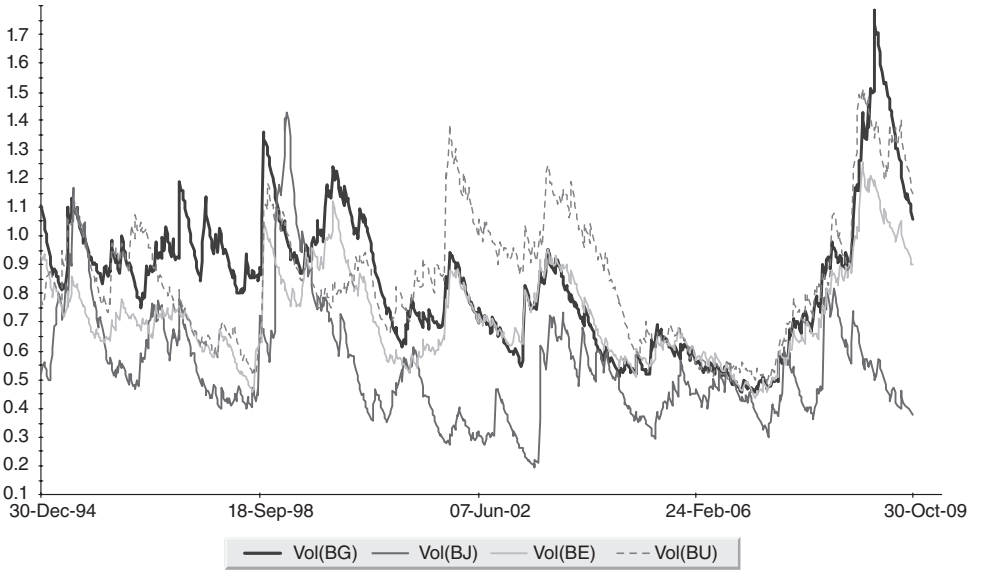


Figure 25.2 Conditional volatilities of weekly bond returns.

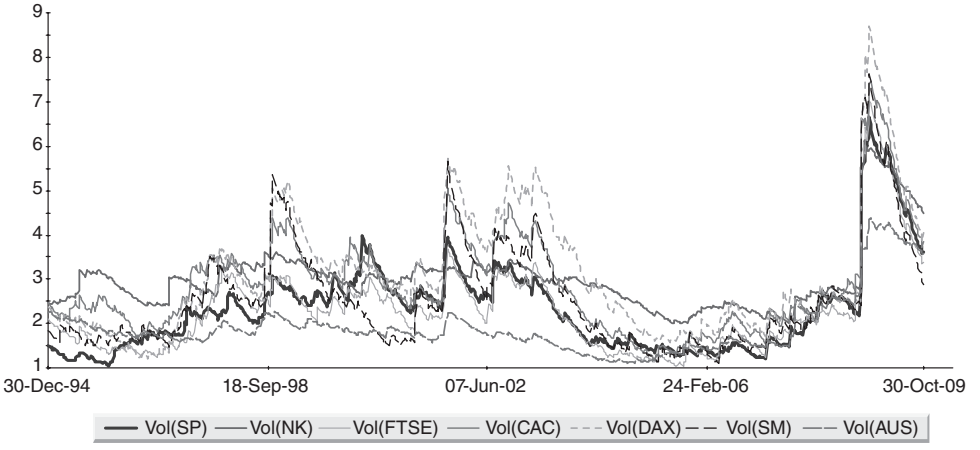


Figure 25.3 Conditional volatilities of weekly equity returns.

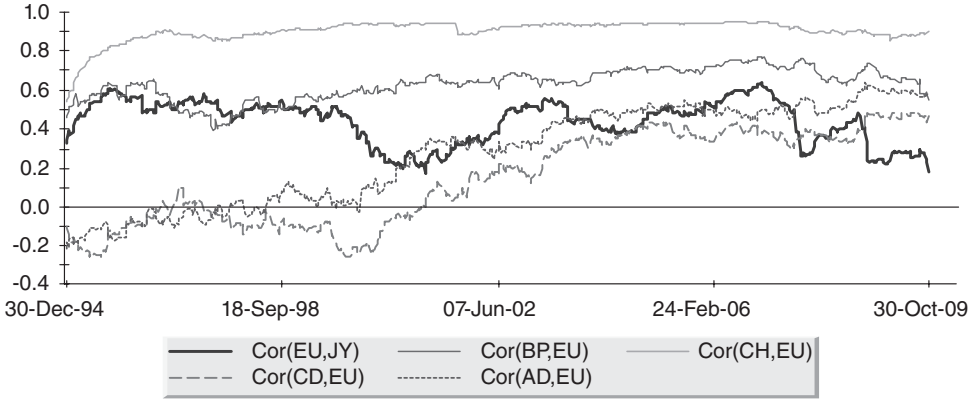


Figure 25.4 Conditional correlations of the euro with other currencies.

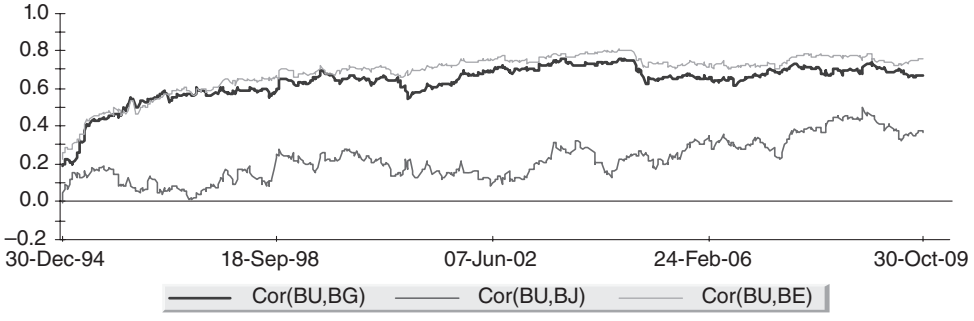


Figure 25.5 Conditional correlations of US 10-year bond with other bonds.

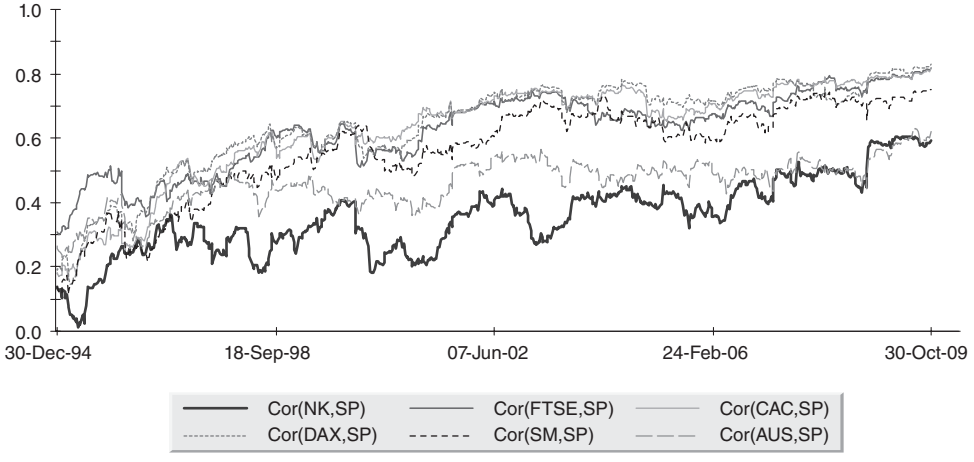


Figure 25.6 Conditional correlations of S&P 500 with other equities.

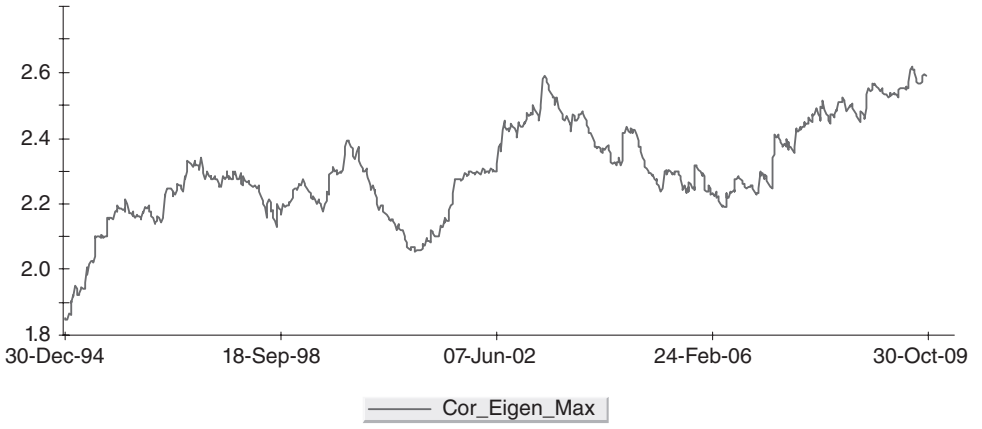


Figure 25.7 Maximum eigenvalue of 17 by 17 matrix of asset return correlations.

25.9 Further reading

See Bauwens, Laurent, and Rombouts (2006) for a review of the existing literature.

25.10 Exercises

1. Consider the $m \times 1$ vector of returns, $\mathbf{r}_t = (r_{1t}, r_{2t}, \ldots, r_{mt})'$, and suppose that

$$r_{it} = \mu_i + u_{it},$$

where

$$u_{it} = \sigma_{it}\varepsilon_{it}, \text{ for } i = 1, 2, \ldots, m,$$
$$\log\left(\sigma_{it}^2\right) = \lambda_i \log\left(\sigma_{i,t-1}^2\right) + \alpha_{0i} + v_{it},$$

$\boldsymbol{\varepsilon}_t = (\varepsilon_{1t}, \varepsilon_{2t}, \ldots, \varepsilon_{mt})' \sim IIDN(\mathbf{0}, \mathbf{R})$, where $\mathbf{R}$ is an $m \times m$ positive definite correlation matrix, and $v_{it} \sim IIDN(0, \omega_i^2)$ for all i.

   (a) Derive $E\left[(\mathbf{r}_t - \boldsymbol{\mu})(\mathbf{r}_t - \boldsymbol{\mu})' | \Omega_{t-1}\right]$, where $\Omega_{t-1} = (\mathbf{r}_{t-1}, \mathbf{r}_{t-2}, \ldots)$.

   (b) Compare the above model with the DCC specification discussed in Section (25.3).

   (c) Discuss the problems of identification and estimation of $\mathbf{R}$ and the parameters of the volatility component, $\log\left(\sigma_{it}^2\right)$.

2. Let $\rho_t(\boldsymbol{\omega}_{t-1}) = \boldsymbol{\omega}_{t-1}'\mathbf{r}_t$ be a portfolio return, where $\boldsymbol{\omega}_{t-1} = (\omega_{1,t-1}, \omega_{2,t-2}, \ldots, \omega_{N,t-1})'$ is the $N \times 1$ vector of weights and $\mathbf{r}_t = (r_{1t}, r_{2t}, \ldots, r_{Nt})'$ is the associated vector of returns. Suppose that $\mathbf{r}_t$ is distributed with the conditional mean, $E(\mathbf{r}_t|\Omega_{t-1})$, and the conditional covariance, $V(\mathbf{r}_t|\Omega_{t-1})$, where Ω_{t-1} is the available information at time $t-1$.

(a) Derive the portfolio weights, $\boldsymbol{\omega}_{t-1}$, assuming the aim is to maximize expected returns subject to a given value for the portfolio variance.

(b) Assume further that $\mathbf{r}_t|\Omega_{t-1}$ is Student t-distributed with $\nu > 2$ degrees of freedom. Derive the portfolio weights subject to the *VaR* constraint given by

$$\Pr\left(\rho_t(\boldsymbol{\omega}_{t-1}) < -L_{t-1}\,|\Omega_{t-1}\right) \leq \alpha, \tag{25.21}$$

where $L_{t-1} > 0$ is a pre-specified maximum daily loss and α is a (small) probability value.

(c) Show that the above two optimization problems can be combined by solving the following mean-variance objective function

$$Q(\omega_{t-1}|\Omega_{t-1}) = \omega'_{t-1}E(\mathbf{r}_t|\Omega_{t-1}) - \frac{\delta_{t-1}}{2}\omega'_{t-1}V(\mathbf{r}_t|\Omega_{t-1})\omega_{t-1},$$

subject to the *VaR* constraint given by (25.21).

(d) Show that the optimal portfolio weights, ω^*_{t-1}, under (c) above can be written as

$$\omega^*_{t-1} = \begin{cases} \frac{1}{\delta_{t-1}}\left[V(\mathbf{r}_t|\Omega_{t-1})\right]^{-1}E(\mathbf{r}_t|\Omega_{t-1}), & \text{if } \delta_{t-1} \geq \delta^*_{t-1}, \\ \frac{1}{\delta^*_{t-1}}\left[V(\mathbf{r}_t|\Omega_{t-1})\right]^{-1}E(\mathbf{r}_t|\Omega_{t-1}), & \text{otherwise,} \end{cases}$$

with

$$\delta^*_{t-1} \equiv \frac{s_{t-1}\left(\sqrt{\frac{\nu-2}{\nu}}c_{\nu,\alpha} - s_{t-1}\right)}{L_{t-1}},$$

where $s_{t-1} = \sqrt{E(\mathbf{r}'_t|\Omega_{t-1})\left[V(\mathbf{r}_t|\Omega_{t-1})\right]^{-1}E(\mathbf{r}_t|\Omega_{t-1})}$, and $c_{\nu,\alpha} > 0$ is the α% left tail of the Student t-distribution with ν degrees of freedom.

Hint: See Pesaran, Schleicher, and Zaffaroni (2009).

3. Use the daily returns data on the equity index futures S&P 500 (SP), FTSE 100 (FTSE), German DAX (DAX), French CAC40 (CAC), Swiss Market Index (SM), Australia SPI200 (AUS), Nikkei 225 (NK) provided in Pesaran and Pesaran (2009) to estimate the conditional covariance of these seven returns using Riskmetrics specification with parameters $\lambda = 0.96$ and $n = 250$, and compare your results (using some suitable diagnostics) with the estimates obtained using the DCC approach.

Part VI

Panel Data Econometrics

26 Panel Data Models with Strictly Exogenous Regressors

26.1 Introduction

Panel data consist of observations on many individual economic units over two or more periods of time. The individual units are usually referred to as cross-sectional units, and in economic and finance applications are typically represented by single individuals, firms, returns on individual securities, industries, regions, or countries.

In recent years, panel data sets have become widely available to empirical researchers. Examples of such data sets in the US include the Panel Study of Income Dynamics (PSID), collected by the Institute for Social Research at the University of Michigan, and the National Longitudinal Surveys of Labor Market Experience (NLS), from the Center for Human Resource Research at Ohio State University. The PSID began in 1968 by collecting of annual economic information from a representative national sample of about 6,000 families and 15,000 individuals. The NLS started in the mid 1960s, and contains five separate annual surveys covering various segments of the labour force. In Europe, many countries have their national annual surveys such as the Netherlands Socioeconomic Panel, the German Social Economics Panel, and the British Household Panel Survey. At aggregated level, the published statistics of the Organisation for Economic Co-operation and Development (OECD) contain numerous series of economic aggregates observed yearly for many countries. New data sources are also emerging through Google search engine and retail scanner datasets. Examples are Google Flu Trends (<http://www.google.org/flutrends/>), Nielson Datasets for consumer marketing (<http://research.chicagobooth.edu/nielsen/>). This increasing availability of panel data sets, while opening up new possibilities for analysis, has also raised a number of new and interesting econometric issues.

Panel data offer several important advantages over data sets with only a temporal or longitudinal dimension. A major motivation for using panel data is the ability to control for possibly correlated, time-invariant heterogeneity without actually observing it. One may be able to identify and measure effects that are otherwise not detectable, as well as to account for latent individual heterogeneity. An additional advantage of panel data, compared to time series data, is the reduction in collinearity among explanatory variables and the increase in efficiency of econometric estimators. Finally, the cross-sectional dimension may also alleviate problems of

aggregation. These benefits do come at a cost. Important difficulties arise when explanatory variables in panel data regression models cannot be assumed strictly exogenous. As we shall see in Chapter 27, standard panel estimators are inconsistent when panel data regression models have weakly exogenous regressors, and their treatment poses a number of methodological challenges. A further complication arises when regression errors attached to different cross-section units are dependent, even after conditioning on variables that are specific to cross-sectional units. In the presence of cross-section dependence, conventional panel estimators can result in misleading inference and even inconsistent estimators (see Chapter 29 on this). Finally, important econometric issues arise when panel data sets involves non-responses and measurement errors.

The literature on panel data can be broadly divided into three categories, depending on their assumptions about the relative magnitudes of the number of cross-sectional units (N) and the number of time periods (T). First, there exists a 'small N, large T' time series literature which closely follows the SURE procedure, due to Zellner (1962) and described in Chapter 19. The main attraction of the SURE approach is that it allows the contemporaneous error covariances to be freely estimated. But this is possible only when N is reasonably small relative to T, while the SURE procedure is not feasible when N is of the same order of magnitude as T. Also the SURE approach assumes that the regressors are uncorrelated with the errors which rules out the error correlation being due to the presence of unobserved common factors. The general problem of error cross-sectional correlation will be discussed in Chapter 29. Second, there is the 'small T, large N' panel literature. The set of econometric models and techniques suggested to carry inference on this type of panel data sets, assuming strictly exogenous regressors, will be the object of this chapter. Next chapter relaxes the exogeneity assumption and allows the regressors to be weakly exogenous. The analysis of 'large T, large N' panels will be covered in Chapters 28 and 31.

26.2 Linear panels with strictly exogenous regressors

Let y_{it} be the observation on the ith cross-sectional unit at time t for $i = 1, 2, \ldots, N; t = 1, 2, \ldots, T$, and assume it is generated by the following panel data regression model

$$y_{it} = \alpha_i + \boldsymbol{\beta}' \mathbf{x}_{it} + u_{it}, \tag{26.1}$$

where $\mathbf{x}_{it}$ is a $k \times 1$ vector of observed individual specific regressors on the ith cross-sectional unit at time t, u_{it} is the error term, $\boldsymbol{\beta}$ is a k-dimensional vector of unknown parameters, and α_i denotes an unobservable, unit-specific effect. Note that α_i is time-invariant, and it accounts for any individual-specific effect that is not included in the regression (Mundlak (1978)).

It is often convenient to rewrite model (26.1) in stacked form using a *unit-specific formulation* as follows

$$\underset{(T \times 1)}{\mathbf{y}_{i.}} = \underset{(1 \times 1)}{\alpha_i} \underset{(T \times 1)}{\boldsymbol{\tau}_T} + \underset{(T \times k)}{\mathbf{X}_{i.}} \underset{(k \times 1)}{\boldsymbol{\beta}} + \underset{(T \times 1)}{\mathbf{u}_{i.}}, \tag{26.2}$$

where

$$\mathbf{y}_{i.} = \begin{pmatrix} y_{i1} \\ y_{i2} \\ \vdots \\ y_{iT} \end{pmatrix}, \ \mathbf{X}_{i.} = \begin{pmatrix} x_{i1}^{(1)} & x_{i1}^{(2)} & \dots & x_{i1}^{(k)} \\ x_{i2}^{(1)} & x_{i2}^{(2)} & \dots & x_{i2}^{(k)} \\ \vdots & \vdots & \vdots & \vdots \\ x_{iT}^{(1)} & x_{iT}^{(2)} & \dots & x_{iT}^{(k)} \end{pmatrix},$$
$$\mathbf{u}_{i.} = \begin{pmatrix} u_{i1} \\ u_{i2} \\ \vdots \\ u_{iT} \end{pmatrix}, \ \boldsymbol{\tau}_T = \begin{pmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{pmatrix}. \tag{26.3}$$

In other cases, it is convenient to rewrite (26.1) in stacked form using a *time-specific formulation*

$$\underset{(N \times 1)}{\mathbf{y}_{.t}} = \underset{(N \times 1)}{\boldsymbol{\alpha}} + \underset{(N \times k)}{\mathbf{X}_{.t}} \underset{(k \times 1)}{\boldsymbol{\beta}} + \underset{(N \times 1)}{\mathbf{u}_{.t}}, \tag{26.4}$$

where

$$\mathbf{y}_{.t} = \begin{pmatrix} y_{1t} \\ y_{2t} \\ \vdots \\ y_{Nt} \end{pmatrix}, \ \mathbf{X}_{.t} = \begin{pmatrix} x_{1t}^{(1)} & x_{1t}^{(2)} & \dots & x_{1t}^{(k)} \\ x_{2t}^{(1)} & x_{2t}^{(2)} & \dots & x_{2t}^{(k)} \\ \vdots & \vdots & \vdots & \vdots \\ x_{Nt}^{(1)} & x_{Nt}^{(2)} & \dots & x_{Nt}^{(k)} \end{pmatrix},$$
$$\mathbf{u}_{.t} = \begin{pmatrix} u_{1t} \\ u_{2t} \\ \vdots \\ u_{Nt} \end{pmatrix}, \ \boldsymbol{\alpha} = \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_N \end{pmatrix}.$$

Finally, equation (26.1) can be expressed in matrix form as

$$\mathbf{y} = (\boldsymbol{\alpha} \otimes \boldsymbol{\tau}_T) + \mathbf{X}\boldsymbol{\beta} + \mathbf{u}. \tag{26.5}$$

where $\mathbf{y} = \left(\mathbf{y}_{1.}', \mathbf{y}_{2.}', \dots, \mathbf{y}_{N.}'\right)'$, $\mathbf{X} = (\mathbf{x}_{1.}, \mathbf{x}_{2.}, \dots, \mathbf{x}_{N.})'$, $\mathbf{u} = \left(\mathbf{u}_{1.}', \mathbf{u}_{2.}', \dots, \mathbf{u}_{N.}'\right)'$, and $\otimes$ is the Kronecker product.

In the rest of this chapter, it is assumed $E(u_{it} \,|\mathbf{X}_{i.}) = 0$, for all i and t, namely, that regressors are strictly exogenous (see Section 9.3 for a discussion of the notion of strict and weak exogeneity). In other words, at each time period, the error term is assumed to be uncorrelated with all lags and leads of the explanatory variables. As we shall see, this is a critical assumption for the methods developed in this chapter. The case of weakly exogenous regressors will be considered in Chapter 27.

26.3 Pooled *OLS* estimator

This estimator assumes that the intercepts are homogeneous, namely $\alpha_i = \alpha$, for all i. In this case the panel data model reduces to

$$y_{it} = \alpha + \boldsymbol{\beta}' \mathbf{x}_{it} + u_{it}, \tag{26.6}$$

and α and $\boldsymbol{\beta}$ can be estimated by the *OLS* procedure. The resultant estimator of $\boldsymbol{\beta}$ is known as pooled *OLS* and is given by

$$\hat{\boldsymbol{\beta}}_{OLS} = \left[\sum_{i=1}^{N} \sum_{t=1}^{T} (\mathbf{x}_{it} - \bar{\mathbf{x}}) (\mathbf{x}_{it} - \bar{\mathbf{x}})' \right]^{-1} \left[\sum_{i=1}^{N} \sum_{t=1}^{T} (\mathbf{x}_{it} - \bar{\mathbf{x}}) (y_{it} - \bar{y}) \right], \tag{26.7}$$

where

$$\bar{\mathbf{x}} = (NT)^{-1} \sum_{i=1}^{N} \sum_{t=1}^{T} \mathbf{x}_{it}, \bar{y} = (NT)^{-1} \sum_{i=1}^{N} \sum_{t=1}^{T} y_{it},$$

and assuming that $\sum_{i=1}^{N} \sum_{t=1}^{T} (\mathbf{x}_{it} - \bar{\mathbf{x}}) (\mathbf{x}_{it} - \bar{\mathbf{x}})'$ is a nonsingular matrix. The pooled estimator is unbiased and consistent if $\mathbf{x}_{it}$ is strictly exogenous and the intercepts are homogeneous. Heteroskedasticity of the errors, u_{it}, and temporal dependence affect inference but does not affect the consistency property of the pooled estimator, when T is fixed and N large. More formally, we make the following assumptions:

Assumption P1: $E(u_{it} | \mathbf{x}_{it'}) = 0$, for all i, t and t'.

Assumption P2 : The regressors, $\mathbf{x}_{it}$, are either deterministic and bounded, namely $\|\mathbf{x}_{it}\| < K < \infty$, or they satisfy the moment conditions $E\| (\mathbf{x}_{it} - \bar{\mathbf{x}}) \left(\mathbf{x}_{jt'} - \bar{\mathbf{x}}\right)' \| < K < \infty$, for all i, j, t and t', where $\|\mathbf{A}\|$ denotes the Frobenius norm of matrix $\mathbf{A}$.

Assumption P3: The $k \times k$ matrix $\mathbf{Q}_{p,NT}$ defined by

$$\mathbf{Q}_{p,NT} = \frac{1}{NT} \sum_{i=1}^{N} \sum_{t=1}^{T} (\mathbf{x}_{it} - \bar{\mathbf{x}}) (\mathbf{x}_{it} - \bar{\mathbf{x}})', \tag{26.8}$$

is positive definite for all N and T, and as N and/or $T \to \infty$.

Assumption P4: The errors, u_{it}, are cross-sectionally independent.

Assumption P5: The errors, u_{it}, could be cross-sectionally heteroskedastic and temporally correlated

$$\begin{aligned}
&E\left(u_{it} u_{jt'} | \mathbf{X}\right) = 0, \text{ if } i \neq j \text{ for all } t \text{ and } t', \\
&E\left(u_{it} u_{jt'} | \mathbf{X}\right) = \gamma_i(t, t'), \text{ if } i = j, \text{ and } t \neq t', \\
&E\left(u_{it} u_{jt'} | \mathbf{X}\right) = \sigma_i^2 < K < \infty, \text{ if } i = j, \text{ and } t = t',
\end{aligned}$$

where $\gamma_i(t,t')$ is the auto-covariance of the u_{it} process, assumed to be bounded, namely $\left|\gamma_i(t,t')\right| < K < \infty$, for all i, t and t'.

Remark 6 *Assumption P2 can be relaxed to allow for trended or unit root processes. The autocovariances $\gamma_i(t,t')$ can be left unrestricted when T is fixed.*

To establish the unbiasedness property we first note that α can be eliminated by demeaning using the grand means, $\bar{\mathbf{x}}$ and $\bar{y}$. We have

$$y_{it} - \bar{y} = \boldsymbol{\beta}'(\mathbf{x}_{it} - \bar{\mathbf{x}}) + (u_{it} - \bar{u}),$$

where $\bar{u} = (NT)^{-1}\sum_{i=1}^{N}\sum_{t=1}^{T} u_{it}$. Using this result in (26.7) we have[1]

$$\hat{\boldsymbol{\beta}}_{OLS} - \boldsymbol{\beta} = \mathbf{Q}_{p,NT}^{-1}\left[\frac{1}{NT}\sum_{i=1}^{N}\sum_{t=1}^{T}(\mathbf{x}_{it} - \bar{\mathbf{x}})\,u_{it}\right].$$

Under the strict exogeneity Assumption P1, we have $E(u_{it}\,|\mathbf{X}) = 0$, for all i and t, where $\mathbf{X} = \{\mathbf{x}_{it}, \text{for } i = 1,2,\ldots,N; t = 1,2,\ldots,T\}$, and it readily follows that

$$E\left(\hat{\boldsymbol{\beta}}_{OLS}\,|\mathbf{X}\right) - \boldsymbol{\beta} = \mathbf{Q}_{p,NT}^{-1}\left[\frac{1}{NT}\sum_{i=1}^{N}\sum_{t=1}^{T}(\mathbf{x}_{it} - \bar{\mathbf{x}})\,E(u_{it}\,|\mathbf{X})\right] = \mathbf{0},$$

and, therefore, unconditionally we also have $E\left[E\left(\hat{\boldsymbol{\beta}}_{OLS}\,|\mathbf{X}\right) - \boldsymbol{\beta}\right] = \mathbf{0}$, or $E\left(\hat{\boldsymbol{\beta}}_{OLS}\right) = \boldsymbol{\beta}$, which establishes that $\hat{\boldsymbol{\beta}}_{OLS}$ is an unbiased estimator of $\boldsymbol{\beta}$.

Consider now the variance of $\hat{\boldsymbol{\beta}}_{OLS}$ and note that

$$Var\left(\hat{\boldsymbol{\beta}}_{OLS}\,|\mathbf{X}\right) = \mathbf{Q}_{p,NT}^{-1}\left[\frac{1}{N^2T^2}\sum_{i=1}^{N}\sum_{j=1}^{N}\sum_{t=1}^{T}\sum_{t'=1}^{T}E\left(u_{it}u_{jt'}\,|\mathbf{X}\right)(\mathbf{x}_{it} - \bar{\mathbf{x}})\left(\mathbf{x}_{jt} - \bar{\mathbf{x}}\right)'\right]\mathbf{Q}_{p,NT}^{-1}. \tag{26.9}$$

Hence, under Assumptions P4 and P5 above we have

$$Var\left(\hat{\boldsymbol{\beta}}_{OLS}\,|\mathbf{X}\right) = \frac{1}{NT}\mathbf{Q}_{p,NT}^{-1}\mathbf{V}_{p,NT}\mathbf{Q}_{p,NT}^{-1}, \tag{26.10}$$

where $\mathbf{Q}_{p,NT}$ is given by (26.8), and

[1] Note that

$$\sum_{i=1}^{N}\sum_{t=1}^{T}(\mathbf{x}_{it} - \bar{\mathbf{x}})\,\bar{u} = \bar{u}\sum_{i=1}^{N}\sum_{t=1}^{T}(\mathbf{x}_{it} - \bar{\mathbf{x}}) = \mathbf{0}.$$

$$\mathbf{V}_{p,NT} = \frac{1}{N}\left[T^{-1}\sum_{i=1}^{N}\sum_{t=1}^{T}\sigma_i^2\left(\mathbf{x}_{it}-\bar{\mathbf{x}}\right)\left(\mathbf{x}_{it}-\bar{\mathbf{x}}\right)' + T^{-1}\sum_{i=1}^{N}\sum_{t\neq t'}^{T}\gamma_i(t,t')\left(\mathbf{x}_{it}-\bar{\mathbf{x}}\right)\left(\mathbf{x}_{it}-\bar{\mathbf{x}}\right)'\right]. \tag{26.11}$$

Also under Assumption P2, it readily follows that

$$\frac{1}{T}\sum_{t=1}^{T}E\left[\left(\mathbf{x}_{it}-\bar{\mathbf{x}}\right)\left(\mathbf{x}_{it}-\bar{\mathbf{x}}\right)'\right] = O_p(1), \text{ for all } i,$$

$$\frac{1}{T}\sum_{t\neq t'}^{T}\gamma_i(t,t')E\left[\left(\mathbf{x}_{it}-\bar{\mathbf{x}}\right)\left(\mathbf{x}_{it}-\bar{\mathbf{x}}\right)'\right] = O(T), \text{ for all } i,$$

and as a result

$$\lim_{N\to\infty} Var\left(\hat{\boldsymbol{\beta}}_{OLS}\,|\mathbf{X}\right) = \mathbf{0}, \text{for a fixed } T.$$

Also, since $\hat{\boldsymbol{\beta}}_{OLS}$ is an unbiased estimator of $\boldsymbol{\beta}$, it then follows that

$$\lim_{N\to\infty} E\left[\left(\hat{\boldsymbol{\beta}}_{OLS}-\boldsymbol{\beta}\right)\left(\hat{\boldsymbol{\beta}}_{OLS}-\boldsymbol{\beta}\right)'\right] = \mathbf{0}, \text{for a fixed } T,$$

which in turn establishes that $\hat{\boldsymbol{\beta}}_{OLS}$ converges in root mean squared error to its true value, and $Plim(\hat{\boldsymbol{\beta}}_{OLS}) \to \mathbf{0}$, as $N \to \infty$.

This is a general result and holds so long as the regressors are strictly exogenous, the errors are cross-sectionally uncorrelated, the individual effects, α_i, are uncorrelated with the errors and the regressors, and T is fixed as $N \to \infty$, or if N and $T \to \infty$, jointly in any order. But when N is fixed and $T \to \infty$, then certain mixing or stationary conditions are required on the autocovariances, $\gamma_i(t,t')$, for the pooled *OLS* estimator to remain consistent. A sufficient condition is given by $T^{-2}\sum_{t=1}^{T}\sum_{t'=1}^{T}\gamma_i^2(t,t') \to 0$, for each i. In a panel data context the most interesting cases are when T is fixed and N large or when both N and T are large. In such cases, under assumptions P1-P5, the pooled *OLS* is robust to any degree of temporal dependence in the errors, u_{it}. It can also account for cross-sectional heteroskedasticity.

Furthermore, as in the case of the classical regression model, if we also assume that u_{it} are normally distributed it then readily follows that

$$\sqrt{NT}\left(\hat{\boldsymbol{\beta}}_{OLS}-\boldsymbol{\beta}\right) \sim N(\mathbf{0}, \boldsymbol{\Omega}_{p,NT}),$$

where

$$\boldsymbol{\Omega}_{p,NT} = \mathbf{Q}_{p,NT}^{-1}\mathbf{V}_{p,NT}\mathbf{Q}_{p,NT}^{-1}.$$

In the case where the errors are not normally distributed, then for any fixed T, $\sqrt{NT}\left(\hat{\boldsymbol{\beta}}_{OLS}-\boldsymbol{\beta}\right) \to_d N(\mathbf{0}, \boldsymbol{\Omega}_{\beta_{ols}})$, as $N \to \infty$, where $\boldsymbol{\Omega}_{\beta_{ols}} = Plim_{N\to\infty}\left(\boldsymbol{\Omega}_{p,NT}\right)$.

All the above results critically depend on the intercept homogeneity assumption. In the case where α_i differ sufficiently across i, the pooled *OLS* estimator could be biased depending on the degree of the heterogeneity of α_i and the extent to which α_i and $\mathbf{x}_{it}$ are correlated. As a simple formulation suppose that

$$\alpha_i = \alpha + \eta_i,$$

where $\eta_i \sim IID(0,\sigma_\eta^2)$, and

$$\mathbf{x}_{it} = \mathbf{g}_t \eta_i + \mathbf{w}_{it}, \tag{26.12}$$

where $\mathbf{g}_t = (g_{1t}, g_{2t}, \ldots, g_{kt})'$, and $\mathbf{w}_{it}$ is a $k \times 1$ vector of strictly exogenous regressors that are uncorrelated with η_i. The degree of correlation between η_i and $\mathbf{x}_{it}$ is given by $\sigma_\eta^2 \mathbf{g}_t$. To derive the asymptotic bias of $\hat{\boldsymbol{\beta}}_{OLS}$ note that under this setup $y_{it} - \bar{y} = \eta_i - \bar{\eta} + (\mathbf{x}_{it} - \bar{\mathbf{x}})' \boldsymbol{\beta} + u_{it} - \bar{u}$, and

$$\hat{\boldsymbol{\beta}}_{OLS} - \boldsymbol{\beta} = \left[\sum_{i=1}^{N} \sum_{t=1}^{T} (\mathbf{x}_{it} - \bar{\mathbf{x}}) (\mathbf{x}_{it} - \bar{\mathbf{x}})' \right]^{-1} \left[\sum_{i=1}^{N} \sum_{t=1}^{T} (\mathbf{x}_{it} - \bar{\mathbf{x}}) (u_{it} - \bar{u} + \eta_i - \bar{\eta}) \right].$$

Now for a fixed T and as $N \to \infty$ we have

$$\underset{N\to\infty}{Plim} \left(\hat{\boldsymbol{\beta}}_{OLS} \right) - \boldsymbol{\beta} = \mathbf{Q}_{T,p}^{-1} \left\{ \lim_{N\to\infty} \frac{1}{NT} \sum_{i=1}^{N} \sum_{t=1}^{T} E \left[(\mathbf{x}_{it} - \bar{\mathbf{x}}) (\eta_i - \bar{\eta}) \right] \right\},$$

where $\mathbf{Q}_{T,p} = Plim_{N\to\infty} \left(\mathbf{Q}_{p,NT}^{-1} \right)$. Also

$$\begin{aligned} E \left[(\mathbf{x}_{it} - \bar{\mathbf{x}}) (\eta_i - \bar{\eta}) \right] &= E \left\{ \left[\mathbf{g}_t (\eta_i - \bar{\eta}) + \left(\mathbf{g}_t - \bar{\mathbf{g}} \right) \bar{\eta} + \mathbf{w}_{it} - \bar{\mathbf{w}} \right] (\eta_i - \bar{\eta}) \right\} \\ &= \sigma_\eta^2 \left(\frac{N-1}{N} \right) \mathbf{g}_t, \end{aligned}$$

and hence

$$\underset{N\to\infty}{Plim} \left(\hat{\boldsymbol{\beta}}_{OLS} \right) = \boldsymbol{\beta} + \sigma_\eta^2 \mathbf{Q}_{T,p}^{-1} \bar{\mathbf{g}}_T,$$

where $\bar{\mathbf{g}}_T = T^{-1} \sum_{t=1}^{T} \mathbf{g}_t$. This bias arises because of the omission of η_i which is correlated with $\mathbf{x}_{it}$. One way of dealing with this bias is to employ the fixed-effects estimator to which we now turn.

26.4 Fixed-effects specification

Under the fixed-effects (FE) specification, α_i are treated as free parameters which are incidental to the analysis, with $\boldsymbol{\beta}$ being the focus of interest. Typically, the only restriction imposed on α_i

is boundedness, namely that $|\alpha_i| < K < \infty$, for all i, where K is a fixed positive constant. Otherwise, α_i is allowed to have any degree of dependence on the regressors, $\mathbf{x}_{it}$ or the error term, u_{it}. This setup does not rule out the possibility that α_i are random draws from a given distribution. In general, one can think of the fixed-effects as draws from a joint probability distribution function over α_i, $\mathbf{x}_{it}$ and u_{it}, where the number of parameters characterizing this distribution is allowed to increase at the same rate as the number of cross-sectional observations, N. For further discussion see Mundlak (1978) and Hausman and Taylor (1981).

Under the FE specification, we assume that conditional on the individual effects, α_i, the regressors, $\mathbf{x}_{it}$, are strictly exogenous, but do not impose any restrictions on the fixed-effects. More formally, we continue to maintain Assumptions P1, P4 and P5 , but replace Assumptions P2 and P3 with the following:

Assumption P2' : The regressors, $\mathbf{x}_{it}$, are either deterministic and bounded, namely $\|\mathbf{x}_{it}\| < K < \infty$, or they satisfy the moment conditions $E\left\|(\mathbf{x}_{it} - \bar{\mathbf{x}}_i)\left(\mathbf{x}_{jt'} - \bar{\mathbf{x}}_j\right)'\right\| < K < \infty$, for all i, j, t and t', where $\bar{\mathbf{x}}_i = T^{-1}\sum_{t=1}^{T}\mathbf{x}_{it}$.

Assumption P3': The $k \times k$ matrix $\mathbf{Q}_{FE,NT}$ defined by

$$\mathbf{Q}_{FE,NT} = \frac{1}{NT}\sum_{i=1}^{N}\sum_{t=1}^{T}(\mathbf{x}_{it} - \bar{\mathbf{x}}_i)(\mathbf{x}_{it} - \bar{\mathbf{x}}_i)', \tag{26.13}$$

is positive definite for all N and T, and as N and/or $T \to \infty$. We denote the (probability) limits of $\mathbf{Q}_{FE,NT}$ as N or T, or both tending to infinity by $\mathbf{Q}_{FE,T}$, $\mathbf{Q}_{FE,N}$ and $\mathbf{Q}_{FE}$, respectively.

The basic idea behind FE estimation is to estimate $\boldsymbol{\beta}$ after eliminating the individual effects, α_i. Averaging over time equation (26.1) yields

$$\bar{y}_{i.} = \alpha_i + \boldsymbol{\beta}'\bar{\mathbf{x}}_{i.} + \bar{u}_{i.}, \tag{26.14}$$

where $\bar{y}_{i.}$, $\bar{\mathbf{x}}_{i.}$ and $\bar{u}_{i.}$ are time averages given by

$$\bar{y}_{i.} = \frac{1}{T}\sum_{t=1}^{T}y_{it}, \quad \bar{\mathbf{x}}_{i.} = \frac{1}{T}\sum_{t=1}^{T}\mathbf{x}_{it}, \quad \bar{u}_{i.} = \frac{1}{T}\sum_{t=1}^{T}u_{it}. \tag{26.15}$$

Subtracting (26.14) from (26.1) yields

$$y_{it} - \bar{y}_i = \boldsymbol{\beta}'(\mathbf{x}_{it} - \bar{\mathbf{x}}_i) + (u_{it} - \bar{u}_i), \tag{26.16}$$

which is known as FE, or within transformation. $\boldsymbol{\beta}$ is now estimated by applying the method of pooled *OLS* to the above transformed relations to obtain

$$\hat{\boldsymbol{\beta}}_{FE} = \left[\sum_{t=1}^{T}\sum_{i=1}^{N}(\mathbf{x}_{it} - \bar{\mathbf{x}}_i)'(\mathbf{x}_{it} - \bar{\mathbf{x}}_i)\right]^{-1}\sum_{t=1}^{T}\sum_{i=1}^{N}(\mathbf{x}_{it} - \bar{\mathbf{x}}_{i.})\left(y_{it} - \bar{y}_i\right). \tag{26.17}$$

The estimator for α_i can be recovered from (26.14). In particular,

$$\hat{\alpha}_i = \bar{y}_i - \hat{\boldsymbol{\beta}}'_{FE}\bar{\mathbf{x}}_i. \tag{26.18}$$

The transformed equation (26.16), and the FE estimator can also be rewritten in a more convenient form using the unit-specific stacked notation (26.2). In particular, let $\mathbf{M}_T = \mathbf{I}_T - \boldsymbol{\tau}_T(\boldsymbol{\tau}'_T\boldsymbol{\tau}_T)^{-1}\boldsymbol{\tau}'_T$. The matrix $\mathbf{M}_T$ is a $T \times T$ idempotent transformation matrix that converts variables in the form of deviations from their mean. Noting that $\mathbf{M}_T\tau_T = \boldsymbol{\tau}_T - \boldsymbol{\tau}_T(\boldsymbol{\tau}'_T\boldsymbol{\tau}_T)^{-1}\boldsymbol{\tau}'_T\boldsymbol{\tau}_T = \mathbf{0}$, and, pre-multiplying both sides of (26.2) by $\mathbf{M}_T$, we obtain

$$\mathbf{M}_T\mathbf{y}_{i.} = \mathbf{M}_T\mathbf{X}_{i.}\boldsymbol{\beta} + \mathbf{M}_T\mathbf{u}_{i.}. \tag{26.19}$$

Applying the *OLS* to equation (26.19) yields (assuming Assumption P3' holds)

$$\hat{\boldsymbol{\beta}}_{FE} = \left(\sum_{i=1}^{N}\mathbf{X}'_{i.}\mathbf{M}_T\mathbf{X}_{i.}\right)^{-1}\sum_{i=1}^{N}\mathbf{X}'_{i.}\mathbf{M}_T\mathbf{y}_{i.}, \tag{26.20}$$

which is identical to (26.17), and can be written more compactly as $\hat{\boldsymbol{\beta}}_{FE} = \mathbf{Q}^{-1}_{FE,NT}\mathbf{q}_{FE,NT}$, where

$$\mathbf{q}_{FE,NT} = (NT)^{-1}\sum_{i=1}^{N}\sum_{t=1}^{T}(\mathbf{x}_{it} - \bar{\mathbf{x}})(y_{it} - \bar{y}) = (NT)^{-1}\sum_{i=1}^{N}\mathbf{X}'_{i.}\mathbf{M}_T\mathbf{y}_{i.}. \tag{26.21}$$

It is now easily seen that, under the above assumptions, $\boldsymbol{\beta}_{FE}$ is unbiased and consistent for any fixed T and as $N \to \infty$. Substituting the expression for $\mathbf{y}_{i.}$ in (26.20), yields

$$\hat{\boldsymbol{\beta}}_{FE} - \boldsymbol{\beta} = \left(\frac{\sum_{i=1}^{N}\mathbf{X}'_{i.}\mathbf{M}_T\mathbf{X}_{i.}}{NT}\right)^{-1}\sum_{i=1}^{N}\frac{\mathbf{X}'_{i.}\mathbf{M}_T\mathbf{u}_{i.}}{NT},$$

and

$$E\left(\hat{\boldsymbol{\beta}}_{FE}\,|\mathbf{X}\right) - \boldsymbol{\beta} = \left(\frac{\sum_{i=1}^{N}\mathbf{X}'_{i.}\mathbf{M}_T\mathbf{X}_{i.}}{NT}\right)^{-1}\sum_{i=1}^{N}\frac{\mathbf{X}'_{i.}\mathbf{M}_T E\left(\mathbf{u}_{i.}\,|\mathbf{X}\right)}{NT}.$$

But under Assumption P1, $E\left(\mathbf{u}_{i.}\,|\mathbf{X}\right) = \mathbf{0}$, and it readily follows that $E\left(\hat{\boldsymbol{\beta}}_{FE}\,|\mathbf{X}\right) = \boldsymbol{\beta}$; and hence unconditionally we also have $E\left(\hat{\boldsymbol{\beta}}_{FE}\right) = \boldsymbol{\beta}$, which establishes that the *FE* estimator of $\boldsymbol{\beta}$ is unbiased under Assumptions P1 and P3'. Consider now the variance of $\hat{\boldsymbol{\beta}}_{FE}$ and note that

$$Var\left(\hat{\boldsymbol{\beta}}_{FE}\,|\mathbf{X}\right) = \frac{1}{NT}\mathbf{Q}^{-1}_{FE,NT}\mathbf{V}_{FE,NT}\mathbf{Q}^{-1}_{FE,NT}, \tag{26.22}$$

where

$$\mathbf{V}_{FE,NT} = \frac{1}{NT}\sum_{i=1}^{N}\sum_{j=1}^{N}\mathbf{X}_{i.}'\mathbf{M}_T E\left(\mathbf{u}_{i.}\mathbf{u}_{j.}'\,|\mathbf{X}\right)\mathbf{M}_T\mathbf{X}_{j.}. \tag{26.23}$$

But under Assumptions P4 and P5, $E\left(\mathbf{u}_{i.}\mathbf{u}_{j.}'\,|\mathbf{X}\right)=\mathbf{0}$, if $i\neq j$ and $E\left(\mathbf{u}_{i.}\mathbf{u}_{i.}\,|\mathbf{X}\right)=\mathbf{\Gamma}_i$ which is a $T\times T$ matrix with $\left(t,t'\right)$ element given by $\gamma_i(t,t')$, and $\gamma_i(t,t)=\sigma_i^2$, and we have

$$\begin{aligned}\mathbf{V}_{FE,NT} &= \frac{1}{N}\sum_{i=1}^{N}\left(\frac{\mathbf{X}_{i.}'\mathbf{M}_T\mathbf{\Gamma}_i\mathbf{M}_T\mathbf{X}_{i.}}{T}\right)\\ &= \frac{1}{NT}\sum_{i=1}^{N}\sum_{t=1}^{T}\sigma_i^2\left(\mathbf{x}_{it}-\bar{\mathbf{x}}_i\right)\left(\mathbf{x}_{it}-\bar{\mathbf{x}}_i\right)' + \frac{1}{NT}\sum_{i=1}^{N}\sum_{t\neq t'}^{T}\gamma_i(t,t')\left(\mathbf{x}_{it}-\bar{\mathbf{x}}_i\right)\left(\mathbf{x}_{it'}-\bar{\mathbf{x}}_i\right)'.\end{aligned}$$

Under Assumptions P2' and P3', $Var\left(\hat{\boldsymbol{\beta}}_{FE}\,|\mathbf{X}\right)\rightarrow\mathbf{0}$, when T is fixed and $N\rightarrow\infty$, which together with $E\left(\hat{\boldsymbol{\beta}}_{FE}\right)=\boldsymbol{\beta}$ establishes the consistency of $\hat{\boldsymbol{\beta}}_{FE}$. In the case where both N and $T\rightarrow\infty$, a sufficient condition for consistency of $\hat{\boldsymbol{\beta}}_{FE}$ is given by

$$\frac{1}{N^2T^2}\sum_{i=1}^{N}\sum_{t=1}^{T}\sum_{t'=1}^{T}\gamma_i^2(t,t')\rightarrow 0,$$

which is met since $\left|\gamma_i(t,t')\right|<K$. But if N is fixed as $T\rightarrow\infty$, then we need

$$T^{-2}\sum_{t=1}^{T}\sum_{t'=1}^{T}\gamma_i^2(t,t')\rightarrow 0, \text{ for each } i,$$

which is the usual time series ergodicity condition and is met if the $T\times T$ autocovariance matrix $\mathbf{\Gamma}_i=(\gamma_i(t,t'))$ has bounded absolute row (column) sum norm. This condition is met, for example, if u_{it} is a stationary process for all i (see Chapter 14).

The asymptotic distribution of $\hat{\boldsymbol{\beta}}_{FE}$ can also be obtained either assuming that the errors, u_{it}, are normally distributed when N and T are fixed, or satisfy certain distributional conditions when N or/and $T\rightarrow\infty$. In the case where u_{it} is normally distributed, under Assumptions P1', P2', P3', P4 and P5 and for any given N and T we have

$$\sqrt{NT}\left(\hat{\boldsymbol{\beta}}_{FE}-\boldsymbol{\beta}\right)\sim N(\mathbf{0},\mathbf{\Omega}_{FE,NT}),$$

where

$$\mathbf{\Omega}_{FE,NT}=\mathbf{Q}_{FE,NT}^{-1}\mathbf{V}_{FE,NT}\mathbf{Q}_{FE,NT}^{-1}.$$

A number of results in the literature can be derived. In the case where the errors are serially uncorrelated, $\mathbf{V}_{FE,NT}$ simplifies to

$$\mathbf{V}_{FE,NT} = \frac{1}{NT}\sum_{i=1}^{N}\sum_{t=1}^{T}\sigma_i^2\left(\mathbf{x}_{it}-\bar{\mathbf{x}}_i\right)\left(\mathbf{x}_{it}-\bar{\mathbf{x}}_i\right)'.$$

If it is further assumed that the errors are homoskedastic, so that $\sigma_i^2 = \sigma^2$, we then have

$$\mathbf{V}_{FE,NT} = \frac{\sigma^2}{NT}\sum_{i=1}^{N}\sum_{t=1}^{T}\left(\mathbf{x}_{it}-\bar{\mathbf{x}}_i\right)\left(\mathbf{x}_{it}-\bar{\mathbf{x}}_i\right)',$$

and $\mathbf{\Omega}_{FE,NT}$ reduces to the familiar variance formula given by

$$\mathbf{\Omega}_{FE,NT} = \sigma^2\mathbf{Q}_{FE,NT}^{-1}.$$

When T is fixed and $N \to \infty$, we have the following limiting distribution

$$\sqrt{NT}\left(\hat{\boldsymbol{\beta}}_{FE}-\boldsymbol{\beta}\right) \sim N(\mathbf{0}, \mathbf{\Omega}_{FE,T}),$$

where

$$\mathbf{\Omega}_{FE,T} = \mathbf{Q}_{FE,T}^{-1}\mathbf{V}_{FE,T}\mathbf{Q}_{FE,T}^{-1},$$

$\mathbf{Q}_{FE,T}$ is defined below by Assumption P3', and

$$\mathbf{V}_{FE,T} = \underset{N\to\infty}{Plim}\left[\frac{1}{N}\sum_{i=1}^{N}\left(\frac{\mathbf{X}_{i.}'\mathbf{M}_T\mathbf{\Gamma}_i\mathbf{M}_T\mathbf{X}_{i.}}{T}\right)\right].$$

An estimator of α_i can be obtained by replacing the expression for $\bar{y}_{i.}$ in (26.18)

$$\hat{\alpha}_i - \alpha_i = \bar{u}_i - \bar{\mathbf{x}}_i'\left(\hat{\boldsymbol{\beta}}_{FE}-\boldsymbol{\beta}\right), \tag{26.24}$$

where $\bar{u}_{i.} = T^{-1}(u_{i1}+u_{i2}+\ldots+u_{iT})$, and the variance of $\hat{\alpha}_i$ is given by

$$Var\left(\hat{\alpha}_i\right) = Var\left(\bar{u}_i\right) + \bar{\mathbf{x}}_i'Var\left(\hat{\boldsymbol{\beta}}_{FE}\right)\bar{\mathbf{x}}_i - 2Cov\left[\bar{u}_i, \bar{\mathbf{x}}_i'\left(\hat{\boldsymbol{\beta}}_{FE}-\boldsymbol{\beta}\right)\right].$$

But it is easily seen that under the above assumptions

$$Var\left(\bar{u}_{i.}\right) = \frac{1}{T^2}\sum_{t=1}^{T}\sum_{t'=1}^{T}\gamma_i^2(t,t'),$$

$$\bar{\mathbf{x}}_i'Var\left(\hat{\boldsymbol{\beta}}_{FE}\right)\bar{\mathbf{x}}_i = O\left(\frac{1}{NT}\right),$$

and

$$Cov\left[\bar{u}_i, \bar{\mathbf{x}}_{i.}'\left(\hat{\boldsymbol{\beta}}_{FE}-\boldsymbol{\beta}\right)|\mathbf{X}\right] = \mathbf{Q}_{FE,NT}^{-1}\left(\frac{\sum_{j=1}^{N}\mathbf{X}_{j.}'\mathbf{M}_T E(\mathbf{u}_{j.}\bar{u}_i)}{NT}\right)$$
$$= \mathbf{Q}_{FE,NT}^{-1}\left(\frac{\mathbf{X}_{i.}'\mathbf{M}_T E(\mathbf{u}_{i.}\bar{u}_i)}{NT}\right).$$

However, $E(\mathbf{u}_{i.}\bar{u}_{i.})$ is a $T \times 1$ vector with its tth element given by $T^{-1}\sum_{\tau=1}^{T}\gamma_i(t,\tau)$. Hence, the last two terms of $Var\left(\hat{\alpha}_i\right)$ vanish as $N \to \infty$. But when T is fixed the first term does not vanish as $N \to \infty$, even if it is assumed that the errors are serially uncorrelated. Therefore, in general, $\hat{\alpha}_i$ is consistent only if $T \to \infty$.

The above results show that the *FE* estimator is fairly robust to temporal dependence and cross-sectional heteroskedasticity. But it is important to note that the robustness of the *FE* estimator to possible correlations between α_i and $\mathbf{x}_{it}$ comes at a cost. Using the *FE* approach we can only estimate the effects of time varying regressors. The effects of non-time varying regressors (such as sex or race) will be unidentified under the within or the *FE* transformation. But with additional assumptions the time-invariant effects can be estimated using time averages of the residuals from fixed-effects regressions. For further details see Section 26.10.

Another important point to bear in mind is that the consistency of $\hat{\boldsymbol{\beta}}_{FE}$ crucially depends on the assumption of strict exogeneity of the explanatory variables. As we shall see in Chapter 27, in the presence of weakly exogenous regressors, since the time averages $\bar{\mathbf{x}}_{i.}$ in (26.15), contain the values of $\mathbf{x}_{it}$ at all time periods, the demeaning operation would introduce a correlation of order $O(T^{-1})$ between the regressors and the error term in the transformed equation (26.16) that renders $\hat{\boldsymbol{\beta}}_{FE}$ biased in small samples. Finally, the *FE* is often not fully efficient since it ignores variation across individuals in the sample (see Hausman and Taylor (1981)).

26.4.1 The relationship between *FE* and least squares dummy variable estimators

The *FE* estimator can also be computed by stacking all the observations for all the cross-sectional units and then adding dummy variables for each unit in the resultant stacked regression. Stacking the N regressions in (26.3) we have

$$\begin{pmatrix}\mathbf{y}_{1.}\\ \mathbf{y}_{2.}\\ \vdots\\ \mathbf{y}_{N.}\end{pmatrix} = \alpha_1\begin{pmatrix}\boldsymbol{\tau}_T\\ \mathbf{0}\\ \vdots\\ \mathbf{0}\end{pmatrix} + \alpha_2\begin{pmatrix}\mathbf{0}\\ \boldsymbol{\tau}_T\\ \vdots\\ \mathbf{0}\end{pmatrix} + \ldots + \alpha_N\begin{pmatrix}\mathbf{0}\\ \mathbf{0}\\ \vdots\\ \boldsymbol{\tau}_T\end{pmatrix}$$
$$+\begin{pmatrix}\mathbf{X}_{1.}\\ \mathbf{X}_{2.}\\ \vdots\\ \mathbf{X}_{N.}\end{pmatrix}\boldsymbol{\beta} + \begin{pmatrix}\mathbf{u}_{1.}\\ \mathbf{u}_{2.}\\ \vdots\\ \mathbf{u}_{N.}\end{pmatrix},$$

which can be written more compactly as

$$\mathbf{y} = \sum_{i=1}^{N} \alpha_i \mathbf{d}_i + \mathbf{X}\boldsymbol{\beta} + \mathbf{u},$$

where $\mathbf{d}_i$ is an $NT \times 1$ vector of a dummy variable with all its elements zero except for the elements associated with the ith cross-sectional unit which are set to unity. It is easily seen that the *OLS* estimator of $\boldsymbol{\beta}$ in this regression is the same as the *FE* estimator. For this reason the *FE* estimator is also known as the least squares dummy variable $(LSDV)$ estimator. However, when N is relatively large it is not computationally efficient as compared to using the pooled formula given by (26.17). Also, especial care needs to be exercised when the *LSDV* approach is used, since the standard errors obtained from such regressions are only valid under the strong assumptions that the errors are homoskedastic and serially uncorrelated. In general it is more appropriate to use (26.17) to compute the *FE* estimator and then compute robust standard errors as set out in Section 26.7.

26.4.2 Derivation of the *FE* estimator as a maximum likelihood estimator

Consider the unit-specific formulation (26.2), and assume that $\mathbf{u}_{i.} \sim N(\mathbf{0}, \boldsymbol{\Gamma}_i)$. Under cross-sectional independence of the errors, the pooled log-likelihood function is given by

$$\ell(\boldsymbol{\theta}) = \sum_{i=1}^{N} \ell_i(\boldsymbol{\theta}_i), \tag{26.25}$$

where $\ell_i(\boldsymbol{\theta}_i)$ is the log-likelihood for the ith unit

$$\ell_i(\boldsymbol{\theta}_i) = -\frac{T}{2}\log(2\pi) - \frac{1}{2}\log|\boldsymbol{\Gamma}_i| - \frac{1}{2}\left(\mathbf{y}_i - \alpha_i\boldsymbol{\tau}_T - \mathbf{X}_{i.}\boldsymbol{\beta}\right)'\boldsymbol{\Gamma}_i^{-1}\left(\mathbf{y}_{i.} - \alpha_i\boldsymbol{\tau}_T - \mathbf{X}_{i.}\boldsymbol{\beta}\right),$$

$\boldsymbol{\theta}_i = \left(\alpha_i, \boldsymbol{\beta}', vech(\boldsymbol{\Gamma}_i)'\right)'$, $\boldsymbol{\theta} = (\boldsymbol{\theta}_1', \boldsymbol{\theta}_2', \ldots, \boldsymbol{\theta}_N')'$. In the special case where $\boldsymbol{\Gamma}_i = \sigma^2\mathbf{I}_T$, it is easily seen that the maximum likelihood estimator for $\boldsymbol{\beta}$ and α_i obtained from the first-order conditions, by maximizing (26.25), is identical to the *FE* estimator for these parameters. However, the estimator for σ^2 does not have the appropriate correction for the degrees of freedom, since from the first-order conditions we obtain

$$\hat{\sigma}^2_{ML} = \frac{1}{NT}\sum_{i=1}^{N}\left(\mathbf{y}_{i.} - \mathbf{X}_{i.}\hat{\boldsymbol{\beta}}_{FE} - \hat{\alpha}_i\boldsymbol{\tau}_T\right)'\left(\mathbf{y}_{i.} - \mathbf{X}_{i.}\hat{\boldsymbol{\beta}}_{FE} - \hat{\alpha}_i\boldsymbol{\tau}_T\right).$$

$\hat{\sigma}^2_{ML}$ is not a consistent estimator of σ^2 when T is fixed and $N \to \infty$. This is due to the dependence of $\hat{\sigma}^2_{ML}$ on $\hat{\alpha}_i$, for $i = 1, 2, \ldots, N$, estimated for each i based on a finite sample of T observations, which is known as the incidental parameters problem discussed by Neyman and Scott (1948).

In the more general case where $\boldsymbol{\Gamma}_i \neq \sigma^2\mathbf{I}_T$, the *FE* and *ML* estimators differ and the latter is only feasible if T is sufficiently large.

26.5 Random effects specification

The random effects (RE) approach assumes that α_i are realizations from a probability distribution function with a fixed number of parameters, distributed independently of the regressors. More formally, under this specification it is assumed that:

Assumption $RE.1$: (a) $E(u_{it}|\mathbf{X}_{i.}, \alpha_i) = 0$. (b) $E(\alpha_i|\mathbf{X}_{i.}) = 0$, for all i and t.

Assumption $RE.2$: (a) $E(\mathbf{u}_{i.}\mathbf{u}_{i.}'|\mathbf{X}_{i.}, \alpha_i) = \sigma^2\mathbf{I}_T$. (b) $E\left(\alpha_i^2|\mathbf{X}_{i.}\right) = \sigma_\alpha^2$, for all i.

As in the FE specification, Assumption $RE.1$(a) implies strict exogeneity of explanatory variables conditional on the individual effects. Assumptions $RE.1$(b) and $RE.2$(b) imply that each group effect, α_i, is a random draw that enters in the regression identically in each time period, and is independent of the explanatory variables, $\mathbf{x}_{it}$, at all time periods.

Contrary to the FE approach, under the random effects formulation, inference pertains to the population from which the sample was randomly drawn.

26.5.1 *GLS* estimator

Let $\mathbf{v}_{i.} = (v_{i1}, v_{i2}, \ldots, v_{iT})'$, with $v_{it} = \alpha_i + u_{it}$. Under the above assumptions,

$$E\left(v_{it}^2\right) = \sigma_\alpha^2 + \sigma^2 + 2Cov\left(\alpha_i, u_{it}\right) = \sigma_\alpha^2 + \sigma^2,$$

and

$$E\left(v_{it}v_{is}\right) = E\left[\left(\alpha_i + u_{it}\right)\left(\alpha_i + u_{is}\right)\right] = \sigma_\alpha^2, \text{ for } t \neq s.$$

It follows that

$$\mathbf{\Sigma}_v = E\left(\mathbf{v}_{i.}\mathbf{v}_{i.}'\right) = \left(\sigma_\alpha^2 + \sigma^2\right)\begin{pmatrix} 1 & \rho & \cdots & \rho \\ \rho & 1 & \cdots & \rho \\ \vdots & \vdots & \ddots & \vdots \\ \rho & \rho & \cdots & 1 \end{pmatrix}, \tag{26.26}$$

where

$$\rho = \frac{\sigma_\alpha^2}{\sigma_\alpha^2 + \sigma^2}. \tag{26.27}$$

Note that the presence of the time-invariant effects, α_i, introduces equi-correlation among regression errors belonging to the same cross-sectional unit, although errors from different cross-sectional units are independent. It follows that the GLS estimator needs to be used to obtain an efficient estimator of $\boldsymbol{\beta}$, which is given by

$$\hat{\boldsymbol{\beta}}_{RE} = \left(\sum_{i=1}^{N}\mathbf{X}_{i.}'\mathbf{\Sigma}_v^{-1}\mathbf{X}_{i.}\right)^{-1}\sum_{i=1}^{N}\mathbf{X}_{i.}'\mathbf{\Sigma}_v^{-1}\mathbf{y}_{i.}. \tag{26.28}$$

We further assume that

Assumption *RE*.3**:** The matrix $(NT)^{-1}\sum_{i=1}^{N}\mathbf{X}_{i.}'\Sigma_v^{-1}\mathbf{X}_{i.}$ is nonsingular for all N, and T and as N and $T \to \infty$.

Under Assumptions *RE*.1–*RE*.3, $\hat{\boldsymbol{\beta}}_{RE}$ is consistent for $\boldsymbol{\beta}$ as N and/or $T \to \infty$. Under these assumptions it is also efficient with the variance

$$Var\left(\hat{\boldsymbol{\beta}}_{RE}\right) = \left(\sum_{i=1}^{N}\mathbf{X}_{i.}'\Sigma_v^{-1}\mathbf{X}_{i.}\right)^{-1}. \tag{26.29}$$

If the variance components σ^2 and σ_α^2 are unknown, a two-step procedure can be used to implement the *GLS*. In the first step, the variance components are estimated using some consistent estimators. In particular, the within-group residuals can be used to estimate σ^2 and σ_α^2

$$\hat{\sigma}^2 = \frac{1}{N(T-1)-k}\sum_{i=1}^{N}\left(\mathbf{y}_{i.} - \mathbf{X}_{i.}\hat{\boldsymbol{\beta}}_{FE}\right)'\mathbf{M}_T\left(\mathbf{y}_{i.} - \mathbf{X}_{i.}\hat{\boldsymbol{\beta}}_{FE}\right),$$

$$\hat{\sigma}_\alpha^2 = \frac{1}{N-k}\sum_{i=1}^{N}\left(\bar{y}_i - \hat{\boldsymbol{\beta}}_{FE}\bar{\mathbf{x}}_i\right)^2 - \frac{1}{T}\hat{\sigma}^2.$$

But care must be exercised since there is no guarantee that $\hat{\sigma}_\alpha^2 > 0$, when T is relatively small. An alternative estimator of σ_α^2 which is ensured to be positive is given by

$$\tilde{\sigma}_\alpha^2 = \frac{\sum_{i=1}^{N}\left(\hat{\alpha}_i - \overline{\hat{\alpha}}\right)^2}{N-1},$$

where $\hat{\alpha}_i$ is the least squares estimate α_i given by (26.24) and $\overline{\hat{\alpha}} = N^{-1}\sum_{i=1}^{N}\hat{\alpha}_i$. However, $\tilde{\sigma}_\alpha^2$ is a consistent estimator of σ_α^2 only if both N and T are large.

Further insights on the RE procedure can be obtained by replacing in (26.28) an explicit expression for the inverse Σ_v^{-1}. To this end, first note that

$$\Sigma_v = \sigma^2\mathbf{I}_T + \sigma_\alpha^2\boldsymbol{\tau}_T\boldsymbol{\tau}_T', \tag{26.30}$$

and, using the fact that $\boldsymbol{\tau}_T'\boldsymbol{\tau}_T = T$, we have

$$\begin{aligned}\Sigma_v &= \sigma^2\mathbf{I}_T + \sigma_\alpha^2 T\boldsymbol{\tau}_T\left(\boldsymbol{\tau}_T'\boldsymbol{\tau}_T\right)^{-1}\boldsymbol{\tau}_T',\\ &= \sigma^2\mathbf{I}_T + \sigma_\alpha^2 T\,\mathbf{P}_T,\\ &= \sigma^2\left(\mathbf{M}_T + \frac{1}{\psi}\mathbf{P}_T\right),\end{aligned}$$

where $\mathbf{P}_T = \mathbf{I}_T - \mathbf{M}_T = \boldsymbol{\tau}_T\left(\boldsymbol{\tau}_T'\boldsymbol{\tau}_T\right)^{-1}\boldsymbol{\tau}_T'$, and

$$\psi = \frac{\sigma^2}{T\sigma_\alpha^2 + \sigma^2} = \frac{1-\rho}{1-\rho+\rho T}, \tag{26.31}$$

where $\rho = \sigma_\alpha^2/(\sigma^2 + \sigma_\alpha^2) = \sigma_\alpha^2/\sigma_\nu^2$, with $0 \leq \psi \leq 1$, and $0 \leq \rho \leq 1$. Noting that $\mathbf{P}_T\mathbf{M}_T = \mathbf{M}_T\mathbf{P}_T = \mathbf{0}$, we have

$$\Sigma_\nu^{-1} = \frac{1}{\sigma^2}\left(\mathbf{M}_T + \psi\mathbf{P}_T\right).$$

Substituting the above expression in (26.28), and making use of the formula for partitioned inverses, it is possible to show that

$$\begin{aligned}\hat{\boldsymbol{\beta}}_{RE} = &\left[\frac{1}{NT}\sum_{i=1}^{N}\mathbf{X}_{i.}'\mathbf{M}\mathbf{X}_{i.} + \frac{\psi}{N}\sum_{i=1}^{N}\left(\bar{\mathbf{x}}_{i.}-\bar{\mathbf{x}}\right)\left(\bar{\mathbf{x}}_{i.}-\bar{\mathbf{x}}\right)'\right]^{-1} \\ &\times\left[\frac{1}{NT}\sum_{i=1}^{N}\mathbf{X}_{i.}'\mathbf{M}\mathbf{y}_{i.} + \frac{\psi}{N}\sum_{i=1}^{N}\left(\bar{\mathbf{x}}_{i.}-\bar{\mathbf{x}}\right)\left(\bar{y}_{i.}-\bar{y}\right)\right].\end{aligned} \tag{26.32}$$

Similarly, the variance of $\hat{\boldsymbol{\beta}}_{RE}$ is given by

$$Var\left(\hat{\boldsymbol{\beta}}_{RE}\right) = \frac{\sigma^2}{NT}\left[N^{-1}\sum_{i=1}^{N}\left(\frac{\mathbf{X}_{i.}'\mathbf{M}\mathbf{X}_{i.}}{T}\right) + \psi N^{-1}\sum_{i=1}^{N}\left(\bar{\mathbf{x}}_{i.}-\bar{\mathbf{x}}\right)\left(\bar{\mathbf{x}}_{i.}-\bar{\mathbf{x}}\right)'\right]^{-1}. \tag{26.33}$$

Because $\psi > 0$, it follows from (26.33) that the difference between the covariance matrices of $\hat{\boldsymbol{\beta}}_{FE}$ and $\hat{\boldsymbol{\beta}}_{RE}$ is a positive semi-definite matrix. Namely, under RE specification the RE estimator is more efficient than the FE estimator.

Further insights into the RE procedure can be obtained by noting that, from (26.30),

$$\Sigma_\nu^{-1/2} = \frac{1}{\sigma^2}\left(\mathbf{I}_T - \phi\mathbf{P}_T\right),$$

where $\phi = 1-\psi^{1/2}$. Hence, the RE estimator is obtained by applying the pooled *OLS* estimator to the transformed equation

$$\mathbf{y}_{i.}^0 = \mathbf{X}_{i.}^0\boldsymbol{\beta} + \mathbf{u}_{i.}^0, \tag{26.34}$$

where

$$\mathbf{y}_{i.}^0 = \left(\mathbf{I}_T - \phi\mathbf{P}_T\right)\mathbf{y}_{i.} = \mathbf{y}_{i.} - \phi\bar{\mathbf{y}}_{i.}, \qquad \mathbf{X}_{i.}^0 = \left(\mathbf{I}_T - \phi\mathbf{P}_T\right)\mathbf{X}_{i.} = \mathbf{X}_{i.} - \phi\bar{\mathbf{X}}_{i.}.$$

Note that errors in (26.34) are serially uncorrelated, and hence the pooled *OLS* in this model is efficient. Seen from this perspective, the RE estimator is obtained by a *quasi-time demeaning* (or *quasi-differencing*) data: rather than removing the time average from the explanatory and dependent variables at each t as in the FE approach, the RE approach removes a fraction of the time

average. If ϕ is close to 1 (which implies ψ close to 0), the random effects and fixed-effects estimates tend to be close.

The above discussion also shows that, similar to FE estimation, the consistency of the RE estimator crucially depends on the assumption of strict exogeneity of the explanatory variables. In the presence of weakly exogenous regressors, the transformation (26.34) to eliminate the individual effects would render the transformed regressors, $\mathbf{X}_{i.}^{0}$, correlated with the new error term, $\mathbf{u}_{i.}^{0}$, thus inducing a small sample bias in the $\hat{\boldsymbol{\beta}}_{RE}$.

There are a number of advantages in using an RE specification. First, it allows the derivation of efficient estimators which, as seen above, make use of both within- and between-group variations. Further, contrary to the FE specification, with an RE specification it is possible to estimate the impact of time-invariant variables. However, the disadvantage is that one has to specify a conditional density of α_i given $\mathbf{X}_{i.}$, which needs to be independent of the explanatory variables. If such an independence assumption does not hold, then the RE estimator would be inconsistent. For further discussion, see Mundlak (1978).

26.5.2 Maximum likelihood estimation of the random effects model

To implement the RE estimation, a suitable estimate of ψ or ρ is required. This is accomplished by the *ML* approach, assuming that $\alpha_i \sim IIDN\left(0, \sigma_\alpha^2\right)$, and $u_{it} \sim IIDN\left(0, \sigma^2\right)$, which yields $\mathbf{v}_i \sim N\left(0, \Sigma_v\right)$, where Σ_v is given by (26.30), which we now write as

$$\Sigma_v = \sigma_v^2 \left[(1-\rho)\,\mathbf{I}_T + \rho \boldsymbol{\tau}_T \boldsymbol{\tau}_T'\right], \tag{26.35}$$

where, as before, $\rho = \sigma_\alpha^2/\sigma_v^2$, and $\sigma_v^2 = \sigma^2 + \sigma_\alpha^2$. Under Pooled *OLS*, we have $\alpha_i = \alpha$ and $\sigma_\alpha^2 = 0$, and thus $\rho = 0$. Under the fixed-effects model, $\rho = 1$. In the case of the random effects model, $0 < \rho < 1$. It is now easily established that

$$\left[(1-\rho)\,\mathbf{I}_T + \rho \boldsymbol{\tau}_T \boldsymbol{\tau}_T'\right]^{-1} = \mathbf{S}'\mathbf{S},$$

where

$$\mathbf{S} = \left(\mathbf{I}_T - \phi \frac{\boldsymbol{\tau}_T \boldsymbol{\tau}_T'}{\boldsymbol{\tau}_T' \boldsymbol{\tau}_T}\right) \frac{1}{\sqrt{1-\rho}},$$

and as before, $\phi = 1 - \sqrt{\frac{1-\rho}{1-\rho+\rho T}}$. Under the cross-sectional independence of the errors we obtain the following log-likelihood function for the RE model

$$\begin{aligned}\ell(\boldsymbol{\theta}) = &-\frac{TN}{2}\log\left(2\pi\sigma_v^2\right) - \frac{TN}{2}\log(1-\rho) \\ &- \frac{1}{2\sigma_v^2}\sum_{i=1}^{N}\left(\mathbf{S}\mathbf{y}_i - \alpha\mathbf{S}\boldsymbol{\tau}_T - \mathbf{S}\mathbf{X}_i\boldsymbol{\beta}\right)'\left(\mathbf{S}\mathbf{y}_i - \alpha\mathbf{S}\boldsymbol{\tau}_T - \mathbf{S}\mathbf{X}_i\boldsymbol{\beta}\right),\end{aligned}$$

where, $\boldsymbol{\theta} = (\sigma_v^2, \rho, \alpha, \boldsymbol{\beta}')'$, $\mathbf{S}\mathbf{y}_i = \tilde{\mathbf{y}}_i = \mathbf{y}_i - \phi\bar{\mathbf{y}}_i$, $\mathbf{S}\mathbf{X}_i = \tilde{\mathbf{X}}_i = \mathbf{X}_i - \phi\bar{\mathbf{X}}_i$, and $\mathbf{S}\boldsymbol{\tau}_T = \frac{1-\phi}{\sqrt{1-\rho}}\boldsymbol{\tau}_T = \frac{1}{\sqrt{1-\rho+\rho T}}\boldsymbol{\tau}_T$. Since α is unrestricted, the above log-likelihood function can be written equivalently as ($\tilde{\alpha} = \alpha/\sqrt{1-\rho+\rho T}$)

$$\ell(\boldsymbol{\theta}) = -\frac{TN}{2}\log\left(2\pi\sigma_\nu^2\right) - \frac{TN}{2}\log(1-\rho) \tag{26.36}$$
$$- \frac{1}{2\sigma_\nu^2}\sum_{i=1}^{N}\left(\tilde{\mathbf{y}}_i - \tilde{\alpha}\boldsymbol{\tau}_T - \tilde{\mathbf{X}}_i\boldsymbol{\beta}\right)'\left(\tilde{\mathbf{y}}_i - \tilde{\alpha}\boldsymbol{\tau}_T - \tilde{\mathbf{X}}_i\boldsymbol{\beta}\right).$$

Also

$$\sum_{i=1}^{N}\left(\tilde{\mathbf{y}}_i - \tilde{\alpha}\boldsymbol{\tau}_T - \tilde{\mathbf{X}}_i\boldsymbol{\beta}\right)'\left(\tilde{\mathbf{y}}_i - \tilde{\alpha}\boldsymbol{\tau}_T - \tilde{\mathbf{X}}_i\boldsymbol{\beta}\right)$$
$$= \sum_{i=1}^{N}\left(\tilde{\mathbf{y}}_i - \tilde{\mathbf{X}}_i\boldsymbol{\beta}\right)'\left(\tilde{\mathbf{y}}_i - \tilde{\mathbf{X}}_i\boldsymbol{\beta}\right) + NT\tilde{\alpha}^2 - 2\tilde{\alpha}\sum_{i=1}^{N}\boldsymbol{\tau}_T'\left(\tilde{\mathbf{y}}_i - \tilde{\mathbf{X}}_i\boldsymbol{\beta}\right).$$

Hence for a given ρ, the *ML* estimators of $\tilde{\alpha}$, $\boldsymbol{\beta}$, and σ_ν^2 are given by

$$\widehat{\tilde{\alpha}}(\rho) = N^{-1}T^{-1}\sum_{i=1}^{N}\boldsymbol{\tau}_T'\left(\tilde{\mathbf{y}}_i - \tilde{\mathbf{X}}_i\hat{\boldsymbol{\beta}}(\rho)\right)$$
$$\hat{\boldsymbol{\beta}}(\rho) = \left(\sum_{i=1}^{N}\tilde{\mathbf{X}}_i'\tilde{\mathbf{X}}_i\right)^{-1}\left(\sum_{i=1}^{N}\tilde{\mathbf{X}}_i'\tilde{\mathbf{y}}_i - \widehat{\tilde{\alpha}}(\rho)\sum_{i=1}^{N}\tilde{\mathbf{X}}_i'\boldsymbol{\tau}_T\right),$$

and

$$\hat{\sigma}_\nu^2(\rho) = \frac{1}{NT}\sum_{i=1}^{N}\left[\tilde{\mathbf{y}}_i - \widehat{\tilde{\alpha}}(\rho)\boldsymbol{\tau}_T - \tilde{\mathbf{X}}_i\hat{\boldsymbol{\beta}}(\rho)\right]'\left[\tilde{\mathbf{y}}_i - \widehat{\tilde{\alpha}}(\rho)\boldsymbol{\tau}_T - \tilde{\mathbf{X}}_i\hat{\boldsymbol{\beta}}(\rho)\right].$$

These *ML* estimators can be substituted back into the log-likelihood function to obtain a concentrated log-likelihood function in terms of ρ. The concentrated log-likelihood function can then be maximized using grid search techniques which can be readily implemented, considering that ρ must lie in the region $0 \leq \rho < 1$. Plotting the profile function of the concentrated log-likelihood function also allows us to check for multiple or local maxima. See Maddala (1971) and Hsiao (2003) for further details.

26.6 Cross-sectional Regression: the between-group estimator of $\boldsymbol{\beta}$

In the case of the random effects model, $\boldsymbol{\beta}$ can also be estimated using a pure cross-sectional regression. Under the random effects specification we have

$$\alpha_i = \alpha + \eta_i, \qquad \eta_i \sim IID\left(0, \sigma_\eta^2\right). \tag{26.37}$$

(Note that $\sigma_\eta^2 = \sigma_\alpha^2$.) A single-period cross-sectional regression is then defined by

$$y_{it} = \alpha + \boldsymbol{\beta}'\mathbf{x}_{it} + v_{it}, \quad i = 1, 2, \ldots, N, \tag{26.38}$$

for a given choice of t, and

$$v_{it} = u_{it} + \eta_i. \tag{26.39}$$

Alternatively, the cross-sectional regression could be based on time averages of y_{it} and $\mathbf{x}_{it}$, for example

$$\bar{y}_i = \alpha + \boldsymbol{\beta}'\bar{\mathbf{x}}_i + \bar{v}_i, \tag{26.40}$$

where as before $\bar{y}_i = T^{-1}\sum_{t=1}^{T} y_{it}$, $\bar{\mathbf{x}}_i = T^{-1}\sum_{t=1}^{T}\mathbf{x}_{it}$, and $\bar{v}_i = \bar{u}_i + \eta_i$. Running the regression of $\bar{y}_i$ on $\bar{\mathbf{x}}_{i\cdot}$ defined by (26.40), we obtain the cross-sectional estimator of $\boldsymbol{\beta}$, which we denote by $\hat{\boldsymbol{\beta}}_b$, namely

$$\hat{\boldsymbol{\beta}}_b = \left[N^{-1}\sum_{i=1}^{N}(\bar{\mathbf{x}}_{i\cdot} - \bar{\mathbf{x}})(\bar{\mathbf{x}}_{i\cdot} - \bar{\mathbf{x}})'\right]^{-1}\left[N^{-1}\sum_{i=1}^{N}(\bar{\mathbf{x}}_{i\cdot} - \bar{\mathbf{x}})(\bar{y}_i - \bar{y})\right]. \tag{26.41}$$

$\hat{\boldsymbol{\beta}}_b$ is also known as the *between* estimator since it only exploits variation between groups, while ignoring the variability of observations within groups. For future reference we also note that

$$\hat{\boldsymbol{\beta}}_b = \mathbf{Q}_{b,NT}^{-1}\mathbf{q}_{b,NT},$$

where

$$\mathbf{Q}_{b,NT} = N^{-1}\sum_{i=1}^{N}(\bar{\mathbf{x}}_{i\cdot} - \bar{\mathbf{x}})(\bar{\mathbf{x}}_{i\cdot} - \bar{\mathbf{x}})', \tag{26.42}$$

and

$$\mathbf{q}_{b,NT} = N^{-1}\sum_{i=1}^{N}(\bar{\mathbf{x}}_{i\cdot} - \bar{\mathbf{x}})(\bar{y}_i - \bar{y}). \tag{26.43}$$

To obtain the variance of the between estimator, since $\bar{y}_i - \bar{y} = (\bar{\mathbf{x}}_i - \bar{\mathbf{x}})'\boldsymbol{\beta} + (\bar{v}_i - \bar{v})$, we note that

$$\hat{\boldsymbol{\beta}}_b - \boldsymbol{\beta} = \mathbf{Q}_{b,NT}^{-1}\left[N^{-1}\sum_{i=1}^{N}(\bar{\mathbf{x}}_i - \bar{\mathbf{x}})\bar{v}_i\right].$$

Therefore, under the assumptions of the RE model we have

$$Var(\hat{\boldsymbol{\beta}}_b) = N^{-1}\left(\sigma_\alpha^2 + \frac{\sigma^2}{T}\right)\mathbf{Q}_{b,NT}^{-1}.$$

Unlike the RE estimator which is consistent in terms of both N and T, the between estimator is consistent only if $N \to \infty$, which is not surprising since it ignores the variability along the time dimension.

26.6.1 Relation between pooled *OLS* and *RE* estimators

We begin with a comparison of pooled *OLS* and *RE* estimators. Although pooled *OLS* is estimated under the assumption that $\alpha_i = \alpha$ for all i, it can also be rationalized under the random effects specification. To see this, note that the *RE* model can be written as

$$y_{it} = \alpha + \boldsymbol{\beta}' \mathbf{x}_{it} + v_{it},$$

where $v_{it} = u_{it} + \eta_i$. Since under *RE* specification $E(v_{it} | \mathbf{X}) = 0$, then Assumption P1 of the pooled *OLS* estimator is satisfied. Also Assumptions P2 and P3 are satisfied under the RE model. Furthermore, Assumption P5 clearly applies to v_{it}, as they allow for the errors of the pooled *OLS* regression to be serially correlated. Finally, Assumption P4, the cross-sectional independence of the errors, is assumed to hold for both pooled *OLS* and *RE* estimators. Therefore, pooled *OLS* continues to be consistent under *RE* specification, although it will be inefficient under the *RE* specification that maintains u_{it} to be serially uncorrelated and homoskedastic. But these assumptions are likely to be quite restrictive in practice, and pooled *OLS* with robust standard errors might be preferable. For estimation of robust standard errors for the pooled *OLS* estimator in the presence of general forms of residual serial correlation and cross-sectional heteroskedasticity see Section 26.7.

26.6.2 Relation between *FE*, *RE*, and between (cross-sectional) estimators

Using (26.8), (26.13), and (26.42) we first note that

$$\mathbf{Q}_{p,NT} = \mathbf{Q}_{FE,NT} + \mathbf{Q}_{b,NT}, \tag{26.44}$$

namely the total variations of the regressors in the case of the pooled *OLS* decomposes into the total variations in the case of within (FE) and between estimators. Also let

$$\mathbf{q}_{p,NT} = (NT)^{-1} \sum_{i=1}^{N} \sum_{t=1}^{T} (\mathbf{x}_{it} - \bar{\mathbf{x}})(y_{it} - \bar{y}),$$

and note that

$$\mathbf{q}_{p,NT} = \mathbf{q}_{FE,NT} + \mathbf{q}_{b,NT}, \tag{26.45}$$

where $\mathbf{q}_{FE,NT}$, and $\mathbf{q}_{b,NT}$ are defined by (26.21) and (26.43), respectively. Using (26.32) and the above notations, the *RE* estimator, $\hat{\boldsymbol{\beta}}_{RE}$, can be rewritten as

$$\hat{\boldsymbol{\beta}}_{RE} = \left(\mathbf{Q}_{FE,NT} + \psi \mathbf{Q}_{b,NT}\right)^{-1} \left(\mathbf{q}_{FE,NT} + \psi \mathbf{q}_{b,NT}\right).$$

Also, since $\mathbf{Q}_{FE,NT}\hat{\boldsymbol{\beta}}_{FE} = \mathbf{q}_{FE,NT}$ and $\mathbf{Q}_{b,NT}\hat{\boldsymbol{\beta}}_{b} = \mathbf{q}_{b,NT}$, then

$$\hat{\boldsymbol{\beta}}_{RE} = \left(\mathbf{Q}_{FE,NT} + \psi\mathbf{Q}_{b,NT}\right)^{-1}\left(\mathbf{Q}_{FE,NT}\hat{\boldsymbol{\beta}}_{FE} + \psi\mathbf{Q}_{b,NT}\hat{\boldsymbol{\beta}}_{b}\right),$$

and upon using (26.44) we have

$$\hat{\boldsymbol{\beta}}_{RE} = \mathbf{W}\hat{\boldsymbol{\beta}}_{b} + (\mathbf{I}_k - \mathbf{W})\,\hat{\boldsymbol{\beta}}_{FE}, \tag{26.46}$$

where

$$\mathbf{W} = \psi\left(\mathbf{Q}_{FE,NT} + \psi\mathbf{Q}_{b,NT}\right)^{-1}\mathbf{Q}_{b,NT}.$$

Expression (26.46) shows that $\hat{\boldsymbol{\beta}}_{RE}$ is a weighted average of the between-group and within-group estimators. If $\psi \to 0$, the RE estimator becomes the FE estimator, while for $\psi \to 1$, it is easy to see from (26.46) that $\hat{\boldsymbol{\beta}}_{RE}$ converges to the *OLS* estimator. The parameter ψ measures the degree of heterogeneity in the intercept; under the pooled *OLS*, we have $\alpha_i = \alpha$ and $\sigma_\alpha^2 = 0$, and thus $\psi = 1$; under the fixed-effects hypothesis, the case of maximum heterogeneity, $\psi = 0$. It also follows from (26.31) that as $T \to \infty$, then $\psi \to 0$ and RE and FE estimators tend to the same value.

26.6.3 Fixed-effects versus random effects

When T is large, whether to treat the group effects as fixed or random makes no difference, because, as seen above, the FE and the RE estimators become identical. When T is finite and N is large, whether to use a fixed-effects or a random effects specification depends on a number of factors, such as the context of the data, the way in which the data were gathered, and the purposes of the analysis. For instance, suppose we are interested in studying the consumption behaviour of a group of people. If an experiment involves hundreds of individuals who are considered a random sample from some larger population, then random effects are more appropriate. Conversely, a fixed-effects specification would be more appropriate if we want to assess differences between specific individuals. As pointed out by Mundlak (1978), the key issue to take into consideration is the degree to which individual effects, α_i, are likely to be correlated to the regressors, $\mathbf{x}_{it}$.

26.7 Estimation of the *variance* of pooled *OLS*, *FE*, and *RE* estimators of β robust to heteroskedasticity and serial correlation

In the case of serially correlated and cross-sectionally heteroskedastic errors, the standard variance formulae that assume serially uncorrelated and homoskedastic errors will be inappropriate and their use can result in spurious inference. In a simulation study, Bertrand, Duflo, and Mullainathan (2004) found that panel data inference procedures which fail to account for within individual serial correlation may be severely size distorted. Arellano (1987) suggests a simple method for obtaining robust estimates of the standard errors for the FE estimator, that allow for

a general covariance matrix of the u_{it}, as in White (1980). In particular, the robust asymptotic covariance matrix of $\hat{\boldsymbol{\beta}}_{FE}$, also known as the 'clustered' covariance matrix (CCM) estimator, is given by

$$\widehat{Var}\left(\hat{\boldsymbol{\beta}}_{FE}\right) = \frac{1}{NT}\mathbf{Q}_{FE,NT}^{-1}\left(\frac{1}{NT}\sum_{i=1}^{N}\mathbf{X}_{i.}'\mathbf{M}_T\hat{\mathbf{u}}_{i.}^{*}\hat{\mathbf{u}}_{i.}^{*\prime}\mathbf{M}_T\mathbf{X}_{i.}\right)\mathbf{Q}_{FE,NT}^{-1},$$

where $\hat{\mathbf{u}}_{i.}^{*} = \mathbf{M}_T\left(\mathbf{y}_{i.} - \mathbf{X}_{i.}\hat{\boldsymbol{\beta}}_{FE}\right)$. To show that $\widehat{Var}\left(\hat{\boldsymbol{\beta}}_{FE}\right)$ is an appropriate estimator of $Var\left(\hat{\boldsymbol{\beta}}_{FE}\right)$, defined by (26.22), we need to establish that

$$\lim_{N,T\to\infty}\frac{1}{NT}\sum_{i=1}^{N}\mathbf{X}_{i.}'\mathbf{M}_T E\left(\hat{\mathbf{u}}_{i.}^{*}\hat{\mathbf{u}}_{i.}^{*\prime}\,|\mathbf{X}\right)\mathbf{M}_T\mathbf{X}_{i.} = \lim_{N,T\to\infty}\frac{1}{NT}\sum_{i=1}^{N}\mathbf{X}_{i.}'\mathbf{M}_T E\left(\mathbf{u}_{i.}\mathbf{u}_{i.}'\,|\mathbf{X}\right)\mathbf{M}_T\mathbf{X}_{i.}.$$

To this end, note that

$$\hat{\mathbf{u}}_{i.}^{*} = \mathbf{M}_T\left(\mathbf{y}_{i.} - \mathbf{X}_{i.}\hat{\boldsymbol{\beta}}_{FE}\right) = \mathbf{M}_T\left[\mathbf{u}_{i.} - \mathbf{X}_i\left(\hat{\boldsymbol{\beta}}_{FE} - \boldsymbol{\beta}\right)\right].$$

and hence (recalling that $\mathbf{M}_T\mathbf{u}_{i.}$ and $\mathbf{X}_i\left(\hat{\boldsymbol{\beta}}_{FE} - \boldsymbol{\beta}\right)$ are uncorrelated)

$$\begin{aligned}E\left(\hat{\mathbf{u}}_{i.}^{*}\hat{\mathbf{u}}_{i.}^{*\prime}\,|\mathbf{X}\right) &= \mathbf{M}_T E\left(\mathbf{u}_{i.}\mathbf{u}_{i.}'\,|\mathbf{X}\right)\mathbf{M}_T \\ &\quad + \mathbf{M}_T\mathbf{X}_{i.}E\left[\left(\hat{\boldsymbol{\beta}}_{FE} - \boldsymbol{\beta}\right)\left(\hat{\boldsymbol{\beta}}_{FE} - \boldsymbol{\beta}\right)'\,|\mathbf{X}\right]\mathbf{X}_i'\mathbf{M}_T.\end{aligned}$$

Using this result we now have

$$\begin{aligned}&\frac{1}{NT}\sum_{i=1}^{N}\mathbf{X}_{i.}'\mathbf{M}_T E\left(\hat{\mathbf{u}}_{i.}^{*}\hat{\mathbf{u}}_{i.}^{*\prime}\,|\mathbf{X}\right)\mathbf{M}_T\mathbf{X}_{i.} \\ &= \frac{1}{NT}\sum_{i=1}^{N}\mathbf{X}_{i.}'\mathbf{M}_T E\left(\mathbf{u}_{i.}\mathbf{u}_{i.}'\,|\mathbf{X}\right)\mathbf{M}_T\mathbf{X}_{i.} \\ &\quad + \frac{T}{N}\sum_{i=1}^{N}\left(\frac{\mathbf{X}_{i.}'\mathbf{M}_T\mathbf{X}_{i.}}{T}\right)E\left[\left(\hat{\boldsymbol{\beta}}_{FE} - \boldsymbol{\beta}\right)\left(\hat{\boldsymbol{\beta}}_{FE} - \boldsymbol{\beta}\right)'\,|\mathbf{X}\right]\left(\frac{\mathbf{X}_i'\mathbf{M}_T\mathbf{X}_{i.}}{T}\right).\end{aligned}$$

Consider now the relevant case where T is fixed as $N \to \infty$. In this case we have already established that

$$E\left[\left(\hat{\boldsymbol{\beta}}_{FE} - \boldsymbol{\beta}\right)\left(\hat{\boldsymbol{\beta}}_{FE} - \boldsymbol{\beta}\right)'\,|\mathbf{X}\right] = Var\left(\hat{\boldsymbol{\beta}}_{FE}\right) = O\left(\frac{1}{N}\right),$$

and hence we have

$$\frac{1}{NT}\sum_{i=1}^{N}\mathbf{X}_{i.}'\mathbf{M}_T E\left(\hat{\mathbf{u}}_{i.}^{*}\hat{\mathbf{u}}_{i.}^{*\prime}\,|\mathbf{X}\right)\mathbf{M}_T\mathbf{X}_{i.}=\frac{1}{NT}\sum_{i=1}^{N}\mathbf{X}_{i.}'\mathbf{M}_T E\left(\mathbf{u}_{i.}\mathbf{u}_{i.}'\,|\mathbf{X}\right)\mathbf{M}_T\mathbf{X}_{i.}+O\left(N^{-1}\right).$$

Namely, for a fixed T

$$\lim_{N\to\infty}E\left[\widehat{Var}\left(\hat{\boldsymbol{\beta}}_{FE}\right)|\mathbf{X}\right]=Var\left(\hat{\boldsymbol{\beta}}_{FE}\right),$$

as desired. But when T is also large we either need to restrict the degree of error serial correlations or assume that $T/N\to 0$, as N and $T\to\infty$, to obtain the consistency of the variance estimator. Clearly $\widehat{Var}\left(\hat{\boldsymbol{\beta}}_{FE}\right)$ is not a consistent estimator if T is large and N is fixed. See also Hansen (2007). In a simulation study, Kezdi (2004) showed that $\widehat{Var}\left(\hat{\boldsymbol{\beta}}_{FE}\right)$ behaves well in finite samples, when N is large and T is fixed.

Similar arguments can also be applied to obtain a consistent estimator of the variance of the pooled *OLS* given by (26.10). Let

$$\hat{u}_{it,OLS}=y_{it}-\bar{y}-\left(\mathbf{x}_{it}-\bar{\mathbf{x}}\right)'\hat{\boldsymbol{\beta}}_{OLS},$$

then an asymptotically unbiased estimator of $Var\left(\hat{\boldsymbol{\beta}}_{OLS}\right)$ is given by

$$\widehat{Var}(\hat{\boldsymbol{\beta}}_{OLS})=\left(\sum_{i=1}^{N}\tilde{\mathbf{X}}_{i.}'\tilde{\mathbf{X}}_{i.}\right)^{-1}\left(\sum_{i=1}^{N}\tilde{\mathbf{X}}_{i.}'\hat{\mathbf{u}}_{i,OLS}\hat{\mathbf{u}}_{i,OLS}'\tilde{\mathbf{X}}_{i.}\right)\left(\sum_{i=1}^{N}\tilde{\mathbf{X}}_{i.}'\tilde{\mathbf{X}}_{i.}\right)^{-1},$$

where $\tilde{\mathbf{X}}_{i.}'=(\mathbf{x}_{i1}-\bar{\mathbf{x}},\mathbf{x}_{i2}-\bar{\mathbf{x}},\ldots,\mathbf{x}_{iT}-\bar{\mathbf{x}})$, and $\hat{\mathbf{u}}_{i,OLS}=\left(\hat{u}_{i1,OLS},\hat{u}_{i2,OLS},\ldots,\hat{u}_{iT,OLS}\right)'$.

In the case of the RE specification, a feasible estimator that allows for an arbitrary error covariance matrix is given by

$$\widehat{Var}\left(\hat{\boldsymbol{\beta}}_{RE}\right)=\left(\sum_{i=1}^{N}\mathbf{X}_{i.}'\hat{\Sigma}_{\nu}^{-1}\mathbf{X}_{i.}\right)^{-1}\left(\sum_{i=1}^{N}\mathbf{X}_{i.}'\hat{\Sigma}_{\nu}^{-1}\hat{\mathbf{v}}_{i.}\hat{\mathbf{v}}_{i.}'\hat{\Sigma}_{\nu}^{-1}\mathbf{X}_{i.}\right)\left(\sum_{i=1}^{N}\mathbf{X}_{i.}'\hat{\Sigma}_{\nu}^{-1}\mathbf{X}_{i.}\right)^{-1},$$

where $\hat{\mathbf{v}}_{i.}=\mathbf{y}_{i.}-\mathbf{X}_{i.}\hat{\boldsymbol{\beta}}_{RE}$.

Example 56 (Agricultural production) *Suppose the production of an agricultural product of farm i at time t, y_{it}, (in logs) follows the Cobb–Douglas production function,*

$$y_{it}=m_i+\beta_1 l_{it}+\beta_2 k_{it}+u_{it},$$

where l_{it} and k_{it} are the logarithm of labour and capital inputs, respectively, and m_i is an input that represents the effect of a set of unobserved inputs such as quality of the soil, or the location of the land. It is realistic to assume that m_i remains constant over time (over a short time period), and that it is known by the farmer, although not observed by the econometrician. If the farmer maximizes his expected profits, then he will choose the observed inputs in $\mathbf{x}_{it}=(l_{it},k_{it})'$, in the light of m_i. Hence, there will be a correlation between the observed and unobserved inputs that renders pooled OLS,

and the RE estimators inconsistent. To avoid inconsistent estimates FE estimates can be used. A similar problem has been considered in Mundlak (1961), who identifies m_i with the effect of an unobserved 'management' activity that influences observed inputs. Mundlak (1961) also suggests how to measure the 'management bias' by comparing the FE regression with the OLS regression from a pooled regression without farm fixed-effects.

Example 57 (Grunfeld's investment equation II) *Following from Examples 37 and 38, we now use data from the study by Grunfeld (1960) and Grunfeld and Griliches (1960) on eleven firms in the US economy over the period 1935–1954. Consider*[2]

$$I_{it} = \alpha_i + \beta_1 F_{it} + \beta_2 C_{it} + u_{it}, i = 1, 2, .., 11; t = 1935, 1936, \ldots, 1954, \tag{26.47}$$

where I_{it} is gross investment, F_{it} is the market value of the firm at the end of the previous year, and C_{it} is the value of the stock of plant and equipment at the end of the previous year. The eleven firms indexed by i are General Motors (GM), Chrysler (CH), General Electric (GE), Westinghouse (WE) and US Steel (USS), Atlantic Refining (AR), IBM, Union Oil (UO), Goodyear (GY), Diamond Match (DM), American Steel (AS). Table 26.1 reports estimation of the above equation using various estimation methods: the pooled OLS estimator given by (26.7), the FE estimator given by (26.17), the RE estimator computed by maximization of the likelihood function, (26.36), using the Newton-Raphson algorithm, and the between (BE) estimator given by (26.41). Note that results from FE and RE (or ML) are very close to each other; this is also confirmed by the Hausman test, which does not reject the null hypothesis that the RE and FE are identical (see Section 26.9.1 for a description of the Hausman test).

Table 26.1 Estimation of the Grunfeld investment equation

Estimation method	$\hat{\beta}_1$	$\hat{\beta}_2$
OLS	0.114 (0.006)	0.227 (0.024)
FE	0.110 (0.011)	0.310 (0.016)
RE	0.109 (0.010)	0.308 (0.016)
RE-MLE	0.109 (0.009)	0.307 (0.016)
BE	0.134 (0.027)	0.029 (0.175)
Hausman test FE vs RE	3.97 [0.137]	

Notes: standard errors in round brackets, and *p*-values in square brackets.

[2] Data can be downloaded from the web page <http://statmath.wu-wien.ac.at/~zeileis/grunfeld/>.

26.8 Models with time-specific effects

Model (26.1) can be generalized to include unobserved, time-specific effects. Consider

$$y_{it} = \alpha_i + d_t + \boldsymbol{\beta}'\mathbf{x}_{it} + u_{it}, \tag{26.48}$$

for $i = 1, 2, \ldots, N$; $t = 1, 2, \ldots, T$. Under the fixed-effects specification, α_i and d_t are assumed to be fixed unknown parameters. The above model is known as two-way fixed-effects specification. Consider model (26.48) expressed in matrix form (see (26.5) for the notation)

$$\mathbf{y} = (\boldsymbol{\alpha} \otimes \boldsymbol{\tau}_T) + (\boldsymbol{\tau}_N \otimes \mathbf{d}) + \mathbf{X}\boldsymbol{\beta} + \mathbf{u}, \tag{26.49}$$

where $\mathbf{d} = (d_1, d_2, \ldots, d_T)'$. Let $\mathbf{P}_T = \boldsymbol{\tau}_T \left(\boldsymbol{\tau}_T'\boldsymbol{\tau}_T\right)^{-1} \boldsymbol{\tau}_T'$, $\mathbf{P}_N = \boldsymbol{\tau}_N \left(\boldsymbol{\tau}_N'\boldsymbol{\tau}_N\right)^{-1} \boldsymbol{\tau}_N'$, and consider the within transformation matrix

$$\mathbf{Q} = \mathbf{I}_N \otimes \mathbf{I}_T - \mathbf{I}_N \otimes \mathbf{P}_T - \mathbf{P}_N \otimes \mathbf{I}_T + \mathbf{P}_N \otimes \mathbf{P}_T.$$

Let $\mathbf{y}^* = \mathbf{Q}\mathbf{y}$, $\mathbf{X}^* = \mathbf{Q}\mathbf{X}$, and $\mathbf{u}^* = \mathbf{Q}\mathbf{u}$. For example, the generic element of $\mathbf{y}^*$ is $y_{it} - \bar{y}_{i.} - \bar{y}_{.t} + \bar{y}_{..}$. Noting that $\mathbf{P}_T\boldsymbol{\tau}_T = \boldsymbol{\tau}_T$, $\mathbf{P}_N\boldsymbol{\tau}_N = \boldsymbol{\tau}_N$, we have $\mathbf{Q}(\boldsymbol{\alpha} \otimes \boldsymbol{\tau}_T) = \mathbf{0}$, and $\mathbf{Q}(\boldsymbol{\tau}_N \otimes \mathbf{d}) = \mathbf{0}$. Hence, the two-way FE estimator of $\boldsymbol{\beta}$ can be obtained by applying *OLS* to the transformed model

$$\mathbf{y}_{i.}^* = \mathbf{X}_{i.}^*\boldsymbol{\beta} + \mathbf{u}_{i.}^*.$$

One important point to bear in mind is that the above transformation wipes out the α_i and d_t effects, as well as the effect of any time-invariant or individual-invariant variables. Therefore, the two-way FE estimator cannot estimate the effect of time-invariant and individual-invariant variables.

If the true model is a two-way fixed-effects model, as in (26.48), then applying the pooled *OLS*, which ignores both time and individual effects, or the one-way FE estimator, which omits the time effects, will yield biased and inconsistent estimates of regression coefficients.

Under the random effects specification, α_i and d_t are assumed to be random draws from a probability distribution, and the *GLS* estimator can be used. Under this specification, it is assumed that $E(u_{it}|\mathbf{X}_{i.}, \alpha_i, d_t) = 0$, $E(\alpha_i|\mathbf{X}_{i.}) = 0$, $E(d_t|\mathbf{X}_{i.}) = 0$, $E(\mathbf{u}_{i.}\mathbf{u}_{i.}'|\mathbf{X}_{i.}, \alpha_i) = \sigma_u^2\mathbf{I}_T$, $E\left(\alpha_i^2|\mathbf{X}_{i.}\right) = \sigma_\alpha^2$, $E\left(d_t^2|\mathbf{X}_{i.}\right) = \sigma_d^2$ for all i and t. In this case, letting $v_{it} = \alpha_i + d_t + u_{it}$, the covariance matrix of $\mathbf{v} = \left(\mathbf{v}_{1.}', \mathbf{v}_{2.}', \ldots, \mathbf{v}_{N.}'\right)'$, where $\mathbf{v}_{i.} = (\mathbf{v}_{i1}, \mathbf{v}_{i2}, \ldots, \mathbf{v}_{iT})'$, is

$$E\left(\mathbf{v}\mathbf{v}'\right) = \boldsymbol{\Sigma}_v = \boldsymbol{\sigma}_\alpha^2\left(\mathbf{I}_N \otimes \boldsymbol{\tau}_T\boldsymbol{\tau}_T'\right) + \sigma_d^2\left(\boldsymbol{\tau}_T\boldsymbol{\tau}_T' \otimes \mathbf{I}_T\right) + \sigma_u^2\left(\mathbf{I}_N \otimes \mathbf{I}_T\right).$$

To obtain the *GLS* estimator, an expression for the inverse of $\boldsymbol{\Sigma}_v$ is needed. It is possible to show that (see Wallace and Hussain (1969))

$$\boldsymbol{\Sigma}_v^{-1} = \frac{1}{\sigma_u^2}\left[\mathbf{I}_{NT} - \psi_1\left(\mathbf{I}_N \otimes \boldsymbol{\tau}_T\boldsymbol{\tau}_T'\right) - \psi_1\left(\boldsymbol{\tau}_N\boldsymbol{\tau}_N' \otimes \mathbf{I}_T\right) + \psi_3\left(\boldsymbol{\tau}_N\boldsymbol{\tau}_N' \otimes \boldsymbol{\tau}_T\boldsymbol{\tau}_T'\right)\right].$$

where

$$\psi_1 = \frac{\sigma_\alpha^2}{\sigma_u^2 + T\sigma_\alpha^2}, \quad \psi_2 = \frac{\sigma_d^2}{\sigma_u^2 + N\sigma_d^2},$$
$$\psi_3 = \frac{\sigma_\alpha^2 \sigma_d^2}{\left(\sigma_u^2 + N\sigma_d^2\right)\left(\sigma_u^2 + T\sigma_\alpha^2\right)} \left(\frac{2\sigma_u^2 + T\sigma_\alpha^2 + N\sigma_d^2}{\sigma_u^2 + T\sigma_\alpha^2 + N\sigma_d^2} \right).$$

See also Section 3.3 of Baltagi (2005) for further details.

Example 58 *In an interesting study, Lillard and Weiss (1979) investigate the sources of variation in the earnings of American scientists reported by a panel of PhDs every two years, and over the decade 1960–1970. The sample is composed of six fields: biology, chemistry, earth sciences, mathematics, physics, and psychology. The earning function has the form*

$$\ln y_{it} = d_t + school_i + male_i + experience_{it} + v_{it},$$

where d_t are year dummies, $male_i$ is a dummy variable equal to 1 if the scientist is a male and 0 otherwise, $school_i$ is a set of schooling related variables, and $experience_{it}$ is a set of experience related variables. The residual earnings variation in v_{it} is decomposed into a random effect individual variance component in the level of earnings, a random effect individual component in earnings growth, and a serially correlated transitory component. Specifically, it is assumed that (see also Exercise 5 below)

$$v_{it} = \alpha_i + u_{it} + \xi_i \left(t - \bar{t}_T\right), \tag{26.50}$$

$\bar{t}_T = T^{-1}(1 + 2 + \ldots + T)$, and

$$u_{it} = \rho u_{i,t-1} + \varepsilon_{it},$$

where

$$\begin{pmatrix} \alpha_i \\ \xi_i \end{pmatrix} \sim \left(\mathbf{0}, \Sigma_{\alpha\xi}\right), \quad \varepsilon_{it} \sim IID\left(0, \sigma_\varepsilon^2\right).$$

The individual-specific term, α_i, represents the effect of unmeasured characteristics such as ability and work-related preferences, on the relative earnings of scientists, while ξ_i represents the effect of omitted variables which influence the growth in earnings such as individual learning ability. It is not unreasonable to expect some of the same unobserved variables to affect both α_i and ξ_i, in which case they will be correlated. The serial correlation coefficient, ρ, represents the rate of deterioration of the effects of random shocks, ε_{it}, which persist for more than a year. The model is estimated by maximum likelihood. One interesting finding is that, during the sample period, divergent patterns in earnings are observed for individuals with similar characteristics. In particular, individuals with greater mean earnings also had greater earnings growth. Further, it is observed that a substantial increase in the variance of individual mean earnings with increased experience, while the variance of the growth component remains constant. These patterns suggest that a substantial amount of

inequality is sustained even if one adopts measures of earnings which are based on longer periods, such as a lifetime or permanent income. See Lillard and Weiss (1979) for further details.

26.9 Testing for fixed-effects

Consider the one-way fixed-effects model, (26.1). In cases where N is small relative to T, one could test the joint significance of the group effects by performing an F-test. The null hypothesis is

$$H_0 : \alpha_1 = \alpha_2 = \ldots = \alpha_N = 0.$$

The F-test is

$$F_1 = \frac{\frac{RRSS - URSS}{N-1}}{\frac{URSS}{N(T-1)-k}}. \tag{26.51}$$

where $RRSS$ denotes the residual sum of squares under the null hypothesis, $URSS$ the residual sum of squares under the alternative. Under H_0, this statistic is distributed as $F_{(N-1),N(T-1)-k}$.

Consider now the two-way fixed-effects specification, (26.48). In this case, it is possible to test for joint significance of the time and group effects

$$H_0 : \alpha_1 = \alpha_2 = \ldots = \alpha_N = 0, \text{ and } d_1 = d_2 = \ldots = d_T = 0.$$

The resulting F-statistic is $F_2 \sim F_{(N+T-2),(N-1)(T-1)-k}$. A further statistic can be considered for testing the null of no group effects in the presence of time effects, i.e.,

$$H_0 : \alpha_1 = \alpha_2 = \ldots = \alpha_N = 0, \text{ and } d_t \neq 0, t = 1, 2, \ldots, T,$$

and the F-statistic is $F_3 \sim F_{(N-1),(N-1)(T-1)-k}$. Finally, one can test the null of no time effects allowing for group effects, namely

$$H_0 : d_1 = d_2 = \ldots = d_T = 0, \text{ and } \alpha_i \neq 0, i = 1, 2, \ldots, N.$$

For this case, the F-statistic is $F_4 \sim F_{(T-1),(N-1)(T-1)-k}$.

In the case where N is large, other testing approaches are needed. One such approach is based on the Hausman misspecification test which we review briefly below.

26.9.1 Hausman's misspecification test

The Hausman principle can be applied to hypothesis testing problems in which two estimators are available, one of which is known to be *consistent and efficient* under the null hypothesis, and inconsistent under the alternative, while the other estimator is consistent under both hypotheses without necessarily being efficient. The idea is to construct a test statistic based on the difference between the two estimators.

Denote the efficient estimator by subscript 'e' and the inefficient but consistent estimator (under the alternative hypothesis) by the subscript 'c'. We then have

$$V(\hat{\boldsymbol{\theta}}_c - \hat{\boldsymbol{\theta}}_e) = V(\hat{\boldsymbol{\theta}}_c) - V(\hat{\boldsymbol{\theta}}_e). \tag{26.52}$$

This is the result used by Hausman (1978) where it is assumed that $\hat{\boldsymbol{\theta}}_e$ is asymptotically the *most* efficient estimator. However, it is easily shown that (26.52) holds under a weaker requirement, namely when the (asymptotic) efficiency of $\hat{\boldsymbol{\theta}}_e$ cannot be enhanced by the information contained in $\hat{\boldsymbol{\theta}}_c$. Consider a third estimator $\hat{\boldsymbol{\theta}}_*$, defined as a convex combination of $\hat{\boldsymbol{\theta}}_c$ and $\hat{\boldsymbol{\theta}}_e$

$$\mathbf{a}'\hat{\boldsymbol{\theta}}_* = (1-\delta)\mathbf{a}'\hat{\boldsymbol{\theta}}_e + \delta\mathbf{a}'\hat{\boldsymbol{\theta}}_c, \tag{26.53}$$

where $\mathbf{a}$ is a vector of constants, and δ is a scalar in the range $0 \leq \delta \leq 1$. Since, by assumption, the asymptotic efficiency of $\hat{\boldsymbol{\theta}}_e$ cannot be enhanced by the knowledge of $\hat{\boldsymbol{\theta}}_c$, then it must be that $Var(\mathbf{a}'\hat{\boldsymbol{\theta}}_*) \geq Var(\mathbf{a}'\hat{\boldsymbol{\theta}}_e)$, and hence the value of δ that minimises $Var(\mathbf{a}'\hat{\boldsymbol{\theta}}_*)$, say δ^*, should be zero. However, using (26.53) directly, we have

$$\delta^* = \frac{\mathbf{a}'[Var(\hat{\boldsymbol{\theta}}_e) - Cov(\hat{\boldsymbol{\theta}}_e, \hat{\boldsymbol{\theta}}_c)]\mathbf{a}}{\mathbf{a}'Var(\hat{\boldsymbol{\theta}}_c - \hat{\boldsymbol{\theta}}_e)\mathbf{a}} = 0, \tag{26.54}$$

and hence $\mathbf{a}'[Var(\hat{\boldsymbol{\theta}}_e) - Cov(\hat{\boldsymbol{\theta}}_e, \hat{\boldsymbol{\theta}}_c)]\mathbf{a} = 0$. But, if this result is to hold for an arbitrary vector, $\mathbf{a}$, we must have

$$Var(\hat{\boldsymbol{\theta}}_e) = Cov(\hat{\boldsymbol{\theta}}_e, \hat{\boldsymbol{\theta}}_c). \tag{26.55}$$

Using this in

$$Var(\hat{\boldsymbol{\theta}}_c - \hat{\boldsymbol{\theta}}_e) = Var(\hat{\boldsymbol{\theta}}_c) + Var(\hat{\boldsymbol{\theta}}_e) - 2\,Cov(\hat{\boldsymbol{\theta}}_e, \hat{\boldsymbol{\theta}}_c),$$

yields (26.52) as desired. Because under the null hypothesis both estimators are consistent, the difference, $\hat{\boldsymbol{\theta}}_c - \hat{\boldsymbol{\theta}}_e$, will converge to zero if the null hypothesis is true, while under the alternative hypothesis it will diverge. The Hausman test based on $\left(\hat{\boldsymbol{\theta}}_c - \hat{\boldsymbol{\theta}}_e\right)'[Var(\hat{\boldsymbol{\theta}}_c) - Var(\hat{\boldsymbol{\theta}}_e)]^{-1}\left(\hat{\boldsymbol{\theta}}_c - \hat{\boldsymbol{\theta}}_e\right)$, will be consistent if $Var(\hat{\boldsymbol{\theta}}_c) - Var(\hat{\boldsymbol{\theta}}_e)$ converges to a positive definite matrix and $\hat{\boldsymbol{\theta}}_c - \hat{\boldsymbol{\theta}}_e$ converges to a non-zero limit under the alternative hypothesis. See Pesaran and Yamagata (2008) for examples of cases where the Hausman test fails to be applicable. See also Chapter 28.

The Hausman testing procedure is quite general and can be applied to a variety of testing problems, and is particularly convenient in the case of panels where N is large and the use of classical tests can encounter the incidental parameter problem. In the context of panels, Hausman and Taylor (1981) consider the hypothesis

$$H_0: \quad E\left(\eta_i | \mathbf{x}_{it}\right) = 0, \tag{26.56}$$

where

$$\alpha_i = \alpha + \eta_i. \tag{26.57}$$

Indeed, under H_0 the RE estimator of $\boldsymbol{\beta}$ achieves the Cramer–Rao lower bound, while under H_1 it is biased. In contrast, the FE estimator of $\boldsymbol{\beta}$ is consistent under both H_0 and H_1, but it is not efficient under H_0. Let

$$\hat{\mathbf{q}} = \hat{\boldsymbol{\beta}}_{RE} - \hat{\boldsymbol{\beta}}_{FE}. \tag{26.58}$$

Hence, the Hausman test examines whether RE and FE estimates are significantly different. We have

$$\widehat{Var}(\hat{\mathbf{q}}) = \widehat{Var}(\hat{\boldsymbol{\beta}}_{FE}) - \widehat{Var}(\hat{\boldsymbol{\beta}}_{RE}), \tag{26.59}$$

where $\widehat{Var}(\hat{\boldsymbol{\beta}}_{FE})$ and $\widehat{Var}(\hat{\boldsymbol{\beta}}_{RE})$ are the estimated covariances of $\hat{\boldsymbol{\beta}}_{FE}$ and $\hat{\boldsymbol{\beta}}_{RE}$ obtained under the assumption that errors, u_{it}, are serially uncorrelated and homoskedastic. Under this setting the *Hausman statistic* is given by

$$H = \hat{\mathbf{q}}' \left[\widehat{Var}(\hat{\mathbf{q}})\right]^{-1} \hat{\mathbf{q}}, \tag{26.60}$$

which is distributed as χ^2_k, for N sufficiently large.

But it is important to note that this test does not apply if u_{it} are serially correlated or cross-sectionally heteroskedastic. This is because in this case the RE estimator is no longer efficient. Nevertheless, it is possible to develop tests of $E(\eta_i|\mathbf{x}_{it}) = 0$ by comparing the pooled *OLS*, $\hat{\boldsymbol{\beta}}_{OLS}$, and the fixed-effects estimator $\hat{\boldsymbol{\beta}}_{FE}$. Under Assumptions P1-P5 both estimators are consistent, but neither is efficient. Therefore the Hausman formula for variance of the difference does not apply, namely $Var(\hat{\boldsymbol{\beta}}_{FE} - \hat{\boldsymbol{\beta}}_{OLS}) \neq Var(\hat{\boldsymbol{\beta}}_{FE}) - Var(\hat{\boldsymbol{\beta}}_{OLS})$. But we note that

$$\begin{aligned}
\hat{\mathbf{q}} &= \hat{\boldsymbol{\beta}}_{FE} - \hat{\boldsymbol{\beta}}_{OLS} \\
&= \mathbf{Q}^{-1}_{FE,NT} \left[(NT)^{-1} \sum_{t=1}^{T} \sum_{i=1}^{N} (\mathbf{x}_{it} - \bar{\mathbf{x}}_{i.})\, u_{it} \right] \\
&\quad - \mathbf{Q}^{-1}_{P,NT} \left[(NT)^{-1} \sum_{t=1}^{T} \sum_{i=1}^{N} (\mathbf{x}_{it} - \bar{\mathbf{x}}) \left(\eta_i + u_{it}\right) \right].
\end{aligned}$$

It is clear that under H_0, and supposing that Assumptions P1–P5 hold, $E(\hat{\boldsymbol{\beta}}_{FE} - \hat{\boldsymbol{\beta}}_{OLS}|\mathbf{X}) = 0$, or $E(\hat{\boldsymbol{\beta}}_{FE} - \hat{\boldsymbol{\beta}}_{OLS}) = \mathbf{0}$. But if H_0 does not hold we then have

$$E\left(\hat{\boldsymbol{\beta}}_{FE} - \hat{\boldsymbol{\beta}}_{OLS}|\mathbf{X}\right) = -\mathbf{Q}^{-1}_{P,NT} \left\{ (NT)^{-1} \sum_{t=1}^{T} \sum_{i=1}^{N} E\left[(\mathbf{x}_{it} - \bar{\mathbf{x}})\, \eta_i\right] \right\} \neq 0.$$

Also, by direct derivations, we have

$$Var\big(\hat{\boldsymbol{\beta}}_{FE} - \hat{\boldsymbol{\beta}}_{OLS} \,|\mathbf{X}\,\big) = Var\big(\hat{\boldsymbol{\beta}}_{FE} \,|\mathbf{X}\,\big) + Var\big(\hat{\boldsymbol{\beta}}_{OLS} \,|\mathbf{X}\,\big) - Cov\big(\hat{\boldsymbol{\beta}}_{FE}, \hat{\boldsymbol{\beta}}_{OLS} \,|\mathbf{X}\,\big) - Cov\big(\hat{\boldsymbol{\beta}}_{OLS}, \hat{\boldsymbol{\beta}}_{FE} \,|\mathbf{X}\,\big).$$

Using (26.10) and (26.22) and noting that (under Assumptions P1-P5) we have

$$Cov\big(\hat{\boldsymbol{\beta}}_{FE}, \hat{\boldsymbol{\beta}}_{OLS} \,|\mathbf{X}\,\big) = \frac{1}{NT}\mathbf{Q}_{FE,NT}^{-1}\mathbf{V}_{FEP,NT}\mathbf{Q}_{P,NT}^{-1},$$

where

$$\mathbf{V}_{FEP,NT} = \left[(NT)^{-1}\sum_{i=1}^{N}\sum_{t=1}^{T}\sum_{t'=1}^{T}\gamma_i(t,t')\,(\mathbf{x}_{it} - \bar{\mathbf{x}}_{i.})\,(\mathbf{x}_{it'} - \bar{\mathbf{x}})'\right].$$

Similarly,

$$Cov\left(\hat{\boldsymbol{\beta}}_{OLS}, \hat{\boldsymbol{\beta}}_{FE} \,|\mathbf{X}\right) = \frac{1}{NT}\mathbf{Q}_{P,NT}^{-1}\mathbf{V}_{PFE,NT}\mathbf{Q}_{FE,NT}^{-1}$$

$$\mathbf{V}_{PFE,NT} = \left[(NT)^{-1}\sum_{i=1}^{N}\sum_{t=1}^{T}\sum_{t'=1}^{T}\gamma_i(t,t')\,(\mathbf{x}_{it} - \bar{\mathbf{x}})\,(\mathbf{x}_{it'} - \bar{\mathbf{x}}_{i.})'\right].$$

Hence

$$Var\big(\hat{\boldsymbol{\beta}}_{FE} - \hat{\boldsymbol{\beta}}_{OLS} \,|\mathbf{X}\,\big) = \frac{1}{NT}\begin{bmatrix} \mathbf{Q}_{FE,NT}^{-1}\mathbf{V}_{FE,NT}\mathbf{Q}_{FE,NT}^{-1} + \mathbf{Q}_{P,NT}^{-1}\mathbf{V}_{P,NT}\mathbf{Q}_{P,NT}^{-1} \\ -\mathbf{Q}_{FE,NT}^{-1}\mathbf{V}_{FEP,NT}\mathbf{Q}_{P,NT}^{-1} - \mathbf{Q}_{P,NT}^{-1}\mathbf{V}_{PFE,NT}\mathbf{Q}_{FE,NT}^{-1} \end{bmatrix}.$$

The above result simplifies if we assume that the errors are serially uncorrelated, and reduces to $Var\big(\hat{\boldsymbol{\beta}}_{FE} \,|\mathbf{X}\,\big) - Var\big(\hat{\boldsymbol{\beta}}_{OLS} \,|\mathbf{X}\,\big)$ if it is further assumed that the errors, u_{it} are homoskedastic. To see this, note that in the case of serially uncorrelated errors, $\gamma_i(t,t') = 0$ if $t \neq t'$, and $\gamma_i(t,t) = \sigma_i^2$, we have

$$\begin{aligned}
\mathbf{V}_{FEP,NT} &= \left[(NT)^{-1}\sum_{i=1}^{N}\sum_{t=1}^{T}\sigma_i^2\,(\mathbf{x}_{it} - \bar{\mathbf{x}}_{i.})\,(\mathbf{x}_{it} - \bar{\mathbf{x}})'\right] \\
&= \left[(NT)^{-1}\sum_{i=1}^{N}\sum_{t=1}^{T}\sigma_i^2\,(\mathbf{x}_{it} - \bar{\mathbf{x}}_{i.})\,[\mathbf{x}_{it} - \bar{\mathbf{x}}_{i.} + (\bar{\mathbf{x}}_{i.} - \bar{\mathbf{x}})]'\right] \\
&= \mathbf{V}_{FE,NT} + \left[(NT)^{-1}\sum_{i=1}^{N}\sum_{t=1}^{T}\sigma_i^2\,(\mathbf{x}_{it} - \bar{\mathbf{x}}_{i.})\,(\bar{\mathbf{x}}_{i.} - \bar{\mathbf{x}})'\right] \\
&= \mathbf{V}_{FE,NT}.
\end{aligned}$$

Similarly, $\mathbf{V}_{PFE,NT} = \mathbf{V}_{FE,NT}$. Therefore, in this case

$$Var\left(\hat{\boldsymbol{\beta}}_{FE} - \hat{\boldsymbol{\beta}}_{OLS}|\mathbf{X}\right) = \frac{1}{NT}\begin{bmatrix} \mathbf{Q}_{FE,NT}^{-1}\mathbf{V}_{FE,NT}\mathbf{Q}_{FE,NT}^{-1} + \mathbf{Q}_{P,NT}^{-1}\mathbf{V}_{P,NT}\mathbf{Q}_{P,NT}^{-1} \\ -\mathbf{Q}_{FE,NT}^{-1}\mathbf{V}_{FE,NT}\mathbf{Q}_{P,NT}^{-1} - \mathbf{Q}_{P,NT}^{-1}\mathbf{V}_{FE,NT}\mathbf{Q}_{FE,NT}^{-1} \end{bmatrix}.$$

If we now further assume that $\sigma_i^2 = \sigma^2$ we have $\mathbf{V}_{FE,NT} = \mathbf{Q}_{FE,NT}$, and $\mathbf{V}_{P,NT} = \mathbf{Q}_{P,NT}$, and we have the further simplification

$$\begin{aligned} Var\left(\hat{\boldsymbol{\beta}}_{FE} - \hat{\boldsymbol{\beta}}_{OLS}|\mathbf{X}\right) &= \frac{1}{NT}\begin{bmatrix} \mathbf{Q}_{FE,NT}^{-1}\mathbf{Q}_{FE,NT}\mathbf{Q}_{FE,NT}^{-1} + \mathbf{Q}_{P,NT}^{-1}\mathbf{Q}_{P,NT}\mathbf{Q}_{P,NT}^{-1} \\ -\mathbf{Q}_{FE,NT}^{-1}\mathbf{Q}_{FE,NT}\mathbf{Q}_{P,NT}^{-1} - \mathbf{Q}_{P,NT}^{-1}\mathbf{Q}_{FE,NT}\mathbf{Q}_{FE,NT}^{-1} \end{bmatrix} \\ &= \frac{1}{NT}\left(\mathbf{Q}_{FE,NT}^{-1} - \mathbf{Q}_{P,NT}^{-1}\right), \end{aligned}$$

which accords with Hausman's variance formula.

For consistent estimation of $Var\left(\hat{\boldsymbol{\beta}}_{FE} - \hat{\boldsymbol{\beta}}_{OLS}|\mathbf{X}\right)$ in the general case see the derivations in Section 26.7.

26.10 Estimation of time-invariant effects

In this section we shall assume there are time-invariant regressors in (26.1), namely

$$y_{it} = \alpha_i + \mathbf{z}_i'\boldsymbol{\gamma} + \mathbf{x}_{it}'\boldsymbol{\beta} + \varepsilon_{it}, \quad i = 1, 2, \ldots, N; t = 1, 2, \ldots, T, \tag{26.61}$$

where

$$\alpha_i = \alpha + \eta_i, \tag{26.62}$$

and $\mathbf{z}_i$ is an $m \times 1$ vector of observed individual-specific variables that only vary over the cross-sectional units, *i*. The focus of the analysis is on estimation and inference involving the elements of $\boldsymbol{\gamma}$. Important examples of time-invariant regressors are sex, ethnicity, and place of birth.

In what follows, we allow for η_i and $\mathbf{x}_{it}$ to have any degree of dependence and distinguish between case 1, where $\mathbf{z}_i$ is assumed to be uncorrelated with η_i, and case 2, where one or more elements of $\mathbf{z}_i$ are allowed to be correlated with η_i. Under case 2, to identify the time-invariant effects we need to assume that there exists a sufficient number of instruments that can be used to deal with the dependence of $\mathbf{z}_i$ and η_i.

26.10.1 Case 1: $\mathbf{z}_i$ is uncorrelated with η_i

When $\mathbf{z}_i$ is uncorrelated with η_i, the fixed-effects filtered (FEF) estimators proposed by Pesaran and Zhou (2014) can be used. The basic idea behind the FEF estimator is to use the residuals from FE estimation of $\boldsymbol{\beta}$ to compute the invariant effects, $\boldsymbol{\gamma}$, by the *OLS* regression of the time averages of the fixed-effects residuals on an intercept and $\mathbf{z}_i$. Let $u_{it} = \alpha + \mathbf{z}_i'\boldsymbol{\gamma} + \eta_i + \varepsilon_{it}$, and note that it can be consistently estimated by

$$\hat{u}_{it} = y_{it} - \hat{\boldsymbol{\beta}}'_{FE}\mathbf{x}_{it}. \tag{26.63}$$

Then FEF estimator of $\boldsymbol{\gamma}$ is computed by regressing $\bar{\hat{u}}_i = \frac{1}{T}\sum_{t=1}^{T}\hat{u}_{it} = \bar{y}_i - \hat{\boldsymbol{\beta}}'_{FE}\bar{\mathbf{x}}_i$ on an intercept and $\mathbf{z}_i$, and is given by

$$\hat{\boldsymbol{\gamma}}_{FEF} = \left[\sum_{i=1}^{N}(\mathbf{z}_i - \bar{\mathbf{z}})(\mathbf{z}_i - \bar{\mathbf{z}})'\right]^{-1}\sum_{i=1}^{N}(\mathbf{z}_i - \bar{\mathbf{z}})\left(\bar{\hat{u}}_i - \bar{\hat{u}}\right), \tag{26.64}$$

where $\bar{\hat{u}} = N^{-1}\sum_{i=1}^{N}\bar{\hat{u}}_i$.

Pesaran and Zhou (2014) also derive the asymptotic distribution of $\hat{\boldsymbol{\gamma}}_{FEF}$ under Assumptions P1–P2, P3, P3' and P4–P5, and the following additional assumptions on the time-invariant regressors:

Assumption P6: Consider the $m \times m$ matrix $\mathbf{Q}_{zz,N}$, and the $m \times k$ matrix $\mathbf{Q}_{z\bar{x},N}$ defined by

$$\mathbf{Q}_{zz,N} = \frac{1}{N}\sum_{i=1}^{N}(\mathbf{z}_i - \bar{\mathbf{z}})(\mathbf{z}_i - \bar{\mathbf{z}})', \tag{26.65}$$

$$\mathbf{Q}_{z\bar{x},N} = \frac{1}{N}\sum_{i=1}^{N}(\mathbf{z}_i - \bar{\mathbf{z}})(\bar{\mathbf{x}}_i - \bar{\mathbf{x}})'. \tag{26.66}$$

Matrix $\mathbf{Q}_{zz,N}$ is nonsingular for all $N > m$ and as $N \to \infty$, namely $\lambda_{\min}\left(\mathbf{Q}_{zz,N}\right) > 1/K$, for all N. Matrices $\mathbf{Q}_{z\bar{x},N}$ and $\mathbf{Q}_{zz,N}$ converge (in probability) to the non-stochastic limits $\mathbf{Q}_{zz}$ and $\mathbf{Q}_{z\bar{x}}$, respectively.

Assumption P7: The time-invariant regressors, $\mathbf{z}_i$, are independently distributed of $v_j = \eta_j + \bar{\varepsilon}_j$, for all i and j, and η_i and $\bar{\varepsilon}_i$ are independently distributed such that $v_i \sim IID(0, \sigma_\eta^2 + \sigma_i^2/T)$, where $E\left(\eta_i^2\right) = \sigma_\eta^2$, and $E\left(\varepsilon_{it}^2\right) = \sigma_i^2$. Also, $\mathbf{z}_i$ are either deterministic or have bounded support, namely $\|\mathbf{z}_i\| < K$, or $\mathbf{z}_i$ satisfy the moment conditions $E\|(\mathbf{z}_i - \bar{\mathbf{z}})\|^4 < K$, for all i.

Under the above assumptions Pesaran and Zhou (2014) show that (for a fixed T and as $N \to \infty$)

$$\sqrt{N}\left(\hat{\boldsymbol{\gamma}}_{FEF} - \boldsymbol{\gamma}\right) \to_d N\left(\mathbf{0}, \boldsymbol{\Omega}_{\hat{\gamma}_{FEF}}\right), \tag{26.67}$$

where

$$\boldsymbol{\Omega}_{\hat{\gamma}_{FEF}} = \mathbf{Q}_{zz}^{-1}\left(\sigma_\eta^2\mathbf{Q}_{zz} + \boldsymbol{\Omega}_{\bar{\xi}}\right)\mathbf{Q}_{zz}^{-1}. \tag{26.68}$$

$$\boldsymbol{\Omega}_{\bar{\xi}} = \lim_{N\to\infty} N^{-1}\sum_{i=1}^{N}\left[T^{-2}\sum_{t,s=1}^{T}\mathbf{d}_{z,it}\mathbf{d}'_{z,is}E\left(\varepsilon_{it}\varepsilon_{is}\right)\right], \tag{26.69}$$

and

$$\mathbf{d}_{z,it} = (\mathbf{z}_i - \bar{\mathbf{z}}) - \frac{1}{N}\sum_{j=1}^{N}\left(\mathbf{z}_j - \bar{\mathbf{z}}\right)w_{ji,t},\ w_{ij,t} = (\bar{\mathbf{x}}_i - \bar{\mathbf{x}})'\mathbf{Q}_{FE,NT}^{-1}\left(\mathbf{x}_{jt} - \bar{\mathbf{x}}_j\right). \tag{26.70}$$

Pesaran and Zhou (2014) propose to estimate $Var\left(\hat{\boldsymbol{\gamma}}_{FEF}\right)$ by

$$\widehat{Var}\left(\hat{\boldsymbol{\gamma}}_{FEF}\right) = N^{-1}\mathbf{Q}_{zz,N}^{-1}\left[\hat{\mathbf{V}}_{zz,N} + \mathbf{Q}_{z\bar{x},N}\left(N\widehat{Var}(\hat{\boldsymbol{\beta}})\right)\mathbf{Q}'_{z\bar{x},N}\right]\mathbf{Q}_{zz,N}^{-1}, \tag{26.71}$$

where

$$\hat{\mathbf{V}}_{zz,N} = \frac{1}{N}\sum_{i=1}^{N}\left(\hat{\varsigma}_i - \bar{\hat{\varsigma}}\right)^2(\mathbf{z}_i - \bar{\mathbf{z}})(\mathbf{z}_i - \bar{\mathbf{z}})', \tag{26.72}$$

$$\hat{\varsigma}_i - \bar{\hat{\varsigma}} = \bar{y}_i - \bar{y} - (\bar{\mathbf{x}}_i - \bar{\mathbf{x}})'\hat{\boldsymbol{\beta}}_{FE} - (\mathbf{z}_i - \bar{\mathbf{z}})'\hat{\boldsymbol{\gamma}}_{FEF},$$

$$\widehat{Var}(\hat{\boldsymbol{\beta}}_{FE}) = \left(\sum_{i=1}^{N}\mathbf{x}'_{i\cdot}\mathbf{x}_{i\cdot}\right)^{-1}\left(\sum_{i=1}^{N}\mathbf{x}'_{i\cdot}\mathbf{e}_i\mathbf{e}'_i\mathbf{x}_{i\cdot}\right)\left(\sum_{i=1}^{N}\mathbf{x}'_{i\cdot}\mathbf{x}_{i\cdot}\right)^{-1}, \tag{26.73}$$

$\mathbf{x}'_{i\cdot} = (\mathbf{x}_{i1} - \bar{\mathbf{x}}_i, \mathbf{x}_{i2} - \bar{\mathbf{x}}_i, \ldots, \mathbf{x}_{iT} - \bar{\mathbf{x}}_i)$ denotes the demeaned vector of $\mathbf{x}_{it}$ and the tth element of $\mathbf{e}_i$ is given by

$$e_{it} = y_{it} - \bar{y}_i - (\mathbf{x}_{it} - \bar{\mathbf{x}}_i)'\hat{\boldsymbol{\beta}}_{FE}.$$

26.10.2 Case 2: $\mathbf{z}_i$ is correlated with η_i

When $\mathbf{z}_i$ is correlated with η_i, then we need instruments for identification and estimation of $\boldsymbol{\gamma}$. With available instruments, two approaches have been proposed to estimate $\boldsymbol{\gamma}$ by Hausman and Taylor (1981) (HT), and by Pesaran and Zhou (2014).

HT estimation procedure

Hausman and Taylor (1981) approach the problem of estimation of the time-invariant effects in the panel data model, (26.61), by assuming that $\mathbf{x}_{it}$ and $\mathbf{z}_i$ can be partitioned into two parts as $\left(\mathbf{x}_{1,it}, \mathbf{x}_{2,it}\right)$ and $\left(\mathbf{z}_{1,i}, \mathbf{z}_{2,i}\right)$, respectively, such that

$$E\left(\mathbf{x}'_{1,it}\eta_i\right) = \mathbf{0}, E\left(\mathbf{z}'_{1,i}\eta_i\right) = \mathbf{0},$$
$$E\left(\mathbf{x}'_{2,it}\eta_i\right) \neq \mathbf{0}, E\left(\mathbf{z}'_{2,i}\eta_i\right) \neq \mathbf{0}.$$

To compute the HT estimator the panel data model is first written as

$$\mathbf{y}_i = \mathbf{X}_i\boldsymbol{\beta} + \left(\mathbf{z}'_i\boldsymbol{\gamma} + \alpha + \eta_i\right)\boldsymbol{\tau}_T + \boldsymbol{\varepsilon}_i, \text{ for } i = 1, 2, \ldots, N, \tag{26.74}$$

where $\mathbf{X}_i = (\mathbf{x}_{i1}, \ldots, \mathbf{x}_{iT})'$, $\mathbf{y}_i = (y_{i1}, y_{i2}, \ldots, y_{iT})'$, and $\boldsymbol{\varepsilon}_i = (\varepsilon_{i1}, \varepsilon_{i2}, \ldots, \varepsilon_{iT})'$. Then the following two-step procedure is used:

Step 1 of HT: $\boldsymbol{\beta}_{FE}$ is estimated by $\hat{\boldsymbol{\beta}}_{FE}$, the FE estimator, and the deviations $\hat{d}_i = \bar{y}_i - \bar{\mathbf{x}}'_i\hat{\boldsymbol{\beta}}_{FE}$, $i = 1, 2, \ldots, N$, are used to compute the $2SLS$ (or IV) estimator

$$\hat{\boldsymbol{\gamma}}_{IV} = \left(\mathbf{Z}'\mathbf{P}_A\mathbf{Z}\right)^{-1}\mathbf{Z}'\mathbf{P}_A\hat{\mathbf{d}}, \tag{26.75}$$

where $\hat{\mathbf{d}} = (\hat{d}_1, \hat{d}_2, \ldots, \hat{d}_N)'$, $\mathbf{Z} = (\mathbf{z}_1, \mathbf{z}_2, \ldots, \mathbf{z}_N)' = (\mathbf{Z}_1, \mathbf{Z}_2)$, and $\mathbf{P}_A = \mathbf{A}\left(\mathbf{A}'\mathbf{A}\right)^{-1}\mathbf{A}'$ is the orthogonal projection matrix of $\mathbf{A} = \left(\boldsymbol{\tau}_N, \bar{\mathbf{X}}_1, \mathbf{Z}_1\right)$, where $\bar{\mathbf{X}} = (\bar{\mathbf{X}}_1, \bar{\mathbf{X}}_2)$, and $\bar{\mathbf{X}} = \left(\bar{\mathbf{x}}_1, \bar{\mathbf{x}}_2, \ldots, \bar{\mathbf{x}}_N\right)'$, $\bar{\mathbf{x}}_i = (\bar{\mathbf{x}}_{i,1}, \bar{\mathbf{x}}_{i,2})$. Using these initial estimates of $\boldsymbol{\beta}$ and $\boldsymbol{\gamma}$, the error variances σ_ε^2 and σ_η^2 are estimated as

$$\begin{aligned}
\hat{\sigma}_\eta^2 &= s^2 - \hat{\sigma}_\varepsilon^2, \\
\hat{\sigma}_\varepsilon^2 &= \frac{1}{N(T-1)} \sum_{i=1}^{N} \left(\mathbf{y}_i - \mathbf{X}_i'\hat{\boldsymbol{\beta}}_{FE}\right)' \mathbf{M}_{\tau_T} \left(\mathbf{y}_i - \mathbf{X}_i'\hat{\boldsymbol{\beta}}_{FE}\right), \\
s^2 &= \frac{1}{NT} \sum_{i=1}^{N} \sum_{t=1}^{T} \left(y_{it} - \hat{\mu} - \mathbf{x}_{it}'\hat{\boldsymbol{\beta}}_{FE} - \mathbf{z}_i'\hat{\boldsymbol{\gamma}}_{IV}\right)^2,
\end{aligned}$$

Step 2 of HT : In the second step the N equations in (26.74) are stacked to obtain

$$\mathbf{y} = \mathbf{W}\boldsymbol{\theta} + (\boldsymbol{\eta} \otimes \boldsymbol{\tau}_T) + \boldsymbol{\varepsilon},$$

where $\mathbf{W} = [(\boldsymbol{\tau}_N \otimes \boldsymbol{\tau}_T), \mathbf{X}, (\mathbf{Z} \otimes \boldsymbol{\tau}_T)]$, $\boldsymbol{\theta} = (\alpha, \boldsymbol{\beta}', \boldsymbol{\gamma}')'$, $\mathbf{y} = (\mathbf{y}_1', \mathbf{y}_2', \ldots, \mathbf{y}_N')'$, $\boldsymbol{\eta} = (\eta_1, \eta_2, \ldots, \eta_N)'$, and $\boldsymbol{\varepsilon} = (\boldsymbol{\varepsilon}_1', \boldsymbol{\varepsilon}_2', \ldots, \boldsymbol{\varepsilon}_N')'$. Under the assumptions that the errors are cross-sectionally independent, serially uncorrelated and homoskedastic we have

$$\boldsymbol{\Omega} = Var\left[(\boldsymbol{\eta} \otimes \boldsymbol{\tau}_T) + \boldsymbol{\varepsilon}\right] = \sigma_\eta^2 \left(\mathbf{I}_N \otimes \boldsymbol{\tau}_T\boldsymbol{\tau}_T'\right) + \sigma_\varepsilon^2 \left(\mathbf{I}_N \otimes \mathbf{I}_T\right),$$

which can be written as $\boldsymbol{\Omega} = \left(\sigma_\varepsilon^2 + T\sigma_\eta^2\right)\mathbf{P}_V + \sigma_\varepsilon^2\mathbf{Q}_V$, where $\mathbf{P}_V = \mathbf{I}_N \otimes (\mathbf{I}_T - \mathbf{M}_T)$ and $\mathbf{Q}_V = \mathbf{I}_N \otimes \mathbf{M}_T$. It is now easily verified that $\boldsymbol{\Omega}^{-1/2} = \frac{1}{\sigma_\varepsilon}\left(\varphi\mathbf{P}_V + \mathbf{Q}_V\right)$, where $\varphi = \sigma_\varepsilon / \sqrt{\sigma_\varepsilon^2 + T\sigma_\eta^2}$. Then the transformed model can be written as

$$\boldsymbol{\Omega}^{-1/2}\mathbf{y} = \boldsymbol{\Omega}^{-1/2}\mathbf{W}\boldsymbol{\theta} + \boldsymbol{\Omega}^{-1/2}\left[(\boldsymbol{\eta} \otimes \boldsymbol{\tau}_T) + \boldsymbol{\varepsilon}\right]. \tag{26.76}$$

To simplify the notation we assume that the first column of $\mathbf{Z}$ is $\boldsymbol{\tau}_N$, and then write the (infeasible) HT estimator as,

$$\hat{\boldsymbol{\theta}}_{HT} = \left(\mathbf{W}'\boldsymbol{\Omega}^{-1/2}\mathbf{P}_A\boldsymbol{\Omega}^{-1/2}\mathbf{W}\right)^{-1}\left(\mathbf{W}'\boldsymbol{\Omega}^{-1/2}\mathbf{P}_A\boldsymbol{\Omega}^{-1/2}\mathbf{y}\right), \tag{26.77}$$

where $\mathbf{P}_A = \mathbf{A}\left(\mathbf{A}'\mathbf{A}\right)^{-1}\mathbf{A}'$ is the projection onto the space of instruments $\mathbf{A} = \left(\boldsymbol{\tau}_N \otimes \boldsymbol{\tau}_T, \mathbf{Q}_V\mathbf{X}, \mathbf{X}^{(1)}, \mathbf{Z}_1 \otimes \boldsymbol{\tau}_T\right)$, where $\mathbf{X}^{(1)} = (\mathbf{x}_{1,1}', \mathbf{x}_{1,2}', \ldots, \mathbf{x}_{1,N}')'$, with $\mathbf{x}_{1,i}' = \left(\mathbf{x}_{1,i1}, \ldots, \mathbf{x}_{1,iT}\right)$, and $\mathbf{x}_{1,it}$ contains the regressors that are uncorrelated with η_i.[3]

The covariance matrix of $\hat{\boldsymbol{\theta}}_{HT}$ is given by

$$\begin{aligned}
Var\left(\hat{\boldsymbol{\theta}}_{HT}\right) = \mathbf{Q}^{-1} + \left(\frac{T}{\sigma_\varepsilon^2 + T\sigma_\eta^2}\right)\mathbf{Q}^{-1} \Bigg\{ &\mathbf{W}'\boldsymbol{\Omega}^{-1/2}\mathbf{P}_A \left[\left(\mathbf{V}_\eta - \sigma_\eta^2\mathbf{I}_N\right) \otimes \frac{1}{T}\boldsymbol{\tau}_T\boldsymbol{\tau}_T'\right] \\
&\mathbf{P}_A\boldsymbol{\Omega}^{-1/2}\mathbf{W}\Bigg\}\mathbf{Q}^{-1},
\end{aligned} \tag{26.78}$$

[3] See Amemiya and MaCurdy (1986) and Breusch, Mizon, and Schmidt (1989) for discussion on the choice of instruments for HT estimation.

where $\mathbf{Q} = \mathbf{W}'\boldsymbol{\Omega}^{-1/2}\mathbf{P}_{\mathbf{A}}\boldsymbol{\Omega}^{-1/2}\mathbf{W}$, and $\mathbf{V}_{\eta}$ represents the covariance matrix of $\boldsymbol{\eta}$. $Var\left(\hat{\boldsymbol{\theta}}_{HT}\right)$ reduces to $\mathbf{Q}^{-1}$ in the standard case where η_i's are assumed to be homoskedastic and cross-sectionally independent, namely when $\mathbf{V}_{\eta} = \sigma_{\eta}^2\mathbf{I}_N$.

Remark 7 *In the case where the effects of the time-invariant regressors are exactly identified, then the HT estimator of $\boldsymbol{\gamma}$, $\hat{\boldsymbol{\gamma}}_{HT}$, is identical to the first stage estimator of $\boldsymbol{\gamma}$, given by (26.75). See Baltagi and Bresson (2012).*

FEF-IV estimation of time-invariant effects

For the FEF estimation discussed above, it is relatively straightforward to modify the FEF estimator to allow for possible endogeneity of the time-invariant regressors, if there exists a sufficient number of valid instruments. Therefore, Pesaran and Zhou (2014) propose to derive an *IV* version of FFE, denoted by FEF-IV, under the following assumptions:

Assumption P8: There exists the $s \times 1$ vector of instruments $\mathbf{r}_i$ for $\mathbf{z}_i$, $i = 1, 2, \ldots, N$, where $\mathbf{r}_i$ is distributed independently of η_j and $\bar{\varepsilon}_j$ for all i and j, $s \geq m$, and $\mathbf{r}_i$ satisfies the moment condition $E\|\mathbf{r}_i - \bar{\mathbf{r}}\|^4 < K < \infty$, if it has unbounded support.

Assumption P9: Let $\mathbf{Z} = (\mathbf{z}_1, \mathbf{z}_2, \ldots, \mathbf{z}_N)'$, $\mathbf{R} = (\mathbf{r}_1, \mathbf{r}_2, \ldots, \mathbf{r}_N)'$, and $\mathbf{M}_{\tau_N} = \mathbf{I}_N - \boldsymbol{\tau}_N\left(\boldsymbol{\tau}_N'\boldsymbol{\tau}_N\right)^{-1}\boldsymbol{\tau}_N'$, with $\boldsymbol{\tau}_N$ being an $N \times 1$ vector of ones. Consider the $s \times m$ matrix $\mathbf{Q}_{rz,N}$, the $s \times k$ matrix $\mathbf{Q}_{rx,N} = N^{-1}\sum_{i=1}^{N}(\mathbf{r}_i - \bar{\mathbf{r}})(\bar{\mathbf{x}}_i - \bar{\mathbf{x}})'$, and the $s \times s$ matrix $\mathbf{Q}_{rr,N}$ defined by

$$\mathbf{Q}_{rz,N} = N^{-1}\sum_{i=1}^{N}(\mathbf{r}_i - \bar{\mathbf{r}})(\mathbf{z}_i - \bar{\mathbf{z}})', \mathbf{Q}_{r\bar{x},N} = N^{-1}\sum_{i=1}^{N}(\mathbf{r}_i - \bar{\mathbf{r}})(\bar{\mathbf{x}}_i - \bar{\mathbf{x}})', \ \mathbf{Q}_{rr,N}$$
$$= N^{-1}\sum_{i=1}^{N}(\mathbf{r}_i - \bar{\mathbf{r}})(\mathbf{r}_i - \bar{\mathbf{r}})', \tag{26.79}$$

where $\bar{\mathbf{r}} = N^{-1}\sum_{i=1}^{N}\mathbf{r}_i$. $\mathbf{Q}_{rz,N}$ and $\mathbf{Q}_{rr,N}$ are full rank matrices for all $N > r$, and have finite probability limits as $N \to \infty$, given by $\mathbf{Q}_{rz}$ and $\mathbf{Q}_{rr}$, respectively. Matrices$\mathbf{Q}_{r\bar{x},N}$ and $\mathbf{Q}_{zz,N}$ have finite probability limits given by $\mathbf{Q}_{r\bar{x}}$ and $\mathbf{Q}_{zz}$, respectively, and in cases where $\mathbf{x}_{it}$ and $\mathbf{z}_i$ are stochastic with unbounded supports, then $\lambda_{\min}(\mathbf{Q}_{rr,N}) > 1/K$, for all N, and as $N \to \infty$, with probability approaching one.

Under the above assumptions (including Assumptions P1–P6) $\boldsymbol{\gamma}$ can be estimated consistently by

$$\hat{\boldsymbol{\gamma}}_{FEF-IV} = \left(\mathbf{Q}_{zr,N}\mathbf{Q}_{rr,N}^{-1}\mathbf{Q}_{zr,N}'\right)^{-1}\left(\mathbf{Q}_{zr,N}\mathbf{Q}_{rr,N}^{-1}\mathbf{Q}_{r\hat{u},N}\right), \tag{26.80}$$

where $\mathbf{Q}_{zr,N}$ and $\mathbf{Q}_{rr,N}$ are defined by (26.79),

$$\mathbf{Q}_{r\hat{u},N} = \frac{1}{N}\sum_{i=1}^{N}(\mathbf{r}_i - \bar{\mathbf{r}})\left(\widehat{\bar{u}}_i - \widehat{\bar{u}}\right),$$

and as before, $\widehat{\bar{u}} = \frac{1}{N}\sum_{i=1}^{N}\widehat{\bar{u}}_i$, and $\widehat{\bar{u}}_i = \bar{y}_i - \bar{\mathbf{x}}_i'\hat{\boldsymbol{\beta}}_{FE}$. It then follows that

$$\sqrt{N}\left(\hat{\boldsymbol{\gamma}}_{FEF-IV} - \boldsymbol{\gamma}\right) \to_d N\left(\mathbf{0}, \boldsymbol{\Omega}_{\hat{\gamma}_{FEF-IV}}\right),$$

where

$$\boldsymbol{\Omega}_{\hat{\gamma}_{FEF-IV}} = \left(\mathbf{Q}_{zr}\,\mathbf{Q}_{rr}^{-1}\mathbf{Q}_{zr}'\right)^{-1}\mathbf{Q}_{zr}\,\mathbf{Q}_{rr}^{-1}\left(\sigma_{\eta}^{2}\mathbf{Q}_{rr}\;+\Omega_{\bar{\psi}}\;\right)\mathbf{Q}_{rr}^{-1}\mathbf{Q}_{zr}'\left(\mathbf{Q}_{zr}\,\mathbf{Q}_{rr}^{-1}\mathbf{Q}_{zr}'\right)^{-1}. \tag{26.81}$$

The variance of $\hat{\boldsymbol{\gamma}}_{FEF-IV}$ can be consistently estimated by

$$\widehat{Var}\left(\hat{\boldsymbol{\gamma}}_{FEF-IV}\right) = N^{-1}\mathbf{H}_{zr,N}\left[\hat{\mathbf{V}}_{rr,N} + \mathbf{Q}_{r\bar{x},N}\left(N\widehat{Var}(\hat{\boldsymbol{\beta}}_{FE})\right)\mathbf{Q}_{r\bar{x},N}'\right]\mathbf{H}_{zr,N}',$$

where

$$\mathbf{H}_{zr,N} = \left(\mathbf{Q}_{zr,N}\mathbf{Q}_{rr,N}^{-1}\mathbf{Q}_{zr,N}'\right)^{-1}\mathbf{Q}_{zr,N}\mathbf{Q}_{rr,N}^{-1},$$

$$\mathbf{Q}_{r\bar{x},N} = \frac{1}{N}\sum_{i=1}^{N}\left(\mathbf{r}_i - \bar{\mathbf{r}}\right)\left(\bar{\mathbf{x}}_i - \bar{\mathbf{x}}\right)',$$

$$\hat{\mathbf{V}}_{rr,N} = \frac{1}{N}\sum_{i=1}^{N}\left(\hat{\upsilon}_i - \bar{\hat{\upsilon}}\right)^2(\mathbf{r}_i - \bar{\mathbf{r}})(\mathbf{r}_i - \bar{\mathbf{r}})'$$

where

$$\hat{\upsilon}_i - \bar{\hat{\upsilon}} = \bar{y}_i - \bar{y} - \left(\bar{\mathbf{x}}_i - \bar{\mathbf{x}}\right)'\hat{\boldsymbol{\beta}} - \left(\mathbf{z}_i - \bar{\mathbf{z}}\right)'\hat{\boldsymbol{\gamma}}_{FEF-IV}.$$

Monte Carlo experiments reported in Pesaran and Zhou (2014) show that $\hat{\boldsymbol{\gamma}}_{FEF}$ and $\hat{\boldsymbol{\gamma}}_{FEF-IV}$ perform well in small samples and are robust to heteroskedasticity and residual serial correlation.

Example 59 (Estimation of return to schooling) *One of the most prominent applications of static panel data techniques is to wage equations estimated across many individuals over a relatively short time period. Here we use data from Vella and Verbeek (1998) which is taken from National Longitudinal Survey (Youth Sample). The sample includes full-time working males who completed their schooling by 1980 and were then followed subsequently over the period 1980 to 1987. The panel data are balanced with* $N = 545$ *and* $T = 8$*, after excluding individuals who failed to provide sufficient information to be included in each year. The wage equation to be estimated is given by*

$$\log(w_{it}) = \alpha_i + \beta_1 Union_{it} + \beta_2 exper_{it} + \beta_3 exper_{it}^2 + \beta_4 Rural_{it} + \beta_5 married_{it} + \gamma_1 educ_i + \gamma_2 black_i + \gamma_3 hisp_i + u_{it},$$

where the time varying regressors are work-experience $\left(exper_{it}\right)$*, marriage status* ($married_{it} = 1$ *if married at time t), union coverage* ($Union_{it} = 1$ *if the individual's wage is set by a union contract), and location* ($Rural_{it} = 1$ *for rural area). The time-invariant variables are years of formal education* ($educ_i$)*, and dummies for race, distinguishing between* $black_i$ *and* $hisp_i$*. These variables were originally used in Cornwell and Rupert (1988). See also a follow up paper by Baltagi and Khanti-Akom (1990). We compute estimates of* $\boldsymbol{\beta} = (\beta_1, \beta_2, \ldots, \beta_5)'$ *and* $\boldsymbol{\gamma} = (\gamma_1, \gamma_2, \gamma_3)'$ *by pooled OLS, FEF, HT and FEF-IV estimators. HT estimates are computed assuming* $x_{1,it} = (married_{it})$ *and* $z_{1,i} = \left(black_i, hisp_i\right)'$ *are exogenous. For the FEF-IV estimates we present three versions that differ in the choice of the instruments. FEF-IV*^{1} *estimates use*

$\overline{married}_i = T^{-1}\sum_{t=1}^{T} married_{it}$ *as an instrument for* $educ_i$, *FEF-IV*^{2} *uses* $black_i$ *as an instrument for* $educ_i$, *and FEF-IV*^{3} *uses* $hisp_i$ *as an instrument for* $educ_i$. *The results are summarized in Table 26.2.*

The results show that the estimates could be quite sensitive to the choice of the estimation procedure and the instrument used for $educ_i$ *which is the time-invariant variable of interest and determines the return to schooling. Estimates of the coefficient of the* $educ_i$ *variable all have the expected positive sign, although they vary widely across the estimation procedures. The pooled OLS and FEF estimates are very close (around* 0.10*), although as to be expected the pooled OLS has a much smaller standard error than that of the FFF estimate (*0.0046 *as compared to* 0.0091*). The FEF estimates are preferable to the pooled OLS estimates since they allow for possible dependence between the time varying regressors and the individual-specific effects, whilst this is ruled out under pooled OLS. Nevertheless, the FEF estimates could still be biased since they ignore possible dependence between* $educ_i$ *and the individual specific effects. To take account of such dependence we need to have suitable instruments. We consider a number of possibilities. Since the model contains three time-invariant variables (*$educ_i$, $black_i$ *and* $hisp_i$*) we need at least three instruments. Initially, following HT, we consider using* $x_{1,it} = (married_{it})$ *and* $\mathbf{z}_{1,i} = \left(black_i, \overline{hisp}_i\right)'$ *as instruments. The corresponding set of instruments for the FEF-IV procedure will be* $\mathbf{r}_i = (\overline{married}_i, black_i, hisp_i)'$. *These associated estimates in Table 26.2 are given under columns HT and FEF-IV*^{1}. *As can be seen, the HT and FEF-IV*^{1} *estimates are very close and differ only marginally in terms of the estimated standard errors. This is partly due to the fact that the parameters are exactly identified and there is little variation in* $married_{it}$ *overtime, and as a result it does not make much difference in using*

Table 26.2 Pooled *OLS*, fixed-effects filter and HT estimates of wage equation

	variables	Pooled OLS	FEF	HT	FEF-IV^1	FEF-IV^2	FEF-IV^3
time varying	$Union_{it}$	0.1801***	0.0815***	0.0815***	0.0815***	0.0815***	0.0815***
		(0.0162)	(0.0227)	(0.0192)	(0.0227)	(0.0227)	(0.0227)
	$exper_{it}$	0.0867***	0.1177***	0.1177***	0.1177***	0.1177***	0.1177***
		(0.0100)	(0.0108)	(0.0084)	(0.0108)	(0.0108)	(0.0108)
	$exper_{it}^2$	-0.0027***	-0.0044***	-0.0043***	-0.0044***	-0.0044***	-0.0044***
		(0.0007)	(0.0007)	(0.0006)	(0.0007)	(0.0007)	(0.0007)
	$rural_{it}$	-0.1411***	0.0493	0.0492*	0.0493	0.0493	0.0493
		(0.0183)	(0.0392)	(0.0290)	(0.0392)	(0.0392)	(0.0392)
	$married_{it}$	0.1258***	0.0453**	0.0453***	0.0453**	0.0453**	0.0453**
		(0.0153)	(0.0210)	(0.0181)	(0.0210)	(0.0210)	(0.0210)
time-invariant	$educ_i$	0.0941***	0.1036***	1.2147	1.2148	0.4691**	0.0711*
		(0.0046)	(0.0091)	(3.1402)	(3.2009)	(0.2158)	(0.0396)
	$hisp_i$	-0.0169	0.0331	1.1629	1.1629	0.4047*	
		(0.0202)	(0.0415)	(3.2012)	(3.2518)	(0.2228)	
	$black_i$	0.0617	-0.1398***	0.2851	0.2852		-0.1522**
		(0.0662)	(0.0508)	(1.2289)	(1.2591)		(0.0516)

Standard error are in parentheses; ***, **, and * denote statistical significance at 99% , 95%,and 90% levels, respectively. For HT, the exogenous variables are $X_1 =$ (married) and $Z_1 =$ (black,hisp). The Pooled LS and HT are calculated using standard command from Stata 13. All other estimates are calculated using Matlab code provided by Pesaran and Zhou (2014). FEF-IV^1 estimates are computed using $\overline{\text{married}}_i$ as the instrument for $educ_i$; FEF-IV^2 uses $black_i$ as the instrument for $educ_i$; and FEF-IV^3 uses $hisp_i$ as the instrument for $educ_i$.

$married_{it}$ or $\overline{married}_i$ as an instrument for $educ_i$. Also, note that $black_i$ and $hisp_i$ are treated as exogenous under both HT and FEF-IV procedures. The HT and FEF-IV[1] estimates of return to schooling are rather disappointing—neither is statistically significant, which seems to be largely due to the quality of $married_{it}$ or $\overline{married}_i$ as an instrument for $educ_i$. As you might recall, it is not sufficient that the instrument is uncorrelated with the individual-specific effects, α_i; a strong instrument should also have a reasonable degree of correlation with the instrumented variable. In the present application the correlation between $\overline{married}_i$ and $educ_i$ is around 0.025, which renders $\overline{married}_i$ a weak instrument for $educ_i$. In view of this, we consider two additional specifications of the wage equation which are reported in Table 26.2 under columns FEF-IV[2] and FEF-IV[3]. Specification FEF-IV[2] excludes $black_i$ from the regression and uses it as the instrument for $educ_i$, whilst FEF-IV[3] excludes $hisp_i$ and uses it as the instrument. In these specifications the estimates of the coefficient $educ_i$ are both positive and statistically significant, although the estimate obtained (namely 0.469) when $black_i$ is used as an instrument is rather large and not very precisely estimated (it has a large standard error of 0.216). Once again this could reflect the poor quality of the instrument ($black_i$) in this specification whose correlation with $educ_i$ is −0.037. In contrast, the correlation between $hisp_i$ and $educ_i$ is around −0.20, and the specification FEF-IV[3] which uses $hisp_i$ as the instrument for $educ_i$ yields a much more reasonable estimate for the schooling variable at 0.071, which is closer to the FEF's estimate of 0.104, although the FEF-IV[3] estimate is much less precisely estimated as compared to the FEF estimate.

26.11 Nonlinear unobserved effects panel data models

A prominent example of a nonlinear panel data model is the binary choice model where the variable to be explained, y_{it}, is a random variable taking one of two different outcomes, $y_{it} = 1$ and $y_{it} = 0$, observed for a random sample of N individuals over T time periods. As in the single cross-section case, the discrete outcomes can be viewed as the observed counterpart of a latent continuous random variable crossing a threshold. Consider the following function for the continuous latent random variable, y_{it}^*,

$$y_{it}^* = \alpha_i + \boldsymbol{\beta}'\mathbf{x}_{it} + u_{it}, \;\; i = 1, 2, \ldots, N; t = 1, 2, \ldots, T,$$

where $\mathbf{x}_{it}$ is a vector of exogenous explanatory variables, and α_i could be treated as fixed or random. Instead of observing y_{it}^*, we observe y_{it}, where

$$y_{it} = \begin{cases} 1, \text{ if } y_{it}^* > 0, \\ 0, \text{ if } y_{it}^* \leq 0. \end{cases}$$

Two widely used parametric specifications are the logit and probit models, based on the logistic and the standard normal distributions, respectively. Specifically, under the unobserved effects logit specification, it is assumed that

$$P\left(y_{it} = 1|\mathbf{x}_{it}, \alpha_i\right) = F\left(\alpha_i + \boldsymbol{\beta}'\mathbf{x}_{it}\right) = \frac{e^{\alpha_i + \boldsymbol{\beta}'\mathbf{x}_{it}}}{1 + e^{\alpha_i + \boldsymbol{\beta}'\mathbf{x}_{it}}},$$

while under the unobserved effects probit model

$$P\left(y_{it}=1|\mathbf{x}_{it},\alpha_i\right)=F\left(\alpha_i+\boldsymbol{\beta}'\mathbf{x}_{it}\right)=\Phi\left(\alpha_i+\boldsymbol{\beta}'\mathbf{x}_{it}\right).$$

In the probit specification, the identification condition that $\text{Var}(u_{\text{it}})=\sigma_u^2=1$ is imposed.

If the individual-specific effects, α_i, are assumed to be fixed, then both α_i and $\boldsymbol{\beta}$ are unknown parameters to be estimated for the model $P\left(y_{it}=1|\mathbf{x}_{it},\alpha_i\right)$. However, when T is small, we have again the incidental-parameter problem (Neyman and Scott (1948)). Contrary to the linear-regression case where individual effects, α_i, can be eliminated by taking a linear transformation, in general there is no simple transformation of data to eliminate the incidental parameters from a nonlinear model. Further, the *MLE*s for α_i and $\boldsymbol{\beta}$ are not independent of each other for the discrete-choice models, and when T is fixed, the inconsistency of α_i is transmitted into the *MLE* for $\boldsymbol{\beta}$. Hence, even if N tends to infinity, the *MLE* of $\boldsymbol{\beta}$ remains inconsistent, if T is fixed.

In the case of the logit model with strictly exogenous regressors, an estimator for $\boldsymbol{\beta}$ can be obtained by adopting a conditional likelihood approach, which consists of finding the joint distribution of $y_{i.}$ conditional on $\mathbf{X}_{i.}$, α_i and the statistic $\tau_i=\sum_{t=1}^{T}y_{it}$. The function τ_i is a minimum sufficient statistic for the incidental parameter, α_i. It turns out that such conditional joint distribution does not depend on α_i, and the estimator obtained by maximizing it, the so called fixed-effects logit estimator, is $\sqrt{N}$-consistent and asymptotically normal. Such a conditional likelihood approach does not work for the probit model, since we cannot find simple functions for the parameters of interest that are independent of the nuisance parameters, α_i. However, it is possible to obtain an estimate of $\boldsymbol{\beta}$ for the probit specification under a random effects framework. Suppose that α_i is an unobservable random variable, and assume that $\alpha_i|\mathbf{X}_{i.}\sim N\left(0,\sigma_\alpha^2\right)$. Then the likelihood function for the ith observation can be integrated against the density of α_i to obtain

$$L_i\left(y_{i1},y_{i2},\ldots,y_{iT};\mathbf{X}_{i.},\boldsymbol{\theta}\right)=\int_{-\infty}^{+\infty}\left\{\prod_{t=1}^{T}\Phi\left(\alpha_i+\mathbf{X}_{.t}\boldsymbol{\beta}\right)^{y_{it}}\left[1-\Phi\left(\alpha_i+\mathbf{X}_{.t}\boldsymbol{\beta}\right)\right]^{1-y_{it}}\right\}\times\left(1/\sigma_\alpha^2\right)\phi\left(\alpha_i/\sigma_\alpha^2\right)d\alpha_i.$$

where $\phi(.)$ is the standard normal density. Hence, the log-likelihood function for the full sample can be computed and maximized with respect to $\boldsymbol{\beta}$ and σ_α^2 to obtain $\sqrt{N}$-consistent asymptotically normal estimators. The conditional *MLE* in this context is typically called the random effects probit estimator. For further details, the reader is referred to Wooldridge (2010).

26.12 Unbalanced panels

Until now we have focused on cases where observations on a cross-section of N individuals, families, firms, school districts, are available over the same time periods, denoted by $t=1,2,\ldots,T$. Such data sets are usually called balanced panels because the same time periods are available for all cross-sectional units. A panel is said to be *unbalanced* or *incomplete* if the observations on different groups do not cover the same time periods. Unbalanced panels can arise for various reasons. For example, a variable is unobserved in certain time periods for a subset of cross-sectional units, or the survey design may rotate individuals out of the sample, according

to some pre-specified rules; individuals initially participating in the panel may not be willing or able to participate in some waves. The way unbalanced panels are treated critically depends on whether the cause of missing observations on individual cross-sectional units is random or systematic.

Let $s_{it} = 1$ if $(y_{it}, \mathbf{x}_{it})$ is observed and zero otherwise. Then s_{it} is an indicator variable telling us which time periods are missing for each i, and $\left(y_{it}, \mathbf{x}_{it}, s_{it}\right)$ can be treated as a random sample from the population. If the indicator variable, $s_{it'}$, is independent of the error term u_{it} for all t and t' (although it may be possibly correlated with α_i and/or $\mathbf{X}_{i,T_i}$), then FE on the unbalanced panel remains consistent and asymptotically normal. The unbalanced panel data model in general can now be written as

$$y_{it}^{*} = \alpha_i s_{it} + \mathbf{x}_{it}^{*\prime}\boldsymbol{\beta} + u_{it}^{*}, \tag{26.82}$$

for $i = 1, 2, \ldots, N$ and $t = 1, 2, \ldots, T$, where $y_{it}^{*} = s_{it} y_{it}$, $\mathbf{x}_{it}^{*} = s_{it}\mathbf{x}_{it}$, and $u_{it}^{*} = s_{it} u_{it}$. The fixed-effects estimation approach can now be applied to the above model. We first note that

$$\sum_{t=1}^{T} y_{it}^{*} = \alpha_i \sum_{t=1}^{T} s_{it} + \sum_{t=1}^{T} \mathbf{x}_{it}^{*\prime}\boldsymbol{\beta} + \sum_{t=1}^{T} u_{it}^{*},$$

and assuming that $\sum_{t=1}^{T} s_{it} \neq 0$, (which will be the case if we have at least one time series observation on unit i), we have

$$\alpha_i = \bar{y}_i^{*} - \bar{\mathbf{x}}_i^{*\prime}\boldsymbol{\beta} - \bar{u}_i^{*},$$

where $\bar{\mathbf{x}}_i^{*} = \sum_{t=1}^{T} \mathbf{x}_{it}^{*} / \sum_{t=1}^{T} s_{it}$, $\bar{y}_i^{*} = \sum_{t=1}^{T} y_{it}^{*} / \sum_{t=1}^{T} s_{it}$, and $\bar{u}_i^{*} = \sum_{t=1}^{T} u_{it}^{*} / \sum_{t=1}^{T} s_{it}$. Using this result to eliminate α_i from (26.82), we have (recalling that $y_{it}^{*} = s_{it} y_{it}$)

$$s_{it}(y_{it} - \bar{y}_i^{*}) = s_{it}\left(\mathbf{x}_{it} - \bar{\mathbf{x}}_i^{*}\right)'\boldsymbol{\beta} + s_{it}(u_{it} - \bar{u}_i^{*}),$$

and hence

$$\hat{\boldsymbol{\beta}}_{FE} = \left[\sum_{i=1}^{N}\sum_{t=1}^{T} s_{it}\left(\mathbf{x}_{it} - \bar{\mathbf{x}}_i^{*}\right)\left(\mathbf{x}_{it} - \bar{\mathbf{x}}_i^{*}\right)'\right]^{-1} \sum_{i=1}^{N}\sum_{t=1}^{T} s_{it}\left(\mathbf{x}_{it} - \bar{\mathbf{x}}_i^{*}\right)\left(y_{it} - \bar{y}_i^{*}\right). \tag{26.83}$$

Alternatively, one could drop all missing observations and consider the remaining T_i observations, so that for unit i (stacking available observations) we have

$$\underset{(T_i \times 1)}{\mathbf{y}_{i,T_i}} = \underset{(1 \times 1)\,(T_i \times 1)}{\alpha_i \boldsymbol{\tau}_{T_i}} + \underset{(T_i \times k)\,(k \times 1)}{\mathbf{X}_{i,T_i}\boldsymbol{\beta}} + \underset{(T_i \times 1)}{\mathbf{u}_{i,T_i}}.$$

The FE estimator in this case is then given by

$$\hat{\boldsymbol{\beta}}_{FE} = \left(\sum_{i=1}^{N} \mathbf{X}_{i.,T_i}' \mathbf{M}_{T_i} \mathbf{X}_{i.,T_i}\right)^{-1} \sum_{i=1}^{N} \mathbf{X}_{i.}' \mathbf{M}_{T_i} \mathbf{y}_{i.,T_i}, \tag{26.84}$$

where $\mathbf{M}_{T_i} = \mathbf{I}_{T_i} - \boldsymbol{\tau}_{T_i}(\boldsymbol{\tau}'_{T_i}\boldsymbol{\tau}_{T_i})^{-1}\boldsymbol{\tau}'_{T_i}$. Note that $T_i = \sum_{t=1}^{T} s_{it}$, it is easily seen that the alternative expressions in (26.83) and (26.84) are the same.

A similar approach can be utilized for pooled *OLS* and RE estimators. The pooled *OLS* estimator will be given by

$$\hat{\boldsymbol{\beta}}_{FE} = \left[\sum_{i=1}^{N}\sum_{t=1}^{T} s_{it}\left(\mathbf{x}_{it} - \bar{\mathbf{x}}^{*}\right)\left(\mathbf{x}_{it} - \bar{\mathbf{x}}^{*}\right)'\right]^{-1} \sum_{i=1}^{N}\sum_{t=1}^{T} s_{it}\left(\mathbf{x}_{it} - \bar{\mathbf{x}}^{*}\right)\left(y_{it} - \bar{y}^{*}\right),$$

where $\bar{\mathbf{x}}^{*} = \sum_{i=1}^{N}\sum_{t=1}^{T}\mathbf{x}^{*}_{it} / \sum_{i=1}^{N}\sum_{t=1}^{T} s_{it}$, and $\bar{y}^{*}_{i} = \sum_{i=1}^{N}\sum_{t=1}^{T} y^{*}_{it} / \sum_{i=1}^{N}\sum_{t=1}^{T} s_{it}$.

For RE specification, the relevant quasi-demeaned observations for the unbalanced panel are

$$\tilde{y}_{it} = y_{it} - \phi_i \bar{y}^{*}_{i}, \text{ and } \tilde{\mathbf{x}}_{it} = \mathbf{x}_{it} - \phi_i \bar{\mathbf{x}}^{*}_{i},$$

where

$$\phi_i = 1 - \sqrt{\frac{\sigma^2}{T_i\sigma^2_{\alpha} + \sigma^2}}.$$

To remain consistent and asymptotically normal, this estimator requires the stronger condition that s_{it} is independent of α_i, as well as of $u_{it'}$ for all t and t'.

Sample selection is likely to be a problem and could induce a bias when selection is related to the idiosyncratic errors. One possible way to test for sample selection bias is to compare estimates of the regression equation based on the balanced sub-panel of complete observations only, with estimates from the unbalanced panel by the means of a Hausman test. Significant differences between the estimates should be caused by a non-random response problem. However, note that, since both estimators are inconsistent under the alternative, the power of this test may be limited. As an alternative, a simple test of selection bias has been suggested by Nijman and Verbeek (1992). It consists of augmenting the regression equation with the lagged selection indicator, $s_{i,t-1}$, estimating the model, and performing a t-test for the significance of $s_{i,t-1}$. Under the null hypothesis, u_{it} is uncorrelated with $s_{it'}$ for all t', and selection in the previous time period should not be significant in the equation at time t.

See Nijman and Verbeek (1992) and Wooldridge (2010) for further discussion.

26.13 Further reading

Early contributions to the methods and applications of panel data econometrics include the works carried during the 1950s and 1960s by Hildreth (1950), Tobin (1950), Mundlak (1961), Balestra and Nerlove (1966) and Maddala (1971). A review of these works, as well as an historical essay on the development of panel data models can be found in Nerlove (2002). Comprehensive textbook treatments of panel data methods can be found in Hsiao (2003) and its latest edition Hsiao (2014), Wooldridge (2010), Arellano (2003) and Baltagi (2005).

26.14 Exercises

1. Show that the maximum likelihood estimator for $\boldsymbol{\beta}$ and α_i of model (26.1) obtained by maximizing (26.25) is identical to the FE estimator for these parameters.
2. Show that the FE estimator of $\boldsymbol{\beta}$ in model (26.1) is identical to the *OLS* estimator of $\boldsymbol{\beta}$ in a regression of y_{it} on $\mathbf{x}_{it}$ and N dummies $d_j, j = 1, 2, \ldots, N$, with $d_j = 1$ if $j = i$ and 0 otherwise.
3. Derive the error covariance structure of v_{it} in (26.50), in Example 58. For further details, see Lillard and Weiss (1979).
4. Consider the RE model $y_{it} = \alpha_i + \boldsymbol{\beta}'\mathbf{x}_{it} + u_{it}$, where $\alpha_i \sim IID(\alpha, \sigma_\alpha^2)$, α_i and u_{it} are independently distributed of each other and of $\mathbf{x}_{it'}$ for all i, t, t'. Derive the bias of estimating σ_α^2 by

$$\tilde{\sigma}_\alpha^2 = \frac{\sum_{i=1}^{N}\left(\hat{\alpha}_i - \overline{\hat{\alpha}}\right)^2}{N-1},$$

   where $\hat{\alpha}_i$ is the least squares estimate α_i given by (26.24) and $\overline{\hat{\alpha}} = N^{-1}\sum_{i=1}^{N}\hat{\alpha}_i$.
5. The panel data model

$$y_{it} = \alpha_i + \beta_0 x_{it} + \beta_1 x_{i,t-1} + \varepsilon_{it},$$

   is defined over the groups $i = 1, 2, \ldots, N$, and the unbalanced periods $t = T_{i0} + 1, T_{i0} + 2, \ldots, T_{i1}$. The unit-specific intercepts are assumed to be random

$$\alpha_i = \alpha + \eta_i,$$

   where $\eta_i \sim IID\left(0, \sigma_\eta^2\right)$ and $\varepsilon_{it} \sim IID\left(0, \sigma_i^2\right)$ are distributed independently of $x_{it'}$ for i, t and t'.

   (a) An investigator estimates the following cross-sectional regression

$$\bar{y}_i = a_0 + a_1\bar{x}_i + \bar{v}_i,$$

   where

$$\bar{y}_i = \frac{1}{T_i}\sum_{t=T_{i0}+1}^{T_{i1}} y_{it}, \text{ and } \bar{x}_i = \frac{1}{T_i}\sum_{t=T_{i0}+1}^{T_{i1}} x_{it},$$

   $T_i = T_{i1} - T_{i0}$, and $\bar{v}_i$ is an error term. Show that $a_1 = \beta_0 + \beta_1$, and

$$\bar{v}_i = \eta_i + \bar{\varepsilon}_i - \beta_1\left[\frac{x_{i,T_{i1}} - x_{i,T_{i0}}}{T_i}\right],$$

where

$$\bar{\varepsilon}_i = \frac{1}{T_i} \sum_{t=T_{i0}+1}^{T_{i1}} \varepsilon_{it}.$$

(b) Under what conditions does the cross-sectional regression in (a) yield a consistent estimate of the long-run relationship between y_{it} and x_{it}?

(c) Assuming the conditions in (b) are satisfied, discuss the efficient estimation of a_1 in view of the unbalanced nature of the underlying panel.

(d) If one is interested in the long-run relations, are there any advantages in cross-sectional estimation over panel estimation?

27 Short *T* Dynamic Panel Data Models

27.1 Introduction

So far we have considered panels with strictly exogenous regressors, but often we also wish to estimate economic relationships that are dynamic in nature, namely, for which the data generating process is a panel containing lagged dependent variables. If lagged dependent variables appear as explanatory variables, strict exogeneity of the regressors does not hold, and the maximum-likelihood estimator or the within estimator under the fixed-effects specification is no longer consistent in the case of panel data models where the number of cross-section units, N, is large and T, the number of time periods, is small. This is due to the presence of the incidental parameters problem discussed earlier (see Section 26.4). In addition, the treatment of initial observations in a dynamic process raise a number of theoretical and practical problems. As we shall see in this chapter, the assumption about initial observations plays a crucial role in interpreting the model and formulating consistent estimators. Solving the initial value problem is even more difficult in nonlinear panels, where misspecification in the distribution of initial values can lead to serious bias in the parameter estimates.

27.2 Dynamic panels with short *T* and large *N*

Consider the following $ARDL(1,0)$ model[1]

$$y_{it} = \alpha_i + \lambda y_{i,t-1} + \boldsymbol{\beta}'\mathbf{x}_{it} + u_{it}, \quad i = 1,2,\ldots,N; t = 1,2,\ldots,T, \tag{27.1}$$

where the k-dimensional vector of regressors, $\mathbf{x}_{it}$, is assumed to be strictly exogenous, namely $E\left(u_{it}|\mathbf{x}_{jt'}\right) = 0$, for all i, j, t and t', α_i are the unit-specific effects, λ is a scalar coefficient of the lagged dependent variable, $\boldsymbol{\beta}$ is a k-dimensional coefficient vector. The dynamic process of y_{it} (conditional on $\mathbf{x}_{it}$) is stable when $|\lambda| < 1$. As in the case of panels with strictly exogenous regressors, α_i can be treated as fixed or random. As compared with the static models considered

[1] An introduction to *ARDL* models is provided in Chapter 6.

in Chapter 26, the presence of lagged values of the dependent variable amongst the regressors, creates two complications. First, $y_{i,t-1}$ can no longer be viewed as a strictly exogenous regressor. By construction, the lagged dependent variable, $y_{i,t-1}$, will be correlated with the unit-specific effects, α_i, whether they are fixed or random, and with lagged u_{it}. The second complication arises due the non-vanishing effects of the initial values, y_{i0}, on y_{it}, in small T panels. More explicitly, using (27.1) to solve for y_{it} recursively from the initial states, y_{i0}, we obtain

$$y_{it} = \lambda^t y_{i0} + \sum_{j=0}^{t} \lambda^j \boldsymbol{\beta}' \mathbf{x}_{i,t-j} + \frac{1-\lambda^t}{1-\lambda}\alpha_i + \sum_{j=0}^{t-1} \lambda^j u_{i,t-j}. \tag{27.2}$$

Each observation on the dependent variable can thus be written as the sum of four components: a term depending on initial observations, a component depending on current and past values of the exogenous variables, a modified intercept term that depends on the unit-specific effects, α_i, and a moving average term in past values of the disturbances. It is firstly clear that y_{it} depends on α_i and the initial values, y_{i0}, and the effects of the latter do not vanish when T is small or when λ is close to unity. In such cases, assumptions on initial observations play an important role in determining the properties of the various estimators proposed in the literature (see Nerlove and Balestra (1992)). At one extreme it can be assumed that the initial observations, y_{i0}, are fixed constants specified independently of the parameters of the model. Under this specification there are no unit-specific effects at the initial period, $t = 0$, which considerably simplifies the analysis. However, as pointed out by Nerlove and Balestra (1992), unless there is a specific argument in favour of treating y_{i0} as fixed (see, e.g., the application in Balestra and Nerlove (1966)), in general such an assumption is not justified and can lead to biased estimates. Alternatively, the initial values can be assumed to be random draws from a distribution with a common mean

$$y_{i0} = \mu + \epsilon_i, \tag{27.3}$$

where ϵ_i is assumed to be independent of α_i and u_{it}. In this case, starting values may be seen as representing the initial individual endowments, and their impact on current observations gradually diminishes and eventually vanishes with t. Finally, in the more general case, y_{i0}, could be specified to depend on the unit-specific effects, α_i, and time averages of the regressors. For example,

$$y_{i0} = \frac{\alpha_i + \boldsymbol{\beta}' \bar{\mathbf{x}}_i}{1-\lambda} + \epsilon_i, \tag{27.4}$$

where $T^{-1}\bar{\mathbf{x}}_i = \Sigma_{t=1}^{T} \mathbf{x}_{it}$, and ϵ_i is independent of α_i. This specification is sufficiently general and encompasses a number of other specifications of the initial values considered in the literature as special cases.

According to Nerlove and Balestra (1992), the data generating process of y_{i0} should be quite similar, if not identical, to the process generating subsequent observations. For further discussion on assumptions concerning initial values, the reader is referred to Anderson and Hsiao (1981) (see, in particular, Table 1 in Anderson and Hsiao (1981)), who show the sensitivity of maximum likelihood estimators to alternative assumptions about initial conditions, and Bhargava and Sargan (1983).

27.3 Bias of the *FE* and *RE* estimators

When T is small, both the fixed-effects (FE) and random effects (RE) estimators of λ introduced in Chapter 26 are biased. To understand the source of this bias, recall that both the FE and RE estimators involve (quasi-) deviation of y_{it} from its individual-specific time average, $\bar{y}_i$. Hence, in the presence of lagged dependent variables, the (quasi-) demeaning operation would introduce a correlation of order $O(1/T)$ between the explanatory variables and the error term in the transformed equation (26.16) that renders FE estimators biased in small (short T) samples. To see this, consider for simplicity the dynamic model with no exogenous regressors

$$y_{it} = \alpha_i + \lambda y_{i,t-1} + u_{it}, \tag{27.5}$$

where we assume that y_{i0} are given (non-stochastic), $|\lambda| < 1$, and $u_{it} \sim IID\left(0, \sigma_u^2\right)$. Using (27.2) and setting $\boldsymbol{\beta} = \mathbf{0}$, we have

$$y_{it} = \alpha_i \left(\frac{1-\lambda^t}{1-\lambda}\right) + \lambda^t y_{i0} + \sum_{j=0}^{t-1} \lambda^j u_{i,t-j}. \tag{27.6}$$

For the fixed-effects estimator of λ we have:

$$\hat{\lambda}_{FE} - \lambda = \frac{\frac{1}{NT}\sum_{i=1}^{N}\sum_{t=1}^{T}\left(y_{i,t-1} - \bar{y}_{i,-1}\right)\left(u_{it} - \bar{u}_i\right)}{\frac{1}{NT}\sum_{i=1}^{N}\sum_{t=1}^{T}\left(y_{i,t-1} - \bar{y}_{i,-1}\right)^2}, \tag{27.7}$$

where $\bar{y}_i = T^{-1}\sum_{t=1}^{T} y_{it}$, $\bar{y}_{i,-1} = T^{-1}\sum_{t=1}^{T} y_{i,t-1}$, and $\bar{u}_i = T^{-1}\sum_{t=1}^{T} u_{it}$. For a fixed T and as $N \to \infty$, we have (by the Slutsky Theorem)

$$\underset{N\to\infty}{Plim}\left(\hat{\lambda}_{FE} - \lambda\right) = \frac{\lim_{N\to\infty}\frac{1}{NT}\sum_{i=1}^{N}\sum_{t=1}^{T} E\left[\left(y_{i,t-1} - \bar{y}_{i,-1}\right)\left(u_{it} - \bar{u}_i\right)\right]}{\lim_{N\to\infty}\frac{1}{NT}\sum_{i=1}^{N}\sum_{t=1}^{T} E\left[\left(y_{i,t-1} - \bar{y}_{i,-1}\right)^2\right]}, \tag{27.8}$$

assuming that the limit of the denominator is finite and non-zero. To derive the above limiting values we first note that

$$\begin{aligned}\bar{y}_{i,-1} = \frac{1}{T}\sum_{t=1}^{T} y_{i,t-1} &= \alpha_i\left[\frac{(T-1) - T\lambda + \lambda^T}{T(1-\lambda)^2}\right] + \frac{y_{i0}}{T}\left(\frac{1-\lambda^T}{1-\lambda}\right) \\ &+ \frac{1}{T}\left(\frac{1-\lambda^{T-1}}{1-\lambda}\right)u_{i1} + \frac{1}{T}\left(\frac{1-\lambda^{T-2}}{1-\lambda}\right)u_{i2} + \ldots + \frac{1}{T}\left(\frac{1-\lambda}{1-\lambda}\right)u_{iT}.\end{aligned} \tag{27.9}$$

Using this result and observing that $E\left(u_{it}y_{i,t-1}\right) = 0$, we have

$$\frac{1}{NT}\sum_{i=1}^{N}\sum_{t=1}^{T}E\left[\left(y_{i,t-1}-\bar{y}_{i,-1}\right)\left(u_{it}-\bar{u}_{i}\right)\right]=\frac{1}{NT}\sum_{i=1}^{N}\sum_{t=1}^{T}E\left[\left(y_{i,t-1}-\bar{y}_{i,-1}\right)u_{it}\right] \quad (27.10)$$
$$=-\frac{1}{NT}\sum_{i=1}^{N}\sum_{t=1}^{T}E\left(u_{it}\bar{y}_{i,-1}\right).$$

But using (27.9) we have

$$E\left(u_{it}\bar{y}_{i,-1}\right)=\frac{E(u_{it}\alpha_{i})}{T}\left[\frac{(T-1)-T\lambda+\lambda^{T}}{(1-\lambda)^{2}}\right]+\frac{E(u_{it}y_{i0})}{T}\left(\frac{1-\lambda^{T}}{1-\lambda}\right)$$
$$+\frac{\sigma_{u}^{2}\left(1-\lambda^{T-t}\right)}{T\left(1-\lambda\right)}.$$

Even if it is assumed that $E(u_{it}\alpha_{i})=0=E(u_{it}y_{i0})$, we still have

$$\frac{1}{NT}\sum_{i=1}^{N}\sum_{t=1}^{T}E\left[\left(y_{i,t-1}-\bar{y}_{i,-1}\right)\left(u_{it}-\bar{u}_{i}\right)\right]=-\frac{\sigma_{u}^{2}}{T\left(1-\lambda\right)}\left(1-\frac{1}{T}\frac{1-\lambda^{T}}{1-\lambda}\right).$$

As for the denominator, using a similar line of reasoning it may be verified that

$$\frac{1}{NT}\sum_{t=1}^{T}\sum_{i=1}^{N}E\left[\left(y_{i,t-1}-\bar{y}_{i,-1}\right)^{2}\right]=\frac{\sigma_{u}^{2}}{1-\lambda^{2}}\left[1-\frac{1}{T}-\frac{2\lambda}{(1-\lambda)\,T}\left(1-\frac{1}{T}\frac{1-\lambda^{T}}{1-\lambda}\right)\right]$$
$$=\frac{\sigma_{u}^{2}}{1-\lambda^{2}}+O\left(T^{-1}\right).$$

Therefore, it follows that the small T bias of the FE estimator is

$$\underset{N\to\infty}{Plim}\left(\hat{\lambda}_{FE}-\lambda\right)=-\frac{(1+\lambda)}{T}\left(1-\frac{1}{T}\frac{1-\lambda^{T}}{1-\lambda}\right)\left[1-\frac{1}{T}-\frac{2\lambda}{(1-\lambda)\,T}\left(1-\frac{1}{T}\frac{1-\lambda^{T}}{1-\lambda}\right)\right]^{-1}. \quad (27.11)$$

This bias is often referred to as the Nickell bias, since it was Nickell (1981) who was one of the first to provide a formal derivation of the bias of the FE estimator of λ. The above result can be written more compactly (assuming $|\lambda|<1$) as

$$\underset{N\to\infty}{Plim}\left(\hat{\lambda}_{FE}-\lambda\right)=-\frac{(1+\lambda)}{T}+O\left(T^{-2}\right).$$

The Nickell bias is of order $1/T$, and disappears only if $T\to\infty$. It could be substantial when T is small and/or λ close to unity, which can arise in the case of microeconomic panels of households and firms where T is typically in the range of 3 to 8, as well as for some cross-country data sets where time series represent averages over sub-periods. Note that, for $\lambda\geq 0$, the bias is negative.

At $\lambda = 0$, the bias is given by $Plim_{N\to\infty}\left(\hat{\lambda}_{FE}\right) = -1/T$. Similar results can be obtained for the RE estimator. The properties of the *ML* estimator under different assumptions on initial values y_{i0} have been investigated by Anderson and Hsiao (1981). Kiviet (1995) derives an approximation for the bias of the FE estimator in a dynamic panel data model with exogenous regressors, and suggests a bias-corrected FE estimator that subtracts a consistent estimator of this bias from the original FE estimator.

To avoid the small T bias, transformations of the regression equation for eliminating the α_i alternative to the within transformation are required.

Example 60 (The demand for natural gas) *One important early application of dynamic panel data methods in economics includes the study by Balestra and Nerlove (1966) on the demand for natural gas. One feature of the proposed model is the distinction between 'captive' and new demand for energy and natural gas, where captive energy consumption depends on the existing stock of energy-consuming equipment. This feature is represented in the model by the following relation*

$$G_t = u \cdot \mathrm{K}_t, \tag{27.12}$$

where G_t is the use of gas, K_t is stock of gas-consuming capital, and u is the utilization rate, assumed to be constant over time. Assuming that the capital stock is depreciated at a constant rate, δ, the following relation holds between the capital stock and new investments I_t

$$\mathrm{K}_t = I_t + (1-\delta)\ \mathrm{K}_{t-1}.$$

Applying (27.12), we obtain a corresponding dynamic equation for the incremental change in consumption of natural gas (G_t^*), given by

$$G_t^* = (G_t - G_{t-1}) + \delta G_{t-1}, \tag{27.13}$$

where $G_t^* = u \cdot I_t$, so that the total new demand appears as the sum of the incremental change in consumption, $(G_t - G_{t-1})$, and a 'replacement' demand term, given by δG_{t-1}. Gross investments in gas-consuming equipment, and thus the new demand for gas, is specified as a function of the relative price of gas, P_g, and new demand for total energy, denoted by E^*, namely

$$G_t^* = f\left(P_{g,t}, E_t^*\right). \tag{27.14}$$

A relation similar to (27.13) can be derived for the increment in total energy use

$$E_t^* = E_t - (1-\delta_e)\, E_{t-1}, \tag{27.15}$$

where E_t is the total use of all fuels in period t, and δ_e is the rate of depreciation for energy-using equipment. The model is then closed by specifying a relation explaining the total consumption of energy

$$E_t = f\left(P_{e,t}, Y_t, \mathbf{H}_t\right), \tag{27.16}$$

where $P_{e,t}$ is the price of energy, Y_t is real income and $\mathbf{H}_t$ is a vector of socioeconomic variables. By combining (27.15) and (27.16), and inserting the expression for E in (27.14), the following linearized dynamic equation for total natural gas demand is obtained

$$G_t = \alpha_0 + \boldsymbol{\alpha}_1' \boldsymbol{P}_t + \boldsymbol{\alpha}_2' \boldsymbol{H}_t + \alpha_3 Y_t + \alpha_4 G_{t-1},$$

where $\mathbf{P}_t = (P_{g,t}, P_{e,t})'$, the coefficient attached to the lagged gas consumption is $\alpha_4 = (1 - \delta)$, and hence can be interpreted as linked to the depreciation rate for gas-consuming equipment. This equation is then estimated by OLS and by FE estimator, using data on 36 US states over 13 years. Results are reported in Table I of Balestra and Nerlove (1966). The OLS yields an estimate for α_4 that is above 1, a result that is incompatible with theoretical expectations as it implies a negative depreciation rate for gas equipment. On the other hand, the FE gives an estimate for α_4 of 0.68. According to the authors, the inclusion of state dummy variables seems to reduce the coefficient of the lagged gas variable to too low a level. Such a result may be explained by the negative bias of the FE estimator as obtained above. For further details, see Balestra and Nerlove (1966) and Nerlove (2002).

A panel data equation with lagged dependent variables among the regressors is a particular case of a panel weakly exogenous regressors (see Section 9.3 for a formal definition of weak exogeneity). Kiviet (1999) discusses the finite sample properties of the FE estimator under model (27.1), where the regressors, $\mathbf{x}_{it}$, are allowed to be weakly exogenous by assuming that

$$\mathbf{x}_{it} = \mathbf{w}_{it} + \boldsymbol{\pi} \alpha_i + \phi u_{i,t-1},$$

where $\mathbf{w}_{it}$ is independent of α_j and $u_{j,t-s}$, for all i, j, t and s. Note that under $\phi = 0$, $\mathbf{x}_{it}$ are strictly exogenous, while if $\phi \neq 0$, $\mathbf{x}_{it}$ are weakly exogeneity, due to feedbacks from $u_{i,t-1}$. Under this specification, Kiviet (1999) shows that weak exogeneity has an effect on the FE bias of similar magnitude as the presence of a lagged dependent variable. Even when no lagged dependent variable is present in the model, weak exogeneity will render the FE estimator inconsistent for a fixed T.

27.4 Instrumental variables and generalized method of moments

Ample literature has been developed on instrumental variables (IV) and the generalized method of moments (*GMM*) estimation of dynamic panel data models. In the following, we discuss the developments of this literature, and refer to Chapter 10 for details on the econometric theory underpinning *GMM*.

27.4.1 Anderson and Hsiao

Consider the model

$$y_{it} = \lambda y_{i,t-1} + \boldsymbol{\beta}' \mathbf{x}_{it} + v_{it}, \tag{27.17}$$

$$v_{it} = \alpha_i + u_{it}, \tag{27.18}$$

where $\alpha_i \sim IID(0, \sigma_\alpha^2)$, and $u_{it} \sim IID(0, \sigma_\alpha^2)$ are assumed to be independent of each other. One possible way of eliminating unit–specific effects is to take first differences:

$$\Delta y_{it} = \lambda \Delta y_{i,t-1} + \boldsymbol{\beta}' \Delta \mathbf{x}_{it} + \Delta u_{it}, \tag{27.19}$$

where α_i are eliminated. However, note that

$$E\left(\Delta y_{i,t-1} \Delta u_{it}\right) = E\left(\lambda \Delta u_{i,t-1} \Delta u_{it}\right) \neq 0.$$

Hence applying *OLS* to the first differenced model (27.19) would yield inconsistent estimates. Also note that, even if the u_{it} is serially uncorrelated, Δu_{it} will be correlated over time since

$$E\left(\Delta u_{it} \Delta u_{i,t-s}\right) = \begin{cases} 2\sigma_u^2, & \text{for } s = 0, \\ -\sigma_u^2, & \text{for } s = 1, \\ 0, & \text{for } s > 1. \end{cases}$$

To deal with the problem of the correlation between $\Delta y_{i,t-1}$ and Δu_{it}, Anderson and Hsiao (1981) suggest using an instrumental variable (IV) approach. They note that since $E\left(y_{i,t-2} \Delta u_{it}\right) = 0$, then $y_{i,t-2}$ is a valid instrument for $\Delta y_{i,t-1}$, since it is correlated with $\Delta y_{i,t-1}$ and not correlated with Δu_{it}, as long as u_{it} are not serially correlated. As an example, suppose that $\boldsymbol{\beta} = \mathbf{0}$, and $T = 3$, then we have (assuming that $E(u_{it}\alpha_i) = E(u_{it}y_{i0}) = 0$),

$$E\left(y_{i,t-2} \Delta y_{i,t-1}\right) = -\sigma_u^2 (1 - \lambda) \left(\frac{1 - \lambda^{2(t-1)}}{1 - \lambda^2}\right),$$

which is non-zero (as required by the IV approach) so long as $|\lambda| < 1$. It is clear that $y_{i,t-2}$ as an instrument for $\Delta y_{i,t-1}$ starts to become rather weak as λ moves closer to unity. The IV approach breaks down for $\lambda = 1$.

The IV estimation method delivers consistent but not necessarily efficient estimates of the parameters in the model because, as we shall see later in the chapter, it does not make use of all the available moment conditions. Furthermore, the suggested IV procedure does not take into account the correlation structure on transformed errors. As noted by Alvarez and Arellano (2003), ignoring autocorrelation in the first differenced errors leads to inconsistency of the IV estimator if $T/N \to c > 0$.

For further discussion, see Anderson and Hsiao (1981), and Anderson and Hsiao (1982).

27.4.2 Arellano and Bond

Arellano and Bond (1991) argue that additional instruments can be obtained in a dynamic panel data model if one exploits the orthogonality conditions that exist between lagged values of y_{it} and the disturbances v_{it} (defined by (27.17)). Hence, the authors suggest using a *GMM* approach based on all available moment conditions. From (27.19),

$$y_{i3} - y_{i2} = \lambda \left(y_{i2} - y_{i1}\right) + \boldsymbol{\beta}' \Delta \mathbf{x}_{i3} + \Delta u_{i3}, \tag{27.20}$$

$$y_{i4} - y_{i3} = \lambda \left(y_{i3} - y_{i2}\right) + \boldsymbol{\beta}' \Delta \mathbf{x}_{i4} + \Delta u_{i4}, \tag{27.21}$$

$$y_{i5} - y_{i4} = \lambda \left(y_{i4} - y_{i3}\right) + \boldsymbol{\beta}' \Delta \mathbf{x}_{i5} + \Delta u_{i5}, \tag{27.22}$$

$$\cdots \tag{27.23}$$

$$y_{iT} - y_{i,T-1} = \lambda \left(y_{i,T-1} - y_{i,T-2}\right) + \boldsymbol{\beta}' \Delta \mathbf{x}_{iT} + \Delta u_{iT}. \tag{27.24}$$

In equation (27.20), the valid instrument for $\left(y_{i2} - y_{i1}\right)$ is y_{i1}; in equation (27.21) valid instruments for $\left(y_{i3} - y_{i2}\right)$ are y_{i1} and y_{i2}; while in (27.22) they are y_{i1}, y_{i2}, and y_{i3}, and so forth until equation (27.24), where the valid instruments are $y_{i1}, y_{i2}, \ldots, y_{i,T-2}$. Hence, an additional valid instrument is added with each additional time period. Clearly, the appropriate instruments for $\Delta \mathbf{x}_{it}$ are themselves, since, by assumption, $\mathbf{x}_{it}$ are strictly exogenous. Hence, there is a total of $T(T-1)/2$ available instruments or moment conditions for $\Delta y_{i,t-1}$ that are given by

$$E\left[y_{is}\left(\Delta y_{it} - \lambda \Delta y_{i,t-1} - \boldsymbol{\beta}' \Delta \mathbf{x}_{it}\right)\right] = 0, s = 0, 1, \ldots, t-2; t = 2, 3, \ldots, T.$$

To deal with the serial correlation in the transformed disturbances, Δu_{it}, Arellano and Bond (1991) apply the *GMM* method to the stacked observations

$$\Delta \mathbf{y}_{i.} = \lambda \Delta \mathbf{y}_{i.,-1} + \Delta \mathbf{X}_{i.} \boldsymbol{\beta} + \Delta \mathbf{u}_{i.}, \tag{27.25}$$

where

$$\Delta \mathbf{y}_{i.} = \begin{pmatrix} \Delta y_{i2} \\ \Delta y_{i3} \\ \vdots \\ \Delta y_{iT} \end{pmatrix}, \quad \Delta \mathbf{X}_{i.} = \begin{pmatrix} \Delta \mathbf{x}'_{i2} \\ \Delta \mathbf{x}'_{i3} \\ \vdots \\ \Delta \mathbf{x}'_{iT} \end{pmatrix},$$

$$\Delta \mathbf{y}_{i.,-1} = \begin{pmatrix} \Delta y_{i1} \\ \Delta y_{i2} \\ \vdots \\ \Delta y_{i,T-1} \end{pmatrix}, \quad \Delta \mathbf{u}_{i.} = \begin{pmatrix} \Delta u_{i2} \\ \Delta u_{i3} \\ \vdots \\ \Delta u_{iT} \end{pmatrix}. \tag{27.26}$$

Let

$$\mathbf{W}_i = \begin{pmatrix} y_{i1} & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{0} & y_{i1}, y_{i2} & \cdots & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{0} & \mathbf{0} & \cdots & y_{i1}, \ldots, y_{i,T-2} \end{pmatrix}, \tag{27.27}$$

be the matrix of instruments. Then moment conditions can be expressed compactly as

$$E\left(\mathbf{W}'_i \Delta \mathbf{u}_{i.}\right) = \mathbf{0}. \tag{27.28}$$

Holtz-Eakin, Newey, and Rosen (1988) have also considered a *GMM* estimator based on similar conditions. Stacking the observations on all the N different groups

$$\Delta \mathbf{y} = \lambda \Delta \mathbf{y}_{-1} + \Delta \mathbf{X}\boldsymbol{\beta} + \Delta \mathbf{u}, \tag{27.29}$$

where

$$\Delta \mathbf{y} = \begin{pmatrix} \Delta \mathbf{y}_{1.} \\ \Delta \mathbf{y}_{2.} \\ \vdots \\ \Delta \mathbf{y}_{N.} \end{pmatrix}, \ \Delta \mathbf{y}_{-1} = \begin{pmatrix} \Delta \mathbf{y}_{1.,-1} \\ \Delta \mathbf{y}_{2.,-1} \\ \vdots \\ \Delta \mathbf{y}_{N.,-1} \end{pmatrix}, \ \Delta \mathbf{X} = \begin{pmatrix} \Delta \mathbf{x}_{1.} \\ \Delta \mathbf{x}_{2.} \\ \vdots \\ \Delta \mathbf{x}_{N.} \end{pmatrix}.$$

Also, let

$$\mathbf{W} = \begin{pmatrix} \mathbf{W}_1 \\ \mathbf{W}_2 \\ \vdots \\ \mathbf{W}_N \end{pmatrix},$$

be the matrix of instruments. Then, pre-multiplying both sides of (27.29) by $\mathbf{W}'$ yields

$$\mathbf{W}'\Delta \mathbf{y} = \lambda \mathbf{W}'\Delta \mathbf{y}_{-1} + \mathbf{W}'\Delta \mathbf{X}\boldsymbol{\beta} + \mathbf{W}'\Delta \mathbf{u}. \tag{27.30}$$

Similarly,

$$(\Delta \mathbf{X})' \Delta \mathbf{y} = \lambda (\Delta \mathbf{X})' \Delta \mathbf{y}_{-1} + (\Delta \mathbf{X})' (\Delta \mathbf{X})\boldsymbol{\beta} + (\Delta \mathbf{X})' \Delta \mathbf{u}. \tag{27.31}$$

Let $\mathbf{Z} = (\mathbf{W}, \Delta \mathbf{X})$, and note that

$$E\left(\mathbf{Z}'\Delta \mathbf{u}\right) = \mathbf{0}. \tag{27.32}$$

However, moments (27.32) still do not account for the serial correlation in the differenced error term. Due to the first-order moving average structure of the error terms we also have

$$E\left(\mathbf{Z}'\mathbf{u}\mathbf{u}'\mathbf{Z}\right) = \mathbf{Z}'\left[\sigma^2 \left(\mathbf{I}_N \otimes \mathbf{A}\right)\right]\mathbf{Z}, \tag{27.33}$$

where

$$\underset{(T-2) \times (T-2)}{\mathbf{A}} = \begin{pmatrix} 2 & -1 & \cdots & 0 & 0 \\ -1 & 2 & \cdots & 0 & 0 \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ 0 & 0 & \cdots & 2 & -1 \\ 0 & 0 & \cdots & -1 & 2 \end{pmatrix}.$$

The *GMM* method can now be applied to the above moment conditions to obtain

$$\hat{\boldsymbol{\gamma}}_{GMM} = \left(\mathbf{G}'\mathbf{Z}\mathbf{S}_N\mathbf{Z}'\mathbf{G}\right)^{-1} \mathbf{G}'\mathbf{Z}\mathbf{S}_N\mathbf{Z}'\boldsymbol{\Delta}\mathbf{y}, \tag{27.34}$$

where $\hat{\boldsymbol{\gamma}}_{GMM} = (\hat{\boldsymbol{\lambda}}_{GMM}, \hat{\boldsymbol{\beta}}'_{GMM})'$, $\mathbf{G} = (\Delta\mathbf{y}_{-1}, \Delta\mathbf{X})$. Alternative choices for the weights $\mathbf{S}_N$ give rise to a set of *GMM* estimators based on the moment conditions in equation (27.32), all of which are consistent for large N and finite T, but which differ in their asymptotic efficiency. It is possible to show that the asymptotically optimal weights are given by

$$\mathbf{S}_N = \left(\sum_{i=1}^{N} \mathbf{Z}'_{i.} \hat{\mathbf{u}}_{i.} \hat{\mathbf{u}}'_{i.} \mathbf{Z}_{i.} \right)^{-1}, \tag{27.35}$$

with $\mathbf{Z}_{i.} = (\mathbf{W}_i, \Delta\mathbf{x}_{i.})$, and $\hat{\mathbf{u}}_{i.}$ are the residuals from a consistent estimate, for example, from preliminary IV estimates of $\boldsymbol{\beta}$ and λ. Such preliminary estimates are given by

$$\hat{\boldsymbol{\gamma}} = \left[\mathbf{G}'\mathbf{Z} \left(\mathbf{Z}'\boldsymbol{\Omega}\mathbf{Z} \right)^{-1} \mathbf{Z}'\mathbf{G} \right]^{-1} \mathbf{G}'\mathbf{Z} \left(\mathbf{Z}'\boldsymbol{\Omega}\mathbf{Z} \right)^{-1} \mathbf{Z}'\Delta\mathbf{y},$$

where $\boldsymbol{\Omega} = (\mathbf{I}_N \otimes \mathbf{A})$. The *GMM* estimator (27.34) with weighting matrix (27.35) is known in the literature as the two-step *GMM* estimator. Note that if any of the $\mathbf{x}_{it}$ variables are predetermined rather than strictly exogenous with $E(\mathbf{x}_{it} v_{is}) = \mathbf{0}$ for $s < t$, and zero otherwise, then only $\left(\mathbf{x}'_{i1}, \mathbf{x}'_{i2}, \ldots, \mathbf{x}'_{i,t-1}\right)'$ are valid instruments for the differenced equation at period t. In this case, the matrix $\mathbf{W}_i$ can be expanded with further columns containing the lagged values $\left(\mathbf{x}'_{i1}, \mathbf{x}'_{i2}, \ldots, \mathbf{x}'_{i,t-1}\right)'$.

In the absence of any additional knowledge about initial conditions for the dynamic processes, the estimator (27.34) with weighting matrix (27.35) is asymptotically normal and is efficient in the class of estimators based on the linear moment conditions (Hansen (1982), Chamberlain (1987)). However, as shown in Blundell and Bond (1998) and Binder, Hsiao, and Pesaran (2005), the performance of the IV and of the one-step and two-step *GMM* estimators deteriorates as the variance of α_i increases relative to the variance of the idiosyncratic error, u_{it}, or when λ is close to 1. Indeed, in these cases it is possible to show that the instruments $y_{i,t-s}$ are only weakly related with the differences Δy_{it}. A further complication with *GMM* arises when T is not small. As $T \to \infty$, the number of *GMM* orthogonality conditions $r = T(T-1)/2$ also tend to infinity. In this case, Alvarez and Arellano (2003) show that the *GMM* remains asymptotically normal, but, unless $lim(T/N) = 0$, it exhibits a bias of order $O(1/N)$. Koenker and Machado (1999) finds conditions on r for the limiting distribution of the *GMM* estimator to remain valid.

Another important point to note is that consistency of the *GMM* estimator relies upon the fact that errors are serially uncorrelated, i.e., that $E\left(\Delta u_{it} \Delta u_{i,t-2}\right) = 0$. In the case of serially correlated errors, the *GMM* estimator would lose its consistency. Hence, Arellano and Bond (1991) suggest testing the hypothesis that the second-order autocovariances for all periods in the sample are zero, based on residuals from first difference equations. See Arellano and Bond (1991, p. 282).

27.4.3 Ahn and Schmidt

Ahn and Schmidt (1995) observe that, under the standard assumptions used in a dynamic panel data, there are $T-2$ additional moment conditions that are ignored by the Anderson and Hsiao (1981) and Arellano and Bond (1991) estimators. In particular, these moment conditions are implied by the assumptions that the u_{it}, apart from being serially uncorrelated, are also assumed

to be uncorrelated with α_i and y_{i0}. As a result, using (27.17), they identify the following additional moment conditions

$$E\left(v_{iT}\Delta v_{it}\right) = 0, t = 1, 3, \ldots, T-1. \tag{27.36}$$

Exploiting the assumption that the error term has constant variance through time

$$E\left(v_{it}^2\right) = \sigma_i^2, \ t = 1, 2, \ldots, T,$$

the authors propose further additional $T-1$ moment conditions given by

$$E\left(y_{i,t-2}\Delta v_{i,t-1} - y_{i,t-1}\Delta v_{it}\right) = 0, \ t = 2, 3, \ldots, T.$$

The above moments can be combined with the moment conditions already introduced by Arellano and Bond (1991), and further columns can be added to the instrument matrix in (27.27). Calculation of the one-step and two-step *GMM* estimators then proceeds exactly as described above.

27.4.4 Arellano and Bover: Models with time-invariant regressors

Arellano and Bover (1995) consider efficient IV estimation within the context of the following general dynamic model, first introduced by Hausman and Taylor (1981),

$$\begin{aligned} y_{it} &= \boldsymbol{\beta}'\mathbf{x}_{it} + \boldsymbol{\gamma}'\mathbf{z}_i + v_{it}, \\ v_{it} &= \alpha_i + u_{it}, \end{aligned}$$

where lagged values of y_{it} are now included in $\mathbf{x}_{it}$, $\mathbf{z}_i$ are time-invariant variables, and α_i are now assumed to be random. More compactly the first equation can be written as

$$y_{it} = \boldsymbol{\delta}'\mathbf{x}_{it}^* + v_{it},$$

where $\mathbf{x}_{it}^* = (\mathbf{x}_{it}, \mathbf{z}_i)$. Arellano and Bover propose the following nonsingular transformation of the above system equations

$$\underset{T\times T}{\mathbf{H}} = \begin{pmatrix} \mathbf{C} \\ \mathbf{1}_T'/T \end{pmatrix}$$

where $\mathbf{1}_T = (1, 1, \ldots, 1)'_{T\times 1}$, $\mathbf{C}$ is any $(T-1)\times T$ matrix of rank $(T-1)$ such that $\mathbf{C}\mathbf{1}_T = \mathbf{0}$. For example, $\mathbf{C}$ could be the first $(T-1)$ rows of the within group transformation or the first difference operator. The transformed disturbances are

$$\mathbf{v}_{i.}^* = \mathbf{H}\mathbf{v}_{i.} = \begin{pmatrix} \mathbf{C}\mathbf{v}_{i.} \\ \bar{u}_{i.} \end{pmatrix}.$$

Since the first $(T-1)$ elements of $\mathbf{v}_{i.}^*$ do not contain α_i, all exogenous variables are valid instruments for the first $(T-1)$ equations of the transformed model. Let $\mathbf{w}_{i.} = \left(\mathbf{x}_{i.}', \mathbf{z}_i'\right)'$ and $\mathbf{m}_{i.}$

be a vector containing a subset of variables of $\mathbf{w}_i$ assumed to be uncorrelated in levels and such that $dim\,(\mathbf{m}_{i.}) > dim\,(\boldsymbol{\gamma})$, a valid matrix of instruments for the complete transformed system, is given by

$$\mathbf{W}_i = \begin{pmatrix} \mathbf{w}_{i.}' & \mathbf{0} & \dots & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{w}_{i.}' & \dots & \mathbf{0} & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ \mathbf{0} & \mathbf{0} & \dots & \mathbf{w}_{i.}' & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \dots & \mathbf{0} & \mathbf{m}_{i.}' \end{pmatrix},$$

and the associated moment conditions are given by

$$E\,(\mathbf{W}_i \mathbf{H} \mathbf{v}_{i.}) = \mathbf{0}.$$

Let $\mathbf{H}^* = (\mathbf{I}_N \otimes \mathbf{H})$, $\boldsymbol{\Omega}^* = (\mathbf{I}_N \otimes \boldsymbol{\Omega})$, $\mathbf{X}^* = \left(\mathbf{X}_{1.}^{*\prime}, \mathbf{X}_{2.}^{*\prime}, \dots, \mathbf{X}_{N.}^{*\prime}\right)'$, and $\mathbf{W} = \left(\mathbf{W}_{1.}', \mathbf{W}_{2.}', \dots, \mathbf{W}_{N.}'\right)'$. The *GMM* estimator based on the above moment conditions is

$$\hat{\boldsymbol{\delta}} = \left[\mathbf{X}^{*\prime}\mathbf{H}^{*\prime}\mathbf{W}\left(\mathbf{W}'\mathbf{H}^*\boldsymbol{\Omega}^*\mathbf{H}^{*\prime}\mathbf{W}\right)^{-1}\mathbf{W}'\mathbf{H}^*\mathbf{X}^*\right]^{-1}\left[\mathbf{X}^{*\prime}\mathbf{H}^{*\prime}\mathbf{W}\left(\mathbf{W}'\mathbf{H}^*\boldsymbol{\Omega}^*\mathbf{H}^{*\prime}\mathbf{W}\right)^{-1}\mathbf{W}'\mathbf{H}^*\mathbf{y}\right].$$

In practice, the covariance matrix of the transformed system, $\mathbf{S} = \mathbf{H}\boldsymbol{\Omega}^*\mathbf{H}'$, will be replaced by a consistent estimator. An unrestricted estimator of $\mathbf{S}$ is

$$\hat{\mathbf{S}} = \frac{1}{N}\sum_{i=1}^{N} \hat{\mathbf{v}}_{i.}^* \hat{\mathbf{v}}_{i.}^{*\prime},$$

where the $\hat{\mathbf{v}}_{i.}^*$ are residuals based on some consistent preliminary estimates. Because the set of instruments, $\mathbf{W}_i$, is block-diagonal, Arellano and Bover show that $\hat{\boldsymbol{\delta}}$ is invariant to the choice of $\mathbf{C}$. Another advantage of their representation is that the form of $\boldsymbol{\Omega}$ need not be known. Further, this approach can be easily extended to the dynamic panel data case. See Arellano and Bover (1995) for details.

Example 61 (Intertemporal dependency in alcohol and tobacco consumption) *In a recent paper, Browning and Collado (2007) study the temporal patterns of consumption for a number of goods categories, using quarterly data on expenditure, income and other characteristic for a set of 3,200 Spanish households over the period 1985Q1 to 1996Q4. The authors distinguish between habits (state-dependence) from correlated heterogeneity in consumption behaviour, by adopting the following dynamic panel with random group effects*

$$w_{iht} = \alpha_i + \gamma_i w_{ih,t-1} + \beta_{i1} \ln x_{ht} + \beta_{i1} (\ln x_{ht})^2 + \sum_k \delta_{ik} z_{kht} + u_{iht},$$

$$u_{iht} = \lambda_{ih} + \varepsilon_{iht},$$

Table 27.1 Arellano-Bover *GMM* estimates of budget shares determinants

	food-in	nds	food-out	alct	clo	sdur
lxtot	−10.4360***	−0.9711	2.4451**	−1.2733***	5.5732***	4.4151***
	(1.9042)	(1.8716)	(1.2255)	(0.4875)	(1.5803)	(1.0275)
lagged budget share	0.0245	0.1468	0.4102***	0.1723**	−0.1132**	−0.3167***
	(0.0459)	(0.0966)	(0.0953)	(0.0677)	(0.0541)	(0.0618)
nch	2.0234***	−0.5354**	−0.3654***	0.0661	−0.2880	−0.6907***
	(0.2412)	(0.2296)	(0.1416)	(0.0636)	(0.1897)	(0.1234)
nad	0.4486	−0.1775	0.4908**	0.3942***	−0.9466***	−0.7361***
	(0.3223)	(0.3326)	(0.2099)	(0.0985)	(0.2871)	(0.1930)
hage	0.5715**	0.6454***	−0.2131	−0.2075***	−0.0706	−0.5410***
	(0.2388)	(0.2425)	(0.1474)	(0.0689)	(0.1898)	(0.1401)
hage2	−0.0041	−0.0070***	0.0019	0.0019***	0.0004	0.0050***
	(0.0026)	(0.0027)	(0.0016)	(0.0007)	(0.0021)	(0.0015)
const	151.5133***	25.5409	−22.1730*	23.6900***	−52.3732***	−33.6156***
	(22.1229)	(20.9289)	(13.2100)	(5.9059)	(17.4310)	(11.3386)
Sargan test	103.09	92.49	75.44	73.15	104.28	93.74
df	81	81	81	81	81	81
p-value	0.0495	0.1800	0.6535	0.7208	0.0418	0.1575

Notes: food-in: food at home; food-out: food outside the home; alct: alcohol and tobacco; clo: clothing; nds: other nondurables and services; sdur: small durables such as books, etc.; nch: number of children, nad: number of adults in the household; hage: age of the husband; hagez: age squared of the husband. Standard errors in parentheses.
*, **, and *** denote significance at 10%, 5% and 1% levels.

where w_{iht} is the budget share for good i, by household h, at time t, x_{ht} is total expenditure deflated by a price index, z_{kht} is a list of demographics and time and seasonal dummies, and ε_{iht} is a possibly autocorrelated error term. In the above specification, the random group effect, λ_{ih}, allows for persistent individual heterogeneity, while the coefficient γ_i captures the state-dependence present in the data. The Arellano and Bover (1995) approach is then adopted to estimate the above model. Results, reported on Table 27.1, show that lagged budget shares are significant for food-out, alcohol and tobacco, clothing and small durables, whereas for food-in and non-durables and services there is no evidence of state dependence once they control for unobserved heterogeneity. The positive coefficient of the lagged budget shares for food-out and alcohol and tobacco is consistent with habit formation in those commodities, while the negative sign for clothing and for small durables reflects the durability of these two goods. The estimated elasticities show that, as expected, food-in and alcohol and tobacco are necessities, whereas food-out, clothing and small durables are luxuries. See Browning and Collado (2007) for further details.

27.4.5 Blundell and Bond

Blundell and Bond (1998) impose restrictions on the distribution of the initial values, y_{i0}, that allow the use of lagged differences of y_{it} as instruments in the levels equations. This further restriction (if valid) is particularly important when λ is close to unity or when $\sigma_\alpha^2/\sigma_u^2$ becomes large, since in these cases lagged levels are weak instruments in the differenced equations.

In particular, Blundell and Bond (1998) consider the following general process for the initial observations in a pure dynamic panel data model with no exogenous variables

$$y_{i0} = \frac{\alpha_i}{1-\lambda} + u_{i0}, \quad i = 1, 2, \ldots, N, \tag{27.37}$$

under the assumption that

$$E\left(\Delta y_{i1}\alpha_i\right) = 0. \tag{27.38}$$

The above condition states that deviations of the initial conditions from $\alpha_i/(1-\lambda)$ are uncorrelated with the level of $\frac{\alpha_i}{1-\lambda}$ itself. To guarantee this, it may be assumed that

$$E\left[\left(y_{i0} - \frac{\alpha_i}{1-\lambda}\right)\alpha_i\right] = 0.$$

If (27.38) holds in addition to other standard assumptions, then the following $T-1$ additional moment conditions are available

$$E\left[\left(y_{it} - \lambda y_{i,t-1}\right)\Delta y_{i,t-1}\right] = 0, \text{ for } t = 2, 3, \ldots, T.$$

Hence, calculation of the one-step and two-step *GMM* estimators proceeds as described above, adding T columns to the matrix of instruments (27.27). This estimator is known as system *GMM*. Blundell and Bond (1998) show that the above moment conditions remain informative when λ is close to unity or when $\sigma_\alpha^2/\sigma_u^2$ is large. A Monte Carlo exercise provided in Blundell, Bond, and Windmeijer (2000) shows that the use of these additional moment conditions yields substantial gains in terms of the properties of the 2-step *GMM* estimators, especially in the 'weak instrument' case. But when using this estimator it is important to bear in mind that its validity critically depends on assumption (27.38).

Example 62 (*GMM* estimation of production functions) *Consider the simple Cobb–Douglas production function expressed in logs for research and development (R&D) firms*

$$y_{it} = \beta_n n_{it} + \beta_k k_{it} + \gamma_t + (\alpha_i + v_{it} + m_{it}), \tag{27.39}$$

$$v_{it} = \rho v_{i,t-1} + \varepsilon_{it}, \tag{27.40}$$

where y_{it} is the log of sales of firm i in year t, n_{it} is the log of employment, k_{it} is the log of capital stock, γ_t is a year-specific intercept, α_i is an unobserved firm-specific effect, v_{it} is an autoregressive productivity shock, and m_{it} reflects serially uncorrelated measurement errors. Constant returns to scale would imply $\beta_n + \beta_k = 1$. Estimation of the above production function is subject to a number of econometric issues, including measurement errors in output and capital, and simultaneity arising from potential correlation between observed inputs and productivity shocks (for example, managerial ability). See, for example, Griliches and Mairesse (1997). GMM methods could be used to control for these sources of bias. To this end, note that model (27.39)–(27.40) has the dynamic common factor representation

$$y_{it} = \pi_1 n_{it} + \pi_2 n_{i,t-1} + \pi_3 k_{it} + \pi_4 k_{i,t-1} + \pi_5 y_{i,t-1} + \gamma_t^* + \left(\alpha_i^* + w_{it}\right),$$

subject to the restrictions $\pi_2 = -\pi_1\pi_5$, *and* $\pi_4 = -\pi_3\pi_5$, $\gamma_t^* = \gamma_t - \rho\gamma_{t-1}$, *and* $\alpha_i^* = \alpha_i(1-\rho)$. *As noted by Blundell and Bond (2000), we have* $w_{it} = \varepsilon_{it}$ *if there are no measurement errors (i.e., if* $m_{it} = 0$*), while* w_{it} *is an MA(1) process under measurement errors. These restrictions can be tested and imposed in the minimum distance estimator, to obtain the restricted parameter vector* (β_n, β_k, ρ). *However, the use of lagged instruments to correct for simultaneity in the first-differenced equations has tended to produce very unsatisfactory results in this context. One possible explanation for these problems is the weak correlation that exists between the current growth rates of firm sales, capital and employment, and the lagged levels of these variables (Blundell and Bond (2000)). Weak instruments could cause large finite-sample biases when using the first differenced GMM procedure to estimate autoregressive models for moderately persistent series from moderately short panels. In this case, the system GMM approach would be appropriate. Under the assumption that* $E(\Delta n_{it}\alpha_i^*) = E(\Delta k_{it}\alpha_i^*) = 0$, *and that the initial conditions satisfy* $E(\Delta y_{i1}\alpha_i^*) = 0$, *the additional moment conditions* $E[\Delta x_{i,t-s}(\alpha_i^* + w_{it})] = 0$ *can be used, where* $x_{it} = (y_{it}, n_{it}, k_{it})$, *and* $s = 1$ *under no measurement errors,* $s = 2$ *otherwise. The above conditions allow the use of suitably lagged first differences of the variables as instruments for the equations in levels. Blundell and Bond (2000) consider GMM estimation of the above production function using a balanced panel of 509 R&D-performing US manufacturing companies observed over the years 1982 to 1989. Table 27.2 reports results for the OLS and FE estimators, the Arellano and Bond (1991) estimator using lagged levels dated* $t-2$ *and* $t-3$ *as instruments, and the corresponding system GMM*

Table 27.2 Production function estimates

	***OLS* Levels**	**Within groups**	**DIF t-2**	**DIF t-3**	**SYS t-2**	**SYS t-3**
n_t	0.479	0.488	0.513	0.499	0.629	0.472
	(.029)	*(.030)*	*(.089)*	*(.101)*	*(.106)*	*(.112)*
n_{t-1}	−0.423	−0.023	0.073	−0.147	−0.092	−0.278
	(.031)	*(.034)*	*(.093)*	*(.113)*	*(.108)*	*(.120)*
k_t	0.235	0.177	0.132	0.194	0.361	0.398
	(.035)	*(.034)*	*(.118)*	*(.154)*	*(.129)*	*(.152)*
k_{t-1}	−0.212	−0.131	−0.207	−0.105	−0.326	−0.209
	(.035)	*(.025)*	*(.095)*	*(.110)*	*(.104)*	*(.119)*
y_{t-1}	0.922	0.404	0.326	0.426	0.462	0.602
	(.011)	*(0.29)*	*(.052)*	*(.079)*	*(.051)*	*(.098)*
m1	−2.60	−8.89	−6.21	−4.84	−8.14	−6.53
m2	−2.06	−1.09	−1.36	−0.69	−0.59	−0.35
Sargan	–	–	.001	.073	.000	.032
Dif Sargan	–	–	–	–	.001	.102
β_n	0.538	0.488	0.583	0.515	0.773	0.479
	(.025)	*(.030)*	*(.085)*	*(.099)*	*(.093)*	*(.098)*
β_k	0.266	0.199	0.062	0.225	0.231	0.492
	(.032)	*(.033)*	*(.079)*	*(.126)*	*(.075)*	*(.074)*
ρ	0.964	0.512	0.377	0.448	0.509	0.565
	(.006)	*(.022)*	*(.049)*	*(.073)*	*(.048)*	*(.078)*
Comfac	.000	.000	.041	.711	.012	.772
CRS	.000	.000	.000	.006	.922	.641

Asymptotic standard errors in parentheses. Year dummies included in all models.

estimator. Estimates are both restricted and unrestricted. As expected in the presence of group effects, the OLS shows an upward-bias in the estimate of π_5, while the FE estimator appears to give a downward-biased estimate of this coefficient. See Blundell and Bond (2000) for further details. We also refer to Levinshon and Petrin (2003) and Ackerberg, Caves, and Frazer (2006) for alternative extended GMM approaches for consistent estimation of production function parameters.

27.4.6 Testing for overidentifying restrictions

The validity of the moment conditions implied by the dynamic panel data model can be tested using the conventional *GMM* test of over-identifying restrictions (see Section 10.7, Sargan (1958), and Hansen (1982))

$$H = \Delta\hat{\mathbf{u}}'\mathbf{W}\left(\sum_{i=1}^{N}\mathbf{W}_i'\Delta\hat{\mathbf{u}}_i\Delta\hat{\mathbf{u}}_i'\mathbf{W}_i\right)^{-1}\mathbf{W}\Delta\hat{\mathbf{u}} \sim \chi^2_{r-k-1},$$

where r (assumed to be greater than k) refers to the number of columns of the matrix $\Delta\mathbf{W}$, containing the instruments, and $\Delta\hat{\mathbf{u}}$ denotes the residuals from the two-step estimation. In a Monte Carlo study, Bowsher (2002) shows that, when T is large, using too many moment conditions causes the above test to have extremely low power.

27.5 Keane and Runkle method

Keane and Runkle (1992) consider the model

$$\begin{aligned} y_{it} &= \boldsymbol{\beta}'\mathbf{x}_{it} + v_{it}, \\ v_{it} &= \alpha_i + u_{it}, \end{aligned} \tag{27.41}$$

where $\mathbf{x}_{it}$ is allowed to contain lagged dependent variables as regressors, and

$$E\left(v_{it}\,|\mathbf{x}_{it}\right) \neq 0. \tag{27.42}$$

Suppose there exists a set of predetermined instruments, $\mathbf{z}_{it}$, such that

$$\begin{aligned} E(v_{it}\,|\mathbf{z}_{is}) &= 0, \quad \text{for} \quad s \leq t, \\ E(v_{it}\,|\mathbf{z}_{is}) &\neq 0, \quad \text{for} \quad s > t. \end{aligned}$$

Hence, $\mathbf{z}_{it}$ may contain lagged values of y_{it}. Keane and Runkle consider the general covariance specification

$$E\left(\mathbf{v}\mathbf{v}'\right) = \mathbf{I}_N \otimes \boldsymbol{\Sigma}, \tag{27.43}$$

where

$$\underset{(NT \times 1)}{\mathbf{v}} = \begin{pmatrix} \mathbf{v}_{1.} \\ \mathbf{v}_{2.} \\ \vdots \\ \mathbf{v}_{N.} \end{pmatrix},$$

$\mathbf{v}_{i.} = (v_{i1}, v_{i2}, \ldots, v_{iT})'$, and $\boldsymbol{\Sigma} = E\left(\mathbf{v}_{i.}\mathbf{v}_{i.}'\right)$ is no longer constrained to be equal to $\mathbf{A}$ as in the case of the Arellano and Bond (1991) estimator. Stacking all the group regressions we have

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{v}, \tag{27.44}$$

where $\mathbf{y} = \left(\mathbf{y}_{1.}, \mathbf{y}_{2.}, \ldots, \mathbf{y}_{N.}\right)'$ and $\mathbf{X} = \left(\mathbf{X}_{1.}', \mathbf{X}_{2.}', \ldots, \mathbf{X}_{N.}'\right)'$. The Keane and Runkle estimator is obtained by applying the forward filtering approach by Hayashi and Sims (1983) to the above stacked form of the panel. Forward filtering eliminates the serial correlation pattern, and yields a more efficient estimator than the standard 2*SLS* estimator. First, $\boldsymbol{\Sigma}^{-1}$ is decomposed using the Cholesky decomposition

$$\boldsymbol{\Sigma}^{-1} = \mathbf{P}'\mathbf{P},$$

where $\mathbf{P}$ is an upper triangular matrix. Hence, both sides of (27.44) are pre-multiplied by $\mathbf{S} = \mathbf{I}_N \otimes \mathbf{P}$, yielding

$$\mathbf{S}\mathbf{y} = \mathbf{S}\mathbf{X}\boldsymbol{\beta} + \mathbf{S}\mathbf{u}.$$

$\boldsymbol{\beta}$ is then estimated by 2*SLS*, using as instruments the matrix $\mathbf{Z}$, namely

$$\hat{\boldsymbol{\beta}}_{KR} = \left(\mathbf{X}'\mathbf{S}'\mathbf{P}_z\mathbf{S}\mathbf{X}\right)^{-1}\mathbf{X}'\mathbf{S}'\mathbf{P}_z\mathbf{S}\mathbf{y}, \tag{27.45}$$

where $\mathbf{P}_z = \mathbf{Z}\left(\mathbf{Z}'\mathbf{Z}\right)^{-1}\mathbf{Z}'$. Matrix $\boldsymbol{\Sigma}$ can be estimated using consistent IV residuals from a preliminary estimation

$$\hat{\boldsymbol{\Sigma}} = \frac{1}{N}\sum_{i=1}^{N}\hat{\mathbf{v}}_{i.}'\hat{\mathbf{v}}_{i.}, \tag{27.46}$$

where $\hat{\mathbf{v}}_i$ are the preliminary IV residuals.

27.6 Transformed likelihood approach

The literature discussed so far has mostly focused on *GMM* as a general method for the estimation of dynamic panel data models, and has by and large neglected the maximum likelihood approach. Indeed, the incidental parameters issue and the initial values problem lead to violations in the standard regularity conditions for the *ML* estimators of the structural parameters. Hsiao, Pesaran, and Tahmiscioglu (2002) develop a transformed likelihood approach to overcome the incidental parameters problem. Consider a dynamic panel data model with

group-specific effects assumed to be fixed unknown parameters. To simplify the analysis, abstract from exogenous regressors and take first differences of (27.5) to obtain

$$\Delta y_{it} = \lambda \Delta y_{i,t-1} + \Delta u_{it}, \quad t = 2, 3, \ldots, T, \tag{27.47}$$

and note that for the initial observations we have

$$\begin{aligned} \Delta y_{i1} &= \lambda^m \Delta y_{i,-m+1} + \sum_{j=0}^{m-1} \lambda^j \Delta u_{i,1-j} \\ &= \lambda^m \Delta y_{i,-m+1} + v_{i1}. \end{aligned}$$

When $|\lambda| < 1$, for any fixed starting values $\Delta y_{i,-m+1}$, $\lim_{m\to\infty} \lambda^m \Delta y_{i,-m+1} = 0$, and, under serial independence of $\Delta u_{i,1-j}$, it is reasonable to assume that $E\left(\Delta y_{i1}\right) = 0$. A non-zero expected value for Δy_{i1} is more reasonable if the process has started from a finite period in the past not too far back, or under non-stationarity. It follows that, depending on whether the process has reached stationarity or not, one of the following two assumptions can be adopted:

MLE.i $|\lambda| < 1$, and the process has been going on for a long time, namely $m \to \infty$, with $E\left(\Delta y_{i1}\right) = 0$, $Var(\Delta y_{i1}) = 2\sigma_u^2/(1+\lambda)$, $Cov(v_{i1}, \Delta u_{i2}) = -\sigma_u^2$, and $Cov\left(v_{i1}, \Delta u_{it}\right) = 0$ for $t = 3, 4, \ldots, T$, $i = 1, 2, \ldots, N$.

MLE.ii The process has started from a finite period in the past not too far back from the 0th period, namely for given values of $y_{i,-m+1}$ with m finite, such that $E(\Delta y_{i1}) = b$, $Var(\Delta y_{i1}) = c\sigma_u^2$, with $c > 0$, $Cov(v_{i1}, \Delta u_{i2}) = -\sigma_u^2$, and $Cov(v_{i1}, \Delta u_{it}) = 0$ for $t = 3, 4, \ldots, T$, $i = 1, 2, \ldots, N$.

Assumption *MLE*.i follows directly from the serial independence of u_{it}, and under $|\lambda| < 1$. Assumption *MLE*.ii imposes the restriction that expected changes in the initial endowments are the same across individuals, but does not require $|\lambda| < 1$, or that all individuals start from the same point in time.

In the case where the dynamic model also contains the regressors, $\mathbf{x}_{it}$, a distinction must be made depending on whether the regressors, $\mathbf{x}_{it}$, are strictly or weakly exogenous (see Assumptions 4.i and 4.ii in Hsiao, Pesaran, and Tahmiscioglu (2002)). In this more general case the first differenced model is

$$\Delta y_{it} = \lambda \Delta y_{i,t-1} + \beta \Delta x_{it} + \Delta u_{it}, \quad t = 2, 3, \ldots, T,$$

with the solution for the initial values

$$\Delta y_{i1} = \lambda^m \Delta y_{i,-m+1} + \beta \sum_{j=0}^{m-1} \lambda^j \Delta x_{i,1-j} + \sum_{j=0}^{m-1} \lambda^j \Delta u_{i,1-j}.$$

To solve the incidental parameters problem associated with the initial conditions, it is also required that $\mathbf{x}_{it}$ follows either

$$x_{it} = \mu_i + g\,t + \sum_{j=0}^{\infty} a_j \varepsilon_{i,t-j}, \quad \sum_{j=0}^{\infty} \mid a_j \mid < \infty, \tag{27.48}$$

or

$$\Delta x_{it} = g + \sum_{j=0}^{\infty} d_j \varepsilon_{i,t-j}, \quad \sum_{j=0}^{\infty} \mid d_j \mid < \infty. \tag{27.49}$$

Using either (27.48) or (27.49), we have

$$\Delta x_{it} = g + \sum_{j=0}^{\infty} d_j^* \varepsilon_{i,t-j}, \quad \sum_{j=0}^{\infty} \mid d_j^* \mid < \infty,$$

where $d_j^* = d_j$ in (27.49), and $d_j^* = a_j - a_{j-1}$ under (27.48), and it is easily seen that

$$E\left(\Delta x_{i,\,1-j} | \Delta \mathbf{x}_i\right) = b_j + \boldsymbol{\pi}_j' \Delta \mathbf{x}_i.$$

Let $\Delta \mathbf{y}_i = \left(\Delta y_{i1}, \Delta y_{i2}, \ldots, \Delta y_{iT}\right)'$ and

$$\Delta \tilde{\mathbf{W}}_i = \begin{pmatrix} 1 & \Delta \mathbf{x}_i' & 0 & 0 \\ 0 & \mathbf{0} & \Delta y_{i1} & \Delta \mathbf{x}_{i2} \\ \vdots & \vdots & \vdots & \vdots \\ 0 & \mathbf{0} & \Delta y_{i,T-1} & \Delta \mathbf{x}_{iT} \end{pmatrix},$$

and $\Delta \mathbf{u}_i = (\Delta u_{i1}, \Delta u_{i2}, \ldots, \Delta u_{iT})'$. Note that under either *MLE*.i or *MLE*.ii we have

$$\Delta \mathbf{y}_i = \Delta \tilde{\mathbf{W}}_i \boldsymbol{\varphi} + \Delta \mathbf{u}_i, \tag{27.50}$$

with $\boldsymbol{\varphi} = \left(b^*, \boldsymbol{\pi}^{*\prime}, \gamma, \beta\right)'$, where $b^* = 0$ under *MLE*.i and if $g = 0$, and $b^* = b$ under *MLE*.ii, and $\boldsymbol{\pi}^*$ is a $T \times 1$ vector of unknown coefficients which in general varies independently of the variations in β and γ.

The transformed disturbances, $\Delta \mathbf{u}_i$, have covariance matrix $\boldsymbol{\Omega} = \sigma_u^2 \boldsymbol{\Omega}^*$, where

$$\boldsymbol{\Omega}^* = \begin{pmatrix} \omega & -1 & \cdots & 0 & 0 \\ -1 & 2 & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 2 & -1 \\ 0 & 0 & \cdots & -1 & 2 \end{pmatrix},$$

where $\omega = \left(1/\sigma_u^2\right) Var(\Delta y_{i1})$. Note that ω is generally unrestricted except for the case when $\beta = 0$ and $y_{it} \sim I(0)$, in which case $\omega = 2/(1+\lambda)$. The determinant of $\boldsymbol{\Omega}^*$ is $\left|\boldsymbol{\Omega}^*\right| = 1 + T(\omega - 1)$.

Let $\boldsymbol{\theta} = \left(\boldsymbol{\varphi}', \omega, \sigma_u^2\right)'$. The likelihood function of the transformed model (27.50) is then

$$\ell(\boldsymbol{\theta}) = -\frac{NT}{2}\ln(2\pi) - \frac{NT}{2}\ln\left(\sigma_u^2\right) - \frac{N}{2}\ln\left[1 + T(\omega - 1)\right] - \frac{1}{2}\sum_{i=1}^{N}\left(\Delta\mathbf{y}_i - \Delta\tilde{\mathbf{W}}_i\boldsymbol{\varphi}\right)\boldsymbol{\Omega}^{-1}\left(\Delta\mathbf{y}_i - \Delta\tilde{\mathbf{W}}_i\boldsymbol{\varphi}\right)'. \tag{27.51}$$

The exact specification of the new parameters b, $\boldsymbol{\pi}^*$, and ω depends on whether y_{it} is I(0), and/or whether $\mathbf{x}_{it}$ is strictly or weakly exogenous. See Hsiao, Pesaran, and Tahmiscioglu (2002) for further details.

The Monte Carlo experiments reported in Hsiao, Pesaran, and Tahmiscioglu (2002) also show that transformed *MLE* performs well in small samples, and tends to dominate *GMM* type estimators. However, the transformed *ML* is based on a number of strong distributional assumptions on the disturbances. In particular, it assumes that u_{it} are cross-sectionally homoskedastic. In a recent paper, Hayakawa and Pesaran (2015) relax this assumption and allow $E\left(u_{it}^2\right) = \sigma_{ui}^2$ to differ across i. This is not a trivial extension, due to the incidental parameters problem that arises, and its implications for estimation and inference. To deal with this problem, Hayakawa and Pesaran (2015) use the pseudo- or quasi-*ML* approach where the error variance heterogeneity is ignored at the estimation stage, but robust standard errors are used at the inference stage.

Let $\boldsymbol{\theta}_*$ denote the pseudo true values obtained by maximizing the pseudo log-likelihood function (27.51) of the misspecified model that ignores the error variance heterogeneity. Hayakawa and Pesaran (2015) establish that in this heteroskedastic case, $\boldsymbol{\theta}_* = \left(\boldsymbol{\varphi}', \omega, \sigma_{u*}^2\right)'$, where $\sigma_{u*}^2 = \lim_{N\to\infty} N^{-1}\sum_{i=1}^{N}\sigma_{ui}^2$. Hence, the *ML* estimator of $\boldsymbol{\varphi}$ and ω by Hsiao, Pesaran, and Tahmiscioglu (2002) continues to be consistent even if cross-sectional heteroskedasticity is present. Hayakawa and Pesaran (2015) also show that as $N \to \infty$, the *ML* estimator $\hat{\boldsymbol{\theta}}$ obtained by maximizing the pseudo log-likelihood function in (27.51) is asymptotically normal

$$\sqrt{N}\left(\widehat{\boldsymbol{\theta}} - \boldsymbol{\theta}_*\right) \xrightarrow{d} \mathcal{N}\left(\mathbf{0}, \mathbf{A}^{*-1}\mathbf{B}^*\mathbf{A}^{*-1}\right) \tag{27.52}$$

where $\boldsymbol{\theta}_* = (\boldsymbol{\varphi}', \omega, \sigma_{u*}^2)'$,

$$\mathbf{A}^* = \lim_{N\to\infty} E\left[-\frac{1}{N}\frac{\partial^2\ell_p(\boldsymbol{\theta}_*)}{\partial\boldsymbol{\theta}\partial\boldsymbol{\theta}'}\right], \quad \textit{and} \quad \mathbf{B}^* = \lim_{N\to\infty} E\left[\frac{1}{N}\frac{\partial\ell_p(\boldsymbol{\theta}_*)}{\partial\boldsymbol{\theta}}\frac{\partial\ell_p(\boldsymbol{\theta}_*)}{\partial\boldsymbol{\theta}'}\right].$$

A covariance matrix estimator robust to cross-sectional heteroskedasticity of unknown form can also be obtained. See Hayakawa and Pesaran (2015) for details on how to obtain consistent estimators of $\mathbf{A}^*$ and $\mathbf{B}^*$. Using Monte Carlo simulations, it is shown that the transformed likelihood estimator outperforms the *GMM* estimators in almost all cases when the model contains an exogenous regressor, and in many cases if we consider pure autoregressive panels. The transformed *ML* approach has also been extended to panel *VAR* (*PVAR*) models by Binder, Hsiao, and Pesaran (2005) and to simultaneous equations models by Hsiao and Zhou (2015).

27.7 Short dynamic panels with unobserved factor error structure

Estimation methods reviewed so far assume that the errors, u_{it}, in (27.1) are distributed independently across i, a cross-sectional error independence assumption which might not hold in many applications where cross-section units are subject to common unobserved effects, or have spatial or network patterns where the error cross dependence could be local. Ignoring cross-sectional dependence can have important consequences for conventional estimators of dynamic panels. Phillips and Sul (2007) study the impact of cross-sectional dependence modelled as a factor structure on the inconsistency of the pooled least squares estimate of a short dynamic panel regression.[2] Sarafidis, Yamagata, and Robertson (2009) investigate the properties of a number of standard widely used *GMM* estimators under cross-sectional dependence, and show that such estimators are inconsistent.

In applications where spatial patterns are important and can be characterized by known spatial weight matrices, error cross-sectional dependence is typically modelled as spatial autoregressions and estimated jointly with the other parameters of the dynamic panel data model. Lee and Yu (2011) provide a review. For small T, Elhorst (2005) and Su and Yang (2015) consider random effects as well as fixed-effects specifications. In the latter case they apply the first-differencing operator to eliminate the fixed-effects and then use the transformed likelihood approach of Hsiao, Pesaran, and Tahmiscioglu (2002) to deal with the initial value problem. The treatment of the initial values in spatial dynamic panel data models poses additional difficulties and requires further investigation.

In addition to spatial effects, it is also likely that the error cross-sectional dependence could be due to omitted unobserved common factor(s). This class of models is referred to as multi-factor error panel data models and is reviewed in Section 29.5 in the case where both N and T are large. The case where T is short and N large (which is the concern of this chapter) is less intensively researched. An early contribution is due to Holtz-Eakin, Newey, and Rosen (1988) and Ahn, Lee, and Schmidt (2001) who employ a quasi-differencing approach (originally proposed by Chamberlain (1984)) to purge the panel data model from the factor structure and then use *GMM* to consistently estimate the model parameters. Nauges and Thomas (2003) follow this approach in addition to prior first-differencing to eliminate the fixed effect, which they consider separately from the single common factor structure assumed for the errors. Ahn, Lee, and Schmidt (2013) extend this approach to the more general case of a multifactor error structure. Robertson and Sarafidis (2013) propose an instrumental variable estimation procedure that introduces new parameters to represent the unobserved covariances between the instruments and the factor component of the errors. They show that the resulting estimator is asymptotically more efficient than the *GMM* estimator based on quasi-differencing as it exploits extra restrictions implied by the model. Elhorst (2010) considers a fixed-effects dynamic panel with contemporaneous endogenous interaction effects under small T. For estimation purposes, he adopts both the maximum likelihood estimator of Hsiao, Pesaran, and Tahmiscioglu (2002) and the *GMM* estimator of Arellano and Bond (1991) reviewed above (see Section 27.4.2). Bai (2013) suggests a quasi-maximum likelihood (*ML*) approach applied to the original dynamic panel without differencing (simple or quasi), and uses the approach of Mundlak (1978) and

[2] The literature on cross-sectional dependence in panels is covered in Chapter 29.

Chamberlain (1982) to deal with the correlation between the factor loadings and the regressors, but continues to assume that all factor loadings (including the one associated with the intercepts) are uncorrelated with the errors. Hayakawa, Pesaran, and Smith (2014), using the transformed *ML* approach of Hsiao, Pesaran, and Tahmiscioglu (2002), propose an alternative quasi-*ML* approach applied to the panel data model after first-differencing. The proposed estimation procedure includes the transformed likelihood procedure of Hsiao, Pesaran, and Tahmiscioglu (2002) as a special case. It allows for both fixed and interactive effects (the latter based on a random coefficient specification), and can be used to test the validity of the fixed-effects specification against the more general model with interactive effects.

In what follows, we give a summary account of the quasi-differencing and the transformed *MLE* approaches, as they represent quite different ways of dealing with the unobserved factors when T is short and N large. Also, to simplify the exposition, we consider a factor error structure with only a single unobserved factor. To this end suppose that u_{it} in the dynamic panel data model, (27.1), is given by the following unobserved factor error structure,

$$u_{it} = \gamma_i f_t + \varepsilon_{it}, \tag{27.53}$$

where f_t is the unobserved common factor, γ_i is the factor loading of ith unit and ε_{it}'s are cross-sectionally independent innovations. (27.53) is an exact factor model.[3] Using (27.53) in (27.1) we have

$$y_{it} = \alpha_i + \lambda y_{i,t-1} + \boldsymbol{\beta}' \mathbf{x}_{it} + \gamma_i f_t + \varepsilon_{it}. \tag{27.54}$$

Ahn, Lee, and Schmidt (2001) employ the quasi-differencing approach of Holtz-Eakin, Newey, and Rosen (1988) to eliminate γ_i. This involves multiplying the equation for $y_{i,t-1}$ by $\phi_t = f_t/f_{t-1}$, and then subtracting it from the equation for y_{it} to obtain

$$y_{it} - \phi_t y_{i,t-1} = \left(1 - \phi_t\right) \alpha_i + \lambda \left(y_{i,t-1} - \phi_t y_{i,t-2}\right) + \boldsymbol{\beta}' \left(\mathbf{x}_{it} - \phi_t \mathbf{x}_{i,t-1}\right) + e_{it}, \tag{27.55}$$

where $e_{it} = \varepsilon_{it} - \phi_t \varepsilon_{i,t-1}$, for $i = 1, 2, \ldots, N$, and $t = 3, 4, \ldots, T$. In the special case of $\phi_t = 1$ for each t, this transformation amounts to simple differencing of equation (27.54). But when $\phi_t \neq 1$, the quasi-differencing eliminates γ_i but not the fixed-effects, α_i. To eliminate the fixed-effects we need to apply the first-differencing operator to eliminate α_i before applying the quasi-differencing to eliminate γ_i. But, to simplify the exposition, we follow Ahn, Lee, and Schmidt (2001) and abstract from the fixed-effects and set $\alpha_i = \alpha$ for all i. Under this assumption, using (27.55), the parameters α, λ and $\boldsymbol{\beta}$ can be estimated consistently along with $\phi_3, \phi_4, \ldots, \phi_T$, using *GMM*. The instruments are obtained from the following moment conditions

$$E\left(y_{is}\varepsilon_{it}\right) = E\left(\gamma_i \varepsilon_{is}\right) = 0, \tag{27.56}$$

$$E\left(\mathbf{x}_{is}\varepsilon_{it}\right) = \mathbf{0}, \text{ for } s < t. \tag{27.57}$$

Conditions (27.56)–(27.57) imply that the error term of the transformed equation (27.55) satisfies the orthogonality conditions

[3] Factor models are discussed in Sections 19.5 and 29.3.

$$E\left(y_{is}e_{it}\right)=0,\ \ E\left(\mathbf{x}_{is}e_{it}\right)=\mathbf{0},\ \text{for}\ s<t-1.$$

Thus, the vector of instrumental variables that is available to identify the parameters of equation (27.55) (under $\alpha_i=\alpha$) is

$$\left(1,y_{i1},\ldots.,y_{i,t-2},\mathbf{x}_{i1},\mathbf{x}_{i2},\ldots,\mathbf{x}_{i,t-2}\right),$$

and the *GMM* estimation based on these instruments is consistent under fixed T and as $N\to\infty$, although with moderate T the number of moments can be large and rises rapidly as T increases. The *GMM* estimation can also be subject to the weak instrument problem. Extension of (27.54) to more than one unobserved factor is attempted by Ahn, Lee, and Schmidt (2007) to estimate a production function for a set of rice farms observed over six seasons, where multiple factors were included to proxy farm-specific, time varying technical inefficiencies. See also Ahn, Lee, and Schmidt (2007, 2013) for details.

Hayakawa, Pesaran, and Smith (2014) consider the dynamic panel data, (27.54), and allow for α_i to differ across *i*. First, they apply the first-differencing operator to eliminate α_i, and then deal with the factor loadings by assuming that γ_i are random draws from a distribution with a fixed number of unknown parameters. For simplicity assume that x_{it} is a scalar, and write the model as

$$y_{it}=\alpha_i+\gamma y_{i,t-1}+\beta x_{it}+\lambda_i f_t+u_{it},\qquad i=1,2,\ldots,N;t=1,2,\ldots,T. \tag{27.58}$$

Taking first differences and assuming that $\lambda_i=\lambda+\eta_i$, $\eta_i\sim IID(0,\sigma_\eta^2)$, we have

$$\Delta y_{it}=\gamma\Delta y_{i,t-1}+\beta\Delta x_{it}+\lambda g_t+\eta_i g_t+\Delta u_{it},\quad t=2,3,\ldots,T, \tag{27.59}$$

where $g_t=\Delta f_t$. The regressor, x_{it}, is assumed to follow

$$\Delta x_{it}=c+\vartheta_i g_t+\sum_{j=0}^{\infty}d_j\varepsilon_{i,t-j},\qquad \sum_{j=0}^{\infty}\left|d_j\right|<\infty, \tag{27.60}$$

which is a generalization of (27.49), where ϑ_i are the random interactive effects distributed independently of u_{it} and f_t. Following similar assumptions as in Hsiao, Pesaran, and Tahmiscioglu (2002), it is further shown that

$$\Delta y_{i1}=b+\boldsymbol{\pi}'\Delta\mathbf{x}_i+v_{i1}, \tag{27.61}$$

where b is a constant, $\boldsymbol{\pi}$ is a T-dimensional vector of constants, $\Delta\mathbf{x}_i=(\Delta x_{i1},\Delta x_{i2},\ldots,\Delta x_{iT})'$ and v_{i1} is distributed independently across i such that $E(v_{i1})=0$ and $E(v_{i1}^2)=\omega\sigma^2$, with $0<\omega<K<\infty$. Under this setup, the panel data model in matrix notation can be written as

$$\Delta\mathbf{y}_i=\Delta\mathbf{W}_i\boldsymbol{\varphi}+\lambda\mathbf{g}+\boldsymbol{\xi}_i, \tag{27.62}$$

where $\Delta\mathbf{y}_i=\left(\Delta y_{i1},\Delta y_{i2},\ldots,\Delta y_{iT}\right)'$, $\boldsymbol{\varphi}=\left(b,\boldsymbol{\pi}',\gamma,\beta\right)'$, $\boldsymbol{\xi}_i=\eta_i\mathbf{g}+\mathbf{r}_i$, $\mathbf{g}=(\tilde{g}_1,g_2,\ldots,g_T)'$, $\mathbf{r}_i=(v_{i1},\Delta u_{i2},\ldots,\Delta u_{iT})'$ and

$$\Delta \mathbf{W}_i = \begin{pmatrix} 1 & \Delta \mathbf{x}_i' & 0 & 0 \\ 0 & \mathbf{0} & \Delta y_{i1} & \Delta x_{i2} \\ \vdots & \vdots & \vdots & \vdots \\ 0 & \mathbf{0} & \Delta y_{i,T-1} & \Delta x_{iT} \end{pmatrix}. \tag{27.63}$$

From Hsiao, Pesaran, and Tahmiscioglu (2002) we have

$$E(\mathbf{r}_i \mathbf{r}_i') = \sigma^2 \begin{pmatrix} \omega & -1 & \cdots & 0 & 0 \\ -1 & 2 & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 2 & -1 \\ 0 & 0 & \cdots & -1 & 2 \end{pmatrix} = \sigma^2 \mathbf{\Omega}. \tag{27.64}$$

Also

$$Var(\boldsymbol{\xi}_i) = \sigma^2 \mathbf{\Omega} + \sigma_\eta^2 \mathbf{g}\mathbf{g}' = \sigma^2 \left(\mathbf{\Omega} + \phi \mathbf{g}\mathbf{g}' \right),$$

where $\phi = \sigma_\eta^2 / \sigma^2$. Hence, the log-likelihood function of the transformed model (27.62) is given by

$$\begin{aligned} \ell_N(\boldsymbol{\psi}) = & -\frac{NT}{2} \ln(2\pi) - \frac{NT}{2} \ln(\sigma^2) - \frac{N}{2} \ln \left| \mathbf{\Omega} + \phi \mathbf{g}\mathbf{g}' \right| \\ & - \frac{1}{2\sigma^2} \sum_{i=1}^{N} \left(\Delta \mathbf{y}_i - \Delta \mathbf{W}_i \gamma - \lambda \mathbf{g} \right)' \left(\mathbf{\Omega} + \phi \mathbf{g}\mathbf{g}' \right)^{-1} \left(\Delta \mathbf{y}_i - \Delta \mathbf{W}_i \gamma - \lambda \mathbf{g} \right). \end{aligned} \tag{27.65}$$

The log-likelihood in (27.65) is a function of a fixed number of unknown parameters, $\boldsymbol{\psi} = (\gamma, \omega, \sigma^2, \phi, \lambda, \mathbf{g}')'$, and standard theory of *ML* estimation applies and the parameters can be estimated by maximizing $\ell_N(\boldsymbol{\psi})$ with respect to $\boldsymbol{\psi}$. For further details and evidence on small sample properties of the *ML* estimator, see Hayakawa, Pesaran, and Smith (2014).

27.8 Dynamic, nonlinear unobserved effects panel data models

Consider the following dynamic panel specification for the latent (unobserved) variable

$$y_{it}^* = \alpha_i + \lambda y_{i,t-1} + \boldsymbol{\beta}' \mathbf{x}_{it} + u_{it}, \tag{27.66}$$

where y_{it}^* is a latent variable, $y_{it} = I(y_{it}^* > d)$ is the discrete observed dependent variable that takes the value of unity if $y_{it}^* > d$ and zero otherwise, d is a threshold value, $\mathbf{x}_{it}$ is a vector of strictly exogenous regressors, $\alpha_i \sim IIDN(0, \sigma_\alpha^2)$, and $u_{it} \sim IIDN(0, \sigma_u^2)$. We are interested in modelling

$$P\left(y_{it}|y_{i0}, y_{i1}, \ldots, y_{i,t-1}, \mathbf{X}_{i.}, \alpha_i\right) = G\left(\alpha_i + \lambda y_{i,t-1} + \boldsymbol{\beta}'\mathbf{x}_{it}\right), \tag{27.67}$$

where G is typically chosen to be logit or the probit functions. Under the above specification the probability of success at time t is allowed to depend on the outcome in the previous period, $t-1$, as well as on unobserved heterogeneity, α_i. Of particular interest is testing the null hypothesis that $\lambda = 0$. Under this hypothesis, the response probability at time t does not depend on past outcomes once controlled for α_i and $\mathbf{X}_{i.}$. As with the linear models specification of the initial values plays an important role in the estimation and inference. A simple approach would be to treat y_{i0} as non-stochastics, and assume that α_i, $i = 1, 2, \ldots, N$ are random and independent of $\mathbf{X}_{i.}$. In such a setting, the density of $\left(y_{i1}, y_{i2}, \ldots, y_{iT}\right)$ given $\mathbf{X}_{i.}$ can be obtained by integrating out α_i's, following a Bayesian approach, along similar lines as those described in Section 26.11. Although treating the y_{i0} as nonrandom simplifies estimation, it is undesirable because it implies that y_{i0} is independent of α_i and of any of the exogenous variables, which is a strong assumption. In a recent paper, using a set of Monte Carlo experiments, Akay (2012) shows that the exogenous initial values assumption, if incorrect, can lead to serious overestimation of the true state dependence and serious underestimation of the variance of unobserved group effects, when T is small. An alternative approach would be to allow the initial condition to be random, and then to use the joint distribution of all outcomes on the responses—including that in the initial time period—conditional on unobserved heterogeneity and observed strictly exogenous explanatory variables. However, as shown by Wooldridge (2005), the main complication with this approach is in specifying the distribution of the initial values given α_i and $\mathbf{x}_{it}$. For the dynamic probit specification, Wooldridge (2005) proposes a very simple approach, which consists of specifying a distribution for α_i conditional on the initial values and on the time averages of the exogenous variables

$$\alpha_i = \pi_0 + \pi_1 y_{i0} + \boldsymbol{\pi}_2'\bar{\mathbf{x}}_{i.} + \eta_i, \tag{27.68}$$

where $\bar{\mathbf{x}}_{i.} = \frac{1}{T}\sum_{t=1}^{T}\mathbf{x}_{it}$, η_i is an unobserved individual-effects, such that $\eta_i|\bar{\mathbf{x}}_{i.} \sim IIDN(0, \sigma_\eta^2)$, and $y_{i0} \sim IIDN(0, \sigma_\eta^2)$. Plugging (27.68) into (27.66), under the probit specification, it is possible to derive the joint distribution of outcomes conditional on the initial values and the strictly exogenous variables. Such a likelihood has exactly the same structure as the standard random effects probit model, except for the regressors, which is now given by $\mathbf{x}_{it}^* = \left(1, \mathbf{x}_{it}', y_{i0}, \bar{\mathbf{x}}_{i.}'\right)'$. Hence, with this approach it is possible to add y_{i0} and $\bar{\mathbf{x}}_{i.}$ as additional explanatory variables in each time period and use standard random effects probit software to estimate $\boldsymbol{\beta}$, λ, π_0, π_1, $\boldsymbol{\pi}_2$ and σ_η^2.

Al-Sadoon, Li, and Pesaran (2012) introduce a binary choice panel data model where the idiosyncratic error term follows an exponential distribution, and derive moment conditions that eliminate the fixed-effect term and at the same time identify the parameters of the model. Appropriate moment conditions are derived both for identification of the state dependent parameter, λ, as well as the coefficients of the exogenous covariates, $\boldsymbol{\beta}$. It is shown that the resultant *GMM* estimators are consistent and asymptotically normally distributed at the $\sqrt{N}$ rate.

We refer to Hsiao (2003, Section 7.5.2), and Wooldridge (2005) for further discussion on the initial conditions problem and on estimation of dynamic nonlinear models. Arellano and Bonhomme (2011) provide a review of recent developments in the econometric analysis of nonlinear panel data models.

27.9 Further reading

Textbook treatment of this topic can be found in Arellano (2003) and Hsiao (2003). In particular, Arellano (2003) offers an exhaustive overview of *GMM* techniques proposed for estimating dynamic panels, while Hsiao (2003) (see, in particular, Ch. 4) provides a discussion of estimation of dynamic models under various assumptions on the distribution of initial values. See also the third edition of Hsiao's pioneering text, Hsiao (2014). For further information on nonlinear dynamic panels, see Chs 15–16 in Wooldridge (2005).

27.10 Exercises

1. Consider model (27.17)–(27.18) and suppose that all variables $\mathbf{x}_{it}$ are predetermined, with $E(\mathbf{x}_{it}v_{is}) = \mathbf{0}$ for $s < t$, and zero otherwise. Write the $\mathbf{W}_i$ matrix with valid instruments for *GMM* estimation and derive the one-step and two-step *GMM* estimator for this case.
2. Consider the dynamic panel data model with no exogenous regressors

$$y_{it} = \alpha_i + \lambda y_{i,t-1} + u_{it}, \quad t = 1, 2, 3, \ldots, T,$$

where $u_{it} \sim IIDN(0, \sigma_u^2)$, and α_i for $i = 1, 2, \ldots, N$ are fixed-effects.

   (a) Transform the model into first differences and write down the log-likelihood function using the transformed likelihood approach.

   (b) Derive the first- and second-order conditions for maximization of the log-likelihood function.

   (c) Distinguish between the stationary and the unit root case and discuss the consistency of the transformed *MLE* estimators under both cases.

3. Consider the dynamic panel data model with a single unobserved factor but without fixed-effects

$$y_{it} = \lambda y_{i,t-1} + \gamma_i f_t + \varepsilon_{it}, \text{ for } t = 1, 2, \ldots, t, \text{ and } i = 1, 2, \ldots, N,$$

where $\lambda_i \sim IID(0, \sigma_\lambda^2)$, with $\sigma_\lambda^2 > 0$, and ε_{it} are $IID(0, \sigma^2)$.

   (a) Derive the bias of the least square estimate of λ, $\hat{\lambda}_{OLS}$, given by

$$\hat{\lambda}_{OLS} = \frac{\sum_{t=1}^{T}\sum_{i=1}^{N} y_{it}y_{i,t-1}}{\sum_{t=1}^{T}\sum_{i=1}^{N} y_{i,t-1}^2}.$$

   (b) Compare this bias with the Nickell bias given by (27.3).

4. Consider the following dynamic panel data model with interactive effects

$$y_{it} = \alpha_i + \lambda y_{i,t-1} + \boldsymbol{\beta}'\mathbf{x}_{it} + \gamma_i f_t + \varepsilon_{it}, \text{ for } t = 1, 2, \ldots, T,$$

where α_i are fixed-effects, f_t is an unobserved factor, and γ_i is the factor loading for the ith unit.

(a) Derive an equation in $y_{it}, \mathbf{x}_{it}$ and their lagged values that do not depend on α_i and γ_i.

(b) Under what conditions can λ and $\boldsymbol{\beta}$ be estimated consistently?

28 Large Heterogeneous Panel Data Models

28.1 Introduction

Panel data models introduced in the previous two chapters, 26 and 27, deal with panels where the time dimension (T) is fixed, and assumes that conditional on a number of observable characteristics, any remaining heterogeneity over the cross-sectional units can be modelled through an additive intercept (assuming either fixed or random), and possibly heteroskedastic errors. This chapter extends the analysis of panels to linear panel data models with slope heterogeneity. It discusses how neglecting such heterogeneities affects the consistency of the estimates and inferences based upon them, and introduces models that explicitly allow for slope heterogeneity both in the case of static and dynamic panel data models. To deal with slope heterogeneity, particularly in the case of dynamic models, it is often necessary to assume that the number of time series observations, T, is relatively large, so that individual equations can be estimated for each unit separately. Models, estimation and inference procedures developed in this and subsequent chapters are more suited to large N and T panels. Such panel data sets are becoming increasingly available and cover countries, regions, industries, and markets over relatively long time periods.

Despite the slope heterogeneity, the cross-sectional units could nevertheless share common features of interest. For example, it is possible for different countries or geographical regions to have different dynamics of adjustments towards equilibrium, due to their historical and cultural differences, but they could all converge to the same economic equilibrium in the very long run, due to forces of arbitrage and interconnections through international trade and cultural exchanges. Other examples include cases where slope coefficients can be viewed as random draws from a distribution with a number of parameters that are bounded in N. Large number of panel data sets fit within this setup, where the cross-sectional units might be industries, regions, or countries, and we wish to identify common patterns of responses across otherwise heterogeneous units. The parameters of interest may be intercepts, short-run coefficients, long-run coefficients or error variances.

This chapter deals with panels with stationary variables. The econometric analyses of panels with unit roots and cointegration is covered in Chapter 31.

28.2 Heterogeneous panels with strictly exogenous regressors

Suppose that the variable y_{it} for the ith unit at time t is specified as a linear function of k strictly exogenous variables, x_{kit}, $k = 1, 2, \ldots, k$, in the form

$$\begin{aligned} y_{it} &= \sum_{k=1}^{k} \beta_{kit} x_{kit} + u_{it}, \\ &= \boldsymbol{\beta}_{it}' \mathbf{x}_{it} + u_{it}, \; i = 1, 2, \ldots, N, \quad t = 1, 2, \ldots, T, \end{aligned} \tag{28.1}$$

where u_{it} denotes the random error term, $\mathbf{x}_{it}$ is a $k \times 1$ vector of exogenous variables and $\boldsymbol{\beta}_{it}$ is the $k \times 1$ vector of coefficients. The above specification is very general and allows the coefficients to vary both across time and over individual units. As it is specified it is too general. It simply states that each individual unit has its own coefficients that are specific to each time period. However, as pointed out by Balestra (1996), this general formulation is, at most, descriptive. It lacks any explanatory power and it is not useful for prediction. Furthermore, it is not estimable, as the number of parameters to be estimated exceeds the number of observations. For a model to become interesting and to acquire explanatory and predictive power, it is essential that some structure is imposed on its parameters.

One way to reduce the number of parameters in (28.1) is to adopt a random coefficient approach, which assumes that the coefficients $\boldsymbol{\beta}_{it}$ are draws from probability distributions with a fixed number of parameters that do not vary with N and/or T. Depending on the type of assumption about the parameter variation, we can further classify the models into one of two categories: stationary and non-stationary random-coefficient models.

The stationary random-coefficient models view the coefficients as having constant means and variance-covariances. Namely, the $k \times 1$ vector $\boldsymbol{\beta}_{it}$ is specified as

$$\boldsymbol{\beta}_{it} = \boldsymbol{\beta} + \boldsymbol{\eta}_{it}, \quad i = 1, 2, \ldots, N, \quad t = 1, 2, \ldots, T, \tag{28.2}$$

where $\boldsymbol{\beta}$ is a $k \times 1$ vector of constants, and $\boldsymbol{\eta}_{it}$ is a $k \times 1$ vector of stationary random variables with zero means and constant variance-covariances. One widely used random coefficient specification is the Swamy (1970) model, which assumes that the randomness is time-invariant

$$\boldsymbol{\beta}_{it} = \boldsymbol{\beta} + \boldsymbol{\eta}_{i}, \; i = 1, 2, \ldots, N, \; t = 1, 2, \ldots, T, \tag{28.3}$$

and

$$E(\boldsymbol{\eta}_i) = \mathbf{0}, \; E(\boldsymbol{\eta}_i \mathbf{x}_{it}') = \mathbf{0}, \tag{28.4}$$

$$E(\boldsymbol{\eta}_i \boldsymbol{\eta}_j) = \begin{cases} \boldsymbol{\Omega}_{\eta}, & \text{if} \quad i = j, \\ \mathbf{0}, & \text{if} \quad i \neq j. \end{cases} \tag{28.5}$$

Estimation and inference in the above specification are discussed in Section 28.4.

Hsiao (1974, 1975) consider the following model

$$\begin{aligned}\boldsymbol{\beta}_{it} &= \boldsymbol{\beta} + \boldsymbol{\xi}_{it} \\ &= \boldsymbol{\beta} + \boldsymbol{\eta}_i + \boldsymbol{\lambda}_t, \quad i = 1, 2, \ldots, N,\ t = 1, 2, \ldots, T,\end{aligned} \tag{28.6}$$

and assume

$$\begin{aligned} E(\boldsymbol{\eta}_i) = E(\boldsymbol{\lambda}_t) &= \mathbf{0}, E\left(\boldsymbol{\eta}_i \boldsymbol{\lambda}_t'\right) = \mathbf{0}, \\ E\left(\boldsymbol{\eta}_i \mathbf{x}_{it}'\right) &= \mathbf{0}, E\left(\boldsymbol{\lambda}_t \mathbf{x}_{it}'\right) = \mathbf{0}, \\ E\left(\boldsymbol{\eta}_i \boldsymbol{\eta}_j'\right) &= \begin{cases} \boldsymbol{\Omega}_{\boldsymbol{\eta}}, & \text{if} \quad i = j, \\ \mathbf{0}, & \text{if} \quad i \neq j, \end{cases} \\ E(\boldsymbol{\lambda}_i \boldsymbol{\lambda}_j') &= \begin{cases} \boldsymbol{\Omega}_{\boldsymbol{\Lambda}}, & \text{if} \quad i = j, \\ \mathbf{0}, & \text{if} \quad i \neq j, \end{cases} \end{aligned} \tag{28.7}$$

Alternatively, a time varying parameter model may be treated as realizations of a stationary stochastic process, thus $\boldsymbol{\beta}_{it}$ can be written in the form,

$$\boldsymbol{\beta}_{it} = \boldsymbol{\beta}_t = \mathbf{H}\boldsymbol{\beta}_{t-1} + \boldsymbol{\delta}_t, \tag{28.8}$$

where all eigenvalues of $\mathbf{H}$ lie inside the unit circle, and $\boldsymbol{\delta}_t$ is a stationary random variable with mean $\boldsymbol{\mu}$. Hence, letting $\mathbf{H} = \mathbf{0}$ and $\boldsymbol{\delta}_t$ be IID we obtain the model proposed by Hildreth and Houck (1968), while for the Pagan (1980) model, $\mathbf{H} = \mathbf{0}$ and

$$\boldsymbol{\delta}_t - \boldsymbol{\mu} = \boldsymbol{\delta}_t - \overline{\boldsymbol{\beta}} = \boldsymbol{A}(L)\boldsymbol{\epsilon}_t, \tag{28.9}$$

where $\overline{\boldsymbol{\beta}}$ is the mean of $\boldsymbol{\beta}_t$ and $\boldsymbol{A}(L)$ is a matrix polynomial in the lag operator L (with $L\boldsymbol{\epsilon}_t = \boldsymbol{\epsilon}_{t-1}$), and $\boldsymbol{\epsilon}_t$ is independent normal. The Rosenberg (1972), Rosenberg (1973) return-to-normality model assumes that the absolute value of the characteristic roots of $\mathbf{H}$ be less than 1, with $\boldsymbol{\eta}_t$ independently normally distributed with mean $\boldsymbol{\mu} = (\mathbf{I}_k - \mathbf{H})\overline{\boldsymbol{\beta}}$.

The non-stationary random coefficients models do not regard the coefficient vector as having constant mean or variances. Changes in coefficients from one observation to the next can be the result of the realization of a nonstationary stochastic process or can be a function of exogenous variables. When the coefficients are realizations of a nonstationary stochastic process, we may again use (28.8) to represent such a process. For instance, the Cooley and Prescott (1976) model can be obtained by letting $\mathbf{H} = \mathbf{I}_k$ and $\boldsymbol{\mu} = \mathbf{0}$. When the coefficients $\boldsymbol{\beta}_{it}$ are functions of individual characteristics or time variables (e.g. see Amemiya (1978), Boskin and Lau (1990)), we can let

$$\boldsymbol{\beta}_{it} = \boldsymbol{\Gamma}\mathbf{q}_{it} + \boldsymbol{\eta}_{it}. \tag{28.10}$$

While the detailed formulation and estimation of the random coefficients model depends on the specific assumptions about the parameter variation, many types of random coefficients models can be conveniently represented using a mixed fixed and random coefficients framework of the form (see, for example, Hsiao, Appelbe, and Dineen (1992))

$$y_{it} = \mathbf{z}_{it}'\overline{\boldsymbol{\gamma}} + \mathbf{w}_{it}'\boldsymbol{\alpha}_{it} + u_{it},\ i = 1, 2, \ldots, N,\ t = 1, 2, \ldots, T, \tag{28.11}$$

where $\mathbf{z}_{it}$ and $\mathbf{w}_{it}$ are vectors of exogenous variables with dimensions ℓ and p respectively, $\overline{\boldsymbol{\gamma}}$ is an $\ell \times 1$ vector of constants, $\boldsymbol{\alpha}_{it}$ is a $p \times 1$ vector of random variables, and u_{it} is the error term. For instance, the Swamy type model, (28.3), can be obtained from (28.11) by letting $\mathbf{z}_{it} = \mathbf{w}_{it} = \mathbf{x}_{it}$, $\overline{\boldsymbol{\gamma}} = \overline{\boldsymbol{\beta}}$, and $\boldsymbol{\alpha}_{it} = \boldsymbol{\eta}_i$; the Hsiao type model (28.6) and (28.7) is obtained by letting $\mathbf{z}_{it} = \mathbf{w}_{it} = \mathbf{x}_{it}$, $\overline{\boldsymbol{\gamma}} = \overline{\boldsymbol{\beta}}$, and $\boldsymbol{\alpha}_{it} = \boldsymbol{\eta}_i + \boldsymbol{\lambda}_t$; the stochastic time varying parameter model (28.8) is obtained by letting $\mathbf{z}_{it} = \mathbf{x}_{it}$, $\mathbf{w}_{it}' = \mathbf{x}_{it}'(\mathbf{H}, \mathbf{I}_k)$, $\overline{\boldsymbol{\gamma}} = \boldsymbol{\mu}$, and $\boldsymbol{\alpha}_{it}' = \boldsymbol{\lambda}_t' = [\boldsymbol{\beta}_{t-1}', (\boldsymbol{\delta}_t - \boldsymbol{\mu})']$; and the model where $\boldsymbol{\beta}_{it}$ is a function of other variables is obtained by letting $\mathbf{z}_{it}' = \mathbf{x}_{it}' \otimes \mathbf{q}_{it}'$, $\overline{\boldsymbol{\gamma}}' = vec(\boldsymbol{\Gamma})$, $\mathbf{w}_{it} = \mathbf{x}_{it}$, $\boldsymbol{\alpha}_{it} = \boldsymbol{\eta}_{it}$, etc.

In this chapter we focus on models with time-invariant slope coefficients that vary randomly or freely over the cross-sectional units. We begin by considering the implications of neglecting such heterogeneity on the consistency and efficiency of the homogenous slope type estimators such as fixed and random effects models.

28.3 Properties of pooled estimators in heterogeneous panels

To understand the consequences of erroneously ignoring slope heterogeneity, consider the following panel data model, where, for simplicity of exposition, we set $k = 1$

$$y_{it} = \mu_i + \beta_i x_{it} + u_{it}, \tag{28.12}$$

$u_{it} \sim IID\left(0, \sigma_u^2\right)$, and μ_i are unknown fixed parameters. The coefficients, β_i, are allowed to vary freely across units but are otherwise assumed to be fixed (over time). It proves useful to decompose β_i into a common component, β, and a remainder term, η_i, that varies across units:

$$\beta_i = \beta + \eta_i. \tag{28.13}$$

The nature of the slope heterogeneity can now be characterized in terms of the properties of η_i, in particular where there is systematic dependence between η_i and the regressors x_{it} and an additional regressor z_{it}.

Consider an investigator that ignores the heterogeneity of the slope coefficients in (28.12), and instead estimates the model

$$y_{it} = \alpha_i + \delta_x x_{it} + \delta_z z_{it} + v_{it}, \tag{28.14}$$

where z_{it} is an additional regressor spuriously thought to be important by the researcher.

To simplify the derivations we make the following assumptions:

Assumption H.1: u_{it} is serially uncorrelated and distributed independently of u_{jt} for all $i \neq j$, with variance $0 < \sigma_i^2 < K$.

Assumption H.2: $\mathbf{w}_{it} = (x_{it}, z_{it})'$ is distributed independently of $u_{it'}$, for all i, t and t'.

Assumption H.3: $\mathbf{w}_{it}$ follows a covariance stationary process with the covariance matrix, $\boldsymbol{\Omega}_i$,

$$\boldsymbol{\Omega}_i = \begin{pmatrix} \omega_{ixx} & \omega_{ixz} \\ \omega_{izx} & \omega_{izz} \end{pmatrix}, \tag{28.15}$$

such that

$$E(\mathbf{\Omega}_i) = \lim_{N\to\infty}\left(\frac{1}{N}\sum_{i=1}^{N}\mathbf{\Omega}_i\right), \tag{28.16}$$

is a positive definite matrix.

Assumption H.4: For each t, $\mathbf{w}_{it}$ is distributed independently across i.

Note that not all the above assumptions are necessary when both N and T are sufficiently large. For example, assumption H.4 is not needed when T is sufficiently large. Assumptions H.1 and H.3 can be relaxed when T is small. It is also worth noting that assumption H.3 does not require the correlation matrix of the regressors for all i to be nonsingular, only that the 'pooled' covariance matrix, $E(\mathbf{\Omega}_i)$, defined by (28.16), should be nonsingular.

In matrix notation, (28.12) and (28.14) can be written as

$$\mathbf{y}_i = \mu_i\boldsymbol{\tau}_T + \beta_i\mathbf{x}_i + \mathbf{u}_i, \tag{28.17}$$

and

$$\mathbf{y}_i = \alpha_i\boldsymbol{\tau}_T + \mathbf{W}_i\boldsymbol{\delta} + \mathbf{v}_i, \tag{28.18}$$

respectively, where

$$\mathbf{y}_i = (y_{i1}, y_{i2}, \ldots, y_{iT})', \boldsymbol{\tau}_T = (1, 1, \ldots, 1)', \mathbf{x}_i = (x_{i1}, x_{i2}, \ldots, x_{iT})',$$
$$\mathbf{u}_i = (u_{i1}, u_{i2}, \ldots, u_{iT})', \boldsymbol{\delta} = (\delta_x, \delta_z)',$$

and

$$\mathbf{W}_i = \begin{pmatrix} x_{i1} & z_{i1} \\ x_{i2} & z_{i2} \\ \vdots & \vdots \\ x_{iT} & z_{iT} \end{pmatrix}, \ \mathbf{v}_i = \begin{pmatrix} v_{i1} \\ v_{i2} \\ \vdots \\ v_{iT} \end{pmatrix}.$$

The fixed-effects (FE) estimators of the slope coefficients in (28.18) can be written as

$$\hat{\boldsymbol{\delta}}_{FE} = \begin{pmatrix} \hat{\boldsymbol{\delta}}_{x,FE} \\ \hat{\boldsymbol{\delta}}_{z,FE} \end{pmatrix} = \left(\sum_{i=1}^{N}\mathbf{W}_i'\mathbf{M}_T\mathbf{W}_i\right)^{-1}\left(\sum_{i=1}^{N}\mathbf{W}_i'\mathbf{M}_T\mathbf{y}_i\right), \tag{28.19}$$

where $\mathbf{M}_T = \mathbf{I}_T - \boldsymbol{\tau}_T(\boldsymbol{\tau}_T'\boldsymbol{\tau}_T)^{-1}\boldsymbol{\tau}_T'$.[1] Under (28.17) we have

$$\hat{\boldsymbol{\delta}}_{FE} = \left(\frac{1}{NT}\sum_{i=1}^{N}\mathbf{W}_i'\mathbf{M}_T\mathbf{W}_i\right)^{-1}\left[\frac{1}{NT}\sum_{i=1}^{N}\left(\mathbf{W}_i'\mathbf{M}_T\mathbf{x}_i\right)\beta_i + \frac{1}{NT}\sum_{i=1}^{N}\mathbf{W}_i'\mathbf{M}_T\mathbf{u}_i\right]. \tag{28.20}$$

[1] The fixed-effects estimator in (28.19) assumes a balanced panel. But the results readily extend to unbalanced panels.

It is now easily seen that under Assumptions H.1–H.4 and for N and/or T sufficiently large

$$\frac{1}{N}\sum_{i=1}^{N}\left(\frac{\mathbf{W}_i'\mathbf{M}_T\mathbf{u}_i}{T}\right)\xrightarrow{p}0, \tag{28.21}$$

where $\xrightarrow{p}$ denotes convergence in probability. To see this, note that since u_{it} are cross-sectionally independent and $\mathbf{w}_{it}$ are strictly exogenous, then we have

$$Var\left[\frac{1}{N}\sum_{i=1}^{N}\left(\frac{\mathbf{W}_i'\mathbf{M}_T\mathbf{u}_i}{T}\right)\right]=\frac{1}{TN}\left[\frac{1}{N}\sum_{i=1}^{N}\sigma_i^2E\left(\frac{\mathbf{W}_i'\mathbf{M}_T\mathbf{W}_i}{T}\right)\right].$$

Also, under Assumptions H.1 and H.3, σ_i^2 and $E\left(T^{-1}\mathbf{W}_i'\mathbf{M}_T\mathbf{W}_i\right)$ are bounded and as a result

$$Var\left[\frac{1}{N}\sum_{i=1}^{N}\left(\frac{\mathbf{W}_i'\mathbf{M}_T\mathbf{u}_i}{T}\right)\right]\rightarrow\mathbf{0},$$

if N and/or $T\rightarrow\infty$. Also, under strict exogeneity of $\mathbf{w}_{it}$, $E\left(T^{-1}\mathbf{W}_i'\mathbf{M}_T\mathbf{u}_i\right)=\mathbf{0}$, for all i, and the desired result in (28.21) follows.

Using (28.21) in (28.20) we now have

$$Plim_{N,T\rightarrow\infty}(\hat{\boldsymbol{\delta}}_{FE})=\left[Plim\left(\sum_{i=1}^{N}\frac{\mathbf{W}_i'\mathbf{M}_T\mathbf{W}_i}{NT}\right)\right]^{-1}Plim\left[\sum_{i=1}^{N}\left(\frac{\mathbf{W}_i'\mathbf{M}_T\mathbf{x}_i}{NT}\right)\beta_i\right]. \tag{28.22}$$

In the case where the slopes are homogenous, namely $\beta_i=\beta$, we have

$$Plim(\hat{\boldsymbol{\delta}}_{FE})=\begin{pmatrix}\beta\\0\end{pmatrix}. \tag{28.23}$$

Consider now the case where the slopes are heterogenous. Using the above results, it is now easily seen that the consistency result in (28.23) will follow if and only if

$$\sum_{i=1}^{N}\left(\frac{\mathbf{W}_i'\mathbf{M}_T\mathbf{x}_i}{NT}\right)\eta_i=\begin{pmatrix}\frac{1}{NT}\sum_{i=1}^{N}\mathbf{x}_i'\mathbf{M}_T\mathbf{x}_i\eta_i\\\frac{1}{NT}\sum_{i=1}^{N}\mathbf{z}_i'\mathbf{M}_T\mathbf{x}_i\eta_i\end{pmatrix}\xrightarrow{p}\mathbf{0}. \tag{28.24}$$

This condition holds under the random coefficient specification where it is assumed that η_i's are distributed independently of $\mathbf{w}_{it}$ for all i and t. (See below and Swamy (1970)). Under Assumption H.3 and as $T\rightarrow\infty$ we have

$$\frac{1}{NT}\sum_{i=1}^{N}\left(\mathbf{x}_i'\mathbf{M}_T\mathbf{x}_i\right)\eta_i-\frac{1}{N}\sum_{i=1}^{N}\omega_{ixx}\eta_i\xrightarrow{p}0,$$

and

$$\frac{1}{NT}\sum_{i=1}^{N}\left(\mathbf{z}_i'\mathbf{M}_T\mathbf{x}_i\right)\eta_i-\frac{1}{N}\sum_{i=1}^{N}\omega_{izx}\eta_i\xrightarrow{p}0.$$

Therefore, for the fixed-effects estimator, (28.19), to be consistent we must have

$$\frac{1}{N}\sum_{i=1}^{N}\omega_{ixx}\eta_i\xrightarrow{p}0,\quad\text{and}\quad\frac{1}{N}\sum_{i=1}^{N}\omega_{izx}\eta_i\xrightarrow{p}0,\tag{28.25}$$

as $N\to\infty$. Namely, any systematic dependence between β_i and the second-order moments of the steady state distribution of the regressors $\mathbf{w}_{it}$ must also be ruled out. When the conditions in (28.25) are not satisfied, the inconsistencies of the fixed-effects estimators (for T and N sufficiently large) are given by[2]

$$Plim(\hat{\delta}_{x,FE}-\beta)=\frac{Cov(\omega_{ixx},\eta_i)E(\omega_{izz})-E(\omega_{ixz})Cov(\omega_{ixz},\eta_i)}{E(\omega_{ixx})E(\omega_{izz})-\left[E(\omega_{ixz})\right]^2},\tag{28.26}$$

$$Plim(\hat{\delta}_{z,FE})=\frac{Cov(\omega_{ixz},\eta_i)E(\omega_{ixx})-E(\omega_{ixz})Cov(\omega_{ixx},\eta_i)}{E(\omega_{ixx})E(\omega_{izz})-\left[E(\omega_{ixz})\right]^2},\tag{28.27}$$

where

$$\begin{aligned}&Cov(\omega_{ixx},\eta_i)=Plim_{N\to\infty}\left(\frac{1}{N}\sum_{i=1}^{N}\omega_{ixx}\eta_i\right),\ Cov(\omega_{ixz},\eta_i)=Plim_{N\to\infty}\left(\frac{1}{N}\sum_{i=1}^{N}\omega_{ixz}\eta_i\right),\\&E(\omega_{ixx})=\lim_{N\to\infty}\left(\frac{1}{N}\sum_{i=1}^{N}\omega_{ixx}\right),\ E(\omega_{izz})=\lim_{N\to\infty}\left(\frac{1}{N}\sum_{i=1}^{N}\omega_{izz}\right),\\&E(\omega_{ixz})=\lim_{N\to\infty}\left(\frac{1}{N}\sum_{i=1}^{N}\omega_{ixz}\right).\end{aligned}\tag{28.28}$$

The above results have a number of interesting implications:

1. The FE estimators, $\hat{\delta}_{x,FE}$ and $\hat{\delta}_{z,FE}$, are both consistent if

$$Cov(\omega_{ixz},\eta_i)=Cov(\omega_{ixx},\eta_i)=0.\tag{28.29}$$

Clearly, these conditions are met under slope homogeneity. In the present application where the regressors are assumed to be strictly exogenous, the fixed-effects estimators converge to their true values under the random coefficient model (RCM) where the slope coefficients and the regressors are assumed to be independently distributed. Notice, however, that since the β_i's are assumed to be fixed over time, then any systematic depen-

[2] Notice that under slope heterogeneity the fixed-effects estimators are inconsistent when N is finite and only $T\to\infty$.

dence of η_i on $\mathbf{w}_{it}$ over time is already ruled out under model (28.12). The random coefficients assumption imposes further restrictions on the joint distribution of η_i and the cross-sectional distribution of $\mathbf{w}_{it}$.

2. The *FE* estimator of δ_z is robust to slope heterogeneity if the incorrectly included regressors, z_{it}, are on average orthogonal to x_{it}, namely when $E(\omega_{ixz}) = 0$, and if $Cov(\omega_{ixz}, \eta_i) = 0$. However, in the presence of slope heterogeneity, the FE estimator of δ_x continues to be inconsistent even if z_{it} and x_{it} are on average orthogonal. The direction of the asymptotic bias of $\hat{\delta}_{x,FE}$ depends on the sign of $Cov(\omega_{ixx}, \eta_i)$. The bias of $\hat{\delta}_{x,FE}$ is positive when $Cov(\omega_{ixx}, \eta_i) > 0$ and *vice versa*.[3]
3. In general, where $E(\omega_{ixz}) \neq 0$ and $Cov(\omega_{ixz}, \eta_i) \neq 0$ and/or $Cov(\omega_{ixx}, \eta_i) \neq 0$, the fixed-effects estimators, $\hat{\delta}_{x,FE}$ and $\hat{\delta}_{z,FE}$, are both inconsistent.

In short, if the slope coefficients are fixed but vary systematically across the groups, the application of the general-to-specific methodology to standard panel data models can lead to misleading results (spurious inference). An important example is provided by the case when attempts are made to check for the presence of nonlinearities by testing the significance of quadratic terms in static panel data models using fixed-effects estimators. In the context of our simple specification, this would involve setting $z_{it} = x_{it}^2$, and a test of the significance of z_{it} in (28.14) will yield sensible results only if the conditions defined by (28.29) are met. In general, it is possible to falsely reject the linearity hypothesis when there are systematic relations between the slope coefficients and the cross-sectional distribution of the regressors. Therefore, results from nonlinearity tests in panel data models should be interpreted with care. The linearity hypothesis may be rejected not because of the existence of a genuine nonlinear relationship between y_{it} and x_{it}, but due to slope heterogeneity.

Finally, it is worth noting that since the β_i's are fixed for each i, the nonlinear specification

$$y_{it} = \alpha_i + \delta_x x_{it} + \delta_z x_{it}^2 + v_{it}, \tag{28.30}$$

cannot be reconciled with (28.29), unless it is assumed that β_i varies proportionately with x_{it}. Clearly, it is possible to allow the slopes, β_i, to vary systematically with some aspect of the cross-sectional distribution of x_{it} without requiring β_i to be proportional to x_{it}, and hence time-varying. For example, it could be that

$$\beta_i = \gamma_0 + \gamma_1 \bar{x}_i, \tag{28.31}$$

where $\bar{x}_i = T^{-1} \sum_{t=1}^{T} x_{it}$. This specification retains the linearity of (28.29) for each i, but can still yield a statistically significant effect for x_{it}^2 in (28.30) if slope heterogeneity is ignored and fixed-effects estimates of (28.30) are used for inference. This feature of fixed-effects regressions under heterogeneous slopes is illustrated in Figure 28.1. The figure shows scatter points and associated regression lines for three countries with slopes that differ systematically with $\bar{x}_i$. It is clear that the pooled regression based on the scatter points from all three countries will exhibit strong nonlinearities, although the country-specific regressions are linear.

[3] Notice that $E(\omega_{ixx})E(\omega_{izz}) - (E(\omega_{ixz}))^2 > 0$, unless x_{it} and z_{it} are perfectly collinear for all i, which we rule out.

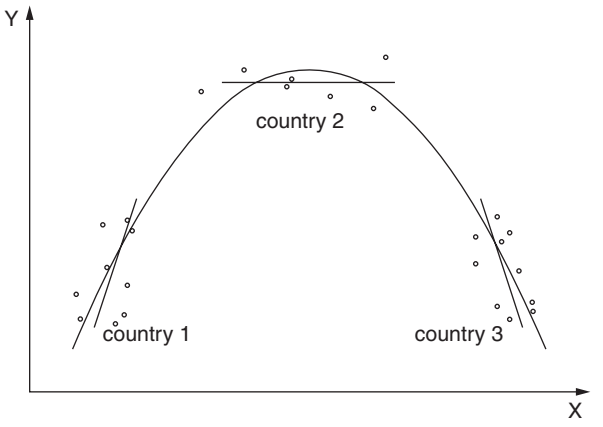


Figure 28.1 Fixed-effects and pooled estimators.

Example 63 *One interesting study illustrating the importance of slope heterogeneity in cross country analysis is the analysis by Haque, Pesaran, and Sharma (2000) on the determinants of cross-country private savings rates, using a subset of data from Masson, Bayoumi, and Samiei (1998) (MBS), on 21 OECD countries over 1971–1993. MBS ran FE regressions of*

PSAV : *the private savings rate, defined as the ratio of aggregate private savings to GDP;*

on the explanatory variables

SUR : *the ratio of general government budget surplus to GDP;*
GCUR : *the ratio of the general government current expenditure to GDP;*
GI : *the ratio of the general government investment to GDP;*
GR : *GDP growth rate;*
RINT : *real interest rate;*
INF : *inflation rate;*
PCTT : *percentage change in terms of trade;*
YRUS : *per capita GDP relative to the U.S.;*
DEP : *dependency ratio, defined as the ratio of those under 20, 65 and over to those aged 20–64;*
W : *ratio of private wealth (measured as the cumulative sum of past nominal private savings) to GDP.*

Table 28.1 contains the FE regression for the industrial countries. We refer to this specification as model M_0. The estimates under 'model M_0' in Table 28.1 are identical to those reported in column 1 of Table 3 in MBS (1998), except for a few typos. Apart from the coefficient of the GDP growth

rate (GR), all the estimated coefficients are statistically (some very highly) significant, and in particular suggest a strong quadratic relationship between saving and per-capita income. However, the validity of these estimates and the inferences based on them critically depend on the extent to which slope coefficients differ across countries, and in the case of static models, whether these differences are systematic. As shown above, one important implication of neglected slope heterogeneity is the possibility of obtaining spurious nonlinear effects. This possibility is explored by adding quadratic terms in W, INF, PCTT, and DEP to the regressors already included in model M_0. Estimation results, reported under 'model M_1' in Table 28.1, show that the quadratic terms are all statistically highly significant. While there may be some a priori argument for a nonlinear wealth effect in the savings equation, the rationale for nonlinear effects in the case of the other three variables seems less clear. The quadratic relationships between the private savings rate and the variables W, PCTT, and DEP are in fact much stronger than the quadratic relationship between savings and per capita income that MBS focus on. The $\bar{R}^2$ of the augmented model, 0.801, is also appreciably larger than that obtained for model M_0, 0.766. A similar conclusion is reached using other model selection criteria such as the Akaike information criterion (AIC) and the Schwarz Bayesian criterion (SBC) also reported in Table 28.1. As an alternative to the quadratic specifications used in model M_1, the authors investigate the possibility that the slope coefficients in each country are fixed over time, but are allowed to vary across countries linearly with the sample means of their wealth to GDP ratio or their per-capita income. More specifically, denote the vector of slope coefficients for country i by $\boldsymbol{\beta}_i$, and define

$$\overline{W}_i = T^{-1}\sum_{t=1}^{T} W_{it}, \text{ and } \overline{YRUS}_i = T^{-1}\sum_{t=1}^{T} YRUS_{it}.$$

Then, slope heterogeneity is modelled by

$$\boldsymbol{\beta}_i = \boldsymbol{\beta}_0 + \boldsymbol{\beta}_{01}\overline{W}_i + \boldsymbol{\beta}_{02}\overline{YRUS}_i. \tag{28.32}$$

Substituting the above expression for $\boldsymbol{\beta}_i$ in the FE specification, yields

$$y_{it} = \mu_i + \boldsymbol{\beta}_0'\mathbf{x}_{it} + \boldsymbol{\beta}_{01}'(\mathbf{x}_{it}\overline{W}_i) + \boldsymbol{\beta}_{02}'(\mathbf{x}_{it}\overline{YRUS}_i) + u_{it},$$

where $y_{it} = PSAV_{it}$,

$$\mathbf{x}_{it} = (SUR_{it}, GCUR_{it}, GI_{it}, GR_{it}, RINT_{it}, W_{it}, INF_{it}, PCTT_{it}, YRUS_{it}, DEP_{it})'.$$

The estimated elements of $\boldsymbol{\beta}_0$, $\boldsymbol{\beta}_{01}$, and $\boldsymbol{\beta}_{02}$ together with their t-ratios are given in Table 28.2. Apart from the coefficient of the SUR variable, all the other coefficients show systematic variation across countries. The coefficient of the SUR variable seems to be least affected by slope heterogeneity, and the hypothesis of slope homogeneity cannot be rejected in the case of this variable. However, none of the other estimates is directly comparable to the FE estimates given in Table 28.1. In particular, the coefficients of output growth variables (GR_{it} and $GR_{it} \times \overline{W}_i$) are both statistically significant, while this was not so in the case of the FE estimates in Table 28.1. Care must also be exercised when interpreting these estimates. For example, the results suggest that the effect of real output growth on the savings rate is likely to be higher in a country with a high wealth–GDP ratio. Similarly, inflation

Table 28.1 Fixed-effects estimates of static private saving equations, models M_0 and M_1 (21 OECD countries, 1971–1993)

	Model M_0		Model M_1	
Regressors	**Linear Terms**	**Quadratic Terms**	**Linear Terms**	**Quadratic Terms**
SUR	−0.574	–	−0.58	–
	(−9.39)	–	(−10.30)	–
GCUR	−0.467	–	−0.521	–
	(−11.30)	–	(−13.39)	–
GI	−0.603	–	−0.701	–
	(−5.71)	–	(−6.92)	–
GR	−0.060	–	−0.065	–
	(−1.14)	–	(−1.33)	
RINT	0.212	–	0.281	
	(4.40)	–	(5.90)	
W	0.023	–	0.175	−0.00025
	(5.11)	–	(8.38)	(−7.69)
INF	0.180	–	−0.041	0.011
	(4.63)		(−0.53)	(3.29)
PCTT	0.047		0.063	−0.0013
	(3.07)		(4.11)	(−2.81)
YRUS	0.586	−0.0048	0.286	−0.0026
	(3.41)	(−3.90)	(1.70)	(−2.15)
DEP	−0.118	–	−1.201	0.0073
	(−4.12)		(−5.25)	(4.85)
$\overline{R}^2$	0.766		0.801	
$\hat{\sigma}$	2.325		2.145	
LL	−1076.4		−1035.3	
AIC	−1108.4		−1071.3	
SBC	−1165.3		−1146.5	

*The dependent variable (PSAV) is the ratio of private savings to GNP. Model M_0 is the specification estimated by Masson et al. (1998), see column 1 of Table 3 in that paper. The figures in brackets are t-ratios. $\overline{R}$ is the adjusted multiple correlation coefficient, $\hat{\sigma}$ is the standard error of the regression; LL is the maximazed value of the log-likehood function; AIC is the Akaike information criterion, and SBC is the Schwarz Bayesian criterion.

effects on the savings rate are estimated to be higher in countries with higher wealth to GDP ratios. However, these results do not predict, for instance, that an individual country's savings rate will necessarily rise with output growth.

For further discussion on the consequences of ignoring parameter heterogeneity see, for example, Robertson and Symons (1992) and Haque, Pesaran, and Sharma (2000).

28.4 The Swamy estimator

Consider the panel data model

$$y_{it} = \boldsymbol{\beta}_i' \mathbf{x}_{it} + u_{it}, \tag{28.33}$$

Table 28.2 Fixed-effects estimates of private savings equations with cross-sectionally varying slopes, (Model M2), (21 OECD countries, 1971–1993)

Regressors	$\hat{\boldsymbol{\beta}}_{\mathbf{0}}$	$\hat{\boldsymbol{\beta}}_{\mathbf{01}}$	$\hat{\boldsymbol{\beta}}_{\mathbf{02}}$
SUR	−0.625	–	–
	(−12.10)		
GCUR	−1.146	0.0022	–
	(−6.91)	(4.26)	
GI	−1.891	0.0039	–
	(−2.44)	(1.60)	
GR	−0.744	0.0023	
	(−2.69)	(2.71)	–
RINT	0.417	–	−0.0052
	(4.36)	–	(−3.53)
W	0.119	−0.00033	–
	(5.28)	(−4.70)	–
INF	−0.860	0.0031	–
	(−5.29)	(6.29)	
PCTT	−0.214	0.00083	–
	(−1.88)	(2.30)	
YRUS	1.435	−0.0046	–
	(6.31)	(−6.72)	
DEP	0.502	−0.0021	–
	(2.54)	(−3.39)	
$\overline{R}^2$	0.838		
$\hat{\sigma}$	1.934		
LL	−982.9		
AIC	−1022.9		
SBC	−1106.5		

*See the notes to Table 28.1

under the Swamy (1970) random coefficient scheme (28.3), where $\boldsymbol{\eta}_i$ satisfies assumptions (28.4)–(28.5). For simplicity, we also assume that u_{it} is *independently* distributed across i and over t with zero mean and $Var\,(u_{it}) = \sigma_i^2$. Substituting $\boldsymbol{\beta}_i = \boldsymbol{\beta} + \boldsymbol{\eta}_i$ into (28.33) we obtain, using stacked form notation,

$$\mathbf{y}_{i.} = \mathbf{X}_{i.}\boldsymbol{\beta} + \mathbf{v}_{i.},$$

where the composite error, $\mathbf{v}_{i.}$, is given by

$$\mathbf{v}_{i.} = \mathbf{X}_{i.}\boldsymbol{\eta}_i + \boldsymbol{\varepsilon}_{i.}.$$

Stacking the regression equations by cross-sectional units we now have

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{v},$$

where

$$\mathbf{y} = \begin{pmatrix} \mathbf{y}_{1.} \\ \mathbf{y}_{2.} \\ \vdots \\ \mathbf{y}_{N.} \end{pmatrix}, \mathbf{X} = \begin{pmatrix} \mathbf{X}_{1.} \\ \mathbf{X}_{2.} \\ \vdots \\ \mathbf{X}_{N.} \end{pmatrix}, \text{ and } \mathbf{v} = \begin{pmatrix} \mathbf{v}_{1.} \\ \mathbf{v}_{2.} \\ \vdots \\ \mathbf{v}_{N.} \end{pmatrix}.$$

Suppose we are interested in estimating the mean coefficient vector, $\boldsymbol{\beta}$, and the covariance matrix of $\mathbf{v}$, $\boldsymbol{\Sigma}$, given by

$$\boldsymbol{\Sigma} = E\left(\mathbf{v}\mathbf{v}'\right) = \begin{pmatrix} \boldsymbol{\Sigma}_1 & \mathbf{0} & \dots & \mathbf{0} \\ \mathbf{0} & \boldsymbol{\Sigma}_2 & \dots & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{0} & \mathbf{0} & \dots & \boldsymbol{\Sigma}_N \end{pmatrix},$$

where

$$\boldsymbol{\Sigma}_i = Var\left(\mathbf{v}_{i.}\right) = \sigma_i^2 \mathbf{I}_T + \mathbf{X}_{i.} \boldsymbol{\Omega}_\eta \mathbf{X}_{i.}'.$$

For known values of $\boldsymbol{\Omega}_\eta$ and σ_i^2, the best linear unbiased estimator of $\boldsymbol{\beta}$ is given by the generalized least squares (GLS) estimator, known in this case as the Swamy estimator

$$\begin{aligned} \hat{\boldsymbol{\beta}}_{SW} &= \left(\mathbf{X}'\boldsymbol{\Sigma}^{-1}\mathbf{X}\right)^{-1}\mathbf{X}'\boldsymbol{\Sigma}^{-1}\mathbf{y}, \\ &= \left(\sum_{i=1}^{N} \mathbf{X}_{i.}'\boldsymbol{\Sigma}_i^{-1}\mathbf{X}_{i.}\right)^{-1} \sum_{i=1}^{N} \mathbf{X}_{i.}'\boldsymbol{\Sigma}_i^{-1}\mathbf{y}_{i.}. \end{aligned}$$

It is easily seen that (under the assumption that $\boldsymbol{\Omega}_\eta$ is nonsingular) (see property (A.9) in Appendix A)

$$\boldsymbol{\Sigma}_i^{-1} = \frac{\mathbf{I}_T}{\sigma_i^2} - \frac{\mathbf{X}_{i.}}{\sigma_i^2}\left(\frac{\mathbf{X}_{i.}'\mathbf{X}_{i.}}{\sigma_i^2} + \boldsymbol{\Omega}_\eta^{-1}\right)^{-1}\frac{\mathbf{X}_{i.}'}{\sigma_i^2}.$$

Note that $\boldsymbol{\Sigma}_i^{-1}$ exists even if $\boldsymbol{\Omega}_\eta$ is singular. In general we can write[4]

$$\boldsymbol{\Sigma}_i^{-1} = \frac{\mathbf{I}_T}{\sigma_i^2} - \frac{\mathbf{X}_{i.}\boldsymbol{\Omega}_\eta}{\sigma_i^2}\left(\mathbf{I}_k + \frac{\mathbf{X}_{i.}'\mathbf{X}_{i.}}{\sigma_i^2}\boldsymbol{\Omega}_\eta\right)^{-1}\frac{\mathbf{X}_{i.}'}{\sigma_i^2},$$

which is valid irrespective of whether $\boldsymbol{\Omega}_\eta$ is singular or not. Let

$$\begin{aligned} \mathbf{Q}_{iT} &= \frac{\mathbf{X}_{i.}'\mathbf{X}_{i.}}{T\sigma_i^2}, \ \mathbf{q}_{iT} = \frac{\mathbf{X}_{i.}'\mathbf{y}_{i.}}{T\sigma_i^2}, \\ \mathbf{H}_{iT} &= \mathbf{Q}_{iT} + \frac{1}{T}\boldsymbol{\Omega}_\eta^{-1}. \end{aligned}$$

[4] In formula (A.9), let $\mathbf{X} = \mathbf{X}_i\boldsymbol{\Omega}_\eta$, $\mathbf{Y} = \mathbf{X}_i'$, $\mathbf{C} = \frac{\mathbf{I}_T}{\sigma_i^2}$, and $\mathbf{D} = \mathbf{I}_k$, then the desired result follows.

Then

$$\frac{\mathbf{X}_{i.}'\boldsymbol{\Sigma}_i^{-1}\mathbf{X}_{i.}}{T} = \mathbf{Q}_{iT} - \mathbf{Q}_{iT}\mathbf{H}_{iT}^{-1}\mathbf{Q}_{iT},$$

and

$$\frac{\mathbf{X}_{i.}'\boldsymbol{\Sigma}_i^{-1}\mathbf{y}_{i.}}{T} = \mathbf{q}_{iT} - \mathbf{Q}_{iT}\mathbf{H}_{iT}^{-1}\mathbf{q}_{iT}.$$

It follows that the Swamy estimator can also be written as

$$\hat{\boldsymbol{\beta}}_{SW} = \left[\sum_{i=1}^{N}\left(\mathbf{Q}_{iT} - \mathbf{Q}_{iT}\mathbf{H}_{iT}^{-1}\mathbf{Q}_{iT}\right)\right]^{-1}\sum_{i=1}^{N}(\mathbf{q}_{iT} - \mathbf{Q}_{iT}\mathbf{H}_{iT}^{-1}\mathbf{q}_{iT}). \tag{28.34}$$

By repeatedly utilizing the identity relation (A.9) in Appendix A, we obtain

$$\bar{\hat{\boldsymbol{\beta}}}_{SW} = \sum_{i=1}^{N}\mathbf{R}_i\hat{\boldsymbol{\beta}}_i,$$

where

$$\mathbf{R}_i = \left[\sum_{i=1}^{N}\left(\boldsymbol{\Omega}_\eta + \boldsymbol{\Sigma}_{\hat{\boldsymbol{\beta}}_i}\right)^{-1}\right]^{-1}\left(\boldsymbol{\Omega}_\eta + \boldsymbol{\Sigma}_{\hat{\boldsymbol{\beta}}_i}\right)^{-1}, \tag{28.35}$$

and

$$\hat{\boldsymbol{\beta}}_i = (\mathbf{X}_{i.}'\mathbf{X}_{i.})^{-1}\mathbf{X}_{i.}'\mathbf{y}_{i.},\ \boldsymbol{\Sigma}_{\hat{\boldsymbol{\beta}}_i} = Var(\hat{\boldsymbol{\beta}}_i) = \sigma_i^2(\mathbf{X}_{i.}'\mathbf{X}_{i.})^{-1}. \tag{28.36}$$

The expression (28.34) shows that the Swamy estimator is a matrix weighted average of the least squares estimator for each cross-sectional unit (28.36), with the weights inversely proportional to their covariance matrices. It also shows that the *GLS* estimator requires only a matrix inversion of order k, and so it is not much more complicated to compute than the sample least squares estimator.

The covariance matrix of the SW estimator is

$$Var\left(\bar{\hat{\boldsymbol{\beta}}}_{SW}\right) = \left(\sum_{i=1}^{N}\mathbf{X}_i'\boldsymbol{\Sigma}_i^{-1}\mathbf{X}_i\right)^{-1} = \left[\sum_{i=1}^{N}\left(\boldsymbol{\Omega}_\eta + \boldsymbol{\Sigma}_{\hat{\boldsymbol{\beta}}_i}\right)^{-1}\right]^{-1}. \tag{28.37}$$

If errors u_{it} and $\boldsymbol{\eta}_i$ are normally distributed, the SW estimator is the same as the maximum likelihood (ML) estimator of $\boldsymbol{\beta}$ conditional on $\boldsymbol{\Omega}_\eta$ and σ_i^2. Without knowledge of $\boldsymbol{\Omega}_\eta$ and σ_i^2, we can estimate $\boldsymbol{\beta}$, $\boldsymbol{\Omega}_\eta$ and σ_i^2, $i = 1, 2, \ldots, N$ simultaneously by the ML method. However, it

can be computationally tedious. A natural alternative is to first estimate $\boldsymbol{\Sigma}_i$, then substitute the estimated $\boldsymbol{\Sigma}_i$ into (28.37).

Swamy proposes using the least squares estimator of $\boldsymbol{\beta}_i$, $\hat{\boldsymbol{\beta}}_i = (\mathbf{X}_{i.}'\mathbf{X}_{i.})^{-1}\mathbf{X}_{i.}'\mathbf{y}_{i.}$ and the residuals $\hat{\mathbf{u}}_{i.} = \mathbf{y}_{i.} - \mathbf{X}_{i.}\hat{\boldsymbol{\beta}}_i$ to obtain consistent estimators of σ_i^2, for $i = 1, \ldots, N$, and $\boldsymbol{\Omega}_\eta$. Noting that

$$\hat{\mathbf{u}}_{i.} = [\mathbf{I}_T - \mathbf{X}_{i.}(\mathbf{X}_{i.}'\mathbf{X}_{i.})^{-1}\mathbf{X}_{i.}']\mathbf{u}_{i.}, \tag{28.38}$$

and

$$\hat{\boldsymbol{\beta}}_i = \boldsymbol{\beta}_i + (\mathbf{X}_{i.}'\mathbf{X}_{i.})^{-1}\mathbf{X}_{i.}'\mathbf{u}_{i.}, \tag{28.39}$$

we obtain the unbiased estimators of σ_i^2 and $\boldsymbol{\Omega}_\eta$ as

$$\begin{aligned}\hat{\sigma}_i^2 &= \frac{\hat{\mathbf{u}}_{i.}'\hat{\mathbf{u}}_{i.}}{T-k}, \qquad (28.40)\\ &= \frac{1}{T-k}\mathbf{y}_i'[\mathbf{I}_T - \mathbf{X}_{i.}(\mathbf{X}_{i.}'\mathbf{X}_{i.})^{-1}\mathbf{X}_{i.}']\mathbf{y}_{i.},\\ \hat{\boldsymbol{\Omega}}_\eta &= \frac{1}{N-1}\sum_{i=1}^{N}\left(\hat{\boldsymbol{\beta}}_i - N^{-1}\sum_{j=1}^{N}\hat{\boldsymbol{\beta}}_j\right)\left(\hat{\boldsymbol{\beta}}_i - N^{-1}\sum_{j=1}^{N}\hat{\boldsymbol{\beta}}_j\right)'\\ &\quad - \frac{1}{TN}\sum_{i=1}^{N}\hat{\sigma}_i^2\left(\frac{\mathbf{X}_{i.}'\mathbf{X}_{i.}}{T}\right)^{-1}. \qquad (28.41)\end{aligned}$$

Just as in the error-components model, the estimator (28.41) is not necessarily non-negative definite. In this situation, Swamy has suggested replacing (28.41) by

$$\hat{\boldsymbol{\Omega}}_\eta^* = \frac{1}{N-1}\sum_{i=1}^{N}\left(\hat{\boldsymbol{\beta}}_i - N^{-1}\sum_{j=1}^{N}\hat{\boldsymbol{\beta}}_j\right)\left(\hat{\boldsymbol{\beta}}_i - N^{-1}\sum_{j=1}^{N}\hat{\boldsymbol{\beta}}_j\right)'. \tag{28.42}$$

This estimator, although biased, is nonnegative definite and consistent when T tends to infinity.

For further discussion on the above estimator see Swamy (1970), and Hsiao and Pesaran (2008).

28.5 The mean group estimator (*MGE*)

One alternative to Swamy's estimator of $\boldsymbol{\beta}$ in equation (28.33) is the mean group (*MG*) estimator, proposed by Pesaran and Smith (1995) for estimation of dynamic random coefficient models. The *MG* estimator is defined as the simple average of the *OLS* estimators, $\hat{\boldsymbol{\beta}}_i$

$$\hat{\boldsymbol{\beta}}_{MG} = \frac{1}{N}\sum_{i=1}^{N}\hat{\boldsymbol{\beta}}_i,$$

where

$$\hat{\boldsymbol{\beta}}_i = \left(\mathbf{X}_{i.}'\mathbf{X}_{i.}\right)^{-1}\mathbf{X}_{i.}'\mathbf{y}_{i.}.$$

MG estimation is possible when both T and N are sufficiently large, and is applicable irrespective of whether the slope coefficients are random (in Swamy's sense), or fixed in the sense that the diversity in the slope coefficients across cross sectional units cannot be captured by means of a finite parameter probability distribution. To compute the variance of the *MG* estimator, first note that

$$\hat{\boldsymbol{\beta}}_i = \boldsymbol{\beta} + \boldsymbol{\eta}_i + \boldsymbol{\xi}_{i.},$$

where

$$\begin{aligned}\boldsymbol{\xi}_{i.} &= \left(\mathbf{X}_{i.}'\mathbf{X}_{i.}\right)^{-1}\mathbf{X}_{i.}'\mathbf{u}_{i.},\\ \hat{\boldsymbol{\beta}}_{MG} &= \boldsymbol{\beta} + \overline{\boldsymbol{\eta}} + \overline{\boldsymbol{\xi}},\end{aligned} \tag{28.43}$$

and

$$\overline{\boldsymbol{\eta}} = \frac{1}{N}\sum_{i=1}^{N}\boldsymbol{\eta}_i, \quad \overline{\boldsymbol{\xi}} = \frac{1}{N}\sum_{i=1}^{N}\boldsymbol{\xi}_{i.}.$$

Hence, when the regressors are strictly exogenous and the errors, u_{it}, are independently distributed, the variance of $\hat{\boldsymbol{\beta}}_{MG}$ is

$$\begin{aligned}Var\left(\hat{\boldsymbol{\beta}}_{MG}\right) &= Var\left(\overline{\boldsymbol{\eta}}\right) + Var\left(\overline{\boldsymbol{\xi}}\right)\\ &= \frac{1}{N}\boldsymbol{\Omega}_{\eta} + \frac{1}{N^2}\sum_{i=1}^{N}\sigma_i^2 E\left[\left(\frac{\mathbf{X}_{i.}'\mathbf{X}_{i.}}{T}\right)^{-1}\right].\end{aligned}$$

An unbiased estimator of the covariance matrix of $\hat{\boldsymbol{\beta}}_{MG}$ can be computed as

$$\widehat{Var}\left(\hat{\boldsymbol{\beta}}_{MG}\right) = \frac{1}{N(N-1)}\sum_{i=1}^{N}\left(\hat{\boldsymbol{\beta}}_i - \hat{\boldsymbol{\beta}}_{MG}\right)\left(\hat{\boldsymbol{\beta}}_i - \hat{\boldsymbol{\beta}}_{MG}\right)'.$$

For a proof, first note that

$$\begin{aligned}\hat{\boldsymbol{\beta}}_i - \hat{\boldsymbol{\beta}}_{MG} &= \left(\boldsymbol{\eta}_i - \overline{\boldsymbol{\eta}}\right) + \left(\boldsymbol{\xi}_{i.} - \overline{\boldsymbol{\xi}}\right),\\ \left(\hat{\boldsymbol{\beta}}_i - \hat{\boldsymbol{\beta}}_{MG}\right)\left(\hat{\boldsymbol{\beta}}_i - \hat{\boldsymbol{\beta}}_{MG}\right)' &= \left(\boldsymbol{\eta}_i - \overline{\boldsymbol{\eta}}\right)\left(\boldsymbol{\eta}_i - \overline{\boldsymbol{\eta}}\right)' + \left(\boldsymbol{\xi}_{i.} - \overline{\boldsymbol{\xi}}\right)\left(\boldsymbol{\xi}_{i.} - \overline{\boldsymbol{\xi}}\right)'\\ &\quad + \left(\boldsymbol{\eta}_i - \overline{\boldsymbol{\eta}}\right)\left(\boldsymbol{\xi}_{i.} - \overline{\boldsymbol{\xi}}\right)' + \left(\boldsymbol{\xi}_{i.} - \overline{\boldsymbol{\xi}}\right)\left(\boldsymbol{\eta}_i - \overline{\boldsymbol{\eta}}\right)',\end{aligned}$$

and

$$\sum_{i=1}^{N} E\left[\left(\hat{\boldsymbol{\beta}}_i - \hat{\boldsymbol{\beta}}_{MG}\right)\left(\hat{\boldsymbol{\beta}}_i - \hat{\boldsymbol{\beta}}_{MG}\right)'\right] = (N-1)\boldsymbol{\Omega}_\eta + \left(1 - \frac{1}{N}\right)\sum_{i=1}^{N}\sigma_i^2 E\left[\left(\mathbf{X}_{i.}'\mathbf{X}_{i.}\right)^{-1}\right].$$

Using the above results it is now easily seen that

$$E\left[\widehat{Var}\left(\hat{\boldsymbol{\beta}}_{MG}\right)\right] = Var\left(\hat{\boldsymbol{\beta}}_{MG}\right),$$

as required. For a further discussion of the mean group estimator, see Pesaran and Smith (1995), and Hsiao and Pesaran (2008).

Example 64 *Continuing from Example 63, Haque, Pesaran, and Sharma (2000) further investigate the determinants of cross country private savings rates by carrying out a country-specific analysis. The FE regression in Table 28.2 assumes that the slope coefficients across countries are exact linear functions of $\overline{W}_i$ and/or $\overline{YRUS}_i$ (see equation 28.32), and that the error variances, $Var(u_{it}) = \sigma_i^2$, are the same across countries. Clearly, these are rather restrictive assumptions, and the consequences of incorrectly imposing them on the parameters of interest need to be examined. Under the alternative assumption of unrestricted slope and error variance heterogeneity, MG estimates can be computed as simple averages of country-specific estimates from country-specific regressions and can then be used to make inferences about $E(\boldsymbol{\beta}_i) = \boldsymbol{\beta}$. Results on country-specific estimates and MG estimates are summarized in Table 28.3. The estimated slope coefficients differ considerably across countries, both in terms of their magnitude and their statistical significance. Some of the coefficients are statistically significant only in the case of 3 or 4 countries and in general are very poorly estimated. This is true of the coefficients of GI, GR, W, PCTT, and YRUS. Also the sign of these estimated coefficients varies quite widely across countries. The coefficients of RINT and INF are better estimated, but still differ significantly both in magnitude and in sign across the countries. Only the coefficients of SUR and GCUR tend to be similar across countries. The coefficient of SUR is estimated to be negative in 19 of the 20 countries, and 13 of these are statistically significant. The positive estimate obtained for New Zealand is very small and not statistically significant. Similarly, 17 out of 20 coefficients estimated for the GCUR variable have a negative sign, with 7 of the 17 negative coefficients statistically significant. None of the three positive coefficients estimated for GCUR is statistically significant. The MG estimates based on the individual country regressions in Table 28.3 support these general conclusions. Only the MG estimates of the SUR and the GCUR variables are statistically significant (see the last two rows of Table 28.3). At −0.671, the MGE of the SUR variable is only marginally higher than the corresponding FE estimate in Table 28.2 that allows for some slope heterogeneity.*

28.5.1 Relationship between Swamy's and *MG* estimators

The Swamy and *MG* estimators are algebraically equivalent when T is sufficiently large. To see this, consider $\hat{\boldsymbol{\beta}}_{SW}$ in equation (28.34), and note that

$$\mathbf{H}_{iT}^{-1} = \left(\mathbf{Q}_{iT} + \frac{1}{T}\boldsymbol{\Omega}_\eta^{-1}\right)^{-1} = \mathbf{Q}_{iT}^{-1}\left(\mathbf{I}_k + \frac{1}{T}\boldsymbol{\Omega}_\eta^{-1}\mathbf{Q}_{iT}^{-1}\right)^{-1}.$$

Table 28.3 Country-specific estimates of 'static' private saving equations (20 OECD countries, 1972–1993)

Country	SUR	CCUR	GI	GR	RINT	W	INF	PCTT	YRUS	DEP
Australia	−0.81	−0.18	−1.00	0.08	0.18	0.06	0.27	0.04	0.42	0.46
	[0.18]	[0.27]	[0.41]	[0.08]	[0.08]	[0.02]	[0.09]	[0.03]	[0.17]	[0.22]
Austria	−0.48	−0.42	0.35	0.06	0.24	0004	0.09	0.11	−0.10	−0.03
	[0.56]	[0.40]	[0.84]	[0.32]	[0.32]	[0.05]	[0.54]	[0.16]	[0.24]	[0.21]
Belgium	−0.68	−0.53	−2.47	0.09	−0.04	−0.02	−0.10	−0.00	0.17	−0.22
	[0.23]	[0.15]	[1.51]	[0.11]	[0.14]	[0.03]	[0.13]	[0.02]	[0.09]	[0.35]
Canada	−1.31	−0.56	1.01	0.24	0.10	−0.03	0.29	0.17	0.07	−0.17
	[0.10]	[0.14]	[1.03]	[0.09]	[0.08]	[0.04]	[0.09]	[0.05]	[0.12]	[0.12]
Denmark	−1.08	−0.64	0.36	0.03	−0.20	−0.01	0.10	0.02	0.14	−1.17
	[0.15]	[0.22]	[0.80]	[0.25]	[0.20]	[0.03]	[0.29]	[0.05]	[0.24]	[0.36]
Finland	−0.70	−0.35	0.87	0.14	0.40	0.03	0.52	0.01	0.02	−0.39
	[0.16]	[0.21]	[1.59]	[0.20]	[0.18]	[0.03]	[0.22]	[0.02]	[0.19]	[0.52]
France	−1.45	−0.78	−3.13	0.10	−0.16	−0.04	−0.22	−0.06	0.12	−0.16
	[0.51]	[0.52]	[2.00]	[0.23]	[0.18]	[0.10]	[0.24]	[0.05]	[0.12]	[0.43]
Germany	−0.80	−0.54	−0.18	0.19	−0.06	0.00	0.02	−0.01	−0.10	−0.28
	[0.35]	[0.28]	[0.71]	[0.18]	[0.17]	[0.03]	[0.25]	[0.05]	[0.20]	[0.11]
Greece	−0.69	−0.29	−1.13	0.15	1.23	0.10	1.05	−0.49	−0.87	1.52
	[0.45]	[0.71]	[1.65]	[0.34]	[0.58]	[0.05]	[0.63]	[0.27]	[1.29]	[1.24]
Ireland	−0.48	−0.50	1.33	−0.08	−0.71	−0.13	−0.88	0.32	0.79	1.14
	[0.29]	[0.14]	[1.18]	[0.14]	[0.28]	[0.06]	[0.22]	[0.11]	[0.24]	[0.35]
Italy	−0.46	0.05	−0.16	0.13	0.12	−0.00	0.09	−0.00	−0.12	0.32
	[0.18]	[0.21]	[0.48]	[0.15]	[0.11]	[0.03]	[0.13]	[0.04]	[0.15]	[0.19]
Japan	−0.58	−0.79	−0.98	−0.14	−0.05	0.04	0.01	0.04	−0.06	0.22
	[0.21]	[0.31]	[0.50]	[0.12]	[0.16]	[0.03]	[0.09]	[0.01]	[0.08]	[0.32]
Netherlands	−0.75	−0.43	−1.50	−0.05	0.09	0.12	−0.37	0.06	0.30	0.22
	[0.33]	[0.33]	[2.64]	[0.20]	[0.28]	[0.05]	[0.27]	[0.15]	[0.26]	[0.39]
New Zealand	[0.02]	−0.54	−1.22	−0.12	−0.07	0.02	−0.20	0.07	−0.46	0.24
	[0.29]	[0.45]	[0.78]	[0.22]	[0.20]	[0.03]	[0.19]	[0.07]	[0.33]	[0.18]
Norway	−0.22	0.13	−0.15	−0.06	0.02	−0.07	−0.04	0.23	0.12	−0.16
	[0.51]	[0.66]	[0.61]	[0.46]	[0.51]	[0.05]	[0.60]	[0.07]	[0.31]	[0.64]
Portugal	−1.00	−0.57	2.91	0.60	0.47	−0.07	0.64	0.21	−0.72	0.16
	[0.20]	[0.32]	[1.64]	[0.24]	[0.20]	[0.05]	[0.19]	[0.13]	[0.37]	[0.41]
Spain	−0 18	−0 06	1.36	−0.01	0.07	−0.09	0.11	0.18	−0.78	−0.28
	[0.55]	[0.59]	[1.58]	[0.31]	[0.38]	[0.05]	[0.42]	[0.12]	[0.32]	[0.57]
Sweden	−0.84	−0.96	−2.54	−0.53	0.24	0.05	−0.02	0.09	0.00	0.22
	[0.11]	[0.20]	[1.49]	[0.30]	[0.23]	[0.05]	[0.23]	[0.10]	[0.22]	[0.81]
Switzerland	−0.22	−0.09	0.36	−0.26	0.02	0.06	0.21	−0.04	−0.06	−0.59
	[0.50]	[0.16]	[0.76]	[0.13]	[0.14]	[0.03]	[0.11]	[0.5]	[0.12]	[0.09]
UK	−0.72	0.03	−0.79	0.37	0.18	−0.04	0.21	0.01	−0.25	0.34
	[0.12]	[0.10]	[0.34]	[0.09]	[0.08]	[0.03]	[0.08]	[0.04]	[0.15]	[0.15]
Average	−0.671	−0.401	−0.335	0.046	0.104	0.001	0.089	0.048	−0.069	0.080
Standard error	[.083]	[.067]	[.332]	[.052]	[.081]	[.014]	[.088]	[.036]	[.127]	[.090]

Table 28.3 Continued

	$\hat{\sigma}^2$	$\chi^2_{SC}(1)$	$\chi^2_{FF}(1)$	$\chi^2_N(2)$	$\chi^2_H(1)$	$\overline{R}^2$	LL
Australia	0.573	0.83	0.29	0.70	1.24	0.90	−11.36
Austria	1.210	0.05	2.69	1.56	0.10	0.28	−27.78
Belgium	0.693	18.10	2.84	1.03	0.81	0.69	−15.51
Canada	0.518	0.01	1.27	0.18	0.22	0.76	−9.10
Denmark	1.197	1.63	0.20	1.56	2.32	0.49	−27.55
Finland	1.079	8.32	2.44	1.78	0.38	0.70	−25.27
France	0.689	1.78	12.51	1.80	1.13	0.54	−15.40
Germany	0.817	10.02	0.00	0.76	0.48	0.16	−19.15
Greece	2.439	6.25	0.09	1.05	0.32	0.53	−43.21
Ireland	1.469	3.04	0.59	1.49	0.01	0.77	−32.06
Italy	0.606	2.80	5.09	0.74	2.76	0.72	−12.57
Japan	0.399	0.39	1.59	4.97	0.12	0.77	−3.37
Netherlands	1.052	3.40	1.57	0.20	2.02	0.52	−24.70
New Zealand	1.743	12.38	8.26	0.68	10.45	0.70	−35.82
Norway	1.622	8.47	2.18	0.81	0.53	0.39	−34.23
Portugal	2.042	0.00	1.67	0.80	0.69	0.86	−39.30
Spain	1.319	8.68	5.47	1.20	0.40	0.58	−29.68
Swedan	1.194	6.97	0.76	0.15	1.67	0.68	−27.49
Switzerland	0.535	3.13	3.40	0.70	7.79	0.44	−9.83
UK	0.541	1.63	2.20	1.38	0.50	0.81	−10.07

**$\hat{\sigma}$ is the standard error of the country specific regressions, $\chi^2_{SC}(1)$, $\chi^2_{FF}(1)$, $\chi^2_N(2)$ and $\chi^2_H(1)$ are chi-aquared statistics for tests of residual serial correlation, functional form mis-specification, non-normal errors and heteroskedasticity. The figures in brackets are their degrees of freedom. $\overline{R}$ is the adjusted multiple correlation coefficient, and LL is the maximized log-likelihood value of the country-specific regressions.

Write

$$\mathbf{\Omega}_{\eta}^{-1} = \lambda \mathbf{A},$$

where λ represents an overall index of parameter heterogeneity, such that

$$\lambda \to 0, \text{ highest degree of heterogeneity,}$$
$$\lambda \to \infty, \text{ homogeneity.}$$

Then $\hat{\boldsymbol{\beta}}_{SW}$ can be written as

$$\hat{\boldsymbol{\beta}}_{SW} = \left\{ \sum_{i=1}^{N} \left[\mathbf{Q}_{iT} - \left(\mathbf{I}_k + \frac{\lambda}{T} \mathbf{A} \mathbf{Q}_{iT}^{-1} \right)^{-1} \mathbf{Q}_{iT} \right] \right\}^{-1} \sum_{i=1}^{N} \left[\mathbf{q}_{iT} - \left(\mathbf{I}_k + \frac{\lambda}{T} \mathbf{A} \mathbf{Q}_{iT}^{-1} \right)^{-1} \mathbf{q}_{iT} \right].$$

For a fixed N and T, and for a sufficiently small λ

$$\left(\mathbf{I}_k + \frac{\lambda}{T}\mathbf{A}\mathbf{Q}_{iT}^{-1}\right)^{-1} = \mathbf{I}_k - \frac{\lambda}{T}\mathbf{G}_{iT} + \left(\frac{\lambda}{T}\right)^2 \mathbf{G}_{iT}^2 - \dots$$

where $\mathbf{G}_{iT} = \mathbf{A}\mathbf{Q}_{iT}^{-1}$. Therefore,

$$\begin{aligned}\hat{\boldsymbol{\beta}}_{SW} = &\left\{\sum_{i=1}^{N}\left[\mathbf{Q}_{iT} - \left(\mathbf{I}_k - \frac{\lambda}{T}\mathbf{G}_{iT} + \left(\frac{\lambda}{T}\right)^2 \mathbf{G}_{iT}^2 + \dots\right)\mathbf{Q}_{iT}\right]\right\}^{-1} \\ &\sum_{i=1}^{N}\left[\mathbf{q}_{iT} - \left(\mathbf{I}_k - \frac{\lambda}{T}\mathbf{G}_{iT} + \left(\frac{\lambda}{T}\right)^2 \mathbf{G}_{iT}^2 + \dots\right)\mathbf{q}_{iT}\right]. \\ = &\left\{\sum_{i=1}^{N}\mathbf{G}_{iT}\mathbf{Q}_{iT} - \frac{\lambda}{T}\sum_{i=1}^{N}\mathbf{G}_{iT}^2\mathbf{Q}_{iT} + O\left[\left(\frac{\lambda}{T}\right)^2\right]\right\}^{-1} \\ &\left\{\sum_{i=1}^{N}\mathbf{G}_{iT}\mathbf{q}_{iT} - \frac{\lambda}{T}\sum_{i=1}^{N}\mathbf{G}_{iT}^2\mathbf{q}_{iT} + O\left[\left(\frac{\lambda}{T}\right)^2\right]\right\}.\end{aligned}$$

Hence for any fixed $T > k$ and for any N, as $\lambda \to 0$,

$$\hat{\boldsymbol{\beta}}_{SW} \to \left(\sum_{i=1}^{N}\mathbf{G}_{iT}\mathbf{Q}_{iT}\right)^{-1}\sum_{i=1}^{N}\mathbf{G}_{iT}\mathbf{q}_{iT}.$$

However, note that

$$\begin{aligned}\left(\sum_{i=1}^{N}\mathbf{G}_{iT}\mathbf{Q}_{iT}\right)^{-1}\sum_{i=1}^{N}\mathbf{G}_{iT}\mathbf{q}_{iT} &= \left(\sum_{i=1}^{N}\mathbf{A}\mathbf{Q}_{iT}^{-1}\mathbf{Q}_{iT}\right)^{-1}\sum_{i=1}^{N}\mathbf{A}\mathbf{Q}_{iT}^{-1}\mathbf{q}_{iT} \\ &= \frac{1}{N}\sum_{i=1}^{N}\mathbf{Q}_{iT}^{-1}\mathbf{q}_{iT} = \frac{1}{N}\sum_{i=1}^{N}\hat{\boldsymbol{\beta}}_i = \hat{\boldsymbol{\beta}}_{MG},\end{aligned}$$

From which it follows that

$$\lim_{\lambda\to 0}\hat{\boldsymbol{\beta}}_{SW}(\lambda) = \hat{\boldsymbol{\beta}}_{MG},$$

and, for all values of N and $\lambda > 0$,

$$\lim_{T\to\infty}\left(\hat{\boldsymbol{\beta}}_{SW}(\lambda) - \hat{\boldsymbol{\beta}}_{MG}\right) = \mathbf{0}.$$

28.6 Dynamic heterogeneous panels

Consider the $ARDL(p, \underbrace{q, q, \ldots, q}_{k\text{-times}})$ model (see Chapter 6 for an introduction to *ARDL* models)

$$y_{it} = \alpha_i + \sum_{j=1}^{p} \lambda_{ij} y_{i,t-j} + \sum_{j=0}^{q} \boldsymbol{\delta}_{ij}' \mathbf{x}_{i,t-j} + u_{it}, \text{ for } i = 1, 2, \ldots, N, \tag{28.44}$$

where $\mathbf{x}_{it}$ is a k-dimensional vector of explanatory variables for group i; α_i represent the fixed-effects; the coefficients of the lagged dependent variables, λ_{ij}, are scalars; and $\boldsymbol{\delta}_{ij}$ are k-dimensional coefficient vectors. In the following, we assume that the disturbances u_{it}, $i = 1, 2, \ldots, N$; $t = 1, 2, \ldots, T$, are independently distributed across i and t, with zero means, variances σ_i^2, and are distributed independently of the regressors $\mathbf{x}_{it}$.

The error correction representation of the above *ARDL* model is:

$$\Delta y_{it} = \alpha_i + \phi_i y_{i,t-1} + \boldsymbol{\beta}_i' \mathbf{x}_{it} + \sum_{j=1}^{p-1} \lambda_{ij}^* \Delta y_{i,t-j} + \sum_{j=0}^{q-1} \boldsymbol{\delta}_{ij}^{*\prime} \Delta \mathbf{x}_{i,t-j} + u_{it}, \tag{28.45}$$

where

$$\phi_i = -(1 - \sum_{j=1}^{p} \lambda_{ij}), \quad \boldsymbol{\beta}_i = \sum_{j=0}^{q} \boldsymbol{\delta}_{ij},$$

$$\lambda_{ij}^* = -\sum_{m=j+1}^{p} \lambda_{im}, \ j = 1, 2, \ldots, p-1,$$

$$\boldsymbol{\delta}_{ij}^* = -\sum_{m=j+1}^{q} \boldsymbol{\delta}_{im}, \ j = 1, 2, \ldots, q-1.$$

If we stack the time series observations for each group, (28.45) can be written as

$$\Delta \mathbf{y}_{i.} = \alpha_i \boldsymbol{\tau}_T + \phi_i \mathbf{y}_{i.,-1} + \mathbf{X}_{i.} \boldsymbol{\beta}_i + \sum_{j=1}^{p-1} \lambda_{ij}^* \Delta \mathbf{y}_{i.,-j} + \sum_{j=0}^{q-1} \Delta \mathbf{X}_{i.,-j} \boldsymbol{\delta}_{ij}^* + \mathbf{u}_{i.},$$

for $i = 1, 2, \ldots, N$, where $\boldsymbol{\tau}_T$ is a $T \times 1$ vector of ones, $\mathbf{y}_{i.,-j}$ and $\mathbf{X}_{i.,-j}$ are j-period lagged values of $\mathbf{y}_{i.}$ and $\mathbf{X}_{i.}$, $\Delta \mathbf{y}_{i.} = \mathbf{y}_{i.} - \mathbf{y}_{i.,-1}$, $\Delta \mathbf{X}_{i.} = \mathbf{X}_{i.} - \mathbf{X}_{i.,-1}$, $\Delta \mathbf{y}_{i.,-j}$ and $\Delta \mathbf{X}_{i.,-j}$ are j-period lagged values of $\Delta \mathbf{y}_{i.}$ and $\Delta \mathbf{X}_{i.}$.

If the roots of the polynomial

$$f_i(z) = 1 - \sum_{j=1}^{p} \lambda_{ij} z^j = 0,$$

for $i = 1, 2, \ldots, N$, fall outside the unit circle, then the $ARDL(p, q, q, \ldots, q)$ model is stable. In this chapter we will take up this assumption, while the non-stationary case will be discussed in Chapter 31. This condition ensures that $\phi_i < 0$, and that there exists a long-run relationship between y_{it} and $\mathbf{x}_{it}$ defined by (see Sections 6.5 and 22.2)

$$y_{it} = \boldsymbol{\theta}_i \mathbf{x}_{it} + \eta_{it},$$

for each $i = 1, 2, \ldots, N$, where η_{it} is $I(0)$, and $\boldsymbol{\theta}_i$ are the long-run coefficients on $\mathbf{X}_{i.}$, $\boldsymbol{\theta}_i = -\boldsymbol{\beta}_i/\phi_i$.

28.7 Large sample bias of pooled estimators in dynamic heterogeneous models

Traditional procedures for estimation of pooled models, such as the FE estimator or the *IV/GMM* approaches reviewed in Chapter 27, can produce inconsistent and potentially misleading estimates of the average value of the parameters in dynamic panel data models unless the slope coefficients are in fact homogeneous. To see this, consider the simple dynamic panel data model $(ARDL(1, 0))$

$$y_{it} = \alpha_i + \lambda_i y_{i,t-1} + \beta_i x_{it} + u_{it}, \tag{28.46}$$

where the slopes, λ_i and β_i, as well as the intercepts, α_i, are allowed to vary across cross-sectional units (groups). Here, for simplicity, x_{it} is a scalar random variable but the analysis can be extended to the case of more than one regressor. We assume that x_{it} is strictly exogenous. Let $\theta_i = \beta_i/(1 - \lambda_i)$ be the long-run coefficient of x_{it} for the ith group and rewrite (28.46) as

$$\Delta y_{it} = \alpha_i - (1 - \lambda_i)\left(y_{i,t-1} - \theta_i x_{it}\right) + u_{it},$$

or

$$\Delta y_{it} = \alpha_i - \phi_i\left(y_{i,t-1} - \theta_i x_{it}\right) + u_{it}.$$

Consider now the random coefficient model

$$\phi_i = \phi + \eta_{i1}, \tag{28.47}$$

$$\theta_i = \theta + \eta_{i2}. \tag{28.48}$$

Hence

$$\beta_i = \theta_i \phi_i = \theta\phi + \eta_{i3}, \tag{28.49}$$

where

$$\eta_{i3} = \phi\eta_{i2} + \theta\eta_{i1} + \eta_{i1}\eta_{i2}, \tag{28.50}$$

$$\begin{pmatrix} \eta_{i1} \\ \eta_{i2} \end{pmatrix} \sim IID\left[\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \omega_{11} & \omega_{12} \\ \omega_{12} & \omega_{22} \end{pmatrix}\right],$$

and

$$\omega_{33} = Var\left(\eta_{i3}\right) = Var(\phi\eta_{i2} + \theta\eta_{i1} + \eta_{i1}\eta_{i2}).$$

Letting $\lambda = 1 - \phi$ and $\beta = \theta\phi$, and using the above in (28.46) we have

$$y_{it} = \alpha_i + \lambda y_{i,t-1} + \beta x_{it} + v_{it}, \tag{28.51}$$

$$v_{it} = u_{it} - \eta_{i1} y_{i,t-1} + \eta_{i3} x_{it}. \tag{28.52}$$

It is now clear that v_{it} and $y_{i,t-1}$ are correlated and the FE or RE estimators will not be consistent. This is not a surprising result in the case where T is small. In Chapter 27 we saw that the FE (and RE) estimators are inconsistent when T is finite and N large when the slopes λ_i and β_i are homogeneous, that is, $\eta_{i1} = \eta_{i3} = 0$. The significant result here is that the inconsistency of the FE and RE estimators *will not disappear* even when both $T \to \infty$ and $N \to \infty$, if the slopes λ_i and/or β_i are heterogenous across groups. In fact, in the relatively simple case where

$$\begin{aligned} \lambda_i &= \lambda, \qquad \left(\text{or } \eta_{i1} = 0\right), \\ \beta_i &= \beta + \eta_{i3}, \end{aligned}$$

namely only the coefficients of x_{it} vary across groups, and

$$\begin{aligned} x_{it} &= \mu_i (1 - \rho) + \rho x_{i,t-1} + v_{it}, \\ |\rho| &< 1, \quad E(x_{it}) = \mu_i, \\ v_{it} &\sim IID\left(0, \tau^2\right), \end{aligned} \tag{28.53}$$

we have[5]

$$\begin{aligned} \underset{N,T\to\infty}{Plim}\left(\hat{\lambda}_{FE}\right) - \lambda &= \frac{\rho(1 - \lambda\rho)\left(1 - \lambda^2\right)\omega_{33}}{\Psi_1}, \\ \underset{N,T\to\infty}{Plim}\left(\hat{\beta}_{FE}\right) - \beta &= -\frac{\beta\rho^2\left(1 - \lambda^2\right)\omega_{33}}{\Psi_1}, \end{aligned} \tag{28.54}$$

where

$$\Psi_1 = \left(\frac{\sigma^2}{\tau^2}\right)\left(1 - \rho^2\right)(1 - \lambda\rho)^2 + \left(1 - \lambda^2\rho^2\right)\omega_{33} + \left(1 - \rho^2\right)\beta^2 > 0,$$

and $\omega_{33} = Var\left(\eta_{i3}\right) = Var\left(\beta_i\right)$ measures the degree of heterogeneity in β_i. It is now clear that when $\rho > 0$,

$$Plim\left(\hat{\lambda}_{FE}\right) > \lambda, \;\; Plim\left(\hat{\beta}_{FE}\right) < \beta.$$

[5] It is interesting that when $\rho > 0$ the heterogeneity bias, given by (28.54), is in the opposite direction to the Nickell bias defined by (27.3).

The bias of the FE estimator of the long-run coefficient, $\hat{\theta}_{FE} = \hat{\beta}_{FE} / \left(1 - \hat{\lambda}_{FE}\right)$, is given by

$$\underset{N,T\to\infty}{Plim}\left(\hat{\theta}_{FE}\right) = \frac{\theta}{1-\rho\Psi_2},$$

where

$$\Psi_2 = \frac{(1+\lambda)\,\omega_{33}}{(1+\rho)\left[\left(\frac{\sigma^2}{\tau^2}\right)(1-\lambda\rho)^2 + \left(\beta^2+\omega_{33}\right)\right]}.$$

Thus note that

$$Plim\left(\hat{\theta}_{FE}\right) > \theta, \quad \text{if} \quad \rho > 0.$$

In the case where x_{it} is trended or if $\rho \to 1$ from below we have

$$\underset{\rho\to 1}{Plim}\left(\hat{\lambda}_{FE}\right) = 1, \text{ and } \underset{\rho\to 1}{Plim}\left(\hat{\beta}_{FE}\right) = 0,$$

irrespective of the true value of λ. See Pesaran and Smith (1995) for further details.

Example 65 *The FE 'static' private savings regressions reported in Tables 28.1 and 28.2 within Example 63 are subject to a substantial degree of residual serial correlation, which can lead to inconsistent estimates even under slope homogeneity since the wealth variable, W, is in fact constructed from accumulation of past savings. The presence of residual serial correlation could be due to a host of factors: omitted variables, neglected slope heterogeneity in the case of serially correlated regressors, and of course neglected dynamics. The diagnostic statistics provided in the second part of Table 28.3, within Example 64, show statistically significant evidence of residual serial correlation in the case of eight of the twenty countries.*[6] *It is clear that, even when the slope coefficients are allowed to be estimated freely across countries, residual serial correlation still continues to be a problem, at least in the case of some, if not all, the countries.*[7] *The usual time series technique for dealing with dynamic misspecification is to estimate error correction models based on ARDL models. ARDL models have the advantage that they are robust to integration and cointegration properties of the regressors, and for sufficiently high lag-orders could be immune to the endogeneity problem, at least as far as the long-run properties of the model are concerned. In the present application, observations for each individual country are available for too short a period to estimate even a first-order ARDL model including all the 10 regressors for each country separately.*[8] *Pooling in the form of FE estimation can compensate for lack of time series observations but, as shown in previous example, this can have its own set of problems. To check the robustness of the 'static' FE estimates presented in Table 28.2 to dynamic misspecification, Haque, Pesaran, and Sharma (2000) estimated the following first-order dynamic panel data model*

[6] The diagnostic statistics are computed using the Lagrange multiplier procedure described in Section 5.8, and are valid irrespective of whether the regressions contain lagged dependent variables, implicitly or explicitly.

[7] Under slope homogeneity restrictions, residual serial correlation is a problem for *all* the countries in the panel.

[8] A first-order *ARDL* model in the private savings rate for each country that contains *all* ten regressors would involve estimating twenty-two unknown parameters with only twenty-two time series observations available per country!

$$y_{it} = \mu_i + \lambda y_{i,t-1} + \boldsymbol{\beta}_0' \mathbf{x}_{it} + \boldsymbol{\beta}_{01}' (\mathbf{x}_{it} \bar{W}_i) + \boldsymbol{\beta}_1' \mathbf{x}_{i,t-1} + u_{it}. \tag{28.55}$$

The country-specific long-run coefficients are given by

$$\boldsymbol{\theta}_i = (\boldsymbol{\beta}_0 + \boldsymbol{\beta}_1 + \boldsymbol{\beta}_{01} \bar{W}_i)/(1 - \lambda). \tag{28.56}$$

The FE estimates computed using all the 21 countries over the period 1972–1993 are given in Table 28.4.[9] *Clearly, there are significant dynamics, particularly in the relationship between changes in the government surplus and expenditure variables (SUR, GCUR, and GI) and the private savings rate. There is also important evidence of cross-sectional variations in the coefficients of wealth, income and demographic variables (W, YRUS and DEP). However, unlike the static estimates in Table 28.2, the coefficients of GDP growth and the real interest rate are no longer statistically significant. Overall, this equation presents a substantial improvement over the static FE estimates. In fact, the estimated standard error of this dynamic regression is 62 percent lower than the standard error of the FE estimates favoured by Masson, Bayoumi, and Samiei (1998), and reproduced in the first column of Table 28.1. Using the formula (28.56) the following estimates of the long-run coefficients are obtained*

SUR	−0.432	
	(−3.11)	
GCUR	−0.398	
	(−4.65)	
GI	−0.202	
	(−0.91)	
GR	−0.004	
	(−0.03)	
RINT	0.154	
	(1.64)	
W	0.224	$-0.00057\bar{W}_i$
	(4.58)	(−3.77)
INF	0.248	
	(3.10)	
PCTT	0.136	
	(4.11)	
YRUS	1.384	$-0.0047\bar{W}_i$
	(2.58)	(−2.92)
DEP	0.708	$-0.0027\bar{W}_i$
	(2.19)	(−2.64)

According to these estimates the long-run coefficients of the SUR and GCUR variables are still statistically significant, although the coefficient of the SUR variable is now estimated to be much lower than the estimate based on the static regressions. The long-run coefficients of the GI, GR and RINT variables are no longer statistically significant. It appears that, in contrast to government consumption expenditures, the effect of changes in government investment expenditures on private savings is temporary and tends to zero in the long run. The inflation and the terms of trade variables (INF

[9] For relatively simple dynamic models where T (= 22) is reasonably large and of the same order of magnitude as N (= 21), the application of the IV type estimators, discussed in Chapter 27, to a first differenced version of (28.55) does not seem necessary and can lead to considerable loss of efficiency.

Table 28.4 Fixed-effects estimates of dynamic private savings equations with cross-sectionally varying slopes (21 OECD countries, 1972–1993)

Regressors	Coefficients	Regressors	Coefficients
$PSAV_{-1}$	0.670	W	0.074
	(20.80)		(4.41)
SUR	−0.771	$W \times \overline{W}_i$	−0.00019
	(−16.28)		(−3.62)
SUR_{-1}	0.628	INE	0.082
	(11.54)		(3.11)
GCUR	−0.544	PCTT	0.045
	(7.78)		(4.54)
$GCUR_{-1}$	0.412	YRUS	0.456
	(6.16)		(2.49)
GI	−0.666	$YRUS \times \overline{W}_i$	−0.00157
	(−5.54)		(−2.81)
GL_1	0.600	DEP	0.233
	(4.80)		(2.12)
GR	−0.0014	$DEP \times \overline{W}_i$	−0.00089
	(−0.03)		(−2.52)
RINT	0.051		
	(1.60)		
$\overline{R}^2$	0.908		
$\tilde{\sigma}$	1.451		
LL	−807.61		
AIC	−845.61		
SBC	924.18		

*The figures in brackets are t-ratios.

and PCTT) have the expected signs and are also statistically significant. The long-run coefficients of the remaining variables vary with country-specific average wealth-GDP ratio and when averaged across countries yield the values of 0.043 [0.026], −0.118 [0.219] *and* −0.148 [0.125] *for W, YRUS, and DEP variables respectively. The cross-sectional standard errors of these estimates are given in square brackets. The average estimate of the coefficient of the relative income variable has the wrong sign, but it is not statistically significant. The average estimates of the other two coefficients have the expected signs, but are not statistically significant either. It seems that the effects of many of the regressors considered in the MBS study are not robust to dynamic misspecifications. However, it would be interesting to examine the consequences of jointly allowing for unrestricted short-run slope heterogeneity and dynamics.*

28.8 Mean group estimator of dynamic heterogeneous panels

Consider a dynamic model of the form

$$y_{it} = \lambda_i y_{i,t-1} + \mathbf{x}_{it}'\boldsymbol{\beta}_i + u_{it},\ i = 1, 2, \ldots, N;\ \ t = 1, 2, \ldots, T, \tag{28.57}$$

where $\mathbf{x}_{it}$ is a $k \times 1$ vector of exogenous variables, and the error term u_{it} is assumed to be independently, identically distributed over t with mean zero and variance σ_i^2, and is independent across i. Let $\boldsymbol{\psi}_i = (\lambda_i, \boldsymbol{\beta}_i')'$. Further assume that $\boldsymbol{\psi}_i$ is independently distributed across i with

$$E(\boldsymbol{\psi}_i) = \boldsymbol{\psi} = (\lambda, \boldsymbol{\beta}')', \tag{28.58}$$

$$E\left[(\boldsymbol{\psi}_i - \boldsymbol{\psi})(\boldsymbol{\psi}_i - \boldsymbol{\psi})'\right] = \boldsymbol{\Delta}. \tag{28.59}$$

Rewriting $\boldsymbol{\psi}_i = \boldsymbol{\psi} + \boldsymbol{\eta}_i$, (28.58) and (28.59) can be equivalently written as

$$E(\boldsymbol{\eta}_i) = \mathbf{0}, \; E\left(\boldsymbol{\eta}_i \boldsymbol{\eta}_j'\right) = \begin{cases} \boldsymbol{\Delta} & \text{if} \quad i = j, \\ \mathbf{0} & \text{if} \quad i \neq j. \end{cases} \tag{28.60}$$

Although we may maintain the Assumption (28.7) that $E\left(\boldsymbol{\eta}_i \mathbf{x}_{it}'\right) = \mathbf{0}$, we can no longer assume that $E\left(\boldsymbol{\eta}_i y_{i,t-1}\right) = \mathbf{0}$. Through continuous substitutions, we have

$$y_{i,t-1} = \sum_{j=0}^{\infty} (\lambda + \eta_{i1})^j \mathbf{x}_{i,t-j-1}' (\boldsymbol{\beta} + \boldsymbol{\eta}_{i2}) + \sum_{j=0}^{\infty} (\lambda + \eta_{i1})^j u_{i,t-j-1}, \tag{28.61}$$

where $\boldsymbol{\eta}_i = \left(\eta_{i1}, \boldsymbol{\eta}_{i2}'\right)'$. It follows that $E(\boldsymbol{\eta}_i y_{i,t-1}) \neq \mathbf{0}$.

The violation of the independence between the regressors and the individual effects, $\boldsymbol{\eta}_i$, implies that the pooled least squares regression of y_{it} on $y_{i,t-1}$, and $\mathbf{x}_{it}$ will yield inconsistent estimates of $\boldsymbol{\psi}$, even for sufficiently large T and N. Pesaran and Smith (1995) have noted that, as $T \to \infty$, the least squares regression of y_{it} on $y_{i,t-1}$ and $\mathbf{x}_{it}$ yields a consistent estimator of $\boldsymbol{\psi}_i$, $\hat{\boldsymbol{\psi}}_i$. Hence, the authors suggest a MG estimator of $\boldsymbol{\psi}$ by taking the average of $\hat{\boldsymbol{\psi}}_i$ across i,

$$\hat{\boldsymbol{\psi}}_{MG} = \frac{1}{N} \sum_{i=1}^{N} \hat{\boldsymbol{\psi}}_i, \tag{28.62}$$

where

$$\hat{\boldsymbol{\psi}}_i = \left(\mathbf{W}_{i.}' \mathbf{W}_{i.}\right)^{-1} \mathbf{W}_{i.}' \mathbf{y}_{i.},$$

$\mathbf{W}_{i.} = (\mathbf{y}_{i.,-1}, \mathbf{X}_{i.})$ with $\mathbf{y}_{i.,-1} = (y_{i0}, y_{i1}, \ldots, y_{iT-1})'$. The variance of $\hat{\boldsymbol{\psi}}_{MG}$ is consistently estimated by

$$\widehat{Var}\left(\hat{\boldsymbol{\psi}}_{MG}\right) = \frac{1}{N(N-1)} \sum_{i=1}^{N} \left(\hat{\boldsymbol{\psi}}_i - \hat{\boldsymbol{\psi}}_{MG}\right)\left(\hat{\boldsymbol{\psi}}_i - \hat{\boldsymbol{\psi}}_{MG}\right)'.$$

Note that, for finite T, $\hat{\boldsymbol{\psi}}_i$ for $\boldsymbol{\psi}_i$ is biased, with a bias of order $1/T$ (Hurwicz (1950), Kiviet and Phillips (1993)). Hsiao, Pesaran, and Tahmiscioglu (1999) have shown that the MG estimator is asymptotically normal for large N, and large T, so long as $\sqrt{N}/T \to 0$ as both N and $T \to \infty$.

28.8.1 Small sample bias

The MG estimator in the case of dynamic panels is biased when T is small, due to the presence of the lagged dependent variable in the model which biases the *OLS* estimator of the short-run coefficients λ_i and $\boldsymbol{\beta}_i$. Pesaran, Smith, and Im (1996) investigate the small sample properties of various estimators of the long-run coefficients for a dynamic heterogeneous panel data model. They find that when T is small the MG estimator can be seriously biased, particularly when N is large relative to T. In particular, for finite T, as $N \to \infty$ (under the usual panel assumption of independence across groups), the MG estimator still converges to a normal distribution, but with a mean which is not the same as the true value of the parameter under consideration, if the underlying equations contain lagged dependent variables or weakly exogenous regressors. To see this, first note that, for a finite T,

$$E\left(\hat{\boldsymbol{\psi}}_{MG}\right) = \boldsymbol{\psi} + \frac{1}{N}\sum_{i=1}^{N} E\left[\left(\mathbf{W}_{i.}'\mathbf{W}_{i.}\right)^{-1}\mathbf{W}_{i.}'\mathbf{u}_{i.}\right]. \tag{28.63}$$

It is easy to see that, due to the presence of lagged dependent variables, $N \to \infty$ is not sufficient for eliminating the second term. One needs large enough T for the bias to disappear. In practice, when the model contains lagged dependent variables, we have

$$E\left[\left(\mathbf{W}_{i.}'\mathbf{W}_{i.}\right)^{-1}\mathbf{W}_{i.}'\mathbf{u}_{i.}\right] = \frac{\mathbf{K}_{iT}}{T} + O\left(T^{-\frac{3}{2}}\right),$$

where $\mathbf{K}_{iT}$ is bounded in T and a function of the unknown underlying parameters. Hence

$$E\left(\hat{\boldsymbol{\psi}}_{MG}\right) = \boldsymbol{\psi} + \frac{1}{T}\sum_{i=1}^{N}\frac{\mathbf{K}_{iT}}{N} + O\left(T^{-\frac{3}{2}}\right).$$

Pesaran and Zhao (1999) propose a number of bias reduction techniques for the MG estimator of the long-run coefficients in dynamic models. Estimation of such coefficients poses additional difficulties due to the nonlinearity of long-run coefficients in terms of the underlying short-run parameters is an additional source of bias for the MG estimation of dynamic models. In a set of Monte Carlo experiments, Hsiao, Pesaran, and Tahmiscioglu (1999) showed that the MG estimator is unlikely to be a good estimator when either N or T is small.

28.9 Bayesian approach

Under the assumption that y_{i0} are fixed and known and $\boldsymbol{\eta}_i$ and u_{it} are independently normally distributed, we can implement the Bayes estimator of $\boldsymbol{\psi}_i$ conditional on σ_i^2 and $\boldsymbol{\Delta}$, namely

$$\hat{\boldsymbol{\psi}}_B = \left\{\sum_{i=1}^{N}\left[\sigma_i^2(\mathbf{W}_i'\mathbf{W}_i)^{-1} + \boldsymbol{\Delta}\right]^{-1}\right\}^{-1}\sum_{i=1}^{N}\left[\sigma_i^2(\mathbf{W}_i'\mathbf{W}_i)^{-1} + \boldsymbol{\Delta}\right]\hat{\boldsymbol{\psi}}_i, \tag{28.64}$$

where $\mathbf{W}_i = (\mathbf{y}_{i,-1}, \mathbf{X}_i)$ with $\mathbf{y}_{i,-1} = (y_{i0}, y_{i1}, \ldots, y_{iT-1})'$. This Bayes estimator is a weighted average of the least squares estimator of individual units with the weights being inversely proportional to individual variances. When $T \to \infty, N \to \infty$, and $N/T^{3/2} \to 0$, the Bayes estimator is asymptotically equivalent to the MG estimator (28.62) (Hsiao, Pesaran, and Tahmiscioglu (1999)).

In practice, the variance components, σ_i^2 and $\boldsymbol{\Delta}$ are rarely known. The Monte Carlo studies conducted by Hsiao, Pesaran, and Tahmiscioglu (1999) show that, following the approach of Lindley and Smith (1972) in assuming that the prior-distributions of σ_i^2 and $\boldsymbol{\Delta}$ are independent and are distributed as

$$P(\boldsymbol{\Delta}^{-1}, \sigma_1^2, \ldots, \sigma_n^2) = W(\boldsymbol{\Delta}^{-1}|(rR)^{-1}, r) \prod_{i=1}^{N} \sigma_i^{-2}, \tag{28.65}$$

yields a Bayes estimator almost as good as the Bayes estimator with known $\boldsymbol{\Delta}$ and σ_i^2, where $W(.)$ represents the Wishart distribution with scale matrix, rR, and degrees of freedom r.

The Hsiao, Pesaran, and Tahmiscioglu (1999) Bayes estimator is derived under the assumption that the initial observation y_{i0} are fixed constants. As discussed in Anderson and Hsiao (1981, 1982), this assumption is clearly unjustifiable for a panel with finite T. However, contrary to the sampling approach where the correct modelling of initial observations is quite important, the Hsiao, Pesaran, and Tahmiscioglu (1999) Bayesian approach appears to perform fairly well in the estimation of the mean coefficients for dynamic random coefficient models as demonstrated in their Monte Carlo studies.

28.10 Pooled mean group estimator

Consider the *ARDL* model (28.44). Pesaran, Shin, and Smith (1999) has proposed an estimation method for *ARDL* models, under the assumption that the long-run coefficients on $\mathbf{X}_i$, defined by $\boldsymbol{\theta}_i = -\boldsymbol{\beta}_i/\phi_i$, are the same across the groups, namely

$$\boldsymbol{\theta}_i = \boldsymbol{\theta}, \quad i = 1, 2, \ldots, N.$$

This estimator, known as the pooled mean group estimator, provides a useful intermediate alternative between estimating separate regressions, which allows all coefficients and error variances to differ across the groups, and standard FE estimators that assume the slope coefficients are the same across i. Under the above assumptions, the error correction model can be written more compactly as

$$\Delta \mathbf{y}_i = \phi_i \boldsymbol{\xi}_i(\boldsymbol{\theta}) + \mathbf{W}_i \boldsymbol{\kappa}_i + \boldsymbol{\varepsilon}_i, \tag{28.66}$$

where

$$\mathbf{W}_i = (\Delta \mathbf{y}_{i,-1}, \Delta \mathbf{y}_{i,-2}, \ldots, \Delta \mathbf{y}_{i,-p+1}, \Delta \mathbf{X}_i, \Delta \mathbf{X}_{i,-1}, \ldots, \Delta \mathbf{X}_{i,-q+1}),$$
$$\boldsymbol{\xi}_i(\boldsymbol{\theta}) = \mathbf{y}_{i,-1} - \mathbf{X}_i \boldsymbol{\theta},$$

is the error correction component, and

$$\boldsymbol{\kappa}_i = (\lambda_{i1}^*, \lambda_{i2}^*, \ldots, \lambda_{i,p-1}^*; \boldsymbol{\delta}_{i0}^{*\prime}, \boldsymbol{\delta}_{i1}^{*\prime}, \ldots, \boldsymbol{\delta}_{i,q-1}^{*\prime})'.$$

There are three issues to be noted in estimating (28.66). First, the regression equations for each group are nonlinear in ϕ_i and $\boldsymbol{\theta}$. A further complication arises from the cross-equation parameter restrictions existing by virtue of the long-run homogeneity assumption. Finally, note that the error variances differ across groups. The log-likelihood function is

$$\ell_T(\boldsymbol{\varphi}) = -\frac{T}{2}\sum_{i=1}^{N} \ln 2\pi\sigma_i^2 - \frac{1}{2}\sum_{i=1}^{N} \sigma_i^{-2} Q_i, \tag{28.67}$$

where

$$Q_i = [\Delta \mathbf{y}_i - \phi_i \boldsymbol{\xi}_i(\boldsymbol{\theta})]' \mathbf{H}_i [\Delta \mathbf{y}_i - \phi_i \boldsymbol{\xi}_i(\boldsymbol{\theta})],$$
$$\mathbf{H}_i = \mathbf{I}_T - \mathbf{W}_i(\mathbf{W}_i'\mathbf{W}_i)^{-1}\mathbf{W}_i',$$

$\mathbf{I}_T$ is an identity matrix of order T, $\boldsymbol{\varphi} = (\boldsymbol{\theta}', \boldsymbol{\phi}', \boldsymbol{\sigma}')'$, $\boldsymbol{\phi} = (\boldsymbol{\varphi}_1, \boldsymbol{\varphi}_2, \ldots, \boldsymbol{\varphi}_N)'$, and $\boldsymbol{\sigma} = (\boldsymbol{\sigma}_1^2, \boldsymbol{\sigma}_2^2, \ldots, \boldsymbol{\sigma}_N^2)'$. In the case where the $\mathbf{x}_{it}^{\prime s}$ are $I(0)$, the pooled observation matrix on the regressors

$$\frac{1}{NT}\sum_{i=1}^{N} \frac{\phi_i^2}{\sigma_i^2} \mathbf{X}_i' \mathbf{H}_i \mathbf{X}_i,$$

converges in probability to a fixed positive definite matrix. In the case where the $\mathbf{x}_{it}$'s are $I(1)$, the matrix

$$\frac{1}{NT^2}\sum_{i=1}^{N} \frac{\phi_i^2}{\sigma_i^2} \mathbf{X}_i' \mathbf{H}_i \mathbf{X}_i,$$

converges to a random positive definite matrix with probability 1. These conditions should hold for all feasible values of ϕ_i and σ_i^2 as $T \to \infty$ either for a fixed N, or for $N \to \infty$ and $T \to \infty$, jointly. See Pesaran, Shin, and Smith (1999) for details.

The *ML* estimates of the long-run coefficients, $\boldsymbol{\theta}$, and the group-specific error-correction coefficients, ϕ_i, can be computed by maximizing (28.67) with respect to $\boldsymbol{\varphi}$. These *ML* estimators are termed pooled mean group (PMG) estimators in order to highlight the pooling effect of the homogeneity restrictions on the estimates of the long-run coefficients, and the fact that averages across groups are used to obtain group-wide mean estimates of the error-correction coefficients and the other short-run parameters of the model.

Pesaran, Shin, and Smith (1999) propose two different likelihood-based algorithms for the computation of the PMG estimators which are computationally less demanding than estimating the pooled regression. The first is a 'back-substitution' algorithm that only makes use of the first derivatives of the log-likelihood function

$$\hat{\boldsymbol{\theta}} = -\left(\sum_{i=1}^{N} \frac{\hat{\phi}_i^2}{\hat{\sigma}_i^2} \mathbf{X}_i' \mathbf{H}_i \mathbf{X}_i\right)^{-1} \left[\sum_{i=1}^{N} \frac{\hat{\phi}_i}{\hat{\sigma}_i^2} \mathbf{X}_i' \mathbf{H}_i \left(\Delta \mathbf{y}_i - \hat{\phi}_i \mathbf{y}_{i,-1}\right)\right], \tag{28.68}$$

$$\hat{\phi}_i = \left(\hat{\boldsymbol{\xi}}_i' \mathbf{H}_i \hat{\boldsymbol{\xi}}_i\right)^{-1} \hat{\boldsymbol{\xi}}_i' \mathbf{H}_i \Delta \mathbf{y}_i, \tag{28.69}$$

$$\hat{\sigma}_i^2 = T^{-1} (\Delta \mathbf{y}_i - \hat{\phi}_i \hat{\boldsymbol{\xi}}_i)' \mathbf{H}_i (\Delta \mathbf{y}_i - \hat{\phi}_i \hat{\boldsymbol{\xi}}_i), \tag{28.70}$$

where $\hat{\boldsymbol{\xi}}_i = \mathbf{y}_{i,-1} - \mathbf{X}_i \hat{\boldsymbol{\theta}}$. Starting with an initial estimate of $\boldsymbol{\theta}$, say $\hat{\boldsymbol{\theta}}^{(0)}$, estimates of ϕ_i and σ_i^2 can be computed using (28.69) and (28.70), which can then be substituted in (28.68) to obtain a new estimate of $\boldsymbol{\theta}$, say $\hat{\boldsymbol{\theta}}^{(1)}$, and so on until convergence is achieved. Alternatively, the PMG estimators can be computed using (a variation of) the Newton-Raphson algorithm which makes use of both the first and the second derivatives. An overview of alternative numerical optimization techniques is provided in Section A.16 of Appendix A.

Note that, for small T, the PMG estimator (as well as the group-specific estimator) will be subject to the familiar downward bias on the coefficient of the lagged dependent variable. Because the bias is in the same direction for each group, averaging or pooling does not reduce this bias. Bias corrections are available in the literature (e.g., Kiviet and Phillips (1993)), but these apply to the short-run coefficients. Because the long-run coefficient is a nonlinear function of the short-run coefficients, procedures that remove the bias in the short-run coefficients can leave the long-run coefficient biased. Pesaran and Zhao (1999) discuss how the bias in the long-run coefficients can be reduced.

Example 66 *Continuing from Example 65, Haque, Pesaran, and Sharma (2000) then allowed for both unrestricted short-run slope heterogeneity and dynamics. To this end, they estimate individual country regressions containing first-order lagged values of the savings rates, $PSAV_{i,t-1}$. The MG and pooled mean group (PMG) estimates of the long-run coefficients based on these dynamic individual country regressions are given in Table 28.5. For ease of comparison, the MG estimator based on a static version of these regressions, as well as the corresponding FE estimates, are reported. Unlike the FE estimates, the consequences of allowing for dynamics on the MG estimates are rather limited. Once again only the coefficients of the SUR and the GCUR variables are statistically significant, although the dynamic MG estimates suggest the coefficient of the PCTT variable to be also marginally significant. Finally, the last column of Table 28.5 provides the pooled mean group estimates of the long-run coefficients, where the short-run dynamics are allowed to differ freely across countries but equality restrictions are imposed on one or more of the long-run coefficients; the rationale being that due to differences in factors such as adjustment costs or the institutional set-up across countries slope homogeneity is more likely to be valid in the long run. The PMG estimates in Table 28.5 impose the slope homogeneity restrictions only on the long-run coefficients of the SUR variable. As expected, the PMG estimates are generally more precisely estimated and confirm that, amongst the various determinants of private savings considered by MBS, only the effects of the SUR and the GCUR variables seem to be reasonably robust to the presence of slope heterogeneity and yield plausible estimates for the offsetting effects of government budget surpluses and government consumption expenditures on private savings across OECD countries.*

Table 28.5 Private saving equations: fixed-effects, mean group and pooled MG estimates (20 OECD countries, 1972–1993)

	FE Estimates		Mean Group Estimates		Pooled MGE
Regressors	**Static**	**Dynamic**	**Static**	**Dynamic**	**Dynamic**
SUR	−0.518	−0.968	−0.671	−0.911	−0.870
	(−8.50)	(−7.76)	(−8.07)	(−5.48)	(−19.81)
GCUR	−0.461	−0.665	−0.401	−0.394	−0.474
	(−10.76)	(−8.17)	(−5.95)	(−4.38)	(−6.88)
GI	−0.555	−0.789	−0.335	−0.109	−0.401
	(−5.28)	(−4.14)	(−1.01)	(−0.22)	(−1.14)
GR	−0.059	0.091	0.046	0.057	0.029
	(−1.09)	(−0.93)	(0.88)	(0.92)	(0.48)
RINT	0.205	0.127	0.104	0.183	0.139
	(4.11)	(1.41)	(1.28)	(1.61)	(1.66)
W	0.020	0.028	0.001	0.002	−0.004
	(4.51)	(3.49)	(0.061)	(0.115)	(−0.21)
INF	0.161	0.069	0.089	0.137	0.103
	(3.91)	(0.93)	(1.02)	(1.18)	(1.11)
PCTT	0.044	0.094	0.048	0.103	0.077
	(2.83)	(3.31)	(1.34)	(2.21)	(2.37)
YP	−0.087	−0.076	−0.069	−0.056	−0.031
	(−2.54)	(−1.23)	(−0.77)	(−0.60)	(−0.35)
DEP	−0.161	−0.241	0.080	0.058	0.050
	(−5.13)	(−4.22)	(0.63)	(0.45)	(0.39)

*The dependent variable is $PSAV_{it}$. The estimates refer to the long-run coefficients. Dynamic fixed-effects (FE) estimates are based on a first-order autoregressive panel data model containing the lagged dependent variables, $PSAV_{i,t-1}$. The dynamic Mean Group (MG) estimates are based on country-specific regressions also containing. $PSAV_{i,t-1}$. The Pooled MG estimates impose the restrictions that the long-run coefficients of the SUR variable is the same across countries, but are otherwise comparable to the dynamic MG estimates. Due to the presence of YRUS variable in the model, country-specific parameters for the U.S. are not identifies, and the U.S. is dropped from the panel.

28.11 Testing for slope homogeneity

Given the adverse statistical consequences of neglected slope heterogeneity, it is important that the assumption of slope homogeneity is tested. To this end, consider the panel data

$$y_{it} = \alpha_i + \boldsymbol{\beta}_i' \mathbf{x}_{it} + u_{it}, \tag{28.71}$$

where α_i are bounded on a compact set, $\mathbf{x}_{it}$ is a k-dimensional vector of regressors, $\boldsymbol{\beta}_i$ is a k-dimensional vector of unknown slope coefficients, and $u_{it} \sim IID(0, \sigma_i^2)$. The null hypothesis of interest is

$$H_0 : \boldsymbol{\beta}_i = \boldsymbol{\beta}, \text{ for all } i, \ \|\boldsymbol{\beta}\| < K < \infty, \tag{28.72}$$

against the alternatives

$$H_1 : \boldsymbol{\beta}_i = \boldsymbol{\beta}, \text{ for a non-zero fraction of slopes.}$$

One assumption underlying existing tests for slope homogeneity is that, under H_1, the fraction of the slopes that are not the same does not tend to zero as $N \to \infty$.

28.11.1 Standard *F*-test

There are a number of procedures that can be used to test H_0, the most familiar of which is the standard F-test defined by

$$F = \left(\frac{N(T-k-1)}{k(N-1)} \right) \frac{RSSR - USSR}{USSR},$$

where $RSSR$ and $USSR$ are restricted and unrestricted residual sum of squares, respectively, obtained under the null $(\boldsymbol{\beta}_i = \boldsymbol{\beta})$ and the alternative hypotheses. This test is applicable when N is fixed as $T \to \infty$, and the error variances are homoskedastic, $\sigma_i^2 = \sigma^2$. But it is likely to perform rather poorly in cases where N is relatively large, the regressors contain lagged values of the dependent variable and/or if the error variances are cross sectionally heteroskedastic.

28.11.2 Hausman-type test by panels

For cases where $N > T$, Pesaran, Smith, and Im (1996) propose using the Hausman (1978) procedure by comparing the fixed-effects (FE) estimator of $\boldsymbol{\beta}$,

$$\hat{\boldsymbol{\beta}}_{FE} = \left(\sum_{i=1}^{N} \mathbf{X}_i' \mathbf{M}_\tau \mathbf{X}_i \right)^{-1} \sum_{i=1}^{N} \mathbf{X}_i' \mathbf{M}_\tau \mathbf{y}_i, \tag{28.73}$$

with the mean group (MG) estimator

$$\hat{\boldsymbol{\beta}}_{MG} = N^{-1} \sum_{i=1}^{N} \hat{\boldsymbol{\beta}}_i, \tag{28.74}$$

where $\mathbf{M}_\tau = \mathbf{I}_T - \boldsymbol{\tau}_T \left(\boldsymbol{\tau}_T' \boldsymbol{\tau}_T \right)^{-1} \boldsymbol{\tau}_T'$, $\boldsymbol{\tau}_T$ is a $\mathbf{T} \times 1$ vector of ones, $\mathbf{I}_T$ is an identity matrix of order T, and

$$\hat{\boldsymbol{\beta}}_i = \left(\mathbf{X}_i' \mathbf{M}_\tau \mathbf{X}_i \right)^{-1} \mathbf{X}_i' \mathbf{M}_\tau \mathbf{y}_i. \tag{28.75}$$

For the Hausman test to have the correct size and be consistent two conditions must be met (see also Section 26.9.1)

(a) Under H_0, $\hat{\boldsymbol{\beta}}_{FE}$ and $\hat{\boldsymbol{\beta}}_{MG}$ must both be consistent for $\boldsymbol{\beta}$, with $\hat{\boldsymbol{\beta}}_{FE}$ being asymptotically more efficient such that

$$AVar\left(\hat{\boldsymbol{\beta}}_{MG} - \hat{\boldsymbol{\beta}}_{FE} \right) = AVar\left(\hat{\boldsymbol{\beta}}_{MG} \right) - AVar\left(\hat{\boldsymbol{\beta}}_{FE} \right) > 0, \tag{28.76}$$

where *AVar* $(\cdot)$ stands for the asymptotic variance operator.

(b) Under H_1, $\hat{\boldsymbol{\beta}}_{MG} - \hat{\boldsymbol{\beta}}_{FE}$ should tend to a non-zero vector.

In the context of dynamic panel data models with exogenous regressors both of these conditions are met, so long as the exogenous regressors are not drawn from the same distribution. In such a case a Hausman-type test based on the difference $\hat{\boldsymbol{\beta}}_{FE} - \hat{\boldsymbol{\beta}}_{MG}$ would be valid and is shown to have reasonable small sample properties. See Pesaran, Smith, and Im (1996) and Hsiao and Pesaran (2008).

However, as is well known, the Hausman procedure can lack power for certain parameter values as its implicit null does not necessarily coincide with the null hypothesis of interest. This problem turns out to be much more serious in the application of the Hausman procedure to the testing problem that concerns us here. For example, in the case of panel data models containing only strictly exogenous regressors, a test of slope homogeneity based on $\hat{\boldsymbol{\beta}}_{FE} - \hat{\boldsymbol{\beta}}_{MG}$ will lack power in *all* directions, if under the alternative hypothesis, the slopes are random draws from the same distribution. To see this, suppose that under H_1 the slopes satisfy the familiar random coefficient specification

$$\boldsymbol{\beta}_i = \boldsymbol{\beta} + \mathbf{v}_i, \mathbf{v}_i \sim IID(\mathbf{0}, \boldsymbol{\Sigma}_v),$$

where $\boldsymbol{\Sigma}_v \neq \mathbf{0}$ is a non-negative definite matrix, and $E(\mathbf{X}_j'\mathbf{v}_i) = \mathbf{0}$ for all i and j. Then

$$\hat{\boldsymbol{\beta}}_{FE} - \hat{\boldsymbol{\beta}}_{MG} = \Big(\sum_{i=1}^{N} \mathbf{X}_i'\mathbf{M}_\tau\mathbf{X}_i\Big)^{-1} \sum_{i=1}^{N} \left(\mathbf{X}_i'\mathbf{M}_\tau\mathbf{X}_i\right)\mathbf{v}_i - N^{-1}\sum_{i=1}^{N}\mathbf{v}_i + \Big(\sum_{i=1}^{N} \mathbf{X}_i'\mathbf{M}_\tau\mathbf{X}_i\Big)^{-1} \sum_{i=1}^{N} \mathbf{X}_i'\mathbf{M}_\tau\boldsymbol{\varepsilon}_i - N^{-1}\sum_{i=1}^{N}\left(\mathbf{X}_i'\mathbf{M}_\tau\mathbf{X}_i\right)^{-1}\mathbf{X}_i'\mathbf{M}_\tau\boldsymbol{\varepsilon}_i,$$

and it readily follows that, under the random coefficients alternatives and strictly exogenous regressors, we have $E\big(\hat{\boldsymbol{\beta}}_{FE} - \hat{\boldsymbol{\beta}}_{MG}\,|H_1\big) = \mathbf{0}$. This result holds for N and T fixed as well as when N and $T \to \infty$, and hence condition (b) of Hausman's procedure is not satisfied.

Another important case where the Hausman test does not apply arises when testing the homogeneity of slopes in pure autoregressive panel data models. To simplify the exposition, consider the following stationary AR(1) panel data model

$$y_{it} = \alpha_i(1 - \beta_i) + \beta_i y_{i,t-1} + \varepsilon_{it}, \text{with } |\beta_i| < 1. \tag{28.77}$$

It is now easily seen that with N fixed and as $T \to \infty$, under H_0 (where $\beta_i = \beta$) we have

$$\sqrt{NT}\left(\hat{\beta}_{FE} - \beta\right) \to_d N\left(0, 1 - \beta^2\right),$$

and

$$\sqrt{NT}\left(\hat{\beta}_{MG} - \beta\right) \to_d N\left(0, 1 - \beta^2\right).$$

Hence the variance inequality part of condition (a), namely (28.76), is not satisfied, and the application of the Hausman test to autoregressive panels will not have the correct size.

28.11.3 *G*-test of Phillips and Sul

Phillips and Sul (2003) propose a different type of Hausman test where, instead of comparing two different pooled estimators of the regression coefficients (as discussed in Section 28.11.2), they propose basing the test of slope homogeneity on the difference between the individual estimates and a suitably defined pooled estimator. In the context of the panel regression model (28.71), their test statistic can be written as

$$G = \left(\hat{\boldsymbol{\beta}} - \boldsymbol{\tau}_N \otimes \hat{\boldsymbol{\beta}}_{FE}\right)' \hat{\boldsymbol{\Sigma}}_g^{-1} \left(\hat{\boldsymbol{\beta}} - \boldsymbol{\tau}_N \otimes \hat{\boldsymbol{\beta}}_{FE}\right),$$

where $\hat{\boldsymbol{\beta}} = (\hat{\boldsymbol{\beta}}_1', \hat{\boldsymbol{\beta}}_2', \ldots, \hat{\boldsymbol{\beta}}_N')'$ is an $Nk \times 1$ stacked vector of all the N individual least square estimates of $\boldsymbol{\beta}_i$, $\hat{\boldsymbol{\beta}}_{FE}$ is a fixed-effect estimator as before, and $\hat{\boldsymbol{\Sigma}}_g$ is a consistent estimator of $\boldsymbol{\Sigma}_g$, the asymptotic variance matrix of $\hat{\boldsymbol{\beta}} - \boldsymbol{\tau}_N \otimes \hat{\boldsymbol{\beta}}_{FE}$, under H_0. Under standard assumptions for stationary dynamic models, and assuming H_0 holds and N is fixed, then $G \rightarrow_d \chi^2(Nk)$ as $T \rightarrow \infty$, so long as $\boldsymbol{\Sigma}_g$ is a non-stochastic positive definite matrix.

As compared to the Hausman test based on $\hat{\boldsymbol{\beta}}_{MG} - \hat{\boldsymbol{\beta}}_{FE}$, the G test is likely to be more powerful; but its use will be limited to panel data models where N is small relative to T. Also, the G test will not be valid in the case of pure dynamic models, very much for the same kind of reasons noted above in relation to the Hausman test based on $\hat{\boldsymbol{\beta}}_{MG} - \hat{\boldsymbol{\beta}}_{FE}$. This is easily established in the case of the stationary first-order autoregressive panel data model considered by Phillips and Sul (2003). In the case of $AR(1)$ panel regressions with $\sigma_i^2 = \sigma^2$, it is easily verified that under H_0

$$\begin{aligned} Avar\left[\sqrt{T}\left(\hat{\beta}_i - \hat{\beta}_{FE}\right)\right] &= Avar\left[\sqrt{T}\left(\hat{\beta}_i - \beta\right) - \sqrt{T}\left(\hat{\beta}_{FE} - \beta\right)\right] \\ &= \left(1 - \beta^2\right) - \left(\frac{1-\beta^2}{N}\right), \\ Acov\left[\sqrt{T}\left(\hat{\beta}_i - \hat{\beta}_{FE}\right), \sqrt{T}\left(\hat{\beta}_j - \hat{\beta}_{FE}\right)\right] &= -\left(\frac{1-\beta^2}{N}\right). \end{aligned}$$

Therefore

$$\boldsymbol{\Sigma}_g = \left(\frac{1-\beta^2}{T}\right)\left(\mathbf{I}_N - N^{-1}\boldsymbol{\tau}_N\boldsymbol{\tau}_N'\right).$$

It is now easily seen that $rank\left(\boldsymbol{\Sigma}_g\right) = N - 1$, and $\boldsymbol{\Sigma}_g$ is non-invertible.

28.11.4 Swamy's test

Swamy (1970) proposes a test of slope homogeneity based on the dispersion of individual slope estimates from a suitable pooled estimator. Like the F-test, Swamy's test is developed for panels

where N is small relative to T, but allows for cross-sectional heteroskedasticity. Swamy's statistic applied to the slope coefficients can be written as

$$\hat{S} = \sum_{i=1}^{N} \left(\hat{\boldsymbol{\beta}}_i - \hat{\boldsymbol{\beta}}_{WFE}\right)' \frac{\mathbf{X}_i'\mathbf{M}_\tau\mathbf{X}_i}{\hat{\sigma}_i^2} \left(\hat{\boldsymbol{\beta}}_i - \hat{\boldsymbol{\beta}}_{WFE}\right), \tag{28.78}$$

where $\hat{\sigma}_i^2$ is an estimator of σ_i^2 based on $\hat{\boldsymbol{\beta}}_{WFE}$, namely

$$\hat{\sigma}_i^2 = \frac{1}{T-k-1}\left(\mathbf{y}_i - \mathbf{X}_i\hat{\boldsymbol{\beta}}_{WFE}\right)' \mathbf{M}_\tau \left(\mathbf{y}_i - \mathbf{X}_i\hat{\boldsymbol{\beta}}_{WFE}\right),$$

and $\hat{\boldsymbol{\beta}}_{WFE}$ is the weighted pooled estimator also computed using $\hat{\sigma}_i^2$, namely

$$\hat{\boldsymbol{\beta}}_{WFE} = \left(\sum_{i=1}^{N} \frac{\mathbf{X}_i'\mathbf{M}_\tau\mathbf{X}_i}{\hat{\sigma}_i^2}\right)^{-1} \sum_{i=1}^{N} \frac{\mathbf{X}_i'\mathbf{M}_\tau\mathbf{y}_i}{\hat{\sigma}_i^2}.$$

In the case where N is fixed and T tends to infinity, under H_0 the Swamy statistic, $\hat{S}$, is asymptotically chi-square-distributed with $k(N-1)$ degrees of freedom.

28.11.5 Pesaran and Yamagata Δ-test

Based on Swamy (1970)'s work, Pesaran and Yamagata (2008) propose a standardized dispersion statistic that is asymptotically normally distributed for large N and T. One version of the dispersion test, denoted by $\hat{\Delta}$, makes use of the Swamy statistic, $\hat{S}$ defined by (28.78), and another version, denoted by $\tilde{\Delta}$, is based on a modified version of the Swamy statistic where regression standard errors for the individual cross-sectional units are computed using the pooled fixed-effects, rather than the ordinary least squares estimator, as proposed by Swamy. It is shown that, in the case of models with strictly exogenous regressors, but with non-normal errors, both versions of the Δ-test tend to the standard normal distribution as $(N,T) \to_j \infty$, subject to certain restrictions on the relative expansion rates of N and T. For the $\hat{\Delta}$-test it is required that $\sqrt{N}/T \to 0$, as $(N,T) \to_j \infty$, whilst for the $\tilde{\Delta}$-test the condition is less restrictive and is given by $\sqrt{N}/T^2 \to 0$. When the errors are normally distributed, mean-variance bias adjusted versions of the Δ-tests, denoted by $\hat{\Delta}_{adj}$ and $\tilde{\Delta}_{adj}$, are proposed that are valid as $(N,T) \to_j \infty$ without any restrictions on the relative expansion rates of N and T.

More specifically, $\hat{\Delta}$ and $\tilde{\Delta}$-tests are defined by

$$\hat{\Delta} = \sqrt{N}\left(\frac{N^{-1}\hat{S}-k}{\sqrt{2k}}\right), \quad \tilde{\Delta} = \sqrt{N}\left(\frac{N^{-1}\tilde{S}-k}{\sqrt{2k}}\right) \tag{28.79}$$

where

$$\tilde{S} = \sum_{i=1}^{N} \left(\hat{\boldsymbol{\beta}}_i - \tilde{\boldsymbol{\beta}}_{WFE}\right)' \frac{\mathbf{X}_i'\mathbf{M}_\tau\mathbf{X}_i}{\tilde{\sigma}_i^2} \left(\hat{\boldsymbol{\beta}}_i - \tilde{\boldsymbol{\beta}}_{WFE}\right), \tag{28.80}$$

$$\tilde{\boldsymbol{\beta}}_{WFE} = \left(\sum_{i=1}^{N} \frac{\mathbf{X}_i'\mathbf{M}_\tau\mathbf{X}_i}{\tilde{\sigma}_i^2}\right)^{-1} \sum_{i=1}^{N} \frac{\mathbf{X}_i'\mathbf{M}_\tau\mathbf{y}_i}{\tilde{\sigma}_i^2},$$

and

$$\tilde{\sigma}_i^2 = \frac{1}{T-1}\left(\mathbf{y}_i - \mathbf{X}_i\hat{\boldsymbol{\beta}}_{FE}\right)' \mathbf{M}_\tau \left(\mathbf{y}_i - \mathbf{X}_i\hat{\boldsymbol{\beta}}_{FE}\right).$$

Although the difference between $\hat{S}$ and $\tilde{S}$ might appear slight at first, the different choices of the estimator of σ_i^2 used in construction of these statistics have important implications for the properties of the two tests as N and T tends to infinity. To see this let

$$\mathbf{Q}_{iT} = T^{-1}\left(\mathbf{X}_i'\mathbf{M}_\tau\mathbf{X}_i\right), \tag{28.81}$$

$$\mathbf{Q}_{NT} = (NT)^{-1}\left(\sum_{i=1}^{N} \mathbf{X}_i'\mathbf{M}_\tau\mathbf{X}_i\right), \tag{28.82}$$

$$\mathbf{P}_i = \mathbf{M}_\tau\mathbf{X}_i\left(\mathbf{X}_i'\mathbf{M}_\tau\mathbf{X}_i\right)^{-1}\mathbf{X}_i'\mathbf{M}_\tau, \tag{28.83}$$

$$\mathbf{M}_i = \mathbf{I}_T - \mathbf{Z}_i(\mathbf{Z}_i'\mathbf{Z}_i)^{-1}\mathbf{Z}_i', \tag{28.84}$$

where $\mathbf{Z}_i = (\boldsymbol{\tau}_T, \mathbf{X}_i)$, and consider the following assumptions:

Assumption H.5:

(i) $\varepsilon_{it}|\mathbf{X}_i \sim IID(0, \sigma_i^2)$, $\sigma^2_{\max} = \max_{1\le i\le N}(\sigma_i^2) < K$, and $\sigma^2_{\min} = \min_{1\le i\le N}(\sigma_i^2) > 0$.

(ii) ε_{it} and ε_{js} are independently distributed for $i \neq j$ and/or $t \neq s$.

(iii) $E(\varepsilon_{it}^9|\mathbf{X}_i) < K$.

Assumption H.6: (i) The $k \times k$ matrices $\mathbf{Q}_{iT}$, $i = 1, 2, \ldots, N$, defined by (28.81) are positive definite and bounded, $\max_{1\le i\le N} E\|\mathbf{Q}_{iT}\| < K$, and $\mathbf{Q}_{iT}$ tends to a non-stochastic positive definite matrix, $\mathbf{Q}_i$, $\max_{1\le i\le N} E\|\mathbf{Q}_i\| < K$, as $T \to \infty$.

(ii) The $k \times k$ pooled observation matrix $\mathbf{Q}_{NT}$ defined by (28.82) is positive definite, and $\mathbf{Q}_{NT}$ tends to a non-stochastic positive definite matrix, $\mathbf{Q} = \lim_{N\to\infty} N^{-1}\sum_{i=1}^{N}\mathbf{Q}_i$, as $(N, T) \xrightarrow{j} \infty$.

Assumption H.7: There exists a finite T_0 such that for $T > T_0$, $E\{[\boldsymbol{v}_i'\mathbf{M}_\tau\boldsymbol{v}_i/(T-1)]^{-4-\epsilon}\} < K$ and $E\{[\boldsymbol{v}_i'\mathbf{M}_i\boldsymbol{v}_i/(T-k-1)]^{-4-\epsilon}\} < K$, for each i and for some small positive constant ϵ, where $\boldsymbol{v}_i = \boldsymbol{\varepsilon}_i/\sigma_i$.

Assumption H.8: Under H_1, the fraction of slopes that are not the same does not tend to zero as $N \to \infty$.

Under Assumptions H.5–H.7 and assuming that H_0 (the null of slope homogeneity) holds, then the dispersion statistics $\hat{S}$ and $\tilde{S}$ defined above can be written as

$$N^{-1/2}\hat{S} = N^{-1/2}\sum_{i=1}^{N}\hat{z}_{iT} + O_p\left(N^{-1/2}\right) + O_p\left(T^{-1/2}\right), \tag{28.85}$$

$$N^{-1/2}\tilde{S} = N^{-1/2}\sum_{i=1}^{N}\tilde{z}_{iT} + O_p\left(N^{-1/2}\right) + O_p\left(T^{-1/2}\right), \tag{28.86}$$

where

$$\hat{z}_{iT} = \frac{(T-k-1)\boldsymbol{v}_i'\mathbf{P}_i\boldsymbol{v}_i}{\boldsymbol{v}_i'\mathbf{M}_i\boldsymbol{v}_i}, \text{ and } \tilde{z}_{iT} = \frac{(T-1)\boldsymbol{v}_i'\mathbf{P}_i\boldsymbol{v}_i}{\boldsymbol{v}_i'\mathbf{M}_\tau\boldsymbol{v}_i}. \tag{28.87}$$

Under Assumptions H.4–H.7, $\hat{z}_{iT}$ and $\tilde{z}_{iT}$ are independently (but not necessarily identically) distributed random variables across i with finite means and variances, and for all i we have

$$E(\hat{z}_{iT}) = k + O(T^{-1}),\ Var(\hat{z}_{iT}) = 2k + O(T^{-1}), \tag{28.88}$$

$$E(\tilde{z}_{iT}) = k + O(T^{-2}),\ Var(\tilde{z}_{iT}) = 2k + O(T^{-1}), \tag{28.89}$$

$$E\left|\hat{z}_{iT}\right|^{2+\epsilon/2} < K, \text{ and } E\left|\tilde{z}_{iT}\right|^{2+\epsilon/2} < K. \tag{28.90}$$

Also under the null hypothesis that the slopes are homogenous, we have

$$\hat{\Delta} \to_d N(0,1), \text{as } (N,T) \xrightarrow{j} \infty, \text{ so long as } \sqrt{N}/T \to 0,$$
$$\tilde{\Delta} \to_d N(0,1), \text{ as } (N,T) \xrightarrow{j} \infty, \text{ so long as } \sqrt{N}/T^2 \to 0,$$

where the standardized dispersion statistics, $\hat{\Delta}$ and $\tilde{\Delta}$ are defined above. Furthermore, if the errors, ε_{it}, are normally distributed, under H_0 we have

$$\hat{\Delta} \to_d N(0,1), \text{as } (N,T) \xrightarrow{j} \infty, \text{ so long as } \sqrt{N}/T \to 0,$$
$$\tilde{\Delta} \to_d N(0,1), \text{ as } (N,T) \xrightarrow{j} \infty.$$

The small sample properties of the dispersion tests can be improved under the normally distributed errors by considering the following mean and variance bias adjusted versions of $\hat{\Delta}$ and $\tilde{\Delta}$

$$\tilde{\Delta}_{adj} = \sqrt{\frac{N(T+1)}{(T-k-1)}}\left(\frac{N^{-1}\tilde{S} - k}{\sqrt{2k}}\right), \tag{28.91}$$

$$\hat{\Delta}_{adj} = \sqrt{N}\left(\frac{N^{-1}\hat{S} - E(\hat{z}_{iT})}{\sqrt{Var(\hat{z}_{iT})}}\right),$$

where

$$E(\hat{z}_{iT}) = \frac{k(T-k-1)}{T-k-3},\ Var(\hat{z}_{iT}) = \frac{2k\,(T-k-1)^2\,(T-3)}{(T-k-3)^2\,(T-k-5)}. \tag{28.92}$$

The Monte Carlo results reported in Pesaran and Yamagata (2008) suggest that the $\tilde{\Delta}_{adj}$ test works well even if there are major departures from normality, and is to be recommended.

28.11.6 Extensions of the Δ-tests

The Δ-tests can be readily extended to test the homogeneity of a subset of slope coefficients. Consider the following partitioned form of (28.71)

$$\underset{T\times 1}{\mathbf{y}_i} = \alpha_i \boldsymbol{\tau}_T + \underset{T\times k_1}{\mathbf{X}_{i1}}\,\boldsymbol{\beta}_{i1} + \underset{T\times k_2}{\mathbf{X}_{i2}}\,\boldsymbol{\beta}_{i2} + \boldsymbol{\varepsilon}_i, i = 1, 2, \ldots, N,$$

or

$$\underset{T\times 1}{\mathbf{y}_i} = \underset{T\times (k_1+1)}{\mathbf{Z}_{i1}}\,\boldsymbol{\delta}_i + \underset{T\times k_2}{\mathbf{X}_{i2}}\,\boldsymbol{\beta}_{i2} + \boldsymbol{\varepsilon}_i,$$

where $\mathbf{Z}_{i1} = (\boldsymbol{\tau}_T, \mathbf{X}_{i1})$ and $\boldsymbol{\delta}_i = \left(\alpha_i, \boldsymbol{\beta}_{i1}'\right)'$. Suppose the slope homogeneity hypothesis of interest is given by

$$H_0:\ \boldsymbol{\beta}_{i2} = \boldsymbol{\beta}_2, \text{for } i = 1, 2, \ldots, N. \tag{28.93}$$

The dispersion test statistic in this case is given by

$$\tilde{S}_2 = \sum_{i=1}^{N}\left(\hat{\boldsymbol{\beta}}_{i2} - \tilde{\boldsymbol{\beta}}_{2,WFE}\right)'\frac{\mathbf{X}_{i2}'\mathbf{M}_{i1}\mathbf{X}_{i2}}{\tilde{\sigma}_i^2}\left(\hat{\boldsymbol{\beta}}_{i2} - \tilde{\boldsymbol{\beta}}_{2,WFE}\right),$$

where

$$\hat{\boldsymbol{\beta}}_{i2} = \left(\mathbf{X}_{i2}'\mathbf{M}_{i1}\mathbf{X}_{i2}\right)^{-1}\mathbf{X}_{i2}'\mathbf{M}_{i1}\mathbf{y}_i,$$

$$\tilde{\boldsymbol{\beta}}_{2,WFE} = \left(\sum_{i=1}^{N}\frac{\mathbf{X}_{i2}'\mathbf{M}_{i1}\mathbf{X}_{i2}}{\tilde{\sigma}_i^2}\right)^{-1}\sum_{i=1}^{N}\frac{\mathbf{X}_{i2}'\mathbf{M}_{i1}\mathbf{y}_i}{\tilde{\sigma}_i^2},$$

$$\mathbf{M}_{i1} = \mathbf{I}_T - \mathbf{Z}_{i1}\left(\mathbf{Z}_{i1}'\mathbf{Z}_{i1}\right)^{-1}\mathbf{Z}_{i1}',$$

$$\tilde{\sigma}_i^2 = \frac{\left(\mathbf{y}_i - \mathbf{X}_{i2}\hat{\boldsymbol{\beta}}_{2,FE}\right)'\mathbf{M}_{i1}\left(\mathbf{y}_i - \mathbf{X}_{i2}\hat{\boldsymbol{\beta}}_{2,FE}\right)}{T-k_1-1},$$

and

$$\hat{\boldsymbol{\beta}}_{2,FE} = \left(\sum_{i=1}^{N} \mathbf{X}_{i2}' \mathbf{M}_{i1} \mathbf{X}_{i2} \right)^{-1} \sum_{i=1}^{N} \mathbf{X}_{i2}' \mathbf{M}_{i1} \mathbf{y}_i.$$

Using a similar line of reasoning as above, it is now easily seen that under H_0 defined by (28.93), and for $(N, T) \xrightarrow{j} \infty$, such that $\sqrt{N}/T^2 \to 0$, then

$$\tilde{\Delta}_2 = \sqrt{N} \left(\frac{N^{-1} \tilde{S}_2 - k_2}{\sqrt{2k_2}} \right) \to_d N(0, 1).$$

In the case of normally distributed errors, the following mean-variance bias adjusted statistic can be used

$$\tilde{\Delta}_{adj} = \sqrt{\frac{N(T - k_1 + 1)}{(T - k - 1)}} \left(\frac{N^{-1} \tilde{S}_2 - k_2}{\sqrt{2k_2}} \right).$$

The Δ-tests can also be extended to unbalanced panels. Denoting the number of time series observations on the ith cross-section by T_i, the standardized dispersion statistic is given by

$$\tilde{\Delta} = \frac{1}{\sqrt{N}} \sum_{i=1}^{N} \left(\frac{\tilde{d}_i - k}{\sqrt{2k}} \right), \tag{28.94}$$

$$\tilde{d}_i = \left(\hat{\boldsymbol{\beta}}_i - \tilde{\boldsymbol{\beta}}_{WFE} \right)' \frac{\mathbf{X}_i' \mathbf{M}_{\tau_i} \mathbf{X}_i}{\tilde{\sigma}_i^2} \left(\hat{\boldsymbol{\beta}}_i - \tilde{\boldsymbol{\beta}}_{WFE} \right),$$

$\mathbf{X}_i = \left(\mathbf{x}_{i1}, \mathbf{x}_{i2}, \ldots, \mathbf{x}_{iT_i} \right)'$, $\mathbf{M}_{\tau_i} = \mathbf{I}_{T_i} - \boldsymbol{\tau}_{T_i} \left(\boldsymbol{\tau}_{T_i}' \boldsymbol{\tau}_{T_i} \right)^{-1} \boldsymbol{\tau}_{T_i}'$ with $\boldsymbol{\tau}_{T_i}$ being a $T_i \times 1$ vector of unity,

$$\hat{\boldsymbol{\beta}}_i = \left(\mathbf{X}_i' \mathbf{M}_{\tau_i} \mathbf{X}_i \right)^{-1} \mathbf{X}_i' \mathbf{M}_{\tau_i} \mathbf{y}_i, \tag{28.95}$$

$$\tilde{\boldsymbol{\beta}}_{WFE} = \left(\sum_{i=1}^{N} \frac{\mathbf{X}_i' \mathbf{M}_{\tau_i} \mathbf{X}_i}{\tilde{\sigma}_i^2} \right)^{-1} \sum_{i=1}^{N} \frac{\mathbf{X}_i' \mathbf{M}_{\tau_i} \mathbf{y}_i}{\tilde{\sigma}_i^2}, \tag{28.96}$$

$\mathbf{y}_i = \left(y_{i1}, y_{i2}, \ldots, y_{iT_i} \right)'$,

$$\tilde{\sigma}_i^2 = \frac{\left(\mathbf{y}_i - \mathbf{X}_i \hat{\boldsymbol{\beta}}_{FE} \right)' \mathbf{M}_{\tau_i} \left(\mathbf{y}_i - \mathbf{X}_i \hat{\boldsymbol{\beta}}_{FE} \right)}{T_i - 1},$$

and

$$\hat{\boldsymbol{\beta}}_{FE} = \left(\sum_{i=1}^{N} \mathbf{X}_i' \mathbf{M}_{\tau_i} \mathbf{X}_i \right)^{-1} \sum_{i=1}^{N} \mathbf{X}_i' \mathbf{M}_{\tau_i} \mathbf{y}_i. \tag{28.97}$$

The $\tilde{\Delta}$-test can also be applied to stationary dynamic models. Pesaran and Yamagata (2008) show that the test will be valid for dynamic panel data models so long as $N/T \to \kappa$, as $(N, T) \to_j \infty$, where $0 \leq \kappa < \infty$. This condition is more restrictive than the one obtained for panels with exogenous regressors, but is the same as the condition required for the validity of the fixed-effects estimator of the slope in $AR(1)$ models in large N and T panels.

Using Monte Carlo experiments it is shown that the $\tilde{\Delta}$-test has the correct size and satisfactory power in panels with strictly exogenous regressors for various combinations of N and T. Similar results are also obtained for dynamic panels, but only if the autoregressive coefficient is not too close to unity and so long as $T \geq N$. See Pesaran and Yamagata (2008) for further discussion.

28.11.7 Bias-corrected bootstrap tests of slope homogeneity for the *AR*(1) model

One possible way of improving on the asymptotic test developed for the *AR* models would be to follow the recent literature and use bootstrap techniques.[10] Here we make use of a bias-corrected version of the recursive bootstrap procedure.[11]

One of the main problems in the application of bootstrap techniques to dynamic models in small T samples is the fact that the *OLS* estimates of the individual coefficients, $\boldsymbol{\lambda}_i$, or their *FE* (or *WFE*) counterparts are biased when T is small; a bias that persists with $N \to \infty$. To deal with this problem we focus on the $AR(1)$ case and use the bias-corrected version of $\tilde{\lambda}_{WFE}$ as proposed by Hahn and Kuersteiner (2002).[12] Denoting the bias-corrected version of $\tilde{\lambda}_{WFE}$ by $\circ$, we have

$$\mathring{\lambda}_{WFE} = \tilde{\lambda}_{WFE} + \frac{1}{T}\left(1 + \tilde{\lambda}_{WFE}\right), \tag{28.98}$$

and estimate the associated intercepts as

$$\mathring{\alpha}_{i,WFE} = \bar{y}_i - \mathring{\lambda}_{WFE}\,\bar{y}_{i,-1},$$

where $\bar{y}_i = T^{-1}\sum_{t=1}^{T} y_{it}$, and $\bar{y}_{i,-1} = T^{-1}\sum_{t=1}^{T} y_{i,t-1}$. The residuals are given by

$$\mathring{e}_{it} = y_{it} - \mathring{\alpha}_{i,WFE} - \mathring{\lambda}_{WFE}\, y_{i,t-1},$$

with the associated bias-corrected estimator of σ_i^2 given by $\mathring{\sigma}_i^2 = (T-1)^{-1}\sum_{t=1}^{T}(\mathring{e}_{it})^2$. The bth bootstrap sample, $y_{it}^{(b)}$ for $i = 1, 2, \ldots, N$ and $t = 1, 2, \ldots, T$ can now be generated as

$$y_{it}^{(b)} = \mathring{\alpha}_{i,WFE} + \mathring{\lambda}_{WFE}\, y_{i,t-1}^{(b)} + \mathring{\sigma}_i \zeta_{it}^{(b)}, \text{ for } t = 1, 2, \ldots, T,$$

[10] For example, see Beran (1988), Horowitz (1994), Li and Maddala (1996), and Bun (2004), although none of these authors makes any bias corrections in their bootstrapping procedures.

[11] Bias-corrected estimates are also used in the literature on the derivation of the bootstrap confidence intervals to generate the bootstrap samples in dynamic $AR(p)$ models. See Kilian (1998), among others.

[12] Bias corrections for the *OLS* estimates of individual λ_i are provided by Kendall (1954) and Marriott and Pope (1954), and further elaborated by Orcutt and Winokur (1969). See also Section 14.5. No bias corrections seem to be available for *FE* or *WFE* estimates of $AR(p)$ panel data models in the case of $p \geq 2$.

where $y_{i0}^{(b)} = y_{i0}$, and $\zeta_{it}^{(b)}$ are random draws with replacements from the set of pooled standardized residuals, $\mathring{e}_{it}/\mathring{\sigma}_i$, $i = 1, 2, \ldots, N$, and $t = 1, 2, \ldots, T$. With $y_{it}^{(b)}$, for $i = 1, 2, \ldots, N$ and $t = 1, 2, \ldots, T$ the bootstrap statistics

$$\tilde{\Delta}^{(b)} = \sqrt{N}\left(\frac{N^{-1}\tilde{S}^{(b)} - 1}{\sqrt{2}}\right), \; b = 1, 2, \ldots, B,$$

where $\tilde{S}^{(b)}$ is the modified Swamy statistic, defined by (28.80), computed using the bth bootstrapped sample. The statistics $\tilde{\Delta}^{(b)}$ for $b = 1, 2, \ldots, B$, can now be used to obtain the bootstrap p-values

$$p_B = \frac{1}{B}\sum_{b=1}^{B} I\left(\tilde{\Delta}^{(b)} - \tilde{\Delta}\right),$$

where B is the number of bootstrap sample, $I(A)$ takes the value of unity if $A > 0$ or zero otherwise, and $\tilde{\Delta}$ is the standardized dispersion statistic applied to the actual observations. If $p_B < 0.05$, say, the null hypothesis of slope homogeneity is rejected at the 5 per cent significance level.

28.11.8 Application: testing slope homogeneity in earnings dynamics

In this section we examine the slope homogeneity of the dynamic earnings equations with the panel study of income dynamics (*PSID*) data set used in Meghir and Pistaferri (2004). Briefly, these authors select male heads aged 25 to 55 with at least nine years of usable earnings data. The selection process leads to a sample of 2, 069 individuals and 31, 631 individual-year observations. To obtain a panel data set with a larger T, only individuals with at least 15 time series observations are included in the panel. This leaves us with 1, 031 individuals and 19, 992 individual-year observations. Following Meghir and Pistaferri (2004), the individuals are categorized into three education groups: High School Dropouts (HSD, those with less than 12 grades of schooling), High School Graduates (HSG, those with at least a high school diploma, but no college degree), and College Graduates (CLG, those with a college degree or more). In what follows, the earning equations for the different educational backgrounds, HSD, HSG, and CLG, are denoted by the superscripts $e = 1, 2$, and 3, and for the pooled sample by 0. The numbers of individuals in the three categories are $N^{(1)} = 249$, $N^{(2)} = 531$, and $N^{(3)} = 251$. The panel is unbalanced with $t = 1, \ldots T_i^{(e)}$ and $i = 1, \ldots, N^{(e)}$, and an average time period of around 18 years.

In the research on earnings dynamics, it is standard to adopt a two-step procedure where in the first stage the log of real earnings is regressed on a number of control variables such as age, race and year dummies. The dynamics are then modelled based on the residuals from this first stage regression. The use of the control variables and the grouping of the individuals by educational backgrounds is aimed at eliminating (minimizing) the effects of individual heterogeneities at the second stage.

It is, therefore, of interest to examine the extent to which the two-step strategy has been successful in dealing with the heterogeneity problem. With this in mind we follow closely the two-step procedure adopted by Meghir and Pistaferri (2004) and first run regressions of log real earn-

ings, $w_{it}^{(e)}$, on the control variables: a square of "age" ($AGE_{it}^{(e)2}$), race ($WHITE_i^{(e)}$), year dummies ($YEAR(t)$), region of residence ($NE_{it}^{(e)}$, $CE_{it}^{(e)}$, $STH_{it}^{(e)}$), and residence in a standard metropolitan statistical area ($SMSA_{it}^{(e)}$), for each education group $e = 0, 1, 2, 3$, separately.[13] The residuals from these regressions, which we denote by $y_{it}^{(e)}$, are then used in the second stage to estimate dynamics of the earnings process.

Specifically,

$$y_{it}^{(e)} = \alpha_i^{(e)} + \lambda^{(e)} y_{it-1}^{(e)} + \sigma_i^{(e)} \varepsilon_{it}^{(e)}, \; e = 0, 1, 2, 3,$$

where within each education group $\lambda^{(e)}$ is assumed to be homogeneous across the different individuals. Our interest is to test the hypothesis that $\lambda^{(e)} = \lambda_i^{(e)}$ for all i in e.

The test results are given in the first panel of Table 28.6. The $\tilde{\Delta}$ statistics and the associated bootstrapped p values by education groups all lead to strong rejections of the homogeneity hypothesis. Judging by the size of the $\tilde{\Delta}$ statistics, the rejection is stronger for the pooled sample as compared with the sub-samples, confirming the importance of education as a discriminatory factor in the characterizations of heterogeneity of earnings dynamics across individuals. The test results also indicate the possibility of other statistically significant sources of heterogeneity within each of the education groups, and casts some doubt on the two-step estimation procedure adopted in the literature for dealing with heterogeneity, a point recently emphasized by Browning, Ejrnæs, and Alvarez (2010).

In Table 28.6 we also provide a number of different *FE* estimates of $\lambda^{(e)}$, $e = 0, 1, 2, 3$, on the assumption of within group slope homogeneity. Given the relatively small number of time series observations available (on average 18), the bias corrections to the *FE* estimates are quite large. The cross-section error variance heterogeneity also plays an important role in this application, as can be seen from a comparison of *FE* and *WFE* estimates with the latter being larger. Focusing on the bias-corrected *WFE* estimates, we also observe that the persistence of earnings dynamics rises systematically from 0.52 in the case of the school drop outs to 0.72 for the college graduates. This seems sensible, and partly reflects the more reliable job prospects that are usually open to individuals with a higher level of education.

The homogeneity test results suggest that further efforts are needed also to take account of within group heterogeneity. One possibility would be to adopt a Bayesian approach, assuming that $\lambda_i^{(e)}$, $i = 1, 2, \ldots, N^{(e)}$ are draws from a common probability distribution and focus attention on the whole posterior density function of the persistent coefficients, rather than the average estimates that tend to divert attention from the heterogeneity problem. Another possibility would be to follow Browning, Ejrnæs, and Alvarez (2010) and consider particular parametric functions, relating $\lambda_i^{(e)}$ to individual characteristics as a way of capturing within group heterogeneity. Finally, one could consider a finer categorization of the individuals in the panel; say by further splitting of the education groups or by introducing new categories such as occupational classifications. The slope homogeneity tests provide an indication of the statistical importance of the heterogeneity problem, but are silent as how best to deal with the problem.

[13] Log real earnings are computed as $w_{it}^{(e)} = \ln\left(LABY_{it}^{(e)}/PCED_t\right)$, where $LABY_{it}^{(e)}$ is earnings in the current US dollar, and $PCED_t$ is the personal consumption expenditure deflator, base year 1992.

Table 28.6 Slope homogeneity tests for the $AR(1)$ model of the real earnings equations

		Pooled Sample $e=0$	High School dropout $e=1$	High school graduate $e=2$	College graduate $e=3$
N		1,031	249	531	251
Average T_i		18.39	18.36	18.22	18.79
Total observations		18,961	4,572	9,673	4,716
Tests for slope homogeneity					
$\tilde{\Delta}$ **test**	Statistic	25.59	7.20	13.65	18.32
	Normal approximation p-value	[0.0000]	[0.0000]	[0.0000]	[0.0000]
	Bias-corrected bootstrap p-value	[0.0000]	[0.0000]	[0.0000]	[0.0000]
Autoregressive coefficient (λ)					
FE estimates ($\hat{\lambda}_{FE}$)		0.4841 (0.0065)	0.4056 (0.0147)	0.4497 (0.0095)	0.5538 (0.0106)
WFE estimates ($\tilde{\lambda}_{WFE}$)		0.5429 (0.0056)	0.4246 (0.0133)	0.5169 (0.0086)	0.6002 (0.0095)
Bias-corrected WFE ($\mathring{\lambda}_{WFE}$)		0.6504 (0.0055)	0.5188 (0.0126)	0.6192 (0.0080)	0.7214 (0.0101)

Notes: The *FE* estimator and the *WFE* estimator are defined by (28.97), and (28.96), respectively, and their associated standard errors (shown in round brackets) are based on $\widehat{Var}\left(\hat{\lambda}_{FE}\right)=\hat{\sigma}^2\left(\sum_{i=1}^{N}\mathbf{y}_{i,-1}'\mathbf{M}_{\tau_i}\mathbf{y}_{i,-1}\right)^{-1}$, where

$$\hat{\sigma}^2=\left(\mathrm{T}-N-1\right)^{-1}\sum_{i=1}^{N}\left(\mathbf{y}_i-\hat{\lambda}_{FE}\mathbf{y}_{i,-1}\right)'\mathbf{M}_{\tau_i}\left(\mathbf{y}_i-\hat{\lambda}_{FE}\mathbf{y}_{i,-1}\right),$$

$$\mathrm{T}=\sum_{i=1}^{N}\mathrm{T}_i, \text{ and } \widehat{Var}\left(\tilde{\lambda}_{WFE}\right)=\left(\sum_{i=1}^{N}\tilde{\sigma}_i^{-2}\mathbf{y}_{i,-1}'\mathbf{M}_{\tau_i}\mathbf{y}_{i,-1}\right)^{-1}.$$

Bias corrected estimates are based on $\mathring{\lambda}_{WFE}=\tilde{\lambda}_{WFE}+(\mathrm{T}/N)\left(1+\tilde{\lambda}_{WFE}\right)$ and $\widehat{Var}\left(\mathring{\lambda}_{WFE}\right)=\mathrm{T}^{-1}\left(1-\mathring{\lambda}_{WFE}^2\right)$. Bias-corrected bootstrapped tests also use $\mathring{\lambda}_{WFE}$ and the associated estimates to generate bootstrap samples (see Section 28.11.7 for further details).

28.12 Further reading

Further details on estimation and inference on large heterogeneous panels can be found in Hsiao (1975), Pesaran and Smith (1995), and Hsiao and Pesaran (2008).

28.13 Exercises

1. Suppose that

$$y_{it}=\boldsymbol{\beta}_{it}'\mathbf{x}_{it}+u_{it},\ i=1,2,\ldots,N,\quad t=1,2,\ldots,T,$$

where $u_{it} \sim IID(0, \sigma_i^2)$, and

$$\boldsymbol{\beta}_{it} = \boldsymbol{\beta} + \boldsymbol{\eta}_i + \boldsymbol{\lambda}_t, \quad i = 1, 2, \ldots, N, \; t = 1, 2, \ldots, T,$$

with

$$\begin{aligned} E(\boldsymbol{\eta}_i) = E(\boldsymbol{\lambda}_t) = \mathbf{0}, E\left(\boldsymbol{\eta}_i \boldsymbol{\lambda}_t'\right) = \mathbf{0}, \\ E\left(\boldsymbol{\eta}_i \mathbf{x}_{it}'\right) = \mathbf{0}, E\left(\boldsymbol{\lambda}_t \mathbf{x}_{it}'\right) = \mathbf{0}, \\ E\left(\boldsymbol{\eta}_i \boldsymbol{\eta}_j'\right) = \begin{cases} \boldsymbol{\Omega}_{\boldsymbol{\eta}}, & \text{if} \quad i = j, \\ \mathbf{0}, & \text{if} \quad i \neq j, \end{cases} \\ E(\boldsymbol{\lambda}_i \boldsymbol{\lambda}_j') = \begin{cases} \boldsymbol{\Omega}_{\boldsymbol{\Lambda}}, & \text{if} \quad i = j, \\ \mathbf{0}, & \text{if} \quad i \neq j. \end{cases} \end{aligned} \tag{28.99}$$

Derive the best linear unbiased estimator of $\boldsymbol{\beta}$ in the above model, for known values of $\boldsymbol{\Omega}_{\eta}$, $\boldsymbol{\Omega}_{\boldsymbol{\Lambda}}$ and σ_i^2.

2. Consider the random coefficient panel data model

$$\begin{aligned} y_{it} &= \alpha_i + \beta_i x_{it} + \varepsilon_{it}, \\ \beta_i &= \beta + \eta_i, \end{aligned} \tag{28.100}$$

where α_i are fixed group-specific effects, ε_{it} are independently distributed across i and t with mean zero and the variance $\sigma_i^2 < \infty$, $\eta_i \sim IID(0, \sigma_\eta^2)$, $\sigma_\eta^2 < \infty$, and x_{it}, $\varepsilon_{it'}$ and η_i are independently distributed for all t, t', and i.

   (a) Show that the standard *FE* estimator of β obtained using

$$y_{it} = \alpha_i + \beta x_{it} + v_{it},$$

   is consistent as N tends to infinity (even if T is fixed).

   (b) Under what conditions is the *FE* estimator of β under (a) also efficient?

   (c) Consider the *MG* estimator of β, $\hat{\beta}_{MG}$. Show that $\hat{\beta}_{MG}$ is a consistent estimator of β (for a finite T and a large enough N) irrespective of whether x_{it} is $I(0)$, or $I(1)$.

3. Consider the pure dynamic panel data model

$$y_{it} = \alpha_i + \lambda_i y_{i,t-1} + u_{it}, \; i = 1, 2, \ldots, N; \;\; t = 1, 2, \ldots, T, \tag{28.101}$$

where the error term u_{it} is assumed to be independently, identically distributed over t with mean zero and variance σ_i^2, is independent across i, and $\lambda_i = \lambda + \eta_i$, with $E\left(\eta_i\right) = 0$, $E\left(\eta_i \eta_j\right) = \Delta$, if $i = j$, and 0 otherwise. Find an expression for the bias of the MG estimator of λ when T is finite (see (28.63)).

4. Consider the random effects dynamic panel data model

$$y_{it} = \alpha_i + \lambda y_{i,t-1} + \beta x_{it} + \varepsilon_{it}, \text{ for } i = 1, 2, \ldots, N, t = 1, 2, \ldots, T, \tag{28.102}$$

where

$$\alpha_i = \alpha + \eta_i, \eta_i \backsim IID(0, \sigma_\eta^2),$$

$| \lambda | < 1$, ε_{it} are independently distributed across i and t with mean zero and the common variance σ^2, and x_{it} are strictly exogenous.

(a) Show that

$$\bar{y}_i = \frac{\alpha}{1-\lambda} + \left(\frac{\beta}{1-\lambda}\right)\bar{x}_i - \left(\frac{\lambda}{1-\lambda}\right)\left(\frac{y_{iT} - y_{i0}}{T}\right) + v_i,$$

where

$$v_i = \frac{\eta_i + \bar{\varepsilon}_i}{1-\lambda},$$

$\bar{y}_i = T^{-1}\sum_{t=1}^{T} y_{it}$, $\bar{x}_i = T^{-1}\sum_{t=1}^{T} x_{it}$ etc.

(b) Derive the conditions under which the cross-sectional regression of $\bar{y}_i$ on $\bar{x}_i$ will yield a consistent estimator of the long-run coefficient, $\theta = \beta/(1-\lambda)$. Is it possible also to obtain consistent estimates of the short-run coefficients, β and λ, from cross-sectional regressions?

(c) How robust are your results under (b) to possible dynamic misspecification of (28.102)?

5. Consider the simple error correction panel data model

$$\begin{aligned} \Delta y_{it} &= \alpha_i - \varphi_i\left(y_{i,t-1} - \theta_i x_{it}\right) + \varepsilon_{it}, \\ i &= 1, 2, \ldots, N;\ T = 1, 2, \ldots, T, \end{aligned} \tag{28.103}$$

where the error correction coefficients, φ_i, and the long-run coefficients, θ_i, as well as the intercepts, α_i, are allowed to vary across the groups, x_{it} is a scalar random variable assumed to be stationary and strictly exogenous, and $\varepsilon_{it} \backsim IID(0, \sigma_i^2)$. Assume also that φ_i and θ_i are generated according to the following random coefficient model

$$\begin{aligned} \varphi_i &= \varphi + \eta_{i1}, \\ \theta_i &= \theta + \eta_{i2}, \end{aligned}$$

where $\boldsymbol{\eta}_i = (\eta_{i1}, \eta_{i2})' \backsim IID(\mathbf{0}, \boldsymbol{\Omega})$, $\boldsymbol{\Omega}$ being a 2×2 nonsingular matrix and $\boldsymbol{\eta}_i$ are distributed independently of x_{jt}, for all i, j, and t.

(a) Show that

$$\Delta y_{it} = \alpha_i - \varphi\left(y_{i,t-1} - \theta x_{it}\right) + v_{it}, \tag{28.104}$$

where

$$v_{it} = \eta_{i2}(\eta_{i1} + \varphi)x_{it} - \eta_{i1}(y_{i,t-1} - \theta x_{it}) + \varepsilon_{it}.$$

Under what conditions (if any) will the fixed-effects estimation of (28.104) yield a consistent estimate of (φ, θ)?

(b) Assuming T and N are sufficiently large, how would you estimate φ and θ?

(c) Suppose $\eta_{i2} = 0$, and slope heterogeneity is confined to the error correction coefficients, φ_i. How would you now estimate θ?

29 Cross-Sectional Dependence in Panels

29.1 Introduction

This chapter reviews econometric methods for large linear panel data models subject to error cross-sectional dependence. Early panel data literature assumed cross-sectionally independent errors and homogeneous slopes. Heterogeneity across units was confined to unit-specific intercepts, treated as fixed or random (see, e.g., the survey by Chamberlain (1984)). Dependence of errors was only considered in spatial models, but not in standard panels. However, with an increasing availability of data (across countries, regions, or industries), the panel literature moved from predominantly micro panels, where the cross dimension (N) is large and the time series dimension (T) is small, to models with both N and T large, and it has been recognized that, even after conditioning on unit-specific regressors, individual units, in general, need not be cross-sectionally independent.

Ignoring cross-sectional dependence of errors can have serious consequences, and the presence of some form of cross-section correlation of errors in panel data applications in economics is likely to be the rule rather than the exception. Cross correlations of errors could be due to omitted common effects, spatial effects, or could arise as a result of interactions within socioeconomic networks. Conventional panel estimators such as fixed or random effects can result in misleading inference and even inconsistent estimators, depending on the extent of cross-sectional dependence and on whether the source generating the cross-sectional dependence (such as an unobserved common shock) is correlated with regressors (Phillips and Sul (2003), Andrews (2005), and Sarafidis and Robertson (2009)). Correlation across units in panels may also have serious drawbacks on commonly used panel unit root tests, since several of the existing tests assume independence. As a result, when applied to cross-sectionally dependent panels, such unit root tests can have substantial size distortions (O'Connell (1998)). This potential problem has recently given major impetus to the research on panel unit root tests that allow for cross unit correlations. These and other related developments are reviewed in Chapter 31. If, however, the extent of cross-sectional dependence of errors is sufficiently weak, or limited to a sufficiently small number of cross-sectional units, then its consequences might be unimportant. Consistency of conventional estimators can be affected only when the factors behind cross-correlations are themselves correlated with regressors. The problem of testing for the extent of cross-section

correlation of panel residuals and modelling the cross-sectional dependence of errors are therefore important issues.

In the case of panel data models where the cross-section dimension is short and the time series dimension is long, the standard approach to cross-sectional dependence is to consider the equations from different cross-sectional units as a system of seemingly unrelated regression equations (*SURE*), and then estimate it by the generalized least squares techniques (see Chapter 19 and Zellner (1962)). This approach assumes that the factors generating the cross-sectional dependence are not correlated with the regressors, an assumption which is required for the consistency of the *SURE* estimator. Also, if the time series dimension is not sufficiently large, and in particular if $N > T$, the *SURE* approach is not feasible either.

Currently, there are two main strands in the literature for dealing with error cross-sectional dependence in panels where N is large, namely the spatial econometric and the residual multifactor approaches. The spatial econometric approach assumes that the structure of cross-sectional correlation is related to location and distance among units, defined according to a pre-specified metric given by a 'connection or spatial' matrix that characterizes the pattern of spatial dependence according to pre-specified rules. Hence, cross-sectional correlation is represented by means of a spatial process, which explicitly relates each unit to its neighbours (see Whittle (1954), Moran (1948), Cliff and Ord (1973, 1981), Anselin (1988, 2001), Haining (2003, Chapter 7), and the recent survey by Lee and Yu (2013)). This approach, however, typically does not allow for slope heterogeneity across the units and requires *a priori* knowledge of the weight matrix. Spatial econometric literature is reviewed in Chapter 30.

The residual multifactor approach assumes that the cross dependence can be characterized by a small number of unobserved common factors, possibly due to economy-wide shocks that affect all units, albeit with different intensities (see Chapter 19 for an introduction to common factor models). Geweke (1977) and Sargent and Sims (1977) introduced dynamic factor models, which have more recently been generalized to allow for weak cross-sectional dependence by Forni and Lippi (2001), Forni et al. (2000, 2004). This approach does not require any prior knowledge regarding the ordering of individual cross-sectional units or a weight matrix used in the spatial econometric literature.

The main focus of this chapter is on estimation and inference in the case of large N and T panel data models with a common factor error structure. We provide a synthesis of the alternative approaches proposed in the literature (such as principal components and common correlated effects approaches), with particular focus on key assumptions and their consequences from the practitioner's view point. In particular, we discuss robustness of estimators to cross-sectional dependence of errors, the consequences of coefficient heterogeneity, panels with strictly or weakly exogenous regressors, including panels with a lagged dependent variable, and highlight how to test for residual cross-sectional dependence.

The outline of the chapter is as follows: an overview of the different types of cross-sectional dependence is provided in Section 29.2. The analysis of cross-sectional dependence using a factor error structure is presented in Section 29.3. A review of estimation and inference in the case of large panels with a multifactor error structure and strictly exogenous regressors is provided in Section 29.4, and its extension to models with lagged dependent variables and weakly exogenous regressors is given in Section 29.5. A review of tests of error cross-sectional dependence in static and dynamics panels is presented in Section 29.7, and Section 29.8 discusses the application of common correlated effects estimators and tests of error cross-sectional dependence to unbalanced panels.

29.2 Weak and strong cross-sectional dependence in large panels

A better understanding of the extent and nature of cross-sectional dependence is an important issue in the analysis of large panels. This section defines and discusses the notions of weak and strong cross-sectional dependence and considers the measurement of the degree of cross-sectional dependence using the exponent of cross-sectional dependence introduced in (Bailey, Kapetanios, and Pesaran 2015).

Consider the double index process $\{z_{it}, i \in \mathbb{N}, t \in \mathbb{Z}\}$, where z_{it} is defined on a suitable probability space, the index t refers to an ordered set such as time, and i refers to units of an unordered population, and suppose that:

Assumption CSD.1: For each $t \in \mathcal{T} \subseteq \mathbb{Z}$, $\mathbf{z}_t = (z_{1t}, z_{2t}, \ldots, z_{Nt})'$ has mean $E(\mathbf{z}_t) = 0$, and variance $Var(\mathbf{z}_t) = \mathbf{\Sigma}_t$, where $\mathbf{\Sigma}_t$ is an $N \times N$ symmetric, nonnegative definite matrix. The $(i,j)^{th}$ element of Σ_t, denoted by $\sigma_{ij,t}$, is bounded such that $0 < \sigma_{ii,t} \leq K$, for $i = 1, 2, \ldots, N$, where K is a finite constant independent of N.

Instead of assuming unconditional mean and variances, one could consider conditioning on a given information set, Ω_{t-1}, for $t = 1, 2, \ldots, T$, as done in Chudik, Pesaran, and Tosetti (2011). The assumption of zero means can also be relaxed to $E(\mathbf{z}_t) = \boldsymbol{\mu}$ [or $E(\mathbf{z}_t | \Omega_{t-1}) = \boldsymbol{\mu}_{t-1}$]. The covariance matrix, $\mathbf{\Sigma}_t$, fully characterizes cross-sectional correlations of the double index process $\{z_{it}\}$, and this section discusses summary measures based on the elements of $\mathbf{\Sigma}_t$ that can be used to characterize the extent of the cross-sectional dependence in $\mathbf{z}_t$.

Summary measures of cross-sectional dependence based on $\mathbf{\Sigma}_t$ can be constructed in a number of different ways. One possible measure, that has received a great deal of attention in the literature, is the largest eigenvalue of $\mathbf{\Sigma}_t$, denoted by $\lambda_1(\mathbf{\Sigma}_t)$ (see, e.g., Bai and Silverstein (1998), Hachem et al. (2005) and Yin et al. (1988).) However, the existing work in this area suggests that the estimates of $\lambda_1(\mathbf{\Sigma}_t)$ based on sample estimates of $\mathbf{\Sigma}_t$ could be very poor when N is large relative to T, and consequently using estimates of $\lambda_1(\mathbf{\Sigma}_t)$ for the analysis of cross-sectional dependence might be problematic, particularly in cases where T is not sufficiently large relative to N. Accordingly, other measures based on matrix norms of $\mathbf{\Sigma}_t$ have also been used in the literature. One prominent choice is the absolute column sum matrix norm, defined by $\|\mathbf{\Sigma}_t\|_1 = \max_{j\in\{1,2,\ldots,N\}} \sum_{i=1}^{N} |\sigma_{ij,t}|$, which is equal to the absolute row sum matrix norm of $\mathbf{\Sigma}_t$, defined by $\|\mathbf{\Sigma}_t\|_\infty = \max_{i\in\{1,2,\ldots,N\}} \sum_{j=1}^{N} |\sigma_{ij,t}|$, due to the symmetry of $\mathbf{\Sigma}_t$. It is easily seen that $|\lambda_1(\mathbf{\Sigma}_t)| \leq \sqrt{\|\mathbf{\Sigma}_t\|_1 \|\Sigma_t\|_\infty} = \|\mathbf{\Sigma}_t\|_1$. See Chudik, Pesaran, and Tosetti (2011). Another possible measure of cross-sectional dependence can be based on the behaviour of (weighted) cross-sectional averages which is often of interest in panel data econometrics, as well as in macroeconomics and finance where the object of the analysis is often the study of aggregates or portfolios of asset returns. In view of this, Bailey, Kapetanios, and Pesaran (2015) and Chudik, Pesaran, and Tosetti (2011) suggest summarizing the extent of cross-sectional dependence based on the behavior of cross-sectional averages, $\bar{z}_{wt} = \sum_{i=1}^{N} w_{it} z_{it} = \mathbf{w}_t' \mathbf{z}_t$, at a point in time t, for $t \in \mathcal{T}$, where $\mathbf{z}_t$ satisfies Assumption CSD.1 and the sequence of weight vectors $\mathbf{w}_t$ satisfies the following assumption.

Assumption CSD.2: Let $\mathbf{w}_t = (w_{1t}, w_{2t}, \ldots, w_{Nt})'$, for $t \in \mathcal{T} \subseteq \mathbb{Z}$ and $N \in \mathbb{N}$, be a vector of non-stochastic weights. For any $t \in \mathcal{T}$, the sequence of weight vectors $\{\mathbf{w}_t\}$ of growing

dimension ($N \to \infty$) satisfies the 'granularity' conditions

$$\|\mathbf{w}_t\| = \sqrt{\mathbf{w}_t'\mathbf{w}_t} = O\left(N^{-\frac{1}{2}}\right), \tag{29.1}$$

$$\frac{w_{jt}}{\|\mathbf{w}_t\|} = O\left(N^{-\frac{1}{2}}\right) \text{ uniformly in } j \in \mathbb{N}. \tag{29.2}$$

Assumption CSD.2, known in finance as the granularity condition, ensures that the weights $\{w_{it}\}$ are not dominated by a few of the cross-sectional units.[1] Although we have assumed the weights to be non-stochastic, this is done for expositional convenience and can be relaxed by allowing the weights, $\mathbf{w}_t$, to be random but distributed independently of $\mathbf{z}_t$. Chudik, Pesaran, and Tosetti (2011) define the concepts of weak and strong cross-sectional dependence based on the limiting behaviour of $\bar{z}_{wt}$ at a given point in time $t \in \mathcal{T}$, as $N \to \infty$.

Definition 29 (Weak and strong cross-sectional dependence) *The process $\{z_{it}\}$ is said to be cross-sectionally weakly dependent (CWD) at a given point in time $t \in \mathcal{T}$, if for* any *sequence of weight vectors $\{\mathbf{w}_t\}$ satisfying the granularity conditions (29.1)–(29.2) we have*

$$\lim_{N\to\infty} Var(\mathbf{w}_t'\mathbf{z}_t) = 0. \tag{29.3}$$

$\{z_{it}\}$ is said to be cross-sectionally strongly dependent (CSD) at a given point in time $t \in \mathcal{T}$, if there exists a sequence of weight vectors $\{\mathbf{w}_t\}$ satisfying (29.1)–(29.2) and a positive constant, K, independent of N such that for any N sufficiently large (and as $N \to \infty$)

$$Var(\mathbf{w}_t'\mathbf{z}_t) \geq K > 0. \tag{29.4}$$

The above concepts can also be defined conditional on a given information set, Ω_{t-1}. The choice of the conditioning set largely depends on the nature of the underlying processes and the purpose of the analysis. For example, in the case of dynamic stationary models, the information set could contain all lagged realizations of the process $\{z_{it}\}$, that is $\Omega_{t-1} = \{\mathbf{z}_{t-1}, \mathbf{z}_{t-2}, \ldots\}$, whilst for dynamic non-stationary models, such as unit root processes, the information included in Ω_{t-1}, could start from a finite past. The conditioning information set could also contain contemporaneous realizations, which might be useful in applications where a particular unit has a dominant influence on the rest of the units in the system. For further details, see Chudik and Pesaran (2013).

The following proposition establishes the relationship between weak cross-sectional dependence and the asymptotic behaviour of the largest eigenvalue of $\mathbf{\Sigma}_t$.

Proposition 46 *The following statements hold:*

(i) The process $\{z_{it}\}$ is CWD at a point in time $t \in \mathcal{T}$, if $\lambda_1(\mathbf{\Sigma}_t)$ is bounded in N or increases at the rate slower than N.

[1] Conditions (29.1)–(29.2) imply existence of a finite constant K (which does not depend on i or N) such that $|w_{it}| < KN^{-1}$ for any $i = 1, 2, \ldots, N$ and any $N \in \mathbb{N}$.

(ii) The process $\{z_{it}\}$ is CSD at a point in time $t \in \mathcal{T}$, if and only if for any N sufficiently large (and as $N \to \infty$), $N^{-1}\lambda_1(\mathbf{\Sigma}_t) \geq K > 0$.

Proof First, suppose $\lambda_1(\mathbf{\Sigma}_t)$ is bounded in N or increases at the rate slower than N. We have

$$Var(\mathbf{w}_t'\mathbf{z}_t) = \mathbf{w}_t'\Sigma_t\mathbf{w}_t \leq \left(\mathbf{w}_t'\mathbf{w}_t\right)\lambda_1(\mathbf{\Sigma}_t), \tag{29.5}$$

and under the granularity conditions (29.1)–(29.2) it follows that

$$\lim_{N\to\infty} Var(\mathbf{w}_t'\mathbf{z}_t) = 0,$$

namely that $\{z_{it}\}$ is CWD, which proves (i). Proof of (ii) is provided in Chudik, Pesaran, and Tosetti (2011).

It is often of interest to know not only whether $\bar{z}_{wt}$ converges to its mean, but also the rate at which this convergence (if at all) takes place. To this end, Bailey, Kapetanios, and Pesaran (2015) propose to characterize the degree of cross-sectional dependence by an exponent of cross-sectional dependence defined by the rate of change of $Var(\bar{z}_{wt})$ in terms of N. Note that in the case where z_{it} are independently distributed across i, we have $Var(\bar{z}_{wt}) = O\left(N^{-1}\right)$, whereas in the case of strong cross-sectional dependence $Var(\bar{z}_{wt}) \geq K > 0$. There is, however, a range of possibilities in between, where $Var(\bar{z}_{wt})$ decays but at a rate slower than N^{-1}. In particular, using a factor framework, Bailey, Kapetanios, and Pesaran (2015) show that in general

$$Var(\bar{z}_{wt}) = \kappa_0 N^{2(\alpha-1)} + \kappa_1 N^{-1} + O(N^{\alpha-2}), \tag{29.6}$$

where $\kappa_i > 0$ for $i = 0$ and 1, are bounded in N, and will be time invariant in the case of stationary processes. Since the rate at which $Var(\bar{z}_{wt})$ tends to zero with N cannot be faster than N^{-1}, the range of α identified by $Var(\bar{z}_{wt})$ lies in the restricted interval $-1 < 2\alpha - 2 \leq 0$ or $1/2 < \alpha \leq 1$. Note that (29.3) holds for all values of $\alpha < 1$, whereas (29.4) holds only for $\alpha = 1$. Hence the process with $\alpha < 1$ is CWD, and a CSD process has the exponent $\alpha = 1$. Bailey, Kapetanios, and Pesaran (2015) show that, under certain conditions on the underlying factor model, α is identified in the range $1/2 < \alpha \leq 1$, and can be consistently estimated. Alternative bias-adjusted estimators of α are proposed and shown by Monte Carlo experiments to have satisfactory small sample properties.

A particular form of a CWD process arises when pair-wise correlations take non-zero values only across finite subsets of units that do not spread widely as the sample size increases. As we shall see in Chapter 30, a similar situation arises in the case of spatial processes, where direct dependence exists only amongst adjacent observations, and indirect dependence is assumed to decay with distance.

Since $\lambda_1(\mathbf{\Sigma}_t) \leq \|\mathbf{\Sigma}_t\|_1$, it follows from (29.5) that both the spectral radius and the column norm of the covariance matrix of a CSD process will be increasing at the rate N. Similar situations also arise in the case of time series processes with long memory or strong temporal dependence where autocorrelation coefficients are not absolutely summable. Along the cross-sectional dimension, common factor models represent examples of strong cross-sectional dependence.

29.3 Common factor models

Consider the m factor model for $\{z_{it}\}$

$$z_{it} = \gamma_{i1} f_{1t} + \gamma_{i2} f_{2t} + \ldots + \gamma_{im} f_{mt} + e_{it}, \quad i = 1, 2, \ldots, N, \tag{29.7}$$

which can be written more compactly as

$$\mathbf{z}_t = \mathbf{\Gamma} \mathbf{f}_t + \mathbf{e}_t, \tag{29.8}$$

where $\mathbf{f}_t = (f_{1t}, f_{2t}, \ldots, f_{mt})'$, $\mathbf{e}_t = (e_{1t}, e_{2t}, \ldots, e_{Nt})'$, and $\mathbf{\Gamma} = (\gamma_{ij})$, for $i = 1, 2, \ldots, N$, $j = 1, 2, \ldots, m$, is an $N \times m$ matrix of fixed coefficients, known as factor loadings (see also Section 19.5). The common factors, $\mathbf{f}_t$, simultaneously affect all cross-sectional units, albeit with different degrees as measured by $\boldsymbol{\gamma}_i = \left(\gamma_{i1}, \gamma_{i2}, \ldots, \gamma_{im}\right)'$. Examples of observed common factors that tend to affect all households' and firms' consumption and investment decisions include interest rates and oil prices. Aggregate demand and supply shocks represent examples of common unobserved factors. In multifactor models, interdependence arises from reaction of units to some external events. Further, according to this representation, correlation between any pair of units does not depend on how far these observations are apart, and violates the distance decay effect that underlies the spatial interaction model.

Assumption CF.1: The $m \times 1$ vector $\mathbf{f}_t$ is a zero mean covariance stationary process, with absolutely summable autocovariances, distributed independently of $e_{it'}$ for all i, t, t', such that $E(f_{\ell t}^2) = 1$ and $E(f_{\ell t} f_{\ell' t}) = 0$, for $\ell \neq \ell' = 1, 2, \ldots, m$.

Assumption CF.2: $Var\,(e_{it}) = \sigma_i^2 < K < \infty$, e_{it} and e_{jt} are independently distributed for all $i \neq j$ and for all t. Specifically, $\max_i \left(\sigma_i^2\right) = \sigma_{\max}^2 < K < \infty$.

Assumption CF.1 is an identification condition, since it is not possible to separately identify $\mathbf{f}_t$ and $\mathbf{\Gamma}$. The above factor model with a fixed number of factors and cross-sectionally independent idiosyncratic errors is often referred to as an exact factor model. Under the above assumptions, the covariance of $\mathbf{z}_t$ is given by

$$E\left(\mathbf{z}_t \mathbf{z}_t'\right) = \mathbf{\Gamma}\mathbf{\Gamma}' + \mathbf{V},$$

where $\mathbf{V}$ is a diagonal matrix with elements σ_i^2 on the main diagonal.

The assumption that the idiosyncratic errors, e_{it}, are cross-sectionally independent is not necessary and can be relaxed. The factor model that allows the idiosyncratic shocks, e_{it}, to be cross-sectionally weakly correlated is known as the approximate factor model (see Chamberlain (1983)). In general, the correlation patterns of the idiosyncratic errors can be characterized by

$$\mathbf{e}_t = \mathbf{R}\boldsymbol{\varepsilon}_t, \tag{29.9}$$

where $\boldsymbol{\varepsilon}_t = (\varepsilon_{1t}, \varepsilon_{2t}, \ldots, \varepsilon_{Nt})' \sim (\mathbf{0}, \mathbf{I}_N)$. In the case of this formulation $\mathbf{V} = \mathbf{R}\mathbf{R}'$, which is no longer diagonal when $\mathbf{R}$ is not diagonal, and further identification restrictions are needed so that the factor specification can be distinguished from the cross-sectional dependence assumed for the idiosyncratic errors. To this end it is typically assumed that $\mathbf{R}$ has bounded row and column

sum matrix norms (so that the cross-sectional dependence of $\mathbf{e}_t$ is sufficiently weak) and the factor loadings are such that $\lim_{N\to\infty}(N^{-1}\mathbf{\Gamma}'\mathbf{\Gamma})$ is a full rank matrix.

A leading example of $\mathbf{R}$ arises in the context of the first-order spatial autoregressive, SAR(1), model, defined by

$$\mathbf{e}_t = \rho\mathbf{W}\mathbf{e}_t + \mathbf{\Lambda}\boldsymbol{\varepsilon}_t, \tag{29.10}$$

where $\mathbf{\Lambda}$ is a diagonal matrix with strictly positive and bounded elements, $0 < \sigma_i < \infty$, ρ is a spatial autoregressive coefficient, and the matrix $\mathbf{W}$ is a 'connection' or 'spatial' weight matrix which is taken as given.[2] Assuming that $(\mathbf{I}_N - \rho\mathbf{W})$ is invertible, we then have $\mathbf{R} = (\mathbf{I}_N - \rho\mathbf{W})^{-1}\mathbf{\Lambda}$. In the spatial literature, $\mathbf{W}$ is assumed to have non-negative elements and is typically row-standardized so that $\|\mathbf{W}\|_\infty = 1$. Under these assumptions, $|\rho| < 1$ ensures that $|\rho|\,\|\mathbf{W}\|_\infty < 1$, and we have

$$\begin{aligned}\|\mathbf{R}\|_\infty &\leq \|\mathbf{\Lambda}\|_\infty \left\|\mathbf{I}_N + \rho\mathbf{W} + \rho^2\mathbf{W}^2 + \ldots\right\|_\infty \\ &\leq \|\mathbf{\Lambda}\|_\infty \left(1 + |\rho|\,\|\mathbf{W}\|_\infty + |\rho|^2\,\|\mathbf{W}\|_\infty^2 + \ldots\right) = \frac{\|\mathbf{\Lambda}\|_\infty}{1 - |\rho|\,\|\mathbf{W}\|_\infty} < K < \infty,\end{aligned}$$

where $\|\mathbf{\Lambda}\|_\infty = \max_i(\sigma_i) < \infty$. Similarly, $\|\mathbf{R}\|_1 < K < \infty$, if it is further assumed that $|\rho|\,\|\mathbf{W}\|_1 < 1$. In general, $\mathbf{R} = (\mathbf{I}_N - \rho\mathbf{W})^{-1}\mathbf{\Lambda}$ has bounded row and column sum matrix norms if $|\rho| < \max(1/\|\mathbf{W}\|_1, 1/\|\mathbf{W}\|_\infty)$. In the case where $\mathbf{W}$ is a row and column stochastic matrix (often assumed in the spatial literature) this sufficient condition reduces to $|\rho| < 1$, which also ensures the invertibility of $(\mathbf{I}_N - \rho\mathbf{W})$. Note that for a doubly stochastic matrix $\rho(\mathbf{W}) = \|\mathbf{W}\|_1 = \|\mathbf{W}\|_\infty = 1$, where $\rho(\mathbf{W})$ is the spectral radius of $\mathbf{W}$. It turns out that almost all spatial models analysed in the spatial econometrics literature characterize weak forms of cross-sectional dependence. See Sarafidis and Wansbeek (2012) for further discussion.

Turning now to the factor representation, to ensure that the factor component of (29.8) represents strong cross-sectional dependence, it is sufficient that the absolute column sum matrix norm of $\|\mathbf{\Gamma}\|_1 = \max_{j\in\{1,2,\ldots,N\}}\sum_{i=1}^N \left|\gamma_{ij}\right|$ rises with N at the rate N, and $\lim_{N\to\infty}(N^{-1}\mathbf{\Gamma}'\Gamma)$ is a full rank matrix, as noted earlier.

The distinction between weak and strong cross-sectional dependence in terms of factor loadings is formalized in the following definition.

Definition 30 (Strong and weak factors) *The factor $f_{\ell t}$ is said to be strong if*

$$\lim_{N\to\infty} N^{-1}\sum_{i=1}^N \left|\gamma_{i\ell}\right| = K > 0. \tag{29.11}$$

The factor $f_{\ell t}$ is said to be weak if

$$\lim_{N\to\infty} \sum_{i=1}^N \left|\gamma_{i\ell}\right| = K < \infty. \tag{29.12}$$

[2] Spatial econometric models are discussed in Chapter 30. In particular, see Section 30.3.

It is also possible to consider intermediate cases of semi-weak or semi-strong factors. In general, let α_ℓ be a positive constant in the range $0 \leq \alpha_\ell \leq 1$ and consider the condition

$$\lim_{N\to\infty} N^{-\alpha_\ell} \sum_{i=1}^{N} \left|\gamma_{i\ell}\right| = K < \infty, \text{ for } K > 0. \tag{29.13}$$

Strong and weak factors correspond to the two values of $\alpha_\ell = 1$ and $\alpha_\ell = 0$, respectively. For any other values of $\alpha_\ell \in (0, 1)$ the factor $f_{\ell t}$ can be said to be semi-strong or semi-weak. It will prove useful to associate the semi-weak factors with values of $0 < \alpha_\ell < 1/2$, and the semi-strong factors with values of $1/2 \leq \alpha_\ell < 1$. In a multi-factor set up the overall exponent can be defined by $\alpha = \max(\alpha_1, \alpha_2, \ldots, \alpha_m)$.

Example 67 *Suppose that z_{it} are generated according to the simple factor model, $z_{it} = \gamma_i f_t + e_{it}$, where f_t is independently distributed of γ_i, and $e_{it} \sim IID(0, \sigma_i^2)$, for all i and t, σ_i^2 is non-stochastic for expositional simplicity and bounded, $E\left(f_t^2\right) = \sigma_f^2 < \infty$, $E\left(f_t\right) = 0$ and f_t is independently distributed of $e_{it'}$ for all i, t and t'. The factor loadings are given by*

$$\gamma_i = \mu + v_i, \text{ for } i = 1, 2, \ldots, [N^{\alpha_\gamma}], \tag{29.14}$$

$$\gamma_i = 0, \text{ for } i = [N^{\alpha_\gamma}] + 1, [N^{\alpha_\gamma}] + 2, \ldots, N, \tag{29.15}$$

for some constant $\alpha_\gamma \in [0, 1]$, where $[N^{\alpha_\gamma}]$ is the integer part of N^{α_γ}, $\mu \neq 0$, and v_i are IID with mean 0 and the finite variance, σ_v^2.[3] *Note that $\sum_{i=1}^{N} \left|\gamma_i\right| = O_p\left([N^{\alpha_\gamma}]\right)$ and the factor f_t with loadings γ_i is strong for $\alpha_\gamma = 1$, weak for $\alpha_\gamma = 0$ and semi-weak or semi-strong for $0 < \alpha_\gamma < 1$. Consider the variance of the (simple) cross-section average $\bar{z}_t = N^{-1} \sum_{i=1}^{N} z_{it}$*

$$Var_N\left(\bar{z}_t\right) = Var\left(\bar{z}_t \left| \left\{\gamma_i\right\}_{i=1}^{N}\right.\right) = \bar{\gamma}_N^2 \sigma_f^2 + N^{-1} \bar{\sigma}_N^2, \tag{29.16}$$

where (dropping the integer part sign, $[.]$, for further clarity)

$$\bar{\gamma}_N = N^{-1} \sum_{i=1}^{N} \gamma_i = N^{-1} \sum_{i=1}^{N^{\alpha_\gamma}} \gamma_i = \mu N^{\alpha_\gamma - 1} + N^{\alpha_\gamma - 1} \left(\frac{1}{N^{\alpha_\gamma}} \sum_{i=1}^{N^{\alpha_\gamma}} v_i \right),$$

$$\bar{\sigma}_N^2 = N^{-1} \sum_{i=1}^{N} \sigma_i^2 > 0.$$

But, noting that

$$E\left(\bar{\gamma}_N\right) = \mu N^{\alpha_\gamma - 1}, \; Var(\bar{\gamma}_N) = N^{\alpha_\gamma - 2} \sigma_v^2,$$

we have

[3] *The assumption of zero loadings for $i > \left[N^{\alpha_\gamma}\right]$ could be relaxed so long as $\sum_{i=[N^{\alpha_\gamma}]+1}^{N} \left|\gamma_i\right| = O_p(1)$. But for expositional simplicity we maintain $\gamma_i = 0$, for $i = \left[N^{\alpha_\gamma}\right] + 1, \left[N^{\alpha_\gamma}\right] + 2, \ldots, N$.*

$$E\left(\bar{\gamma}_N^2\right) = \left[E\left(\bar{\gamma}_N\right)\right]^2 + Var(\bar{\gamma}_N) = \mu^2 N^{2(\alpha_\gamma - 1)} + N^{\alpha_\gamma - 2}\sigma_v^2.$$

Therefore, using this result in (29.16), we have

$$Var\left(\bar{z}_t\right) = E\left[Var_N\left(\bar{z}_t\right)\right] = \sigma_f^2 \mu^2 N^{2(\alpha_\gamma - 1)} + \bar{\sigma}_N^2 N^{-1} + \sigma_v^2 \sigma_f^2 N^{\alpha_\gamma - 2} \tag{29.17}$$

$$= \sigma_f^2 \mu^2 N^{2(\alpha_\gamma - 1)} + \bar{\sigma}_N^2 N^{-1} + O\left(N^{\alpha_\gamma - 2}\right). \tag{29.18}$$

Thus the exponent of cross-sectional dependence of z_{it}, denoted as α_z, and the exponent α_γ coincide in this example, so long as $\alpha_\gamma > 1/2$. When $\alpha_\gamma = 1/2$, one cannot use Var $(\bar{z}_t)$ to distinguish the factor effects from those of the idiosyncratic terms. Of course, this does not necessarily mean that other more powerful techniques cannot be found to distinguish such weak factor effects from the effects of the idiosyncratic terms. Finally, note also that in this example $\sum_{i=1}^{N} \gamma_i^2 = O_p\left(N^{\alpha_\gamma}\right)$, and the largest eigenvalue of the $N \times N$ covariance matrix, Var $(\mathbf{z}_t)$, also rises at the rate of N^{α_γ}.

The relationship between the notions of CSD and CWD and the definitions of weak and strong factors are explored in the following theorem.

Theorem 47 *Consider the factor model (29.8) and suppose that Assumptions CF.1-CF.2 hold, and there exists a positive constant $\alpha = \max(\alpha_1, \alpha_2, \ldots, \alpha_m)$ in the range $0 \leq \alpha \leq 1$, such that condition (29.13) is met for any $\ell = 1, 2, \ldots, m$. Then the following statements hold:*

(i) The process $\{z_{it}\}$ is cross-sectionally weakly dependent at a given point in time $t \in \mathcal{T}$, if $\alpha < 1$, which includes cases of weak, semi-weak or semi-strong factors, $f_{\ell t}$, for $\ell = 1, 2, \ldots, m$.

(ii) The process $\{z_{it}\}$ is cross-sectionally strongly dependent at a given point in time $t \in \mathcal{T}$, if and only if there exists at least one strong factor.

Proof is provided in Chudik, Pesaran, and Tosetti (2011).

Since a factor structure can lead to strong as well as weak forms of cross-sectional dependence, cross-sectional dependence can also be characterized more generally by the following N-factor representation

$$z_{it} = \sum_{j=1}^{N} \gamma_{ij} f_{jt} + \varepsilon_{it}, \text{ for } i = 1, 2, \ldots, N,$$

where ε_{it} is independently distributed across i. Under this formulation, to ensure that the variance of z_{it} is bounded in N, we also require that

$$\sum_{\ell=1}^{N} \left|\gamma_{i\ell}\right| \leq K < \infty, \text{for } i = 1, 2, \ldots, N. \tag{29.19}$$

z_{it} can now be decomposed as

$$z_{it} = z_{it}^s + z_{it}^w, \tag{29.20}$$

where

$$z_{it}^{s} = \sum_{\ell=1}^{m} \gamma_{i\ell} f_{\ell t}; \; z_{it}^{w} = \sum_{\ell=m+1}^{N} \gamma_{i\ell} f_{\ell t} + \varepsilon_{it}, \tag{29.21}$$

and $\gamma_{i\ell}$ satisfy conditions (29.11) for $\ell = 1, \ldots, m$, where m must be finite in view of the absolute summability condition (29.19) that ensures finite variances. Remaining loadings $\gamma_{i\ell}$ for $\ell = m+1, m+2, \ldots, N$ must satisfy either (29.12) or (29.13) for some $\alpha < 1$.[4] In the light of Theorem 47, it can be shown that z_{it}^{s} is CSD and z_{it}^{w} is CWD. Also, notice that when z_{it} is CWD, we have a model with no strong factors and potentially an infinite number of weak or semi-strong factors. Seen from this perspective, spatial models considered in the literature can be viewed as an N weak factor model.

Consistent estimation of factor models with weak or semi-strong factors may be problematic, as evident from the following example.

Example 68 *Consider the single factor model with known factor loadings*

$$z_{it} = \gamma_i f_t + \varepsilon_{it}, \quad \varepsilon_{it} \sim IID\left(0, \sigma^2\right).$$

The least squares estimator of f_t, which is the best linear unbiased estimator, is given by

$$\hat{f}_t = \frac{\sum_{i=1}^{N} \gamma_i z_{it}}{\sum_{i=1}^{N} \gamma_i^2}, \; Var\left(\hat{f}_t\right) = \frac{\sigma^2}{\sum_{i=1}^{N} \gamma_i^2}.$$

In the weak factor case where $\sum_{i=1}^{N} \gamma_i^2$ is bounded in N, then $Var\left(\hat{f}_t\right)$ does not vanish as $N \to \infty$, and $\hat{f}_t$ need not be a consistent estimator of f_t. See also Onatski (2012).

The presence of weak or semi-strong factors in errors does not affect consistency of conventional panel data estimators, but affects inference, as is evident from the following example.

Example 69 *Consider the following panel data model*

$$y_{it} = \beta x_{it} + u_{it}, u_{it} = \gamma_i f_t + \varepsilon_{it},$$

where

$$x_{it} = \delta_i f_t + v_{it}.$$

To simplify the exposition we assume that, ε_{it}, v_{js} and $f_{t'}$ are independently, and identically distributed across all i, j, t,s and t', as $\varepsilon_{it} \sim IID(0, \sigma_{\varepsilon}^2)$, $v_{it} \sim IID(0, \sigma_v^2)$, and $f_t \sim IID(0, 1)$. The pooled estimator of β satisfies

[4] Note that the number of factors with $\alpha_\ell > 0$ is limited by the absolute summability condition (29.19).

$$\sqrt{NT}\left(\hat{\beta}_P - \beta\right) = \frac{\frac{1}{\sqrt{NT}}\sum_{i=1}^{N}\sum_{t=1}^{T} x_{it}u_{it}}{\frac{1}{NT}\sum_{i=1}^{N}\sum_{t=1}^{T} x_{it}^2}, \tag{29.22}$$

where the denominator converges in probability to $\sigma_v^2 + \lim_{N\to\infty} N^{-1}\sum_{i=1}^{N}\delta_i^2 > 0$*, while the numerator can be expressed, after substituting for* x_{it} *and* u_{it}*, as*

$$\frac{1}{\sqrt{NT}}\sum_{i=1}^{N}\sum_{t=1}^{T} x_{it}u_{it} = \frac{1}{\sqrt{NT}}\sum_{i=1}^{N}\sum_{t=1}^{T}\gamma_i\delta_i f_t^2 + \frac{1}{\sqrt{NT}}\sum_{i=1}^{N}\sum_{t=1}^{T}\left(\delta_i f_t\varepsilon_{it} + \gamma_i v_{it}f_t + v_{it}\varepsilon_{it}\right). \tag{29.23}$$

Under the above assumptions it is now easily seen that the second term in the above expression is $O_p(1)$*, but the first term can be written as*

$$\begin{aligned}\frac{1}{\sqrt{NT}}\sum_{i=1}^{N}\sum_{t=1}^{T}\gamma_i\delta_i f_t^2 &= \frac{1}{\sqrt{N}}\sum_{i=1}^{N}\gamma_i\delta_i \cdot \frac{1}{\sqrt{T}}\sum_{t=1}^{T} f_t^2 \\ &= \frac{1}{\sqrt{N}}\sum_{i=1}^{N}\gamma_i\delta_i \cdot O_p\left(T^{1/2}\right).\end{aligned}$$

Suppose now that f_t *is a factor such that loadings* γ_i *and* δ_i *are given by (29.14)–(29.15) with the exponents* α_γ *and* α_δ $(0 \leq \alpha_\gamma, \alpha_\delta \leq 1)$*, respectively, and let* $\alpha = \min\left(\alpha_\gamma, \alpha_\delta\right)$*. It then follows that* $\sum_{i=1}^{N}\gamma_i\delta_i = O_p\left(N^{\alpha}\right)$*, and*

$$\frac{1}{\sqrt{NT}}\sum_{i=1}^{N}\sum_{t=1}^{T}\gamma_i\delta_i f_t^2 = O_p(N^{\alpha-1/2}T^{1/2}).$$

Therefore, even if $\alpha < 1$ *the first term in (30.9) diverges, and overall we have* $\hat{\beta}_P - \beta = O_p(N^{\alpha-1}) + O_p(T^{-1/2}N^{-1/2})$*. It is now clear that even if* f_t *is not a strong factor, the rate of convergence of* $\hat{\beta}_P$ *and its asymptotic variance will still be affected by the factor structure of the error term. In the case where* $\alpha = 0$*, and the errors are spatially dependent, the variance matrix of the pooled estimator also depends on the nature of the spatial dependence which must be taken into account when carrying out inference on* β*. See Pesaran and Tosetti (2011) for further results and discussions. See also Section 30.7.*

Weak, strong and semi-strong common factors may be used to represent very general forms of cross-sectional dependence. For example, as we will see in Chapter 30, a factor process with an infinite number of weak factors, and no idiosyncratic errors can be used to represent spatial processes. In particular, the spatial model (29.9) can be represented by $e_{it} = \sum_{j=1}^{N}\gamma_{ij}f_{jt}$, where $\gamma_{ij} = r_{ij}$ and $f_{jt} = \varepsilon_{jt}$. Strong factors can be used to represent the effect of the cross-sectional units that are "dominant" or pervasive, in the sense that they impact all the other units in the sample and their effect does not vanish as N tends to infinity (Chudik and Pesaran (2013)). As outlined in Example 70 below, a large city may play a dominant role in determining house prices nationally (Holly, Pesaran, and Yamagata (2011)). Semi-strong factors may exist if there

is a cross-sectional unit or an unobserved common factor that affects only a subset of the units and the number of affected units rise more slowly than the total number of units. Estimates of the exponent of cross-sectional dependence reported by Bailey, Kapetanios, and Pesaran (2015) suggest that for typical large macroeconomic data sets the estimates of α fall in the range of 0.77–0.92, which fall short of 1 assumed in the factor literature. For cross-country quarterly real GDP growth, inflation and real equity prices the estimates of α are much closer to unity and tend to be around 0.97.

Example 70 (The diffusion of UK house prices) *Holly, Pesaran, and Yamagata (2011) study the diffusion of (log) of quarterly house prices, p_{it}, over time and across London and 11 UK regions in the years from 1973q4 to 2008q2. The authors assume that one of the regions, the London area, to be denoted as region 0, is dominant in the sense that shocks to it propagate to other regions simultaneously and over time. Conversely, shocks to the remaining regions are assumed to have little immediate impact on region 0—although there may be some lagged effects of shocks from the other regions onto region 0. Hence, the following first-order linear error correction specification for region 0 is specified by*

$$\Delta p_{0t} = \phi_{0s}\left(p_{0,t-1} - \bar{p}^{s}_{0,t-1}\right) + a_0 + a_{01}\Delta p_{0,t-1} + b_{01}\Delta \bar{p}^{s}_{0,t-1} + \varepsilon_{0t}. \tag{29.24}$$

While for the remaining regions the price equation is specified to be

$$\begin{aligned}\Delta p_{it} = {} & a_i + \phi_{is}\left(p_{i,t-1} - \bar{p}^{s}_{i,t-1}\right) + \phi_{i0}\left(p_{i,t-1} - p_{0,t-1}\right) \\ & + a_{i1}\Delta p_{i,t-1} + b_{i1}\Delta \bar{p}^{s}_{i,t-1} + c_{i0}\Delta p_{0,t-1} + \varepsilon_{it}.\end{aligned} \tag{29.25}$$

In the above equations, $\bar{p}^{s}_{0t}$ and $\bar{p}^{s}_{it}$ are the spatial lags of prices defined as

$$\bar{p}^{s}_{0t} = \sum_{j=1}^{N} s_{0j}\bar{p}^{s}_{j,t-1}, \quad \bar{p}^{s}_{it} = \sum_{j=0, j\neq i}^{N} s_{ij}\bar{p}^{s}_{j,t-1},$$

where $s_{ij} = 1/n_i$ if i and j share a border and zero otherwise, with n_i being the number of neighbours of region i. From the above specifications, London prices are assumed to be cointegrating with average prices in the neighbourhood of London. At the same time, prices in other regions are allowed to cointegrate with London as well as with their neighbouring regions. The assumption that p_{0t} is weakly exogenous in the equations for p_{it}, $i = 1, 2, \ldots, N$, can be tested using the procedure advanced by Wu (1973). OLS estimation of equations (29.24)–(29.25), and the Wu (1973) statistic are reported in Table 29.1. The error correction term measured relative to London is statistically significant in five regions (East Anglia, East Midlands, West Midlands, South West, and North), while the error correction term measured relative to neighboring regions is statistically significant only in the price equation for Scotland. The estimates of short-term dynamics show a considerable degree of heterogeneity in lag lengths and short-term dynamics. Surprisingly, the own lag effect (a_{i1}) is rather weak and generally statistically insignificant, except for the region 'North'. Finally, the contemporaneous effect of London house prices (c_{i0}) is sizeable and statistically significant in all regions. Figure 29.1 shows the effect of a unit shock to London house prices on London over time, compared to the impact effects of the same shock on regions ordered by their distance from London, for different horizons, $h = 0, 1, \ldots, 11$. This figure clearly shows the levelling off of the effect of shocks over time and across regions, indicating that the decay along the geographical dimension

Table 29.1 Error correction coefficients in cointegrating bivariate $VAR(4)$ of log of real house prices in London and other UK regions (1974q4-2008q2)

Regions	Lag-orders $\{\hat{k}_{ia}, \hat{k}_{ib}, \hat{k}_{ic}\}$ selected by *SBC*									
	EC1 $(\hat{\phi}_{i0})$	**EC2** $(\hat{\phi}_{is})$	**Own Lag Effects**	**Neighbour Lag Effects**	**London Lag Effects**	**London Contemporaneous Effects**	**Wu-Hausman Statistics**	$\hat{k}_{ia}$	$\hat{k}_{ib}$	$\hat{k}_{ic}$
London	–	–	0.036 (0.246)	0.666*** (4.314)	–	–		1	1	-
Outer Metropolitan	–	–	−0.071 (−0.708)	0.319*** (3.422)	–	0.655*** (13.987)	0.274	1	1	0
Outer South East	–	–	−0.158 (−1.349)	0.423*** (3.290)	–	0.746*** (15.017)	0.821	1	1	0
East Anglia	−0.045** (−2.002)	–	−0.033 (−0.320)	0.271** (2.158)	–	0.653*** (9.085)	−0.903	1	1	0
East Midlands	−0.057*** (−3.475)	–	−0.029 (−0.279)	0.808*** (5.184)	−0.501*** (−4.459)	0.523*** (8.525)	−0.694	1	2	2
West Midlands	−0.061*** (−3.770)	–	−0.203* (−1.933)	0.791*** (4.952)	−0.442*** (−3.524)	0.498*** (7.043)	0.032	1	1	2
South West	−0.113*** (−4.557)	–	−0.026 (−0.249)	0.371*** (3.095)	−0.326*** (−2.744)	0.670*** (10.813)	−1.240	1	1	2
Wales	–	–	−0.137 (−1.414)	1.319*** (7.777)	−0.757*** (−6.645)	0.661*** (9.455)	−0.895	1	3	3
Yorkshire & Humberside	–	–	0.180 (1.338)	0.561*** (3.834)	−0.333*** (−3.047)	0.577*** (7.252)	−1.874*	2	1	2
North West	–	–	0.061 (0.470)	0.918*** (6.399)	−0.452*** (−5.757)	0.423*** (7.751)	0.054	3	2	2
North	−0.039*** (−2.984)	–	−0.213** (−2.150)	0.750*** (5.074)	−0.235** (−2.248)	0.266*** (3.078)	0.610	1	1	1
Scotland	–	−0.098*** (−4.232)	0.019 (0.202)	0.050 (0.640)	–	0.326*** (5.266)	−1.174	1	1	0

Notes: This table reports estimates based on the price equations $\Delta p_{it} = \phi_{is}(p_{i,t-1} - \bar{p}^{s}_{i,t-1}) + \phi_{i0}(p_{i,t-1} - p_{0,t-1}) + \Sigma_{\ell=1}^{k_{ia}} a_{i\ell}\Delta p_{i,t-\ell} + \Sigma_{\ell=1}^{k_{ib}} b_{i\ell}\Delta \bar{p}^{s}_{i,t-\ell} + \Sigma_{\ell=1}^{k_{ic}} c_{i\ell}\Delta p_{0,t-\ell} + c_{i0}\Delta p_{0,t} + \varepsilon_{it}$, for $i = 1, 2, \ldots, N$. For $i = 0$, denoting the London equation, we have the additional *a priori* restrictions, $\phi_{00} = c_{00} = 0$. 'EC1', 'EC2', 'Own lag effects', 'Neighbour lag effects', 'London lag effects', and 'London contemporaneous effects' relate to the estimates of ϕ_{i0}, ϕ_{is}, $\Sigma_{\ell=1}^{k_{ia}} a_{i\ell}$, $\Sigma_{\ell=1}^{k_{ib}} b_{i\ell}$, $\Sigma_{\ell=1}^{k_{ic}} c_{i\ell}$, and c_{i0}, respectively. t-ratios are shown in parentheses. *** signifies that the test rejects the null at the 1% level , ** at the 5% level, and * at the 10% level. The error correction coefficients (ϕ_{is} and ϕ_{i0}) are restricted such that at least one of them is statistically significant at the 5% level. Wu-Hausman is the t-ratio for testing $H_0 : \lambda_i = 0$ in the augmented regression $\Delta p_{it} = \phi_{is}(p_{i,t-1} - \bar{p}^{s}_{i,t-1}) + \phi_{i0}(p_{i,t-1} - p_{0,t-1}) + \Sigma_{\ell=1}^{k_{ia}} a_{i\ell}\Delta p_{i,t-\ell} + \Sigma_{\ell=1}^{k_{ib}} b_{i\ell}\Delta \bar{p}^{s}_{i,t-\ell} + \Sigma_{\ell=0}^{k_{ic}} c_{i\ell}\Delta p_{0,t-\ell} + \lambda_i \hat{\varepsilon}_{0t} + \varepsilon_{it},$, where $\hat{\varepsilon}_{0t}$ is the residual of the London house price equation, and the error correction coefficients are restricted as described above. In selecting the lag-orders, k_{ia}, k_{ib}, and k_{ic} the maximum lag-order is set to 4. All regressions include an intercept term.

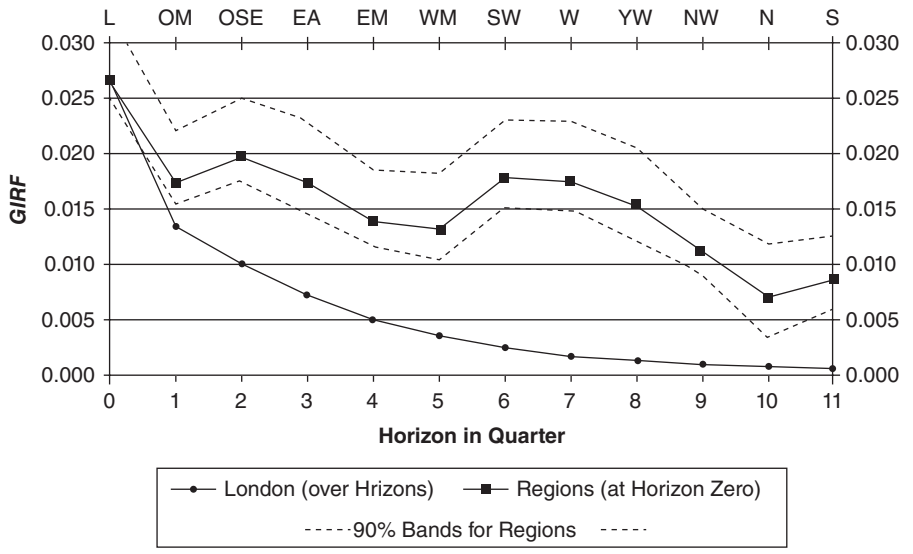


Figure 29.1 *GIRF*s of one unit shock (+ s.e.) to London on house price changes over time and across regions.

Notes: Broken lines are bootstrap 90% confidence band of the *GIRF*s for the regions, based on 10,000 bootstrap samples.

seems to be slower as compared with the decay along the time dimension. The effects of a shock to London on itself, die away and are largely dissipated after two years. By contrast, the effects of the same shock on other regions takes much longer to dissipate, the further the region is from London. This finding is in line with other empirical evidence on the rate of spatial as compared to temporal decay discussed in Whittle (1954). For further details see Holly, Pesaran, and Yamagata (2011).

29.4 Large heterogeneous panels with a multifactor error structure

Consider the following heterogeneous panel data model (see Chapter 28 for details on heterogeneous panels)

$$y_{it} = \boldsymbol{\alpha}_i' \mathbf{d}_t + \boldsymbol{\beta}_i' \mathbf{x}_{it} + u_{it}, \tag{29.26}$$

where $\mathbf{d}_t$ is an $n \times 1$ vector of observed common effects (including deterministics such as intercepts or seasonal dummies), $\mathbf{x}_{it}$ is a $k \times 1$ vector of observed individual-specific regressors on the ith cross-sectional unit at time t, and disturbances, u_{it}, have the following common factor structure

$$u_{it} = \gamma_{i1} f_{1t} + \gamma_{i2} f_{2t} + \ldots + \gamma_{im} f_{mt} + e_{it} = \boldsymbol{\gamma}_i' \mathbf{f}_t + e_{it}, \tag{29.27}$$

in which $\mathbf{f}_t = (f_{1t}, f_{2t}, \ldots, f_{mt})'$ is an m-dimensional vector of unobservable common factors, and $\boldsymbol{\gamma}_i = (\gamma_{i1}, \gamma_{i2}, \ldots, \gamma_{im})'$ is the associated $m \times 1$ vector of factor loadings. The number of factors, m, is assumed to be fixed relative to N, and in particular $m << N$. The idiosyncratic errors, e_{it}, could be CWD, for example, being generated by a spatial process, or, more generally, by a weak factor structure. For estimation purposes, as in the case of panels with group effects, the factor loadings, $\boldsymbol{\gamma}_i$, could be either random or fixed unknown coefficients. We distinguish between the homogeneous coefficient case where $\boldsymbol{\beta}_i = \boldsymbol{\beta}$ for all i, and the heterogeneous case where $\boldsymbol{\beta}_i$ are random draws from a given distribution. In the latter case, we assume that the object of interest is the mean coefficients, $\boldsymbol{\beta} = E(\boldsymbol{\beta}_i)$, for all i. When the regressors, $\mathbf{x}_{it}$, are strictly exogenous and the deviations $\boldsymbol{\upsilon}_i = \boldsymbol{\beta}_i - \boldsymbol{\beta}$ are distributed independently of the errors and the regressors, the mean coefficients, $\boldsymbol{\beta}$, can be consistently estimated using pooled as well as mean group estimation procedures. But only mean group estimation will be consistent if the regressors are weakly exogenous and/or if the deviations are correlated with the regressors/errors.[5]

The assumption of slope homogeneity is also crucially important for the derivation of the asymptotic distribution of the pooled or the mean group estimators of $\boldsymbol{\beta}$. Under slope homogeneity, the asymptotic distribution of the estimator of $\boldsymbol{\beta}$ typically converges at the rate of $\sqrt{NT}$, whilst under slope heterogeneity the rate is $\sqrt{N}$. In view of the uncertainty regarding the assumption of slope heterogeneity, non-parametric estimators of the variance matrix of the pooled and mean group estimators are proposed.[6] In the following sub-sections we review a number of different estimators of $\boldsymbol{\beta}$ proposed in the literature.

29.4.1 Principal components estimators

The principal components (PC) approach proposed by Coakley, Fuertes, and Smith (2002) and Bai (2009), by requiring that $N^{-1}\boldsymbol{\Gamma}'\boldsymbol{\Gamma}$ tends to a positive definite matrix, implicitly assumes that all the unobserved common factors in (29.27) are strong. Coakley, Fuertes, and Smith (2002) consider the panel data model with strictly exogenous regressors and homogeneous slopes (i.e., $\boldsymbol{\beta}_i = \boldsymbol{\beta}$), and propose a two-stage estimation procedure. In the first stage, PCs are extracted from the OLS residuals as proxies for the unobserved variables, and in the second step the estimated factors are treated as observable and the following augmented regression is estimated

$$y_{it} = \boldsymbol{\alpha}_i'\mathbf{d}_t + \boldsymbol{\beta}'\mathbf{x}_{it} + \boldsymbol{\gamma}_i'\hat{\mathbf{f}}_t + \varepsilon_{it}, \text{ for } i = 1, 2, \ldots, N;\ t = 1, 2, \ldots, T, \tag{29.28}$$

where $\hat{\mathbf{f}}_t$ is an $m \times 1$ vector of principal components of the residuals computed in the first stage. The resultant estimator of $\boldsymbol{\beta}$ is consistent for N and T large, so long as $\mathbf{f}_t$ and the regressors, $\mathbf{x}_{it}$, are uncorrelated. However, if the factors and the regressors are correlated, as is likely to be the case in practice, the two-stage estimator becomes inconsistent (Pesaran (2006)).

[5] Pooled estimation is carried out assuming that $\boldsymbol{\beta}_i = \boldsymbol{\beta}$ for all i, whilst mean group estimation allows for slope heterogeneity and estimates $\boldsymbol{\beta}$ by the average of the individual estimates of $\boldsymbol{\beta}_i$. See Chapter 28.

[6] Tests of slope homogeneity hypothesis in static and dynamic panels are discussed in Pesaran and Yamagata (2008) and in Section 28.11.

Building on Coakley, Fuertes, and Smith (2002), Bai (2009) has proposed an iterative method which consists of alternating the PC method applied to *OLS* residuals and the least squares estimation of (29.28), until convergence. In particular, to simplify the exposition suppose $\boldsymbol{\alpha}_i = \mathbf{0}$. Then the least squares estimator of $\boldsymbol{\beta}$ and $\mathbf{F}$ is the solution of the following set of nonlinear equations

$$\hat{\boldsymbol{\beta}}_{PC} = \left(\sum_{i=1}^{N} \mathbf{X}_i' \mathbf{M}_{\hat{F}} \mathbf{X}_i \right)^{-1} \sum_{i=1}^{N} \mathbf{X}_i' \mathbf{M}_{\hat{F}} \mathbf{y}_i,$$

$$\frac{1}{NT} \sum_{i=1}^{N} \left(\mathbf{y}_i - \mathbf{X}_i \hat{\boldsymbol{\beta}}_{PC} \right) \left(\mathbf{y}_i - \mathbf{X}_i \hat{\boldsymbol{\beta}}_{PC} \right)' \hat{\mathbf{F}} = \hat{\mathbf{F}} \hat{\mathbf{V}},$$

where $\mathbf{X}_i = \left(\mathbf{x}_{i1}, \mathbf{x}_{i2}, \ldots, \mathbf{x}_{iT} \right)'$ is the matrix of observations on $\mathbf{x}_{it}$, $\mathbf{y}_i = \left(y_{i1}, y_{i2}, \ldots, y_{iT} \right)'$ is the vector of observations on y_{it}, $\mathbf{M}_{\hat{F}} = \mathbf{I}_T - \hat{\mathbf{F}} \left(\hat{\mathbf{F}}' \hat{\mathbf{F}} \right)^{-1} \hat{\mathbf{F}}'$, $\hat{\mathbf{F}} = \left(\hat{\mathbf{f}}_1, \hat{\mathbf{f}}_2, \ldots, \hat{\mathbf{f}}_T \right)'$, and $\hat{\mathbf{V}}$ is a diagonal matrix with the m largest eigenvalues of the matrix $(NT)^{-1} \sum_{i=1}^{N} \left(\mathbf{y}_i - \mathbf{X}_i \hat{\boldsymbol{\beta}}_{PC} \right) \left(\mathbf{y}_i - \mathbf{X}_i \hat{\boldsymbol{\beta}}_{PC} \right)'$ arranged in a decreasing order. The solution $\hat{\boldsymbol{\beta}}_{PC}$, $\hat{\mathbf{F}}$ and $\hat{\boldsymbol{\gamma}}_i = \left(\hat{\mathbf{F}}' \hat{\mathbf{F}} \right)^{-1} \hat{\mathbf{F}}' \left(\mathbf{y}_i - \mathbf{X}_i \hat{\boldsymbol{\beta}}_{PC} \right)$ minimizes the sum of squared residuals function,

$$SSR_{NT} \left(\boldsymbol{\beta}, \left\{ \boldsymbol{\gamma}_i \right\}_{i=1}^{N}, \left\{ \mathbf{f}_t \right\}_{t=1}^{T} \right) = \sum_{i=1}^{N} \left(\mathbf{y}_i - \mathbf{X}_i \boldsymbol{\beta} - \mathbf{F} \boldsymbol{\gamma}_i \right)' \left(\mathbf{y}_i - \mathbf{X}_i \boldsymbol{\beta} - \mathbf{F} \boldsymbol{\gamma}_i \right),$$

where $\mathbf{F} = (\mathbf{f}_1, \mathbf{f}_2, \ldots, \mathbf{f}_T)'$. This function is a Gaussian quasi-maximum likelihood function of the model and, in this respect, Bai's iterative principal components estimator can also be seen as a quasi-maximum likelihood estimator, since it minimizes the quasi-likelihood function.

Bai (2009) shows that such an estimator is consistent even if common factors are correlated with the explanatory variables. Specifically, the least square estimator of $\boldsymbol{\beta}$ obtained from the above procedure, $\hat{\boldsymbol{\beta}}_{PC}$, is consistent if both N and T tend to infinity, without any restrictions on the ratio T/N. When in addition $T/N \to K > 0$, $\hat{\boldsymbol{\beta}}_{PC}$ converges at the rate $\sqrt{NT}$, but the limiting distribution of $\sqrt{NT} \left(\hat{\boldsymbol{\beta}}_{PC} - \boldsymbol{\beta} \right)$ does not necessarily have a zero mean. Nevertheless, Bai shows that the asymptotic bias can be consistently estimated and proposes a bias corrected estimator.

It is important to bear in mind that PC-based estimators generally require the determination of the unknown number of strong factors (PCs), m, to be included in the second stage of estimation, and this can introduce some degree of sampling uncertainty into the analysis. There is now an extensive literature that considers the estimation of m, assuming all the m factors to be strong. See, for example, Bai and Ng (2002, 2007), Kapetanios (2004, 2010), Amengual and Watson (2007), Hallin and Liska (2007), Onatski (2009, 2010), Ahn and Horenstein (2013), Breitung and Pigorsch (2013), Choi and Jeong (2013), and Harding (2013). There are also a number of useful surveys by Bai and Ng (2008), Stock and Watson (2011), and Breitung and Choi (2013), amongst others, that can be consulted for detailed discussions of these methods and additional references. An extensive Monte Carlo investigation into the small sample performance of different selection/estimation methods is provided in Choi and Jeong (2013). See also Section 19.5.2.

29.4.2 Common correlated effects estimator

Pesaran (2006) suggests the common correlated effects (*CCE*) estimation procedure that consists of approximating the linear combinations of the unobserved factors by cross-section averages of the dependent and explanatory variables, and then running standard panel regressions augmented with these cross-section averages. Both pooled and mean group versions are proposed, depending on the assumption regarding the slope homogeneity.

Under slope heterogeneity the *CCE* approach assumes that $\boldsymbol{\beta}_i's$ follow the random coefficient model

$$\boldsymbol{\beta}_i = \boldsymbol{\beta} + \boldsymbol{\upsilon}_i, \;\; \boldsymbol{\upsilon}_i \sim IID(\mathbf{0}, \boldsymbol{\Omega}_{\boldsymbol{\upsilon}}) \;\; \text{for } i = 1, 2, \ldots, N,$$

where the deviations, $\boldsymbol{\upsilon}_i$, are distributed independently of e_{jt}, $\mathbf{x}_{jt}$, and $\mathbf{d}_t$, for all i, j and t. To allow for possible dependence of the regressors and the factors, the following model for the individual-specific regressors in (29.26) is adopted

$$\mathbf{x}_{it} = \mathbf{A}_i'\mathbf{d}_t + \boldsymbol{\Gamma}_i'\mathbf{f}_t + \mathbf{v}_{it}, \tag{29.29}$$

where $\mathbf{A}_i$ and $\boldsymbol{\Gamma}_i$ are $n \times k$ and $m \times k$ factor loading matrices with fixed components, $\mathbf{v}_{it}$ is the idiosyncratic component of $\mathbf{x}_{it}$ distributed independently of the common effects $\mathbf{f}_{t'}$ and errors $e_{jt'}$ for all i, j, t and t'. However, $\mathbf{v}_{it}$ is allowed to be serially correlated, and cross sectionally weakly correlated.

Equations (29.26), (29.27) and (29.29) can be combined to yield the following system of equations

$$\mathbf{z}_{it} = \begin{pmatrix} y_{it} \\ \mathbf{x}_{it} \end{pmatrix} = \mathbf{B}_i'\mathbf{d}_t + \mathbf{C}_i'\mathbf{f}_t + \boldsymbol{\xi}_{it}, \tag{29.30}$$

where

$$\boldsymbol{\xi}_{it} = \begin{pmatrix} e_{it} + \boldsymbol{\beta}_i'\mathbf{v}_{it} \\ \mathbf{v}_{it} \end{pmatrix},$$

$$\mathbf{B}_i = \begin{pmatrix} \boldsymbol{\alpha}_i & \mathbf{A}_i \end{pmatrix}\begin{pmatrix} 1 & \mathbf{0} \\ \boldsymbol{\beta}_i & \mathbf{I}_k \end{pmatrix}, \mathbf{C}_i = \begin{pmatrix} \boldsymbol{\gamma}_i & \boldsymbol{\Gamma}_i \end{pmatrix}\begin{pmatrix} 1 & 0 \\ \boldsymbol{\beta}_i & \mathbf{I}_k \end{pmatrix}.$$

Consider the weighted average of $\mathbf{z}_{it}$ using the weights w_i satisfying the granularity conditions (29.1)–(29.2)

$$\bar{\mathbf{z}}_{wt} = \bar{\mathbf{B}}_w'\mathbf{d}_t + \bar{\mathbf{C}}_w'\mathbf{f}_t + \bar{\boldsymbol{\xi}}_{wt},$$

where

$$\bar{\mathbf{z}}_{wt} = \sum_{i=1}^{N} w_i\mathbf{z}_{it},$$

$$\bar{\mathbf{B}}_w = \sum_{i=1}^{N} w_i\mathbf{B}_i\,, \; \bar{\mathbf{C}}_w = \sum_{i=1}^{N} w_i\mathbf{C}_i, \;\; \text{and} \;\; \bar{\boldsymbol{\xi}}_{wt} = \sum_{i=1}^{N} w_i\boldsymbol{\xi}_{it}.$$

Assume that[7]

$$Rank(\bar{\mathbf{C}}_w) = m \leq k + 1, \tag{29.31}$$

we have

$$\mathbf{f}_t = (\bar{\mathbf{C}}_w \bar{\mathbf{C}}_w')^{-1} \bar{\mathbf{C}}_w \left(\bar{\mathbf{z}}_{wt} - \bar{\mathbf{B}}_w' \mathbf{d}_t - \bar{\boldsymbol{\xi}}_{wt} \right). \tag{29.32}$$

Under the assumption that e_{it}'s and $\mathbf{v}_{it}$'s are CWD processes, it is possible to show that (see Pesaran and Tosetti (2011))

$$\bar{\boldsymbol{\xi}}_{wt} \xrightarrow{q.m.} \mathbf{0}, \tag{29.33}$$

which implies

$$\mathbf{f}_t - (\bar{\mathbf{C}}_w \bar{\mathbf{C}}_w')^{-1} \bar{\mathbf{C}}_w \left(\bar{\mathbf{z}}_{wt} - \bar{\mathbf{B}}' \mathbf{d}_t \right) \xrightarrow{q.m.} 0, \text{ as } N \to \infty, \tag{29.34}$$

where

$$\mathbf{C} = \lim_{N \to \infty} (\bar{\mathbf{C}}_w) = \tilde{\boldsymbol{\Gamma}} \begin{pmatrix} 1 & \mathbf{0} \\ \boldsymbol{\beta} & \mathbf{I}_k \end{pmatrix}, \tag{29.35}$$

$\tilde{\boldsymbol{\Gamma}} = [E(\boldsymbol{\gamma}_i), E(\boldsymbol{\Gamma}_i)]$, and $\boldsymbol{\beta} = E(\boldsymbol{\beta}_i)$. Therefore, the unobservable common factors, $\mathbf{f}_t$, can be well approximated by a linear combination of observed effects, $\mathbf{d}_t$, the cross-section averages of the dependent variable, $\bar{y}_{wt}$, and those of the individual-specific regressors, $\bar{\mathbf{x}}_{wt}$.

When the parameters of interest are the cross-section means of the slope coefficients, $\boldsymbol{\beta}$, we can consider two alternative estimators, the CCE Mean Group (CCEMG) estimator, originally proposed by Pesaran and Smith (1995), and the CCE Pooled (CCEP) estimator. Let $\bar{\mathbf{M}}_w$ be defined by

$$\bar{\mathbf{M}}_w = \mathbf{I}_T - \bar{\mathbf{H}}_w (\bar{\mathbf{H}}_w' \bar{\mathbf{H}}_w)^{+} \bar{\mathbf{H}}_w', \tag{29.36}$$

where $\mathbf{A}^{+}$ denotes the Moore–Penrose inverse of matrix $\mathbf{A}$, $\bar{\mathbf{H}}_w = (\mathbf{D}, \bar{\mathbf{Z}}_w)$, and $\mathbf{D}$ and $\bar{\mathbf{Z}}_w$ are, respectively, the matrices of the observations on $\mathbf{d}_t$ and $\bar{\mathbf{z}}_{wt} = (\bar{y}_{wt}, \bar{\mathbf{x}}_{wt}')'$.

The CCEMG is a simple average of the estimators of the individual slope coefficients[8]

$$\hat{\boldsymbol{\beta}}_{CCEMG} = N^{-1} \sum_{i=1}^{N} \hat{\boldsymbol{\beta}}_{CCE,i}, \tag{29.37}$$

where

$$\hat{\boldsymbol{\beta}}_{CCE,i} = (\mathbf{X}_i' \bar{\mathbf{M}}_w \mathbf{X}_i)^{-1} \mathbf{X}_i' \bar{\mathbf{M}}_w \mathbf{y}_i. \tag{29.38}$$

[7] This assumption can be relaxed. See the discussions at the end of this Section and examples 71 and 72.

[8] Pesaran (2006) also considered a weighted average of individual $\hat{\mathbf{b}}_i$, with weights inversely proportional to the individual variances.

Under some general conditions Pesaran (2006) shows that $\hat{\boldsymbol{\beta}}_{CCEMG}$ is asymptotically unbiased for $\boldsymbol{\beta}$, and as $(N, T) \to \infty$,

$$\sqrt{N}\left(\hat{\boldsymbol{\beta}}_{CCEMG} - \boldsymbol{\beta}\right) \xrightarrow{d} N(\mathbf{0}, \boldsymbol{\Sigma}_{CCEMG}), \tag{29.39}$$

where $\boldsymbol{\Sigma}_{CCEMG} = \boldsymbol{\Omega}_v$. A consistent estimator of the variance of $\hat{\boldsymbol{\beta}}_{CCEMG}$, denoted by $Var\left(\hat{\boldsymbol{\beta}}_{CCEMG}\right)$, can be obtained by adopting the non-parametric estimator

$$\widehat{Var}\left(\hat{\boldsymbol{\beta}}_{CCEMG}\right) = N^{-1}\hat{\Sigma}_{CCEMG} = \frac{1}{N(N-1)}\sum_{i=1}^{N}\left(\hat{\boldsymbol{\beta}}_{CCE,i} - \hat{\boldsymbol{\beta}}_{CCEMG}\right)\left(\hat{\boldsymbol{\beta}}_{CCE,i} - \hat{\boldsymbol{\beta}}_{CCEMG}\right)'. \tag{29.40}$$

The CCEP estimator is given by

$$\hat{\boldsymbol{\beta}}_{CCEP} = \left(\sum_{i=1}^{N} w_i \mathbf{X}_i' \bar{\mathbf{M}}_w \mathbf{X}_i\right)^{-1} \sum_{i=1}^{N} w_i \mathbf{X}_i' \bar{\mathbf{M}}_w \mathbf{y}_i. \tag{29.41}$$

It is now easily seen that $\hat{\boldsymbol{\beta}}_{CCEP}$ is asymptotically unbiased for $\boldsymbol{\beta}$, and, as $(N, T) \to \infty$,

$$\left(\sum_{i=1}^{N} w_i^2\right)^{-1/2}\left(\hat{\boldsymbol{\beta}}_{CCEP} - \boldsymbol{\beta}\right) \xrightarrow{d} N(0, \Sigma_{CCEP}),$$

where

$$\begin{aligned}
\boldsymbol{\Sigma}_{CCEP} &= \Psi^{*-1}\mathbf{R}^*\Psi^{*-1}, \\
\Psi^* &= \lim_{N\to\infty}\left(\sum_{i=1}^{N} w_i \boldsymbol{\Sigma}_i\right), \quad \mathbf{R}^* = \lim_{N\to\infty}\left[N^{-1}\sum_{i=1}^{N}\tilde{w}_i^2(\boldsymbol{\Sigma}_i\boldsymbol{\Omega}_v\boldsymbol{\Sigma}_i)\right], \\
\boldsymbol{\Sigma}_i &= \underset{T\to\infty}{Plim}\left(T^{-1}\mathbf{X}_i'\bar{\mathbf{M}}_w\mathbf{X}_i\right), \text{ and } \tilde{w}_i = \frac{w_i}{\sqrt{N^{-1}\sum_{i=1}^{N} w_i^2}}.
\end{aligned}$$

A consistent estimator of $Var\left(\hat{\boldsymbol{\beta}}_{CCEP}\right)$, denoted by $\widehat{Var}\left(\hat{\boldsymbol{\beta}}_{CCEP}\right)$, is given by

$$\widehat{Var}\left(\hat{\boldsymbol{\beta}}_{CCEP}\right) = \left(\sum_{i=1}^{N} w_i^2\right)\hat{\Sigma}_{CCEP} = \left(\sum_{i=1}^{N} w_i^2\right)\hat{\Psi}^{*-1}\hat{\mathbf{R}}^*\hat{\Psi}^{*-1}, \tag{29.42}$$

where

$$\begin{aligned}
\hat{\Psi}^* &= \sum_{i=1}^{N} w_i\left(\frac{\mathbf{X}_i'\bar{\mathbf{M}}_w\mathbf{X}_i}{T}\right), \\
\hat{\mathbf{R}}^* &= \frac{1}{N-1}\sum_{i=1}^{N}\tilde{w}_i^2\left(\frac{\mathbf{X}_i'\bar{\mathbf{M}}_w\mathbf{X}_i}{T}\right)\left(\hat{\boldsymbol{\beta}}_{CCE,i} - \hat{\boldsymbol{\beta}}_{CCEMG}\right)\left(\hat{\boldsymbol{\beta}}_{CCE,i} - \hat{\boldsymbol{\beta}}_{CCEMG}\right)'\left(\frac{\mathbf{X}_i'\bar{\mathbf{M}}_w\mathbf{X}_i}{T}\right).
\end{aligned}$$

The rate of convergence of $\hat{\boldsymbol{\beta}}_{CCEMG}$ and $\hat{\boldsymbol{\beta}}_{CCEP}$ is $\sqrt{N}$ when $\boldsymbol{\Omega}_{\upsilon} \neq \mathbf{0}$. Note that even if $\boldsymbol{\beta}_i$ were observed for all i, the estimate of $\boldsymbol{\beta} = E(\boldsymbol{\beta}_i)$ cannot converge at a faster rate than $\sqrt{N}$. If the individual slope coefficients $\boldsymbol{\beta}_i$ are homogeneous (namely if $\boldsymbol{\Omega}_{\upsilon} = \mathbf{0}$), $\hat{\boldsymbol{\beta}}_{CCEMG}$ and $\hat{\boldsymbol{\beta}}_{CCEP}$ are still consistent and converge at the rate $\sqrt{NT}$ rather than $\sqrt{N}$.

The advantage of the non-parametric estimators $\hat{\Sigma}_{CCEMG}$ and $\hat{\Sigma}_{CCEP}$ is that they do not require knowledge of the form of weak cross-sectional dependence of e_{it}, nor the knowledge of serial correlation of e_{it}. An important question is whether the non-parametric variance estimators $\widehat{Var}(\hat{\boldsymbol{\beta}}_{CCEMG})$ and $\widehat{Var}(\hat{\boldsymbol{\beta}}_{CCEP})$ can be used in both cases of homogeneous and heterogeneous slopes. As established in Pesaran and Tosetti (2011), the asymptotic distribution of $\hat{\boldsymbol{\beta}}_{CCEMG}$ and $\hat{\boldsymbol{\beta}}_{CCEP}$ depends on nuisance parameters when slopes are homogeneous ($\boldsymbol{\Omega}_{\upsilon} = \mathbf{0}$), including the nature of cross-section correlations of e_{it} and their serial correlation structure. However, it can be shown that the robust non-parametric estimators $\widehat{Var}(\hat{\boldsymbol{\beta}}_{CCEMG})$ and $\widehat{Var}(\hat{\boldsymbol{\beta}}_{CCEP})$ are consistent when the regressor-specific components, $\mathbf{v}_{it}$, are independently distributed across i.

The *CCE* continues to be applicable even if the rank condition (29.31) is not satisfied. Failure of the rank condition can occur if there is an unobserved factor for which the average of the loadings in the y_{it} and $\mathbf{x}_{it}$ equations tends to a zero vector. This could happen if, for example, the factor in question is weak, in the sense defined above. Another possible reason for failure of the rank condition is if the number of unobservable factors, m, is larger than $k + 1$, where k is the number of the unit-specific regressors included in the model. In such cases, common factors cannot be estimated from cross-section averages. However, it is possible to show that the cross-section means of the slope coefficients, $\boldsymbol{\beta}_i$, can still be consistently estimated, under the additional assumption that the unobserved factor loadings, $\boldsymbol{\gamma}_i$, in equation (29.27) are independently and identically distributed across i, and of e_{jt}, $\mathbf{v}_{jt}$, and $\mathbf{g}_t = (\mathbf{d}_t', \mathbf{f}_t')'$ for all i, j and t, are uncorrelated with the loadings attached to the regressors, $\boldsymbol{\Gamma}_i$. The consequences of the correlation between loadings $\boldsymbol{\gamma}_i$ and $\boldsymbol{\Gamma}_i$ for the performance of CCE estimators in the rank deficient case are documented in Sarafidis and Wansbeek (2012). The following example illustrates the implications of such correlations.

Example 71 *Consider the simple panel data model*

$$y_{it} = \alpha_i + \beta_i x_{it} + \gamma_i f_t + \varepsilon_{it},\ i = 1, 2, \ldots, N; t = 1, 2, 3, \ldots, T,$$
$$x_{it} = \delta_i f_t + v_{it},\ \beta_i = \beta + \upsilon_i,\ \gamma_i = \gamma + \eta_i, \delta_i = \delta + \xi_i,$$

where υ_i, η_i and ξ_i are distributed with zero means and constant variances. Suppose also that ε_{it} and v_{it} are cross-sectionally and serially uncorrelated, and v_{it} is uncorrelated with f_t. But allow υ_i, η_i and ξ_i to be correlated with one another. Let $w_i = N^{-1}$ and note that for $i = 1, 2, \ldots, N$, we have

$$\hat{\beta}_{CCEP} - \beta = \left(N^{-1} \sum_{i=1}^{N} \frac{\mathbf{x}_i' \bar{\mathbf{M}} \mathbf{x}_i}{T} \right)^{-1} \left[\frac{1}{N} \sum_{i=1}^{N} \frac{\mathbf{x}_i' \bar{\mathbf{M}} (\mathbf{x}_i \upsilon_i + \boldsymbol{\varepsilon}_i)}{T} + q_{NT} \right], \tag{29.43}$$

$$q_{NT} = \frac{1}{N} \sum_{i=1}^{N} \frac{\left(\mathbf{x}_i' \bar{\mathbf{M}} \mathbf{f} \right) \gamma_i}{T}, \tag{29.44}$$

where $\bar{\mathbf{M}} = \mathbf{I}_T - \bar{\mathbf{H}}(\bar{\mathbf{H}}'\bar{\mathbf{H}})^{-1}\bar{\mathbf{H}}$, *with* $\bar{\mathbf{H}} = (\tau_T, \bar{\mathbf{x}}, \bar{\mathbf{y}})$, $\boldsymbol{\tau}_T = (1, 1, \ldots, 1)'$, $\bar{\mathbf{x}} = (\bar{x}_1, \bar{x}_2, \ldots, \bar{x}_T)'$, $\bar{x}_t = N^{-1}\sum_{i=1}^{N} x_{it}$, $\bar{\mathbf{y}} = \left(\bar{y}_1, \bar{y}_2, \ldots, \bar{y}_N\right)'$, *and* $\bar{y}_t = N^{-1}\sum_{i=1}^{N} y_{it}$. *Note that*

$$q_{NT} = N^{-1}\sum_{i=1}^{N}\left(\frac{\mathbf{x}_i'\bar{\mathbf{M}}\mathbf{f}}{T}\right)\gamma + N^{-1}\sum_{i=1}^{N}\left(\frac{\mathbf{x}_i'\bar{\mathbf{M}}\mathbf{f}}{T}\right)\eta_i. \tag{29.45}$$

But $N^{-1}\sum_{i=1}^{N}\left(\frac{\mathbf{x}_i'\bar{\mathbf{M}}\mathbf{f}}{T}\right)\gamma = \left(\frac{\bar{\mathbf{x}}'\bar{\mathbf{M}}\mathbf{f}}{T}\right)\gamma$, *and since* $\bar{\mathbf{x}}$ *is in* $\bar{\mathbf{H}}$, *then* $N^{-1}\sum_{i=1}^{N}\left(\frac{\mathbf{x}_i'\bar{\mathbf{M}}\mathbf{f}}{T}\right)\gamma = 0$ *irrespective of whether* $\gamma = 0$ *or not. Also, irrespective of whether* $\upsilon_i = 0$ *(the homogeneous slope case) or* $\upsilon_i \neq 0$ *(the heterogeneous slope case), it is easily seen that under standard assumptions* $\frac{1}{N}\sum_{i=1}^{N}\frac{\mathbf{x}_i'\bar{\mathbf{M}}(\mathbf{x}_i\upsilon_i+\boldsymbol{\varepsilon}_i)}{T}$ *tends to zero as N and T tends to infinity. Hence the bias of* $\hat{\beta}_{CCEP}$ *is governed by the second term of (29.45) which we write as (using the expression for* x_{it}*)*

$$\begin{aligned} d_{NT} &= N^{-1}\sum_{i=1}^{N}\left[\frac{(\delta_i\mathbf{f}+\boldsymbol{v}_i)'\bar{\mathbf{M}}\mathbf{f}}{T}\right]\eta_i \\ &= \left(N^{-1}\sum_{i=1}^{N}\delta_i\eta_i\right)\left(\frac{\mathbf{f}'\bar{\mathbf{M}}\mathbf{f}}{T}\right) + N^{-1}\sum_{i=1}^{N}\left(\frac{\boldsymbol{v}_i'\bar{\mathbf{M}}\mathbf{f}}{T}\right)\eta_i. \end{aligned}$$

The second term tends to zero as N and T tend to infinity since by assumption v_{it} *and* f_t *are independently distributed. It is clear that the first term also tends to zero if* γ_i *and* δ_i *are uncorrelated. So far none of these results requires* γ_i *and/or* δ_i *to have non-zero mean. Consider now the case where* $N^{-1}\sum_{i=1}^{N}\delta_i\eta_i \to \rho_{\gamma\delta} \neq 0$, *then the asymptotic bias of* $\hat{\beta}_{CCEP}$ *depends on the limiting property of* $T^{-1}\mathbf{f}'\bar{\mathbf{M}}\mathbf{f}$. *It is now clear that if either of* γ *or* δ *are non-zero then* $\bar{\mathbf{y}}$ *or* $\bar{\mathbf{x}}$ *can be written as a function of* $\mathbf{f}$ *even as* $N \to \infty$, *and* $T^{-1}\mathbf{f}'\bar{\mathbf{M}}\mathbf{f} \to 0$. *It is only in the case where* $\gamma = \delta = 0$ *that neither* $\bar{\mathbf{y}}$ *nor* $\bar{\mathbf{x}}$ *have any relationship to* $\mathbf{f}$ *in the limit and as a result* $T^{-1}\mathbf{f}'\bar{\mathbf{M}}\mathbf{f}$ *does not tend to zero. Further, in the case where* $\gamma = 0$ *but* $\delta \neq 0$, $\hat{\beta}_{CCEP}$ *will be consistent even if* $\rho_{\gamma\delta} \neq 0$. *In the case where* $\gamma = \delta = 0$, $\bar{y}_t \to E(\alpha_i)$ *and* $\bar{x}_t \to 0$. *Also* $Var(\bar{y}_t) \to 0$ *and* $Var(\bar{x}_t) \to 0$ *as* $N \to \infty$ *. In economic applications such non-stochastic limits do not seem plausible. For example, when factor loadings have zero means, then variances of per capita consumption, output, and investment will all tend to zero, which seems unlikely. Similarly, in the case of capital asset pricing models,* $r_{it} = \alpha_i + \beta_i f_t + \varepsilon_{it}$, *the assumption that* β_i *have a zero mean is equivalent to saying that there exist risk free portfolios that have positive excess returns—again a very unlikely scenario.*

An advantage of the *CCE* approach is that it yields consistent estimates under a variety of situations. Kapetanios, Pesaran, and Yamagata (2011) consider the case where the unobservable common factors follow unit root processes and could be cointegrated. They show that the asymptotic distribution of panel estimators in the case of *I(1)* factors is similar to that in the stationary case. Pesaran and Tosetti (2011) prove consistency and asymptotic normality for *CCE* estimators when $\{e_{it}\}$ are generated by a spatial process. Chudik, Pesaran, and Tosetti (2011) prove consistency and asymptotic normality of the *CCE* estimators when errors are subject to a finite number of unobserved strong factors and an infinite number of weak and/or semi-strong unobserved common factors as in (29.20)–(29.21), provided that certain conditions on the loadings

of the infinite factor structure are satisfied. A further advantage of the CCE approach is that it does not require an *a priori* knowledge of the number of unobserved common factors.

In a Monte Carlo (MC) study, Coakley, Fuertes, and Smith (2006) compare ten alternative estimators for the mean slope coefficient in a linear heterogeneous panel regression with strictly exogenous regressors and unobserved common (correlated) factors. Their results show that, overall, the mean group version of the *CCE* estimator stands out as the most efficient and robust. These conclusions are in line with those in Kapetanios and Pesaran (2007) and Chudik, Pesaran, and Tosetti (2011), who investigate the small sample properties of *CCE* estimators and the estimators based on principal components. The *MC* results show that *PC* augmented methods do not perform as well as the *CCE* approach, and can lead to substantial size distortions, due, in part, to the small sample errors in the number of factors selection procedure. In a theoretical study, Westerlund and Urbain (2011) investigate the merits of the *CCE* and *PC* estimators in the case of homogeneous slopes and known number of unobserved common factors and find that, although the PC estimates of factors are more efficient than the cross-sectional averages, the CCE estimators of slope coefficients generally perform the best.

Example 72 *There exists extensive literature on the relationship between long-run economic growth and investment in physical capital. Simple exogenous growth theories, such as the Solow model, predict a positive association between investment and the level of per capita GDP, but no relation between investment and steady-state growth rates (Barro and Sala-i-Martin (2003)). This conclusion has been supported by a number of empirical studies (see, e.g., the review of empirical literature in Easterly and Levine (2001)). Bond, Leblebicioglu, and Schiantarelli (2010) reconsider this problem, using data for seventy-five economies over the period 1960–2005, distinguishing between OECD and non-OECD countries. They adopt a model that allows for country-specific heterogeneity, endogeneity of investments and cross-sectional dependence. Let* y_{it} *denote the logarithm of GDP per worker in country* i *in year* t*, and* x_{it} *denote the logarithm of the investment to GDP ratio. The authors consider the ARDL*(p, p) *model*

$$y_{it} = g_{it} + \sum_{s=1}^{p} \alpha_s y_{i,t-s} + \sum_{s=1}^{p} \beta_s x_{i,t-s} + \eta_i + u_{it}, \tag{29.46}$$

where g_{it} *is a non-stationary process that determines the behavior of the growth rate of* y_{it} *in the long-run. The long-run growth rate is modelled as*

$$\Delta g_{it} = \theta_0 + \theta_1 x_{it} + d_i + \gamma_i f_t + v_{it}, \tag{29.47}$$

where d_i *is a country-specific effect,* f_t *and* v_{it} *are permanent shocks, common to all countries* (f_t)*, and country-specific* (v_{it})*. The main object of the analysis is* θ_1*. Under* $\theta_1 = 0$*, there is no long-run relationship between investment as a share of GDP and long-run growth rate, while under* $\theta_1 > 0$*, a permanent increase in investment predicts a higher long-run growth rate. Taking first differences of equation (29.46) and substituting for* Δg_{it} *from equation (29.47), yields*

$$\Delta y_{it} = \theta_0 + \theta_1 x_{it} + \sum_{s=1}^{p} \alpha_s \Delta y_{i,t-s} + \sum_{s=1}^{p} \beta_s \Delta x_{i,t-s} + d_i + \gamma_i f_t + v_{it} + \Delta u_{it}. \tag{29.48}$$

Table 29.2 Mean group estimates allowing for cross-sectional dependence

		(i)	*(ii)*	*(iii)*	*(iv)*	*(v)*
		Full sample	**OECD** (with full sample means)	**Non-OECD** (with full sample means)	**OECD)** (with OECD means)	**Non-OECD** (with Non-OECD means)
θ_1	Mean	0.0164	0.0078	0.0198	0.0040	0.0150
		(3.07)	(0.83)	(3.04)	(0.38)	(1.95)
	Median	0.0207	0.0095	0.0250	0.0085	0.0138
$\beta_0 + \beta_1$	Mean	0.0470	0.1511	0.0064	0.1196	0.0121
		(4.24)	(7.27)	(0.49)	(5.63)	(0.91)
	Median	0.0385	0.1334	–0.0054	0.1103	0.0042
$\alpha_1 + \alpha_2 - 1$	Mean	–0.7992	–0.6788	–0.8397	–0.6569	–0.7907
		(–25.29)	(–12.89)	(–21.82)	(–11.51)	(–20.69)
	Median	–0.8091	–0.7495	–0.8202	–0.6915	–0.7603
Growth effect	Mean	0.0288	0.0109	0.0360	0.0081	0.0237
		(2.25)	(0.57)	(2.12)	(0.49)	(1.82)
	Median	0.0295	0.0217	0.0408	0.0119	0.0189
Level effect	Mean	0.0734	0.2041	0.0209	0.1804	0.0342
		(4.10)	(4.85)	(1.15)	(5.22)	(1.60)
	Median	0.0664	0.2167	–0.0037	0.1692	0.0042

One interesting point to observe is that, from (29.47), g_{it} can be written as

$$g_{it} = g_{i0} + (\theta_0 + d_i)\, t + \theta_1 \sum_{s=1}^{t} x_{is} + \gamma_i \sum_{s=1}^{t} f_s + \sum_{s=1}^{t} v_{is}.$$

Hence, substituting g_{it} in (29.46) yields a model for the level of y_{it} in which the error term has an I(1) component $\sum_{s=1}^{t} v_{is}$ if these idiosyncratic permanent shocks to income levels are present, implying that the I(1) series y_{it} and $\sum_{s=1}^{t} x_{is}$ are not cointegrated among themselves. Equation (29.48) is then estimated separately for each country, by the IV method using as instruments lagged observations dated from $t-2$ to $t-6$ on y_{it} and x_{it}, and lagged observations dated $t-2$ and $t-3$ on a set of additional instruments (inflation, trade as a share of GDP, and government spending as a share of GDP). Following the CCE approach, Bond, Leblebicioglu, and Schiantarelli (2010) approximate the unobserved common factor, f_t, by including $\bar{y}_t$ and $\bar{x}_t$ in the regression specification. Estimation results for the mean and median estimated coefficients are reported in Table 29.2. Results show that investment as a share of GDP has a large and statistically significant effect on long-run growth rates, using the full sample of seventy-five countries and sub-sample of non-OECD countries. However, this evidence is weaker for OECD countries, for which the estimated coefficient θ_1 is not statistically significant. This result may reflect important differences across countries in the growth process.

29.5 Dynamic panel data models with a factor error structure

The problem of estimation of panels subject to cross-section error dependence becomes much more complicated once the assumption of strict exogeneity of the unit-specific regressors is

relaxed. One important example is the panel data model with lagged dependent variables and unobserved common factors (possibly correlated with the regressors)[9]

$$y_{it} = \lambda_i y_{i,t-1} + \boldsymbol{\beta}_i' \mathbf{x}_{it} + u_{it}, \tag{29.49}$$

$$u_{it} = \boldsymbol{\gamma}_i' \mathbf{f}_t + e_{it}, \tag{29.50}$$

for $i = 1, 2, \ldots, N$; $t = 1, 2, \ldots, T$. It is assumed that $|\lambda_i| < 1$, and the dynamic processes have started a long time in the past. As in Section 29.4, we distinguish between the case of homogeneous coefficients, where $\lambda_i = \lambda$ and $\boldsymbol{\beta}_i = \boldsymbol{\beta}$ for all i, and the heterogeneous case, where λ_i and $\boldsymbol{\beta}_i$ are randomly distributed across units and the object of interest are the mean coefficients $\lambda = E(\lambda_i)$ and $\boldsymbol{\beta} = E(\boldsymbol{\beta}_i)$. This distinction is more important for dynamic panels, since not only is the rate of convergence affected by the presence of coefficient heterogeneity, but, as shown by Pesaran and Smith (1995), pooled least squares estimators are no longer consistent in the case of dynamic panel data models with heterogeneous coefficients. See also Section 28.6.

It is convenient to define the vector of regressors $\boldsymbol{\zeta}_{it} = \left(y_{i,t-1}, \mathbf{x}_{it}'\right)'$ and the corresponding parameter vector $\boldsymbol{\pi}_i = \left(\lambda_i, \boldsymbol{\beta}_i'\right)'$ so that (29.49) can be written as

$$y_{it} = \boldsymbol{\pi}_i' \boldsymbol{\zeta}_{it} + u_{it}. \tag{29.51}$$

29.5.1 Quasi-maximum likelihood estimator

Moon and Weidner (2015) assume $\boldsymbol{\pi}_i = \boldsymbol{\pi}$ for all i and develop a Gaussian quasi-maximum likelihood estimator ($QMLE$) of the homogeneous coefficient vector $\boldsymbol{\pi}$. The $QMLE$ of $\boldsymbol{\pi}$ is

$$\hat{\boldsymbol{\pi}}_{QMLE} = \underset{\boldsymbol{\pi} \in \mathbb{B}}{\operatorname{argmin}} \left[L_{NT}(\boldsymbol{\pi}) \right],$$

where $\mathbb{B}$ is a compact set assumed to contain the true parameter values, and

$$L_{NT}(\boldsymbol{\pi}) = \min_{\{\boldsymbol{\gamma}_i\}_{i=1}^N, \{\mathbf{f}_t\}_{t=1}^T} \frac{1}{NT} \sum_{i=1}^{N} \left(\mathbf{y}_i - \boldsymbol{\Xi}_i \boldsymbol{\pi} - \mathbf{F} \boldsymbol{\gamma}_i\right)' \left(\mathbf{y}_i - \boldsymbol{\Xi}_i \boldsymbol{\pi} - \mathbf{F} \boldsymbol{\gamma}_i\right),$$

where $\mathbf{y}_i = (y_{i1}, y_{i2}, \ldots, y_{iT})$,

$$\boldsymbol{\Xi}_i = \begin{pmatrix} y_{i0} & \mathbf{x}_{i,1}' \\ y_{i,1} & \mathbf{x}_{i,2}' \\ \vdots & \vdots \\ y_{i,T-1} & \mathbf{x}_{iT}' \end{pmatrix}.$$

Both $\hat{\boldsymbol{\pi}}_{QMLE}$ and $\hat{\boldsymbol{\beta}}_{PC}$ minimize the same objective function and therefore, when the same set of regressors is considered, these two estimators are numerically the same. But there are important

[9] Fixed-effects and observed common factors (denoted by $\mathbf{d}_t$ previously) can also be included in the model. They are excluded here to simplify the exposition.

differences in their bias-corrected versions also considered in Bai (2009) and Moon and Weidner (2015). The latter paper allows for more general assumptions on regressors, including the possibility of weak exogeneity, and adopts a quadratic approximation of the profile likelihood function, which allows the authors to work out the asymptotic distribution and to conduct inference on the coefficients.

Moon and Weidner (MW) show that $\hat{\boldsymbol{\pi}}_{QMLE}$ is a consistent estimator of $\boldsymbol{\pi}$, as $(N,T) \to \infty$ without any restrictions on the ratio T/N. To derive the asymptotic distribution of $\hat{\boldsymbol{\pi}}_{QMLE}$, MW require $T/N \to \varkappa$, $0 < \varkappa < \infty$, as $(N,T) \to \infty$, and assume that the idiosyncratic errors, e_{it}, are cross-sectionally independent. Under certain high level assumptions, they show that $\sqrt{NT}\left(\hat{\boldsymbol{\pi}}_{QMLE} - \boldsymbol{\pi}\right)$ converges to a normal distribution with a non-zero mean, which is due to two types of asymptotic bias. The first follows from the heteroskedasticity of the error terms, as in Bai (2009), and the second is due to the presence of weakly exogenous regressors. The authors provide consistent estimators of these two components, and propose a bias-corrected *QMLE*.

There are two important considerations that should be borne in mind when using the *QMLE* proposed by MW. First, it is developed for the case of slope homogeneity, namely under $\boldsymbol{\pi}_i = \boldsymbol{\pi}$ for all i. This assumption, for example, rules out the inclusion of fixed-effects into the model, which can be quite restrictive in practice, although the unobserved factor component, $\boldsymbol{\gamma}_i'\mathbf{f}_t$, does in principle allow for fixed-effects if the first element of $\mathbf{f}_t$ can be constrained to be unity at the estimation stage. A second consideration is the small sample properties of *QMLE* in the case of models with fixed-effects, which are of primary interest in empirical applications. Simulations reported in Chudik and Pesaran (2015a) suggest that the bias correction does not go far enough and the *QMLE* procedure could yield tests which are grossly over-sized. To check the robustness of the *QMLE* to the presence of fixed-effects, we carried out a small Monte Carlo experiment in the case of a homogeneous AR(1) panel data model with fixed-effects, $\lambda_i = 0.70$, and $N = T = 100$. Using $R = 2,000$ replications, the bias of the bias-corrected *QMLE*, $\hat{\lambda}_{QMLE}$, turned out to be -0.024, and tests based on $\hat{\lambda}_{QMLE}$ were grossly oversized with size exceeding 60 per cent.

29.5.2 *PC* estimators for dynamic panels

Song (2013) extends Bai (2009)'s approach to dynamic panels with heterogeneous coefficients. The focus of Song's analysis is on the estimation of unit-specific coefficients $\boldsymbol{\pi}_i = \left(\lambda_i, \boldsymbol{\beta}_i'\right)'$. In particular, Song proposes an iterated least squares estimator of $\boldsymbol{\pi}_i$, and shows as in Bai (2009) that the solution can be obtained by alternating the *PC* method applied to the least squares residuals and the least squares estimation of (29.49) until convergence. In particular, the least squares estimators of $\boldsymbol{\pi}_i$ and $\mathbf{F}$ are the solution to the following set of nonlinear equations

$$\hat{\boldsymbol{\pi}}_{i,PC} = \left(\boldsymbol{\Xi}_i'\mathbf{M}_{\hat{F}}\boldsymbol{\Xi}_i\right)^{-1}\boldsymbol{\Xi}_i'\mathbf{M}_{\hat{F}}\mathbf{y}_i, \text{ for } i = 1,2,\ldots,N, \tag{29.52}$$

$$\frac{1}{NT}\sum_{i=1}^{N}\left(\mathbf{y}_i - \boldsymbol{\Xi}_i\hat{\boldsymbol{\pi}}_{i,PC}\right)\left(\mathbf{y}_i - \boldsymbol{\Xi}_i\hat{\boldsymbol{\pi}}_{i,PC}\right)'\hat{\mathbf{F}} = \hat{\mathbf{F}}\hat{\mathbf{V}}. \tag{29.53}$$

Song (2013) establishes consistency of $\hat{\boldsymbol{\pi}}_{i,PC}$ when $(N,T) \to \infty$ without any restrictions on T/N. If in addition $T/N^2 \to 0$, Song (2013) shows that $\hat{\boldsymbol{\pi}}_{i,PC}$ is $\sqrt{T}$ consistent, and derives the asymptotic distribution under some additional requirements including the cross-section

independence of e_{it}. Song (2013) does not provide theoretical results on the estimation of the mean coefficients $\boldsymbol{\pi} = E(\boldsymbol{\pi}_i)$, but he considers the following mean group estimator based on the individual estimates $\hat{\boldsymbol{\pi}}_{i,PC}$,

$$\hat{\boldsymbol{\pi}}^{s}_{PCMG} = \frac{1}{N}\sum_{i=1}^{N}\hat{\boldsymbol{\pi}}_{i,PC},$$

in a Monte Carlo study and finds that $\hat{\boldsymbol{\pi}}^{s}_{PCMG}$ has satisfactory small sample properties in terms of bias and root mean squared error. But he does not provide any results on the asymptotic distribution of $\hat{\boldsymbol{\pi}}^{s}_{PCMG}$. However, results of a Monte Carlo study presented in Chudik and Pesaran (2015a) suggest that $\sqrt{N}\left(\hat{\boldsymbol{\pi}}^{s}_{PCMG} - \boldsymbol{\pi}\right)$ is asymptotically normally distributed with mean zero and a covariance matrix that can be estimated by (as in the case of the CCEMG estimator),

$$\widehat{Var}\left(\hat{\boldsymbol{\pi}}^{s}_{PCMG}\right) = \frac{1}{N(N-1)}\sum_{i=1}^{N}\left(\hat{\boldsymbol{\pi}}^{s}_{i} - \hat{\boldsymbol{\pi}}^{s}_{MG}\right)\left(\hat{\boldsymbol{\pi}}^{s}_{i} - \hat{\boldsymbol{\pi}}^{s}_{MG}\right)'.$$

The test results based on this conjecture tend to perform well so long as T is sufficiently large. However, as with the other PC based estimators, knowledge of the number of factors and the assumption that the factors under consideration are strong continue to play an important role in the small sample properties of the tests based on $\hat{\boldsymbol{\pi}}^{s}_{MGPC}$.

29.5.3 Dynamic *CCE* estimators

The *CCE* approach as it was originally proposed in Pesaran (2006) does not cover the case where the panel includes a lagged dependent variable or weakly exogenous regressors.[10] Extension of the *CCE* approach to dynamic panels with heterogeneous coefficients and weakly exogenous regressors is proposed by Chudik and Pesaran (2015a). In what follows we refer to this extension as 'dynamic *CCE*'.

The inclusion of a lagged dependent variable amongst the regressors has three main consequences for the estimation of the mean coefficients. The first is the well-known time series bias, which affects the individual specific estimates and is of order $O\left(T^{-1}\right)$. The second consequence is that the full rank condition becomes necessary for consistent estimation of the mean coefficients unless the $\mathbf{f}_t$ is serially uncorrelated. The third complication arises from the interaction of dynamics and coefficient heterogeneity, which leads to infinite lag-order relationships between unobserved common factors and cross-section averages of the observables when N is large. This issue also arises in cross-sectional aggregation of heterogeneous dynamic models. See Section 15.8 and Chapter 32 and references cited therein.

To illustrate these complications, using (29.49) and recalling assumption $|\lambda_i| < 1$, for all i, then we have

$$y_{it} = \sum_{\ell=0}^{\infty}\lambda_i^{\ell}\boldsymbol{\beta}_i'\mathbf{x}_{i,t-\ell} + \sum_{\ell=0}^{\infty}\lambda_i^{\ell}\boldsymbol{\gamma}_i'\mathbf{f}_{t-\ell} + \sum_{\ell=0}^{\infty}\lambda_i^{\ell}e_{i,t-\ell}. \tag{29.54}$$

[10] See Everaert and Groote (2011) who derive the asymptotic bias of the *CCE* pooled estimator in the case of dynamic *homogeneous* panels.

Taking weighted cross-sectional averages, and assuming independence of λ_i, $\boldsymbol{\beta}_i$, and $\boldsymbol{\gamma}_i$, strict exogeneity of $\mathbf{x}_{it}$, and weak cross-sectional dependence of $\{e_{it}\}$, we obtain (see Section 32.6 and Pesaran and Chudik (2014)),

$$\bar{y}_{wt} = a(L)\,\boldsymbol{\gamma}'\mathbf{f}_t + a(L)\,\boldsymbol{\beta}'\bar{\mathbf{x}}_{wt} + \xi_{wt}, \tag{29.55}$$

where $a(L) = \sum_{\ell=0}^{\infty} a_\ell L^\ell$, with $a_\ell = E\left(\lambda_i^\ell\right)$, $\boldsymbol{\beta} = E(\boldsymbol{\beta}_i)$, and $\boldsymbol{\gamma} = E(\boldsymbol{\gamma}_i)$. Under the assumption that the idiosyncratic errors are cross-sectionally weakly dependent, we have $\xi_{wt} \xrightarrow{p} 0$, as $N \to \infty$, with the rate of convergence depending on the degree of cross-sectional dependence of $\{e_{it}\}$ and the granularity of $\mathbf{w}$. In the case where $\mathbf{w}$ satisfies the usual granularity conditions (29.1)–(29.2), and the exponent of cross-sectional dependence of e_{it} is $\alpha_e \leq 1/2$, we have $\xi_{wt} = O_p\left(N^{-1/2}\right)$. In the special case where $\boldsymbol{\beta} = 0$ and $m = 1$, (29.55) reduces to

$$\bar{y}_{wt} = \gamma a(L) f_t + O_p\left(N^{-1/2}\right).$$

The extent to which f_t can be accurately approximated by $\bar{y}_{wt}$ and its lagged values depends on the rate at which, $a_\ell = E\left(\lambda_i^\ell\right)$, the coefficients in the polynomial lag operator, $a(L)$, decay with ℓ, and the size of the cross-sectional dimension, N. The coefficients in $a(L)$ are given by the moments of λ_i and therefore these coefficients need not be absolute summable if the support of λ_i is not sufficiently restricted in the neighborhood of the unit circle (see Section 15.8 and Chapter 32). Assuming that for all i the support of λ_i lies strictly within the unit circle, it is easily seen that α_ℓ will then decay exponentially and for N sufficiently large, f_t can be well approximated by $\bar{y}_{wt}$ and a number of its lagged values.[11] The number of lagged values of $\bar{y}_{wt}$ needed to approximate f_t rises with T but at a slower rate.

In the general case where $\boldsymbol{\beta}$ is nonzero, $\mathbf{x}_{it}$ are weakly exogenous, and $m \geq 1$, Chudik and Pesaran (2015a) show that there exists the following large N distributed lag relationship between the unobserved common factors and cross-sectional averages of the dependent variable and the regressors, $\bar{\mathbf{z}}_{wt} = \left(\bar{y}_{wt}, \bar{\mathbf{x}}_{wt}'\right)'$,

$$\boldsymbol{\Lambda}(L)\,\tilde{\boldsymbol{\Gamma}}'\mathbf{f}_t = \bar{\mathbf{z}}_{wt} + O_p\left(N^{-1/2}\right),$$

where as before $\tilde{\boldsymbol{\Gamma}} = E\left(\boldsymbol{\gamma}_i, \boldsymbol{\Gamma}_i\right)$ and the decay rate of the matrix coefficients in $\boldsymbol{\Lambda}(L)$ depends on the heterogeneity of λ_i and $\boldsymbol{\beta}_i$ and other related distributional assumptions. The existence of a large N relationship between the unobserved common factors and cross-sectional averages of variables is not surprising considering that only the components with the largest exponents of cross-sectional dependence can survive cross-sectional aggregation with granular weights. Assuming $\tilde{\boldsymbol{\Gamma}}$ has full row rank, namely $rank\left(\tilde{\boldsymbol{\Gamma}}\right) = m$, and the distributions of coefficients are such that $\boldsymbol{\Lambda}^{-1}(L)$ exists and has exponentially decaying coefficients, yields the following unit-specific dynamic CCE regressions,

[11] For example, if λ_i is distributed uniformly over the range $(0, b)$ where $0 < b < 1$, we have $\alpha_\ell = E(\lambda_i^\ell) = b^\ell/(1+\ell)$, which decays exponentially with ℓ. See also Section 15.8.3.

$$y_{it} = \lambda_i y_{i,t-1} + \boldsymbol{\beta}_i' \mathbf{x}_{it} + \sum_{\ell=0}^{p_T} \boldsymbol{\delta}_{i\ell}' \bar{\mathbf{z}}_{w,t-\ell} + e_{yit}, \tag{29.56}$$

where $\bar{\mathbf{z}}_{wt}$ and its lagged values are used to approximate $\mathbf{f}_t$. The error term e_{yit} consists of three parts: an idiosyncratic term, e_{it}, an error component due to the truncation of possibly infinite distributed lag function, and an $O_p\left(N^{-1/2}\right)$ error component due to the approximation of unobserved common factors based on large N relationships.

Chudik and Pesaran (2015a) consider the least squares estimates of $\boldsymbol{\pi}_i = \left(\lambda_i, \boldsymbol{\beta}_i'\right)'$ based on the above dynamic *CCE* regressions, denoted as $\hat{\boldsymbol{\pi}}_i = \left(\hat{\lambda}_i, \hat{\boldsymbol{\beta}}_i'\right)'$, and the mean group estimate of $\boldsymbol{\pi} = E\left(\boldsymbol{\pi}_i\right)$ based on $\hat{\boldsymbol{\pi}}_i$. To define these estimators, we introduce the following data matrices

$$\tilde{\boldsymbol{\Xi}}_i = \begin{pmatrix} y_{ip_T} & \mathbf{x}_{i,p_T+1}' \\ y_{i,p_T+1} & \mathbf{x}_{i,p_T+2}' \\ \vdots & \vdots \\ y_{i,T-1} & \mathbf{x}_{iT}' \end{pmatrix}, \; \bar{\mathbf{Q}}_w = \begin{pmatrix} \bar{\mathbf{z}}_{w,p_T+1}' & \bar{\mathbf{z}}_{w,p_T}' & \cdots & \bar{\mathbf{z}}_{w,1}' \\ \bar{\mathbf{z}}_{w,p_T+2}' & \bar{\mathbf{z}}_{w,p_T+1}' & \cdots & \bar{\mathbf{z}}_{w,2}' \\ \vdots & \vdots & & \vdots \\ \bar{\mathbf{z}}_{w,T}' & \bar{\mathbf{z}}_{w,T-1}' & \cdots & \bar{\mathbf{z}}_{w,T-p_T}' \end{pmatrix}, \tag{29.57}$$

and the projection matrix $\bar{\mathbf{M}}_q = \mathbf{I}_{T-p_T} - \bar{\mathbf{Q}}_w \left(\bar{\mathbf{Q}}_w' \bar{\mathbf{Q}}_w\right)^+ \bar{\mathbf{Q}}_w'$, where $\mathbf{I}_{T-p_T}$ is a $\left(T-p_T\right) \times \left(T-p_T\right)$ dimensional identity matrix.[12] p_T should be set such that p_T^2/T tends to zero as p_T and T both tend to infinity. The number of lags cannot increase too fast, otherwise there will not be a sufficient number of observations to accurately estimate the parameters, whilst at the same time a sufficient number of lags is needed to ensure that the factors are well approximated. Setting the number of lags equal to $T^{1/3}$ seems to be a good choice, balancing the effects of the above two opposing considerations.[13]

The individual estimates, $\hat{\boldsymbol{\pi}}_i$, can now be written as

$$\hat{\boldsymbol{\pi}}_i = \left(\tilde{\boldsymbol{\Xi}}_i' \bar{\mathbf{M}}_q \tilde{\boldsymbol{\Xi}}_i\right)^{-1} \tilde{\boldsymbol{\Xi}}_i' \bar{\mathbf{M}}_q \tilde{\mathbf{y}}_i, \tag{29.58}$$

where $\tilde{\mathbf{y}}_i = \left(y_{i,p_T+1}, y_{i,p_T+2}, \ldots, y_{i,T}\right)'$. The mean group estimator of $\boldsymbol{\pi} = E\left(\boldsymbol{\pi}_i\right) = \left(\lambda, \boldsymbol{\beta}'\right)'$ is given by

$$\hat{\boldsymbol{\pi}}_{MG} = \frac{1}{N} \sum_{i=1}^{N} \hat{\boldsymbol{\pi}}_i. \tag{29.59}$$

Chudik and Pesaran (2015a) show that $\hat{\boldsymbol{\pi}}_i$ and $\hat{\boldsymbol{\pi}}_{MG}$ are consistent estimators of $\boldsymbol{\pi}_i$ and $\boldsymbol{\pi}$, respectively, assuming that the rank condition is satisfied and $\left(N, T, p_T\right) \to \infty$ such that $p_T^3/T \to \varkappa, 0 < \varkappa < \infty$, but without any restrictions on the ratio N/T. The rank condition is necessary for the consistency of $\hat{\boldsymbol{\pi}}_i$ because the unobserved factors are allowed to be correlated with the regressors. If the unobserved common factors were serially uncorrelated (but still

[12] Matrices $\boldsymbol{\Xi}_i$, $\bar{\mathbf{Q}}_w$, and $\bar{\mathbf{M}}_q$ depend also on p_T, N and T, but these subscripts are omitted to simplify notations.

[13] See Berk (1974), Said and Dickey (1984), and Pesaran and Chudik (2014) for a related discussion on the choice of lag truncation for estimation of infinite-order autoregressive models.

correlated with $\mathbf{x}_{it}$), then $\hat{\boldsymbol{\pi}}_{MG}$ is consistent also in the rank deficient case, despite the inconsistency of $\hat{\boldsymbol{\pi}}_i$, so long as factor loadings are independently, identically distributed across i. The convergence rate of $\hat{\boldsymbol{\pi}}_{MG}$ is $\sqrt{N}$ due to the heterogeneity of the slope coefficients. Chudik and Pesaran (2015a) show that $\hat{\boldsymbol{\pi}}_{MG}$ converges to a normal distribution as $\left(N, T, p_T\right) \xrightarrow{j} \infty$ such that $p_T^3/T \to \varkappa_1$ and $T/N \to \varkappa_2$, $0 < \varkappa_1, \varkappa_2 < \infty$. The ratio N/T needs to be restricted for conducting inference, due to the presence of small time series bias. In the full rank case, the asymptotic variance of $\hat{\boldsymbol{\pi}}_{MG}$ is given by the variance of $\boldsymbol{\pi}_i$ alone. When the rank condition does not hold, but factors are serially uncorrelated, then the asymptotic variance depends also on other parameters, including the variance of factor loadings. In both cases the asymptotic variance can be consistently estimated non-parametrically, as in (29.40).

Monte Carlo experiments in Chudik and Pesaran (2015a) show that the dynamic *CCE* approach performs reasonably well (in terms of bias, RMSE, size and power). This is particularly the case when the parameter of interest is the average slope of the regressors ($\boldsymbol{\beta}$), where the small sample results are quite satisfactory even if N and T are relatively small (around 40). But the situation is different if the parameter of interest is the mean coefficient of the lagged dependent variable (λ). In the case of λ, the *CCEMG* estimator suffers from the well known time series bias and tests based on it tend to be over-sized, unless T is sufficiently large. To reduce this bias, Chudik and Pesaran (2015a) consider application of the half-panel jackknife procedure (Dhaene and Jochmans (2012)), and the recursive mean adjustment procedure (So and Shin (1999)), both of which are easy to implement.[14] The proposed jackknife bias-corrected *CCEMG* estimator is found to be more effective in mitigating the time series bias, but it cannot deal fully with the size distortion when T is relatively small. Improving the small T sample properties of the CCEMG estimator of λ in the heterogeneous panel data models still remains a challenge.

29.5.4 Properties of *CCE* in the case of panels with weakly exogenous regressors

The application of the *CCE* approach to static panels with weakly exogenous regressors (namely without lagged dependent variables) has not yet been investigated in the literature. In order to investigate whether the standard *CCE* mean group and pooled estimators could be applied in this setting, we conducted some additional Monte Carlo experiments. We used the following data generating process

$$y_{it} = c_{yi} + \beta_{0i} x_{it} + \beta_{1i} x_{i,t-1} + u_{it}, u_{it} = \boldsymbol{\gamma}_i' \mathbf{f}_t + \varepsilon_{it}, \tag{29.60}$$

and

$$x_{it} = c_{xi} + \alpha_{xi} y_{i,t-1} + \boldsymbol{\gamma}_{xi}' \mathbf{f}_t + v_{it}, \tag{29.61}$$

for $i = 1, 2, \ldots, N$, and $t = -99, \ldots, 0, 1, 2, \ldots, T$ with the starting values $y_{i,-100} = x_{i,-100} = 0$. This set up allows for feedbacks from $y_{i,t-1}$ to the regressors, thus rendering x_{it} weakly exogenous. The size of the feedback is measured by α_{xi}. The unobserved common factors in $\mathbf{f}_t$ and the unit-specific components v_{it} are generated as independent stationary $AR(1)$ processes

[14] The jackknife bias-correction procedure was first proposed by Quenouille (1949). See also Section 14.5.

$$f_{t\ell} = \rho_{f\ell} f_{t-1,\ell} + \varsigma_{ft\ell}, \varsigma_{ft\ell} \sim IIDN\left(0, 1 - \rho_{f\ell}^2\right),$$
$$v_{it} = \rho_{xi} v_{i,t-1} + \varsigma_{it}, \varsigma_{it} \sim IIDN\left(0, \sigma_{vi}^2\right), \tag{29.62}$$

for $i = 1, 2, \ldots, N$, $\ell = 1, 2, \ldots, m$, and for $t = -99, \ldots, 0, 1, 2, \ldots, T$ with the starting values $f_{\ell,-100} = 0$ and $v_{i,-100} = 0$. The first 100 time observations ($t = -99, -98, \ldots, 0$) are discarded. We generate ρ_{xi}, for $i = 1, 2, \ldots .N$ as $IIDU[0, 0.95]$, and set $\rho_{f\ell} = 0.6$, for $\ell = 1, 2, \ldots, m$. We also set $\sigma_{vi} = \sqrt{1 - \left[E\left(\rho_{xi}\right)\right]^2}$ for all i.

The fixed-effects are generated as $c_{yi} \sim IIDN(1, 1)$, $c_{xi} = c_{yi} + \varsigma_{c_x i}$, where $\varsigma_{c_x i} \sim IIDN(0, 1)$, thus allowing for dependence between x_{it} and c_{yi}. We set $\beta_{1i} = -0.5$ for all i, and generate β_{0i} as $IIDU(0.5, 1)$. We consider two possibilities for the feedback coefficients α_{xi}: weakly exogenous regressors where we generate α_{xi} as draws from $IIDU(0, 1)$ (in which case $E(\alpha_{xi}) = 0.5$); and strictly exogenous regressors where we set $\alpha_{xi} = 0$ for all i. We consider $m = 3$ unobserved common factors, with all factor loadings generated independently in the same way as in Chudik and Pesaran (2015a). Similarly, the idiosyncratic errors, ε_{it}, are generated to be heteroskedastic and weakly cross-sectionally dependent. We consider the following combinations of sample sizes: $N \in \{40, 50, 100, 150, 200\}$, $T \in \{20, 50, 100, 150, 200\}$, and set the number of replications to $R = 2,000$.

The small sample results for the *CCE* mean group and pooled estimators (with lagged augmentations) in the case of these experiments with weakly exogenous regressors are presented in Panel A of Table 29.3. The rank condition in these experiment does not hold, but this does not seem to cause any major problems for the *CCE* mean group estimator, which performs very well (in terms of bias and *RMSE*) for $T > 50$ and for all values of N. Also tests based on this estimator are correctly sized and have good power properties. When $T \leq 50$, we observe a negative bias and the tests are oversized. The *CCE* pooled estimator, however, is no longer consistent in the case of weakly exogenous regressors with heterogeneous coefficients, due to the bias caused by the correlation between the slope coefficients and the regressors. For comparison, we also provide, in panel B of Table 29.3, the results of the same experiments but with strictly exogenous regressors ($\alpha_{xi} = 0$), where the bias is negligible and all tests are correctly sized.

29.6 Estimating long-run coefficients in dynamic panel data models with a factor error structure

The previous section focused on the estimation of short-run coefficients λ_i and $\boldsymbol{\beta}_i$ in the dynamic panel data model (29.49)–(29.50). However, in many empirical applications the objective of interest is often the long-run coefficients defined by

$$\boldsymbol{\theta}_i = \frac{\boldsymbol{\beta}_i}{1 - \lambda_i}, \text{for } i = 1, 2, \ldots, N. \tag{29.63}$$

Long-run relationships are of great importance in economics. The concept of 'long-run relations' is typically associated with the steady-state solution of a structural macroeconomic model. Often the same long-run relations can also be obtained from arbitrage conditions within and

Table 29.3 Small sample properties of *CCEMG* and *CCEP* estimators of mean slope coefficients in panel data models with weakly and strictly exogenous regressors

	Bias (x100)					**RMSE** (x100)					**Size** (x100)					**Power** (x100)				
(N,T)	20	50	100	150	200	20	50	100	150	200	20	50	100	150	200	20	50	100	150	200
Panel A: Experiments with weakly exogenous regressors																				
	CCEMG																			
40	−5.70	−1.46	-0.29	0.00	0.11	7.82	3.65	2.80	2.67	2.61	23.70	9.35	6.20	6.05	6.25	86.80	94.05	96.00	96.30	96.95
50	−5.84	−1.56	−0.39	0.04	0.11	7.56	3.43	2.56	2.41	2.33	29.50	9.30	7.00	6.70	6.20	93.40	96.75	98.75	98.70	99.20
100	−5.88	−1.50	−0.41	−0.05	0.07	6.82	2.63	1.83	1.70	1.64	46.70	13.10	6.00	5.75	5.25	99.75	99.95	100.00	100.00	100.00
150	−6.11	−1.59	−0.45	−0.11	0.08	6.73	2.36	1.53	1.35	1.30	66.05	16.15	6.60	4.75	4.80	100.00	100.00	100.00	100.00	100.00
200	−6.04	−1.55	−0.43	−0.12	0.01	6.54	2.17	1.37	1.18	1.18	74.65	19.70	7.35	4.50	6.10	100.00	100.00	100.00	100.00	100.00
	CCEP																			
40	−3.50	−0.09	0.76	0.98	1.23	6.58	3.71	3.33	3.24	3.35	14.80	6.75	7.50	7.55	9.85	72.30	78.45	80.55	82.70	82.55
50	−3.55	−0.27	0.70	1.08	1.19	6.07	3.31	2.96	3.00	2.96	14.00	5.70	6.20	8.65	8.80	79.70	86.90	88.55	88.70	90.90
100	−3.56	−0.10	0.76	1.08	1.17	5.11	2.42	2.22	2.27	2.26	21.75	5.50	6.75	9.10	10.45	96.05	97.80	98.80	98.95	99.30
150	−3.78	−0.10	0.74	1.10	1.16	4.86	1.98	1.87	1.99	1.98	30.45	5.85	7.60	11.45	12.60	99.15	99.75	99.95	99.95	100.00
200	−3.66	−0.19	0.80	1.08	1.13	4.56	1.77	1.67	1.78	1.77	35.65	6.25	8.35	12.50	12.45	100.00	100.00	100.00	100.00	100.00

Table 29.3 Continued

	Bias (x100)					**RMSE** (x100)					**Size** (x100)					**Power** (x100)				
(N,T)	20	50	100	150	200	20	50	100	150	200	20	50	100	150	200	20	50	100	150	200
Panel B: Experiments with strictly exogenous regressors																				
	CCEMG																			
40	0.19	−0.05	0.02	0.07	0.04	6.43	3.91	3.06	2.91	2.75	6.20	6.40	4.60	6.40	5.55	36.20	74.40	89.95	93.90	95.60
50	−0.02	0.08	0.11	−0.05	−0.02	5.72	3.48	2.83	2.68	2.46	5.25	6.10	5.90	6.75	5.75	43.90	82.20	93.70	96.80	98.05
100	−0.06	0.01	0.02	−0.05	−0.01	4.13	2.52	2.02	1.79	1.78	5.55	6.45	4.90	4.95	6.20	69.95	97.60	99.75	99.95	100.00
150	0.06	0.03	0.00	0.02	0.01	3.29	2.03	1.62	1.50	1.42	5.40	6.00	5.50	5.05	5.30	85.65	99.95	100.00	100.00	100.00
200	−0.06	0.03	−0.02	−0.03	−0.01	2.87	1.75	1.39	1.33	1.23	4.50	5.30	4.85	6.50	5.15	94.10	100.00	100.00	100.00	100.00
	CCEP																			
40	0.21	0.17	0.02	−0.01	−0.02	5.78	3.85	3.16	3.08	2.85	6.40	6.45	5.95	7.10	6.35	74.55	72.90	88.10	92.15	93.50
50	0.03	−0.01	−0.13	0.02	−0.02	5.20	3.48	2.84	2.59	2.54	5.60	6.25	6.25	6.00	5.95	83.35	83.30	94.80	96.30	97.30
100	−0.01	-0.06	0.05	−0.04	0.07	3.67	2.56	2.03	1.89	1.76	5.60	6.15	5.00	5.35	5.65	98.50	97.75	99.85	100.00	100.00
150	0.05	0.02	0.02	0.01	0.01	2.95	2.02	1.65	1.52	1.49	4.50	5.20	5.50	4.95	5.60	99.80	99.95	100.00	100.00	100.00
200	−0.09	−0.04	−0.06	0.03	0.02	2.57	1.74	1.43	1.38	1.28	6.05	5.75	5.15	5.75	4.95	100.00	100.00	100.00	100.00	100.00

Notes: Observations are generated as $y_{it} = c_{yi} + \beta_{0i}x_{it} + \beta_{1i}x_{i,t-1} + u_{it}$, $u_{it} = \boldsymbol{\gamma}_i'\mathbf{f}_t + \varepsilon_{it}$, and $x_{it} = c_{xi} + \alpha_{xi}y_{i,t-1} + \boldsymbol{\gamma}_{xi}'\mathbf{f}_t + v_{it}$, (see (29.60)–(29.61)), where $\beta_{0i} \sim IIDU(0.5, 1)$, $\beta_{1i} = -0.5$ for all i, and $m = 3$ (number of unobserved common factors). Fixed-effects are generated as $c_{yi} \sim IIDN(1, 1)$, and $c_{xi} = c_{yi} + IIDN(0, 1)$. In the case of weakly exogenous regressors, $\alpha_{xi} \sim IIDU(0, 1)$ (with $E(\alpha_{xi}) = 0.5$), and under the case of strictly exogenous regressors $\alpha_{xi} = 0$ for all i. The errors are generated to be heteroskedastic and weakly cross-sectionally dependent. See Section 29.5.3 for a more detailed description of the MC design.

across markets. As a result, many long-run relationships in economics are free of particular model assumptions; examples include purchasing power parity, uncovered interest parity and the Fisher inflation parity.

Estimation of long-run relations in the case of pure time series models has been discussed in Section 6.5, and for dynamic panel data models without cross-sectional dependence has been considered in Chapter 28 (Sections 28.6–28.10). This section extends the estimation of long-run effects to dynamic panels with multifactor error structure.

There are two approaches to estimating long-run coefficients. One approach, is to estimate the individual short-run coefficients λ_i and $\boldsymbol{\beta}_i$ in the *ARDL* relation (29.49) and then compute the estimates of long-run effects using formula (29.63) with the short-run coefficients replaced by their estimates ($\hat{\lambda}_i$ and $\hat{\boldsymbol{\beta}}_i$) discussed in Section 29.5. This is the '*ARDL* approach to the estimation of long-run effects.' This approach is consistent irrespective of whether the underlying variables are $I(0)$ or $I(1)$, and whether the regressors in $\mathbf{x}_{it}$ are strictly or weakly exogenous. These robustness properties are clearly important in empirical research. However, the *ARDL* approach also has its own drawbacks. Most importantly, the sampling uncertainty could be large, especially when the speed of convergence towards the long-run relation is rather slow and the time dimension is not sufficiently long. This is readily apparent from (29.63), since even a small change to $1 - \hat{\lambda}_i$ could have a large impact on the estimates of $\boldsymbol{\theta}_i$, when $\hat{\lambda}_i$ is close to unity. In this respect, a correct specification of lag-orders could be quite important for the performance of the *ARDL* estimates. Moreover, the estimates of the short-run coefficients are subject to small T bias.

An alternative approach, proposed by Chudik et al. (2015), is to estimate the long-run coefficients $\boldsymbol{\theta}_i$ *directly,* without first estimating the short run coefficients. This is possible by observing that the *ARDL* model (29.49) can be written as

$$y_{it} = \boldsymbol{\theta}_i \mathbf{x}_{it} + \boldsymbol{\alpha}_i'(L)\,\Delta\mathbf{x}_{it} + \tilde{u}_{it}, \tag{29.64}$$

where $\tilde{u}_{it} = \lambda_i(L)^{-1} u_{it}$, $\lambda_i(L) = 1 - \lambda_i L^{\ell}$, and $\boldsymbol{\alpha}_i(L) = \sum_{\ell=0}^{\infty}\sum_{s=\ell+1}^{\infty}\lambda_i^s \boldsymbol{\beta}_i L^{\ell}$. We shall refer to the direct estimation of $\boldsymbol{\theta}_i$ based on the distributed lag (DL) representation (29.64) as the 'DL approach to the estimation of long-run effects'. Under the usual assumptions $|\lambda_i| < 1$ (the roots of $\lambda_i(L)$ fall strictly outside the unit circle), the coefficients of $\boldsymbol{\alpha}_i(L)$ are exponentially decaying, and in the absence of feedback effects from lagged values of y_{it} onto the regressors $\mathbf{x}_{it}$, a consistent estimate of $\boldsymbol{\theta}_i$ can be obtained directly based on the least squares regression of y_{it} on $\mathbf{x}_{it}$, $\{\Delta\mathbf{x}_{it-\ell}\}_{\ell=0}^{p_T}$ and a set of cross-sectional averages that deals with the effects of unobserved common factors in u_{it}. The truncation lag-order $p_T = p(T)$ is chosen as a non-decreasing function of T such that $0 \leq p_T < T$.

The cross-section augmented distributed lag (CS-DL) mean group estimator of the long-run coefficients is given by

$$\hat{\boldsymbol{\theta}}_{MG} = \frac{1}{N}\sum_{i=1}^{N}\hat{\boldsymbol{\theta}}_i, \tag{29.65}$$

where

$$\hat{\boldsymbol{\theta}}_i = \left(\tilde{\mathbf{X}}_i'\mathbf{M}_{qi}\tilde{\mathbf{X}}_i'\right)^{-1}\tilde{\mathbf{X}}_i'\mathbf{M}_{qi}\tilde{\mathbf{y}}_i. \tag{29.66}$$

The *CS-DL* pooled estimator of the long-run coefficients is

$$\hat{\boldsymbol{\theta}}_P = \left(\sum_{i=1}^{N} w_i \tilde{\mathbf{X}}_i' \mathbf{M}_{qi} \tilde{\mathbf{X}}_i \right)^{-1} \sum_{i=1}^{N} w_i \tilde{\mathbf{X}}_i' \mathbf{M}_{qi} \tilde{\mathbf{y}}_i. \tag{29.67}$$

Estimators $\hat{\boldsymbol{\theta}}_{MG}$ and $\hat{\boldsymbol{\theta}}_P$ differ from the mean group and pooled *CCE* estimator developed in Pesaran (2006) (see (29.37)-(29.38)), which only allows for the inclusion of a fixed number of regressors, whilst the CS-DL type estimators include p_T lags of $\Delta \mathbf{x}_{it}$ and their cross-section averages, where p_T increases with T, albeit at a slower rate. Specifically, when $\left(N, T, p_T\right) \xrightarrow{j} \infty$ such that $\sqrt{N} p_T \rho^p \to 0$, for any constant $0 < \rho < 1$ and $p_T^3/T \to \varkappa$, $0 < \varkappa < \infty$, Chudik et al. (2015) establish asymptotic normality of $\hat{\boldsymbol{\theta}}_{MG}$ and $\hat{\boldsymbol{\theta}}_P$ under the assumption of the random coefficient model,

$$\boldsymbol{\theta}_i = \boldsymbol{\theta} + \boldsymbol{\upsilon}_i, \boldsymbol{\upsilon}_i \sim IID(\mathbf{0}, \boldsymbol{\Omega}_\theta), \text{for } i = 1, 2, \ldots, N, \tag{29.68}$$

where $\|\boldsymbol{\theta}\| < K, \|\boldsymbol{\Omega}_\theta\| < K, \boldsymbol{\Omega}_\theta$ is a $k \times k$ symmetric nonnegative definite matrix. The rate of convergence is $\sqrt{N}$ (due to coefficient heterogeneity) and the asymptotic variance can be estimated non-parametrically along similar lines as in Section 29.4.2 (see (29.40) and (29.42)).

Monte Carlo evidence presented in Chudik et al. (2015) suggests that the DL approach has often better small sample performance when T is in the range $30 \leq T < 100$, compared to the *ARDL* approach. The advantage of the DL approach is its robustness to residual serial correlation, breaks in error processes, and dynamic misspecifications. However, unlike the *ARDL* approach, the DL procedure could be subject to simultaneity bias (when there are feedbacks from lagged values of y_{it} to the regressors in $\mathbf{x}_{it}$). Nevertheless, the extensive Monte Carlo experiments reported in Chudik et al. (2015) suggest that the endogeneity bias of the DL approach is more than compensated for by its better small sample performance as compared with the *ARDL* procedure when the time dimension is not very large. *ARDL* seems to dominate DL only if the time dimension is sufficiently large and the underlying *ARDL* model is correctly specified.

29.7 Testing for error cross-sectional dependence

In this section we provide an overview of alternative approaches to testing cross-sectional independence or weak dependence of the errors in the following panel data model[15]

$$y_{it} = a_i + \boldsymbol{\beta}_i' \mathbf{x}_{it} + u_{it}, \tag{29.69}$$

where a_i and $\boldsymbol{\beta}_i$ for $i = 1, 2, \ldots, N$, are assumed to be fixed unknown coefficients, and $\mathbf{x}_{it}$ is a k-dimensional vector of regressors. We consider both cases where the regressors are strictly and weakly exogenous, as well as when they include lagged values of y_{it}.

[15] Strictly speaking, the hypothesis being tested is zero cross-section correlations rather than independence of the errors, although the two notions coincide in the case of linear or Gaussian models.

The literature on testing for error cross-sectional dependence in large panels follows two separate strands, depending on whether the cross-sectional units are ordered or not. In the case of ordered data sets (which could arise when observations are spatial or belong to given economic or social networks) tests of cross-sectional independence that have high power with respect to such ordered alternatives have been proposed in the spatial econometrics literature. A prominent example of such tests is Moran's I test. See Moran (1948) with further developments by Anselin (1988), Anselin and Bera (1998), Haining (2003), and Baltagi, Song, and Koh (2003).

In the case of cross-section observations that do not admit an ordering, tests of cross-sectional dependence are typically based on estimates of pair-wise error correlations (ρ_{ij}) and are applicable when T is sufficiently large so that relatively reliable estimates of ρ_{ij} can be obtained. An early test of this type is the Lagrange multiplier (LM) test of Breusch and Pagan (1980) which tests the null hypothesis that *all* pair-wise correlations are zero, namely that $\rho_{ij} = 0$ for all $i \neq j$. This test is based on the average of the *squared* estimates of pair-wise correlations, and under standard regularity conditions it is shown to be asymptotically (as $T \to \infty$) distributed as χ^2 with $N(N-1)/2$ degrees of freedom. The LM test tends to be highly over-sized in the case of panels with relatively large N.

In what follows, we review the various attempts made in the literature to develop tests of cross-sectional independence when N is large and the cross-sectional units are unordered. But before proceeding further, we first need to consider the appropriateness of the null hypothesis of cross-sectional 'independence' or 'uncorrelatedness', that underlies the LM test of Breusch and Pagan (1980), namely that all ρ_{ij} are zero for all $i \neq j$, when N is large. The null that underlies the LM test is sensible when N is small and fixed as $T \to \infty$. But when N is relatively large and rising with T, it is unlikely to matter if, out of the total $N(N-1)/2$ pair-wise correlations, only a few are non-zero. Accordingly, Pesaran (2015) argues that the null of cross-sectionally uncorrelated errors, defined by

$$H_0 : E\left(u_{it}u_{jt}\right) = 0, \text{for all } t \text{ and } i \neq j, \tag{29.70}$$

is restrictive for large panels and the null of a sufficiently weak cross-sectional dependence could be more appropriate since the mere incidence of isolated error dependencies is of little consequence for estimation or inference about the parameters of interest, such as the individual slope coefficients, $\boldsymbol{\beta}_i$, or their average value, $E(\boldsymbol{\beta}_i) = \boldsymbol{\beta}$.

Consider the panel data model (29.69), and let $\hat{u}_{it}$ be the *OLS* estimator of u_{it} defined by

$$\hat{u}_{it} = y_{it} - \hat{a}_i - \hat{\boldsymbol{\beta}}_i'\mathbf{x}_{it}, \tag{29.71}$$

with $\hat{a}_i$, and $\hat{\boldsymbol{\beta}}_i$ being the *OLS* estimates of a_i and $\boldsymbol{\beta}_i$, based on the T sample observations, $y_t, \mathbf{x}_{it}$, for $t = 1, 2, \ldots, T$. Consider the sample estimate of the pair-wise correlation of the residuals, $\hat{u}_{it}$ and $\hat{u}_{jt}$, for $i \neq j$

$$\hat{\rho}_{ij} = \hat{\rho}_{ji} = \frac{\sum_{t=1}^{T} \hat{u}_{it}\hat{u}_{jt}}{\left(\sum_{t=1}^{T} \hat{u}_{it}^2\right)^{1/2}\left(\sum_{t=1}^{T} \hat{u}_{jt}^2\right)^{1/2}}.$$

In the case where the u_{it} is symmetrically distributed and the regressors are strictly exogenous, then under the null hypothesis of no cross-sectional dependence, $\hat{\rho}_{ij}$ and $\hat{\rho}_{is}$ are cross-sectionally uncorrelated for all i, j and s such that $i \neq j \neq s$. This follows since

$$E\left(\hat{\rho}_{ij}\hat{\rho}_{is}\right) = \sum_{t=1}^{T}\sum_{t'=1}^{T} E\left(\hat{\eta}_{it}\hat{\eta}_{it'}\hat{\eta}_{jt}\hat{\eta}_{st'}\right) = \sum_{t=1}^{T}\sum_{t'=1}^{T} E\left(\hat{\eta}_{it}\hat{\eta}_{it'}\right)E\left(\hat{\eta}_{jt}\right)E\left(\hat{\eta}_{st'}\right) = 0, \tag{29.72}$$

where $\hat{\eta}_{it} = \hat{u}_{it} \Big/ \left(\sum_{t=1}^{T} \hat{u}_{it}^2\right)^{1/2}$. Note that when $\mathbf{x}_{it}$ is strictly exogenous for each i, $\hat{u}_{it}$, being a linear function of u_{it}, for $t = 1, 2, \ldots, T$, will also be symmetrically distributed with zero means, which ensures that η_{it} is also symmetrically distributed around its mean which is zero. Further, under (29.70) and when N is finite, it is known that (see Pesaran (2004))

$$\sqrt{T}\hat{\rho}_{ij} \overset{a}{\sim} N(0, 1), \tag{29.73}$$

for a given i and j, as $T \to \infty$. The above result has been widely used for constructing tests based on the sample correlation coefficient or its transformations. Noting that, from (29.73), $T\hat{\rho}_{ij}^2$ is asymptotically distributed as a χ_1^2, it is possible to consider the following test statistic

$$CD_{LM} = \sqrt{\frac{1}{N(N-1)}} \sum_{i=1}^{N-1}\sum_{j=i+1}^{N} \left(T\hat{\rho}_{ij}^2 - 1\right). \tag{29.74}$$

Based on the Euclidean norm of the matrix of sample correlation coefficients, (29.74) is a version of the Lagrange multiplier test statistic due to Breusch and Pagan (1980). Frees (1995) first explored the finite sample properties of the *LM* statistic, calculating its moments for fixed values of T and N, under the normality assumption. He advanced a non-parametric version of the *LM* statistic based on the Spearman rank correlation coefficient. Dufour and Khalaf (2002) have suggested to apply Monte Carlo exact tests to correct the size distortions of CD_{LM} in finite samples. However, these tests, being based on the bootstrap method applied to the CD_{LM}, are computationally intensive, especially when N is large.

An alternative adjustment to the *LM* test is proposed by Pesaran, Ullah, and Yamagata (2008), where the *LM* test is centered to have a zero mean for a fixed T. These authors also propose a correction to the variance of the *LM* test. The basic idea is generally applicable, but analytical bias corrections can be obtained only under the assumption that the regressors, $\mathbf{x}_{it}$, are strictly exogenous and the errors, u_{it} are normally distributed. Under these assumptions, Pesaran, Ullah, and Yamagata (2008) show that the exact mean and variance of $(N-k)\,\hat{\rho}_{ij}^2$ are given by:

$$\mu_{Tij} = E\left[(N-k)\,\hat{\rho}_{ij}^2\right] = \frac{1}{T-k} Tr\left[E\left(\mathbf{M}_i\mathbf{M}_j\right)\right],$$
$$v_{Tij}^2 = Var\left[(N-k)\,\hat{\rho}_{ij}^2\right] = \left\{Tr\left[E\left(\mathbf{M}_i\mathbf{M}_j\right)\right]\right\}^2 a_{1T} + 2\left\{Tr\left[E\left(\mathbf{M}_i\mathbf{M}_j\right)^2\right]\right\} a_{2T},$$

where

$$a_{1T} = a_{2T} - \left(\frac{1}{T-k}\right)^2,\ a_{2T} = 3\left[\frac{(T-k-8)(T-k+2)+24}{(T-k+2)(T-k-2)(T-k-4)}\right]^2,$$

$\mathbf{M}_i = \mathbf{I}_T - \tilde{\mathbf{X}}_i \left(\tilde{\mathbf{X}}_i'\tilde{\mathbf{X}}_i\right)^{-1} \tilde{\mathbf{X}}_i'$, and $\tilde{\mathbf{X}}_i$ is $T \times (k+1)$ matrix of observations on $\left(1, \mathbf{x}_{it}'\right)'$. The adjusted *LM* statistic is now given by

$$LM_{Adj} = \sqrt{\frac{2}{N(N-1)}} \sum_{i=1}^{N-1} \sum_{j=i+1}^{N} \frac{(T-k)\,\hat{\rho}_{ij}^2 - \mu_{Tij}}{v_{Tij}}, \tag{29.75}$$

which is asymptotically $N(0, 1)$ under H_0, $T \to \infty$ followed by $N \to \infty$. The asymptotic distribution of LM_{Adj} is derived under sequential asymptotics, but it might be possible to establish it under the joint asymptotics following the method of proof in Schott (2005) or Pesaran (2015).

The application of the LM_{Adj} test to dynamic panels or panels with weakly exogenous regressors is further complicated by the fact that the bias corrections depend on the true values of the unknown parameters and will be difficult to implement. The implicit null of *LM* tests when T and $N \to \infty$, jointly rather than sequentially could also differ from the null of uncorrelatedness of all pair-wise correlations. To overcome some of these difficulties, Pesaran (2004) has proposed a test that has exactly mean zero for fixed values of T and N. This test is based on the average of pair-wise correlation coefficients

$$CD_P = \sqrt{\frac{2T}{N(N-1)}} \left(\sum_{i=1}^{N-1} \sum_{j=i+1}^{N} \hat{\rho}_{ij}\right). \tag{29.76}$$

As it is established in (29.72), under the null hypothesis $\hat{\rho}_{ij}$ and $\hat{\rho}_{is}$ are uncorrelated for all $i \neq j \neq s$, but they need not be independently distributed when T is finite. Therefore, the standard central limit theorems cannot be applied to the elements of the double sum in (29.76) when $(N, T) \to \infty$ jointly, and as shown in Pesaran (2015, Theorem 2) the derivation of the limiting distribution of the CD_P statistic involves a number of complications. It is also important to bear in mind that the implicit null of the test in the case of large N depends on the rate at which T expands with N. Indeed, as argued in Pesaran (2004), under the null hypothesis of $\rho_{ij} = 0$ for all $i \neq j$, we continue to have $E\left(\hat{\rho}_{ij}\right) = 0$, even when T is fixed, so long as u_{it} are symmetrically distributed around zero, and the CD_P test continues to hold.

Pesaran (2015) extends the analysis of the CD_P test and shows that the implicit null of the test is weak cross-sectional dependence. In particular, the implicit null hypothesis of the test depends on the relative expansion rates of N and T.[16] Using the exponent of cross-sectional dependence, α, developed in Bailey, Kapetanios, and Pesaran (2015) and discussed above, Pesaran (2015) shows that when $T = O\left(N^{\epsilon}\right)$ for some $0 < \epsilon \leq 1$, the implicit null of the CD_P test is given by $0 \leq \alpha < (2-\epsilon)/4$. This yields the range $0 \leq \alpha < 1/4$ when $(N, T) \to \infty$ at the same rate such that $T/N \to \varkappa$ for some finite positive constant $\varkappa$, and the range $0 \leq \alpha < 1/2$ when T is

[16] Pesaran (2015) also derives the exact variance of the CD_P test under the null of cross-sectional independence and proposes a slightly modified version of the CD_P test distributed exactly with mean zero and a unit variance.

small relative to N. For larger values of α, as shown by Bailey, Kapetanios, and Pesaran (2015), α can be estimated consistently using the variance of the cross-section averages.

Monte Carlo experiments reported in Pesaran (2015) show that the CD_P test has good small sample properties for values of α in the range $0 \leq \alpha \leq 1/4$, even in cases where T is small relative to N, as well as when the test is applied to residuals from pure autoregressive panels so long as there are no major asymmetries in the error distribution.

Other statistics have also been proposed in the literature to test for zero contemporaneous correlation in the errors of panel data model (29.69).[17] Using results from the literature on *spacing* discussed in Pyke (1965), Ng (2006) considers a statistic based on the qth differences of the cumulative normal distribution associated to the $N(N-1)/2$ pair-wise correlation coefficients ordered from the smallest to the largest, in absolute value. Building on the work of John (1971), and under the assumption of normal disturbances, strictly exogenous regressors, and homogeneous slopes, Baltagi, Feng, and Kao (2011) propose a test of the null hypothesis of sphericity, defined by

$$H_0^{BFK} : \mathbf{u}_t \sim IIDN\left(\mathbf{0}, \sigma_u^2 \mathbf{I}_N\right),$$

based on the statistic

$$J_{BFK} = \frac{T\left(tr(\hat{\mathbf{S}})/N\right)^{-2} tr(\hat{\mathbf{S}}^2)/N - T - N}{2} - \frac{1}{2} - \frac{N}{2\,(T-1)}, \tag{29.77}$$

where $\hat{\mathbf{S}}$ is the $N \times N$ sample covariance matrix, computed using the fixed-effects residuals under the assumption of slope homogeneity, $\boldsymbol{\beta}_i = \boldsymbol{\beta}$. Under H_0^{BFK}, errors u_{it} are cross-sectionally independent and homoskedastic and the J_{BFK} statistic converges to a standardized normal distribution as $(N, T) \to \infty$ such that $N/T \to \varkappa$ for some finite positive constant $\varkappa$. The rejection of H_0^{BFK} could be caused by cross-sectional dependence, heteroskedasticity, slope heterogeneity, and/or non-normal errors. Simulation results reported in Baltagi, Feng, and Kao (2011) show that this test performs well in the case of homoskedastic, normal errors, strictly exogenous regressors, and homogeneous slopes, although it is oversized for panels with large N and small T, and is sensitive to non-normality of disturbances. Joint assumption of homoskedastic errors and homogeneous slopes is quite restrictive in applied work, and therefore the use of the J_{BFK} statistics as a test of cross-sectional dependence should be approached with care.

A slightly modified version of the CD_{LM} statistic, given by

$$LM_S = \sqrt{\frac{T+1}{N(N-1)\,(T-2)}} \sum_{i=1}^{N-1} \sum_{j=i+1}^{N} \left[(T-1)\,\hat{\rho}_{ij}^2 - 1\right] \tag{29.78}$$

has also been considered by Schott (2005), who shows that when the LM_S statistic is computed based on normally distributed observations, as opposed to panel residuals, it converges to $N(0, 1)$ under $\rho_{ij} = 0$ for all $i \neq j$ as $(N, T) \to \infty$ such that $N/T \to \varkappa$ for some $0 < \varkappa < \infty$. Monte Carlo simulations reported in Jensen and Schmidt (2011) suggest that the LM_S test has good size properties for various sample sizes when applied to panel residuals in the case when

[17] A review is provided by Moscone and Tosetti (2009).

slopes are homogeneous and estimated using the fixed-effects approach. However, the LM_S test can lead to severe over-rejection when the slopes are in fact heterogeneous and the fixed-effects estimators are used. Over-rejection of the LM_S test could persist even if mean group estimates are used in the computation of the residuals to take care of slope heterogeneity. This is because for relatively small values of T, unlike the LM_{Adj} statistic defined by (29.75), the LM_S statistic defined by (29.78) is not guaranteed to have a zero mean exactly.

The problem of testing for cross-sectional dependence in limited dependent variable panel data models with strictly exogenous covariates has also been investigated by Hsiao, Pesaran, and Pick (2012). In this paper the authors derive an *LM* test and show that in terms of the generalized residuals of Gourieroux et al. (1987), the test reduces to the *LM* test of Breusch and Pagan (1980). However, not surprisingly as with the linear panel data models, the *LM* test based on generalized residuals tends to over-reject in panels with large N. They then develop a *CD* type test based on a number of different residuals, and using Monte Carlo experiments they find that the *CD* test preforms well for most combinations of N and T.

The existing literature on testing for error cross-sectional dependence, with the exception of Sarafidis et al. (2009), has mostly focused on the case of strictly exogenous regressors. This assumption is required for both LM_{Adj} and J_{BFK} tests, while Pesaran (2004) shows that the CD_P test is also applicable to autoregressive panel data models so long as the errors are symmetrically distributed. The properties of the CD_P test for dynamic panels that include weakly or strictly exogenous regressors have not yet been investigated.

We conduct Monte Carlo experiments to investigate the performance of these tests in the case of dynamic panels and to shed light also on the performance of LM_S test in the case of panels with heterogeneous slopes. We generate the dependent variable and the regressors in the same way as described in Section 29.5.3 with the following two exceptions. First, we introduce lags of the dependent variable in (29.74)

$$y_{it} = c_{yi} + \lambda_i y_{i,t-1} + \beta_{0i} x_{it} + \beta_{1i} x_{i,t-1} + u_{it}, \tag{29.79}$$

and generate λ_i as $IIDU\,(0, 0.8)$. As discussed in Chudik and Pesaran (2015a) the lagged dependent variable coefficients, λ_i, and the feedback coefficients, α_{xi}, in (29.61) need to be chosen such as to ensure the variances of y_{it} remain bounded. We generate α_{xi} as $IIDU\,(0, 0.35)$, which ensures that this condition is met and $E(\alpha_{xi}) = 0.35/2$. For comparison purposes, we also consider the case of strictly exogenous regressors where we set $\lambda_i = \alpha_{xi} = 0$ for all i. The second exception is the generation of the reduced form errors. In order to consider different options for cross-sectional dependence, we use the following residual factor model to generate the errors u_{it}

$$u_{it} = \gamma_i g_t + \varepsilon_{it}, \tag{29.80}$$

where $\varepsilon_{it} \sim IIDN\left(0, \frac{1}{2}\sigma_i^2\right)$, with $\sigma_i^2 \sim \chi^2\,(2)$, $g_t \sim IIDN\,(0, 1)$ and the factor loadings are generated as

$$\gamma_i = v_{\gamma i}, \text{for } i = 1, 2, \ldots, M_\alpha,$$
$$\gamma_i = 0, \text{for } i = M_\alpha + 1, M_\alpha + 2, \ldots, N,$$

where $M_\alpha = [N^\alpha]$, $v_{\gamma i} \sim IIDU\,[\mu_v - 0.5, \mu_v + 0.5]$. We set $\mu_v = 1$, and consider four values for the exponent of the cross-sectional dependence of the errors, namely $\alpha = 0, 0.25, 0.5$, and

0.75. We also consider the following combinations of $N \in \{40, 50, 100, 150, 200\}$, and $T \in \{20, 50, 100, 150, 200\}$, and use 2,000 replications for all experiments.

Table 29.4 presents the findings for the CD_p, LM_{Adj} and LM_S tests. The rejection rates for J_{BFK} in all cases, including the cross-sectionally independent case of $\alpha = 0$, were all close to 100 per cent, in part due to the error variance heteroskedasticity, and are not included in Table 29.4. Panel A of Table 29.4 reports the test results for the case of strictly exogenous regressors, and Panel B gives the results for the case of weakly exogenous regressors. We see that the CD_P test continues to perform well even when the panel data model contains a lagged dependent variable and other weakly exogenous regressors, for the combination of N and T samples considered. The results also confirm the theoretical finding discussed above that shows the implicit null of the CD_P test to be $0 \leq \alpha \leq 0.25$. In contrast, the LM_{Adj} test tends to over-reject when the panel includes dynamics and T is small compared with N. For $N = 200$ and $T = 20$ the rejection rate is 14.25 per cent.[18] Furthermore, the *MC* results also suggest that the LM_{Adj} test has power when the cross-sectional dependence is very weak, namely in the case when the exponent of cross-sectional dependence is $\alpha = 0.25$. LM_S also over-rejects when T is small relative to N, but the over-rejection is much more severe as compared with the LM_{adj} test since in the weakly exogenous regressor case the test statistic is not centered at zero for a fixed T.

The over-rejection of the J_{BFK} test in these experiments is caused by a combination of several factors, including heteroskedastic errors and heterogeneous coefficients. In order to distinguish between these effects, we also conducted experiments with homoskedastic errors where we set $Var(\varepsilon_{it}) = \sigma_i^2 = 1$, for all i, and strictly exogenous regressors (by setting $\alpha_{xi} = 0$ for all i), and consider two cases for the coefficients: heterogeneous and homogeneous (we set $\beta_{i0} = E(\beta_{i0}) = 0.75$, for all i). The results under homoskedastic errors and homogeneous slopes are summarized in the upper part of Table 29.5. As to be expected, the J_{BFK} test has good size and power when $T > 20$ and $\alpha = 0$. But the test tends to over-reject when $T = 20$ and N is relatively large even under these restrictions. The bottom part of Table 29.5 presents findings for the experiments with slope heterogeneity, whilst maintaining the assumptions of homoskedastic errors and strictly exogenous regressors. We see that even a small degree of slope heterogeneity can cause the J_{BFK} test to over-reject badly .

Finally, it is important to bear in mind that even the CD_P test is likely to over-reject in the case of models with weakly exogenous regressors if N is much larger than T. Only in the case of models with strictly exogenous regressors, and pure autoregressive models with symmetrically distributed disturbances, we would expect the CD_P test to perform well even if N is much larger than T. To illustrate this property we provide empirical size and power results when $N = 1,000$ and $T = 10$ in Table 29.6. As can be seen, the CD_P test has the correct size when we consider panel data models with strictly exogenous regressors or in the case of pure $AR(1)$ models. This is in contrast to the case of panels with weakly exogenous regressors where the size of the CD_P test is close to 70 per cent. It is clear that the small sample properties of the CD_p test for very large N and small T panel very much depend on whether the panel includes weakly exogenous regressors.

[18] The rejection rates based on the LM_{Adj} test were above 90 per cent for the sample sizes $N = 500$, 1000 and $T = 10$.

Table 29.4 Size and power of *CD* and *LM* tests in the case of panels with weakly and strictly exogenous regressors (nominal size is set to 5 per cent)

	$\alpha = 0$					$\alpha = 0.25$					$\alpha = 0.5$					$\alpha = 0.75$				
(N,T)	20	50	100	150	200	20	50	100	150	200	20	50	100	150	200	20	50	100	150	200
Panel A: Experiments with strictly exogenous regressors																				
	CD_P **test**																			
40	5.65	6.25	5.15	5.15	4.95	6.20	6.50	6.20	6.25	6.75	24.60	46.70	74.60	86.35	91.95	99.50	100.00	100.00	100.00	100.00
50	5.40	4.80	5.30	5.15	5.40	5.05	5.15	6.85	7.25	6.50	28.55	52.60	78.85	90.45	96.10	99.70	100.00	100.00	100.00	100.00
100	5.45	5.45	5.40	4.40	5.45	5.10	6.15	6.75	6.60	8.20	32.50	60.15	82.55	92.95	97.45	99.95	100.00	100.00	100.00	100.00
150	4.80	4.75	4.65	4.95	5.05	5.05	5.70	5.85	5.15	6.10	31.90	56.45	83.45	92.80	97.50	100.00	100.00	100.00	100.00	100.00
200	5.85	4.70	5.25	6.60	4.50	6.00	5.80	5.30	5.55	6.40	30.00	57.60	83.65	94.15	97.95	100.00	100.00	100.00	100.00	100.00
	LM_{Adj} **test**																			
40	4.75	5.25	5.50	4.30	5.20	6.80	7.65	15.95	28.30	36.55	43.05	93.35	99.80	100.00	100.00	99.70	100.00	100.00	100.00	100.00
50	6.05	5.25	4.00	4.95	4.95	6.05	6.45	12.40	19.70	31.50	47.85	95.85	100.00	100.00	100.00	99.70	100.00	100.00	100.00	100.00
100	7.00	5.10	4.75	4.70	4.80	7.35	8.80	18.40	34.30	46.25	53.75	98.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
150	6.55	4.85	5.10	5.15	5.40	7.45	6.70	11.95	18.55	28.65	49.85	98.55	99.95	100.00	100.00	100.00	100.00	100.00	100.00	100.00
200	7.75	4.95	5.15	3.90	5.10	8.75	6.45	8.50	13.25	19.00	52.05	98.70	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
	LM_S **test**																			
40	11.35	5.30	4.70	5.70	5.30	15.75	10.85	20.10	28.15	41.50	63.35	95.65	99.70	99.95	100.00	99.80	100.00	100.00	100.00	100.00
50	17.65	6.70	5.90	5.40	4.90	18.90	11.40	15.65	21.05	31.45	73.85	97.05	99.95	99.90	100.00	99.95	100.00	100.00	100.00	100.00
100	44.70	9.40	5.80	6.15	6.15	49.70	19.80	24.80	40.00	51.05	88.65	99.20	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
150	67.25	14.30	7.25	7.30	5.45	70.40	25.85	19.35	25.20	35.40	94.15	99.70	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
200	85.10	24.45	9.45	6.55	6.60	85.75	31.10	19.85	22.05	26.05	98.20	99.85	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00

Table 29.4 Continued

	$\alpha = 0$					$\alpha = 0.25$					$\alpha = 0.5$					$\alpha = 0.75$				
(N,T)	20	50	100	150	200	20	50	100	150	200	20	50	100	150	200	20	50	100	150	200
Panel B: Experiments with lagged dependent variable and weakly exogenous regressors																				
	CD_P **test**																			
40	5.90	4.75	5.55	5.05	4.90	6.00	6.10	5.70	6.40	5.90	26.00	49.75	72.55	84.80	93.10	99.80	100.00	100.00	100.00	100.00
50	6.35	5.00	5.15	5.60	4.40	5.90	6.75	6.00	5.95	6.45	28.90	55.00	77.90	90.50	96.00	99.75	100.00	100.00	100.00	100.00
100	6.55	5.50	5.05	5.10	3.95	6.85	6.85	6.80	6.75	8.15	31.75	59.15	81.30	93.65	97.10	100.00	100.00	100.00	100.00	100.00
150	7.55	5.90	4.50	5.60	4.35	8.30	5.75	6.80	5.25	6.45	34.30	56.15	80.95	94.15	97.00	100.00	100.00	100.00	100.00	100.00
200	8.10	4.75	5.05	5.65	4.60	10.30	6.00	7.25	6.40	6.45	35.75	61.10	83.20	94.05	98.45	100.00	100.00	100.00	100.00	100.00
	LM_{Adj} **test**																			
40	5.05	4.70	5.25	4.10	5.85	6.40	6.85	15.25	26.70	38.65	32.60	92.00	99.80	99.95	100.00	99.40	100.00	100.00	100.00	100.00
50	5.80	4.90	4.70	4.90	4.90	5.30	6.15	12.10	20.80	30.85	35.65	95.55	99.85	100.00	100.00	99.65	100.00	100.00	100.00	100.00
100	6.45	5.60	5.05	4.70	4.80	7.85	7.50	18.45	31.05	47.00	36.40	98.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
150	11.55	5.95	5.30	5.20	3.85	10.35	6.50	10.35	19.65	28.60	31.60	97.65	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
200	14.25	5.50	5.35	4.90	5.25	12.85	6.00	8.20	11.55	19.25	31.55	98.80	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
	LM_S **test**																			
40	15.40	6.10	5.40	4.50	5.10	17.00	10.90	19.45	28.35	41.70	61.30	94.85	99.85	99.85	100.00	99.65	100.00	100.00	100.00	100.00
50	18.60	6.55	5.60	4.25	5.05	22.25	10.25	14.25	22.80	31.70	72.80	97.35	99.95	100.00	100.00	99.80	100.00	100.00	100.00	100.00
100	50.25	10.55	6.60	5.10	5.55	55.60	21.65	26.35	37.95	54.40	88.20	99.25	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
150	77.65	18.25	7.70	6.25	6.95	77.70	28.20	18.75	27.70	34.95	95.40	99.55	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
200	89.60	29.55	11.90	6.90	6.30	87.95	36.65	19.95	23.40	25.85	98.70	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00

Notes: Observations are generated using the equations $y_{it} = c_{yi} + \lambda_i y_{i,t-1} + \beta_{0i} x_{it} + \beta_{1i} x_{i,t-1} + u_{it}$, $x_{it} = c_{xi} + \alpha_{xi} y_{i,t-1} + \boldsymbol{\gamma}'_{xi} \mathbf{f}_t + v_{it}$, (see (29.79) and (29.61), respectively), and $u_{it} = \gamma_i g_t + \varepsilon_{it}$, (see (29.80)). Four values of $\alpha = 0, 0.25, 0.5$ and 0.75 are considered. Null of weak cross-sectional dependence is characterized by $\alpha = 0$ and $\alpha = 0.25$. In the case of panels with strictly exogenous regressors $\lambda_i = \alpha_{xi} = 0$, for all i. For a more detailed account of the MC design see Section 29.7. LM_S test statistic is computed using the fixed-effects estimates.

Table 29.5 Size and power of the J_{BFK} test in the case of panel data models with strictly exogenous regressors and homoskedastic idiosyncratic shocks (nominal size is set to 5 per cent)

	$\alpha = 0$					$\alpha = 0.25$					$\alpha = 0.5$					$\alpha = 0.75$				
(N,T)	20	50	100	150	200	20	50	100	150	200	20	50	100	150	200	20	50	100	150	200
										Experiments with homogeneous slopes										
40	7.85	5.60	5.60	5.20	5.90	21.85	53.80	79.40	86.65	92.50	82.70	99.30	100.00	100.00	100.00	99.70	100.00	100.00	100.00	100.00
50	8.90	5.90	6.00	6.10	4.20	17.85	44.90	73.75	83.75	89.10	84.35	99.90	100.00	100.00	100.00	99.85	100.00	100.00	100.00	100.00
100	9.70	6.10	5.65	5.30	5.50	19.35	52.30	81.30	91.90	95.55	88.25	100.00	100.00	100.00	100.00	99.95	100.00	100.00	100.00	100.00
150	15.00	5.90	5.30	5.10	5.60	14.65	39.60	69.80	83.95	91.00	87.95	99.95	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
200	21.30	6.60	5.30	4.60	5.60	15.90	27.45	58.70	75.45	84.55	87.10	99.95	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
										Experiments with heterogeneous slopes										
40	7.30	9.10	13.70	22.10	31.80	22.15	55.15	83.90	93.30	96.45	81.40	99.60	100.00	100.00	100.00	99.75	100.00	100.00	100.00	100.00
50	7.60	8.80	18.20	30.90	40.45	18.65	53.25	80.95	92.45	96.95	85.45	99.85	100.00	100.00	100.00	99.85	100.00	100.00	100.00	100.00
100	9.40	16.85	42.05	65.20	83.10	21.40	65.65	94.80	99.20	99.90	88.75	100.00	100.00	100.00	100.00	99.90	100.00	100.00	100.00	100.00
150	12.65	24.70	60.80	86.25	96.25	17.35	62.30	94.10	99.60	99.95	88.45	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
200	15.20	36.80	78.90	95.65	99.00	16.05	64.85	96.15	99.75	99.90	87.75	99.95	100.00	100.00	100.00	99.95	100.00	100.00	100.00	100.00

Notes: The data generating process is the same as the one used to generate the results in Table 29.4 with strictly exogenous regressors, but with two exceptions: error variances are assumed homoskedastic ($Var\left(\varepsilon_{it}\right) = \sigma_i^2 = 1$, for all i) and two possibilities are considered for the slope coefficients: heterogeneous and homogeneous (in the latter case $\beta_{i0} = E\left(\beta_{i0}\right) = 0.75$, for all i). Null of weak cross-sectional dependence is characterized by $\alpha = 0$ and $\alpha = 0.25$. See also the notes to Table 29.4. The J_{BFK} test statistic is computed using the fixed-effects estimates.

Table 29.6 Size and power of the *CD* test for large N and short T panels with strictly and weakly exogenous regressors (nominal size is set to 5 per cent)

	$\alpha = 0$	$\alpha = 0.25$	$\alpha = 0.5$	$\alpha = 0.75$
(N,T)	10	10	10	10
Panel with strictly exogenous regressors				
1,000	5.10	6.30	20.50	99.90
Pure AR(1) panel				
1,000	5.50	6.05	22.10	100.00
Dynamic panel with weakly exogenous regressors				
1,000	69.45	70.70	73.95	100.00

Notes: See the notes to Tables 29.3 and 29.4, and Section 29.7 for further details. In particular, note that null of weak cross-sectional dependence is characterized by $\alpha = 0$ and $\alpha = 0.25$, with alternatives of semi-strong and strong cross-sectional dependence given by values of $\alpha \geq 1/2$.

29.8 Application of *CCE* estimators and *CD* tests to unbalanced panels

CCE estimators can be readily extended to unbalanced panels, a situation which frequently arises in practice. Denote the set of cross-sectional units with the available data on y_{it} and $\mathbf{x}_{it}$ in period t as $\mathcal{N}_t$ and the number of elements in the set by $\#\mathcal{N}_t$. Initially, we suppose that data coverage for the dependent variables and regressors is the same and later we relax this assumption. The main complication of applying *CCE* estimator to the case of unbalanced panels is the inclusion of cross-section averages in the individual regressions. There are two possibilities regarding the units to include in the computation of cross-section averages, either based on the same number of units or based on a varying number of units. In both cases, cross-section averages should be constructed using at least a minimum number of units ($N > 20$). If the same units are used, we have

$$\bar{y}_t = \frac{1}{\#\mathcal{N}} \sum_{i \in \mathcal{N}} y_{it}, \text{ and similarly } \bar{\mathbf{x}}_t = \frac{1}{\#\mathcal{N}} \sum_{i \in \mathcal{N}} \mathbf{x}_{it},$$

for $t = \underline{t}, \underline{t}+1, \ldots, \bar{t}$ where $\mathcal{N} = \bigcap_{t=\underline{t}}^{\bar{t}} \mathcal{N}_t$ and the starting and ending points of the sample $\underline{t}$ and $\bar{t}$ are chosen to maximize the use of data subject to the constraint $\#\mathcal{N} \geq N_{\min}$.[19] The second possibility utilizes data in a more efficient way,

$$\bar{y}_t = \frac{1}{\#\mathcal{N}_t} \sum_{i \in \mathcal{N}_t} y_{it}, \text{ and } \bar{\mathbf{x}}_t = \frac{1}{\#\mathcal{N}_t} \sum_{i \in \mathcal{N}_t} \mathbf{x}_{it},$$

[19] Based on Monte Carlo experiments $N_{\min} = 20$ seems a sensible choice.

for $t = \underline{t}, \underline{t}+1, \ldots, \bar{t}$, where $\underline{t}$ and $\bar{t}$ are chosen such that $\#\mathcal{N}_t \geq N_{\min}$ for all $t = \underline{t}, \underline{t}+1, \ldots, \bar{t}$. Both procedures are likely to perform similarly when $\#\mathcal{N}$ is reasonably large, and the occurrence of missing observations is random. In cases where new cross-sectional units are added to the panel over time and such additions can have systematic influences on the estimation outcomes, it might be advisable to de-mean or de-trend the observations for individual cross-sectional units before computing the cross-section averages to be used in the *CCE* regressions.

Now suppose that the cross-section coverage differs for each variable. For example, the dependent variable can be available only for *OECD* countries, whereas some of the regressors could be available for a larger set of countries. Then it is preferable to utilize also data on non-*OECD* countries to maximize the number of units for the computation of cross-section averages for each of the individual variables.

The *CD* and *LM* tests can also be readily extended to unbalanced panels. Denote by $\mathcal{T}_i$, the set of dates over which time series observations on y_{it} and $\mathbf{x}_{it}$ are available for the ith individual, and denote the number of elements in the set by $\#\mathcal{T}_i$. For each i compute the *OLS* residuals based on the full set of available time series observations. As before, denote these residuals by $\hat{u}_{it}$, for $t \in \mathcal{T}_i$, and compute the pair-wise correlations of $\hat{u}_{it}$ and $\hat{u}_{jt}$ using the common set of data points in $\mathcal{T}_i \cap \mathcal{T}_j$. Since in such cases the estimated residuals need not sum to zero over the common sample period, ρ_{ij} should be estimated by

$$\hat{\rho}_{ij} = \frac{\sum_{t\in\mathcal{T}_i\cap\mathcal{T}_j}\left(\hat{u}_{it}-\bar{\hat{u}}_i\right)\left(\hat{u}_{jt}-\bar{\hat{u}}_j\right)}{\left[\sum_{t\in\mathcal{T}_i\cap\mathcal{T}_j}\left(\hat{u}_{it}-\bar{\hat{u}}_i\right)^2\right]^{1/2}\left[\sum_{t\in\mathcal{T}_i\cap\mathcal{T}_j}\left(\hat{u}_{jt}-\bar{\hat{u}}_j\right)^2\right]^{1/2}},$$

where

$$\bar{\hat{u}}_i = \frac{\sum_{t\in\mathcal{T}_i\cap\mathcal{T}_j}\hat{u}_{it}}{\#\left(\mathcal{T}_i\cap\mathcal{T}_j\right)}.$$

The *CD* (similarly the *LM* type) statistics for the unbalanced panel can then be computed as usual by

$$CD_P = \sqrt{\frac{2}{N(N-1)}}\left(\sum_{i=1}^{N-1}\sum_{j=i+1}^{N}\sqrt{T_{ij}}\hat{\rho}_{ij}\right), \tag{29.81}$$

where $T_{ij} = \#\left(\mathcal{T}_i\cap\mathcal{T}_j\right)$.

29.9 Further reading

Further discussion on econometric methods for panel data under error cross-sectional dependence can be found in Andrews (2005), Pesaran (2006), and Chudik, Pesaran, and Tosetti (2011).

29.10 Exercises

1. Consider the following 'star' model

$$\mathbf{z}_t = \mathbf{R}\boldsymbol{\varepsilon}_t,$$

where $\boldsymbol{\varepsilon}_t = (\varepsilon_{1t}, \ldots, \varepsilon_{Nt})', \varepsilon_{it} \sim IID\left(0, \sigma_\varepsilon^2\right)$,

$$\mathbf{R} = \begin{pmatrix} 1 & 0 & \cdots & 0 & 0 \\ r_{21} & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ r_{N-1,1} & 0 & \cdots & 1 & 0 \\ r_{N1} & 0 & \cdots & & 1 \end{pmatrix},$$

and $\frac{1}{N-1}\sum_{i=2}^{N}|r_{i1}|$ is bounded away from zero. Prove that $Var\left(\mathbf{w}'\mathbf{z}_t\right) > 0$ for any N and as $N \to \infty$, where $\mathbf{w} = (w_{i1}, w_{i2}, \ldots, w_{iN})'$ satisfies granularity conditions (29.1)–(29.2).

2. Consider the single factor model

$$u_{it} = \gamma_i f_t + \varepsilon_{it}, i = 1, \ldots, N; t = 1, 2, \ldots, T, \tag{29.82}$$

with $\varepsilon_{it} \sim IID(0, \sigma_\varepsilon^2)$, $f_t \sim IID(0, 1)$, and γ_i are fixed coefficients. Write down $E\left(u_{it}^2\right)$ and $E\left(u_{it}u_{jt}\right)$ the elements of the covariance matrix, $\boldsymbol{\Sigma}$, of $\mathbf{u}_{.t} = (u_{1t}, u_{2t}, \ldots, u_{Nt})'$. Hence, derive the largest eigenvalue of $\boldsymbol{\Sigma}$ and check conditions for the $\{u_{it}\}$ process to be CSD.

3. Consider the single factor model (29.82), with $\varepsilon_{it} \sim IID(0, \sigma_\varepsilon^2)$ and $f_t \sim IID(0, 1)$. Assume that γ_i for $i = 1, 2, \ldots, N$ are fixed coefficients.

   (a) Find a set of weights, $\mathbf{w} = (w_1, w_2, \ldots, w_N)'$, such that $Var\left(\mathbf{w}'\mathbf{u}_{.t}\right) > 0$, for all N.

   (b) Derive the correlation matrix of $\mathbf{u}_{.t} = (u_{i1}, u_{i2}, \ldots, u_{iN})'$, and use the elements of this correlation matrix to write an expression for the statistics CD_P and $\bar{\rho}$ (see (29.76)).

   (c) Find conditions on the loadings, γ_i, which ensure $\bar{\rho} \neq 0$, even if $N \to \infty$.

4. Consider the following panel data model

$$y_{it} = \beta x_{it} + u_{it},\ i = 1, 2, \ldots, N;\ t = 1, 2, \ldots, T,$$

where

$$\begin{aligned} x_{it} &= \delta_i f_t + v_{it} \\ u_{it} &= \gamma_i f_t + \varepsilon_{it}, \end{aligned}$$

f_t is a covariance stationary unobserved common factor, and the errors u_{it}, v_{it} and ε_{it} are serially and cross-sectionally independently distributed with zero means and finite variances σ_{iu}^2, σ_{iv}^2, and $\sigma_{i\varepsilon}^2$, respectively. Further assume that f_t is distributed independently of these errors.

(a) Discuss the statistical properties of the pooled *OLS* estimator of β in the case where $\delta_i = 0$. In particular show that the pooled *OLS* is unbiased but inefficient if $\delta_i = 0$.

(b) Derive an expression for the probability limit of the pooled *OLS* in the general case where $\delta_i \neq 0$, as N and T tend to infinity either sequentially or jointly. Hence or otherwise show that the pooled *OLS* estimator of β is inconsistent.

(c) Is SURE estimation likely to help under case (b)?

30 Spatial Panel Econometrics

30.1 Introduction

This chapter reviews econometric methods for linear panel data models that exhibit spatial dependence. Spatial dependence may arise from local interaction of individuals, or from unobserved characteristics that show persistence across space or over a network. In order to measure spatial correlation, we need to specify the modalities of how agents interact and define a metric of distance between individuals. 'Local' does not need to be specified in terms of physical space, but can be related to other types of metrics, such as the economic, policy, or social distance.

A sizeable literature has analysed the role of interactions and externalities in several different branches of economics, both at a theoretical and at an empirical level. For instance, a rich literature in microeconomics explores the decision-making process of an agent embedded in a system of social relations, where he/she can watch other agents' actions (Bala and Goyal (2001), Brock and Durlauf (2001)). A key finding is that local interaction may allow some forms of behaviour to propagate to the entire population (Ellison (1993)). Recent studies in macroeconomics have theorized the existence of strategic complementaritis that produce aggregate fluctuations in industrial market economies (Cooper and Haltiwanger (1996); Binder and Pesaran (1998)). Factors such as input–output linkages, demand complementarities and human capital spillovers have been used to explain observed comovements not attributable to aggregate shocks (Aoki (1996)). Finally, literature on endogenous growth has emphasized the importance of linkages between countries in the analysis of regional income growth (Rivera-Batiz and Romer (1991); Barro and Sala-i-Martin (2003); Arbia (2006)). According to this literature, relations established with neighbouring regions, in the form of demand linkages, interacting labour markets and knowledge spillovers, also due to the increased economic integration between developed economies, are determinants of regional economic growth.

Spatial correlation can also be caused by a variety of measurement problems often encountered in applied work, or by the particular sampling scheme used to select units. An example is the lack of concordance between the delineation of observed spatial units, such as the region or the country, and the spatial scope of the phenomenon under study (Anselin (1988)). When the sampling scheme is clustered, potential correlation may also arise between respondents

belonging to the same cluster. Indeed, units sharing observable characteristics such as location or industry, may also have similar unobservable characteristics that would cause the regression disturbances to be correlated (Moulton (1990), Pepper (2002)).

This chapter provides a survey of econometric methods proposed to deal with spatial dependence in the context of linear panel data regression models.

30.2 Spatial weights and the spatial lag operator

In spatial econometrics, neighbourhood effects are typically characterized by means of a non-negative *spatial weights matrix*. The rows and columns of this matrix, often denoted by $\mathbf{W} = (w_{ij})$, correspond to the cross-section observations (e.g., individuals, regions, or countries), and the generic element, w_{ij}, can be interpreted as the strength of potential interaction between units i and j. The specification of $\mathbf{W}$ is typically based on some measure of distance between units, using for example contiguity or geographic proximity, or more general metrics, such as economic (Conley (1999), Pesaran, Schuermann, and Weiner (2004)), political (Baicker (2005)), or social distance (Conley and Topa (2002)). The weights, w_{ij}, are set exogenously, although there are some recent attempts at endogenizing the determination of the weights. By convention, the diagonal elements of the weighting matrix are set to zero, implying that an observation is not a neighbour to itself. To facilitate the interpretation of the estimates, $\mathbf{W}$ is typically row-standardized so that the sum of the weights for each row is one, ensuring that all the weights are between 0 and 1. Finally, although most empirical works assume that weights are time-invariant, these can vary over time (see, for example, Druska and Horrace (2004) and Cesa-Bianchi et al. (2012)).

An important role in spatial econometrics is played by the notion of *spatial lag operator*. Let $\mathbf{W} = \left(w_{ij}\right)$ be a time-invariant $N \times N$ spatial weights matrix. The spatial lag of $\mathbf{z}_{.t} = (z_{1t}, z_{2t}, \ldots, z_{Nt})'$ is defined by $\mathbf{W}\mathbf{z}_{.t}$, with generic i^{th} element given by $\mathbf{w}_i'\mathbf{z}_{.t} = \sum_{j=1}^{N} w_{ij} z_{jt}$, where $\mathbf{w}_i'$ is the ith row of $\mathbf{W}$. Hence, a spatial lag operator constructs a new variable that is a weighted average of neighbouring observations, with weights reflecting distance among units. The incorporation of these spatial lags into a regression equation is considered in the next section.

30.3 Spatial dependence in panels

30.3.1 Spatial lag models

Spatial dependence can be incorporated into a panel data model by including spatial lags of the dependent variable among the regressors. Under this specification,

$$y_{it} = \alpha_i + \rho \sum_{j=1}^{N} w_{ij} y_{jt} + \boldsymbol{\beta}' \mathbf{x}_{it} + u_{it}, \quad i = 1, 2, \ldots, N; t = 1, 2, \ldots, T, \tag{30.1}$$

where $\mathbf{x}_{it}$ is a $k \times 1$ vector of observed regressors on the ith cross-sectional unit at time t, u_{it} is the error term, and ρ and $\boldsymbol{\beta}$ are unknown parameters to be estimated. The group or individual

effects, α_i, could be either considered fixed, unknown parameters to be estimated, or draws from a probability distribution. For the time being, it is assumed that regressors are strictly exogenous. The above specification is typically considered for representation of the equilibrium outcome of a spatial or social interaction process in which the value of the dependent variable for one individual is jointly determined with that of the neighbouring individual (Anselin, Le Gallo, and Jayet (2007)).

It is now easily seen that estimation of ρ and $\boldsymbol{\beta}$ by least squares applied to (30.1) can lead to biased and inconsistent estimates. It is sufficient to show that $Cov\left(\sum_{j=1}^{N} w_{ij}y_{jt}, u_{it}\right) \neq 0$ when $\rho \neq 0$. To see this, it is convenient to rewrite the model in stacked form as

$$\mathbf{y}_{.t} = \boldsymbol{\alpha} + \rho\mathbf{W}\mathbf{y}_{.t} + \mathbf{X}_{.t}\boldsymbol{\beta} + \mathbf{u}_{.t}, \tag{30.2}$$

where $\mathbf{y}_{.t} = \left(y_{1t}, y_{2t}, \ldots, y_{Nt}\right)'$, $\boldsymbol{\alpha} = (\alpha_1, \alpha_2, \ldots, \alpha_N)'$, $\mathbf{X}_{.t} = (\mathbf{x}_{1t}, \mathbf{x}_{2t}, \ldots, \mathbf{x}_{Nt})'$, $\mathbf{u}_{.t} = \left(u_{1t}, u_{2t}, \ldots, u_{Nt}\right)' \sim IID(\mathbf{0}, \mathbf{D})$, and $\mathbf{D}$ is an $N \times N$ diagonal matrix with elements $0 < \sigma_i^2 < K$.

To solve the above model, we first need to establish conditions under which $\mathbf{I}_N - \rho\mathbf{W}$ is invertible. To this end note that the eigenvalues of $\mathbf{I}_N - \rho\mathbf{W}$ are given by $1 - \lambda(\rho\mathbf{W})$, and $\mathbf{I}_N - \rho\mathbf{W}$ is invertible if $|\lambda_{\max}(\rho\mathbf{W})| < 1$, where $\lambda_{\max}(\mathbf{A})$ denotes the largest eigenvalue of matrix $\mathbf{A}$. This condition can also be written in terms of column and row norms of $\mathbf{W}$. Since $|\lambda_{\max}(\rho\mathbf{W})| \leq |\rho| \, \|\mathbf{W}\|$, where $\|\mathbf{W}\|$ is any matrix norm of $\mathbf{W}$, then we also have that $|\lambda_{\max}(\rho\mathbf{W})| \leq |\rho| \, \|\mathbf{W}\|_1$ and $|\lambda_{\max}(\rho\mathbf{W})| \leq |\rho| \, \|\mathbf{W}\|_\infty$, where $\|\mathbf{W}\|_1$ and $\|\mathbf{W}\|_\infty$ are, respectively, the column and row matrix norms of $\mathbf{W}$. Therefore, invertibility of $\mathbf{I}_N - \rho\mathbf{W}$ is ensured if $|\rho| < \max\left(1/\|\mathbf{W}\|_1, 1/\|\mathbf{W}\|_\infty\right)$. This condition can also be written equivalently as $|\rho| < 1/\tau^*$, where $\tau^* = \min\left(\|\mathbf{W}\|_1, \|\mathbf{W}\|_\infty\right)$ (see Kelejian and Prucha (2010)). Under this condition we have

$$\mathbf{y}_{.t} = \left(\mathbf{I}_N - \rho\mathbf{W}\right)^{-1}\left(\boldsymbol{\alpha} + \mathbf{X}_{.t}\boldsymbol{\beta} + \mathbf{u}_{.t}\right).$$

Also, let $\mathbf{w}_i' \, \mathbf{y}_{.t} = \sum_{j=1}^{N} w_{ij}y_{jt}$ and since $\mathbf{w}_i$ is exogenously given we have

$$\begin{aligned} Cov\left(\mathbf{w}_i'\mathbf{y}_{.t}, u_{it}\right) &= Cov\left[\mathbf{w}_i'\left(\mathbf{I}_N - \rho\mathbf{W}\right)^{-1}\left(\boldsymbol{\alpha} + \mathbf{X}_{.t}\boldsymbol{\beta} + \mathbf{u}_{.t}\right), \, u_{it}\right] \\ &= Cov\left[\mathbf{w}_i'\left(\mathbf{I}_N - \rho\mathbf{W}\right)^{-1}\mathbf{u}_{.t}, \, \mathbf{e}_i'\mathbf{u}_{.t}\right], \end{aligned}$$

where $\mathbf{e}_i$ is an $N \times 1$ selection vector such that $\mathbf{e}_i'\mathbf{u}_{.t} = u_{it}$. Hence

$$Cov\left(\mathbf{w}_i'\mathbf{y}_{.t}, u_{it}\right) = \mathbf{w}_i'\left(\mathbf{I}_N - \rho\mathbf{W}\right)^{-1}\mathbf{D}\mathbf{e}_i.$$

First, it is easily verified that $Cov\left(\mathbf{w}_i'\mathbf{y}_{.t}, u_{it}\right) = 0$, if $\rho = 0$. In this case, $Cov\left(\mathbf{w}_i'\mathbf{y}_{.t}, u_{it}\right) = \mathbf{w}_i'\mathbf{D}\mathbf{e}_i = w_{ii}\sigma_i^2 = 0$, since $w_{ii} = 0$, by assumption. But when $\rho \neq 0$, we have[1]

$$\begin{aligned} \mathbf{w}_i'\left(\mathbf{I}_N - \rho\mathbf{W}\right)^{-1}\mathbf{D}\mathbf{e}_i &= \mathbf{w}_i'\mathbf{D}\mathbf{e}_i + \rho\mathbf{w}_i'\mathbf{W}\mathbf{D}\mathbf{e}_i + \rho^2\mathbf{w}_i'\mathbf{W}^2\mathbf{D}\mathbf{e}_i + \ldots \\ &= \rho\sigma_i^2\left(\mathbf{w}_i'\mathbf{W}\mathbf{e}_i + \rho\mathbf{w}_i'\mathbf{W}^2\mathbf{e}_i + \ldots\right). \end{aligned}$$

[1] Under condition $|\rho| < \max\left(1/\|\mathbf{W}\|_1, 1/\|\mathbf{W}\|_\infty\right)$, we have $\left(\mathbf{I}_N - \rho\mathbf{W}\right)^{-1} = \mathbf{I}_N + \rho\mathbf{W} + \boldsymbol{\rho}^2\mathbf{W}^2 + \ldots$. See also Section 29.3.

But $\mathbf{w}_i'\mathbf{W}\mathbf{e}_i = \sum_{\ell=1}^{N} w_{i\ell}w_{\ell i}$ and given that $w_{ij} \geq 0$, then $\mathbf{w}_i'\mathbf{W}^j\mathbf{e}_i \geq 0$ for $j = 1, 2, \ldots$. Hence, it must follow that $Cov\left(\mathbf{w}_i'\mathbf{y}_{.t}, u_{it}\right) > 0$ if $\rho > 0$ and $\sum_{\ell=1}^{N} w_{i\ell}w_{\ell i} \neq 0$. The last condition holds if there are non-zero elements in the ith column and row of $\mathbf{W}$. Also asymptotically (as $N \to \infty$) we need to have $\lim_{N\to\infty}\left(\sum_{\ell=1}^{N} w_{i\ell}w_{\ell i}\right) > 0$, which rules out the possibility of spatial weights to be granular, namely $w_{ij} = O(N^{-1})$. In such a case, $\sum_{\ell=1}^{N} w_{i\ell}w_{\ell i} = O(N^{-1})$ and $\lim_{N\to\infty} Cov(\mathbf{w}_i'\mathbf{y}_{.t}, u_{it}) = 0$, for each i.

We can therefore conclude that, in the case of non-granular spatial weights and assuming that $|\rho| < \max\left(1/\left\|\mathbf{W}\right\|_1, 1/\left\|\mathbf{W}\right\|_\infty\right)$, conventional estimators of parameters ρ and $\boldsymbol{\beta}$ (such as pooled *OLS* or *FE*) are inconsistent, and alternative estimation approaches, such as maximum likelihood and generalized method of moments, are needed for consistent estimation of spatial lag models.

30.3.2 Spatial error models

Another way to include spatial dependence in the regression equation is to allow the disturbances to be spatially correlated. Consider the simple linear regression in stacked form

$$\mathbf{y}_{.t} = \boldsymbol{\alpha} + \mathbf{X}_{.t}\boldsymbol{\beta} + \mathbf{u}_{.t}, \tag{30.3}$$

where the notation is as above. There exist few main approaches to assign a spatial structure to the error term, $\mathbf{u}_{.t}$;[2] the intent is to represent the covariance as a simpler, lower dimensional matrix than the unconstrained version

One way is to define the covariance between two observations directly as a function of the distance between them. Accordingly, the covariance matrix for the cross-section at time t is $E(\mathbf{u}_{.t}\mathbf{u}_{.t}') = f(\boldsymbol{\theta}, \mathbf{W})$, where $\boldsymbol{\theta}$ is a parameter vector, and f is a suitable distance decay function, such as the negative exponential (Dubin (1988), Cressie (1993); see also Example 75). The decaying function suggests that the disturbances should become uncorrelated when the distance separating the observations is sufficiently large. One shortcoming of this method is that it requires the specification of a functional form for the distance decay, which is subject to a degree of arbitrariness.

An alternative strategy consists of specifying a spatial process for the error term, which relates each unit to its neighbours through $\mathbf{W}$. The most widely used model is the spatial autoregressive (SAR) specification. Proposed by Cliff and Ord (1969) and Cliff and Ord (1981), the SAR process is a variant of the model introduced by Whittle (1954)

$$\mathbf{u}_{.t} = \delta\mathbf{W}\mathbf{u}_{.t} + \boldsymbol{\varepsilon}_{.t}, \tag{30.4}$$

where δ is a scalar parameter, and $\boldsymbol{\varepsilon}_{.t} = (\varepsilon_{1t}, \varepsilon_{2t}, \ldots, \varepsilon_{Nt})'$, with $\boldsymbol{\varepsilon}_{.t} \sim IID(\mathbf{0}, \sigma_\varepsilon^2\mathbf{I}_N)$. Assuming that the matrix $\mathbf{I}_N - \delta\mathbf{W}$ is invertible, (30.4) can be rewritten in reduced form as

$$\mathbf{u}_{.t} = (\mathbf{I}_N - \delta\mathbf{W})^{-1}\boldsymbol{\varepsilon}_{.t},$$

so that $\mathbf{u}_{.t}$ has the covariance matrix

[2] As we shall see in Section 30.4.3, if $\boldsymbol{\alpha}$ is assumed to be random, then a spatial structure could also be assigned to it.

$$\boldsymbol{\Sigma}_{SAR} = \sigma_{\varepsilon}^2(\mathbf{I}_N - \delta\mathbf{W})^{-1}(\mathbf{I}_N - \delta\mathbf{W}')^{-1}.$$

Other spatial processes suggested to model spatial error dependence, although less used in the empirical literature, are the spatial moving average (SMA) and the spatial error component (SEC) specifications. The first, proposed by Haining (1978) (see also Huang (1984)), assumes that

$$\mathbf{u}_{.t} = \delta\mathbf{W}\boldsymbol{\varepsilon}_{.t} + \boldsymbol{\varepsilon}_{.t}, \tag{30.5}$$

where $\boldsymbol{\varepsilon}_{.t}$ is as above. Its covariance matrix is

$$\boldsymbol{\Sigma}_{SMA} = \sigma_{\varepsilon}^2(\mathbf{I}_N + \delta\mathbf{W})(\mathbf{I}_N + \delta\mathbf{W}').$$

According to the SEC specification, introduced by Kelejian and Robinson (1995),

$$\mathbf{u}_{.t} = \delta\mathbf{W}\boldsymbol{\psi}_{.t} + \boldsymbol{\varepsilon}_{.t}, \tag{30.6}$$

where $\boldsymbol{\psi}_{.t} = \left(\psi_{1t}, \psi_{2t}, \ldots, \psi_{Nt}\right)'$ and $\psi_{it} \sim IID\left(0, \sigma_{\Psi}^2\right)$. The covariance matrix induced by this model is

$$\boldsymbol{\Sigma}_{SEC} = \delta^2\sigma_{\Psi}^2\mathbf{W}\mathbf{W}' + \sigma_{\varepsilon}^2\mathbf{I}_N.$$

A major distinction between the SAR and the other two specifications is that under SAR there is an inverse involved in the covariance matrix. This has important consequences on the range of dependence implied by its covariance matrix. Indeed, even if $\mathbf{W}$ contains few non-zero elements, the covariance structure induced by the SAR is not sparse, linking all the units in the system to each other, so that a perturbation in the error term of one unit will be ultimately transmitted to all other units. Conversely, for the SMA and SEC, the off-diagonal non-zero elements of the covariance matrix are those corresponding to the non-zero elements in $\mathbf{W}$.

Conventional panel estimators introduced in Chapter 26, such as the fixed-effects (FE) or random effects (RE) estimators of slope coefficients in equation (30.3) with spatially dependent errors, are $\sqrt{NT}$-consistent under broad regularity conditions and strictly exogenous regressors. However, these estimators are in general not efficient since the covariance of errors is non-diagonal and the elements along its main diagonal are in general not constant.

30.3.3 Weak cross-sectional dependence in spatial panels

Spatial models are often formulated in such a way that each cross-sectional unit has a limited number of neighbours regardless of the sample size. To see this, first note that, under certain invertibility conditions, the spatial processes (30.4)–(30.6) can all be written as special cases of the following general form

$$\mathbf{u}_{.t} = \mathbf{R}\boldsymbol{\varepsilon}_{.t}, \tag{30.7}$$

where $\mathbf{R} = (r_{ij})$ is an $N \times N$ matrix, and $\boldsymbol{\varepsilon}_{.t} \sim IID(\mathbf{0}, \mathbf{D})$, with $\mathbf{D} = diag(\sigma^2_{\varepsilon 1}, \sigma^2_{\varepsilon 2}, \ldots, \sigma^2_{\varepsilon N})$, $\sigma^2_{\max} = \sup_i \left(\sigma^2_{\varepsilon i}\right) < K$. For example, for an invertible SAR process $\mathbf{R} = (\mathbf{I}_N - \rho \mathbf{W})^{-1}$, while in the case of a SMA process, we have $\mathbf{R} = \mathbf{I}_N + \delta \mathbf{W}$. It is now easily seen that (see Section A.10 in Appendix A)

$$\begin{aligned}\lambda_1(\boldsymbol{\Sigma}) = \lambda_1\left(\mathbf{R}\mathbf{D}\mathbf{R}'\right) &\leq \sup_i \left(\sigma^2_{\varepsilon i}\right) \left\|\mathbf{R}\mathbf{R}'\right\|_1 \\ &\leq \sigma^2_{\max} \|\mathbf{R}\|_1 \left\|\mathbf{R}'\right\|_1 = \sigma^2_{\max} \|\mathbf{R}\|_1 \|\mathbf{R}\|_\infty .\end{aligned}$$

Hence assuming that $\mathbf{R}$ has bounded row and column matrix norms, namely $\|\mathbf{R}\|_\infty$ and $\|\mathbf{R}\|_1 < K$, then $\lambda_1(\boldsymbol{\Sigma})$ is also bounded in N. Under these conditions spatial processes lead to cross-sectional dependence that is weak (see Chudik, Pesaran, and Tosetti (2011), and Section 29.2). For SMA process $\|\mathbf{R}\|_\infty$ and $\|\mathbf{R}\|_1 < K$ if the spatial weights, $\mathbf{W}$, have bounded row and column matrix norms. For SAR models it is further required that $|\rho| < \max\left(1/\|\mathbf{W}\|_1, 1/\|\mathbf{W}\|_\infty\right)$. In the case where $\mathbf{W}$ is row and column standardized the latter condition reduces to $|\rho| < 1$.

It is also interesting to observe that, under these conditions, the above process can be represented by a factor process with an infinite number of weak factors, and no idiosyncratic error, by setting $u_{it} = \sum_{j=1}^N \gamma_{ij} f_{jt}$, where $\gamma_{ij} = r_{ij}$, and $f_{jt} = \varepsilon_{jt}$, for $i, j = 1, \ldots, N$. Under the bounded column and row norms of $\mathbf{R}$, the loadings in the above factor structure satisfy condition (29.12) in Chapter 29, and hence u_{it} will be a cross-sectionally weakly dependent (CWD) process.

30.4 Estimation

30.4.1 Maximum likelihood estimator

The theoretical properties of quasi-maximum likelihood (*ML*) estimator in a single cross-sectional framework have been studied by Ord (1975), Anselin (1988), and Lee (2004), among others. More recently, considerable work has been undertaken to investigate the properties of *ML* estimators in panel data contexts, in the presence of spatial dependence and unobserved, time-invariant heterogeneity (Elhorst (2003); Baltagi, Song, and Koh (2003); Baltagi, Egger, and Pfaffermayr (2013); and Lee and Yu (2010a)).

30.4.2 Fixed-effects specification

For *ML* estimation of spatial regression models, it is convenient to consider the general case of a spatial lag model having *SAR* errors

$$\mathbf{y}_{.t} = \boldsymbol{\alpha} + \rho \mathbf{W}_1 \mathbf{y}_{.t} + \mathbf{X}_{.t} \boldsymbol{\beta} + \mathbf{u}_{.t}, \tag{30.8}$$

$$\mathbf{u}_{.t} = \delta \mathbf{W}_2 \mathbf{u}_{.t} + \boldsymbol{\varepsilon}_{.t}, \tag{30.9}$$

where the spatial lags in the dependent variable and in the error term are constructed using two (possibly different) spatial weights matrices, $\mathbf{W}_1$ and $\mathbf{W}_2$. Suppose that the group effects, α_i, are treated as fixed and unknown parameters, and that $\varepsilon_{it} \sim IID(0, \sigma^2_\varepsilon)$. Lee and Yu (2010a)

propose a transformation of the above model to get rid of the fixed-effects, and then use *ML* to estimate the remaining parameters, ρ, $\boldsymbol{\beta}, \delta$ and σ_ε^2. Specifically, the authors suggest multiplying all variables by a $T \times (T-1)$ matrix, $\mathbf{P}$, having as columns the $(T-1)$ eigenvectors associated to the non-zero eigenvalues of the deviation from the mean transformation, $\mathbf{M}_T = \mathbf{I}_T - \boldsymbol{\tau}_T \left(\boldsymbol{\tau}_T' \boldsymbol{\tau}_T\right)^{-1} \boldsymbol{\tau}_T'$, where $\boldsymbol{\tau}_T$ is a T-dimensional vector of ones. Let $\mathbf{Z} = (\mathbf{z}_{.1}, \mathbf{z}_{.2}, \ldots, \mathbf{z}_{.T})$ be an $N \times T$ matrix of variables and let $\mathbf{Z}^* = \mathbf{ZP}$, with $\mathbf{Z}^* = \left(\mathbf{z}_{.1}^*, \mathbf{z}_{.2}^*, \ldots, \mathbf{z}_{.T}^*\right)$, be the corresponding transformed matrix of variables. It is easily seen that $\boldsymbol{\tau}_T' \mathbf{P} = \mathbf{0}$, so that such a transformation removes the individual-specific intercepts. The transformed model is

$$\mathbf{y}_{.t}^* = \rho \mathbf{W}_1 \mathbf{y}_{.t}^* + \mathbf{X}_{.t}^* \boldsymbol{\beta} + \mathbf{u}_{.t}^*, \tag{30.10}$$

$$\mathbf{u}_{.t}^* = \delta \mathbf{W}_2 \mathbf{u}_{.t}^* + \boldsymbol{\varepsilon}_{.t}^*, \quad t = 1, 2, \ldots, T-1. \tag{30.11}$$

After the transformation, the effective sample size reduces to $N(T-1)$, and, since $\mathbf{P}'\mathbf{P} = \mathbf{I}_{T-1}$, the new error term, $\boldsymbol{\varepsilon}_{.t}^*$, has uncorrelated elements, that is, $E(\boldsymbol{\varepsilon}_{.t}^* \boldsymbol{\varepsilon}_{.t}^{*\prime}) = \sigma_\varepsilon^2 \mathbf{I}_N$. The log-likelihood function associated with the equations (30.10) and (30.11) is given by

$$\begin{aligned} \ell(\boldsymbol{\theta}) = &-\frac{N(T-1)}{2} \ln(2\pi\sigma_\varepsilon^2) + (T-1)\left[\ln|\mathbf{I}_N - \rho\mathbf{W}_1|\right. \\ &\left. + \ln|\mathbf{I}_N - \delta\mathbf{W}_2|\right] - \frac{1}{2\sigma_\varepsilon^2} \sum_{t=1}^{T-1} \boldsymbol{\varepsilon}_{.t}^{*\prime} \boldsymbol{\varepsilon}_{.t}^{*\prime}, \end{aligned} \tag{30.12}$$

where $\boldsymbol{\theta} = \left(\rho, \boldsymbol{\beta}', \delta, \sigma_\varepsilon^2\right)'$, and $\boldsymbol{\varepsilon}_{.t}^* = (\mathbf{I}_N - \delta\mathbf{W}_2)\left[(\mathbf{I}_N - \rho\mathbf{W}_1)\,\mathbf{y}_{.t}^* - \mathbf{X}_{.t}^* \boldsymbol{\beta}\right]$. Subject to some identification conditions, the estimator of $\boldsymbol{\theta}$ obtained by maximizing (30.12) is consistent and asymptotically normal when either N and/or $T \to \infty$. See Lee and Yu (2010a).

30.4.3 Random effects specification

This formulation assumes that the group effects, α_i, are random and independent of the exogenous regressors. In this case, following Baltagi, Egger, and Pfaffermayr (2013), a general specification can be suggested by assuming that spatial processes apply both to the random group effects and the remainder disturbances

$$\mathbf{y}_{.t} = \rho \mathbf{W}_1 \mathbf{y}_{.t} + \mathbf{X}_{.t} \boldsymbol{\beta} + \mathbf{v}_{.t}, \tag{30.13}$$

$$\mathbf{v}_{.t} = \boldsymbol{\alpha} + \mathbf{u}_{.t}, \tag{30.14}$$

$$\boldsymbol{\alpha} = \gamma \mathbf{W}_2 \boldsymbol{\alpha} + \boldsymbol{\mu}, \tag{30.15}$$

$$\mathbf{u}_{.t} = \delta \mathbf{W}_3 \mathbf{u}_{.t} + \boldsymbol{\varepsilon}_{.t}, \tag{30.16}$$

where $\boldsymbol{\mu} = \left(\mu_1, \mu_2, \ldots, \mu_N\right)'$, and it is assumed that $\mu_i \sim IID(0, \sigma_\mu^2)$, and $\varepsilon_{it} \sim IID(0, \sigma_\varepsilon^2)$. The above model, by distinguishing between time-invariant spatial error spillovers and spatial spillovers of transitory shocks, encompasses various econometric specifications proposed in the literature as special cases. If the same spatial process applies to $\boldsymbol{\alpha}$ and $\mathbf{u}_{.t}$ (i.e., $\delta = \gamma$ and $\mathbf{W}_2 = \mathbf{W}_3$), this model reduces to that proposed by Kapoor, Kelejian, and Prucha (2007); if $\gamma = 0$, it simplifies to that considered by Anselin (1988) and Baltagi, Song, and Koh (2003).

In matrix form, equations (30.13)–(30.16) can be rewritten more compactly as

$$\mathbf{y} = \rho(\mathbf{I}_T \otimes \mathbf{W}_1)\mathbf{y} + \mathbf{X}\boldsymbol{\beta} + \mathbf{v}, \tag{30.17}$$

$$\mathbf{v} = (\boldsymbol{\tau}_T \otimes \mathbf{A}^{-1})\boldsymbol{\mu} + (\mathbf{I}_T \otimes \mathbf{B}^{-1})\boldsymbol{\varepsilon}, \tag{30.18}$$

where $\mathbf{y} = \left(\mathbf{y}'_{.1}, \mathbf{y}'_{.2}, \ldots, \mathbf{y}'_{.T}\right)'$, $\mathbf{X} = \left(\mathbf{X}'_{.1}, \mathbf{X}'_{.2}, \ldots, \mathbf{X}'_{.T}\right)'$, $\mathbf{v} = \left(\mathbf{v}'_{.1}, \mathbf{v}'_{.2}, \ldots, \mathbf{v}'_{.T}\right)'$, $\boldsymbol{\varepsilon} = \left(\boldsymbol{\varepsilon}'_{.1}, \boldsymbol{\varepsilon}'_{.2}, \ldots, \boldsymbol{\varepsilon}'_{.T}\right)'$, $\mathbf{A} = (\mathbf{I}_N - \gamma\mathbf{W}_2)$, $\mathbf{B} = (\mathbf{I}_N - \delta\mathbf{W}_3)$. The covariance matrix of $\mathbf{v}$ is

$$\boldsymbol{\Sigma}_v = \sigma_\mu^2 \left[\left(\boldsymbol{\tau}_T \boldsymbol{\tau}'_T\right) \otimes (\mathbf{A}'\mathbf{A})^{-1}\right] + \sigma_\varepsilon^2 \left[\mathbf{I}_T \otimes (\mathbf{B}'\mathbf{B})^{-1}\right], \tag{30.19}$$

and applying a set of lemmas by Magnus (1982), the inverse and determinant of $\boldsymbol{\Sigma}_v$ are

$$\boldsymbol{\Sigma}_v^{-1} = \left\{\left(\frac{1}{T}\boldsymbol{\tau}_T \boldsymbol{\tau}'_T\right) \otimes \left[T\sigma_\mu^2 (\mathbf{A}'\mathbf{A})^{-1} + \sigma_\varepsilon^2 (\mathbf{B}'\mathbf{B})^{-1}\right]^{-1}\right\} + \frac{1}{\sigma_\varepsilon^2}\left[\mathbf{M} \otimes (\mathbf{B}'\mathbf{B})\right],$$

$$|\boldsymbol{\Sigma}_v| = \left|T\sigma_\mu^2 (\mathbf{A}'\mathbf{A})^{-1} + \sigma_\varepsilon^2 (\mathbf{B}'\mathbf{B})^{-1}\right| \left|\sigma_\varepsilon^2 (\mathbf{B}'\mathbf{B})^{-1}\right|^{T-1}.$$

Thus, the log-likelihood function of the random effects (RE) model (30.13)-(30.16) is given by

$$\begin{aligned}\ell(\boldsymbol{\theta}) = &-\frac{NT}{2}\ln(2\pi) - \frac{1}{2}\ln\left|T\sigma_\mu^2 (\mathbf{A}'\mathbf{A})^{-1} + \sigma_\varepsilon^2 (\mathbf{B}'\mathbf{B})^{-1}\right| \\ &- \frac{T-1}{2}\ln\left|\sigma_\varepsilon^2 (\mathbf{B}'\mathbf{B})^{-1}\right| + T\ln|\mathbf{I}_N - \rho\mathbf{W}_1| - \frac{1}{2\sigma_\varepsilon^2}\mathbf{v}'\boldsymbol{\Sigma}_v^{-1}\mathbf{v},\end{aligned} \tag{30.20}$$

where $\boldsymbol{\theta} = \left(\rho, \boldsymbol{\beta}', \gamma, \sigma_\mu^2, \delta, \sigma_\varepsilon^2\right)'$, and $\mathbf{v} = [\mathbf{I}_T \otimes (\mathbf{I}_N - \rho\mathbf{W}_1)]\,\mathbf{y} - \mathbf{X}\boldsymbol{\beta}$. Consistency of the *ML* estimator of $\boldsymbol{\theta}$ is established in Baltagi, Egger, and Pfaffermayr (2013). Under the Kapoor, Kelejian, and Prucha (2007) RE specification, $\mathbf{A} = \mathbf{B}$, and the covariance matrix (30.19) reduces to

$$\boldsymbol{\Sigma}_v = \left[\left(\sigma_\varepsilon^2 + T\sigma_\mu^2\right)\frac{1}{T}\boldsymbol{\tau}_T \boldsymbol{\tau}'_T + \sigma_\varepsilon^2 \mathbf{M}\right] \otimes (\mathbf{A}'\mathbf{A})^{-1},$$

and its inverse and determinant simplify considerably. When some observations are missing at random, selection matrices excluding missing observations may be used to obtain $|\boldsymbol{\Sigma}_v|$ and $\boldsymbol{\Sigma}_v^{-1}$. However, the computational burden in this case may be considerable even at medium-sized N and small T (Pfaffermayr (2009)). A set of joint and conditional specification Lagrange multiplier (*LM*) tests for spatial effects within the *RE* framework are proposed by Baltagi, Egger, and Pfaffermayr (2013). These statistics allow for testing of the model (30.13)–(30.16) against its restricted counterparts: the Anselin model, the Kapoor, Kelejian, and Prucha model, and the random effects model without spatial correlation (see also Baltagi, Song, and Koh (2003); Baltagi and Liu (2008); and Baltagi and Yang (2013)).

As in the non-spatial case, the choice between the *FE* spatial model and its *RE* counterpart can be based on the Hausman test. The properties of a Hausman type specification test and an *LM* statistic for testing the *FE* versus *RE* specification are studied by Lee and Yu (2010c).

Example 73 *One of the earliest applications of the spatial RE model is by Case (1991), who studies households' demand for rice in Indonesia, using data on households within a set of districts. The author considers a regression model with district-specific random errors, which is a special case of model (30.13)–(30.16). Specifically, let $y_{i\ell}$ be the log of quantity of rice purchased by household i living in district ℓ, with $i = 1, 2, \ldots, N$, $\ell = 1, 2, \ldots, M$, and $\mathbf{y} = \left(y_{11}, y_{12}, \ldots, y_{1N}, \ldots, y_{M1}, y_{M2}, \ldots, y_{MN}\right)'$. Then the following specification is assumed for* $\mathbf{y}$

$$\mathbf{y} = \phi \mathbf{W}_{NM}\mathbf{y} + \mathbf{X}\boldsymbol{\beta} + \mathbf{u}, \tag{30.21}$$

$$\mathbf{u} = \lambda \mathbf{W}_{NM}\mathbf{u} + (\mathbf{1}_N \otimes \boldsymbol{\varphi}) + \boldsymbol{\varepsilon}, \tag{30.22}$$

where $\boldsymbol{\varepsilon}$ is an NM-dimensional vector with $\varepsilon_{it} \sim IID(0, \sigma_\varepsilon^2)$, and $\boldsymbol{\varphi} = \left(\varphi_1, \varphi_2, \ldots, \varphi_N\right)'$ is a vector of district-specific random effects uncorrelated with $\mathbf{X}$*. The log-likelihood function for the above model is*

$$\ell(\boldsymbol{\theta}) = -\frac{NT}{2}\ln(2\pi) + \ln|\mathbf{A}| + \ln|\mathbf{B}| - \ln|\boldsymbol{\Sigma}_v| - \frac{1}{2\sigma_\varepsilon^2}\mathbf{v}'\boldsymbol{\Sigma}_v^{-1}\mathbf{v},$$

where $\boldsymbol{\theta} = \left(\rho, \boldsymbol{\beta}', \delta, \sigma_\varepsilon^2\right)'$, $\mathbf{A} = \left(\mathbf{I}_{NM} - \lambda \mathbf{W}_{NM}\right)$, $\mathbf{B} = (\mathbf{I}_{NM} - \phi \mathbf{W}_{NM})$ and $\mathbf{v} = \mathbf{A}\left(\mathbf{B}\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\right)$, and $\boldsymbol{\Sigma}_v$ is the covariance matrix of the error term $(\boldsymbol{\tau}_N \otimes \boldsymbol{\varphi}) + \boldsymbol{\varepsilon}$. The model is fitted to data, using a sample of 2089 households across 141 districts, and including as exogenous regressors household expenditure per household member, the size of the household, the number of its members above the age of 10, and the mean village log price of rice, fish, housing, and fuel. Results are reported in Table 30.1. Note that empirical evidence strongly supports the presence of spatial error correlation, while it is weaker when compared with the spatial lag effect model.

Example 74 (Prediction in spatial panels) *Baltagi and Li (2006) consider the problem of prediction in a panel data regression model with spatial correlation in the context of a simple demand equation for liquor in the US, at state level. The authors consider the following panel data model with SAR errors for real per capita consumption of liquor (expressed in logs), for $t = 1, 2, \ldots, T$,*

$$\mathbf{y}_{\cdot t} = \mathbf{X}_{\cdot t}\boldsymbol{\beta} + \mathbf{v}_{\cdot t}, \tag{30.23}$$

$$\mathbf{v}_{\cdot t} = \boldsymbol{\alpha} + \mathbf{u}_{\cdot t},$$

$$\mathbf{u}_{\cdot t} = \delta \mathbf{W}\mathbf{u}_{\cdot t} + \boldsymbol{\varepsilon}_{\cdot t}, \tag{30.24}$$

where the explanatory variables include the average retail price of a 750 ml of Seagram's seven (a blended American whiskey) expressed in real terms, real per capita disposable income, and a time trend. It is assumed that $\varepsilon_{it} \sim IID(0, \sigma_\varepsilon^2)$. The authors estimate the above model both under RE and FE specifications. Under the RE hypothesis, $\alpha_i \sim IID(0, \sigma_\alpha^2)$, and it is easily seen that the covariance matrix of $\mathbf{v} = \left(\mathbf{v}_{\cdot 1}', \mathbf{v}_{\cdot 2}', \ldots, \mathbf{v}_{\cdot T}'\right)'$ is

$$\boldsymbol{\Sigma}_v = \sigma_\alpha^2\left(\boldsymbol{\tau}_T\boldsymbol{\tau}_T' \otimes \mathbf{I}_N\right) + \sigma_\varepsilon^2\left[\mathbf{I}_T \otimes (\mathbf{B}'\mathbf{B})^{-1}\right],$$

where $\mathbf{B} = \mathbf{I}_N - \delta\mathbf{W}$. The parameters $\boldsymbol{\beta}, \sigma_\alpha^2, \delta, \sigma_\varepsilon^2$ are then estimated by ML. Under the FE framework, $\alpha_1, \alpha_2, \ldots, \alpha_N$ are treated as fixed unknown parameters and the authors estimate

Table 30.1 *ML* estimates of spatial models for household rice consumption in Indonesia

Model (4.1): No district specific effects or spatial correlation ($\varphi = \tau = \phi = 0$).
Model (4.2): District specific effects, but no spatial correlation ($\varphi \neq 0, \tau = \phi = 0$).
Model (4.3): Spatial correlation in dependent variable ($\varphi \neq 0, \tau = 0, \phi \neq 0$).
Model (4.4): Spatial correlation in errors ($\varphi \neq 0, \tau \neq 0, \phi = 0$).
Model (4.5): Spatial correlation in both ($\varphi \neq 0, \tau \neq 0, \phi \neq 0$).

	Model Estimates				
Explanatory variables[a] (standard errors)	(4.1)	(4.2)	(4.3)	(4.4)	(4.5)
Log expend per household member	0.1259	0.1111	0.1101	0.1094	0.1095
	(.0173)	(.0154)	(.0154)	(.0158)	(.0154)
Number of household members	0.1762	0.1679	0.1670	0.1667	0.1667
	(.0064)	(.0060)	(.0060)	(.0056)	(.0060)
Number of adults in household	0.0195	0.0314	0.0323	0.0329	0.0329
	(.0081)	(.0080)	(.0078)	(.0071)	(.0078)
Village log price market rice	–0.4786	–0.3978	-0.4607	–0.4190	–0.4210
	(.0830)	(.1049)	(.1042)	(.1073)	(.1064)
Village log price fish	0.0018	0.0343	0.0293	0.0313	0.0314
	(.0256)	(.0296)	(.0292)	(.0295)	(.0295)
Village log price fuel	0.2631	0.0605	0.0512	0.0477	0.0479
	(.0334)	(.0369)	(.0372)	(.0352)	(.0374)
Village log price housing	0.0295	0.0605	0.0512	0.0477	0.0479
	(.0095)	(.0126)	(.0124)	(.0123)	(.0128)
τ—coefficient of spatial correlation in errors	–	–	–	0.4529	0.4401
				(.0804)	(.0956)
ϕ—coefficient of spatial correlation in dep var	–	–	0.3970	–	0.0152
			(.0737)		(.0420)
σ^2_ε—household variance	0.1124	0.1088	0.1088	0.1088	0.1088
σ^2_φ—district variance	–	0.0958	0.0687	0.0646	0.0649
Chi-square test statistic[b]	1161.26	41.58	5.70	0.03	
	(0.99)	(0.99)	(0.91)	(0.09)	

[a] Intercept not reported. Standard errors are estimated using the outer-product of first partial derivatives of the log likelihood function.
[b] LR test of equality in log likelihood between each column and column 5. Probability of correctly rejecting null hypothesis of equality in likelihoods is presented in parentheses.

them by ML, jointly with other parameters ($\boldsymbol{\beta}$, δ, and σ^2_ε). Hence, they consider the following best linear unbiased predictor for the ith state at a future period $T + S$ under the RE framework

$$\hat{y}_{i,T+S} = \hat{\boldsymbol{\beta}}'_{RE}\mathbf{x}_{i,T+S} + T\theta \sum_{j=1}^{N} v^{ij}\bar{\hat{\varepsilon}}_{j.},$$

where $\theta = \hat{\sigma}^2_\alpha/\hat{\sigma}^2_\varepsilon$, v^{ij} *is the* $(i,j)^{th}$ *element of* $\mathbf{V}^{-1}$, *with* $\mathbf{V} = T\theta\mathbf{I}_N + (\mathbf{B}'\mathbf{B})^{-1}$, *and* $\bar{\hat{\varepsilon}}_{j.} = T^{-1}\sum_{t=1}^{T}\hat{\varepsilon}_{jt}$. *Under the FE framework, the*

$$\hat{y}_{i,T+S} = \hat{\boldsymbol{\beta}}'_{FE}\mathbf{x}_{i,T+S} + \hat{\alpha}_{i,FE},$$

Table 30.2 Estimation and RMSE performance of out-of-sample forecasts (estimation sample of twenty-five years; prediction sample of five years)

Estimation results

	Price	Income	Year
Poded *OLS*	−0.774 (0.088)	1.468 (0.065)	−0.062 (0.004)
Pooled spatial	−0.819 (0.093)	1.605 (0.070)	−0.067 (0.004)
Average heterogeneous *OLS*	−0.584 (0.064)	1.451 (0.041)	
Average spatial *MLE*	−0.766 (0.062)	1.589 (0.044)	
FE	−0.679 (0.044)	0.938 (0.063)	−0.049 (0.002)
FE-spatial	−0.314 (0.044)	0.612 (0.075)	−0.029 (0.002)
RE	−0.682 (0.044)	0.959 (0.062)	−0.049 (0.002)
RE-spadal	−0.317 (0.045)	0.654 (0.075)	−0.030 (0.002)

Notes:
[a] The numbers in parentheses are standard errors.
[b] The F-test for H_0; $\mu = 0$ in the FE model is $F(42, 1029) = 165.79$, with $p = 0.000$.
[c] The Breusch–Pagan test for H_0; $\sigma_\mu^2 = 0$ in the RE model is 97.30, with $p = 0.000$.
[d] The Hausman test based on FE and RE yields a χ_3^2 of 3.36, with $p = 0.339$.

RMSE of forecasts

	1990	1991	1992	1993	1994	5 years
Poded *OLS*	0.2485	0.2520	0.2553	0.2705	0.2678	0.2590
Pooled spatial	0.2548	0.2594	0.2638	0.2816	0.2783	0.2678
Average heterogeneous *OLS*	0.7701	0.8368	0.8797	0.9210	0.9680	0.8781
Average spatial *MLE*	0.8516	0.9237	0.9715	1.0142	1.0640	0.9678
FE	0.1232	0.1351	0.1362	0.1486	0.1359	0.1360
FE-spatial	0.1213	0.1532	0.1529	0.1655	0.1605	0.1515
RE	0.1239	0.1356	0.1368	0.1493	0.1366	0.1367
RE-spadal	0.1207	0.1517	0.1513	0.1633	0.1581	0.1497

where $\hat{\alpha}_{i,FE}$ and $\hat{\boldsymbol{\beta}}_{FE}$ are estimated by ML. The predictive performance is then compared using data on forty-three States over the period 1965–1994. ML estimates and the RMSE of out-of-sample forecasts of various estimators are reported in Table 30.2. Note that overall, both the FE and RE estimators perform well in predicting liquor demand, while adding spatial correlation in the model does not improve prediction except for the first year. See Baltagi and Li (2006) and Baltagi, Bresson, and Pirotte (2012) for further discussion of forecasting with spatial panel data models.

30.4.4 Instrumental variables and *GMM*

In the presence of heteroskedastic disturbances, the *ML* estimator for spatial models under the assumption of homoskedastic innovations is generally inconsistent (Lin and Lee (2010)). As an alternative, instrumental variables (*IV*) and generalized method of moments (*GMM*) techniques have been suggested.

In a single cross-sectional setting, Kelejian and Robinson (1993) and Kelejian and Prucha (1998) propose a simple IV strategy to deal with the endogeneity of the spatially lagged dependent variable, $\mathbf{Wy}_{.t}$, that consists of using as instruments the spatially lagged (exogenous) explanatory variables, $\mathbf{WX}_{.t}$ (see Section 10.8 and the example therein). As shown by Mutl and Pfaffermayr (2011), the *IV* approach can be easily adapted to spatial panel data either with fixed or random effects (Wooldridge (2003)). The reader is also referred to Lee (2003) for a discussion on the choice of optimal instruments.

GMM estimation of spatial regression models in a single cross-sectional setting was originally advanced by Kelejian and Prucha (1999). The authors focus on a regression equation with SAR disturbances, and suggest the use of three moment conditions that exploit the properties of disturbances implied by a standard set of assumptions. Estimation consists of solving a nonlinear optimization problem, which yields a consistent estimator under a number of regularity conditions. Considerable work has been carried to extend this procedure in various directions. Liu, Lee, and Bollinger (2006) and Lee and Liu (2006) suggest a set of moments that encompass Kelejian and Prucha conditions as special cases. They focus on a spatial lag model with SAR disturbances (30.4) and $T = 1$, and consider a vector of linear and quadratic conditions in the error term, where the matrices appearing in the quadratic forms have bounded row and column norms (see also Lee (2007)). In panel (30.3), assuming α_i are fixed parameters, consider r quadratic moments of the type

$$\mathcal{M}_\ell(\delta) = \frac{1}{NT}\sum_{t=1}^{T} E\left(\boldsymbol{\varepsilon}_{.t}'\mathbf{A}_\ell\boldsymbol{\varepsilon}_{.t}\right), \quad \ell = 1, 2, \ldots, r, \tag{30.25}$$

where $\boldsymbol{\varepsilon}_{.t} = (\mathbf{I}_N - \delta\mathbf{W})\,\mathbf{u}_{.t}$, and $\mathbf{A}_\ell$, for $\ell = 1, 2, \ldots, r$ are non-stochastic matrices having bounded row and column sum matrix norms. Lee and Liu (2006) note that the matrices $\mathbf{A}_\ell$ have zero diagonal elements, so that $\mathcal{M}_\ell(\delta) = 0$. Interestingly, this assumption renders the *GMM* procedure robust to unknown, cross-sectional heteroskedasticity. The empirical counterpart of (30.25) is obtained by dropping the expectation operator, replacing $\mathbf{u}_{.t}$ by a consistent estimator (e.g., the IV estimator).

Lee and Liu (2006) focus on the problem of selecting the matrices appearing in the vector of linear and quadratic moment conditions, in order to obtain the lowest variance for the *GMM* estimator. Lee and Liu (2010) extend this framework to estimate the *SAR* model with higher-order spatial lags. Kelejian and Prucha (2010) generalize their original work to include spatial lags in the dependent variable and allow for heteroskedastic disturbances. This setting is extended by Kapoor, Kelejian, and Prucha (2007) to estimate a spatial panel regression model with group error components, and by Moscone and Tosetti (2011) for a panel with fixed-effects. Druska and Horrace (2004) have introduced the Keleijan and Prucha *GMM* within the framework of a panel with SAR disturbances, time dummies and time varying spatial weights, while Fingleton (2008a, 2008b) has extended it to the case of a regression with spatial moving average disturbances. Egger, Larch, Pfaffermayr, and Walde (2009) compare the small sample properties of *ML* and *GMM* estimators, observing that they perform similarly under the assumption of normally and non-normally distributed homoskedastic disturbances. However, one advantage of the *GMM* procedure over *ML* is that it is computationally simpler, especially when dealing with unbalanced panels (Egger, Pfaffermayr, and Winner (2005)).

Example 75 (Spatial price competition) *Spatial methods have been widely used to study firm competition across space, under the assumption that markets are limited in extent. One important example is the study by Pinkse, Slade, and Brett (2002) on competition in petrol prices in the US. The authors consider the following model for the price of the ith product*

$$p_i = \sum_{j=1}^{N} g\left(d_{ij}\right) p_j + \boldsymbol{\beta}'\mathbf{x}_i + \varepsilon_i, \;\; i = 1, 2, \ldots, N,$$

where $g(.)$ is a function of distance d_{ij}, measuring the influence of distance on the strength of competition between products i and j, and $\mathbf{x}_i$ is an h-dimensional vector of observed demand and cost variables. It is further assumed that d_{ij} depends on a discrete measure, d_{ij}^D, taking a finite number of different values, D, and a vector of continuous distance measures, d_{ij}^C, so that

$$g\left(d_{ij}\right) = \sum_{r=1}^{D} I_{d_{ij}^D = r} g_r(d_{ij}^C),$$

where $I_{d_{ij}^D = r}$ is an indicator function, and it is assumed that $g_r(d_{ij}^C) = \sum_{\ell=1}^{\infty} \alpha_{r\ell} e_{r\ell}\left(d_{ij}\right)$, where $\alpha_{r\ell}$ are unknown coefficients, and $e_{r\ell}(.)$ form a basis of the function space to which $g_r(.)$ belongs. Setting $e_\ell(d_{ij}) = \sum_{r=1}^{D} I_{d_{ij}^D = r} e_{r\ell}\left(d_{ij}\right)$, and α_ℓ the corresponding coefficients, it follows that

$$g\left(d_{ij}\right) = \sum_{\ell=1}^{\infty} \alpha_\ell e_\ell(d_{ij}).$$

The model estimated is

$$p_i = \sum_{\ell=1}^{L_N} \alpha_\ell \sum_{j=1}^{N} e_\ell(d_{ij}) p_j + \boldsymbol{\beta}'\mathbf{x}_i + v_i,$$

$$v_i = \varepsilon_i + \sum_{\ell=L_N+1}^{\infty} \alpha_\ell \sum_{j=1}^{N} e_\ell(d_{ij}) p_j,$$

where L_N denotes the number of expansion terms to be estimated. In matrix form,

$$\mathbf{p} = \mathbf{Z}\boldsymbol{\alpha} + \mathbf{X}\boldsymbol{\beta} + \mathbf{v}, \tag{30.26}$$

where $\mathbf{Z}$ is an $N \times L_N$ matrix with a generic $(i, \ell)^{th}$ element given by $\sum_{j=1}^{N} e_\ell(d_{ij}) p_j$. Note that the dimension of $\mathbf{Z}$ increases with N, and the disturbances $\mathbf{v}$, containing neglected expansion terms, are correlated with the dependent variable, $\mathbf{p}$. Hence, Pinkse, Slade, and Brett (2002) suggest using an IV approach to estimate $\boldsymbol{\theta} = \left(\boldsymbol{\alpha}', \boldsymbol{\beta}'\right)'$ in equation (30.26). The authors propose using, as instruments for $\sum_{j=1}^{N} e_\ell(d_{ij}) p_j$, $\ell = 1, 2, \ldots, L_N$, the spatial variables $\sum_{j=1}^{N} e_\ell(d_{ij}) x_{jh}$, for $\ell = 1, 2, \ldots, L_N$ and $h = 1, 2, \ldots, H$, where x_{jh} is the observation on the hth regressor. Each

exogenous regressor provides an additional L_N instruments. Let $\mathbf{B}$ *be the* $N \times b_N$ *matrix of instruments, and* $\mathbf{Q} = \mathbf{B}\left(\mathbf{B}'\mathbf{B}\right)^{-1}\mathbf{B}'$ *be the orthogonal projection matrix. The suggested IV estimator is then given by*

$$\hat{\boldsymbol{\theta}}_{IV} = \left(\mathbf{W}'\mathbf{P}\mathbf{W}\right)^{-1}\mathbf{W}'\mathbf{Q}\,\mathbf{p},$$

where $\mathbf{W} = (\mathbf{Z}, \mathbf{X})$*, so that*

$$\hat{g}\left(d_{ij}\right) = \sum_{\ell=1}^{L_N} \hat{\alpha}_\ell e_\ell(d_{ij}),$$

Pinkse, Slade, and Brett (2002) establish consistency and asymptotic normality of the above estimator, and provide OLS and IV estimates of the above model using data on prices at 312 terminals in the US in 1993.

30.5 Dynamic panels with spatial dependence

Considerable work has been undertaken on estimation of panel data models that feature both spatial dependence and temporal dynamics. A variety of spatiotemporal models have been proposed in the literature (see, for example, Anselin, Le Gallo, and Jayet (2007) and Lee and Yu (2010b)), most of which can be cast as

$$\mathbf{y}_{.t} = \boldsymbol{\alpha} + \gamma\mathbf{y}_{.,t-1} + \rho\mathbf{W}\mathbf{y}_{.t} + \lambda\mathbf{W}\mathbf{y}_{.,t-1} + \mathbf{X}_{.t}\boldsymbol{\beta} + \mathbf{u}_{.t}. \tag{30.27}$$

This model is stable if $|\gamma| + |\rho| + |\lambda| < 1$ assuming that the spatial weight matrix, $\mathbf{W}$, is row and column standardized. Yu, de Jong, and Lee (2008) derive *ML* estimators for the fixed-effects specification of the above model and show that when T is large relative to N, the *ML* estimators are consistent and asymptotically normal. But if $\lim_{N,T\to\infty} N/T > 0$, the limit distribution of the *ML* estimators is not centered around 0, in which case the authors propose a bias corrected estimator. See Yu, de Jong, and Lee (2008); and Lee and Yu (2010b) for further details. IV and *GMM* estimation of a stationary spatiotemporal model is considered in Kukenova and Monteiro (2009).

30.6 Heterogeneous panels

Consider the following panel specification with heterogeneous slopes (see also Chapter 28 on heterogeneous panels)

$$y_{it} = \boldsymbol{\alpha}_i'\mathbf{d}_t + \boldsymbol{\beta}_i'\mathbf{x}_{it} + u_{it}, \tag{30.28}$$

where $\mathbf{d}_t = (d_{1t}, d_{2t}, \ldots, d_{nt})'$ is an $n \times 1$ vector of observed common effects, $\mathbf{x}_{it}$ is a k-dimensional vector of strictly exogenous regressors, and $\boldsymbol{\beta}_i$ follow the random coefficient model $\boldsymbol{\beta}_i = \boldsymbol{\beta} + \mathbf{v}_i$, with $\mathbf{v}_i \sim IID(\mathbf{0}, \boldsymbol{\Omega}_v)$, and $\boldsymbol{\Omega}_v \neq \mathbf{0}$. It is further assumed that errors are generated

by a spatial process of the form (30.7), such as SAR, SMA, or SEC, where $\boldsymbol{\varepsilon}_{.t}$ follows a covariance stationary process. Pesaran and Tosetti (2011) focus on estimation of the cross-section means of parameters, $\boldsymbol{\beta} = E(\boldsymbol{\beta}_i)$, in model (30.28), by fixed-effects (FE) and mean group (MG) estimators, introduced in Chapters 26 and 28, respectively. In the general case of equation (30.28) these estimators are

$$\hat{\boldsymbol{\beta}}_{FE} = \left(\sum_{i=1}^{N} \mathbf{X}_{i.}' \mathbf{M}_D \mathbf{X}_{i.} \right)^{-1} \sum_{i=1}^{N} \mathbf{X}_{i.}' \mathbf{M}_D \mathbf{y}_{i.} \tag{30.29}$$

$$\hat{\boldsymbol{\beta}}_{MG} = N^{-1} \sum_{i=1}^{N} \left(\mathbf{X}_{i.}' \mathbf{M}_D \mathbf{X}_{i.} \right)^{-1} \mathbf{X}_{i.}' \mathbf{M}_D \mathbf{y}_{i.}, \tag{30.30}$$

where $\mathbf{M}_D = \mathbf{I}_T - \mathbf{D} \left(\mathbf{D}'\mathbf{D} \right)^{-1} \mathbf{D}'$, and $\mathbf{D} = (\mathbf{d}_1, \mathbf{d}_2, \ldots, \mathbf{d}_n)$. Pesaran and Tosetti (2011) show that, under general regularity conditions, as $(N, T) \xrightarrow{j} \infty$, for the FE estimator, $\hat{\boldsymbol{\beta}}_{FE}$, we have

$$\sqrt{N} \left(\hat{\boldsymbol{\beta}}_{FE} - \boldsymbol{\beta} \right) \xrightarrow{d} N(\mathbf{0}, \boldsymbol{\Sigma}_{FE}),$$

where

$$\boldsymbol{\Sigma}_{FE} = \mathbf{Q}^{-1} \boldsymbol{\Lambda} \mathbf{Q}^{-1}, \tag{30.31}$$

with

$$\mathbf{Q} = \underset{N,T \to \infty}{Plim} \; N^{-1} \sum_{i=1}^{N} \left(\frac{\mathbf{X}_{i.}' \mathbf{M}_D \mathbf{X}_{i.}}{T} \right), \tag{30.32}$$

$$\boldsymbol{\Lambda} = \underset{N,T \to \infty}{Plim} \left[N^{-1} \sum_{i=1}^{N} \left(\frac{\mathbf{X}_{i.}' \mathbf{M}_D \mathbf{X}_{i.}}{T} \right) \boldsymbol{\Omega}_{\boldsymbol{\upsilon}} \left(\frac{\mathbf{X}_{i.}' \mathbf{M}_D \mathbf{X}_{i.}}{T} \right) \right].$$

While for the *MG* estimator, $\hat{\boldsymbol{\beta}}_{MG}$, given by (30.30), as $(N, T) \xrightarrow{j} \infty$ we have

$$\sqrt{N} \left(\hat{\boldsymbol{\beta}}_{MG} - \boldsymbol{\beta} \right) \xrightarrow{d} N(\mathbf{0}, \boldsymbol{\Sigma}_{MG}),$$

where $\boldsymbol{\Sigma}_{MG} = \boldsymbol{\Omega}_{\upsilon}$. Therefore, the asymptotic distribution of *FE* and *MG* estimators does not depend on the particular spatial structure of the error, u_{it}, but only on $\boldsymbol{\Omega}_{v}$. This result follows from the random coefficients hypothesis, since the time-invariant variability of $\boldsymbol{\beta}_i$ dominates the other sources of randomness in the model. Robust estimators for the variances of $\hat{\boldsymbol{\beta}}_{FE}$ and $\hat{\boldsymbol{\beta}}_{MG}$ can be obtained following the non-parametric approach employed in Pesaran (2006), which makes use of estimates of $\boldsymbol{\beta}$ computed for different-cross-sectional units

$$\widehat{Asy.Var}\left(\hat{\boldsymbol{\beta}}_{FE}\right) = \frac{1}{N}\mathbf{Q}_{NT}^{-1}\boldsymbol{\Lambda}_{NT}\mathbf{Q}_{NT}^{-1}, \tag{30.33}$$

$$\widehat{Asy.Var}\left(\hat{\boldsymbol{\beta}}_{MG}\right) = \frac{1}{N\left(N-1\right)}\sum_{i=1}^{N}\left(\hat{\boldsymbol{\beta}}_i - \hat{\boldsymbol{\beta}}_{MG}\right)\left(\hat{\boldsymbol{\beta}}_i - \hat{\boldsymbol{\beta}}_{MG}\right)'. \tag{30.34}$$

where

$$\mathbf{Q}_{NT} = \frac{1}{N}\sum_{i=1}^{N}\left(T^{-1}\mathbf{X}_{i.}'\mathbf{M}_D\mathbf{X}_{i.}\right), \tag{30.35}$$

$$\boldsymbol{\Lambda}_{NT} = \frac{1}{N-1}\sum_{i=1}^{N}\left[\left(\frac{\mathbf{X}_{i.}'\mathbf{M}_D\mathbf{X}_{i.}}{T}\right)\left(\hat{\boldsymbol{\beta}}_i - \hat{\boldsymbol{\beta}}_{MG}\right)\left(\hat{\boldsymbol{\beta}}_i - \hat{\boldsymbol{\beta}}_{MG}\right)'\left(\frac{\mathbf{X}_{i.}'\mathbf{M}_D\mathbf{X}_{i.}}{T}\right)\right].$$

One advantage of the above non-parametric variance estimators is that their computation does not require *a priori* knowledge of the spatial arrangement of cross-sectional units. In a set of Monte Carlo experiments, Pesaran and Tosetti (2011) show that misspecification of the spatial weights matrix may lead to substantial size distortions in tests based on the *ML* or quasi-*ML* estimators of $\boldsymbol{\beta}_i$ (or $\boldsymbol{\beta}$). Another advantage of using the above approach over standard spatial techniques is that, while allowing for serially correlated errors, it does not entail information on the time series processes underlying ε_{it}, so long as these processes are covariance stationary.

30.6.1 Temporal heterogeneity

Temporal heterogeneity may be incorporated in a spatial version of the seemingly unrelated regression equations (*SURE*) approach, as suggested by Anselin (1988). See also Mur, López, and Herrera (2010). This approach, suitable when N greatly exceeds T, permits slope parameters to vary over time, and errors are allowed to be both spatially and serially correlated. In its more general form the spatial SURE is

$$\mathbf{y}_{.t} = \rho_t\mathbf{W}_1\mathbf{y}_{.t} + \mathbf{X}_{.t}\boldsymbol{\beta}_t + \mathbf{u}_{.t}, \tag{30.36}$$

$$\mathbf{u}_{.t} = \delta_t\mathbf{W}_2\mathbf{u}_{.t} + \boldsymbol{\varepsilon}_{.t}, \tag{30.37}$$

where $\boldsymbol{\beta}_t$, ρ_t and δ_t are time varying parameters, and $\boldsymbol{\varepsilon}_{.t}$ satisfies $E(\boldsymbol{\varepsilon}_{.t}\boldsymbol{\varepsilon}_{.s}') = \sigma_{ts}\mathbf{I}_N$. Let $\boldsymbol{\Omega}$ be a $T\times T$ positive definite matrix with elements σ_{ts}. *ML* or *GMM* techniques can be used to estimate the above model. The log-likelihood function of (30.36)-(30.37) is

$$\ell\left(\boldsymbol{\theta}\right) = -\frac{NT}{2}\ln 2\pi - \frac{N}{2}\ln\left|\boldsymbol{\Omega}\right| + \sum_{t=1}^{T}\ln\left|\mathbf{I}_N - \rho_t\mathbf{W}_1\right| + \sum_{t=1}^{T}\ln\left|\mathbf{I}_N - \delta_t\mathbf{W}_2\right|$$
$$-\frac{1}{2}\mathbf{u}'\left(\boldsymbol{\Omega}^{-1}\otimes\mathbf{I}_N\right)\mathbf{u},$$

where $\boldsymbol{\theta} = \left(\rho_1,\ldots,\rho_T,\boldsymbol{\beta}_1',\ldots,\boldsymbol{\beta}_T',\delta_1,\ldots,\delta_T,vech(\boldsymbol{\Omega})'\right)'$, and $\mathbf{u} = \left(\mathbf{u}_{.1}',\mathbf{u}_{.2}',\ldots,\mathbf{u}_{.T}'\right)'$, with $\mathbf{u}_{.t} = \left(\mathbf{I}_N - \delta_t\mathbf{W}_2\right)\left[\left(\mathbf{I}_N - \rho_t\mathbf{W}_1\right)\mathbf{y}_{.t} - \mathbf{X}_{.t}\boldsymbol{\beta}_t\right]$. A number of *LM* tests for the presence of

spatial effects in the above specification are proposed by Mur, López, and Herrera (2010). See also Baltagi and Pirotte (2010), who consider *ML* and *GMM* estimation of a SURE model with spatial error of the *SAR* or *SMA* type, assuming that the remainder term of the spatial process follows an error component structure.

30.7 Non-parametric approaches

Non-parametric regression techniques based on certain mixing conditions applied to space and time have also been considered in the literature, as robust alternatives to *ML* and *GMM* estimation techniques. Variants of the Newey and West (1987) heteroskedasticity autocorrelation consistent (*HAC*) estimators employed in the time series literature are adapted to spatial models by Conley (1999) and Driscoll and Kraay (1998) in the context of *GMM* estimators of spatial panel data regression models (see also Pinkse, Slade, and Brett (2002)).[3] More recently, Kelejian and Prucha (2007) have proposed a spatial heteroskedasticity autocorrelation consistent (*SHAC*) estimator for a single cross-sectional regression with spatially correlated errors. This approach approximates the true covariance matrix with a weighted average of cross products of regression errors, where each element is weighted by a function of (possible multiple) distances between cross-sectional units. This procedure can be extended for estimation of the covariance of the FE and MG estimators for a panel of type (30.3), with fixed-effects and spatial errors (30.7), where $\boldsymbol{\varepsilon}_{.t}$ is allowed to be serially correlated. Let $\phi_N > 0$ be an arbitrary scalar function of N, m the window size for the time series dimension, and $K(.)$ a kernel function such that

$$K\left(\frac{\phi_{ij}}{\phi_N}, \frac{|t-s|}{m+1}\right) = K_1\left(\frac{\phi_{ij}}{\phi_N}\right) K_2\left(\frac{|t-s|}{m+1}\right),$$

where $K_1(.)$ and $K_2(.)$ satisfy a set of regularity conditions (see Kelejian and Prucha (2007)). A Newey-West type *SHAC* estimator of the variance of the classic FE estimator (30.29) is given by

$$\widehat{Asy.Var}\left(\hat{\boldsymbol{\beta}}_{FE}\right) = \mathbf{Q}_{NT}^{-1}\left[\frac{1}{(NT)^2}\sum_{i,j=1}^{N}\sum_{t,s=1}^{T} K\left(\frac{\phi_{ij}}{\phi_N}, \frac{|t-s|}{m+1}\right)\tilde{\mathbf{x}}_{it}\tilde{\mathbf{x}}_{js}'\hat{u}_{it}\hat{u}_{js}\right]\mathbf{Q}_{NT}^{-1}, \tag{30.38}$$

where $\mathbf{Q}_{NT} = \frac{1}{NT}\sum_{t=1}^{T}\left(\mathbf{X}_{.t}'\mathbf{M}\mathbf{X}_{.t}\right)$, $\hat{u}_{it} = y_{it} - \hat{\alpha}_i' - \hat{\boldsymbol{\beta}}_{FE}'\mathbf{x}_{it}$, and $\widetilde{\mathbf{X}}_{i.}' = \mathbf{X}_{i.}'\mathbf{M}$, with $\mathbf{X}_{i.} = (\mathbf{x}_{i1}, \mathbf{x}_{i2}, \ldots, \mathbf{x}_{iT})'$, $\tilde{\mathbf{X}}_{i.} = (\tilde{\mathbf{x}}_{i1}, \tilde{\mathbf{x}}_{i2}, \ldots, \tilde{\mathbf{x}}_{iT})'$. For the MG estimator we have

$$\widehat{Asy.Var}\left(\hat{\boldsymbol{\beta}}_{MG}\right) = \frac{1}{(NT)^2}\sum_{i,j=1}^{N}\sum_{t,s=1}^{T} K\left(\frac{\phi_{ij}}{\phi_N}, \frac{|t-s|}{m+1}\right)\mathbf{w}_{it}\mathbf{w}_{js}'\hat{u}_{it}\hat{u}_{js}, \tag{30.39}$$

where $\mathbf{w}_{it}$ is the tth column of $\mathbf{W}_{i.}' = \left(T^{-1}\mathbf{X}_{i.}'\mathbf{M}_D\mathbf{X}_{i.}\right)^{-1}\mathbf{X}_{i.}'\mathbf{M}_D$. One shortcoming of this method is that its finite sample properties may be quite poor when N or T are not sufficiently large (Pesaran and Tosetti (2011)). An alternative strategy has been suggested by Bester, Conley, and

[3] See Section 5.9 for a discussion of HAC estimators in the context of the time series literature.

Hansen (2011), who, using results taken from Ibragimov and Müller (2010), propose dividing the sample in groups so that group-level averages are approximately independent, and accordingly suggest an *HAC* estimator based on a discrete group-membership metric. However, the validity of this approach relies on the capacity of the researcher to construct groups whose averages are approximately uncorrelated.

Robinson (2007) considers smoothed nonparametric kernel regression estimation. Under this approach, rather than employing mixing conditions, it is assumed that regression errors follow a general linear process representation covering both weak (spatial) dependence as well as dependence at longer ranges. Robinson (2007) establishes consistency of the Nadaraya-Watson kernel estimate and derives its asymptotic distribution (see also Hallin, Lu, and Tran (2004)).

Spatial filtering techniques can also be used to control for spatial effects (Tiefelsdorf and Griffith (2007)). Under this framework, spatial dependence in the regression is proxied by a linear combination of a subset of eigenvectors of a matrix function of $\mathbf{W}$. Hence, estimation is carried out by least squares applied to auxiliary regressions where the observed regressors are augmented with these artificial variables. The reader is referred to Tiefelsdorf and Griffith (2007) and Griffith (2010) for further discussion.

30.8 Testing for spatial dependence

Spatial econometrics literature proposes a number of statistics for testing the null hypothesis of spatial independence, i.e., $H_0 : E\left(u_{it}u_{jt}\right) = 0, i \neq j$ in model (30.3). Tests of cross-sectional dependence in the absence of ordering (i.e., when $\mathbf{W}$ is not known) are reviewed in Section 29.7. The majority of the tests in the spatial literature have been studied only in the case of a single cross-section. One of the most commonly used is the Moran statistic (Moran (1950); Pinkse (1999); Kelejian and Prucha (2001)), which, extended to a panel setup, takes the form

$$CD_{Moran} = \frac{\sum_{t=1}^{T} \hat{\mathbf{u}}_{.t}' \mathbf{W} \hat{\mathbf{u}}_{.t}}{\left(T\hat{\sigma}_{\varepsilon}^{4} \sum_{i=1}^{N} \sum_{j=1}^{i-1} \left(w_{ij} + w_{ji} \right)^2 \right)^{1/2}}, \tag{30.40}$$

where $\hat{\sigma}_{\varepsilon}^{2}$ is a consistent estimator of σ_{ε}^{2}, and $\hat{\boldsymbol{u}}_{.t}$ is a consistent estimator of regression errors. The CD_{Moran} is asymptotically normally distributed (see Kelejian and Prucha (2001)), and, for large N, is equivalent to the Burridge (1980) *LM* statistic. Another test that has attracted considerable interest in the spatial literature is the Geary (1954)'s *c*. This statistic, based on the average squared differences of residuals, shares very similar characteristics with the CD_{Moran}. See Hepple (1998) for a comparison and discussion on the properties of Geary's *c* and Moran's statistics.

The information on the distance among units can also be used to obtain 'local' versions of some statistics proposed in the panel literature to test against generic forms of cross-sectional dependence. For example, the local CD_P test proposed by Pesaran (2004) is

$$CD_{P,Local} = \sqrt{\frac{T}{S_0}} \left(\sum_{i=1}^{N} \sum_{j=1}^{N} w_{ij} \hat{\rho}_{ij} \right), \tag{30.41}$$

where $S_0 = \sum_{i=1}^{N} \sum_{j=1}^{N} w_{ij}$, and $\hat{\rho}_{ij}$ is the sample pair-wise correlation coefficient computed between fixed-effects residuals of units i and j. The $CD_{P,Local}$ test is asymptotically normally distributed. Similarly, it is possible to derive local versions of other tests proposed in the panel literature such as the *LM* test given by (29.74). See also Pesaran, Ullah, and Yamagata (2008) and Moscone and Tosetti (2009).

Robinson (2008) has proposed a general class of statistics that, like CD_{Moran}, is based on quadratic forms in *OLS* regression residuals, where the matrices appearing in the quadratic forms satisfy certain sparseness conditions. The author shows that these statistics have a limiting chi-square distribution under the null hypothesis of error independence. Special cases of this class of statistics can be interpreted as *ML* tests directed against specific alternative hypotheses.

30.9 Further reading

Textbook treatment of the spatial econometrics literature can be found in Anselin (1988); Arbia (2006); and LeSage and Pace (2009), while a description of recent developments of these techniques can be found in Anselin, Le Gallo, and Jayet (2007), Lee and Yu (2010b) and Lee and Yu (2011). The reader is referred to McMillen (1995); Fleming (2004); and LeSage and Pace (2009) for an extension of spatial techniques to nonlinear regression models.

30.10 Exercises

1. Consider the simple *SAR* process

$$u_{it} = 0.5\rho \left(u_{i-1,t} + u_{i+1,t}\right) + \varepsilon_{it}, i = 2, \ldots, N-1; t = 1, 2, \ldots, T, \tag{30.42}$$

with the end points

$$u_{1t} = 0.5\rho \left(u_{Nt} + u_{2t}\right) + \varepsilon_{1t}, \tag{30.43}$$

$$u_{Nt} = 0.5\rho \left(u_{N-1,t} + u_{1t}\right) + \varepsilon_{Nt} \tag{30.44}$$

where $\varepsilon_{it} \sim IID(0, \sigma_{\varepsilon}^2)$. Write down the spatial weights matrix for the above process and derive the covariance matrix of $\mathbf{u}_{.t} = (u_{1t}, u_{2t}, \ldots, u_{Nt})'$.

2. Derive the conditions under which the spatial process

$$\mathbf{y}_{it} = \rho_i \mathbf{w}_i' \mathbf{y}_t + \varepsilon_{it}, \text{ for } i = 1, 2, \ldots, N,$$

where $\varepsilon_{it} \sim IID(0, \sigma_i^2)$, $\mathbf{w}_i = (w_{i1}, w_{i2}, \ldots, w_{iN})'$, and $\mathbf{y}_t = (y_{1t}, y_{2t}, \ldots, y_{Nt})'$, is cross-sectionally weakly dependent.

3. Consider the simple *SAR* process (30.42)-(30.44), and assume that $\varepsilon_{it} \sim IIDN(0, \sigma_{\varepsilon}^2)$. Write down the log-likelihood function for this process and derive the first- and second-order conditions for estimation of ρ and σ_{ε}^2.

4. Derive an expression for the conventional *FE* estimator for $\boldsymbol{\beta}$ in model (30.3) with *SAR* errors, (30.4). Obtain its covariance matrix. Derive a set of conditions under which this estimator is $\sqrt{NT}$-consistent.

5. Consider equation (30.1) and assume that $\alpha_i = 0, i = 1, 2, \ldots, N$. Derive an expression for the instrumental variable estimator of ρ and $\boldsymbol{\beta}$, using as instruments the variables $(\mathbf{X}_{.t}, \mathbf{WX}_{.t})$, $t = 1, 2, \ldots, T$.
6. Consider model (30.23)–(30.24), where it is assumed that $\alpha_i \sim IID(0, \sigma_\alpha^2)$, and $\varepsilon_{it} \sim IID(0, \sigma_\varepsilon^2)$. Write down the log-likelihood function for this model and obtain the first-order conditions for maximization of the log-likelihood function.

31 Unit Roots and Cointegration in Panels

31.1 Introduction

This chapter provides a review of the theoretical literature on testing for unit roots and cointegration in panels where the time dimension (T), and the cross-section dimension (N) are relatively large. In cases where N is large (say over 100) and T small (less than 50) the analysis can proceed only under restrictive assumptions such as dynamic homogeneity and/or local cross-sectional dependence as in spatial autoregressive or moving average models. In cases where N is small (say less than ten) and T is relatively large, standard time series techniques applied to systems of equations, such as the seemingly unrelated regression equations (*SURE*), can be used and the panel aspect of the data should not pose new technical difficulties.

One of the primary reasons behind the application of unit root and cointegration tests to a panel of cross-section units was to gain statistical power and to improve on the poor power of their univariate counterparts. This was supported by the application of what might be called the first generation panel unit root tests to real exchange rates, output and inflation. For example, the augmented Dickey–Fuller (1979) test, reviewed in Chapter 15, is typically not able to reject the hypothesis that the real exchange rate is nonstationary. By contrast, panel unit root tests applied to a collection of industrialized countries generally find that real exchange rates are stationary, thereby lending empirical support to the purchasing power parity hypothesis (e.g., Coakley and Fuertes (1997) and Choi (2001)).

Testing the unit root and cointegration hypotheses by using panel data instead of individual time series involves several additional complications. First, as seen in previous chapters, the analysis of panel data generally involves a substantial amount of unobserved heterogeneity, rendering the parameters of the model cross-section specific. Second, the assumption of cross-sectional independence is inappropriate in many empirical applications, particularly in the analysis of real exchange rates mentioned above. To overcome these difficulties, variants of panel unit root tests are developed that allow for different forms of cross-sectional dependence (see Section 31.4). Third, the panel test outcomes are often difficult to interpret if the null of the unit root or cointegration is rejected. The best that can be concluded is that 'a significant fraction of the cross-section units is stationary or cointegrated'. Conventional panel tests do not provide explicit guidance as to the size of this fraction or the identity of the cross-section units that are stationary

or cointegrated. To deal with issue, recent studies have proposed methods for estimating the fraction of non-stationary series in the panel, and for classifying the individual series into stationary and non-stationary sets (see (Pesaran 2012)). Fourth, with unobserved $I(1)$ common factors affecting some or all of the variables in the panel, it is also necessary to consider the possibility of cointegration between the variables across the groups (cross-section cointegration) as well as within group cointegration (see Section 31.5). Finally, the asymptotic theory is considerably more complicated due to the fact that the sampling design involves a time as well as a cross-section dimension. For example, applying the usual Dickey–Fuller test to a panel data set introduces a bias that is not present in the case of a univariate test. Furthermore, a proper limit theory has to take account of the relationship between the increasing number of time periods and cross-section units (see Phillips and Moon (1999)).

In comparison with panel unit root tests, the analysis of cointegration in panels is still at an early stages of its development. So far the focus of the panel cointegration literature has been on residual based approaches, although there has been a number of attempts at the development of system approaches as well. As in the case of panel unit root tests, such tests are developed based on homogenous and heterogeneous alternatives. The residual based tests were developed to ward against the 'spurious regression' problem that can also arise in panels when dealing with $I(1)$ variables. Such tests are appropriate when it is known *a priori* that at most there can be only one within group cointegration in the panel. System approaches are required in more general settings where more than one within group cointegrating relation might be present, and/or there exist unobserved common $I(1)$ factors.

Having established a cointegration relationship, the long-run parameters can be estimated efficiently using techniques similar to those proposed in the case of single time series models. Specifically, fully-modified *OLS* procedures, the dynamic *OLS* estimator and estimators based on a vector error correction representation were adapted to panel data structures. Most approaches employ a homogeneous framework, that is, the cointegration vectors are assumed to be identical for all panel units, whereas the short-run parameters are panel specific. Although such an assumption seems plausible for some economic relationships (like the *PPP* hypothesis mentioned above) there are other behavioural relationships (like the consumption function or money demand), where a homogeneous framework seems overly restrictive. On the other hand, allowing all parameters to be individual specific would substantially reduce the appeal of a panel data study. It is therefore important to identify parameters that are likely to be similar across panel units whilst at the same time allowing for sufficient heterogeneity of other parameters. This requires the development of appropriate techniques for testing the homogeneity of a sub-set of parameters across the cross-section units. When N is small relative to T, standard likelihood ratio based statistics can be used. Groen and Kleibergen (2003) provide an application. Testing for parameter homogeneity in the case of large panels poses new challenges that require further research (see Section 28.11).

31.2 Model and hypotheses to test

Assume that time series $\{y_{i0}, \ldots, y_{iT}\}$ on the cross-section units $i = 1, 2, \ldots, N$ are generated for each i by a simple first-order autoregressive, $AR(1)$, process

$$y_{it} = (1 - \alpha_i)\mu_i + \alpha_i y_{i,t-1} + \varepsilon_{it}, \tag{31.1}$$

where the initial values, y_{i0}, are given, and the errors ε_{it} are identically, independently distributed across i and t with $E(\varepsilon_{it}) = 0, E(\varepsilon_{it}^2) = \sigma_i^2 < \infty$ and $E(\varepsilon_{it}^4) < \infty$. These processes can also be written equivalently as simple Dickey–Fuller (DF) regressions

$$\Delta y_{it} = -\phi_i \mu_i + \phi_i y_{i,t-1} + \varepsilon_{it}\,, \tag{31.2}$$

where $\Delta y_{it} = y_{it} - y_{i,t-1}, \phi_i = \alpha_i - 1$. In further developments of the model it is also helpful to write (31.1) or (31.2) in mean-deviations forms $\tilde{y}_{it} = \alpha_i \tilde{y}_{i,t-1} + \varepsilon_{it}$, where $\tilde{y}_{it} = y_{it} - \mu_i$. The corresponding DF regression in $\tilde{y}_{it}$ is given by

$$\Delta \tilde{y}_{it} = \phi_i \tilde{y}_{i,t-1} + \varepsilon_{it}. \tag{31.3}$$

Most panel unit root tests are designed to test the null hypothesis of a unit root for each individual series in a panel. Accordingly, the null hypothesis of interest is

$$H_0 : \phi_1 = \phi_2 = \cdots = \phi_N = 0, \tag{31.4}$$

that is, all time series are independent random walks. The formulation of the alternative hypothesis is instead a controversial issue that critically depends on which assumptions one makes about the nature of the homogeneity/heterogeneity of the panel. First, under the assumption that the autoregressive parameter is identical for all cross-section units, we can consider

$$H_1^a : \phi_1 = \phi_2 = \cdots = \phi_N \equiv \phi \text{ and } \phi < 0.$$

The panel unit root statistics motivated by H_1^a pools the observations across the different cross-section units before forming the 'pooled' statistic (see, e.g., Harris and Tzavalis (1999) and Levin, Lin, and Chu (2002)). One drawback of tests based on such alternative hypotheses is that they tend to have power even if only a few of the units are stationary; hence a rejection of the null hypothesis, H_0, is not convincing evidence that a significant proportion of the series are indeed stationary. In particular, Westerlund and Breitung (2014) show that the local power of the Levin, Lin, and Chu (2002) test is greater than that of the Im, Pesaran, and Shin (2003) test, based on a less restrictive alternative, also when not all individual series are stationary. A further drawback in using H_1^a is that this is likely to be unduly restrictive, particularly for cross-country studies involving differing short-run dynamics. For example, such a homogeneous alternative seems particularly inappropriate in the case of the PPP hypothesis, where y_{it} is taken to be the real exchange rate. There are no theoretical grounds for the imposition of the homogeneity hypothesis, $\phi_i = \phi$, under PPP. At the other extreme, there is the alternative hypothesis stating that at least one of the series in the panel is generated by a stationary process

$$H_1^b : \phi_i < 0, \text{ for one or more } i.$$

Such an alternative hypothesis is at the basis of panel unit root tests proposed by Chang (2002) and Chang (2004). We observe that H_1^b is only appropriate when N is finite, namely within the multivariate model with a fixed number of variables analyzed in the time series literature. On the contrary, in the case of large N and T, panel unit root tests will lack power if the alternative, H_1^b,

is adopted. For large N and T panels it is reasonable to entertain alternatives that lie somewhere between the two extremes of H_1^a and H_1^b. In this context, a more appropriate alternative is given by the heterogeneous alternative

$$H_1^c : \phi_i < 0,\ i = 1, 2, \ldots, N_1, \phi_i = 0,\ i = N_1 + 1, N_1 + 2, \ldots, N, \tag{31.5}$$

such that

$$\lim_{N\to\infty} \frac{N_1}{N} = \delta,\ 0 < \delta \leq 1. \tag{31.6}$$

Using the above specification the null hypothesis is $H_0 : \delta = 0$, while H_1^c can be written as

$$H_1^c : \delta > 0.$$

In other words, rejection of the unit root null hypothesis can be interpreted as providing evidence in favour of rejecting the unit root hypothesis for a non-zero fraction of panel members as $N \to \infty$. The tests developed against the above heterogeneous alternatives, H_1^c, operate directly on the test statistics for the individual cross-section units using (standardized) simple averages of the underlying individual statistics or their suitable transformations such as rejection probabilities (see, among others, Choi (2001), Im, Pesaran, and Shin (2003), and Pesaran (2007b)).

Remark 8 *The heterogeneity of panel data models used in cross-country analysis introduces a new kind of asymmetry in the way the null and the alternative hypotheses are treated, which is not usually present in the univariate time series (or cross-sectional) models. This is because the same null hypothesis is imposed across all i but the specification of the alternative hypothesis is allowed to vary with i. This asymmetry is assumed away in homogeneous panels. However, as demonstrated in Pesaran and Smith (1995), neglected heterogeneity (even if purely random) can lead to spurious results in dynamic panels. Therefore, in cross-country analysis where slope heterogeneity is a norm, the asymmetry of the null and the alternative hypotheses has to be taken into account. The appropriate response critically depends on the relative size of N and T. In large N-heterogeneous panel data models with small T (say around 15) it is only possible to devise sufficiently powerful unit root tests which are informative in some average sense, namely whether the null of a unit root can be rejected in the case of a significant fraction of the countries in the panel.*[1] *To identify the exact proportion of the sample for which the null hypothesis is rejected, one requires country-specific data sets with T sufficiently large. But if T is large enough for reliable country-specific inferences to be made, then there seems little rationale in pooling countries into a panel.*

In the rest of the chapter we will focus on panel unit root tests designed for one of the alternative hypotheses H_1^a or H_1^c. However, we observe that, despite the differences in the way the two classes of test view the alternative hypothesis, both types of test can be consistent against both types of the alternative. See, for example, the discussion in Westerlund and Breitung (2014).

[1] Some of these difficulties can be circumvented if slope heterogeneity can be modelled in a sensible and parsimonious manner.

31.3 First generation panel unit root tests

The various first generation panel unit roots proposed in the literature can be obtained using the pooled log-likelihood function of the individual Dickey–Fuller regressions given by (31.2)

$$\ell_{NT}(\boldsymbol{\phi},\boldsymbol{\theta}) = \sum_{i=1}^{N}\left[-\frac{T}{2}\log 2\pi\sigma_i^2 - \frac{1}{2\sigma_i^2}\sum_{t=1}^{T}\left(\Delta y_{it} + \phi_i\mu_i - \phi_i y_{i,t-1}\right)^2\right], \tag{31.7}$$

where $\boldsymbol{\phi} = (\phi_1,\phi_2,\ldots,\phi_N)'$, $\boldsymbol{\theta}_i = (\mu_i,\sigma_i^2)'$ and $\boldsymbol{\theta} = (\boldsymbol{\theta}_1',\boldsymbol{\theta}',\ldots,\boldsymbol{\theta}_N')'$. In the case of the homogeneous alternatives, H_1^a, where $\phi_i = \phi$, the maximum likelihood estimator of ϕ is given by

$$\hat{\phi}(\boldsymbol{\theta}) = \frac{\sum_{i=1}^{N}\sum_{t=1}^{T}\sigma_i^{-2}\Delta y_{it}\left(y_{i,t-1}-\mu_i\right)}{\sum_{i=1}^{N}\sum_{t=1}^{T}\sigma_i^{-2}\left(y_{i,t-1}-\mu_i\right)^2}. \tag{31.8}$$

The nuisance cross-section specific parameters $\boldsymbol{\theta}_i$ can be estimated either under the null or the alternative hypothesis. Under the null hypothesis μ_i is unidentified, but as we shall see it is often replaced by y_{i0}, on the implicit (identifying) assumption that $\tilde{y}_{i0} = 0$ for all i. For this choice of μ_i the effective number of time periods used for estimation of ϕ_i is reduced by one. Under the alternative hypothesis the particular estimates of μ_i and σ_i^2 chosen naturally depend on the nature of the alternatives envisaged. Under homogeneous alternatives, $\phi_i = \phi < 0$, the *ML* estimates of μ_i and σ_i^2 are given as nonlinear functions of $\hat{\phi}$. Under heterogeneous alternatives ϕ_i and σ_i^2 can be treated as free parameters and estimated separately for each i.

Levin, Lin, and Chu (2002) avoid the problems associated with the choice of the estimators for μ_i and base their tests on the t-ratio of ϕ in the pooled fixed-effects regression

$$\Delta y_{it} = a_i + \phi y_{i,t-1} + \varepsilon_{it},\ \varepsilon_{it} \backsim IID(0,\sigma_i^2).$$

The t-ratio of the *FE* estimator of ϕ is given by

$$\mathrm{t}_{\phi} = \frac{\sum_{i=1}^{N}\hat{\sigma}_i^{-2}\Delta\mathbf{y}_i'\mathbf{M}_{\tau}\mathbf{y}_{i,-1}}{\sqrt{\sum_{i=1}^{N}\hat{\sigma}_i^{-2}\left(\mathbf{y}_{i,-1}'\mathbf{M}_{\tau}\mathbf{y}_{i,-1}\right)}}, \tag{31.9}$$

where $\Delta\mathbf{y}_i = \left(\Delta y_{i1},\Delta y_{i2},\ldots,\Delta y_{iT}\right)'$, $\mathbf{y}_{i,-1} = \left(y_{i0},y_{i1},\ldots,y_{i,T-1}\right)'$, $\mathbf{M}_{\tau} = \mathbf{I}_T - \boldsymbol{\tau}_T(\boldsymbol{\tau}_T'\boldsymbol{\tau}_T)^{-1}\boldsymbol{\tau}_T'$, $\boldsymbol{\tau}_T$ is a T-dimensional vector of ones,

$$\hat{\sigma}_i^2 = \frac{\Delta\mathbf{y}_i'\mathbf{M}_i\Delta\mathbf{y}_i}{T-2}, \tag{31.10}$$

$\mathbf{M}_i = \mathbf{I}_T - \mathbf{X}_i(\mathbf{X}_i'\mathbf{X}_i)^{-1}\mathbf{X}_i'$, and $\mathbf{X}_i = \left(\boldsymbol{\tau}_T,\mathbf{y}_{i,-1}\right)$.

The construction of a unit root test against H_1^c is less clear because the alternative consists of a set of inequality conditions. Im, Pesaran, and Shin (2003) suggest the mean of the individual specific t-statistics[2]

$$\bar{\mathrm{t}} = \frac{1}{N}\sum_{i=1}^{N} \mathrm{t}_i\,,$$

where

$$\mathrm{t}_i = \frac{\Delta \mathbf{y}_i' \mathbf{M}_\tau \mathbf{y}_{i,-1}}{\hat{\sigma}_i \left(\mathbf{y}_{i,-1}' \mathbf{M}_\tau \mathbf{y}_{i,-1}\right)^{1/2}},$$

is the Dickey–Fuller t-statistic of cross-sectional unit i.[3] *LM* versions of the t-ratios of ϕ and ϕ_i, that are analytically more tractable, can also be used which are given by

$$\tilde{\mathrm{t}}_\phi = \frac{\sum_{i=1}^{N} \tilde{\sigma}_i^{-2} \Delta \mathbf{y}_i' \mathbf{M}_\tau \mathbf{y}_{i,-1}}{\sqrt{\sum_{i=1}^{N} \tilde{\sigma}_i^{-2} \left(\mathbf{y}_{i,-1}' \mathbf{M}_\tau \mathbf{y}_{i,-1}\right)}}, \tag{31.11}$$

and

$$\tilde{\mathrm{t}}_i = \frac{\Delta \mathbf{y}_i' \mathbf{M}_\tau \mathbf{y}_{i,-1}}{\tilde{\sigma}_i \left(\mathbf{y}_{i,-1}' \mathbf{M}_\tau \mathbf{y}_{i,-1}\right)^{1/2}}, \tag{31.12}$$

where $\tilde{\sigma}_i^2 = (T-1)^{-1} \Delta \mathbf{y}_i' \mathbf{M}_\tau \Delta \mathbf{y}_i$. It is easily established that the panel unit root tests based on t_ϕ and $\tilde{\mathrm{t}}_\phi$ in the case of the pooled versions, and those based on $\bar{\mathrm{t}}$ and

$$\bar{\tilde{\mathrm{t}}} = N^{-1} \sum_{i=1}^{N} \tilde{\mathrm{t}}_i, \tag{31.13}$$

in the case of their mean group versions are asymptotically equivalent.

31.3.1 Distribution of tests under the null hypothesis

To establish the distribution of $\tilde{\mathrm{t}}_\phi$ and $\bar{\tilde{\mathrm{t}}}$, we first note that under $\phi_i = 0$, $\Delta \mathbf{y}_i = \sigma_i \mathbf{v}_i = \sigma_i (v_{i1}, v_{i2}, \ldots, v_{iT})'$, where $\mathbf{v}_i \backsim (\mathbf{0}, \mathbf{I}_T)$ and $\mathbf{y}_{i,-1}$ can be written as

[2] Andrews (1998) has considered optimal tests in such situations. His directed Wald statistic that gives a high weight to alternatives close to the null (i.e., parameter c in Andrews (1998) tends to zero) is equivalent to the mean of the individual specific test statistics.

[3] The mean of other unit root test statistics may be used as well. For example, Smith, Leybourne, Kim, and Newbold (2004) suggest using the mean of the weighted symmetric test statistic proposed for single time series by Park and Fuller (1995) and Fuller (1996) (see Section 10.1.3 in Fuller (1996)), or the Max-*ADF* test proposed by Leybourne (1995) based on the maximum of the original and the time-reversed Dickey–Fuller test statistics. See also Section 15.7.

$$\mathbf{y}_{i,-1} = y_{i0}\boldsymbol{\tau}_T + \sigma_i \mathbf{s}_{i,-1}, \tag{31.14}$$

where y_{i0} is a given initial value (fixed or random), $\mathbf{s}_{i,-1} = \left(s_{i0}, s_{i1}, \ldots, s_{i,T-1}\right)'$, with $s_{it} = \sum_{j=1}^{t} v_{ij}$, $t = 1, 2, \ldots, T$, and $s_{i0} = 0$. Using these results in (31.11) and (31.12) we have

$$\tilde{t}_\phi = \frac{\sum_{i=1}^{N}\left(\frac{\sqrt{T-1}\mathbf{v}_i'\mathbf{M}_\tau \mathbf{s}_{i,-1}}{\mathbf{v}_i'\mathbf{M}_\tau \mathbf{v}_i}\right)}{\sqrt{\sum_{i=1}^{N}\left(\frac{\mathbf{s}_{i,-1}'\mathbf{M}_\tau \mathbf{s}_{i,-1}}{\mathbf{v}_i'\mathbf{M}_\tau \mathbf{v}_i}\right)}},$$

and

$$\bar{\tilde{t}} = N^{-1}\sum_{i=1}^{N}\frac{\sqrt{T-1}\mathbf{v}_i'\mathbf{M}_\tau \mathbf{s}_{i,-1}}{\left(\mathbf{v}_i'\mathbf{M}_\tau \mathbf{v}_i\right)^{1/2}\left(\mathbf{s}_{i,-1}'\mathbf{M}_\tau \mathbf{s}_{i,-1}\right)^{1/2}}.$$

It is clear that under the null hypothesis both test statistics are free of nuisance parameters and their critical values can be tabulated for all combinations of N and T assuming, for example, that ε_{it} (or v_{it}) are normally distributed. Therefore, in the case where the errors, ε_{it}, are serially uncorrelated, an exact sample panel unit root test can be developed using either of the test statistics and no adjustments to the test statistics are needed. The main difference between the two tests lies in the way information on individual units is combined and their relative small sample performance would naturally depend on the nature of the alternative hypothesis being considered.

Asymptotic null distributions of the tests can also be derived depending on whether $(T, N) \to \infty$, sequentially, or when both N and $T \to \infty$, jointly. To derive the asymptotic distributions we need to work with the standardized versions of the test statistics

$$Z_{LL} = \frac{\tilde{t}_\phi - E\left(\tilde{t}_\phi\right)}{\sqrt{Var\left(\tilde{t}_\phi\right)}}, \tag{31.15}$$

and

$$Z_{IPS} = \frac{\sqrt{N}\left[\bar{t} - E(t_i)\right]}{\sqrt{Var(t_i)}}, \tag{31.16}$$

assuming that T is sufficiently large such that the second-order moments of t_i and t_ϕ exist. The conditions under which t_i has a second-order moment are discussed in IPS and it is shown that, when the underlying errors are normally distributed, the second-order moments exist for $T > 5$. For non-normal distributions, the existence of the moments can be ensured by basing the IPS test on suitably truncated versions of the individual t-ratios (see Pesaran (2007b) for further details). The exact first- and second-order moments of t_i and $\tilde{t}_i$ for different values of T are given in Im, Pesaran, and Shin (2003, Table 1). Using these results it is also possible to generalize the IPS test for unbalanced panels. Suppose the number of time periods available on the ith cross-sectional unit is T_i, the standardized IPS statistics will now be given by

$$Z_{IPS} = \frac{\sqrt{N}\left[\bar{t} - N^{-1}\sum_{i=1}^{N} E(t_{iT_i})\right]}{\sqrt{N^{-1}\sum_{i=1}^{N} Var(t_{iT_i})}}, \tag{31.17}$$

where $E(t_{iT_i})$ and $Var(t_{iT_i})$ are, respectively, the exact mean and variance of the DF statistics based on T_i observations. IPS show that for all finite $T_i > 6$, $Z_{IPS} \xrightarrow{d} \mathcal{N}(0,1)$ as $N \to \infty$. Similar results follow for the LL test.

To establish the asymptotic distribution of the panel unit root tests in the case of $T \to \infty$, we first note that for each i

$$t_i \xrightarrow{d} \eta_i = \frac{\int_0^1 \widetilde{W}_i(a)d\widetilde{W}_i(a)}{\int_0^1 \widetilde{W}_i(a)^2 da},$$

where $\widetilde{W}_i(a)$ is a demeaned Brownian motion defined as $\widetilde{W}_i(a) = W_i(a) - \int_0^1 W_i(a)da$ and $W_1(a), W_2(a), \ldots, W_N(a)$ are independent standard Brownian motions. The existence of the moments of η_i are established in Nabeya (1999) who also provides numerical values for the first six moments of the DF-distribution for the three standard specifications; namely models with and without intercepts and linear trends. Therefore, since the individual Dickey–Fuller statistics $t_1, t_2, \ldots, t_N$ are independent, it follows that $\eta_1, \eta_2, \ldots \eta_N$ are also independent with finite moments. Hence, by standard central limit theorems we have

$$Z_{IPS} \xrightarrow[T\to\infty]{d} \frac{\sqrt{N}\left[\bar{\eta} - E(\eta_i)\right]}{\sqrt{Var(\eta_i)}} \xrightarrow[N\to\infty]{d} \mathcal{N}(0,1),$$

where $\bar{\eta} = N^{-1}\sum_{i=1}^{N}\eta_i$. Similarly,

$$Z_{LL} = \frac{t_\phi - E(t_\phi)}{\sqrt{Var(t_\phi)}} \xrightarrow[(T,N)\to\infty]{d} \mathcal{N}(0,1).$$

To simplify the exposition, the above asymptotic results are derived using a sequential limit theory, where $T \to \infty$ is followed by $N \to \infty$. However, Phillips and Moon (1999) show that sequential convergence does not imply joint convergence so that in some situations the sequential limit theory may break down. In the case of models with serially uncorrelated errors, IPS (2003) show that the t-bar test is in fact valid for N and $T \to \infty$ jointly. Furthermore, as we shall see, the IPS test is valid for the case of serially correlated errors as N and $T \to \infty$ so long as $N/T \to k$ where k is a finite non-zero constant.

Maddala and Wu (1999) and Choi (2001) independently suggested a test against the heterogenous alternative H_1^c that is based on the p-values of the individual statistic as originally suggested by Fisher (1932). Let π_i denote the p-value of the individual specific unit root test applied to cross-sectional unit i. The combined test statistic is

$$\overline{\pi} = -2\sum_{i=1}^{N}\log(\pi_i). \tag{31.18}$$

Another possibility would be to use the inverse normal test defined by

$$Z_{INV} = \frac{1}{\sqrt{N}} \sum_{i=1}^{N} \Phi^{-1}(\pi_i), \tag{31.19}$$

where $\Phi(\cdot)$ denotes the cdf of the standard normal distribution. An important advantage of this approach is that it is possible to allow for different specifications (such as different deterministic terms and lag-order) for each panel unit.

Under the null hypothesis $\overline{\pi}$ is χ^2 distributed with $2N$ degrees of freedom. For large N the transformed statistic

$$\bar{\pi}^* = -\frac{1}{\sqrt{N}} \sum_{i=1}^{N} [\log(\pi_i) + 1], \tag{31.20}$$

is shown to have a standard normal limiting null distribution as $T, N \to \infty$, sequentially.

31.3.2 Asymptotic power of tests

It is interesting to compare the asymptotic power of test statistics against the sequence of local alternatives

$$H_\ell \quad : \quad \alpha_{i,NT} = 1 - \frac{c_i}{T\sqrt{N}}. \tag{31.21}$$

Following Breitung (2000) and Moon, Perron, and Phillips (2007), the asymptotic distribution under H_ℓ is obtained as $Z_j \xrightarrow{d} \mathcal{N}(-\bar{c}\,\theta_j, 1), j =$LL,IPS, where $\bar{c} = \lim_{N\to\infty} N^{-1} \sum_{i=1}^{N} c_i$ and

$$\theta_1 = \sqrt{E\left(\int_0^1 \widetilde{W}_i(a)^2 da\right)}, \; \theta_2 = \frac{E\left(\sqrt{\int_0^1 \widetilde{W}_i(a)^2 da}\right)}{\sqrt{Var(\mathrm{t}_i)}}.$$

It is interesting to note that the local power of both test statistics depends on the mean $\bar{c}$. Accordingly, the test statistics do not exploit the deviations from the mean value of the autoregressive parameter.

Moon, Perron, and Phillips (2007) derive the most powerful test statistic against the local alternative (31.21). Assume that we (randomly) choose the sequence $c_1^*, c_2^*, \ldots, c_N^*$ instead of the unknown values $c_1, c_2, \ldots, c_N$. The point optimal test statistic is constructed using the (local-to-unity) pseudo differences

$$\Delta_{c_i^*} y_{it} = y_{it} - \left(1 - c_i^*/T\sqrt{N}\right) y_{i,t-1} \text{ for } t = 1, 2, \ldots, T.$$

For the model without individual constants and homogeneous variances the point optimal test results in the statistic

$$V_{NT} = \frac{1}{\hat{\sigma}^2}\left[\sum_{i=1}^{N}\sum_{t=1}^{T}(\Delta_{c_i^*} y_{it})^2 - (\Delta y_{it})^2\right] - \frac{1}{2}\kappa^2,$$

where $E(c_i^*)^2 = \kappa^2$. Under the sequence of local alternatives (31.21), Moon, Perron, and Phillips (2007) derive the limiting distribution as

$$V_{NT} \xrightarrow{d} \mathcal{N}\left(-E(c_i c_i^*), 2\kappa^2\right).$$

The upper bound of the local power is achieved with $c_i = c_i^*$, that is, if the local alternatives used to construct the test coincide with the actual alternative. Unfortunately, in practice it seems extremely unlikely that one could select values of c_i^* that are perfectly correlated with the true values, c_i. If, on the other hand, the variates c_i^* are independent of c_i, then the power is smaller than the power of a test using identical values $c_i^* = c^*$ for all i. This suggests that if there is no information about the variation of c_i, then a test cannot be improved by taking into account a possible heterogeneity of the alternative.

31.3.3 Heterogeneous trends

To allow for more general mean functions we consider the model

$$y_{it} = \boldsymbol{\delta}_i' \mathbf{d}_{it} + \tilde{y}_{it}, \tag{31.22}$$

where $\mathbf{d}_{it}$ represents the deterministics and $\Delta\tilde{y}_{it} = \phi_i \tilde{y}_{i,t-1} + \varepsilon_{it}$. For the model with a constant mean we let $\mathbf{d}_{it} = 1$ and the model with individual specific time trends $\mathbf{d}_{it}$ is given by $\mathbf{d}_{it} = (1, t)'$. Furthermore, structural breaks in the mean function can be accommodated by including (possibly individual specific) dummy variables in the vector $\mathbf{d}_{it}$. The parameter vector $\boldsymbol{\delta}_i$ is assumed to be unknown and has to be estimated. For the Dickey–Fuller test statistic, the mean function is estimated under the alternative, that is, for the model with a time trend, $\hat{\boldsymbol{\delta}}_i' \mathbf{d}_{it}$ can be estimated from a regression of y_{it} on a constant and t ($t = 1, 2, \ldots, T$). Alternatively, the mean function can also be estimated under the null hypothesis (see Schmidt and Phillips (1992)) or under a local alternative (Elliott, Rothenberg, and Stock (1996)).[4]

Including deterministic terms may have an important effect on the asymptotic properties of the test. Let $\Delta\hat{\tilde{y}}_t$ and $\hat{\tilde{y}}_{i,t-1}$ denote estimates for $\Delta\tilde{y}_{it} = \Delta y_{it} - E(\Delta y_{it})$ and $\tilde{y}_{i,t-1} = y_{i,t-1} - E(y_{i,t-1})$. In general, running the regression

$$\Delta\hat{\tilde{y}}_{it} = \phi\hat{\tilde{y}}_{i,t-1} + e_{it}$$

does not render a t-statistic with a standard normal limiting distribution due to the fact that $\hat{\tilde{y}}_{i,t-1}$ is correlated with e_{it}. For example, if $\mathbf{d}_{it}$ is an individual specific constant such that $\hat{\tilde{y}}_{it} = y_{it} - T^{-1}(y_{i0} + \cdots + y_{i,T-1})$, we obtain under the null hypothesis

[4] See, e.g. Choi (2002) and Harvey, Leybourne, and Sakkas (2006).

$$\lim_{T\to\infty} \frac{1}{T} E\left(\sum_{t=1}^{T} e_{it}\hat{\tilde{y}}_{i,t-1}\right) = -\sigma_i^2/2\,.$$

It follows that the t-statistic of $\phi = 0$ tends to $-\infty$ as N or T tends to infinity.

To correct for the bias, Levin, Lin, and Chu (2002) suggested using the correction terms

$$a_T(\hat{\boldsymbol{\delta}}) = E\left(\frac{1}{\sigma_i^2 T}\sum_{t=1}^{T}\Delta\hat{\tilde{y}}_{it}\hat{\tilde{y}}_{i,t-1}\right), \tag{31.23}$$

$$b_T^2(\hat{\boldsymbol{\delta}}) = \frac{Var\left(T^{-1}\sum_{t=1}^{T}\Delta\hat{\tilde{y}}_{it}\hat{\tilde{y}}_{i,t-1}\right)}{\sigma_i^2 E\left(T^{-1}\sum_{t=1}^{T}\hat{\tilde{y}}_{i,t-1}^2\right)}, \tag{31.24}$$

where $\hat{\boldsymbol{\delta}} = (\hat{\boldsymbol{\delta}}_1', \hat{\boldsymbol{\delta}}_2', \ldots, \hat{\boldsymbol{\delta}}_N')'$, and $\hat{\boldsymbol{\delta}}_i$ is the estimator of the coefficients of the deterministics, $\mathbf{d}_{it}$, in the *OLS* regression of y_{it} on $\mathbf{d}_{it}$. The corrected, standardized statistic is given by

$$Z_{LL}(\hat{\boldsymbol{\delta}}) = \frac{\left(\sum_{i=1}^{N}\sum_{t=1}^{T}\Delta\hat{\tilde{y}}_{it}\hat{\tilde{y}}_{i,t-1}/\hat{\sigma}_i^2\right) - NTa_T(\hat{\boldsymbol{\delta}})}{b_T(\hat{\boldsymbol{\delta}})\sqrt{\sum_{i=1}^{N}\sum_{t=1}^{T}\hat{\tilde{y}}_{i,t-1}^2/\hat{\sigma}_i^2}}\,.$$

Levin, Lin, and Chu (2002) present simulated values of $a_T(\hat{\boldsymbol{\delta}})$ and $b_T(\hat{\boldsymbol{\delta}})$ for models with constants, time trends and various values of T. A problem is, however, that for unbalanced data sets no correction terms are tabulated.

Alternatively, the test statistic may be corrected such that the adjusted t-statistic

$$Z_{LL}^*(\hat{\boldsymbol{\delta}}) = [Z_{LL}(\hat{\boldsymbol{\delta}}) - a_T^*(\hat{\boldsymbol{\delta}})]/b_T^*(\hat{\boldsymbol{\delta}})$$

is asymptotically standard normal. Harris and Tzavalis (1999) derive the small sample values of $a_T^*(\hat{\boldsymbol{\delta}})$ and $b_T^*(\hat{\boldsymbol{\delta}})$ for T fixed and $N \to \infty$. Therefore, their test statistic can be applied for small values of T and large values of N.

An alternative approach is to avoid the bias—and hence the correction terms—by using alternative estimates of the deterministic terms. Breitung and Meyer (1994) suggest using the initial value y_{i0} as an estimator of the constant term. As argued by Schmidt and Phillips (1992), the initial value is the best estimate of the constant given the null hypothesis is true. Using this approach, the regression equation for a model with a constant term becomes

$$\Delta y_{it} = \phi^*(y_{i,t-1} - y_{i0}) + v_{it}\,.$$

Under the null hypothesis, the pooled t-statistic of $H_0 : \phi^* = 0$ has a standard normal limit distribution.

For a model with a linear time trend, a minimal invariant statistic is obtained by the transformation (see Ploberger and Phillips (2002))

$$x_{it}^{*} = y_{it} - y_{i0} - \frac{t}{T}(y_{iT} - y_{i0}) .$$

In this transformation, subtracting y_{i0} eliminates the constant and $(y_{iT} - y_{i0})/T = (\Delta y_{i1} + \cdots + \Delta y_{iT})/T$ is an estimate of the slope of the individual trend function.

A Helmert transformation can be used to correct for the mean of Δy_{it},

$$\Delta y_{it}^{*} = s_t \left[\Delta y_{it} - \frac{1}{T-t}(\Delta y_{i,t+1} + \cdots + \Delta y_{iT}) \right], t = 1, \ldots, T-1,$$

where $s_t^2 = (T-t)/(T-t+1)$ (see Arellano (2003), p. 17). Using these transformations, the regression equation becomes

$$\Delta y_{it}^{*} = \phi^{*} x_{i,t-1}^{*} + v_{it} . \tag{31.25}$$

It is not difficult to verify that, under the null hypothesis we have $E(\Delta y_{it}^{*} x_{i,t-1}^{*}) = 0$, and thus the t-statistic for $\phi^{*} = 0$ is asymptotically standard normally distributed (see Breitung (2000)).

It is important to note that including individual specific time trends substantially reduces the (local) power of the test. This was first observed by Breitung (2000) and studied more rigorously by Ploberger and Phillips (2002) and Moon, Perron, and Phillips (2007). Specifically, the latter two papers show that a panel unit root test with incidental trends has non-trivial asymptotic power only for local alternatives with rate $T^{-1}N^{-1/4}$. A similar result is found by Moon, Perron, and Phillips (2006) for the test suggested by Breitung (2000).

The test against heterogeneous alternatives, H_1^c, can easily be adjusted for individual specific deterministic terms such as linear trends or seasonal dummies. This can be done by computing IPS statistics, defined by (31.16) and (31.17) for the balanced and unbalanced panels, using Dickey–Fuller t-statistics based on DF regressions including the deterministics $\boldsymbol{\delta}_i' \mathbf{d}_{it}$, where $\mathbf{d}_{it} = 1$ in the case of a constant term, $\mathbf{d}_{it} = (1, t)'$ in the case of models with a linear time trend and so on. The mean and variance corrections should, however, be computed to match the nature of the deterministics. Under a general setting IPS (2003) have shown that the Z_{IPS} statistic converges in distribution to a standard normal variate as $N, T \to \infty$, jointly.

In a straightforward manner it is possible to include dummy variables in the vector $\mathbf{d}_{it}$ that accommodate structural breaks in the mean function (see, e.g., Murray and Papell (2002); Tzavalis (2002); Carrion-i-Sevestre, Del Barrio, and Lopez-Bazo (2005); Breitung and Candelon (2005); Im, Lee, and Tieslau (2005)).

31.3.4 Short-run dynamics

If it is assumed that the error in the autoregression (31.1) is a serially correlated stationary process, the short-run dynamics of the errors can be accounted for by including lagged differences

$$\Delta y_{it} = \boldsymbol{\delta}_i' \mathbf{d}_{it} + \phi_i y_{i,t-1} + \gamma_{i1} \Delta y_{i,t-1} + \cdots + \gamma_{i,p_i} \Delta y_{i,t-p_i} + \varepsilon_{it} . \tag{31.26}$$

For example, the IPS statistics (31.16) and (31.17) developed for balanced and unbalanced panels can now be constructed using the ADF(p_i) statistics based on the above regressions. As noted in IPS (2003), small sample properties of the test can be much improved if the standardization of the IPS statistic is carried out using the simulated means and variances of $t_i(p_i)$, the t-ratio of ϕ_i computed based on ADF(p_i) regressions. This is likely to yield better approximations, since $E[t_i(p_i)]$, for example, makes use of the information contained in p_i while $E[t_i(0)] = E(t_i)$ does not. Therefore, in the serially correlated case, IPS propose the following standardized t-bar statistic

$$Z_{IPS} = \frac{\sqrt{N}\left\{\bar{t} - \frac{1}{N}\sum_{i=1}^{N} E[t_i(p_i)]\right\}}{\sqrt{\frac{1}{N}\sum_{i=1}^{N} Var[t_i(p_i)]}} \xrightarrow[(T,N)\to\infty]{d} \mathcal{N}(0,1). \tag{31.27}$$

The values of $E[t_i(p)]$ and $Var[t_i(p)]$ simulated for different combinations of T and p, are provided in Table 3 of IPS. These simulated moments also allow the IPS panel unit root test to be applied to unbalanced panels with serially correlated errors.

For tests against the homogeneous alternatives, $\phi_1 = \phi_2 = \cdots = \phi_N = \phi < 0$, Levin, Lin, and Chu (2002) suggest removing all individual specific parameters within a first step regression such that e_{it} ($v_{i,t-1}$) are the residuals from a regression of Δy_{it} ($y_{i,t-1}$) on $\Delta y_{i,t-1}, \ldots, \Delta y_{i,t-p_i}$ and d_{it}. In the second step the common parameter ϕ is estimated from a pooled regression

$$(e_{it}/\hat{\sigma}_i) = \phi(v_{i,t-1}/\hat{\sigma}_i) + \nu_{it},$$

where $\hat{\sigma}_i^2$ is the estimated variance of e_{it}. Unfortunately, the first step regressions are not sufficient to remove the effect of the short-run dynamics on the null distribution of the test. Specifically,

$$\lim_{T\to\infty} E\left[\frac{1}{T-p}\sum_{t=p+1}^{T} e_{it}v_{i,t-1}/\sigma_i^2\right] = \frac{\bar{\sigma}_i}{\sigma_i}\, a_\infty(\hat{\boldsymbol{\delta}}),$$

where $\bar{\sigma}_i^2$ is the long-run variance and $a_\infty(\hat{\boldsymbol{\delta}})$ denotes the limit of the correction term given in (31.23). Levin, Lin, and Chu (2002) propose a nonparametric (kernel based) estimator for $\bar{\sigma}_i^2$

$$\bar{s}_i^2 = \frac{1}{T}\left[\sum_{t=1}^{T}\Delta\hat{\tilde{y}}_{it}^2 + 2\sum_{l=1}^{K}\left(\frac{K+1-l}{K+1}\right)\left(\sum_{t=l+1}^{T}\Delta\hat{\tilde{y}}_{it}\Delta\hat{\tilde{y}}_{i,t-l}\right)\right], \tag{31.28}$$

where $\Delta\hat{\tilde{y}}_{it}$ denotes the demeaned difference and K denotes the truncation lag. As noted by Breitung and Das (2005), in a time series context the estimator of the long-run variance based on differences is inappropriate since under the stationary alternative $\bar{s}_i^2 \xrightarrow{p} 0$; thus the use of this estimator yields an inconsistent test. In contrast, in the case of panels the use of $\bar{s}_i^2$ improves the power of the test, since with $\bar{s}_i^2 \xrightarrow{p} 0$ the correction term drops out and the test statistic tends to $-\infty$.

It is possible to avoid the use of a kernel based estimator of the long-run variance by using an alternative approach suggested by Breitung and Das (2005). Under the null hypothesis we have

$$\gamma_i(L)\Delta y_{it} = \boldsymbol{\delta}_i'\mathbf{d}_{it} + \varepsilon_{it}\,,$$

where $\gamma_i(L) = 1-\gamma_{i1}L-\cdots-\gamma_{i,p_i}L^p$ and L is the lag operator. It follows that $\widetilde{g}_t = \gamma_i(L)[y_{it}-E(y_{it})]$ is a random walk with uncorrelated increments. Therefore, the serial correlation can be removed by replacing y_{it} by the pre-whitened variable $\hat{y}_{it} = \hat{\gamma}_i(L)y_{it}$, where $\hat{\gamma}_i(L)$ is an estimator of the lag polynomial obtained from the least-square regression

$$\Delta y_{it} = \boldsymbol{\delta}_i'\mathbf{d}_{it} + \gamma_{i1}\Delta y_{i,t-1} + \cdots + \gamma_{i,p_i}\Delta y_{i,t-p_i} + \varepsilon_{it}\,. \tag{31.29}$$

This approach may also be used for modifying the 'unbiased statistic' based on the t-statistic of $\phi^* = 0$ in (31.25). The resulting t-statistic has a standard normal limiting distribution if $T \to \infty$ is followed by $N \to \infty$.

A related approach is proposed by Westerlund (2009), who suggests testing the unit root hypothesis by running a modified ADF regression of the form

$$\Delta y_{it} = \boldsymbol{\delta}_i'\mathbf{d}_{it} + \phi_i y^*_{i,t-1} + \gamma_{i1}\Delta y_{i,t-1} + \cdots + \gamma_{i,p_i}\Delta y_{i,t-p_i} + \varepsilon_{it}\,, \tag{31.30}$$

where $y^*_{i,t-1} = (\hat{\sigma}_i/\overline{s}_i)y_{i,t-1}$ and $\overline{s}_i^2$ is a consistent estimator of the long-run variance, $\overline{\sigma}_i^2$. Westerlund (2009) recommends using a parametric estimate of the long-run variance based on an autoregressive representation. This transformation of the lagged dependent variable eliminates the nuisance parameters in the asymptotic distribution of the ADF statistic and, therefore, the correction for the numerator of the corrected t-statistic of Levin, Lin, and Chu (2002) is the same as in the case without short-run dynamics.

Pedroni and Vogelsang (2005) have proposed a test statistic that avoids the specification of the short-run dynamics by using an autoregressive approximation. Their test statistic is based on the pooled variance ratio statistic

$$Z^w_{NT} = \frac{Tc_i(0)}{N\hat{s}_i^2},$$

where $c_i(\ell) = T^{-1}\sum_{t=\ell+1}^{T}\hat{\tilde{y}}_{it}\hat{\tilde{y}}_{i,t-\ell}$, $\hat{\tilde{y}}_{it} = y_{it} - \hat{\boldsymbol{\delta}}_i'\mathbf{d}_{it}$, and $\hat{s}_i^2$ is the untruncated Bartlett kernel estimator defined as $\hat{s}_i^2 = \sum_{\ell=-T+1}^{T+1}(1-|\ell|/T)c_i(\ell)$. As has been shown by Kiefer and Vogelsang (2002) and Breitung (2002), the limiting distribution of such 'non-parametric' statistics does not depend on nuisance parameters involved by the short run dynamics of the processes. Accordingly, no adjustment for short-run dynamics is necessary.

31.3.5 Other approaches to panel unit root testing

An important problem of combining Dickey–Fuller type statistics in a panel unit root test is that they involve a nonstandard limiting distribution. If the panel unit root statistic is based on a standard normally distributed test statistic z_i, then $N^{-1/2}\sum_{i=1}^{N} z_i$ has a standard normal limiting

distribution even for a finite N. In this case no correction terms need to be tabulated to account for the mean and the variance of the test statistic.

Chang (2002) proposes a nonlinear instrumental variable (IV) approach, where the transformed variable

$$w_{i,t-1} = y_{i,t-1} e^{-c_i |y_{i,t-1}|},$$

with $c_i > 0$ is used as an instrument for estimating ϕ_i in the regression $\Delta y_{it} = \phi_i y_{i,t-1} + \varepsilon_{it}$ (which may also include deterministic terms and lagged differences). Since $w_{i,t-1}$ tends to zero as $y_{i,t-1}$ tends to $\pm\infty$ the trending behaviour of the nonstationary variable $y_{i,t-1}$ is eliminated. Using the results of Chang, Park, and Phillips (2001), Chang (2002) showed that the Wald test of $\phi = 0$ based on the nonlinear IV estimator possesses a standard normal limiting distribution. Another important property of the test is that the nonlinear transformation also takes account of possible contemporaneous dependence among the cross-sectional units. Accordingly, Chang's panel unit root test is also robust against cross-sectional dependence.

It should be noted that $w_{i,t-1} \in [-(c_i e)^{-1}, (c_i e)^{-1}]$ with a maximum (minimum) at $y_{i,t-1} = 1/c_i$ ($y_{i,t-1} = -1/c_i$). Therefore, the choice of the parameter c_i is crucial for the properties of the test. First, the parameter should be proportional to the inverse of the standard deviations of Δy_{it}. Chang notes that, if the time dimension is short, the test slightly over-rejects the null and therefore she proposes the use of a larger value of K to correct for the size distortion.

An alternative approach to obtain an asymptotically standard normal test statistic is to adjust the given samples in all cross-sections so that they all have sums of squares $y_{i1}^2 + \cdots + y_{ik_i}^2 = \sigma_i^2 c T^2 + h_i$, where $h_i \xrightarrow{p} 0$ as $T \to \infty$. In other words, the panel data set becomes an unbalanced panel with k_i time periods in the ith unit. Chang calls this setting the 'equi-squared sum contour', whereas the traditional framework is called the 'equi-sample-size contour'. The nice feature of this approach is that it yields asymptotically standard normal test statistics. An important drawback is, however, that a large number of observations may be discarded by applying this contour which may result in a severe loss of power.

Testing the null of stationarity in panels

As in the time series case (see Section 15.7.5), it is possible to test the null hypothesis that the series are stationary against the alternative that (at least some of) the series are nonstationary. The test suggested by Tanaka (1990) and Kwiatkowski et al. (1992) is designed to test the hypothesis $H_0^* : \theta_i = 0$ in the model

$$y_{it} = \boldsymbol{\delta}_i' \mathbf{d}_{it} + \theta_i r_{it} + u_{it}\,,\ t = 1, \ldots, T, \tag{31.31}$$

where Δr_{it} is white noise with unit variance and u_{it} is stationary. The cross-sectional specific KPSS statistic is

$$\kappa_i = \frac{1}{T^2 \bar{\sigma}_{T,i}^2} \sum_{t=1}^{T} \hat{S}_{it}^2\,,$$

where $\bar{\sigma}_{T,i}^2$ denotes a consistent estimator of the long-run variance of Δy_{it} and $\hat{S}_{it} = \sum_{\ell=1}^{t} \left(y_{i\ell} - \hat{\boldsymbol{\delta}}_i' \mathbf{d}_{i\ell} \right)$ is the partial sum of the residuals from a regression of y_{it} on the deterministic

terms (a constant or a linear time trend). The individual test statistics can be combined as in the test suggested by IPS (2003) yielding

$$\bar{\kappa} = N^{-1/2} \frac{\sum_{i=1}^{N} [\kappa_i - E(\kappa_i)]}{\sqrt{Var(\kappa_i)}},$$

where asymptotic values of $E(\kappa_i)$ and $Var(\kappa_i)$ are derived in Hadri (2000) and values for finite T and $N \to \infty$ are presented in Hadri and Larsson (2005).

The test of Harris, Leybourne, and McCabe (2004) is based on the stationarity statistic

$$Z_i(k) = \sqrt{T}\, \hat{c}_i(k) / \hat{\omega}_{zi}(k),$$

where $\hat{c}_i(k)$ denotes the usual estimator of the covariance at lag k of cross-sectional unit i and $\hat{\omega}^2_{zi}(k)$ is an estimator of the long-run variance of $z^k_{it} = (y_{it} - \hat{\boldsymbol{\delta}}_i' \mathbf{d}_{it})(y_{i,t-k} - \hat{\boldsymbol{\delta}}_i' \mathbf{d}_{i,t-k})$. The intuition behind this test statistic is that for a stationary and ergodic time series we have $E[\hat{c}_i(k)] \to 0$ as $k \to \infty$. Since $\hat{\omega}^2_{zi}$ is a consistent estimator for the variance of $\hat{c}_i(k)$ it follows that $Z_i(k)$ converges to a standard normally distributed random variable as $k \to \infty$ and $k/\sqrt{T} \to \delta < \infty$.

31.3.6 Measuring the proportion of cross-units with unit roots

A strand of literature proposes methods for estimating the proportion of stationary units, δ (or, equivalently, the fraction of the panel having unit roots, $1 - \delta$) in the panel, rather than looking at the non-stationarity properties of individual series in the panel (see Pesaran (2012)). In the context of testing for output and growth convergence, Pesaran (2007a) suggests using the proportion of unit root tests applied to pairs of log per capita output gaps across N economies, for which the null hypothesis of non-stationarity is rejected at a given significance level, α. He shows that, although the underlying individual unit root tests are not cross-sectionally independent, under the null hypothesis of non-stationarity such average rejection statistic converges to α, as N and T jointly tend to infinity. Similarly, Ng (2008) shows that, if a fraction, δ, of the panel is made of stationary units, with the remaining series having unit roots, the cross-sectional variance of the panel will have a linear trend that increases exactly at rate $1 - \delta$. Hence, she suggests a statistic for the proportion of non-stationary units $(1 - \delta)$ based on the time average of the sample cross-sectional variance. While these procedures deliver an estimate of the fraction of (non)stationary units, they are not designed to identify which units are stationary. After rejection of the null hypothesis of unit roots for each individual series in a panel, it is often of interest to identify which series can be considered stationary and which can be deemed non-stationary. Kapetanios (2003) and Chortareas and Kapetanios (2009) propose a sequential panel selection method that consists of applying the Im, Pesaran, and Shin (2003) panel unit root test sequentially on progressively smaller fractions of the original data set, where the reduction is carried out by dropping series for which there is evidence of stationarity, signalled by low individual t-statistics. A similar approach is taken by Smeeks (2010), who proposes testing on user-defined fractions of the panel, using panel unit root tests based on order statistics and computing the corresponding critical values by block bootstrap. Hanck (2009) and Moon and Perron (2012) apply methods from the literature on multiple testing to classify the individual series into stationary and non-stationary sets. In particular, Moon and Perron (2012) suggest the use of the

so-called false discovery rate (FDR), given by the expected fraction of series classified as I(0) that are in fact I(1), as a useful diagnostic on the aggregate decision. In the computation of the FDR, the authors estimate the fraction of true null hypotheses by applying the Ng (2008) approach described above.

31.4 Second generation panel unit root tests

31.4.1 Cross-sectional dependence

So far we have assumed that the time series $\{y_{it}\}_{t=0}^{T}$ are independent across i. However, as discussed in Chapter 29, in many macroeconomic applications using country or regional data it is found that the time series are contemporaneously correlated. Prominent examples are the analysis of purchasing power parity and output convergence.[5]

Abstracting from common observed effects and residual serial correlation, a general specification for cross-sectional error dependence can be written as

$$\Delta y_{it} = -\mu_i \phi_i + \phi_i y_{i,t-1} + u_{it},$$

where

$$u_{it} = \boldsymbol{\gamma}_i' \mathbf{f}_t + \xi_{it}, \tag{31.32}$$

or

$$\mathbf{u}_t = \boldsymbol{\Gamma} \mathbf{f}_t + \boldsymbol{\xi}_t, \tag{31.33}$$

$\mathbf{u}_t = (u_{1t}, u_{2t}, \ldots, u_{Nt})'$, $\mathbf{f}_t$ is an $m \times 1$ vector of serially uncorrelated unobserved common factors, and $\boldsymbol{\xi}_t = (\xi_{1t}, \xi_{2t}, \ldots, \xi_{Nt})'$ is an $N \times 1$ vector of serially uncorrelated errors with mean zero and the positive definite covariance matrix $\boldsymbol{\Sigma}_\xi$, and $\boldsymbol{\Gamma}$ is an $N \times m$ matrix of factor loadings defined by $\boldsymbol{\Gamma} = (\boldsymbol{\gamma}_1, \boldsymbol{\gamma}_2, \ldots, \boldsymbol{\gamma}_N)'$.[6] Without loss of generality, the covariance matrix of $\mathbf{f}_t$ is set to $\mathbf{I}_m$, and it is assumed that $\mathbf{f}_t$ and $\boldsymbol{\xi}_t$ are independently distributed. If $\boldsymbol{\gamma}_1 = \cdots = \boldsymbol{\gamma}_N$, then $\theta_t = \boldsymbol{\gamma}' \mathbf{f}_t$ is a conventional 'time effect' that can be removed by subtracting the cross-section means from the data. In general it is assumed that $\boldsymbol{\gamma}_i$, the factor loading for the ith cross-sectional unit, differs across i and represents draws from a given distribution.

Under the above assumptions and conditional on $\boldsymbol{\gamma}_i, i = 1, 2, \ldots, N$, the covariance matrix of the composite errors, $\mathbf{u}_t$, is given by $\boldsymbol{\Sigma} = \boldsymbol{\Gamma}\boldsymbol{\Gamma}' + \boldsymbol{\Sigma}_\xi$. It is clear that without further restrictions the matrices $\boldsymbol{\Gamma}$ and $\boldsymbol{\Sigma}_\xi$ are not separately identified. The properties of $\boldsymbol{\Sigma}$ also crucially depend on the relative eigenvalues of $\boldsymbol{\Gamma}\boldsymbol{\Gamma}'$ and $\boldsymbol{\Sigma}_\xi$, and their limits as $N \to \infty$. A general discussion of the concepts of weak and strong cross-sectional dependence is provided in Chapter 29, where it is shown that all spatial econometric models considered in the literature are examples of weak cross-sectional dependence.

[5] See, for example, O'Connell (1998) and Phillips and Sul (2003).

[6] The case where $\mathbf{f}_t$ and/or ξ_{it} might be serially correlated will be considered below.

A simple example of panel data models with weak cross-sectional dependence is given by

$$\begin{pmatrix} \Delta y_{1t} \\ \Delta y_{2t} \\ \vdots \\ \Delta y_{Nt} \end{pmatrix} = \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_N \end{pmatrix} + \phi \begin{pmatrix} \Delta y_{1,t-1} \\ \Delta y_{2,t-2} \\ \vdots \\ \Delta y_{N,t-1} \end{pmatrix} + \begin{pmatrix} u_{1t} \\ u_{2t} \\ \vdots \\ u_{Nt} \end{pmatrix} \tag{31.34}$$

or

$$\Delta \mathbf{y}_t = \mathbf{a} + \phi \mathbf{y}_{t-1} + \mathbf{u}_t, \tag{31.35}$$

where $a_i = -\phi\mu_i$, $\Delta\mathbf{y}_t$, $\mathbf{y}_{t-1}$, $\mathbf{a}$ and $\mathbf{u}_t$ are $N \times 1$ vectors, and the cross-sectional correlation is represented by a non-diagonal matrix

$$\mathbf{\Sigma} = E(\mathbf{u}_t \mathbf{u}_t'), \text{ for all } t,$$

with bounded eigenvalues. For the model without constants, Breitung and Das (2005) showed that the regression t-statistic of $\phi = 0$ in (31.35) is asymptotically distributed as $\mathcal{N}(0, \nu)$ where

$$\nu = \lim_{N\to\infty} \frac{tr(\mathbf{\Sigma}^2/N)}{(tr\mathbf{\Sigma}/N)^2}. \tag{31.36}$$

Note that $tr(\mathbf{\Sigma})$ and $tr(\mathbf{\Sigma}^2)$ are $O(N)$ and, thus, ν converges to a constant that can be shown to be larger than one. This explains why the test ignoring the cross-correlation of the errors has a positive size bias.

31.4.2 Tests based on *GLS* regressions

Since (31.35) can be seen as a seemingly unrelated regression system, O'Connell (1998) suggests estimating the system by using a *GLS* estimator (see also Flores et al. (1999)). Let $\hat{\mathbf{\Sigma}} = T^{-1}\sum_{t=1}^{T} \hat{\mathbf{u}}_t \hat{\mathbf{u}}_t'$ denote the sample covariance matrix of the residual vector. The *GLS* t-statistic is given by

$$t_{gls}(N) = \frac{\sum_{t=1}^{T} \Delta\widetilde{\mathbf{y}}_t' \hat{\mathbf{\Sigma}}^{-1} \widetilde{\mathbf{y}}_{t-1}}{\sqrt{\sum_{t=1}^{T} \widetilde{\mathbf{y}}_{t-1}' \hat{\mathbf{\Sigma}}^{-1} \widetilde{\mathbf{y}}_{t-1}}},$$

where $\widetilde{\mathbf{y}}_t$ is the vector of demeaned variables. Harvey and Bates (2003) derive the limiting distribution of $t_{gls}(N)$ for a fixed N and as $T \to \infty$, and tabulate its asymptotic distribution for various values of N. Breitung and Das (2005) show that if $\widetilde{\mathbf{y}}_t = \mathbf{y}_t - \mathbf{y}_0$ is used to demean the variables and $T \to \infty$ is followed by $N \to \infty$, then the *GLS* t-statistic possesses a standard normal limiting distribution.

The *GLS* approach cannot be used if $T < N$ since in this case the estimated covariance matrix $\hat{\Sigma}$ is singular. Furthermore, Monte Carlo simulations suggest that for reasonable size properties

of the *GLS* test, T must be substantially larger than N (e.g., Breitung and Das (2005)). Maddala and Wu (1999) and Chang (2004) have suggested a bootstrap procedure that improves the size properties of the *GLS* test.

31.4.3 Tests based on *OLS* regressions

An alternative approach based on 'panel corrected standard errors' (*PCSE*) is considered by Jönsson (2005) and Breitung and Das (2005). In the model with weak dependence, the variance of the *OLS* estimator $\hat{\phi}$ is consistently estimated by

$$\widehat{Var}(\hat{\phi}) = \frac{\sum_{t=1}^{T} \widetilde{\mathbf{y}}'_{t-1} \hat{\boldsymbol{\Sigma}} \widetilde{\mathbf{y}}_{t-1}}{\left(\sum_{t=1}^{T} \widetilde{\mathbf{y}}'_{t-1} \widetilde{\mathbf{y}}_{t-1} \right)^2}.$$

If $T \to \infty$ is followed by $N \to \infty$ the robust t statistic $t_{rob} = \hat{\phi} / \sqrt{\widehat{Var}(\hat{\phi})}$ is asymptotically standard normally distributed (Breitung and Das (2005)).

If it is assumed that the cross-correlation is due to common factors, then the largest eigenvalue of the error covariance matrix, $\boldsymbol{\Sigma}$, is $O_p(N)$ and the robust PCSE approach breaks down. Specifically, Breitung and Das (2008) show that in this case t_{rob} is distributed as the ordinary Dickey–Fuller test applied to the first principal component.

In the case of a single unobserved common factor, Pesaran (2007b) suggests a simple modification of the usual test procedure. Let $\bar{y}_t = N^{-1} \sum_{i=1}^{N} y_{it}$ and $\Delta \bar{y}_t = N^{-1} \sum_{i=1}^{N} \Delta y_{it} = \bar{y}_t - \bar{y}_{t-1}$. The cross-section augmented Dickey–Fuller (CADF) test is based on the following regression

$$\Delta y_{it} = a_i + \phi_i y_{i,t-1} + b_i \bar{y}_{t-1} + c_i \Delta \bar{y}_t + e_{it}\,.$$

In this regression the additional variables $\Delta \bar{y}_t$ and $\bar{y}_{t-1}$ are $\sqrt{N}$-consistent estimators for the rescaled factors $\bar{\gamma} f_t$ and $\bar{\gamma} \sum_{j=0}^{t-1} f_j$, where $\bar{\gamma} = N^{-1} \sum_{i=1}^{N} \gamma_i$. Pesaran (2007b) shows that the distribution of the regression t-statistic for $\phi_i = 0$ is free of nuisance parameters. To test the unit root hypothesis in a heterogenous panel, the average of the N individual CADF t-statistics (or suitably truncated version of them) can be used. Coakley, Kellard, and Snaith (2005) apply the CADF test to real exchange rates of fifteen OECD countries.

Pesaran, Smith, and Yamagata (2013) have extended the cross-sectionally augmented panel unit root test proposed by Pesaran (2007b) to the case of a multifactor error structure. They propose utilizing the information contained in a number of k additional variables, $\mathbf{x}_{it}$, that together are assumed to share the common factors of the series of interest, y_{it}. The basic idea is to exploit information regarding the m unobserved factors that are shared by k observed time series (or covariates) in addition to the series under consideration. The requirement of finding such additional variables seems quite plausible in the case of panel data sets from economics and finance where economic agents often face common economic environments. Most macroeconomic theories postulate the presence of the same unobserved common factors (such as shocks to technology, tastes and fiscal policy), and it is therefore natural to expect that many macroeconomic

variables, such as interest rates, inflation and output share the same factors. If anything, it would be difficult to find macroeconomic time series that do not share one or more common factors. For example, in testing for unit roots in a panel of real outputs one would expect the unobserved common shocks to output (that originate from technology) to also manifest themselves in employment, consumption and investment. In the case of testing for unit roots in inflation across countries, one would expect the unobserved common factors that correlate inflation rates across countries to also affect short-term and long-term interest rates across markets and economies. The basic idea of using covariates to deal with a multiple factor structure is intuitive and easy to implement—the ADF regression for y_{it} is simply augmented with cross-section averages of y_{it} and $\mathbf{x}_{it}$.[7] Pesaran, Smith, and Yamagata (2013) show that the extended version of the Pesaran (2007b) test, denoted as *CIPS*, is valid so long as $k = m_{\max} - 1$, where $m_{\max}$ is the assumed maximum number of factors. Importantly, the estimation of the true number of factors, m, is not needed so long as $m \leq m_{\max}$. Furthermore, it is not required that all of the factors be strong. Following Bai and Ng (2010), Pesaran, Smith, and Yamagata (2013) also consider a panel unit root test based on simple averages of cross-sectionally augmented Sargan-Bhargava-type statistics, denoted as *CSB*. Monte Carlo simulations reported by these authors suggest that both *CIPS* and *CSB* tests have the correct size across different experiments and with various combinations of N and T being considered. The experimental results also show that the proposed *CSB* test has satisfactory power, which for some combinations of N and T tends to be higher than that of the *CIPS* test.

31.5 Cross-unit cointegration

As argued by Banerjee, Marcellino, and Osbat (2005) panel unit root tests may be severely biased if the panel units are cross-cointegrated, namely if under the null hypothesis (of unit roots) one or more linear combinations of $\mathbf{y}_t$ are stationary. This needs to be distinguished from the case where the errors are cross-correlated without necessarily involving cointegration across the cross-section units. Under the former, two or more cross-sectional units must share at least one common stochastic trend. Such a situation is likely to occur if the PPP hypothesis is examined (see Lyhagen (2008); Banerjee, Marcellino, and Osbat (2005); and Wagner (2008)).

The tests proposed by Moon and Perron (2004) and Pesaran (2007b) are based on the model

$$y_{it} = \left(1 - \phi_i\right)\mu_i + \phi_i y_{i,t-1} + \gamma_i f_t + \varepsilon_{it}. \tag{31.37}$$

Under the unit root hypothesis, $\phi_i = 1$, this equation yields

$$y_{it} = y_{i0} + \gamma_i s_{ft} + s_{it},$$

[7] The idea of augmenting ADF regressions with other covariates has been investigated in the unit root literature by Hansen (1995) and Elliott and Jansson (2003). These authors consider the additional covariates in order to gain power when testing the unit root hypothesis in the case of a single time series. Pesaran, Smith, and Yamagata (2013) augment ADF regressions with cross-section averages to eliminate the effects of unobserved common factors in the case of panel unit root tests.

where

$$s_{ft} = f_1 + f_2 + \ldots + f_t,$$
$$s_{it} = \varepsilon_{i1} + \varepsilon_{i2} + \ldots + \varepsilon_{it}.$$

Clearly, under the null hypothesis all cross-section units are related to the common stochastic component, s_{ft}, albeit with varying effects, γ_i. This framework rules out cross-unit cointegration as under the null hypothesis there does not exist a linear combination of $y_{1t}, \ldots, y_{Nt}$ that is stationary. Therefore, tests based on (31.37) are designed to test the joint null hypothesis: 'All time series are I(1) and not cointegrated'.

To allow for cross-unit cointegration, Bai and Ng (2004) propose analyzing the common factors and idiosyncratic components separately. A simple multi-factor example of the Bai and Ng framework is given by

$$y_{it} = \mu_i + \boldsymbol{\gamma}_i' \mathbf{g}_t + e_{it},$$
$$\Delta \mathbf{g}_t = \boldsymbol{\Lambda} \mathbf{g}_{t-1} + \mathbf{v}_t,$$
$$e_{it} = \rho_i e_{i,t-1} + \varepsilon_{it},$$

where $\mathbf{g}_t$ is the $m \times 1$ vector of unobserved components, $\mathbf{v}_t$ and ε_{it} are stationary common and individual specific shocks, respectively. Two different sets of null hypotheses are considered: H_{0a}: (testing the $I(0)/I(1)$ properties of the common factors) $Rank(\boldsymbol{\Lambda}) = r \leq m$, and H_{0b}: (panel unit root tests) $\rho_i = 1$, for all i. A test of H_{0a} is based on common factors estimated by principal components and cointegration tests are used to determine the number of the common trends, $m - r$. Panel unit root tests are then applied to the idiosyncratic components. The null hypothesis that the time series have a unit root is rejected if either the test of the common factors or the test for the idiosyncratic component reject the null hypothesis of nonstationary components.[8] As has been pointed out by Westerlund and Larsson (2009), replacing the unobserved idiosyncratic components by estimates introduces an asymptotic bias when pooling the t-statistic (or p-values) of the panel units, which renders the pooled tests in Bai and Ng (2004) asymptotically invalid. Bai and Ng (2010) provide an alternative panel unit root test based on Sargan and Bhargava (1983) that has much better small sample properties.

To allow for short-run and long-run dependencies, Chang and Song (2005) suggest a nonlinear instrument variable test procedure. As the nonlinear instruments suggested by Chang (2002) are invalid in the case of cross-unit cointegration, panel specific instruments based on the Hermite function of different order are used as nonlinear instruments. Chang and Song (2005) show that the t-statistic computed from the nonlinear IV statistic are asymptotically standard normally distributed and, therefore, a panel unit root statistics against the heterogeneous alternative H_1^c can be constructed that has an standard normal limiting distribution.

Choi and Chue (2007) employ a subsampling procedure to obtain tests that are robust against a wide range of cross-sectional dependence such as weak and strong correlation as well as cross-unit cointegration. To this end, the sample is grouped into a number of overlapping blocks of b

[8] An alternative factor extraction method is suggested by Kapetanios (2007) who also provides detailed Monte Carlo results on the small sample performance of panel unit root tests based on a number of alternative estimates of the unobserved common factors. He shows that the factor-based panel unit root tests tend to perform rather poorly when the unobserved common factor is serially correlated.

time periods. Using all $(T - b + 1)$ possible overlapping blocks, the critical value of the test is estimated by the respective quantile of the empirical distribution of the $(T-b+1)$ test statistics computed. The advantage of this approach is that the null distribution of the test statistic may depend on unknown nuisance parameters. Whenever the test statistics converge in distribution to some limiting null distribution as $T \to \infty$ and N fixed, the sub-sample critical values converge in probability to the true critical values. Using Monte Carlo simulations Choi and Chue (2007) demonstrate that the size of the subsample test is indeed very robust against various forms of cross-sectional dependence. But such tests are only appropriate in the case of panels where N is small relative to T.

31.6 Finite sample properties of panel unit root tests

It has become standard to distinguish first generation panel unit root tests that are based on the assumption of independent cross-sectional units and second generation tests that allow for some kind of cross-sectional dependence. Maddala and Wu (1999) compared several first generation tests. For the heterogeneous alternative under consideration they found that in most cases the Fisher test (31.18) performs similarly or slightly better than the IPS statistic with respect to size and power. The Levin and Lin statistic (in the version of the 1993 paper) performs substantially worse. Similar results are obtained by Choi (2001). Madsen (2010) derived the local power function against homogeneous alternatives under different detrending procedures. Her Monte Carlo simulations support her theoretical findings that the test based on estimating the mean under the null hypothesis (i.e., the initial observation is subtracted from the time series) outperforms tests based on alternative demeaning procedures. Similar findings are obtained by Bond, Nauges, and Windmeijer (2002).

Moon, Perron, and Phillips (2007) compare the finite sample powers of alternative tests against the homogeneous alternative. They find that the point-optimal test of Moon, Perron, and Phillips (2007) performs best and show that the power of this test is close to the power envelope. Another important finding from these simulation studies is the observation that the power of the test drops dramatically if a time trend is included. This confirms theoretical results on the local power of panel unit root tests derived by Breitung (2000), Ploberger and Phillips (2002) and Moon, Perron, and Phillips (2007).

Hlouskova and Wagner (2006) compare a large number of first generation panel unit root tests applied to processes with MA(1) errors. Not surprisingly, all tests are severely biased as the root of the MA process approaches unity. Overall, the tests of Levin, Lin, and Chu (2002) and Breitung (2000) have the smallest size distortions. These tests also perform best against the homogeneous alternative, where the autoregressive coefficient is the same for all panel units. Of course, this is not surprising as these tests are optimal under homogeneous alternatives. Furthermore, it turns out that the stationarity tests of Hadri (2000) perform very poorly in small samples. This may be due to the fact that asymptotic values for the mean and variances of the KPSS statistics are used, whereas Levin, Lin, and Chu (2002) and IPS (2003) provide values for small T as well.

The relative performance of several second generation tests has been studied by Gutierrez (2006), and Gengenbach, Palm, and Urbain (2010), where the cross-sectional dependence is assumed to follow a factor structure. The results very much depend on the underlying model. The simulations carried out by Gengenbach, Palm, and Urbain (2010) show that, in general,

the mean CADF test has better size properties than the test of Moon and Perron (2004), which tends to be conservative in small samples. However the latter test appears to have more power against stationary idiosyncratic components. Since these tests remove the common factors, they will eventually indicate stationary time series in cases where the series are actually nonstationary due to a common stochastic trend. The results of Gengenbach, Palm, and Urbain (2010) also suggest that the approach of Bai and Ng (2004) is able to cope with this possibility although the power of the unit test applied to the nonstationary component is not very high.

In general, the application of factor models in the case of weak cross sectional dependence does not yield valid test procedures. Alternative unit root tests that allow for weak cross-sectional dependence are considered in Breitung and Das (2005). They find that the *GLS* t-statistic may have a severe size bias if T is only slightly larger than N. In these cases, the Chang (2004) bootstrap procedure is able to substantially improve the size properties. The robust *OLS* t-statistic performs slightly worse but outperforms the nonlinear IV test of Chang (2002). However, Monte Carlo simulations carried out by Baltagi, Bresson, and Pirotte (2007) show that there can be considerable size distortions even in panel unit root tests that allow for weak dependence. Interestingly enough Pesaran's test, which is not designed for weak cross-sectional dependence, tends to be the most robust to spatial type dependence.

31.7 Panel cointegration: general considerations

We now consider the panel counterpart of the methods developed in the time series literature for investigating the existence and the nature of long-run relations in the case of variables on a single cross-sectional unit, introduced in Chapter 22.

Consider the n_i time series variables $\mathbf{z}_{it} = (z_{i1t}, z_{i2t}, \ldots, z_{in_it})'$ observed on the ith cross-sectional unit over the period $t = 1, 2, \ldots, T$, and suppose that for each i

$$z_{ijt} \sim I(1), \qquad j = 1, 2, \ldots, n_i.$$

Then $\mathbf{z}_{it}$ is said to form one or more cointegrating relations if there are linear combinations of z_{ijt}'s for $j = 1, 2, \ldots, n_i$ that are $I(0)$, i.e. if there exists an $n_i \times r_i$ matrix ($r_i \geq 1$) such that

$$\underset{r_i \times n_i}{\boldsymbol{\beta}_i'} \quad \underset{n_i \times 1}{\mathbf{z}_{it}} \quad = \quad \underset{r_i \times 1}{\boldsymbol{\xi}_{it}} \quad \sim I(0).$$

r_i denotes the number of cointegrating (or long-run) relations. The residual-based tests are appropriate when $r_i = 1$, and $\mathbf{z}_{it}$ can be partitioned such that $\mathbf{z}_{it} = (y_{it}, \mathbf{x}_{it}')'$ with no cointegration amongst the $k_i \times 1$ ($k_i = n_i - 1$) variables, $\mathbf{x}_{it}$. The system cointegration approaches are much more generally applicable and allow for $r_i > 1$ and do not require any particular partitioning of the variables in $\mathbf{z}_{it}$. Another main difference between the two approaches is the way the stationary component of $\boldsymbol{\xi}_{it}$ is treated in the analysis. Most of the residual-based techniques employ non-parametric (spectral density) procedures to model the residual serial correlation in the error correction terms, $\boldsymbol{\xi}_{it}$, whilst vector autoregressions (VAR) are utilized in the development of system approaches.

In panel data models, the analysis of cointegration is further complicated by heterogeneity, unbalanced panels, cross-sectional dependence, cross unit cointegration and the N and T asymptotics. But in cases where n_i and N are small, such that $\Sigma_{i=1}^{N} n_i$ is less than 10, and T is relatively

large ($T > 100$), as noted by Banerjee, Marcellino, and Osbat (2004), many of these problems can be avoided by applying the system cointegration techniques discussed in Chapter 22 to the pooled vector, $\mathbf{z}_t = (\mathbf{z}'_{1t}, \mathbf{z}'_{2t}, \ldots, \mathbf{z}'_{Nt})'$. In this setting, cointegration will be defined by the relationships $\boldsymbol{\beta}'\mathbf{z}_t$ that could contain cointegration between variables from different cross-section units as well as cointegration amongst the different variables specific to a particular cross-sectional unit. This framework can also deal with residual cross-sectional dependence since it allows for a general error covariance matrix that covers all the variables in the panel.

Despite its attractive theoretical features, the 'full' system approach to panel cointegration is not feasible even in the case of panels with moderate values of N and n_i. In practice, cross-sectional cointegration can be accommodated using common factors as in the work of Bai and Ng (2004), Pesaran (2006), Pesaran, Schuermann, and Weiner (2004) (PSW) and its subsequent developments in Dées et al. (2007) (DdPS). Bai and Ng (2004) consider the simple case where $n_i = 1$ but allow N and T to be large. But their setup can be readily generalized so that cointegration within each cross-sectional unit as well as across the units can be considered. Following DdPS suppose that[9]

$$\mathbf{z}_{it} = \boldsymbol{\Gamma}_{id}\mathbf{d}_t + \boldsymbol{\Gamma}_{if}\mathbf{f}_t + \boldsymbol{\xi}_{it}, \tag{31.38}$$

for $i = 1, 2, \ldots, N$; $t = 1, 2, \ldots, T$, and to simplify the exposition assume that $n_i = n$, where as before $\mathbf{d}_t$ is the $s \times 1$ vector of deterministics $(1, t)$ or observed common factors such as oil prices, $\mathbf{f}_t$ is an $m \times 1$ vector of unobserved common factors, $\boldsymbol{\Gamma}_{id}$ and $\boldsymbol{\Gamma}_{if}$ are $n \times s$ and $n \times m$ associated unknown coefficient matrices, $\boldsymbol{\xi}_{it}$ is an $n \times 1$ vector of error terms.

Unit root and cointegration properties of $\mathbf{z}_{it}$, $i = 1, 2, \ldots, N$, can be analyzed by allowing the common factors, $\mathbf{f}_t$, and/or the country-specific factors, $\boldsymbol{\xi}_{it}$, to have unit roots. To see this suppose

$$\Delta\mathbf{f}_t = \boldsymbol{\Lambda}(L)\,\boldsymbol{\eta}_t, \boldsymbol{\eta}_t \sim IID(\mathbf{0}, \mathbf{I}_m), \tag{31.39}$$

$$\Delta\boldsymbol{\xi}_{it} = \boldsymbol{\Psi}_i(L)\,\mathbf{v}_{it}, \ \ \mathbf{v}_{it} \sim IID(\mathbf{0}, \mathbf{I}_n), \tag{31.40}$$

where L is the lag operator and

$$\underset{m\times m}{\boldsymbol{\Lambda}(L)} = \sum_{\ell=0}^{\infty} \boldsymbol{\Lambda}_\ell L^\ell, \ \underset{n\times n}{\boldsymbol{\Psi}_i(L)} = \sum_{\ell=0}^{\infty} \boldsymbol{\Psi}_{i\ell} L^\ell. \tag{31.41}$$

The coefficient matrices, $\boldsymbol{\Lambda}_\ell$ and $\boldsymbol{\Psi}_{i\ell}, i = 1, 2, \ldots, N$, are absolute summable, so that $Var(\Delta\mathbf{f}_t)$ and $Var\left(\Delta\boldsymbol{\xi}_{it}\right)$ are bounded and positive definite, and $[\boldsymbol{\Psi}_i(L)]^{-1}$ exists. In particular we require that

$$\left\| \sum_{\ell=0}^{\infty} \boldsymbol{\Psi}_{i\ell}\boldsymbol{\Psi}'_{i\ell} \right\| \leq K < \infty, \tag{31.42}$$

[9] DdPS also allow for common observed macro factors (such as oil prices), but they are not included to simplify the exposition. Also see Chapter 33.

where K is a fixed constant. A sufficient condition is given by $\left\| \boldsymbol{\Psi}_{i\ell} \boldsymbol{\Psi}'_{i\ell} \right\| < 1$, for all i and ℓ.

Using the familiar decomposition (see Chapter 22):

$$\boldsymbol{\Lambda}(L) = \boldsymbol{\Lambda}(1) + (1-L)\boldsymbol{\Lambda}^*(L), \text{ and } \boldsymbol{\Psi}_i(L) = \boldsymbol{\Psi}_i(1) + (1-L)\boldsymbol{\Psi}_i^*(L),$$

the common stochastic trend representations of (31.39) and (31.40) can now be written as

$$\mathbf{f}_t = \mathbf{f}_0 + \boldsymbol{\Lambda}(1)\mathbf{s}_t + \boldsymbol{\Lambda}^*(L)\left(\boldsymbol{\eta}_t - \boldsymbol{\eta}_0\right),$$

and

$$\boldsymbol{\xi}_{it} = \boldsymbol{\xi}_{i0} + \boldsymbol{\Psi}_i(1)\mathbf{s}_{it} + \boldsymbol{\Psi}_i^*(L)\left(\mathbf{v}_{it} - \mathbf{v}_{i0}\right),$$

where

$$\mathbf{s}_t = \sum_{j=1}^{t} \boldsymbol{\eta}_j, \text{ and } \mathbf{s}_{it} = \sum_{j=1}^{t} \mathbf{v}_{ij}.$$

Using the above results in (31.38) now yields

$$\begin{aligned}\mathbf{z}_{it} = \mathbf{a}_i + \boldsymbol{\Gamma}_{id}\mathbf{d}_t + \boldsymbol{\Gamma}_{if}\boldsymbol{\Lambda}(1)\mathbf{s}_t + \boldsymbol{\Psi}_i(1)\mathbf{s}_{it} \\ + \boldsymbol{\Gamma}_{if}\boldsymbol{\Lambda}^*(L)\boldsymbol{\eta}_t + \boldsymbol{\Psi}_i^*(L)\mathbf{v}_{it},\end{aligned}$$

where[10]

$$\mathbf{a}_i = \boldsymbol{\Gamma}_{if}\left[\mathbf{f}_0 - \boldsymbol{\Lambda}^*(L)\boldsymbol{\eta}_0\right] + \boldsymbol{\xi}_{i0} - \boldsymbol{\Psi}_i^*(L)\mathbf{v}_{i0}.$$

In this representation $\boldsymbol{\Lambda}(1)\mathbf{s}_t$ and $\boldsymbol{\Psi}_i(1)\mathbf{s}_{it}$ can be viewed as common global and individual-specific stochastic trends, respectively; whilst $\boldsymbol{\Lambda}^*(L)\boldsymbol{\eta}_t$ and $\boldsymbol{\Psi}_i^*(L)\mathbf{v}_{it}$ are the common and individual-specific stationary components. From this result it is clear that, in general, it will not be possible to simultaneously eliminate the two types of common stochastic trends (global and individual-specific) in $\mathbf{z}_{it}$.

Specific cases of interest where it would be possible for $\mathbf{z}_{it}$ to form a cointegrating vector are when $\boldsymbol{\Lambda}(1) = \mathbf{0}$ or $\boldsymbol{\Psi}_i(1) = \mathbf{0}$. Under the former, panel cointegration exists if $\boldsymbol{\Psi}_i(1)$ is rank deficient. The number of cointegrating relations could differ across i and is given by $r_i = n - Rank\left[\boldsymbol{\Psi}_i(1)\right]$. Note that even in this case $\mathbf{z}_{it}$ can be cross-sectionally correlated through the common stationary components, $\boldsymbol{\Lambda}^*(L)\boldsymbol{\eta}_t$. Under $\boldsymbol{\Psi}_i(1) = \mathbf{0}$ for all i with $\boldsymbol{\Lambda}(1) \neq \mathbf{0}$, we will have panel cointegration if there exists $n \times r_i$ matrices $\boldsymbol{\beta}_i$ such that $\boldsymbol{\beta}_i'\boldsymbol{\Gamma}_{if}\boldsymbol{\Lambda}(1) = \mathbf{0}$.

Turning to the case where $\boldsymbol{\Lambda}(1)$ and $\boldsymbol{\Psi}_i(1)$ are both non-zero, panel cointegration could still exist but must involve both $\mathbf{z}_{it}$ and $\mathbf{f}_t$. But since $\mathbf{f}_t$ is unobserved it must be replaced by a suitable estimate. The global *VAR* (*GVAR*) approach of Pesaran, Schuermann, and Weiner (2004) and Dées et al. (2007) implements this idea by replacing $\mathbf{f}_t$ with the (weighted) cross-section averages of $\mathbf{z}_{it}$ (see also Chapter 33). To see how this can be justified, first differencing (31.38) and

[10] In usual case where $\mathbf{d}_t$ is specified to include an intercept, 1, $\mathbf{a}_i$ can be absorbed into the deterministics.

using (31.40), note that

$$[\mathbf{\Psi}_i(L)]^{-1}(1-L)\left(\mathbf{z}_{it}-\mathbf{\Gamma}_{id}\mathbf{d}_t-\mathbf{\Gamma}_{if}\mathbf{f}_t\right)=\mathbf{v}_{it}.$$

Using the approximation

$$(1-L)[\mathbf{\Psi}_i(L)]^{-1}\approx\sum_{\ell=0}^{p}\mathbf{\Phi}_{i\ell}L^{\ell}=\mathbf{\Phi}_i\left(L,p\right),$$

we obtain the following approximate $VAR(p)$ model

$$\mathbf{\Phi}_i\left(L,p\right)\left(\mathbf{z}_{it}-\mathbf{\Gamma}_{id}\mathbf{d}_t-\mathbf{\Gamma}_{if}\mathbf{f}_t\right)\approx\mathbf{v}_{it}. \tag{31.43}$$

When the common factors, $\mathbf{f}_t$, are observed the model for the ith cross-sectional unit decouples from the rest of the units and can be estimated using the econometric techniques developed in Pesaran, Shin, and Smith (2000), reviewed in Chapter 23, with $\mathbf{f}_t$ treated as weakly exogenous. But in general where the common factors are unobserved appropriate proxies for the common factors can be used. There are two possible approaches, one could either use the principal components of the observables, $\mathbf{z}_{it}$, or alternatively, following Pesaran (2006), $\mathbf{f}_t$ can be approximated in terms of $\bar{\mathbf{z}}_t=N^{-1}\Sigma_{i=1}^{N}\mathbf{z}_{it}$, the cross-section averages of the observables. To see how this procedure could be justified in the present context, average the individual equations given by (31.38) over i to obtain

$$\bar{\mathbf{z}}_t=\bar{\mathbf{\Gamma}}_d\mathbf{d}_t+\bar{\mathbf{\Gamma}}_f\mathbf{f}_t+\bar{\boldsymbol{\xi}}_t, \tag{31.44}$$

where $\bar{\mathbf{\Gamma}}_d=N^{-1}\Sigma_{i=1}^{N}\mathbf{\Gamma}_{id}$, $\bar{\mathbf{\Gamma}}_f=N^{-1}\Sigma_{i=1}^{N}\mathbf{\Gamma}_{if}$, and $\bar{\boldsymbol{\xi}}_t=N^{-1}\Sigma_{i=1}^{N}\boldsymbol{\xi}_{it}$. Also, note from (31.40) that

$$\bar{\boldsymbol{\xi}}_t-\bar{\boldsymbol{\xi}}_{t-1}=N^{-1}\sum_{j=1}^{N}\mathbf{\Psi}_j(L)\mathbf{v}_{jt}. \tag{31.45}$$

But using results in Pesaran (2006), for each t and as $N\rightarrow\infty$ we have $\bar{\boldsymbol{\xi}}_t-\bar{\boldsymbol{\xi}}_{t-1}\overset{q.m.}{\rightarrow}0$, and hence $\bar{\boldsymbol{\xi}}_t\overset{q.m.}{\rightarrow}\bar{\boldsymbol{\xi}}$, where $\bar{\boldsymbol{\xi}}$ is a time-invariant random variable. Using this result in (31.44) and assuming that the $n\times m$ average factor loading coefficient matrix, $\bar{\mathbf{\Gamma}}_f$, has full column rank (with $n\geq m$) we obtain

$$\mathbf{f}_t\overset{q.m.}{\rightarrow}\left(\bar{\mathbf{\Gamma}}_f'\bar{\mathbf{\Gamma}}_f\right)^{-1}\bar{\mathbf{\Gamma}}_f\left(\bar{\mathbf{z}}_t-\bar{\mathbf{\Gamma}}_d\mathbf{d}_t-\bar{\boldsymbol{\xi}}\right),$$

which justifies using the observable vector $\{\mathbf{d}_t,\bar{\mathbf{z}}_t\}$ as proxies for the unobserved common factors.

The various contributions to the panel cointegration literature will now be reviewed in the context of the above general set up. First-generation literature on panel cointegration tends to ignore the possible effects of global unobserved common factors, or attempts to account for them either by cross-section de-meaning or by using observable common effects such as oil

prices or US output. This literature also focusses on residual based approaches where it is often assumed that there exists at most one cointegrating relation in the individual specific models. Notable contributions to this strand of the literature include Kao (1999), Pedroni (1999, 2001, 2004), and more recently Westerlund (2005b). System approaches to panel cointegration that allow for more than one cointegrating relation include the work of Larsson, Lyhagen, and Lothgren (2001), Groen and Kleibergen (2003) and Breitung (2005) who generalized the likelihood approach introduced in Pesaran, Shin, and Smith (1999). Like the second generation panel unit root tests, recent contributions to the analysis of panel cointegration have also emphasized the importance of allowing for cross-sectional dependence which, as we have noted above, could be due to the presence of common stationary or non-stationary components or both. The importance of allowing for the latter has been emphasized in Banerjee, Marcellino, and Osbat (2004) through the use of Monte Carlo experiments in the case of panels where N is very small, at most 8 in their analysis. But to date a general approach that is capable of addressing all the various issues involved does not exist if N is relatively large.

We now consider in some further detail the main contributions, beginning with a brief discussion of the spurious regression problem in panels.

31.8 Residual-based approaches to panel cointegration

Under this approach $\mathbf{z}_{it}$ is partitioned as $\mathbf{z}_{it} = \left(y_{it}, \mathbf{x}'_{it}\right)'$ and the following regressions

$$y_{it} = \boldsymbol{\delta}'_i \mathbf{d}_{it} + \mathbf{x}'_{it}\boldsymbol{\beta} + u_{it},\ i = 1, 2, \ldots, N, \tag{31.46}$$

are considered, where as before $\boldsymbol{\delta}'_i \mathbf{d}_{it}$ represent the deterministics and the $k \times 1$ vector of regressors, $\mathbf{x}_{it}$, are assumed to be $I(1)$ and not cointegrated. However, the innovations in $\Delta\mathbf{x}_{it}$, denoted by $\boldsymbol{\varepsilon}_{it} = \Delta\mathbf{x}_{it} - E(\Delta\mathbf{x}_{it})$, are allowed to be correlated with u_{it}. Residual-based approaches to panel cointegration focus on testing for unit roots in *OLS* or panel estimates of u_{it}.

31.8.1 Spurious regression

Let $\mathbf{w}_{it} = \left(u_{it}, \boldsymbol{\varepsilon}'_{it}\right)'$ and assume that the conditions for the functional central limit theorem are satisfied such that

$$\frac{1}{\sqrt{T}} \sum_{t=1}^{[T]} \mathbf{w}_{it} \xrightarrow{d} \boldsymbol{\Sigma}_i^{1/2} \mathbf{W}_i(\cdot),$$

where $\mathbf{W}_i$ is a $(k+1) \times 1$ vector of standard Brownian motions, $\xrightarrow{d}$ denotes weak convergence on $D[0, 1]$ and

$$\boldsymbol{\Sigma}_i = \begin{pmatrix} \sigma^2_{i,u} & \boldsymbol{\sigma}_{i,u\varepsilon} \\ \boldsymbol{\sigma}'_{i,u\varepsilon} & \boldsymbol{\Sigma}_{i,\varepsilon\varepsilon} \end{pmatrix}.$$

Kao (1999) showed that in the homogeneous case with $\boldsymbol{\Sigma}_i = \boldsymbol{\Sigma}$, $i = 1, \ldots, N$, and abstracting from the deterministics, the *OLS* estimator $\hat{\boldsymbol{\beta}}$ converges in probability to the limit $\boldsymbol{\Sigma}^{-1}_{\varepsilon\varepsilon}\boldsymbol{\sigma}_{\varepsilon u}$, where it is assumed that $\mathbf{w}_{it}$ is identically independently distributed across i. In the heterogeneous

case $\boldsymbol{\Sigma}_{\varepsilon\varepsilon}$ and $\boldsymbol{\sigma}_{\varepsilon u}$ are replaced by the means $\overline{\boldsymbol{\Sigma}}_{\varepsilon\varepsilon} = N^{-1}\sum_{i=1}^{N}\boldsymbol{\Sigma}_{i,\varepsilon\varepsilon}$ and $\overline{\boldsymbol{\sigma}}_{\varepsilon u} = N^{-1}\sum_{i=1}^{N}\boldsymbol{\sigma}_{i,\varepsilon u}$, respectively (see Pedroni (2000)). In contrast, the *OLS* estimator of $\boldsymbol{\beta}$ fails to converge within a pure time series framework. On the other hand, if $\mathbf{x}_{it}$ and y_{it} are independent random walks, then the *t*-statistics for the hypothesis that one component of $\boldsymbol{\beta}$ is zero is $O_p(T^{1/2})$ and, therefore, the *t*-statistic has similar properties as in the time series case. As demonstrated by Entorf (1997) and Kao (1999), the tendency for spuriously finding a relationship among y_{it} and $\mathbf{x}_{it}$ may be even stronger in panel data regressions than in the pure time series case. Therefore, it is important to test whether the errors in a panel data regression such as (31.46) are stationary.

Example 76 (House prices in the US) *Holly, Pesaran, and Yamagata (2010) investigate the extent to which real house prices at state level in the US are driven by fundamentals such as real per capita disposable income, as well as by common shocks, and determine the speed of adjustment of real house prices to macroeconomic and local disturbances. Economic theory suggests that real house prices and incomes are cointegrated with cointegrating vector* $(1, -1)$. *Let* p_{it} *be the logarithm of the real price of housing in the* i^{th} *state during year t, and* y_{it} *be the logarithm of the real per capita personal disposable income. Table 31.1 reports CIPS panel unit root tests for these variables, using data on forty-nine US states followed over the years 1975 to 2003. Results show that the unit root hypothesis cannot be rejected for* p_{it} *and* y_{it}*, if the trended nature of these variables are taken into account. This conclusion seems robust to the choice of the augmentation order of the underlying CADF regressions. Hence, the analysis proceeds taking* y_{it} *and* p_{it} *as* $I(1)$. *To test for possible cointegration between* p_{it} *and* y_{it}*, the authors estimate the following model*

$$p_{it} = \alpha_i + \beta_i y_{it} + u_{it}, i = 1, 2, \ldots, N; t = 1, 2, \ldots, T, \tag{31.47}$$

Table 31.1 Pesaran's CIPS panel unit root test results

	With an Intercept			
	CADF(1)	**CADF(2)**	**CADF(3)**	**CADF(4)**
Δy_{it}	-2.61^{*}	-2.39^{*}	-2.42^{*}	-2.34^{*}
Δp_{it}	-2.28^{*}	-1.86	-1.76	-1.81
y_{it}	-2.52^{*}	-2.44^{*}	-2.39^{*}	-2.49^{*}
p_{it}	-2.56^{*}	-2.44^{*}	-2.83^{*}	-2.84^{*}
	With an intercept and a linear trend			
	CADF(1)	CADF(2)	CADF(3)	CADF(4)
y_{it}	-2.51	-2.22	-2.24	-2.09
p_{it}	-2.18	-2.02	-2.27	-2.30

Notes: The reported values are CIPS(*s*) statistics, computed as the average of cross-sectionally augmented Dickey–Fuller (CADF(*s*)) test statistics ((Pesaran 2007b)). The relevant lower 5% (10%) critical values for the CIPS statistics are -2.11 (-2.03) with an intercept case, and -2.62 (-2.54) with an intercept and a linear trend case. $c_{it} = r_{it} - \Delta p_{it}$, which is the real cost of borrowing net of real house price appreciation/depreciation. The superscripts '*' and '†' signify the test is significant at the 5 and 10 per cent levels, respectively.

where, to allow for possible error cross-sectional dependence u_{it} is assumed to have the multi-factor error structure

$$u_{it} = \sum_{\ell=1}^{m} \gamma_{i\ell} f_{\ell t} + \varepsilon_{it}. \tag{31.48}$$

As explained in Section 29.4.2, the common correlated effects (CCE) estimators are consistent regardless of whether the unobserved factors $\{f_{\ell t}\}$ are stationary or non-stationary and/or cointegrated, so long as ε_{it} is stationary and m (the number of factors) is a finite fixed number. Table 31.2 presents the MG and CCE type estimates together with the average pair-wise correlation coefficients of the residuals, $\overline{\hat{\rho}}$. The first column gives the mean group estimates, which yields a small coefficient on the income variable of 0.30 (0.09), and a large estimate of $\overline{\hat{\rho}}$ (0.38), which is highly significant.[11] *The associated CD test statistic is* 71.03 *to be compared to the 95% critical value of the test which is* 1.96. *The other two columns report the common correlated effects mean group (CCEMG) and the common correlated effects pooled (CCEP) estimates. The coefficient on income is now significantly larger and the residual cross-sectional dependence has been purged with the average error cross-correlation coefficient, $\overline{\hat{\rho}}$, reduced from 0.38 for the MG estimates to 0.024 and 0.003 for the CCEMG and CCEP estimates, respectively. The CCEMG and CCEP estimates of β (the mean of β_i) are 1.14 (0.20) and 1.20 (0.21), respectively, and the hypothesis that $\beta = 1$ cannot be rejected. Therefore, the long-run relation to be tested for cointegration is given by*

$$\hat{u}_{it} = p_{it} - y_{it} - \hat{\alpha}_i,$$

where $\hat{\alpha}_i = T^{-1} \sum_{t=1}^{T} (p_{it} - y_{it})$. The above residuals can now be used to test the null of non-cointegration between p_{it} and y_{it}. The possible dependence of u_{it} on common factors, $\mathbf{f}_t$, requires

Table 31.2 Estimation result: income elasticity of real house prices: 1975–2003

	MG	CCEMG	CCEP
$\hat{\alpha}$	3.85 (0.20)	−0.11 (0.26)	0.00 (0.24)
$\hat{\beta}$	0.30 (0.09)	1.14 (0.20)	1.20 (0.21)
Average pair-wise cross-correlation coefficient $\left(\overline{\hat{\rho}}\right)$	0.38	0.024	0.003
CD test statistic	71.03	4.45	0.62

Notes: Estimated model is $p_{it} = \alpha_i + \beta_i y_{it} + u_{it}$. MG stands for mean group estimates. CCEMG and CCEP denote the common correlated effects mean group and pooled estimates, respectively. $\hat{\alpha} = N^{-1} \sum_{i=1}^{N} \hat{\alpha}_i$ for all estimates, and $\hat{\beta} = N^{-1} \sum_{i=1}^{N} \hat{\beta}_i$ for MG and CCEMG estimates. Standard errors are given in parentheses. The average cross-correlation coefficient is computed as the simple average of the pair-wise cross-sectional correlation coefficients of the regression residuals, namely $\overline{\hat{\rho}} = [2/N(N-1)] \sum_{i=1}^{N-1} \sum_{j=i+1}^{N} \hat{\rho}_{ij}$, with $\hat{\rho}_{ij}$ being the correlation coefficient of the regression residuals of the i and j cross-section units. The CD test statistic is $[TN(N-1)/2]^{1/2} \overline{\hat{\rho}}$, which tends to $N(0, 1)$ under the null hypothesis of no error cross-sectional dependence. See Section 29.7.

[11] The standard errors are in brackets.

that the panel unit root tests applied to $\hat{u}_{it}$ should also allow for the cross-sectional dependence of the residuals. Computing CIPS(s) panel unit root test statistics for $p_{it} - y_{it}$, including state-specific intercepts, for different augmentation and lag-orders, $s = 1, 2, 3$ and 4, yields the results, -2.16, -2.39, -2.45, and -2.29, respectively. The 5 per cent and 1 per cent critical values of the CIPS statistic for the intercept case with $N = 50$ and $T = 30$ are -2.11 and -2.23, respectively. The results suggest rejection of a unit root in $p_{it} - y_{it}$ for all the augmentation orders at 5 per cent level and rejection at 1 per cent level in the case of the augmentation orders 2 and more. Therefore, one could conclude that p_{it} and y_{it} are cointegrated for a sufficiently large number of States. Having established panel cointegration between p_{it} and y_{it}, Holly, Pesaran, and Yamagata (2010) turn their attention to the dynamics of the adjustment of real house prices to real incomes and estimate the panel error correction model

$$\Delta p_{it} = \alpha_i + \phi_i(p_{i,t-1} - y_{i,t-1}) + \delta_{1i}\Delta p_{i,t-1} + \delta_{2i}\Delta y_{it} + \upsilon_{it}. \tag{31.49}$$

The coefficient ϕ_i provides a measure of the speed of adjustment of house prices to a shock. The half-life of a shock to p_{it} is approximately $-ln(2)/ln(1 + \phi_i)$. To allow for possible cross-sectional dependence in the errors, υ_{it}, the authors compute CCEMG and CCEP estimators, and compare these estimates with the mean group (MG) estimates, which do not take account of cross-sectional dependence, as a benchmark. The former estimates are computed by the OLS regressions of Δp_{it} on 1, $(p_{i,t-1} - y_{i,t-1})$, $\Delta p_{i,t-1}$, Δy_{it}, and the associated cross-section averages, $(\bar{p}_{t-1} - \bar{y}_{t-1})$, $\Delta\bar{y}_t$, $\Delta\bar{p}_t$, and $\Delta\bar{p}_{t-1}$. The results are summarized in Table 31.3. The coefficients are all correctly signed. The CCEMG and CCEP estimators are very close and yield error correction coefficients given by $-0.183(0.016)$ and $-0.171(0.015)$ that are reasonably large and statistically highly significant. The average half-life estimates are around 3.5 years, much smaller than the half-life estimates of 6.3 years obtained using the MG estimators. But the MG estimators are likely to be biased, since the residuals from these estimates show a high degree of cross-sectional dependence. The same is not true of the CCE type estimators. This analysis suggests that, even if house prices deviate from the equilibrating relationship because of state-specific or common shocks, they will eventually revert. If

Table 31.3 Panel error correction estimates: 1977–2003

Δp_{it}	MG	CCEMG	CCEP
$p_{i,t-1} - y_{i,t-1}$	−0.105 (0.008)	−0.183 (0.016)	−0.171 (0.015)
$\Delta p_{i,t-1}$	0.524 (0.030)	0.449 (0.038)	0.518 (0.065)
Δy_{it}	0.500 (0.040)	0.277 (0.059)	0.227 (0.063)
Half life	6.248	3.429	3.696
$\bar{R}^2$	0.54	0.70	0.66
Average pair-wise cross-correlation coefficients $(\overline{\hat{\rho}})$	0.284	−0.005	−0.016
CD test statistics	50.60	−0.84	−2.80

Notes: The state-specific intercepts are estimated but not reported. MG stands for Mean Group estimates. CCEMG and CCEP denote the common correlated effects mean group and pooled estimates, respectively. Standard errors are given in parentheses. The half life of a shock to p_{it} is approximated by $-ln(2)/ln(1 + \hat{\phi})$ where $\hat{\phi}$ is the pooled estimates for the coefficient on $p_{i,t-1} - y_{i,t-1}$.

house prices are above equilibrium they will tend to fall relative to income, and vice versa if they are above equilibrium. Of course, because there is heterogeneity across states, a particular state need not be in the same disequilibrium position as other states. But on average the change in the ratio of house prices to per capita incomes should be zero, consistent with a cointegrating relationship, for T sufficiently large. In their conclusions, Holly, Pesaran, and Yamagata (2010) also examine the temporal pattern of the differences, $p_{it} - y_{it}$, since 2003. The process of house price boom that started in the US in early 2000 accelerated during 2003–06 and some have interpreted this as a bubble. Over the period 2000 to 2006 the average (unweighted) rise in US house prices was 46 per cent, as compared with a 25 per cent rise in income per capita. However, the price increases relative to per capita incomes have been quite heterogeneous. While house prices over the period 2000 to 2006 rose by 67 per cent in Virginia, 73 per cent in Arizona and 92 per cent in the District of Columbia, they rose by only 20 per cent in Indiana and 21 per cent in Ohio. These differences were much more pronounced than the rise in income per capita in these states (respectively 26 per cent, 23 per cent, 40 per cent, 20 per cent, and 19 per cent). Individual states can move about the average because the loading of the driving variables differ across states or because the initial disequilibrium is different. The extent of the heterogeneity in the disequilibrium, as measured by the time profile of the logarithm of price-income per capita over the full sample, 1976–2007, for all the 49 states is displayed in Figure 31.1. It is interesting that the excess rise in house prices tends to be associated with increased dispersion in the log price-income ratios, which begin to decline with moderation of house price rises relative to incomes. This fits well with the development of house prices in 2007, where prices rose only by 4 per cent as compared with a rise in per capita income of 5 per cent. The range of house price changes across states was also narrowed down substantially. In fact, in the case

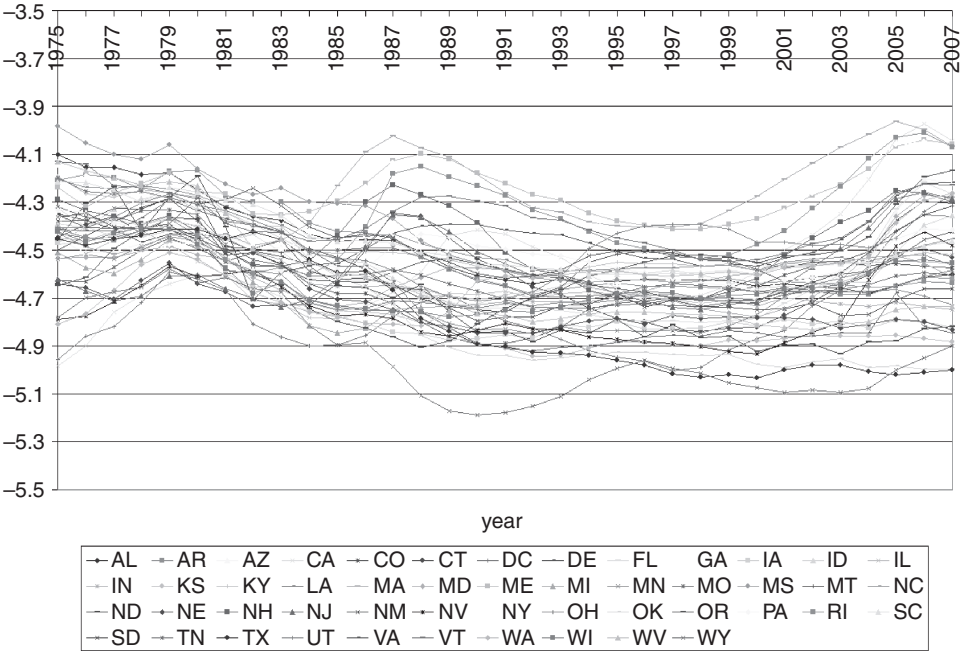


Figure 31.1 Log ratio of house prices to per capita incomes over the period 1976–2007 for the 49 states of the US.

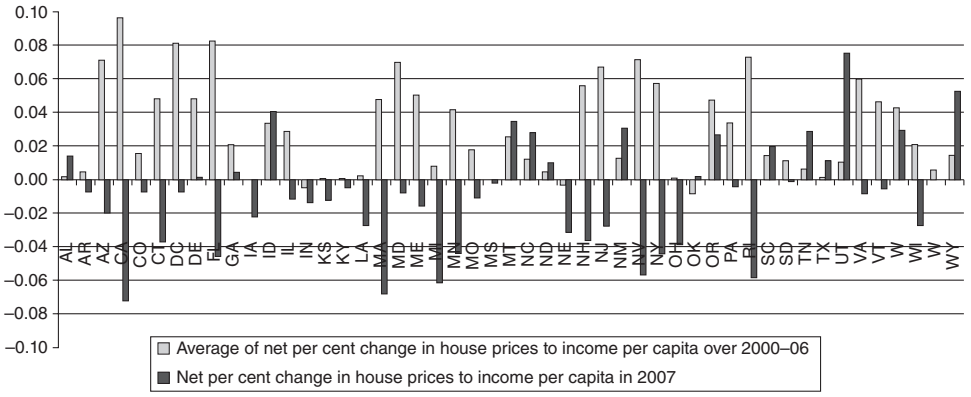


Figure 31.2 Percent change in house prices to per capita incomes across the US states over 2000–06 as compared with the corresponding ratios in 2007.

of the five states mentioned above, the price-income ratio declined by 1 per cent in Virginia, 2 per cent in Arizona, 0 per cent in District of Columbia, −2 per cent in Indiana, and 4 per cent in Ohio. If we calculate the average change in the log ratio of house prices to per capita income for each state over the period 2000–06, and compare it to the average change in the ratio for 2007, it is to be expected that, if a state on average is above its equilibrium before 2006, the average change after 2006 should be negative, and vice versa otherwise. The results are plotted in Figure 31.2, and show that of 49 states, 32 states have an average rate of price change in 2007 with the opposite sign to the average price changes for 2000–06. Moreover, we note that the correlation coefficient between the change in the price-income ratio in 2007, when the house price boom began to unwind, and the average change in the same ratio over the preceding price boom period, 2000–2006, is negative and quite substantial, around −0.42.

31.8.2 Tests of panel cointegration

As in the pure time series framework, the variables in a regression function can be tested for cointegration by applying unit roots tests of the sort suggested in the previous sections to the residuals of the estimated regression. Unfortunately, panel unit root tests cannot be applied to the residuals in (31.46) if $\mathbf{x}_{it}$ is endogenous, that is, if $\boldsymbol{\sigma}_{\varepsilon u} \neq 0$. Letting $T \to \infty$ followed by $N \to \infty$, Kao (1999) shows that the limiting distribution of the DF t-statistic applied to the residuals of a pooled *OLS* regression of (31.46) is

$$(t_\phi - \sqrt{N}\,\mu_K)/\sigma_K \xrightarrow{d} \mathcal{N}(0,1), \tag{31.50}$$

where the values of μ_K and σ_K depend on the kind of deterministics included in the regression, the contemporaneous covariance matrix $E(\mathbf{w}_{it}\mathbf{w}_{it}')$ and the long-run covariance matrix, $\boldsymbol{\Sigma}_i$. Kao (1999) proposes adjusting t_ϕ by using consistent estimates of μ_K and σ_K, where he assumes that the nuisance parameters are the same for all units in the panel.

Pedroni (2004) suggests two different test statistics for the models with heterogeneous cointegration vectors. Let $\hat{u}_{it} = y_{it} - \hat{\boldsymbol{\delta}}_i'\mathbf{d}_{it} - \hat{\boldsymbol{\beta}}_i'\mathbf{x}_{it}$ denote the *OLS* residual of the cointegration

regression. Pedroni considers two different classes of test statistics: (i) the 'panel statistic' that is equivalent to the unit root statistic against homogeneous alternatives and (ii) the 'Group Mean statistic' which is analogous to the panel unit root tests against heterogeneous alternatives. The two versions of the t statistic are defined as

$$\text{panel } Z_{Pt} = \left(\tilde{\sigma}^2_{NT}\sum_{i=1}^{N}\sum_{t=1}^{T}\hat{u}^2_{i,t-1}\right)^{-1/2}\left(\sum_{i=1}^{N}\sum_{t=1}^{T}\hat{u}_{i,t-1}\hat{u}_{it} - T\sum_{i=1}^{N}\hat{\lambda}_i\right),$$

$$\text{group-mean } \widetilde{Z}_{Pt} = \sum_{i=1}^{N}\left(\hat{\sigma}^2_{ie}\sum_{t=1}^{T}\hat{u}^2_{i,t-1}\right)^{-1/2}\left(\sum_{t=1}^{T}\hat{u}_{i,t-1}\hat{u}_{it} - T\hat{\lambda}_i\right),$$

where $\hat{\lambda}_i$ is a consistent estimator of the one-sided long run variance $\lambda_i = \sum_{j=1}^{\infty} E(e_{it}e_{i,t-j})$, $e_{it} = u_{it} - \delta_i u_{i,t-1}$, $\delta_i = E(u_{it}u_{i,t-1})/E(u^2_{i,t-1})$, $\hat{\sigma}^2_{ie}$ denotes the estimated variance of e_{it} and $\tilde{\sigma}^2_{NT} = N^{-1}\sum_{i=1}^{N}\hat{\sigma}^2_{ie}$. Pedroni presents values of μ_p, σ^2_p and $\tilde{\mu}_p, \tilde{\sigma}^2_p$ such that $(Z_{Pt} - \mu_p\sqrt{N})/\sigma_p$ and $(\widetilde{Z}_{Pt} - \tilde{\mu}_p\sqrt{N})/\tilde{\sigma}_p$ have standard normal limiting distributions under the null hypothesis.

Other residual-based panel cointegration tests include the recent contribution of Westerlund (2005b) which is based on variance ratio statistics and does not require corrections for the residual serial correlations.

The finite sample properties of some residual based tests for panel cointegration are discussed in Baltagi and Kao (2000). Gutierrez (2006) compares the power of various panel cointegration test statistics. He shows that in homogeneous panels with a small number of time periods Kao's tests tend to have higher power than Pedroni's tests, whereas in panels with large T the latter tests performs best. Both tests outperform the system test suggested by Larsson, Lyhagen, and Lothgren (2001). Wagner and Hlouskova (2010) compare various panel cointegration tests in a large scale simulation study. They found that the Pedroni (2004) test based on *ADF* regressions performs best, whereas all other tests tend to be severely undersized and have very low power in many cases. Furthermore, the system tests suffer from large small sample distortions and are unreliable tools for finding out the correct cointegration rank. Gengenbach, Palm, and Urbain (2006) investigate the performance of Pedroni's tests in cross-dependent models with a factor structure.

31.9 Tests for multiple cointegration

It is also possible to adapt the Johansen (1995) multivariate test based on a *VAR* representation of the variables in a panel context. Let $\Lambda_i(r)$ denote the cross-section specific likelihood-ratio ('trace') statistic of the hypothesis that there are (at most) r stationary linear combinations in the cointegrated *VAR* system given by $\mathbf{z}_{it} = \left(y_{it}, \mathbf{x}'_{it}\right)'$. Following the unit root test proposed in *IPS* (2003), Larsson, Lyhagen, and Lothgren (2001) suggested the standardized *LR*-bar statistic

$$\widetilde{\Lambda}(r) = \frac{1}{\sqrt{N}}\sum_{i=1}^{N}\frac{\Lambda_i(r) - E[\lambda_i(r)]}{\sqrt{Var[\lambda_i(r)]}},$$

to test the null hypothesis that $r = 0$ against the alternative that at most $r = r^0 \geq 1$. Using a sequential limit theory it can be shown that $\widetilde{\Lambda}(r)$ is asymptotically standard normally distributed. Asymptotic values of $E[\lambda_i(r)]$ and $Var[\lambda_i(r)]$ are tabulated in Larsson, Lyhagen, and Lothgren (2001) for the model without deterministic terms and Breitung (2005) for models with a constant and a linear time trend. Unlike the residual-based tests, the LR-bar test allows for the possibility of multiple cointegration relations in the panel.

It is also possible to test the null hypothesis that the errors of the cointegration regression are stationary. That is, under the null hypothesis it is assumed that $y_{it}, \mathbf{x}_{it}$ are cointegrated with cointegration rank $r = 1$. McCoskey and Kao (1998) suggest a panel version of the Shin (1994) cointegration test based on the residuals of a fully modified *OLS* regression. Westerlund (2005c) suggests a related test procedure based on the *CUSUM* statistic.

31.10 Estimation of cointegrating relations in panels

31.10.1 Single equation estimators

First, we consider a single-equation framework where it is assumed that y_{it} and the $k \times 1$ vector of regressors $\mathbf{x}_{it}$ are $I(1)$ with at most one cointegrating relation amongst them, namely that there exists a linear relationship of the form (31.46) such that the error u_{it} is stationary. As before, it is assumed that $\mathbf{z}_{it} = \left(y_{it}, \mathbf{x}_{it}'\right)'$ is independently and identically distributed across i, and the regressors, $\mathbf{x}_{it}$, are not cointegrated. We do not explicitly consider deterministic terms like individual specific constants or trends as the asymptotic theory applies to mean- or trend-adjusted variables as well.

It is assumed that the vector of coefficients, $\boldsymbol{\beta}$, is the same for all cross-sectional units, that is, a homogeneous cointegration relationship is assumed. Alternatively, it may be assumed that the cointegration parameters are cross-section specific (heterogenous cointegration).

By applying a sequential limit theory it can be shown that the *OLS* estimator of $\boldsymbol{\beta}$ is $T\sqrt{N}$ consistent and, therefore, the time series dimension is more informative on the long-run coefficients than the cross-section dimension. Furthermore, it is important to notice that—as in the time series framework—the *OLS* estimator is consistent but inefficient in the model with endogenous regressors.

Pedroni (2004) and Phillips and Moon (1999) propose a 'fully-modified *OLS*' (*FM-OLS*) approach to obtain an asymptotically efficient estimator for homogeneous cointegration vectors. This estimator adjusts for the effects of endogenous regressors and short-run dynamics of the errors (see Phillips and Hansen (1990) and Section 22.3.2). To correct for the effect of (long-run) endogeneity of the regressors, the dependent variable is adjusted for the part of the error that is correlated with the regressor

$$y_{it}^{+} = y_{it} - \boldsymbol{\sigma}_{i,\varepsilon u}' \boldsymbol{\Sigma}_{i,\varepsilon\varepsilon}^{-1} \Delta \mathbf{x}_{it}. \tag{31.51}$$

A second correction is necessary when computing the *OLS* estimator

$$\hat{\boldsymbol{\beta}}_{FM} = \left(\sum_{i=1}^{N} \sum_{t=1}^{T} \mathbf{x}_{it} \mathbf{x}_{it}' \right)^{-1} \left[\sum_{i=1}^{N} \sum_{t=1}^{T} (\mathbf{x}_{it} y_{it}^{+} - \boldsymbol{\lambda}_{i,\varepsilon u}) \right], \tag{31.52}$$

where

$$\boldsymbol{\lambda}_{i,\varepsilon u} = E\left(\sum_{j=0}^{\infty} \boldsymbol{\varepsilon}_{i,t-j} u_{it}\right).$$

The nuisance parameters can be estimated consistently using familiar nonparametric procedures.

An alternative approach is the 'Dynamic *OLS*' (*DOLS*) estimator suggested by Saikkonen (1991). This estimator is based on the error decomposition

$$u_{it} = \sum_{k=-\infty}^{\infty} \boldsymbol{\gamma}_k' \Delta \mathbf{x}_{i,t+k} + v_{it}, \tag{31.53}$$

where v_{it} is orthogonal to all leads and lags of $\Delta \mathbf{x}_{it}$. Inserting (31.53) in the regression (31.46) yields

$$y_{it} = \boldsymbol{\beta}' \mathbf{x}_{it} + \sum_{k=-\infty}^{\infty} \boldsymbol{\gamma}_k' \Delta \mathbf{x}_{i,t+k} + v_{it}\,. \tag{31.54}$$

In practice the infinite sums are truncated at some small numbers of leads and lags (see Kao and Chiang (2001), Mark and Sul (2003)). Westerlund (2005a) considers data dependent choices of the truncation lags. Kao and Chiang (2001) show that, in the homogeneous case with $\boldsymbol{\Sigma}_i = \boldsymbol{\Sigma}$ and individual specific intercepts, the limiting distribution of the *DOLS* estimator $\hat{\boldsymbol{\beta}}_{DOLS}$ is given by

$$T\sqrt{N}(\hat{\boldsymbol{\beta}}_{DOLS} - \boldsymbol{\beta}) \xrightarrow{d} \mathcal{N}(0, 6\,\sigma^2_{u|\varepsilon} \boldsymbol{\Sigma}_{\varepsilon\varepsilon}^{-1}),$$

where

$$\sigma^2_{u|\varepsilon} = \sigma^2_u - \sigma'_{\varepsilon u} \boldsymbol{\Sigma}_{\varepsilon\varepsilon}^{-1} \sigma_{\varepsilon u}.$$

Furthermore, the *FM-OLS* estimator possesses the same asymptotic distribution as the *DOLS* estimator. In the heterogeneous case $\boldsymbol{\Sigma}_{\varepsilon\varepsilon}$ and $\sigma^2_{u|\varepsilon}$ are replaced by $\overline{\boldsymbol{\Sigma}}_{\varepsilon\varepsilon} = N^{-1} \sum_{i=1}^{N} \boldsymbol{\Sigma}_{i,\varepsilon\varepsilon}$ and $\overline{\sigma}^2_{u|\varepsilon} = N^{-1} \sum_{i=1}^{N} \sigma^2_{i,u|\varepsilon}$, respectively (see Phillips and Moon (1999)). Again, the matrix $\boldsymbol{\Sigma}_i$ can be estimated consistently (for $T \to \infty$) by using a non-parametric approach.

In many applications the number of time periods is smaller than 20 and, therefore, the kernel based estimators of the nuisance parameters may perform poorly in such small samples. In these cases, the pooled mean group estimator introduced by Pesaran, Shin, and Smith (1999) and discussed in Section 28.10 may be used. This method assumes that the long-run parameters are identical across the cross-section units. Economic theory often predicts the same cointegration relation(s) across the cross-section units, although it is often silent on the magnitude of short-run dynamics, across *i*. For example, the long-run relationships predicted by the PPP, the uncovered interest parity, or the Fisher equation are the same across countries, although the speed of

convergence to these long-run relations can differ markedly over countries due to differences in economic and political institutions. For further discussion see, for example, Pesaran (1997).

31.10.2 System estimators

Single equation estimators have several drawbacks that can be avoided by using a system approach. First, these estimators assume that all regressors are I(1) and not cointegrated. If there is more than one cointegration relationship, then the matrix $\mathbf{\Sigma}_{\varepsilon\varepsilon}$ is singular and the asymptotic results are no longer valid. Second, the cointegration relationship has to be normalized such that the variable y_{it} enters with unit coefficient. As has been argued by Boswijk (1995), this normalization is problematic if the original coefficient of the variable y_{it} tends to zero.

In the case of short panels with T fixed and N large, Binder, Hsiao, and Pesaran (2005) consider estimation and inference in panel vector autoregressions (*PVAR*s) with homogeneous slopes where (i) the individual effects are either random or fixed, (ii) the time-series properties of the model variables are unknown *a priori* and may feature unit roots and cointegrating relations. generalized method of moments (*GMM*) and quasi-maximum likelihood (*QML*) estimators are obtained and compared in terms of their asymptotic and finite sample properties. It is shown that the asymptotic variances of the *GMM* estimators that are based on levels as well as on first differences of the model variables depend on the variance of the individual effects; whereas by construction, the fixed-effects *QML* estimator is not subject to this problem. Monte Carlo evidence is provided showing that the fixed-effects *QML* estimator tends to outperform the various *GMM* estimators in finite sample under both normal and non-normal errors. The paper also shows how the fixed-effects *QML* estimator can be successfully used for unit root and cointegration tests in short panels.

In the case of panels with large N and T, Larsson and Lyhagen (1999), Groen and Kleibergen (2003), and Breitung (2005) consider the vector error correction model (*VECM*) for the $k+1$ dimensional vector $\mathbf{z}_{it} = \left(y_{it}, \mathbf{x}_{it}'\right)'$ given by

$$\Delta \mathbf{z}_{it} = \boldsymbol{\alpha}_i \boldsymbol{\beta}_i' \mathbf{z}_{i,t-1} + \mathbf{w}_{it}, \tag{31.55}$$

where $\mathbf{w}_{it} = (u_{it}, \boldsymbol{\varepsilon}_{it}')'$. Once again we leave out deterministic terms and lagged differences. To be consistent with the approaches considered above, we confine ourselves to the case of homogeneous cointegration, that is, we let $\boldsymbol{\beta}_i = \boldsymbol{\beta}$ for $i = 1, 2, \ldots, N$. Larsson and Lyhagen (1999) propose an *ML* estimator, whereas the estimator of Groen and Kleibergen (2003) is based on a nonlinear *GMM* approach.

It is well known that the *ML* estimator of the cointegration parameters for a single series may behave poorly in small samples. Phillips (1994) has shown that the finite sample moments of the estimator do not exist. Using Monte Carlo simulations Hansen, Kim, and Mittnik (1998) and Brüggemann and Lütkepohl (2005) found that the *ML* estimator may produce implausible estimates far away from the true parameter values. Furthermore the asymptotic χ^2 distribution of the likelihood ratio test for restrictions on the cointegration parameters may be a poor guide for small sample inference (e.g., Gredenhoff and Jacobson (2001)).

To overcome these problems, Breitung (2005) proposes a computationally convenient two-step estimator, which is adapted from Ahn and Reinsel (1990) . This estimator is based on the fact that the Fisher information is block-diagonal with respect to the short- and long-run parameters. Accordingly, an asymptotically efficient estimator can be constructed by estimating the

short- and long-run parameters in separate steps. Suppose that the $n \times r$ matrix of cointegrating vectors is 'normalized' as $\boldsymbol{\beta} = (\mathbf{I}_r, \mathbf{B})'$, where $\mathbf{I}_r$ is the identity matrix of order r and $\mathbf{B}$ is the $(n-r) \times r$ matrix of unknown coefficients.[12] Then $\boldsymbol{\beta}$ is exactly identified and the Gaussian *ML* estimator of $\mathbf{B}$ is equivalent to the *OLS* estimator of $\mathbf{B}$ in

$$\mathbf{z}_{it}^{*} = \mathbf{B}\mathbf{z}_{i,t-1}^{(2)} + \mathbf{v}_{it}\,, \tag{31.56}$$

where $\mathbf{z}_{it}^{(2)}$ is the $r \times 1$ vector defined by $\mathbf{z}_{it} = \left(\mathbf{z}_{it}^{(1)'}, \mathbf{z}_{it}^{(2)'}\right)'$, and

$$\mathbf{z}_{it}^{*} = (\boldsymbol{\alpha}_i'\boldsymbol{\Sigma}_i^{-1}\boldsymbol{\alpha}_i)^{-1}\boldsymbol{\alpha}_i'\boldsymbol{\Sigma}_i^{-1}\Delta\mathbf{z}_{it} - \mathbf{z}_{i,t-1}^{(1)}.$$

The matrices $\boldsymbol{\alpha}_i$ and $\boldsymbol{\Sigma}_i$ can be replaced by $\sqrt{T}$-consistent estimates without affecting the limiting distribution. Accordingly, these matrices can be estimated for each panel unit separately, for example by using the Johansen (1991) *ML* estimator. To obtain the same normalization as in (31.56) the estimator for $\boldsymbol{\alpha}_i$ is multiplied with the $r \times r$ upper block of the *ML* estimator of $\boldsymbol{\beta}$.

Breitung (2005) shows that the limiting distribution of the *OLS* estimator of $\mathbf{B}$ is asymptotically normal. Therefore, tests of restrictions on the cointegration parameters have the standard limiting distributions (i.e. a χ^2 distribution for the usual Wald tests).

Some Monte Carlo experiments are performed by Breitung (2005) to compare the small sample properties of the two-step estimator with the *FM-OLS* and *DOLS* estimators. The results suggest that the latter two tests may be severely biased in small samples, whereas the bias of the two-step estimator is relatively small. Furthermore, the standard errors (and hence the size properties of the *t*-statistics) of the two-step procedure are more reliable than the ones of the semi-parametric estimation procedures. In a large scale simulation study, Wagner and Hlouskova (2010) found that the *DOLS* estimator outperforms all other estimators, whereas the *FM-OLS* and the two-step estimator perform similarly.

31.11 Panel cointegration in the presence of cross-sectional dependence

As discussed in Chapter 29, an important limitation of the econometric approaches discussed so far is that they assume that all cross-sectional units are independent. In many applications based on multi-country data sets this assumption is clearly unrealistic. To accommodate cross-dependence among panel units Mark, Ogaki, and Sul (2005) and Moon and Perron (2005) propose a dynamic seemingly unrelated regression (DSUR) estimator. Their approach is based on a *GLS* estimator of the dynamic representation (31.54) when there exists a single cointegrating relation between y_{it} and $\mathbf{x}_{it}$, and does not allow for the possibility of cross unit cointegration. Let $\mathbf{h}_{it}(p) = \left(\Delta\mathbf{x}_{i,t-p}', \ldots, \Delta\mathbf{x}_{i,t+p}'\right)'$ and $\mathbf{h}_{pt} = \left(\mathbf{h}_{1t}(p)', \ldots, \mathbf{h}_{Nt}(p)'\right)'$. To correct for endogeneity of the regressors, first y_{it} and $\mathbf{x}_{it}$ are regressed on $\mathbf{h}_{pt}$. Let $\tilde{y}_{it}$ and $\tilde{\mathbf{x}}_{it}$ denote the resulting

[12] The analysis can be readily modified to take account of other types of exact identifying restrictions on $\boldsymbol{\beta}$ that might be more appropriate from the viewpoint of long-run economic theory. See Pesaran and Shin (2002) for a general discussion of identification and testing of cointegrating relations in the context of a single cross-sectional unit.

regression residuals. Furthermore, define $\tilde{\mathbf{y}}_t = \left(\tilde{y}_{1t}, \tilde{y}_{2t}, \ldots, \tilde{y}_{Nt}\right)'$ and $\widetilde{\mathbf{X}}_t = (\tilde{\mathbf{x}}_{1t}, \tilde{\mathbf{x}}_{2t}, \ldots, \tilde{\mathbf{x}}_{Nt})'$. The DSUR estimator of the (homogeneous) cointegration vector is

$$\hat{\boldsymbol{\beta}}_{dsur} = \left(\sum_{t=p+1}^{T-p} \widetilde{\mathbf{X}}_t' \boldsymbol{\Sigma}_{uu}^{-1} \widetilde{\mathbf{X}}_t \right) \sum_{t=p+1}^{T-p} \widetilde{\mathbf{X}}_t' \boldsymbol{\Sigma}_{uu}^{-1} \tilde{\mathbf{y}}_t, \tag{31.57}$$

where $\boldsymbol{\Sigma}_{uu}$ denotes the long-run covariance matrix of $\mathbf{u}_t = (u_{1t}, u_{2t}, \ldots, u_{Nt})'$, namely

$$\boldsymbol{\Sigma}_{uu} = \lim_{T\to\infty} \frac{1}{T} E\left[\left(\sum_{t=1}^{T} \mathbf{u}_t \right) \left(\sum_{t=1}^{T} \mathbf{u}_t' \right) \right],$$

for a fixed N. This matrix is estimated by using an autoregressive representation of $\mathbf{u}_t$. See also (31.53). An alternative approach is suggested by Breitung (2005), where an SUR procedure is applied in the second step of the two-step estimator.

Bai and Kao (2005), Westerlund (2007), and Bai, Kao, and Ng (2009) suggest estimators for the cointegrated panel data model given by

$$y_{it} = \boldsymbol{\beta}' \mathbf{x}_{it} + \boldsymbol{\gamma}_i' \mathbf{f}_t + e_{it}, \tag{31.58}$$

where f_t is an $r \times 1$ vector of common factors and e_{it} is the idiosyncratic error. Bai and Kao (2005) and Westerlund (2007) assume that f_t is stationary. They suggest an *FM-OLS* cointegration regression that accounts for the cross-correlation due to the common factors. Bai, Kao, and Ng (2009) consider a model with non-stationary factors. Their estimation procedure is based on a sequential minimization of the criterion function

$$S_{NT}(\boldsymbol{\beta}, \mathbf{f}_1, \ldots, \mathbf{f}_T, \boldsymbol{\gamma}_1, \ldots, \boldsymbol{\gamma}_N) = \sum_{i=1}^{N} \sum_{t=1}^{T} (y_{it} - \boldsymbol{\beta}' \mathbf{x}_{it} - \boldsymbol{\gamma}_i' \mathbf{f}_t)^2, \tag{31.59}$$

subject to the constraint $T^{-1} \sum_{t=1}^{T} \mathbf{f}_t \mathbf{f}_t' = \mathbf{I}_r$ and $\sum_{i=1}^{N} \boldsymbol{\gamma}_i \boldsymbol{\gamma}_i'$ being diagonal. The asymptotic bias of the resulting estimator is corrected for by using an additive bias adjustment term or by using a procedure similar to the *FM-OLS* estimator suggested by Phillips and Hansen (1990).

A common feature of these approaches is that cross-sectional dependence can be represented by a contemporaneous correlation of the errors, and does not allow for the possibility of cross-unit cointegration. In many applications it is more realistic to allow for some form of dynamic cross-sectional dependence. A general model to accommodate cross-section cointegration and dynamic links between panel units is the panel *VECM* model considered by Groen and Kleibergen (2003) and Larsson and Lyhagen (1999). As in Section 31.7, let $\mathbf{z}_{it}$ denote an n-dimensional vector of times series on the ith cross-sectional unit. Consider the $nN \times 1$ vector $\mathbf{z}_t = \left(\mathbf{z}_{1t}', \mathbf{z}_{2t}' \ldots, \mathbf{z}_{Nt}'\right)'$ of all available time series in the panel data set. The *VECM* representation of this time series vector is

$$\Delta \mathbf{z}_t = \boldsymbol{\Pi} \mathbf{z}_{t-1} + \boldsymbol{\Gamma}_1 \Delta \mathbf{z}_{t-1} + \cdots + \boldsymbol{\Gamma}_p \Delta \mathbf{z}_{t-p} + \mathbf{u}_t \,. \tag{31.60}$$

For cointegrated systems $rank(\Pi) < nN$. It is obvious that such systems typically involve a large number of parameters as the number of parameters increases with N^2. Therefore, to obtain reliable estimates of the parameters T must be considerably larger than N. In many macroeconomic applications, however, the number of time periods is roughly as large as the number of cross-section units. Therefore, a simple structure must be imposed on the matrices $\mathbf{\Pi}, \mathbf{\Gamma}_1, \ldots, \mathbf{\Gamma}_p$ that yields a reasonable approximation to the underlying dynamic system.

31.12 Further reading

Further discussion on unit roots and cointegration in panels can be found in Banerjee (1999), Baltagi and Kao (2000), and Choi (2006).

31.13 Exercises

1. Let y_{it} be the real exchange rate (in logs) of country $i = 1, 2, \ldots, N$, observed over the period $t = 1, 2, \ldots, T$. Suppose that y_{it} is generated by the first-order autoregressive process

$$y_{it} = \left(1 - \phi_i\right)\mu_i + \phi_i y_{i,t-1} + \varepsilon_{it},\ i = 1, 2, \ldots, N;\ t = 1, 2, \ldots, T,$$

where initial values, y_{i0}, are given, and ε_{it} are serially uncorrelated and distributed independently across i, $\varepsilon_{it} \sim IIDN(0, \sigma_i^2)$.

(a) Show that the *OLS* estimator of ϕ_i is given by

$$\hat{\phi}_i = \frac{\mathbf{y}_i' \mathbf{M}_\tau \mathbf{y}_{i,-1}}{\mathbf{y}_{i,-1}' \mathbf{M}_\tau \mathbf{y}_{i,-1}},$$

where $\mathbf{y}_{i,-1} = \left(y_{i0}, y_{i1}, \ldots, y_{i,T-1}\right)'$, $\Delta\mathbf{y}_i = \left(\Delta y_{i1}, \Delta y_{i2}, \ldots, \Delta y_{iT}\right)'$, $\mathbf{M}_\tau = \mathbf{I}_T - \boldsymbol{\tau}_T(\boldsymbol{\tau}_T'\boldsymbol{\tau}_T)^{-1}\boldsymbol{\tau}_T'$, $\boldsymbol{\tau}_T' = (1, 1, \ldots, 1)$. Hence, or otherwise establish that under $\phi_i = 1$

$$\hat{\phi}_i = 1 + \frac{\boldsymbol{\varepsilon}_i' \mathbf{M}_\tau \mathbf{s}_{i,-1}}{\mathbf{s}_{i,-1}' \mathbf{M}_\tau \mathbf{s}_{i,-1}}$$

where $\mathbf{s}_{i,-1} = \left(0, s_{i1}, \ldots, s_{i,T-1}\right)'$, with $s_{it} = \sum_{j=1}^{t} \varepsilon_{ij}$, and for a fixed $T\ (> 3)$ show that

$$E\left(\hat{\phi}_i\right) = 1 + bias,$$

where

$$Bias = E\left(\frac{\mathbf{x}'\mathbf{M}_\tau \mathbf{H}\mathbf{x}}{\mathbf{x}'\mathbf{H}'\mathbf{M}_\tau \mathbf{H}\mathbf{x}}\right),$$

$\mathbf{x} \sim N(\mathbf{0}, \mathbf{I}_T)$, and $\mathbf{H}$ is the $T \times T$ matrix given by

$$\mathbf{H} = \begin{pmatrix} 0 & 0 & 0 & \cdots & 0 & 0 \\ 1 & 0 & 0 & \cdots & 0 & 0 \\ 1 & 1 & 0 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & 1 & 1 & \cdots & 0 & 0 \\ 1 & 1 & 1 & \cdots & 1 & 0 \end{pmatrix}.$$

(b) Consider now the mean group estimator defined by

$$\hat{\phi}_{MGE} = N^{-1} \sum_{i=1}^{N} \hat{\phi}_i.$$

Show that if $T > 5$ then as $N \to \infty$, $\sqrt{N}\left(\hat{\phi}_{MGE} - 1 - Bias\right)$ will be normally distributed with mean zero and a finite variance.

(c) Discuss the relevance of the above results for the development of unit root tests in panels. Are panel unit root tests useful in the empirical analysis of the purchasing power parity hypothesis?

Hint: Let $\mathbf{x}$ be an $m \times 1$ vector of independent normal variables, and $\mathbf{A}$ be an $m \times m$ positive semi-definite symmetric matrix of rank g. Then the rth moment of the inverse of $\mathbf{x}'\mathbf{A}\mathbf{x}$ exists if $g > 2r$.

2. Consider the following panel data model

$$y_{it} = \alpha_i(1 - \phi) + \phi y_{i,t-1} + u_{it}, \tag{31.61}$$

for $i = 1, 2, \ldots, N$ and $t = 1, 2, \ldots, T$, where $u_{it} \sim IID(0, \sigma_i^2)$.

(a) Suppose that $|\phi| < 1$, $E(\alpha_i u_{it}) = 0$ for all i and t, and the processes in (31.61) started a long time ago. Show that the IV estimator of ϕ given by

$$\hat{\phi}_{IV} = \frac{\sum_{t=3}^{T} \sum_{i=1}^{N} \Delta y_{it} y_{i,t-2}}{\sum_{t=3}^{T} \sum_{i=1}^{N} \Delta y_{i,t-1} y_{i,t-2}},$$

is a consistent estimator of ϕ for a fixed $T > 2$, and as $N \to \infty$. Make sure you establish that $N^{-1} \sum_{i=1}^{N} \Delta y_{i,t-1} y_{i,t-2}$ tends to a non-zero value.

(b) Consider now the case where $\phi = 1$. Are there any conditions under which $\hat{\phi}_{IV}$ is a consistent estimator?

3. Consider the following fixed-effects model with a linear trend

$$y_{it} = \mu_i + gt + v_{it},$$

for $i = 1, 2, \ldots, N$ and $t = 1, 2, \ldots, T$, where

$$v_{it} = \phi v_{i,t-1} + u_{it},$$

and $u_{it} \sim IID(0, \sigma_i^2)$.

(a) Show that $E(\Delta y_{it}) = g$, irrespective of whether $|\phi| < 1$ or $\phi = 1$. What is the implication of this observation for robust estimation of g?

(b) Let

$$\Delta \mathbf{y}_t = \begin{pmatrix} \Delta y_{1t} \\ \Delta y_{2t} \\ \vdots \\ \Delta y_{Nt} \end{pmatrix}, \ \Delta \mathbf{u}_t = \begin{pmatrix} \Delta u_{1t} \\ \Delta u_{2t} \\ \vdots \\ \Delta u_{Nt} \end{pmatrix},$$

$$\mathbf{W}_t = \begin{pmatrix} 1 & \Delta y_{1,t-1} \\ 1 & \Delta y_{2,t-1} \\ \vdots & \vdots \\ 1 & \Delta y_{N,t-1} \end{pmatrix}, \ \mathbf{Z}_t = \begin{pmatrix} 1 & y_{1,t-2} \\ 1 & y_{2,t-2} \\ \vdots & \vdots \\ 1 & y_{N,t-2} \end{pmatrix},$$

and

$$\Delta \mathbf{y}_t = \mathbf{W}_t \boldsymbol{\psi} + \Delta \mathbf{u}_t,$$

where $\boldsymbol{\psi} = (a, \phi)'$. Show that

$$\hat{\boldsymbol{\psi}}_{IV} = \left(\sum_{t=3}^{T} \mathbf{Z}_t' \mathbf{W}_t \right)^{-1} \left(\sum_{t=3}^{T} \mathbf{Z}_t' \Delta \mathbf{y}_t \right),$$

converges in probability to $[(1 - \phi)g, \phi]'$ if $|\phi| < 1$ and $E(u_{it}) = 0 = E(u_{it}\mu_i)$.

(c) Consider the case where $\phi = 1$ and analyse the asymptotic properties of $\hat{\boldsymbol{\psi}}_{IV}$ when $\phi = 1$.

4. Consider the dynamic factor model

$$\begin{aligned} y_{it} &= \alpha_i + \lambda y_{i,t-1} + \gamma_{i1} f_{1t} + \gamma_{i2} f_{2t} + u_{it}, \ u_{it} \sim IID(0, \sigma_i^2), \\ x_{it} &= \theta_i f_{1t} + v_{it}, \\ v_{it} &= \rho_i v_{i,t-1} + \sqrt{1 - \rho_i^2}\, \varepsilon_{it}, \ \varepsilon_{it} \sim IID(0, 1), \end{aligned}$$

for $i = 1, 2, \ldots, N$, $t = 1, 2, \ldots, T$, where f_{jt} for $j = 1, 2$, are unobserved factors, the factor loadings are either fixed constants or are random draws with non-zero means, and $|\rho_i| < 1$, for all i.

(a) Show that

$$E\left|\bar{y}_t - \bar{\alpha} - \lambda \bar{y}_{t-1} - \bar{\gamma}_1 f_{1t} - \bar{\gamma}_2 f_{2t}\right| = O_p(N^{-1/2}),$$
$$E\left|\bar{x}_t - \bar{\theta} f_{1t}\right| = O_p(N^{-1/2}),$$

where $\bar{y}_t = N^{-1}\sum_{i=1}^{N} y_{it}$, $\bar{x}_t = N^{-1}\sum_{i=1}^{N} x_{it}$, $\bar{\alpha} = N^{-1}\sum_{i=1}^{N} \alpha_i$, $\bar{\gamma}_j = N^{-1}\sum_{i=1}^{N} \gamma_{ij}$, and $\bar{\theta} = N^{-1}\sum_{i=1}^{N} \theta_i$.

(b) Using the results in (a) derive a panel unit root test of $\lambda = 1$, against the alternative of $\lambda < 1$, assuming that f_{jt} follows stationary processes.

(c) Consider now the case where $\gamma_{i2} = 0$, and y_{it} and x_{it} are determined by the same factor f_{1t} and assume that f_{1t} is I(1). Discuss the conditions under which $\bar{y}_t$ and $\bar{x}_t$ are cointegrated. How do you test and estimate such a cointegrating relationship, assuming that it does exist?

(d) How do you interpret a test of $\lambda = 1$ if one of the factors is I(1)?

32 Aggregation of Large Panels

32.1 Introduction

The aggregation problem is an inevitable aspect of applied research in economics. Nearly every study in economics implicitly or explicitly involves aggregation over time, individuals (consumers or firms), products, or space, and usually over most of these dimensions. Naturally, the problem is more pervasive in the case of macroeconomic research, which is primarily concerned with the analysis of the relationships amongst the aggregates. For example, in a typical small open macroeconomic model, the 'rest of the world' is aggregated into a single foreign economy, domestic agents such as households are often represented by a single representative counterpart, and, when it comes to estimation and inference, the available macro data, such as gross domestic product, are typically available only in the form of temporally aggregated data at monthly or quarterly frequencies. Aggregation also tends to be present in applied microeconomic analysis. For example, in the case of microeconometric studies of household consumption, the issues of commodity aggregation and the associated index number problem have been the subject of substantial research in the past. See Gorman (1953) and Muellbauer (1975). Similar considerations also arise in the microeconometric analysis of households' labour supply, firms' investment and employment decisions, and governments' expenditure decisions.

Since aggregation is prevalent in economics, it is important that the consequences of aggregation for the analysis of economic problems of interest are adequately understood. It is widely acknowledged that aggregation can be problematic, but its implications for empirical research are often ignored either by resorting to the concept of a 'representative agent', or by arguing that 'aggregation errors' are of second-order importance. However, there are empirical studies where aggregation errors are shown to be quite important, including the contributions by Hsiao et al. (2005), Altissimo et al. (2009), and Imbs et al. (2005). In addition to the empirical studies, Geweke (1985) develops a theoretical example, where he argues that ignoring the sensitivity of the aggregates to policy changes seems no more compelling than the Lucas critique of ignoring the dependence of expectations on the policy regime.

Aggregation can be broadly divided into two categories: aggregation across time and over cross-sectional units. The concept of cross-sectional units in this chapter is broadly defined, and includes geographical dimensions (e.g., regions or countries), individuals (e.g., firms or households), industries and products (e.g., the consumer price index basket of goods and services). There are two leading examples of aggregation across time: sequential sampling and temporal aggregation. The former method sequentially samples data from a higher frequency to a lower frequency. One example of sequential sampling is market closing prices or end-of-period prices. The latter method, on the other hand, combines the data typically by using period averages, to convert the series from a higher frequency to a lower frequency. Examples of temporally aggregated data are data measuring economic activity (gross domestic product, industrial output, retail sales) or consumer price data, where prices are collected repeatedly over a number of days within a month and then averaged across collection days to obtain monthly price indices for individual goods and/or services.

Aggregation over cross-sectional units (or 'cross-sectional aggregation') can also be divided into two categories, depending on the number of units, whether aggregation is carried out across a finite number of units (N) and/or over a large number of units ($N \to \infty$). This distinction is important for theoretical analysis, where taking limits ($N \to \infty$) often simplifies the analysis. Large N asymptotics often seem a reasonable approximation when it comes to macroeconomic data, where the number of cross-sectional units (households, products, firms, etc.) can be very large.

The focus of this chapter is predominantly on large N aggregation. It first briefly reviews the main aggregation problems studied in the literature (Section 32.2). Then it presents a general framework for micro/disaggregate behavioral relationships (Section 32.3) and develops a forecasting approach to derive the optimal aggregate function (Section 32.4). This approach is applied to a large cross-section aggregation of panel ARDL models (Section 32.5) and to the case of large factor-augmented VAR models in N cross-sectional units, where each micro unit is potentially related to all other micro units, and where micro innovations are allowed to be cross sectionally dependent (Section 32.6). The optimal aggregate function is used to examine the relationship between micro and macro parameters to show which distributional features of micro parameters can be identified from the aggregate model (Section 32.7). This chapter also derives and contrasts impulse response functions for the aggregate variables, distinguishing between the effects of composite macro and aggregated idiosyncratic shocks (Section 32.8). Some of these findings are illustrated by Monte Carlo experiments (Section 32.9) and two applications are presented. The first application investigates the aggregation of life-cycle consumption decision rules under habit formation (Section 32.10). The second application investigates the sources of persistence of consumer price inflation in Germany, France, and Italy, and re-examines the extent to which 'observed' inflation persistence at the aggregate level is due to aggregation and/or common unobserved factors (Section 32.11).

32.2 Aggregation problems in the literature

There is a number of questions that have been studied in the aggregation literature. Early surveys are provided by Granger (1990) and Stoker (1993). One important question is how aggregation affects time series properties. One key property is the persistence effects of shocks to the aggregate variables. For example, the extent to which real exchange rates or inflation rates react

to shocks is of considerable interest in policy making. Both inflation and real exchange rates are aggregates from data on a large number of goods and services.

The problem of aggregating a large number of independent time series processes was first addressed by Robinson (1978) and Granger (1980). Granger shows that aggregate variables can have fundamentally different time series properties as compared with those of the underlying micro units. Focusing on autoregressive models of order 1, $AR(1)$, he shows that aggregation can generate long memory even if the micro units follow stochastic processes with exponentially decaying autocovariances (see also Section 15.8.3). Consider the following $AR(1)$ disaggregate relations,

$$y_{it} = \lambda_i y_{i,t-1} + u_{it},$$

for $i = 1, 2, \ldots, N$, and $t = \ldots -1, 0, 1, 2, \ldots$, where $|\lambda_i| < 1$. Suppose these relations are independent, and in addition λ_i and $Var\,(u_{it}) = \sigma_i^2$ are independently and identically distributed (IID) random draws with the distribution function $F(\lambda)$ for λ on the range $[0, 1)$. Granger's objective is the memory properties of the aggregate variable $S_{t,N}\left(y\right) = \sum_{i=1}^{N} y_{it}$. The same setup is considered also in an earlier work by Robinson (1978), but with a focus on the estimation of the moments of $F(\lambda)$. To study the persistence properties of the aggregates, Granger considers the spectrum of $\bar{y}_t = N^{-1} \sum_{i=1}^{N} y_{it}$,

$$\bar{f}_N(\omega) = N^{-1} \sum_{i=1}^{N} f_i(\omega) \approx \frac{1}{2\pi} E\left[Var\,(u_{it})\right] \int \frac{1}{\left|1 - \lambda e^{-i\omega}\right|^2} dF(\lambda).$$

Then assuming that λ is type II beta distributed with parameters $p > 0$ and $q > 0$, he shows that for sufficiently large N, the sth-order autocovariance of $S_{t,N}\left(y\right) = N\bar{y}_t$, is $O(s^{1-q})$, and therefore the aggregate variable behaves as a fractionally integrated process of order $1 - q/2$. In fact the long memory property holds more generally, so long as the support of the distribution of λ covers 1. The problem of aggregation of a finite number of independent autoregressive moving average ($ARMA$) processes is considered, for example, by Granger and Morris (1976), Rose (1977), and Lütkepohl (1984). Aggregation across time as opposed to cross-sectional units also changes the time series properties. The persistence of temporally aggregated data has been investigated, for instance, by Chambers (2005).

A second, closely related problem is derivation of an optimal aggregate function (Theil (1954), Lewbel (1994), and Pesaran (2003)), which we review in more detail in Section 32.4 below. The optimal aggregate function can be used for various purposes such as to compare parameter values or impulse responses of the effects of shocks (common or unit specific) when aggregate and disaggregated models are used. The original 'aggregation problem' discussed by Theil (1954) is concerned with the possibility of deriving a relationship between the parameters of the aggregate model and the parameters of the underlying micro relations. Theil was the first to consider the problem of identification and estimation of micro parameters or some of their distributional features from aggregate relations in the context of static micro relations. Robinson (1978) considers the problem of estimating moments of the distribution of AR(1) micro coefficients. He identifies moments of the cross-section distribution of λ_i in the AR(1) model in terms of the autocovariances $\gamma_\ell = E(y_{it} y_{i,t+\ell})$, and establishes necessary and sufficient conditions for $y_t = E(y_{it})$ to have continuous spectral density. He also considers the problem of estimation

of the moments of $F(\lambda)$ using disaggregate data, abstracting from cases where $\bar{y}_t$ turns out to be a long-memory process. In particular, he requires $E\left(y_{it}^2\right)$ to exist for consistency, and $E\left(y_{it}^4\right)$ to exist for asymptotic normality of his proposed estimator. See Section 32.7 for further analysis and discussions of the aggregation problem in the case of long-run effects and estimation of mean lags.

A third issue of importance concerns the role of common factors and cross-sectional dependence in aggregation, which was first highlighted by Granger (1987), and further developed and discussed in Forni and Lippi (1997) and Zaffaroni (2004). Granger (1987) shows that the strength and pattern of cross-sectional dependence plays a central role in aggregation. He considers a simple factor model to illustrate the main issues,

$$y_{it} = x_{it} + \gamma_i f_t,$$

where x_{it} is unit-specific explanatory variable, f_t is a common factor with loadings, γ_i, and y_{it} is the observation for the unit i at time t. Suppose x_{it} and f_t have zero means, bounded variances, and x_{it} is independently distributed of f_t and of x_{jt} for all $j \neq i$. Consider the variance of the aggregate variable $N\bar{y}_t = \sum_{i=1}^N y_{it}$,

$$Var\left(N\bar{y}_t\right) = \sum_{i=1}^{N} Var\left(x_{it}\right) + N^2\overline{\gamma}^2 Var\left(f_t\right),$$

where $\bar{y}_t = N^{-1}\sum_{i=1}^N y_{it}$ and $\overline{\gamma} = N^{-1}\sum_{i=1}^N \gamma_i$. The first summand is at most of order N, denoted as $O(N)$, and, provided that $\lim_{N\to\infty}\overline{\gamma} \neq 0$, the second summand is of order N^2. The second term will therefore generally dominate the aggregate relationship. Granger demonstrates striking implications of this finding in terms of the fit of the aggregate (macro) relationship, where the common factor prevails when N is sufficiently large, and disaggregate (micro) relationships, where the micro regressor could play a leading role. If the common factor was unobserved, then the aggregate relation would have zero fit (for N large) whereas the fit of disaggregate relations could be quite high, being driven by the micro regressor, x_{it}. On the other hand, if f_t was observed and x_{it} was unobserved then the macro relation would have a perfect fit (for N large), whereas the micro relation may have a very poor fit due to the missing micro regressor, x_{it}. Hence variables that may have very good explanatory power at the micro level might be unimportant at the macro level, and *vice versa*. Granger shows that the strength and pattern of cross-sectional dependence thus play a central role in aggregation and components with weaker cross-sectional dependence typically do not matter for the behaviour of aggregate variables.

Aggregation has also been studied from the perspective of forecasting: is it better to forecast using aggregate or disaggregate data, if the primary objective is to forecast the aggregates? Pesaran, Pierse, and Kumar (1989) and Pesaran, Pierse, and Lee (1994), building on Grunfeld and Griliches (1960), develop selection criteria for a choice between aggregate and disaggregate specifications. Giacomini and Granger (2004) discuss forecasting of aggregates in the context of space-time autoregressive models.

Other contributions to the theory of aggregation include the contributions of Kelejian (1980), Stoker (1984, 1986), and Garderen et al. (2000), on aggregation of static nonlinear micro models; Pesaran and Smith (1995), Phillips and Moon (1999), and Trapani and Urga (2010) on the effects of aggregation on cointegration.

32.3 A general framework for micro (disaggregate) behavioural relationships

Aggregation of behavioural or technical relations across individuals becomes a problem when there is some form of heterogeneity across individuals' relations. When individuals are identical in every respect and the associated micro relations are homogeneous, aggregation will not be a problem. In practice, however, this is extremely unlikely to be the case. Sources of heterogeneity include

- input variables (heterogeneous initial endowments)
- micro parameters (heterogeneous coefficients)
- micro functionals (heterogeneous preferences and/or production functions).

Let the micro behavioural relationship be represented as

$$\mathbf{y}_{it} = \mathbf{f}_i\left(\mathbf{x}_{it}, \mathbf{u}_{it}, \boldsymbol{\theta}_i\right), \qquad i = 1, 2, \ldots, N; \quad t = 1, 2, \ldots, T, \tag{32.1}$$

where $\mathbf{y}_{it}$ denotes the vector of decision variables, $\mathbf{x}_{it}$ is a vector of observable variables, $\mathbf{u}_{it}$ is a vector of unobservable variables, and $\boldsymbol{\theta}_i$ denotes the vector of unknown parameters.

Example 77 *When the source of heterogeneity is different inputs (or endowments) across individuals only, we have*

$$\mathbf{y}_{it} = \mathbf{f}\left(\mathbf{x}_{it}, \mathbf{u}_{it}, \boldsymbol{\theta}\right), \qquad i = 1, 2, \ldots, N; \quad t = 1, 2, \ldots, T. \tag{32.2}$$

Such a scenario may arise in the analysis of nonlinear Engel curves,

$$w_{it} = a_0 + a_1 \log x_{it} + a_2 \left(\log x_{it}\right)^2 + u_{it}, \tag{32.3}$$

or in the analysis of (Cobb–Douglas) production functions of the form

$$y_{it} = AL_{it}^{\alpha} K_{it}^{1-\alpha} e^{u_{it}}. \tag{32.4}$$

For this type of heterogeneity, aggregation clearly will not be a problem when the micro relations are linear.

Example 78 *When the input variables as well as the parameters differ across individuals, we have*

$$\mathbf{y}_{it} = \mathbf{f}\left(\mathbf{x}_{it}, \mathbf{u}_{it}, \boldsymbol{\theta}_i\right), \qquad i = 1, 2, \ldots, N; \quad t = 1, 2, \ldots, T. \tag{32.5}$$

In the analysis of nonlinear Engel curves, such a scenario arises, for example, if the model is given by

$$w_{it} = a_{0i} + a_{1i} \log x_{it} + a_{2i} \left(\log x_{it}\right)^2 + u_{it}. \tag{32.6}$$

Example 79 *It is also possible that there is heterogeneity in the functional form of the micro relations, for example a production function of the form*

$$y_{it} = \left(\lambda_i L_{it}^{-\delta_i} + (1 - \lambda_i) K_{it}^{-\delta_i}\right)^{-1/\delta_i} e^{u_{it}}. \tag{32.7}$$

In this chapter, we consider the case where $\mathbf{f}(\cdot)$ is the same across individuals, but the input variables $\mathbf{x}_{it}$ and $\mathbf{u}_{it}$, and/or the parameters $\boldsymbol{\theta}_i$ differ across individuals. The analysis can also be easily extended to account for observed and unobserved macro (or aggregate) effects on individual behaviour, namely

$$\mathbf{y}_{it} = \mathbf{f}(\mathbf{x}_{it}, \mathbf{z}_t, \mathbf{u}_{it}, \mathbf{v}_t, \boldsymbol{\theta}_i), \qquad i = 1, 2, \ldots, N; \quad t = 1, 2, \ldots, T, \tag{32.8}$$

where $\mathbf{z}_t$ represents a vector of observed macro effects, and $\mathbf{v}_t$ represents a vector of unobserved macro effects.

32.4 Alternative notions of aggregate functions

32.4.1 Deterministic aggregation

This approach, employed for example by Gorman (1953) and Theil (1957), treats all the input variables and parameters as given and asks whether an aggregate function exists which is identical to the function that results from the aggregation of the micro relations. Let $\bar{\mathbf{y}}_t = N^{-1} \sum_{i=1}^{N} \mathbf{y}_{it}$. Then aggregating (32.1) under $\mathbf{f}_i(\cdot) = \mathbf{f}(\cdot)$ across all i, taking $\mathbf{x}_{it}$, $\mathbf{u}_{it}$, and $\boldsymbol{\theta}_i$ as given, we have

$$\bar{\mathbf{y}}_t = N^{-1} \sum_{i=1}^{N} \mathbf{f}(\mathbf{x}_{it}, \mathbf{u}_{it}, \boldsymbol{\theta}_i). \tag{32.9}$$

An aggregation problem is said to be present if the aggregate function $\mathbf{F}(\bar{\mathbf{x}}_t, \bar{\mathbf{u}}_t, \boldsymbol{\theta}_a)$ (with $\bar{\mathbf{x}}_t = N^{-1} \sum_{i=1}^{N} \mathbf{x}_{it}$, $\bar{\mathbf{u}}_t = N^{-1} \sum_{i=1}^{N} \mathbf{u}_{it}$, and where $\boldsymbol{\theta}_a$ is the vector of parameters of the aggregate function) differs from $N^{-1} \sum_{i=1}^{N} \mathbf{f}(\mathbf{x}_{it}, \mathbf{u}_{it}, \boldsymbol{\theta}_i)$. Perfect aggregation holds if

$$\left\| \mathbf{F}(\bar{\mathbf{x}}_t, \bar{\mathbf{u}}_t, \boldsymbol{\theta}_a) - N^{-1} \sum_{i=1}^{N} \mathbf{f}(\mathbf{x}_{it}, \mathbf{u}_{it}, \boldsymbol{\theta}_i) \right\| = 0, \tag{32.10}$$

for all $\mathbf{x}_{it}$, $\mathbf{u}_{it}$, and $\boldsymbol{\theta}_i$, where $\|\mathbf{a} - \mathbf{b}\|$ denotes a suitable norm discrepancy measure between $\mathbf{a}$ and $\mathbf{b}$. This requirement turns out to be extremely restrictive and is rarely met in applied economic analysis, except for linear models with identical coefficients. Condition (32.10) is not satisfied when $\mathbf{f}(\cdot)$ is a nonlinear function of $\mathbf{x}_{it}$ and $\mathbf{u}_{it}$, even if $\boldsymbol{\theta}_i$ is identical across individuals.

32.4.2 A statistical approach to aggregation

The restrictive nature of the deterministic aggregation condition (32.10) arises primarily because it requires the condition to be satisfied for all realizations of $\mathbf{x}_{it}$, $\mathbf{u}_{it}$, and $\boldsymbol{\theta}_i$, no matter how

remote the possibility of their occurrence. An alternative and less restrictive approach would be to require that (32.10) holds 'on average'. More precisely, let $\boldsymbol{\mu}_{\mathbf{y}}(t)$ and $\boldsymbol{\mu}_{\mathbf{x}}(t)$ be the means of $\mathbf{y}_{it}$ and $\mathbf{x}_{it}$ across individuals at a point in time or over a given period of time (depending on whether the variables are stocks or flows) and define a macro (or aggregate) relation as one that links $\boldsymbol{\mu}_{\mathbf{y}}(t)$ to $\boldsymbol{\mu}_{\mathbf{x}}(t)$ at a point in time t. This approach is suggested by Kelejian (1980) and rigorously formalized by Stoker (1984). It treats $\mathbf{x}_{it}$, $\mathbf{u}_{it}$, and $\boldsymbol{\theta}_i$ across individuals as stochastic, having a joint probability distribution function $P\left(\mathbf{x}_t, \mathbf{u}_t, \boldsymbol{\theta}; \boldsymbol{\phi}_t\right)$ with parameter vector $\boldsymbol{\phi}_t$ that can vary over time, but not across individuals. Then

$$\boldsymbol{\mu}_{\mathbf{y}}(t) = \Psi_{\mathbf{y}}\left(\boldsymbol{\phi}_t\right) = \int \mathbf{f}\left(\mathbf{x}_t, \mathbf{u}_t, \boldsymbol{\theta}\right) P\left(\mathbf{x}_t, \mathbf{u}_t, \boldsymbol{\theta}; \boldsymbol{\phi}_t\right) d\mathbf{x}_t \, d\mathbf{u}_t \, d\boldsymbol{\theta}, \tag{32.11}$$

and

$$\boldsymbol{\mu}_{\mathbf{x}}(t) = \Psi_{\mathbf{x}}\left(\boldsymbol{\phi}_t\right) = \int \mathbf{x}_t P\left(\mathbf{x}_t, \mathbf{u}_t, \boldsymbol{\theta}; \boldsymbol{\phi}_t\right) d\mathbf{x}_t \, d\mathbf{u}_t \, d\boldsymbol{\theta}. \tag{32.12}$$

Let $\boldsymbol{\phi}_t = \left(\boldsymbol{\phi}_{1t}, \boldsymbol{\phi}_{2t}\right)'$, where $\boldsymbol{\phi}_{2t}$ has the same dimension as $\mathbf{x}_{it}$, for all i, and suppose that for a given $\boldsymbol{\phi}_{1t}$ there is a one-to-one relationship between $\boldsymbol{\phi}_{2t}$ and $\boldsymbol{\mu}_{\mathbf{x}}(t)$. Then

$$\boldsymbol{\phi}_{2t} = \Psi_{\mathbf{x}}^{-1}\left(\boldsymbol{\phi}_{1t}, \boldsymbol{\mu}_{\mathbf{x}}(t)\right), \tag{32.13}$$

and

$$\boldsymbol{\mu}_{\mathbf{y}}(t) = \Psi_{\mathbf{y}}\left[\boldsymbol{\phi}_{1t}, \Psi_{\mathbf{x}}^{-1}\left(\boldsymbol{\phi}_{1t}, \boldsymbol{\mu}_{\mathbf{x}}(t)\right)\right] = \mathbf{F}\left(\boldsymbol{\mu}_{\mathbf{x}}(t), \boldsymbol{\phi}_{1t}\right). \tag{32.14}$$

The relationship between $\boldsymbol{\mu}_{\mathbf{y}}(t)$ and $\boldsymbol{\mu}_{\mathbf{x}}(t)$ is then defined as the exact aggregate equation.

This is clearly an improvement over the deterministic approach, but it is still rather removed from direct empirical analysis and does not adequately focus on the inevitably approximate nature of econometric analysis. Moreover, perhaps more importantly, due to its reliance on unconditional means, this approach is not suitable for the analysis of dynamic systems.

32.4.3 A forecasting approach to aggregation

Once again, consider the exact aggregation condition (32.10) specified for all $\mathbf{x}_{it}$, $\mathbf{u}_{it}$, and $\boldsymbol{\theta}_i$, but now require that conditional on the aggregate information set $\Omega_t = \{\bar{\mathbf{y}}_{t-1}, \bar{\mathbf{y}}_{t-2}, \ldots; \bar{\mathbf{x}}_t, \bar{\mathbf{x}}_{t-1}, \ldots\}$,

$$E\left\| \mathbf{F}\left(\Omega_t, \boldsymbol{\theta}_{at}\right) - N^{-1} \sum_{i=1}^{N} \mathbf{f}\left(\mathbf{x}_{it}, \mathbf{u}_{it}, \boldsymbol{\theta}_i\right) \right\|,$$

be as small as possible over $\mathbf{F}(\cdot)$. For expositional simplicity denote the aggregate function $\mathbf{F}\left(\Omega_t, \boldsymbol{\theta}_{ta}\right)$ by $\mathbf{F}_t$, and $\mathbf{f}\left(\mathbf{x}_{it}, \mathbf{u}_{it}, \boldsymbol{\theta}_i\right)$ by $\mathbf{f}_{it}$. Also note that the parameters of the aggregate function, $\boldsymbol{\theta}_{ta}$, will typically include first and higher moments of the joint distribution of $(\mathbf{x}_{it}, \mathbf{u}_{it}, \boldsymbol{\theta}_i)$ across i, and could be time-dependent. To simplify the exposition in what follows we assume $\mathbf{F}(\cdot)$ is a scalar function.

Suppose that $\|a-b\|$ is quadratic, namely $\|a-b\| = (a-b)^2$, where a and b are scalars. Then,

$$\begin{aligned} E\left[\left(\mathbf{F}_t-\bar{\mathbf{y}}_t\right)^2|\Omega_t\right] &= E\left\{\left[\left(\mathbf{F}_t-E\left(\bar{\mathbf{y}}_t|\Omega_t\right)\right)-\left(\bar{\mathbf{y}}_t-E\left(\bar{\mathbf{y}}_t|\Omega_t\right)\right)\right]^2|\Omega_t\right\} \\ &= E\left\{\left[\mathbf{F}_t-E\left(\bar{\mathbf{y}}_t|\Omega_t\right)\right]^2|\Omega_t\right\}+E\left\{\left[\bar{\mathbf{y}}_t-E\left(\bar{\mathbf{y}}_t|\Omega_t\right)\right]^2|\Omega_t\right\} \\ &\quad -2E\left\{\left[\mathbf{F}_t-E\left(\bar{\mathbf{y}}_t|\Omega_t\right)\right]\left[\bar{y}_t-E\left(\bar{\mathbf{y}}_t|\Omega_t\right)\right]|\Omega_t\right\}, \end{aligned} \tag{32.15}$$

and therefore the function that minimizes $E\left[\left(\mathbf{F}_t-\bar{\mathbf{y}}_t\right)^2|\Omega_t\right]$ is given by

$$\mathbf{F}_t = E\left(\bar{\mathbf{y}}_t|\Omega_t\right) = N^{-1}\sum_{i=1}^{N} E\left[\mathbf{f}\left(\mathbf{x}_{it},\mathbf{u}_{it},\boldsymbol{\theta}_i\right)|\Omega_t\right]. \tag{32.16}$$

This function will be referred to as the 'optimal aggregator function' (in the mean squared error sense). The orthogonal projection used (implicitly or explicitly) by Granger (1980), Lütkepohl (1984), and Lippi (1988) for aggregation of linear time series is a special case of this optimal aggregator which is more widely applicable. For an application to aggregation of static nonlinear models see Garderen et al. (2000).

This choice of $\mathbf{F}_t$ globally minimizes $E\left[\left(\mathbf{F}_t-\bar{\mathbf{y}}_t\right)^2|\Omega_t\right]$, but does not reduce it to zero, which is what (32.10) requires. We have

$$E\left[\left(\mathbf{F}_t-\bar{\mathbf{y}}_t\right)^2|\Omega_t\right] = Var\left(\bar{\mathbf{y}}_t|\Omega_t\right) \neq 0, \tag{32.17}$$

unless, of course, $E\left(\bar{\mathbf{y}}_t|\Omega_t\right) = \bar{\mathbf{y}}_t$.

It is also possible to define an aggregate prediction function, based on individual prediction of $\mathbf{y}_{it}$, conditional on information on *all* the observed disaggregate variables at time t. Let

$$\Phi_{it} = \left\{\mathbf{y}_{it-1},\mathbf{y}_{it-2},\ldots;\mathbf{x}_{it},\mathbf{x}_{it-1},\ldots\right\}, \tag{32.18}$$

denote the information set specific to individual i, and as before denote the information common to all individuals by

$$\Omega_t = \left\{\bar{\mathbf{y}}_{t-1},\bar{\mathbf{y}}_{t-2},\ldots;\bar{\mathbf{x}}_t,\bar{\mathbf{x}}_{t-1},\ldots\right\}. \tag{32.19}$$

Then

$$\Psi_{it} = \Phi_{it}\cup\Omega_t, \tag{32.20}$$

contains the information on the variables in the ith equation, and

$$\Psi_t = \cup_{i=1}^{N}\Psi_{it}, \tag{32.21}$$

all information available in the disaggregate model. Then the aggregate forecast, $\bar{\mathbf{y}}_{td}$, based on the universal information set, Ψ_t, is given by

$$\bar{\mathbf{y}}_{td} = N^{-1} \sum_{i=1}^{N} E\left[\mathbf{f}\left(\mathbf{x}_{it}, \mathbf{u}_{it}, \boldsymbol{\theta}_i\right) | \Psi_t\right], \tag{32.22}$$

which in most cases simplifies to

$$\bar{\mathbf{y}}_{td} = N^{-1} \sum_{i=1}^{N} E\left[\mathbf{f}\left(\mathbf{x}_{it}, \mathbf{u}_{it}, \boldsymbol{\theta}_i\right) | \Psi_{it}\right]. \tag{32.23}$$

Then we have

$$E\left[\left(\bar{\mathbf{y}}_t - \bar{\mathbf{y}}_{td}\right)^2 | \Psi_t\right] \leq E\left\{\left(\bar{\mathbf{y}}_t - E\left(\bar{\mathbf{y}}_t | \Omega_t\right)\right)^2 | \Psi_t\right\}, \tag{32.24}$$

and hence

$$E\left(\bar{\mathbf{y}}_t - \bar{\mathbf{y}}_{td}\right)^2 \leq E\left(\bar{\mathbf{y}}_t - E\left(\bar{\mathbf{y}}_t | \Omega_t\right)\right)^2, \tag{32.25}$$

which is basically saying that the optimal predictors $\bar{\mathbf{y}}_{td}$ which utilize information on micro variables on average are expected to do better than the optimal predictors based on the aggregate information only.

32.5 Large cross-sectional aggregation of *ARDL* models

Consider the simple autoregressive-distributed lag $(ARDL)$ model

$$y_{it} = \lambda_i y_{i,t-1} + \beta_i x_{it} + u_{it}, \qquad i = 1, 2, \ldots, N, \quad t = 1, 2, \ldots, T, \tag{32.26}$$

and assume that N is large.[1]

Assumption A.1: $\left(\lambda_i, \beta_i\right)$ are identically and independently distributed of x_{jt} and u_{jt}, for all i, j and t.

Assumption A.2: $|\lambda_i| < 1$ for all i, and the micro processes, (32.26), have been initialized in the past $(t \to -\infty)$.

Assumption A.3: x_{is}'s have finite second-order moments and are distributed independently of u_{jt} for all i, j, t, and $s \leq t$.

Assumption A.4: micro disturbances, u_{it}, are serially uncorrelated with mean zero and a finite variance, and admit the following decomposition

$$u_{it} = \varphi_i \eta_t + \xi_{it}, \tag{32.27}$$

[1] A unit-specific intercept term can also be included in (32.26) without affecting the main results.

where η_t is the component which is common across all micro units, and ξ_{it} is the idiosyncratic component assumed to be distributed independently across i, with a mean zero and a finite variance.

Assumption A.1 is standard in the aggregation and panel literature with random coefficients. The stability conditions, $|\lambda_i| < 1$, for all i, can be relaxed at the expense of additional assumptions in the way the micro processes are initialized. Assumption A.3 is required for consistent estimation of the parameters of the aggregate equation and can be relaxed. Assumption A.4 is quite general and allows a considerable degree of dependence across the micro disturbances, u_{it}. Nor does it require ξ_{it} and $\varphi_i\eta_t$ to be independently distributed.

To derive the optimal aggregator function, $E(\bar{y}_t|\Omega_t)$, one possibility will be to work with the autoregressive distributed lag representations, (32.26). But this will involve deriving expectations such as $E(\lambda_i y_{i,t-j}|\Omega_t)$ which is complicated by the fact that λ_i and $y_{i,t-j}$ are not independently distributed. To see this notice that under Assumption A.2, (32.26) may be solved for

$$y_{it} = \beta_i \sum_{j=0}^{\infty} \lambda_i^j x_{i,t-j} + \sum_{j=0}^{\infty} \lambda_i^j u_{i,t-j},\ i = 1, 2, \ldots, N, \tag{32.28}$$

which makes the dependence of $y_{i,t-j}$ on λ_i and β_i explicit, and suggests that it might be more appropriate to work directly with the distributed lag representations, (32.28). This is the approach followed by Pesaran (2003).

Aggregating (32.28) across all i, we have

$$\bar{y}_t = N^{-1} \sum_{j=0}^{\infty} \sum_{i=1}^{N} \beta_i \lambda_i^j x_{i,t-j} + N^{-1} \sum_{j=0}^{\infty} \sum_{i=1}^{N} \lambda_i^j u_{i,t-j}, \tag{32.29}$$

where as before $\bar{y}_t = N^{-1} \sum_{i=1}^{N} y_{it}$. Introduce the new information set $\boldsymbol{\Upsilon}_{it} = \{x_{it}, x_{i,t-1}, \ldots\} \cup \Omega_t$ which excludes the individual-specific information on lagged values of y_{it}, and let $\boldsymbol{\Upsilon}_t = \cup_{i=1}^{N} \boldsymbol{\Upsilon}_{it}$. Suppose also that N is large enough so that $y_{i,t-j}, j = 1, 2, \ldots$, cannot be revealed from the aggregates $\bar{y}_{t-1}, \bar{y}_{t-2}, \ldots$. Now, under Assumptions A.1 and A.4

$$E\left(\beta_i \lambda_i^j \mid \boldsymbol{\Upsilon}_t\right) = E\left(\beta \lambda^j\right) = a_j, \tag{32.30}$$

$$E\left(\lambda_i^j \mid \boldsymbol{\Upsilon}_t\right) = E\left(\lambda^j\right) = b_j, \tag{32.31}$$

and

$$E\left(\lambda_i^j u_{i,t-j} | \boldsymbol{\Upsilon}_t\right) = E\left(\lambda_i^j \mid \boldsymbol{\Upsilon}_t\right) E\left(u_{i,t-j} | \boldsymbol{\Upsilon}_t\right).$$

Taking conditional expectations of both sides of (32.29) with respect to $\boldsymbol{\Upsilon}_t$ we now have

$$E(\bar{y}_t|\boldsymbol{\Upsilon}_t) = N^{-1} \sum_{j=0}^{\infty} \sum_{i=1}^{N} x_{i,t-j} E\left[\left(\beta_i \lambda_i^j\right) | \boldsymbol{\Upsilon}_t\right] + N^{-1} \sum_{j=0}^{\infty} \sum_{i=1}^{N} E\left(\lambda_i^j \mid \boldsymbol{\Upsilon}_t\right) E\left(u_{i,t-j} | \boldsymbol{\Upsilon}_t\right).$$

Hence, using (32.30) and (32.31) we have

$$E(\bar{y}_t|\boldsymbol{\Upsilon}_t) = \sum_{j=0}^{\infty} a_j \bar{x}_{t-j} + \sum_{j=0}^{\infty} b_j E(U_{t-j}|\boldsymbol{\Upsilon}_t), \tag{32.32}$$

where $\bar{x}_t = N^{-1}\sum_{i=1}^{N} x_{it}$ and $U_t = N^{-1}\sum_{i=1}^{N} u_{it}$. This result provides the forecast of the aggregate series $\{\bar{y}_t\}$ conditional on $\boldsymbol{\Upsilon}_t$ that involves disaggregated observations on $x_{it}'s$. To obtain the aggregate forecast function we need to take expectations of both sides of (32.32) with respect to Ω_t. Noting that Ω_t is contained in $\boldsymbol{\Upsilon}_t$ we now have

$$E\left(\bar{y}_t|\Omega_t\right) = \sum_{j=0}^{\infty} a_j \bar{x}_{t-j} + \sum_{j=0}^{\infty} b_j E\left(U_{t-j}|\Omega_t\right). \tag{32.33}$$

The aggregate predictor function, $E\left(\bar{y}_t|\Omega_t\right)$, is composed of a predetermined component, $\sum_{j=0}^{\infty} a_j \bar{x}_{t-j}$, and a random component, $\sum_{j=0}^{\infty} b_j E\left(U_{t-j}|\Omega_t\right)$. To learn more about the random component, using (32.27) first note that

$$U_t = \varphi\, \eta_t + \mathcal{Z}_t,$$

where

$$\varphi = N^{-1}\sum_{i=1}^{N} \varphi_i, \text{ and } \mathcal{Z}_t = N^{-1}\sum_{i=1}^{N} \xi_{it}.$$

Namely, the aggregate error term, U_t, is itself composed of a common component, η_t, and an aggregate of the idiosyncratic shocks, $\mathcal{Z}_t$. Under Assumptions A.3 and A.4, η_t and $\mathcal{Z}_t$ are serially uncorrelated and independently distributed of x_{it}'s, and hence (noting that $\bar{y}_t$ is not contained in Ω_t) we have

$$E\left(U_t|\Omega_t\right) = \varphi\, E\left(\eta_t|\Omega_t\right) + E\left(\mathcal{Z}_t|\Omega_t\right) = 0. \tag{32.34}$$

Using this result in (32.33) now yields

$$E\left(\bar{y}_t|\Omega_t\right) = \sum_{j=0}^{\infty} a_j \bar{x}_{t-j} + \sum_{j=1}^{\infty} b_j V_{t-j}, \tag{32.35}$$

where

$$V_{t-j} = E\left(U_{t-j}|\Omega_t\right) = \varphi\, E\left(\eta_{t-j}|\Omega_t\right) + E\left(\mathcal{Z}_{t-j}|\Omega_t\right),\ j = 1, 2, \ldots. \tag{32.36}$$

The optimal aggregate dynamic model corresponding to the micro relations, (32.26), is now given by

$$\bar{y}_t = \sum_{j=0}^{\infty} a_j \bar{x}_{t-j} + \sum_{j=1}^{\infty} b_j V_{t-j} + \varepsilon_t, \tag{32.37}$$

or

$$\bar{y}_t = \sum_{j=0}^{\infty} a_j \bar{x}_{t-j} + \varphi \sum_{j=1}^{\infty} b_j E\left(\eta_{t-j}|\Omega_t\right) + \sum_{j=1}^{\infty} b_j E\left(\mathcal{Z}_{t-j}|\Omega_t\right) + \varepsilon_t, \tag{32.38}$$

where $\varepsilon_t = \bar{y}_t - E\left(\bar{y}_t|\Omega_t\right)$. By construction ε_t is orthogonal to $\{\bar{x}_t, \bar{x}_{t-1,\ldots}\}$ and $\{V_{t-1}, V_{t-2}, \ldots\}$. But, as in the static case, the contemporaneous errors of the aggregate equation, ε_t, are likely to be heteroskedastic. The above aggregate specification is optimal in the sense that $E\left(\bar{y}_t|\Omega_t\right)$ minimizes $E[\bar{y}_t - E\left(\bar{y}_t|\Omega_t\right)]^2$ with respect to the aggregate information set, Ω_t.[2]

The terms $V_{t-1}, V_{t-2}, \ldots$ in addition to being orthogonal to the aggregate disturbances, ε_t, are in fact serially uncorrelated with zero means and a finite variance. First, it is easily seen that

$$E(V_{t-j}) = E\left[E\left(U_{t-j}|\Omega_t\right)\right] = E(U_{t-j}) = 0.$$

Also, for $j > 0$

$$\begin{aligned} E(V_{t-j}V_{t-j-1}|\Omega_{t-j-1}) &= V_{t-j-1}E(V_{t-j}|\Omega_{t-j-1}) \\ &= V_{t-j-1}E\left[E\left(U_{t-j}|\Omega_t\right)|\Omega_{t-j-1}\right] \\ &= V_{t-j-1}E(U_{t-j}|\Omega_{t-j-1}). \end{aligned}$$

But U_{t-j} is a serially uncorrelated process with zero mean. Hence, $E(V_{t-j}V_{t-j-1}|\Omega_{t-j-1}) = 0$, which also implies that $E(V_{t-j}V_{t-j-1}) = 0$. Using a similar line of reasoning it is also easily established that $E(V_{t-j}V_{t-j-s}) = 0$, for all $s \geq 0$. Finally, since by Assumptions A.3 and A.4, x_{is} and u_{it} have finite variances, the random variables $V_{t-1}, V_{t-2}, \ldots$, being linear functions of x_{is} and u_{it}, will also have finite variances. Clearly, the same arguments also apply to the components of V_{t-j}, namely $V^{\eta}_{t-j} = E(\eta_{t-j}|\Omega_t)$ and $V^{z}_{t-j} = E\left(\mathcal{Z}_{t-j}|\Omega_t\right)$, namely V^{η}_{t-j} and V^{z}_{t-j} have zero means, are serially uncorrelated with finite variances.

The aggregate function, (32.37), holds irrespective of whether the shocks to the underlying micro relations contain a common component. But the contribution of the idiosyncratic shocks, $\mathcal{Z}_t$, to the aggregate function will depend on the rate at which the distributed lag coefficients, b_j, decay as $j \to \infty$. Although, under Assumption A.4 $\mathcal{Z}_t \xrightarrow{p} 0$, this does not necessarily mean that the contribution of the idiosyncratic shocks, given by $\sum_{j=1}^{\infty} b_j E\left(\mathcal{Z}_{t-j}|\Omega_t\right)$, will also tend to zero as $N \to \infty$. Heuristically, this is due to the fact that, under Assumptions A.3 and A.4, the variance of V^{z}_{t-j} is of order of $\sum_{j=1}^{\infty} b_j^2/N$ and need not tend to zero if the coefficients, b_j, do not decay sufficiently fast. An example of such a possibility was first discussed by Granger (1980). We now turn to this and other examples and show how a number of results in the literature can be obtained from the optimal aggregator function given by (32.37). In the general case where micro relations are subject to both common and idiosyncratic shocks, the effect of the common

[2] Notice that $\{\bar{x}_t, \bar{x}_{t-1}, \ldots\}$ and $\{V_{t-1}, V_{t-2}, \ldots\}$ are contained in Ω_t.

shocks on the aggregate forecast, $E\left(\bar{y}_t|\Omega_t\right)$, will dominate as $N \to \infty$. Hence, for forecasting purposes, the effects of idiosyncratic shocks can be ignored.

The analysis of aggregation of *ARDL* models by Lewbel (1994) can also be related to the optimal aggregate function, (32.37). Lewbel considers the relatively simple case where the coefficients β_i and λ_i are independently and identically distributed across i, and makes the additional assumption that the distributions of β_i and x_{it} are uncorrelated and that λ_i and $\beta_i x_{it} + u_{it}$ are independently distributed.[3] Under these assumptions and adopting the statistical approach described in Section 32.4.2, Lewbel derives the following aggregate infinite-order autoregressive specification

$$\mu_y(t) = \sum_{j=1}^{\infty} c_j \mu_y(t-j) + \beta \mu_x(t) + \mu_u(t), \tag{32.39}$$

where $\mu_y(t)$, $\mu_x(t)$, and $\mu_u(t)$ are the cross-section means of y_{it}, x_{it}, and u_{it}, respectively.

Assuming the above infinite-order autoregressive representation exists, Lewbel shows that the coefficients c_s satisfy the recursions

$$b_s = \sum_{r=0}^{s-1} b_r c_{s-r}, \tag{32.40}$$

with $b_j = E\left(\lambda^j\right)$, as before. It is then easily seen that $c_1 = b_1 = E(\lambda)$, $c_2 = E(\lambda - b_1)^2 = Var(\lambda)$, which establishes that the autoregressive component of the aggregate specification must at least be of second-order; otherwise the distribution of λ will be degenerate with all agents having the same lag coefficient.

Lewbel's result and a number of its generalizations can be derived from the optimal aggregate specification given by (32.37). Our approach also provides the conditions that ensure the existence of Lewbel's infinite-order autoregressive representation. In the simple case considered by Lewbel, where β_i and λ_i are assumed to be independently distributed, we have $E\left(\beta_i \lambda_i^j\right) = E(\beta_i) E\left(\lambda_i^j\right) = \beta b_j$, and (32.37) simplifies to[4]

$$\bar{y}_t = \beta \sum_{j=0}^{\infty} b_j \bar{x}_{t-j} + \sum_{j=1}^{\infty} b_j V_{t-j} + \varepsilon_t. \tag{32.41}$$

To see the relationship between (32.41) and Lewbel's result, (32.39) first note that

$$\bar{y}_t = B(L)\left[\beta \bar{x}_t + V_t\right] + \varepsilon_t - V_t, \tag{32.42}$$

where $B(L) = \sum_{j=0}^{\infty} b_j L^j$. Whether it is possible to write (32.42) as an infinite-order autoregressive specification in $\bar{y}_t$, depends on whether $B(L)$ is invertible and this in turn depends on

[3] The consequences of relaxing some of these assumptions are briefly discussed by Lewbel (1994, Section 4).

[4] Recall that here we are assuming that there are no common components in the micro shocks, u_{it}, and hence $V_t \xrightarrow{p} 0$.

the probability distribution of λ. It is, for example, clear from our discussion in the previous section that if λ has a beta distribution of the second type with $0 < q \leq 1$, then $\{b_j\}$ will not be absolute summable and $B(L) = \sum_{j=0}^{\infty} b_j L^j$ may not be invertable. Therefore, under this distributional assumption, Lewbel's autoregressive representation may not exist. But if $\{b_j\}$ is absolute summable, $B(L)$ can be inverted and (32.42) can be written as

$$\bar{y}_t = \sum_{j=1}^{\infty} c_j \bar{y}_{t-j} + \beta \bar{x}_t + \sum_{j=1}^{\infty} c_j V_{t-j} + C(L)\varepsilon_t, \tag{32.43}$$

where $C(L) = 1 - \sum_{j=1}^{\infty} c_j L^j$. The coefficients c_j are obtainable from the polynomial identity $B(L)C(L) \equiv 1$, and it is easily verified that they in fact satisfy the recursive relations (32.40) derived by Lewbel (1994).

In the more general case where β_i and λ_i are allowed to be statistically dependent, the optimal aggregate specification does not simplify to (32.43) and will be given by (32.37). In this more general setting there seems little gain in rewriting the resultant distributed lag model in the infinite-order autoregressive form used by Lewbel (1994).

32.6 Aggregation of factor-augmented *VAR* models

Consider the following factor-augmented *VAR* model in N cross-sectional units

$$\mathbf{y}_t = \mathbf{\Phi}\mathbf{y}_{t-1} + \mathbf{B}\mathbf{x}_t + \mathbf{\Gamma}\mathbf{f}_t + \boldsymbol{\varepsilon}_t, \text{ for } t = 1, 2, \ldots, T, \tag{32.44}$$

where $\mathbf{x}_t = (x_{1t}, x_{2t}, \ldots, x_{Nt})'$ is an $N \times 1$ vector of cross-section specific regressors, $\mathbf{f}_t$ is an $m \times 1$ vector of common factors, $\mathbf{\Phi}$ and $\mathbf{B}$ are $N \times N$ matrices of randomly distributed coefficients, and $\mathbf{\Gamma}$ is an $N \times m$ matrix of randomly distributed factor loadings with elements γ_{ij}, for $i = 1, 2, \ldots, N$, and $j = 1, 2, \ldots, m$. We denote the elements of $\mathbf{\Phi}$ by ϕ_{ij}, for $i, j = 1, 2, \ldots, N$, and assume that $\mathbf{B}$ is a diagonal matrix with elements β_i, also collected in the $N \times 1$ vector $\boldsymbol{\beta} = (\beta_1, \beta_2, \ldots, \beta_N)'$.[5] The objective is to derive an optimal aggregate function for $\bar{y}_{wt} = \mathbf{w}'\mathbf{y}_t$ in terms of its lagged values, and current and lagged values of $\bar{x}_{wt} = \mathbf{w}'\mathbf{x}_t$ and $\mathbf{f}_t$, where $\mathbf{w} = (w_1, w_2, \ldots, w_N)'$ is a set of predetermined aggregation weights such that $\Sigma_{i=1}^{N} w_i = 1$. Throughout, it is assumed that $\mathbf{w}$ is known and the weights are granular, in the sense that

$$\frac{|w_i|}{\|\mathbf{w}\|} = O\left(N^{-1/2}\right), \text{ for any } i, \text{ and } \|\mathbf{w}\| = O\left(N^{-1/2}\right). \tag{32.45}$$

Denote the aggregate information set by $\Omega_t = (\bar{y}_{w,t-1}, \bar{y}_{w,t-2}, \ldots; \bar{x}_{wt}, \bar{x}_{w,t-1}, \ldots; \mathbf{f}_t, \mathbf{f}_{t-1}, \ldots)$. When $\mathbf{f}_t$ is not observed, the current and lagged values of $\mathbf{f}_t$ in Ω_t must be replaced by their fitted or forecast values obtained from an auxiliary model for $\mathbf{f}_t$, and possibly other variables, not included in (32.44). Consider the augmented information set $\Upsilon_t = (\mathbf{y}_{t-M}; \mathbf{w}; \mathbf{x}_t, \mathbf{x}_{t-1}, \ldots;$

[5] This specification can be readily generalized to allow for more than one cross-section specific regressor, by replacing $\mathbf{B}\mathbf{x}_t$ with $\mathbf{B}_1\mathbf{x}_{1t} + \mathbf{B}_2\mathbf{x}_{2t} + \ldots + \mathbf{B}_k\mathbf{x}_{kt}$.

$\mathbf{f}_t, \mathbf{f}_{t-1}, \ldots; \bar{y}_{w,t-1}, \bar{y}_{w,t-2}, \ldots)$, that includes the weights, $\mathbf{w}$, and the disaggregate observations on the regressors, x_{it}. Note that Ω_t is contained in Υ_t.

Now introduce the following assumptions on the eigenvalues of $\mathbf{\Phi}$ and the idiosyncratic errors, $\boldsymbol{\varepsilon}_t = (\varepsilon_{1t}, \varepsilon_{2t}, \ldots, \varepsilon_{Nt})'$.

Assumption A.5. The coefficient matrix, $\mathbf{\Phi}$, of the *VAR* model in (32.44) has distinct eigenvalues $\lambda_i(\mathbf{\Phi})$, for $i = 1, 2, \ldots, N$, and satisfies the following cross-sectionally invariant conditional moments

$$\left.\begin{array}{l} E\left(\lambda_i^s(\mathbf{\Phi}) \left| \Upsilon_t, \mathbf{P}, \boldsymbol{\varepsilon}_{t-s}\right.\right) = a_s, \\ E\left(\lambda_i^s(\mathbf{\Phi}) \left| \Upsilon_t, \mathbf{P}, \boldsymbol{\beta}\right.\right) = b_s(\boldsymbol{\beta}), \\ E\left(\lambda_i^s(\mathbf{\Phi}) \left| \Upsilon_t, \mathbf{P}, \mathbf{\Gamma}\right.\right) = c_s(\mathbf{\Gamma}), \end{array}\right\} \tag{32.46}$$

for all $s = 1, 2, \ldots,$ and $i = 1, 2, \ldots, N$, where $\Upsilon_t = (\mathbf{y}_{t-M}; \mathbf{w}; \mathbf{x}_t, \mathbf{x}_{t-1}, \ldots; \mathbf{f}_t, \mathbf{f}_{t-1}, \ldots; \bar{y}_{w,t-1}, \bar{y}_{w,t-2}, \ldots)$, and $\mathbf{P}$ is an $N \times N$ matrix containing the eigenvectors of $\mathbf{\Phi}$ as column vectors.

Assumption A.6 The idiosyncratic shocks, $\boldsymbol{\varepsilon}_t = (\varepsilon_{1t}, \varepsilon_{2t}, \ldots, \varepsilon_{Nt})'$, in (32.44) are serially uncorrelated with zero means and finite variances.

Remark 10 *Assumption A.5 is analytically convenient and can be viewed as a natural generalization of the simple AR(1) specifications considered by Robinson (1978), Granger (1980) and others. Using the spectral decomposition of* $\mathbf{\Phi} = \mathbf{P}\mathbf{\Lambda}\mathbf{P}^{-1}$, *where* $\mathbf{\Lambda} = diag[\lambda_1(\mathbf{\Phi}), \lambda_2(\mathbf{\Phi}), \ldots, \lambda_N(\mathbf{\Phi})]$ *is a diagonal matrix with eigenvalues of* $\mathbf{\Phi}$ *on its diagonal, the factor-augmented VAR model can be written as*

$$y_{it}^* = \lambda_i(\mathbf{\Phi})\, y_{i,t-1}^* + z_{it}^*, \; i = 1, 2, \ldots, N, \text{ and } t = 1, 2, \ldots, T; \tag{32.47}$$

where y_{it}^* *is the* i^{th} *element of* $\mathbf{y}_t^* = \mathbf{P}^{-1}\mathbf{y}_t$, *and* z_{it}^* *is the* i^{th} *element of* $\mathbf{z}_t^* = \mathbf{P}^{-1}(\mathbf{B}\mathbf{x}_t + \mathbf{\Gamma}\mathbf{f}_t + \boldsymbol{\varepsilon}_t)$. *Consider now the conditions under which an optimal aggregate function exists for* $\bar{y}_{wt}^* = \mathbf{w}'\mathbf{y}_t^* = \mathbf{w}'\mathbf{P}^{-1}\mathbf{y}_t$. *We know from the existing literature that such an aggregate function exists if* $E\left(\lambda_i^s(\mathbf{\Phi}) \left| z_{it}^*\right.\right) = a_s^*$, *for all i. Seen from this perspective, our assumption that conditional on* $\mathbf{P}$ *the eigenvalues have moments that do not depend on i seems sensible, and is likely to be essential for the validity of Granger's conjecture.*

Remark 11 *It is also worth noting that Assumption A.5 does allow for possible dependence of* $\lambda_i(\mathbf{\Phi})$ *on the coefficients* β_i *and* γ_{ij}.

As already shown, the optimal aggregate function (in the mean squared error sense) is given by

$$\bar{y}_{wt} = E\left(\mathbf{w}'\mathbf{y}_t \left| \Omega_t\right.\right) + v_{wt}, \tag{32.48}$$

where $\bar{y}_{wt} = \mathbf{w}'\mathbf{y}_t$, and by construction $E(v_{wt}|\Omega_t) = 0$, and v_{wt}, $t = 1, 2, \ldots$, are serially uncorrelated, although they could be conditionally heteroskedastic. Solving (32.44) recursively forward from the initial state, $\mathbf{y}_{-M}$, we have

$$\mathbf{y}_t = \mathbf{\Phi}^{t+M}\mathbf{y}_{-M} + \sum_{s=0}^{t+M-1} \mathbf{\Phi}^s \left(\mathbf{B}\mathbf{x}_{t-s} + \mathbf{\Gamma}\mathbf{f}_{t-s} + \boldsymbol{\varepsilon}_{t-s}\right). \tag{32.49}$$

Hence, using the spectral decomposition of $\mathbf{\Phi} = \mathbf{P}\mathbf{\Lambda}\mathbf{P}^{-1}$, we obtain

$$\bar{y}_{wt} = \mathbf{w}'\mathbf{P}\mathbf{\Lambda}^{t+M}\mathbf{P}^{-1}\mathbf{y}_{-M} + \sum_{s=0}^{t+M-1} \mathbf{w}'\mathbf{P}\mathbf{\Lambda}^{s}\mathbf{P}^{-1} \left(\mathbf{B}\mathbf{x}_{t-s} + \mathbf{\Gamma}\mathbf{f}_{t-s} + \boldsymbol{\varepsilon}_{t-s}\right). \tag{32.50}$$

It is now possible to show that (see Pesaran and Chudik (2014) for details)

$$\begin{aligned} E\left(\bar{y}_{wt} \,|\Omega_t\right) = \left(\mathbf{w}'\mathbf{y}_{-M}\right) E\left(a_{t+M} \,|\Omega_t\right) + \sum_{s=0}^{t+M-1} \mathbf{w}' E\left[b_s(\boldsymbol{\beta})\mathbf{B}\mathbf{x}_{t-s} \,|\Omega_t\right] \\ + \sum_{s=0}^{t+M-1} \mathbf{w}' E\left[c_s(\Gamma)\mathbf{\Gamma} \,|\Omega_t\right] \mathbf{f}_{t-s} + \sum_{s=1}^{t+M-1} a_s E\left(\bar{\varepsilon}_{w,t-s} \,|\Omega_t\right). \end{aligned} \tag{32.51}$$

where $\bar{\varepsilon}_{wt} = \mathbf{w}'\boldsymbol{\varepsilon}_t$.

32.6.1 Aggregation of stationary micro relations with random coefficients

The optimal aggregate function derived in (32.51) is quite general and holds for any N, and does not require the underlying micro processes to be stationary. But its use in empirical applications is limited as it depends on unobserved initial state, $\mathbf{w}'\mathbf{y}_{-M}$, and the micro variables, $\mathbf{x}_t$. To derive empirically manageable aggregate functions, in what follows we assume that the underlying processes are stationary and the micro parameters, β_i and γ_{ij}, are random draws from a common distribution. More specifically, we make the following assumptions:

Assumption A.7: The micro coefficients, β_i and γ_{ij}, are random draws from common distributions with finite moments such that

$$E\left[b_s(\boldsymbol{\beta})\mathbf{B} \,|\Omega_t\right] = b_s\mathbf{I}_N, \tag{32.52}$$

$$E\left[c_s(\mathbf{\Gamma})\mathbf{\Gamma} \,|\Omega_t\right] = \boldsymbol{\tau}_N\mathbf{c}_s', \tag{32.53}$$

where $b_s(\boldsymbol{\beta})$ and $c_s(\mathbf{\Gamma})$ are defined in Assumption A.5, $b_s = E\left[b_s(\boldsymbol{\beta})\beta_i\right]$, $\mathbf{c}_s = E\left[c_s(\mathbf{\Gamma})\boldsymbol{\gamma}_i\right]$, and $\boldsymbol{\tau}_N$ is an $N \times 1$ vector of ones.

Assumption A.8: The eigenvalues of $\mathbf{\Phi}$, $\lambda_i(\mathbf{\Phi})$, are draws from a common distribution with support over the range $(-1, 1)$.

Under Assumption A.7, (32.51) simplifies to

$$E\left(\bar{y}_{wt}\,|\Omega_t\right)=\left(\mathbf{w}'\mathbf{y}_{-M}\right)E\left(a_{t+M}\,|\Omega_t\right)+\sum_{s=0}^{t+M-1}b_s\bar{x}_{w,t-s}+\sum_{s=0}^{t+M-1}\mathbf{c}_s'\mathbf{f}_{t-s}+\sum_{s=1}^{t+M-1}a_sE\left(\bar{\varepsilon}_{w,t-s}\,|\Omega_t\right),$$

where $\bar{x}_{wt}=\mathbf{w}'\mathbf{x}_t$, and $E\left(\bar{y}_{wt}\,|\Omega_t\right)$ no longer depends on the individual specific regressors. Under the additional Assumption A.8, and for M sufficiently large, the initial states are also eliminated and we have

$$E\left(\bar{y}_{wt}\,|\Omega_t\right)=\sum_{s=0}^{\infty}b_s\bar{x}_{w,t-s}+\sum_{s=0}^{\infty}\mathbf{c}_s'\mathbf{f}_{t-s}+\sum_{s=1}^{\infty}a_s\eta_{t-s},$$

where $\eta_{t-s}=E\left(\bar{\varepsilon}_{w,t-s}\,|\Omega_t\right)$. Note that $\sum_{s=1}^{\infty}a_s\eta_{t-s}=E\left[\sum_{s=1}^{\infty}a_s\bar{\varepsilon}_{w,t-s}\,|\Omega_t\right]$. Using this result in (32.48) we obtain the optimal aggregate function

$$\bar{y}_{wt}=\sum_{s=0}^{\infty}b_s\bar{x}_{w,t-s}+\sum_{s=0}^{\infty}\mathbf{c}_s'\mathbf{f}_{t-s}+\sum_{s=1}^{\infty}a_s\eta_{t-s}+\nu_{wt},\tag{32.54}$$

which holds for any finite N.

The dynamic properties of $\bar{y}_{wt}$ and its persistence to shocks depend on the decay rates of the distributed lag coefficients, $\{a_s\}$, $\{b_s\}$ and $\{\mathbf{c}_s\}$. If $|\lambda_i(\mathbf{\Phi})|<1-\epsilon$, for some strictly positive constant $\epsilon>0$, then the distributed lagged coefficients, $\{a_s\}$, $\{b_s\}$ and $\{\mathbf{c}_s\}$ decay exponentially fast and the aggregate function will not exhibit long memory features. However, in the case where $\lambda_i(\mathbf{\Phi})'s$ are draws from distributions with supports covering -1 and/or 1, the rate of decay of the distributed lagged coefficients will be slower than exponential, and the resultant aggregate function will be subject to long memory effects. This result confirms Granger's conjecture in the case of large dimensional *VAR* models, and establishes sufficient conditions for its validity.

It is also worth noting that, in general, $\bar{y}_{wt}$ has an infinite-order distributed lag representation even if the underlying micro relations have finite lag-orders. This is an important consideration in empirical macro economic analysis where the macro variables under consideration are often constructed as aggregates of observations on a large number of micro units.

32.6.2 Limiting behaviour of the optimal aggregate function

The aggregate function in (32.54) continues to hold even if $N\to\infty$, so long as the degree of cross-sectional dependence in the idiosyncratic errors, ε_{it}, $i=1,2,\ldots,N$, is sufficiently weak; otherwise there is no guarantee for the aggregation error, $\sum_{s=1}^{\infty}a_s\eta_{t-s}+\nu_{wt}$, to vanish as $N\to\infty$. To this end we introduce the following assumption that governs the degree of error cross-section dependence.

Assumption A.9: The idiosyncratic errors, $\boldsymbol{\varepsilon}_t=(\varepsilon_{1t},\varepsilon_{2t},\ldots,\varepsilon_{Nt})'$ in (32.44) are cross-sectionally weakly dependent in the sense that

$$\|\mathbf{\Sigma}_{\varepsilon}\|_1 = \|\mathbf{\Sigma}_{\varepsilon}\|_{\infty} = O\left(N^{\alpha_{\varepsilon}}\right),$$

where $\mathbf{\Sigma}_{\varepsilon} = E(\boldsymbol{\varepsilon}_t \boldsymbol{\varepsilon}_t')$, for some constant $0 \leq \alpha_{\varepsilon} < 1$.

Remark 12 *Condition $0 \leq \alpha_{\varepsilon} < 1$ in Assumption A.9 is sufficient and necessary for weak cross-section dependence of micro innovations. See Chudik, Pesaran, and Tosetti (2011). Following Bailey, Kapetanios, and Pesaran (2015) we shall refer to the constant α_{ε} as the exponent of cross-sectional dependence of the idiosyncratic shocks. See also Section 29.2.*

Since under Assumption A.6 the errors, $\boldsymbol{\varepsilon}_t$, are serially uncorrelated, we have

$$Var\left(\sum_{s=1}^{\infty} a_s \bar{\varepsilon}_{w,t-s}\right) = \sum_{s=1}^{\infty} a_s^2 Var\left(\bar{\varepsilon}_{w,t-s}\right) \leq \left(\sum_{s=1}^{\infty} a_s^2\right) \sup_t \left[Var(\bar{\varepsilon}_{wt})\right].$$

Furthermore

$$Var(\bar{\varepsilon}_{wt}) = \mathbf{w}' \mathbf{\Sigma}_{\varepsilon} \mathbf{w} \leq \|\mathbf{w}\|^2 \varrho\left(\mathbf{\Sigma}_{\varepsilon}\right),$$

and by Assumption A.9, and the granularity conditions (32.45), we have[6]

$$\sup_t \left[Var(\bar{\varepsilon}_{wt})\right] = O\left(N^{\alpha_{\varepsilon}-1}\right),$$

and $\sum_{s=1}^{\infty} a_s \bar{\varepsilon}_{w,t-s} \xrightarrow{q.m} 0$, so long as $\sum_{s=1}^{\infty} a_s^2 < K$, for some positive constant K.[7] Recall that under Assumption A.9, $\alpha_{\varepsilon} < 1$, and $\sup_t \left[Var(\bar{\varepsilon}_{wt})\right] \to 0$, as $N \to \infty$. Moreover, since $\sum_{s=1}^{\infty} a_s \eta_{t-s} = E\left(\sum_{s=1}^{\infty} a_s \bar{\varepsilon}_{w,t-s} | \Omega_t\right)$, it follows that

$$\sum_{s=1}^{\infty} a_s \eta_{t-s} \xrightarrow{q.m} 0, \tag{32.55}$$

and hence for each t we have

$$\bar{y}_{wt} - \sum_{s=0}^{\infty} b_s \bar{x}_{w,t-s} - \sum_{s=0}^{\infty} \mathbf{c}_s' \mathbf{f}_{t-s} - v_{wt} \xrightarrow{q.m} 0, \text{ as } N \to \infty.$$

The limiting behaviour of v_{wt}, as $N \to \infty$, depends on the nature of the processes generating x_{it}, $\mathbf{f}_t$, and ε_{it}, as well as the degree of cross-section dependence that arises from the non-zero off-diagonal elements of $\mathbf{\Phi}$. Sufficient conditions for $v_{wt} \xrightarrow{q.m} 0$ are not presented here due to space constraints, but can be found in Pesaran and Chudik (2014, Proposition 1). The key conditions

[6] Note that $\varrho\left(\mathbf{\Sigma}_{\varepsilon}\right) \leq \|\mathbf{\Sigma}_{\varepsilon}\|_1 = O\left(N^{\alpha_{\varepsilon}}\right)$.

[7] A sufficient condition for $\sum_{s=1}^{\infty} a_s^2$ to be bounded is $|\lambda_i| < 1 - \epsilon$, where ϵ is a small, strictly positive number.

for $v_{wt} \xrightarrow{q.m} 0$ are weak error cross-sectional dependence and sufficiently bounded dynamic interactions across the units. These conditions are satisfied, for example, if $\|\mathbf{\Sigma}_{\varepsilon}\| = \left\|E\left(\boldsymbol{\varepsilon}_t \boldsymbol{\varepsilon}_t'\right)\right\| < K$, and $\sum_{s=1}^{\infty} E\|\mathbf{\Phi}^s\| \leq \sum_{s=1}^{\infty} E\|\mathbf{\Phi}\|^s < K$, for some finite positive constant, K. If, on the other hand, $\sum_{s=1}^{\infty} E\|\mathbf{\Phi}\|^s$ is not bounded as $N \to \infty$, or $\boldsymbol{\varepsilon}_t$ is strongly cross-sectionally dependent, then the aggregation error, v_{wt}, does not necessarily converge to zero and could be sizeable.

32.7 Relationship between micro and macro parameters

In this section we discuss the problem of identification of micro parameters, or some of their distributional features, from the aggregate function given by (32.54). Although it is not possible to recover all of the parameters of micro relations, there are a number of notable exceptions. An important example is the average long-run impact defined by,

$$\bar{\theta} = \frac{1}{N}\sum_{i=1}^{N}\theta_i = \frac{1}{N}\boldsymbol{\tau}_N'\boldsymbol{\theta} = \frac{1}{N}\boldsymbol{\tau}_N'\left(\mathbf{I}_N - \mathbf{\Phi}\right)^{-1}\boldsymbol{\beta}, \tag{32.56}$$

where $\boldsymbol{\theta} = \left(\mathbf{I}_N - \mathbf{\Phi}\right)^{-1}\boldsymbol{\beta} = \left(\boldsymbol{\beta} + \mathbf{\Phi}\boldsymbol{\beta} + \mathbf{\Phi}^2\boldsymbol{\beta} + \ldots\right)$ is the $N \times 1$ vector of individual long-run coefficients, and as before $\boldsymbol{\tau}_N$ is an $N \times 1$ vector of ones. Suppose that Assumptions A.7 and A.8 are satisfied and denote the common mean of β_i by β. Using (32.52), we have $E\left(\mathbf{\Phi}^s\boldsymbol{\beta}\right) = E\left\{E\left[b_s(\boldsymbol{\beta})\mathbf{B}\,|\Omega_t\right]\right\} = b_s\mathbf{I}_N$ for $s = 0, 1, \ldots$. Hence, the elements of $\boldsymbol{\theta}$ have a common mean, $E(\theta_i) = \theta = \sum_{\ell=0}^{\infty} b_s$, which does not depend on the elements of $\mathbf{P}$. If, in addition, the sequence of random variables θ_i is ergodic in mean, then for sufficiently large N, $\bar{\theta}$ is well approximated by its mean, $\sum_{\ell=0}^{\infty} b_s$, and the cross-section mean of the micro long-run effects can be estimated by the long-run coefficient of the associated optimal aggregate model. This result holds even if β_i and $\lambda_i(\mathbf{\Phi})$ are not independently distributed, and irrespective of whether micro shocks contain a common factor.

Whether $\bar{\theta} \to_p \theta$ deserves a comment. A sufficient condition for $\bar{\theta}$ to converge to its mean (in probability) is given by

$$\|Var(\boldsymbol{\theta})\| = O\left(N^{1-\epsilon}\right), \text{for some } \epsilon > 0, \tag{32.57}$$

in which case $\left\|Var\left(\bar{\theta}\right)\right\| \leq N^{-1}\|Var(\boldsymbol{\theta})\| = O\left(N^{-\epsilon}\right) \to 0$ as $N \to \infty$ and $\bar{\theta} \xrightarrow{q.m.} \theta$. Condition (32.57) need not always hold. This condition can be violated if there is a high degree of dependence of micro coefficients β_i across i, or if there is a dominant unit in the underlying model in which case the column norm of $\mathbf{\Phi}$ becomes unbounded in N.

The mean of β_i is straightforward to identify from the aggregate relation since $E\left(\beta_i\right) = b_0$. But further restrictions are needed for identification of $E\left[\lambda_i(\mathbf{\Phi})\right]$ from the aggregate model. As with Pesaran (2003) and Lewbel (1994), the independence of β_i and $\lambda_i(\mathbf{\Phi})$ will be sufficient for the identification of the moments of $\lambda_i(\mathbf{\Phi})$. Under the assumption that β_i and $\lambda_i(\mathbf{\Phi})$ are independently distributed, all moments of $\lambda_i(\mathbf{\Phi})$ can be identified by

$$E\left[\lambda_i^s(\mathbf{\Phi})\right] = \frac{b_s}{b_0}. \tag{32.58}$$

Another possibility is to adopt a parametric specification for the distribution of the micro coefficients and then identify the unknown parameters of the cross-sectional distribution of micro coefficients from the aggregate specification. For example, suppose β_i is independently distributed of $\lambda_i(\boldsymbol{\Phi})$, and $\lambda_i(\boldsymbol{\Phi})$ has a beta distribution over (0, 1),

$$f(\lambda) = \frac{\lambda^{p-1}\left(1-\lambda^{q-1}\right)}{B(p,q)}, p > 0,\ q > 0, 0 < \lambda < 1.$$

Then, as discussed in Robinson (1978) and Pesaran (2003), we have

$$p = \frac{b_1(b_1 - b_2)}{b_2 b_0 - b_1^2}, q = \frac{(b_0 - b_1)(b_1 - b_2)}{b_2 b_0 - b_1^2},$$

and $\theta = b_0\left(p+q-1\right)/\left(q-1\right)$. Another example is uniform distribution for $\lambda_i(\boldsymbol{\Phi})$ on interval $[\lambda_{\min}, \lambda_{\max}]$, $\lambda_{\min} > -1$, $\lambda_{\max} < 1$. Equation (32.58) can be solved to obtain (see Robinson, 1978),

$$\lambda_{\min} = \frac{b_1 - \sqrt{3\left(b_0 b_2 - b_1^2\right)}}{b_0}, \text{and } \lambda_{\max} = \frac{b_1 + \sqrt{3\left(b_0 b_2 - b_1^2\right)}}{b_0}.$$

32.8 Impulse responses of macro and aggregated idiosyncratic shocks

For the analysis of impulse responses we assume that the common factors in (32.44) follow the *VAR*(1) model

$$\mathbf{f}_t = \boldsymbol{\Psi}\mathbf{f}_{t-1} + \mathbf{v}_t, \tag{32.59}$$

where $\boldsymbol{\Psi}$ is an $m \times m$ matrix of coefficients, and $\mathbf{v}_t = (v_{1t}, v_{2t}, \ldots, v_{mt})'$ is the $m \times 1$ vector of macro shocks. To simplify the analysis we also set $\boldsymbol{\beta} = \mathbf{0}$, and write the micro relations as

$$\mathbf{y}_t = \boldsymbol{\Phi}\mathbf{y}_{t-1} + \mathbf{u}_t,\ \mathbf{u}_t = \boldsymbol{\Gamma}\mathbf{f}_t + \boldsymbol{\varepsilon}_t. \tag{32.60}$$

Including the exogenous variables, $\mathbf{x}_t$, in the model is relatively straightforward and does not affect the impulse responses of the shocks to macro factors, $\mathbf{v}_t$, or to the idiosyncratic errors. The lag-orders of the *VAR* models in (32.59) and (32.60) are set to unity only for expositional convenience.

We make the following additional assumption.

Assumption A.10: The $m \times 1$ macro shocks, $\mathbf{v}_t$, are distributed independently of $\boldsymbol{\varepsilon}_{t'}$, for all t and t'. They are also serially uncorrelated, with zero means, and a diagonal variance matrix, $\boldsymbol{\Sigma}_v = Diag(\sigma_{v_1}^2, \sigma_{v_2}^2, \ldots, \sigma_{v_m}^2)$, where $0 < \sigma_{v_j}^2 < \infty$, for all j.

We are interested in the effects of two types of shocks on the aggregate variable $\bar{y}_{wt} = \mathbf{w}'\mathbf{y}_t$, namely the composite macro shock, defined by $\bar{v}_{\bar{\gamma}t} = \mathbf{w}'\boldsymbol{\Gamma}\mathbf{v}_t = \bar{\boldsymbol{\gamma}}_w'\mathbf{v}_t$, and the aggregated

idiosyncratic shock, defined by $\bar{\varepsilon}_{wt} = \mathbf{w}'\boldsymbol{\varepsilon}_t$. We shall also consider the combined aggregate shock defined by

$$\bar{\xi}_{wt} = \mathbf{w}'\boldsymbol{\Gamma}\mathbf{v}_t + \mathbf{w}'\boldsymbol{\varepsilon}_t = \bar{\boldsymbol{\gamma}}_w'\mathbf{v}_t + \bar{\varepsilon}_{wt} = \bar{v}_{\bar{\gamma}t} + \bar{\varepsilon}_{wt},$$

and investigate the time profiles of the effects of these shocks on $\bar{y}_{w,t+s}$, for $s = 0, 1, \ldots$. The combined aggregate shock, $\bar{\xi}_{wt}$, can be identified from the aggregate equation in $\bar{y}_{wt}$, so long as an $AR(\infty)$ approximation for $\bar{y}_{wt}$ exists. Since by assumption $\boldsymbol{\varepsilon}_t$ and $\mathbf{v}_t$ are distributed independently then

$$Var\left(\bar{\xi}_{wt}\right) = \bar{\boldsymbol{\gamma}}_w'\boldsymbol{\Sigma}_v\bar{\boldsymbol{\gamma}}_w + \mathbf{w}'\boldsymbol{\Sigma}_\varepsilon\mathbf{w} = \sigma_{\bar{v}}^2 + \sigma_{\bar{\varepsilon}}^2 = \sigma_{\bar{\xi}}^2,$$

where $\sigma_{\bar{v}}^2 = \bar{\boldsymbol{\gamma}}_w'\boldsymbol{\Sigma}_v\bar{\boldsymbol{\gamma}}_w$ is the variance of the composite macro shock, and $\sigma_{\bar{\varepsilon}}^2 = \mathbf{w}'\boldsymbol{\Sigma}_\varepsilon\mathbf{w}$ is the variance of the aggregated idiosyncratic shock. Note that when $\mathbf{f}_t$ is unobserved, the separate effects of the composite macro shock, $\bar{v}_{\bar{\gamma}t}$, and the aggregated idiosyncratic shock, $\bar{\varepsilon}_{wt}$, can only be identified under the disaggregated model (32.60). Only the effects of $\bar{\xi}_{wt}$ on $\bar{y}_{w,t+h}$ can be identified if the aggregate specification is used.

Using the disaggregate model we obtain the following generalized impulse response functions (*GIRF*s)[8]

$$g_{\bar{\varepsilon}}(s) = E\left(\bar{y}_{w,t+s}\,|\bar{\varepsilon}_{wt} = \sigma_{\bar{\varepsilon}}, \mathcal{I}_{t-1}\right) - E\left(\bar{y}_{w,t+s}\,|\mathcal{I}_{t-1}\right) = \frac{\mathbf{w}'\boldsymbol{\Phi}^s\boldsymbol{\Sigma}_\varepsilon\mathbf{w}}{\sqrt{\mathbf{w}'\boldsymbol{\Sigma}_\varepsilon\mathbf{w}}}, \tag{32.61}$$

$$g_{v_j}(s) = E\left(\bar{y}_{w,t+s}\,\middle|\,v_{jt} = \sigma_{v_j}, \mathcal{I}_{t-1}\right) - E\left(\bar{y}_{w,t+s}\,|\mathcal{I}_{t-1}\right) = \frac{\mathbf{w}'\mathbf{C}_s\boldsymbol{\Sigma}_v\mathbf{e}_{j,v}}{\sqrt{\mathbf{e}_{j,v}'\boldsymbol{\Sigma}_v\mathbf{e}_{j,v}}}, \tag{32.62}$$

for $j = 1, 2, \ldots, m$, where $\mathcal{I}_t$ is an information set consisting of all current and past available information at time t,

$$\mathbf{C}_s = \sum_{j=0}^{s}\boldsymbol{\Phi}^{s-j}\boldsymbol{\Gamma}\boldsymbol{\Psi}^j, \tag{32.63}$$

and $\mathbf{e}_{j,v}$ is an $m \times 1$ selection vector that selects the jth element of $\mathbf{v}_t$. Hence

$$g_{\bar{v}}(s) = E\left(\bar{y}_{w,t+s}\,\middle|\bar{v}_{\bar{\gamma}t} = \sigma_{\bar{v}}, \mathcal{I}_{t-1}\right) - E\left(\bar{y}_{w,t+s}\,|\mathcal{I}_{t-1}\right) = \frac{\mathbf{w}'\mathbf{C}_s\boldsymbol{\Sigma}_v\bar{\boldsymbol{\gamma}}_w}{\sqrt{\bar{\boldsymbol{\gamma}}_w'\boldsymbol{\Sigma}_v\bar{\boldsymbol{\gamma}}_w}}. \tag{32.64}$$

Finally,

$$\begin{aligned} g_{\bar{\xi}}(s) &= E\left(\bar{y}_{w,t+s}\,\middle|\,\bar{\xi}_{wt} = \sigma_{\bar{\xi}}, \mathcal{I}_{t-1}\right) - E\left(\bar{y}_{w,t+s}\,|\mathcal{I}_{t-1}\right) \\ &= \frac{\mathbf{w}'\mathbf{C}_s\boldsymbol{\Sigma}_v\bar{\boldsymbol{\gamma}}_w + \mathbf{w}'\boldsymbol{\Phi}^s\boldsymbol{\Sigma}_\varepsilon\mathbf{w}}{\sqrt{\bar{\boldsymbol{\gamma}}_w'\boldsymbol{\Sigma}_v\bar{\boldsymbol{\gamma}}_w + \mathbf{w}'\boldsymbol{\Sigma}_\varepsilon\mathbf{w}}}. \end{aligned} \tag{32.65}$$

Note that $\mathbf{C}_0 = \boldsymbol{\Gamma}$, and we have $g_{\bar{\xi}}(0) = \sqrt{\bar{\boldsymbol{\gamma}}_w'\boldsymbol{\Sigma}_v\bar{\boldsymbol{\gamma}}_w + \mathbf{w}'\boldsymbol{\Sigma}_\varepsilon\mathbf{w}} = \sigma_{\bar{\xi}}$, as to be expected.

[8] See Chapter 24 for an account of impulse response analysis where the notion of *GIRF* is also discussed.

When N is finite, both, the combined aggregated idiosyncratic shock $(\bar{\varepsilon}_{wt})$ and the composite macro shock $(\bar{v}_{\bar{\gamma}t})$ are important; and the impulse response of the combined aggregate shock on the aggregate variable, given by (32.65), is a linear combination of $g_{\bar{\varepsilon}}(s)$ and $g_{\bar{v}}(s)$, namely

$$g_{\bar{\xi}}(s) = \omega_{\bar{v}} g_{\bar{v}}(s) + \omega_{\bar{\varepsilon}} g_{\bar{\varepsilon}}(s), \tag{32.66}$$

where $\omega_{\bar{\varepsilon}} = \sigma_{\bar{\varepsilon}}/\sigma_{\bar{\xi}}$, $\omega_{\bar{v}} = \sigma_{\bar{v}}/\sigma_{\bar{\xi}}$, and $\omega_{\bar{\varepsilon}}^2 + \omega_{\bar{v}}^2 = 1$.

When $N \to \infty$, it is not necessarily true that both shocks are important, and $\lim_{N\to\infty} \sigma_{\bar{v}}^2/\sigma_{\bar{\xi}}^2$, if it exists, could be any value on the unit interval, including one or zero. We first consider the impulse responses of the aggregated idiosyncratic shock on the aggregate variable in the next proposition.

Proposition 48 *Suppose that* $\|\mathbf{\Sigma}_{\varepsilon}\|_1 = O(N^{\alpha_{\varepsilon}})$, *for some constant* $0 \leq \alpha_{\varepsilon} < 1$, $E\|\mathbf{\Phi}\|$ *is bounded in* N, *where* $\|\mathbf{\Phi}\| = \varrho\left(\mathbf{\Phi}\mathbf{\Phi}'\right)$, *and the aggregation weights satisfy* $\|\mathbf{w}\| = O\left(N^{-1/2}\right)$. *Then, for any given* $s = 0, 1, 2, \ldots$, *we have*

$$E\left|g_{\bar{\varepsilon}}(s)\right| = O\left(N^{(\alpha_{\varepsilon}-1)/2}\right). \tag{32.67}$$

For a proof see Pesaran and Chudik (2014).

The aggregated idiosyncratic shock and its corresponding impulse response function vanishes as $N \to \infty$ at the rate which depends on the degree of cross-sectional dependence of idiosyncratic shocks. This rate could be very slow; and if the condition $\|\mathbf{w}\| = O\left(N^{-1/2}\right)$ is not satisfied, then the rate of convergence would also depend on the degree of granularity of the weights, w_i. The composite macro shock and its corresponding impulse-response function, on the other hand, does not necessarily vanish as $N \to \infty$, depending on the factor loadings. For ease of exposition, we focus on the following model for factor loadings

$$\begin{aligned} \boldsymbol{\gamma}_i &= \boldsymbol{\varkappa}_i, \text{ for } i = 1, 2, \ldots, [N^{\alpha_{\gamma}}], \\ \boldsymbol{\gamma}_i &= \mathbf{0}, \text{ for } i = [N^{\alpha_{\gamma}}] + 1, \ldots, N, \end{aligned}$$

where $\boldsymbol{\varkappa}_i \sim IID(\boldsymbol{\mu}_{\varkappa}, \mathbf{\Sigma}_{\varkappa})$, $[N^{\alpha_{\gamma}}]$ denotes the integer part of $N^{\alpha_{\gamma}}$, constant α_{γ} $(0 < \alpha_{\gamma} \leq 1)$ is the exponent of cross-section dependence of y_{it} due to factors, see Bailey, Kapetanios, and Pesaran (2015) and Section 29.2. Note that the aggregated factor loadings satisfy $Plim_{N\to\infty} N^{1-\alpha_{\gamma}} \bar{\boldsymbol{\gamma}}_w = \boldsymbol{\mu}_{\varkappa}$, and the variance of the composite macro shock, $\sigma_{\bar{v}}^2 = \bar{\boldsymbol{\gamma}}_w' \mathbf{\Sigma}_v \bar{\boldsymbol{\gamma}}_w$, satisfies

$$\underset{N\to\infty}{Plim}\, N^{2(1-\alpha_{\gamma})} \sigma_{\bar{v}}^2 = \boldsymbol{\mu}_{\varkappa}' \mathbf{\Sigma}_v \boldsymbol{\mu}_{\varkappa}. \tag{32.68}$$

The variance of the aggregated idiosyncratic shock, on the other hand, is bounded by

$$\sigma_{\bar{\varepsilon}}^2 = \mathbf{w}' \mathbf{\Sigma}_{\varepsilon} \mathbf{w} \leq O\left(N^{\alpha_{\varepsilon}-1}\right). \tag{32.69}$$

It follows from (32.68)–(32.69) that only when $\alpha_{\gamma} > (\alpha_{\varepsilon} + 1)/2$ and $\boldsymbol{\mu}_{\varkappa} \neq \mathbf{0}$, the variance of the composite macro shock dominates, in which case $Plim_{N\to\infty} \sigma_{\bar{v}}^2/\sigma_{\bar{\xi}}^2 = 1$, and the combined

aggregate shock, $\bar{\xi}_{wt} = \bar{v}_{\bar{\gamma}t} + \bar{\varepsilon}_{wt}$ converges in quadratic mean to the composite macro shock as $N \to \infty$. It is then possible to scale $g_{\bar{\xi}}(s)$ by $\sigma_{\bar{v}}^{-1}$, and for any given $s = 0, 1, 2, \ldots$, we can obtain

$$\underset{N\to\infty}{Plim}\left[\sigma_{\bar{v}}^{-1} g_{\bar{\xi}}(s)\right] = \underset{N\to\infty}{Plim}\left[\sigma_{\bar{v}}^{-1} g_{\bar{v}}(s)\right].$$

When $\alpha_\gamma \leq (\alpha_\varepsilon + 1)/2$ and/or $\boldsymbol{\mu}_{\varkappa} = \mathbf{0}$, the macro shocks do not necessarily dominate the aggregated idiosyncratic shock (as $N \to \infty$), and the latter shock can be as important as macro shocks, or even dominate the macro shocks as $N \to \infty$.

32.9 A Monte Carlo investigation

We consider a first-order *VAR* model with a single unobserved factor to examine the response of $\bar{y}_t = N^{-1}\sum_{i=1}^{N} y_{it}$, to the combined aggregate shock, $\bar{\xi}_t = \bar{\gamma} v_t + \bar{\varepsilon}_t$, where $\bar{\gamma} = N^{-1}\sum_{i=1}^{N}\gamma_i$ and $\bar{\varepsilon}_t = N^{-1}\sum_{i=1}^{N}\varepsilon_{it}$. As before, we decompose the effects into the contribution due to a macro shock, v_t, and the aggregated idiosyncratic shock, $\bar{\varepsilon}_t$. Using (32.66), we have

$$g_{\bar{\xi}}^{d}(s) = m_{v}^{d}(s) + m_{\bar{\varepsilon}}^{d}(s), \tag{32.70}$$

where $m_{v}^{d}(s) = \omega_v g_{v}^{d}(s)$, and $m_{\bar{\varepsilon}}^{d}(s) = \omega_{\bar{\varepsilon}} g_{\bar{\varepsilon}}^{d}(s)$ are the respective contributions of the macro and aggregated idiosyncratic shocks, and the weights ω_v and $\omega_{\bar{\varepsilon}}$ are defined below (32.66).

Aggregation weights are set equal to N^{-1} in all simulations. The subscript d is introduced to highlight the fact that these impulse responses are based on the disaggregate model. We know from theoretical results that, in cases where the optimal aggregate function exists, the common factor is strong (i.e., $\alpha_\gamma = 1$), and the idiosyncratic shocks are weakly correlated (i.e., $\alpha_\varepsilon = 0$), then $g_{\bar{\xi}}^{d}(s)$ converges to $g_{v}^{d}(s)$ as $N \to \infty$, for all s. But it would be of interest to investigate the contributions of macro and aggregated idiosyncratic shocks to the aggregate impulse response functions, when N is finite, as well as when α_γ takes intermediate values between 0 and 1.

We also use Monte Carlo experiments to investigate the persistence properties of the aggregate variable. The degree and sources of persistence in macro variables, such as consumer price inflation, output and real exchange rates, have been of considerable interest in economics. We know from the theoretical results that there are two key components affecting the persistence of the aggregate variables: distribution of the eigenvalues of lagged micro coefficients matrix, $\boldsymbol{\Phi}$, which we refer to as dynamic heterogeneity; and the persistence of the common factor itself, which we refer to as the factor persistence. Our aim is to investigate how these two sources of persistence combine and get amplified in the process of aggregation.

Finally, a related issue of practical significance is the effect of estimation uncertainty on the above comparisons. To this end, we estimate disaggregated models using observations on individual micro units, y_{it}, as well as an aggregate model that only makes use of the aggregate observations, $\bar{y}_t$. We denote the estimated impulse responses of the combined aggregate shock on the aggregate variable by $\hat{g}_{\bar{\xi}}^{d}(s)$ when based on the disaggregate model, and by $\hat{g}_{\bar{\xi}}^{a}(s)$ when based on an aggregate autoregressive model fitted to $\bar{y}_t$. It is important to recall that, in general, the effects of macro and aggregated idiosyncratic shocks cannot be identified from the aggregate model.

The remainder of this section is organized as follows. The next sub-section outlines the Monte Carlo design. Section 32.9.2 describes the estimation of $g_{\bar{\xi}}^{d}(s)$ using aggregate and disaggregate data, and the last sub-section discusses the main findings.

32.9.1 Monte Carlo design

To allow for neighbourhood effects as well as an unobserved common factor we used the following data generating process (DGP)

$$y_{it} = \lambda_i y_{i,t-1} + \gamma_i f_t + \varepsilon_{it}, \text{ for } i = 1, \tag{32.71}$$

and

$$y_{it} = d_i y_{i-1,t-1} + \lambda_i y_{i,t-1} + \gamma_i f_t + \varepsilon_{it}, \text{ for } i = 2, 3, \ldots, N, \tag{32.72}$$

where each unit, except the first, has one left neighbour $(y_{i-1,t-1})$. The micro model given by (32.71)-(32.72) can be written conveniently in vector notations as

$$\mathbf{y}_t = \mathbf{\Phi}\mathbf{y}_{t-1} + \boldsymbol{\gamma} f_t + \boldsymbol{\varepsilon}_t, \tag{32.73}$$

where $\mathbf{y}_t = \left(y_{1t}, y_{2t}, \ldots, y_{Nt}\right)'$, $\boldsymbol{\gamma} = \left(\gamma_1, \gamma_2, \ldots, \gamma_N\right)'$, $\boldsymbol{\varepsilon}_t = (\varepsilon_{1t}, \varepsilon_{2t}, \ldots, \varepsilon_{Nt})'$, and

$$\mathbf{\Phi} = \begin{pmatrix} \lambda_1 & 0 & 0 & \cdots & 0 \\ d_2 & \lambda_2 & 0 & \cdots & 0 \\ 0 & d_3 & \lambda_3 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & d_N & \lambda_N \end{pmatrix}.$$

The autoregressive micro coefficients, λ_i, are generated as $\lambda_i \sim IIDU(0, \lambda_{\max})$, for $i = 1, 2, \ldots, N$, with $\lambda_{\max} = 0.9$ or 1. Recall that $\bar{y}_t$ will exhibit long memory features when $\lambda_{max} = 1$, but not when $\lambda_{\max} = 0.9$. The neighbourhood coefficients, d_i, are generated as $IIDU(0, 1 - \lambda_i)$, for $i = 2, 3, \ldots, N$, to ensure bounded variances as $N \to \infty$. Specifically, $\|\mathbf{\Phi}\|_{\infty} \leq \max_i \{|\lambda_i| + |d_i|\} < 1$, see Chudik and Pesaran (2011).

The idiosyncratic errors, $\boldsymbol{\varepsilon}_t$, are generated according to the following spatial autoregressive process,

$$\boldsymbol{\varepsilon}_t = \delta \mathbf{S}\boldsymbol{\varepsilon}_t + \boldsymbol{\varsigma}_t, 0 < \delta < 1,$$

where $\boldsymbol{\varsigma}_t = \left(\varsigma_{1t}, \varsigma_{2t}, \ldots, \varsigma_{Nt}\right)'$, $\boldsymbol{\varsigma}_t \sim IIDN\left(\mathbf{0}, \sigma_{\varsigma}^2 \mathbf{I}_N\right)$, and the $N \times N$ dimensional spatial weights matrix $\mathbf{S}$ is given by

$$\mathbf{S} = \begin{pmatrix} 0 & 1 & 0 & \cdots & 0 \\ \frac{1}{2} & 0 & \frac{1}{2} & \cdots & 0 \\ 0 & \frac{1}{2} & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 & 0 \end{pmatrix}.$$

To ensure that the idiosyncratic errors are weakly correlated, the spatial autoregressive parameter, δ, must lie in the range $[0, 1)$. We set $\delta = 0.4$. The variance σ_ς^2 is set equal to $N/(\boldsymbol{\tau}_N'\mathbf{R}\mathbf{R}'\boldsymbol{\tau}_N)$, where $\boldsymbol{\tau}_N = (1, 1, \ldots, 1)'$ and $\mathbf{R} = (\mathbf{I}_N - \delta\mathbf{S})^{-1}$, so that $Var(\bar{\varepsilon}_t) = N^{-1}$.

The common factor, f_t, is generated as

$$f_t = \psi f_{t-1} + v_t, v_t \sim IIDN\left(0, 1 - \psi^2\right), |\psi| < 1,$$

for $t = -49, -48, \ldots, 1, 2, \ldots, T$, with $f_{-50} = 0$. We consider three values for ψ, namely 0, 0.5 and 0.8. By construction, $Var\left(f_t\right) = 1$.

Finally, the factor loadings are generated as

$$\begin{aligned}\gamma_i &= \varkappa_i, \text{for } i = 1, 2, \ldots, [N^{\alpha_\gamma}],\\ \gamma_i &= 0, \text{for } i = [N^{\alpha_\gamma}] + 1, [N^{\alpha_\gamma}] + 2, \ldots, N,\end{aligned}$$

where $[N^{\alpha_\gamma}]$ denotes the integer part of N^{α_γ}, $0 < \alpha_\gamma \leq 1$ is the exponent of cross-section dependence of y_{it} due to the common factor and $\varkappa_i \sim IIDN\left(1, 0.5^2\right)$. The unobserved common factor therefore affects a fraction $[N^{\alpha_\gamma}]/N$ of the units, with this fraction tending to zero if $\alpha_\gamma < 1$. It is easily seen that $\bar{\gamma} = N^{-1}\sum_{i=1}^{N}\gamma_i = O(N^{\alpha_\gamma})$. We consider four values for $\alpha_\gamma \in \{0.25, 0.5, 0.75, 1\}$, representing different degrees of cross-section dependence due to the common factor. Note that for $\alpha_\gamma = 1$, we have $Plim_{N\to\infty}\bar{\gamma} = 1$, whereas $Plim_{N\to\infty}\bar{\gamma} = 0$ for $\alpha_\gamma < 1$. Note also that $\lim_{N\to\infty} NVar\left(\bar{\gamma}f_t\right) = 1$ for $\alpha_\gamma = 0.5$, in which case we would expect the macro shock and the aggregated idiosyncratic shock to be of equal importance for $g_{\bar{\xi}}^d(s)$.

32.9.2 Estimation of $g_{\bar{\xi}}(s)$ using aggregate and disaggregate data

The estimate of $g_{\bar{\xi}}(s)$ based on the aggregate data, which we denote by $\hat{g}_{\bar{\xi}}^a(s)$, is straightforward to compute and can be based on the following autoregression (intercepts are included in all regressions below, but not shown)

$$\bar{y}_t = \sum_{\ell=1}^{p_a}\pi_\ell\bar{y}_{t-\ell} + \zeta_{at}.$$

To estimate $g_{\bar{\xi}}(s)$ using disaggregated data is much more complicated and requires estimates of the micro coefficients. In terms of the micro parameters, using (32.65), we have

$$\begin{aligned}g_{\bar{\xi}}^d(s) &= E\left(\bar{y}_{w,t+s}\middle|\bar{\xi}_{wt} = \sigma_{\bar{\xi}}, \mathcal{I}_{t-1}\right) - E\left(\bar{y}_{w,t+s}\middle|\mathcal{I}_{t-1}\right)\\ &= \sum_{\ell=0}^{s}\boldsymbol{\Phi}^\ell[E\left(\mathbf{u}_{t+s-\ell}\middle|\bar{\xi}_{wt} = \sigma_{\bar{\xi}}, \mathcal{I}_{t-1}\right) - E\left(\mathbf{u}_{t+s-\ell}\middle|\mathcal{I}_{t-1}\right)].\end{aligned} \tag{32.74}$$

Following Chudik and Pesaran (2011), we first estimate the non-zero elements of $\boldsymbol{\Phi}$, namely λ_i and d_i, using the cross-section augmented least squares regressions,

$$y_{it} = \lambda_i y_{i,t-1} + d_i y_{i-1,t-1} + h_i\left(L, p_{hi}\right)\bar{y}_t + \zeta_{it}, \text{for } i = 2, 3, \ldots, N, \tag{32.75}$$

where $h_i\left(L, p_i\right) = \sum_{\ell=0}^{p_{hi}} h_{i\ell} L^{\ell}$, and p_{hi} is the lag-order. The equation for the first micro unit is the same except that it does not feature any neighbourhood effects.[9] These estimates are denoted by $\hat{\lambda}_i$ and $\hat{d}_i$, and an estimate of u_{it} is computed as

$$\hat{u}_{it} = y_{it} - \hat{\lambda}_i y_{i,t-1}, \text{for } i = 1, \text{and} \tag{32.76}$$

$$\hat{u}_{it} = y_{it} - \hat{\lambda}_i y_{i,t-1} - \hat{d}_i y_{i-1,t-1}, \text{for } i = 2, 3, \ldots, N. \tag{32.77}$$

To obtain an estimate of $\xi_{it} = \gamma_i v_t + \varepsilon_{it}$, we fit the following conditional models

$$\hat{u}_{it} = r_i \bar{\hat{u}}_t + \epsilon_{it}, \text{for } i = 1, 2, \ldots, N, \tag{32.78}$$

where $\bar{\hat{u}}_t = N^{-1} \sum_{i=1}^{N} \hat{u}_{it}$; and the following marginal model,

$$\bar{\hat{u}}_t = \psi_{\bar{u}} \bar{\hat{u}}_{t-1} + \vartheta_t. \tag{32.79}$$

An estimate of ξ_{it} is computed as $\hat{\xi}_{it} = \hat{u}_{it} - \hat{r}_i \hat{\psi}_{\bar{u}} \bar{\hat{u}}_{t-1}$, for $i = 1, 2, \ldots, N$, where $\hat{r}_i$ and $\hat{\psi}_{\bar{u}}$ are the estimates of r_i and $\psi_{\bar{u}}$, respectively. When $\alpha_\gamma = 1$, $\hat{\psi}_{\bar{u}}$ is a consistent estimator (as $N, T \underset{j}{\to} \infty$) of the autoregressive parameter ψ that characterizes the persistence of the factor, $\hat{r}_i$ is a consistent estimator of the scaled factor loading, $\gamma_i/\bar{\gamma}$, and the regression residuals from (32.79), denoted by $\hat{\vartheta}_t$, are consistent estimates of the macro shock, v_t. But, when $\boldsymbol{\gamma} = \mathbf{0}$, $\bar{u}_t = N^{-1} \sum_{i=1}^{N} u_{it}$ is serially uncorrelated and $\hat{\psi}_{\bar{u}} \to_p 0$ as $N, T \overset{j}{\to} \infty$.

To compute the remaining terms in (32.74), we note that for $s = \ell = 0$, $E\left(\mathbf{u}_t \middle| \bar{\xi}_{wt} = \hat{\sigma}_{\bar{\xi}}, \mathcal{I}_{t-1}\right) - E\left(\mathbf{u}_t \middle| \mathcal{I}_{t-1}\right) = E\left(\boldsymbol{\xi}_t \middle| \bar{\xi}_{wt} = \hat{\sigma}_{\bar{\xi}}, \mathcal{I}_{t-1}\right)$ can be consistently estimated by $\hat{\boldsymbol{\Sigma}}_\xi \mathbf{w} / \hat{\sigma}_{\bar{\xi}}$, where $\hat{\sigma}_{\bar{\xi}} = \left(\mathbf{w}' \hat{\boldsymbol{\Sigma}}_\xi \mathbf{w}\right)^{1/2}$, $\hat{\boldsymbol{\Sigma}}_\xi = T^{-1} \sum_{t=p_h+1}^{T} \hat{\boldsymbol{\xi}}_t \hat{\boldsymbol{\xi}}_t'$, $\hat{\boldsymbol{\xi}}_t = \left(\hat{\xi}_{1t}, \hat{\xi}_{2t}, \ldots, \hat{\xi}_{Nt}\right)'$, and $p_h = \max_i p_{hi}$. Similarly, for $s - \ell > 0$, $E\left(\mathbf{u}_{t+s-\ell} \middle| \bar{\xi}_{wt} = \hat{\sigma}_{\bar{\xi}}, \mathcal{I}_{t-1}\right) - E\left(\mathbf{u}_{t+s-\ell} \middle| \mathcal{I}_{t-1}\right)$ can be consistently estimated by $\hat{\psi}_u^{s-\ell} \hat{\sigma}_\vartheta^2 \hat{\mathbf{r}} / \left(\mathbf{w}' \hat{\boldsymbol{\Sigma}}_\xi \mathbf{w}\right)^{1/2}$, where $\hat{\mathbf{r}} = \left(\hat{r}_1, \hat{r}_2, \ldots, \hat{r}_N\right)'$, and $\hat{\sigma}_\vartheta^2 = T^{-1} \sum_{t=p_h+1}^{T} \hat{\vartheta}_t^2$. All lag-orders are selected by AIC with the maximum lag-order set to $[T^{1/2}]$.

32.9.3 Monte Carlo results

Figure 32.1 plots the relative contributions of macro and aggregated idiosyncratic shocks to the *GIRF* of the aggregate variable for the sample of $N = 200$ micro units (see (32.70)). There are four panels, corresponding to different choices of cross-section exponents, α_γ, with the plots on the left of each panel relating to $\lambda_{\max} = 0.9$, and those on the right to $\lambda_{\max} = 1$. As expected, when $\alpha_\gamma = 0.25$ the macro shock is not 'strong enough' and the aggregated idiosyncratic shock dominates. When $\alpha_\gamma = 0.5$ (Panel B), the macro shock is as equally important as the aggregated idiosyncratic shock. As α_γ is increased to 0.75 (Panel C), the aggregated idiosyncratic shock starts to play only a minor role; and when $\alpha_\gamma = 1$ (Panel D), the macro shock

[9] Chudik and Pesaran (2011) show that if $\|\boldsymbol{\Phi}\|_\infty < 1$, these augmented least squares estimates of the micro lagged coefficients are consistent and asymptotically normal when $\alpha_\gamma = 1$ (as $N, T \overset{j}{\to} \infty$), and also when there is no factor, i.e., $\boldsymbol{\gamma} = \mathbf{0}$.

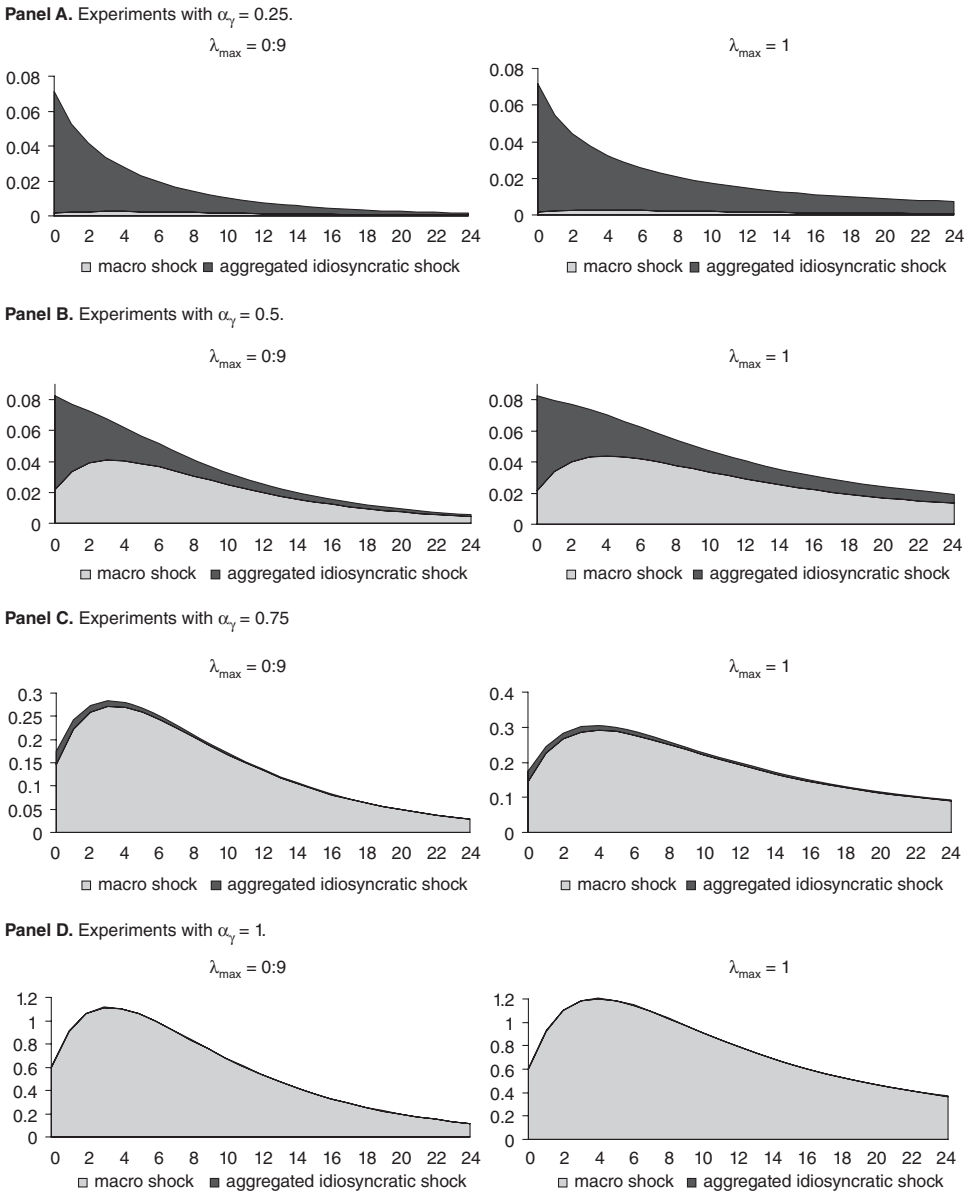


Figure 32.1 Contribution of the macro and aggregated idiosyncratic shocks to *GIRF* of one unit (1 s.e.) combined aggregate shock on the aggregate variable; $N = 200$.

completely dominates the aggregate relationship. Similar results are obtained for N as small as 25 (not reported). Whether the support of the distribution of the eigenvalues λ_i covers unity or not does not seem to make any difference to the relative importance of the macro shock. Table 32.1 reports the weights ω_ν and $\omega_{\bar{\varepsilon}}$ for different values of N, and complements what can be seen from the plots in Figure 32.1. Note that these weights do not depend on the choice of $\lambda_{\max}$ and,

Table 32.1 Weights ω_v and $\omega_{\bar{\varepsilon}}$ in experiments with $\psi = 0.5$

	$\alpha_\gamma = 0.25$		$\alpha_\gamma = 0.5$		$\alpha_\gamma = 0.75$		$\alpha_\gamma = 1$	
N	ω_v	$\omega_{\bar{\varepsilon}}$	ω_v	$\omega_{\bar{\varepsilon}}$	ω_v	$\omega_{\bar{\varepsilon}}$	ω_v	$\omega_{\bar{\varepsilon}}$
25	0.33	0.93	0.63	0.76	0.88	0.47	0.97	0.23
50	0.24	0.96	0.63	0.76	0.90	0.42	0.99	0.16
100	0.25	0.96	0.64	0.76	0.93	0.35	0.99	0.12
200	0.18	0.98	0.64	0.76	0.95	0.30	1.00	0.08

Notes: Weights $\omega_v = \sigma_v/\sigma_{\bar{\xi}}$ and $\omega_{\bar{\varepsilon}} = \sigma_{\bar{\varepsilon}}/\sigma_{\bar{\xi}}$ do not depend on the parameter $\lambda_{\max}$.

by construction $\omega_v^2 + \omega_{\bar{\varepsilon}}^2 = 1$. We see in Table 32.1 that for $\alpha_\gamma = 1$, ω_v is very close to unity for all values of N considered, and $g_{\bar{\xi}}^d(s)$ is mainly explained by the macro shock, regardless of the shape of the impulse response functions.

Next we examine how dynamic heterogeneity and factor persistence affect the persistence of the aggregate variable. Figure 32.2 plots the *GIRF* of the combined aggregate shock on the aggregate variable, $g_{\bar{\varepsilon}}^d(s)$, for $N = 200$ and different values of $\lambda_{\max}$ and ψ, that control the dynamic heterogeneity and the persistence of the factor, respectively. The plot on the left of the figure relates to $\lambda_{\max} = 0.9$ and the one on the right to $\lambda_{\max} = 1$. It is interesting that $g_{\bar{\xi}}^d(s)$ looks very different when we allow for serial correlation in the common factor. Even for a moderate value of ψ, say 0.5, the factor contributes significantly to the overall persistence of the aggregate. By contrast, the effects of long memory on persistence (comparing the plots on the left and the right of the panels in Figure 32.2), are rather modest. Common factor persistence tends to become accentuated by the individual-specific dynamics.

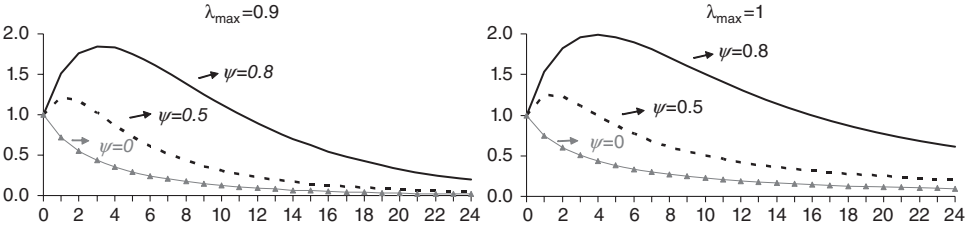


Figure 32.2 *GIRF*s of one unit combined aggregate shock on the aggregate variable, $g_{\bar{\xi}}(s)$, for different persistence of common factor, $\psi = 0, 0.5$, and 0.8.

Finally, we consider the estimates of $g_{\bar{\xi}}(s)$ based on the disaggregate and the aggregate models, namely $\hat{g}_{\bar{\xi}}^d(s)$ and $\hat{g}_{\bar{\xi}}^a(s)$. Table 32.2 reports the root mean square error ($RMSE \times 100$) of these estimates averaged over horizons $s = 0$ to 12 and $s = 13$ to 24, for the parameter values $\alpha_\gamma = 0.5$, 1, and $\psi = 0.5$, using 2,000 Monte Carlo replications.[10] The estimator based on the disaggregate model, $\hat{g}_{\bar{\xi}}^d(s)$, performs much better (in some cases by twice as much) than its counterpart based on the aggregate model. The difference between the two estimators is slightly smaller when $\alpha_\gamma = 0.5$. As to be expected, an increase in the time dimension considerably improves

[10] The bias statistics are not reported due to space constraints.

Table 32.2 *RMSE* ($\times 100$) of estimating *GIRF* of one unit (1 s.e.) combined aggregate shock on the aggregate variable, averaged over horizons $s = 0$ to 12 and $s = 13$ to 24

	Estimates averaged over horizons from $s = 0$ to 12				Estimates averaged over horizons from $s = 13$ to 24			
N\T	**100**		**200**		**100**		**200**	
	$\hat{g}_{\bar{\xi}}^{a}$	$\hat{g}_{\bar{\xi}}^{d}$	$\hat{g}_{\bar{\xi}}^{a}$	$\hat{g}_{\bar{\xi}}^{d}$	$\hat{g}_{\bar{\xi}}^{a}$	$\hat{g}_{\bar{\xi}}^{d}$	$\hat{g}_{\bar{\xi}}^{a}$	$\hat{g}_{\bar{\xi}}^{d}$
	Experiments with $\alpha_\gamma = 1$							
	(a) $\lambda_{\max} = 0.9$							
50	20.18	12.81	13.50	8.70	10.39	4.38	8.22	3.20
100	20.00	12.41	13.49	8.32	10.76	3.89	8.39	2.76
200	20.45	12.39	13.61	8.30	10.27	3.61	8.17	2.62
	(b) $\lambda_{\max} = 1$							
50	24.13	15.23	15.95	10.41	21.15	12.55	16.34	8.66
100	23.92	14.76	16.44	9.96	20.36	11.37	16.96	7.34
200	24.34	14.65	15.99	9.70	20.75	10.58	16.36	6.56
	Experiments with $\alpha_\gamma = 0.5$							
	(c) $\lambda_{\max} = 0.9$							
50	3.24	2.21	2.31	1.57	1.87	0.96	1.48	0.72
100	2.24	1.50	1.62	1.06	1.24	0.59	1.02	0.45
200	1.55	0.99	1.11	0.72	0.88	0.36	0.69	0.28
	(d) $\lambda_{\max} = 1$							
50	3.66	2.86	2.84	1.99	3.38	2.86	2.64	2.04
100	2.71	1.96	1.96	1.30	2.54	1.77	1.90	1.25
200	1.78	1.27	1.36	0.88	1.56	1.09	1.29	0.78

Notes: Experiments with $\psi = 0.5$.

the precision of the estimates. Also, $\hat{g}_{\bar{\xi}}^{d}(s)$ improves with an increase in N, whereas the *RMSE* of $\hat{g}_{\bar{\xi}}^{a}(s)$ is little affected by increasing N when $\alpha_\gamma = 1$, but improves with N when $\alpha_\gamma = 0.5$.

32.10 Application I: aggregation of life-cycle consumption decision rules under habit formation

In the life-cycle literature, habit formation has been emphasized as a potentially important factor that may help resolve a number of empirical puzzles. Deaton (1987), among others, argues that habit formation could help explain 'excess smoothness' and 'excess sensitivity' of aggregate consumption expenditures. 'Excess smoothness' refers to the situation where, contrary to the prediction of the permanent income hypothesis, changes in aggregate consumption do not vary closely with unanticipated changes in labour income. 'Excess sensitivity' refers to the situa-

tion where changes in aggregate consumption respond to anticipated changes in labour income, whilst the theory predicts otherwise. For a review of the empirical literature on excess smoothness and excess sensitivity see, for example, Muellbauer and Lattimore (1995). Carroll and Weil (1994) suggest that the reverse causality between growth and saving often observed in aggregate data could be due to the neglect of habit formation in consumption behaviour. Fuhrer (2000) maintains that the dynamics of aggregate consumption decisions as represented by autocovariance functions can be much better understood using a model with habit formation than using a model with standard time-separable preferences. A problem common to all these studies using representative agent frameworks is that the coefficient of habit formation needed to reconcile the model with the data is typically deemed implausibly high. In this section we consider the aggregate implications of allowing for heterogeneity in habit formation coefficients across individuals and investigate the extent to which empirical puzzles observed in aggregate consumption data are due to the aggregation problem. Using stochastic simulations, Pesaran (2003) shows that the estimates of the habit persistence coefficient are likely to be seriously biased downward if they are based on analogue aggregate consumption functions, which could partly explain the excess smoothness and excess sensitivity puzzles in terms of neglected heterogeneity.[11]

Consider an economy composed of a large number of consumers, where each consumer indexed by i, $i = 1, 2, \ldots, N$, at the beginning of period t is endowed with an initial level of financial wealth, $a_{i,t-1}$. His/her labour income over the period $t-1$ to t, y_{it}, is generated according to the following geometric random walk model

$$\log y_{it} = \alpha_i + \mu t + \sum_{s=1}^{t} v_s + \xi_{it}, \tag{32.80}$$

where α_i is the time-invariant individual-specific component, μ is an economy-wide drift term, v_t is the economy-wide random component, and ξ_{it} is the residual random component. The random components α_i, v_t, and ξ_{it} are assumed to be mutually independent, $i = 1, 2, \ldots, N$; $t = 1, 2, \ldots$, and distributed identically as normal variates with zero means and constant variances

$$\alpha_i \sim IIDN\left(\alpha, \sigma_\alpha^2\right), \quad v_t \sim IIDN\left(0, \sigma_v^2\right), \quad \text{and} \quad \xi_{it} \sim IIDN\left(0, \sigma_\xi^2\right). \tag{32.81}$$

This formulation allows labour incomes at the individual and the economy-wide levels to exhibit geometric growth and at the same time yields a plausible steady state size distribution for labour incomes. Each individual solves the following inter-temporal optimization problem

$$\max_{\{c_{i,t+s}\}_{s=0}^{\infty}} E\left[\sum_{s=0}^{\infty} \delta^s u(c_{i,t+s}, c_{i,t+s-1}) | \Phi_{it}\right] \tag{32.82}$$

subject to the period-by-period budget constraints,

$$a_{i,t+s} = (1+r)a_{i,t+s-1} + y_{i,t+s} - c_{i,t+s}, \qquad s = 0, 1, \ldots \tag{32.83}$$

[11] In a different attempt at resolving the excess smoothness and excess sensitivity puzzles, Binder and Pesaran (2002) argue that social interactions when combined with habit formation can also help.

the transversality condition,

$$\lim_{s\to\infty}(1+r)^{-s}E(a_{i,t+s}|\Phi_{it})=0, \tag{32.84}$$

and given initial consumption levels, $c_{i,t-1}$, as well as initial wealth levels, $a_{i,t-1}$, for all i. In equations (32.82)–(32.84) $u_{it}=u(c_{it},c_{i,t-1})$ represents individual i^{th} current-period utility function for period t, $\delta=1/(1+\rho)$ represents a constant discount factor, r is the constant real rate of interest, and $E(\cdot|\Phi_{it})$ denotes the mathematical conditional expectations operator with respect to the information set available to the individual at time t

$$\Phi_{it}=\{c_{it},c_{i,t-1},\ldots;y_{it},y_{i,t-1},\ldots;a_{it},a_{i,t-1},\ldots\}. \tag{32.85}$$

Given the focus of our analysis on aggregation of linear models, we consider the case where the current period utility function is quadratic, namely

$$u_{it}=\frac{-1}{2}(c_{it}-\lambda_i c_{i,t-1}-\bar{c}_i)^2,\ 0<\lambda_i<1, \tag{32.86}$$

λ_i is the habit formation coefficient, and $\bar{c}_i$ is the saturation coefficient. For simplicity we also assume that $\rho=r$, so that individuals are time-indifferent. For each individual the consumption decision rule for time period t that solves the above inter-temporal optimization problem is given by

$$\Delta c_{it}=\lambda_i\Delta c_{i,t-1}+\beta_i y_{it}+\gamma_i\exp(\alpha_i+\frac{1}{2}\sigma_\xi^2)\left[\tilde{y}_t-(1+r)\tilde{y}_{t-1}\right], \tag{32.87}$$

where $\tilde{y}_t$ is the economy-wide component of labour income,

$$\tilde{y}_t=\exp(\mu t+\sum_{s=1}^{t}v_s), \tag{32.88}$$

$$\beta_i=\frac{r(1+r-\lambda_i)}{(1+r)^2}, \tag{32.89}$$

$$\gamma_i=\frac{r(1+r-\lambda_i)(1+g)}{(1+r)^2(r-g)}, \tag{32.90}$$

and g is the rate of growth of labour income

$$g=\exp(\mu+\frac{1}{2}\sigma_v^2)-1. \tag{32.91}$$

Notice that the labour income of individual i can be decomposed as

$$y_{it}=\tilde{y}_t\exp(\alpha_i+\xi_{it}). \tag{32.92}$$

Defining economy-wide average labour income as $\bar{y}_t = (1/N)\sum_{i=1}^{N} y_{it}$, then under (32.81) as $N \to \infty$ we have

$$\bar{y}_t \xrightarrow{p} \tilde{y}_t \exp(\alpha + \frac{\sigma_\alpha^2}{2} + \frac{\sigma_\xi^2}{2}). \tag{32.93}$$

Further, aggregating the budget constraints, (32.83), yields

$$A_{t+s} = (1+r)A_{t+s-1} + \bar{y}_{t+s} - \bar{c}_{t+s}, \qquad s = 0, 1, \ldots,$$

where $A_t = (1/N)\sum_{i=1}^{N} a_{it}$, and $\bar{c}_t = (1/N)\sum_{i=1}^{N} c_{it}$.

There will be an aggregation problem only when the habit formation coefficients, λ_i, differ across individuals. In the case where $\lambda_i = \lambda$ for all i we have

$$\Delta c_{it} = \lambda \Delta c_{i,t-1} + \beta y_{it} + \gamma \exp(\alpha_i + \frac{1}{2}\sigma_\xi^2)\left[\tilde{y}_t - (1+r)\tilde{y}_{t-1}\right], \tag{32.94}$$

$$\beta = \frac{r(1+r-\lambda)}{(1+r)^2}, \tag{32.95}$$

$$\gamma = \frac{r(1+r-\lambda)(1+g)}{(1+r)^2(r-g)}, \tag{32.96}$$

and using (32.93) and noting that $\frac{1}{N}\sum_{i=1}^{N} \exp(\alpha_i) \xrightarrow{p} \exp(\alpha + \frac{1}{2}\sigma_\alpha^2)$ yields the perfect aggregate model

$$\Delta \bar{c}_t = \lambda \Delta \bar{c}_{t-1} + \beta \bar{y}_t + \frac{r(1+r-\lambda)(1+g)}{(1+r)^2(r-g)}\left[\bar{y}_t - (1+r)\bar{y}_{t-1}\right], \tag{32.97}$$

or equivalently

$$\Delta \bar{c}_t = \lambda \Delta \bar{c}_{t-1} + \frac{r(1+r-\lambda)}{(1+r)(r-g)}\left[\bar{y}_t - (1+g)\bar{y}_{t-1}\right]. \tag{32.98}$$

This specification is perfect in the sense that it yields aggregate forecasts of $\Delta\bar{c}_t$ (or $\bar{c}_t$) based only on aggregate time series observations $\Omega_t = \left\{\bar{c}_{t-1}, \bar{c}_{t-2}, \ldots; \bar{y}_t, \bar{y}_{t-1}, \ldots\right\}$ that have zero mean-squared errors and are indistinguishable from forecasts of aggregate consumption based on the individual-specific decision rules, (32.94) (using individual-specific consumption and labour income data).

Consider now the empirically more interesting case where λ_i's are allowed to vary across consumers. Since $|\lambda_i| < 1$ for all i, then

$$\Delta c_{it} = \beta_i \sum_{j=0}^{\infty} \lambda_i^j y_{i,t-j} + \gamma_i \exp(\alpha_i + \frac{1}{2}\sigma_\xi^2) \sum_{j=0}^{\infty} \lambda_i^j (\tilde{y}_{t-j} - (1+r)\tilde{y}_{t-j-1}). \tag{32.99}$$

Aggregating across i, we have

$$\Delta\bar{c}_t = \frac{1}{N}\sum_{j=0}^{\infty}\sum_{i=1}^{N}\beta_i\lambda_i^j y_{i,t-j} + \frac{1}{N}\sum_{j=0}^{\infty}\sum_{i=1}^{N}\gamma_i\lambda_i^j \exp(\alpha_i + \frac{1}{2}\sigma_\xi^2)\left[\tilde{y}_{t-j} - (1+r)\tilde{y}_{t-j-1}\right]. \tag{32.100}$$

Assuming that λ_i's are *IID* draws from a distribution with finite moments of all orders defined on the unit interval, and taking conditional expectations of both sides of (32.100) with respect to $\boldsymbol{\Upsilon}_t = \cup_{i=1}^N \boldsymbol{\Upsilon}_{it}$, where $\boldsymbol{\Upsilon}_{it} = \{y_{it}, y_{i,t-1}, \ldots\} \cup \{\bar{y}_t, \bar{y}_{t-1}, \ldots\}$, we have

$$E(\Delta\bar{c}_t|\boldsymbol{\Upsilon}_t) = \frac{1}{N}\sum_{j=0}^{\infty}\sum_{i=1}^{N}E\left(\beta_i\lambda_i^j|\boldsymbol{\Upsilon}_t\right)y_{i,t-j} + \frac{1}{N}\sum_{j=0}^{\infty}\sum_{i=1}^{N}E\left(\gamma_i\lambda_i^j|\boldsymbol{\Upsilon}_t\right)\exp(\alpha_i + \frac{1}{2}\sigma_\xi^2)\left[\tilde{y}_{t-j} - (1+r)\tilde{y}_{t-j-1}\right].$$

Since $E\left(\beta_i\lambda_i^j|\boldsymbol{\Upsilon}_t\right) = E\left(\beta_i\lambda_i^j\right) = a_j$ and $E\left(\gamma_i\lambda_i^j|\boldsymbol{\Upsilon}_t\right) = E\left(\gamma_i\lambda_i^j\right) = b_j$, for all i, then we have[12]

$$E(\Delta\bar{c}_t|\boldsymbol{\Upsilon}_t) = \sum_{j=0}^{\infty}a_j\bar{y}_{t-j} + \left(\frac{1}{N}\sum_{i=1}^{N}\exp(\alpha_i + \frac{1}{2}\sigma_\xi^2)\right)\sum_{j=0}^{\infty}b_j\left[\tilde{y}_{t-j} - (1+r)\tilde{y}_{t-j-1}\right]. \tag{32.101}$$

But, as noted earlier, for N sufficiently large

$$\frac{1}{N}\sum_{i=1}^{N}\exp(\alpha_i + \frac{1}{2}\sigma_\xi^2) \xrightarrow{p} \exp(\alpha + \frac{\sigma_\alpha^2}{2} + \frac{\sigma_\xi^2}{2}),$$

and in view of (32.93) we have

$$E(\Delta\bar{c}_t|\boldsymbol{\Upsilon}_t) = \sum_{j=0}^{\infty}a_j\bar{y}_{t-j} + \sum_{j=0}^{\infty}b_j\left[\bar{y}_{t-j} - (1+r)\bar{y}_{t-j-1}\right].$$

Also using (32.89) it is easily seen that

$$a_j = E\left(\beta_i\lambda_i^j\right) = E\left[\frac{r(1+r-\lambda_i)\lambda_i^j}{(1+r)^2}\right] = \frac{r}{1+r}m_j - \frac{r}{(1+r)^2}m_{j+1}, \tag{32.102}$$

[12] Recall that $\frac{1}{N}\sum_{i=1}^{N}\exp(\alpha_i) \xrightarrow{p} \exp(\alpha + \frac{1}{2}\sigma_\alpha^2)$.

and $m_j = E(\lambda^j)$ is the jth-order moment of λ_i. Similarly, using (32.90) and (32.102) we have

$$b_j = E\left(\gamma_i \lambda_i^j\right) = \frac{(1+g)}{(r-g)} a_j. \tag{32.103}$$

Now taking conditional expectations of (32.101) with respect to the aggregate information set $\Omega_t = \left\{\bar{y}_t, \bar{y}_{t-1}, \ldots; \bar{c}_{t-1}, \bar{c}_{t-2}, \ldots\right\}$

$$\begin{aligned} &E(\Delta \bar{c}_t | \Omega_t) \\ &= \sum_{j=0}^{\infty} a_j \bar{y}_{t-j} + \frac{1+g}{r-g} \sum_{j=0}^{\infty} a_j \left[\bar{y}_{t-j} - (1+r)\bar{y}_{t-j-1}\right] \\ &= \left(\frac{1+r}{r-g}\right) \left\{ a_0 \bar{y}_t + \sum_{j=1}^{\infty} [a_j - (1+g)a_{j-1}] \bar{y}_{t-j} \right\}. \end{aligned}$$

The optimal aggregate consumption function can therefore be written as

$$\Delta \bar{c}_t = \left(\frac{1+r}{r-g}\right) \sum_{j=0}^{\infty} a_j \left[\bar{y}_{t-j} - (1+g)\bar{y}_{t-j-1}\right] + \varepsilon_t, \tag{32.104}$$

where ε_t is the aggregation error and by construction satisfies the orthogonality condition

$$E(\varepsilon_t \mid \Omega_t) = 0.$$

The aggregation errors are serially uncorrelated with zero means, but in general are not homoskedastic. The above optimal aggregate function is directly comparable to the aggregate model, (32.98), obtained under homogeneous habit formation coefficients. It is easily seen that (32.104) reduces to (32.98) if $\lambda_i = \lambda$ for all i. The aggregation errors, ε_t's, also vanish if and only if $\lambda_i = \lambda$. Finally, unless the habit formation coefficients are homogeneous, the optimal aggregate model cannot be written as a finite-order *ARDL* model in $\Delta \bar{c}_t$ and $\bar{y}_t - (1+g)\bar{y}_{t-1}$. See Pesaran (2003) for an illustrative numerical result on the extent of the aggregation bias.

32.11 Application II: inflation persistence

Persistence of aggregate inflation and its sources have attracted a great deal of attention in the literature. Prices at the micro level are known to be relatively flexible, whereas at the aggregate level the overall rate of inflation seems to be quite persistent. Using individual category price series, Altissimo et al. (2009) conclude that 'the aggregation mechanism explains a significant amount of aggregate inflation persistence.' (p. 231). Pesaran and Chudik (2014) investigate the robustness of this conclusion by estimating a factor-augmented high dimensional *VAR* model in disaggregate inflation series, where the relative contribution of aggregation and common factor persistence is evaluated. The analysis is based on the same data set as that used by Altissimo et al. (2009), so that the results can be compared more readily. It is found that the persistence

due to dynamic heterogeneity alone does not explain the persistence of the aggregate inflation, rather it is the combination of factor persistence and dynamic heterogeneity that is responsible for the high persistence of aggregate inflation as compared to the persistence of the underlying individual inflation series.

32.11.1 Data

The inflation series for the ith price category is computed as $y_{it} = 400 \times \left[\ln\left(q_{it}\right) - \ln\left(q_{i,t-1}\right)\right]$, where q_{it} is the seasonally adjusted consumer price index of unit i at time t.[13] Units are individual categories of the consumer price index (e.g. bread, wine, medical services, ...) and the time dimension is quarterly covering the period 1985Q1 to 2004Q2; altogether 78 observations per price category. There are 85 categories in Germany, 145 in France, and 168 in Italy. The aggregate inflation measure is computed as $\bar{y}_{wt} = \sum_{i=1}^{N} w_i y_{it}$, where N is the number of price categories and w_i is the weight of the ith category in the consumer price index. Pesaran and Chudik (2014) conduct their empirical analysis for each of the three countries separately.

32.11.2 Micro model of consumer prices

Following Chudik and Pesaran (2011), the possibility that there are unobserved factors or neighbourhood effects in the micro relations are investigated. Individual items are categorized into a small sets of products that are close substitutes. For example, spirits, wine and beer are assumed to be 'neighbours'. A complete list of 'neighbours' for Germany is provided in Pesaran and Chudik (2014). Let $\mathbb{C}_i$ be the index set defining the neighbours of unit i, and consider the following local averages

$$y_{it}^{\diamond} = \frac{1}{|\mathbb{C}_i|} \sum_{j \in \mathbb{C}_i} y_{jt} = \mathbf{s}_i' \mathbf{y}_t, \; i = 1, 2, \ldots, N,$$

where $|\mathbb{C}_i|$ is the number of neighbours of unit i, assumed to be small and fixed as $N \to \infty$, $\mathbf{s}_i$ is the corresponding $N \times 1$ sparse weights vector with $|\mathbb{C}_i|$ nonzero elements. $y_{it}^{\diamond}$ represents the local average of unit i. No unit is assumed to be dominant in the sense discussed by Chudik and Pesaran (2011).

Following Pesaran (2006) and Chudik and Pesaran (2015a), economy wide average, $\bar{y}_t = N^{-1} \sum_{j=1}^{N} y_{jt}$, and the three sectoral averages

$$\bar{y}_{kt} = \frac{1}{|\mathbb{Q}_k|} \sum_{j \in \mathbb{Q}_k} y_{jt} = \mathbf{w}_k' \mathbf{y}_t, \text{ for } k \in \{f, g, s\},$$

are used in estimation, where $\mathbb{Q}_k$ for $k = \{f, g, s\}$ defines the set of units belonging to the food and beverages sector (f), the goods sector (g), and the services sector (s). $|\mathbb{Q}_k|$ is the number of units in sector k, and $\mathbf{w}_k$ is the corresponding vector of sectoral weights. The following cross-section augmented regressions are estimated by least squares for the price category i belonging to sector k (intercepts are included but not shown)[14]

[13] Descriptive statistics of the individual price categories are provided in Altissimo et al. (2009, Table 2).

[14] The estimates are dynamic CCE discussed in Section 29.5.3.

$$y_{it} = \sum_{\ell=1}^{p_{i\phi}} \phi_{ii\ell} y_{i,t-\ell} + \sum_{\ell=1}^{p_{id}} d_{i\ell} y^{\diamond}_{i,t-\ell} + \sum_{\ell=0}^{p_{ih}} h_{i\ell} \bar{y}_{t-\ell} + \sum_{\ell=0}^{p_{ik}} h_{ki\ell} \bar{y}_{k,t-\ell} + \zeta_{it}, \text{ for } i \in \mathbb{Q}_k \text{ and } k \in \{f, g, s\}. \tag{32.105}$$

The same equations are also estimated for the energy price category, but without sectoral averages. The estimates are dynamic CCE discussed in Section 29.5.3. Impulse response function of the combined aggregate shock on the aggregate variable in a disaggregate model is computed in the same way as in Section 32.9. The lag-orders for the individual price equations are chosen by AIC with the maximum lag-order set to 2. In line with the theoretical derivations, a higher maximum lag-order is selected when estimating the aggregate inflation equations.

32.11.3 Estimation results

Table 32.3 summarizes the statistical significance of the various coefficients in the price equations, (32.105), for Germany, France, and Italy. The parameters are grouped into own lagged effects ($\phi_{ii\ell}$), lagged neighbourhood effects ($d_{i\ell}$), country effects ($h_{i\ell}$), and sectoral effects ($h_{ki\ell}$, for $k = f, g, s$). All four types of effects are statistically important, although own lagged effects, perhaps not surprisingly, are more important statistically as compared to the other effects. At the 5 per cent significance level, own lagged effects are significant in 90 cases out of 112 in Germany, 111 cases out of 169 in France, and 158 out of 209 cases in Italy. Local and cross-section averages are statistically significant in about 12 and 25 per cent of cases. These results suggest that micro relations that ignore common factors and the neighbourhood effects are most likely misspecified. Idiosyncratic shocks are likely to dominate the micro relations, which could explain

Table 32.3 Summary statistics for individual price relations for Germany, France, and Italy (equation (32.105))

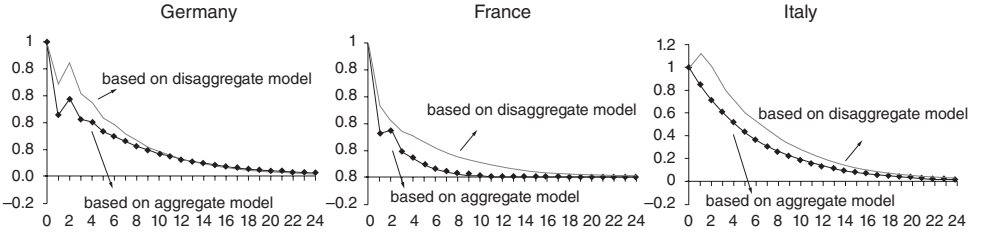
	No. of estimated coef.	No. of significant coef. (*at the 5% nominal level*)	Share
	Results for Germany		
Own lagged effects	112	90	80.4%
Lagged neighbourhood effects	66	16	24.2%
Sectoral effects	182	34	18.7%
Country effects	190	33	17.4%
	Results for France		
Own lagged effects	169	111	65.7%
Lagged neighbourhood effects	166	23	13.9%
Sectoral effects	302	57	18.9%
Country effects	314	38	12.1%
	Results for Italy		
Own lagged effects	209	158	75.6%
Lagged neighbourhood effects	173	38	22.0%
Sectoral effects	335	54	16.1%
Country effects	345	73	21.2%

the lower rejection rate for the cross-section averages, compared to the own lagged coefficients. The fit is relatively high in most cases. The average $\overline{R}^2$ is 56 per cent in Germany, 48 per cent in France, and 51 per cent in Italy (median values are 61 per cent, 52 per cent, and 54 per cent, respectively).

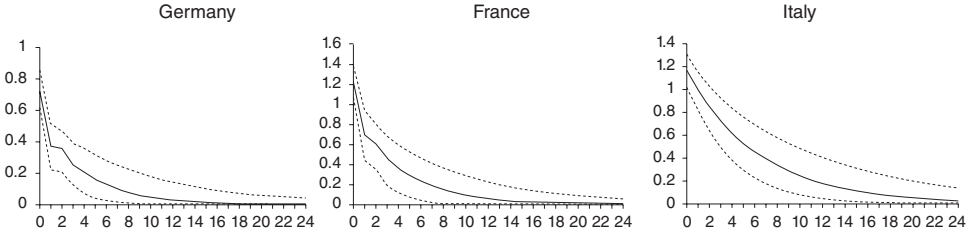
32.11.4 Sources of aggregate inflation persistence

For each of the three countries, Pesaran and Chudik (2014) compute and report the *GIRF* of a unit combined aggregate shock on the aggregate variable, using aggregate and disaggregate models, as explained in Section 32.8. (see Figure 32.3). They also provide 90 per cent bootstrap confidence bounds together with the bootstrap means. These impulse responses are quite persistent. The estimates based on the disaggregate model show a higher degree of persistence in the case of France and Italy.

Panel A. Point estimates, y-axis shows units of the shock.



Panel B. Bootstrap means and 90% confidence bounds based on aggregate model; y-axis shows the estimated size of the shock.



Panel C. Bootstrap means and 90% confidence bounds based on disaggregate model; y-axis shows the estimated size of the shock.

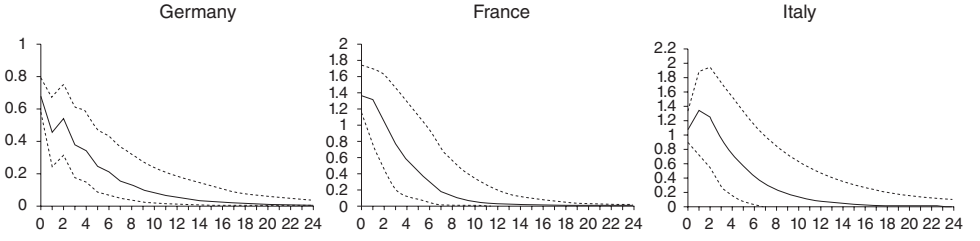


Figure 32.3 *GIRF*s of one unit combined aggregate shock on the aggregate variable.

Using the estimates of micro lagged coefficients in (33.36), for $i = 1, 2, \ldots, N$, Pesaran and Chudik (2014) compute eigenvalues of the companion matrix corresponding to the *VAR* polynomial matrix $\hat{\mathbf{\Phi}}(L)$,

$$\hat{\mathbf{\Phi}}(L) = \begin{pmatrix} \hat{\phi}_{11}(L) & 0 & \cdots & 0 \\ 0 & \hat{\phi}_{22}(L) & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \hat{\phi}_{NN}(L) \end{pmatrix} + \begin{pmatrix} \hat{d}_1(L)\mathbf{s}_1' \\ \hat{d}_2(L)\mathbf{s}_2' \\ \vdots \\ \hat{d}_N(L)\mathbf{s}_N' \end{pmatrix},$$

where $\hat{\phi}_{ii}(L) = \sum_{\ell=1}^{p_{i\phi}} \phi_{ii\ell} L^{\ell-1}$, $\hat{d}_i(L) = \sum_{\ell=1}^{p_{id}} d_{i\ell} L^{\ell-1}$, and $\hat{\phi}_{ii\ell}$ and $\hat{d}_{i\ell}$ denote estimates of $\phi_{ii\ell}$ and $d_{i\ell}$, respectively. The modulus of the largest eigenvalue is 0.94 for Germany and Italy, and 0.89 for France, and do not cover unity. The authors therefore conclude that it is unlikely that dynamic heterogeneity alone could generate the degree of persistence observed in Figure 32.3.

This conclusion is further investigated in Figure 32.4, which compares the estimates of *GIRF*s for the combined aggregate shock on the aggregate variable with $\hat{a}_s = \mathbf{w}'\hat{\mathbf{G}}_s\boldsymbol{\tau}_N$ at horizons $s = 6$, 12 and 24, where the matrix $\hat{\mathbf{G}}_s$is defined by $\hat{\mathbf{\Phi}}^{-1}(L) = \hat{\mathbf{G}}(L) = \sum_{s=0}^{\infty} \hat{\mathbf{G}}_s L^{\ell}$. $\hat{a}_s$ shows the effects of dynamic heterogeneity on the persistence of the aggregate variable, whereas the *GIRF*s of the combined aggregate shock on the aggregate variable is determined by factor persistence as well as dynamic heterogeneity. In the case of all the three countries, $\hat{a}_s$ is found to decline with s much faster when compared to the effects of the combined aggregate shock . It therefore seems that dynamic heterogeneity alone does not sufficiently explain the observed persistence of the aggregate inflation.

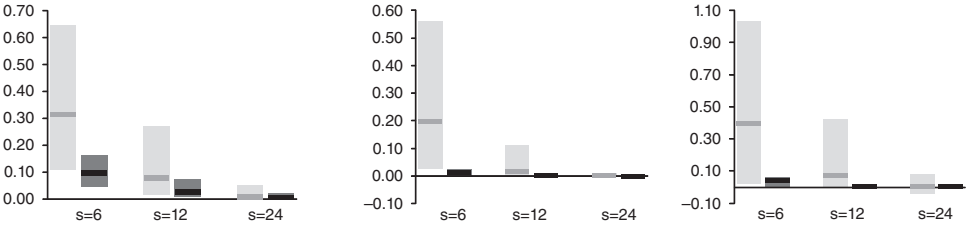


Figure 32.4 *GIRF*s of one unit combined aggregate shocks on the aggregate variable (light-grey colour) and estimates of a_s (dark-grey colour); bootstrap means and 90% confidence bounds, $s = 6$, 12, and 24.

32.12 Further reading

Further discussion on aggregation in econometrics can be found in Robinson (1978), Granger (1980), Pesaran (2003), and Pesaran and Chudik (2014).

32.13 Exercises

1. Consider the following dynamic factor models for the n cross-sectional units

$$y_{it} = \alpha_i(L) f_t, \text{ for } i = 1, 2, \ldots, n,$$

where

$$\alpha_i(L) = \alpha_{i0} + \alpha_{i1} L + \alpha_{i2} L^2 + \ldots.$$

(a) Show that

$$\bar{y}_t = \bar{\alpha}_n(L) f_t,$$

where $\bar{y}_t = n^{-1} \sum_{i=1}^{n} y_{it}$, $\bar{\alpha}_n(L) = \bar{\alpha}_{0n} + \bar{\alpha}_{1n} L + \bar{\alpha}_{2n} L^2 + \ldots$, and

$$\bar{\alpha}_{jn} = n^{-1} \sum_{i=1}^{n} \alpha_{ij}.$$

(b) Discuss the conditions under which

$$\sum_{j=0}^{\infty} \left| \bar{\alpha}_{jn} \right| < K,$$

where K is a fixed positive constant.

(c) As an example, suppose that

$$\alpha_i(L) = \frac{1 - \theta_i L}{1 - \phi_i L}, \text{ with } \left| \phi_i \right| < 1 \text{ and } |\theta_i| < 1,$$

and θ_i and ϕ_i are random draws from uniform distribution over the ranges $[a_\theta, b_\theta]$, and $\left[a_\phi, b_\phi\right]$. Under what values of these ranges is the absolute summability condition in (b) satisfied?

2. Suppose that

$$y_{it} = \left(\frac{1 - \theta_i L}{1 - \phi_i L} \right) f_t + u_{it},$$

where

$$u_{it} = \rho \sum_{j=1}^{n} w_{ij} u_{jt} + \varepsilon_{it},$$

$$\varepsilon_{it} \sim IID(0, \sigma_i^2), w_{ij} > 0, \sum_{j=1}^{n} w_{ij} = 1 = \sum_{i=1}^{n} w_{ij}, \text{ and } |\rho| < 1.$$

(a) Show that

$$E\left|\bar{y}_t - \bar{\alpha}_n(L) f_t\right| = O\left(n^{-1/2}\right),$$

where

$$\bar{\alpha}_n(L) = n^{-1} \sum_{i=1}^{n} \left(\frac{1 - \theta_i L}{1 - \phi_i L}\right).$$

(b) Suppose that θ_i and ϕ_i are independently uniformly distributed over the ranges $[0, 1]$. Show that the limit of $\bar{y}_t$ as $n \to \infty$, is a long memory process.

(c) Derive the autocorrelation function of $\bar{y}_t$ for n sufficiently large.

(d) Discuss the relevance of the above for the analysis of the relationship between macro and micro relationships in economics.

3. Consider the factor-augmented panel data model for $i = 1, 2, \ldots, n$

$$\begin{aligned} y_{it} &= \lambda_i x_{it} + \gamma_i f_t + u_{it}, \\ x_{it} &= \theta_i f_t + v_{it}, \end{aligned}$$

where f_t is an unobserved factor following the $AR(1)$ process,

$$f_t = \rho f_{t-1} + v_t,$$

u_{it} and v_{it} are defined by the following linear stationary processes

$$u_{it} = \sum_{j=0}^{\infty} a_j \varepsilon_{t-j}, v_{it} = \sum_{j=0}^{\infty} b_j \xi_{t-j},$$

where ε_t and ξ_t are $IID(0, 1)$. The coefficients, λ_i, γ_i and θ_i are either fixed constants or, if stochastic, are independently distributed. Further f_t follows the AR(1) process.

(a) Suppose that u_{it} and v_{it} are weakly cross-sectionally uncorrelated. Derive the correlation between f_t and $\bar{y}_t = n^{-1} \sum_{i=1}^{n} y_{it}$, and show that this correlation tends to unity as $n \to \infty$.

(b) How do you forecast $\bar{y}_t$ based on (a) the aggregate information set $\left\{\bar{y}_{t-1}, \bar{x}_t; \bar{y}_{t-2}, \bar{x}_{t-1}; \ldots.\right\}$, (b) disaggregated information set $\left\{y_{i,t-1} x_{it}; y_{i,t-2}, x_{i,t-1}; \ldots\ldots \text{ for } i = 1, 2, \ldots., n\right\}$, distinguishing between cases when n is small and when n is large?

(c) What additional information/restrictions are required if the object of interest is to forecast y_{it} for a particular cross-sectional unit i?

4. Consider the disaggregated rational expectations model

$$y_{it} = \alpha_i E\left(y_{i,t+1} \,|\Omega_t\right) + \theta_i x_{it} + u_{it},$$

for $i = 1, 2, \ldots, n$, where u_{it} and x_{it} are independently distributed, $u_{it} \sim IID(0, \sigma_u^2)$, $\Omega_t = \cup_i^n \Omega_{it}$, $\Omega_{it} = (y_{it}, x_{it}; y_{i,t-1} x_{i,t-1}; \ldots)$,

$$\begin{aligned} x_{it} &= \gamma_i f_t + v_{it}, \\ v_{it} &= \rho_i v_{i,t-1} + \varepsilon_{it} \end{aligned}$$

$\gamma_i \sim IID(\mu_\gamma, \sigma_\gamma^2)$, $\mu_\gamma \neq 0$, $\theta_i \sim IID(\mu_\theta, \sigma_\theta^2)$, $\mu_\theta \neq 0$, γ_i and θ_i are independently distributed,

$$f_t = \rho f_{t-1} + \varepsilon_t,$$

with ε_{it} and ε_t being random draws with zero means and constant variances, $\left|\rho_i\right| \leq 1$, and $|\rho| < 1$.

(a) Assuming that $|\alpha_i| < 1$ for all i, show that the above disaggregated rational expectations model has a unique solution.

(b) Derive an expression for the aggregates $\bar{y}_t$ and $\bar{x}_t$ constructed as simple averages of y_{it} and x_{it} over i.

(c) Suppose that u_{it} and ε_{it} are cross-sectionally weakly correlated. Derive the limiting properties of $\bar{y}_t$ and $\bar{x}_t$ as $n \to \infty$, and show that they are cointegrated when $\rho = 1$.

33 Theory and Practice of *GVAR* Modelling

33.1 Introduction

Individual economies in the global economy are interlinked through many different channels in a complex way. These include sharing scarce resources (such as oil and other commodities), political and technological developments, cross-border trade in financial assets as well as trade in goods and services, labour and capital movement across countries. Even after allowing for such effects, there might still be residual interdependencies due to unobserved interactions and spillover effects not taken properly into account by using the common channels of interaction. Taking account of these channels of interaction poses a major challenge to modelling the global economy and conducting policy simulations and counterfactual scenario analyses.

The global *VAR* (*GVAR*) approach, originally proposed in Pesaran et al. (2004), provides a relatively simple yet effective way of modelling complex high-dimensional systems such as the global economy. Although *GVAR* is not the first large global macroeconomic model of the world economy, its methodological contributions lie in dealing with the curse of dimensionality (i.e., the proliferation of parameters as the dimension of the model grows) in a theoretically coherent and statistically consistent manner. Other existing large models are often incomplete and do not present a closed system, which is required for simulation analysis. See Granger and Jeon (2007) for a recent overview of global models.

The *GVAR* approach was developed in the aftermath of the 1997 Asian financial crisis to quantify the effects of macroeconomic developments on the losses of major financial institutions. It was clear then that all major banks are exposed to risk from adverse global or regional shocks, but quantifying these effects required a coherent and simple-to-simulate global macroeconomic model. The *GVAR* approach provides a useful and practical way of building such a model, and, although developed originally as a tool for credit risk analysis, it soon became apparent that it has numerous other applications. This chapter surveys the *GVAR* approach, focusing on the theoretical foundations of the approach as well as its empirical applications.

The *GVAR* can be briefly summarized as a two-step approach. In the first step, small-scale country-specific models are estimated conditional on the rest of the world. These models are represented as augmented *VAR* models, denoted as *VARX** and feature domestic variables and weighted cross-section averages of foreign variables, also commonly referred to as 'star variables', which are treated as weakly exogenous (or long-run forcing). In the second step, individual country *VARX** models are stacked and solved simultaneously as one large global *VAR* model. The solution can be used for shock scenario analysis and forecasting as is usually done with standard low-dimensional *VAR* models.

The simplicity and usefulness of this approach has proved to be quite attractive and there are numerous applications of the *GVAR* approach. Individual units need not necessarily be countries, but could be regions, industries, goods categories, banks, municipalities, or sectors of a given economy, just to mention a few notable examples. Mixed cross-section *GVAR* models, for instance linking country data with firm-level data, have also been considered in the literature. The *GVAR* approach is conceptually simple, although it requires some programming skills since it handles large data sets, and it is not yet incorporated in any of the mainstream econometric software packages. Fortunately, an open source toolbox developed by Smith and Galesi (2014) together with a global macroeconomic data set, covering the period 1979–2013, can be obtained from the web at <https://sites.google.com/site/gvarmodelling/>. This toolbox has greatly facilitated empirical research using *GVAR* methodology.

We start with methodological issues, considering large linear dynamic systems. We suppose that the large set of variables under consideration are all endogenously determined in a factor-augmented high-dimensional *VAR* model. This model allows for a very general pattern of interlinkages among variables, but, as is well known, it cannot be estimated consistently due to the curse of dimensionality when the cross-section dimension (N) is large. *GVAR* is one of the common solutions to the curse of dimensionality, alongside popular factor-based modelling approaches, large-scale Bayesian *VAR*s and panel *VAR*s. We introduce the *GVAR* approach as originally proposed by Pesaran et al. (2004) and then review conditions (on the underlying unobserved high-dimensional *VAR* data generating process) that justify the individual equations estimated in the *GVAR* approach when N and T (the time dimension) are large, and of the same order of magnitude. Next, we survey the impulse response analysis, forecasting, analysis of long-run and specification tests in the *GVAR* approach. Last but not least, we review empirical *GVAR* applications. We separate forecasting from non-forecasting applications, and divide the latter group of empirical papers into global applications (featuring countries) and the remaining sectoral/other applications, where cross-section units represent sectors, industries or regions within a given economy.

33.2 Large-scale *VAR* reduced form representation of data

Consider a panel of N cross-sectional units, each featuring k_i variables observed during the time periods $t = 1, 2, \ldots, T$. Let $\mathbf{x}_{it}$ denote a $k_i \times 1$ vector of variables specific to cross-sectional unit i in time period t, and let $\mathbf{x}_t = \left(\mathbf{x}'_{1t}, \mathbf{x}'_{2t}, \ldots, \mathbf{x}'_{Nt}\right)'$ denote a $k \times 1$ vector of all variables in the panel, where $k = \sum_{i=1}^{N} k_i$. Suppose that $\mathbf{x}_t$ is generated according to the following factor-augmented $VAR(p)$ model,

$$\mathbf{\Theta}\left(L, p\right)\mathbf{x}_t = \mathbf{\Gamma}_f\left(L, s_f\right)\mathbf{f}_t + \mathbf{\Gamma}_\omega\left(L, s_\omega\right)\boldsymbol{\omega}_t + \mathbf{u}_t, \tag{33.1}$$

where L is the time lag operator, $\mathbf{\Theta}\left(L,p\right)=\mathbf{I}_k-\sum_{\ell=1}^{p}\mathbf{\Theta}_\ell L^\ell$ is a matrix lag polynomial in L, $\mathbf{\Theta}_\ell$ for $\ell=1,2,\ldots,p$ are $k\times k$ matrices of unknown coefficients, $\mathbf{\Gamma}_a\left(L,s_\alpha\right)=\sum_{\ell=1}^{s_a}\mathbf{\Gamma}_{a\ell}L^\ell$, for $a=f,\omega$, $\mathbf{\Gamma}_{a\ell}$ for $\ell=1,2,\ldots,s$ and $a=f,\omega$ are $k\times m_a$ matrices of factor loadings, $\mathbf{f}_t$ is the $m_f\times 1$ vector of unobserved common factors, $\boldsymbol{\omega}_t$ is the $m_\omega\times 1$ vector of observed common effects, $\mathbf{u}_t$ is a $k\times 1$ vector of reduced form errors with zero means, and the $k\times k$ covariance matrix, $\mathbf{\Sigma}_u=E\left(\mathbf{u}_t\mathbf{u}_t'\right)$. We abstract from deterministic terms to keep the exposition simple, but such terms can be easily incorporated in the analysis. *GVAR* allows for very general forms of interdependencies across individual variables within a given unit and/or across units, since lags of all k variables enter individual equations, and the reduced form errors are allowed to be cross-sectionally dependent. *GVAR* can also be extended to allow for time varying parameters, nonlinearities, or threshold effects. But such extensions are not considered in this Chapter.[1]

VAR models provide a rather general description of linear dynamic systems, but their number of unknown parameters to be estimated grows at a quadratic rate in the dimension of the model, k. We are interested in applications where the cross-section dimension, N, as well as the time series dimension, T, can both be relatively large, while k_i, for $i=1,2,\ldots,N$, are small, so that $k=O\left(N\right)$. A prominent example arises in the case of global macroeconomic modelling, where the number of cross-section units is relatively large but the number of variables considered within each cross-sectional unit (such as real output, inflation, stock prices and interest rates) is small. Understanding the transmission of shocks across economies (space) and time is a key question in this example. Clearly, in such settings unrestricted *VAR* models cannot be estimated due to the proliferation of unknown parameters (often referred to as the curse of dimensionality). The main problem is how to impose a plethora of restrictions on the model (33.1) so that the parameters can be consistently estimated as $N,T\to_j\infty$, while still allowing for a general pattern of interdependencies between the individual variables.

There are several approaches developed for modelling data sets with a large number of variables: models that utilize common factors (see Chapters 19 and 29 on factor models), large Bayesian *VAR*s, Panel *VAR*s, and global *VAR*s. Factor models can be interpreted as data shrinkage procedures, where a large set of variables is shrunk into a small set of factors.[2] Estimated factors can be used together with the vector of domestic variables to form a small-scale model, as in factor-augmented *VAR* models (Bernanke, Bovian, and Eliasz (2005) and Stock and Watson (2005)).[3] Large-scale Bayesian *VAR*s, on the other hand, explicitly shrink the parameter space by imposing tight priors on all or a sub-set of parameters. Such models have been explored, among others, by Giacomini and White (2006), De Mol, Giannone, and Reichlin (2008), Carriero, Kapetanios, and Marcellino (2009), and Banbura, Giannone, and Reichlin (2010). Large Bayesian *VAR*s share many similarities with Panel *VAR*s. The difference between the two is that, while large Bayesian *VAR*s typically treat each variable symmetrically, Panel *VAR*s take account of the structure of the variables, namely the division of the variables into different cross-section groups and variable types. Parameter space is shrunk in the Panel *VAR* literature by assuming that

[1] Extensions of the linear setting to allow for nonlinearities could also be considered, but most of the *GVAR* papers in the literature are confined to a linear framework. The few exceptions include Binder and Gross (2013), who develop a regime-switching *GVAR* model, and *GVAR* papers that consider time varying weights.

[2] Stock and Watson (1999, 2002), and Giannone, Reichlin, and Sala (2005) conclude that only a few, perhaps two, factors explain much of the predictable variation, while Bai and Ng (2007) estimate four factors and Stock and Watson (2005) estimate as many as seven factors.

[3] Dynamic factor models were introduced by Geweke (1977) and Sargent and Sims (1977), which have more recently been generalized to allow for weak cross-sectional dependence by Forni and Lippi (2001), Forni et al. (2000), and Forni et al. (2004).

the unknown coefficients can be decomposed into a component that is common across all variables, a cross-section specific component, a variable-specific component, lag-specific component, and idiosyncratic effects; see Canova and Ciccarelli (2013) for a survey. Last but not least, the *GVAR* approach solves the dimensionality problem by decomposing the underlying large dimensional *VAR*s into a smaller number of conditional models, which are linked together *via* cross-sectional averages. The *GVAR* approach imposes an intuitive structure on cross-country interlinkages and no restrictions are imposed on the dynamics of the individual country sub-models. In the case where the number of lags is relatively large (compared with the time dimension of the panel) and/or the number of country specific variables is moderately large, it is possible to combine the *GVAR* structure with shrinkage estimation approaches in light of the usual bias-variance trade-offs. Bayesian estimation of country-specific sub-models that feature in the *GVAR* approach have been considered, for instance in Feldkircher et al. (2014).

33.3 The *GVAR* solution to the curse of dimensionality

The *GVAR* approach was originally proposed by Pesaran et al. (2004) (PSW) as a pragmatic approach to building a coherent global model of the world economy. We follow the exposition of PSW and introduce the *GVAR* approach initially without the inclusion of common variables.

Consider a panel of N cross-section units, each featuring k_i variables observed during the time period $t = 1, 2, \ldots, T$. Let $\mathbf{x}_{it}$ denote a $k_i \times 1$ vector of variables specific to cross-sectional unit i in time period t, and let $\mathbf{x}_t = \left(\mathbf{x}'_{1t}, \mathbf{x}'_{2t}, \ldots, \mathbf{x}'_{Nt}\right)'$ denote a $k \times 1$ vector of all variables in the panel, where $k = \sum_{i=1}^{N} k_i$. At the core of the *GVAR* approach are small-scale country specific conditional models that can be estimated separately. These individual country models explain the domestic variables of a given economy, collected in the $k_i \times 1$ vector $\mathbf{x}_{it}$, in terms of the country-specific cross-section averages of foreign variables, collected in the $k^* \times 1$ vector

$$\mathbf{x}^*_{it} = \tilde{\mathbf{W}}'_i \mathbf{x}_t, \tag{33.2}$$

for $i = 1, 2, \ldots, N$, where $\tilde{\mathbf{W}}_i$ is $k \times k^*$ matrix of country-specific weights typically constructed using data on bilateral foreign trade or capital flows.[4] Both k_i and k^* are assumed to be small (typically 4 to 6). A larger number of domestic variables can easily be incorporated within the *GVAR* framework as well by using shrinkage methods applied to the country-specific sub-models. $\mathbf{x}_{it}$ is modelled as a *VAR* augmented by the vector of the 'star' variables $\mathbf{x}^*_{it}$, and their lagged values,

$$\mathbf{x}_{it} = \sum_{\ell=1}^{p_i} \mathbf{\Phi}_{i\ell} \mathbf{x}_{i,t-\ell} + \mathbf{\Lambda}_{i0} \mathbf{x}^*_{it} + \sum_{\ell=1}^{q_i} \mathbf{\Lambda}_{i\ell} \mathbf{x}^*_{i,t-\ell} + \boldsymbol{\varepsilon}_{it}, \tag{33.3}$$

for $i = 1, 2, \ldots, N$, where $\mathbf{\Phi}_{i\ell}$, for $\ell = 1, 2, \ldots, p_i$, $\mathbf{\Lambda}_{i\ell}$, for $\ell = 0, 1, 2, \ldots q_i$, are $k_i \times k_i$ and $k_i \times k^*$ matrices of unknown parameters, respectively, and $\boldsymbol{\varepsilon}_{it}$ are $k_i \times 1$ error vectors. We continue to abstract from the deterministic terms and observed common effects from the country-specific conditional *VARX** models in (33.3). Star variables $\mathbf{x}^*_{it}$ in country-specific models (33.3) can,

[4] It is straightforward to accommodate a different number of star variables across countries (k^*_i instead of k^*), if desired.

under conditions reviewed in Section 33.4, be treated as weakly exogenous for the purpose of estimating the unknown coefficients of the conditional country models.

Let $\mathbf{z}_{it} = \left(\mathbf{x}_{it}', \mathbf{x}_{it}^{*\prime}\right)'$ be the $k_i + k^*$ dimensional vector of domestic and country-specific foreign variables included in the sub-model of country i and rewrite (33.3) as

$$\mathbf{A}_{i0}\mathbf{z}_{it} = \sum_{\ell=1}^{p} \mathbf{A}_{i\ell}\mathbf{z}_{it-\ell} + \boldsymbol{\varepsilon}_{it}, \tag{33.4}$$

where

$$\mathbf{A}_{i0} = \left(\mathbf{I}_{k_i}, -\boldsymbol{\Lambda}_{i0}\right), \mathbf{A}_{i\ell} = \left(\boldsymbol{\Phi}_{i\ell}, \boldsymbol{\Lambda}_{i\ell}\right) \text{ for } \ell = 1, 2, \ldots, p,$$

$p = \max_i \left(p_i, q_i\right)$, and define $\boldsymbol{\Phi}_{i\ell} = \mathbf{0}$ for $\ell > p_i$, and similarly $\boldsymbol{\Lambda}_{i\ell} = 0$ for $\ell > q_i$. Individual country-models in (33.4) can be equivalently written in the form of error-correction representation,

$$\Delta\mathbf{x}_{it} = \boldsymbol{\Lambda}_{i0}\Delta\mathbf{x}_{it}^{*} - \boldsymbol{\Pi}_i\mathbf{z}_{i,t-1} + \sum_{\ell=1}^{p} \mathbf{H}_{i\ell}\Delta\mathbf{z}_{i,t-1} + \boldsymbol{\varepsilon}_{it}, \tag{33.5}$$

where $\Delta = 1 - L$ is the usual first difference operator, and

$$\boldsymbol{\Pi}_i = \mathbf{A}_{i0} - \sum_{\ell=1}^{p} \mathbf{A}_{i\ell}, \text{ and } \mathbf{H}_{i\ell} = -\left(\mathbf{A}_{i,\ell+1} + \mathbf{A}_{i,\ell+2} + \ldots + \mathbf{A}_{i,\ell+p}\right).$$

Star variables $\mathbf{x}_{it}^{*}$ are treated as weakly exogenous for the purpose of estimating (33.5). Econometric theory for estimating $VARX^*\left(p_i, q_i\right)$ models with weakly exogenous $I(1)$ regressors have been developed by Harbo et al. (1998) and Pesaran et al. (2000) and discussed in Section 23.2. The assumption of weak exogeneity can be easily tested as outlined in Section 7.1 of *PSW*, and typically is not rejected when the economy under consideration is small relative to the rest of the world and the weights used in the construction of the star variables are granular.

It is clear from (33.5) that country specific models allow for cointegration both amongst domestic variables as well as between domestic and foreign (star) variables.[5] In particular, assuming $\mathbf{z}_{it}$ is $I(1)$, the rank of $\boldsymbol{\Pi}_i$, denoted as $r_i = rank\left(\boldsymbol{\Pi}_i\right) \leq k_i$, specifies the number of cointegrating relationships that exist among the domestic and country-specific foreign variables in $\mathbf{z}_{it}$; and $\boldsymbol{\Pi}_i$ can be decomposed as

$$\boldsymbol{\Pi}_i = \boldsymbol{\alpha}_i\boldsymbol{\beta}_i',$$

where $\boldsymbol{\alpha}_i$ is the $k_i \times r_i$ full column rank loading matrix and $\boldsymbol{\beta}_i$ is the $(k_i + k^*) \times r_i$ full column rank matrix of cointegrating vectors. It is well known that this decomposition is not unique and the identification of long-run relationships requires theory-based restrictions (see Sections 23.6 and 33.7).

[5] See Chapter 22 for an introduction to cointegration analysis.

Country models in (33.3) resemble the small open economy (*SOE*) macroeconomic models in the literature, where domestic variables are modelled conditional on the rest of the world. The data shrinkage given by (33.2) solves the dimensionality problem. The conditions under which it is valid to specify (33.3) are reviewed in Section 33.4. The estimation of country models in (33.3), which allows for cointegration within and across countries (*via* the star variables), is the first step of the *GVAR* approach.

The second step of the *GVAR* approach consists of stacking estimated country models to form one large global *VAR* model. Using the $(k_i + k^*) \times k$ dimensional 'link' matrices $\mathbf{W}_i = \left(\mathbf{E}_i', \tilde{\mathbf{W}}_i'\right)$, where $\mathbf{E}_i$ is the $k \times k_i$ dimensional selection matrix that selects $\mathbf{x}_{it}$, namely $\mathbf{x}_{it} = \mathbf{E}_i'\mathbf{x}_t$, and $\tilde{\mathbf{W}}_i'$ is the weight matrix introduced in (33.2) to define country-specific foreign star variables, we have

$$\mathbf{z}_{it} = \left(\mathbf{x}_{it}', \mathbf{x}_{it}^{*\prime}\right)' = \mathbf{W}_i\mathbf{x}_t. \tag{33.6}$$

Using (33.6) in (33.4) yields

$$\mathbf{A}_{i0}\mathbf{W}_i\mathbf{x}_t = \sum_{\ell=1}^{p} \mathbf{A}_{i\ell}\mathbf{W}_i\mathbf{x}_{t-\ell} + \boldsymbol{\varepsilon}_{it},$$

and stacking these models for $i = 1, 2, \ldots, N$, we obtain

$$\mathbf{G}_0\mathbf{x}_t = \sum_{\ell=1}^{p} \mathbf{G}_\ell\mathbf{x}_{t-\ell} + \boldsymbol{\varepsilon}_t, \tag{33.7}$$

where $\boldsymbol{\varepsilon}_t = \left(\boldsymbol{\varepsilon}_{1t}', \boldsymbol{\varepsilon}_{2t}', \ldots, \boldsymbol{\varepsilon}_{Nt}'\right)'$, and

$$\mathbf{G}_\ell = \begin{pmatrix} \mathbf{A}_{1,\ell}\mathbf{W}_1 \\ \mathbf{A}_{2,\ell}\mathbf{W}_2 \\ \vdots \\ \mathbf{A}_{N,\ell}\mathbf{W}_N \end{pmatrix}, \qquad \text{for } \ell = 0, 1, 2, \ldots, p.$$

If matrix $\mathbf{G}_0$ is invertible, then by multiplying (33.7) by $\mathbf{G}_0^{-1}$ from the left we obtain the *GVAR* model

$$\mathbf{x}_t = \sum_{\ell=1}^{p} \mathbf{F}_\ell\mathbf{x}_{t-\ell} + \mathbf{G}_0^{-1}\boldsymbol{\varepsilon}_t, \tag{33.8}$$

where $\mathbf{F}_\ell = \mathbf{G}_0^{-1}\mathbf{G}_\ell$ for $\ell = 1, 2, \ldots, p$. *PSW* established that the overall number of cointegrating relationships in the *GVAR* model (33.8) cannot exceed the total number of long-run relations $\sum_{i=1}^{N} r_i$ that exist in country-specific models.

33.3.1 Case of rank deficient $\mathbf{G}_0$

The *GVAR* model (33.8) is derived under the assumption that the contemporaneous coefficient matrix, $\mathbf{G}_0$, is full rank. To clarify the role of this assumption and to illustrate the consequences of possible rank deficiency of $\mathbf{G}_0$, consider the following illustrative *GVAR* model,

$$\mathbf{x}_{it} = \mathbf{\Lambda}_{i0}\mathbf{x}_{it}^{*} + \boldsymbol{\varepsilon}_{it}, \text{ for } i = 1, 2, \ldots, N, \tag{33.9}$$

where we abstract from lags of $\left(\mathbf{x}_{it}', \mathbf{x}_{it}^{*\prime}\right)'$. Let $\mathbf{\Lambda}_0$ be the $k \times k$ block diagonal matrix defined by $\mathbf{\Lambda}_0 = diag\left(\mathbf{\Lambda}_{i0}\right)$, and let $\tilde{\mathbf{W}}' = (\tilde{\mathbf{W}}_1, \tilde{\mathbf{W}}_2, \ldots., \tilde{\mathbf{W}}_N)$. Write (33.9) as

$$\mathbf{x}_t = \mathbf{\Lambda}_0\tilde{\mathbf{W}}x_t + \boldsymbol{\varepsilon}_t,$$

or

$$\mathbf{G}_0\mathbf{x}_t = \boldsymbol{\varepsilon}_t, \tag{33.10}$$

where $\mathbf{G}_0 = \mathbf{I}_N - \mathbf{\Lambda}_0\tilde{\mathbf{W}}$. Suppose that $\mathbf{G}_0$ is rank deficient, namely $rank\left(\mathbf{G}_0\right) = k - m$, for some $m > 0$. Then the solution of (33.10) exists only if $\boldsymbol{\varepsilon}_t$ lies in the range of $\mathbf{G}_0$, denoted as $Col\left(\mathbf{G}_0\right)$. Assuming this is the case, system (33.10) does not uniquely determine $\mathbf{x}_t$, and the set of all its possible solutions can be characterized as

$$\mathbf{x}_t = \mathbf{\Gamma}\tilde{\mathbf{f}}_t + \mathbf{G}_0^{+}\boldsymbol{\varepsilon}_t, \tag{33.11}$$

where $\tilde{\mathbf{f}}_t$ is any $m \times 1$ arbitrary stochastic process, $\mathbf{\Gamma}$ is a $k \times m$ matrix which is a basis of the null space of $\mathbf{G}_0$, namely $\mathbf{G}_0\mathbf{\Gamma} = 0$, $rank\left(\mathbf{\Gamma}'\mathbf{\Gamma}\right) = m$, and $\mathbf{G}_0^{+}$ is the Moore–Penrose inverse of $\mathbf{G}_0$.[6] To verify that (33.11) maps all possible solutions of (33.10), note that $\mathbf{G}_0^{+}\boldsymbol{\varepsilon}_t$ is the particular solution of (33.10) and $\mathbf{\Gamma}'\tilde{\mathbf{f}}_t$ is a general solution of the homogeneous counterpart of (33.10), given by $\mathbf{G}_0\mathbf{x}_t = \mathbf{0}$. To prove the former, we note from the property of Moore–Penrose inverses that $\mathbf{G}_0\mathbf{G}_0^{+}\mathbf{G}_0 = \mathbf{G}_0$ and $\mathbf{G}_0\mathbf{x}_t = \mathbf{G}_0\mathbf{G}_0^{+}\mathbf{G}_0\mathbf{x}_t$, or $\boldsymbol{\varepsilon}_t = \mathbf{G}_0\mathbf{G}_0^{+}\boldsymbol{\varepsilon}_t$, which establishes that $\mathbf{G}_0^{+}\boldsymbol{\varepsilon}_t$ is indeed a solution of $\mathbf{G}_0\mathbf{x}_t = \boldsymbol{\varepsilon}_t$. To prove the latter, we note that $\mathbf{\Gamma}$ is a basis of the null space of $\mathbf{G}_0$ and therefore $\mathbf{G}_0\mathbf{\Gamma}\tilde{\mathbf{f}}_t = \mathbf{0}$ for any $m \times 1$ arbitrary stochastic process $\tilde{\mathbf{f}}_t$, and the set of solutions must be complete since the dimension of $Col\left(\mathbf{\Gamma}\right)$ is m.

Let $\mathbf{f}_t = \tilde{\mathbf{f}}_t - E\left(\tilde{\mathbf{f}}_t \middle| \boldsymbol{\varepsilon}_t\right) = \tilde{\mathbf{f}}_t - \mathbf{M}'\boldsymbol{\varepsilon}_t$. Then (33.11) can also be written as an approximate factor model, namely

$$\mathbf{x}_t = \mathbf{\Gamma}\mathbf{f}_t + \mathbf{R}\boldsymbol{\varepsilon}_t,$$

where $\mathbf{f}_t$ is uncorrelated with $\boldsymbol{\varepsilon}_t$ by construction,[7] and

$$\mathbf{R} = \mathbf{\Gamma}\mathbf{M}' + \mathbf{G}_0^{+}.$$

[6] For a description of Moore-Penrose inverses see Section A.7 in Appendix A.

[7] Note that $E(\mathbf{f}_t|\boldsymbol{\varepsilon}_t) = E(\tilde{\mathbf{f}}_t|\boldsymbol{\varepsilon}_t) - E(\tilde{\mathbf{f}}_t|\boldsymbol{\varepsilon}_t) = \mathbf{0}$.

Without any loss of generality, it is standard convention to use the normalization $Var(\mathbf{f}_t) = \mathbf{I}_m$, and to set the first non-zero element in each of the m column vectors of $\mathbf{\Gamma}$ to be positive. These normalization conditions ensure that $\mathbf{\Gamma}$ is unique, in which case $\mathbf{R}$ is unique up to the rotation matrix, $\mathbf{M}$. Note also that all of the findings above hold for any N.

Therefore, the full rank condition, $rank(\mathbf{G}_0) = k$, is necessary and sufficient for $\mathbf{x}_t$, given by (33.9), to be uniquely determined. If $\mathbf{G}_0$ is known to be rank deficient with rank $k - m$, and $m > 0$, then the *GVAR* model (33.9) would need to be augmented by m equations that determine the m cross-section averages defined by $\mathbf{\Gamma}'\mathbf{x}_t$ in order for $\mathbf{x}_t$ to be uniquely determined. We provide further clarification on the rank of $\mathbf{G}_0$ in Section 33.6, where we review conditions under which the individual equations estimated in the *GVAR* approach can lead to a singular $\mathbf{G}_0$ as $N \to \infty$.

33.3.2 Introducing common variables

When common variables are present in the country models ($m_\omega > 0$), either as observed common factors or as dominant variables as defined in Chudik and Pesaran (2013), then the conditional country models need to be augmented by $\boldsymbol{\omega}_t$ and its lagged values, in addition to the country-specific vector of cross-section averages of the foreign variables, namely

$$\mathbf{x}_{it} = \sum_{\ell=1}^{p_i} \mathbf{\Phi}_{i\ell}\mathbf{x}_{i,t-\ell} + \mathbf{\Lambda}_{i0}\mathbf{x}_{it}^* + \sum_{\ell=1}^{q_i} \mathbf{\Lambda}_{i\ell}\mathbf{x}_{i,t-\ell}^* + \mathbf{D}_{i0}\boldsymbol{\omega}_t + \sum_{\ell=1}^{s_i} \mathbf{D}_{i\ell}\boldsymbol{\omega}_{t-\ell} + \boldsymbol{\varepsilon}_{it}, \tag{33.12}$$

for $i = 1, 2, \ldots, N$. Both types of variables (common variables $\boldsymbol{\omega}_t$ and cross-section averages $\mathbf{x}_{it}^*$) can be treated as weakly exogenous for the purpose of estimation. As noted above, the weak exogeneity assumption is testable. Also not all of the coefficients $\{\mathbf{D}_{i\ell}\}$ associated with the common variables need be significant and, in the case when they are not significant, they could be excluded for the sake of parsimony.[8] The marginal model for the dominant variables can be estimated with or without the feedback effects from $\mathbf{x}_t$. In the latter case, we have the following marginal model,

$$\boldsymbol{\omega}_t = \sum_{\ell=1}^{p_\omega} \mathbf{\Phi}_{\omega\ell}\boldsymbol{\omega}_{t-\ell} + \boldsymbol{\eta}_{\omega t}, \tag{33.13}$$

which can be equivalently written in the error-correction form as

$$\Delta\boldsymbol{\omega}_t = -\boldsymbol{\alpha}_\omega\boldsymbol{\beta}_\omega'\boldsymbol{\omega}_{t-1} + \textstyle\sum_{\ell=1}^{p_\omega-1} \mathbf{H}_{\omega\ell}\Delta\boldsymbol{\omega}_{t-\ell} + \boldsymbol{\eta}_{\omega t}, \tag{33.14}$$

where $\boldsymbol{\alpha}_\omega\boldsymbol{\beta}_\omega' = \sum_{\ell=1}^{p_\omega} \mathbf{\Phi}_{\omega\ell}$, $\mathbf{H}_{\omega\ell} = -\left(\mathbf{\Phi}_{\omega,\ell+1} + \mathbf{\Phi}_{\omega,\ell+2} + \ldots + \mathbf{\Phi}_{\omega,\ell+p_\omega-1}\right)$, for $\ell = 1, 2, \ldots, p_\omega - 1$. In the case of $I(1)$ variables, representation (33.14) clearly allows for cointegration among the dominant variables. To allow for feedback effects from the variables in the *GVAR* model back to the dominant variables *via* cross-section averages, the *VAR* model (33.13) can be augmented by lags of $\mathbf{x}_{\omega t}^* = \tilde{\mathbf{W}}_\omega\mathbf{x}_t$, where $\tilde{\mathbf{W}}_\omega$ is a $k^* \times k$ dimensional weight matrix defining k^* global cross-section averages,

[8] Chudik and Smith (2013) find that contemporaneous US variables are significant in individual non-US country models in about a quarter of cases. Moreover, weak exogeneity of the US variables is not rejected by the data.

$$\boldsymbol{\omega}_t = \sum_{\ell=1}^{p_\omega} \boldsymbol{\Phi}_{\omega\ell}\boldsymbol{\omega}_{i,t-\ell} + \sum_{\ell=1}^{q_\omega} \boldsymbol{\Lambda}_{\omega\ell}\mathbf{x}^*_{i,t-\ell} + \boldsymbol{\eta}_{\omega t}. \tag{33.15}$$

Assuming there is no cointegration among the common variables, $\boldsymbol{\omega}_t$, and the cross-section averages, $\mathbf{x}^*_{i,t-\ell}$, (33.15) can be written as

$$\Delta\boldsymbol{\omega}_t = -\boldsymbol{\alpha}_\omega\boldsymbol{\beta}'_\omega\boldsymbol{\omega}_{t-1} + \textstyle\sum_{\ell=1}^{p_\omega-1} \mathbf{H}_{\omega\ell}\Delta\boldsymbol{\omega}_{t-\ell} + \sum_{\ell=1}^{q_\omega-1} \mathbf{B}_{\omega\ell}\Delta\mathbf{x}^*_{\omega,t-\ell} + \boldsymbol{\eta}_{\omega t}, \tag{33.16}$$

where $\mathbf{B}_{\omega\ell} = -\left(\boldsymbol{\Lambda}_{\omega,\ell+1} + \boldsymbol{\Lambda}_{\omega,\ell+2} + \ldots + \boldsymbol{\Lambda}_{\omega,\ell+q_\omega-1}\right)$. Different lag-orders for the dominant variables (p_ω) and cross-section averages (q_ω) can be considered. Note that contemporaneous values of star variables do not feature in (33.16), and its unknown parameters can be estimated consistently using least squares or reduced rank regression techniques depending on the assumed rank of $\boldsymbol{\alpha}_\omega\boldsymbol{\beta}'_\omega$. Similar equations are estimated in Holly, Pesaran, and Yamagata (2011), and in a stationary setting in Smith and Yamagata (2011).

Conditional models (33.12) and the marginal model (33.16) can be combined and solved as a complete global *VAR* model in the usual way. Specifically, let $\mathbf{y}_t = \left(\boldsymbol{\omega}'_t, \mathbf{x}'_t\right)'$ be the $(k + m_\omega) \times 1$ vector of all observable variables. Using (33.6) in (33.12) and stacking country-specific conditional models (33.12) together with the model for common variables (33.15) yields

$$\mathbf{G}_{y,0}\mathbf{y}_t = \sum_{\ell=1}^{p} \mathbf{G}_{y,\ell}\mathbf{y}_{t-\ell} + \boldsymbol{\varepsilon}_{yt}, \tag{33.17}$$

where $\boldsymbol{\varepsilon}_{yt} = \left(\boldsymbol{\varepsilon}'_t, \boldsymbol{\eta}'_{\omega t}\right)'$,

$$\mathbf{G}_{y,0} = \begin{pmatrix} \mathbf{I}_{m_\omega} & \mathbf{0}_{m_\omega\times k} \\ \mathbf{D}_0 & \mathbf{G}_0 \end{pmatrix}, \mathbf{G}_{y,\ell} = \begin{pmatrix} \boldsymbol{\Phi}_{\omega\ell} & \boldsymbol{\Lambda}_{\omega\ell}\tilde{\mathbf{W}}_\omega \\ \mathbf{D}_\ell & \mathbf{G}_\ell \end{pmatrix}, \text{for } \ell = 1, 2, \ldots, p,$$

$\mathbf{D}_\ell = \left(\mathbf{D}'_{1\ell}, \mathbf{D}'_{2\ell}, \ldots, \mathbf{D}'_{N\ell}\right)'$ for $\ell = 0, 1, \ldots, p$, $p = \max_i \left\{p_i, q_i, s_i, p_\omega, q_\omega\right\}$, and we define $\mathbf{D}_{i\ell} = \mathbf{0}$ for $\ell > s_i$, $\boldsymbol{\Phi}_{\omega\ell} = \mathbf{0}$ for $\ell > p_\omega$, and $\boldsymbol{\Lambda}_{\omega\ell} = \mathbf{0}$ for $\ell > q_\omega$. Matrix $\mathbf{G}_{y,0}$ is invertible if and only if $\mathbf{G}_0$ is invertible. Assuming $\mathbf{G}_0^{-1}$ exists, the inverse of $\mathbf{G}_{y,0}$ is

$$\mathbf{G}_{y,0}^{-1} = \begin{pmatrix} \mathbf{I}_{m_\omega} & \mathbf{0}_{m_\omega\times k} \\ -\mathbf{G}_0^{-1}\mathbf{D}_0 & \mathbf{G}_0^{-1} \end{pmatrix},$$

which is a block lower triangular matrix, showing the long-run causal nature of the common (dominant) variables, $\boldsymbol{\omega}_t$. Multiplying both sides of (33.17) by $\mathbf{G}_{y,0}^{-1}$ we now obtain the following *GVAR* model for $\mathbf{y}_t$

$$\mathbf{y}_t = \sum_{\ell=1}^{p} \mathbf{F}_\ell\mathbf{y}_{t-\ell} + \mathbf{G}_{y,0}^{-1}\boldsymbol{\varepsilon}_{yt}, \tag{33.18}$$

where $\mathbf{F}_\ell = \mathbf{G}_{y,0}^{-1}\mathbf{G}_{y,\ell}$, for $\ell = 1, 2, \ldots, p$.

33.4 Theoretical justification of the *GVAR* approach

The *GVAR* approach as proposed by *PSW* builds on separate estimation of country-specific *VARX** models based on the assumption that foreign variables are weakly exogenous. However, *PSW* did not provide a theoretical justification and it was left to the future research to derive conditions under which the weak exogeneity assumptions underlying the *GVAR* approach can be maintained. An overview of the subsequent literature is now provided.

33.4.1 Approximating a global factor model

A first attempt at a theoretical justification of the *GVAR* approach was provided by Dées et al. (2007) (DdPS), who derive (33.3) as an approximation to a global factor model.[9] Their starting point is the following canonical global factor model (abstracting again from deterministic terms and observed factors)

$$\mathbf{x}_{it} = \mathbf{\Gamma}_i \mathbf{f}_t + \boldsymbol{\xi}_{it}, \text{ for } i = 1, 2, \ldots, N. \tag{33.19}$$

For each i, $\mathbf{\Gamma}_i$ is a $k_i \times m$ matrix of factor loadings, assumed to be uniformly bounded ($\|\mathbf{\Gamma}_i\| < K < \infty$), and $\boldsymbol{\xi}_{it}$ is a $k_i \times 1$ vector of country-specific effects. Factors and the country effects are assumed to satisfy

$$\Delta \mathbf{f}_t = \mathbf{\Lambda}_f(L)\, \boldsymbol{\eta}_{ft}, \boldsymbol{\eta}_{ft} \sim IID(\mathbf{0}, \mathbf{I}_m), \tag{33.20}$$

$$\Delta \boldsymbol{\xi}_{it} = \mathbf{\Xi}_i(L)\, \mathbf{u}_{it}, \mathbf{u}_{it} \sim IID(\mathbf{0}, \mathbf{I}_{k_i}), \text{ for } i = 1, 2, \ldots, N, \tag{33.21}$$

where $\mathbf{\Lambda}_f(L) = \sum_{\ell=0}^{\infty} \mathbf{\Lambda}_{f\ell} L^{\ell}$, $\mathbf{\Xi}_i(L) = \sum_{\ell=0}^{\infty} \mathbf{\Xi}_{i\ell} L^{\ell}$, and the coefficient matrices $\mathbf{\Lambda}_{f\ell}$ and $\mathbf{\Xi}_{i\ell}$, for $i = 1, 2, \ldots, N$, are uniformly absolute summable, which ensures the existence of $Var(\Delta \mathbf{f}_t)$ and $Var\left(\Delta \boldsymbol{\xi}_{it}\right)$. In addition, $[\mathbf{\Xi}_i(L)]^{-1}$ is assumed to exist.

Under these assumptions, after first differencing (33.19) and using (33.21), DdPS obtain

$$[\mathbf{\Xi}_i(L)]^{-1}(1-L)(\mathbf{x}_{it} - \mathbf{\Gamma}_i \mathbf{f}_t) = \mathbf{u}_{it}.$$

Using the approximation

$$(1-L)[\mathbf{\Xi}_i(L)]^{-1} \approx \sum_{\ell=0}^{p_i} \mathbf{\Phi}_{i\ell} L^{\ell} = \mathbf{\Phi}_i\left(L, p_i\right),$$

DdPS further obtain the following approximate $VAR\left(p_i\right)$ model with factors

$$\mathbf{\Phi}_i\left(L, p_i\right) \mathbf{x}_{it} \approx \mathbf{\Phi}_i\left(L, p_i\right) \mathbf{\Gamma}_i \mathbf{f}_t + \mathbf{u}_{it}, \tag{33.22}$$

for $i = 1, 2, \ldots, N$, which is a special case of (33.1). Model (33.22) is more restrictive than (33.1) because lags of other units do not feature in (33.22), and the errors, $\mathbf{u}_{it}$, are assumed to be cross-sectionally independently distributed.

[9] See Chapters 19 and 29 for an introduction to factor models.

Unobserved common factors in (33.22) can be estimated by linear combinations of cross-section averages of observable variables, $\mathbf{x}_{it}$. As before, let $\tilde{\mathbf{W}}_i$ be the $k \times k^*$ matrix of country-specific weights and assume that it satisfies the usual granularity conditions

$$\left\|\tilde{\mathbf{W}}_i\right\| < KN^{-\frac{1}{2}}, \text{ for all } i \tag{33.23}$$

$$\frac{\left\|\tilde{\mathbf{W}}_{ij}\right\|}{\left\|\tilde{\mathbf{W}}_i\right\|} < KN^{-\frac{1}{2}}, \text{ for all } i, j, \tag{33.24}$$

where $\tilde{\mathbf{W}}_{ij}$ are the blocks in the partitioned form of $\tilde{\mathbf{W}}_i = \left(\tilde{\mathbf{W}}'_{i1}, \tilde{\mathbf{W}}'_{i2}, \ldots, \tilde{\mathbf{W}}'_{iN}\right)'$, and the constant $K < \infty$ does not depend on i, j or N. Taking cross-section averages of $\mathbf{x}_{it}$ given by (33.19) yields

$$\mathbf{x}^*_{it} = \tilde{\mathbf{W}}'_i \mathbf{x}_t = \mathbf{\Gamma}^*_i \mathbf{f}_t + \boldsymbol{\xi}^*_{it},$$

where $\left\|\mathbf{\Gamma}^*_i\right\| = \left\|\tilde{\mathbf{W}}'_i \mathbf{\Gamma}\right\| \leq \left\|\tilde{\mathbf{W}}'_i\right\| \|\mathbf{\Gamma}\| < K$, $\mathbf{\Gamma} = \left(\mathbf{\Gamma}'_1, \mathbf{\Gamma}'_2, \ldots, \mathbf{\Gamma}'_N\right)'$, and $\boldsymbol{\xi}^*_{it}$ satisfies

$$\Delta\boldsymbol{\xi}^*_{it} = \sum_{j=1}^{N} \tilde{\mathbf{W}}'_{ij} \Delta\boldsymbol{\xi}_{it} = \sum_{j=1}^{N} \tilde{\mathbf{W}}'_{ij} \mathbf{\Xi}_i (L)\, \mathbf{u}_{it}.$$

Assuming that $\Delta\boldsymbol{\xi}_{it}$, $i = 1, 2, \ldots, N$, are covariance stationary and weakly cross-sectionally dependent, DdPS show that for each t, $\Delta\boldsymbol{\xi}^*_{it} \overset{q.m.}{\to} \mathbf{0}$ as $N \to \infty$, which implies $\boldsymbol{\xi}^*_{it} \overset{q.m.}{\to} \boldsymbol{\xi}^*_i$. Under the additional condition that $\mathbf{\Gamma}^*_i$ has a full column rank, it then follows that

$$\mathbf{f}_t \overset{q.m.}{\to} \left(\mathbf{\Gamma}^{*\prime}_i \mathbf{\Gamma}^*_i\right)^{-1} \mathbf{\Gamma}^*_i \left(\mathbf{x}^*_{it} - \boldsymbol{\xi}^*_i\right)$$

as $N \to \infty$, which justifies using $\left(1, \mathbf{x}^{*\prime}_{it}\right)'$ as proxies for the unobserved common factors. Thus, for N sufficiently large, DdPS obtain the following country-specific *VAR* models augmented with $\mathbf{x}^*_{it}$,

$$\mathbf{\Phi}_i \left(L, p_i\right) \left(\mathbf{x}_{it} - \tilde{\boldsymbol{\delta}}_i - \tilde{\mathbf{\Gamma}}_i \mathbf{x}^*_{it}\right) \approx \mathbf{u}_{it}, \tag{33.25}$$

where $\tilde{\boldsymbol{\delta}}_i$ and $\tilde{\mathbf{\Gamma}}_i$ are given in terms of $\boldsymbol{\xi}^*_i$ and $\mathbf{\Gamma}^*_i$. (33.25) motivates the use of *VARX** conditional country models in (33.3) as an approximation to a global factor model.

Note that the weights $\left\{\tilde{\mathbf{W}}_i\right\}_{i=1}^N$ used in the construction of cross-sectional averages only need to satisfy the granularity conditions (33.23) and (33.24), and for large N asymptotics one might as well use equal weights, namely replace all cross-sectional averages by simple averages. For the theory to work, it is only needed that $\Delta\boldsymbol{\xi}^*_{it} \overset{q.m.}{\to} \mathbf{0}$ at a sufficiently fast rate as $N \to \infty$. For example, the weights could also be time varying without any major consequences so long as the granularity conditions are met in each period. In practice, where the number of countries (N) is moderate and spillover effects could also be of importance, it is advisable to use trade weights

that also capture cultural and political interlinkages across countries.[10] Trade weights can also be used to allow for time variations in the weights used when constructing the star variables. This is particularly important in cases where there are important shifts in the trade weights, as has occurred in the case of China and its trading partners. Allowing for such time variations is also important in analyzing the way shocks transmit across the world economy. We review some of the empirical applications of the *GVAR* that employ time varying weights below.

The analysis of DdPS has been further extended by Chudik and Pesaran (2011) and Chudik and Pesaran (2013) to allow for joint asymptotics (i.e., as N and $T \rightarrow \infty$, jointly), and weak cross-sectional dependence in the errors in the case of stationary variables.

33.4.2 Approximating factor-augmented stationary high dimensional *VAR*s

Chudik and Pesaran (2011) (CP) consider the conditions on the unknown parameters of the *VAR* model (33.1) that would deliver individual country models (33.3) when N is large. CP consider the following factor-augmented high dimensional *VAR* model,

$$(\mathbf{x}_t - \mathbf{\Gamma}\mathbf{f}_t) = \mathbf{\Theta}\left(\mathbf{x}_{t-1} - \mathbf{\Gamma}\mathbf{f}_{t-1}\right) + \mathbf{u}_t, \tag{33.26}$$

where $\mathbf{x}_t$ is a $k \times 1$ vector of endogenous variables, $\mathbf{\Gamma}$ is a $k \times m$ matrix of factor loadings, and $\mathbf{f}_t$ is an $m \times 1$ covariance stationary process of unobserved common factors. To simplify the exposition the lag-order, p, is set to unity. CP assume that $\varrho\left(\mathbf{\Theta}\mathbf{\Theta}'\right) < 1 - \epsilon$, where $\epsilon > 0$ is an arbitrary small constant that does not depend on N, and $\mathbf{u}_t$ is weakly cross-sectionally dependent such that $\left\|E\left(\mathbf{u}_t\mathbf{u}_t'\right)\right\| = \|\mathbf{\Sigma}_u\| < K$. The condition that the spectral radius of $\mathbf{\Theta}\mathbf{\Theta}'$ is below and bounded away from unity is a slightly stronger requirement than the usual stationarity condition that assumes the eigenvalues of $\mathbf{\Theta}$ lie within the unit circle. The stronger condition is needed to ensure that variances exist when $N \rightarrow \infty$, as can be seen from the following illustrative example.

Example 80 *Consider the following simple VAR(1) model,*

$$\mathbf{x}_t = \mathbf{\Theta}\mathbf{x}_{t-1} + \mathbf{u}_t.$$

Let

$$\underset{N\times N}{\mathbf{\Theta}} = \begin{pmatrix} \alpha & 0 & 0 & \cdots & 0 \\ \beta & \alpha & 0 & \cdots & 0 \\ 0 & \beta & \alpha & \cdots & 0 \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ 0 & 0 & 0 & \beta & \alpha \end{pmatrix},$$

and suppose that $\mathbf{u}_t \sim IID(0, \mathbf{I}_N)$. *Hence, we have*

$$\begin{aligned} x_{1t} &= \alpha x_{1,t-1} + u_{1t} \\ x_{it} &= \beta x_{i-1,t-1} + \alpha x_{i,t-1} + u_{it}, \text{for } i = 2, 3, \ldots, N. \end{aligned}$$

[10] Data-dependent rules to construct weights $\{\tilde{\mathbf{w}}_i\}$ are considered in Gross (2013).

This model is stationary for any given $N \in \mathbb{N}$, if and only if $|\alpha| < 1$. Nevertheless, the stationarity condition $|\alpha| < 1$ is not sufficient to ensure that the variance of x_{Nt} is bounded in N, and without additional conditions Var (x_{Nt}) can rise with N. To see this, note that

$$\begin{aligned} x_{1t} &= (1-\alpha L)^{-1} u_{1t}, \\ x_{2t} &= (1-\alpha L)^{-2} \beta L u_{1t} + (1-\alpha L)^{-1} u_{2t}, \\ &\vdots \\ x_{Nt} &= \sum_{j=1}^{N} (1-\alpha L)^{-N-1+j} \beta^{N-j} L^{N-j} u_{jt}. \end{aligned}$$

Let $\lambda = \beta^2/(1-\alpha^2)$, and note that

$$\begin{aligned} Var(x_{1t}) &= 1/(1-\alpha^2), \\ Var(x_{2t}) &= \frac{1}{1-\alpha^2}(\lambda+1), \\ &\vdots \\ Var(x_{Nt}) &= \frac{1}{1-\alpha^2}\left(\lambda^{N-1} + \lambda^{N-2} + \ldots + \lambda + 1\right). \end{aligned}$$

The necessary and sufficient condition for $Var(x_{Nt})$ to be bounded in N is given by $\alpha^2 + \beta^2 < 1$. Therefore, the condition $|\alpha| < 1$ is not sufficient if $N \to \infty$. The condition $\varrho\left(\boldsymbol{\Theta}\boldsymbol{\Theta}'\right) < 1 - \epsilon$ implies $\alpha^2 + \beta^2 < 1$, and is therefore sufficient (and in this example it is also necessary) for $Var(x_{Nt})$ to be bounded in N.

Similarly, as in DdPS, it is assumed in (33.26) that factors are included in the *VAR* model in an additive way so that $\mathbf{x}_t$ can be written as

$$\mathbf{x}_t = \boldsymbol{\Gamma}\mathbf{f}_t + \boldsymbol{\xi}_t, \tag{33.27}$$

where $\boldsymbol{\xi}_t = (\mathbf{I}_k - \boldsymbol{\Theta}L)^{-1}\mathbf{u}_t$, and the existence of the inverse of $(\mathbf{I}_k - \boldsymbol{\Theta}L)$ is ensured by the assumption on $\varrho\left(\boldsymbol{\Theta}\boldsymbol{\Theta}'\right)$ above. One can also consider the alternative factor augmentation setup,

$$\mathbf{x}_t = \boldsymbol{\Theta}\mathbf{x}_{t-1} + \boldsymbol{\Gamma}\mathbf{f}_t + \mathbf{u}_t, \tag{33.28}$$

where factors are added to the errors of the *VAR* model, instead of (33.26), where deviations of $\mathbf{x}_t$ from the factors are modelled as a *VAR*. But it is important to note that both specifications, (33.26) and (33.28), yield similar asymptotic results. The main difference between the two formulations lies in the fact that the factor error structure in (33.28) results in infinite-order distributed lag polynomials (as large *N* representation for cross-section averages and individual units), whilst the specification (33.26) yields finite-order lag representations. In the case of (33.28), the infinite lag-order polynomials must be appropriately truncated for the purposes of

consistent estimation and inference, as in Berk (1974), Said and Dickey (1984) and Chudik and Pesaran (2013, 2015a).

For any set of weights represented by the $k \times k^*$ matrix $\tilde{\mathbf{W}}_i$ we obtain (using (33.27))

$$\mathbf{x}_{it}^* = \tilde{\mathbf{W}}_i' \mathbf{x}_t = \mathbf{\Gamma}_i^* \mathbf{f}_t + \boldsymbol{\xi}_{it}^*,$$

where $\mathbf{\Gamma}_i^* = \tilde{\mathbf{W}}_i' \mathbf{\Gamma}$ and

$$\boldsymbol{\xi}_{it}^* = \tilde{\mathbf{W}}_i' \left(\mathbf{I}_k - \mathbf{\Theta} L\right)^{-1} \mathbf{u}_t.$$

CP show that if $\tilde{\mathbf{W}}_i$ satisfies (33.23), then

$$\begin{aligned} \left\| E\left(\boldsymbol{\xi}_{it}^* \boldsymbol{\xi}_{it}^{*\prime}\right) \right\| &= \left\| \sum_{\ell=0}^{\infty} \tilde{\mathbf{W}}_i' \mathbf{\Theta}^{\ell} E\left(\mathbf{u}_{t-\ell} \mathbf{u}_{t-\ell}'\right) \mathbf{\Theta}^{\ell} \tilde{\mathbf{W}}_i \right\| \\ &\leq \left\| \tilde{\mathbf{W}}_i \right\|^2 \left\| \mathbf{\Sigma}_u \right\| \sum_{\ell=0}^{\infty} \left\| \mathbf{\Theta}^{\ell} \right\|^2 \\ &= O\left(N^{-1}\right), \end{aligned} \tag{33.29}$$

where $\left\| \tilde{\mathbf{W}}_i \right\|^2 = O\left(N^{-1}\right)$ by (33.23), $\left\| \mathbf{\Sigma}_u \right\| < K$ by the weak cross-sectional dependence assumption, and $\sum_{\ell=0}^{\infty} \left\| \mathbf{\Theta}^{\ell} \right\|^2 < K$ by the assumption on spectral radius of $\varrho\left(\mathbf{\Theta}\mathbf{\Theta}'\right)$. (33.29) establishes that $\boldsymbol{\xi}_{it}^* \overset{q.m.}{\to} \mathbf{0}$ (uniformly in i and t) as $N, T \to_j \infty$. It now follows that

$$\mathbf{x}_{it}^* - \mathbf{\Gamma}_i^* \mathbf{f}_t \overset{q.m.}{\to} \mathbf{0}, \text{ as } N, T \to_j \infty, \tag{33.30}$$

which confirms the well-known result that only strong cross-sectional dependence can survive large N aggregation with granular weights (see Section 32.5). Therefore, the unobserved common factors can be approximated by cross-section averages $\mathbf{x}_{it}^*$ in this dynamic setting, provided that $\mathbf{\Gamma}_i^*$ has full column rank.

It is now easy to see what additional requirements are needed on the coefficient matrix $\mathbf{\Theta}$ to obtain country *VARX** models in (33.3) when N is large. The model for the country specific variables, $\mathbf{x}_{it}$, from the system (33.26) is given by

$$\mathbf{x}_{it} = \mathbf{\Theta}_{ii} \mathbf{x}_{it-1} + \sum_{j=1, j \neq i} \mathbf{\Theta}_{ij} \left(\mathbf{x}_{j,t-1} - \mathbf{\Gamma}_j \mathbf{f}_t\right) + \mathbf{\Gamma}_i \mathbf{f}_t - \mathbf{\Theta}_i' \mathbf{\Gamma}_i f_{t-1} + \mathbf{u}_{it}, \tag{33.31}$$

where $\mathbf{\Theta}_{ij}$ are appropriate partitioned sub-matrices of

$$\mathbf{\Theta} = \begin{pmatrix} \mathbf{\Theta}_{11} & \mathbf{\Theta}_{12} & \cdots & \mathbf{\Theta}_{1N} \\ \mathbf{\Theta}_{21} & \mathbf{\Theta}_{22} & \cdots & \mathbf{\Theta}_{2N} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{\Theta}_{N1} & \mathbf{\Theta}_{N2} & \cdots & \mathbf{\Theta}_{NN} \end{pmatrix}.$$

Suppose now that

$$\left\|\mathbf{\Theta}_{ij}\right\| < \frac{K}{N}, \text{for all } i \neq j. \tag{33.32}$$

This assumption implies that the matrix $\mathbf{\Theta}_{-i} = \left(\mathbf{\Theta}_{i1}, \mathbf{\Theta}_{i2}, \ldots, \mathbf{\Theta}_{i,i-1}, 0, \mathbf{\Theta}_{i,i+1}, \ldots, \mathbf{\Theta}_{iN}\right)'$ satisfies the granularity condition (33.23), in particular $\|\mathbf{\Theta}_{-i}\|^2 < KN^{-1}$, and using (33.29) but with $\mathbf{\Theta}_{-i}$ instead of $\tilde{\mathbf{W}}_i$, we obtain

$$\sum_{j=1,j\neq i} \mathbf{\Theta}_{ij}\left(\mathbf{x}_{j,t-1} - \mathbf{\Gamma}_j \mathbf{f}_t\right) \xrightarrow{q.m.} \mathbf{0}, \text{ as } N \to \infty. \tag{33.33}$$

Finally, substituting (33.30) and (33.33) in (33.31) we obtain the country-specific *VARX** (1, 1) model

$$\mathbf{x}_{it} - \mathbf{\Theta}_{ii}\mathbf{x}_{it-1} - \mathbf{\Lambda}_{i0}\mathbf{x}_{it}^* - \mathbf{\Lambda}_{i1}\mathbf{x}_{i,t-1}^* - \mathbf{u}_{it} \xrightarrow{q.m.} \mathbf{0} \text{ uniformly in } i, \text{ and as } N \to \infty, \tag{33.34}$$

where

$$\mathbf{\Lambda}_{i0} = \mathbf{\Gamma}_i\left(\mathbf{\Gamma}^{*\prime}\mathbf{\Gamma}^*\right)^{-1}\mathbf{\Gamma}^*, \text{and } \mathbf{\Lambda}_{i1} = \mathbf{\Theta}_i'\mathbf{\Gamma}_i\left(\mathbf{\Gamma}^{*\prime}\mathbf{\Gamma}^*\right)^{-1}\mathbf{\Gamma}^*.$$

Requirement (33.32) together with the remaining assumptions in this sub-section, is thus sufficient to obtain (33.3) when N is large. In addition to the derivations of large N representations of the individual country models, CP also show that the coefficient matrices $\mathbf{\Theta}_{ii}$, $\mathbf{\Lambda}_{i0}$ and $\mathbf{\Lambda}_{i1}$ can be consistently estimated under the joint asymptotics when N and $T \to \infty$, jointly, plus a number of further assumptions as set out in CP.

It is also important to consider the consequences of relaxing the restrictions in (33.32). One interesting case is when units have 'neighbours' in the sense that there exist some country pairs $j \neq i$ for which $\left\|\mathbf{\Theta}_{ij}\right\|$ remains non-negligible as $N \to \infty$. Another interesting departure from the above assumptions is when $\|\mathbf{\Sigma}_u\|$ is not bounded in N, and there exists a dominant unit j for which $\left\|\mathbf{\Theta}_{ij}\right\|$ is non-negligible for the other units, $i \in S_j \subseteq \{1, 2, \ldots, N\}$. These scenarios are investigated in Chudik and Pesaran (2011, 2013), and they lead to different specifications of the country-specific models featuring additional variables. To improve estimation and inference in such cases one can combine the *GVAR* approach with various penalized shrinkage methods such as Bayesian shrinkage (Ridge), Lasso or other related techniques where the estimation is subject to penalty, which becomes increasingly more binding as the number of parameters is increased.[11]

33.5 Conducting impulse response analysis with *GVAR*s

We have seen that under plausible conditions country-specific models can be obtained as large N approximations to global factor-augmented models of different forms. Moreover, individual

[11] LASSO and Ridge regressions are discussed in Sections 11.9 and C.7 in Appendix C. Feldkircher et al. (2014) implement a number of Bayesian priors (the normal-conjugate prior, a non-informative prior on the coefficients and the variance, the inverse Wishart prior, the Minnesota prior, the single-unit prior, which accommodates potential cointegration relationships, and the stochastic search variable selection prior) in estimating country-specific models in the *GVAR*.

country-specific models can be consistently estimated. In this section, we discuss conducting impulse response analysis with *GVAR*s. The analysis of impulse responses is subject to the same issues as in the small-scale *VAR*s discussed in Chapter 24, but is further complicated due to the dimensionality of the *GVAR* model.

For expositional convenience initially suppose that the *DGP* is given by (33.8). This model features $k = \sum_{i=1}^{N} k_i$ country-specific errors collected in the vector $\boldsymbol{\varepsilon}_t = \left(\boldsymbol{\varepsilon}'_{1t}, \boldsymbol{\varepsilon}'_{2t}, \ldots, \boldsymbol{\varepsilon}'_{Nt}\right)'$, and there are no common variables included in the model. Suppose also that there are k distinct *structural* (orthogonal) shocks. Identification of structural shocks, defined by $\mathbf{v}_t = \mathbf{P}^{-1}\boldsymbol{\varepsilon}_t$, requires finding the $k \times k$ matrix of contemporaneous dependence, $\mathbf{P}$, such that

$$\boldsymbol{\Sigma} = E\left(\boldsymbol{\varepsilon}_t \boldsymbol{\varepsilon}'_t\right) = \mathbf{P}\mathbf{P}'. \tag{33.35}$$

Therefore, by construction, we have $E\left(\mathbf{v}_t \mathbf{v}'_t\right) = \mathbf{I}_k$, and the $k \times 1$ vector of structural impulse response functions is given by

$$\begin{aligned} \mathbf{g}_{vj}(h) &= E\left(\mathbf{x}_{t+h} | \, v_{jt} = 1, \, \mathcal{I}_{t-1}\right) - E\left(\mathbf{x}_{t+h} | \, \mathcal{I}_{t-1}\right), \\ &= \frac{\mathbf{R}_h \mathbf{G}_0^{-1} \mathbf{P} \mathbf{e}_j}{\sqrt{\mathbf{e}'_j \boldsymbol{\Sigma} \mathbf{e}_j}}, \end{aligned} \tag{33.36}$$

for $j = 1, 2, \ldots, k$, where $\mathcal{I}_t = \{\mathbf{x}_t, \mathbf{x}_{t-1}, \ldots\}$ is the information set consisting of all available information at time t, and $\mathbf{e}_j$ is a $k \times 1$ selection vector that selects the variable j, and the $k \times k$ matrices, $\mathbf{R}_h$, are obtained recursively as (see (33.18))

$$\mathbf{R}_h = \sum_{\ell=1}^{p} \mathbf{F}_\ell \mathbf{R}_{h-\ell} \text{ with } \mathbf{R}_0 = \mathbf{I}_k \text{ and } \mathbf{R}_\ell = \mathbf{0} \text{ for } \ell < 0.$$

Expectation operators in (33.36) are taken assuming that the *GVAR* model (33.8) is the *DGP*. Decomposition (33.35) is not unique and identification of shocks requires $k(k-1)/2$ restrictions, which is of order $O\left(k^2\right)$.[12] Even for moderate values of k, motivating such a large number of restrictions is problematic, especially given that the existing macroeconomic literature focuses mostly on distinguishing between different types of shocks (e.g., monetary policy shocks, fiscal shocks, technology shocks, etc.), and does not provide a thorough guidance on how to identify country origins of shocks, which is necessary to identify all the shocks in the *GVAR* model.

One possible approach to the identification of the shocks is orthogonalized IR analysis of Sims (1980), who consider setting $\mathbf{P}$ to the Choleski factor of $\boldsymbol{\Sigma}$ (see Section 24.4). But, as is well known, the choice of the Choleski factor is not unique and depends on the ordering of variables in the vector $\mathbf{x}_t$. Such an ordering is clearly difficult to entertain in the global setting, but partial ordering could be considered to identify a single shock or a subset of shocks. This is, for example, accomplished by Dées et al. (2007) who identify the US monetary policy shock (by assuming that the US variables come first, and two different orderings for the vector of the US variables are considered). Another well-known possibility to identify shocks in reduced-form *VAR*s includes the work of Bernanke (1986), Blanchard and Watson (1986), and Sims (1986) who consider

[12] This corrects the statement in Pesaran et al. (2004, p. 136).

a priori restrictions on the contemporaneous covariance matrix of shocks; Blanchard and Quah (1989) and Clarida and Gali (1994) who consider restrictions on the long-run impact of shocks to identify the impulse responses; and the sign-restriction approach considered, among others, in Faust (1998), Canova and Pina (1999), Canova and de Nicolò (2002), Uhlig (2005), Mountford and Uhlig (2009), and Inoue and Kilian (2013). Identification of shocks in a *GVAR* is subject to the same issues as in standard *VAR*s (see Chapter 24), but is further complicated due to the cross-country interactions and the high dimensionality of the model. Dées et al. (2014) provide a detailed discussion of the identification and estimation of the *GVAR* model subject to theoretical constraints.

In view of these difficulties, Pesaran et al. (2004), Pesaran and Smith (2006), Dées et al. (2007) and the subsequent literature mainly adopt the generalized *IRF* (*GIRF*) approach, advanced in Koop et al. (1996), Pesaran and Shin (1998) and Pesaran and Smith (1998) (see also Section 24.5). The *GIRF* approach does not aim at identification of shocks according to some canonical system or *a priori* economic theory, but considers a counterfactual exercise where the historical correlations of shocks are assumed as given. In the context of the *GVAR* model (33.8) the $k \times 1$ vector of *GIRF*s is given by

$$\begin{aligned}\mathbf{g}_{\varepsilon j}(h) &= E(\mathbf{x}_{t+h} |\, \varepsilon_{jt} = \sqrt{\sigma_{jj}},\, \mathcal{I}_{t-1}) - E(\mathbf{x}_{t+h} |\, \mathcal{I}_{t-1}), \\ &= \frac{\mathbf{R}_h \mathbf{G}_0^{-1} \boldsymbol{\Sigma} \mathbf{e}_j}{\sqrt{\mathbf{e}_j' \boldsymbol{\Sigma} \mathbf{e}_j}},\end{aligned} \tag{33.37}$$

for $j = 1, 2, \ldots, k$, $h = 0, 1, 2, \ldots$, where $\sigma_{jj} = E\left(\varepsilon_{jt}^2\right)$ is the size of the shock, which is set to one standard deviation (s.d.) of ε_{jt}.[13] The *GIRF*s can also be obtained for (synthetic) 'global' or 'regional' shocks, defined by $\varepsilon_{m,t}^g = \mathbf{m}' \boldsymbol{\varepsilon}_t$, where the vector of weights, $\mathbf{m}$, relates to a global aggregate or a particular region. The vector of *GIRF* for the global shock, $\varepsilon_{m,t}^g$, is

$$\begin{aligned}\mathbf{g}_m(h) &= E\left(\mathbf{x}_{t+h} |\, \varepsilon_{m,t}^g = \sqrt{\mathbf{m}' \boldsymbol{\Sigma} \mathbf{m}},\, \mathcal{I}_{t-1}\right) - E(\mathbf{x}_{t+h} |\, \mathcal{I}_{t-1}), \\ &= \frac{\mathbf{R}_h \mathbf{G}_0^{-1} \boldsymbol{\Sigma} \mathbf{m}}{\sqrt{\mathbf{m}' \boldsymbol{\Sigma} \mathbf{m}}}.\end{aligned} \tag{33.38}$$

Closely related to the impulse-response analysis is the forecast-error variance decomposition, which shows the relative contributions of the shocks to reducing the mean square error of forecasts of individual endogenous variables at a given horizon h (see also Section 24.7). In the case of orthogonalized shocks, $\mathbf{v}_t = \mathbf{P}^{-1} \boldsymbol{\varepsilon}_t$, and assuming for simplicity of exposition that $m_\omega = 0$, the contribution of the jth innovation, v_{jt}, to the mean square error of the h-step ahead forecast of x_{it} is

$$\mathcal{SFEVD}\left(x_{it}, v_{jt}, h\right) = \frac{\sum_{\ell=0}^{h} \left[\mathbf{e}_i' \mathbf{R}_\ell \mathbf{G}_0^{-1} \mathbf{P} \mathbf{e}_j\right]^2}{\sum_{\ell=0}^{h} \mathbf{e}_i' \mathbf{R}_\ell \mathbf{G}_0^{-1} \boldsymbol{\Sigma} \mathbf{G}_0^{\prime -1} \mathbf{R}_\ell' \mathbf{e}_i},$$

[13] Estimation and inference on impulse responses can be conducted by bootstrapping, see Dées et al. (2007) for details.

and since the shocks are orthogonal, it follows that $\sum_{j=1}^{N} \mathcal{SFEVD}\left(x_{it}, v_{jt}, h\right) = 1$ for any i and h. In the case of non-orthogonal shocks, the forecast-error variance decompositions need not sum to unity. Analogously to the *GIRF*s, generalized forecast error variance decomposition of generalized shocks can be obtained as

$$\mathcal{GFEVD}\left(x_{it}, \varepsilon_{jt}, h\right) = \frac{\sigma_{jj}^{-1} \sum_{\ell=0}^{h} \left[\mathbf{e}_i' \mathbf{R}_\ell \mathbf{G}_0^{-1} \boldsymbol{\Sigma} \mathbf{e}_j\right]^2}{\sum_{\ell=0}^{h} \mathbf{e}_i' \mathbf{R}_\ell \mathbf{G}_0^{-1} \boldsymbol{\Sigma} \mathbf{G}_0^{\prime -1} \mathbf{R}_\ell' \mathbf{e}_i}.$$

33.6 Forecasting with *GVAR*s

Forecasting is another important application of the *GVAR* approach, which provides a viable alternative to other methods developed for data-sets with a large number of predictors. A difference between *GVAR* and other data-rich forecasting methods is that *GVAR* utilizes the structure of the panel, which is assumed to consist of many cross-section units (e.g., countries) with each cross-sectional unit consisting of a small number of variables. Other data-rich methods, such as Lasso, Ridge, or elastic net (see for instance Tibshirani (1996), De Mol et al. (2008), and Hastie et al. (2009)), popular factor models (Geweke (1977), Sargent and Sims (1977), and other contributions),[14] or partial least squares (Wold (1982)) do not typically utilize such a structure. See Eklund and Kapetanios (2008) and Groen and Kapetanios (2008) for recent surveys of data-rich forecasting methods and Chapter 17 for an introduction to forecasting.

As in Section 33.5, we shall assume that the *DGP* is given by *GVAR* model (33.8). Taking expectations of both sides of (33.8) for $t = t_0 + h$, conditional on the information set Ω_{t_0}, we obtain

$$E\left(\mathbf{x}_{t_0+h} \middle| \Omega_{t_0}\right) = \sum_{\ell=1}^{p} \mathbf{F}_\ell E\left(\mathbf{x}_{t_0+h-\ell} \middle| \Omega_{t_0}\right) + \mathbf{G}_0^{-1} E\left(\boldsymbol{\varepsilon}_{t_0+h} \middle| \Omega_{t_0}\right), \tag{33.39}$$

for any $h = 0, 1, 2, \ldots$. In the case when the conditioning information set Ω_{t_0} is given by all available information up to the period t_0, $\Omega_{t_0} = \mathcal{I}_{xt_0} \equiv \left\{\mathbf{x}_{t_0}, \mathbf{x}_{t_0-1}, \ldots\right\}$ we have

$$E\left(\boldsymbol{\varepsilon}_{t_0+h} \middle| \mathcal{I}_{xt_0}\right) = \mathbf{0}, \text{for } h > 0, \tag{33.40}$$

and standard forecasts $E\left(\mathbf{x}_{t_0+h} \middle| \mathcal{I}_{xt_0}\right)$ can be easily computed from (33.8) recursively using the estimates of $\mathbf{F}_\ell$ and $\mathbf{G}_0^{-1}$, and noting that (33.40) holds and $E\left(\mathbf{x}_{t'} \middle| \mathcal{I}_{xt_0}\right) = \mathbf{x}_{t'}$ for all $t' \leq t_0$. Forecasts from model (33.18) featuring observed common variables can be obtained in a similar way.

Generating conditional forecasts for non-standard conditioning information sets with mixed information on (future, present, and past values of) variables in the panel is more challenging. This situation could arise, for instance, in the case where data for different variables are released at

[14] See also Forni and Lippi (2001), Forni et al. (2000, 2004), Stock and Watson (1999, 2002, 2005), Giannone, Reichlin, and Sala (2005), and Bai and Ng (2007).

different dates, or when mixed information sets are intentionally considered to answer specific questions as in Bussière et al. (2012). Without loss of generality, and for expositional convenience, suppose, for some date t', that the first k_a variables in the vector $\mathbf{x}_{t'}$ belong to Ω_{t_0} and the remaining $k_b = k - k_a$ variables do not, and partition $\boldsymbol{\varepsilon}_t$ as $\boldsymbol{\varepsilon}_t = \left(\boldsymbol{\varepsilon}'_{at}, \boldsymbol{\varepsilon}'_{bt}\right)'$, and the associated covariance matrix, $\boldsymbol{\Sigma} = E\left(\boldsymbol{\varepsilon}_t \boldsymbol{\varepsilon}'_t\right)$ as

$$\boldsymbol{\Sigma} = \begin{pmatrix} \boldsymbol{\Sigma}_{aa} & \boldsymbol{\Sigma}_{ab} \\ \boldsymbol{\Sigma}_{ba} & \boldsymbol{\Sigma}_{bb} \end{pmatrix}. \tag{33.41}$$

It then follows that $E\left(\boldsymbol{\varepsilon}_{at'} | \, \Omega_{t_0}\right) = \boldsymbol{\varepsilon}_{at'}$, whereas $E\left(\boldsymbol{\varepsilon}_{bt'} | \, \Omega_{t_0}\right) = \boldsymbol{\Sigma}_{ba}\boldsymbol{\Sigma}_{aa}^{-1}\boldsymbol{\varepsilon}_{at'}$. Let $\hat{\boldsymbol{\Sigma}}$ be an estimate of $\boldsymbol{\Sigma}$, then an estimate of $E\left(\boldsymbol{\varepsilon}_{t'} | \, \Omega_{t_0}\right)$ can be computed as

$$\hat{E}\left(\boldsymbol{\varepsilon}_{t'} | \, \Omega_{t_0}\right) = \begin{pmatrix} \hat{\boldsymbol{\varepsilon}}_{at'} \\ \hat{\boldsymbol{\Sigma}}_{ba}\hat{\boldsymbol{\Sigma}}_{aa}^{-1}\hat{\boldsymbol{\varepsilon}}_{at'} \end{pmatrix}.$$

for any given $t' \leq t_0 + h$. The conditional forecasts $E\left(\mathbf{x}_{t_0+h} \middle| \, \Omega_{t_0}\right)$ can then be computed recursively as in (33.39). One problem is that $\boldsymbol{\Sigma}$ and its four sub-matrices in (33.41) can have large dimensions relative to the available number of time series observations, and therefore it is not guaranteed that $\hat{\boldsymbol{\Sigma}}_{aa}$ will be invertible. Even if it were, the inverse of the traditional estimate of variance-covariance matrices does not necessarily have good small sample properties when the number of variables is large. For these reasons, it is desirable to make use of other covariance matrix estimators with better small sample properties. There are several estimators proposed in the literature for estimation of high-dimensional covariance matrices, including Ledoit and Wolf (2004), Bickel and Levina (2008), Fan et al. (2008), Friedman et al. (2008), the shrinkage estimator considered in Dées et al. (2014), and the multiple testing approach by Bailey et al. (2015).

The implicit assumption in construction of the *GVAR* model (33.8) is invertibility of $\mathbf{G}_0$, which ensures that the model is complete as discussed in Section 33.3.1. If $\mathbf{G}_0$ is not invertible, then the system of country-specific equations is incomplete and it needs to be augmented with additional equations. This possibility is considered in Chudik, Grossman, and Pesaran (2014) who consider forecasting with *GVAR*s in the case when $N, T \underset{j}{\to} \infty$, and the *DGP* is given by a factor-augmented infinite-dimensional *VAR* model considered by *CP* and outlined above in Section 33.4.2. For simplicity of exposition, consider a large dimensional *VAR* with one variable per country ($k_i = 1$) and one unobserved common factor ($m = 1$) generated as

$$f_t = \rho f_{t-1} + \eta_{ft}, \tag{33.42}$$

in which $|\rho| < 1$ and the macro shock, η_{ft}, is serially uncorrelated and distributed with zero mean and variance σ_η^2. Let the factor loadings be denoted by $\boldsymbol{\gamma} = \left(\gamma_1, \gamma_2, \ldots, \gamma_N\right)'$, and consider the granular weights vector $\mathbf{w} = (w_1, w_2, \ldots, w_N)'$ that defines the cross-section averages $x_{it}^* = x_t^* = \mathbf{w}'\mathbf{x}_t$ (assumed to be identical across countries). In this simple setting the *GVAR* model can be written as (see (33.34))

$$x_{it} = \phi_{ii}x_{i,t-1} + \lambda_{i0}x_t^* + \lambda_{i1}x_{t-1}^* + u_{it} + O_p\left(N^{-1/2}\right), \text{for } i \in \{1, 2, \ldots, N\}, \tag{33.43}$$

where $\lambda_{i0} = \gamma_i/\gamma^*, \lambda_{i1} = -\phi_{ii}\gamma_i/\gamma^*$, and $\gamma^* = \mathbf{w}'\boldsymbol{\gamma}$. Denote the corresponding least squares estimates of the unknown coefficients by hats, namely $\hat{\phi}_{ii}$, $\hat{\lambda}_{i0}$ and $\hat{\lambda}_{i1}$. These estimates are consistent and asymptotically normally distributed (see CP). Note that (33.43) consists of N different equations. Therefore, using the estimates $\hat{\phi}_{ii}$, $\hat{\lambda}_{i0}$ and $\hat{\lambda}_{i1}$, for $i = 1, 2, \ldots, N$, and provided that matrix $\hat{\mathbf{G}}_0 = \mathbf{I}_N - \hat{\boldsymbol{\Lambda}}_0\tilde{\mathbf{W}}'$ is invertible, where $\hat{\boldsymbol{\Lambda}}_0 = diag\left(\hat{\lambda}_{10}, \hat{\lambda}_{20}, \ldots, \hat{\lambda}_{N0}\right)$, $\tilde{\mathbf{W}} = \boldsymbol{\tau}\mathbf{w}'$, $\boldsymbol{\tau}$ is $N \times 1$ vector of ones, one can obtain the following *GVAR* model

$$\mathbf{x}_t = \hat{\mathbf{F}}\mathbf{x}_{t-1} + \hat{\mathbf{G}}_0^{-1}\hat{\boldsymbol{\varepsilon}}_t, \tag{33.44}$$

where $\hat{\mathbf{F}} = \hat{\mathbf{G}}_0^{-1}\hat{\mathbf{G}}_1, \hat{\mathbf{G}}_1 = \hat{\boldsymbol{\Phi}} + \hat{\boldsymbol{\Lambda}}_1\tilde{\mathbf{W}}', \hat{\boldsymbol{\Lambda}}_1 = diag\left(\hat{\lambda}_{11}, \hat{\lambda}_{21}, \ldots, \hat{\lambda}_{N1}\right)$, and $\hat{\boldsymbol{\Phi}} = diag\left(\hat{\phi}_{11}, \hat{\phi}_{22}, \ldots, \hat{\phi}_{NN}\right)$. However, in this setup it is not optimal to use (33.44) for forecasting for the following two reasons. First, $\mathbf{G}_0 = \mathbf{I}_N - \hat{\boldsymbol{\Lambda}}_0\tilde{\mathbf{W}}'$ is by construction rank deficient; to see this note that

$$\begin{aligned}\mathbf{w}'\mathbf{G}_0 &= \mathbf{w}'\left(\mathbf{I}_N - \hat{\boldsymbol{\Lambda}}_0\tilde{\mathbf{W}}'\right)\\ &= \mathbf{w}' - \mathbf{w}'\hat{\boldsymbol{\Lambda}}_0\boldsymbol{\tau}\mathbf{w}',\end{aligned}$$

and recalling that $\sum_{i=1}^{N} w_i\gamma_i = \gamma^*$, we have

$$\mathbf{w}'\mathbf{G}_0 = \mathbf{w}' - \sum_{i=1}^{N}\left(\frac{w_i\gamma_i}{\gamma^*}\right)\mathbf{w}' = \mathbf{w}' - \mathbf{w}' = \mathbf{0}',$$

which establishes that $\mathbf{G}_0$ has a zero eigenvalue. Since $\mathbf{G}_0$ is singular, the system of equations (33.43) is not complete and it is unclear what the properties of $\hat{\mathbf{G}}_0^{-1}$ are, given that the individual elements of $\hat{\mathbf{G}}_0$ are consistent estimates of the elements of $\mathbf{G}_0$. Second, the parameters in the conditional models $\left\{\phi_{ii}, \lambda_{i0}, \lambda_{i1}\right\}_{i=1}^{N}$ do not contain information about the persistence of unobserved common factor, ρ, due to the conditional nature of these models.

Chudik, Grossman, and Pesaran (2014) consider augmenting (33.43) with a set of equations for the cross-section averages. In the present example we consider augmenting the *GVAR* model, (33.43), with the following equation

$$x_t^* = \rho x_{t-1}^* + \gamma\eta_{ft} + O_p\left(N^{-1/2}\right), \tag{33.45}$$

where x_t^* is treated as a proxy for the (scaled) unobserved common factor. See (33.30). Combining (33.43) and (33.45), the following augmented *VAR* model in $\mathbf{z}_t = \left(\mathbf{x}_t', x_t^*\right)'$ is obtained

$$\mathbf{B}_0\mathbf{z}_t = \mathbf{B}_1\mathbf{z}_{t-1} + \mathbf{u}_{zt} + O_p\left(N^{-1/2}\right), \tag{33.46}$$

where $\mathbf{u}_{zt} = \left(\mathbf{u}_t', \gamma\eta_{ft}\right)'$,

$$\mathbf{B}_0 = \begin{pmatrix}\mathbf{I}_N & -\boldsymbol{\lambda}_0\\ \mathbf{0}' & 1\end{pmatrix}, \mathbf{B}_1 = \begin{pmatrix}\boldsymbol{\Phi} & \boldsymbol{\lambda}_1\\ \mathbf{0}' & \rho\end{pmatrix},$$

and $\mathbf{\Phi}$ is an $N \times N$ diagonal matrix with elements ϕ_{ii}, for $i = 1, 2, \ldots, N$, on the diagonal. The matrix $\mathbf{B}_0$ is by construction invertible. The feasible optimal forecast based on (33.46) and conditional on $\mathcal{I}_{\mathbf{x}t} = \{\mathbf{x}_t, \mathbf{x}_{t-1}, \ldots\}$ is given by

$$\mathbf{x}_{t+h}^f = \mathbf{B}^h \mathbf{z}_t, \tag{33.47}$$

where

$$\mathbf{B} = \mathbf{B}_0^{-1}\mathbf{B}_1 = \begin{pmatrix} \mathbf{I}_N & \boldsymbol{\lambda}_0 \\ \mathbf{0}' & 1 \end{pmatrix} \begin{pmatrix} \mathbf{\Phi} & \boldsymbol{\lambda}_1 \\ \mathbf{0}' & \rho \end{pmatrix} = \begin{pmatrix} \mathbf{\Phi} & \boldsymbol{\lambda}_1 + \rho\boldsymbol{\lambda}_0 \\ \mathbf{0}' & \rho \end{pmatrix}.$$

Consider now the infeasible optimal forecasts obtained using the factor-augmented infinite-dimensional *VAR* model, (33.26), one factor given by (33.42), and conditional on the combined information set $\mathcal{I}_{\mathbf{x}t} \cup \mathcal{I}_f = \left\{\mathbf{x}_t, f_t; \mathbf{x}_{t-1}, f_{t-1}; \ldots\right\}$

$$E\left(\mathbf{x}_{t+h} | \mathcal{I}_{\mathbf{x}t} \cup \mathcal{I}_f\right) = \mathbf{\Theta}^h \mathbf{x}_t + \left(\rho^h \mathbf{I}_N - \mathbf{\Theta}^h\right) \boldsymbol{\gamma} f_t. \tag{33.48}$$

Chudik, Grossman, and Pesaran (2014) show that

$$\mathbf{x}_{t+h}^f = \mathbf{B}^h \mathbf{z}_t = \mathbf{\Theta}^h \mathbf{x}_t + \left(\rho^h \mathbf{I}_N - \mathbf{\Theta}^\ell\right) \boldsymbol{\gamma} f_t + O_p\left(N^{-1/2}\right),$$

namely $\mathbf{B}^h \mathbf{z}_t \to E\left(\mathbf{x}_{t+h} | \mathcal{I}_{\mathbf{x}t} \cup \mathcal{I}_f\right)$, the infeasible optimal forecasts, as $N \to \infty$.

Even when $\mathbf{G}_0$ is invertible, it is possible that augmentation of the *GVAR* by equations for cross-section averages leads to forecast improvements. Note that the *GVAR* model (33.8) does not feature an unobserved factor error structure. We have seen that a sufficient number of cross-section averages in the individual country-specific conditional models in (33.3) takes care of the effects of any strongly cross-sectionally dependent processes that enter as unobserved common factors for the purpose of estimation of country-specific coefficients. Inclusion of a sufficient number of cross-section averages will also lead to weak cross-section dependence of the vector of errors $\boldsymbol{\varepsilon}_t$ in the country-specific models. But since the reduced form innovations $\mathbf{G}_0^{-1}\boldsymbol{\varepsilon}_t$ must be strongly cross-sectionally dependent when a strong factor is present in $\mathbf{x}_t$, then it follows that $\mathbf{G}_0^{-1}$ (if it exists) cannot have bounded spectral matrix norm in N. Forecasts based on the augmented *GVAR* model avoid the need for inversion of high-dimensional matrices. Monte Carlo findings reported in Chudik, Grossman, and Pesaran (2014) suggest that augmentation of the *GVAR* by equations for cross-section averages does not hurt when $\mathbf{G}_0$ is invertible, while it can considerably improve forecasting performance when $\mathbf{G}_0$ is singular.

The majority of applications of the *GVAR* approach in the literature are concerned with modelling of the global economy. Therefore, a brief discussion of important issues in forecasting the global economy is in order. There are two important issues in particular: the presence of structural breaks and model uncertainty. Structural breaks are quite likely, considering the diverse set of economies and the time period spanning three or more decades, which covers a lot of historical events (financial crises, wars, regime changes, natural disasters, etc.). The timing and the magnitude of breaks and the underlying *DGP* are not exactly known, which complicates the forecasting problem. Pesaran, Schuermann, and Smith (2009a) address both problems by

using a forecast combination method. They considered simple averaging across selected models (AveM) and estimation windows (AveW) as well as across both dimensions, models and windows (AveAve); and obtain evidence of superior performance for their double-average (AveAve) forecasts. These and other forecasting evidence are reviewed in more detail in the next section. Forecast evaluation in the *GVAR* model is also challenging due to the fact that the multi-horizon forecasts obtained from the *GVAR* model could be cross-sectionally as well as serially dependent. One test statistic to evaluate forecasting performance of the *GVAR* model is proposed by Pesaran, Schuermann, and Smith (2009a) who develop a panel version of the Diebold and Mariano (1995, DM) DM test assuming cross-sectional independence.

33.7 Long-run properties of *GVAR*s

33.7.1 Analysis of the long run

Individual country *VARX** models in (33.3) allow for cointegration among domestic variables as well as between domestic and country-specific cross-sectional averages of foreign variables. Let $\mathbf{z}_{it} = (\mathbf{x}_{it}', \mathbf{x}_{it}^{*\prime})'$ be a $(k_i + k^*) \times 1$ vector of domestic and country-specific foreign variables for country i, and denote r_i cointegrating relations among the variables in the vector $\mathbf{z}_{it}$ as $\boldsymbol{\beta}_i'\mathbf{z}_{it}$, where $\boldsymbol{\beta}_i$ is a $(k_i + k^*) \times r_i$ dimensional matrix consisting of r_i cointegrating vectors. The overall number of cointegrating vectors in the stacked *GVAR* model is naturally reflected in the eigenvalues of the companion representation of the *GVAR* model. These eigenvalues characterize the dynamic properties of the model which can also be used to examine the overall stability of the *GVAR*. In particular, when the overall number of cointegrating relations is $r = \Sigma_{i=1}^{N} r_i$, then $k-r$ eigenvalues of the *GVAR* model fall on the unit circle, and the remaining eigenvalues fall within the unit circle for the model to be stable.

Testing for the number of cointegrating vectors

Testing for the number of cointegrating relations can be conducted using Johansen's trace and maximum eigenvalue test statistics as set out in Pesaran et al. (2000) for models with weakly exogenous $I(1)$ regressors (see Chapter 23). Small sample evidence typically suggests that the trace test performs better than the maximum eigenvalue test, but both are subject to the usual size distortions when the time dimension is not sufficiently large. Selecting the number of cointegrating vectors is important, since misspecification of the rank of the cointegrating space can have a severe impact on the performance of the resulting *GVAR* model, with adverse implications for stability, persistence profiles, and impulse responses.

Identification of long-run relations

Once the number of cointegrating vectors is determined, it is possible to proceed with the identification of long-run structural relations and, if desired, to impose over-identifying restrictions, See Chapter 23 for details. These restrictions can then be tested using the log-likelihood ratio test statistics. See Garratt et al. (2006) for a comprehensive review of long-run identification methods in macroeconometric literature. The first contribution on the identification of long-run relations in the *GVAR* literature is Dées et al. (2007b) who used bootstrapping to compute critical values for the likelihood ratio tests of over-identifying restrictions on the long-run relations of country-specific models.

Persistence profiles

The speed of convergence with which the adjustment to long-run relations takes place in the global model can be examined by persistence profiles (*PP*s). *PP*s refer to the time profiles of the effects of system or variable-specific shocks on the cointegrating relations, and they provide additional valuable evidence on the validity of long-run relations. In particular, when the speed of convergence towards a cointegrating relation turns out to be very slow, then this is an important indication of misspecification in the cointegrating vector under consideration. See Chapter 24 and Pesaran and Shin (1996) for a discussion of *PP*s in cointegrated *VAR* models, and Dées et al. (2007b) for implementation of *PP*s in the *GVAR* context.

33.7.2 Permanent/transitory component decomposition

Given that the *GVAR* model provides a coherent description of the short-run as well as long-run relations in the global economy, it can be used to provide estimates of steady states or the permanent components of the variables in the *GVAR* model.[15] Assuming no deterministic components are present, then the vector of permanent components is simply defined as long-horizon expectations

$$\mathbf{x}_t^P = \lim_{h\to\infty} E_t\left(\mathbf{x}_{t+h}\right). \tag{33.49}$$

When the *GVAR* contains deterministic components, $\mathbf{x}_t^P$ will be given by the sum of the deterministic components and long-horizon expectations of de-trended variables. The vector of deviations from steady states in both cases is given by

$$\tilde{\mathbf{x}}_t = \mathbf{x}_t - \mathbf{x}_t^P.$$

Assuming that the information set is non-decreasing over time, it follows from (33.49) that $\mathbf{x}_t^P = \lim_{h\to\infty} E_t\left(\mathbf{x}_{t+h}^P\right)$, which ensures that the steady states are time consistent, in the sense that

$$E_t\left(\mathbf{x}_{t+s}^P\right) = \lim_{h\to\infty} E_t\left(\mathbf{x}_{t+s+h}^P\right) = \mathbf{x}_t^P \text{ for any } s = 0, 1, 2, \ldots,$$

and, in the absence of deterministic components, $\mathbf{x}_t^P$ satisfies the martingale property, $E_t\left(\mathbf{x}_{t+1}^P\right) = \mathbf{x}_t^P$. Such a property is a natural requirement of any coherent definition of steady states, but this property is not satisfied for the commonly used Hodrick–Prescott (HP) filter and some of the other statistical measures of steady states.

Permanent components can be easily obtained from the estimated *GVAR* model using the Beveridge-Nelson decomposition, as illustrated in detail by Dées et al. (2007) and Dées et al. (2009). Estimates of steady states are crucial for the mainstream macroeconomic literature, which focuses predominantly on modelling the business cycle, that is explaining the behaviour of deviations from the steady states. The *GVAR* provides a coherent method for constructing steady states that reflect global influences and long-run structural relationships within, as well as across, countries in the global economy.

[15] See Chapter 13 for an introduction to trend and cycle decompositions. A multivariate analysis is provided in Section 22.15.

33.8 Specification tests

It has become a norm in applied work to perform a number of specification tests and robustness checks. DdPS apply a suite of residual-based break tests to test for the stability of coefficients and/or breaks in error variances. Although, in the context of cointegrated models, the possibility of a structural break is relevant for both long-run as well as short-run coefficients, the focus is on the stability of short-run coefficients, as the availability of data hinders any meaningful tests of the stability of cointegrating vectors. In particular, DdPS perform the following tests: Ploberger and Krämer (1992) maximal *OLS* cumulative sum (*CUSUM*) statistics; its mean square variant; Nyblom's (1989) tests for the parameter constancy against non-stationary alternatives; the Wald form of Quandt's (1960) likelihood ratio statistics; the mean Wald statistics of Hansen; and Andrews and Ploberger (1994) Wald statistics based on exponential average. The last three tests are Wald-type tests considering a single break at an unknown point. The heteroskedasticity-robust version of the tests are also conducted. Stability tests performed are based on residuals of the individual country models, which depend on the dimension of the cointegrating space, and do not require the cointegrating relationships to be identified. The critical values of the tests, computed under the null of parameter stability, can again be calculated using the sieve bootstrap samples. The detail of the bootstrap procedure is given in DdPS (2007, Supplement A). In the context of global macroeconomic modelling, DdPS and other applied papers typically find, perhaps surprisingly, relatively small rejection rates, and the main reason for rejection seems to be breaks in the error variances as opposed to coefficient instability. Once breaks in error variances are allowed for, the remaining parameters are typically reasonably stable.

A number of robustness checks can also be performed to test the sensitivity of the findings to variations of different modelling assumptions. For example, the sensitivity of findings to the number of lags selected in individual country-specific models can be investigated. Selecting shorter lags than required will also manifest itself in serial correlation of residuals. Sensitivity of findings can also be investigated with respect to the choice of the aggregation weights. While weights based on bilateral trade are employed in most applications, the weights based on other measures, such as cross-border financial data, can also be considered, depending on the application in hand. Time varying predetermined weights can be considered as well to take into account shifts in bilateral trade over the last couple of decades.

33.9 Empirical applications of the *GVAR* approach

Since the introduction of the *GVAR* model by Pesaran et al. (2004), there have been numerous applications of the *GVAR* approach developed over the past decade in the academic literature. The *GVAR* approach has also found its way into policy institutions, including the International Monetary Fund (IMF) and the European Central Bank (ECB), where this approach is one of the main techniques used to understand interlinkages across individual countries.[16]

[16] See the following IMF policy publications for examples of the use of *GVAR* approach by fund staff: 2011 and 2014 Spillover Reports; 2006 World Economic Outlook; October 2010 and April 2014 Regional Economic Outlook: Asia and Pacific Department; April 2014 Regional Economic Outlook: Western Hemisphere Department; November 2012 Regional Economic Outlook: Middle East and Central Asia Department; October 2008 Regional Economic Outlook: Europe; April and October 2012 Regional Economic Outlook: Sub-Saharan Africa; and IMF country reports for Algeria, India, Italy, Russia, Saudi Arabia, South Africa, and Spain.

The *GVAR* handbook edited by di Mauro and Pesaran (2013) provides an interesting collection of a number of *GVAR* empirical applications from 27 contributors. The *GVAR* handbook is a useful non-technical resource aimed at a general audience and/or practitioners interested in the *GVAR* approach. This handbook provides a historical background of the *GVAR* approach (Chapter 1), describes an updated version of the basic DdPS model (Chapter 2), and then provides seven applications of the *GVAR* approach on international transmission of shocks and forecasting (Chapters 3–9), three finance applications (Chapters 10–12), and 5 regional applications. The applications in the handbook span various areas of the empirical literature. Chapters on international transmission on forecasting investigate, among others, the problem of measuring output gaps across countries, structural modelling, the role of financial markets in the transmission of international business cycles, international inflation interlinkages, and forecasting the global economy. Finance applications include a macroprudential application of the *GVAR* approach, a model of sovereign bond spreads, and an analysis of cross-country spillover effects of fiscal spending on financial variables. Regional applications investigate the increasing importance of the Chinese economy, forecasting of the Swiss economy, imbalances in the euro area, regional and financial spillovers across Europe, and modelling interlinkages in the West African Economic and Monetary Union. We refer the reader to this handbook for further details on these interesting applications. In what follows we provide an overview of a number of more recent applications, starting with forecasting.

33.9.1 Forecasting applications

Pesaran, Schuermann, and Smith (2009a) is the first *GVAR* forecasting application to the global economy. These authors utilize the version of the *GVAR* model developed in DdPS and focus on forecasting real as well as financial variables at one and four quarters ahead. They consider forecasting real output, inflation, real equity prices, exchange rates, and interest rates. As we mentioned earlier in Section 33.6, forecasting the global economy is challenging due to the likely presence of multiple structural breaks and model uncertainty. The main finding of Pesaran, Schuermann, and Smith (2009a) is that simple averaging of forecasts across model specifications and estimation windows can make a significant difference. In particular, the double-averaged *GVAR* forecasts (across windows and models) perform better than the typical univariate benchmark competitors, especially for output, inflation, and real equity prices. Further forecasting results and discussions are presented in a rejoinder, Pesaran, Schuermann, and Smith (2009b).

Ericsson and Reisman (2012) provide an empirical assessment of the DdPS version of *GVAR* using the impulse indicator saturation technique, which is a new generic procedure for evaluating parameter constancy. Their results indicate the potential for an improved, more robust specification of the *GVAR* model.

de Waal and van Eyden (2013a) develop two versions of the *GVAR* model for South Africa (a small and a large version), and compare forecasts from these models with those from a vector error correction model (*VECM*) augmented with foreign variables as well as with univariate benchmark forecasts. The authors find that modelling the rest-of-the-world economies in a coherent way using the *GVAR* model can be useful for forecasting macro-variables in South Africa. In particular, they find that the forecast performance of the large version of the *GVAR* model is generally superior to the performance of the customized small *GVAR*, and that forecasts of both *GVAR* models tend to be better than the forecasts of the augmented *VECM*, especially at longer forecast horizons.

Schanne (2011) estimates a *GVAR* model applied to German regional labour market data, and uses the *GVAR* to forecast different labour market indicators. The author finds that including information about labour market policies and vacancies, and accounting for lagged and contemporaneous spatial dependence, can improve the forecasts relative to a simple bivariate benchmark model. On the other hand, business cycle indicators seem to help little with labour market predictions.

Forecasting using a mixed conditional information set is considered in Bussière, Chudik, and Sestieri (2012), who develop a *GVAR* model to analyse global trade imbalances. In particular, they compare the growth rates of exports and imports of 21 countries during the Great Trade Collapse of 2008–09 with the model's prediction, conditioning on the observed values of real output and real exchange rates. The objective of this exercise is to assess whether the collapse in world trade that took place during 2008–09 can be rationalized by standard macro explanatory variables (such as domestic and foreign output variables and real exchange rates) alone, or if other factors may have played a role. Standard macro explanatory variables alone are found to be quite successful in explaining the collapse of global trade for most of the economies in the sample. This exercise also reveals that it is easier to reconcile the Great Trade Collapse of 2008–09 in the case of advanced economies as opposed to emerging economies.

Forecasting of trade imbalances is also considered in Greenwood-Nimmo, Nguyen, and Shin (2012b). These authors compute both central forecasts and scenario-based probabilistic forecasts for a range of events and account for structural instability by the use of country-specific intercept shifts. They find that the predictive accuracy of the *GVAR* model is broadly comparable to that of standard benchmark models over short horizons and superior over longer horizons. Similarly to Bussière, Chudik, and Sestieri (2012), they conclude *GVAR* models may be useful forecasting tools for policy analysis.

Forecasting of global output growth with *GVAR*s is considered in a number of papers. Chudik, Grossman, and Pesaran (2014) focus on the information content of purchasing manager indices (*PMI*s) for nowcasting and for forecasting of real output growth. Feldkircher et al. (2014) present Bayesian estimates of the *GVAR*, and report improved forecasts when the *GVAR* model is based on country models estimated with shrinkage estimators. Garratt, Lee, and Shields (2014) model real output growth for G7 economies using survey output expectations, and find that both cross-country interdependencies and survey data are important for density forecasts of real output growth of G7 economies. Forecasting with a regime-switching *GVAR* model is considered in Binder and Gross (2013) who find that combining the regime-switching and the *GVAR* methodology significantly improves out-of-sample forecast accuracy in the case of real GDP, inflation, and stock prices.

33.9.2 Global finance applications

The first *GVAR* model in the literature, developed by PSW, is applied to the problem of credit risk modelling with a global perspective. PSW investigate the effects of various global risk scenarios on a bank's loan portfolio. The *GVAR* approach for modelling credit risk has also been explored in Pesaran, Schuermann, and Treutler (2007) who investigate the potential for portfolio diversification across industry sectors and across different countries. They find that the simulated credit loss distribution is largely explained by firm-level parameter heterogeneity and credit rating information. Further results on the modelling of credit risk with a global perspective are provided by Pesaran et al. (2006). The *GVAR*-based conditional credit loss distribution

is used, for example, to compute the effects of a hypothetical negative equity price shock in Southeast Asia on the loss distribution of a typical credit portfolio of a private bank with global exposures over one or more quarters ahead. The authors find that the effects of such shocks on losses are asymmetric and non-proportional, reflecting the highly nonlinear nature of the credit risk model. de Wet, van Eyden, and Gupta (2009) develop a South African-specific component of the *GVAR* model for the purpose of credit portfolio management in South Africa. Castrén, Dées, and Zaher (2010) use a *GVAR* model to analyse the behaviour of euro area corporate sector probabilities of default under a wide range of shocks. They link the core *GVAR* model with a satellite equation for firm-level expected default frequencies (EDFs) and find that, at the aggregate level, the median EDFs react most to shocks to GDP, exchange rate, oil and equity prices.

A number of other empirical *GVAR* papers focus on modelling various types of risk (sovereign, non-financial corporate or banking sector risks). Favero (2013) uses the *GVAR* approach to model sovereign risk, particularly time varying interdependence among ten-year sovereign bond spreads of the euro area member states. Gray et al. (2013) analyse interactions between banking sector risk, sovereign risk, corporate sector risk, real economic activity, and credit growth for 15 European countries and the United States. The goal is to analyze the impact and spillover effects of shocks and to help identify policies that could mitigate banking system failures, and sovereign credit risk. Alessandri et al. (2009) develop a quantitative framework which evaluates systemic risk due to banks' balance sheets which also allows for macro credit risk, interest income risk, market risk, and asset side feedback effects. These authors show that a combination of extreme credit and trading losses can precipitate widespread defaults and trigger contagious default associated with network effects and fire sales of distressed assets. Chen et al. (2010) investigate how bank and corporate default risks are transmitted internationally. They find strong macro-financial linkages within domestic economies as well as globally, and report significant global spillover effects when the shock originates from an important economy.

Dreger and Wolters (2011) investigate the implications of an increase in liquidity in the years preceding the global financial crises on the formation of price bubbles in asset markets. They find that the link between liquidity and asset prices seems fragile and far from being obvious. Implications of liquidity shocks and their transmission are also investigated in Chudik and Fratzscher (2011). In addition to liquidity shocks, Chudik and Fratzscher (2011) identify risk shocks and find that, while liquidity shocks have had a more severe impact on advanced economies during the recent global financial crisis, it was mainly the decline in risk appetite that affected emerging market economies. Effects of risk shocks are also scrutinized in Bussière, Chudik, and Mehl (2011) for a monthly panel of real effective exchange rates featuring 62 countries. Bussière, Chudik, and Mehl (2011) find that the responses of real effective exchange rates of euro area countries to a global risk aversion shock after the creation of the euro have been similar to the effects of such shocks on Italy, Portugal, or Spain before the European Monetary Union, that is, of economies in the euro area's periphery. Moreover, their findings suggest that the divergence in external competitiveness among euro area countries over the past decade, which is at the core of today's debate on the future of the euro area, is more likely due to country-specific shocks rather than to global shocks. Dovern and van Roye (2013) use a *GVAR* model to study the international transmission of financial stress and its effects on economic activity and find that financial stress is quickly transmitted internationally. Moreover, they find that financial stress has a lagged but persistent negative effect on economic activity, and that economic slowdowns tend to limit financial stress.

Gross and Kok (2013) use a mixed cross-section (23 countries and 41 international banks) *GVAR* specification to investigate contagion among sovereigns and private banks. They find that the potential for spillovers in the credit default swap market was particularly pronounced in 2008 and again in 2011–12. Moreover, contagion primarily tends to move from banks to sovereigns in 2008, whereas the direction seems to have been reversed in 2011–12 in the course of the sovereign debt crisis.

Interrelation between volatility in financial markets and macroeconomic dynamics is investigated in Cesa-Bianchi, Pesaran, and Rebucci (2014), who augment the *GVAR* model of DdPS with a global volatility module. They find a statistically significant and economically sizable impact of future output growth on current volatility, and no effect of an exogenous change in volatility on the business cycle over and above those driven by the common factors. They interpret this evidence as suggesting that volatility is a symptom rather than a cause of economic instability.

Implication of global financial conditions on individual economies is also the object of a study by Georgiadis and Mehl (2015), but with a very different focus from earlier studies, which mostly concentrate on transmission of financial risk. These authors investigate the hypothesis that global financial cycles determine domestic financial conditions regardless of an economy's exchange rate regime. Using a quarterly sample of 59 economies spanning 1999Q1:2009Q4 period, the authors reject this hypothesis and find that the classic Mundell-Flemming trilemma (namely that an economy cannot simultaneously maintain a fixed exchange rate, free capital movement, and an independent monetary policy) remains valid, despite the significant rise in financial globalization since the 1990s.

33.9.3 Global macroeconomic applications

DdPS update the PSW *GVAR* model by expanding the country coverage as well as the time coverage, and provide further theoretical results, some of which are reviewed above. Their focus is on the enhancement of the global model and its use in analysing transmission of shocks across countries with particular attention on the implications for the euro area economy. Using a variety of shocks, including shocks to US real equity prices, oil prices, US short-term interest rates, as well as US monetary policy shocks (identified by using partial ordering of variables), DdPS find that financial shocks are transmitted relatively rapidly and often get amplified as they travel from the US to the euro area. The impact of US monetary policy shocks on the euro area is, however, rather limited.

Global inflation

Galesi and Lombardi (2009) study the effects of oil and food price shocks on inflation. They find that the inflationary effects of oil price shocks are felt mostly in the developed countries while less sizeable effects are observed in the case of emerging economies. Moreover, food price increases also have significant inflationary direct effects, especially for emerging economies, and significant second-round effects are reported in a number of other countries. Inflation is also the focus of Anderton et al. (2010) who construct a *GVAR* model to examine oil price shocks and other key factors affecting global inflation. They consider calculating the impact of increased imports from low-cost countries on manufacturing import prices and estimate Phillips curves in order to shed light on whether the inflationary process in OECD countries has changed over time. They find that there seem to be various significant pressures on global trade prices and labour

markets associated with structural factors, and argue that these are partly due to globalization which, in addition to changes in monetary policy, seem to be behind some of the changes in the inflationary process over the period under consideration.

Using the *GVAR* model, Dées et al. (2009) provide estimates of new Keynesian phillips Curves (*NKPC*) for eight developed industrial countries and discuss the weak instrument problem and the characterization of the steady states. It is shown that the *GVAR* generates global factors that are valid instruments and help alleviate the weak instrument problem. The use of foreign variables as instruments is found to substantially increase the precision of the estimates of the output coefficient in the NKPC equations. Moreover, it is argued that the *GVAR* steady states perform better than the Hodrick–Prescott (HP) measure. Unlike HP, the *GVAR* measures of the steady states are coherent and reflect long-run structural relationships within as well as across countries.

Global imbalances and exchange rate misalignments

The effects of demand shocks and shocks to relative prices on global imbalances are examined in Bussière, Chudik, and Sestieri (2012), using a *GVAR* model of global trade flows. Their results indicate that changes in domestic and foreign demand have a much stronger effect on trade flows as compared to changes in relative trade prices. Using the *GVAR* approach, global imbalances are also investigated by Bettendorf (2012), although with a different focus. Estimating exchange rate misalignments using a *GVAR* model is undertaken in Marçal et al. (2014). This paper contrasts *GVAR*-based measures of misalignment with traditional time series estimates that treat individual countries as separate units. Large differences between a *GVAR* and more traditional time series estimates are reported, especially for small and developing countries.

Role of the US as a dominant economy

The role of the US as a dominant economy in the global economy is examined in Chudik and Smith (2013) by comparing two models: one that treats the US as a globally dominant economy, and a standard version of the *GVAR* model that does not separate the impact of US variables from the cross-section averages of foreign economies, as is done in DdPS, for example. They find some support for the extended version of the *GVAR* model, with the US treated as a dominant economy. A similar approach is also adopted by Dées and Saint-Guilhem (2011) who find that the role of the US is somewhat diminished over time.

Business cycle synchronization and the rising role of China in the world economy

Dreger and Zhang (2013) investigate interdependence of business cycles in China and industrial countries and study the effects of shocks to the Chinese economy. Cesa-Bianchi et al. (2012) investigate the interdependence between China, Latin America, and the world economy. Feldkircher and Korhonen (2012) consider the effects of the rise of China on emerging markets. All these studies find a significant degree of business cycle synchronization in the world economy with the importance of the Chinese economy increasing for both advanced and emerging economies. Cesa-Bianchi et al. (2012), using a *GVAR* model with time varying trade weights, find that the long-term impact of a China GDP shock on typical Latin American economies has increased threefold since the mid-1990s, and the long-term impact of a US GDP shock has halved. Feldkircher and Korhonen (2012) find that a 1 per cent shock to Chinese output translates to a 1.2 per cent increase in Chinese real GDP and 0.1 to 0.5 per cent rise in real output in the case of large economies. The countries of Central Eastern Europe and the former

Commonwealth of Independent States also experience a rise of 0.2 per cent in their real output. By contrast, China seems to be little affected by shocks to the US economy.

Boschi and Girardi (2011) investigate the business cycle in Latin America using a nine country/region version of the *GVAR*, and quantify the relative contribution of domestic, regional, and international factors to the fluctuation of domestic output in Latin American economies. In particular, they find that only a modest proportion of Latin American domestic output variability is explained by industrial countries' factors and that domestic and regional factors account for the main share of output variability at all simulation horizons.

International linkages of the Korean economy are investigated in Greenwood-Nimmo, Nguyen, and Shin (2012a). They find that the real economy and financial markets are highly sensitive to oil price changes even though they have little effect on inflation. They also show that the interest rate in Korea is set largely without recourse to overseas conditions except to the extent that these influences are captured by the exchange rate. They find that the Korean economy is most affected by the US, the euro area, Japan, and China.

Understanding interlinkages between emerging Europe and the global economy is investigated in Feldkircher (2013) who develops a *GVAR* model covering 43 countries. The main findings are that emerging Europe's real economy reacts to a US output shock as strongly as it does to a corresponding euro area shock. Moreover, Feldkircher (2013) uncovers a negative effect of tightening in the euro area's short-term interest rate on output in the long-run throughout Central, Eastern, and Southeastern Europe and the Commonwealth of Independent States.

Sun, Heinz, and Ho (2013) use the *GVAR* approach with combined trade and financial weights to investigate cross-country linkages in Europe. Their findings show strong co-movements in output growth and interest rates but weaker linkages between inflation and real credit growth within Europe.

The impact of foreign shocks on South Africa is studied in de Waal and van Eyden (2013b). Using time varying weights they show the increasing role of China and the decreasing role of the US in South African economy, reflecting the substantial increase in South Africa's trade with China since the mid-1990s. The impact of a US shock on South African GDP is found to be insignificant by 2009, whereas the impact of a shock to Chinese GDP on South African GDP is found to be three times stronger in 2009 than in 1995. These findings are in line with the way the global crisis of 2007-09 affected South Africa, and highlight increased risk to the South African economy from shocks to the Chinese economy.

Spillover effects of shocks in large economies (such as China, euro area, and the US) to the Middle East and North Africa (MENA) region, as well as the effects of shocks originating in the MENA oil exporters and Gulf Cooperation Countries to the rest of the world, are investigated using a *GVAR* model by Cashin, Mohaddes, and Raissi (2014b). The results are as to be expected with shocks from China playing an increasingly more important role for the MENA countries.

Impact of EMU membership

Two papers, Pesaran, Smith, and Smith (2007) and Dubois, Hericourt, and Mignon (2009), investigate counterfactual scenarios regarding monetary union membership. Pesaran, Smith, and Smith (2007) analyse counterfactual scenarios using a *GVAR* macroeconometric model and empirically investigate 'what if the UK had joined the Euro in 1999'. They report probability estimates that output could have been higher and prices lower in the UK and in the euro area as a result of the entry. They also examine the sensitivity of these results to a variety of assumptions about the UK entry. The aim of Dubois, Hericourt, and Mignon (2009) is to answer the

counterfactual question of the consequences of no euro launch in 1999. They find that monetary unification promoted lower interest rates and higher output in most euro area economies, relative to a situation where national monetary policies would have followed a German-type monetary policy. An opposite picture emerges if national monetary policies had adopted British monetary preferences after September 1992.

Commodity price models

Gutierrez and Piras (2013) construct a *GVAR* model of the global wheat market, where the feedback between the real and the financial sectors, and also the link between food and energy prices, are taken into account. Their impulse response analysis reveals that a negative shock to wheat consumption, an increase in oil prices, and real exchange rate devaluation all have inflationary effects on wheat export prices, although their impacts are different across the main wheat export countries.

While oil prices are included in the majority of *GVAR* models as an important observed common factor, these studies do not generally focus on the nature of oil shocks and their effects. Identification of oil price shocks is attempted in Chudik and Fidora (2012) and Cashin et al. (2014). Both papers argue that the cross-section dimension can help in the identification of (global) oil shocks and exploit sign restrictions for identification. The former paper investigates the effects of supply-induced oil price increases on aggregate output and real effective exchange rates. It finds that adverse oil supply shocks have significant negative impacts on the real output growth of oil importers, with emerging markets being more affected as compared to the more mature economies. Moreover, oil supply shocks tend to cause an appreciation (depreciation) of oil exporters' (oil importers') real effective exchange rates, but they also lead to an appreciation of the US dollar. Cashin et al. (2014) identify demand as well as supply shocks and find that the economic consequences of the two types of shocks are very different. They also find negative impacts of adverse oil supply shocks for energy importers, while the impacts on oil exporters that possess large proven oil/gas reserves is positive. A positive oil-demand shock, on the other hand, is found to be associated with long-run inflationary pressures, an increase in real output, a rise in interest rates, and a fall in real equity prices.

Impact of the commodity price boom and bust over the period 1980–2010 on output growth in Latin America and the Caribbean is estimated in a *GVAR* model by Gruss (2014). It is found that, even if commodity prices remain unchanged at their high levels, the growth in the commodity exporting region would be significantly lower than during the commodity price boom period.

Housing

Hiebert and Vansteenkiste (2009) adopt the *GVAR* approach to investigate the spillover effects of house price changes across euro area economies, using three housing demand variables: real house prices, real per capita disposable income, and the real interest rate for ten euro area countries. Their results suggest limited house price spillovers in the euro area, in contrast to the impacts of a shock to domestic long term interest rates, with the latter causing a permanent shift in house prices after around three years. Moreover, they find the effects of house price spillover to be quite heterogeneous across countries.

Jannsen (2010) investigates the international effects of the 2008–10 housing crises, focusing on the US, Great Britain, Spain, and France. Among other findings, Jannsen's results show that the adverse effects of housing crisis tend to be greatest during the first two years—particularly between the fifth and the seventh quarter after house prices have reached their peak. It is also

found that when several important industrial countries face a housing bust at the same time, economic activity in other countries is likely to be dampened *via* international transmission effects, leading to significant losses of GDP growth in a number of countries, notably in Europe.

Effects of fiscal and monetary policy

There are a number of studies that use the *GVAR* approach to examine the international effects of fiscal policy shocks. Favero, Giavazzi, and Perego (2011) highlight the heterogeneous nature of fiscal policy multipliers across countries, and show that the effects of fiscal shocks on output differ according to the nature of the debt dynamics, the degree of openness of the economies under consideration, and the fiscal reaction functions across countries. Hebous and Zimmermann (2013) estimate spillovers of a fiscal shock in one euro area member country on the rest, and find that the positive effects of area-wide fiscal shocks are larger than those of the domestic shocks of comparable magnitude, thus showing that coordinated fiscal action is likely to be more effective.

Cross-country effects of monetary policy shocks are investigated by Georgiadis (2014a) and Georgiadis (2014b). These papers investigate the global spillover effects of monetary policy shocks to US and euro area, respectively. In both papers, monetary policy shocks are identified by sign restrictions. Georgiadis finds that the effects of US monetary policy shocks on aggregate output are heterogeneous across countries with the foreign output effects being larger than the domestic effects for many of the economies in the global economy. Substantial heterogeneity is also observed in the transmission of euro area monetary policy shocks, where countries with more wage and fewer unemployment rigidities are found to exhibit stronger output effects.

The role of US monetary policy shocks is also examined in Feldkircher and Huber (2015) who, in addition to monetary policy shocks, also identify the US aggregate demand and supply shocks within a Bayesian version of the *GVAR* model. Among the variety of interesting findings reported in Feldkircher and Huber (2015), is the fact that US monetary policy shocks are found to have most pronounced effects on real output internationally.

Labour market

The *GVAR* model developed by Hiebert and Vansteenkiste (2010) is used to analyse spillovers in the labour market in the US. Using data on 12 manufacturing industries over the period 1977–2003, Hiebert and Vansteenkiste (2010) analyse responses of a standard set of labour-market related variables (employment, real compensation, productivity and capital stock) to exogenous factors (such as a sector-specific measure of trade openness or a common technology shock), along with industry spillovers using sector-specific manufacturing-wide measures. Their findings suggest that increased trade openness negatively affects real compensation, has negligible employment effects, and leads to higher labour productivity. Technology shocks are found to have significantly positive effects on both real compensation and employment.

Role of credit

The role of credit in the international business cycles is investigated using a *GVAR* approach by Eickmeier and Ng (2011), Xu (2012) and Konstantakis and Michaelides (2014). Eickmeier and Ng focus on the transmission of credit supply shocks in the US, the euro area and Japan, using sign restrictions to identify the shocks. They find that negative US credit supply shocks have stronger negative effects on domestic and foreign GDP, as compared with credit supply shocks from the euro area and Japan. Xu (2012) investigates the effects of US credit shocks

and the importance of credit in explaining business cycle fluctuations. Her findings reveal the importance of bank credit in explaining output growth, changes in inflation, and long term interest rates in countries with a developed banking sector. Using *GIRF*s she finds strong evidence of spillovers from US credit shocks to the UK, the euro area, Japan, and other industrialized economies. Konstantakis and Michaelides (2014) use the *GVAR* approach to model output and debt fluctuations in the US and the EU15 economies. Konstantakis and Michaelides analyse the transmission of shocks to debt and real output using *GIRF*s and find that the EU15 economy is more vulnerable than the US to foreign shocks. Moreover, the effects of a shock to the US debt has a significant and persistent impact on the EU15 and US economies, whereas a shock to EU15 debt does not have a statistically significant impact on the US economy.

Macroeconomic effects of weather shocks

In a unique study, Cashin, Mohaddes, and Raissi (2014a) investigate macroeconomic impacts of El Niño weather shocks measured by the Southern Oscillation Index (SOI). Arguably, El Niño weather events are exogenous in nature, and can have important consequences for economic activity worldwide. SOI is added to a standard *GVAR* framework as an observable common factor and the effects of a shock to SOI on economic variables across the globe are investigated. The authors find considerable heterogeneities in responses to El Niño weather shocks: some countries experience a short-lived fall in economic activity (Australia, Chile, Indonesia, India, Japan, New Zealand, and South Africa), while others experience a growth-enhancing effect (the US and the European region). Some inflationary pressures are also observed in response to El Niño weather shocks, due to short-lived commodity price increases.

33.9.4 Sectoral and other applications

The *GVAR* approach does not necessarily need to have a country dimension; other cross-sectional units could be considered. Holly and Petrella (2012) adopt the *GVAR* approach to model highly disaggregated manufacturing sectors within the UK. They show that factor demand linkages can be important for the transmission of both sectoral and aggregate shocks.

Vansteenkiste (2007) models regional housing market spillovers in the US. Using state-level data on the 31 largest US states she finds strong interregional linkages for both real house prices and real income per capita. Vansteenkiste (2007) also considers the effects of real interest rates shocks on house prices and finds that an increase of 100 basis points in the real ten-year government bond yield results in a relatively small long-run fall in house prices of between 0.5 and 2.5 per cent. Holly, Pesaran, and Yamagata (2011) investigate adjustment to shocks in a system of UK regional house prices, treating London as a dominant region and linking UK house prices to international developments *via* New York house price changes. They show that shocks to house prices in the London region impact other UK regions with a delay, and these lagged effects then echo back to the London housing market as the dominant region. They also show that, due to close financial inter-linkages between London and New York, house price changes in New York tend to pre-date house price changes in London.

33.10 Further reading

Further discussion on the *GVAR* approach can be found in di Mauro and Pesaran (2013).

33.11 Exercises

1. Consider the following first-order *GVAR* model,

$$\Delta y_{it} = -\gamma (y_{i,t-1} - \bar{y}_{t-1}) + \varepsilon_{it},\ 0 < \gamma < 1,$$

for $i = 1, 2, \ldots, N$, and $t = 1, 2, \ldots, T$, where $\Delta y_{it} = y_{it} - y_{i,t-1}$,

$$\begin{aligned} \varepsilon_{it} &\sim IID(0, \sigma_i^2), 0 < \sigma_i^2 < \infty, \\ y_{i0} &\sim IID(\mu_i, \omega_i^2), 0 < \omega_i^2 < \infty, \\ \bar{y}_{t-1} &= N^{-1} \sum_{i=1}^{N} y_{i,t-1}. \end{aligned}$$

   (a) Show that for a fixed N and for each i the variables y_{it}, $i = 1, 2, \ldots, N$, are integrated of order one (i.e. $I(1)$) and pair-wise cointegrated.

   (b) Suppose T is fixed and N is allowed to increase without bounds. Derive the integration/coinetgration properties of y_{it}, $i = 1, 2, \ldots, N$.

   (c) Derive optimal forecasts of $y_{i,T+h}$, $h > 0$, conditional on $y_{i,T-\ell}$, for $\ell = 0, 1, 2, \ldots$.

   (d) Consider now the forecast problem if $y_{i,t-1} - \bar{y}_{t-1}$ is replaced by $y_{i,t-1} - \bar{y}_t$.

2. Consider the following factor-augmented *VAR* models for the N countries comprising the world economy

$$\mathbf{x}_{it} = \mathbf{\Phi} \mathbf{x}_{i,t-1} + \mathbf{\Lambda}_i \mathbf{f}_t + u_{it}, \text{ for } i = 1, 2, \ldots, N,$$

where $\mathbf{x}_{it}$ is a $k \times 1$ vector of endogenous variables specific to country i, $\mathbf{f}_t$ is an $m \times 1$ vector of unobserved common variables, and u_{it} is the $k \times 1$ vector of country-specific shocks that are weakly cross-sectionally correlated. Also, without loss of generality, assume that u_{it} and $\mathbf{f}_t$ are uncorrelated with zero means.

   (a) Let $\bar{\mathbf{\Lambda}} = N^{-1} \sum_{i=1}^{N} \mathbf{\Lambda}_i$ and $\bar{\mathbf{x}}_t = N^{-1} \sum_{i=1}^{N} \mathbf{x}_{it}$, and suppose that $\bar{\mathbf{\Lambda}}' \bar{\mathbf{\Lambda}}$ is a positive definite matrix for N including as $N \to \infty$, and let $\mathbf{S} = \left(\bar{\mathbf{\Lambda}}' \bar{\mathbf{\Lambda}} \right)^{-1} \bar{\mathbf{\Lambda}}'$. Then

$$E \left\| \mathbf{f}_t - \mathbf{S} \left(\bar{\mathbf{x}}_t - \mathbf{\Phi} \bar{\mathbf{x}}_{t-1} \right) \right\| = O\left(N^{-1/2} \right),$$

   and

$$\mathbf{x}_{it} = \mathbf{\Phi} \mathbf{x}_{i,t-1} + \mathbf{\Lambda}_i \mathbf{S} \left(\bar{\mathbf{x}}_t - \mathbf{\Phi} \bar{\mathbf{x}}_{t-1} \right) + u_{it} + O_p(N^{-1/2}), \text{ for } i = 1, 2, \ldots, N.$$

   (b) Using the above results discuss the problem of identification and estimation of the country-specific shocks.

(c) Consider now the case where $\mathbf{\Phi}$ is replaced by $\mathbf{\Phi}_i$, thus allowing for dynamic heterogeneity across the countries. How are the above results affected by this generalization?

3. Suppose that we are interested in identifying the possible links between uncertainty (as measured by asset price volatility) and the macro economy. To this end an investigator considers the following bivariate relations for country 1 (say the US)

$$\begin{aligned}\mathbf{x}_{1t} &= \mathbf{\Phi}_1\mathbf{x}_{1,t-1} + \mathbf{\Lambda}_1\mathbf{f}_t + \mathbf{u}_{1t},\\ v_t &= \phi\, v_{t-1} + \boldsymbol{\lambda}'\mathbf{f}_t + \varepsilon_t,\end{aligned}$$

where, as in the above question, $\mathbf{x}_{1t}$ is the vector of macro-economic variables for country 1, $\mathbf{f}_t$ is the $m \times 1$ vector of unobserved common factors (shocks), and u_{1t} is the vector of shocks specific to country 1. Also, v_t is a measure of uncertainty which is assumed to be affected by common factors. ε_t represents the uncertainty specific shock.

(a) Discuss conditions under which $Cov(\mathbf{u}_{1t}, \varepsilon_t)$ can be identified.

(b) In an attempt to relax the conditions under (*a*) above, the investigator considers a multi-country approach to the problem and considers the following system of equations

$$\begin{aligned}\mathbf{x}_{it} &= \mathbf{\Phi}_i\mathbf{x}_{1,t-1} + \mathbf{\Lambda}_i\mathbf{f}_t + \mathbf{u}_{it},\ i = 1, 2, \ldots, N\\ v_t &= \phi\, v_{t-1} + \boldsymbol{\lambda}'\mathbf{f}_t + \varepsilon_t.\end{aligned}$$

Compare the conditions for identification of $Cov(\mathbf{u}_{1t}, \varepsilon_t)$ in the above multi-country setting with the single country framework considered above.

(c) How would you estimate $Cov(\mathbf{u}_{it}, \varepsilon_t)$ for $i = 1, 2, \ldots, N$?

4. Consider the following large dimensional factor-augmented model

$$\mathbf{y}_t = \mathbf{\Phi}\mathbf{y}_{t-1} + \boldsymbol{\gamma} f_t + \boldsymbol{\varepsilon}_t,$$

where $\mathbf{y}_t = \left(y_{1t}, y_{2t}, \ldots, y_{Nt}\right)'$, $\boldsymbol{\gamma} = \left(\gamma_1, \gamma_2, \ldots, \gamma_N\right)'$, f_t is an unobserved common factor, $\mathbf{\Phi}$, is an $N \times N$ matrix of unknown coefficients, and $\boldsymbol{\varepsilon}_t = (\varepsilon_{1t}, \varepsilon_{2t}, \ldots, \varepsilon_{Nt})'$, is an $N \times 1$ vector of idiosyncratic shocks. It is assumed that the common factor follows the covariance stationary AR(1) process

$$f_t = \rho f_{t-1} + v_t.$$

The errors $\boldsymbol{\varepsilon}_t$ and v_t are uncorrelated and serially independent with zero means. Further, it is assumed that

$$\boldsymbol{\varepsilon}_t = \mathbf{R}\boldsymbol{\eta}_t,$$

where the $N \times N$ matrix $\mathbf{R}$ has bounded row and column matrix norms (in N), and $\boldsymbol{\eta}_t \sim IID(\mathbf{0}, \mathbf{I}_N)$.

(a) Show that

$$\mathbf{y}_{t+h|t} = E\left(\mathbf{y}_{t+h} \middle| \mathbf{y}_t, \mathbf{y}_{t-1}, \ldots; f_t, f_{t-1}, \ldots\right) = \mathbf{\Phi}^h \mathbf{y}_t + \mathbf{a}_h f_t, \tag{33.50}$$

for $h = 1, 2, \ldots$, where $\mathbf{a}_h = \sum_{\ell=0}^{h-1} \rho^{h-\ell} \mathbf{\Phi}^\ell \boldsymbol{\gamma}$.

(b) Show that

$$\bar{y}_t = \mathbf{w}' \mathbf{\Phi} \mathbf{y}_{t-1} + \bar{\gamma} f_t + O_p\left(N^{-1/2}\right),$$

and

$$\mathbf{w}' \mathbf{\Phi} \mathbf{y}_{t-1} = \sum_{j=0}^{\infty} \mathbf{w}' \mathbf{\Phi}^{j+1} \boldsymbol{\varepsilon}_{t-j-1} + \sum_{j=0}^{\infty} \mathbf{w}' \mathbf{\Phi}^{j+1} \boldsymbol{\gamma} f_{t-j-1}.$$

where $\bar{y}_t = N^{-1} \sum_{i=1}^{N} y_{it}$, $\bar{\gamma} = N^{-1} \sum_{i=1}^{N} \gamma_i$, and $\mathbf{w}' = N^{-1}(1, 1, \ldots, 1)$.

(c) Denoting the spectral norm of $\mathbf{\Phi}$, by $\|\mathbf{\Phi}\|$, show that if $\|\mathbf{\Phi}\| < 1$, then

$$\bar{y}_t = \left(\sum_{\ell=0}^{\infty} d_\ell L^\ell\right) f_t + O_p\left(N^{-1/2}\right),$$

and $|d_\ell| = O\left[(1-\epsilon)^\ell\right]$, for some small positive ϵ.

(d) Use the above results to obtain h-step forecasts of y_{it} and $\bar{y}_t$ based on the observables, $\mathbf{y}_t, \mathbf{y}_{t-1}, \ldots$, for N sufficiently large.

Appendices

Appendix A: Mathematics

This appendix reviews background material on complex numbers, trigonometry, matrix algebra, calculus, and linear difference equations. Further information on the topics covered can be found in Horn and Johnson (1985), Hamilton (1994), Lütkepohl (1996), Magnus and Neudecker (1999), Golub and Van Loan (1996), and Bernstein (2005).

A.1 Complex numbers and trigonometry

A.1.1 Complex numbers

Complex numbers are composed of a real part plus an imaginary part,

$$z = a + bi, \tag{A.1}$$

where a, b are real numbers, and $i = \sqrt{-1}$, is the imaginary unit. The *length* (or norm) of a complex number $z = a + bi$ is defined as

$$|z| = \sqrt{a^2 + b^2}.$$

Standard arithmetic operations for complex numbers are defined as follows

1. Sum:

$$(a + bi) + (c + di) = (a + c) + (b + d)\, i.$$

2. Product:

$$(a + bi)\,(c + di) = (ac - bd) + (bc + ad)\, i.$$

   In particular, the square of the imaginary unit is -1.

3. Division:

$$(a + bi) \,/\, (c + di) = \frac{ac + bd}{c^2 + d^2} + \frac{bc - ad}{c^2 + d^2} i,$$

   provided that $c^2 + d^2 > 0$.

4. Square root: the square root of $(a + bi)$ is given by $\pm(\gamma + \delta i)$, with

$$\gamma = \sqrt{\frac{1}{2}\left(a + \sqrt{a^2 + b^2}\right)},$$
$$\delta = \frac{|b|}{b}\sqrt{\frac{1}{2}\left(-a + \sqrt{a^2 + b^2}\right)}.$$

We define the *complex exponential function* as:

$$e^{a+bi} = \sum_{j=0}^{\infty} \frac{(a+bi)^j}{j!} = e^a \left[\sum_{j=0}^{\infty} \frac{(-1)^j b^{2j}}{(2j)!} + i \sum_{j=0}^{\infty} \frac{(-1)^j b^{2j+1}}{(2j+1)!} \right]. \tag{A.2}$$

For a review of complex analysis see Bierens (2005).

A.1.2 Trigonometric functions

Consider any right triangle containing the angle θ, having hypotenuse h, opposite side a and adjacent side b, as visualized in Figure A. We measure angles in *radians*.

We define

$$\sin(\theta) = a/h, \ \cos(\theta) = b/h, \ \tan\theta = a/b.$$

The functions sine and cosine are called trigonometric or *sinusoidal* functions. Viewed as a function of θ, we have $\sin(0) = 0$. As θ increases to $\pi/2$, the sine function increases to 1 and then falls back to zero as θ rises further to π. The function then reaches its minimum of -1

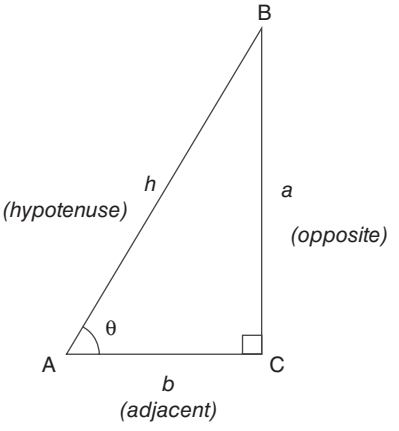


Figure A

when $\theta = 3\pi/2$ and then begins climbing back to zero. The function is periodic with period 2π, since

$$\sin(\theta + 2\pi) = \sin\theta,$$

for all values of θ, and, more generally,

$$\sin\left(\theta + 2\pi j\right) = \sin\theta, \text{ for any integer } j.$$

The cosine can be seen as a horizontal shift of the sine function since

$$\cos\theta = \sin\left(\theta + \frac{\pi}{2}\right).$$

Hence, the cosine will also be a periodic function, starting out at 1 (i.e., $\cos(0) = 1$), and falling to zero as θ increases to $\pi/2$. More generally, any linear combinations of sinusoidal functions, of the type

$$f(\theta) = \sum_{j=0}^{\infty} \left[a_j \cos\left(j\theta\right) + b_j \sin\left(j\theta\right)\right], \tag{A.3}$$

where $\{a_j\}$ and $\{b_j\}$ are arbitrary sequences of constants, is a periodic function with period 2π.

Some important identities

$$\begin{aligned}
\sin^2(\theta) + \cos^2(\theta) &= 1,\\
\sin(\theta \pm \phi) &= \sin(\theta)\cos(\phi) \pm \cos(\theta)\sin(\phi),\\
\cos(\theta \pm \phi) &= \cos(\theta)\cos(\phi) \mp \sin(\theta)\sin(\phi),\\
\sin(-\theta) &= -\sin\theta,\ \ \cos(-\theta) = \cos\theta,\\
\cos\theta &= \left(e^{i\theta} + e^{-i\theta}\right)/2,\\
\sin\theta &= \left(e^{i\theta} - e^{-i\theta}\right)/2i,
\end{aligned}$$

where $i = \sqrt{-1}$, and $e^{a+ib} = e^a\left[\cos b + i\sin b\right]$.

See, for example, Hamilton (1994) for more details on trigonometric identities useful for time series analysis.

A.1.3 Fourier analysis

The Fourier analysis is essentially concerned with the approximation of a function by the sum of sine and cosine terms, called the Fourier series representation. Any periodic function $f(\theta)$ can be expressed in the form of a *Fourier series*

$$f(\theta) = \frac{1}{2}a_0 + \sum_{j=1}^{\infty} \left[a_j \cos\left(j\theta\right) + b_j \sin\left(j\theta\right)\right], \tag{A.4}$$

where

$$a_j = \frac{1}{\pi}\int_{-\pi}^{+\pi} f(\omega)\cos(j\omega)\,d\omega, \quad j = 0, 1, 2, \ldots,$$
$$b_j = \frac{1}{\pi}\int_{-\pi}^{+\pi} f(\omega)\sin(j\omega)\,d\omega, \quad j = 1, 2, \ldots.$$

Let

$$f_m(\theta) = \sum_{j=0}^{m}\left[a_j\cos(j\theta) + b_j\sin(j\theta)\right],$$

under some general conditions on $f(\theta)$, it is possible to show that, as $m \to \infty$, $f_m(\theta)$ converges to $f(\theta)$ in mean square, that is,

$$\int_{-\pi}^{+\pi}[f_m(\omega) - f(\omega)]^2\,d\omega \to 0, \text{ as } m \to \infty.$$

Consider now any function $f(\theta)$ defined over the interval $(-\infty, +\infty)$ and such that

$$\int_{-\infty}^{+\infty}\left|f(\omega)\right|d\omega < \infty.$$

Under some general conditions it is possible to show that

$$f(\theta) = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{+\infty} G(x)\,e^{i\theta x}dx, \tag{A.5}$$
$$G(x) = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{+\infty} f(\theta)\,e^{-i\theta x}d\theta. \tag{A.6}$$

Equation (A.5) is the *Fourier integral representation* of $f(\theta)$, while equation (A.6) is known as *Fourier transform* of $f(\theta)$.

See Priestley (1981, Ch. 4), and Chatfield (2003) for further details on Fourier analysis applied to time series.

A.2 Matrices and matrix operations

An $m \times n$ matrix $\mathbf{A}$ is

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & \ldots & a_{1n} \\ a_{21} & a_{22} & \ldots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \ldots & a_{mn} \end{pmatrix},$$

a_{ij} is the element in the ith row and j^{th} column of $\mathbf{A}$, and is a real number. In the following, we indicate by $\mathbb{M}^{m\times n}$ the space of real $m \times n$ matrices. We indicate by $\mathbf{a}_{.j}$ the jth column of $\mathbf{A}$, namely

$$\mathbf{a}_{.j} = \begin{pmatrix} a_{1j} \\ a_{2j} \\ \vdots \\ a_{mj} \end{pmatrix},$$

$\mathbf{a}_{.j}$ is an m-dimensional vector. We indicate by $\mathbf{a}'_{i.}$ the ith row of $\mathbf{A}$, namely

$$\mathbf{a}_{i.} = \begin{pmatrix} a_{i1} \\ a_{i2} \\ \vdots \\ a_{in} \end{pmatrix},$$

$\mathbf{a}_{i.}$ is an n-dimensional vector. An $n \times n$ matrix $\mathbf{A} \in \mathbb{M}^{m\times n}$ is a *square* matrix. Its elements a_{ii}, for $i = 1, 2, \ldots, n$, are called diagonal elements and the elements $a_{11}, a_{22}, \ldots, a_{nn}$ constitute the main diagonal of $\mathbf{A}$.

A.2.1 Matrix operations

We can define the following matrix operations:

Matrix addition: Let $\mathbf{A}, \mathbf{B} \in \mathbb{M}^{m\times n}$. Let $\mathbf{C} = \mathbf{A} + \mathbf{B}$. $\mathbf{C}$ has its $(i, j)^{th}$ generic element

$$c_{ij} = a_{ij} + b_{ij}.$$

Scalar multiplication: Let $\mathbf{A} \in \mathbb{M}^{m\times n}$. Let $\mathbf{B} = \beta\mathbf{A}$. $\mathbf{B}$ has its $(i, j)^{th}$ generic element

$$b_{ij} = \beta a_{ij}.$$

Matrix multiplication: Let $\mathbf{A} \in \mathbb{M}^{m\times n}$, $\mathbf{B} \in \mathbb{M}^{n\times p}$. Let $\mathbf{C} = \mathbf{AB}$. $\mathbf{C}$ has its $(i, j)^{th}$ generic element

$$c_{ij} = \sum_{h=1}^{n} a_{ih} b_{hj}.$$

Let $\mathbf{A} \in \mathbb{M}^{m\times n}$. The transpose of $\mathbf{A}$, indicated by $\mathbf{A}'$, with the generic (i, j) element, a_{ji}. A matrix $\mathbf{A} \in \mathbb{M}^{n\times n}$ is symmetric if $a_{ij} = a_{ji}$ for all i, j, namely for symmetric matrices we have $\mathbf{A} = \mathbf{A}'$. The following basic properties hold:

1. $(\mathbf{A} + \mathbf{B}) + \mathbf{C} = \mathbf{A} + (\mathbf{B} + \mathbf{C}) = \mathbf{A} + \mathbf{B} + \mathbf{C}$.
2. $(\mathbf{AB})\,\mathbf{C} = \mathbf{A}\,(\mathbf{BC}) = \mathbf{ABC}$.
3. $\mathbf{A}\,(\mathbf{B} + \mathbf{C}) = \mathbf{AB} + \mathbf{AC}$.

4. $\left(\mathbf{A}'\right)' = \mathbf{A}$.
5. $(\mathbf{A} + \mathbf{B})' = \mathbf{A}' + \mathbf{B}'$.
6. $(\mathbf{AB})' = \mathbf{B}'\mathbf{A}'$.
7. $\mathbf{A}^2 = \mathbf{AA}$.

A.2.2 Trace

The *trace* of a square matrix $\mathbf{A} \in \mathbb{M}^{n\times n}$ is the sum of the diagonal elements of $\mathbf{A}$, i.e.

$$Tr(\mathbf{A}) = \sum_{i=1}^{n} a_{ii}. \tag{A.7}$$

The trace operator satisfies the following basic properties:

1. $Tr(\mathbf{A}) = Tr\left(\mathbf{A}'\right)$.
2. $Tr(\mathbf{A} + \mathbf{B}) = Tr(\mathbf{A}) + Tr(\mathbf{B})$.
3. $Tr(\beta\mathbf{A}) = \beta Tr(\mathbf{A})$, for every scalar β.
4. $Tr(\mathbf{AB}) = Tr(\mathbf{BA})$.
5. $Tr(\mathbf{ABC}) = Tr(\mathbf{CAB}) = Tr(\mathbf{BCA})$.
6. $Tr(\mathbf{ABC}) = Tr(\mathbf{ACB})$, if $\mathbf{A}, \mathbf{B}, \mathbf{C}$ are symmetric matrices.

A.2.3 Rank

The *rank* of a matrix $\mathbf{A} \in \mathbb{M}^{m\times n}$, indicated by $rank(\mathbf{A})$, is the maximum number of linearly independent columns of $\mathbf{A}$. We have:

1. $rank(\mathbf{AB}) \leq \min\{rank(\mathbf{A})rank(\mathbf{B})\}$.
2. $rank(\mathbf{AA}') = rank(\mathbf{A})$.

A.2.4 Determinant

The *determinant* of a matrix $\mathbf{A} \in \mathbb{M}^{n\times n}$, indicated by $\det(\mathbf{A})$, is a scalar that can be defined iteratively as follows:

$$\begin{aligned}\det(\mathbf{A}) &= a_{11}, \text{ for } \mathbf{A} \in \mathbb{M}^{1\times 1}\\ &= a_{11}a_{22} - a_{12}a_{21}, \text{ for } \mathbf{A} \in \mathbb{M}^{2\times 2}\\ &= \sum_{j=1}^{n} (-1)^{i+j} a_{ij} \det\left(\mathbf{A}^{ij}\right), \text{ for } \mathbf{A} \in \mathbb{M}^{n\times n}, \text{ with } n > 2,\end{aligned}$$

where $\mathbf{A}^{ij}$ is the matrix obtained by deleting the ith row and jth column of $\mathbf{A}$. The inverse $\mathbf{A}^{-1}$ exists if and only if $\mathbf{A}$ has full rank. The determinant is also often indicated by $|\mathbf{A}|$. The determinant satisfies the following properties:

1. Let $\mathbf{A} \in \mathbb{M}^{n\times n}$, and β be a scalar, we have $\det(\beta\mathbf{A}) = \beta^n \det(\mathbf{A})$.
2. Let $\mathbf{A}, \mathbf{B} \in \mathbb{M}^{n\times n}$, then $\det(\mathbf{AB}) = \det(\mathbf{A})\det(\mathbf{B})$.
3. Let $\mathbf{A}, \mathbf{B} \in \mathbb{M}^{n\times n}$, with $\mathbf{B}$ nonsingular then $\det\left(\mathbf{B}^{-1}\mathbf{AB}\right) = \det(\mathbf{A})$.
4. $\det(\mathbf{I}_n) = 1$.
5. Let $\mathbf{A} \in \mathbb{M}^{n\times n}$, $rank(\mathbf{A}) = n$ if and only if $\det(\mathbf{A}) \neq \mathbf{0}$.
6. Let $\mathbf{A} \in \mathbb{M}^{n\times n}$, $rank(\mathbf{A}) < n$ if and only if $\det(\mathbf{A}) = \mathbf{0}$.
7. Let $\mathbf{A} \in \mathbb{M}^{n\times n}$, $\mathbf{B}, \mathbf{C} \in \mathbb{M}^{n\times k}$, $k < n$, and suppose that $\mathbf{A}$ is invertible. Then

$$\left|\mathbf{A} + \mathbf{BC}'\right| = |\mathbf{A}| \left|\mathbf{I}_k + \mathbf{C}'\mathbf{A}^{-1}\mathbf{B}\right|.$$

A.3 Positive definite matrices and quadratic forms

Let $\mathbf{x} = (x_1, x_2 \ldots, x_n)'$, be an n-dimensional vector with real entries, and $\mathbf{A} \in \mathbb{M}^{n\times n}$ be symmetric. The matrix $\mathbf{A}$ is *non-negative definite*, denoted as $\mathbf{A} \geq 0$, if the quadratic form

$$\mathbf{x}'\mathbf{Ax} \geq \mathbf{0},$$

for all $\mathbf{x} \neq \mathbf{0}$. If $\mathbf{x}'\mathbf{Ax} > \mathbf{0}$,for all $\mathbf{x} \neq \mathbf{0}$, then $\mathbf{A}$ is *positive definite*, and write $\mathbf{A} > 0$. Positive definite matrices satisfy the following properties:

1. If $\mathbf{A} > 0$ then also $\mathbf{A}^{-1} > 0$.
2. If $\mathbf{A} > 0$ and $\mathbf{B} > 0$, then also $\mathbf{A} + \mathbf{B} > 0$, $\mathbf{ABA} > 0$, $\mathbf{BAB} > 0$.
3. If $\mathbf{A} > 0$ then $Tr(\mathbf{A}) > 0$.
4. If $\mathbf{A} > 0$ then $\det(\mathbf{A}) > 0$.

A.4 Properties of special matrices

A.4.1 Triangular matrices

An *upper triangular* matrix $\mathbf{A}$ is square and has zeros below the main diagonal, that is, $a_{ij} = 0$ for all $i > j$. A *lower triangular* matrix is square and has zeros above the diagonal, that is, $a_{ij} = 0$ for all $i < j$. A *triangular* matrix is either upper or lower triangular. Properties of triangular matrices are:

1. If $\mathbf{A}$ is upper (lower) triangular then $\mathbf{A}'$ is lower (upper) triangular.
2. If $\mathbf{A}$ is upper (lower) triangular then $\mathbf{A}^{-1}$ is also upper (lower) triangular.
3. If $\mathbf{A}$, $\mathbf{B}$ are upper (lower) triangular then the matrices $\mathbf{A} + \mathbf{B}$ and $\mathbf{AB}$ are also upper (lower) triangular.
4. If $\mathbf{A}$ is upper (lower) triangular then $\det(\mathbf{A}) = \prod_{i=1}^{n} a_{ii}$.

A.4.2 Diagonal matrices

A *diagonal* matrix is a square matrix with non-zero elements only on the main diagonal, that is $a_{ij} = 0$ for all $i \neq j$. The *identity* matrix $\mathbf{I}_n$ is an $n \times n$ matrix having 1 on the main diagonal and zero elsewhere. The identity matrix satisfies

$$\mathbf{A}\mathbf{I}_n = \mathbf{A},$$

for all $\mathbf{A} \in \mathbb{M}^{n\times n}$.

A.4.3 Orthogonal matrices

An *orthogonal* matrix $\mathbf{A} \in \mathbb{M}^{n\times n}$ is such that $\mathbf{A}\mathbf{A}' = \mathbf{A}'\mathbf{A} = \mathbf{I}_n$. An orthogonal matrix $\mathbf{A}$ satisfies:

1. If $\mathbf{A}$, $\mathbf{B}$ are orthogonal then also $\mathbf{AB}$ is orthogonal.
2. If $\mathbf{A} \in \mathbb{M}^{n\times n}$ is orthogonal then $\det(\mathbf{A}) = 1$, and $rank\,(\mathbf{A}) = n$.

A.4.4 Idempotent matrices

A matrix $\mathbf{A} \in \mathbb{M}^{n\times n}$ is *idempotent* if $\mathbf{A}^2 = \mathbf{A}$. Idempotent matrices satisfy the following properties:

1. If $\mathbf{A} \in \mathbb{M}^{n\times n}$ is idempotent then all eigenvalues of $\mathbf{A}$ are 0 or 1.
2. If $\mathbf{A} \in \mathbb{M}^{n\times n}$ is idempotent then $rank(\mathbf{A}) = Tr(\mathbf{A})$.
3. If $\mathbf{A} \in \mathbb{M}^{n\times n}$ is idempotent and nonsingular then $\mathbf{A} = \mathbf{I}_n$.
4. If $\mathbf{A} \in \mathbb{M}^{n\times n}$ is diagonal, then all elements of $\mathbf{A}$ are either 0 or 1.

Further results on idempotent matrices can be found in Lütkepohl (1996).

A.5 Eigenvalues and eigenvectors

Let $\mathbf{A} \in \mathbb{M}^{n\times n}$. An *eigenvalue* and *eigenvector* of $\mathbf{A}$ are a scalar, λ, and a non-zero vector, $\mathbf{x}$, such that

$$\mathbf{A}\mathbf{x} = \lambda\mathbf{x},$$

or

$$(\mathbf{A} - \lambda\mathbf{I}_n)\,\mathbf{x} = \mathbf{0}, \;\; \mathbf{x} \neq \mathbf{0}.$$

This implies that $(\mathbf{A}-\lambda\mathbf{I}_n)$ is singular and hence that

$$\det(\mathbf{A} - \lambda\mathbf{I}_n) = 0. \tag{A.8}$$

The above expression is called the characteristic equation or characteristic polynomial of $\mathbf{A}$. Let $\lambda_1(\mathbf{A}), \lambda_2(\mathbf{A}), \ldots, \lambda_n(\mathbf{A})$ be the eigenvalues of $\mathbf{A}$. The following properties hold:

1. $\lambda_i(\mathbf{A}) = \lambda_i\left(\mathbf{A}'\right)$.
2. $Tr(\mathbf{A}) = \sum_{i=1}^{n} a_{ii} = \sum_{i=1}^{n} \lambda_i(\mathbf{A})$.
3. $\det(\mathbf{A}) = \prod_{i=1}^{n} \lambda_i(\mathbf{A})$.
4. If $\lambda_i(\mathbf{A}) \geq 0$ for all i, then $\mathbf{A} \geq 0$.
5. If $\lambda_i(\mathbf{A}) > 0$ for all i, then $\mathbf{A} > 0$.
6. If $\mathbf{A} \geq 0$ then $\lambda_i(\mathbf{A})$ for all i are real eigenvalues.
7. If $\mathbf{B}$ is positive semi-definite, $Tr(\mathbf{AB}) \leq \sqrt{\lambda_{\max}(\mathbf{A}'\mathbf{A})}\, Tr(\mathbf{B})$.

Let $\lambda_{\min}(\mathbf{A})$ and $\lambda_{\max}(\mathbf{A})$ be the minimum and maximum eigenvalues of a symmetric matrix $\mathbf{A}$, respectively. Then the *Rayleigh–Ritz theorem* states that

$$\lambda_{\min}(\mathbf{A}) = \min_{\mathbf{x}\neq\mathbf{0}} \left\{ \frac{\mathbf{x}'\mathbf{A}\mathbf{x}}{\mathbf{x}'\mathbf{x}} \right\},$$
$$\lambda_{\max}(\mathbf{A}) = \max_{\mathbf{x}\neq\mathbf{0}} \left\{ \frac{\mathbf{x}'\mathbf{A}\mathbf{x}}{\mathbf{x}'\mathbf{x}} \right\}.$$

The *Courant–Fischer theorem* states that

$$\lambda_i(\mathbf{A}) = \min_{\mathbf{y}_1,\mathbf{y}_2,\ldots,\mathbf{y}_{n-i}} \max_{\substack{\mathbf{x}\neq\mathbf{0},\\ \mathbf{x}'\mathbf{y}_j=0, j=1,2,\ldots,n-i}} \left\{ \frac{\mathbf{x}'\mathbf{A}\mathbf{x}}{\mathbf{x}'\mathbf{x}} \right\},$$

and

$$\lambda_i(\mathbf{A}) = \max_{\mathbf{y}_1,\mathbf{y}_2,\ldots,\mathbf{y}_{i-1}} \min_{\substack{\mathbf{x}\neq\mathbf{0},\\ \mathbf{x}'\mathbf{y}_j=0, j=1,2,\ldots,i-1}} \left\{ \frac{\mathbf{x}'\mathbf{A}\mathbf{x}}{\mathbf{x}'\mathbf{x}} \right\}.$$

See Horn and Johnson (1985) and Bernstein (2005) for further properties of eigenvalues.

A.6 Inverse of a matrix

Let $\mathbf{A} \in \mathbb{M}^{n\times n}$ with $det(\mathbf{A}) \neq 0$. The inverse of $\mathbf{A}$ is the unique $n \times n$ matrix, $\mathbf{A}^{-1}$, such that $\mathbf{A}^{-1}\mathbf{A} = \mathbf{A}\mathbf{A}^{-1} = \mathbf{I}_n$. The inverse $\mathbf{A}^{-1}$ exists if and only if $\mathbf{A}$ has full rank. If $\mathbf{A}^{-1}$ exists we say that $\mathbf{A}$ is nonsingular.

The inverse satisfies the following properties:

1. Let $\mathbf{A} \in \mathbb{M}^{n\times n}$ be nonsingular. Then $(c\mathbf{A})^{-1} = c^{-1}\mathbf{A}^{-1}$.
2. Let $\mathbf{A}, \mathbf{B} \in \mathbb{M}^{n\times n}$ be nonsingular. Then $(\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}$.
3. Let $\mathbf{A} \in \mathbb{M}^{n\times n}$ be nonsingular matrix with eigenvalues $\lambda_1(\mathbf{A}), \lambda_2(\mathbf{A}), \ldots, \lambda_n(\mathbf{A})$. Then $\mathbf{A}^{-1}$ has eigenvalues $\lambda_1(\mathbf{A})^{-1}, \lambda_2(\mathbf{A})^{-1}, \ldots, \lambda_n(\mathbf{A})^{-1}$.
4. Let $\mathbf{A}, \mathbf{B} \in \mathbb{M}^{n\times n}$. Then $\mathbf{B}^{-1} = \mathbf{A}^{-1} - \mathbf{B}^{-1}(\mathbf{B} - \mathbf{A})\mathbf{A}^{-1}$.

5. Let $\mathbf{A}, \mathbf{B} \in \mathbb{M}^{n\times n}$, and $\mathbf{U}, \mathbf{V} \in \mathbb{M}^{n\times k}$, with $\mathbf{A}, \mathbf{B}$ nonsingular. Then

$$\left(\mathbf{A} + \mathbf{UBV}'\right)^{-1} = \mathbf{A}^{-1} - \mathbf{A}^{-1}\mathbf{U}\left(\mathbf{B}^{-1} + \mathbf{V}'\mathbf{A}^{-1}\mathbf{U}\right)^{-1}\mathbf{V}'\mathbf{A}^{-1}, \tag{A.9}$$

   This is known as the Woodbury matrix identity, which is a generalization of the Sherman–Morrison formula. The latter obtains if $\mathbf{B} = \mathbf{I}_n$, and $k = 1$. Also see Dhrymes (2000, p. 44).
6. Let $\mathbf{A} \in \mathbb{M}^{n\times n}$. If $\lim_{k\to\infty} \mathbf{A}^k = \mathbf{0}$ then $\mathbf{I} - \mathbf{A}$ is nonsingular and

$$(\mathbf{I} - \mathbf{A})^{-1} = \sum_{k=0}^{\infty} \mathbf{A}^k.$$

   This is known as *the Neumann series*.

A.7 Generalized inverses

Let $\mathbf{A} \in \mathbb{M}^{m\times n}$ of any rank. A generalized inverse of $\mathbf{A}$ is an $n \times m$ matrix $\mathbf{A}^-$ such that $\mathbf{AA}^-\mathbf{A} = \mathbf{A}$. The matrix $\mathbf{A}^-$ is not unique in general. It satisfies the following properties:

1. $\mathbf{A}^-$ exists and $rank\left(\mathbf{A}^-\right) \geq rank\,(\mathbf{A})$.
2. The matrices $\mathbf{AA}^-$ and $\mathbf{A}^-\mathbf{A}$ are idempotent matrices.
3. If $\mathbf{A}$ is idempotent, then $\mathbf{A}$ is also the generalized inverse of itself.
4. A general solution of $\mathbf{AX} = \mathbf{0}$ is $(\mathbf{A}^-\mathbf{A} - \mathbf{I}_n)\mathbf{Z}$ where $\mathbf{Z}$ is arbitrary.
5. $Tr\left(\mathbf{AA}^-\right) = rank(\mathbf{A})$.

A.7.1 Moore–Penrose inverse

Let $\mathbf{A} \in \mathbb{M}^{m\times n}$. The $n \times m$ matrix $\mathbf{A}^+$ is the Moore–Penrose generalized inverse of $\mathbf{A}$ if it satisfies the following four conditions:

(i) $\mathbf{AA}^+\mathbf{A} = \mathbf{A}$.
(ii) $\mathbf{A}^+\mathbf{AA}^+ = \mathbf{A}^+$.
(iii) $\left(\mathbf{A}^+\mathbf{A}\right)' = \mathbf{A}^+\mathbf{A}$.
(iv) $\left(\mathbf{AA}^+\right)' = \mathbf{AA}^+$.

The matrix $\mathbf{A}^+$ exists and is unique.

Further properties of inverse matrices, the generalized inverse and the Moore–Penrose inverse can be found in Lütkepohl (1996).

A.8 Kronecker product and the vec operator

Let $\mathbf{A} \in \mathbb{M}^{m\times n}$, $\mathbf{B} \in \mathbb{M}^{p\times q}$. The Kronecker product of $\mathbf{A}$ and $\mathbf{B}$ is defined by

$$\underset{mp\times nq}{\mathbf{A}\otimes\mathbf{B}}=\begin{pmatrix} a_{11}\mathbf{B} & a_{12}\mathbf{B} & \dots & a_{1n}\mathbf{B} \\ a_{21}\mathbf{B} & a_{22}\mathbf{B} & \dots & a_{2n}\mathbf{B} \\ \vdots & \vdots & \vdots & \vdots \\ a_{m1}\mathbf{B} & a_{m2}\mathbf{B} & \dots & a_{mn}\mathbf{B} \end{pmatrix}. \tag{A.10}$$

The Kronecker product satisfies the following properties:

1. $(\mathbf{A}\otimes\mathbf{B})'=\left(\mathbf{A}'\otimes\mathbf{B}'\right)$.
2. $Tr\,(\mathbf{A}\otimes\mathbf{B})=Tr\,(\mathbf{A})\,Tr\,(\mathbf{B})$.
3. $(\mathbf{A}\otimes\mathbf{B})^{-1}=\left(\mathbf{A}^{-1}\otimes\mathbf{B}^{-1}\right)$, for $\mathbf{A}$, $\mathbf{B}$ nonsingular.
4. $(\mathbf{A}\otimes\mathbf{B})^{-}=\left(\mathbf{A}^{-}\otimes\mathbf{B}^{-}\right)$.
5. Let $\mathbf{A}\in\mathbb{M}^{n\times n}$, $\mathbf{B}\in\mathbb{M}^{m\times m}$, we have $\det(\mathbf{A}\otimes\mathbf{B})=[\det(\mathbf{A})]^{m}[\det(\mathbf{B})]^{n}$.
6. Let $\mathbf{A}\in\mathbb{M}^{m\times n}$, $\mathbf{B}\in\mathbb{M}^{p\times q}$, $\mathbf{C}\in\mathbb{M}^{n\times s}$, $\mathbf{D}\in\mathbb{M}^{q\times g}$, we have

$$(\mathbf{A}\otimes\mathbf{B})(\mathbf{C}\otimes\mathbf{D})=(\mathbf{AC}\otimes\mathbf{BD}). \tag{A.11}$$

Let $\mathbf{A}\in\mathbb{M}^{n\times m}$. Then $vec\,(\mathbf{A})$ is defined to be the mn-dimensional vector formed by stacking the columns of $\mathbf{A}$ on top of each other, that is,

$$vec\,(\mathbf{A})=(a_{11},a_{21},\dots,a_{n1},a_{12},a_{22},\dots,a_{n2},\dots,a_{1m},a_{2m},\dots.,a_{nm})'.$$

The *vec* operator satisfies

1. Let $\mathbf{A},\mathbf{B}\in\mathbb{M}^{m\times n}$ then

$$vec\,(\mathbf{A}+\mathbf{B})=vec\,(\mathbf{A})+vec\,(\mathbf{B}).$$

2. Let $\mathbf{A}\in\mathbb{M}^{m\times n}$, $\mathbf{B}\in\mathbb{M}^{n\times p}$ then

$$vec\,(\mathbf{AB})=\left(\mathbf{I}_p\otimes\mathbf{A}\right)vec\,(\mathbf{B})=\left(\mathbf{B}'\otimes\mathbf{I}_m\right)vec\,(\mathbf{A})=\left(\mathbf{B}'\otimes\mathbf{A}\right)vec\,(\mathbf{I}_n).$$

3. Let $\mathbf{A}\in\mathbb{M}^{m\times n}$, $\mathbf{B}\in\mathbb{M}^{n\times r}$, $\mathbf{C}\in\mathbb{M}^{r\times s}$, we have

$$vec(\mathbf{ABC})=(\mathbf{C}'\otimes\mathbf{A})vec(\mathbf{B}). \tag{A.12}$$

4. Let $\mathbf{x},\mathbf{y}$ be two n-dimensional vectors, we have

$$vec\left(\mathbf{xy}'\right)=\mathbf{x}\otimes\mathbf{y}. \tag{A.13}$$

5. Let $\mathbf{A}\in\mathbb{M}^{m\times n}$, $\mathbf{B}\in\mathbb{M}^{n\times p}$, $\mathbf{C}\in\mathbb{M}^{p\times q}$, $\mathbf{D}\in\mathbb{M}^{q\times m}$, we have

$$vec(\mathbf{D}')'(\mathbf{C}'\otimes\mathbf{A})vec(\mathbf{B})=Tr\,(\mathbf{ABCD}). \tag{A.14}$$

6. Let $\mathbf{A} \in \mathbb{M}^{m\times n}$, $\mathbf{B} \in \mathbb{M}^{n\times p}$, $\mathbf{C} \in \mathbb{M}^{p\times q}$, $\mathbf{D} \in \mathbb{M}^{q\times m}$, we have

$$vec(\mathbf{D}')'(\mathbf{C}' \otimes \mathbf{A})vec(\mathbf{B}) = vec(\mathbf{A}')'(\mathbf{D}' \otimes \mathbf{B})vec(\mathbf{C}). \tag{A.15}$$

Let $\mathbf{A} \in \mathbb{M}^{n\times n}$. Then $vech(\mathbf{A})$ is defined to be the $n\,(n+1)\,/2$-dimensional vector with the elements on and below the principal diagonal $\mathbf{A}$ stacked on top of each other. In other words, $vech(\mathbf{A})$ is given by the vectorization of $\mathbf{A}$ using only the elements on and below the principal diagonal:

$$vech(\mathbf{A}) = (a_{11}, a_{21}, \ldots, a_{n1}, a_{22}, \ldots, a_{n2}, \ldots, a_{nn})'.$$

Further results on Kronecker and *vec* operators can be found in Bernstein (2005).

A.9 Partitioned matrices

A partitioned, or block, matrix is a matrix $\mathbf{A} \in \mathbb{M}^{m\times n}$ consisting of sub-matrices $\mathbf{A}_{ij} \in \mathbb{M}^{m_i\times n_j}$, $i = 1, 2, \ldots, p, j = 1, 2, \ldots, q$. A *block diagonal* matrix $\mathbf{A} \in \mathbb{M}^{m\times n}$ is such that

$$\mathbf{A} = \begin{pmatrix} \mathbf{A}_1 & \mathbf{0} & \ldots & \mathbf{0} \\ \mathbf{0} & \mathbf{A}_2 & \ldots & \mathbf{0} \\ \vdots & \vdots & \vdots & \vdots \\ \mathbf{0} & \mathbf{0} & \ldots & \mathbf{A}_p \end{pmatrix},$$

with sub-matrices $\mathbf{A}_i$ $i = 1, 2, \ldots, p$. Its inverse is

$$\mathbf{A}^{-1} = \begin{pmatrix} \mathbf{A}_1^{-1} & \mathbf{0} & \ldots & \mathbf{0} \\ \mathbf{0} & \mathbf{A}_2^{-1} & \ldots & \mathbf{0} \\ \vdots & \vdots & \vdots & \vdots \\ \mathbf{0} & \mathbf{0} & \ldots & \mathbf{A}_p^{-1} \end{pmatrix}.$$

Let

$$\mathbf{A} = \begin{pmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} \\ \mathbf{A}_{21} & \mathbf{A}_{22} \end{pmatrix},$$

where $\mathbf{A}_{11}$ is $m_1 \times m_1$, $\mathbf{A}_{12}$ is $m_1 \times n_1$, $\mathbf{A}_{21}$ is $n_1 \times m_1$, and $\mathbf{A}_{22}$ is $n_1 \times n_1$. The following properties can be derived:

1. Its transpose is

$$\begin{pmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} \\ \mathbf{A}_{21} & \mathbf{A}_{22} \end{pmatrix}' = \begin{pmatrix} \mathbf{A}'_{11} & \mathbf{A}'_{21} \\ \mathbf{A}'_{12} & \mathbf{A}'_{22} \end{pmatrix}.$$

2. $Tr\,(\mathbf{A}) = Tr\,(\mathbf{A}_{11}) + Tr(\mathbf{A}_{22})$.

3. If $\mathbf{A}_{11}$ and $\mathbf{A}_{22} - \mathbf{A}_{21}\mathbf{A}_{11}^{-1}\mathbf{A}_{12}$ are nonsingular, then

$$\mathbf{A}^{-1} = \begin{pmatrix} \mathbf{A}_{11}^{-1} + \mathbf{A}_{11}^{-1}\mathbf{B}^{-1}\mathbf{A}_{11} & -\mathbf{A}_{11}^{-1}\mathbf{A}_{12}\mathbf{B}^{-1} \\ -\mathbf{B}^{-1}\mathbf{A}_{21}\mathbf{A}_{11}^{-1} & \mathbf{B}^{-1} \end{pmatrix}, \tag{A.16}$$

where

$$\mathbf{B} = \mathbf{A}_{22} - \mathbf{A}_{21}\mathbf{A}_{11}^{-1}\mathbf{A}_{12}.$$

4. If $\mathbf{A}_{22}$ and $\mathbf{A}_{11} - \mathbf{A}_{12}\mathbf{A}_{22}^{-1}\mathbf{A}_{21}$ are nonsingular, then

$$\mathbf{A}^{-1} = \begin{pmatrix} \mathbf{C}^{-1} & -\mathbf{C}^{-1}\mathbf{A}_{12}\mathbf{A}_{22}^{-1} \\ -\mathbf{A}_{22}^{-1}\mathbf{A}_{21}\mathbf{C}^{-1} & \mathbf{A}_{22}^{-1} + \mathbf{A}_{22}^{-1}\mathbf{A}_{21}\mathbf{C}^{-1}\mathbf{A}_{12}\mathbf{A}_{22}^{-1} \end{pmatrix}, \tag{A.17}$$

where

$$\mathbf{C} = \mathbf{A}_{11} - \mathbf{A}_{12}\mathbf{A}_{22}^{-1}\mathbf{A}_{21}.$$

5. If $\mathbf{A}_{11}$ is nonsingular then $\det(\mathbf{A}) = \det(\mathbf{A}_{11}) \det\left(\mathbf{A}_{22} - \mathbf{A}_{21}\mathbf{A}_{11}^{-1}\mathbf{A}_{12}\right)$.
6. If $\mathbf{A}_{22}$ is nonsingular then $\det(\mathbf{A}) = \det(\mathbf{A}_{22}) \det\left(\mathbf{A}_{11} - \mathbf{A}_{12}\mathbf{A}_{22}^{-1}\mathbf{A}_{21}\right)$.

Further properties of partitioned matrices can be found in Bernstein (2005).

A.10 Matrix norms

The norm of a matrix $\mathbf{A} \in \mathbb{M}^{m\times n}$, $\|\mathbf{A}\|$, is a scalar function satisfying the following properties:

1. *Positivity*: $\|\mathbf{A}\| > 0$ if $\mathbf{A} \neq \mathbf{0}$, and $\|\mathbf{A}\| = 0$ if and only if $\mathbf{A} = \mathbf{0}$.
2. *Homogeneity*: $\|\beta\mathbf{A}\| = |\beta| \, \|\mathbf{A}\|$, for every scalar β.
3. *Triangle inequality*: $\|\mathbf{A} + \mathbf{B}\| \leq \|\mathbf{A}\| + \|\mathbf{B}\|$, for all $\mathbf{B} \in \mathbb{M}^{m\times n}$.

The following matrix norms are often used:
The *column norm* of $\mathbf{A}$ is

$$\|\mathbf{A}\|_1 = \max_{1\leq j\leq n} \sum_{i=1}^{m} \left|a_{ij}\right| \tag{A.18}$$

The *row norm* of $\mathbf{A}$ is

$$\|\mathbf{A}\|_\infty = \max_{1\leq i\leq m} \sum_{j=1}^{n} \left|a_{ij}\right|. \tag{A.19}$$

The *Euclidean norm* (or *Frobenius norm*) of $\mathbf{A}$ is

$$\|\mathbf{A}\|_2 = \left[Tr(\mathbf{A}'\mathbf{A})\right]^{1/2} = \left(\sum_{i=1}^{m}\sum_{j=1}^{n} a_{ij}^2\right)^{1/2}. \tag{A.20}$$

The matrix $\mathbf{A}$ is *square summable* if $\|\mathbf{A}\|_2 \leq K < \infty$.

The *spectral norm* of $\mathbf{A}$ is

$$\|\mathbf{A}\|_{spec} = \rho_{\max}(\mathbf{A}) = \max_{1\leq i\leq n}\left\{\sqrt{\lambda_i(\mathbf{A}'\mathbf{A})}\right\},$$

where $\rho_{\max}(\mathbf{A})$ is also known as the maximum singular value of $\mathbf{A}$.

A norm is called multiplicative if

$$\|\mathbf{AB}\| \leq \|\mathbf{A}\|\,\|\mathbf{B}\|,$$

for all matrices $\mathbf{A}, \mathbf{B} \in \mathbb{M}^{n\times n}$.

Some properties of matrix norms are:

1. $\frac{1}{\sqrt{m}}\|\mathbf{A}\|_1 \leq \|\mathbf{A}\|_2 \leq \sqrt{n}\,\|\mathbf{A}\|_1$.
2. $\frac{1}{\sqrt{n}}\|\mathbf{A}\|_\infty \leq \|\mathbf{A}\|_2 \leq \sqrt{m}\,\|\mathbf{A}\|_\infty$.
3. $\|\mathbf{A}\|_2 \leq \sqrt{\|\mathbf{A}\|_1\,\|\mathbf{A}\|_\infty}$.
4. $\|\mathbf{A}\|_2 \leq \min\left(\sqrt{m}, \sqrt{n}\right)\|\mathbf{A}\|_{spec}$, for $\mathbf{A} \in \mathbb{M}^{n\times n}$.
5. $\left\|\mathbf{A}'\mathbf{B}\right\|_2 \leq \|\mathbf{A}\|_2\,\|\mathbf{B}\|_2$, for $\mathbf{A}, \mathbf{B} \in \mathbb{M}^{m\times n}$.
6. $\left|Tr\left(\mathbf{A}'\mathbf{B}\right)\right| \leq \|\mathbf{A}\|_2\,\|\mathbf{B}\|_2$, for $\mathbf{A}, \mathbf{B} \in \mathbb{M}^{m\times n}$.
7. $\|\mathbf{A}\|_1 \leq m\,\|\mathbf{A}\|_\infty$.
8. $\|\mathbf{A}\|_\infty \leq n\,\|\mathbf{A}\|_1$.
9. $\|\mathbf{A}\|_1 \leq \sqrt{n}\,\|\mathbf{A}\|_{spec}$, for $\mathbf{A} \in \mathbb{M}^{n\times n}$.
10. $\|\mathbf{A}\|_\infty \leq \sqrt{n}\,\|\mathbf{A}\|_{spec}$, for $\mathbf{A} \in \mathbb{M}^{n\times n}$.

A wide discussion on matrix norms and their properties can be found in Bernstein (2005).

A.11 Spectral radius

Let λ_i be the ith eigenvalue of $\mathbf{A} \in \mathbb{M}^{n\times n}$. The spectral radius of $\mathbf{A}$ denoted by $\rho(\mathbf{A})$ is given by $\rho(\mathbf{A}) = \max_i |\lambda_i|$.

Suppose now $\|\mathbf{A}\|$ is a matrix norm of $\mathbf{A}$. Then

$$\rho(\mathbf{A}) \leq \left\|\mathbf{A}^s\right\|^{1/s},$$

where s is a positive integer. In particular, $\rho(\mathbf{A}) \leq \|\mathbf{A}\|$.

$$\lim_{s\to\infty} \mathbf{A}^s = 0, \quad \Longleftrightarrow \quad \rho(\mathbf{A}) < 1.$$
$$\rho(\mathbf{A}) = \lim_{s\to\infty} \left\|\mathbf{A}^s\right\|^{1/s}.$$

A.12 Matrix decompositions

A.12.1 Schur decomposition

Let $\mathbf{A} \in \mathbb{M}^{n\times n}$ be a matrix with real eigenvalues. There exists a real orthogonal, $n \times n$ matrix $\mathbf{U}$ and a real upper triangular matrix $\mathbf{\Lambda}$ with the eigenvalues of $\mathbf{A}$ on the principal diagonal, such that

$$\mathbf{A} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}'.$$

A.12.2 Generalized Schur decomposition

Let $\mathbf{A}, \mathbf{B} \in \mathbb{M}^{n\times n}$ be two matrices. There exist two unitary, $n \times n$ matrices $\mathbf{Q}$ and $\mathbf{Z}$ such that

$$\mathbf{Q}'\mathbf{A}\mathbf{Z} = \mathbf{T},$$
$$\mathbf{Q}'\mathbf{B}\mathbf{Z} = \mathbf{S},$$

Where $\mathbf{T}$ is an upper quasi-triangular (namely, a block matrix with non-zero elements on the blocks along the main diagonal and on the entries above the main diagonal), while $\mathbf{S}$ is an upper triangular matrix. If the diagonal elements of $\mathbf{T}$ and $\mathbf{S}$, namely t_{kk} and s_{kk} are non-zero then

$$\lambda_i(\mathbf{A}, \mathbf{B}) = \frac{t_{ii}}{s_{ii}},$$

are the so-called generalized eigenvalues and are solutions of the equation $\mathbf{A}\mathbf{x} - \lambda\mathbf{B}\mathbf{x} = \mathbf{0}$. The columns of the matrices $\mathbf{Q}$ and $\mathbf{Z}$ are called generalized Schur vectors.

For further details on the Generalized Schur decomposition see Golub and Van Loan (1996, p. 377).

A.12.3 Spectral decomposition

Let $\mathbf{A} \in \mathbb{M}^{n\times n}$ be a real symmetric matrix with eigenvalues $\lambda_1(\mathbf{A}), \lambda_2(\mathbf{A}), \ldots, \lambda_n(\mathbf{A})$. Then $\mathbf{A}$ can be decomposed as follows

$$\mathbf{A} = \mathbf{C}\mathbf{\Lambda}\mathbf{C}',$$

where $\mathbf{C} \in \mathbb{M}^{n\times n}$ is a real orthogonal matrix, whose columns are the n eigenvectors of $\mathbf{A}$, and $\mathbf{\Lambda} = diag\{\lambda_1(\mathbf{A}), \lambda_2(\mathbf{A}), \ldots, \lambda_n(\mathbf{A})\}$.

A.12.4 Jordan decomposition

Let $\mathbf{A} \in \mathbb{M}^{n\times n}$ be a matrix with $m < n$ distinct eigenvalues $\lambda_1(\mathbf{A}), \lambda_2(\mathbf{A}), \ldots, \lambda_m(\mathbf{A})$. There exists a nonsingular matrix $\mathbf{T}$ such that

$$\mathbf{A} = \mathbf{T}\boldsymbol{\Lambda}\mathbf{T}^{-1},$$

where

$$\boldsymbol{\Lambda} = \begin{pmatrix} \boldsymbol{\Lambda}_1 & \mathbf{0} & \ldots & \mathbf{0} \\ \mathbf{0} & \boldsymbol{\Lambda}_2 & \ldots & \mathbf{0} \\ \vdots & \vdots & \vdots & \vdots \\ \mathbf{0} & \mathbf{0} & \ldots & \boldsymbol{\Lambda}_p \end{pmatrix}$$

is block diagonal with

$$\boldsymbol{\Lambda}_i = \begin{pmatrix} \lambda_i(\mathbf{A}) & 1 & 0 & \ldots & 0 & 0 \\ 0 & \lambda_i(\mathbf{A}) & 1 & \ldots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \ddots & \vdots \\ 0 & 0 & 0 & \ldots & \lambda_i(\mathbf{A}) & 1 \end{pmatrix}.$$

If $\mathbf{A} \in \mathbb{M}^{n\times n}$ has n distinct eigenvalues $\lambda_1(\mathbf{A}), \lambda_2(\mathbf{A}), \ldots, \lambda_n(\mathbf{A})$, then

$$\mathbf{A} = \mathbf{T}\boldsymbol{\Lambda}\mathbf{T}^{-1},$$

where $\boldsymbol{\Lambda} = diag\{\lambda_1(\mathbf{A}), \lambda_2(\mathbf{A}), \ldots, \lambda_n(\mathbf{A})\}$ and $\mathbf{T}$ is a nonsingular $n \times n$ matrix whose columns are the n eigenvectors associated to the eigenvalues of $\mathbf{A}$.

A.12.5 Cholesky decomposition

Let $\mathbf{A} \in \mathbb{M}^{n\times n}$ be positive definite. There exists a unique lower (upper) triangular matrix $\mathbf{L}$ with real positive diagonal entries such that

$$\mathbf{A} = \mathbf{L}\mathbf{L}'.$$

See Lütkepohl (1996) for further discussion on the above matrix decompositions.

A.13 Matrix calculus

Let $f(\mathbf{x})$ be a differentiable real valued function of the real n-dimensional vector, $\mathbf{x} = (x_1, x_2, \ldots, x_n)'$. Then

$$\frac{\partial f(\mathbf{x})}{\partial \mathbf{x}} = \begin{pmatrix} \dfrac{\partial f(\mathbf{x})}{\partial x_1} \\ \dfrac{\partial f(\mathbf{x})}{\partial x_2} \\ \vdots \\ \dfrac{\partial f(\mathbf{x})}{\partial x_n} \end{pmatrix}, \tag{A.21}$$

is the vector of first-order partial derivatives. $\frac{\partial f(\mathbf{x})}{\partial \mathbf{x}}$ is sometimes called the *gradient vector* of $f(\mathbf{x})$.

$$\left.\frac{\partial f}{\partial \mathbf{x}}\right|_{\mathbf{x}=\mathbf{x}_0} = \frac{\partial f(\mathbf{x}_0)}{\partial \mathbf{x}} = \begin{pmatrix} \left.\dfrac{\partial f(\mathbf{x})}{\partial x_1}\right|_{\mathbf{x}=\mathbf{x}_0} \\ \left.\dfrac{\partial f(\mathbf{x})}{\partial x_2}\right|_{\mathbf{x}=\mathbf{x}_0} \\ \vdots \\ \left.\dfrac{\partial f(\mathbf{x})}{\partial x_n}\right|_{\mathbf{x}=\mathbf{x}_0} \end{pmatrix}, \tag{A.22}$$

is the vector of first-order partial derivatives evaluated at $\mathbf{x} = \mathbf{x}_0$. The *Hessian matrix* of second-order partial derivatives of $f(\mathbf{x})$ is

$$\frac{\partial^2 \mathbf{f}(\mathbf{x})}{\partial \mathbf{x} \partial \mathbf{x}'} = \begin{pmatrix} \dfrac{\partial^2 f(\mathbf{x})}{\partial x_1 \partial x_1} & \dfrac{\partial^2 f(\mathbf{x})}{\partial x_1 \partial x_2} & \cdots & \dfrac{\partial^2 f(\mathbf{x})}{\partial x_1 \partial x_n} \\ \dfrac{\partial^2 f(\mathbf{x})}{\partial x_2 \partial x_1} & \dfrac{\partial^2 f(\mathbf{x})}{\partial x_2 \partial x_2} & \cdots & \dfrac{\partial^2 f(\mathbf{x})}{\partial x_2 \partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \dfrac{\partial^2 f(\mathbf{x})}{\partial x_n \partial x_1} & \dfrac{\partial^2 f(\mathbf{x})}{\partial x_n \partial x_2} & \cdots & \dfrac{\partial^2 f(\mathbf{x})}{\partial x_n \partial x_n} \end{pmatrix}.$$

The *Hessian matrix* of second-order partial derivatives of $f(\mathbf{x})$ evaluated at $\mathbf{x} = \mathbf{x}_0$ is

$$\left.\frac{\partial^2 \mathbf{f}}{\partial \mathbf{x} \partial \mathbf{x}'}\right|_{\mathbf{x}=\mathbf{x}_0} = \frac{\partial^2 \mathbf{f}(\mathbf{x}_0)}{\partial \mathbf{x} \partial \mathbf{x}'} = \begin{pmatrix} \left.\dfrac{\partial^2 f}{\partial x_1 \partial x_1}\right|_{\mathbf{x}=\mathbf{x}_0} & \left.\dfrac{\partial^2 f}{\partial x_1 \partial x_2}\right|_{\mathbf{x}=\mathbf{x}_0} & \cdots & \left.\dfrac{\partial^2 f}{\partial x_1 \partial x_n}\right|_{\mathbf{x}=\mathbf{x}_0} \\ \left.\dfrac{\partial^2 f}{\partial x_2 \partial x_1}\right|_{\mathbf{x}=\mathbf{x}_0} & \left.\dfrac{\partial^2 f}{\partial x_2 \partial x_2}\right|_{\mathbf{x}=\mathbf{x}_0} & \cdots & \left.\dfrac{\partial^2 f}{\partial x_2 \partial x_n}\right|_{\mathbf{x}=\mathbf{x}_0} \\ \vdots & \vdots & \ddots & \vdots \\ \left.\dfrac{\partial^2 f}{\partial x_n \partial x_1}\right|_{\mathbf{x}=\mathbf{x}_0} & \left.\dfrac{\partial^2 f}{\partial x_n \partial x_2}\right|_{\mathbf{x}=\mathbf{x}_0} & \cdots & \left.\dfrac{\partial^2 f}{\partial x_n \partial x_n}\right|_{\mathbf{x}=\mathbf{x}_0} \end{pmatrix}.$$

Let $f(\mathbf{X})$ be a differentiable real valued function of the matrix $\mathbf{X} \in \mathbb{M}^{m \times n}$. The matrix of first-order partial derivatives is

$$\frac{\partial f(\mathbf{X})}{\partial \mathbf{X}} = \begin{pmatrix} \frac{\partial f(\mathbf{x})}{\partial x_{11}} & \frac{\partial f(\mathbf{x})}{\partial x_{12}} & \cdots & \frac{\partial f(\mathbf{x})}{\partial x_{1n}} \\ \frac{\partial f(\mathbf{x})}{\partial x_{21}} & \frac{\partial f(\mathbf{x})}{\partial x_{22}} & \cdots & \frac{\partial f(\mathbf{x})}{\partial x_{2n}} \\ \vdots & & \ddots & \vdots \\ \frac{\partial f(\mathbf{x})}{\partial x_{m1}} & \frac{\partial f(\mathbf{x})}{\partial x_{m2}} & \cdots & \frac{\partial f(\mathbf{x})}{\partial x_{mn}} \end{pmatrix}. \tag{A.23}$$

and the Hessian matrix of second-order partial derivatives of $f(\mathbf{X})$ is the $mn \times mn$ matrix

$$\frac{\partial^2 f(\mathbf{X})}{\partial vec(\mathbf{X})\, \partial vec(\mathbf{X})'}.$$

The following properties of matrix derivatives hold:

1. Let $\mathbf{A} \in \mathbb{M}^{n\times n}$, and $\mathbf{x}$ be an n-dimensional vector then $\frac{\partial \mathbf{x}'\mathbf{A}\mathbf{x}}{\partial \mathbf{x}} = \left(\mathbf{A} + \mathbf{A}'\right)\mathbf{x}$.
2. Let $\mathbf{A} \in \mathbb{M}^{n\times n}$, and $\mathbf{x}$ be an n-dimensional vector then $\frac{\partial^2 \mathbf{x}'\mathbf{A}\mathbf{x}}{\partial \mathbf{x}\partial \mathbf{x}'} = \mathbf{A} + \mathbf{A}'$.
3. Let $\mathbf{X} \in \mathbb{M}^{n\times n}$ then $\frac{\partial Tr(\mathbf{X})}{\partial \mathbf{X}} = \frac{\partial Tr(\mathbf{X}')}{\partial \mathbf{X}} = \mathbf{I}_n$.
4. Let $\mathbf{X} \in \mathbb{M}^{m\times n}$, $\mathbf{A} \in \mathbb{M}^{n\times m}$ then $\frac{\partial Tr(\mathbf{A}\mathbf{X})}{\partial \mathbf{X}} = \frac{\partial Tr(\mathbf{X}\mathbf{A})}{\partial \mathbf{X}} = \mathbf{A}'$.
5. Let $\mathbf{X} \in \mathbb{M}^{m\times n}$, $\mathbf{A} \in \mathbb{M}^{p\times m}$, $\mathbf{B} \in \mathbb{M}^{n\times p}$ then $\frac{\partial Tr(\mathbf{A}\mathbf{X}\mathbf{B})}{\partial \mathbf{X}} = \mathbf{A}'\mathbf{B}'$.
6. Let $\mathbf{X} \in \mathbb{M}^{m\times n}$, $\mathbf{A} \in \mathbb{M}^{n\times n}$ then $\frac{\partial Tr(\mathbf{X}\mathbf{A}\mathbf{X}')}{\partial \mathbf{X}} = \mathbf{X}\left(\mathbf{A} + \mathbf{A}'\right)$.
7. Let $\mathbf{X} \in \mathbb{M}^{m\times m}$ then $\frac{\partial \det(\mathbf{X})}{\partial \mathbf{X}} = [\det(\mathbf{X})]\left(\mathbf{X}'\right)^{-1}$.
8. Let $\mathbf{X} \in \mathbb{M}^{m\times n}$, $\mathbf{A} \in \mathbb{M}^{p\times m}$, $\mathbf{B} \in \mathbb{M}^{n\times p}$, and $\mathbf{B}'\mathbf{X}'\mathbf{A}'$ is nonsingular, then $\frac{\partial \det(\mathbf{A}\mathbf{X}\mathbf{B})}{\partial \mathbf{X}} = \det(\mathbf{A}\mathbf{X}\mathbf{B})\,\mathbf{A}'\left(\mathbf{B}'\mathbf{X}'\mathbf{A}'\right)^{-1}\mathbf{B}'$.
9. Let $\mathbf{X} \in \mathbb{M}^{m\times m}$ then $\frac{\partial \ln[\det(\mathbf{X})]}{\partial \mathbf{X}} = \left(\mathbf{X}'\right)^{-1}$.
10. Let $\mathbf{X} \in \mathbb{M}^{m\times n}$, $\mathbf{A} \in \mathbb{M}^{m\times m}$ positive definite, then $\frac{\partial \ln\left[\det\left(\mathbf{X}'\mathbf{A}\mathbf{X}\right)\right]}{\partial \mathbf{X}} = 2\mathbf{A}\mathbf{X}\left(\mathbf{X}'\mathbf{A}\mathbf{X}\right)^{-1}$.

See Lütkepohl (1996, Ch. 10), and Magnus and Neudecker (1999) for further details.

A.14 The mean value theorem

Let $f(x)$ be a continuous function on the closed interval $[a, b]$, and differentiable on the open interval (a, b), where $a < b$, with derivative $f'(x)$. Then there is at least one number c in the interval (a, b) such that

$$f(b) - f(a) = (b - a) f'(c).$$

In the case of a real function of more than one variable, suppose $f(\mathbf{x})$ be a differentiable real function of $\mathbf{x} = (x_1, x_2, \ldots, x_k)'$, defined on the open subset $\mathbf{C}$ of $\mathbb{R}^k$. Assume that the line

segment from **a** to **b** is contained in **C**, and suppose that $f(\mathbf{x})$ is continuous along the segment and differentiable between **a** and **b**. Then there exists **c** on the line segment **a** to **b** such that

$$f(\mathbf{b}) - f(\mathbf{a}) = (\mathbf{b} - \mathbf{a})' \left. \frac{\partial f(\mathbf{u})}{\partial \mathbf{u}'} \right|_{\mathbf{u}=\mathbf{c}}.$$

A.15 Taylor's theorem

The mean value theorem implies that if, for two points $a < b$ we have $f(a) = f(b)$, then there exists a point $c \in [a, b]$ such that $f'(c) = 0$. This fact is the core of Taylor's theorem. Let $f(x)$ be an n-times, continuously differentiable real function on an interval $[a, b]$ with the nth derivative denoted by $f^{(n)}(x)$. For any pair of points $x, x_0 \in [a, b]$ there exists a $\lambda \in [0, 1]$ such that

$$f(x) = f(x_0) + \sum_{k=1}^{n-1} \frac{(x - x_0)^k}{k!} f^{(k)}(x_0) + \frac{(x - x_0)^n}{n!} f^{(n)}(x_0 + \lambda(x - x_0)). \tag{A.24}$$

The above result carries over to real functions of more than one variable. We provide here the second-order Taylor expansion:

$$f(\mathbf{x}) = f(\mathbf{x}_0) + (\mathbf{x} - \mathbf{x}_0)' \left. \frac{\partial f(\mathbf{u})}{\partial \mathbf{u}'} \right|_{\mathbf{u}=\mathbf{x}_0} \tag{A.25}$$

$$+ (\mathbf{x} - \mathbf{x}_0)' \left. \frac{\partial^2 f(\mathbf{u})}{\partial \mathbf{u} \partial \mathbf{u}'} \right|_{\mathbf{u}=\mathbf{x}_0+\lambda(\mathbf{x}-\mathbf{x}_0)} (\mathbf{x} - \mathbf{x}_0). \tag{A.26}$$

A.16 Numerical optimization techniques

Consider the general problem of maximizing a function of several variables

$$\hat{\boldsymbol{\theta}} = \underset{\boldsymbol{\theta}}{\operatorname{argmax}} F(\boldsymbol{\theta}),$$

where, for example, $F(\boldsymbol{\theta})$ may be the log-likelihood or the generalized method of moments objective function, and $\boldsymbol{\theta}$ is a p-dimensional vector of unknown parameters.

A.16.1 Grid search methods

One simple method for obtaining the maximizer, $\hat{\boldsymbol{\theta}}$, is to make use of a grid search method. In grid search methods, the procedure is to select many values of $\boldsymbol{\theta}$ along a grid, compute $F(\boldsymbol{\theta})$ for each of these values, and choose as the estimator the value of $\boldsymbol{\theta}$ that provides the largest value of $F(\boldsymbol{\theta})$. This procedure is convenient when the dimension of the unknown parameters is small, say one or two, and the range of variation of these parameters is known *a priori*. The grid search procedure might, however, be very cumbersome when doing Monte Carlo studies or in empirical work

where a large number of parameters needs to be estimated. Further, the amount of computations involved when using the search procedure to obtain an estimate of $\boldsymbol{\theta}$ which is accurate up to three or four significant figures can be considerable.

A.16.2 Gradient methods

Iterative methods consist of starting from an initial (or 'guess estimate') value of the parameters and then updating it according to an iterative formula. Most iterative methods are gradient methods that change in a direction determined by the gradient. The updated formula is a weighted average of the gradient

$$\hat{\boldsymbol{\theta}}^{(i+1)} = \hat{\boldsymbol{\theta}}^{(i)} - \mathbf{A}^{(i)}\mathbf{g}^{(i)},$$

where $\mathbf{A}^{(i)}$ is a matrix that depends on $\hat{\boldsymbol{\theta}}^{(i)}$, and $\mathbf{g}^{(i)}$ is the gradient vector evaluated at $\hat{\boldsymbol{\theta}}^{(i)}$, given by

$$\mathbf{g}^{(i)} = \left.\frac{\partial F(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}\right|_{\boldsymbol{\theta}=\hat{\boldsymbol{\theta}}^{(i)}}.$$

The choice of the weighting matrix, $\mathbf{A}^{(i)}$, leads to different gradient methods. One common modification to gradient methods is to include a 'damping factor' to prevent possible overshooting or undershooting, so that

$$\hat{\boldsymbol{\theta}}^{(i+1)} = \hat{\boldsymbol{\theta}}^{(i)} - \lambda^{(i)}\mathbf{A}^{(i)}\mathbf{g}^{(i)}, \tag{A.27}$$

The iterations need to be started with a guess-estimate, $\hat{\boldsymbol{\theta}}^{(0)}$. Iterations usually stops when one or more of the following convergence criteria are satisfied: (i) a small relative change occurs in the objective function, $F(\boldsymbol{\theta})$; (ii) a small change occurs in the gradient vector, $\mathbf{g}^{(i)}$, relative to $\mathbf{A}^{(i)}$; and (iii) a small relative change occurs in the parameter estimates, $\hat{\boldsymbol{\theta}}^{(i)}$. Normally, there is a maximum number of iterations that will be attempted, and if such a maximum is reached, then estimates should not be used, unless convergence has been achieved. Note that a poor choice of starting values can lead to exiting at the maximum number of iterations, and general failure of iterative methods.

Newton–Raphson method

The most frequently used method is the Newton–Raphson technique that makes use of the second-order Taylor series expansion. This method works especially well when the function is globally concave in $\boldsymbol{\theta}$. The Newton–Raphson iteration is

$$\hat{\boldsymbol{\theta}}^{(i+1)} = \hat{\boldsymbol{\theta}}^{(i)} - \left[\mathbf{H}^{(i)}\right]^{-1}\mathbf{g}^{(i)}, \tag{A.28}$$

for $i = 1, 2, \ldots$ where

$$\mathbf{H}^{(i)} = \left[\frac{\partial^2 F(\boldsymbol{\theta})}{\partial \boldsymbol{\theta} \partial \boldsymbol{\theta}'}\right]_{\boldsymbol{\theta}=\hat{\boldsymbol{\theta}}^{(i)}}, \tag{A.29}$$

A modification of the Newton–Raphson technique is the *method of scoring*, which consists of replacing the Hessian matrix in (A.28) by its expected value

$$\mathbf{H}^{(i)} = \left[E\left(\frac{\partial^2 F(\boldsymbol{\theta})}{\partial \boldsymbol{\theta} \partial \boldsymbol{\theta}'}\right)\right]_{\boldsymbol{\theta}=\hat{\boldsymbol{\theta}}^{(i)}}. \tag{A.30}$$

This modification is particularly useful in maximum likelihood estimation, because in this case, by information matrix inequality, $\mathbf{H}^{(i)}$ is positive definite.

See Cameron and Trivedi (2005, Ch. 10), for further discussion on gradient methods. See also Boyd and Vandenberghe (2004) for a textbook treatment of optimization algorithms.

Method of steepest ascent

The method of steepest ascent sets $\mathbf{A}^{(i)} = \mathbf{I}_p$. It then usually employs the modified method (A.27), using as damping factor

$$\lambda^{(i)} = -\mathbf{g}^{(i)\prime}\mathbf{g}^{(i)} / \mathbf{g}^{(i)\prime}\mathbf{H}^{(i)}\mathbf{g}^{(i)},$$

where $\mathbf{H}^{(i)}$ is the Hessian matrix, (A.29). The advantage of this method over the Newton–Raphson is that it works even in the case when $\mathbf{H}^{(i)}$ is singular.

A.16.3 Direct search methods

Gradient methods require the objective function to be sufficiently smooth to ensure the existence of the gradient. However, when there is no gradient or when the function has multiple local optima, alternative methods are required. A number of derivative-free methods of searching for a function optimum have been proposed in the literature, such as the simulated annealing or genetic algorithms. These methods are often very effective in problems with many variables in the objective function. However, they usually require far more function evaluations than the methods based on derivatives that were considered above.

Simulated annealing

Suppose we have a value $\hat{\boldsymbol{\theta}}^{(i)}$ at the ith iteration. The simulated annealing consists of perturbing the jth component of $\hat{\boldsymbol{\theta}}^{(i)}$ to obtain a new trial value of

$$\hat{\boldsymbol{\theta}}^{*} = \hat{\boldsymbol{\theta}}^{(i)} + \left(0, 0, \ldots .0, \lambda_j r_j, 0, \ldots, 0\right)',$$

where λ_j is a pre-specified step length and r_j is a draw from a uniform distribution on $(-1, 1)$. Hence, the method sets $\hat{\boldsymbol{\theta}}^{(i+1)} = \hat{\boldsymbol{\theta}}^{*}$ if it increases the objective function, or if it does not increase the value of the objective function but does pass the Metropolis criterion that

$$\exp\left\{\left[F\left(\hat{\boldsymbol{\theta}}^{(i)}\right) - F\left(\hat{\boldsymbol{\theta}}^{*}\right)\right]/T_i\right\} > u,$$

where $u \sim U(0,1)$, and T_i is a scaling parameter called *temperature*. Thus, the method accepts both uphill and downhill moves, with a probability that decreases with the difference $F\left(\hat{\boldsymbol{\theta}}^{(i)}\right) - F\left(\hat{\boldsymbol{\theta}}^{*}\right)$ and that increases with the temperature.

See Goffe, Ferrier, and Rogers (1994, Ch. 10) and Cameron and Trivedi (2005) for further details of simulated annealing procedure.

A.17 Lag operators

If $x_t = x(t)$ is a function of time, the *lag operator*, L, is defined by

$$Lx_t = x_{t-1}.$$

Powers of the lag operator are defined as successive applications of L, that is,

$$L^2 x_t = L\left(Lx_t\right) = Lx_{t-1} = x_{t-2},$$

and, in general, for any integer k,

$$L^k x_t = x_{t-k}.$$

The lag operator satisfies

$$\begin{aligned} L\left(ax_t\right) &= aLx_t = ax_{t-1}, \\ \left(a_1 L^p + a_2 L^q\right) x_t &= a_1 x_{t-p} + a_2 x_{t-q}. \end{aligned}$$

A *lag polynomial* is defined by

$$a(L) = a_0 + a_1 L + a_2 L^2 + \ldots.$$

Lag polynomials satisfy the following properties

1. $a(L)b(L) = \left(a_0 + a_1 L + \ldots\right)\left(b_0 + b_1 L + \ldots\right) = a_0 b_0 + \left(a_0 b_1 + a_1 b_0\right) L + \ldots.$
2. $a(L)b(L) = b(L)a(L).$
3. $\left[a(L)\right]^2 = a(L)a(L).$
4. The lag polynomial $a(L)$ can be *factorized* as follows:

$$a(L) = \left(1 - \lambda_1 L\right)\left(1 - \lambda_2 L\right)\left(1 - \lambda_3 L\right)\ldots,$$

   where $\lambda_1, \lambda_2, \ldots$ are coefficients.
5. The inverse of the lag polynomial $a(L)$ is:

$$a(L)^{-1} = (1 - \lambda_1 L)^{-1} (1 - \lambda_2 L)^{-1} (1 - \lambda_3 L)^{-1} \ldots.$$

See Griliches (1967) for further discussion on the properties of lag polynomials.
Let

$$P(L) = 1 - \lambda L.$$

Under the condition that $|\lambda| < 1$, we have

$$[P(L)]^{-1} = \sum_{j=0}^{\infty} \lambda^j L^j.$$

Further, we have

$$(1 - \lambda L)^{-1} (1 - \lambda L) = 1.$$

The inverse of the lag operator is defined by

$$L^{-k} x_t = F^k x_t = x_{t+k},$$

which is known as the *lead* (or forward) operator.
See Hamilton (1994, Ch. 2) for further details.

A.18 Difference equations

A difference equation relates consecutive terms in a sequence of numbers.

A.18.1 First-order difference equations

Let y_t be a real valued function defined for $t = -1, 0, 1, 2, \ldots$. A *linear first-order difference equation* for y_t is

$$y_t = \phi y_{t-1} + w_t, \tag{A.31}$$

where w_t is a real valued function defined for $t \geq 0$. By applying recursive substitution we can rewrite (A.31) as a function of its initial value at date $t = 0$, y_0, and of the sequence of values of the variable w_t in dates between 1 and t

$$y_t = \phi^t y_0 + \sum_{s=0}^{t-1} \phi^s w_{t-s}. \tag{A.32}$$

Equation (A.32) is the unique solution for (A.31).

A.18.2 p^{th}-difference equations

Consider now the linear p^{th}-order difference equation for y_t

$$y_t = \phi_1 y_{t-1} + \phi_2 y_{t-2} + \ldots + \phi_p y_{t-p} + w_t, \tag{A.33}$$

where w_t is an exogenous 'forcing' variable, in the sense that w_t does not depend on y_t or its lagged values. Using the lag operator, we note that (A.33) can also be expressed as follows

$$\left(1 - \phi_1 L - \phi_2 L^2 \ldots - \phi_p L^p\right) y_t = \phi(L) y_t = w_t,$$

where $\phi(L) = 1 - \phi_1 L - \phi_2 L^2 - \ldots - \phi_p L^p$. It is often convenient to rewrite the pth-order equation in the scalar y_t, as a first-order difference equation in a p-dimensional vector, $\boldsymbol{\xi}_t$. Define

$$\boldsymbol{\xi}_t = \begin{pmatrix} y_t \\ y_{t-1} \\ \vdots \\ y_{t-p+1} \end{pmatrix}, \mathbf{F} = \begin{pmatrix} \phi_1 & \phi_2 & \ldots & \phi_{p-1} & \phi_p \\ 1 & 0 & \ldots & 0 & 0 \\ 0 & 1 & \ldots & 0 & 0 \\ \vdots & \vdots & \ldots & \vdots & \vdots \\ 0 & 0 & \ldots & 0 & 0 \\ 0 & 0 & \ldots & 1 & 0 \end{pmatrix}, \mathbf{v}_t = \begin{pmatrix} w_t \\ 0 \\ \vdots \\ 0 \end{pmatrix}.$$

We have that

$$\boldsymbol{\xi}_t = \mathbf{F}\boldsymbol{\xi}_{t-1} + \mathbf{v}_t, \tag{A.34}$$

is a system of p equations where the first equation is given by (A.33), and the remaining equations are simple identities. By recursive substitution we can rewrite (A.34) as a function of p initial values $y_0, y_{-1}, \ldots, y_{-p+1}$, and of the sequence of values for the variable w_t in dates between 1 and t

$$\boldsymbol{\xi}_t = \mathbf{F}^t \boldsymbol{\xi}_0 + \sum_{s=0}^{t-1} \mathbf{F}^s \mathbf{v}_{t-s},$$

where $\boldsymbol{\xi}_0 = (y_0, y_{-1}, \ldots, y_{-p+1})$. A solution for y_t can now be obtained as $\mathbf{e}_1' \boldsymbol{\xi}_t$, where $\mathbf{e}_1$ is a $p \times 1$ selection vector, namely $\mathbf{e}_1' = (1, 0, \ldots, 0)$.

The limit properties of y_t (as $t \to \infty$) depend on the eigenvalues of $\mathbf{F}$, which are given as the roots of the following p^{th}-order polynomial equation, also known as the auxiliary equation associated with (A.33)

$$\lambda^p - \phi_1 \lambda^{p-1} - \phi_2 \lambda^{p-2} \ldots - \phi_{p-1} \lambda - \phi_p = 0. \tag{A.35}$$

The solution for y_t is stable if all eigenvalues of the above auxiliary equation lie inside the unit circle. In that case, the limit solution is given by

$$\lim_{t\to\infty} y_t = \lim_{t\to\infty}\left(\sum_{s=0}^{t-1} \mathbf{e}_1' \mathbf{F}^s \mathbf{v}_{t-s}\right).$$

In the case that the process has started a long time ago, namely from an initial value y_{-M}, we have

$$\boldsymbol{\xi}_t = \mathbf{F}^{t+M}\boldsymbol{\xi}_{-M} + \sum_{s=0}^{t+M-1} \mathbf{F}^s \mathbf{v}_{t-s},$$

and the limit value of y_t when $M \to \infty$ exists and does not depend on the initial values if all the roots of the auxiliary equation falls inside the unit circle. Under these conditions we have

$$\lim_{M\to\infty} y_t = \sum_{s=0}^{\infty} \mathbf{e}_1' \mathbf{F}^s \mathbf{v}_{t-s}.$$

Therefore, in terms of the forcing variables we have

$$\lim_{M\to\infty} y_t = \left(1 - \phi_1 L - \phi_2 L^2 \ldots . - \phi_p L^p\right)^{-1} w_t.$$

The inverse polynomial exists since

$$1 - \phi_1 z - \phi_2 z^2 \ldots . - \phi_p z^p = (1 - \lambda_1 z)(1 - \lambda_2 z) \ldots .(1 - \lambda_p z),$$

where λ_i is the ith root of the auxiliary equation, and assuming that the underlying difference equation is stable, namely that $\rho = \max_k(|\lambda_k|) < 1$. Under this condition $|\lambda_i z| < 1$, for all i, and there exist p constants, A_k, $(|A_k| < K < \infty)$ such that

$$\frac{1}{(1 - \lambda_1 z)(1 - \lambda_2 z) \ldots .(1 - \lambda_p z)} = \sum_{k=1}^{p} \frac{A_k}{1 - \lambda_k z},$$

and

$$\begin{aligned}\lim_{M\to\infty} y_t &= \sum_{k=1}^{p}\sum_{s=0}^{\infty} A_k \lambda_k^s w_{t-s} = \sum_{s=0}^{\infty}\left(\sum_{k=1}^{p} A_k \lambda_k^s\right) w_{t-s} \\ &= \sum_{s=0}^{\infty} \alpha_s w_{t-s} = \alpha(L) w_t,\end{aligned}$$

where

$$\alpha_s = \sum_{k=1}^{p} A_k \lambda_k^s, \text{ and } \alpha(L) = \sum_{s=0}^{\infty} \alpha_s L^s.$$

It is interesting to note that $\alpha_s < K\left[\max_k(\phi_k)\right]^s = K\rho^s$, which establishes that $\{\alpha_s\}$ is an absolute summable sequence. Furthermore,

$$\sum_{s=0}^{\infty}\alpha_s^2 < K < \infty$$
$$\sum_{s=0}^{\infty}s\alpha_s^2 < K < \infty,\ \sum_{s=0}^{\infty}s^2\alpha_s^2 < K < \infty,$$
$$\sum_{s=0}^{\infty}s\,|\alpha_s| < K < \infty, \sum_{s=0}^{\infty}s^2\,|\alpha_s| < K < \infty.$$

It is also convenient to derive α_s directly in terms of ϕ_i. Since $\alpha(L)$ is the inverse of $\phi(L)$. We note that we must have

$$\phi(L)\alpha(L) = \alpha(L)\phi(L) = 1.$$

Multiplying the two polynomials and equating the coefficients of the non zero powers of L to zero we have

$$\begin{aligned}
\alpha_0 &= 1\\
\alpha_1 &= \phi_1\\
\alpha_2 &= \alpha_1\phi_1 + \alpha_0\phi_2\\
&\vdots\\
\alpha_p &= \alpha_{p-1}\phi_1 + \alpha_{p-2}\phi_2 + \ldots + \alpha_0\phi_p,\\
\alpha_s &= \alpha_{s-1}\phi_1 + \alpha_{s-2}\phi_2 + \ldots + \alpha_{s-p}\phi_p, \text{ for } s = p+1, p+2, \ldots.
\end{aligned}$$

Further details on solution of difference equations can be found in Agarwal (2000), and in Hamilton (1994, Ch. 1).

Appendix B: Probability and Statistics

This appendix covers key concepts from probability theory and statistics that are used in the book. We refer to Rao (1973), Billingsley (1995), Zwillinger and Kokoska (2000), Bierens (2005) and Durrett (2010) for further details.

B.1 Probability space and random variables

The set of possible outcomes (events) of an experiment is called the *sample space* and is usually denoted by Ω. A σ-algebra is a collection, $\mathcal{F}$, of subsets of a non-empty set Ω satisfying the following conditions:

(i) If $A \in \mathcal{F}$ then $A^c \in \mathcal{F}$, where A^c is the complement set of A, i.e., the set of all elements of Ω not belonging to A.

(ii) If $A_j \in \mathcal{F}$, for $j = 1, 2, \ldots$, then $\cup_{j=1}^{\infty} A_j \in \mathcal{F}$.

A mapping $P : \mathcal{F} \to [0, 1]$ is a *probability measure* on $\{\Omega, \mathcal{F}\}$ if it satisfies the following three conditions:

1. For all $A \in \mathcal{F}$, $P(A) \geq 0$.
2. $P(\Omega) = 1$.
3. For disjoint sets $A_j \in \mathcal{F}$, $P\left(\cup_{j=1}^{\infty} A_j\right) = \sum_{j=1}^{\infty} P\left(A_j\right)$.

Recall that sets are disjoint if they have no elements in common.

An important special case occurs when $\Omega = \mathbb{R}$ and we consider a collection of subsets of all open intervals (a, b), with $a < b$, and $a, b \in \mathbb{R}$. The σ-algebra generated by the collection of all open intervals in $\mathbb{R}$ is called the Euclidean Borel field, denoted by $\mathcal{B}$, and its members are called the *Borel sets*.

A *probability space* consists of the triple $\{\Omega, \mathcal{F}, P\}$, namely, the sample space (i.e., the set of all possible outcomes of the statistical experiment involved), a σ-algebra of events (i.e., a collection of subsets of the sample space such that conditions (i) and (ii) are satisfied), and a probability measure $P(.)$ satisfying the conditions 1–3 above.

A *random variable* is a variable whose value is not known and depends on the outcome of a statistical experiment. More formally, let $\{\Omega, \mathcal{F}, P\}$ be a probability space. A mapping $X{:}\Omega \to \mathbb{R}$ is called a *random variable* defined on $\{\Omega, \mathcal{F}, P\}$ if X is measurable, which means that for every Borel set B, $\{\omega : X(\omega) \in B\} \in \mathcal{F}$.

See, for example, Bierens (2005).

B.2 Probability distribution, cumulative distribution, and density function

A random variable X is said to be discrete if it can only assume a finite or a countable infinity number of values. The probability distribution of X is a set of numbers that gives the probability of each outcome,

$$p(x) = P(X = x), \tag{B.1}$$

and such that:

1. $p(x) \geq 0$, for all x.
2. $\sum_x p(x) = 1$.

We have

$$P\left(X \in A\right) = \sum_{x \in A} p(x).$$

The cumulative distribution function $F_X(x)$ for a discrete random variable is

$$F_X\left(x\right) = P\left(X \leq x\right) = \sum_{y \leq x} p(y). \tag{B.2}$$

A random variable X is continuous if its set of possible values is an interval of numbers. The probability density function, $f_X(x)$, of X is a real-valued function such that

$$P\left(a \leq X \leq b\right) = \int_a^b f_X(x)dx. \tag{B.3}$$

The density function $f_X(x)$ satisfies:

1. $f\left(x\right) \geq 0, -\infty < x < \infty$.
2. $\int_{-\infty}^{+\infty} f_X(x)dx = 1$.

The cumulative distribution function, $F_X(x)$, for a continuous random variable X is defined by

$$F_X\left(x\right) = P\left(X \leq x\right) = \int_{-\infty}^{x} f_X(u)du. \tag{B.4}$$

B.3 Bivariate distributions

Let X and Y be discrete random variables. The joint probability distribution of (X, Y) is

$$p\left(x, y\right) = P\left(X = x, Y = y\right), \tag{B.5}$$

for each outcome (x, y), satisfying:

1. $p\left(x, y\right) \geq 0$.
2. $\sum_x \sum_y p\left(x, y\right) = 1$.

For any subset A containing pairs (x, y) we have

$$P\left[(X, Y) \in A\right] = \sum_{(x,y)\in A} p\left(x, y\right).$$

If X and Y are continuous random variables, then for any subset A containing values of (X, Y), the joint density function is defined *via* the double integral

$$P\left[(X, Y) \in A\right] = \int\int_A f_{XY}(u, v)dudv. \tag{B.6}$$

We have:

1. $f_{XY}(x, y) \geq 0, -\infty < x, y < \infty$.
2. $\int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} f_{XY}(x, y)dxdy = 1$.

The cumulative distribution function of (X, Y) is

$$F_{XY}\left(x, y\right) = \int_{-\infty}^{x} \int_{-\infty}^{y} f_{XY}(u, v)dudv. \tag{B.7}$$

The marginal distribution of $X, f_X(x)$, is defined by

$$f_X\left(x\right) = \int_{-\infty}^{+\infty} f_{XY}(y)dy.$$

The conditional density of Y given that $X = x$, denoted as $f_{Y|X}\left(y|x\right)$ is given by

$$f_{Y|X}\left(y|x\right) = \frac{f_{XY}\left(x, y\right)}{f_X\left(x\right)}, \tag{B.8}$$

where $f_{XY}\left(x, y\right)$ is the joint probability density function of (X, Y).

B.4 Multivariate distribution

Let $(X_1, X_2, \ldots, X_n)$ be an n-dimensional, discrete random variable. The joint probability distribution of $(X_1, X_2, \ldots, X_n)$ is given by

$$p\left(x_1, x_2, \ldots, x_n\right) = P\left(X_1 = x_1, X_2 = x_2, \ldots, X_n = x_n\right). \tag{B.9}$$

For any subset A containing the values that the random variable may assume, we have

$$P\left[\left(X_1, X_2, \ldots, X_n\right) \in A\right] = \sum_{(x_1, x_2, \ldots, x_n) \in A} p\left(x_1, x_2, \ldots, x_n\right).$$

If $(X_1, X_2, \ldots, X_n)$ is an n-dimensional, continuous random variable, then the joint density function is defined *via* the integral

$$P\left[\left(X_1, X_2, \ldots, X_n\right) \in A\right] = \int_A f_{X_1, X_2, \ldots, X_n}(x_1, x_2, \ldots, x_n) dx_1 dx_2 \ldots dx_n.$$

The cumulative distribution function of $(X_1, X_2, \ldots, X_n)$ is

$$F_{X_1, X_2, \ldots, X_n}(x_1, x_2, \ldots, x_n) = \int_{-\infty}^{x_1} \int_{-\infty}^{x_2} \ldots \int_{-\infty}^{x_n} f_{X_1, X_2, \ldots, X_n}(u_1, u_2, \ldots, u_n) du_1 du_2 \ldots du_n. \tag{B.10}$$

B.5 Independent random variables

Let X and Y be two discrete random variables with joint probability distribution, $p\left(x, y\right)$. The random variables X and Y are said to be *statistically independent* if and only if

$$p\left(x, y\right) = p\left(x\right) p(y), \text{ for all } (x, y) \in \mathbb{R}^2. \tag{B.11}$$

Hence, under independence the joint probability is equal to the product of the two marginal probability distributions. If X and Y are continuous with joint density function, $f_{XY}\left(x, y\right)$, then independence occurs if and only if

$$f_{XY}\left(x, y\right) = f_X(x) f_Y(y). \tag{B.12}$$

We have:

1. If X and Y are independent we have

$$f_{Y|X}\left(y|x\right) = f_Y\left(y\right), \tag{B.13}$$

   and, equivalently, $f_{X|Y}\left(x|y\right) = f_X(x)$. See (B.8).
2. If X and Y are independent then $f(X)$ and $g(Y)$ are also independent, where f (.) and g (.) are two functions.

B.6 Mathematical expectations and moments of random variables

Let X be a discrete random variable. The expected value (or expectation, mean) of X is given by

$$E(X) = \sum_x xp(x), \tag{B.14}$$

The variance of X is given by

$$Var(X) = E\left[(X - E(X))^2\right] = \sum_x [x - E(X)]^2 p(x), \tag{B.15}$$

The rth central moment of X is given by

$$\mu_r = E\left[(X - E(X))^r\right] = \sum_x [x - E(X)]^r p(x). \tag{B.16}$$

In the case X is a continuous random variable with probability density function, $f_X(x)$, the expressions corresponding to (B.14)-(B.16) are

$$E(X) = \int_{-\infty}^{+\infty} x f_X(x)\, dx, \tag{B.17}$$

$$Var(X) = \int_{-\infty}^{+\infty} [x - E(X)]^2 f_X(x)\, dx, \tag{B.18}$$

$$\mu_r = \int_{-\infty}^{+\infty} [x - E(X)]^r f_X(x)\, dx, \tag{B.19}$$

provided that the integrals exist. The following properties for the expectation operator (we now only focus on the continuous case) are easy to verify:

1. Let $g(\cdot)$ be a function. Then $E[g(X)] = \int_{-\infty}^{+\infty} g(x) f_X(x)\, dx$.
2. $E(a + bX) = a + bE(X)$, where a, b are two constants.
3. $E(aX + bY) = aE(X) + bE(Y)$, for any two random variables X and Y.
4. Let $g(\cdot)$ be a function. Then $Var[g(X)] = \int_{-\infty}^{+\infty} \left\{g(X) - E[g(X)]\right\}^2 f_X(x)\, dx$.
5. $Var(a + bX) = b^2 Var(X)$.
6. $Var(X) = E\left(X^2\right) - [E(X)]^2$.

Let X, Y be two random variables, the conditional expectation of Y given that X takes on the particular value x is

$$E(Y|X = x) = \int_{-\infty}^{+\infty} y f_{Y|X}(y|x)\, dy, \tag{B.20}$$

where $f_{Y|X}\left(y|x\right)$ is the conditional density of Y given that $X = x$ (see equation (B.8)).

The following propositions hold.

Proposition 49 (Law of iterated expectations) *Let X and Y be two random variables, then we have*

$$E\left(Y\right) = E\left[E\left(Y|X\right)\right]. \tag{B.21}$$

Using the law of iterated expectations it is possible to prove the following proposition.

Proposition 50 (Law of total variance) *Let Y, X be two random variables, and assume that the variance of Y is finite, then*

$$Var\left(Y\right) = E\left[Var\left(Y|X\right)\right] + Var\left[E\left(Y|X\right)\right]. \tag{B.22}$$

Two other measures are often used to describe a probability distribution. These are the coefficients of skewness and the coefficient of kurtosis. The coefficient of skewness is a measure of the asymmetry of a probability distribution and is defined as

$$\sqrt{b_1} = \frac{\mu_3}{\left[Var\left(X\right)\right]^{3/2}}. \tag{B.23}$$

The excess kurtosis is a measure of the thickness of the tails of a distribution

$$b_2 = \frac{\mu_4}{\left[Var\left(X\right)\right]^2}. \tag{B.24}$$

In some cases, the degree of excess kurtosis, given by

$$\frac{\mu_4}{\left[Var\left(X\right)\right]^2} - 3, \tag{B.25}$$

is used. In particular, this measure is adopted to characterize departures from the normal distribution, which has excess of zero.

B.7 Covariance and correlation

Let X and Y be two discrete random variables. The covariance between X and Y is

$$\begin{aligned} Cov(X,Y) &= E\left\{\left[X - E\left(X\right)\right]\left[Y - E\left(Y\right)\right]\right\} \\ &= \sum_x \sum_y \left[x - E\left(X\right)\right]\left[y - E\left(Y\right)\right] p\left(x,y\right). \end{aligned} \tag{B.26}$$

If X and Y are continuous then

$$Cov(X,Y) = \int_{-\infty}^{+\infty} \left[x - E\left(X\right)\right]\left[y - E\left(Y\right)\right] f_{XY}\left(x,y\right) dxdy. \tag{B.27}$$

The sign of the covariance indicates the direction of covariation of X and Y. The following properties of the covariance can be verified:

1. $Cov(X, Y) = Cov(Y, X)$.
2. $Cov(X, Y) = E(XY) - E(X)E(Y)$.
3. $Cov(aX + bY, Z) = aCov(X, Z) + bCov(Y, Z)$.
4. $Var(aX + bY) = a^2 Var(X) + b^2 Var(Y) + 2abCov(X, Y)$.

Since the magnitude of the covariance depends on the scale of measurement of the variables, a preferable measure is the correlation coefficient

$$\rho_{XY} = \frac{Cov(X, Y)}{[Var(X)]^{1/2}[Var(Y)]^{1/2}}. \tag{B.28}$$

If X and Y are independent then the expectation operator has the property $E(XY) = E(X)E(Y)$. As a consequence, two independent random variables satisfy

$$Cov(X, Y) = 0,$$

and, consequently, $\rho_{XY} = 0$. It follows that for two independent random variables, X and Y, we have

$$Var(aX + bY) = a^2 Var(X) + b^2 Var(Y),$$

where a and b are fixed constants.

B.8 Correlation versus independence

A pair of uncorrelated random variables need not be independent. For example, consider the following random variables

$$\tilde{Y} = \left(\frac{k-2}{\chi^2_k}\right)^{1/2} Y, \text{ and } \tilde{X} = \left(\frac{k-2}{\chi^2_k}\right)^{1/2} X,$$

where Y and X are independent random variables with zero means and unit variances, and χ^2_k is a chi-squared random variate with $k > 4$ degrees of freedom, distributed independent of Y and X. It is now easily seen that

$$E(\tilde{Y}) = E\left[\left(\frac{k-2}{\chi^2_k}\right)^{1/2} Y\right] = E\left[\left(\frac{k-2}{\chi^2_k}\right)^{1/2}\right] E(Y) = 0,$$

$$E\left(\tilde{X}\right) = E\left[\left(\frac{k-2}{\chi_k^2}\right)^{1/2} X\right] = E\left[\left(\frac{k-2}{\chi_k^2}\right)^{1/2}\right] E(X) = 0,$$

and

$$E\left(\tilde{Y}\tilde{X}\right) = E\left(\frac{k-2}{\chi_k^2} YX\right) = E\left(\frac{k-2}{\chi_k^2}\right) E\left(Y\right) E\left(X\right) = 0,$$

which yields $Cov\left(\tilde{Y}, \tilde{X}\right) = 0$. Yet $\tilde{Y}$ and $\tilde{X}$ are not independent. To see this note that

$$E\left(\tilde{Y}^2\right) = E\left[\left(\frac{k-2}{\chi_k^2}\right) Y^2\right] = E\left(\frac{k-2}{\chi_k^2}\right) E\left(Y^2\right) = 1.$$

The result $E\left(\frac{k-2}{\chi_k^2}\right) = 1$, follows since the first moment of the inverse-chi-squared distribution is given by $E(1/\chi_k^2) = 1/(k-2) > 0$.[1] Similarly, $E\left(\tilde{X}^2\right) = 1$. But

$$E\left(\tilde{Y}^2\tilde{X}^2\right) = E\left[\left(\frac{k-2}{\chi_k^2}\right)^2 X^2Y^2\right] = E\left[\left(\frac{k-2}{\chi_k^2}\right)^2\right] E\left(X^2\right) E\left(Y^2\right).$$

Furthermore, using the results for second-order moment of the inverse-chi-squared distribution we have $E\left[\left(\frac{k-2}{\chi_k^2}\right)^2\right] = (k-2)/(k-4)$. Hence,

$$Cov\left(\tilde{Y}^2, \tilde{X}^2\right) = E\left(\tilde{Y}^2\tilde{X}^2\right) - E\left(\tilde{Y}^2\right) E\left(\tilde{X}^2\right) = (k-2)/(k-4) - 1 = \frac{2}{k-4},$$

which is non-zero for any finite k. But $Cov\left(Y^2, X^2\right) = 0$, since Y and X are independently distributed. In the case where Y and X are normally distributed, it follows that $\tilde{Y}$ and $\tilde{X}$ are distributed as a multi-variate t with k degrees of freedom. See also Section B.10.3.

B.9 Characteristic function

For a random variable X the characteristic function is defined as

$$\varphi_X(\theta) = E\left(e^{i\theta X}\right) = \int_{-\infty}^{+\infty} e^{i\theta x} f_X(x)\, dx, \tag{B.29}$$

[1] For an account of the properties of inverse-chi-squared distribution, see <http://en.wikipedia.org/wiki/Inverse-chi-squared_distribution>.

where $i = \sqrt{-1}$ is the imaginary number, and $\theta \in \mathbb{R}$.

The characteristic function of the sum of two independent random variables X and Y satisfies

$$\varphi_{X+Y}(\theta) = \varphi_X(\theta)\,\varphi_Y(\theta). \tag{B.30}$$

Namely, the characteristic function of their sum is the product of their marginal characteristic functions.

B.10 Useful probability distributions

We now summarize some of the more common probability distributions used in probability and statistics, and refer to Zwillinger and Kokoska (2000) for further details.

B.10.1 Discrete probability distributions

Bernoulli

The random variable X has a Bernoulli distribution if it takes the value 1 with success probability p, and the value 0 with failure probability $q = 1 - p$. If so we have

$$P(X = 1) = 1 - P(X = 0) = 1 - q = p. \tag{B.31}$$

The Bernoulli distribution is used to describe an experiment, the so-called Bernoulli trial, where the outcome is random and can be either of two possible outcomes, typically a 'success' and a 'failure'. A sequence of Bernoulli trials is referred to as repeated trials.

Binomial

The random variable X has a binomial distribution with parameters n and p, denoted by $X \sim Bi(n, p)$ if

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}, \tag{B.32}$$

for $k = 0, 1, 2, \ldots, n$, where

$$\binom{n}{k} = \frac{n!}{k!\,(n-k)!}, \tag{B.33}$$

is the binomial coefficient. Expression (B.32) gives the probability of getting exactly k successes in n Bernoulli trials. Note that, for $n = 1$, X reduces to a Bernoulli random variable. The expected value and variance of $X \sim Bi(n, p)$ are

$$E(X) = np, \;\; Var(X) = np(1-p).$$

Poisson

The random variable X has a Poisson distribution with parameter λ, denoted by $X \sim Poisson\ (\lambda)$, if

$$P(X = k) = \frac{\lambda^k}{k!} e^{-\lambda}, \tag{B.34}$$

for $k = 0, 1, 2, \ldots$, and λ is a positive real number. Let $p(n) = \lambda/n$ for some positive λ. Then the Binomial distribution approaches the Poisson with parameter λ as $n \to \infty$, that is, binomial with small parameter and large number of draws is like a Poisson. The expected value and variance of $X \sim Poisson\ (\lambda)$ are

$$E(X) = \lambda, \ \ Var(X) = \lambda.$$

B.10.2 Continuous distributions

Uniform

The random variable X has a uniform distribution between a and b, denoted by $X \sim U(a, b)$, if its probability density function is

$$f_X(x) = \begin{cases} \frac{1}{a-b}, & \text{for } a \leq x \leq b \\ 0, & \text{for } x < a \text{ or } x > b \end{cases}. \tag{B.35}$$

Hence, the uniform distribution has constant probability within the interval $[a, b]$. The expected value and variance of $X \sim U(a, b)$ are

$$E(X) = \frac{1}{2}(a + b), \ \ Var(X) = \frac{1}{12}(b - a)^2.$$

Normal

The random variable X has a normal distribution (or Gaussian distribution), denoted by $X \sim N\left(\mu, \sigma^2\right)$, if its probability density function is

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}. \tag{B.36}$$

Its expected value and variance are

$$E(X) = \mu, \ Var(X) = \sigma^2.$$

One important property of the Gaussian distribution is that

$$E\left[(X - \mu)^r\right] = 0, \text{for } r = 1, 3, 5, 7, \ldots,$$

namely, centered, odd-ordered moments are zero. Further,

$$E\left[(X-\mu)^4\right] = 3\sigma^4.$$

One useful property of the normal distribution is its preservation under linear transformation. If $X \sim N\left(\mu, \sigma^2\right)$ then $(a+bX) \sim N\left(a+b\mu, b^2\sigma^2\right)$. One convenient transformation is obtained by setting $a = -\mu/\sigma, b = 1/\sigma$. The resulting variable $Z = \frac{X-\mu}{\sigma} \sim N(0,1)$, namely it has a standard normal distribution. The density distribution of Z is often indicated as $\phi(z)$ and the cumulative density function as $\Phi(z)$.

Chi-square

The random variable X has a central chi-square distribution (or simply chi-square distribution) with k degrees of freedom, denoted by $X \sim \chi^2_k$, if its probability density function is

$$f_X(x) = \frac{1}{2^{k/2}\Gamma(k/2)} x^{(k/2)-1} e^{-x/2}, \tag{B.37}$$

for $x \geq 0$, and k is a positive integer. In the expression above, $\Gamma(.)$ denotes the gamma function, which, if n is a positive integer, is given by

$$\Gamma(n) = (n-1)! = (n-1)\cdot(n-2)\cdot\ldots\cdot 2\cdot 1.$$

More generally, the gamma function can be defined as

$$\Gamma(z) = \int_0^\infty t^{z-1} e^{-t} dt. \tag{B.38}$$

The expected value and variance of $X \sim \chi^2_k$ are

$$E(X) = k,\ Var(X) = 2k.$$

The chi-square distribution has the following properties:

1. If two independent random variables X_1 and X_2 have $\chi^2_{n_1}$ and $\chi^2_{n_2}$ distributions, respectively, then $(X_1 + X_2) \sim \chi^2_{n_1+n_2}$.
2. Given three random variables X, X_1 and X_2, such that $X = X_1 + X_2, X \sim \chi^2_n, X_1 \sim \chi^2_{n_1}$, for $n_1 < n$, then X_2 is independently distributed of X_1, and $X_2 \sim \chi^2_{n_2}$, with $n = n_1 + n_2$.
3. If $X_1, X_2, \ldots, X_n$ are independent and $N(0,\sigma^2)$, then $\sum_{i=1}^n X_i^2/\sigma^2 \sim \chi^2_n$.

A related distribution is the non-central chi-square distribution, which often arises in the power analysis of statistical tests. Indicated as $\chi^2_k(\lambda)$, its probability density function is

$$f_X(x) = \frac{e^{-(x+\lambda)/2}}{2^{k/2}} \sum_{j=1}^\infty \frac{x^{(k/2)+j-1}\lambda^j}{\Gamma\left(\frac{k}{2}+j\right) 2^{2j} j!},$$

for $x > 0$, where k is the number of degrees of freedom, and $\lambda > 0$ is the non-centrality parameter.

Student's t

The random variable X has a Student's t-distribution (or simply t-distribution) with ν degrees of freedom, and denoted by $X \sim t_\nu$, if its probability density function is

$$f_X(x) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\sqrt{\nu\pi}\,\Gamma\left(\frac{\nu}{2}\right)}\left(1+\frac{x^2}{\nu}\right)^{-\frac{(\nu+1)}{2}}, \tag{B.39}$$

where $\Gamma(.)$ is the Gamma function and ν is a positive integer. Let $Z \sim N(0,1)$ and $V \sim \chi^2_k$, with Z and V independent, then

$$\frac{Z}{\sqrt{V/k}} \sim t_k.$$

The expected value and variance of $X \sim t_\nu$ are

$$\begin{aligned} E(X) &= 0, \text{ for } \nu > 1, \\ Var(X) &= \frac{\nu}{\nu-2}, \text{ for } \nu > 2. \end{aligned}$$

Fisher–Snedecor or F-distribution

The random variable X has a central F-distribution (also known as the Fisher–Snedecor distribution) with d_1 and d_2 degrees of freedom, denoted by $X \sim F(d_1, d_2)$, if its probability density function is

$$f_X(x) = \frac{1}{B\left(\frac{d_1}{2}, \frac{d_2}{2}\right)}\left(\frac{d_1}{d_2}\right)^{\frac{d_1}{2}} x^{\frac{d_1}{2}-1}\left(1+\frac{d_1}{d_2}x\right)^{-\frac{d_1+d_2}{2}}, \tag{B.40}$$

for $x \geq 0$, where d_1 and d_2 are positive integers and $B(.,.)$ is the beta function defined by

$$B(x,y) = \int_0^1 t^{x-1}(1-t)^{y-1}\,dt.$$

The expected value and variance of $X \sim F(d_1, d_2)$ are

$$\begin{aligned} E(X) &= \frac{d_2}{d_2-2}, \text{ for } d_2 > 2, \\ Var(X) &= \frac{2d_2^2(d_1+d_2-2)}{d_1(d_2-2)^2(d_2-4)}, \text{ for } d_2 > 4. \end{aligned}$$

Some properties of the F-distribution are

1. If $X_1 \sim \chi^2_{d_1}$, and $X_2 \sim \chi^2_{d_2}$, with X_1 and X_2 independent, then

$$\frac{X_1/d_1}{X_2/d_2} \sim F(d_1, d_2).$$

2. If $X \sim F(d_1, d_2)$, let $Y = \lim_{d_2 \to \infty} d_1 X$ then $Y \sim \chi^2_{d_1}$.
3. If $X \sim F(d_1, d_2)$ then $\frac{1}{X} \sim F(d_2, d_1)$.

A related distribution is the non-central F-distribution, for which

$$\begin{aligned} f_X(x) &= \frac{1}{B\left(\frac{d_1}{2}, \frac{d_2}{2}\right)} e^{-\lambda/2} d_1^{d_1/2} d_2^{d_2/2} x^{\frac{1}{2}(d_1-2)} (d_1 x + d_2)^{-\frac{1}{2}(d_1+d_2)} \\ &\quad \cdot F_{11}\left[\frac{d_1+d_2}{2}, \frac{d_1}{2}, \frac{d_1 \lambda x}{2(d_1 x + d_2)}\right], \end{aligned}$$

for $x > 0$, $\lambda > 0$, where F_{1q} is the generalized hypergeometric function (see Zwillinger and Kokoska (2000) for details). If $X_1 \sim \chi^2_{d_1}(\lambda)$, and $X_2 \sim \chi^2_{d_2}$, with X_1 and X_2 independent, then the random variable

$$\frac{X_1/d_1}{X_2/d_2},$$

is a non-central F-distributed random variable with λ as the non-centrality parameter.

B.10.3 Multivariate distributions

Multinomial

The multinomial distribution is a generalization of the binomial to the multivariate case. Suppose that there are k independent trials and that each trial results in one of n possible distinct outcomes. For $i = 1, 2, \ldots, n$, let p_i be the probability that outcome i occurs on any given trial, with $\sum_{i=1}^n p_i = 1$. The multinomial random variable is the n-dimensional vector of random variables, $\mathbf{X} = (X_1, X_2, \ldots, X_n)'$, where X_i is the number of times outcome i occurs. Its probability distribution is:

$$P(X_1 = x_1, X_2 = x_2, \ldots, X_n = x_n) = \begin{cases} k! \prod_{i=1}^n \frac{p_i^{x_i}}{x_i!}, \text{ when } \sum_{i=1}^n x_i = k, \\ 0, \text{ otherwise.} \end{cases} \tag{B.41}$$

The elements of the mean and covariance matrix of $\mathbf{X}$ are given by

$$\begin{aligned} E(X_i) &= k p_i, \\ Var(X_i) &= k p_i (1 - p_i), \; Cov(X_i X_j) = -k p_i p_j, \text{ for } i \neq j. \end{aligned}$$

If $n = 2$ and $p_1 = p$, then the multinomial corresponds to the binomial random variable with parameters k and p.

Multivariate normal

The n-dimensional vector of random variables, $\mathbf{X} = (X_1, X_2, \ldots, X_n)'$, has a multivariate normal distribution, and denoted by $\mathbf{X} \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, if its probability density function is

$$f_{\mathbf{X}}(\mathbf{x}) = \frac{1}{(2\pi)^{\frac{n}{2}} |\boldsymbol{\Sigma}|^{\frac{1}{2}}} e^{-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})' \boldsymbol{\Sigma}^{-1} (\mathbf{x}-\boldsymbol{\mu})}, \tag{B.42}$$

where $\boldsymbol{\mu}$ is an n-dimensional vector, $\boldsymbol{\Sigma}$ is an $n \times n$, symmetric, positive definite matrix, and $|\boldsymbol{\Sigma}|$ is the determinant of $\boldsymbol{\Sigma}$. $\boldsymbol{\Sigma}$ is the covariance matrix of $\mathbf{x}$. The multivariate normal distribution satisfies the following properties:

1. If $\mathbf{X} \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, then its individual marginal distributions are univariate normals.
2. Suppose $\mathbf{X} = \left(\mathbf{X}_1', \mathbf{X}_2'\right)'$, with $\mathbf{X} \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$. If $\mathbf{X}_1$ and $\mathbf{X}_2$ are uncorrelated, that is, $Cov(\mathbf{X}_1, \mathbf{X}_2) = \mathbf{0}$, then $\mathbf{X}_1$ and $\mathbf{X}_2$ are independent.
3. Suppose $\mathbf{X} = \left(\mathbf{X}_1', \mathbf{X}_2'\right)'$, with $\mathbf{X} \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, and $\mathbf{X}_1$ and $\mathbf{X}_2$ being two vectors of dimension n_1 and n_2, respectively. Partition $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$ accordingly, as follows:

$$\boldsymbol{\mu} = \begin{pmatrix} \boldsymbol{\mu}_1 \\ \boldsymbol{\mu}_2 \end{pmatrix}, \boldsymbol{\Sigma} = \begin{pmatrix} \boldsymbol{\Sigma}_{11} & \boldsymbol{\Sigma}_{12} \\ \boldsymbol{\Sigma}_{22} & \boldsymbol{\Sigma}_{22} \end{pmatrix},$$

   where $\boldsymbol{\Sigma}_{11}$ is $n_1 \times n_1$, $\boldsymbol{\Sigma}_{12}$ is $n_1 \times n_2$, $\boldsymbol{\Sigma}_{21}$ is $n_2 \times n_1$, and $\boldsymbol{\Sigma}_{22}$ is $n_2 \times n_2$. The conditional distribution of $\mathbf{X}_1$ given $\mathbf{X}_2$ has normal distribution, namely $(\mathbf{X}_1 | \mathbf{X}_2 = \mathbf{x}_2) \sim N\left(\boldsymbol{\mu}_c, \boldsymbol{\Sigma}_c\right)$ with expected value

$$\boldsymbol{\mu}_c = \boldsymbol{\mu}_1 + \boldsymbol{\Sigma}_{12} \boldsymbol{\Sigma}_{22}^{-1} \left(\mathbf{x}_2 - \boldsymbol{\mu}_2\right), \tag{B.43}$$

   and covariance matrix

$$\boldsymbol{\Sigma}_c = \boldsymbol{\Sigma}_{11} - \boldsymbol{\Sigma}_{12} \boldsymbol{\Sigma}_{22}^{-1} \boldsymbol{\Sigma}_{21}. \tag{B.44}$$

   For further details see, for example, Bierens (2005).
4. If $\mathbf{X} \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, then $\boldsymbol{\Sigma}^{-1/2} (\mathbf{X} - \boldsymbol{\mu}) \sim N(\mathbf{0}, \mathbf{I}_n)$.
5. Let $\mathbf{X}$ be an n-dimensional random vector with $\mathbf{X} \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, where $\boldsymbol{\Sigma}$ nonsingular. Then $(\mathbf{X} - \boldsymbol{\mu})' \boldsymbol{\Sigma}^{-1} (\mathbf{X} - \boldsymbol{\mu}) \sim \chi_n^2$.
6. Let $\mathbf{X}$ be an n-dimensional random vector with $\mathbf{X} \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, where $\boldsymbol{\Sigma}$ nonsingular. Then $\mathbf{X}' \boldsymbol{\Sigma}^{-1} \mathbf{X} \sim \chi_n^2 \left(\boldsymbol{\mu}' \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu} / 2\right)$.

Multivariate Student's t

The n-dimensional vector of random variables, $\mathbf{X}$, has a multivariate t-distribution with parameters ν, $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$, and written as $\mathbf{X} \sim t_\nu(\boldsymbol{\mu}, \boldsymbol{\Sigma}, n)$, if its probability density function is

$$f_{\mathbf{X}}(\mathbf{x}) = \frac{\Gamma\left(\frac{n+\nu}{2}\right)}{\Gamma\left(\frac{\nu}{2}\right)(\nu\pi)^{n/2}} |\mathbf{S}|^{-1/2} \left[1 + \frac{1}{\nu}(\mathbf{x}-\boldsymbol{\mu})' \mathbf{S}^{-1}(\mathbf{x}-\boldsymbol{\mu})\right]^{-(n+\nu)/2}, \tag{B.45}$$

where ν is the degrees of freedom, $\boldsymbol{\mu} = \left(\mu_1, \mu_2, \ldots, \mu_n\right)'$ is an n-dimensional vector with real entries, and $\mathbf{S}$ is the scale matrix which is $n \times n$ symmetric and positive definite. $\boldsymbol{\mu}$ is defined if $\nu > 1$. The covariance of $\mathbf{x}$ is given by $\boldsymbol{\Sigma} = \left(\frac{\nu}{\nu-2}\right)\mathbf{S}$, if $\nu > 2$. The Student's t is said to be central if $\boldsymbol{\mu} = \mathbf{0}$, otherwise it is said to be non-central. Note that when $n = 1$, $t_\nu(0, 1, 1) \equiv t_\nu$.

B.11 Cochran's theorem and related results

In this section we provide some results on the distribution of quadratic forms in normal variables, and refer to Cochran (1934), Styan (1970), Tian and Styan (2006) for further details. We first state some propositions giving conditions for a quadratic form to be distributed as a chi-square, we then provide the well-known Cochran's theorem and some extensions. In this section, when stating $\mathbf{X} \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, the matrix $\boldsymbol{\Sigma}$ is allowed to be singular.

Proposition 51 *Let $\mathbf{X}$ be an n-dimensional random vector with $\mathbf{X} \sim N(\mathbf{0}, \boldsymbol{\Sigma})$, and let $q = \mathbf{X}'\mathbf{A}\mathbf{X}$, where $\mathbf{A}$ is a symmetric $n \times n$ matrix. Then $q \sim \chi^2_k$ if and only if $\mathbf{A}\boldsymbol{\Sigma}$ has k eigenvalues equal to 1, the rest being zero.*

Proposition 52 *Let $\mathbf{X}$ be an n-dimensional random vector with $\mathbf{X} \sim N(\mathbf{0}, \boldsymbol{\Sigma})$, and let $q = \mathbf{X}'\mathbf{A}\mathbf{X}$, where $\mathbf{A}$ is a symmetric $n \times n$ matrix. Then $q \sim \chi^2_k$ if and only if*

(i) $\boldsymbol{\Sigma}\mathbf{A}\boldsymbol{\Sigma}\mathbf{A}\boldsymbol{\Sigma} = \boldsymbol{\Sigma}\mathbf{A}\boldsymbol{\Sigma}$.

(ii) $rank(\boldsymbol{\Sigma}\mathbf{A}\boldsymbol{\Sigma}) = Tr(\mathbf{A}\boldsymbol{\Sigma}) = k$.

From the above propositions it follows that if $\mathbf{X} \sim N(\mathbf{0}, \boldsymbol{\Sigma})$ and $\mathbf{A}\boldsymbol{\Sigma}$ is an idempotent matrix then $\mathbf{X}'\mathbf{A}\mathbf{X} \sim \chi^2_k$, with $k = Tr(\mathbf{A}\boldsymbol{\Sigma})$.

Proposition 53 *Let $\mathbf{X}$ be an n-dimensional random vector with $\mathbf{X} \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, and let $q = \mathbf{X}'\mathbf{A}\mathbf{X}$, where $\mathbf{A}$ is a symmetric $n \times n$ matrix. Then $q \sim \chi^2_k(\lambda)$ if and only if*

(i) $\boldsymbol{\Sigma}\mathbf{A}\boldsymbol{\Sigma}\mathbf{A}\boldsymbol{\Sigma} = \boldsymbol{\Sigma}\mathbf{A}\boldsymbol{\Sigma}$.

(ii) $rank(\boldsymbol{\Sigma}\mathbf{A}\boldsymbol{\Sigma}) = Tr(\mathbf{A}\boldsymbol{\Sigma}) = k$.

(iii) $\boldsymbol{\mu}'(\mathbf{A}\boldsymbol{\Sigma})^2 = \boldsymbol{\mu}'(\mathbf{A}\boldsymbol{\Sigma})$

(iv) $\lambda = \boldsymbol{\mu}'\mathbf{A}\boldsymbol{\Sigma}\mathbf{A}\boldsymbol{\mu} = \boldsymbol{\mu}'\mathbf{A}\boldsymbol{\mu}$.

Proposition 54 *Let $\mathbf{X}$ be an n-dimensional random vector with $\mathbf{X} \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, and let $q_1 = \mathbf{X}'\mathbf{A}\mathbf{X}$, $q_2 = \mathbf{X}'\mathbf{B}\mathbf{X}$, where $\mathbf{A}, \mathbf{B}$ are two symmetric $n \times n$ matrices. Then q_1 and q_2 are independently distributed if and only if:*

(i) $\boldsymbol{\Sigma}\mathbf{A}\boldsymbol{\Sigma}\mathbf{B}\boldsymbol{\Sigma} = \mathbf{0}$.

(ii) $\boldsymbol{\Sigma}\mathbf{A}\boldsymbol{\Sigma}\mathbf{B}\boldsymbol{\mu} = \boldsymbol{\Sigma}\mathbf{B}\boldsymbol{\Sigma}\mathbf{A}\boldsymbol{\mu} = \mathbf{0}$.

(iii) $\boldsymbol{\mu}'\mathbf{A}\boldsymbol{\Sigma}\mathbf{B}\boldsymbol{\mu} = \mathbf{0}$.

From the above theorem it follows that, if $\mathbf{X} \sim N(\mathbf{0}, \mathbf{I}_n)$, and $\mathbf{M}_1$, and $\mathbf{M}_2$ are two idempotent matrices, Then $\mathbf{X}'\mathbf{M}_1\mathbf{X}$ and $\mathbf{X}'\mathbf{M}_2\mathbf{X}$ are two independent central chi-square matrices if $\mathbf{M}_1\mathbf{M}_2 = \mathbf{0}$, or equivalently if $\mathbf{M}_2\mathbf{M}_1 = \mathbf{0}$. For further details, see Styan (1970).

The following theorem is due to Cochran (1934).

Proposition 55 *Let $\mathbf{X}$ be an n-dimensional random vector with $\mathbf{X} \sim N(\mathbf{0}, \mathbf{I}_n)$, and let $q_1, q_2, \ldots, q_k$ be quadratic forms in $\mathbf{X}$ with ranks $r_1, r_2, \ldots, r_k$, respectively, and suppose that $q_1 + q_2 + \ldots + q_k = \mathbf{X}'\mathbf{X}$. Then $q_1, q_2, \ldots, q_k$ are independently distributed as $\chi^2_{r_i}$ if and only if $\sum_{i=1}^k r_i = n$.*

Cochran's theorem has been widely investigated in the literature due to its importance in the distribution theory for quadratic forms in normal random variables and in the analysis of variance. The following theorem extends Cochran's theorem to the case of multivariate normal random variables with non-diagonal covariance matrix, possibly singular.

Proposition 56 *Let $\mathbf{X}$ be an n-dimensional random vector with $\mathbf{X} \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$. Further, let $q = \mathbf{X}'\mathbf{A}\mathbf{X}$ and $q_i = \mathbf{X}'\mathbf{A}_i\mathbf{X}$ be quadratic forms such that $q = \sum_{i=1}^k q_i$, $r = rank(\boldsymbol{\Sigma}\mathbf{A}\boldsymbol{\Sigma})$ and $r_i = rank(\boldsymbol{\Sigma}\mathbf{A}_i\boldsymbol{\Sigma})$, with $\mathbf{A}_1, \mathbf{A}_2 \ldots, \mathbf{A}_k$ be $n \times n$ symmetric matrices, and $\mathbf{A} = \sum_{i=1}^k \mathbf{A}_i$. Consider the following statements:*

(a) $q \sim \chi^2_r(\boldsymbol{\mu}'\mathbf{A}\boldsymbol{\mu})$.

(b) $q_i \sim \chi^2_{r_i}(\boldsymbol{\mu}'\mathbf{A}_i\boldsymbol{\mu})$.

(c) q_i and q_j are independently distributed, for $i \neq j = 1, 2, \ldots, k$.

(d) $r = \sum_{i=1}^k r_i$.

If $\boldsymbol{\Sigma}$ is nonsingular or if $\boldsymbol{\Sigma}$ is singular and $\boldsymbol{\mu} = \mathbf{0}$, or if $\boldsymbol{\Sigma}$ is singular, $\boldsymbol{\mu}$ is not necessarily $\mathbf{0}$ and $\mathbf{A}_i$ are positive semidefinite ($i = 1, 2, \ldots, k$), then: (a),(d) $\Longrightarrow$ (b),(c); (a),(b) $\Longrightarrow$ (c),(d); (a),(c) $\Longrightarrow$ (b),(d); (b),(c) $\Longrightarrow$ (a),(d).

B.12 Some useful inequalities

We now provide some useful inequalities involving expectations, and refer to Billingsley (1995) for further details.

B.12.1 Chebyshev's inequality

Proposition 57 *Let X be a random variable with mean μ and variance σ^2, then*

$$\Pr(|X - \mu| \geq \lambda\sigma) \leq \frac{1}{\lambda^2}. \tag{B.46}$$

Proof We have

$$\begin{aligned}\sigma^2 &= \int_{-\infty}^{+\infty} (x-\mu)^2 \, dF_X(x) \\ &= \int_{-\infty}^{\mu-\lambda\sigma} (x-\mu)^2 \, dF_X(x) + \int_{\mu-\lambda\sigma}^{\mu+\lambda\sigma} (x-\mu)^2 \, dF_X(x) + \int_{\mu+\lambda\sigma}^{+\infty} (x-\mu)^2 \, dF_X(x) \\ &\geq \int_{-\infty}^{-\lambda\sigma} (x-\mu)^2 \, dF_X(x-\mu) + \int_{\lambda\sigma}^{+\infty} (x-\mu)^2 \, dF_X(x-\mu).\end{aligned}$$

Noting that

$$\int_{-\infty}^{-\lambda\sigma} (x-\mu)^2 \, dF_X(x-\mu) \geq \lambda^2\sigma^2 \int_{-\infty}^{-\lambda\sigma} dF_X(x-\mu),$$

and

$$\int_{\lambda\sigma}^{+\infty} (x-\mu)^2 \, dF_X(x-\mu) \geq \lambda^2\sigma^2 \int_{\lambda\sigma}^{\infty} dF_X(x-\mu),$$

we have

$$\begin{aligned}\sigma^2 &\geq \lambda^2\sigma^2 \left(\int_{-\infty}^{-\lambda\sigma} dF_X(x-\mu) + \int_{\lambda\sigma}^{\infty} dF_X(x-\mu) \right) \\ &\geq \lambda^2\sigma^2 \Pr(|X-\mu| \geq \lambda\sigma).\end{aligned}$$

Hence

$$\Pr(|X-\mu| \geq \lambda\sigma) \leq \frac{1}{\lambda^2},$$

or

$$\Pr(|X-\mu| \geq \varepsilon) \leq \frac{\sigma^2}{\varepsilon^2},$$

if we set $\varepsilon = \lambda\sigma$.

When X has s^{th}-order moments $(s > 0)$ we have the following generalization of Chebyshev's inequality:

$$\Pr\left(|X-\mu|^s \geq \varepsilon\right) \leq \frac{E\{|X-\mu|^s\}}{\varepsilon^s}.$$

B.12.2 Cauchy–Schwarz's inequality

Proposition 58 *Let X and Y be two random variables. Then*

$$|E(XY)| \leq E(|XY|) \leq E\left(X^2\right)^{\frac{1}{2}} E\left(Y^2\right)^{\frac{1}{2}}. \tag{B.47}$$

Proof Consider linear combination of X and Y, given by $aX + bY$, where a, b are two non-zero constants. We have

$$E\left[(aX + bY)^2\right] = a^2 E\left(X^2\right) + b^2 E\left(Y^2\right) + 2abE(XY) \geq 0.$$

The above can be viewed as a quadratic form in a, and will be non-negative if

$$\Delta = \left[2bE(XY)^2\right]^2 - 4b^2 E\left(X^2\right) E\left(Y^2\right) \leq 0.$$

The equality holds only when $E\left[(aX + bY)^2\right] = 0$. Hence,

$$b^2 \left[E(XY)\right]^2 \leq b^2 E\left(X^2\right) E\left(Y^2\right).$$

Hence we have

$$|E(XY)| \leq E\left(X^2\right)^{1/2} E\left(Y^2\right)^{1/2}.$$

B.12.3 Holder's inequality

Holder's inequality is a generalization of Cauchy–Schwarz's inequality.

Proposition 59 *Let X and Y be two random variables such that $E(|X|^p) < \infty$ and $E(|Y|^q) < \infty$, where $1 < p < \infty$, and $1 < q < \infty$, with $\frac{1}{p} + \frac{1}{q} = 1$. Then*

$$E(XY) \leq E\left(|X|^p\right)^{\frac{1}{p}} E\left(|Y|^q\right)^{\frac{1}{q}}. \tag{B.48}$$

B.12.4 Jensen's inequality

Proposition 60 *Suppose that $f(X)$ is a convex twice differentiable function on an open interval and X is a random variable such that*

$$\begin{aligned} E(|X|) &< \infty, P(X \in \Omega) = 1, \\ E\left|f(X)\right| &< \infty, \end{aligned}$$

then

$$E\left[f(X)\right] \geq f\left[E(X)\right]. \tag{B.49}$$

Proof Consider the following mean value expansion of $f(X)$ around $E(X) = \mu$

$$f(X) = f(\mu) + (X - \mu) f'(X) + \frac{1}{2}\left(X - \bar{X}\right)^2 f''\left(\bar{X}\right),$$

where the random variable $\bar{X}$ lies in the range (X, μ). Then

$$E\left[f\left(X\right)\right] = f\left(\mu\right) + \frac{1}{2}E\left[\left(X - \bar{X}\right)^2 f''\left(\bar{X}\right)\right].$$

Since $f\left(X\right)$ is convex then $f''\left(\bar{X}\right) \geq 0$ for all $\bar{X}$. Hence

$$E\left[f\left(X\right)\right] \geq f\left[E\left(X\right)\right].$$

B.13 Brownian motion

We now introduce some definitions and results on Brownian motions, and refer to Billingsley (1995, 1999), and Mörters and Peres (2010) for further details.

A standard Brownian motion $b\left(.\right)$ is a continuous-time stochastic process associating each date $a \in \left[0, 1\right]$ with the scalar $b\left(a\right)$ such that:

(i) $b\left(0\right) = 0$,

(ii) For any dates $0 \leq a_1 \leq a_2 \leq \ldots \leq a_k \leq 1$ the changes $\left[b\left(a_2\right) - b\left(a_1\right)\right]$, $\left[b\left(a_3\right) - b\left(a_2\right)\right], \ldots, \left[b\left(a_k\right) - b\left(a_{k-1}\right)\right]$ are independent multivariate Gaussian with $b\left(a\right) - b\left(s\right) \sim N(0, a - s)$,

(iii) For any given realization, $b\left(a\right)$ is continuous in a with probability 1.

Other continuous time processes can be generated from the standard Brownian motion. For example, a Brownian motion with variance σ^2 can be obtained as

$$w\left(a\right) = \sigma b\left(a\right), \tag{B.50}$$

where $b\left(a\right)$ is a standard Brownian motion.

An m-dimensional standard Brownian motion $b(.)$ is a continuous-time stochastic process associating each date $a \in \left[0, 1\right]$ with the $m \times 1$ vector $b(a)$ such that:

(i) $\mathbf{b}(0) = \mathbf{0}$,

(ii) For any dates $0 \leq a_1 \leq a_2 \leq \ldots \leq a_k \leq 1$ the changes $\left[\mathbf{b}\left(a_2\right) - \mathbf{b}\left(a_1\right)\right]$, $\left[\mathbf{b}\left(a_3\right) - \mathbf{b}\left(a_2\right)\right], \ldots, \left[\mathbf{b}\left(a_k\right) - \mathbf{b}\left(a_{k-1}\right)\right]$ are independent multivariate Gaussian with $\mathbf{b}(a) - \mathbf{b}(s) \sim N(\mathbf{0}, (a - s)\,\mathbf{I}_m)$,

(iii) For any given realization, $\mathbf{b}(a)$ is continuous in a with probability 1.

The continuous time process

$$\mathbf{w}\left(a\right) = \boldsymbol{\Sigma}^{\frac{1}{2}}\mathbf{b}\left(a\right), \tag{B.51}$$

is a Brownian motion with covariance matrix $\boldsymbol{\Sigma}$.

B.13.1 Probability limits involving unit root processes

Theorem 61 *Let*

$$\mathbf{s}_{[Ta]} = \sum_{j=1}^{[Ta]} \mathbf{u}_j,$$

where $[Ta]$ *is the largest integer part of* Ta, *and* $\mathbf{u}_t = (u_{1t}, u_{2t}, \ldots, u_{mt})'$ *is an* $m \times 1$ *random vector satisfying:*

(i) $E(\mathbf{u}_t|\Omega_{t-1}) = 0$, *and* $Var(\mathbf{u}_t|\Omega_{t-1}) = \mathbf{\Sigma}$ *for all* t, *where* Ω_{t-1} *is a non-decreasing information set, and* $\mathbf{\Sigma}$ *is a positive definite symmetric matrix.*

(ii) $\sup_t E(\|\mathbf{u}_t\|^s) < \infty$, *for some* $s > 2$.
The following results hold

$$T^{-\frac{1}{2}}\mathbf{s}_{[Ta]} \Rightarrow \mathbf{w}(a), \ a \in [0,1], \tag{B.52}$$

$$T^{-\frac{3}{2}}\sum_{t=1}^{T}\mathbf{s}_t \Rightarrow \int_0^1 \mathbf{w}(a)da, \tag{B.53}$$

$$T^{-2}\sum_{t=1}^{T}\mathbf{s}_t\mathbf{s}_t' \Rightarrow \int_0^1 \mathbf{w}(a)\mathbf{w}'(a)da, \tag{B.54}$$

$$\frac{1}{T}\sum_{t=1}^{T}\mathbf{u}_t\mathbf{s}_{t-1}' \Rightarrow \int_0^1 \mathbf{w}(a)d\mathbf{w}(a), \tag{B.55}$$

where $\mathbf{w}(a)$ *is an* $m \times 1$ *vector of Brownian motion with covariance matrix* $\mathbf{\Sigma}$, *and* $\int_0^1 \mathbf{w}(a)\mathbf{w}'(a)da$ *is a stochastic matrix which is positive definite with probability 1.*

See Phillips and Durlaf (1986) for the proof of the above results.

Appendix C: Bayesian Analysis

C.1 Introduction

The statistical approach adopted in this volume is primarily classical, but a Bayesian approach is also considered in the analysis of DSGE models, forecast combination and panel data modelling. This appendix provides an overview of the Bayesian approach and formally introduces the Bayesian concepts and results used in various parts of the book. A full treatment of Bayesian analysis can be found in Geweke (2005), Greenberg (2013), Koop (2003), and Geweke, Koop, and van Dijk (2011).

C.2 Bayes theorem

Consider events $\mathcal{A}$ and $\mathcal{B}$ such that $0 < P(\mathcal{A}), P(\mathcal{B}) < 1$, and suppose that the conditional probabilities $P(\mathcal{A}|\mathcal{B})$, and $P(\mathcal{B}|\mathcal{A})$ exist. Then using standard results from calculus of probability we have

$$P(\mathcal{A} \cap \mathcal{B}) = P(\mathcal{A})\,P(\mathcal{B}|\mathcal{A}) = P(\mathcal{B} \cap \mathcal{A}) = P(\mathcal{B})\,P(\mathcal{A}|\mathcal{B}).$$

Hence

$$P(\mathcal{A}|\mathcal{B}) = \frac{P(\mathcal{A})\,P(\mathcal{B}|\mathcal{A})}{P(\mathcal{B})}. \tag{C.1}$$

Bayes theorem provides a rule for updating the probability of an event (such as $\mathcal{A}$) in the light of observing another event such as $\mathcal{B}$. The theorem is named after Reverend Thomas Bayes whose work was posthumously published in 1763 by the Royal Society as 'An Essay towards solving a Problem in the Doctrine of Chances', *Philosophical Transactions* (1683–1775).

C.2.1 Prior and posterior distributions

Suppose that observations $\mathbf{y} = (y_1, y_2, \ldots, y_T)'$ are random draws from the joint probability distribution, $f(\mathbf{y}|\boldsymbol{\theta})$, where $\boldsymbol{\theta}$ is a p-dimensional vector of unknown parameters. The objective is to learn about $\boldsymbol{\theta}$ having observed the data, $\mathbf{y}$. Within a Bayesian context, the investigator begins with a *subjective prior* about $\boldsymbol{\theta}$ which is characterized by the probability distribution, $\pi(\boldsymbol{\theta})$, known as the prior distribution of $\boldsymbol{\theta}$. The next step is to derive the posterior distribution of $\boldsymbol{\theta}$, having

observed the data, $\mathbf{y}$. This is achieved by using the Bayes rule, (C.1), setting $\mathcal{A} \equiv \boldsymbol{\theta}$ and $\mathcal{B} \equiv \mathbf{y}$, which gives (assuming that $P(\mathbf{y}) > 0$)

$$\pi(\boldsymbol{\theta}|\mathbf{y}) = \frac{\pi(\boldsymbol{\theta})P(\mathbf{y}|\boldsymbol{\theta})}{P(\mathbf{y})} = \frac{\pi(\boldsymbol{\theta})f(\mathbf{y}|\boldsymbol{\theta})}{\int \pi(\boldsymbol{\theta})f(\mathbf{y}|\boldsymbol{\theta})d\boldsymbol{\theta}}, \tag{C.2}$$

where the integral in the denominator is taken over the range of variation of $\boldsymbol{\theta}$. $\pi(\boldsymbol{\theta}|\mathbf{y})$ is known as the posterior distribution of $\boldsymbol{\theta}$. Note also that $f(\mathbf{y}|\boldsymbol{\theta})$ is the likelihood function and

$$\ln \pi(\boldsymbol{\theta}|\mathbf{y}) \propto \ln \pi(\boldsymbol{\theta}) + \ln f(\mathbf{y}|\boldsymbol{\theta}).$$

C.3 Bayesian inference

The focus of the Bayesian inference is on the posterior distribution, $\pi(\boldsymbol{\theta}|\mathbf{y})$. A point estimate of $\boldsymbol{\theta}$, say $\hat{\boldsymbol{\theta}}_T$, is obtained based on a loss function such as

$$L\left(\boldsymbol{\theta}, \hat{\boldsymbol{\theta}}_T\right) = c\left\|\boldsymbol{\theta} - \hat{\boldsymbol{\theta}}_T\right\|^2, \quad c > 0,$$

$\hat{\boldsymbol{\theta}}_T$ is then derived by minimizing the risk function, $E_{\boldsymbol{\theta}}[L(\boldsymbol{\theta}, \hat{\boldsymbol{\theta}}_T)]$ where the expectation, $E_{\boldsymbol{\theta}}(\cdot)$, is taken with respect to the posterior distribution, $\pi(\boldsymbol{\theta}|\mathbf{y})$. Under a quadratic loss function the Bayes estimator of $\boldsymbol{\theta}$ is given by the mean of the posterior distribution, namely

$$\hat{\boldsymbol{\theta}}_T = \int \boldsymbol{\theta}\, \pi(\boldsymbol{\theta}|\mathbf{y})\, d\boldsymbol{\theta}.$$

Other Bayes estimates, such as mode or the median of the posterior distribution, can also be motivated using other loss functions.

When the focus of the analysis is on one of the elements of $\boldsymbol{\theta}$, say θ_1, the marginal posterior distribution is considered. For θ_1 the marginal posterior distribution is obtained by integrating out all the other elements of $\boldsymbol{\theta}$ from $\pi(\boldsymbol{\theta}|\mathbf{y})$, namely

$$\pi(\theta_1|\mathbf{y}) = \int \pi(\boldsymbol{\theta}|\mathbf{y})\, d\theta_2 d\theta_3 \ldots d\theta_p.$$

The mean and variance of θ_1 are then computed as

$$E(\theta_1|\mathbf{y}) = \int \theta_1 \pi(\boldsymbol{\theta}|\mathbf{y})\, d\theta_1 d\theta_2 d\theta_3 \ldots d\theta_p$$

$$Var(\theta_1|\mathbf{y}) = \int \theta_1^2 \pi(\boldsymbol{\theta}|\mathbf{y})\, d\theta_1 d\theta_2 d\theta_3 \ldots d\theta_p - \left[E(\theta_1|\mathbf{y})\right]^2.$$

The precision of θ_1, defined as the inverse of its variance, is given by

$$h\left(\theta_1 \left| \mathbf{y}\right.\right) = \frac{1}{Var\left(\theta_1 | \mathbf{y}\right)}.$$

Computations of $E\left(\theta_1 | \mathbf{y}\right)$ and $h\left(\theta_1 | \mathbf{y}\right)$ are often quite complicated and time consuming, particularly when p is relatively large. But, thanks to recent advances in computing technology, such computations are carried out reasonably fast using Markov Chain Monte Carlo (MCMC) simulation techniques such as Metropolis–Hastings and Gibbs algorithms. An overview of alternative Monte Carlo techniques used in the literature is provided by Chib (2011). A more accessible textbook account is given in Greenberg (2013, Ch. 7).

C.3.1 Identification

In the case where $\boldsymbol{\theta}$ is identified, namely $f(\mathbf{y}; \boldsymbol{\theta}_1) = f(\mathbf{y}; \boldsymbol{\theta}_2)$ if and only if $\boldsymbol{\theta}_1 = \boldsymbol{\theta}_2$, the posterior distribution gets dominated by likelihood function, and the precision of the individual elements of $\boldsymbol{\theta}$ rises with T. This follows since $\ln \pi(\boldsymbol{\theta})$ is fixed as $T \to \infty$, but $\ln f(\mathbf{y} | \boldsymbol{\theta})$ rises with T when $\boldsymbol{\theta}$ is identified. Bayesian inference in the case of non-identified or weakly identified parameters is discussed in Koop, Pesaran, and Smith (2013), where it is shown that if θ_1 is non-identified then its precision does not rise with T, and $\lim_{T\to\infty} T^{-1} h\left(\theta_1 | \mathbf{y}\right) = 0$.

C.3.2 Choice of the priors

The choice of the priors can be quite important if T is not sufficiently large or if one of the parameters is non-identified or weakly identified. As a result there is a large literature on how best to specify priors such that they are least 'informative'. In cases where the parameters are defined on finite intervals or regions it is reasonable to use uniform priors as characterizations of non-informativeness. But it may not generally be possible to specify non-informative priors. For example, if one of the parameters of interest ranges from $-\infty$ to $+\infty$, specification of a uniform prior probability distribution will not be proper in the sense that its integral over $-\infty$ to $+\infty$ will be unbounded and cannot be normalized to unity. Such priors that do not integrate to unity are referred to as 'improper'. In cases where informative priors are used, it is important that the prior distributions cover the true values of the unknown parameters.

Typical choices of proper priors are conjugate priors and hierarchical (or multilevel) priors. Conjugate priors are specified so that to ensure that the posterior and the prior distributions have the same form. For example, in the case of the independent Bernoulli processes

$$f(y_i | \theta) = \theta^{y_i}(1-\theta)^{1-y_i}, \text{for } i = 1, 2, \ldots N,$$

the likelihood is given by

$$f(\mathbf{y} | \theta) = \theta^{\Sigma_{i=1}^N y_i}(1-\theta)^{1-\Sigma_{i=1}^N y_i},\ 0 \leq \theta \leq 1.$$

Using the beta-distributed prior ($\Gamma(\alpha)$ denotes a gamma function defined by (B.38))

$$\pi(\theta) = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\alpha)}\theta^{\alpha-1}(1-\theta)^{\beta-1}, \text{ for } \alpha,\beta > 0,$$

yields a beta-distributed posterior with a posterior mean given by a weighted average of $\bar{y} = N^{-1}\Sigma_{i=1}^{N} y_i$ and the prior mean, $\alpha/(\alpha+\beta)$. The parameters of the prior, in this example α and β, are known as the hyper-parameters.

Hierarchical priors are intended to render the posterior distribution less sensitive to the choice of the priors and involve placing priors on priors, the so-called hyper-priors. In the above example, $\pi(\theta)$ is replaced by $\pi(\boldsymbol{\phi})\pi(\theta|\boldsymbol{\phi})$ with $\boldsymbol{\phi} = (\alpha,\beta)'$.

C.4 Posterior predictive distribution

Within the Bayesian framework, prediction of new data values (either across time or over cross-section units) is carried out using the *posterior predictive distribution*. Suppose $\mathbf{y} = (y_1, y_2, \ldots, y_T)'$ is the observed data and the aim is to predict y_{T+1} (which is not yet observed) conditional on model M defined by $f(\mathbf{y}|\boldsymbol{\theta},M)$, with the prior distribution of $\boldsymbol{\theta}$ given by $\pi(\boldsymbol{\theta})$.

$$f(y_{T+1}|\mathbf{y}\,M) = \int \pi(\boldsymbol{\theta}|\mathbf{y}\,M) f(y_{T+1}|\boldsymbol{\theta},\mathbf{y}\,M)\,d\boldsymbol{\theta},$$

where $\pi(\boldsymbol{\theta}|\mathbf{y})$ is the posterior probability distribution of $\boldsymbol{\theta}$, defined by (C.2). This result can be obtained by application of the following Bayes rule

$$f(y_{T+1},\mathbf{y},\boldsymbol{\theta}|M) = \pi(\boldsymbol{\theta}|M) f(\mathbf{y}|\boldsymbol{\theta},M) f(y_{T+1}|\mathbf{y},\boldsymbol{\theta},M).$$

Marginalizing with respect to $\boldsymbol{\theta}$

$$f(y_{T+1},\mathbf{y}|M) = \int \pi(\boldsymbol{\theta}|M) f(\mathbf{y}|\boldsymbol{\theta},M) f(y_{T+1}|\mathbf{y},\boldsymbol{\theta},M)\,d\boldsymbol{\theta},$$

and applying the Bayes rule now to $f(y_{T+1},\mathbf{y})$ conditional on model M, we have

$$\begin{aligned} f(y_{T+1}|\mathbf{y},M) &= \frac{f(y_{T+1},\mathbf{y}|M)}{f(\mathbf{y}|M)} = \frac{\int \pi(\boldsymbol{\theta}|M) f(\mathbf{y}|\boldsymbol{\theta}\,M) f(y_{T+1}|\mathbf{y},\boldsymbol{\theta}\,M)\,d\boldsymbol{\theta}}{f(\mathbf{y}|M)} \\ &= \int \frac{\pi(\boldsymbol{\theta}|M) f(\mathbf{y}|\boldsymbol{\theta}\,M)}{f(\mathbf{y}|M)} f(y_{T+1}|\mathbf{y},\boldsymbol{\theta},M)\,d\boldsymbol{\theta} \\ &= \int \pi(\boldsymbol{\theta}|\mathbf{y}\,M) f(y_{T+1}|\boldsymbol{\theta},\mathbf{y}\,M)\,d\boldsymbol{\theta}, \end{aligned}$$

as required.

The posterior predictive distribution can also be extended to allow for multiple models. See Section 17.9.

C.5 Bayesian model selection

Suppose it is known that observations, $\mathbf{y} = (y_1, y_2, \ldots, y_T)$, are draws either from model M_1 or model M_2, characterized by the probability distributions $f(y\,|\boldsymbol{\theta}_1, M_1)$ and $f(y\,|\boldsymbol{\theta}_2, M_2)$, respectively. Denote the priors for $\boldsymbol{\theta}_1$ and $\boldsymbol{\theta}_2$, conditional on models M_1 and M_2, by $\pi(\boldsymbol{\theta}_1\,|M_1)$ and $\pi(\boldsymbol{\theta}_2\,|M_2)$, respectively. Also denote the priors on models M_1 and M_2 by $\pi(M_1)$ and $\pi(M_2)$. Then applying the Bayes rule to the joint distribution of $\mathbf{y}$ and M_1 we have

$$P\left(M_1\,|\mathbf{y}\right) = \frac{\pi(M_1)P(\mathbf{y}\,|M_1)}{P(\mathbf{y})},$$

which is the posterior of model M_1. But

$$P(\mathbf{y}\,|M_1) = \int f(\mathbf{y}, \boldsymbol{\theta}_1\,|M_1)d\boldsymbol{\theta} = \int \pi(\boldsymbol{\theta}_1\,|M_1)f(y\,|\boldsymbol{\theta}_1, M_1)d\boldsymbol{\theta}_1.$$

Note that $\pi(\boldsymbol{\theta}_1\,|M_1)f(y\,|\boldsymbol{\theta}_1, M_1) = \pi\left(\boldsymbol{\theta}_1\,|\mathbf{y}, M_1\right)$ is the posterior probability distribution of $\boldsymbol{\theta}_1$ under M_1. $P(\mathbf{y}\,|M_1)$ is also known as the 'marginal likelihood' of model M_1, and can be viewed as the expected value of the likelihood with respect to the prior distribution. It can also be viewed as an 'average likelihood' where the averaging is carried out with respect to the priors. Similarly

$$P(\mathbf{y}\,|M_2) = \int \pi(\boldsymbol{\theta}_2\,|M_2)f(y\,|\boldsymbol{\theta}_2, M_2)d\boldsymbol{\theta}_2 = \int \pi\left(\boldsymbol{\theta}_2\,|\mathbf{y}, M_2\right)d\boldsymbol{\theta}_2.$$

Also, since M_1 and M_2 are assumed to be mutually exclusive and exhaustive, in the sense that one or the other model holds, we have

$$P(\mathbf{y}) = \pi(M_1)\int \pi\left(\boldsymbol{\theta}_1\,|\mathbf{y}, M_1\right)d\boldsymbol{\theta}_1 + \pi(M_2)\int \pi\left(\boldsymbol{\theta}_2\,|\mathbf{y}, M_2\right)d\boldsymbol{\theta}_2.$$

Finally, the posterior ratio of model M_1 to model M_2 (also known as the 'posterior odds' ratio) is given by

$$\frac{P\left(M_1\,|\mathbf{y}\right)}{P\left(M_2\,|\mathbf{y}\right)} = \frac{\pi(M_1)}{\pi(M_2)} \times \frac{\int \pi(\boldsymbol{\theta}_1\,|M_1)f(y\,|\boldsymbol{\theta}_1, M_1)d\boldsymbol{\theta}_1}{\int \pi(\boldsymbol{\theta}_2\,|M_2)f(y\,|\boldsymbol{\theta}_2, M_2)d\boldsymbol{\theta}_2}.$$

In words, the posterior odds ratio of model M_1 to model M_2 is equal to the prior odds ratio multiplied by the ratio of the marginal likelihoods, also known as the 'Bayes factor'

$$\text{posterior odds} = \text{prior odds} \times \text{Bayes factor}.$$

It is important to note that the Bayes factor is only well defined when the priors $\pi(\boldsymbol{\theta}_1|M_1)$ and $\pi(\boldsymbol{\theta}_2|M_2)$ are proper.

For large values of the sample size T, the logarithm of the posterior odds will be dominated by the Bayes factor. Under standard regularity conditions, and assuming that $\boldsymbol{\theta}_i$ is identified under M_i, we obtain the familiar Schwarz model selection criterion

$$\ln P\left(M_1\left|\mathbf{y}\right.\right) - \ln P\left(M_2|\mathbf{y}\right) = \ln f(y|\hat{\boldsymbol{\theta}}_{1,ML}, M_1) - \ln f(y|\hat{\boldsymbol{\theta}}_{2,ML}, M_2) - \frac{1}{2}(p_1 - p_2)\ln T + O(1),$$

where $\hat{\boldsymbol{\theta}}_{i,ML}$ is the ML estimator of $\boldsymbol{\theta}_i$ under M_i, and $p_i = \dim(\boldsymbol{\theta}_i)$.

C.6 Bayesian analysis of the classical normal linear regression model

Consider the classical regression model

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{u},$$

where $\mathbf{y}$ is a vector of $T \times 1$ observations on the dependent variable, $\mathbf{X}$ is the $T \times k$ matrix of observations on the k regressors, $\boldsymbol{\beta}$ is the $k \times 1$ vector of unknown regression coefficients, and $\mathbf{u}$ is the $T \times 1$ vector of disturbances assumed to be distributed as $N(\mathbf{0},\sigma^2\mathbf{I}_T)$, where σ^2 is the unknown error variance.[1] Denote the prior probability distribution for this problem by $\pi(\boldsymbol{\beta},\sigma^2)$ and note that

$$\pi(\boldsymbol{\beta},\sigma^2) = \pi\left(\sigma^2\right)\pi\left(\boldsymbol{\beta}|\sigma^2\right).$$

The posterior distribution of $\boldsymbol{\theta} = (\boldsymbol{\beta}', \sigma^2)'$ is given by

$$\pi\left(\boldsymbol{\theta}|\mathbf{y},\mathbf{X}\right) = \pi\left(\sigma^2\right)\pi\left(\boldsymbol{\beta}|\sigma^2\right)P\left(\mathbf{y}|\mathbf{X},\boldsymbol{\theta}\right),$$

where $P\left(\mathbf{y}\,|\mathbf{X},\boldsymbol{\theta}\right)$ is the likelihood function given by (2.9), namely

$$P\left(\mathbf{y}|\mathbf{X},\boldsymbol{\theta}\right) = \left(2\pi\sigma^2\right)^{-T/2}\exp\left[-\frac{1}{2\sigma^2}\left(\mathbf{y}-\mathbf{X}\boldsymbol{\beta}\right)'\left(\mathbf{y}-\mathbf{X}\boldsymbol{\beta}\right)\right].$$

The conjugate priors for the regression model are the inverse gamma distribution for σ^2, and the normal distribution for $\boldsymbol{\beta}|\sigma^2$. More specifically,

$$\pi\left(\sigma^2\right) = \left(\frac{1}{\sigma^2}\right)^{(\underline{a}/2)+1}\exp\left(\frac{-\underline{d}}{2\sigma^2}\right),$$

[1] For further details of the regression model and the underlying assumptions see Section 2.2.

and

$$\pi\left(\boldsymbol{\beta}\left|\sigma^2\right.\right)=\left(2\pi\sigma^2\right)^{-k/2}\left|\underline{\mathbf{H}}\right|^{1/2}\exp\left[-\frac{1}{2\sigma^2}\left(\boldsymbol{\beta}-\underline{\mathbf{b}}\right)'\underline{\mathbf{H}}\left(\boldsymbol{\beta}-\underline{\mathbf{b}}\right)\right],$$

where $\underline{a}$ and $\underline{d}$ are the prior hyperparameter of the inverse-gamma distribution, and $\underline{\mathbf{b}}$ and $\sigma^{-2}\underline{\mathbf{H}}$ are the prior mean and precision of $\boldsymbol{\beta}|\sigma^2$. Recall that $\sigma^2\underline{\mathbf{H}}^{-1}$ is the prior variance of $\boldsymbol{\beta}|\sigma^2$. Combining the above results we have

$$\begin{aligned}\pi\left(\boldsymbol{\theta}\left|\mathbf{y},\mathbf{X}\right.\right)=&\left(2\pi\sigma^2\right)^{-(T+k)/2}\left|\underline{\mathbf{H}}\right|^{1/2}\left(\frac{1}{\sigma^2}\right)^{\underline{a}/2+1}\exp\left(\frac{-\underline{d}}{2\sigma^2}\right)\\&\times\exp\left[-\frac{1}{2\sigma^2}\left(\boldsymbol{\beta}-\underline{\mathbf{b}}\right)'\underline{\mathbf{H}}\left(\boldsymbol{\beta}-\underline{\mathbf{b}}\right)-\frac{1}{2\sigma^2}\left(\mathbf{y}-\mathbf{X}\boldsymbol{\beta}\right)'\left(\mathbf{y}-\mathbf{X}\boldsymbol{\beta}\right)\right].\end{aligned}\tag{C.3}$$

Also $\mathbf{y}-\mathbf{X}\boldsymbol{\beta}=\mathbf{y}-\mathbf{X}\hat{\boldsymbol{\beta}}-\mathbf{X}(\boldsymbol{\beta}-\hat{\boldsymbol{\beta}})=\hat{\mathbf{u}}-\mathbf{X}(\boldsymbol{\beta}-\hat{\boldsymbol{\beta}})$, and using (2.11) we have

$$\left(\mathbf{y}-\mathbf{X}\boldsymbol{\beta}\right)'\left(\mathbf{y}-\mathbf{X}\boldsymbol{\beta}\right)=\hat{\mathbf{u}}'\hat{\mathbf{u}}+(\beta-\hat{\beta})'\left(\mathbf{X}'\mathbf{X}\right)(\boldsymbol{\beta}-\hat{\boldsymbol{\beta}}).$$

Hence

$$\begin{aligned}&\left(\boldsymbol{\beta}-\underline{\mathbf{b}}\right)'\underline{\mathbf{H}}\left(\boldsymbol{\beta}-\underline{\mathbf{b}}\right)+\left(\mathbf{y}-\mathbf{X}\boldsymbol{\beta}\right)'\left(\mathbf{y}-\mathbf{X}\boldsymbol{\beta}\right)\\&=\hat{\mathbf{u}}'\hat{\mathbf{u}}+\left(\boldsymbol{\beta}-\hat{\boldsymbol{\beta}}\right)'\left(\mathbf{X}'\mathbf{X}\right)\left(\boldsymbol{\beta}-\hat{\boldsymbol{\beta}}\right)+\left(\boldsymbol{\beta}-\underline{\mathbf{b}}\right)'\underline{\mathbf{H}}\left(\boldsymbol{\beta}-\underline{\mathbf{b}}\right).\end{aligned}$$

The term $\hat{\mathbf{u}}'\hat{\mathbf{u}}$ does not depend on $\boldsymbol{\theta}$ and can be ignored. Further

$$\begin{aligned}&\left(\boldsymbol{\beta}-\hat{\boldsymbol{\beta}}\right)'\left(\mathbf{X}'\mathbf{X}\right)\left(\boldsymbol{\beta}-\hat{\boldsymbol{\beta}}\right)+\left(\boldsymbol{\beta}-\underline{\mathbf{b}}\right)'\underline{\mathbf{H}}\left(\boldsymbol{\beta}-\underline{\mathbf{b}}\right)\\&=\hat{\boldsymbol{\beta}}'\left(\mathbf{X}'\mathbf{X}\right)\hat{\boldsymbol{\beta}}+(\boldsymbol{\beta}-\bar{\boldsymbol{\beta}})'\bar{\mathbf{H}}(\boldsymbol{\beta}-\bar{\boldsymbol{\beta}}),\end{aligned}\tag{C.4}$$

where

$$\bar{\boldsymbol{\beta}}=\left(\mathbf{X}'\mathbf{X}+\underline{\mathbf{H}}\right)^{-1}\left[\left(\mathbf{X}'\mathbf{X}\right)\hat{\boldsymbol{\beta}}+\underline{\mathbf{H}}\underline{\mathbf{b}}\right],\tag{C.5}$$

and

$$\bar{\mathbf{H}}=\mathbf{X}'\mathbf{X}+\underline{\mathbf{H}}.\tag{C.6}$$

In the case where σ^2 is known or when the analysis is done conditional on σ^2 (the case of conditional conjugate priors), it readily follows that distribution of $\boldsymbol{\beta}|\sigma^2$ is $N(\bar{\boldsymbol{\beta}},\sigma^2\bar{\mathbf{H}}^{-1})$, where $\bar{\boldsymbol{\beta}}$ is the posterior mean and $\bar{\mathbf{H}}$ is the posterior precision of $\boldsymbol{\beta}$. It is easily seen that $\bar{\boldsymbol{\beta}}$ is a matrix weighted average of the *OLS* estimator, $\hat{\boldsymbol{\beta}}$, and the prior mean, $\underline{\mathbf{b}}$. The weights

$$\mathbf{W}_{OLS}=\left(\mathbf{X}'\mathbf{X}+\underline{\mathbf{H}}\right)^{-1}\left(\mathbf{X}'\mathbf{X}\right),\text{ and }\mathbf{W}_{\mathrm{Prior}}=\left(\mathbf{X}'\mathbf{X}+\underline{\mathbf{H}}\right)^{-1}\underline{\mathbf{H}},$$

add up to $\mathbf{I}_k$. In the case where the regression coefficients are identified and $T^{-1}\mathbf{X}'\mathbf{X} \to_p \mathbf{\Sigma}_{xx} > 0$, then

$$\mathbf{W}_{Prior} = \left(T^{-1}\mathbf{X}'\mathbf{X} + T^{-1}\underline{\mathbf{H}}\right)^{-1} T^{-1}\underline{\mathbf{H}} \to_p \mathbf{0},$$

and

$$\mathbf{W}_{OLS} = \left(T^{-1}\mathbf{X}'\mathbf{X} + T^{-1}\underline{\mathbf{H}}\right)^{-1} \left(T^{-1}\mathbf{X}'\mathbf{X}\right) \to_p \mathbf{I}_k,$$

and $\bar{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}} \to_p \mathbf{0}$.

When the Bayesian analysis is carried out jointly in $\boldsymbol{\beta}$ and σ^2, we first need to use (C.3) to integrate out $\boldsymbol{\beta}$ to obtain the posterior distribution of σ^2. This yields the following inverse-gamma posterior for σ^2

$$\pi\left(\sigma^2 \left| \mathbf{y}, \mathbf{X}\right.\right) = \left(\frac{1}{\sigma^2}\right)^{\bar{a}/2+1} \exp\left(\frac{-\bar{d}}{2\sigma^2}\right),$$

where

$$\bar{a} = T + \underline{a}, \text{ and } \bar{d} = \underline{d} + \underline{\mathbf{b}}'\underline{\mathbf{H}}\underline{\mathbf{b}} + \left(\mathbf{y}'\mathbf{y} - \bar{\boldsymbol{\beta}}'\bar{\mathbf{H}}\bar{\boldsymbol{\beta}}\right).$$

It is then easily seen that posterior distribution of $\boldsymbol{\beta}$ is a multivariate t with degrees of freedom $\bar{a}$, mean $\bar{\boldsymbol{\beta}}$, and the scale matrix $\left(\bar{d}/\bar{a}\right)\bar{\mathbf{H}}^{-1}$. Using (B.45), more specifically we have

$$\pi\left(\boldsymbol{\beta} \left| \mathbf{y}, \mathbf{X}\right.\right) = \frac{\Gamma\left(\frac{k+\bar{a}}{2}\right)\left(\bar{d}/\bar{a}\right)^{-k/2}}{\Gamma\left(\frac{\bar{a}}{2}\right)(\bar{a}\pi)^{k/2}} \left|\bar{\mathbf{H}}\right|^{1/2} \left[1 + \frac{1}{\bar{d}}\left(\boldsymbol{\beta} - \bar{\boldsymbol{\beta}}\right)'\bar{\mathbf{H}}\left(\boldsymbol{\beta} - \bar{\boldsymbol{\beta}}\right)\right]^{-(k+\bar{a})/2},$$

or after some simplification

$$\pi\left(\boldsymbol{\beta} \left| \mathbf{y}, \mathbf{X}\right.\right) = \frac{\Gamma\left(\frac{k+\bar{a}}{2}\right)\left(\bar{d}\right)^{\bar{a}/2}}{\pi^{k/2}\Gamma\left(\frac{\bar{a}}{2}\right)} \left|\bar{\mathbf{H}}\right|^{1/2} \left[\bar{d} + \left(\boldsymbol{\beta} - \bar{\boldsymbol{\beta}}\right)'\bar{\mathbf{H}}\left(\boldsymbol{\beta} - \bar{\boldsymbol{\beta}}\right)\right]^{-(k+\bar{a})/2}. \tag{C.7}$$

The posterior precision of $\boldsymbol{\beta}$ is given by $\left(\frac{\bar{a}-2}{\bar{d}}\right)\bar{\mathbf{H}}$.

For further details see Zellner (1971, Ch. 3) and Greenberg (2013, Ch. 4).

C.7 Bayesian shrinkage (ridge) estimator

The Bayesian shrinkage estimator is obtained from the posterior mean of $\boldsymbol{\beta}$ given by (C.5) by setting its prior mean, $\underline{\mathbf{b}}$, to zero, and its prior precision, $\sigma^{-2}\underline{\mathbf{H}}$, to $\theta\mathbf{I}_k$, where $\theta > 0$ is known as the shrinkage parameter. Using (C.5) we have

$$\begin{aligned}\hat{\boldsymbol{\beta}}_{Shrinkage} &= \left(\mathbf{X}'\mathbf{X} + \sigma^2\theta\mathbf{I}_k\right)^{-1}\left(\mathbf{X}'\mathbf{X}\right)\hat{\boldsymbol{\beta}} \\ &= \left(\mathbf{X}'\mathbf{X} + \sigma^2\theta\mathbf{I}_k\right)^{-1}\mathbf{X}'\mathbf{y}.\end{aligned} \tag{C.8}$$

A similar estimator (known as the ridged estimator) can also be derived using penalized regression with an L_2 penalty norm. The criterion function for this penalized regression is given by

$$\mathbf{Q}(\boldsymbol{\beta},\lambda) = \left(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\right)'\left(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\right) + \lambda\left(\boldsymbol{\beta}'\boldsymbol{\beta} - K\right),$$

where $\lambda > 0$ and K is a positive constant such that $\boldsymbol{\beta}'\boldsymbol{\beta} \leq K$. The first-order condition for this optimization problem is

$$-2\mathbf{X}'\left(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\right) + 2\lambda\boldsymbol{\beta} = \mathbf{0},$$

which yields

$$\hat{\boldsymbol{\beta}}_{Ridge} = \left(\mathbf{X}'\mathbf{X} + \lambda\mathbf{I}_k\right)^{-1}\mathbf{X}'\mathbf{y}. \tag{C.9}$$

It is clear that the shrinkage and the ridge estimators coincide when $\lambda = \sigma^2\theta$. The main difference between the Bayesian and the penalized regression approaches lies in the way the shrinkage (or penalty) parameter is chosen. Under the Bayesian approach the choice of λ must be *a priori*, whilst under the penalized regression approach the choice is often made by cross validation. See also Section 11.9 and Hastie, Tibshirani, and Friedman (2009) and Buhlmann and van de Geer (2012).

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